Program Operation Manual
PSS®E 35.2.0

November 2020

SIEMENS

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Preface

The Siemens PTI Power System Simulator (PSS®E) is a package of programs for studies of power system transmission network and generation performance in both steady-state and dynamic conditions. PSS®E handles power flow, fault analysis (balanced and unbalanced), network equivalent construction, and dynamic simulation.

PSS®E is bundled with a number of program sections that vary from user to user. The base configuration for all installations of PSS®E is made up of the Power Flow and Libraries program sections. Other available program sections are:

- Optimal Power Flow (OPF)
- Fault Analysis (Short Circuit)
- Dynamic Simulation
- Dynamics Model Library Source
- Transmission Line Constant Calculation (TMLC) and Line Properties (LINEPROP)
- Linear System Dynamic Analysis (LSYSAN)
- WECC Data Conversion Programs
- Graphic Model Builder (GMB) - installed separately
- NEVA Eigenvalue Analysis - installed separately

PSS®E achieves its broad capabilities by a highly modular structure and, in dynamic simulation, by encouraging the engineer to introduce user-written subroutines describing the problem of interest whenever the standard calculation procedures are not appropriate. PSS®E is not set up to solve *any* specific problem. Rather, it is a carefully optimized data structure associated with a comprehensive array of computational tools that are directed by the user in an interactive manner. By applying these tools in the appropriate sequence, the engineer can handle a wide range of variations on the basic power flow and stability theme.

PSS®E is designed on the premise that the engineer can derive the greatest benefit from computational tools by retaining the most intimate control over their application. The interactive structure of PSS®E, therefore, encourages the user to examine the results of each step in the computation process before proceeding to the next. This assists the engineer in understanding the engineering capabilities of these tools without having to become a master of the mathematical fine points of computation. The execution of standard studies such as power flow and basic transient stability on PSS®E requires no programming expertise. An engineer who is able to translate a problem formulation into simple FORTRAN statements will find, however, that PSS®E can handle virtually any system dynamics problem for which the requisite equipment models and input data can be produced.

The standard maximum capacities of PSS®E in terms of buses, branches, generators, and other system components are the same in all of its activities. Table 3-1, Standard Maximum PSS®E Program Capacities lists the dimensional capacities of the various PSS®E arrays at the standard size levels (1,000, 4,000, 12,000 and 50,000 buses) and at the largest size level (200,000 buses). Section 3.3 Environment provides details setting dimensional capacities of the various PSS®E arrays at size levels other than the four standard size levels and

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the interpolation, extrapolation, and rounding down (the I/E or S column of Table 3-1) calculations made at non-standard bus size levels.

All PSS®E manuals and reference guides are in PDF format and readily available and viewable with the Adobe Acrobat Reader (freely downloadable from the Adobe web site).

The PSS®E manuals and reference guides are located on the installation CD in the DOCS folder. If the option to install documentation with PSS®E is selected during the PSS®E installation, a shortcut to the PSS®E documentation will be placed on the PSS®E Start menu.

Chapters 1 through 4 of this manual describe the interface of PSS®E and the user with the computer's operating system. Chapters 5 through 17 contain detailed information on the operation of each PSS®E activity. Chapter 18 defines the procedures involved in establishing a system model for dynamic simulation runs. Chapter 19 contains samples of some of the more commonly used PSS®E data input files. Chapter 20 documents the requirements of user-written dynamic models. Chapter 21 describes the use of event studies in which the series of events (e.g., faults, switchings, etc.) to be employed in a dynamics or power flow study are specified, and Chapter 22 describes the use of scenarios to identify the set of data files to be used in a given study.

The *PSS®E Program Operation Manual* assumes throughout that the reader is familiar with the content of the *PSS®E Program Application Guide*. This manual and all its references to the *PSS®E Program Application Guide* apply only to the standard form of the PSS®E package.

Contacting Siemens PTI for Support

If after consulting the documentation and online help you find that additional assistance with PSS®E is needed, you may contact Siemens PTI via any of the following methods (please send a thorough description of the problem or question, including files):

- Send an e-mail to technical support at pti-psse-support.ptd@siemens.com.
- Send a fax to (518) 346-2777, attention PSS®E Support.
- Call the telephone support number between the hours of 8:00 a.m. and 5:00 p.m. Eastern Time, Monday through Friday. Dial (518) 395-5075 and select option 1 for PSS®E support.
- Visit the Siemens PTI Web Site at www.siemens.com/power-technologies and the PSS®E User Support Web Page.

Please note that access to the PSS®E User Support Web Page and to technical PSS®E support is only available to those users whose companies have purchased or renewed their PSS®E maintenance and support agreement for the current calendar year.

Submitting Bug Reports and Feature Requests

Bug reports and feature requests should be submitted directly to PSS®E support at pti-psse-support.ptd@siemens.com. Please provide as much detail as possible. If submitting a report for a potential bug please include the steps taken, along with pertinent data files and scripts so that we may accurately reproduce the problem. If an issue does turn out to be a program bug, one of the following priorities will be assigned to it:

- High priority is given to issues that cause the program to crash or produce incorrect results with no published work around.

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- Medium priority is given to issues that cause incorrect functionality, however the problem can be remedied by a published work around.
- Low priority is given to those issues that do not fall into the above categories, or are purely cosmetic in nature.

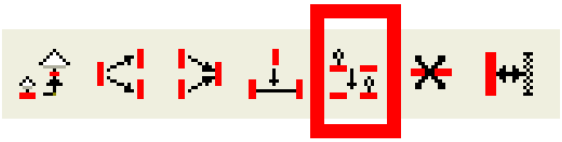
Every effort is made to address high and medium issues in a PSS®E point release. Some low priority items may also be include in a point release although this cannot be guaranteed.

Feature requests are evaluated on the basis of whether it benefits a wide majority of PSS®E users. If it does, it will likely be considered for implementation in a point release or major release of PSS®E. If several users have the same feature request, the feature may sit higher in the queue than others for implementation in PSS®E.

Document Conventions

The following conventions are used in PSS®E manuals:

Table 1.

Examples	Description
Power Flow > Solution > Parameters...	Navigation path in a PSS®E GUI menu
[Solution Parameters] dialog	Interactive dialog in Graphical User Interface (GUI)
[Spreadsheet] [Diagram]	Interface windows and views
SOLV,OPT LIST	User entry the line mode in the Command Line Interface (CLI) dialog
[F10], [Enter]	Keys found on a standard computer keyboard
[OK], [...], [Close]	Action buttons available on a dialog
	GUI toolbar button that starts an activity sequence
Set tap ratios to Unity	Activity option
Print outaged branches	Program option
GENERATOR CONVERSION COMPLETED	Message sent to Progress tab
<quantity>, <bus number>	Variable in message sent to Progress tab
ENTER OUTPUT DEVICE CODE:	Line mode dialog request/response
Progress tab, Report tab	Activity results displayed in [Output] view
OPEN	PSS®E activity name with hyperlink to source information
Section 5.15.2 Removing Outaged Equipment in a Subsystem, Figure 12-4	Cross-reference with hyperlink to source information
Creating a Power Flow Raw Data File	Short reference with hyperlink to source information
PSS®E Program Operation Manual, Section 20.14, Listing Saved Case or Snapshot Files	Full reference with hyperlink to source information
	Indicates report output example clipped to fit the page
PSS®E Program Application Guide	PSS®E manual reference

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Examples	Description
	Indicates additional information of interest.
	Indicates important information.
n.a.	Abbreviation meaning "not applicable".

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Chapter 1
Overview

1.1. PSSE Functional Description

PSS®E is an integrated set of computer programs that handles the following power system analysis calculations:

- Power flow and related network analysis functions.
- Balanced and unbalanced fault analysis.
- Network equivalent construction.
- Dynamic simulation.

PSS®E is structured around its own array address space (called the *working case*) along with a carefully designed set of temporary data files (called the *temporary files)*. These data structures are set up in a way that optimizes the computational aspects of the key power system simulation functions: network solution and equipment dynamic modeling. The working case and/or one or more of the temporary files are operated upon by a comprehensive set of functional program modules called *activities*. Each activity performs a distinct computational, input, output, or data manipulation function needed in the course of power flow, short circuit, equivalent construction, or dynamic simulation work.

PSS®E is entered by starting its master program module (see Section 3.2, "Starting PSSE"). The master module immediately invites the user to select an activity. The selected activity is immediately executed, performing its processing operation on the working case and/or one or more of the temporary files. The activity retains control until either:

- Its processing is successfully completed.
- Its processing encounters an error condition.
- It is interrupted and terminated by the user.

Each of these conditions returns control back to the master module, leaving the working case and the temporary files such that they reflect the results of the processing by this activity. The master module again invites the user to select another activity. Any activity may be selected at any time, but the determination of which activities are meaningful must be made by the user on the basis of the recent sequencing of activities and the present condition of the working case and the temporary files.

Complementing the working case and the set of temporary files, each user will generate a library of power flow *Saved Cases* and dynamic simulation *Snapshots*. These, in conjunction with the temporary files, form a working database. The PSS®E user quickly becomes adept at manipulating these database elements (i.e., files), resulting in the efficient management and execution of any power system analysis studies.

Refer to *Additional Resources for PSS®E*for details on PSS®E auxiliary programs.

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1.2. PSSE Operations

The use of PSS®E requires several types of operation on its host computer. Among these are:

- Creating and filling files to be used as input files to PSS®E.
- Compiling and linking the connection subroutines, CONEC and CONET, and any user-written dynamic simulation equipment models into the main body of PSS®E.
- Running the PSS®E programs.
- Running the auxiliary programs.

The first two of these operations require the use of commands and functional programs of the host computer's operating system. The user of PSS®E does not need to be an expert in the use of the host's operating system, but a brief study of those computer system manuals covering basic terminal keyboard and text editor operations will generally be beneficial.

The PSS®E user needs only a very minimal knowledge of computer system protocol in order to make productive use of the PSS®E package. The user does not need to become proficient with a complicated computer command language and is therefore able to apply effort to the task at hand: solving the engineering problem. The casual PSS®E user will need to become comfortable with the basic capabilities of the text editor and a small subset of system keyboard commands. More advanced users will find it to their advantage to familiarize themselves with the various disk file and related utility programs; a fundamental understanding of the host computer's file system organization will also be useful.

The purpose of this manual is to specify input data formats for PSS®E, to present detailed descriptions on the input requirements and operation of each PSS®E activity, and to assist the user in interpreting PSS®E's output messages.

1.3. Extending PSSE with User DLLs

PSS®E advanced features involving user-written code are implemented through the use of user- written dynamically linked libraries, or DLLs. When the user starts PSS®E, the program searches for the user-written DLL by name. These library names are:

DSUSR.DLL

This library may contain CONEC, CONET, and/or user-written dynamics models.

IPLUSR.DLL

This library may contain user-defined routines for IPLAN (see *IPLAN Program Manual*).

PSSUSR.DLL

This library may contain a user-written USERAC subroutine that will be called the USER activity.

If you have a user-written DLL in your directory, or anywhere on the load path, PSS®E will automatically find it and load it. For example, say you build a DSUSR.DLL in C:\WORKING\ONE. If you start PSS®E with C:\WORKING \ONE set as your working directory, your custom DSUSR.DLL will be loaded. If, on the other hand, you start

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PSS®E with C:\WORKING\TWO set as your working directory (and C:\WORKING\ONE is not on the load path), then DSUSR.DLL will not be loaded.

You can modify your PSS®E installation to always have your customized DLL's available by placing them in the PSSBIN directory of your PSS®E installation. Files that are in PSSBIN will always be located by PSS®E, and always located first should a DLL with the same name be anywhere else on the load search paths.

If your results are not what you expect, first make certain that you are loading the correct copies of the DLL(s) you wish to use.

If you create a customized DLL for the use of PSS®E, you can create a corresponding icon or shortcut in order to use that DLL. The new icon or shortcut should identify as the working directory the directory that contains the customized DLL. Alternatively, if you choose to start programs from the PSS®E 33 Command Prompt, simply change directories into your working directory before starting the program and the correct DLL will be loaded.

Instead of creating new shortcuts, it is possible to change the working directory (i.e., the Start In directory) of the existing PSS®E 33 shortcuts. PSS®E 33 includes a program called STARTIN, which should be run from the PSS®E 33 Command Line. When executed, this program (after prompting) will change the working directories of all standard (non-custom) shortcuts associated with PSS®E 33 to be the current directory (i.e., the directory from which the STARTIN command was given.) The STARTIN command may be used to switch the working directories as frequently as desired. To return the working directories to the original settings, use the shortcut *PSSE-33 Example Directory* in the PSS®E Utilities menu, to bring up a command prompt and give the STARTIN command.

1.4. PSSE Support for Building User DLLs

COMPILE	Batch file used to compile the connection subroutines CONEC and CONET.
CLOAD4	Batch file used to link the compiled code of the user's connection subroutines into DSUSR.DLL in the user's directory, used by the PSSDS4, the Dynamics entry point.
CreateUsrDll	A program, with a GUI, used for building a User DLL. By default it combines the functionality of COMPILE and CLOAD4, below, but can be used to create any DLL.
CLIPLU	Batch file used to compile and link user-written routines into IPLUSR.DLL in the user's directory, used by IPLAN and by power flow and dynamics entry points.
CLPSSUSR	Batch file used to compile and link a user activity into PSSUSR.DLL in the user's directory, used by the power flow and dynamics entry points.
DSUSR Visual Studio solution file	Microsoft Visual Studio solutions and projects to build the DSUSR.DLL, and allow the use of Visual Studio's interactive debugging tools.
PARSE32	Utility program used by CLOAD4.BAT

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PARSEXT	Utility program used by CMDLUSR.BAT.

1.4.1. COMPILE

The compiling file generated by activities DYRE and SRRS for compiling the connection subroutines CONEC and CONET is in the form of a BATCH file. The file can be named anything, but the sample provided in the EXAMPLE directory is named COMPILE; so we will use that name here. The name of this file must have the extension .BAT. If no extension is specified when this file is created, the extension .BAT is automatically appended to the filename. The compiling file is executed by entering the command:

```
filename.bat
```

which will compile the CONEC and CONET subroutines. The compilation must be run from a PSS®E-33 Command Prompt. When the compilation is complete, you must execute CLOAD4 at the PSS®E-33 Command Prompt to create a new DSUSR.DLL file.

In addition to compiling CONEC.FLX and CONET.FLX, the PSS®E-generated COMPILE.BAT file will allow a single user written-model to be compiled at the same time. If you have such a model, which must be written in the FLECS language, simply specify it on the COMPILE command line, e.g.:

```
COMPILE MY_MODEL.FLX
```

The result will be a correctly compiled MY_MODEL.OBJ, ready to process with the CLOAD4 step.

The CMDLUSR command can be used if you have more than one user-written model, or if some of the files are not in FLECS. This command, new for PSS®E-27, can compile any number of FLX, FOR, F90, F, C, and CPP files at one time. For example, the command:

CMDLUSR MY_MODEL.FLX EXTRA1.FLX EXTRA2.FOR

will FLECS and compile MY_MODEL.FLX and EXTRA1.FLX, and then compile EXTRA2.FOR. The object files MY_MODEL.OBJ, EXTRA1.OBJ and EXTRA2.OBJ should be generated in the current directory.

When compiling FLX, FOR, F90, and F files, CMDLUSR will search the current directory and the PSSLIB directory in an attempt to find any INCLUDEd files or MOD files. If other directories are to be searched, make sure that your INCLUDE environment variable defines those extra directories.

When compiling C and CPP files, only the current directory will be automatically searched for #include files. Therefore, you will almost certainly need to set your INCLUDE variable appropriately before compiling such files!

The top section of the CMDLUSR.BAT file discusses the INCLUDE variable in more detail.

1.4.2. CLOAD4

CLOAD4 is a batch program that is delivered with PSS®E. It can be run from the PSS®E-33 Command Prompt. The CLOAD4 linking procedures create a custom DSUSR.DLL, which allows for inclusion of user- written models in the dynamics program. The CLOAD4 command will automatically link in the CONEC.OBJ and CONET.OBJ files. Up to 35 additional object files may be specified on the CLOAD4 command line, for example:

```
CLOAD4 MYMDL.OBJ E:\PSSE31W\PSSLIB\MOREMDLS.OBJ
```


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The result of running CLOAD4 is a new DSUSR.DLL.

1.4.3. CreateUsrDll

CreateUsrDll is a GUI driven application that is delivered with PSS®E. It can be run from the shortcut in the Start Menu.

CreateUsrDll combines the functionality of COMPILE and CLOAD4. It can handle several types of source files, and includes a file browser so that input files can be located in any directory. It can create a dll of any name, although it defaults to DSUSR.DLL. It analyses the input files for external names and exports them for you (CLOAD4 only exports the CONEC and CONET program names); names must be exported in order to be referenced directly from outside of the DLL.

Use the Browse buttons to locate the files you want to include. These will be added to a list that is shown to you. You can remove files from that list later if you wish. If you wish to create a file other than DSUSR.DLL, the output file name is shown at the top of the GUI and can be changed. Once all your files are listed, use the “Compile + CreateDLL” button at the bottom of the GUI to build the file.

1.4.4. CLIPLU

CLIPLU provides a means for incorporating user-written Fortran routines into IPLAN. If your license includes IPLAN, nine dummy files (USREX1.FOR through USREX9.FOR) are placed in the IPLUSR subdirectory of PSS®E. You may edit these files to incorporate your own routines and, in addition, you may include routines on the command line as with CLOAD4. CLIPLU will compile and link these routines to create a new IPLUSR.DLL.

1.4.5. CLPSSUSR

CLPSSUSR enables the user to include a new activity in PSS®E. By editing USERAC.FOR (found in the PSSLIB subdirectory of PSS®E) and then running CLPSSUSR, you can compile and link your new activity into a new PSSUSR.DLL. Starting PSS®E from the directory containing PSSUSR.DLL will automatically include the new user function, which can then be executed by the API routine USER or the line mode command USER.

1.4.6. DSUSR Visual Studio solution file

Interactive debugging can be a very helpful tool when developing any program. The simple tools described above do not provide access to interactive debugging.

The project supplied, and described below, can be used if the version of Microsoft Visual Studio and Intel Visual Fortran used to build PSSE are installed. A knowledgeable person can modify this project for other versions, for other dll's, or copy the files to use in another directory location. The instructions provided here only account for the circumstance where the project is used as is where it is installed and with the expected versions of the software. This documentation does not explain how to perform debugging tasks using Microsoft Visual Studio.

All directory references are relative to the installed location of PSSE, usually "C:\program files\pti\pssexx" where "xx" is the major release number of PSSE.

To use the DSUSR solution

1. In the following, xxxx (as in dsusr-xxxx) refers to the version of Microsoft Visual Studio for which the files in that directory were prepared (e.g. 2013 for Visual Studio 213). You should be able to use those files in that or later versions of Visual Studio.

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Open dsusr-xxxx\dsusr.sln (either double click on the file if you have the filetype associations established, or start up Microsoft Visual Studio, select File > Open > Project/Solution and navigate to the file).

2. The Microsoft Visual Studio GUI is composed of several windows. You should see one titled "Solution Explorer". If you do not, go to the menu and select View > Solution Explorer.

3. In the Solution Explorer window you should see two projects, dsusr and dsusrlib. Select the dsusr project.

   a. Select "Project > Properties" from the menu.

   b. Select "Debugging" under "Configuration properties".

   c. In the field "Command" enter ..\pssbin\PSSE34.exe

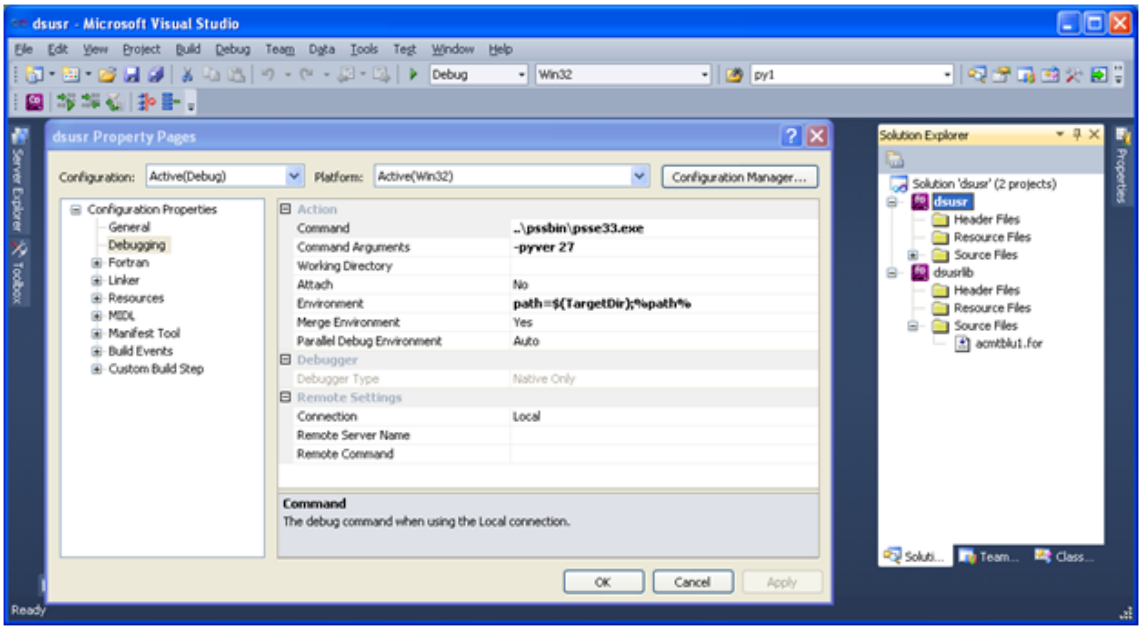
   d. In Environment enter the following: path=$(TargetDir);%path%

   e. Check that Merge Environment is set to Yes.

   f. In Command Arguments you can enter command line arguments, if you wish, such as -buses 50000

   g. In Working Directory you can set the working, or start-up, directory, if you wish (by default it is the project directory). If you have data files that you plan to use in testing, you may want to set this value to that directory.

   h. Save changes by clicking OK.



4. In the Solution Explorer window now select the dsusrlib project.

   a. Select the Source Files folder, and bring up the context menu (click the secondary mouse button).

   b. Select Add, then either:

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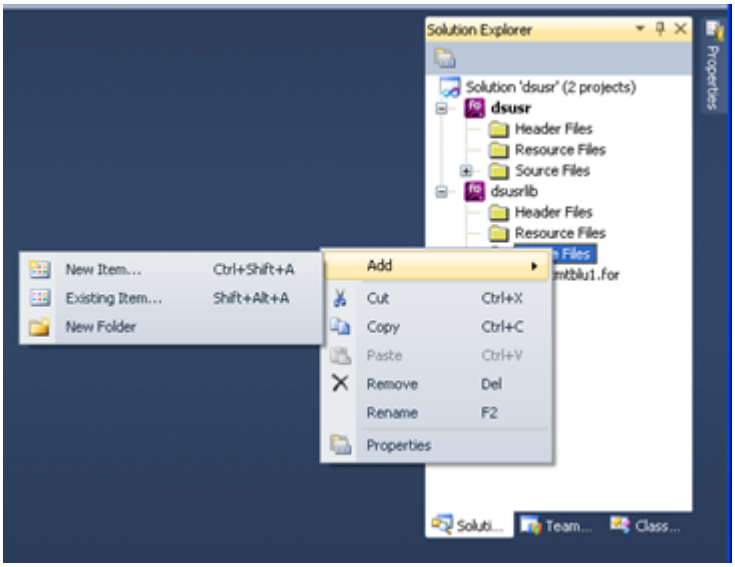
Select Existing Item.



i. Navigate to your source file and select it.

--- or ---

Select New Item

i. Select source form (free or fixed).

ii. Modify the Name and Location fields at the bottom of the form.

c. Repeat step (a) as many times as needed.

5. If necessary, make any code modifications needed.

6. Select "Build > Configuration Manager..." from the menu.

a. Select "Debug" in the "Active solution configuration" pull-down list. (there is a pull-down list in the menu that is a shortcut to this).

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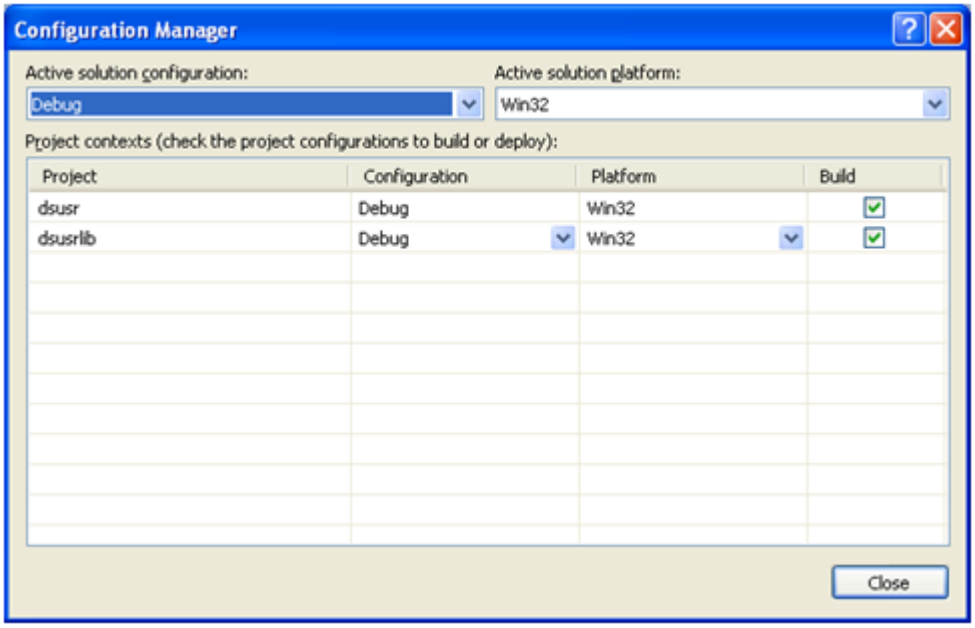

b. Click Close.

7. Select Build > Build Solution (usually F7 will do this).

8. If there are errors, make corrections and repeat step (7).

9. Select Debug > Start Debugging from the menu (usually F5 will do this). This will cause the program specified in Command (step 3c above) to run.

Notes:

- You may find, in some cases, that the above instructions tell you to set a value that is already the current value of that field. This is not a problem; it is simply that those values could have been changed by a prior use of Visual Studio.
- It is not necessary to copy your source files from wherever you would otherwise have liked to keep them. In step 4 above you can navigate to wherever the files are.
- It is not necessary to copy your data files from wherever you would otherwise have liked to keep them. In step 3g above you can set the working directory to wherever you like.
- If you have standard tests that you want to run, if you set them up in an automation file, you can specify that as a command argument (3f, above)
- You will only be able to debug your code, i.e. the code that you compile and build in your library.
- When looking at variable values in the debugger, you will not be able to see allocated arrays that are exported from PSSE. You can, of course, write code to copy them to local variables to assist in the debugging process.
- You are debugging, but starting an application that was not built for debugging (only your new library is built for debugging, not PSSE itself). Therefore Visual Studio will put up a messageBox and tell you that

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"Debugging information for 'PSSE34.exe' cannot be found [...]"; this is informational and not an error - just hit Yes.

- As you work on your project you can add or remove source files, change the working directory (step 3g), change the command line arguments (step 3f), or modify your code (step 5). If you make changes or change files, rebuild your project (step 7). If you only change parameters (working directory, command line, etc) you can run the program (step 9) without rebuilding.

Common Questions

1. Can I make a copy of the dsusr project so that I can work with more than one dynamics set-up?

    a. Yes. Copy the entire dsusr tree. In fact, it is a fine idea to do that all the time and leave the original dsusr tree in its original state for comparison purposes. It is easiest if you copy it within the PSSE installation. The projects do not care what the name of the directory tree is (i.e. whether it is dsusr or not). If you move the project outside of the PSSE installation tree then there will several directory settings that you will need to find and update.

2. I follow the directions but PSSE doesn't stop at a breakpoint in my program.

    a. There are several possibilities:

        i. You are looking at the wrong source. Check this by opening your source program by double clicking on the file name as listed in Solution Explorer.

        ii. PSSE has loaded a different dsusr.dll. Check this by looking at Help > Loaded Libraries in PSSE. You should see an entry to dsusr.dll, and the directory reference should be to dsusr\debug in your PSSE installation tree (or, if you have copied the dsusr tree to, say, mydsusr, then mydsusr\debug should be there). If it is different, you need to move/rename the file that it found, or change your system PATH.

        iii. Are you running a function in PSSE that you are certain will call your model program? Similarly, is your breakpoint set in a location that must be executed if the model program is called?

3. My source file is FLECS (*.flx). How do I use that in the dsusr project?

    a. We recommend converting your program to Fortran. If that is not practical, you can do one of the following:

        i. Use the CMDLUSR command (see Section 1.4.1, “COMPILE”)

        ii. Use CreateUsrDll (Section 1.4.3, “CreateUsrDll”), enter your FLECS program under "Compile Files" and select Compile.

        iii. In each case add the resulting .f file to your project (see step 4 above). If you subsequently change the FLECS source you will have to repeat the conversion (i.e. recreate the .f file) but you will not have to add the .f file to the project again.

4. What is the significance of the settings in step 3 of the instructions?

    a. The Command field (c) tells studio what program to run when instructed to "run" or "debug". By default it attempts to run the product of the build, but dsusr.dll cannot be run directly. Instead we instruct it to run the PSSE GUI.

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- The Environment field can be used to override or set Environment variables when "running" or "debugging". In this case, we set the PATH variable so that PSSE will find the dsusr.dll file we just created.
- The Merge Environment field determines whether the variables that are specified in the Environment box will be merged with the environment that is defined by the operating system.
- The Command Arguments specify the Command Line Options for PSSE (see Appendix D, *Appendix D - Command Line Options*).
- The Working Directory specifies the working directory of the program being debugged, relative to the project directory where your EXE is located. If you leave this blank, the working directory is the project directory. This will be the current directory when PSSE starts running.

5. Do we need to add all the custom code or only the code we are debugging into source folder under dsuserlib? What about conec and conet?

   a. No, you do not need to add all the custom code (i.e. user written models, possibly supporting functions, maybe conec and conet) to dsusrlib, although it will probably be simplest if you do. You can build any number of user libraries and load them all in PSSE. If your dynamics data indicates a model is needed then it will search all loaded libraries for it. However (1) all libraries other than dsusr.dll will have to be loaded explicitly, i.e. you will have to tell PSSE to load the library whereas it looks for and loads dsusr.dll automatically; (2) if you have custom code that you want PSSE to use then you have to build it into a library - if not dsusr.dll then some other library. Unless you already have a library of your other code, or have some other reason to build them separately, you might have fewer steps if you just build them all in dsusr.

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Chapter 2
File System

The power system calculations applied by the PSS®E program system require large quantities of different types of data. Consequently, the user of PSS®E must have some familiarity with the file structure of the host computer. The purpose of this chapter is to describe the relationship between the PSS®E programs and the file system in sufficient detail to allow the engineer to effectively perform studies using PSS®E without having to handle cumbersome program/file details.

2.1. Directories and Files Overview

To execute PSS®E (or any other program), the user must have a path to a *directory* or *folder*. A directory may be viewed as a catalog of files that are somehow related (e.g., they apply to a specific study). A directory may contain an unlimited number of files and subdirectories. PSS®E always operates out of a working *directory*. By default this is the EXAMPLE subdirectory in the main PSS®E directory.

It is perfectly reasonable, for example, to set up one directory for running the PSS®E programs (a working directory) and another for the storage of base case power flow and dynamics data files. Various other working directories may also be established in which to run PSS®E on a variety of different studies or investigations, each with its own set of data files (see Section 2.7.1, "Multiple Working Directories").

The files containing input data for PSS®E and output, Saved Case and Snapshot Files generated by PSS®E, may be located in the directory in which PSS®E is started up. However, this is not required; they may be placed in another directory. See Section 2.4.2, "Specifying Filenames" for further details.

If the user specifies a filename without an extension, PSS®E automatically appends a default filename extension to the specified filename. The three-character combination comprising the extension is dependent upon the type of file being accessed and/or the host computer's conventions. See Section 2.4.1, "File Usage " for further details.

There are several types of files accessed by PSS®E for which a directory search sequence is automatically applied if the file to be opened is not found in the user's current directory. In these cases, PSS®E then searches the user's home directory and, if the file is not found there, it looks in the appropriate subdirectory of the PSS®E master directory. Reference is made to this directory search sequence throughout the remainder of this manual where the files to which it applies are discussed. See Section 2.7.3, "File Search Path Rules" for further details.

2.2. The PSS®E Working Case and Temporary Files

Each PSS®E activity may operate upon the PSS®E *working case* and/or one or more of its *temporary files*.

A set of arrays in PSS®E's address space contains a complete set of power flow data. The content of these arrays is referred to throughout this manual as the working case. The working case is modified by the power flow activities and is always a valid working case even though its voltages may not represent a solution of Kirchhoff's laws.

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Several PSS®E activities create and access one or more temporary files that are used strictly as scratch files; each such file is deleted by the activity that created it prior to its termination. The user of PSS®E does not need to be concerned with names and contents of these files.

There are, however, three temporary files that are often used to preserve the results of certain activities for subsequent use by other activities. Contents of the files are variable depending upon the recent sequencing and context of activity executions. The user of PSS®E does not need to be concerned with the specific contents of these files as long as the prerequisites listed for each activity are observed. These temporary files are an integral part of the PSS®E package. The user never needs to reference these files by name but must be aware that these files may be processed every time PSS®E is used. The names and general functions of these temporary files are:

YMATnnn	Created by the triangular factorization activity FACT, and used by the triangularized Y matrix network solution activity TYSL, and the dynamic simulation activities. It is also used as a scratch file by the Newton-Raphson based power flow solution activities and the circuit breaker duty analysis activity BKDY.
SEQDnnn	Created by the sequence network setup activity SEQD, and used by the multiply unbalanced network solution activity SCMU, and the separate pole circuit breaker duty activity SPCB. It is also used as a scratch file by the automatic sequencing short circuit calculation activity ASCC, by the ANSI and IECS fault current calculation activities, and by the three sequence equivalencing activity SCEQ.
SCMUnnn	Created by the multiply unbalanced network solution activity SCMU, and used by the unbalanced network solution output activity SCOP, and by the short circuit data retrieval routines SCINIT, SC3WND,SCBRN2, SCBUS2 and SCMAC2 described in *PSS®E Application Programming Interface (API)*.

When temporary files are created, < nnn > is set to a three-digit integer such that the filename is unique in the directory in which it is created. See also Section 2.3.11, “Temporary Files”.

2.3. File Classes

The user of PSS®E must understand the categories of files that are used by PSS®E. PSS®E takes full advantage of the file management capabilities of the host operating system. This allows PSS®E to be used at all times without the need to select file assignments before it is started up. As a result, the user has a great degree of flexibility in the use of files during an interactive problem-solving session with PSS®E. The user can instruct PSS®E to read from or write into virtually any existing file at any time, and can have PSS®E create a new file at any time that it is needed. The prerequisite for full use of this capability is a sound understanding of the distinct classes of files used by PSS®E.

Note that a file class is a distinction imposed by PSS®E, not by the computer’s file management system.

The two major types of files are:

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1. Files created by the user – These include: files specifying data that is required by one or more of the PSS®E activities; and batch and program control files used for directing the execution of PSS®E.

2. Files created by PSS®E – These include: files generated when the user's working case and dynamics working memory are preserved; and output and results files that are either requested by the user or automatically generated.

The classes of files used by PSS®E are summarized in Table 2.1, "PSS®E File Classes". With the exception of the temporary files and the *PSS®E* and *OPF Options Files*, the user may assign any name to any file. The maximum filename or pathname length that PSS®E can handle is determined by the Windows® operating system. Both the computer's file management system and PSS®E activities always identify a file by this user-assigned name. Section 2.2, "The PSS®E Working Case and Temporary Files" and the following subsections discuss these file classes in some detail.

Table 2.1. PSS®E File Classes

File Class	Created By	Type	Accessible To
Input data files	User via text editor or auxiliary program	Source	PSS®E and user
Saved Case and Snapshot Files	PSS®E	Binary	PSS®E
Output listing files	PSS®E	Source	User
Channel Output Files	PSS®E	Binary	PSS®E and PSSPLT
Response Files	User via text editor or PSS®E activities ECHO, PSEB, and PSAS	Source	PSS®E and user
PSEB and PSAS Command Files	User via text editor	Source	PSS®E and user
IPLAN Source Program Files	User via text editor	Source	IPLAN compiler and user
IPLAN Executable Program Files	IPLAN compiler	Binary	PSS®E
Python Program Files	User via text editor	Source	PSS®E and other Python supported applications (e.g., IDLE)
Options Files	PSS®E	Binary	PSS®E
Results files	PSS®E	Source or Binary	PSS®E and other programs
Temporary files	PSS®E	Source or Binary	PSS®E

2.3.1. Input Data Files

PSS®E must, from time to time, accept large volumes of data from external sources. Such large volumes of data could be entered directly into the PSS®E working case using the Spreadsheet View. For other than very small systems, this is not practical; voluminous data is best assembled in an *input data file* independently of PSS®E before PSS®E is started up. This file may then be used as the input source for PSS®E to feed the data through the appropriate input activity into the PSS®E working case.

Input data files may be obtained by reading from storage mediums (e.g., CDs) or e-mail attachments from external sources (e.g., other computer installations), or by the typing and file editing facilities of the host

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computer. In the case of power flow and dynamics data input, the input data files may often be created by reading and reformatting data obtained from other computer installations. While they are not accessible from the PSS®E master program module, reformatting programs are available for translating several widely used power flow and stability data formats into the PSS®E input format. Section 2.4.2, "Specifying Filenames" discusses the alternative methods of creating input data files.

Table 2.2, "Summary of PSS®E Input Data Files" defines the principal PSS®E input data files.

Table 2.2. Summary of PSS®E Input Data Files

File Type	Description
Power Flow Raw Data Files	Power flow system specification data for the establishment of an initial working case via activity READ. Several of these files may have to be read by activities READ and/or RDCH when a new power flow case is being built from subsystem data being provided by several different power companies or organizations.
Machine Impedance Data Files	Data describing the unit configurations at generator buses (plants) for input into the working case via activity MCRE.
Dynamics Data Files	Files that contain dynamics data on synchronous machines and other system components for input to the PSS®E dynamic simulation working memory via activity DYRE.
Sequence Data Files	Negative and zero sequence data needed by activity RESQ to build the working case from a basic positive sequence model into a full symmetrical component transmission system model for unbalanced fault analysis.
Optimal Power Flow Raw Data Files	Data on constraints and controls for input into the working case via activity ROPF. When added to the power flow model, this data is used for defining and solving the optimal power flow problem.
Economic Dispatch Data Files	Machine incremental heat rate data, fuel costs, and other data used by activity ECDI in applying its unit commitment and economic dispatch calculation. These files are also used by the auxiliary program PLINC to plot incremental heat-rate curves (see *Additional Resources for PSS® E* in reference manual , PLINC).
Inertia and Governor Response Data Files	Machine inertia and governor response data for the generator redispatch used in the inertial and governor response power flow activity INLF.
Drawing Coordinate Data Files	Coordinates of busbars, line ends, generators, loads, etc. for the construction of a one-line diagram via the obsolete activities DRAW, GDIF and SCGR.

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File Type	Description
Graphical Report Definition Files	Positional data for geometric elements and PSS®E power flow data and solution results used for the construction of a user-defined graphics display via the obsolete activity GRPG.
Contingency Analysis Data Files	Data defining monitored elements, contingencies, and subsystems used by activity DFAX in constructing a Distribution Factor Data File. A Distribution Factor Data File is required by activities OTDF, DCCC, TLTG, SPIL,POLY,IMPC, LLRF, the various ac contingency calculation functions (e.g., ACCC, AC Corrective Actions, etc.), the PV Analysis function, the QV Analysis function, the Probabilistic Reliability Assessment function, and the Substation Reliability Assessment function.
Tripping Element Data Files	Tripping event specifications that are required when multiple level contingency analysis is initiated with the tripping simulation option enabled.
Outage Statistics Data Files	Outage statistics data in the form of frequency and duration required by the Probabilistic Reliability Assessment function and the Substation Reliability Assessment function.
Subsystem Participation Data Files	Data defining participation blocks for one or more of the subsystems defined in the Distribution Factor Data File that is being used in the transfer limit analysis of activity SPIL.
Breaker Duty Data Files	Machine parameters used by the circuit breaker current interrupting duty analysis activity BKDY.
Fault Specification Data Files	Fault locations and fault duty times used by the circuit breaker current interrupting duty analysis activity BKDY.
ANSI Fault Specification Data Files	Fault locations, maximum operating voltages, and contact parting times used by the ANSI fault current calculation activity, ANSI.
IEC Short Circuit Data	Additional data for the modeling of machines and transformers used by the IEC fault current calculation activity IECS.
Fault Control Data Files	Data specifying various faulting and reporting options for use by the fault calculation activities ASCC and IECS.
Machine Capability Curve Data Files	Machine capability curves for use by the reactive power limit checking and updating activity GCAP.
Load Throwover Data Files	Bus load throwover data for use by the various ac contingency calculation functions (e.g., ACCC, AC Corrective Actions, etc.), the PV Analysis function, and the QV Analysis function.
Transactions Raw Data Files	Transaction event definitions that are entered into PSS®E working memory by activity REMM for use in

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File Type	Description
	Transmission Access studies by activities ALOC, IMPC, and LLRF.
Bus Location Data FIles	Bus location data that is used in creating or modifying a one-line diagram by Specifying Bus Locations from a File.

Reading basic power flow and stability data from input data files should be a relatively infrequent occurrence in PSS®E. When an initial working case has been built, the input data files should be set aside and all data changes and small additions should be made directly on the working case from the Spreadsheet View and the data modification activities such as CHNG, RDCH, SQCH, ALTR, and CCON. Attempts to keep a large input data file up to date with an ongoing power system study are usually both error prone and time consuming. PSS®E Saved Cases and Snapshots, as described below, are far more efficient vehicles for maintaining the power flow and dynamic system database of a study. Table 2.3, "Restorable Data Input Files" shows PSS®E activities that may be used to write portions of PSS®E's working memory in the corresponding Input Data File format.

Table 2.3. Restorable Data Input Files

Input Data File	Read by PSS®E Activity	Written by PSS®E Activity
Power Flow Raw Data File	READ	RAWD
Machine Impedance Data File	MCRE	RWMA
Sequence Data File	RESQ	RWSQ
Dynamics Data File	DYRE	DYDA
Optimal Power Flow Raw Data File	ROPF	RWOP

2.3.2. Saved Case and Snapshot Files

Because all PSS®E activities always operate upon the working case, it is necessary to load the required data into the working case before starting any sequence of simulation work. Even though it is possible to load the working case on a routine basis by using activities such as READ, RESQ, and DYRE to read input data files, this is discouraged because it would:

1. Be grossly inefficient because input data files are organized for people and must be reorganized by the data input activities to match the computational data structure of PSS®E.

2. Require continuous updating of the input data files as interactive data changes are made in PSS®E to ensure that the problem data is available for future use.

PSS®E overcomes these problems by using Saved Case and Snapshot Files. These files are binary images of the power flow working case and dynamics data memory, respectively. To conserve disk space and minimize the time required to store and retrieve these files, Saved Cases and Snapshots are compressed in the sense that they do not record unoccupied parts of the data structure if the system model is smaller than the capacity limits of the program.

Saved Case and Snapshot Files are loaded with copies of these memory images by activities SAVE and SNAP respectively. Activities SAVE and SNAP require a filename at the time they are selected.

The user may create as many Saved Cases and Snapshots as desired. Each Saved Case is a complete power flow description that may be returned to the working case as a new base case at any time with activity CASE. Each Snapshot File records the exact instantaneous condition of all dynamic simulation models at the time activity SNAP is executed. Returning a Snapshot to dynamics working memory *and its corresponding Saved*

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Case to the power flow working case allows an interrupted dynamic simulation run to be continued from the time point at which activity SNAP was executed, just as if the run had not been stopped in the first place.

2.3.3. Output Listing Files

The majority of PSS®E report-generating activities, such as LIST and POUT, may direct output to the user terminal, to a high-speed printing unit, or to a named file. When an output listing file is selected, the report is written into that file in exactly the same format as if it was being printed directly to a printing device.

The output listing file is available to all standard file manipulation functions; it may be printed, transferred to a backup medium, examined with a text editor, or simply discarded.

The user may instruct PSS®E to create as many output listing files as are needed. It is good practice, however, to limit the number of such files because they can use a large amount of disk storage capacity. Soon after being created, output listing files should be processed (e.g., copied from disk to a backup medium for archival purposes) and then deleted from the disk (see Section 2.8, "Deleting Files").

2.3.4. Channel Output Files

Channel Output Files are produced by the PSS®E dynamic simulation activities. They are named and created in activities STRT, ESTR,GSTR, MSTR, and ALTR in response to a user name specification and receive the values of the PSS®E output channels at regular time intervals during a simulation run. Unlike output listing files, Channel Output Files are of binary type and cannot be intelligibly printed by the standard file management functions of the computer. Rather, the Channel Output Files are organized as required for input to the plot page processing capability of the PSS®E master program module, and to the Channel Output File processing program, PSSPLT.

As with output listing files, it is advisable to minimize the number of Channel Output Files by copying them to an off-line bulk storage medium if necessary for archival purposes and deleting or reusing them as soon as the plotting of a simulation run has been completed.

2.3.5. Response Files

Response Files allow the PSS®E user to automate the execution of a sequence of activities. A Response File is an ordinary source file that the user creates with a text editor *before* starting up PSS®E. A Response File for the performance of a given calculation contains either or both of the following:

1. An *exact* image of the line mode activity command and all of its inputs just as they would be entered by the user at the console in executing the activity through PSS®E's command line interface (see Section 4.2.2, "Command Line Interface").

2. The batch commands (BAT_), as described in *PSS®E Application Program Interface (API)*, corresponding to activities and menu bar functions executed from the menu bar of the Graphical User Interface (GUI) (see Section 4.2.1, "GUI Windows").

Response Files may be used to execute an entire program run where the requirements are able to be determined completely in advance. They may also be used as a convenience to save typing of a specific subgrouping of activity commands that is found by a user to be needed frequently in the application of PSS®E.

Constructing a Response File requires familiarity with details of the PSS®E BAT_ commands and/or its line mode dialog. Response files in batch command format can be recorded in the PSS®E GUI. Terminal input, including line mode commands, can be recorded using activity ECHO in PSSECMD.

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2.3.6. PSEB and PSAS Command Files

PSEB and PSAS Command Files allow the PSS®E user to specify many routine power flow and dynamic simulation runs, respectively, in English sentence form. These files are ordinary source files that the user creates with a text editor *before* starting up PSS®E. Each record is in the form of a command starting with a verb from a defined vocabulary.

The PSS®E run assembler activities, PSEB and PSAS, accept such commands, either from a data input file or directly from the user's terminal, and translate them into a PSS®E Response File for immediate execution or for subsequent processing as a standard PSS®E Response File (see Section 2.3.5, "Response Files").

2.3.7. IPLAN Program Files

The user may define and implement processing and/or reporting functions in PSS®E via the IPLAN programming language. Programs written in this language are created by the user with a text editor and compiled with the IPLAN compiler program *before* starting up PSS®E. The resulting IPLAN Executable Program File may then be specified to the PSS®E activity EXEC for execution.

For details on the IPLAN language, its hooks into the PSS®E working case, and the use of its compiler, refer to the *IPLAN Program Manual*.

2.3.8. Python Program Files

The user may define and implement processing and/or reporting functions in PSS®E via the Python programming language. Python scripts or programs written in this language are generated by using a text editor or created by PSS®E's program automation capability. Python scripts may be executed from the PSS®E menu option *I/O Control > Run Program Automation File...*. Python commands may also be entered directly through PSS®E's command line interface.

2.3.9. PSS®E Options Files

When PSS®E is installed on the system, a set of default run-time options is established (refer to Section 3.3.3, "Program Run-Time Option Settings"). Activity OPTN may be used to modify the current option settings. Activity OPTN may then be instructed to write the current option settings into a file; this options file is given the name PSSE with the extension OPT (i.e., PSSE.OPT).

Similarly, an Optimal Power Flow Options File, PSSOPF.OPT, may be used to preserve the current optimal power flow solution parameter settings (see Section 15.10, "Optimal Power Flow Solution Options"). The OPF options settings may be modified by selecting the *OPF > Parameters...* menu entry or by executing the functions (i.e., the APIs; see Optimal Power Flow Program Operation APIs of *PSS®E Application Program Interface (API)*) called from the *[OPF - Change Parameters]* dialog.

Then, whenever PSS®E is subsequently started up, it searches for options files of these names using the directory search sequence described in Section 2.1, "Directories and Files Overview". If either options file is found, it is read in, establishing new default options settings.

2.3.10. Results Files

Several PSS®E activities produce binary files intended to be used either by other PSS®E activities (e.g., the Distribution Factor Data File of activity DFAX for use by activity DCCC) or by other programs (e.g., the Matrix Output File of activity ASTR for use by LSYSAN).

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Similarly, several PSS®E activities produce source files intended to be used either by other PSS®E activities (e.g., the Bus Number Translation File of activity BSNM for use by activity RNFI) or by other programs (e.g., the Relay Output File of activity ASCC for use by an external relay coordination program).

2.3.11. Temporary Files

Any temporary file that is preserved following the completion of the activity that created and filled it (see Section 2.2, "The PSS®E Working Case and Temporary Files") is similar in purpose to the Results Files. The differences are that: the temporary files are volatile (i.e., they are automatically deleted when PSS®E is terminated); and the user has no naming control over the temporary files.

2.4. PSS®E Data Files

2.4.1. File Usage

Before PSS®E can be started up, the user must create and fill those files needed for its input. Files needed for PSS®E output, Saved Cases, and Snapshots will be created by PSS®E as needed. The importance of planning and noting the contents of PSS®E-created files, as well as user input files, cannot be overemphasized. It is strongly recommended that, along with study notes, the engineer keep a record identifying the names and contents of all significant files used in the study. A PSS®E File Planning Sheet is suggested.

The definitions of the various input data files used by PSS®E are contained in this manual in the activity descriptions of the PSS®E activities that utilize them. On completion of problem data collection, the user will have a completed file planning sheet giving the names of all files to be used in the initial phases of a study. The user will also have written out (in study notes, in spreadsheets, etc.) the input data that must be placed into the PSS®E input files. A typical completed PSS®E File Planning Sheet and a sample of the data to be placed in the user-supplied files is given in Chapter 24, *Example Data Files* of this manual.

The types of data files used by PSS®E, the activities that use them, and the default extensions used when specifying filenames are summarized in Table 2.4, "PSS®E Data File Summary". Except as described below, whenever a file type listed in Table 2-4 PSS®E Data File Summary is being specified to PSS®E, the extension designated in the table is automatically appended to the filename if the user specifies a filename without an extension. For file types not listed in Table 2.4, "PSS®E Data File Summary", as well as for those activities listed below, a default extension compatible with the Windows® operating system is used.

The following activities write various types of data in the form of input files and use Windows® compatible default extensions: activities BSNM, DYDA, RAWD, RNFI, RWCM, RWDY, RWMA, RWMM, RWOP, and RWSQ.

Some input files need not be created if the activities that use them are not going to be executed.

Table 2.4. PSS®E Data File Summary

PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
Saved Case	sav	SAV	Binary	One is essential. More are optional.	CASE, DIFF, GDIF, DFTI, and CMPR. Built by SAVE.
Snapshot	snp	SNP	Binary	For dynamics, one is essential.	RSTR. Built by SNAP.

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
				More are optional.	
Source Form Snapshot	srs	SRS	Source	Optional	SRRS. Built by SRSN of PSS®E-26 or earlier.
Channel Output	out	OUT	Binary	For dynamics, one is essential. More are optional.	Plot page processing capability of PSS®E and program PSSPLT. Built by dynamic simulation.
Power Flow Raw Data	raw	RAW	Source	Optional	READ. Built by RAWD or format conversion program.
Dynamics Data	dyr	DYR	Source	Optional	DYRE. Built by DYDA or format conversion program.
Machine Impedance Data	rwm	RWM	Source	Optional	MCRE. Built by RWMA or format conversion program.
Sequence Data	seq	SEQ	Source	Optional	RESQ. Built by RWSQ or format conversion program.
Optimal Power Flow Raw Data	rop	ROP	Source	Optional	ROPF. Built by RWOP.
Drawing Coordinate Data	drw	DRW	Source	Optional	Obsolete one-line diagram definition format file that may be imported into the Diagram View.
Drawing Coordinate Binary Data	drb	DRB	Binary	Optional	
Graphical Report Data	grp	GRP	Source	Optional	GRPG.
Binary Graphical Report Definition Data	grb	GRB	Binary	Optional	GRPG.
GRED Library	sgf	GRL	Binary	Optional	GRPG.

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
Inertia and Governor Response Data	inl	INL	Source	Optional	INLF. Built by RWDY.
Breaker Duty Data	bkd	BKD	Source	Optional	BKDY. Built by RWDY.
Fault Specification Data	bkf	BKF	Source	Optional	BKDY.
Fault Control Data	fcd	FCD	Source	Optional	ASCC, ASCC2 and IECS.
Relay Output	rel	REL	Source	Optional	Built by ASCC, ASCC2 and IECS.
Bus Subsystem Selection Data	sbs	SUB	Source	Optional	Recall function. Built by memorize function of the bus subsystem selector.
Subsystem Description Data	sub	MON	Source	Optional	RANK and DFAX.
Monitored Element Data	mon	CON	Source	Optional	RANK and DFAX.
Contingency Description Data	con		Source	Optional	DFAX. Built by RANK.
Distribution Factor Data	dfx	DFX	Binary	Optional	OTDF, DCCC, TLTG, SPIL, POLY, IMPC, LLRF, the various ac contingency calculation functions (e.g., ACCC, ac corrective actions, etc.), the PV Analysis function, the QV Analysis function, and the Probabilistic Reliability Assessment function and the Substation Reliability Assessment function. Built by DFAX.
Tripping Element Data	trp		Source	Optional	Multi-level ac contingency analysis.

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
Reliability Outage Statistics Data	prb		Source	Optional	The probabilistic reliability function and the substation reliability function.
Load Throwover Data	thr	THR	Source	Optional	The various ac contingency calculation functions, the PV Analysis function, and the QV Analysis function.
Subsystem Participation Data	prt	PRT	Source	Optional	SPIL.
Machine Capability Curve	gcp	GCP	Source	Optional	GCAP.
Transactions Raw Data	mwm	MWM	Source	Optional	REMM. Built by RWMM.
Bus Renumbering Translation	trn	TRN	Source	Optional	BSNM and RNFI. Built by BSNM.
Relay Characteristic Data	rlc	RLC	Source	Optional	Dynamics channel output file processing functions. Built by RWDY.
Response	idv	IDV	Source	Optional	Built by ECHO, PSEB and PSAS.
PSEB Command	pse	PSE	Source	Optional	PSEB.
PSAS Command	psa	PSA	Source	Optional	PSAS.
IPLAN Source Program	ipl		Source	Optional	IPLAN compiler.
IPLAN Executable Program	irf	IRF	Binary	Optional	EXEC. Built by IPLAN compiler.
Matrix Output	lsa	LSA	Binary	Optional	Program LSYSAN. Built by ASTR.
Economic Dispatch Data	ecd	ECD	ECD	Source	ECDI
Bus Subsystem Selection Data	sbs	SBS	SBS	Optional	Recall function. Built by memorize function of the bus subsystem selector.

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
AC Contingency Solution Output	acc	ACC	ACC	Optional	The various ac contingency calculation reporting functions. Built by the various ac contingency calculation functions.
Conversion Program Log	log	LOG	Source	Optional	Several data conversion programs.
ANSI Fault Specification Data	ans	ANS	ANS	Optional	ANSI
Miscellaneous Output Data	dat	DAT	Source	Optional	Built by various functions (e.g., RWCM, OUTPUT_Y_MATRIX, section 1.190*API Guide*).
Python Program	py	PYT	Source	Optional	Run program automation file function. Built by recording function.
PV Solution Output	pv	PVA	Binary	Optional	The various PV analysis reporting functions. Built by PV Analysis function.
QV Solution Output	qv	QVA	Binary	Optional	The various QV analysis reporting functions. Built by QV Analysis function.
POLY Results Output	pol	POL	Binary	Optional	section / POLY and POLY previous results plotting. Built by POLY.
Bus Location Data	loc	LOC	Source	Optional	GUI one-line diagram.

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
Tripping Element Data	trp	TRP	Source	Optional	Multi-level ac contingency analysis.
FLECS Output	flx	FLX	Source	Optional	Built by section / DYRE.
Compiling Command	bat	COM	Source	Optional	Built by section / DYRE and section / SRRS.
LNTSA Solution Output	lnt	LNT	Binary	Optional	
IEC Short Circuit Data	iec	IEC	Source	Optional	section / IECS.
Reliability Outage Statistics Data	prb	PRB	Source	Optional	The probabilistic reliability function and the substation reliability function.
Event Study	evs	EVS	Binary	Optional	Event study functions.
Short Circuit Results	sc	OSC	Binary	Optional	Slider diagram. Built by ASCC2 and section / IECS.
Slider Template Data	slt	SLT	Binary	Optional	
EPC Data	epc	EPC	Source	Optional	
Dynamic Model Library	dll	DLL	Binary	Optional	
Slider Data	sld	SLD	Binary	Optional	Slider diagram. Built by slider.
Power Flow Change Data	raw	RAC	Source	Optional	section / RDCH. Built by section / RAWD or format conversion program.
Slider XML Data	sldxml	SLX	Source	Optional	Slider diagram in readable XML format. Built by slider.
Scenario Data	pssxml	PEX	Source	Optional	Scenario Manager. Built by Scenario Manager.
Dynamics Add Data	dyr	DYA	Source	Optional	section / DYRE, ADD. Built by section / DYDA or

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
					format conversion program.
Compiled Python Program	pyc	PYC	Binary	Optional	Run program automation file function. Built by compiling a python file.
Python Program (no console)	pyw	PYW	Source	Optional	Run program automation file function.
Zipped Scenario Archive	zip	ZIP	Binary	Optional	Scenario Manager. Built by Scenario manager
Google Earth Data	kml	KML	Binary	Optional	
Plot Attributes Data	plt	PLT	Binary	Optional	
Text	txt	TXT	Source	Optional	Built by various functions
Database Data	pdb	PDB	Binary	Optional	
Aspen Raw Data	dxt	DXT	Source	Optional	
Coordinate Pair Data	gcp	COP	Source	Optional	
MWMI Data	dat	MMI	Source	Optional	
Transactions Data	dat	TRA	Source	Optional	
Detailed Branch Data	log	DBR	Source	Optional	
Long Title Data	txt	RET	Source	Optional	
Short Circuit Summery Data	dat	SCM	Source	Optional	
Comma Separated Value Data	csv	CVS	Source	Optional	
Python Dynamic Library	pyd	PYD	Binary	Optional	
PSSE XML Configuration Data	xml	XML	Source	Optional	PSS®E Import options. Built by PSS®E Export options.
PSS®E Options	OPT	CFG	Binary	Optional	PSS®E start-up. Built by OPTN.
Optimal Power Flow Options	OPT		Binary	Optional	PSS®E start-up. Built from the [OPF - Change

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PSS®E File Designation	Extension		File Type	Essential/ Optional	Used by Activities/ Functions
					Parameters] dialog.
Temporary Files	*		Binary	Essential	Created and used as needed.

2.4.2. Specifying Filenames

In specifying filenames to PSS®E, entering a simple filename always results in PSS®E attempting to access the specified file in the user's current directory; specifying the complete pathname of a file allows the user access to other directories for which appropriate privileges are assigned. In addition, PSS®E provides a shorthand method of designating the directory path component of a complete pathname. Activity PATH allows the user to specify such a path. Then, anytime a simple filename preceded by an ampersand (&) is specified to PSS®E (e.g., &case1), the ampersand is replaced by the specified directory path component and the resulting pathname is used as the filename.

PSS®E files over which the user has no naming control are generally accessed from the current directory or from some other system default directory (e.g., the Temp directory on Windows systems). This includes files such as the PSS®E temporary files (see Section 2.2, "The PSS®E Working Case and Temporary Files"), temporary files used during graphics-related activities, and so on. (PSS®E also looks to the user's home directory and to its own master directories for PSS®E Options Files, for output device parameter files, and for several other types of files if they are not found in the user's current directory; see Section 2.7.3, "File Search Path Rules").

2.5. Creating Input Files

As is seen from Table 2.4, "PSS®E Data File Summary", there are several source files that are designated as input files to various PSS®E activities. Such files must be created prior to executing the PSS®E activities that utilize them. A source file must be filled with some data at the time it is created. The initial fill of data need not be the final content of the file because additional data and corrections may be placed in the file by a text editor at a later time if desired. The content and formats of the various PSS®E data files are described in this manual in the descriptions of the activities that use them.

Input records may not exceed 256 characters.

2.5.1. Via a Text Editor

The primary tool for creating and preparing input data files for PSS®E is a host computer's text editor. A detailed description of text editors is beyond the scope of this manual. The user is referred to the appropriate computer system documentation.

2.5.2. Via Format Conversion Programs

It is often desirable to be able to utilize data for a system model that might have been set up for another program. To that end, a group of auxiliary programs is supplied with PSS®E which, though not accessible from the PSS®E master program module, are valuable data preparation tools. Details on format conversion programs may be found in *Additional Resources for PSS®E*, Auxiliary Program Descriptions.

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2.5.3. Via PSS®E

As shown in Table 2.4, "PSS®E Data File Summary", several PSS®E activities replicate portions of working memory in the form of PSS®E input data files. Details on the applications of such activities are provided by individual descriptions contained in this manual.

2.6. Files Created By PSS®E

Files into which PSS®E is to write its output need not be created or specified prior to initiating PSS®E. The name of an output file is requested at the time the activity utilizing it is ready for it. If the specified file does not already exist, it is created and used.

For the case in which the selected file already exists, the user may elect to have PSS®E operate in either of two modes:

1. The specified file is utilized.
2. The user is informed that the file exists and is given the option of either specifying another filename or using the designated file.

Generally, whenever an existing file is used for output, its previous contents are overwritten and the file is lengthened or truncated as required.

It is the responsibility of the user to ensure that files that are needed for future reference are not overwritten.

2.7. File System Details

2.7.1. Multiple Working Directories

A simple way of setting up PSS®E to run in a variety of working directories is to create a shortcut icon for each one and alter the *[PSS®E Properties]* dialog for the shortcut to indicate the directory in which the program is to start. Refer to *PSS®E GUI Users Guide,* Properties Dialog; the directory path specified in the *Start in:* field may be set to the desired directory.

2.7.2. PSS®E Directory Organization

The PSS®E program is usually installed in your Program Files directory; the default installation directory is C:\Program Files\PTI\PSSE35. However, PSS®E may be installed in any user-named directory. If you choose to install PSS®E in other than the default directory, substitute your installation directory for any reference of C:\Program Files\PTI\PSSE35 found in this manual.

This top level PSS®E Master Directory consists of subdirectories that contain the files required for executing PSS®E; these subdirectories are shown in Table 2.5, "PSS®E Subdirectories".

Table 2.5. PSS®E Subdirectories

Name	Description
DSUSR	MicroSoft Visual Studio project to build DSUSR.dll.
DVRM Templates	DVRM Workbook template files used in data visualizations

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Name	Description
EXAMPLE	Sample PSS®E data files and batch files for building user versions of the PSS®E programs.
IPLUSR	User-written IPLAN files.
PSSBIN	PSS®E executable files.
PSSHLP	PSS®E help files in binary format.
PSSLIB	Contains several different types of files: Files necessary for rebuilding the DLLs: • Batch files (*.BAT) • Link libraries (*.LIB) • Fortran include files (*.INS, *.COD) • Fortran module files (*.MOD) • C header files (*.H) • Some source files (*.FOR) • Other miscellaneous files
PSSMAP	Library of Graphics Report Definition Data Files (*.GRP) for use with activity GRPG.
PSSPRM	Parameter files that are used to override the default characteristics of output devices supported by PSS®E and the PSS®E Windows environment.
PSSPYxx	PSS®E Python files used to support program functionality for Pythonxx.
SETUP	Programs and data used in setting up your installation. In general, you should not need to use the files in this directory unless you are doing a node installation from a server. If you call Siemens PTI for assistance, support personnel may ask you to use programs in this directory to assist in diagnosing and resolving installation-related issues.
Shortcuts	Windows shortcuts used in Start Menu.
TMLC	Conductor data files used by TMLC (only for installations where the lease includes the Transmission Line Constants program section).

2.7.3. File Search Path Rules

PSS®E programs use a variety of parameter and binary files from which to gather specific information.

Parameter files are standard text files that can be customized for either individual or system-wide use. These files typically have the extensions .DAT or .PRM such as those for printers, plotters, and the PSS®E Windows® environment. Binary files are created by PSS®E programs from previously developed input files or from specific user input.

When attempting to open a file, PSS®E tries to locate files by sequentially checking the following locations:

Current Working Directory. When started via a shortcut, PSS®E uses as its working directory the Start In directory associated with the shortcut. When started from a command prompt, PSS®E uses the current directory as the working directory. When a file is opened, the current working directory will change to the directory

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the file came from. Refer to Section 2.1, "Directories and Files Overview" for details on how the working directory is used.

Current Study. If a study is open, PSS®E will see if the desired file can be found in the study.

Home Directory. By default, this directory is under the personal Documents folder of the current user, [%USERPROFILE%\Documents\PTI\PSSE35]. Setting this variable does not change the program working directory (see above).

Master Directories. If the type of file being opened has an associated master directory list, that list of directories will be searched for the specified file. Refer to Section 2.7.6, "Master Directories"for detail on how master directories can be created and used.

Example Directory. By default, this directory is the Example directory under the default Home Directory.

2.7.4. Temporary Files

The temporary files have the extension BIN and are created in the directory specified by the Temp parameter in the PSS®E .INI file.

2.7.5. Reserved Filenames

There are a number of files that PSS®E and its auxiliary programs access automatically. You must avoid assigning any of the following reserved filenames to your own files in any directory from which PSS®E is to be executed. In addition, the filenames ending with .EXE and .BAT located in the subdirectory PSSBIN should be treated as reserved.

Table 2.6. Reserved Filenames

Filename	Description
MODELS.BIN	Binary data file for activity CCON.
PARMdv.DAT	Printing and plotting parameter files; <dv> is either PR for the Printer Definition Parameter File or a two character device code assigned to any of the various graphics output devices supported by PSS®E on the PC: HP, PS, etc.
PSCRIPT.DAT	PostScript parameter file for tabular output.
IMD3500.INI	IMD settings file.
PSSPLT3500.INI	PSSPLT settings file.
PSSE.OPT	PSS®E options file.
PSSERAS.PRM	PSS®E RAS Aux file color coding.
PSSOPF.OPT	PSS®E OPF options file.
PSSPLT.OPT	PSSPLT options file.
SMRYSC.DAT	Fault analysis summary file written to by activity SCMU.
WINDOWS.PRM	General Graphical User Interface (GUI) parameter file.
WINIMD.PRM	IMD GUI parameter file.
WINPLT.PRM	PSSPLT GUI parameter file.

2.7.6. Master Directories

Any number of master directories can be used to search for files on a per file type basis. Refer to Table 2.4, "PSS®E Data File Summary" for a list of all file types. Each file type has its own master directory list which

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will be searched when looking for a file type associated with that master directory list. The master directory entry for a specific file type can also incorporate macro variables. Refer to Section 2.7.7, "Macros" for details on how macros can be created and used. By default all master directory lists are empty.

Master directories are stored in the registry under the key "Master Paths". Each entry in a master directory list must be separated by a semicolon, ';'. If the directory contains a semicolon, the directory must be quoted.

Some programs rely on an older master directories structure. As such, "hidden" master directory lists are created for backwards compatibility. Refer to Table 2.7, "PSS®E Paths and Default Master Directory" Directory for a list of file and their corresponding master directory name and list.

Table 2.7. PSS®E Paths and Default Master Directory

File(s)	Used By	Default Setting
PARMxx	graphics and printing	Pssprm=x:\Program Files\PTI\PSSE35\35.x\pssprm
WINIMD.PRM	IMD (GUI only)	Pssprm=x:\Program Files\PTI\PSSE35\35.x\pssprm
WINPLT.PRM	PSSPLT (GUI only)	Pssprm=x:\Program Files\PTI\PSSE35\35.x\pssprm

2.7.7. Macros

Macros are provided to give the user a single place to modify directory variables. Macros are denoted inside master directories by prefixing the string '$(' and postfixing the string ')' to the macro name. If the macro name contains any of these special characters, it should be quoted.

Some macros are generated automatically. Refer to Table 2.8, "PSS®E Default Macros" for a list of the automatically generated macros. A handful of macros are read only as they refer to values that are defined when the program is installed and cannot be changed without reinstalling the program.

All non-read-only macros are saved in the registry under the key "Macros".

Table 2.8. PSS®E Default Macros

Name	Default Value
$(PsseInstallPath)	C:\Program Files\PTI\PSSE35\ (default, read only)
$(HELP)	PSSHLP (read only)
$(PSSLIB)	PSSLIB (read only)
$(PROGRAMS)	PROGRAMS
$(GRPG)	PSSMAP
$(PSSPRM)	PSSPRM
$(RSPFIL)	EXAMPLE
$(PYFILE)	EXAMPLE
$(HomePath)	C:\Users\[User Name]\Documents\PTI\PSSE35\
$(ExamplePath)	C:\Users\[User Name]\Documents\PTI\PSSE35\EXAMPLE\

2.8. Deleting Files

Any file created by the user or by PSS®E may be deleted when it is no longer required. A file should be purged as a matter of course when its contents have become obsolete or invalid.

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Although an unlimited number of files may be contained in a directory, the user should be careful not to accumulate unneeded files in the working directory. If the user does not exercise discipline in this regard, confusion could result even if the user is careful in noting the significance of each file on a File Planning Sheet. Furthermore, disk space is unnecessarily wasted and, although disks have large capacities, they are finite.

Files that are not current but need to be retained for documentation or archival purposes should be kept on some off-line bulk storage medium.

The user is referred to the appropriate computer system documentation for details on the mechanics of deleting files.

2.9. File Usage Summary

The use of files by PSS®E gives the user great freedom in adapting the handling of input, the recording of cases, and the output to suit the work as observed during its progress. As with all systems that give a user great flexibility and many options, PSS®E also gives the user the responsibility of managing the options, in this case the files.

PSS®E is able, at any time, to write over the contents of a file that had previously been created. PSS®E does not generally append to files; each time that a user specifies a filename to a PSS®E file writing activity, such as SAVE or POUT, the writing commences at the *start* of that file, destroying the previous contents. Activity OPTN does offer a File overwrite option that can be set to either *Overwrite without asking* or *Ask first*.

This mode of operation has been found in thousands of man-years of use of PSS®E and its predecessors to be the preferred way of handling files. It simply requires that the user keep effective records of the files created. The best form of record is a concise written catalog to which the user can refer during the use of PSS®E, together with a systematic way of assigning filenames. The catalog may take the form of the File Planning Sheet.

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Chapter 3
Startup

3.1. Preliminaries

When PSS®E is installed on your workstation (refer to the *PSS®E Installation Guide*), the following steps are recommended prior to starting up PSS®E:

- Establish a working directory from which PSS®E is to be executed (see Section 2.1, "Directories and Files Overview"). It is strongly recommended that the default working directory (the EXAMPLE subdirectory of the main PSS®E directory) be used only for post-installation testing and demonstration purposes, and not for your engineering studies and investigations.

- Establish in your working directory those initial input data files required for your PSS®E work session (refer to Section 2.3.1, "Input Data Files" and Section 2.5, "Creating Input Files").

3.2. Starting PSSE

PSS®E may be started through one of the following methods:

- From the Windows® Start menu, select *Programs > Siemens PTI > PSSE 33 > PSS® E*.

  This is the default location as established during program installation. Other applications in the PSS®E suite of tools may also be initiated from this PSS®E Windows® Start menu.

- From the Windows® Explorer application, double-click the psse33.exe file.

  C:\Program Files\PTI\PSSE33\PSSBIN is the default directory location of psse33.exe. An alternate directory location may be specified during program installation.

- From the Windows® Start menu, select *Programs > Siemens PTI > PSSE 33 > PSS® E-33 Command Prompt* and enter `psse33.exe` (or simply `psse33`) at the DOS prompt, followed by *[Enter]*.

- Start a PSSE-33 Command Prompt and enter prog params where:

  prog = name of program (e.g., IPLAN33) that is to be executed.

  params = whatever start-up parameters are appropriate for the program.

- Double-click a previously defined PSS®E shortcut icon located on the desktop.

Do not try to close the copyright start-up window that displays during the GUI version of a PSS®E-33 program.

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3.2.1. Startup Command Options

Additional command options may be appended to the psse33.exe startup command to further direct and customize program execution. These optional arguments take the form of a keyword, sometimes followed by a value. All keywords begin with a hyphen (-).

The following command line options may be specified:

- -inifile filename

Override the default *.ini file with the one specified by filename.

- -pyfile filename

Begin the program by running the Python program specified.

- -rspfile filename

Begin the program by running the Response File specified.

- -argstr string

When -rspfile filename or -pyfile filename is also specified, use string as the argument for filename; otherwise the -argstr option is ignored.

- -buses buses

Set the program bus size level. Buses will be rounded to the next multiple of 1000, up to a maximum of 200,000. If this command line option is omitted or if *buses* is specified as zero, the current default size level is used.

- -embed

When -rspfile or -pyfile is also specified, do not display the user interface until that file has completed its execution; otherwise the -embed option is ignored.

- -resetwrk

Reset the program workspace to system defaults.

The above keywords may be abbreviated to any unique string. The options may be specified in any order. If the same option is specified more than once, the last one specified is used. The -pyfile and -rspfile options are mutually exclusive; if both are used, the -rspfile option is ignored.

Filename is interpreted as a pathname relative to the current working directory.

String and filename should be quoted if they contain embedded blanks - use the quote character ("). When the string value from -argstr is passed to the Python program or Response File, the surrounding quotes are removed. Also, any embedded pairs of quotes will be interpreted as a single quote character. For example, "abc""def" will appear to be the single 7-character string "abc def". When used with -rspfile, that resulting string will then be parsed according to the normal free-format parsing rules to assign the individual response file arguments. When used with -pyfile, that resulting string will be the value of sys.argv[1].

The above keywords may be appended to the PSS®E startup command by appending the argument to the psse33.exe path name specified in the Target field of the *[PSS®E Properties]* dialog associated with the PSSE

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shortcut icon or the Windows® Start menu *Programs > Siemens PTI > PSSE 33 > PSS® E* (see *PSS®E GUI Users Guide*, Properties Dialog).

The keywords may also be appended to the psse command prompt by selecting *Programs > Siemens PTI > PSSE 33 > PSSE-33 Command Prompt* from the Windows® Start menu and appending the command keyword to the psse33.exe command (see Figure 3.1, "Specifying the -buses Option in the PSSE-33 Command Prompt Window").

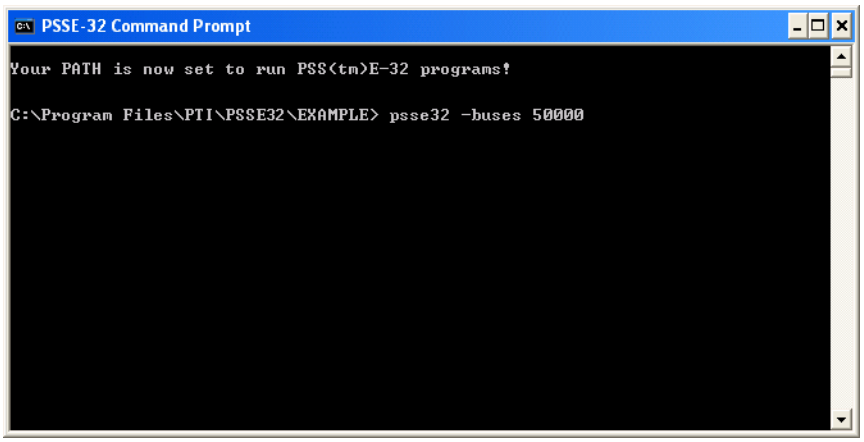

Figure 3.1. Specifying the -buses Option in the PSSE-33 Command Prompt Window

3.3. Environment

The principal power system analysis calculations of PSS®E (power flow, equivalent construction, switching studies, unbalanced fault analysis calculations, linear network analysis, and dynamic simulation) are all accessible from the main PSS®E entry point. In addition, a set of auxiliary programs (or utilities) for handling related calculations and various data format conversion applications are supplied with PSS®E (see *Additional Resources for PSS® E*, Auxiliary Program Descriptions).

3.3.1. Specifying the Maximum Number of Buses

Specifying the token -buses on the command that initiates PSS®E, followed by a number indicating the maximum number of buses that PSS®E is able to handle, determines the dimensional capacity of PSS®E. The numeric value specified must be an integer multiple of 1,000, between 1,000 and 200,000 inclusive. If this token and its associated numeric value are omitted, PSS®E is started up at the size level defined in the PSS®E Options File if one is found (see Section 2.3.9, "PSS®E Options Files", Section 3.3.3, "Program Run-Time Option Settings", and PSS®E activity OPTN), or at the default bus size level determined during program installation if no PSS®E Options File is found.

When the specified maximum bus size level is one of the standard sizes of 1,000, 4,000, 12,000 or 50,000 buses, or the largest size level of 200,000 buses, the maximum dimensional capacities are as shown in Table 3.1, "Standard Maximum PSS®E Program Capacities". When a bus size level between a pair of the standard sizes is specified, the dimensional capacities of some elements are determined by interpolating between the adjacent standard size dimensions; other elements take on the dimension of the nearest smaller standard size level. When a bus size level greater than 50,000 buses is specified, the dimensional capacities of some elements are determined by extrapolating and others take on the dimension at the 50,000 bus size level. The *I/E or S* column of Table 3.1, "Standard Maximum PSS®E Program Capacities" contains IE if interpolation or extrapolation is used or an S if the dimension of the nearest smaller standard size level is taken. As an

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example, if PSS®E is started up at 10,000 buses, the maximum number of loads, transformers, and CONs are determined by interpolating between the corresponding capacities at the 4,000 and 12,000 bus size levels, and the maximum number of areas, dc lines, and zero sequence mutuals are the same as at the 4,000 bus size level.

Table 3.1. Standard Maximum PSS®E Program Capacities

	I/E or S	1,000 Buses	4,000 Buses	12,000 Buses	50,000 Buses	200,000 Buses
TRANSMISSION NETWORK COMPONENTS						
Buses (including star point buses of three-winding transformers)	-	1,000	4,000	12,000	50,000	200,000
Loads	I/E	2,000	8,000	24,000	100,000	400,000
Plants	I/E	300	1,200	3,600	10,000	35,260
Machines	I/E	360	1,440	4,000	12,000	43,580
Non-conventional (Renewable and in-feed type) machines	I/E	20	60	120	5,600	27,230
Induction machines	I/E	20	60	120	240	710
Fixed shunts	I/E	1,000	4,000	12,000	50,000	200,000
Switched shunts	I/E	126	500	1,500	4,000	13,870
Branches (including transformers and zero impedance lines)	I/E	2,500	10,000	24,000	100,000	400,000
Two-winding transformers (including three-winding transformer members)	I/E	400	1,600	4,800	20,000	80,000
Three-winding transformers	I/E	100	400	1,200	5,000	20,000
Transformer impedance correction tables	S	16	32	64	96	96
Zero impedance lines	I/E	500	2,000	6,000	25,000	100,000
Multisection line groupings	I/E	100	400	800	1,600	4,760
Multisection line sections	I/E	250	1,000	2,000	4,000	11,890
Two-terminal dc transmission lines	S	20	40	60	100	100
Voltage source converter (VSC) dc lines	S	10	20	30	40	40
Multiterminal dc lines	S	5	5	5	20	20
Converters per multiterminal dc line	S	12	12	12	12	12
dc buses per multiterminal dc line	S	20	20	20	20	20
dc circuits per multiterminal dc line	S	20	20	20	20	20
FACTS devices	S	20	50	100	250	250
GNE devices	S	10	20	30	40	40
Interchange control areas	S	100	250	500	1,200	1,200
Interarea transfers	S	300	500	1,000	2,000	2,000
Zones	S	999	999	999	9,999	9,999
Owners	S	999	999	999	9,999	9,999
Machine owner specifications	I/E	720	2,880	8,000	24,000	87,160
Branch owner specifications	I/E	5,000	20,000	48,000	200,000	800,000

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	I/E or S	1,000 Buses	4,000 Buses	12,000 Buses	50,000 Buses	200,000 Buses
TRANSMISSION NETWORK COMPONENTS						
Zero sequence mutual couplings	I/E	500	2,000	3,000	6,000	17,840
Substations	I/E	500	2,000	6,000	25,000	100,000
Substation nodes	I/E	5,000	20,000	60,000	250,000	1,000,000
Substation switching devices	I/E	5,000	20,000	60,000	250,000	1,000,000
DYNAMIC SIMULATION ELEMENTS						
Constant parameters (CONs)	I/E	12,500	40,000	80,000	671,000	3,003,900
Algebraic variables (VARs)	I/E	7,500	16,000	32,000	650,000	3,089,500
State variables (STATEs)	I/E	5,000	20,000	40,000	442,000	2,028,800
Integer parameters (ICONs)	I/E	10,000	20,000	40,000	442,000	2,028,800
Output channels	I/E	750	3,000	8,000	30,000	116,840
User model definitions	S	500	500	500	3000	3000
Activity CHAN/CHSB model entries	S	500	2,000	2,000	10,000	10,000
Load model table entries	I/E	1,000	4,000	12,000	50,000	200,000
Bus type load models	I/E	500	2,000	6,000	25,000	100,000
Owner type load models	S	999	999	999	9,999	9,999
Zone type load models	S	999	999	999	9,999	9,999
Area type load models	S	100	250	500	1,200	1,200
All type load models	S	5	10	15	25	25
Line relay model connection table entries	I/E	25	100	240	1,000	4,000
Auxiliary signal models	S	20	30	40	50	50
Auxiliary signal injection point per two-terminal dc line	S	4	4	4	4	4
Auxiliary signal injection point per multi-terminal dc line	S	12	12	12	12	12
Auxiliary signal injection point per VSC dc line	S	2	2	2	2	2
Auxiliary signal injection point per FACTS device	S	1	1	1	1	1
Maximum number of CCT type dynamic models	S	100	250	1000	150,000	150,000

3.3.2. Setting Program Preferences

The *[Program Preferences]* dialog, which is accessible from the *Edit* > *Preferences...* menu entry, allows for the selection of variety of program options regarding its operation from the Graphical User Interface (GUI). Refer to *PSS®E GUI Users Guide*, Changing Program Preferences.

3.3.3. Program Run-Time Option Settings

Most PSS®E calculation and reporting functions recognize one or more program run-time option settings. When PSS®E is installed on the system, default settings for the PSS®E run-time options are established.

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The user may override most of the option settings described below during a PSS®E work session via activity OPTN. Furthermore, the user may change a given option setting several times during a PSS®E work session with repeated executions of activity OPTN. These option settings may then be preserved in a PSS®E Options File (refer to activity OPTN). Such an Options File may then be accessed on subsequent initiations of PSS®E as described below.

Each time PSS®E is initiated, the option settings are initialized according to:

- An options file if one is found in the user's current directory; otherwise,
- An options file if one is found in the user's home directory; otherwise,
- An options file if one is found in the PSS®E master directory PSSLIB; otherwise,
- The defaults established when PSS®E was compiled and linked.

Saved Case Specific Option Settings

The option settings described in Table 3.2, "Saved Case Specific PSS®E Program Option Settings" are preserved with the working case when it is saved with activity SAVE. When a power flow case is retrieved with CASE, the option settings contained in it override those then set in PSS®E working memory. Whenever an option setting is changed in this manner, an appropriate message is printed.

As described above, these option settings may also be preserved in a PSS®E Options File with activity OPTN.

PSS®E option settings that pertain only to the Optimal Power Flow program section are described in Sections Section 15.8.3, "Optimal Power Flow Options File: PSSOPF.OPT" and Section 15.10, "Optimal Power Flow Solution Options". They are preserved with Saved Cases and may also be preserved in an Optimal Power Flow Options File.

Table 3.2. Saved Case Specific PSS®E Program Option Settings

Option	Default Value	Description
Bus input	Numbers	Buses are selected in the interactive dialog portions of PSS®E either by bus number (n*umbers* option) or extended bus name consisting of the twelve-character alphanumeric name plus the bus base voltage (*names* option). This option *does not* apply to the bulk data input activities such as READ, TREA, RDCH, MCRE, RESQ, and TRSQ.
Bus output	Numbers	Buses are ordered in PSS®E reports in either ascending bus number order (n*umbers* option) or alphabetical extended bus name order (*names* option).
Power output	MVA	Power is reported in either MVA or kVA. This option is recognized by the following power flow reporting activities and functions: AREA,

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Option	Default Value	Description
		DRAW, GENS, GEOL, GOUT, INTA, INTZ,LAMP, LOUT, OLTL,OLTR, OWNR, POUT,RAT3, RATE, SHNT, SUBS,TIES, TIEZ,ZONE, the load reduction report (Section 12.6, "Summarizing Load Reduction"). It is also honored by the single element and subsystem data retrieval subroutines (refer to *PSS®E Application Program Interface (API)*).
Voltage output	pu	Voltages are tabulated in either per unit or kV. This applies only to activities LIST, EXAM, GENS, TLST, and Listing Control and Constraint Data
Voltage input	pu	Voltages are reported and entered in the *[Spreadsheet]*s and activity CHNG in either per unit or kV.
Transmission line data input	pu	Transmission line (*not* transformer or generator) impedances are entered and/or reported in either per unit or ohms; line capacitances are in per unit or microfarads. This option is recognized in activities CHNG, LIST, EXAM, SQLI, SQEX and the nontransformer branch *[Spreadsheet]*.
Fault analysis output units	pu	Fault analysis results are tabulated in either physical units or per unit. This option is recognized by activities SCMU, SCOP, ASCC, BKDY, ANSI and IECS.
Fault analysis output coordinates	Rectangular	Fault analysis results are tabulated in either rectangular (e.g., MW, Mvar) or polar (e.g., MVA, angle) coordinates. This option is recognized by activities SCMU, SCOP, ASCC, BKDY, ANSI and IECS.
Fault analysis modeling	3-phase	The fault analysis activities SEQD, SCMU, SCOP, ASCC and IECS can handle either 3-phase systems modeled by positive, negative, and zero sequence networks, or center-tapped 2-phase systems used in some electric traction systems modeled by positive and zero sequence networks.

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Option	Default Value	Description
Base frequency	60.0	The system base frequency in Hertz (usually 50 or 60 Hz).
Default rating set	Rate A	Establishes the default rating used by activities POUT, LOUT, and LAMP. It also is used as the default value for those activities that allow the user to select a rating set.
Tap adjustment	Disabled	Sets the mode of tap adjustment in power flow solutions. • The *Disabled* mode suppresses transformer adjustments. • In the *Step* mode of tap adjustment, transformer taps are adjusted in multiples of a user-specified tap step. • In the *Direct* mode of tap adjustment, a simultaneous continuous adjustment is made of *all* voltage controlling transformers as well as all Mvar controlling transformers and all bus voltage magnitudes. Upon convergence of the main power flow iteration, tap ratios of controlling transformers are moved to the nearest step and the solution refined with tap ratios locked at those positions.
Area interchange adjustment	Disabled	Sets the mode of area interchange control in power flow solutions. • The *Disabled* mode suppresses area interchange control. • If *Tie lines only* is selected, area swings are adjusted with an area's net interchange defined as the sum of the flows on all of its tie lines. • If *Tie lines and loads* is selected, area swings are adjusted where an area's net interchange includes tie line flows as well as contributions from loads connected to area buses that are assigned to areas other than the

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Option	Default Value	Description
		bus' area, and from loads assigned to the area that are connected to buses assigned to other areas.
Phase shift adjustment	Disabled	Enables or disables phase shift adjustment during power flow solutions.
Switched shunt adjustment	Enable all	Sets the mode of switched shunt adjustment during power flow solutions. • The *Lock all* mode suppresses all switched shunt adjustments • The *Enable all* mode allows adjustment of all in-service switched shunts that are not locked • The *Enable continuous, disable discrete* mode allows adjustment only of continuous mode switched shunts
DC tap adjustment	Enabled	Enables or disables the adjustment of dc transformer taps and of ac transformers controlling a converter of a two-terminal dc line during power flow solutions.
Non-divergent Newton power flow solution	Disabled	Activities FDNS, FNSL and the various ac contingency calculation functions may optionally be executed in a mode in which the voltage vector of diverging power flow iterations is captured and the solution terminated prior to blowing up.
Solution connectivity check	Enabled	The power flow solution activities may be instructed to perform a network connectivity check before beginning the voltage solution iterations. This ensures that all Type 1 and Type 2 buses are connected back to a swing (Type 3) bus by in-service ac branches.
Three-winding transformer to bus reporting	Disabled	When flow into three-winding transformer windings is reported in activities POUT, LOUT, LAMP, and DCLF, the other two buses

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Option	Default Value	Description
		connected to the transformer may optionally be listed.
Out-of-service branch reporting	Disabled	In the bus output blocks of activities POUT, LOUT, and LAMP, out-of-service branches connected to the bus may optionally be listed.
Line shunt reporting	Disabled	Line shunt powers may optionally be reported in activities POUT, LOUT, and LAMP.
Non-transformer branch percent loading units	Current expressed as MVA	Percent loadings on non-transformer branches may be either percent MVA loadings or percent current loadings in the reports of activities POUT, LOUT, LAMP, RATE, RAT3, OLTR, and the various ac contingency calculation reporting functions. It is also honored by the single element and subsystem data retrieval subroutines (refer to *PSS®E Application Program Interface (API)*).
Transformer percent loading units	MVA	Transformer percent loadings may be either percent MVA loadings or percent current loadings in the reports of activities POUT, LOUT, LAMP, RATE, RAT3, OLTR, and the various ac contingency calculation reporting functions. It is also honored by the single element and subsystem data retrieval subroutines (refer to *PSS®E Application Program Interface (API)*).
Multi-section line reporting	Enabled	Multisection line groupings are either recognized or ignored in activities POUT, LOUT, LAMP,OUTS, GEXM, GOUT, BKDY , DCLF, RANK, DFAX, OTDF, DCCC, TLTG, SPIL, POLY, AREA, ZONE, INTA, INTZ,TIES, TIEZ, SUBS, CMPR, CHAN, and CHSB, in the various ac contingency calculation functions, in the area interchange control option of the power flow solutions, and in the interchange subroutines of the single element data retrieval subroutines.
Fault analysis warning	Enabled	Certain warning messages can be either printed or suppressed in the

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Option	Default Value	Description
		fault analysis solution activities (sequence network isolated buses in activities SEQD, SCMU, ASCC, ANSI and IECS).
Sequence network setup	Disabled	The automatic execution of activity SEQD by activity SCMU is either enabled or disabled.
Matrix growth factor	2.0	The matrix growth factor is used in allocating working arrays by activities involving the ordering, factorization, or triangularization of network matrices. PSS®E updates this value as required, so that users do not normally need to be concerned with it.
Graphics output device		Beginning with PSS®E-31, this option setting is no longer used.

User Specified Option Settings

The option settings described in Table 3.3, "User Specific PSS®E Program Option Settings" may be preserved in a PSS®E Options File, but *are not preserved* in any PSS®E Saved Case File.

Table 3.3. User Specific PSS®E Program Option Settings

Option	Default Value	Description
Default Newton power flow solution tolerance	0.1	The Newton convergence tolerance (MW/Mvar) is initialized to this value when a new case is read into PSS®E with activity READ.
File overwrite	Overwrite without asking	PSS®E may operate in either an *Overwrite without asking* mode or an *Ask first* mode when a file that already exists is specified to a PSS®E file writing activity.
Default PSS®E size level	12000	Defines the default size level of PSS®E in terms of the maximum number of buses allowed. The value specified must be an integer multiple of 1,000, between 1,000 and 200,000 inclusive. When a new bus size level is specified and the options setting then saved to an options file, the new dimensional capacities take effect when PSS®E is restarted. This setting may be overridden by the user at the time PSS®E is initiated

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Option	Default Value	Description
		by specifying the -buses command line token followed by the desired bus dimension as described in Section 3.3.1, "Specifying the Maximum Number of Buses".
Output device page lengths	24, 58, 60 and 60, respectively	The number of lines per page on the user's terminal (in the Report tab), on the two primary hard copy output devices, and in disk files are recognized when these devices are used as the output destination of PSS®E reports.

Installation Dependent Option Settings

The option settings described in Table 3.4, "Installation Dependent Option Settings" *are not preserved* in any PSS®E Saved Case File or in a PSS®E Options File. To change these requires recompiling the PSS®E option subroutine PSSOPT.

Table 3.4. Installation Dependent Option Settings

Highest Allowed Bus Number	Defines the largest permissible number that may be assigned to a bus. This is set to 999997 at all size levels of PSS®E as supplied by Siemens PTI. The use of 5 and 6-digit bus numbers should be avoided in cases that are subsequently processed as special formats where only 4-digit bus numbers can be accommodated (e.g., IEEE Common Format).
Screen Clearing Codes	These character codes are written to the user's terminal at the beginning of each page of PSS®E output reports when the CRT terminal is selected as the output destination. Where such capability exists, these terminal-dependent codes should correspond to the character sequence required to clear the screen and position the cursor at the home position.

3.4. Startup Parameter Files

Some PSS® E programs will search for and use certain parameter files at program startup. Those programs and their parameter files are listing in Table 3.5, "Program Parameter Files" below.

Table 3.5. Program Parameter Files

Program	Parameter File
IMD	WINIMD.PRM
PSSPLT	WINPLT.PRM
PSSE33	PSSE3300.INI

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Program	Parameter File
IMD33	IMD3300.INI
PSSPLT33	PSSPLT3300.INI

3.4.1. WINDOWS.PRM Files

The term WINDOWS.PRM is a generic name for files of the form WIN*.PRM in Table 3.6, "WINDOWS.PRM Parameter Descriptions" above. The files are used to define certain parameters for the GUI of those programs, including the toolbar values. Table 3.6, "WINDOWS.PRM Parameter Descriptions" describes the parameters in the WINDOWS.PRM file.

Table 3.6. WINDOWS.PRM Parameter Descriptions

Parameter	Description
DEBUG = ON/OFF	Used to display command line input and output while using the GUI
XSCALE = real_number	Used to define x-axis scale factor to adjust for low resolution displays
YSCALE = real_number	Used to define y-axis scale factor to adjust for low resolution displays
TOOLBAR = label,command	Used to define a toolbar command for PSSPLT. Buttons may be defined for executing either activity commands or Response Files.

Any parameter can be defined in WINDOWS.PRM. Any TOOLBAR definition will start an entirely new toolbar definition.

PSS®E programs search for the parameter files using the standard search path rules as described in Section 2.7.3, "File Search Path Rules". You can copy the files from the PSSPRM directory to one of the alternate search path directories for your own customizing.

Node users may wish to modify the PSSPRM line in corresponding *.INI so that it points to a local directory where they can store copies of the parameter files in order to allow customizing their operations.

3.4.2. INI Files

The INI files are installed in the Windows directory. They contain all of the start-up and formatting instructions such as directory location, report window font, and window placement. Use this file to change such settings as fonts, printer page margins, or positions of frequently used windows. The format of the file is: parameter=value, divided into sections whose names are contains in brackets. The order in which the parameters are located in a given section of the file is unimportant. Any parameter misspelled or incorrectly specified is either ignored or, as with fonts, partly used. Setup automatically sets the start-up paths at installation time, but you may change them at any time to suit your particular directory structure. You can override the default INI file and directory location by specifying the argument, -INI pathname on the target path found in the properties shortcut field of the program's Windows shortcut.

The following sections are valid in all the INI files listed in Table 3.6, "WINDOWS.PRM Parameter Descriptions":

[Startup] Section: This section is used to define startup parameters for PSS®E. The following parameters are valid:

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Home

Defines one of the alternate directories used to automatically search for program, user, and PSS®E system files.

Path

Defines the directory path where PSS®E has been installed (required).

Pssmap

Defines the directory path for GRPG include files.

Psspgm

Defines the directory path for IPLAN programs used by PSS ®E.

Pssprm

Defines the directory path for PSS®E parameter files such as:

- PARMPR.DAT - the printer definition parameter file.
- PARMPS.DAT - the PostScript definition parameter file.

  Initial value is the subdirectory in the installation path.

Temp

Defines the directory path for temporary files created by PSS®E. If this parameter is not defined, the Windows temporary directory will be used.

Unit

Used to specify the unit used for defining other values in this file. Valid values are: inch or mm.

[FileMapping] Section: Filename: This is a special assignment that allows most files opened internally to PSS®E to be redirected to another location.

The following sections are used by PSSPLT and IMD:

[TextMargins] Section: This section is used to define margins associated with all Windows printing from PSS®E. Remember that the units are defined in the [Startup] section. The following parameters are valid in this section:

LeftMargin

Defines the left margin.

RightMargin

Defines the right margin.

TopMargin

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Defines the top margin.

BottomMargin

Defines the bottom margin.

[GraphicsMargins] Section: This section is used to define margins associated with all graphics output using a Windows driver. Remember that the units are defined in the [Startup] section. The following parameters are valid in this section:

LeftMargin

Defines the left margin.

RightMargin

Defines the right margin.

TopMargin

Defines the top margin.

BottomMargin

Defines the bottom margin.

Font Definition Sections: There are several font sections used by PSS®E. Each can be controlled by defining the appropriate section. The following table lists the valid font sections:

Table 3.7. Font Definitions

Font	Description
[PrinterFont]	Defines the font when printer output is sent using a Windows driver.
[GraphicsFont]	Defines the fixed spacing font used by graphics activities such as DRAW menus and GEXM.
[DefaultFont]	Default font used in the GUI interface for most controls such as push buttons.
[InputFont]	Defines the font used for text input fields in the GUI interface.
[ReportFont]	Defines the fixed spacing font used for report tabs in the GUI interface, including the *Progress* tab and help window.
[FormFont]	Defines the fixed spacing font used by our forms package.

Each of the font sections can contain one or more of the parameters from Table 3.8, "Font Parameters" to define the font characteristics (Windows selects the most suitable font based on the information).

You may use either PointSize or CharsPerInch to specify the font size.

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Table 3.8. Font Parameters

Font Parameter	Description	Valid Values
PointSize	Defines the font size using point size. The larger the point size, the larger the font.	Depends on the font chosen; 10 or 12 are good initial values.
CharsPerInch	Defines the font size using characters per inch (cpi). The larger the cpi number, the smaller the font.	Depends on the font chosen; 10 or 12 are good initial values.
Pitch	Specifies the pitch of the font.	Fixed, variable.
Family	Specifies the font family.	Decorative, modern, roman, script, swiss.
Quality	Specifies the output quality.	Draft, proof.
Weight	Specifies the thickness of the character to be drawn.	Thin, extralight, light, normal, medium, semibold, bold, extrabold, heavy.
FaceName	Specifies the font name such as Arial, Courier New, Helvetica, Times.	Depends on the fonts installed on your system. A good initial choice for a variable- pitch font is Arial; a good choice for a fixed-pitch font is Courier New. These fonts are installed with Windows and are available on all machines.

[ReportBufferSize] Section: This section is used to define the size, in bytes, of each report tab used by the program. The memory allocated for this is the same for other items dynamically allocated within the program and the total is limited to a size defined when the program is linked. The following parameters with default values are:

Activity Selector:	5000000
Reports:	5000000
Help:	61440

Making these numbers too large could cause the program to crash at startup or fail to allocate memory later on in its execution.

When examining the *.INI files, keep in mind that anything preceded by a semi-colon (;) represents a comment and is not read by the program.

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Chapter 4
Operating Modes

This chapter describes those procedures necessary to access PSS®E from a terminal workstation. Included are the methods used to run PSS®E using the graphical user interface (GUI), interactive command line, or batch mode, to specify the disposition of PSS®E reports, and to incorporate the user's connection subroutines into the PSS®E activity structure.

4.1. How to Approach PSS®E

The new user, after reading the preceding sections, may form the impression that PSS®E is a complex and difficult program. It must be recognized, however, that PSS®E is an extremely powerful tool that allows its user to handle a very broad range of power system calculations using a single integrated program system. Its apparent complexity is, accordingly, more a function of the many different types of data needed in power system work than it is a function of PSS®E itself. The user will find that, after a very few work sessions with PSS®E, driving PSS®E will have become quite comfortable; and attention is refocused from concern over which buttons to push to solving the engineering problem.

The new user approaching PSS®E should recognize the need to gain a firm understanding of three separate aspects of PSS®E:

1. The roles of the files in handling problem input data and in building up a library of working cases that represent system conditions at selected stages of a study.

2. The capabilities of the individual PSS®E activities and the sequencing of these activities to achieve a desired power flow, simulation, or short circuit calculation.

3. The process of setting up the dynamic model of the system to be simulated.

The role of the PSS®E files has been discussed in Chapter 2, *File System*. The detailed format requirements of each user-filled file are described in Chapter 5, *Power Flow Data Entry, Modification, and Display Activities* through Chapter 22, *Miscellaneous Activity Descriptions* in the activity description of the activity that reads it. The mechanics involved in the second and third aspects listed above are described in the remainder of this manual, and the corresponding engineering basis is described in the *PSS®E Program Application Guide*.

After an understanding of PSS®E has been established, the new user should recognize the following basic steps in the initiation of any PSS®E dynamics study:

1. Identify all the files that will be needed including input data files, Saved Case Files, Snapshot Files, and Channel Output Files. Write out a summary listing the names and content of all of these files using a PSS®E File Planning Sheet such as that suggested in Section 2.9, "File Usage Summary".

2. Collect all power flow data and establish one or more valid power flow Saved Case Files before proceeding to dynamic simulation setup.

3. Develop the dynamic simulation data and equipment modeling subroutines using the methodology given in Chapter 23, *Dynamic Simulation Setup and Procedures*.

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4. Link the connection subroutines into the PSS®E dynamic simulation activities as described in Section 4.10, "Loading the Connection Subroutines".

5. Read the dynamic simulation data into PSS®E working memory and establish one or more valid initial condition Snapshot Files corresponding to the Saved Case Files from step 2.

6. Use the auxiliary programs VCV and IMD and the PSS®E activities ESTR, ERUN, GSTR, and GRUN to verify the dynamic simulation data. See also *Additional Resources for PSS®E*, V Curves and Motor Parameters, respectively.

7. Make power flow and simulation runs, creating additional Saved Case and Snapshot Files as needed to accommodate significant system conditions developed during the study.

A typical PSS®E File Planning Sheet for the initial stages of a PSS®E study is provided. The importance of this file planning cannot be overemphasized.

4.2. Activity Selection

There are two methods of interactive operation of PSS®E: GUI Windows and Command Line Interface. In both of these modes of operation, the user specifies the PSS®E activity or function to be performed, along with any input data required by it. The manual *PSS®E Application Program Interface (API) documents the calling sequences and data requirements of all* PSS®E API routines.

When operating PSS®E via the GUI, a single dialog box often provides the equivalent of multiple API routines or calls to API routines. It can also provide alternatives of several API routines. The question and answer dialog of the line mode may require several prompts and user responses for the same functionality. The correspondence between input fields contained in these customized dialogs to the arguments of the API calls and to the prompts issued when using the line mode should be self-evident.

When an activity is initiated using either the GUI or command line, it must be completed in the same mode. The next activity to be executed may then be initiated in either mode.

4.2.1. GUI Windows

The default set of windows displayed when PSS®E is started up is the highest level window in PSS®E. These windows provide the following options and program feedback:

- Select activities and other functions to be executed (see below).
- Receive progress messages and reports from PSS®E (see Section 4.4, "Virtual Output Devices").
- Issue interrupt control codes (see Section 4.3, "Interruption of PSS®E by the User").

Across the top of the PSS®E interface is a menu bar containing several categories of activities. Activities are grouped together according to the general type of function they perform. Selecting one of these menu bar entries causes a pull-down menu to appear. Each pull-down menu contains a list of submenus and/or a list of application functions and/or activities. Refer to *PSS®E GUI Users Guide* for details on activity execution in windows mode.

4.2.2. Command Line Interface

Access to the PSS®E Command Line Interface (CLI) is available by selecting the *View > CLI Window* menu entry. Select either PSS®E Response or Python from the Command language: drop-down list next to the CLI window. When PSS®E Response is selected, either batch commands or line mode commands may be

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entered in the Enter command: field. The command is executed by pressing the [Enter] key. Activities initiated in this manner execute using the command line, with dialog prompts displayed in the Progress tab of the output view and additional commands or user responses (in the case of the line mode) entered at the Enter command: field. See Section 17.11, "Line Mode Commands" and Section 17.5, "Batch Commands" for more details.

Refer to PSS®E Command Line Interface (CLI) Users Guide for details on activity execution using the command line.

4.2.3. Automation Methods

PSS®E may also be operated using any of several program automation methods as described in Chapter 17, *Program Automation*.

4.3. Interruption of PSS®E by the User

As an interactive program system, PSS®E activities are sensitive to an interrupt from the user's terminal for the purpose of redirecting the course of the computation. This facility gives the user the ability to terminate nonconvergent iterations, suppress unwanted listings, select convergence monitor options, and, in general, direct the sequencing of activities based upon the user's observation of processing results to date.

PSS®E interrupts are triggered by pressing the *[F10]* function key on the keyboard.

Pressing the *[F10]* function key while using the PSS®E GUI brings up the *Interrupt Handler* window, which contains:

- A toggle button box containing all the interrupt control codes to which PSS®E activities are sensitive (see Table 4.1, "Interrupt Codes").
- A [*Resume]* pushbutton through which PSS®E execution is continued with any selected interrupt control codes passed to the current activity.

The *[Interrupt Processing]* dialog contains all of the interrupt control codes used throughout PSS®E. The interrupt control codes to which an individual activity is sensitive are given in the activity descriptions in Chapter 5, *Power Flow Data Entry, Modification, and Display Activities* through Chapter 22, *Miscellaneous Activity Descriptions* of this manual; any other interrupt control codes that are specified to the activity are ignored.

Pressing the *[F10]* function key while using the PSS®E console interface, PSSECMD, will generate the following prompt:

```
Yes?
```

Acceptable responses are:

- *[Enter]* - Resume PSS®E execution is continued with any selected interrupt control codes passed to the current activity.
- *QUIT* - Stop the program.
- *START*, followed by a space delimited set of the two letter interrupt codes that you want to enter (see Table 4.1, "Interrupt Codes").

Note that QUIT and START can be abbreviated to as little as one letter, and that START without any following interrupt codes just resumes the program.

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In working with large cases, there may be a delay before an *[F10]* function keystroke is acknowledged by PSS®E. During this delay, there is no visible evidence that the interrupt request has been received by PSS®E.

In addition, pressing the *[F10]* function key before running a PSS®E activity will have no effect. For an activity to acknowledge an interrupt and bring up the *Interrupt Handler* window, the *[F10]* key must be pressed *after* the activity is invoked.

Table 4.1. Interrupt Codes

Interrupt Code	Action
AB	Depending on type of activity: • End the activity • End activity input/output • Abandon solution activity following completion of the next iteration • Suppress initial conditions summary/reports • Suppress messages • Force a pause by setting TPAUSE = now.
CH	Tabulate output channel values at each time step.
CL	Close all active Response Files.
CM	Print the network solution convergence monitor.

DC	Tabulate dc line conditions after each power flow iteration (activity SOLV):
	The data printed for each noncapacitor commutative two-terminal dc line includes the dc line name followed by two lines including the following quantities:
	`IPR IPI ALPHA GAMMA VDCR VDCI`
	`TAPR TAPI DCCUR`
	`PACR QACR PACI QACI`
	The information printed for each capacitor commutated two-terminal dc line includes the dc line name, followed by an iteration report of the Newton solution run for one or more operating modes, followed by:
	`ALPHA_R MU_R VDC_R(kV) TAP_R PAC_R`

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Interrupt Code	Action

	`QAC_R DC_CURRENT GAMA_I MU_I VDC_I(kV) TAP_I PAC_I QAC_I ALPHA_I`
	
	For multi-terminal lines, the dc line number is followed by conditions at each converter bus:
	
	`bus # ANGLE TAP VDC DCCUR PAC QAC`
	
	For VSC dc lines with both converters in-service, the dc line name is followed by line and converter bus conditions as follows:
	
	• dc voltages at the power and voltage controlling converters and dc current
	
	• if a limit violation is relieved at this iteration, the converter bus at which it occurs, the present ac injections at that bus, and the reduced values are tabulated, followed by the new dc voltages and the dc current
	
	• dc power, converter losses, and ac power at the power controlling converter
	
	• dc line losses
	
	• dc power, converter losses, and ac power at the voltage controlling converter
	
	For VSC dc lines with only one converter in-service, the dc line name is followed by the dc power, converter losses, and ac power at the in-service converter.

FD	Tabulate FACTS device conditions after each power flow iteration. The data printed for each FACTS device includes conditions at the sending and terminal buses, series voltage and current in polar coordinates, series voltage with respect to both series current and sending end voltage in rectangular coordinates, and an indication of those quantities currently at a limit.
MO	Tabulate the conditions for each in-service induction machine after each iteration. The data printed for each induction machine includes the bus number and machine identifier, the voltage magnitude at the bus, the voltage error in the induction machine iteration, slip, mechanical torque, mechanical power, the magnitude of the voltage across the mutual reactance, the saturation multiplier ALFA, the partial

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Interrupt Code	Action
	derivative of mechanical output power with respect to slip, the partial derivative of mechanical output power with respect to terminal voltage, the partial derivative of mechanical load with respect to slip, and the apparent active and reactive power flowing into the induction machine at the ac bus.
IP	End execution of the IPLAN program.
MO	Print induction motor local iteration convergence monitor.
NC	Suppress convergence monitor.
NM	Suppress automatic adjustment monitors.
NV	Suppress incorrect plant VAR limit report.
TI	Print the present value of simulation TIME.

4.4. Virtual Output Devices

PSS®E categorizes all text output into four types:

Progress	Informational messages. This can include intermediate values of solutions, identifying data from a file, copies of commands entered in the CLI, and warning about data conditions that may or may not warrant further investigation by the user.
Alerts	Error messages. Something didn't work, or some condition was encountered that prevented the program from performing the task it was asked to do.
Reports	Results of calculations and or analysis, or summaries of data from the program's memory.
Prompts	The program is asking you a question. The PSS®E GUI doesn't use prompts, but they can be generated by automation files.

In concept, PSS®E decides what kind of output it is producing, and the user decides where that kind of output goes. By default, in the PSS®E GUI, the four output types are handled as follows:

Progress	These messages appear in the Progress tab of the output bar. Messages internally marked as warnings are copied to the Alerts/Warnings tab.
Alerts	These messages appear in MsgBoxes to which the user must respond. They are copied to the Progress tab and to the Alerts/Warnings tab.
Reports	This output is sent to the Report tab(s) of the output bar (whether one or multiple tabs is used is a preferences setting).
Prompts	These will not be shown until a request for input is generated. Normally a special input window will then be created showing the prompt(s) and request-

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	ing a response. This normal operation is suppressed for line mode dialog, which is strictly directed to the Progress window.

If there is no GUI, such as in PSSECMD or when using PSS®E's API from Python, all the virtual devices default to the console. If there is no console (such as with Pythonw) the output will appear in a series of MsgBoxes.

Any of the four output types can be redirected in the PSS®E GUI by selecting I/O Control > Direct xxx output, where xxx is the output type. This functionality can be accomplished via the PSS®E API as well. The table below lists the API routine to redirect each type of output, and other commands that pertain to each output type.

	Redirection API routine	Redirection Line Mode Command	API routine to write a line of text	IPLAN command to write a line of text
Progress	progress_output	PDEV	progress	PRINT
Alert	alert_output	ODEV	alert	
Report	report_output	OPEN	report	REPORT
Prompt	prompt_output	ODEV	prompt	ASK

When virtual devices are redirected to files, they can each be directed the same file. Output will appear in the file in the order in which it was generated by the program.

Activities that write data files (e.g. RAWD) will write to either a named file (i.e. named by the user) or the current reporting device.

The Progress and the Report virtual devices can be destinations for any of the other devices, including each other (circular redirection is checked for and disallowed). One way to collect all the output in one place is the redirect the other devices to the Progress device, and then redirect the Progress device to where you want it all to go.

One destination that can be used for any of the devices is POM-Data-ReadingPowerFlowDataAdditionsFrom-TheTerminal, in which case no output of that type takes place at all.

4.4.1. User-Written Dynamics Models

Dynamics model programs generally write their output to either the unit number that is stored in the variable IPRT, or in LPDEV, or in ITERM. This output actually goes to temporary files and then is read back in and written to the Report device in the case of IPRT, and else to the Progress device.

An alternative method, one that involves less I/O, is to use the Fortran routines defined for the API routines that write a line of text to a virtual device.

4.4.2. Graphical Output

Graphics are a feature of GUI, and require the GUI to be created. Diagrams and Plots must be created in the GUI after which they can be copied to a file or to a printer. These operations can be recorded and controlled through automation files.

4.5. Line Mode Output Commands

When using the GUI or using the API directly report output goes to the current report device. The line mode, however, recognizes whether the report device has ever been set by the user and, if it has not, every activity

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that writes to the report device will explicitly ask where the output should go. When operating in this mode it is equivalent to setting the report device, running the activity, and resetting the report device back to the default.

Once the report device has been set explicitly this mode of operation terminates. OPEN is the line mode command to set the report device.

Line mode activity CLOS is provided to return to the original initial state where each activity in the line mode will ask for the report device.

PDEV is the line mode activity to redirect the progress device.

The prompt device and the alert device are not distinguished in the line mode, and both are controlled by the activity ODEV.

Activities that write data files will present the following choices for directing output:

```
0 FOR NO OUTPUT                1 FOR REPORT DEVICE
               2 FOR A FILE:
```

Output redirection activities will present the following choices in addition to those above:

```
3 FOR DEFAULT PRINTER
               6 FOR ALTERNATE SPOOL DEVICE
```

When OPEN has not been run, other reporting activities will display an output selection menu and will have the following additional choice:

```
7 FOR PROGRESS DEVICE
```

In all cases entries of 0 through 7 will be accepted, but values that do not correspond to choices displayed will default to other valid choices. The values that are skipped represent choices that are no longer meaningful in the current program, but may still exist in recorded automation files.

4.6. Automation

PSS®E offers several methods of automation, i.e. mechanisms to control PSS®E execution other than by direct user interaction. These can be used to develop macro-type applications, i.e. files of set procedures to be used interactively, or for unattended operation (batch), where the PSS®E can run a pre-set series of operations to completion with no further interaction from the user. The operations can be fixed, or can be implemented in programming languages to whatever level of complexity is desired. Chapter 17, *Program Automation* describes these features in detail.

4.7. Extended Modeling in Dynamics Simulation

PSS®E's Dynamics Simulation module comes with hundreds of available models that can be used for network equipment built into the program. No special steps need to be taken by the user beyond the data requirements for the models.

Additional models can be provided in user-created libraries. Dsusr.dll is assumed as the default name for a user-created library and is automatically loaded if it is found. Additional model libraries can be identified to PSS®E via the ADDMODELLIBRARY API routine (see PSS®E Application Program Interface (API)).

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Not every model can be identified to PSS®E by its data requirements. For those models, PSS®E uses connection routines. These subroutines are called CONEC and CONET, and are normally built as part of a dsusr.dll file.

Procedures to build user libraries are discussed in Section 1.3, "Extending PSSE with User DLLs" and Section 1.4, "PSSE Support for Building User DLLs". Model writing is discussed in Chapter 25, *Model Writing*.

4.8. Subsystem Selection

4.8.1. Windows Mode

Most PSS®E activities allow the user to specify processing the current subsystem. In addition, the appropriate *[Subsystem Selector]* dialog may be opened from the PSS®E interface menu and the current subsystem redefined before the activity is initiated. When a subsystem definition is modified, it becomes the current subsystem and remains the current subsystem until it is changed.

There are four types of subsystem: bus, area, owner, and zone. Menu items in the *Subsystem* menu allow access to the four *[Subsystem Selector]* dialogs.

Most reporting and processing activities that operate on a subsystem of the working case operate on a bus subsystem. The selection criteria available for bus subsystems are buses, areas, zones, owners, and/or base voltages; these are handled in a manner that is logically equivalent to the OPT suffix using line mode. Area, owner, and zone subsystems are used for those reports which, by definition, are naturally restricted to areas, owners, or zones (e.g., GUI menu selection *Power Flow* > *Reports* > *Area* / *owner* / *zone totals*...). The four types of subsystems exist simultaneously, and changing one has no effect on any of the others; e.g., changing the current area subsystem does not affect the area selections of the current bus subsystem.

4.8.2. Command Line

Generally, when no suffix is specified in selecting many of PSS®E's reporting and processing line mode activities, either all buses are processed or the user specifies the buses on which the processing is to take place. For details, refer to the individual activity descriptions in *PSS®E Command Line Interface (CLI) Users Guide*. When using PSS®E API routines directly through Batch or Python commands, setting the subsystem is a separate API call which the API routine that uses the subsystem refers to by number.

4.8.3. Subsystem Numbers

Four different subsystems, of each type, may be defined in PSS®E, numbered 0 through 3. The Subsystem menu entry in the GUI always sets subsystem 0 and the spreadsheet displays react to that. Some dialogs require a temporary subsystem that is different from the primary subsystem; subsystem 1 is used for that purpose. The line mode always uses subsystem 1. API routines can refer to any subsystem number 0 through 3.

4.9. Subsystem Reporting

When buses are selected by interchange area, buses are displayed in groups that are ordered by ascending area number, and in numerical or alphabetical order within each selected area grouping, according to the bus output option in effect (see Section 3.3.3, "Program Run-Time Option Settings"). Similarly, when buses are selected by zone or owner, buses are displayed in groups that are ordered by ascending zone or owner number, and in numerical or alphabetical order within each selected zone or owner grouping, according to the bus output option in effect.

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If both area and zone selection are specified, buses are displayed grouped in ascending area number order. Within each area grouping, buses are grouped by ascending zone number, and in numerical or alphabetical order within each selected zone subgrouping, according to the bus output option in effect.

If both area (or zone) and owner selection are specified, buses are displayed grouped in ascending area (or zone) number order. Within each area (or zone) grouping, buses are grouped by ascending owner number, and in numerical or alphabetical order within each selected owner subgrouping, according to the bus output option in effect.

When area, zone, and owner are all used as subsystem selection criteria, the grouping hierarchy is area, then zone, then owner.

When selection by bus is specified, buses are displayed in the order specified by user input. If bus selection has been specified as well as area, zone, and/or owner selection, the buses within each area, zone, and/or owner groupings are output in the order specified by user input.

4.10. Loading the Connection Subroutines

To utilize the dynamic simulation activities of PSS®E, two connection subroutines, CONEC and CONET, must be provided. The function of these two FORTRAN subroutines is to associate (connect) dynamic equipment and monitoring models with network buses. In addition, six user model calling routines, USRXXX, USRLOD, USRREL, USRAUX, USRDCL, and USRFCT are required for the calling of user-written plant, load, line relay, auxiliary signal, dc line, and FACTS device models, respectively. After being prepared using the methodology given in Chapter 23, *Dynamic Simulation Setup and Procedures*, these subroutines must be incorporated into the PSS®E structure. The procedure is as follows:

1. Compile these subroutines. It is strongly recommended that the command file, which may optionally be generated by activities DYRE and SRRS, be used for this purpose so that the correct compiler options are used and the object code is stored in the files expected in step 2.

2. Having compiled error-free object code of these subroutines, link them into PSS®E. Issuing the command CLOAD4 (for those user models for which the user would write a model code) executes the linking command procedure CLOAD4, which builds an executable program file in the user's directory.

4.11. Activity Suffixes

Command line only. Many output reporting activities and several processing activities make use of the optional suffixes to enable the user to restrict the activity's reporting or processing to a selected subset of the working case. These are described in detail in *PSS®E Command Line Interface (CLI) Users Guide*.

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Chapter 5
Power Flow Data Entry, Modification, and Display Activities

5.1. Retrieving a Power Flow Saved Case File

CASE

Run Activity CASE - GUI
File > Open... *[Open]* Saved Case file (*.sav)
Interrupt Control Codes
None

The case retrieval activity CASE restores the contents of a previously saved power flow Saved Case File into the working case. The contents of the working case are overwritten while the contents of the specified Saved Case File are unchanged.

The CASE API routine requires that the user designate the name of the Saved Case File to be opened; refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity CASE.

If an asterisk (*) is specified as the name of the Saved Case file, the last Saved Case file accessed by either activity SAVE or activity CASE in the current session of PSS®E is used.

If the file to be used is not in a Saved Case File format recognized by this release of PSS®E, the following error message is generated:

```
FILE filename NOT IN SAVED CASE FORMAT
```

Then activity CASE ends. To recover, simply select activity CASE again, specifying a valid Saved Case File filename.

If the Saved Case exceeds the current capacity limits of PSS ®E, an appropriate message is printed along with the case title and dimensional information. PSS®E then redimensions itself to a size level sufficient to accommodate the Saved Case. If the dynamics data arrays had previously been allocated, dynamics data capacity limits are also increased. Refer to Section 3.3.1, "Specifying the Maximum Number of Buses".

Generally, activity CASE is able to access Saved Case Files filled by activity SAVE of the current and previous releases of PSS®E. However, a Saved Case File written by the current version of PSS®E is normally not able to be read by earlier releases of the program.

Activity CASE is able to access Saved Case Files written on other computer platforms.

Activity CASE is designed to access only Saved Case Files written by PSS®E's activity SAVE; it *should not* be used to attempt to read files written by any other programs.

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When picking up Saved Case Files created prior to PSS®E-33, the solution parameters MXTPSS and MXSWIM are set to 99 and 10, respectively. For those Saved Case Files with sequence data, any buses with non-zero exceptional negative and zero sequence bus shunt loads have these quantities shared among the in-service loads at the bus. If there are no in-service loads present at the bus, a new load is introduced at the bus, with negative and zero sequence loads corresponding to the bus shunt load quantities, and no positive sequence load; this new load is assigned the load identifier "SC" unless an out-of-service load with that identifier is already present at the bus.

When picking up Saved Case Files created prior to PSS®E-32, branch circuit identifiers are examined to check that none contain an at sign (@) as their first character (indicating a breaker; refer to Non-Transformer Branch Data). If any such branches are detected, they are tabulated and the user is instructed to specify a character to which these at signs are to be changed. A check is made to ensure that this substitution does not result in two or more branches with the same circuit identifiers connecting a pair of buses; if it does the user specifies another character to be used. Then a similar check is made for circuit identifiers with an asterisk (*) as their first character (indicating a switch; refer to Non-Transformer Branch Data); if any are detected, the user is instructed to specify a substitute character as described above.

When picking up Saved Case Files created prior to PSS®E-30.3, the solution parameters VCTOLQ and VCTOLV are set to default values. This results in matching the convergence criteria used by PSS®E-30.0 through PSS®E-30.2 (refer to Section 6.5.1, "Characteristics of Activity FNSL").

When picking up Saved Case Files created prior to PSS®E-27, transformers with non-zero values of charging and winding 2 side (previously referred to as the untapped side) line shunt in the positive sequence are logged at the *Progress* device (refer to Section 4.4, "Virtual Output Devices"); charging is set to zero, and the winding 2 side line shunt is added to the winding 1 side line shunt and then the winding 2 side line shunt set to zero. In the zero sequence, zero sequence transformer line shunt is converted to the PSS®E-31 transformer model wherever possible; non-zero transformer charging, and non-zero line shunts that were not converted, generate an alarm and are set to zero. Non-zero values of line length for transformers are set to zero but are not logged.

When picking up Saved Case Files created prior to PSS®E-20, zone assignments are checked and any bus assigned to zone zero is reassigned to the zone with the largest permissible zone number. The number of buses for which zone assignments are so changed is tabulated at the *Progress* device.

When picking up Saved Case Files created prior to PSS®E-18, branch circuit identifiers are examined to check that none contain an ampersand (&) as their first character (refer to Non-Transformer Branch Data and Multi-Section Line Grouping Data). If any such branches are detected, they are tabulated and the user specifies a character to which these ampersands are to be changed. A check is made to ensure that this substitution does not result in two or more branches with the same circuit identifiers connecting a pair of buses; if it does the user specifies another character to be used.

Very old Saved Case Files created prior to PSS®E-7 cannot be accessed directly by activity CASE. Such files must be converted to PSS®E-7 Saved Case File format by the auxiliary program CASCNV, which was supplied with PSS®E-7 through PSS®E-10.

Following the successful restoration of the Saved Case into the working case, the two-line case title is printed followed by the time and date at which the case was last saved.

Whenever any of the PSS®E run-time option settings contained in the Saved Case File differ from those set in PSS®E's working memory, the settings from the Saved Case are established and an appropriate message is printed (refer to Section 3.3.3, "Program Run-Time Option Settings").

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Additional Information
PSS®E GUI Users Guide,Retrieving a Power Flow Saved Case File *PSS®E Command Line Interface (CLI) Users Guide*, Retrieving a Power Flow Saved Case File *PSS®E Application Program Interface (API)*, CASE See also: Section 5.45, "Creating a Saved Case File" Section 22.14, "Listing Saved Case or Snapshot Files"

5.2. Reading Power Flow Raw Data into the Working Case

READ

Run Activity READ - GUI
File > Open... *[Open]* Power Flow Raw Data File (*.raw) Power Flow Data File, Options (*.raw)
Interrupt Control Codes
AB

The bulk power flow data input activities READ and TREA read hand-typed power flow source data and enter it into the power flow working case, rearranging it from its original format into a computationally oriented data structure in the process. The source data records are read from a Power Flow Raw Data File.

One of several PSS®E API routines is used to implement these activities, depending on which of their various options the user selects (see Section 5.2.2, "Operation of Activity READ"). While several of them require additional data, each of these API routines requires that the following two data input items be specified:

- the name of the Power Flow Raw Data File that is to be read.
- a flag for bus number or bus name specification on input data records. When 1 is specified for this data item, those data items that designate ac buses on all records except bus data records may be specified as either Extended Bus Names enclosed in single quotes or bus numbers. Otherwise, bus numbers *must* be used to designate ac buses on these data records.

5.2.1. Power Flow Raw Data File Contents

For more information please see *Power Flow Raw Data File*.

5.2.2. Operation of Activity READ

The following API routines are used to implement activity READ. Each of them includes among its input data items the name of the Power Flow Raw Data File to be read and the flag for selecting the bus *names* input

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option of activity READ. (see Section 5.2, "Reading Power Flow Raw Data into the Working Case" and Extended Bus Names).

Read	Standard READ of a Power Flow Raw Data File in the format of the PSS®E release indicated in the file. No other inputs.
ReadRawVersion	Standard READ of a Power Flow Raw Data File in the format of the current or a prior release of PSS®E. It accepts as input a character string indicating the PSS®E release.
ReadSub	Subsystem READ of a Power Flow Raw Data File in the format of the PSS®E release indicated in the file. It accepts as input several data items defining the subsystem for which the data records are to be read and other processing options.
ReadSubRawVersion	Subsystem READ of a Power Flow Raw Data File in the format of the current or a prior release of PSS®E. It accepts as input a character string indicating the PSS®E release along with several data items defining the subsystem for which the data records are to be read and other processing options.

As data records are read, a message is displayed at the *Progress* device at the start of each new category of data.

Before it has completed reading its input data, activity READ may be ended by entering the AB interrupt control code (refer to Section 4.3, "Interruption of PSS®E by the User"). Activity READ checks for an interrupt following processing of each group of data records corresponding to 50 equipment items.

Bus Names Input Option

When the bus *names* input option of activity READ is enabled, data fields designating ac buses on load, fixed shunt, generator, induction machine, non-transformer branch, transformer, area, two-terminal dc line, VSC dc line, multi-terminal dc line, multi-section line, FACTS device, switched shunt, and GNE device data records may be specified as either Extended Bus Names enclosed in single quotes or as bus numbers. Otherwise, bus numbers *must* be used to designate ac buses on these records.

Use of the bus *names* input option of activity READ requires that all buses be assigned unique extended bus names. While reading each bus data record with this option enabled, if a bus with the same extended bus name but a different bus number is present in the working case, an error message is printed, the record is ignored, and processing continues.

Bus Sequence Numbers

As each bus data record is read, activity READ assigns to each new bus (i.e., a bus not previously read) a *bus sequence number,* which defines the location of data for the bus in the various bus data arrays. Bus sequence numbers are assigned sequentially starting with 1 in the order in which bus data records are read.

Plant and Machine Sequence Numbers

Each bus for which a generator data record is read is assigned a *plant sequence number*, which defines the location of its plant-related generator data in the plant data arrays. Data for each plant contains a *machine*

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sequence number assigned for each machine for which a generator data record is read. This number defines the location of its machine-specific data in the machine data arrays. Plant and machine sequence numbers are assigned sequentially starting with 1 in the order in which generator data records are read. It is permissible to enter a generator data record for a bus that was assigned a type code of 1 or 4 during bus data input. (Refer to Generator Data and activity MCRE.)

Load Sequence Numbers

Each load introduced into PSS®E is assigned a *load sequence number*, which defines the location of its data in the load data arrays. Load sequence numbers are assigned sequentially starting with 1 in the order in which load data records are read.

Fixed Shunt Sequence Numbers

Each fixed shunt introduced into PSS®E is assigned a fixed shunt *sequence number*, which defines the location of its data in the fixed shunt data arrays. Fixed shunt sequence numbers are assigned sequentially starting with 1 in the order in which fixed shunt data records are read.

Induction Machine Sequence Numbers

Each induction machine introduced into PSS®E is assigned an induction machine sequence number, which defines the location of its data in the induction machine data arrays. Induction machine sequence numbers are assigned sequentially starting with 1 in the order in which induction machine data records are read.

Branch Sequence Numbers

Each ac branch introduced into PSS®E is assigned a *branch sequence number*, which defines the location of its data in the branch data arrays. Branch sequence numbers are assigned sequentially starting with 1 in the order in which branch data and transformer data records are read.

Transformer Sequence Numbers

Each two-winding transformer is assigned a *two-winding transformer sequence number*, which defines the location of its data in the two-winding transformer data arrays; it is also assigned a branch sequence number. Similarly, each three-winding transformer is assigned a three-winding transformer sequence number, as well as three two-winding transformer sequence numbers and three branch sequence numbers (see Three-Winding Transformer Notes). Transformer sequence numbers are assigned sequentially starting with 1 in the order in which transformer data record blocks are read.

5.2.3. Change Case Data in a Standard READ

When data is to be added to the network contained in the working case, the IC data item on the first input record must be set to 1 (refer to Case Identification Data). New buses, loads, fixed shunts, generators, induction machines, branches, transformers, and other equipment items are treated in the same manner as in base case data input. Bus sequence numbers, machine sequence numbers, and so on are assigned starting with the next available location in the respective data arrays.

When entering data for an existing piece of equipment in activity READ, complete data records *must be entered*; omitted data items take on their default values rather than retaining their previous values. Activity RDCH should normally be used for this function rather than activity READ or TREA (refer to Section 5.8, "Reading or Changing Power Flow Data").

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When the bus names input option of activity READ is enabled, activity READ checks for the presence of duplicate extended bus names in the working case *before* it starts to read bus data records. Any violations are logged at the *Progress* device and activity READ is prohibited from executing.

In the change case mode, after reading the case identification data, activity READ prints a warning message at the *Progress* device if generators have been converted (see Section 13.2, "Converting Generators"). After generators are converted, machine impedance data (MBASE, ZSORCE, XTRAN, and GTAP; refer to Generator Data) *must not* be changed.

Merging Cases

When two or more power utility organizations, or any owners of power flow information, wish to create a jointly representative power flow case, it is likely that they are using the same numbers in their independent files to represent buses, areas, zones, owners and transformer impedance correction tables unique to each case. Merging such system models requires the avoidance of overlapping or conflicting numeric identifiers, as well as conflicting names of dc lines, FACTS devices, and GNE devices. A pre-requisite, therefore, is to ensure that the system models to be merged are examined to verify the absence of such data conflicts. If conflicts exist, it is necessary to apply renumbering operations and/or other data changes in at least one of the cases to be merged. Refer to activities BSNM, ARNM, OWNM, and ZONM on the methods to apply the renumbering function. Similar operations may be required for auxiliary files dependent on bus numbering (see activity RNFI).

To merge two power flow Saved Cases, designated Case A and Case B, first resolve any numbering and naming conflicts. Then open Case B (see Section 5.1, "Retrieving a Power Flow Saved Case File"), and, using activity RAWD, create a Power Flow Raw Data File representation of it. This Power Flow Raw Data File must be formatted so as to add data to the working case rather than to initialize the working case; that is, IC on the first data record must be 1 (see Case Identification Data).

Next, open Case A and enter the Power Flow Raw Data representing Case B. At the completion of activity READ, the newly introduced data from Case B is contained in the working case together with the existing data of Case A.

5.2.4. Subsystem READ

Activity READ has a supplementary mode of operation used to add to the working case a subsystem of the network where the complete representation is contained in a Power Flow Raw Data File. The subsystem to be read may be defined by area, zone, owner, base voltage, or a combination of two or more of these subsystem selection criteria.

In a subsystem READ, the IC data item on the first record of the Power Flow Raw Data File is ignored. Rather, one of the data items entered at the API routine is an append flag that overrides the IC value specified in the file. Like IC, this flag indicates that either: the working case is to be cleared and initialized before reading data; or data is to be appended to that already in the working case.

An input data item allows selection of one of the following data input functions:

1. Add only data from within the subsystem: all buses and their connected equipment in a specified subsystem contained in the Power Flow Raw Data File are appended to the working case.

2. Add only the tie branches connected to the subsystem: all branches in the Power Flow Raw Data File for which both buses are in the working case and only one bus is in a specified subsystem (ties) are appended to the working case.

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3. Add subsystem data and its ties: all buses and their connected equipment in a specified subsystem as described in (1), along with its ties as described in (2), are appended to the working case.

An input data item may be used to enable a *boundary bus* identification option. When this option is enabled, any in-service bus that is in the specified subsystem and connected to a bus that is not in the working case may be identified as a boundary bus; the type code of each boundary bus is changed from 1, 2, or 3 to 5, 6, or 7, respectively. Refer to activity EEQV for further discussion of boundary buses.

When the subsystem to be processed is specified by voltage level, it is not possible to process buses at two different voltage levels while omitting those buses at intervening levels. Multiple executions of activity READ are required to accomplish this.

When appending tie branches to the working case, one (and only one) of the two subsystems that are to be joined must be specified. Any branch contained in the Power Flow Raw Data File for which the following conditions are met is appended to the working case:

- Both buses are in the working case, regardless of whether they were placed there during this execution of READ or were already there.
- One of the buses is in the specified subsystem and the other is not.
- There is not already a branch in the working case between the two buses with the same circuit identifier or with circuit identifier 99.

This has the effect of joining the separate subsystems contained in the working case by adding all tie branches running between them.

Data on an area interchange data record is added to the working case if either:

- At least one data record for a bus, load, or induction machine residing in the area was read during the current execution of activity READ.
- The area subsystem option was specified in selecting activity READ and the area was one of those specified by the user.

Data on zone and owner records is handled using similar criteria.

Data on interarea transfer data records is added to the working case if *both* the from and to areas satisfy the criteria above.

5.2.5. Reading Power Flow Raw Data Files Created by Previous Releases of PSS®E

A Power Flow Raw Data File in the format required for a prior release of PSS®E is able to be processed by activity READ. Both the standard and subsystem READ functions are able to handle Power Flow Raw Data Files from as far back as PSS®E-15.

Upon being presented with such a file, activity READ first converts the file from its original format to the format required by the current release of PSS®E. It then reads the converted file just as it would any file in the format of the current release.

Two methods are available to identify the format in which the records of the Power Flow Raw Data File had been written.

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- From the Power Flow Raw Data File itself

  In this method, the information identifying the file format is contained with the data records to which it applies. Therefore, the user does not need to remember the PSS®E revision numbers corresponding to the various Power Flow Raw Data Files being used in a study.

  At PSS®E-31, a third data item containing the revision number of the release of PSS®E to which the file format corresponds was added to the first data record (refer to Case Identification Data). For files in the format required by PSS®E-15 through PSS®E-30, the user may add this revision number to the first record in the data file so that it contains the first three data items of the current Power Flow Raw Data File:

  ```
  IC, SBASE, REV
  ```

  Thus, the use of this method requires that the REV data item be correctly specified.

- From an input data item of the appropriate API routine

  In this method, the old Power Flow Raw Data File does not require any editing. Rather, the VERNUM input data item of the ReadRawVersion and ReadSubRawVersion API routines is a character string designating the PSS®E release (e.g., 30.3.3).

When the revision number is specified in the file, use of the Read and ReadSub API routines is recommended. If either the ReadRawVersion or ReadSubRawVersion API routine is used, specify the current version for the VERNUM input data item; if some other version is specified, it will take precedence over that specified as REV on the first record of the Power Flow Raw Data File.

Additional Information
PSS®E GUI Users Guide, Reading Power Flow Raw Data into the Working Case *PSS® E Command Line Interface (CLI) Users Guide*, Reading Power Flow Raw Data into the Working Case *PSS®E Application Program Interface (API)*, READ READRAWVERSION READSUB READSUBRAWVERSION See also: Section 5.8, "Reading or Changing Power Flow Data" Section 5.49, "Creating a Power Flow Raw Data File"

5.3. Reading Power Flow Data Additions from the Terminal

TREA

Run Line Mode Activity TREA - CLI
ACTIVITY? >>**TREA** ENTER BUS DATA I, 'BUS NAME', BASKV, IDE, AREA, ZONE, OWNER, VM, VA >>
Interrupt Control Codes
AB

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Activity TREA is special purpose version of activity READ available only in line mode. It is designed for the purpose of adding a limited amount of equipment to the system contained in the working case, with data records read from the dialog input device (the terminal keyboard, a Response File, or an IPLAN program).

Additional Information
PSS® E Command Line Interface (CLI) Users Guide, Reading Power Flow Data Additions from the Terminal *PSS®E Application Program Interface (API)*, READ See also: Section 5.2.1, "Power Flow Raw Data File Contents"

5.4. Adding Machine Impedance Data

MCRE

Run Activity MCRE - GUI
File > Open... *[Open]* Machine Impedance Data File (*.rwm)
Interrupt Control Codes
None

When establishing an initial working case in PSS®E for basic power flow studies, it is not necessary to provide detailed modeling of generating plants (see Generator Data). A single equivalent machine specifying plant totals is sufficient for basic power flow modeling. However, some analytical activities, including more advanced power flow analysis involving the dispatching or outaging of individual machines, fault analysis, balanced switching, and dynamic simulation require more detailed information on generating plants. That data can be entered from a Machine Impedance Data File.

Activity MCRE can be used to:

- Add machines at an existing generator bus (i.e., at a plant).
- Enter the machine quantities MBASE, ZSORCE, XTRAN, and GENTAP into the working case.
- Apportion the total plant output and power limits, as contained in the working case, among the machines at the plant.

5.4.1. Machine Impedance Data File Contents

The machine impedance data input activity MCRE enters source data records from a Machine Impedance Data File into the power flow working case. The Machine Impedance Data File consists of a series of free format records with data items separated by a comma or one or more blanks. Each record is in the following format:

```
I, ID, FP, FQ, MBASE, ZR, ZX, RT, XT, GENTAP, STAT
```

where:

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I	Bus number. Bus I must be specified in the working case with a plant sequence number assigned to it (refer to Plant and Machine Sequence Numbers). No default is allowed.
ID	One- or two-character machine identifier used to distinguish among multiple machines at a plant (i.e., at a generator bus). ID = 1 by default.
FP,FQ	Fractions of total plant active and reactive power output, respectively, to be assigned to this machine. FP and FQ are 1.0 by default.
MBASE	Total MVA base of the units represented by this machine; entered in MVA. This quantity is not needed in normal power flow and equivalent construction work, but is required for switching studies, fault analysis, and dynamic simulation. MBASE = system base MVA by default.
ZR,ZX	Complex machine impedance, ZSORCE = ZR+jZX; entered in pu on MBASE base. This data is not needed in normal power flow and equivalent construction work, but is required for switching studies, fault analysis, and dynamic simulation. For dynamic simulation, this impedance must be set equal to the unsaturated subtransient impedance for those generators to be modeled by subtransient level machine models, and to unsaturated transient impedance for those to be modeled by classical or transient level models. ZR = 0.0 and ZX = 1.0 by default.
RT,XT	Complex step-up transformer impedance, XTRAN = RT+jXT; entered in pu on MBASE base. XTRAN should be entered as zero if the step-up transformer is explicitly modeled as a network branch and bus I is the terminal bus. RT = 0.0 and XT = 0.0 by default.
GENTAP	Step-up transformer off-nominal turns ratio, GTAP; entered in pu. GENTAP is used only if XTRAN is nonzero. GENTAP = 1.0 by default.
STAT	Machine status of one for in-service and zero for out-of-service. STAT = 1 by default.

Data records may be entered in any order. Input is terminated with a record specifying an I value of zero.

5.4.2. Operation of Activity MCRE

The MCRE API routine requires the following as input:

- The name of the Machine Impedance Data File that is to be read.
- A flag indicating how the status of new machines is to be set. These are machines that are not in the working case at the time activity MCRE is selected, but are added because of the presence of data records for them in the Machine Impedance Data File.

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- A flag indicating the treatment of machines for which no data record is specified and that are at a bus with at least one machine for which a data record is specified.

As activity MCRE processes data records, If a record is encountered on which bus I is not in the working case, if it does not have a generator (i.e., plant) slot assigned to it, or if an invalid machine identifier is specified, an appropriate message is printed at the *Progress* device, the record is ignored, and processing continues.

If, in attempting to add a machine to the machine data arrays, the maximum number of machines or machine ownership specifications for which PSS®E is dimensioned is exceeded, an error message is printed at the *Progress* device and the record is ignored. Processing of records successfully entered is completed prior to terminating activity MCRE.

When entering data records for machines already contained in the working case, complete data records must be entered. Omitted data items take on their default values rather than retaining their previous values.

If sequence data is contained in the working case (i.e., activity RESQ had previously been executed), for all machines being added to the working case, the three sequence machine impedances used in the fault analysis activities are set to the impedance ZSORCE specified on the data input record.

After all records have been read, activity MCRE sets the status of each machine that was added to the working case. The status of these machines is set in accordance with the value specified as the new machine status input data item:

- When 0 is specified for the new machine status option, the status of each new machine is set to the value specified as the STAT data item on its data record (refer to Section 5.4.1, "Machine Impedance Data File Contents").

- When 1 is specified for the new machine status option, the STAT data item is ignored and the status of any machines added to a bus is set as follows:

- if the plant bus has any in-service machines for which no data record was read, the status of any new machines at the bus is set to in-service.

- otherwise, the status of any new machines at the bus is set to out-of-service.

Activity MCRE cycles through all plants for which at least one machine data record had been successfully read in the current execution of activity MCRE. The sums of the active and reactive power split fractions of all the plant's machines for which a data record was read with a status flag of one are calculated. Then the plant totals of machine powers and power limits for those machines with a status flag of one which were initially in the case are calculated. (If any of the above sums are zero, the corresponding quantities of the out-of-service machines are used.) Finally, the machine power outputs and limits of all machines at the bus for which a data record was read are set to the product of the corresponding plant quantity and the ratio of the machine fraction to the plant's sum of machine fractions. Plant totals are then updated as the sum of the corresponding machine quantities of its in-service machines.

In processing each such plant, if any machine is encountered that existed at the plant prior to entering activity MCRE and for which no data was read, such machine generates an alarm and is either placed out-of-service with its data items in the working case unchanged or deleted from the working case, in accordance with the value specified treatment of machines with no data record input data item. Furthermore, any machine for which the status flag is changed is tabulated. In either of these cases, the plant totals could be changed and the plant configuration should be examined to verify that it is as intended.

Prior to terminating, the machine arrays are compacted to eliminate holes resulting from machine deletions.

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5.4.3. Application Notes

The sum of split fractions of all machines at a plant need not sum to unity because the factor used in setting each machine's powers is normally taken as the ratio of the machine's fraction to the sum of the fractions of all in-service machines at the plant. This is convenient, for example, if machine outputs are to be set according to their ratings. In this case, each machine's MBASE may be specified as its split fractions.

The power split fractions are used in setting the machine power limits as well as the initial power output of the machine. Note that the machine split fractions *are not* retained in the working case or subsequently written Saved Case Files following termination of activity MCRE.

When a machine is added to the working case by activity MCRE, it inherits the ownership assignment of the bus to which it is attached.

The introduction of multiple machines at a plant into the working case may be accomplished either via activity MCRE or during the initial input of the network model into the PSS®E working case via activity READ (refer to Multiple Machine Plants).

Activity MCRE must be executed *before* any execution of activity CONG.

Additional Information
PSS®E GUI Users Guide, Adding Machine Impedance Data *PSS®E Command Line Interface (CLI) Users Guide*, Adding Machine Impedance Data *PSS®E Application Program Interface (API)*, MCRE

5.5. Reading Capability Curve Data

REGC

Run Activity REGB - GUI
File > Open... *[Open]* Machine Capability Data file (*.gcp)
Interrupt Control Codes
None

The machine capability curve data input activity REGC appends machine capability curve data to the working case. The source data records are read from a Capability Curve Data File for the system contained in the working case.

After a set of capability curve data has been read into the working case, it is carried along with the network as the case is saved and retrieved with activities SAVE and CASE, respectively. Capability curve data may be examined (with activities LIST and EXAM) and modified (with the *[Spreadsheet]*) in a manner similar to that of other power flow data.

5.5.1. Capability Curve Data File Contents

For more information please see *Machine Capability Curve Data File*.

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5.5.2. Operation of Activity REGC

The ReadCapCurves API routine requires as an input the name of the Capability Curve Data File to be read.

The working case must contain the network for which the capability curve data is to be read. If the change code parameter IC in the first record of the Capability Curve Data File is set to 1 and capability curve data had not previously been read for the system in the working case, the data is processed as if IC was set to zero.

5.6. Reading Sequence Data

RESQ

Run Activity RESQ - GUI
File > Open... *[Open]* Sequence Data file (*.seq)
Run Line Mode Activity RESQ - CLI
ACTIVITY? >>**RESQ** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>
Interrupt Control Codes
AB

The sequence data input activity RESQ appends negative and zero sequence data to the working case in preparation for unbalanced network solutions (i.e., fault analysis). The source data records are read from a Sequence Data File for the system where the positive sequence representation is contained in the working case.

Most of the other unbalanced network analysis activities do not permit themselves to be executed unless the sequence data arrays in the working case have previously been initialized via the execution of activities RESQ or TRSQ. This *does not* imply that activities RESQ or TRSQ must be executed during any PSS®E work session in which the unbalanced network analysis activities are to be used; after a set of sequence data has been read into the working case, it is carried along with the network as the case is saved and retrieved with activities SAVE and CASE, respectively. Sequence data may be examined (with activities SQLI and SQEX) and modified (with activity SQCH or the *[Spreadsheet]*) in a manner similar to that of standard (positive sequence) power flow data.

5.6.1. Sequence Data File Contents

For more information please see *Sequence Data File*.

5.6.2. Operation of Activity RESQ

The RESQ API routine requires as an input the name of the Sequence Data File to be read.

The working case must contain the network for which the sequence data is to be read. If the change code parameter IC in the first record of the Sequence Data File is set to 1 and sequence data had not previously

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been read for the system in the working case, a message is printed at the *Progress* device and the data is processed as if IC was set to zero.

As data records are read, a message is displayed at the *Progress* device at the start of each new category of data.

Except for the change code parameter IC in the first data record read by activity RESQ, specifying a data record with a Q in column one is used to indicate that no more data records are to be supplied to activity RESQ.

When the fault analysis warning option is enabled (refer to Saved Case Specific Option Settings), activity RESQ produces the following tabulations:

- When IC is zero, a listing of all in-service machines at Type 2 and 3 buses for which no negative sequence generator impedance is entered. The negative sequence generator impedance, ZNEG, is set to the positive sequence value, ZPOS.
- When IC is zero, a listing of all in-service machines at Type 2 and 3 buses for which no zero sequence generator impedance is entered. The zero sequence generator impedance, ZZERO, is set to the positive sequence value, ZPOS.

Each of these tabulations may be individually suppressed by entering the AB interrupt control code (see Section 4.3, "Interruption of PSS®E by the User").

5.7. Reading Transactions Raw Data

REMM

Run Activity REMM - GUI
File > Open... *[Open]* Transactions Raw Data File (*.mwm)
Run Line Mode Activity REMM - CLI
ACTIVITY? >>**REMM** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Transaction data is introduced into working memory using activity REMM. This information consists of data records presented in a Transactions Raw Data File.

5.7.1. Transactions Raw Data File Contents

The input stream to activity REMM consists of two groups of records. All data is read in free format with data items separated by a comma or one or more blanks. Each category of data except the participation flag data is terminated by a record specifying an initial field value of zero.

An initial record presents a participation flag, a single integer field that sets the treatment for some data values contained in the second data group. The second data group defines transaction events.

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Participation Flag

The initial data record consists of a single integer field.

```
IPF
```

where:

IPF	Participation flag: 0 for bus load/generation fractions. 1 for bus participation factors.

This value is employed to determine the treatment of data fields, contained in the second data group, which define the participation of buses with transaction events. A value of zero for this field indicates that the later fields shall be interpreted as load and generation fractions. A non-zero value (the preferred usage being a value of 1) indicates that the later fields shall be interpreted as bus participation factors.

Transaction Events

Each transaction event consists of a single data record that assigns a set of transaction attributes, followed by one or more data records that associate participating buses with the transaction event. The set of transaction attributes are introduced using data records of the form:

```
ID, ISTAT, MAG, 'LABEL', IPR, CURT
```

where:

ID	Transaction event number. ID = 0 by default.
ISTAT	Transaction event status: 0 for out-of-service. 1 for in-service. ISTAT = 0 by default.
MAG	Transaction event magnitude in MW. MAG = 0.0 by default.
LABEL	Alphanumeric label assigned to the transaction event. The label may be up to twenty four characters and must be enclosed in single quotes. LABEL may contain any combination of blanks, uppercase letters, numbers and special characters. LABEL is twenty four blanks by default.
IPR	Transaction event priority. This integer value priority number is used to group transactions into sets. IPR = 0 by default.
CURT	Transaction event curtailment magnitude in MW. CURT = 0.0 by default.

Each transaction attribute record is followed with a group of records that identify the buses participating in the transaction event. The bus participation records take the form:

```
I, LV, GV
```

where:

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I	Bus number. I = 0 by default.
LV	Participating bus load value. LV = 0.0 by default.
GV	Participating bus generation value. GV = 0.0 by default.

Negative data may be presented for load or generation values. The interpretation of the participating bus load and generation values is a function of the participation flag, IPF, presented on the initial data record. The consequence of the differing treatment is discussed in Section 5.7.3, "Implications of the Participation Flag".

Participating bus data input is terminated with a record specifying a bus number of zero.

Transaction event data input is terminated with a record specifying a transaction event number of zero.

The following is an example consisting of two transaction events,

```
1
    6 1     200.0000 'UPSTART                '  1 0.
   3008  1.0000   .0000
    102   .0000   .2667
  0
    5 1     354.0000 'WORLD                  '  1 0.
   3008   .0000  1.0000
   3007  1.0000  1.0000
   3006  1.0000  1.0000
   3005  1.0000  1.0000
   3004   .0000  1.0000
   3003   .0000  1.0000
   3002   .0000  1.0000
   3001   .0000  1.0000
  0
0
```

5.7.2. Operation of Activity REMM

The user selects the desired Transaction Raw Data File (refer to Section 2.3.1, "Input Data Files"). If the file does not exist or some other file system related error occurs, an appropriate error message is printed. Using line mode the user may enter data directly from the dialog input device. This applies to both interactive and Response File operation of PSS®E.

Activity REMM processes data records from the selected source input stream. REMM prints a message each time that it starts processing a new category of data.

5.7.3. Implications of the Participation Flag

Each transaction event includes a magnitude and participating load and generation buses. The participation flag, IPF, affects both the interpretation of the transaction event participating bus load and generation values and the interpretation of a zero value transaction magnitude.

The transaction magnitude is apportioned among the participating buses, power injections at participating buses with positive generation value or negative load value, and power demand at participating buses with positive load value or negative generation value. The magnitude of power injection or demand at each participating bus is in proportion to the transaction magnitude and the bus participation factor, r:

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$$P_i = MAG * \frac{r_i}{\Sigma r}$$

where:

P_i	Active power injection (demand) at bus i in MW.
MAG	Transaction event magnitude in MW.
r_i	Participating bus i participation factor.
Σr	Sum of participation factors for injection (demand) buses associated with the transaction event.

When IPF = 0, the load and generation values are interpreted as fractions of bus total MW (load or generation). The bus participation factors are then the product of these fractions with the total bus load or generation as appropriate. Also when IPF = 0, a transaction magnitude value of zero indicates that the transaction magnitude shall take a value equal to the sum of participating power demand.

When IPF is non-zero (e.g., IPF = 1) the load and generation values are interpreted as participation factors and no alternate interpretation is implied for a zero value transaction magnitude.

Activity REMM introduces the transaction data into working memory, but this data is not retained in the working case. Transaction data is not restored to the working memory by use of activity CASE.

The transaction data makes reference to participating buses by bus number. These references are not altered or affected by bus modifications made through program dialog. For example, if a transaction participating bus is eliminated from the working case by dialog with activity JOIN then the participating bus reference persists in the working memory but is corrupt. The user should ensure that transaction data which is consistent with the network model is introduced to working memory before invoking any open access related procedures.

Additional Information
PSS®E GUI Users Guide, Reading Transactions Raw Data *PSS®E Command Line Interface (CLI) Users Guide*, Entering Transactions Raw Data See also: Section 5.52, "Creating a Transactions Raw Data File"

5.8. Reading or Changing Power Flow Data

RDCH

Run Activity RDCH - GUI
File > Open *[Open]* Power Flow Data File, Options (*.raw)
Run Line Mode Activity RDCH - CLI
ACTIVITY? >>**RDCH** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>

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Interrupt Control Codes
AB

The bulk power flow data input and modification activity RDCH picks up manually-entered power flow source data and enters it into the power flow working case.

The following API routines are used to implement activity RDCH. Each of them includes among its input data items the name of the Power Flow Raw Data File to be read and the flag for selecting the bus names input option of activity RDCH (see Section 5.2, "Reading Power Flow Raw Data into the Working Case" and Extended Bus Names).

Rdch	Input records are in the format of the PSS®E release indicated in the file. No other inputs.
RdchRawVersion	Input records are in the format of the current or a prior release of PSS®E. It accepts as input a character string indicating the PSS®E release.

5.8.1. Operation of Activity RDCH

The input stream to activity RDCH is in the form of a Power Flow Raw Data File (refer to Section 5.2.1, "Power Flow Raw Data File Contents"), *except* that the Case Identification Data records are omitted (i.e., the first data record is expected to be a Bus Data record). All data is read in free format with data items separated by a comma or one or more blanks. All data categories except the case identification data must be specified. The end of each category of data is indicated by a record specifying a value of zero; FACTS device, GNE device, and DC line data input is terminated by specifying record with a NAME value of either blanks or 0.

Activity RDCH adds a new component to the working case whenever it encounters a data record for which the corresponding component is not found in the working case. In this case, it behaves identically to activity READ, and data items omitted on the data record take on the default values defined in Section 5.2.1, "Power Flow Raw Data File Contents".

For all record types except the multi-section line grouping data, when reading a data record corresponding to an existing component, data items that are specified replace those contained in the working case; data items omitted on the data record have their values unchanged in the working case (specifically, they are *not* set to default values as in activity READ).

Activity RDCH prints a warning message at the *Progress* device if generators have been converted (see Section 13.2, "Converting Generators"). After generators are converted, machine impedance data (MBASE, ZSORCE, XTRAN, and GTAP; refer to Generator Data) *must not* be changed.

As data records are read, a message is displayed at the *Progress* device at the start of each new category of data.

Before it has completed reading its input data, activity RDCH may be ended by entering the AB interrupt control code (refer to Section 4.3, "Interruption of PSS®E by the User"). Activity RDCH checks for an interrupt following processing of each group of data records corresponding to 50 equipment items.

Bus Names Option

When the bus names input option of activity RDCH is enabled, data fields designating ac buses on load, fixed shunt, generator, non-transformer branch, transformer, area, two-terminal dc line, VSC dc line, multi-terminal dc line, multi-section line, FACTS device, switched shunt, GNE device, and induction machine data records

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may be specified as either Extended Bus Names enclosed in single quotes or as bus numbers. Otherwise, bus numbers *must* be used to designate ac buses on these records.

Use of the bus names input option of activity RDCH requires that all buses be assigned unique extended bus names. Activity RDCH checks for the presence of duplicate extended bus names in the working case *before* it starts to read bus data records. Any violations are logged at the *Progress* device and activity RDCH is prohibited from executing.

While reading each bus data record with this option enabled, if a bus with the same extended bus name but a different bus number is present in the working case, an error message is printed, the record is ignored, and processing continues.

5.8.2. Reading RDCH Data Files Created by Previous Releases of PSS®E

Power Flow Raw Data input in the format required for a prior release of PSS®E is able to be processed by activity RDCH. The RdchRawVersion API routine is able to handle Power Flow Raw Data input from as far back as PSS®E-15. The revision number of the release of PSS®E to which the format of the input records corresponds is specified as an input data item to RdchRawVersion.

Upon being presented with such a file, activity RDCH first converts the file from its original format to the format required by the current release of PSS®E. It then reads the converted file just as it would any input stream in the format of the current release.

5.8.3. Application Notes

Activity RDCH always modifies or adds to the working case; it cannot be used to read a new network model into the working case.

The user has the opportunity to enter all categories of data defined in Section 5.2.1, "Power Flow Raw Data File Contents" with the exception of the case identification data.

When specifying a non-transformer branch between buses I and J with circuit identifier CKT, if a two-winding transformer between buses I and J with a circuit identifier of CKT is already present in the working case, it is replaced (i.e., the transformer is deleted from the working case and the newly specified branch is then added to the working case). In this case, omitted data items take on the standard default values (i.e., none of the data from the deleted transformer is inherited by the non-transformer branch).

When specifying a two-winding transformer between buses I and J with circuit identifier CKT, if a non-transformer branch between buses I and J with a circuit identifier of CKT is already present in the working case, it is replaced (i.e., the non-transformer branch is deleted from the working case and the newly specified two-winding transformer is then added to the working case). In this case, omitted data items take on the standard default values (i.e., none of the data from the deleted non-transformer branch is inherited by the transformer).

In specifying the data records for a multi-terminal dc line I which already exists in the working case, the values entered for NCONV, NDCBS, and NDCLN on the first data record are the number of converter, dc bus, and dc link records, respectively, to be read for multi-terminal dc line NAME. If a component record is then read for a component (e.g., a dc link), which has already been defined as a part of multi-terminal dc line NAME, the record modifies existing data; if a record is read for a component not previously included in multi-terminal dc line NAME, the component is added to the line.

For any multi-section line grouping data records entered in activity RDCH, the complete definition of the multi-section line must be entered.

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In preparing an automation file (Python program, IPLAN program or Response File) that includes the addition of power flow data to the working case, if long term use of the automation file is anticipated, the use of the RdchRawVersion API routine rather than RDCH is recommended. In this case, no modification of the automation file (at least this portion of it) will be necessary for it to be used in future releases of PSS®E.

Additional Information
PSS®E GUI Users Guide, Reading / Changing Power Flow Data *PSS® E Command Line Interface (CLI) Users Guide* Reading / Changing Power Flow Data *PSS®E Application Program Interface (API)*, RAWD_2 RDCHRAWVERSION See also: Section 5.2.1, "Power Flow Raw Data File Contents" Section 5.49, "Creating a Power Flow Raw Data File"

5.9. Producing Power Flow and Fault Analysis Data Checking Reports

PFCHK

SCCHK

Run Activity PFCHK and SCCHK - GUI
Power Flow > Check Data > Data checking report *[Power Flow and Fault Analysis Data Checking]*
Run Line Mode Activities PFCHK and SCCHK - CLI
ACTIVITY? >>**PFCHK** ACTIVITY? >>**SCCHK**
Interrupt Control Codes
None

When power flow data is input from a Power Flow Raw Data File, as well as when power flow data is added or modified via the *[Spreadsheet]* (refer to *PSS® E GUI Users Guide*, Spreadsheet View), many data checks are performed and exceptions reported to the Progress or Alert device. The CHECK_POWERFLOW_DATA API routine may be used to repeat these checks on power flow data contained in the working case and provide a report.

Similarly, the CHECK_SEQUENCE_DATA API routine may be used to repeat the checks that are performed when sequence data is input from a Sequence Data File or when sequence data is modified via the *[Spreadsheet]*.

This pair of API routines includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".

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- A designation of the data category where data values are to be checked

The following data categories are available in activity PFCHK:

• all data categories	• bus data
• load data	• plant data
• machine data	• induction machine data
• fixed bus shunt data	• switched shunt data
• non-transformer branch data	• two-winding transformer data
• three-winding transformer data	• transformer impedance correction data
• multi-section line data	• two-terminal dc line data
• multi-terminal dc line data	• VSC dc line data
• FACTS device data	• GNE device data
• area interchange data	• owner data
• zone data	

The following data categories are available in activity SCCHK:

• all data categories	• load data
• machine data	• induction machine data
• fixed bus shunt data	• switched shunt data
• non-transformer branch data	• two-winding transformer data
• three-winding transformer data	• zero sequence mutual data

The report of these activities is directed to the *Report* device.

When the data checking function is run from the GUI, either or both of these checking API routines is run, as appropriate.In line mode, the reports are selected independently.

5.10. Changing Service Status and Power Flow Parametric Data

CHNG

Change Data - GUI
Use *[Spreadsheet]*
Run Line Mode Activity CHNG - CLI
ACTIVITY? >>**CHNG**

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Run Line Mode Activity CHNG - CLI

```
ENTER CHANGE CODE:
  0 = EXIT ACTIVITY                    1 = BUS DATA
  2 = GENERATOR DATA                   3 = BRANCH DATA
  4 = 2 WINDING TRANSFORMER DATA       5 = AREA INTERCHANGE DATA
  6 = TWO-TERMINAL DC LINE DATA        7 = SOLUTION PARAMETERS
  8 = CASE HEADING                     9 = SWITCHED SHUNT DATA
 10 = IMPEDANCE CORRECTION TABLES     11 = MULTI-TERMINAL DC DATA
 12 = ZONE NAMES                      13 = INTER-AREA TRANSFER DATA
 14 = OWNER NAMES                     15 = MACHINE OWNERSHIP DATA
 16 = BRANCH OWNERSHIP DATA           17 = FACTS CONTROL DEVICE DATA
 18 = 3 WINDING TRANSFORMER DATA      19 = VSC DC LINE DATA
 20 = FIXED BUS SHUNT DATA            21 = INDUCTION MACHINE DATA:
>>
```

Interrupt Control Codes

POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

PSS®E permits the user to change all service status, control mode, and other parametric data associated with equipment represented in the working case. The appropriate tool for changing working case data is dependent upon the interface through which PSS®E is being operated.

In the GUI, the primary means of changing individual data items is the *[Spreadsheet]* (refer to *PSS® E GUI Users Guide*, Spreadsheet View). The *[Spreadsheet]* may also be used to add new elements to the working case.

In line mode, the power flow data modification activity CHNG is the primary tool for making data changes to the working case (refer to *PSS® E Command Line Interface (CLI) Users Guide*, Changing Service Status and Power Flow Parametric Data).

In Python programs, the data entry/changing API routines described in Chapter 2, Power Flow Data Changing of the *PSS® E Application Program Interface (API)* manual are used to change data associated with existing elements in the working case (these routines may also be used to add new elements to the working case).

In Response Files and IPLAN programs, activity CHNG and/or the data entry/changing API routines may be used to change power flow data in the working case.

All of these methods modify the working case but do not affect any file. In particular, they do not affect any Saved Case File or Power Flow Raw Data File. To have data changes incorporated into a Saved Case File, activity SAVE must be executed following the implementation of changes in the working case.

5.10.1. Equipment Status Changes

Associated with each ac series branch is a status flag, which is initially set to one (for in-service) or zero (for out-of-service) as specified on the branch data record or in the transformer record block when a new branch is introduced into the working case with activities READ, TREA, or RDCH. The status of a branch may be modified by changing this flag. Out-of-service branches and their data remain with the case as it is saved and retrieved with activities SAVE and CASE, and are simply ignored during the execution of other PSS®E activities such as FNSL or POUT.

If branch status changes isolate a bus, the user *must* change the bus type code to 4. Activities TREE and OUTS and the power flow solution network connectivity check option (refer to Saved Case Specific Option Settings and activity SOLV) aid the user in identifying isolated buses and islands.

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When the specified branch is a member of a multi-section line grouping (refer to Multi-Section Line Grouping Data), the same status change is automatically implemented for all branches in the multi-section line. In addition, the bus type codes of the dummy buses are changed to:

4	When the branch is being removed from service.
1	If the branch is being placed in-service and the bus has no generator data associated with it.
2	If the branch is being placed in-service and the bus has generator data associated with it. In this case, the individual machine status flags, which are not changed in this status processing of multi-section lines, determine which machines are actually returned to service.

The bus type codes of the endpoint buses of the multi-section line grouping are not changed in the branch status processing. As with single section lines, it is the user's responsibility to ensure that branch statuses and bus type codes are coordinated.

When specifying a branch subject to branch data changes, if a multi-section line grouping is specified, (i.e., the circuit identifier has an ampersand as its first character; refer to Multi-Section Line Grouping Data), the user may change the status (and metered end) of the specified multi-section line grouping. Status changes are implemented as described in the preceding paragraph.

When changing the status of a three-winding transformer, status values of 0 through 4 are permitted (refer to Transformer Data); the status flags of the three two-winding transformers that comprise the three-winding transformer are set to the appropriate values. As with multi-section line groupings and two terminal branches, the bus type codes of the endpoint buses of the three-winding transformer are not changed in the branch status processing.

FACTS devices, GNE devices, and dc transmission lines may be blocked in a similar manner. A value of zero for MODE, the control mode flag, indicates a blocked FACTS device (refer to FACTS Device Data). A value of zero for STATUS indicates an out-of-service GNE device (refer to GNE Device Data). A value of zero for MDC, the control mode flag, indicates a blocked dc line (refer to Two-Terminal DC Transmission Line Data, Multi-Terminal DC Transmission Line Data, and Voltage Source Converter (VSC) DC Transmission Line Data).

The status of an individual machine at a generator bus may be modified by changing its status flag; plant total power output and power limits are automatically updated following a change of machine power or status. A plant may be taken out-of-service *either* by changing the status of all units at the bus to zero *or* by changing the bus type code to 1 or 4 (this is the recommended approach because power flow solution activities will run slightly faster). Conversely, for a machine to be treated as in-service, *both* its status must have the value 1 *and* the bus type code must be either 2 or 3.

Similarly, the status of an individual load, induction machine fixed shunt, or switched shunt at a bus may be modified by changing its status. For one of these elements to be treated as in-service, *both* its status must have the value 1 *and* the bus type code must be either 1, 2 or 3. Thus, disconnecting the bus to which it is connected (i.e., changing the bus type code to 4) also outages any load, induction machine, fixed shunt, or switched shunt connected to it; it is not necessary to set the device status to zero.

5.10.2. Solution Parameter Changes

The convergence tolerance used by activities SOLV and MSLV (TOL), as well as that of activities TYSL and BKDY (TOLTY), specifies largest voltage magnitude change in per unit. In activities FNSL, FDNS, NSOL, and

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INLF, the mismatch convergence tolerances TOLN and VCTOLQ specify largest mismatch in MW and Mvar, and the controlled bus voltage error convergence tolerance, VCTOLV, specifies largest voltage magnitude change in per unit. The blowup threshold (BLOWUP) is in units of largest voltage magnitude change in per unit and angle change in radians.

5.10.3. Interarea Transfer Data Changes

When modifying interarea transfer data, changes in transfer MW may optionally be applied to the desired area net interchange values of both the from and to areas (refer to Area Interchange Data). The new desired net interchange of the from area is set to its old value plus the change in transfer MW. Similarly, the to area's desired net interchange is set to its old value minus the transfer MW change.

Additional Information
PSS®E GUI Users Guide, Changing Service Status and Power Flow Parametric Data *PSS®E Command Line Interface (CLI) Users Guide*, Changing Service Status and Power Flow Parametric Data See also: Section 5.2, "Reading Power Flow Raw Data into the Working Case"

5.11. Scaling Loads, Generators, and Shunts

SCAL

Run Activity SCAL - GUI
Power Flow > Changing > Scale generation, load, shunt (SCAL)... *[Scale Powerflow Data]*
Run Line Mode Activity SCAL - CLI
ACTIVITY? >>**SCAL** ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
None

System planners and operators need the ability to analyze a wide range of demand and generation scenarios. Often the need is to study a variety of demand levels that cover a daily, weekly or a seasonal profile. To facilitate simple and rapid changes in load level, PSS®E provides a means by which not only loads but also generation and fixed bus shunts can be scaled.

The load, generation and shunt scaling activity SCAL enables the user to uniformly increase or decrease any or all of the following quantities for a specified grouping of loads, fixed shunts, and synchronous machines:

- Load active power.
- Load reactive power.
- Active component of fixed bus shunt admittance.
- Positive reactive component of fixed bus shunt admittance (capacitors).

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- Negative reactive component of fixed bus shunt admittance (reactors).
- Generator active power output (positive generation); excludes induction machines.
- Motor active power output (negative generation); excludes induction machines.

The SCAL_4 API routine includes the following among its inputs:

- A designation of the subsystem in which the generation, load and/or shunt is to be processed; refer to Section 4.8, "Subsystem Selection".
- An interruptible load scaling option.
- A load component scaling option.
- A base loaded generation scaling option.
- The method used to specify scaling targets.
- A flag to either enforce or ignore machine active power limits.
- A reactive load scaling option.
- The scaling targets.

5.11.1. Operation of Activity SCAL

The user specifies either that the entire working case is to be processed or that the generation, load and/or shunt in a designated bus subsystem is to be processed. Any machine in the specified subsystem where the active power output is outside of its active power limits generates an alarm.

The reactive components of in-service fixed bus shunts are separated into capacitors and reactors (inductors), and different scaling may be applied to each category.

Any in-service machine in the subsystem being processed where the active power output is negative is treated as a motor. Motor totals are excluded from the generator totals; rather, motors are handled separately and the user is able to apply different scaling to generators and motors. If no motors are contained in the subsystem being processed, the motor tabulations are suppressed.

When scaling to new total powers, the following rule is applied in determining the individual bus, load, or machine powers:

> Adjust the power such that, at each load, fixed shunt, or machine, the ratio of individual load, fixed shunt, or machine power to the total power of all loads, fixed shunts, or machines being processed remains unchanged.

Details on the scaling of fixed shunts, loads, and generation are given in Section 5.11.2, "Scaling Fixed Shunts", Section 5.11.3, "Scaling Loads" and Section 5.11.4, "Scaling Generation", respectively.

Pre- and post-scaling subsystem totals are tabulated as in the following example:

```
Present totals:
    Load-MW 3200.0 (    Scaling    0.0:    0.0 Interruptible +    0.0 Non-interruptible
                    + Unchanged 3200.0:    0.0 Interruptible + 3200.0 Non-interruptible )
  Load-Mvar 1950.0 (    Scaling    0.0:    0.0 Interruptible +    0.0 Non-interruptible
                    + Unchanged 1950.0:    0.0 Interruptible + 1950.0 Non-interruptible )
```


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```
 Generation 3258.6 (    Scaling 2408.6: 1808.6 Non-base loaded +   0.0 Down only +   0.0 Neither up nor down + 600.0 Up only
                    + Unchanged  850.0:    0.0 Non-base loaded + 750.0 Down only + 100.0 Neither up nor down +   0.0 Up only )
    Shunt-MW    0.0
    Reactors -600.0
  Capacitors  950.0

New totals:
     Load-MW 3200.0 (    Scaling    0.0:    0.0 Interruptible +    0.0 Non-interruptible
                    + Unchanged 3200.0:    0.0 Interruptible + 3200.0 Non-interruptible )
   Load-Mvar 1950.0 (    Scaling    0.0:    0.0 Interruptible +    0.0 Non-interruptible
                    + Unchanged 1950.0:    0.0 Interruptible + 1950.0 Non-interruptible )
 Generation 3278.6 (    Scaling 2428.6: 1823.7 Non-base loaded +   0.0 Down only +   0.0 Neither up nor down + 605.0 Up only
                    + Unchanged  850.0:    0.0 Non-base loaded + 750.0 Down only + 100.0 Neither up nor down +   0.0 Up only )
    Shunt-MW    0.0
    Reactors -600.0
  Capacitors  950.0
```

5.11.2. Scaling Fixed Shunts

Shunt totals include only those fixed shunts at buses in the specified subsystem which would be modeled during network solutions (i.e., in-service fixed bus shunts at non-Type 4 buses). Shunt quantities are tabulated and scaled as nominal values (i.e., at unity voltage).

The reactive components of in-service fixed bus shunts are separated into capacitors and inductors, and different scaling may be applied to each category.

5.11.3. Scaling Loads

Only scalable loads that are in-service at non-Type 4 buses are included in the load totals and processing of activity SCAL. Both interruptible and non-interruptible loads may be scaled, or load scaling may be restricted to either only interruptible loads or only non-interruptible loads. Either the net of load and its associated in-service distributed generation may be scaled, or load scaling may be restricted to either only loads or only in-service distributed generation.

When the entire working case is being processed by activity SCAL, all in-service loads in the working case are processed.

When the subsystem being processed is selected only by bus, all in-service loads at subsystem buses for which the Type code is not 4 are included in the load totals and subsequent scaling.

When the subsystem being processed is selected only by base voltage level, activity SCAL processes all in-service loads at each non-Type 4 bus for which the base voltage falls within the specified base voltage band.

When a single subsystem selection criterion of area, zone, or owner is used, each in-service load assigned to one of the areas, zones, or owners specified is processed (see Load Data). The area, zone, or owner assignments of buses (see Bus Data) are not considered in these subsystem selection modes

When multiple subsystem selection criteria are enabled, the base voltage and/or bus selection criteria, if enabled, dictate those buses where the loads are candidates for processing; otherwise, all buses are candidates for processing. The area, zone, and/or owner selection criteria, if enabled, dictate which loads at candidate buses are to be included; otherwise, all loads at candidate buses are included.

Load totals include voltage dependent effects (e.g., PQBRAK is recognized; see Constant Power Load Characteristic).

The load totals are the sums of any constant power, current, and admittance components of those loads being scaled. When both load and its associated distributed generation are being scaled, any in-service distributed generation associated with any of the loads being scaled is subtracted from the load totals, and this net load

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is scaled. When either only load or only distributed generation is being scaled, the total shown is that of the corresponding component. All of the load and distributed generation components being scaled are scaled by the same factor.

Subsystem active and reactive load are scaled separately. Reactive load may be handled in one of the following ways:

- Modify the reactive component of each load such that the existing P/Q ratio is retained (i.e., retain the existing load power factor for each load).
- Specify a new total reactive load.
- Specify the percent by which the reactive loads are to be changed.
- Specify a new load power factor to be used at all loads being processed.
- Specify the amount of reactive load by which the total reactive load is to be increased.
- Leave it unchanged.

5.11.4. Scaling Generation

Machines with positive active power generation are included in the generator totals, and those with negative active power generation are included in the motor totals. The discussion below is in terms of generator totals, but the same approach is used for motor totals.

Only machines that are in-service at Type 2 and 3 buses are included in the generator totals and processing of activity SCAL. Both baseloaded and non-baseloaded generation may be scaled, or generation scaling may be restricted to either only baseloaded generators or only non-baseloaded generators. When baseloaded generators are included in the scaling, the change restrictions imposed by their baseload flags (i.e., up only, neither up nor down, or down only) may be either honored or ignored.

When the entire working case is being processed by activity SCAL, all in-service generators in the working case are processed.

When the subsystem being processed is selected only by bus, all in-service generators at subsystem buses for which the Type code is 2 or 3 are included in the generator totals and subsequent scaling.

When the subsystem being processed is selected only by base voltage level, activity SCAL processes all in-service generators at each Type 2 or 3 bus where the base voltage falls within the specified base voltage band.

When a single subsystem selection criterion of area or zone is used, each in-service generator connected to a bus assigned to one of the areas or zones specified is processed (see Bus Data).

When the single subsystem selection criterion of owner is used, each such machine wholly or partly owned by any of the owners specified is processed (see Generator Data). The owner assignments of buses are not considered.

When multiple selection criteria are enabled, the area, zone, base voltage, and/or bus selection criteria, if enabled, dictate those buses for which the machines are candidates for processing; otherwise, all buses are candidates for processing. The owner selection criterion, if enabled, dictates which machines at candidate buses are to be included; otherwise, all machines with positive active power generation at candidate buses are included.

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When active power limits are enforced, any active power limits of machines being scaled that are negative are temporarily set to zero. When reducing the subsystem's active power generation by a large amount, this prevents any such machines from having their active power settings set to a negative value as other machines are set at their lower active power limits. The active power limit totals shown in the display of activity SCAL are calculated using these modified limits.

In scaling generation, it is useful to remember the following points:

- Activity SCAL is *not* a dispatch activity; it merely scales the existing active power output of in-service machines such that the ratio of machine to total power is retained. Machine power limits are recognized only if the user selects that option.
- If a system swing (Type 3) bus is in the subsystem being scaled, the working case should be solved so that the swing power, which is included in the total generation, is reasonable.
- If the total generation is being changed by a large amount, the new generator outputs, as set by activity SCAL, should be examined to verify that machines are set at realistic operating points. Activities LIST, GENS, and GEOL are helpful for this purpose. (For these situations, the unit commitment/economic dispatch activity, ECDI may be more appropriate.)

5.11.5. Scaling Example

In this example, all active power load is scaled by + 2% while maintaining a constant P/Q ratio. No scaling is be imposed on active power generation. Consequently, the swing bus will need to generate the increased power as well any additional system losses.

Figure 5.1, "Pre-Scaling Load, Generation, Losses and Swing Bus Output"shows the total loads, the swing bus power and the system losses in the *savnw.sav* Saved Case File prior to scaling. That data includes the total real load (3,200 MW), the total generation real power (3,258.7 MW) and the total reactive power for bus connected reactors and capacitor (600 Mvar and 950 Mvar, respectively). In addition the total reactive component of load is indicated to be 1,950 Mvar. Figure 5.2, "Post-Scaling Load, Generation, Losses and Swing Bus Output" shows the same data after scaling and resolving the case.

Pload (MW)	Qload (MVAR)	Iload (MW)	Iload (MVAR)	Yload (MW)	Yload (MVAR)
200.0	100.0	0.0	0.0	0.0	0.0
600.0	450.0	0.0	0.0	0.0	0.0
400.0	350.0	0.0	0.0	0.0	0.0
300.0	150.0	0.0	0.0	0.0	0.0
1200.0	700.0	0.0	0.0	0.0	0.0
100.0	50.0	0.0	0.0	0.0	0.0
200.0	75.0	0.0	0.0	0.0	0.0
200.0	75.0	0.0	0.0	0.0	0.0

TOTAL GENERATION	PQLOAD	I LOAD	Y LOAD	SHUNTS	CHARGING	LOSSES	SWING
MW 3258.7	3200.0	0.0	0.0	0.0	0.0	58.7	258.7
MVAR 964.2	1950.0	0.0	0.0	-291.5	1810.1	1115.7	104.0

Figure 5.1. Pre-Scaling Load, Generation, Losses and Swing Bus Output

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Pload (MW)	Qload (MVAR)	Iload (MW)	Iload (MVAR)	Yload (MW)	Yload (MVAR)
204.0	102.0	0.0	0.0	0.0	0.0
612.0	459.0	0.0	0.0	0.0	0.0
408.0	357.0	0.0	0.0	0.0	0.0
306.0	153.0	0.0	0.0	0.0	0.0
1224.0	714.0	0.0	0.0	0.0	0.0
102.0	51.0	0.0	0.0	0.0	0.0
204.0	76.5	0.0	0.0	0.0	0.0
204.0	76.5	0.0	0.0	0.0	0.0

TOTAL GENERATION	PQLOAD	I LOAD	Y LOAD	SHUNTS	CHARGING	LOSSES	SWING
MW 3326.1	3264.0	0.0	0.0	0.0	0.0	62.1	326.1
MVAR 1087.5	1989.0	0.0	0.0	-276.1	1790.9	1165.5	131.8

Figure 5.2. Post-Scaling Load, Generation, Losses and Swing Bus Output

Additional Information
PSS®E GUI Users Guide, Scaling Loads, Generators, and/or Shunts *PSS®E Command Line Interface (CLI) Users Guide*, Scaling Loads, Generators, and/or Shunts *PSS®E Application Program Interface (API)*, SCAL See also: Load Data Fixed Bus Shunt Data Generator Data

5.12. Electrically Disconnecting a Bus

DSCN

Run Activity DSCN - GUI
Power Flow > Changing > Disconnect / Reconnect bus (DCSN/RECN)... *[Disconnect / Reconnect Bus]* OPTION: Disconnect bus
Run Line Mode Activity DSCN - CLI
ACTIVITY? >>**DSCN** ENTER BUS NUMBER: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The bus disconnection activity DSCN automates the data changes required to electrically isolate a bus.

The only input to the DSCN API routine is the number of the bus to be disconnected. If the specified bus is not contained in the working case, an appropriate message is printed at the *Alert* device.

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For each bus processed, its Type code is set to 4 and all ac branches, dc lines, series FACTS devices, and series GNE devices connected to the bus are set to out-of-service. The status flags of bus connected equipment (e.g., loads, FACTS devices without a series element, etc.) is not changed by activity DSCN; they are, however, treated as out-of-service because such elements are in-service only if the Type code of the bus to which they are connected is 1, 2 or 3 (refer to Section 5.10.1, "Equipment Status Changes").

If the bus being processed is the endpoint bus of a multi-section line grouping (refer to Multi-Section Line Grouping Data), the entire multi-section line grouping is removed from service; i.e., each line section is set to out-of-service and each dummy bus has its type code set to 4.

Similarly, if the bus being processed is a dummy bus of a multi-section line grouping, the multi-section line is removed from service (refer to Section 5.10.1, "Equipment Status Changes").

If a three-winding transformer (refer to Transformer Data) is connected to the bus being processed, all three windings of the transformer are removed from service.

For each bus disconnected, a summary of bus type code and branch status changes is printed at the *Progress* device. Figure 5.3, "Bus Disconnect Progress Output" shows an example in which five circuits have been removed from service as a result of disconnecting a bus. This summary, however, does not list the bus connected equipment that is taken out-of-service as a result of the setting of the bus Type code to 4.

```
TYPE CODE OF BUS    154 [DOWNTN      230.00] SET TO  4
AND STATUS OF FOLLOWING CONNECTED BRANCHES SET TO 0:
 CIRCUIT 1  TO BUS    153 [MID230      230.00]
 CIRCUIT 2  TO BUS    153 [MID230      230.00]
 CIRCUIT 1  TO BUS    203 [EAST230     230.00]
 CIRCUIT 1  TO BUS    205 [SUB230      230.00]
 CIRCUIT 1  TO BUS   3008 [CATDOG      230.00]
```

Figure 5.3. Bus Disconnect Progress Output

Additional Information
PSS®E GUI Users Guide, Changing Service Status and Power Flow Parametric Data
PSS®E Command Line Interface (CLI) Users Guide, Electrically Disconnecting a Bus
PSS®E Application Program Interface (API), DSCN
See also: Section 5.10.1, "Equipment Status Changes" Section 5.13, "Electrically Reconnecting a Bus"

5.13. Electrically Reconnecting a Bus

RECN

Run Activity RECN - GUI
Power Flow > Changing > Disconnect / Reconnect bus (DCSN/RECN)...
[Disconnect / Reconnect Bus]
OPTION: Connect bus

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Run Line Mode Activity RECN - CLI
ACTIVITY? >>**RECN** ENTER BUS NUMBER: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The bus reconnection activity RECN automates the data changes required to electrically reconnect a disconnected (i.e., Type 4) bus. That is, activity RECN is the logical inverse of activity DSCN.

The only input to the RECN API routine is the number of the bus to be reconnected. If the specified bus is not contained in the working case, an appropriate message is printed at the *Alert* device.

For each bus processed, if its Type code is 4, it is set to 2 if there is a plant table entry assigned to the bus or to 1 if no generator data is associated with the bus. Then all branches and other series elements connected to the bus for which the to bus is not a Type 4 bus are set to in-service. All two-terminal and multi-terminal dc lines returned to service are placed in power control mode. All series FACTS devices returned to service have their control modes set to a positive value as described below. A summary of bus type code and branch status changes is printed at the *Progress* tab.

The status flags of bus connected equipment (e.g., loads, FACTS devices without a series element, etc.) is not changed by activity RECN. If the bus had been disconnected by activity DSCN, they retain the service status that they had before they were disconnected because DSCN did not change their status flags (refer to Section 5.12, "Electrically Disconnecting a Bus").

If the bus being processed is the endpoint bus of a multi-section line grouping (refer to Multi-Section Line Grouping Data), the entire multi-section line grouping is returned to service as long as the Type code of the other endpoint bus is not 4; i.e., each line section is set to in-service and each dummy bus has its type code set to 1 (if there is no generator data for the dummy bus) or 2 (if there is a generator slot for the dummy bus).

Similarly, if the bus being processed is a dummy bus of a multi-section line grouping, the multi-section line is set to in-service as long as the Type codes of neither of the two end point buses is 4 (refer to Section 5.10.1, "Equipment Status Changes").

If a three-winding transformer (refer to Transformer Data) is connected to the bus being processed, all three windings of the transformer are set to in-service as long as the Type codes of neither of the other two buses connected to the transformer is 4; otherwise, the three-winding transformer remains out-of-service.

Any series FACTS device that is returned to service has its control mode set as follows:

3	if it was previously holding constant series impedance and then was outaged via activity DSCN.
4	if it was previously holding constant series voltage and then was outaged via activity DSCN.
5 through 8	as appropriate, if it was previously a series element of an IPFC and then was outaged via activity DSCN.
1	in all other cases.

Additional Information
PSS®E GUI Users Guide, Changing Service Status and Power Flow Parametric Data

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Electrically Reconnecting a Bus *PSS®E Application Program Interface (API)*, RECN See also: Section 5.10.1, "Equipment Status Changes" Section 5.12, "Electrically Disconnecting a Bus"

5.14. Outage Stalled and Tripped Induction Machines

ImOutage

Run *Activity ImOutage* - GUI
Power Flow > *Changing* > *Outage "stalled" / "tripped"* induction machines... *[Outage "stalled" / "tripped" Induction Machines]*
Interrupt Control Codes
None

The induction machine outaging activity ImOutage allows the user to change the status of induction machines that are in-service, but that have been set to the "stalled" (for motors) or "tripped" (for generators) state by the power flow solution, to out-of-service (refer to Section 6.3.14, "Induction Machines").

The ImOutage API routine includes the following among its inputs:

- A designation of the subsystem in which "stalled" and/or "tripped" induction machines are to be removed from service.
- An option to outage "stalled" induction motors.
- An option to outage "tripped" induction generators.

When processing induction machines in a bus subsystem, the bus and/or base voltage selection criteria, if enabled, determine those buses whose connected induction machines are candidates for outaging; otherwise, all buses are candidate buses. The area, owner, and/or zone selection criterion, if enabled, determine which induction machines at candidate buses are checked for outaging; otherwise, all induction machines at candidate buses are checked for outaging.

Following the processing of activity ImOutage, if the option to outage "stalled" induction motors was enabled, the number of motors that have been placed out-of-service is printed at the *Progress* device. Similarly, if the option to outage "tripped" induction generators was enabled, the number of generators that have been placed out-of-service is printed at the *Progress* device.

Additional Information
PSS® E Application Program Interface (API) ImOutage

5.15. Removing Specified Buses and Connected Equipment

EXTR

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Run Activity EXTR - GUI
Power Flow > Changing > Delete network elements (PURG/EXTR)... *[Delete Buses and Outaged Network Elements]* OPTION: Remove Buses from Case
Run Line Mode Activity EXTR - CLI
ACTIVITY? >>**EXTR** ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The bus removal activity EXTR removes subsystem buses and all equipment connected to them from the working case, regardless of their service status. Because EXTR irrevocably deletes data from the working case, it is *strongly* recommended that a Saved Case File of the working case be made with activity SAVE prior to using activity EXTR.

The EXTR API routine includes the following among its inputs:

- A designation of the subsystem from which buses and their connected equipment are to be deleted; refer to RECN.
- An option to remove generator table entries for out-of-service plants. When enabled, this option results in the removal of plant and machine table entries for all Type 1 buses in the case (i.e., for all buses in the case at which the plant is currently out-of-service). The proper selection for this option is dependent upon the application at hand. When activity EXTR is being used to set up the working case in preparation for building an equivalent of a subsystem contained within it, the usual procedure is to enable this option. For other applications, it is often desirable to retain these plant table allocations if the machines may subsequently be returned to service.
- An option to change the type codes of boundary buses. When enabled, this option results in adding the value 4 to the bus type codes of boundary buses, which are retained in the working case following the processing of activity EXTR. A boundary bus is defined as a bus that is retained in the working case that is connected to a bus that is deleted. The normal procedure is to disable this option unless some advanced equivalencing operation is to be applied to the resultant working case.

Activity EXTR rebuilds all data arrays in the working case, compacting them to eliminate holes created by the deleted equipment. New bus sequence numbers, generator sequence numbers, and so on are assigned.

A three-winding transformer is removed if any of the buses it connects are removed.

A dc transmission line is removed if any of the buses designated as its converter station buses are deleted.

A FACTS device is removed if either its sending end bus or terminal end bus is deleted.

A GNE device is removed if any bus to which it is connected is deleted.

A multi-section line grouping is deleted if at least one of its line sections is removed. This does *not* mean that the remaining line sections and dummy buses are deleted from the working case; only the information defining the multi-section line grouping itself is removed.

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For any bus where voltage is being controlled by a remote generator, switched shunt, FACTS device shunt element, or VSC dc line converter, an alarm is printed if the controlled bus is deleted while the voltage controlling equipment is retained. The controlling equipment is then set to control local bus voltage and the scheduled voltage setpoint or band is *not* changed.

For any bus with voltage controlling equipment that is being controlled by a remote switched shunt, an alarm is printed if the controlled bus is deleted while the controlling switched shunt bus is retained. The switched shunt's control mode is *not* changed.

Following the processing of activity EXTR, the number of buses that have been deleted is printed at the *Progress* device.

Additional Information
PSS®E GUI Users Guide, Removing Buses and Connected Equipment *PSS®E Command Line Interface (CLI) Users Guide*, Removing Specified Buses and Connected Equipment *PSS®E Application Program Interface (API)*, EXTR

5.16. Deleting Equipment

PURG

Run Activity PURG - GUI
Power Flow > Changing > Delete network elements (PURG/EXTR)... *[Delete Buses and Outaged Network Elements]* OPTION: Remove outaged equipment
Delete Individual Equipment Elements - GUI
Use *[Spreadsheet]*
Run Line Mode Activity PURG - CLI

```
ACTIVITY?
>>PURG
ENTER UP TO 20 BUS NUMBERS
>>


ACTIVITY?
>>PURG,SINGLE

 EQUIPMENT ITEMS WHICH MAY BE REMOVED ARE:
  1 = AC LINES                    2 = SWITCHED SHUNTS
  3 = MACHINES                    4 = PLANTS
  5 = 2-TERMINAL DC LINES         6 = MULTI-TERMINAL DC LINES
  7 = MULTI-SECTION LINE          8 = ZERO SEQUENCE MUTUALS
  9 = INTER-AREA TRANSFERS       10 = LOADS
 11 = FACTS DEVICES              12 = THREE-WINDING TRANSFORMERS
 13 = VSC DC LINES               14 = FIXED BUS SHUNTS
 ENTER EQUIPMENT CATEGORY CODE (0 TO EXIT):
>>
```


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Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The equipment removal activity PURG deletes designated equipment items from the working case. Activity PURG may be used to remove either individually specified equipment items or all outaged items of a specified equipment category that are contained in a specified subsystem of the working case.

Because PURG irrevocably deletes data from the working case, it is *strongly* recommended that a Saved Case File of the working case be made with activity SAVE prior to using activity PURG.

5.16.1. Removing Individual Equipment Items

In the GUI, the primary means of deleting individual equipment items is the *[Spreadsheet]* (refer to *PSS® E GUI Users Guide*,Spreadsheet View).

In line mode, the equipment removal activity PURG selected with the suffix `SINGLE` (or simply `SI`) is the primary tool for deleting individual equipment items (refer to *PSS® E Command Line Interface (CLI) Users Guide*, Deleting Equipment).

In Python programs, over 20 PURGaaa API routines described in Chapter 1, Power Flow Operation of the *PSS® E Application Program Interface (API)* manual are used to remove individual items from the working case.

In Response Files and IPLAN programs, activity PURG,SINGLE and/or the equipment removal API routines may be used to delete individual items from the working case.

In removing multi-section line groupings, the branches and dummy buses that are members of a deleted multi-section line grouping *are not* removed from the working case; only the definition of the grouping is deleted.

5.16.2. Removing Outaged Equipment in a Subsystem

Activity PURG may be used to remove from the working case all outaged equipment of a designated equipment category connected to buses within a specified subsystem.

The PURG API routine includes the following among its inputs:

- A designation of the subsystem from which out-of-service equipment is to be deleted; refer to Section 4.8, "Subsystem Selection".
- An integer indicating the type of equipment to be removed. The following equipment types are allowed:
- ac branches (non-transformer branches and two-winding transformers)
- Three-winding transformers
- Loads
- Synchronous
- Machines
- Plants
- Induction machines

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- Fixed bus shunts
- Switched shunts
- FACTS devices
- Two-terminal dc lines
- Multi-terminal dc lines
- VSC dc lines
- GNE devices
- An integer indicating the treatment of out-of-service tie branches to other subsystems. Such branches may either be kept in the working case or deleted.

When the entire working case is selected for processing, all out-of-service elements of the selected category are deleted, and the out-of-service tie branches option is ignored.

Otherwise, the area, zone, and/or owner assignments of buses are used to determine which equipment items are to be processed when subsystem selection by area, zone, and/or owner is enabled. The owner assignments of synchronous machines, ac branches, three-winding transformers, FACTS devices, and GNE devices, and the area, zone, and owner assignments of loads, and induction machines are not considered.

The option indicating the treatment of outaged subsystem ties applies in the processing of ac branches, three-winding transformers, dc lines, series FACTS devices, and series GNE devices. When choosing to remove out-of-service ties, any outaged branch with at least one bus in the specified subsystem is deleted. Otherwise, a branch is deleted only if all buses connected by the branch are in the specified subsystem. In the case of multi-terminal dc lines, only the ac converter buses are considered in determining if all buses are in the specified subsystem; specifically, the area, zone, and owner assignments of the dc buses are not considered.

Each deleted item is reported at the *Progress* device. As the processing of an equipment category is completed, the number of items of that category that were deleted is tabulated.

Activity PURG deletes subsystem equipment items as follows:

ac branches	The branch status flag is zero or negative.
Switched shunts	The switched shunt status flag is zero.
Synchronous Machines	The machine status flag is zero or negative.
Plants	The bus type code is 1 or greater than 3.
Induction machines	The machine status flag is zero or negative.
Loads	The load status flag is zero or negative.
Fixed bus shunts	The shunt status flag is zero or negative.
dc lines	The control mode is zero.
FACTS devices	The control mode is zero.
Three-winding transformers	The status flag is zero.
GNE devices	The device status is zero.

5.16.3. Example of Activity PURG

Figure 5.4, “Purging the Branches Data Category” displays the output of activity PURG for ac branches.

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```
X------- FROM BUS --------X X-------- TO BUS ---------X CKT
   151 [NUCPANT     500.00]    152 [MID500      500.00]  1
   151 [NUCPANT     500.00]    152 [MID500      500.00]  2
   151 [NUCPANT     500.00]    201 [HYDRO       500.00]  1

    3 BRANCHES DELETED
```

Figure 5.4. Purging the Branches Data Category

5.16.4. Application Notes

The user must understand the difference between the use of activity PURG and the changing of the service status of an equipment item as may be done in activity CHNG (refer to Section 5.10.1, "Equipment Status Changes" and Section C.8.1, "Equipment Status Changes and the Spreadsheet View") or the *[Spreadsheet]*. Changing the service status flag simply indicates that the equipment item is to be ignored during PSS®E network solution and output activities. Activity PURG permanently removes the equipment item and all its data from the working case.

In deleting loads from the working case, if all loads at a bus are removed, an appropriate message is printed. Similarly, in deleting fixed bus shunts from the working case, if all fixed shunts at a bus are removed, an appropriate message is printed. Similarly, in deleting induction machines from the working case, if all induction machines at a bus are removed, an appropriate message is printed.

In deleting machines from the working case, if all machines at a bus (plant) are removed, an appropriate message is printed and the plant data at the bus is also deleted.

If the plant entry of any area swing bus is deleted, an appropriate message is printed and the area swing bus number is set to zero (refer to Area Interchange Data and Area Interchange Control).

If a branch that is a member of a multi-section line grouping is deleted, an appropriate message is printed and the multi-section line grouping is deleted.

Activity PURG *does not* check that the working case is in the form required by the power flow solution activities. The user is advised to use activities such as TREE (to check for islands and system swing bus specification) and LIST (to check the area interchange control parameters) before attempting to solve the modified case.

Additional Information
PSS®E GUI Users Guide, Removing Buses and Connected Equipment *PSS®E Command Line Interface (CLI) Users Guide*, Deleting Equipment *PSS®E Application Program Interface (API)*, PURG PURG2DC PURG3WND PURGAREA PURGATRN PURGBRN PURGFACTS PURGGNE PURGLOAD PURGLOADS PURGMAC PURGMDC PURGMSL PURGMUT PURGOWNER PURGPLNT PURGSHUNT PURGSHUNTS PURGSWS PURGVSC PURGZONE PURGE_MULTI_TERM_DC_BUS PURGE_MULTI_TERM_DC_CONVR PURGE_MULTI_TERM_DC_LINK

5.17. Joining Buses

JOIN

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Run Activity JOIN - GUI
Power Flow > Changing > Join buses (JOIN)... *[Join Buses]*
Run Line Mode Activity JOIN - CLI
ACTIVITY? >>**JOIN** ENTER 1 TO ADD LINE SHUNTS OF DELETED IN-SERVICE BRANCHES TO RETAINED BUS SHUNT: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The bus joining activity JOIN enables the user to combine pairs of buses, retaining the identity of one of the two buses. It is intended primarily to bolt together buses that are connected by a low impedance jumper branch. However, activity JOIN allows any pair of buses to be joined together, regardless of the impedance of any connecting branches or whether there is a branch between them at all.

The JOIN API routine includes the following among its inputs:

- The number of the bus to be retained.
- The number of the bus to be joined to the retained bus; this bus is removed from the working case.
- An integer indicating the treatment of non-zero line shunts on branches and magnetizing admittance on transformers that connect the specified buses; these series elements are removed from the working case. Line shunts of deleted in-service branches may be handled using one of the following methods:
- They are added to the retained bus as a fixed bus shunt with the identifier LS; if a shunt with the identifier LS already exists at either of the buses, an available identifier is assigned.
- They are neglected.

5.17.1. Operation of Activity JOIN

The following conditions generate an alarm at the *Alert* device and the buses are not joined:

- Either of the specified buses is not contained in the working case.
- The resulting retained bus would have more synchronous machines, loads, induction machines, or fixed shunts than are able to be accommodated with unique identifiers.
- Either of the buses has a type code of 4 or greater.
- The two buses are connected by a three-winding transformer.
- There is not enough room in the bus tables to handle the processing of three-winding transformers connected to the bus to be deleted

If both buses are accepted, activity JOIN applies the required changes to the working case.

In joining buses together, activity JOIN retains the identity of the first bus entered. This includes the bus attributes of number, name, base voltage, area, zone, and owner. Any branches between the buses being

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joined are removed from the working case, with line connected shunts and magnetizing admittances handled in accordance with the line shunt treatment input option.

All other branches originally emanating from the bus being removed are rerouted to the retained bus. If there are branches from the two specified buses to a common third bus with identical circuit identifiers, branches originally connected to the retained bus keep their original circuit identifiers and new identifiers are assigned to the rerouted circuits.

All loads at the bus being removed are transferred to the retained bus. In the case of conflicting load identifiers, those loads originally at the retained bus keep their original identifiers and new identifiers are assigned to the transferred loads.

Similarly, all fixed bus shunts at the bus being removed are transferred to the retained bus. In the case of conflicting shunt identifiers, those shunts originally at the retained bus keep their original identifiers and new identifiers are assigned to the transferred shunts.

Similarly, all induction machines at the bus being removed are transferred to the retained bus. In the case of conflicting machine identifiers, those induction machines originally at the retained bus keep their original indetifiers and new identifiers are assigned to the transferred induction machines.

If the retained bus is not a generator bus and the deleted one is, the plant and machine data of the removed bus is transferred to the retained bus. If both buses are generator buses, machines from the removed bus are transferred to the retained bus and plant totals are updated. In the case of conflicting machine identifiers, those machines originally at the retained bus keep their original machine identifiers and new identifiers are assigned to the transferred machines.

All changes to branch, load, shunt, and machine identifiers are reported at the *Progress* device (refer to Section 4.4, “Virtual Output Devices”). Alternative identifier settings may be implemented via activity MBID either before or after the execution of activity JOIN.

If the retained bus does not have a switched shunt and the deleted one does, the switched shunt of the removed bus is transferred to the retained bus.

If both buses have switched shunts, the switched shunts are merged, with blocks being assigned in the following order:

1. Reactors from the removed bus.
2. Reactors from the retained bus.
3. Capacitors from the retained bus.
4. Capacitors from the removed bus.

Control parameters from the retained bus are kept unless the switched shunt at the retained bus controls the admittance setting of the switched shunt at the removed bus. In this case, the control parameter data items (MODSW, VSWHI, VSWLO, SWREM, RMPCT and RMIDNT) of the removed bus are transferred to the retained bus.

If either the retained bus or removed bus is involved in multi-section line groupings, each such grouping is either:

- Redefined if the topology change results in a valid multi-section line grouping.

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- Deleted if the redefined grouping violates any of the requirements for multi-section line groupings (refer to Multi-Section Line Grouping Data).

If sequence data is contained in the working case (refer to RESQ), it is handled appropriately, including the rerouting or removal of zero sequence mutuals. If the option to move line shunts of deleted branches to the retained bus is enabled, such zero sequence line shunts are handled in the same manner as they are in the positive squence. Zero sequence switched shunts are treated in the same manner as they are in the positive sequence.

5.17.2. Example Progress Output

```
DUPLICATE LOAD IDENTIFIER 1
IDENTIFIER OF TRANSFERRED LOAD CHANGED TO 3
```

Figure 5.5. Join Buses Summary Report on Relocation of Load

The example summary report (Figure 5.5, "Join Buses Summary Report on Relocation of Load") indicates that the load on Bus 3008, with identifier 1, was moved to bus 154 and its identifier changed to 3 so as to avoid conflict with two existing loads on that bus.

Additional Information
PSS®E GUI Users Guide, Joining Buses *PSS®E Command Line Interface (CLI) Users Guide*, Joining Buses *PSS®E Application Program Interface (API)*, JOIN

5.18. Splitting Buses

SPLT

Run Activity SPLT - GUI
Power Flow > Changing > Split buses (SPLT)... *[Split Buses]*
Run Line Mode Activity SPLT - CLI
ACTIVITY? >>**SPLT** ENTER BUS NUMBER: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The bus sectionalizing activity SPLT enables the user to split a bus into two buses connected by a branch.

The SPLT API routine includes the following among its inputs:

- The number of the bus to be split.

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- The number to be assigned to the new bus; the bus number specified must not already be assigned to another bus in the working case.
- The name to be assigned to the new bus.
- The base voltage to be assigned to the new bus.

5.18.1. Operation of Activity SPLT

The new bus is given the same area, zone, and owner assignments as the original bus. When generation is moved from the original to the new bus, the new bus is assigned the bus type code of the original bus. If all machines are moved to the new bus, the original bus becomes a Type 1 bus. If no machines are moved to the new bus, the original bus type code is unchanged and the new bus becomes a Type 1 bus with no generator entry.

A new jumper branch is created connecting the original and new buses. If the zero impedance line threshold tolerance, THRSHZ, is greater than zero, this branch is created as a zero impedance line (refer to Zero Impedance Lines). Otherwise, a branch with an impedance of j0.0001 is introduced between the original and new buses. The new branch is given the same owner as the original bus.

If the original bus is an endpoint bus of one or more multi-section line groupings and the adjacent line section of a multi-section line grouping is rerouted to the new bus, the new bus becomes the endpoint bus of the multi-section line.

Activity SPLT optionally allows some or all of each of the following equipment items connected to the bus being split to be transferred to the new bus:

- Synchronous machines.
- Induction machines.
- Loads.
- Fixed bus shunts (available only from the GUI; when running activity SPLT in line mode, any fixed shunts connected to the bus being split remain there).
- Switched shunt.
- Ac branches.

If the original bus is a dummy bus of a multi-section line grouping, the multi-section line grouping is either:

- Redefined if exactly one of the two branches connected to the original bus is rerouted to the new bus.
- Deleted if neither or both are rerouted to the new bus.

The new bus is given a type code of 1 (or 2 or 3 if generation is moved to it) and the jumper branch between the original bus and the new bus is set to in-service *unless* the new bus becomes a dummy bus of an out-of-service multi-section line grouping; in this case, the bus type code is set to 4 and the jumper branch is set to out-of-service.

If sequence data is contained in the working case (refer to RESQ), it is handled appropriately, including the rerouting of zero sequence mutuals. Branches introduced by activity SPLT have their zero sequence impedances set to j0.0001.

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After they have been created with activity SPLT, the data associated with the new bus and branch may be modified with activity CHNG or with the *[Spreadsheet]* (and with activity SQCH if sequence data is contained in the working case) just as any other bus and branch in the working case.

5.18.2. Example of Activity SPLT

Using the *savnw.sav* Saved Case File, bus 3003 can be split and circuit 2 from bus 3005 reassigned to the new bus 3020. In the example, the option to move circuit 2 from bus 3005 to the new bus has been selected. A summary of the changes is shown in the *Progress* tab (Figure 5.6, "Output Report Summary fo Bus Split").

```
SPLITTING BUS    3003 [S. MINE     230.00]

CIRCUIT 2  FROM   3005 [WEST        230.00] TO   3003 [S. MINE     230.00] MOVED--IS NOW
CIRCUIT 2  FROM   3005 [WEST        230.00] TO   3020 [NEWBUS      230.00]
```

Figure 5.6. Output Report Summary fo Bus Split

Additional Information
PSS®E GUI Users Guide, Splitting Buses *PSS®E Command Line Interface (CLI) Users Guide*, Splitting Buses *PSS® E Application Program Interface (API)* SPLT MOVE3WND MOVEBRN MOVELOAD MOVELOADS MOVEMAC MOVEPLNT MOVESHUNT MOVESHUNTS MOVESWS

5.19. Tapping a Line

LTAP

Run Activity LTAP - GUI
Power Flow > Changing > Tap line (LTAP)... *[Tap Line]*
Run Line Mode Activity LTAP - CLI
ACTIVITY? >>**LTAP** ENTER FROM BUS, TO BUS, CIRCUIT IDENTIFIER   (FROM BUS = 0 TO EXIT): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The line tapping activity LTAP enables the user to introduce a new bus into the working case at a designated location along a specified ac branch. Any non-transformer branch may be tapped with activity LTAP.

The LTAP API routine includes the following among its inputs:

- The two buses connected by the branch to be tapped.
- The circuit identifier of the branch to be tapped.
- The location of the new bus relative to the first bus specified in per unit of total line length.

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- The number to be assigned to the new bus; the bus number specified must not already be assigned to another bus in the working case.
- The name to be assigned to the new bus.
- The base voltage to be assigned to the new bus.

If the specified branch is not contained in the working case, or if it is a transformer, an appropriate error message is printed.

The new bus is then introduced into the working case and assigned to the area, zone, and owner of the nearer bus. The new bus is given a bus type code of 1 if the branch being tapped is in-service, or a type code of 4 if the branch is out-of-service. It is made a passive node with no load, generation, or shunt elements.

The original branch is split into two branches: one from the original from bus to the new bus, and one from the new bus to the original to bus, both with appropriate fractions of the original branch impedance and charging. Line connected shunts on the new branches are set such that the line connected shunts from the original branch remain at the original from and to bus. Circuit identifiers, ratings, status, metered ends, and ownership follow from the original branch.

If the original branch was a member of a multi-section line grouping, the grouping's definition is modified to include the two new branches in place of the original branch.

If the working case contains sequence data (refer to RESQ), zero sequence branch parameters are handled in the same manner as their positive sequence counterparts. If the original branch is involved in a zero sequence mutual coupling, the mutual data arrays are modified and extended as appropriate. All mutual data changes are displayed at the *Progress* device (refer to Section 4.4, "Virtual Output Devices").

In the example (Figure 5.7, "Output Summary of Line Tap"), the *Progress* summary report registers the tapping process and also indicates that the mutual coupling has been modified to take into account the new topology.

```
TAPPING CIRCUIT 1  FROM BUS    3003 [S. MINE     230.00] TO BUS    3005 [WEST

MUTUAL COUPLING RECORD:
  FROM      TO CKT    FROM      TO CKT  MUTUAL IMPEDANCE  BIJ1    BIJ2    BKL1    BKL2
  3003    3005  1     3003    3005  2    0.00250  0.02500 0.0000 1.0000 0.0000 1.0000
CHANGED TO:
  3003    3020  1     3003    3005  2    0.00100  0.01000 0.0000 1.0000 0.0000 0.4000
  3005    3020  1     3003    3005  2   -0.00150 -0.01500 0.0000 1.0000 0.4000 1.0000
```

Figure 5.7. Output Summary of Line Tap

Additional Information
PSS®E GUI Users Guide, Tapping a Line
PSS®E Command Line Interface (CLI) Users Guide, Tapping a Line
PSS®E Application Program Interface (API), LTAP

5.20. Changing Equipment Identifiers

MBID

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Change Equipment Identifiers - GUI
Use *[Spreadsheet]*
Run Line Mode Activity MBID - CLI
ACTIVITY? >>**MBID** ENTER 0 TO EXIT 1 FOR MACHINE ID's 2 FOR CIRCUIT ID's 3 FOR INTER-AREA TRANSFER ID's 4 FOR LOAD ID's 5 FOR TWO-TERMINAL DC LINE NAMES 6 FOR MULTI-TERM DC LINE NAMES 7 FOR FACTS DEVICE NAMES 8 FOR 3-WINDING TRANSFORMER IDs 9 FOR VSC DC LINE NAMES 10 FOR SHUNT ID's: >>
Interrupt Control Codes
None

PSS®E permits the user to change the following types of alphanumeric identifiers associated with equipment represented in the working case:

- Synchronous machine, induction machine, load, and fixed shunt identifiers.
- Circuit identifiers of non-transformer branches, two-winding transformers, three-winding transformers, and multi-section line groupings.
- Two-terminal, multi-terminal, and VSC dc line names.
- FACTS device names.
- GNE device names.
- Inter-area transfer identifiers.

The appropriate tool for changing working case equipment identifiers is dependent upon the interface through which PSS®E is being operated.

In the GUI, the primary means of changing individual data items is the *[Spreadsheet]* (refer to *PSS® E GUI Users Guide*, Spreadsheet View).

In line mode, the equipment identifier modification activity MBID is the primary tool for making these data changes to the working case (refer to *PSS® E Command Line Interface (CLI) Users Guide*, Changing Equipment Identifiers).

In Python programs, the MBIDaaa identifier changing API routines described in Chapter 1, Power Flow Operation of the *PSS® E Application Program Interface (API)* manual are used to change data associated with existing elements in the working case.

In Response Files and IPLAN programs, activity MBID and/or the identifier changing API routines may be used to change equipment identifiers in the working case.

Each non-transformer branch is categorized as either a normal branch, a breaker, or a switch by the first character of its circuit identifier: an at sign (@) for a breaker, an asterisk (*) for a switch, and any other character other than an ampersand (&) for a normal branch (refer to Non-Transformer Branch Data and Multi-Section Line Grouping Data). Making appropriate changes to its identifier allows the user to change the characterization of a non-transformer branch from its present category to a different category. For example, a

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branch with identifier @1 is characterized as a breaker; changing its identifier to 1 changes its characterization to a normal branch.

In changing the identifier of a multi-section line grouping, the circuit identifier *must* have an ampersand (&) as its first character; refer to Multi-Section Line Grouping Data. Similarly, in changing the circuit identifier of a non-transformer branch, two-winding transformer, or three-winding transformer, the new identifier may *not* have an ampersand specified as its first character.

In changing the identifier of an equipment item, if there already exists an item of the same type with the new identifier, an appropriate message is printed and the user is instructed to provide a different identifier.

Additional Information
PSS®E GUI Users Guide, Changing Equipment Identifiers *PSS®E Command Line Interface (CLI) Users Guide*, Changing Equipment Identifiers *PSS®E Application Program Interface (API)* MBID2DC , MBID3WND , MBIDATRN , MBIDBRN , MBIDFACTS , MBIDGNE , MOVELOAD , MBIDMAC , MBIDMDC , MBIDMSL , MBIDSHUNT , MBIDVSC

5.21. Moving Equipment

MOVE

Run Activity MOVE - GUI
Power Flow > Changing > Move network elements (MOVE)... *[Move Network Elements]*
Run Line Mode Activity MOVE - CLI
ACTIVITY? >>**MOVE** EQUIPMENT ITEMS WHICH MAY BE MOVED ARE: 1 = AC LINES 2 = SWITCHED SHUNTS 3 = MACHINES 4 = PLANTS 5 = INDIVIDUAL LOADS 6 = ALL LOADS AT A BUS 7 = THREE-WINDING TRANSFORMERS 8 = INDIVIDUAL FIXED SHUNTS 9 = ALL FIXED SHUNTS AT A BUS ENTER EQUIPMENT CATEGORY CODE: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The equipment transferal activity MOVE allows the user to move specified fixed shunts, switched shunts, loads, synchronous machines, plants, and induction machines from one bus to another. It also provides for connecting the far end of specified branches and one winding of a three-winding transformer to different to buses.

5.21.1. Operation of Activity MOVE

Following the selection of the equipment category, the user specifies the equipment to be moved and the bus to which the equipment is to be transferred. Following each move, a summary of the action taken is printed at the *Progress* device.

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When rerouting a branch for which the circuit identifier matches that of a branch already existing between the from bus and the new to bus, the existing branch keeps its original identifier and the user specifies a new identifier for the rerouted branch. A similar approach is used in the rerouting of three-winding transformers.

The terms from and to in this network element moving process are not related to the transformer data attributes that relate to windings and taps. Here the from and to are used merely to select branches, recognizing that the to end is the end that will be relocated. For three-winding transformers there are three buses to be selected; the from, to and a third bus. It is the third bus that can be moved.

In rerouting branches and three-winding transformers, if the new to bus is a dummy bus of a multi-section line grouping, that line grouping is deleted because the new to bus will have three branches connected to it following the rerouting.

If the branch being rerouted is a member of a multi-section line grouping and the original to bus is a dummy bus, the multi-section line grouping is deleted; if the original to bus is an endpoint bus of the multi-section line grouping, either the grouping is deleted if the new to bus is the other endpoint bus, or the grouping is redefined.

In rerouting a branch that is involved in zero sequence mutual couplings (refer to Zero Sequence Mutual Impedance Data), the mutual arrays are updated under the assumption that the rerouted branch retains the same couplings as the original branch.

In moving switched shunts, if the destination bus does not have a switched shunt and the original one does, the switched shunt of the original bus is transferred to the destination bus. If both buses have switched shunts, the switched shunts are merged, with blocks being assigned in the following order:

1. Reactors from the original bus.
2. Reactors from the destination bus.
3. Capacitors from the destination bus.
4. Capacitors from the original bus.

Control parameters from the destination bus are kept unless the switched shunt at the destination bus controls the admittance setting of the switched shunt at the original bus. In this case, the control parameter data items (MODSW, VSWHI, VSWLO, SWREM, RMPCT and RMIDNT) of the original bus are transferred to the destination bus.

When moving synchronous machines, if the destination bus is not a generator bus, the plant data of the original bus is copied to the destination bus. If both buses are generator buses, the machine is transferred to the designated bus and plant totals of both buses are updated. In the case of conflicting machine identifiers, those machines already at the destination bus keep their original machine identifiers and the user must provide a new identifier for the moved machine. If all machines at a bus are moved to other buses, its plant entry is removed.

In moving plants, the destination bus must not be a generator bus; all machines at a plant may be moved to another plant only by moving each individual machine.

In moving machines and plants, bus type codes are updated appropriately. If any moves result in the plant entry of an area swing being deleted, an appropriate message is printed. If the current destination bus is in the same area as the former area swing bus, it becomes the new area swing. Otherwise, the area swing bus number is set to zero (refer to Area Interchange Data and Area Interchange Control).

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When moving individual loads, if both buses are load buses and a load with the same identifier as the load being moved is already present at the destination bus, the load previously at the destination bus keeps its original load identifier and the user must provide a new identifier for the moved load. If all loads at a bus are moved to other buses, an appropriate message is printed. In moving all loads at a bus to another bus, the destination bus must not have any other loads; all loads at a bus may be moved to another load bus only by moving each individual load.

When moving individual fixed bus shunts, if both buses are fixed shunt buses and a fixed shunt with the same identifier as the shunt being moved is already present at the destination bus, the shunt previously at the destination bus keeps its original shunt identifier and the user must provide a new identifier for the moved shunt. If all fixed shunts at a bus are moved to other buses, an appropriate message is printed. In moving all fixed shunts at a bus to another bus, the destination bus must not have any other fixed shunts; all fixed shunts at a bus may be moved to another fixed shunt bus only by moving each individual shunt.

When moving individual induction machines, if both buses are induction machine buses and an induction machine with the same identifier as the induction machine being moved is already present at the destination bus, the induction machine previously at the destination bus keeps its original machine identifier and the user must provide a new identifier for the moved induction machine. If all induction machines at a bus are moved to other buses, an appropriate message is printed. In moving all induction machines at a bus to another bus, the destination bus must not have any other induction machines; all induction machines at a bus may be moved to another induction machine bus only by moving each individual induction machine.

The example displays the *savnw.sav* Saved Case File. The 500 kV line from bus 151, which terminates at bus 201 will be moved to terminate at bus 202. A summary of the moving action is printed at the *Progress* device (Figure 5.8, "Branch Move Output Report").

```
CIRCUIT 1  FROM    151 [NUCPANT     500.00] TO    201 [HYDRO       500.00] MOVED--IS NOW
CIRCUIT 1  FROM    151 [NUCPANT     500.00] TO    202 [EAST500     500.00]
```

Figure 5.8. Branch Move Output Report

5.21.2. Application Notes

Activity MOVE *does not* check that the working case is in the form required by the power flow solution activities. The user is advised to use activities such as TREE (to check for islands and system swing bus specification) and CNTB and EXAM (to check the control parameters of merged plants and switched shunts) before attempting to solve the modified case.

Additional Information
PSS®E GUI Users Guide, Moving Equipment
PSS®E Command Line Interface (CLI) Users Guide, Moving Equipment
PSS®E Application Program Interface (API) MOVE3WND , MOVEBRN , MOVELOAD , MOVELOADS , MOVEMAC , MOVEPLNT , MOVESHUNT , MOVESHUNTS , MOVESWS

5.22. Renumbering Buses in Auxiliary Files

RNFI

Run Activity RNFI - GUI
File > Renumber buses in auxiliary files (RNFI)...

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Run Activity RNFI - GUI
[Renumber Buses in Auxiliary Files]
Run Line Mode Activity RNFI - CLI
ACTIVITY? >>**RNFI** ENTER BUS TRANSLATION FILE NAME (FROM BSNM): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The auxiliary data input file bus renumbering activity RNFI reflects changes in bus numbering in auxiliary data input files. It is used primarily in conjunction with activity BSNM to coordinate bus renumbering implemented in a Saved Case with auxiliary data input files associated with the Saved Case and read by other PSS®E activities.

The RNFI API routine includes the following among its inputs:

- The type of data file to be processed.
- The name of the auxiliary data input file to be processed.
- The name of the output file reflecting the bus renumbering.
- The name of a Bus Renumbering Translation Data File.

5.22.1. Bus Renumbering Translation Data File

During the bus renumbering process, PSS®E provides for the creation of an optional Bus Renumbering Translation Data File. This file contains a record of which bus numbers were changed and what the new bus numbers are. The advantage of directing the output of activity BSNM to a file is that this file may be specified to activity RNFI for the purpose of implementing these bus number changes in one or more of the auxiliary data input files associated with the original case. The files constructed by activity RNFI are then compatible with the renumbered case.

Thus, the usual input to activity RNFI is the output file created by activity BSNM. However, a user-created data input file containing data records of the following form may also be used as the input to activity RNFI:

```
old bus number   new bus number
```

5.22.2. Operation of Activity RNFI

The user specifies the Bus Renumbering Translation File and (optionally) the name of an output file into which activity RNFI is to place data records containing the new bus numbering scheme. Although the output data file is optional, it is the primary output of activity RNFI. This file will contain a copy of the processed auxiliary data file reconstructed with the new numbering. Consequently it will be immediately available for use with the renumbered case. The alternative is to allow the reconstructed file to be printed at the *Report* device for previewing purposes.

The same file may not be specified for both the auxiliary data input file and the output data file. That is, activity RNFI is not able to replace the input file with a modified version of the file that reflects the bus renumbering.

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Activity RNFI is able to process the following auxiliary data files:

- Sequence Data File (used by activity RESQ)
- Drawing Coordinate Data File (obsolete one-line diagram definition format file that may be imported into the *[Diagram]* view)
- Dynamics Data File (used by activity DYRE)
- Machine Impedance Data File (used by activity MCRE)
- Economic Dispatch Data File (used by activity ECDI)
- Inertia and Governor Response Data File (used by activity INLF)
- Breaker Duty Data File (used by activity BKDY)
- Fault Specification Data File (used by activity BKDY)
- Fault Control Data (used by activity ASCC and IECS)
- Machine Capability Curve Data (used by activity GCAP)
- Load Throwover Data File (used by the various ac contingency calculation functions, the P-V Analysis function, and the Q-V analysis function)
- ANSI Fault Specification Data File (used by activity ANSI)
- Subsystem Description Data File (used by activities DFAX and RANK)
- Monitored Element Data File (used by activities DFAX and RANK)
- Contingency Description Data File (used by activity DFAX)
- Tripping Element Data File (see Section 6.15, "Performing Multi-Level AC Contingency Solution")

In processing Dynamics Data Files, data records for User-Written Models are copied to the destination file without change.

As with all files used in the course of a PSS®E engineering study, the user is urged to note the relationship between power flow Saved Cases and their various related data files used as input to, and generated as output by, activity RNFI. Refer to Section 2.5, "Creating Input Files", Section 2.6, "Files Created By PSS®E"and Section 2.8, "Deleting Files".

Additional Information
PSS®E GUI Users Guide, Renumbering Buses in Auxiliary Data Files
PSS®E Command Line Interface (CLI) Users Guide, Renumbering Buses in Auxiliary Files
PSS®E Application Program Interface (API), RNFI
See also: Section 5.46, "Bus Renumbering"

5.23. Changing Owner Assignments

OWNM

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Run Activity OWNM - GUI
Power Flow > Renumbering Areas / Owners / Zones ... *[Renumbering Areas / Owners / Zones]* *Owner assignments* Tab
Run Line Mode Activity OWNM - CLI
ACTIVITY? >>**OWNM** ENTER 1 TO CHANGE OWNERSHIP OF BUSES: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The owner renumbering activity OWNM reassigns buses, loads, synchronous machines, induction machines, branches, FACTS devices, VSC dc lines, and/or GNE devices in the working case from their original owners to a designated owner. All elements of the selected equipment types in the working case may be reassigned to a designated owner; more typically, all elements of the selected equipment types in a specified subsystem of the working case (refer to Section 4.8, "Subsystem Selection") are assigned to a designated owner.

The OWNM_3 API routine includes the following among its inputs:

- A designation of the subsystem in which the owner assignments of selected equipment types are to be reassigned; refer to Section 4.8, "Subsystem Selection".
- Eight flags to indicate whether owner assignments of subsystem buses, loads, synchronous machines, induction machines, branches, FACTS devices, VSC dc lines, and GNE devices are to be changed.
- A flag indicating which subsystem series elements (i.e., branches, FACTS devices with a series element, VSC dc lines, and series GNE devices) are to be processed:
- Subsystem branches only (all of its terminals are in the specified subsystem).
- Subsystem ties only (at least one of its terminals is in the specified subsystem, and at least one is in a neighboring subsystem).
- Both subsystem branches and ties.
- The owner to which subsystem elements are to be assigned.

5.23.1. Operation of Activity OWNM

Following the reassignment process, activity OWNM logs the following occurrences and conditions at the *Progress* device (refer to Section 4.4, "Virtual Output Devices"):

- The destination owner has a blank name and activity OWNM assigns to it the owner name from an owner that has a non-blank name and that is emptied during the owner reassignment process.
- An owner no longer has any equipment assigned to it.

Then activity OWNM then lists the number of buses, loads, synchronous machines, induction machines, branches, FACTS devices, VSC dc lines, and/or GNE devices transferred to the destination owner, and the total number of buses, loads, synchronous machines, induction machines, branches, dc buses, FACTS devices,

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VSC dc lines, and GNE devices now assigned to the owner (see Figure 5.9, "Example Output Report of Owner Reassignment"). If no changes were made, activity OWNM displays:

```
 No owner assignment changes are implemented for
the selected subsystem

 ALL EQUIPMENT REMOVED FROM OWNER   22 [GEN 2       ]

 OWNERSHIP CHANGED FOR:
      3 BUSES
      3 MACHINES
      3 BRANCHES

 OWNER    2 [TRAN 2      ] CONTAINS:
   BUSES     LOADS   MACHINES   BRANCHES   DC BUSES   FACTS   VSC DC   GNE
       7         2          5          8          0       0        0     0
```

Figure 5.9. Example Output Report of Owner Reassignment

5.23.2. Application Notes

To determine if an equipment item is processed in subsystem mode, for any active attribute (e.g., area), if the equipment item possesses the attribute (e.g., loads have an area attribute), it is used; otherwise (e.g., machines do not have an area attribute), the corresponding attribute of the bus is used.

When reassigning ownership for synchronous machines, branches, and VSC dc lines previously designated as having multiple owners, each ownership block is assigned to the new owner unless subsystem specification by owner is enabled. When using subsystem specification by owner, only those ownership blocks previously assigned to the specified owners are reassigned.

When reassigning ownership for FACTS and GNE devices, devices connected to a single bus are not candidates for processing when using series element renumbering option 2 (subsystem tie branches). Series FACTS and GNE devices are processed using the same criteria as ac branches and VSC dc lines (only subsystem branches, only subsystem ties, or both subsystem branches and ties).

Activity OWNM does not change the owner assignments of dc buses internal to multi-terminal dc lines (refer to Multi-Terminal DC Transmission Line Data).

Additional Information
Areas, Zones and Owners
PSS®E GUI Users Guide, Changing Owner Assignments
PSS®E Command Line Interface (CLI) Users Guide, Changing Owner Assignments
PSS®E Application Program Interface (API), OWNM_3

5.24. Changing Zone Assignments

ZONM

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LDZO

Run Activity ZONM - GUI
Power Flow > Renumbering Areas / Owners / Zones ... *[Renumbering Areas / Owners / Zones]* *Zone assignments* Tab
Run Line Mode Activities ZONM and LDZO - CLI
ACTIVITY? >>**ZONM** ENTER UP TO 20 BUS NUMBERS >> ACTIVITY? >>**LDZO** ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The zone renumbering activity ZONM reassigns buses, loads, and/or induction machines in the working case from their original zones to a designated zone. All buses, loads, and/or induction machines in the working case may be reassigned to a designated zone; more typically, buses, loads, and/or induction machines in a specified subsystem of the working case (refer to Section 4.8, "Subsystem Selection") are assigned to a designated zone.

The ZONM_2 API routine includes the following among its inputs:

- A designation of the subsystem in which the zone assignments of buses and/or loads are to be reassigned; refer to Section 4.8, "Subsystem Selection".
- A flag to indicate whether zone assignments of subsystem buses are to be changed.
- A flag to indicate whether zone assignments of subsystem loads are to be changed.
- A flag to indicate whether zone assignments of subsystem induction machines are to be changed.
- The zone to which subsystem elements are to be assigned.

In the GUI, the ZONM_2 API routine is accessed from the *Zone assignments* tab of the *[Renumbering Area / Owners / Zones]* dialog for modifying the zone assignments of buses, loads and induction machines.

In line mode, zone assignments of buses are changed using activity ZONM, and zone assignments of loads are modified using activity LDZO.

5.24.1. Operation of Activity ZONM

Following the reassignment process, activity ZONM logs the following occurrences and conditions at the *Progress* device (refer to Section 4.4, "Virtual Output Devices"):

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- The destination zone has a blank name and activity ZONM assigns to it the zone name from a zone that has a non-blank name and that is emptied during the zone reassignment process.
- A zone no longer has any equipment assigned to it.

Then activity ZONM lists the number of buses, loads and induction machines transferred to the destination zone, and the total number of buses, induction machines, loads, and dc buses now assigned to the zone. If no changes were made, activity ZONM displays:

```
No zone assignment changes are implemented for the selected subsystem
```

5.24.2. Application Notes

Activity ZONM *does not* change the zone assignments of dc buses internal to multi-terminal dc lines (refer to Multi-Terminal DC Transmission Line Data).

Additional Information
Areas, Zones and Owners *PSS®E GUI Users Guide*, Changing Zone Assignments *PSS®E Command Line Interface (CLI) Users Guide*, Changing Zone Assignments of Buses *PSS®E Application Program Interface (API)*,ZONM_2

5.25. Reading Sequence Data Additions from the Terminal

TRSQ

Run Line Mode Activity TRSQ - CLI
ACTIVITY? >>**TRSQ** ENTER POSITIVE SEQUENCE MACHINE IMPEDANCES I, ID, R, X >>
Interrupt Control Codes
AB

Activity TRSQ is special purpose version of activity RESQ available only in line mode. It is designed for the purpose of adding a limited amount of sequence data to the system contained in the working case, with data records read from the dialog input device (the terminal keyboard, a Response File, or an IPLAN program).

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Reading Sequence Data Additions from the Terminal *PSS®E Application Program Interface (API)*, RESQ See also: Section 5.6.1, “Sequence Data File Contents”

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5.26. Changing Sequence Data

SQCH

Change Sequence Data - GUI
Use *[Spreadsheet]*
Run Line Mode Activity SQCH - CLI
ACTIVITY? >>**SQCH** ENTER CHANGE CODE: 0 = EXIT ACTIVITY 1 = BUS SHUNT LOADS (ZERO & NEG SEQS) 2 = GENERATOR IMPEDANCES 3 = ZERO SEQ NONTRANSFORMER BRANCH DATA 4 = ZERO SEQ SWITCHED SHUNTS 5 = ZERO SEQ 2 WINDING TRANSFORMER DATA 6 = ZERO SEQ 3 WINDING TRANSFORMER DATA 7 = ZERO SEQ FIXED SHUNTS: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The sequence data modification activity SQCH allows the user to make changes to the various sequence data arrays in the working case. The appropriate tool for changing working case data is dependent upon the interface through which PSS®E is being operated.

In the GUI, the primary means of changing individual data items is the *[Spreadsheet]* (refer to *PSS® E GUI Users Guide*,Spreadsheet View). The *[Spreadsheet]* may also be used to add new zero sequence mutual elements to the working case.

In line mode, the sequence data modification activity SQCH is the primary tool for making sequence data changes to the working case (refer to *PSS® E Command Line Interface (CLI) Users Guide*, Changing Sequence Data).

In Python programs, the data entry/changing API routines described in Chapter 2, Power Flow Data Changing of the *PSS® E Application Program Interface (API)* manual are used to change data associated with existing elements in the working case (the SEQ_MUTUAL_DATA routine may also be used to add new zero sequence mutuals to the working case). The name of each of the API routines associated with sequence data starts with SEQ_.

In Response Files and IPLAN programs, activity SQCH and/or the data entry/changing API routines may be used to change sequence data in the working case.

Activity SQCH allows access to the following items of data:

- Zero and negative sequence load elements. Note that the ground paths corresponding to fixed shunt elements *are not* included as part of the zero and negative sequence shunt elements (refer to Zero Sequence Fixed Shunt Data). In the case of zero sequence loads, note that the ground ties created by grounded transformer windings *are not* included as part of the load elements (refer to Load Sequence Data and ???). Note also that care is required in manipulating exceptional (i.e., non-zero) negative and zero sequence shunt loads (refer to Load Sequence Data). In particular, when changing the positive sequence loading at a bus (with activity CHNG or the *[Spreadsheet]*), it is the user's responsibility to ensure that any corresponding exceptional negative and zero sequence load is coordinated with it.
- Zero, positive, and negative sequence machine impedances (refer to Generator Sequence Data).

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- Zero sequence data for non-transformer branches (refer to Zero Sequence Non-Transformer Branch Data).
- Zero sequence switched shunt step admittances (refer to Zero Sequence Switched Shunt Data).
- Zero sequence data for transformer branches (refer to Zero Sequence Transformer Data).
- Zero sequence fixed shunt elements (refer to Zero Sequence Fixed Shunt Data). Note that the ground ties created by grounded transformer windings *are not* included as part of the fixed shunt elements (refer to Load Sequence Data and ???).
- Zero sequence mutual impedance data (refer to Zero Sequence Mutual Impedance Data).

Network topological data (e.g., branch status, bus type codes, etc.), positive sequence loading data, positive sequence branch data, and positive sequence fixed and switched shunt data changes are handled with the power flow data changing activity CHNG or the *[Spreadsheet]*.

Additional Information
PSS®E GUI Users Guide, Changing Sequence Data *PSS®E Command Line Interface (CLI) Users Guide*, Changing Sequence Data See also: Section 5.6, "Reading Sequence Data"

5.27. Checking Branch Parameters

BRCH

Run Activity BRCH - GUI
Power Flow > Check Data > Branch parameters (BRCH)... *[Check Branch Parameters]*
Run Line Mode Activity BRCH - CLI
ACTIVITY? >>**BRCH** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The branch parameter checking activity BRCH tabulates those branches where impedances or other characteristics are such that they may be detrimental to the rate of convergence of one or more of the power flow solution activities (refer to SOLV, MSLV, FNSL, NSOL, FDNS, INLF, and ACCC).

The BRCH_2 API routine includes the following among its inputs:

- A designation of the subsystem in which branch parameters are to be checked; refer to Section 4.8, "Subsystem Selection".
- The selection of one or more checks to be performed.
- Threshold or limits values used by several of the checks.

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5.27.1. Branch Parameter Checks Available

Table 5.1, "Branch Parameter Data Check Options" describes the branch parameter checks that can be enabled in activity BRCH. For those checks that use a limit or threshold value, the default value is listed.

Table 5.1. Branch Parameter Data Check Options

Data Check	Default Limit or Threshold Value	Description
Small Reactance	0.0005 pu	Any branch where reactance magnitude is less than threshold value will be listed. Very small impedance branches not being treated as zero impedance lines may result in a slowing of the rate of convergence of the power flow solution activities, and their inability to reach the default convergence tolerances.
Large Reactance	1.0 pu	Any branch where reactance magnitude exceeds this value will be listed. Large impedance branches in themselves do not cause problems with the power flow solution methods. However, when relatively small impedance branches are also connected to these buses, convergence may be slowed and the solution activities may not be able to reach the default convergence tolerances. In addition, very large impedances are often present in inefficient network equivalents with some buses having an unusually large number of branches connected to them. This will generally result in the loss of the beneficial sparsity characteristics present with real system when the system matrices are processed, with a corresponding increase in solution times.
High Resistance to Reactance Ratio	0.66667	Any branch where resistance magnitude exceeds reactance magnitude times the threshold will be listed. The decoupled Newton-Raphson power flow solution, activity NSOL, will diverge if there are any branches where resistance is greater than reactance. The other solution methods are not par-

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Data Check	Default Limit or Threshold Value	Description
		ticularly sensitive to this ratio. The default value generates an alarm for branches with a resistance magnitude greater than 2/3 of reactance magnitude.
Negative Reactance	n.a.	Branches with a negative reactance are listed. If either of the buses for such a branch is a generator bus (i.e., its type code is 2 or 3), an asterisk (*) is printed before the branch circuit identifier. The Gauss-Seidel power flow solution, activity SOLV usually diverges if the working case contains any in-service negative reactance branches. The modified Gauss-Seidel power flow solution, activity MSLV, is able to handle negative reactance branches between Type 1 buses, but negative reactances connected to Type 2 or 3 buses usually result in its divergence.
High Reactance Ratio	500.0	Activity BRCH scans each bus and tabulates all lines from any bus for which the ratio of largest to smallest branch reactance magnitude exceeds a specified threshold. In this check, buses connected by Zero Impedance Lines are treated as the same bus in determining the largest and smallest connected reactance; zero impedances are neglected in determining the smallest reactance. The presence of a bus with a wide range of reactances connected to it may slow the rate of convergence of the power flow solution activities and result in failure to reach the default convergence tolerances.
High or Negative Charging	5.0 pu	The line charging entry for each branch is checked and those branches where line charging is greater than the threshold or negative are tabulated. High values of line charging do not necessarily cause problems with the power flow solution methods ex-

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Data Check	Default Limit or Threshold Value	Description
		cept in cases of data entry errors (for example, all charging values entered in Mvar rather than per unit resulting in abnormally high voltages). Negative values of line charging are data errors.
Parallel Transformers	n.a.	Activity BRCH scans parallel branches between pairs of buses and generates an alarm for: • Non-transformer and transformer branches in parallel. • Parallel two-winding transformers with different Winding 1 complex tap ratios. • Parallel two-winding transformers with different Winding 2 tap ratios. • Parallel two-winding transformers with their Winding 1/Winding 2 side relationship reversed. As in the reactance ratio check, buses connected by zero impedance lines are treated as the same bus. Transformers listed in this check have an F or T following the Winding 1 tap ratio to indicate the from bus or to bus as the Winding 1 side bus.
High Tap Ratio	1.1	Any transformer branch for which the tap ratio of any winding is beyond the specified limit is tabulated. Transformers listed in this check have an F or T following the Winding 1 tap ratio to indicate the from bus or to bus as the Winding 1 side bus.
Low Tap Ratio	0.9	
Transformer Loss Data Inconsistencies	n.a.	Any transformer branch for which its copper loss (load loss) is less than its no-load loss is listed. No-load loss is derived from the transformer's per unit magnetizing conductance, the nominal winding 1 voltage base, the base

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Data Check	Default Limit or Threshold Value	Description
		voltage of the winding 1 bus, and the system MVA base. For two-winding transformers, copper loss is derived from the transformer's per unit winding resistance, the three phase winding 1 to winding 2 base MVA, and the system MVA base. For three-winding transformers, copper loss is derived from the three winding-to-winding per-unit resistances, the three winding-to-winding three phase MVA bases, and the system MVA base.
Missing Zero Sequence Impedance	n.a.	Non-transformer branches with a zero sequence impedance of (0.0 + j 0.0) are listed. This test is bypassed if sequence data is not contained in the working case.

When branches are reported by activity BRCH, this does not necessarily mean that their data is incorrect. Rather, it indicates the potential for difficulties in obtaining a power flow solution which converges to the specified tolerances.

5.27.2. Operation of Activity BRCH

The user specifies either that the entire working case is to be processed, or that branch data checks are to be performed for branches that are connected to at least one bus in the specified subsystem.

When a subsystem is being processed and ownership is the only selection criterion, each branch wholly or partly owned by any of the owners specified is included in subsequent checks (refer to Non-Transformer Branch Data and Transformer Data). The owner assignments of the branch's endpoint buses are not considered.

When one or more of the other subsystem selection criteria are enabled, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected branches are candidates for processing; otherwise, all buses are candidate buses. The owner selection criterion, if enabled, determines which such branches at candidate buses are to be processed; otherwise, all such branches at candidate buses are processed.

For each of the checks, the report contains the from bus and to bus identifiers and areas, and the circuit identifier and status of each branch tabulated.

For the parallel transformer and high and low tap ratio checks, the transformer's per unit impedance is printed, along with the Winding 1 tap ratio and phase shift angle, and the Winding 2 tap ratio. In the parallel transformer check, transformer ratings are also tabulated.

For the transformer loss data consistency check, separate tabulations are produced for two-winding and three-winding transformers. In the two-winding transformer report, copper loss at 1.0 per unit current and

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no-load loss at 1.0 per unit voltage are listed in watts, along with the per unit values of transformer resistance and magnetizing conductance from which they are derived. In the three-winding transformer report, copper loss at 1.0 per unit current and no-load loss at 1.0 per unit voltage are listed in watts, along with the per unit values of three bus-to-bus resistances and the magnetizing conductance from which they are derived; the status shown is the status of the three-winding transformer (0 through 4).

For each of the remaining checks, the report includes the per unit impedance and, for non-transformer branches, line charging. Whenever a branch that is treated as a zero impedance line is tabulated, it is so noted.

Except in the transformer loss data consistency check, each winding of a three-winding transformer is checked and reported separately. For each such winding reported, the bus to which the winding is connected is shown as the from bus, and the to bus identifiers include the transformer name in the bus name column, the winding number in the base voltage column, and the string 3WNDTR in the bus number column; the charging and Winding 2 tap ratio fields are left blank. The status shown is the status of the winding (0 or 1).

Except for the reactance ratio check, which is reported in double entry format, branches are listed in single entry format (refer to Branch Data) and are listed in ascending numerical (using the *numbers* output option) or alphabetical (using the *names* option) order by from bus, and, for each from bus, in ascending order by to bus and circuit identifier.

Additional Information
PSS®E GUI Users Guide, Checking Branch Parameters *PSS®E Command Line Interface (CLI) Users Guide*, Checking/Changing Branch Parameters *PSS®E Application Program Interface (API)*, BRCH

5.28. Check for Islands Without a Swing Bus

TREE

Run Activity TREE - GUI
Power Flow > Check Data > Buses not in swing bus tree (TREE)
Run Line Mode Activity TREE - CLI
ACTIVITY? >>**TREE**
Interrupt Control Codes
AB

The network continuity checking activity TREE enables the user to identify buses not connected back to a Type 3 (swing) bus through the in-service ac network. It also tabulates in-service branches connected to Type 4 (disconnected) buses. Each swingless island may optionally be disconnected.

The TREE API routine includes the following among its inputs:

- A flag indicating the calculation phase to be performed during this call of the TREE API routine:
- Initialization and check for the presence of a swingless island.

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- Process the previously detected island as indicated by another input value; then check for the presence of another swingless island.
- An option describing the handling of the previously detected swingless island.

5.28.1. Operation of Activity TREE

The report produced by activity TREE first lists any Type 4 bus with one or more in-service branches connected to it. Such error conditions usually result from manually isolating a bus by changing its bus type code and the branch status flags to the appropriate values (with activity CHNG or in the *[Spreadsheet]* view) but overlooking one or more branches connected to it. This error may be avoided by using activity DSCN to electrically disconnect a bus.

Activity TREE then tabulates all Type 3 buses in the working case, followed by a listing of buses that do not have a type code of 4 and are not looped back to a swing bus. These buses are grouped by island, and, within each island, in ascending bus number (using the *numbers* output option) or alphabetical (using the *names* output option) order. The tabulation of each island is followed by a summary of the number of buses and plants in the island along with total island load, shunt, generation, and var limits.

The tabulations of activity TREE are printed at the *Progress* device (refer to Section 4.4, "Virtual Output Devices").

For each swingless island detected in the working case, user input is required to select one of the following actions:

- Disconnect this island, then check for another swingless island.
- Leave this island unchanged and check for another swingless island.
- Leave this island unchanged and exit activity TREE.

When activity TREE disconnects an island, it performs the required bus type code and branch status flag changes to disconnect *all* buses that were listed as not tied back to a swing bus. Any in-service dc lines connected to such buses are blocked. Any in-service series FACTS devices and series GNE devices connected to such buses are placed out-of-service.

5.28.2. Application Notes

Activity OUTS flags buses not connected to other buses by in-service branches and for which the type code is not 4, but is not able to identify islands; activity TREE tabulates both isolated buses and islands not containing a swing bus.

Activity TREE always scans the entire working case.

Activity TREE permits itself to be executed after generators have been converted with activity CONG. In this case, at least one island is always present because there should be no Type 3 buses following the execution of activity CONG.

Additional Information
PSS®E GUI Users Guide, Check for Islands Without a Swing Bus *PSS®E Command Line Interface (CLI) Users Guide*, Check for Islands Without a Swing Bus

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Additional Information
PSS®E Application Program Interface (API), TREE

5.29. Checking or Changing Controlled Bus Scheduled Voltage

CNTB

Run Activity CNTB - GUI
Power Flow > Check Data > Check / Change controlled bus scheduled voltages (CNTB)... *[Check / Change Controlled Bus Scheduled Voltages]*
Run Line Mode Activity CNTB - CLI
ACTIVITY? >>**CNTB** ENTER 0 TO INCLUDE ALL ELEMENTS, 1 FOR IN-SERVICE ELEMENTS ONLY: >>
Interrupt Control Codes
AB

The bus scheduled voltage checking activity CNTB tabulates the voltage setpoints and desired voltage bands of voltage controlling equipment in the working case, and, optionally, allows the user to specify new scheduled voltages. It also performs certain checks on voltage controlling buses that are not themselves voltage controlled buses. It may be instructed to process all such buses, or only those with suspect or conflicting voltage schedules or other errors.

The CNTB API routine includes the following among its inputs:

- A designation of the subsystem in which the voltage controlled buses are to be processed; refer to Section 4.8, “Subsystem Selection”.
- A flag indicating the calculation phase to be performed during this call of the CNTB API routine:
- Initialization.
- Voltage setpoint change for a voltage controlled bus.
- Voltage band change for a load drop compensating transformer.
- In the initialization phase, the following control flags are specified:
- Process either all voltage controlled and voltage controlling buses in the specified subsystem, or only those with apparent conflicting voltage objective data or other errors.
- Report only, or allow scheduled voltage changes.
- Process either all subsystem controlling elements and controlled buses, or only in-service elements.
- In the voltage setpoint change phase, the following are specified:

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- A flag to indicate either no change for this voltage setpoint, or that a change is to be implemented.
- When a change is to be implemented, the new voltage setpoint value.
- In the voltage band change phase for a load drop compensating transformer, the following are specified:
- A flag to indicate either no change for this transformer, or that a change is to be implemented.
- When a change is to be implemented, the new voltage setpoint band.

5.29.1. Operation of Activity CNTB

Activity CNTB may be instructed to either include or omit Type 4 controlled buses and out-of-service or disabled controlling equipment in its checking and reporting.

If only in-service elements are to be processed, any bus that is designated as the controlled bus of some voltage controlling equipment item, but for which the type code is greater than 3, is excluded from processing by activity CNTB. In addition, the following voltage controlling equipment items are ignored by activity CNTB:

- Generation at a bus for which the type code is not 2 or 3.
- A switched shunt at a bus for which the type code is greater than 3 or for which the switched shunt control mode is not 1 (discrete voltage control) or 2 (continuous voltage control).
- A voltage controlling transformer that is out-of-service or for which the adjustment control mode is –1 (i.e., control is disabled).
- The shunt element of a FACTS device for which the control mode is zero.
- The series element of a FACTS device for which the control mode is zero.
- A converter of a VSC dc line for which the VSC dc line control mode (MDC) is zero or for which the converter dc control code (TYPE) is zero.

Activity CNTB requires that the user select the processing of either all voltage controlled buses and their controlling equipment in the specified subsystem, or only those with apparent conflicting voltage objective data or other errors.

Activity CNTB may be operated in either an interactive mode in which new voltage schedules may be specified, or in a reporting mode.

When voltage schedule changes option is enabled, the user responds to prompts, displayed for one controlled bus or load drop compensating transformer at a time, to specify new voltage schedules. In this mode, these reports are directed to the *Progress* device (refer to Section 4.4, "Virtual Output Devices"); in reporting mode, these reports are directed to the *Report* device.

The user specifies either that the entire working case is to be processed, or that voltage schedules of voltage controlled buses in a designated bus subsystem are to be reported and, if the option is enabled, modified (refer to Section 4.8, "Subsystem Selection").

For each bus tabulated, if it has a generator, switched shunt, VSC dc line converter, or FACTS device shunt element (again, either all or only in-service and enabled devices) that is controlling the voltage at some other bus or the reactive power output of a remote device, an appropriate message is printed. Then activity CNTB

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lists equipment for which control parameter data is such that it is controlling the voltage at the bus being processed. Such equipment includes:

- Generation at the bus itself, if it is not remotely controlling the voltage at some other bus. The type code of the generator bus is printed in the STATUS column.
- Generation at a remote bus, which is controlling the voltage at this bus. The type code of the remote generator bus is printed in the STATUS column.
- A switched shunt at the bus itself, if it is not remotely controlling the voltage at some other bus or the reactive power output of some other voltage controlling device. The control mode of the switched shunt is printed in the STATUS column.
- A switched shunt at a remote bus, which is controlling the voltage at this bus. The type code of the bus to which the switched shunt is connected and the control mode of the switched shunt are printed in the STATUS column.
- A FACTS device connected to the bus. The device's control mode is printed in the STATUS column.
- A FACTS device at a remote bus, which is controlling the voltage at this bus. The type code of the bus to which the sending end bus of the FACTS device is connected and the device's control mode are printed in the STATUS column.
- A converter of a VSC dc line connected to the bus itself, if the converter is in ac voltage control mode and is not remotely controlling the voltage at some other bus. The VSC dc line's control mode (MDC) and the converter's dc control code (TYPE) are printed in the STATUS column.
- A converter of a VSC dc line connected to a remote bus, if the converter is in ac voltage control mode and is remotely controlling the voltage at this bus. The type code of the converter bus, the VSC dc line's control mode (MDC), and the converter's dc control code (TYPE) are printed in the STATUS column.
- A voltage controlling transformer without load drop compensation. For two-winding transformers, transformer status is printed; for three-winding transformers, winding status is printed in the STATUS column.

For each such equipment, the desired voltage setpoint or voltage band, as appropriate, is tabulated. Activity CNTB lists any suspect voltage control specifications.

Then, if the voltage schedule changing option was selected, the user has the option of specifying a new scheduled voltage. If a new value is entered, the control parameters are modified accordingly. For those devices controlling to a voltage band, the band retains its previous voltage spread with the designated voltage as the midpoint. These data changes are then echoed for verification.

Each voltage controlling transformer with load drop compensation (refer to Transformer Data and Voltage Control) for which the controlled bus is being processed is then listed, along with the compensated voltage it would sense and its voltage limits. If the voltage schedule changing option was selected, the user then has the option of specifying a new voltage band for each such transformer.

5.29.2. Application Notes

Buses connected together by zero impedance lines (refer to Zero Impedance Lines) are treated as the same bus. Controlling equipment, along with any error and warning messages, apply to the combined bus. In interactive mode, each group of buses connected together by zero impedance lines that has at least one of

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the buses in the specified subsystem is tabulated once, even if more than one of its buses is in the specified subsystem. In reporting mode, each such bus is reported in its usual position in the bus collating sequence.

Additional Information
PSS®E GUI Users Guide, Checking/Changing Controlled Bus Scheduled Voltage
PSS®E Command Line Interface (CLI) Users Guide, Checking/Changing Controlled Bus Scheduled Voltage
PSS®E Application Program Interface (API), CNTB

5.30. Checking or Changing Transformer Adjustment Data

TPCH

Run Activity TPCH - GUI
Power Flow > Check Data > Check / Change transformer adjustment data (TPCH)... *[Check / Change Transformer Adjustment Data]*
Run Line Mode Activity TPCH - CLI
ACTIVITY? >>**TPCH** ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
AB

The controlling transformer parameter checking activity TPCH performs several checks on the adjustment data (refer to Transformer Data) associated with voltage and flow controlling transformers.

The TPCH API routine includes the following among its inputs:

- A flag indicating the calculation phase to be performed during this call of the CNTB API routine:
- Initialization.
- Perform the selected check of transformer adjustment data.
- Modify adjustment data of transformers failing the current check.
- Exit TPCH.
- In the initialization phase, the following is specified:
- A designation of the subsystem in which regulating transformers are to be processed; refer to Section 4.8, “Subsystem Selection”.
- In the data checking phase, the following are specified:
- A flag to indicate the type of data check to be performed.

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- The threshold value to be used in the specified data check.
- In the adjustment data modification phase, the following are specified:
- For tests which can modify either tap steps or voltage bands, a flag to indicate which of these are to be modified.
- For tests for which both voltage controlling and Mvar controlling transformers were tabulated, a flag to indicate which types of transformers are to have their data modified:
- both types
- only voltage controlling transformers
- only Mvar controlling transformers
- For tests for which both MW controlling and Mvar controlling transformers were tabulated, a flag to indicate which types of transformers are to have their data modified:
- both types
- only MW controlling transformers
- only Mvar controlling transformers
- The new tap step, voltage band, or flow band, as appropriate.

5.30.1. Operation of Activity TPCH

The user specifies either that the entire working case is to be processed, or that controlling transformer windings connected to buses in a designated bus subsystem are to be checked (refer to Section 4.8, "Subsystem Selection").

When a subsystem is being processed and ownership is the only selection criterion, each voltage or flow controlling transformer wholly or partly owned by any of the owners specified is included in subsequent checks (refer to Transformer Data). The owner assignments of the transformer's endpoint buses are not considered.

When one or more of the other subsystem selection criteria are enabled, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected transformer branches are candidates for processing; otherwise, all buses are candidate buses. The owner selection criterion, if enabled, determines which controlling transformers whose controlling winding sides are connected to candidate buses are to be processed; otherwise, all controlling transformers whose controlling winding sides are connected to candidate buses are processed.

The following types of checks may be performed by activity TPCH:

- Three tests check the tap ratio step increment (derived from the ratio limits and the number of tap positions) of voltage and Mvar controlling transformers:
- Tap step = 0.0
- Small tap step (0.0 < tap step < user specified threshold)

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- Large tap step (tap step > user specified threshold)
- Three tests check the voltage band of voltage controlling transformers:
- Voltage band narrow relative to the tap step (voltage band < two tap steps)
- Narrow voltage band (voltage band < user specified threshold)
- Wide voltage band (voltage band > user specified threshold)
- Two tests check the flow band of MW and Mvar controlling transformers:
- Narrow flow band (flow band < user specified threshold)
- Wide flow band (flow band > user specified threshold)

For each check performed, an appropriate heading is printed followed by a tabulation of any controlling transformers failing the selected check. If any transformers have failed the check, the user has the option of modifying the data of all transformers tabulated.

Two-winding transformers are listed in the report with their Winding 1 bus as the from bus and with their Winding 2 bus as the to bus. Three-winding transformers are listed in the report with their controlling winding bus as the from bus and with their transformer name and winding number as the to bus. The table is ordered in ascending numerical (using the *numbers* output option) or alphabetical (using the *names* option) order by from bus, and, for each from bus, in ascending order by to bus and circuit identifier, with two-winding transformers first followed by three-winding transformers. The number of controlling transformer windings failing the check is also reported.

For the tap ratio step increment tests, if any transformers are found failing the test, the user may elect to modify the steps, and therefore the number of tap positions, of these transformers. If this option is enabled and there are both voltage and Mvar controlling transformers listed, the user may have both types changed, only the voltage controlling transformers changed, or only the Mvar controlling transformers changed to a specified new tap step. From the specified tap step and the ratio limits of each such transformer, a new value for the number of tap positions is determined.

The remaining checks of activity TPCH are handled in a similar manner. The check for voltage band less than two tap steps provides for modification of either the voltage band (widened to two tap steps) or the number of tap positions (increased such that each tap step is no more than half of the voltage band) of each voltage controlling transformer tabulated.

The tests of the voltage band of voltage controlling transformers against a user specified threshold provide for the modification of the voltage band of each listed transformer to a specified new band. For each transformer that has its voltage band adjusted, its voltage limits are set such that the midpoint of its band is unchanged.

The test of the flow band of MW or Mvar controlling transformers against a user specified threshold provide for the modification of the flow band of each listed transformer to a specified new band. If this option is enabled and there are both MW and Mvar controlling transformers listed, the user may have both types changed, only the MW controlling transformers changed, or only the Mvar controlling transformers changed to a specified new flow band. For each transformer that has its flow band adjusted, its flow limits are set such that the midpoint of its band is unchanged.

As an example use of this function, assume we need to check the controlling Flow band for all MW controlling transformers in the *savnw.sav* Saved Case File. Use a threshold Flow band of 5.0 MW. The result of

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this example check shows one power controlling transformer with a Flow band of 10 MW compared to the threshold check of 5.0 MW (see Figure 5.10, "Results of Example Transformer Flow Band Check").

```
MW OR MVAR CONTROLLING TRANSFORMERS WITH FLOW BAND >    5.00
X--- ADJUSTABLE SIDE ---X  X--------- TO ----------X
  BUS# X-- NAME --X BASKV    BUS# X-- NAME --X BASKV CKT  FLOMAX  FLOMIN    BAND  CONTROLLING
   202 EAST500     500.00     203 EAST230     230.00  1   555.00  545.00   10.00  MW

    1 TRANSFORMERS FOUND
```

Figure 5.10. Results of Example Transformer Flow Band Check

5.30.2. Application Notes

Transformers that control dc line quantities are excluded from the parameter checking of activity TPCH.

When transformers are reported by activity TPCH, this does not necessarily mean that their data is incorrect.

Additional Information
PSS®E GUI Users Guide, Checking/Changing Transformer Adjustment Data
PSS®E Command Line Interface (CLI) Users Guide, Checking/Changing Transformer Adjustment Data
PSS®E Application Program Interface (API), TPCH

5.31. Changing Adjustment Enable Flags of Transformers

TFLG

Run Activity TFLG - GUI
Power Flow > Changing > Transformer adjustment flags (TFLG)... *[Transformer Adjustment Flags]*
Run Line Mode Activity TFLG - CLI
ACTIVITY? >>**TFLG** ENTER 0 TO DISABLE, 1 TO ENABLE: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The transformer adjustment enable flag setting activity TFLG allows the user to either enable or disable the adjustment status of all automatically adjustable transformer windings contained in the subsystem specified by the user.

The TFLG API routine includes the following among its inputs:

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- A designation of the subsystem in which the adjustment enable flag of adjustable transformer windings is to be modified; refer to Section 4.8, "Subsystem Selection".

- The setting to be assigned to the adjustment enable flags of subsystem adjustable transformer windings (refer to Transformer Data).

5.31.1. Operation of Activity TFLG

The user specifies either that the entire working case is to be processed, or that adjustable transformers connecting buses in a designated bus subsystem are to be processed (refer to Section 4.8, "Subsystem Selection").

When a subsystem is being processed and ownership is the only selection criterion, each adjustable transformer wholly or partly owned by any of the owners specified has its adjustment enable flag set to the designated value. The owner assignments of the transformer's endpoint buses are not considered.

When one or more of the other subsystem selection criteria are enabled, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected transformer branches are candidates for processing; otherwise, all buses are candidate buses. The owner selection criterion, if enabled, determines which controlling transformers connecting candidate buses are to be processed; otherwise, all controlling transformers connecting candidate buses are processed.

Note that only transformers connecting subsystem buses are processed by activity TFLG; transformers that are ties from the specified subsystem are not processed.

Activity TFLG sets the sign of the adjustment control mode of those adjustable transformer windings being processed to the appropriate value.

Activity TFLG logs the total number of transformers in the working case that had their adjustment enable flags changed:

```
ADJUSTMENT CONTROL MODE OF nn TRANSFORMERS CHANGED
```

This log is printed at the *Progress* device.

Refer to Transformer Data, Automatic Transformer Adjustments, and Phase Shift Angle Adjustment for additional details on the transformer adjustment control flag.

Additional Information
PSS®E GUI Users Guide, *Changing Adjustment Enable Flags of Transformers*
PSS®E Command Line Interface (CLI) Users Guide, Changing Adjustment Enable Flags of Transformers
PSS®E Application Program Interface (API), TFLG

5.32. Performing Unit Commitment and Economic Dispatch

ECDI

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Run Activity ECDI - GUI
Power Flow > Changing > Economic Dispatch (ECDI)... *[Economic Dispatch]*
Run Line Mode Activity ECDI - CLI
ACTIVITY? >>**ECDI** ENTER ECONOMIC DISPATCH DATA FILE NAME: >>
Interrupt Control Codes
AB abandon activity ECDI following the completion of the next dispatch iteration NC suppress the dispatch convergence monitor

The unit commitment/economic dispatch activity ECDI places machines in a specified subsystem in- or out-of-service to satisfy a given subsystem minimum capacity. The in-service machines in the subsystem are then dispatched on the basis of equal incremental cost to meet a specified total subsystem generation.

Activity ECDI requires minimum and maximum outputs, incremental heat rates, fuel costs, and start-up priority rankings for all machines to be scheduled. This data exists in source file form in an Economic Dispatch Data File (*.ecd) that is read during the execution of activity ECDI.

The ECDI API routine includes the following among its inputs:

- A flag indicating the calculation phase to be performed during this call of the ECDI API routine:
- Initialization.
- Specify subsystem.
- Unit commitment and economic dispatch calculation.
- Perform housekeeping and exit the ECDI API.
- In the initialization phase, the following is specified:
- The name of the Economic Dispatch Data File to be used.
- In the subsystem specification phase, the following is specified:
- A designation of the subsystem in which the generation is to be dispatched; refer to Section 4.8, "Subsystem Selection".
- An initial commitment flag indicating one of the following:
- Begin the unit commitment calculation from the current commitment profile.
- Determine a new commitment profile (i.e., begin the unit commitment with all dispatchable units out-of-service).
- Disable the unit commitment (i.e., no machine status changes allowed); normally used only in P-V analysis calculations.

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- In the calculation phase, the following is specified:
- The desired loading of the units being dispatched.
- The desired minimum capacity of the units being dispatched.

5.32.1. Economic Dispatch Data File Contents

Each machine to be dispatched by activity ECDI must have its dispatch data specified in a data record in an Economic Dispatch Data File. Activity ECDI recognizes two formats of data records in this file. The majority of machines are specified on records of the form shown in Standard Record Format. A special record format is used for the supplementary units that are dispatched as part of a dispatch group; this format is described in Supplementary Units of Dispatch Groups, and dispatch groups are described in Section 5.32.4, "Dispatch Groups".

Records may be in any order, but the last record in the file must be a record with its first data item (I) specified as zero.

Standard Record Format

Each machine to be dispatched by activity ECDI, except for those machines that are the supplementary units of a dispatch group, must have a data record specified for it in the Economic Dispatch Data File in the following format:

```
I,ID,PRIOR,FUELCO,PMAX,PMIN,HEMIN,X1,Y1,X2,Y2,X3,Y3,X4,Y4,X5,Y5,X6,Y6
```

where:

I	Bus number; bus I must be present in the working case with one or more machines connected to it.
ID	One- or two-character machine identifier of the machine at bus I for which data is specified by this record. ID = 1 by default.
PRIOR	Priority ranking code (≥ 0). Machines with a priority ranking of zero do not have their status changed during the unit commitment portion of activity ECDI. Machines with lowest positive priority ranking codes are switched in-service first even though machines with higher priority rankings may be more economical to run. Machines with highest priority ranking codes are switched out-of-service first even though machines with lower priority rankings may be more expensive to run. PRIOR = 0 by default.
FUELCO	Fuel cost for the machine in dollars per MBtu. No default is allowed.
PMAX	Maximum machine active power output; entered in MW. If this machine is the principal unit of a dispatch group, this is the sum of the maximum outputs of *all* machines in the dispatch group. If PMAX is not specified, PMAX *and PMIN* for this machine are set to the

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	power limits contained in the working case (refer to Generator Data).
PMIN	Minimum machine active power output; entered in MW. If this machine is the principal unit of a dispatch group, this is the sum of the minimum outputs of *all* machines in the dispatch group. If PMAX is not specified, the value specified for PMIN is ignored and the one from the working case is used; otherwise, no default is allowed.
HEMIN	Minimum heat input required by the machine when in-service; entered in MBtu/hr. If X_1 (see below) is greater than zero, HEMIN should be specified as the heat input required at X_1 MW. No default is allowed.
X_i,Y_i	Points on the incremental heat rate curve; X_i values are entered in MW and Y_i values are entered in Btu/kWh. At least two points, and up to six points, may be entered. Both X and Y must be in ascending order, with $X_1 \leq$ PMIN and $X_n \geq$ PMAX. If this machine is the principal unit of a dispatch group, this curve is the combined curve of *all* machines in the dispatch group.

The auxiliary program PLINC may be used to plot the incremental heat rate curves of specified machines. Refer to *Additional Resources for PSS® E*, PLINC for details on its use.

The PSS®E EXAMPLE directory contains an Economic Dispatch Data File, *savnw.ecd*, which can be used in conjunction with the *savnw.sav* Saved Case File. A plot of those incremental heat rate curves is shown in Figure 5.11, "Incremental Heat Rate Curves Provided in the PSS®E savnw.sav Case".

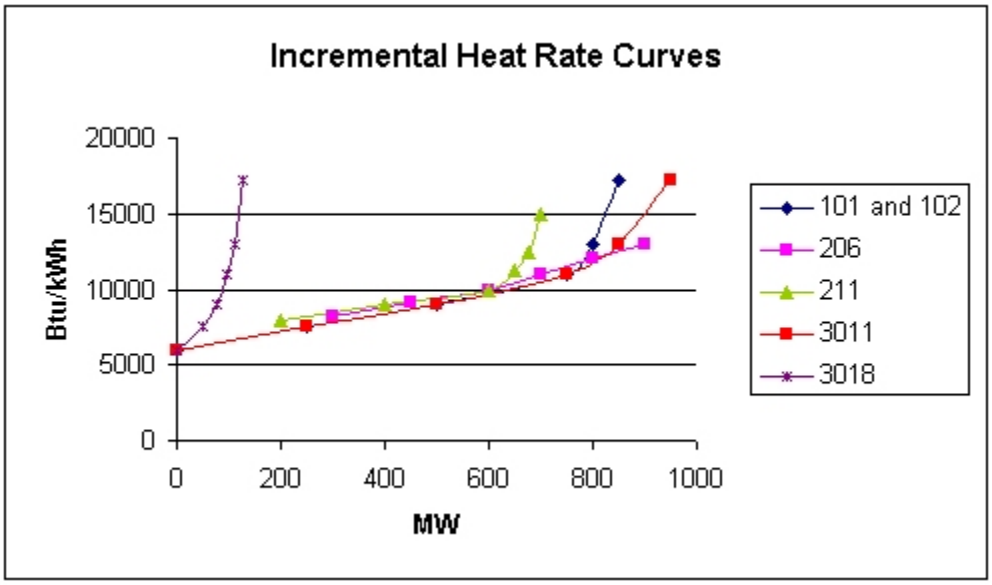

Figure 5.11. Incremental Heat Rate Curves Provided in the PSS®E savnw.sav Case

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Supplementary Units of Dispatch Groups

A special record format is recognized on which a supplementary unit of a dispatch group is specified. This record has the following format:

```
I,ID,PRIOR,FRACT,PRNBUS,PRNMAC,CODFRC
```

where:

I	Bus number; bus I must be present in the working case with one or more machines connected to it.
ID	One- or two-character machine identifier of the machine at bus I for which data is specified by this record. ID = 1 by default.
PRIOR	Priority ranking code (< 0). A negative priority ranking code signifies that this machine is a supplementary unit of a dispatch group. No default is allowed.
FRACT	When CODFRC is 0 or negative, the fraction of the total power dispatched for the dispatch group to be assigned to the principal unit. When CODFRC is positive, the fraction of the total power dispatched for the dispatch group to be assigned to this supplementary unit. FRACT must be greater than 0.0 and less than 1.0. No default is allowed.
PRNBUS	Bus number of the principal machine associated with the dispatch group of which this supplementary unit is a member. No default is allowed.
PRNMAC	One- or two-character machine identifier of the principal machine at bus PRNBUS associated with the dispatch group of which this supplementary unit is a member. PRNMAC = 1 by default.
CODFRC	Defines the interpretation of FRACT on this record by activity ECDI. When CODFRC is 0 or negative, FRACT is the fraction of the total power dispatched for the dispatch group to be assigned to the principal unit. When CODFRC is positive, FRACT is the fraction of the total power dispatched for the dispatch group to be assigned to this supplementary unit. CODFRC = 0 by default.

A standard economic dispatch data record (refer to Standard Record Format) must be included in the Economic Dispatch Data File for machine PRNMAC at bus PRNBUS. Its power output limits and incremental heat rate curve must be that corresponding to the combination of *all* machines in the dispatch group.

5.32.2. Operation of Activity ECDI

If the specified Economic Dispatch Data File does not exist or some other file system error occurs, an appropriate error message is printed. Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity ECDI.

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If any data inconsistencies are detected during the processing of the Economic Dispatch Data File, they are logged at the *Progress* device and the corresponding record is ignored. Activity ECDI performs its unit commitment and dispatch calculations using only the valid data records.

The user specifies either that the entire working case is to be processed, or that machines in a designated bus subsystem are to be scheduled and dispatched (refer to Section 4.8, "Subsystem Selection").

When a subsystem is being processed and ownership is included among the selection criteria, the owners to which machines (rather than buses) are assigned are used in defining the subsystem of machines to be processed. If a machine has multiple owners (refer to Generator Data), the owner specified in the owner/fraction pair with the largest ownership fraction is assumed to have dispatch responsibility. If this largest ownership fraction is identical for two or more owners of a machine, the first one in the tables, as shown in the machine data listings, in the *[Spreadsheet]* View, and in activities such as LIST and EXAM, is used.

If the option for a new commitment profile is specified, all machines in the specified subsystem with a non-zero priority ranking for which dispatch data had been read are placed out-of-service prior to commencing the unit commitment calculation. If starting from the current commitment profile, the current unit status profile is used as the starting point. If the unit commitment calculation is disabled, all positive priority rankings are set to zero.

Activity ECDI processes only those machines in the specified subsystem for which valid dispatch data had been read. Other machines retain their initial status and power outputs. Activity ECDI summarizes the pre-dispatch condition by tabulating the following:

- the pre-dispatch production cost/hour.
- the minimum generation, maximum generation, and present total generation of in-service units that are to be included in the dispatch.
- the present generation of subsystem being dispatched; this includes active power generation subsystem machines that are to be omitted from the dispatch calculation.

Figure 5.12, "Summary of the Predispatch Condition" shows the result obtained using the *savnw.sav* Saved Case File and *savnw.ecd* Economic Dispatch Data File.

```
PRE-DISPATCH PRODUCTION COST IS  15031.86 $/HR

PRESENT TOTAL GENERATION OF UNITS BEING DISPATCHED IS    3258.670
MAXIMUM GENERATION OF DISPATCHED UNITS NOW ON-LINE IS    4375.000
MINIMUM GENERATION OF DISPATCHED UNITS NOW ON-LINE IS    1800.000

PRESENT GENERATION OF SUBSYSTEM BEING DISPATCHED IS      3258.670
```

Figure 5.12. Summary of the Predispatch Condition

The first four quantities in the above summary are totals based upon the present power outputs and dispatch data specified in the Economic Dispatch Data File for those machines initially in-service in the subsystem to be processed. The final total is the sum of the present power outputs of:

- those machines to be included in the dispatch calculation; and
- those machines in the specified subsystem for which no dispatch data was provided (the outputs of these machines will not be changed by activity ECDI).

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The values specified for the desired loading and minimum capacity of the units to be dispatched should be entered as values applying to the machines to be dispatched. Specifically, they *must not* include the power outputs assigned to those machines in the specified subsystem for which dispatch data was not read; the outputs of such machines are not changed by activity ECDI.

Activity ECDI then commences the unit commitment portion of its calculation. Note again that the status of any machine with a priority ranking of zero is not changed during the unit commitment portion of activity ECDI, but such machines are included in the economic dispatch calculation. Neither the status nor power outputs of machines for which either no dispatch data was provided or for which a data error message was generated, and of machines outside of the specified subsystem, is changed during the execution of activity ECDI.

If the specified desired minimum capacity is greater than the sum of the maximum power outputs of the machines currently in-service, activity ECDI places additional units in-service. Machines with lowest priority rankings are connected first, and, within a given priority ranking, units with the lowest full load average cost per MW are placed in-service first.

If the specified desired loading is less than the sum of the minimum power outputs of the machines currently in-service, activity ECDI places units out-of-service. Machines with highest priority rankings are disconnected first, and, within a given priority ranking, units with the highest full load average cost per MW are placed out-of-service first.

Any time activity ECDI changes a machine status, a message is printed at the *Progress* device. Any processing errors in the unit commitment portion of activity ECDI generate a message at the *Alert* device; different values of desired loading and minimum capacity must be specified.

After a commitment profile satisfying the desired loading and capacity constraints has been successfully determined, activity ECDI calculates the power outputs of the machines to be dispatched. Power outputs are set on the basis of equal incremental costs subject to the machine power output limits.

The dispatch calculation uses an iterative approach and, at the end of each iteration, activity ECDI prints a convergence monitor, which tabulates:

- The iteration number.
- The total power mismatch.
- The incremental cost.
- The change in incremental cost.

Following the dispatch calculation, activity ECDI tabulates the production cost and the incremental cost along with the post-dispatch totals of dispatched power, capacity limits of the dispatched machines, and subsystem generation; the format is identical to the pre-dispatch tabulation (see Figure 5.12, “Summary of the Predispatch Condition”). Plant power outputs and reactive power limits are updated to reflect the commitment and dispatch calculation results. Results are summarized at the *Progress* device.

The example in Figure 5.13, “Summary Results of the Economic Dispatch Process” uses the *savnw.sav* Saved Case FIle, with the desired loading specified as 4000. The convergence monitor and results summary are shown.

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```
  1      239.876        7.982        2.04675
  2     -364.735        5.935        1.02338
  3       73.860        6.958        0.51169
  4     -127.351        6.446        0.25584
  5      -20.627        6.702        0.12792
  6       26.617        6.830        0.06396
  7        2.995        6.766        0.03198
  8       -8.816        6.734        0.01599
  9       -2.911        6.750        0.00800
 10        0.042        6.758        0.00400
 11       -1.435        6.754        0.00200
 12       -0.696        6.756        0.00100
 13       -0.327        6.757        0.00050
 14       -0.143        6.758        0.00025
 15       -0.050        6.758        0.00012
 16       -0.004        6.758        0.00006

REACHED TOLERANCE IN  16 ITERATIONS

TOTAL MISMATCH IS    -0.0042 MEGAWATTS
PRODUCTION COST IS  19483.91 $/HR
INCREMENTAL COST IS     6.76 $/MW-HR

PRESENT TOTAL GENERATION OF UNITS BEING DISPATCHED IS    3999.996
MAXIMUM GENERATION OF DISPATCHED UNITS NOW ON-LINE IS    4375.000
MINIMUM GENERATION OF DISPATCHED UNITS NOW ON-LINE IS    1800.000

PRESENT GENERATION OF SUBSYSTEM BEING DISPATCHED IS      3999.996
```

Figure 5.13. Summary Results of the Economic Dispatch Process

5.32.3. Viewing Economic Dispatch Results

Following completion of the dispatch calculation, plant power outputs and reactive power limits are updated to reflect the commitment and dispatch calculation results.

One method of examining the results is to use activity GEOL (refer to Section 12.23, "Producing a Machine Terminal Limits Report") to look at machine terminal conditions. In Figure 5.14, "Before and After Machine Terminal Conditions for Economic Dispatch Example" those conditions can be seen for the *savnw.sav* Saved Case File before and after the economic dispatch calculation example described here.

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

MACHINE SUMMARY:
  BUS# X-- NAME --X BASKV ID     MW     MVAR    QMAX
   101 NUC-A       21.600 1    750.0    81.2   600.0
   102 NUC-B       21.600 1    750.0    81.2   600.0
   206 URBGEN      18.000 1    800.0   600.0   600.0
   211 HYDRO_G     20.000 1    600.0    17.7   400.0
  3011 MINE_G      13.800 1    258.7   104.0   600.0
  3018 CATDOG_G    13.800 1    100.0    80.0    80.0
SUBSYSTEM TOTALS              3258.7   964.2  2880.0
```

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

MACHINE SUMMARY:
  BUS# X-- NAME --X BASKV ID     MW     MVAR    QMAX
   101 NUC-A       21.600 1    808.3    81.2   600.0
   102 NUC-B       21.600 1    808.3    81.2   600.0
   206 URBGEN      18.000 1    870.5   600.0   600.0
   211 HYDRO_G     20.000 1    666.0    17.7   400.0
  3011 MINE_G      13.800 1    847.0   104.0   600.0
SUBSYSTEM TOTALS              4000.0   884.2  2800.0
```

Figure 5.14. Before and After Machine Terminal Conditions for Economic Dispatch Example

Another method for observing the changes resulting from the execution of activity ECDI is to use activity DIFF (refer to Section 12.27, "Comparing Power Flow Cases"). Compare generator active powers in the working case with those in the Saved Case File which contains the pre-dispatch Saved Case.

5.32.4. Dispatch Groups

A dispatch group provides a means of handling a group of machines that are to be treated as a single entity in the unit commitment and economic dispatch calculations. A combined cycle plant, in which multiple steam units and combustion turbines are to be dispatched as one unit, may be handled as a dispatch group in activity ECDI. The high pressure, low pressure pair of a cross compound unit may also be treated as a dispatch group.

To specify a dispatch group, any one (and only one) of the machines in the dispatch group must be designated on a standard Economic Dispatch Data File record (refer to Standard Record Format); this machine is referred to as the principal unit of the dispatch group. The dispatch data specified on this data record must be that for the combination of *all* machines that are members of the dispatch group. Each of the remaining units in the dispatch group must be specified on a supplementary Economic Dispatch Data File record (refer to Supplementary Units of Dispatch Groups); these machines are referred to as the supplementary units of the dispatch group.

The sum of the FRACT values on all of the supplementary data records of a dispatch group *must* be less than one. When a dispatch group contains more than two machines, the data record for each of its supplementary

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machines *must* have CODFRC specified as a positive number, and FRACT as the fraction of total group power assigned to the supplementary machine. When a dispatch group contains exactly two machines, CODFRC for the supplementary machine may be specified as a positive or negative number or zero, and FRACT must be specified accordingly (refer to Figure 5.15, "Assignments for Supplementary Machine Economic Dispatch").

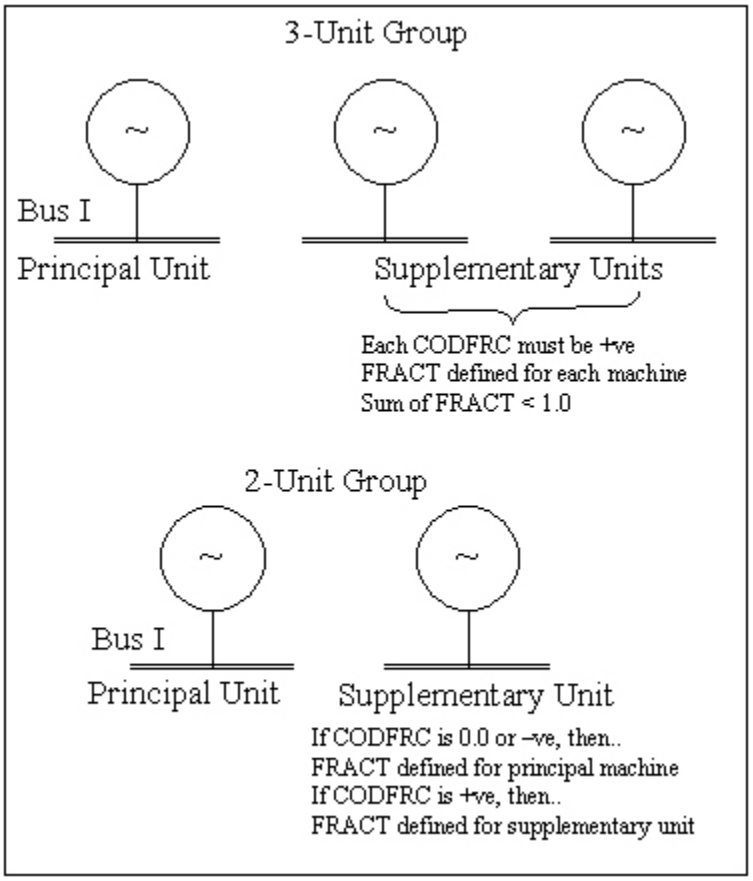

Figure 5.15. Assignments for Supplementary Machine Economic Dispatch

The initial service status of each machine in a dispatch group, as determined by the combination of the type code of the bus to which it is connected and its machine status flag, *must* be the same: either all in-service or all out-of-service (refer to Section 5.10.1, "Equipment Status Changes"). If the status of a dispatch group is switched during the unit commitment calculation, the status of each machine in the dispatch group is changed.

At the completion of the dispatch calculation, each dispatch group's total power output is split among the machines in the group according to the split fractions FRACT specified on the data records of the supplementary machines in the group (refer to Supplementary Units of Dispatch Groups).

Recall that, when no value of PMAX is specified in the Economic Dispatch Data File, the machine limits contained in the working case are used as the dispatch limits (refer to Standard Record Format). For a dispatch group, the sums of the limits of all machines in the group are used as the dispatch limits. In this case, it is possible to violate the individual machine limits as contained in the working case if FRACT is not coordinated with the individual machine limits.

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When a subsystem of the working case is being processed by activity ECDI, the subsystem assignment of the principal unit of a dispatch group is taken as the subsystem assignment of the group. If the principal unit is in the subsystem being processed, its dispatch group is processed even if one or more of its supplementary units is not in the subsystem. Conversely, if the principal unit is not in the subsystem being processed, the dispatch group is not processed even if some or all of its supplementary units are in the subsystem.

5.32.5. Application Notes

If the value specified for the desired minimum capacity is less than the value specified for the desired loading, it is assumed to be equal to the specified loading by the unit commitment calculation.

In specifying the desired loading and minimum capacity, the quantity (minimum capacity - loading) may be thought of as spinning reserve.

The iterative dispatch calculation in activity ECDI uses a binary search technique with an upper limit of 40 iterations. Convergence is assumed when the difference between dispatched power and desired power is less than 0.00001 times the desired power. Convergence failure, which is usually the result of precision limitations, generates an alarm and activity ECDI continues as if convergence had been achieved.

The dispatch data as read from the Economic Dispatch Data File is not retained in the working case following termination of activity ECDI. An Economic Dispatch Data File is required for each execution of activity ECDI.

Additional Information
PSS®E GUI Users Guide, Performing Unit Commitment and Economic Dispatch *PSS®E Command Line Interface (CLI) Users Guide*, Running Unit Commitment and Economic Dispatch *PSS®E Application Program Interface (API)*, ECDI

5.33. Displaying the Long Case Title

PRTI

Run Line Mode Activity PRTI - CLI
ACTIVITY? >>**PRTI** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The long title output activity PRTI prints the 16-line long title at the *Report* device and requires no additional inputs.

Activity PRTI is available only in line mode.

Lines of the long title for which no text has been specified appear as blank lines in the output of activity PRTI.

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Long Case Title *PSS®E Application Program Interface (API)*, PRTI

5.34. Managing Case Titles

During the initial input of data into PSS®E for a new network model, via either a Power Flow Raw Data File (see Case Identification Data) or the NEWCAS_2 API routine (see NEWCASE_2 of the *PSS® E Application Program Interface (API)* manual), two 60 character case title lines are specified. These lines are printed whenever the Saved Case containing this system model is brought into the working case; they are also included in the reports produced by the various reporting activities of PSS®E.

In addition, the PSS®E working case provides for a long title containing up to 16 lines of alphanumeric data, each of which may contain up to 72 characters. The user may employ the long title to enter more detailed case descriptive information than the two 60 character case title lines can accommodate. During the initial input of data into PSS®E for a new network model, the long title lines are initialized to blanks.

PSS®E includes functions for reading the long title, for modifying the both the case title and the long title, and for printing the long title.

Both the two line case title and the long title are carried along with the working case as it is saved and retrieved with activities SAVE and CASE, respectively.

5.34.1. Importing a Long Title

RETI

Run Activity RETI - GUI
File > Import > Long Title (RETI)... *[Select file containing long case title]*
Run Line Mode Activity RETI - CLI
ACTIVITY? >>**RETI** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL):
Interrupt Control Codes
None

The long title data input activity RETI reads a data file containing up to 16 lines of alphanumeric data and places them into the long title. The previous content of the long title is overwritten.

The RETI API routine requires as input the name of a data file containing the lines to be placed into the long title.

Records read for activity RETI are placed into the long title array in the exact format in which they are entered. If any input record contains more than 72 characters, those beyond column 72 are ignored. If a record contains fewer than 72 characters, the corresponding title line is blank filled.

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Activity RETI accepts data records until either 16 records have been entered or a data record is entered containing the characters /E in columns one and two. In this case, the remaining lines of the long title are blank filled and activity RETI ends.

If the data file contains more than 16 lines, only the first 16 are recognized. If the data file contains fewer than 16 lines and the last line does not contain the characters /E or /e in columns one and two, the following message is displayed at the *Progress* device:

```
OUT OF FILE DATA--SWITCH TO TERMINAL INPUT MODE
```

Then those title lines for which no data was read are blank filled and activity RETI ends.

Additional Information
PSS®E GUI Users Guide, Importing a Long Title *PSS®E Command Line Interface (CLI) Users Guide*, Importing a Long Title *PSS®E Application Program Interface (API)*,RETI

5.34.2. Changing the Long Title

CHTI

Run Activity CHTI - GUI
File > Case titles, short & long (CHTI)... *[Case Titles]*
Run Line Mode Activity CHTI - CLI
ACTIVITY? >>**CHTI** ENTER LINE NUMBER (CARRIAGE RETURN FOR LINE 1, 0 TO EXIT): >>
Interrupt Control Codes
None

The long title data change activity CHTI enables the user to modify the contents of individual lines of the 16-line long title without having to reread the entire set of lines with activity RETI.

Line mode activity CHTI is similar to activity CHNG, involving a prompt and response conversational dialog.

The LongTitleData API routine is the functional equivalent of activity CHTI.

Both interfaces provide the following capabilities:

- Editing the text of one or more title lines while leaving the others unchanged.
- Entering the entire long title manually without prior preparation of a Long Title File.
- Adding additional long title lines if only the first few title lines had previously been specified.

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Additional Information
PSS®E GUI Users Guide, Changing the Case Title and the Long Title *PSS®E Command Line Interface (CLI) Users Guide*, Changing the Long Title *PSS®E Application Program Interface (API)*, LONG_TITLE_DATA

5.34.3. Editing the Two-Line Case Title or the Long Title

EDTR

Run Line Mode Activity EDTR - CLI
ACTIVITY? >>**EDTR** EDITOR OPTIONS ARE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Activity EDTR is a title editing tool available only in line mode. It provides for changing part of a title line without having to enter the entire line.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Editing the Two-Line Case Title or the Long Title

5.35. Counting System Components in the Working Case

SIZE

Run Activity SIZE - GUI
File > File Information (SIZE/SHOW/BUSN)... *[File Information]* OPTION: List the number of system components
Run Line Mode Activity SIZE - CLI
ACTIVITY? >>**SIZE**
Interrupt Control Codes
None

The case size summary activity SIZE tabulates the number of components in the working case, along with the maximum number permitted at the current size level of PSS®E working memory.

The SIZE API routine prints its summary at the *Report* device (refer to Section 4.4, "Virtual Output Devices"), and requires no additional inputs.

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The following network elements are included in the report of activity SIZE:

• Buses, including generator buses and star point buses of three-winding transformers. • Plants. • Synchronous machines, including wind machines. • Wind machines. • Machine ownership specifications. • Fixed shunts. • Switched shunts. • Loads. • Induction machines. • Areas. • Interarea transfers. • Owners zones.s • AC branches, including two-winding transformers (both those that were entered as two-winding transformers and those that are members of three-winding transformers) and zero impedance lines.	• Two-winding transformers (both those that were entered as two-winding transformers and those that are members of three-winding transformers). • Three-winding transformers. • Branches treated as zero impedance lines. • Branch ownership specifications. • Multi-section line groupings. • Branches designated as line sections in multi-section line groupings. • Zero sequence mutual couplings. • Two-terminal dc transmission lines. • Multi-terminal dc transmission lines. • Voltage source converter (VSC) dc lines. • FACTS devices. • GNE devices.

An example of the output of activity SIZE is shown in Figure 5.16, “Example Output of Activity SIZE”.

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```
PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      FRI, APR 08 2011   9:40
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

            BUSES         PLANTS       MACHINES  WIND MACHINES  MACHINE OWNERS
TOTAL          23              6              6              0              14
MAXIMUM      1000            300            360             20             720

               S H U N T S                          INDUCTION
            FIXED       SWITCHED          LOADS       MACHINES
TOTAL           5              0              8              0
MAXIMUM      1000            126           2000             20


PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      FRI, APR 08 2011   9:40
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

            BUSES         PLANTS       MACHINES  WIND MACHINES  MACHINE OWNERS
TOTAL          23              6              6              0              14
MAXIMUM      1000            300            360             20             720

               S H U N T S                          INDUCTION
            FIXED       SWITCHED          LOADS       MACHINES
TOTAL           5              0              8              0
MAXIMUM      1000            126           2000             20

                        T R A N S F O R M E R S
       BRANCHES    TWO-WINDING  THREE-WINDING  ZERO IMPEDANCE BRANCH OWNERS MUTUALS
TOTAL         34           11              0              0           35           6
MAXIMUM     2500          400            100            500         5000         500

             MULTI-SECTION LINE
         GROUPINGS       SECTIONS       AREAS        ZONES       OWNERS    TRANSFERS
TOTAL           2              4              3              4            7            4
MAXIMUM       100            250            100            999          999          300


       2-TERM. DC   N-TERM. DC       VSC DC FACTS DEVICES  GNE DEVICES
TOTAL           0            0              0              0            0
MAXIMUM        20            5             10             20           10

       2-TERM. DC   N-TERM. DC       VSC DC FACTS DEVICES  GNE DEVICES
TOTAL           0            0              0              0            0
MAXIMUM        20            5             10             20           10
```

Figure 5.16. Example Output of Activity SIZE

Additional Information
PSS®E GUI Users Guide, System Components

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, System Components

5.36. Listing Unused Bus Numbers

BUSN

Run Activity BUSN - GUI
File > File information (SIZE/SHOW/BUSN)... *[File Information]* OPTION: List unused bus numbers in a range
Run Line Mode Activity BUSN - CLI
ACTIVITY? >>**BUSN** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The unused bus number summary activity BUSN tabulates those numbers, from within a user specified bus number range, which are not assigned to buses in the working case.

The inputs to the BUSN API routine define the desired bus number range (starting and ending bus numbers).

Activity BUSN lists those numbers in the designated range that are not assigned to a bus in the working case. It lists single numbers and ranges of numbers (e.g., 1208 THROUGH 1219), as appropriate. It then reports the number of numbers used and the number of numbers available from within the designated range. The report of activity BUSN is directed to the *Report* device.

An example of the output of activity BUSN using the default range is shown in Figure 5.17, "Example Output of Activity BUSN". The default range is the complete set of valid bus numbers available in PSS®E.

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSSE      TUE, FEB 03 2009  11:49
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA
                UNUSED BUS NUMBERS BETWEEN      1 AND 999997
     1  THROUGH    100
   103  THROUGH    150
   155  THROUGH    200
   207  THROUGH    210
   212  THROUGH   3000
  3009  THROUGH   3010
  3012  THROUGH   3017
  3019  THROUGH 999997

    23 NUMBERS USED AND 999974 NUMBERS AVAILABLE BETWEEN      1 AND 999997
```

Figure 5.17. Example Output of Activity BUSN

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Additional Information
PSS®E GUI Users Guide, Unused Bus Numbers *PSS®E Command Line Interface (CLI) Users Guide*, Unused Bus Numbers

5.37. Displaying Power Flow Data

LIST

Run Activity LIST - GUI
Power Flow > List Data... *[List Data]* OPTION: Powerflow
Run Line Mode Activity LIST - CLI
ACTIVITY? >>**LIST** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The data listing activity LIST tabulates the power flow working case in a form suitable for problem data documentation. The report generated by activity LIST is separated into several categories of data.

The LIST API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".
- A designation of the power flow data category to be tabulated.
- A designation of the units to be used to display ac voltages:
- The units (per unit or kV) indicated by the voltage output program option setting (see Saved Case Specific Option Settings).
- The alternate units.

5.37.1. Operation of Activity LIST

The user specifies either that the entire working case is to be processed, or that data items for equipment in a designated bus subsystem are to be tabulated.

If output for a bus subsystem is being produced, only equipment in the designated subsystem is included in the report *except* for the following data categories: the case summary; area data; inter-area transfer data; owner data; and zone data. Any subsystem filtering applied to a data category is described in the data category descriptions given in Section 5.37.2, "Listing Format".

Ac voltages are normally tabulated in the units indicated by the voltage output program option setting (either per unit or kV). An input of the LIST API routine provides for the tabulation of ac voltages using the other

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units. If voltages are to be listed in kV but the base voltage at the bus has not been specified, its voltage is listed in per unit.

The report of activity LIST is directed to the *Report* device.

The following data categories are available in activity LIST:

• case summary	• induction machine miscellaneous data
• all data (single entry branch)	• load data
• all data (double entry branch)	• multi-section line grouping data
• area interchange data	• owner data
• bus data	• plant data
• branch data (single entry) [a]	• switched shunt data
• branch data (double entry) [c]	• synchronous machine data
• dc line data	• 2-winding transformer impedance data
• FACTS device sending end data	• 2-winding transformer winding data
• FACTS device terminal end data	• 2-winding transformer control data
• fixed shunt data	• 3-winding transformer impedance data
• GNE device data	• 3-winding transformer winding data
• impedance correction data	• 3-winding transformer control data
• inter-area transfer data	• 3-winding transformer general data
• line shunt data	• zone data
• induction machine circuit data	

ab

cd

If all data categories are to be tabulated, branch-related sections may be in either single or double entry format (refer to Branch Data). This format selection applies to the following data categories: non-transformer branch data; line shunt data; the three categories of two-winding transformer data; and multi-section line grouping data.

5.37.2. Listing Format

Most of the data items tabulated by activity LIST correspond to data entered in activities READ, TREA, RDCH, CHNG, and the *[Spreadsheet]*. Refer to Section 5.2.1, "Power Flow Raw Data File Contents" for detailed descriptions of these quantities.

The following paragraphs describe those data items printed that do not correspond to data previously described. Unless otherwise stated, data in the various categories are tabulated in ascending bus number order

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when the numbers output option is in effect and in alphabetical order when the names output option is in effect.

Case Summary

The case summary consists of three pages. The first page tabulates various system totals and solution parameters. The BUSES summary items are interpreted as follows:

TOTAL	Total number of buses in the case, including star point buses of three-winding transformers.
PQ<>0.	Total number of Type 1 buses with non-zero load.
PQ=0.	Total number of Type 1 buses with no load, including star point buses of in-service three-winding transformers.
PE/E	Total number of Type 2 buses that are not at a reactive power limit.
PE/Q	Total number of Type 2 buses that are at a reactive power limit (each of these buses has its type code set to -2 by the power flow solution activities).
SWING	Total number of Type 3 buses.
OTHER	Total number of buses having other type codes, including star point buses of out-of-service three-winding transformers; these are usually Type 4 (i.e., disconnected) buses and/or those buses identified as boundary buses by activities READ and EXTR.

The BRANCHES summary tabulates ac branches. Its data items are interpreted as follows:

TOTAL	Total number of ac branches in the case; includes two-winding transformers (both those that were specified as two-winding transformers and those that are members of three-winding transformers), branches treated as zero impedance lines, and branches that are members of multi-section line groupings.
RXB	Total number of non-transformer branches with charging (i.e., transmission lines); excludes zero impedance lines.
RX	Total number of non-transformer branches with no charging (e.g., series capacitors, equivalent branches); excludes zero impedance lines.
RXT	Total number of two-winding transformer branches; includes those at nominal ratio and those that are members of three-winding transformers.
RX=0.	Total number of branches treated as zero impedance lines.
IN	Total number of in-service branches.

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OUT	Total number of out-of-service branches.

The number of multi-section line groupings and the number of ac branches that are members of multi-section line groupings are tabulated separately. The total numbers of three-winding transformers, dc transmission lines, FACTS devices, and GNE devices are tabulated separately. Totals by service status of three-winding transformers, multi-section line groupings, dc lines, FACTS devices, and GNE devices are not tabulated. The number of loads, plants, synchronous machines, wind machines, fixed shunts, switched shunts, induction generators, and induction motors is reported, as are the number of areas, zones, and owners that have been defined, and the number of inter-area transactions.

System totals of generation, load, and so on are tabulated. The quantity shown as SHUNTS is the sum of fixed bus shunts, switched shunts, line connected shunts, magnetizing admittance of transformers, and the shunt elements of FACTS devices. Losses are the sum of I^2R and I^2X losses for in-service ac branches plus dc line and FACTS device losses, and do not include charging and line shunt contributions or contributions from GNE devices. The loss and swing bus power totals are meaningful only if the working case represents a solved system condition. The highest and lowest pu bus voltages are listed along with the largest individual bus and total system mismatch. Up to twelve swing buses are tabulated in ascending bus number order; if more than twelve Type 3 buses exist in the case, only twelve of these are listed.

The second page of the case summary tabulates those buses in the working case with the largest mismatches. At least one, and up to 25, buses are listed, but not more than the number that can fit on one page. The list is terminated at a largest mismatch of 0.005 MVA.

The third page of the case summary lists rating set information. The column labeled "# NON-ZERO RATINGS" lists the number of ac branches for which the value in the corresponding rating set is non-zero.

The case summary always applies to the complete system, even if activity LIST is reporting for a bus subsystem.

Figure 5.18, “Case Summary for the savnw.sav Saved Case File” shows the case summary for the *savnw.sav* Saved Case File.

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```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     THU, JUN 20 2019  15:21
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                      SYSTEM SUMMARY
BASE CASE INCLUDING SEQUENCE DATA

--------------------BUSES----------------------        -----GENERATION----- ----SHUNTS----- -IND MACHS- FACTS   GNE
 TOTAL  PQ<>0.  PQ=0.  PE/E   PE/Q  SWING  OTHER  LOADS PLANTS MACHNS   WIND  FIXED SWITCHED GENS MOTORS  DEVS  DEVS
    23      7     10      3      2      1      0      8      6      6      0      5      0      0      0     0     0
------------------AC BRANCHES------------------ 3WIND MULTI-SECTION ---DC LINES--                    AREA X------- SWING BUSES -------X
 TOTAL     RXB     RX     RXT   RX=0.    IN     OUT XFORM  LINES SECTNS 2TRM MTRM VSC AREAS ZONES OWNRS TRANS   3011     MINE_G      13.800
    34      22      1      11      0     34      0      0      2      4    0    0   0      3     4     7     4
     ----GENERATION---- INDUCTION                                    GNE
TOTAL   SYNCHR  INDUCTN    MOTORS   PQLOAD   I LOAD   Y LOAD   SHUNTS  DEVICES CHARGING   LOSSES    SWING
MW     3258.6      0.0       0.0   3200.0      0.0      0.0      0.0      0.0      0.0     58.6    258.6
MVAR    963.5      0.0       0.0   1950.0      0.0      0.0   -291.8      0.0   1810.1   1115.4    103.6

TOTAL MISMATCH =    0.00 MVA X--------- AT BUS ----------X   THRSHZ  PQBRAK BLOWUP  SBASE MXTPSS IsolLvl
 MAX. MISMATCH =    0.00 MVA    201     HYDRO       500.00  0.000100  0.700   5.00 100.00   99       4
  HIGH VOLTAGE = 1.04034 PU     211     HYDRO_G     20.000   ADJTHR  ACCTAP TAPLIM SWVBND MXSWIM  CAMax
  LOW  VOLTAGE = 0.93918 PU     154     DOWNTN      230.00   0.0050  1.0000 0.0500  100.0   10   ReptSln
                                                                                                  20
X------ SOLV AND MSLV ------X  X----------------- NEWTON ------------------X  X----- TYSL -----X ChkDupl
 ACCP  ACCQ  ACCM   TOL  ITER  ACCN  TOLN  VCTOLQ  VCTOLV ITER  DVLIM NDVFCT  ACCTY    TOL  ITER CntLabl
1.600 1.600 1.000 0.00010 100  1.00 0.1000 0.1000 0.000010  20 0.9900 0.9900  1.000 0.000010  20    NO

--------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     THU, JUN 20 2019  15:21
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                        WORST
BASE CASE INCLUDING SEQUENCE DATA                              MISMATCHES
  BUS#-SCT X-- NAME --X BASKV       MW       MVAR       MVA
   201     HYDRO       500.00      0.00     -0.00      0.00

--------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     THU, JUN 20 2019  15:21
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                       RATING SET
BASE CASE INCLUDING SEQUENCE DATA                              UTILIZATION
RATING SET  -NAME-  X---------DESCRIPTION----------X  # NON-ZERO RATINGS
     1      RATE1   RATING SET 1                          24
     2      RATE2   RATING SET 2                          24
     3      RATE3   RATING SET 3                          34
     4      RATE4   RATING SET 4                           0
     5      RATE5   RATING SET 5                           0
     6      RATE6   RATING SET 6                           0
     7      RATE7   RATING SET 7                           0
     8      RATE8   RATING SET 8                           0
     9      RATE9   RATING SET 9                           0
    10      RATE10  RATING SET 10                          0
    11      RATE11  RATING SET 11                          0
    12      RATE12  RATING SET 12                          0
```

Figure 5.18. Case Summary for the savnw.sav Saved Case File

Bus Data

A bus type CODE of -2 indicates a Type 2 bus for which generation is at a reactive power limit. This is set by the various power flow solution activities. The column labeled OR is the original type code of the bus whenever the current type code of the bus has been changed as the result of operations of substation breakers and/or switches; it will differ from the CODE value. The OR entry will be blank for buses which have not been affected by breaker or switch operations.

The number of loads independently modeled at the bus is tabulated in the column labeled LOADS. Similarly, the number of fixed and switched shunts modeled at the bus is tabulated in the columns labeled FIXED and SWITCHED, respectively.

When reporting for a bus subsystem, the bus data tabulation is restricted to subsystem buses.

Load Data

Data for loads is tabulated in the load data category. Loads are listed in ascending bus order (numeric or alphabetic), and loads at each bus are in ascending load identifier order.

Loads tabulated are at nominal values (i.e., at 1.0 pu voltage). The nominal constant MVA load component tabulated is not adjusted by any load multiplier associated with an Optimal Power Flow adjustable bus load table; the load multiplier is tabulated in the column labeled PSI. Constant admittance loads do not include any shunt elements modeled at the bus (refer to Fixed Bus Shunt Data and Switched Shunt Data).

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The column labeled SCALE reflects the setting of the load scaling flag, and the column labeled INTRPT reflects the setting of the interruptible load flag (refer to Load Data).

When reporting for a bus subsystem, the bus and/or base voltage selection criteria, if enabled, determine those buses whose connected loads are candidates for reporting; otherwise, all buses are candidate buses. The area, owner, and/or zone selection criterion, if enabled, determine which loads at candidate buses are to be reported; otherwise, all loads at candidate buses are reported.

Plant Data

Buses that have plant data associated with them (refer to Plant and Machine Sequence Numbers) are tabulated in the plant data listing. This tabulation includes plants that are out-of-service (i.e., with a type code of 1 or 4). The number of machines independently modeled at the plant is tabulated in the column labeled MCNS.

Any plant that regulates the voltage at a remote bus has the remote bus listed at the right-hand side of the report. Any plant regulating its own terminal voltage has no entry in the REMOTE BUS field.

When reporting for a bus subsystem, the plant data tabulation is restricted to subsystem buses.

Generator Unit Data

Data for conventional (synchronous) machines is tabulated in the generator unit data category. Machines are listed in ascending bus order (numeric or alphabetic), and machines in each plant are in ascending machine identifier order.

For each machine, the bus type code, machine identifier and machine status flag are listed in the columns labelled CD, ID, and ST, respectively. Recall that the actual status of a machine is determined *both* by its status flag *and* the bus type code (refer to Section 5.10.1, "Equipment Status Changes"). For machines that are not designated as wind machines, the columns labelled WMOD and WPF are blank.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected machines are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which machines at candidate buses are to be reported; otherwise, all machines at candidate buses are reported.

Induction Machine Data

Data for induction machines is tabulated in the induction machine circuit data listing and in the induction machine miscellaneous data listing. Machines are listed in ascending bus order (numeric or alphabetic), and machines at the same bus are in ascending machine identifier order. In both of these listings, the machine identifier and machine status flag are listed in the columns labelled ID and ST, respectively.

In the induction machine circuit data listings, when the machine design code is zero, CUSTOM is printed in the column labelled DESIGN, and the remaining data in the listing corresponds to the circuit and saturation data specified by the user. When one of the NEMA or IEC designs had been specified (a machine design code of one through 5 was specified), the design description (e.g., "NEMA B") is printed in the DESIGN column, and the circuit and saturation data values listed correspond to the pre-programmed values for that machine design.

In the induction machine miscellaneous data listings, an M (for mechanical) or E (for electrical) indicates the machine base power code and is printed following the machine base power. Similarly, an M or E follows the power setpoint and indicates the scheduled power code. In addition to the induction machine input data (refer to Induction Machine Data), solution results are listed. These include the complex power flowing into the machine from the network, and the machine slip.

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When reporting for a bus subsystem, the bus and/or base voltage selection criteria, if enabled, determine those buses whose connected induction machines are candidates for reporting; otherwise, all buses are candidate buses. The area, owner, and/or zone selection criterion, if enabled, determine which induction machines at candidate buses are to be reported; otherwise, all induction machines at candidate buses are reported.

Fixed Bus Shunt Data

Data for fixed bus shunts is tabulated in the fixed shunt data category. Shunts are listed in ascending bus order (numeric or alphabetic), and shunts at each bus are in ascending shunt identifier order.

Shunt data tabulated are at nominal values (i.e., at 1.0 pu voltage), and do not include any constant admittance load modeled at the bus (refer to Load Data).

When reporting for a bus subsystem, the fixed bus shunt data tabulation is restricted to subsystem buses.

Switched Shunt Data

Data for switched shunts is tabulated in the switched shunt data category. The value listed as SHUNT is the admittance presently switched on at the bus.

Any switched shunt that regulates the voltage at a remote bus or the reactive power output of another voltage controlling device has the controlled bus listed in the REMOTE BUS columns at the right-hand side of the report; those regulating the reactive power output of a VSC dc line converter or of the shunt element of a FACTS device also have the device name listed. Switched shunts regulating their own terminal voltage have no entry in these fields.

When reporting for a bus subsystem, the switched shunt data tabulation is restricted to subsystem buses.

Branch Data

Data for non-transformer ac branches is tabulated in the branch data category. Branches are ordered in ascending numerical or alphabetical order by from bus, and, for each from bus, in ascending order by to bus. Parallel circuits between any pair of buses are tabulated in ascending circuit identifier order.

The branch data listing may be in either single or double entry format. In single entry format, each branch that is tabulated appears once in the listing: with the lower ordered bus (number or name) listed as the from bus. In double entry format, each branch is listed in both directions.

An asterisk (*) follows the number, name and base voltage of the bus designated as the metered end.

The column labeled ZI contains a Z if the branch is treated as a zero impedance line; for other branches, the ZI column is blank. The column labeled ST indicates the current status of the branch, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the branch whenever the current status of the branch has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for branches which have not been affected by breaker or switch operations. Columns for rating sets that are unused (i.e., all elements of the rating set are 0.0) are omitted from the report. The remaining data items in this report category correspond to data items specified in branch data input records (refer to Non-Transformer Branch Data).

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected branches are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which branches at candidate buses

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are to be reported; otherwise, all non-transformer branches at candidate buses, including branches to non-candidate buses (i.e., ties), are reported.

Line Shunt Data

Data for line connected shunts on non-transformer ac branches is tabulated in the line shunt data category. This report tabulates branches with a non-zero line connected shunt at either end; branches with line shunt values of (0.0 + j 0.0) at both ends are omitted from the report.

The line shunt tabulation is ordered in the same manner as is branch data report (refer to Branch Data). Branches are tabulated in single entry format as in the branch data category *except* when full listing (double entry branch) is specified.

The column labeled ST indicates the current status of the branch, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the branch whenever the current status of the branch has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for branches which have not been affected by breaker or switch operations.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected branches are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which branches at candidate buses are to be reported; otherwise, all non-transformer branches at candidate buses with a non-zero line shunt at either end, including branches to non-candidate buses (i.e., ties), are reported.

Two-Winding Transformer Data

Data for two-winding transformers is tabulated in the two-winding transformer impedance data listing, in the two-winding transformer winding data listing, and in the two-winding transformer control data listing. The majority of data items in these report categories correspond to data items specified in the two-winding transformer data input block (refer to Transformer Data).

The two-winding transformer tabulations are ordered in the same manner as is branch data report (refer to Branch Data). Transformer branches are tabulated in single entry format as in the branch data category *except* when full listing (double entry branch) is specified.

In the transformer impedance and control data listings, the column labeled W1 contains an F if the bus listed as the from bus is the bus to which Winding 1 of the transformer is connected, or a T if the to bus is the bus to which Winding 1 is connected.

In the transformer impedance data listing, the column labeled ST indicates the current status of the transformer, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the transformer whenever the current status of the transformer has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for transformers which have not been affected by breaker or switch operations. The column labeled MT contains an F if the bus listed as the from bus is the metered end, or a T if the to bus is the metered end. For any transformer with an impedance correction table assigned to it, both specified (i.e., nominal) and corrected (i.e., as modified by application of the transformer impedance correction table) impedances are tabulated.

In the winding data listing, columns for rating sets that are unused (i.e., all elements of the rating set are 0.0) are omitted from the report.

In the transformer control data listing, the controlled bus number is preceded by a minus sign if it is on the Winding 1 side of the transformer.

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When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected two-winding transformers are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which transformers at candidate buses are to be reported; otherwise, all two-winding transformer branches at candidate buses, including transformers to non-candidate buses (i.e., ties), are reported.

Three-Winding Transformer Data

Data for three-winding transformers is tabulated in the three-winding transformer general data listing, in the three-winding transformer impedance data listing, in the three-winding transformer winding data listing, and in the three-winding transformer tap and control data listing. The majority of data items in these report categories correspond to data items specified in the three-winding transformer data input block (refer to Transformer Data).

The three-winding transformer tabulations are ordered in ascending numerical or alphabetical order by Winding 1 bus, and, for each Winding 1 bus, in ascending order by Winding 2 bus, and for each Winding 1, Winding 2 bus combination, in ascending order by Winding 3 bus. Transformers connecting the same three buses are tabulated in ascending circuit identifier order. Single entry format is always used.

In the general and impedance data listings, the column labeled ST indicates the current status of the transformer, with values of 0 through 4 as defined in Transformer Data. The column labeled OR is the original status of the transformer whenever the current status of the transformer has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for transformers which have not been affected by breaker or switch operations.

The winding data and the tap and control data listings are comprised of three lines of data for each three-winding transformer: one for each of the two-winding transformers in the star model used to model a three-winding transformer.

In the impedance data listing, measured impedances are listed in the units indicated by CZ. For any transformer with an impedance correction table assigned to one or more of its windings, both specified (i.e., nominal) and corrected (i.e., as modified by application of the transformer impedance correction table) measured impedances are tabulated.

In the winding data listing, the winding bus identifiers of the two metered windings are followed by an asterisk (*). The column labeled ST indicates the current status of the transformer winding, as derived from the current status of the three-winding transformer, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the transformer winding whenever the current status of the three-winding transformer has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for transformer windings which have not been affected by breaker or switch operations. The winding impedances are derived from the values entered as measured impedances, and are listed in per unit on system MVA base and winding voltage base. For any winding with an impedance correction table assigned to it, both nominal (i.e., as derived from the measured impedances) and corrected (i.e., as modified by application of the transformer impedance correction table) impedances are tabulated. Columns for rating sets that are unused (i.e., all elements of the rating set are 0.0) are omitted from the report.

In the control data listing, the controlled bus number is preceded by a minus sign if it is on the same side of the transformer as the bus connected to the controlling winding.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected three-winding transformers are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which transformers

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at candidate buses are to be reported; otherwise, all three-winding transformer branches at candidate buses, including transformers to non-candidate buses (i.e., ties), are reported.

Transformer Impedance Correction Table Data

Each transformer impedance correction table in the working case is tabulated in ascending table number order. Each table is followed by a listing of those transformers assigned to the table, their winding off-nominal turns ratio or phase shift angle as appropriate, and their winding actual and nominal impedances. Transformers are listed in single entry format except when full listing (double entry branch) is specified.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected transformers are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which transformers at candidate buses are to be reported; otherwise, all transformers at candidate buses that are assigned to the table, including transformers to non-candidate buses (i.e., ties), are reported.

Multi-Section Line Grouping Data

Data for multi-section line groupings, along with the ac branches that are their members, is tabulated in the multi-section line grouping data listing. The multi-section line groupings are tabulated in single entry format as in the branch data category (refer to Branch Data) except when full listing (double entry branch) is specified.

An asterisk (*) follows the identifiers of the bus designated as the metered end.

The branches comprising each grouping are listed in series order starting at the bus listed as the from bus.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those endpoint buses whose connected multi-section lines are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, is applied to each member of the multi-section lines at candidate buses and determines which multi-section lines are to be reported; otherwise, all multi-section lines at candidate buses, including those to non-candidate endpoint buses (i.e., ties), are reported.

DC Line Data

Data for dc lines is tabulated in the dc line data listing. The dc line listing has two-terminal lines in dc line name alphabetical order listed first, followed by multi-terminal lines in dc line name alphabetical order, followed by VSC dc lines in dc line name alphabetical order.

The column labeled MDC indicates the current control mode of the dc line. The column labeled OR (ORG for two-terminal dc lines) is the original control mode of the dc line whenever the current control mode of the dc line has been changed as the result of operations of substation breakers and/or switches; it will differ from the MDC value. The OR or ORG entry will be blank for dc lines which have not been affected by breaker or switch operations.

In the converter bus block for each VSC dc line, the column labeled TYP indicates the current dc control code of the converter. The column labeled OR is the original dc control code of the converter whenever its current dc control code has been changed as the result of operations of substation breakers and/or switches; it will differ from the TYP value. The OR entry will be blank for converters which have not been affected by breaker or switch operations.

In addition to the dc line input data (refer to Two-Terminal DC Transmission Line Data, Voltage Source Converter (VSC) DC Transmission Line Data and Multi-Terminal DC Transmission Line Data), solution results are

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listed. For each converter of a two-terminal or multi-terminal line, the alpha or gamma angle is printed along with the apparent ac system complex load and the tap setting.

For each two-terminal line, the dc line current (DCAMPS) and the compounded dc line voltage (VCOMP) are printed on the first line of the output block, and the dc voltage at each end of the line is printed with the converter data. The Winding 1 tap ratio (RATIO) of an ac transformer that is controlling a dc line quantity is shown following the ac transformer identifiers in the output block of the appropriate dc line.

For each multi-terminal line, the voltage at each dc bus and the current on the dc side of each converter transformer is printed. For each dc link in the line, the metered end dc bus is listed in the column labeled MET.

For each VSC dc line, the dc line current (DCAMPS) is printed. For each converter, the apparent ac system complex generation, the ac current in amps (ACAMPS), and the voltage on the dc side of the converter are listed.

When reporting for a bus subsystem, the dc line data tabulation is restricted to those dc lines that are connected to at least one subsystem bus.

Area Data

Data for areas that are defined is listed in the area data listing. The tabulation is in ascending area number order.

If an area has an area slack bus designated for it, its number, name, base voltage, and active power output are tabulated in the listing. This is followed by the plant active power limits, which are taken as the sums of these limits for the in-service machines at the plant. If the total power output is outside of these limits, an asterisk (*) is printed following the generator power.

For each area that is listed, the tabulation includes the number of buses, loads, induction machines, and dc buses assigned to the area.

The final line tabulates the summation of the listed desired interchanges.

When reporting for a bus subsystem for which area is one of the subsystem selection criteria, the area tabulation is restricted to those areas specified.

Interarea Transfer Data

Data defining transfers between pairs of areas are listed in the interarea transfer data listing. Interarea transfers are ordered in ascending order by from area number, and, for each from area, in ascending order by to area number. Multiple transfers between any pair of areas are tabulated in ascending interarea transfer identifier order. Interarea transfers are listed in double entry format, with each interarea transfer listed in both directions.

For each from area block, the line reporting the last transfer includes the algebraic sum of all transfers involving the from area, as well as the area's desired net interchange (refer to Area Interchange Data and Area Data).

When reporting for a bus subsystem for which area is one of the subsystem selection criteria, the interarea transfer tabulation is restricted to the from area blocks of those areas specified.

Zone Data

Data for zones that are defined is listed in the zone data listing. The tabulation is in ascending zone number order.

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For each zone that is listed, the tabulation includes the zone name along with the number of buses, loads, induction machines, and dc buses assigned to the zone.

When reporting for a bus subsystem for which zone is one of the subsystem selection criteria, the zone tabulation is restricted to those zones specified.

Owner Data

Data for owners that are defined is listed in the owner data listing. The tabulation is in ascending owner number order.

For each owner that is listed, the tabulation includes the owner name along with the number of buses, loads, synchronous machines, induction machines, branches, dc buses, FACTS devices, VSC dc lines, and GNE devices assigned to the owner.

When reporting for a bus subsystem for which owner is one of the subsystem selection criteria, the owner tabulation is restricted to those owners specified.

FACTS Device Data

Data for FACTS devices is tabulated in the FACTS device sending end data listing and in the FACTS device terminal end data listing. FACTS devices are ordered in ascending FACTS device name alphabetical order.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected FACTS devices are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which FACTS devices at candidate buses are to be reported; otherwise, all FACTS devices at candidate buses, including those with a series element connected to a non-candidate bus (i.e., ties), are reported.

The column labeled MODE indicates the current control mode of the FACTS device. The column labeled OR is the original control mode of the FACTS device whenever the current control mode of the FACTS device has been changed as the result of operations of substation breakers and/or switches; it will differ from the MODE value. The OR entry will be blank for FACTS devices which have not been affected by breaker or switch operations.

GNE Device Data

Data for GNE devices is tabulated in the GNE device data listing. GNE devices are ordered in ascending GNE device name alphabetical order.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected GNE devices are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which GNE devices at candidate buses are to be reported; otherwise, all GNE devices at candidate buses, including those with a series element connected to a non-candidate bus (i.e., ties), are reported.

Additional Information
PSS®E GUI Users Guide, Working Case Data Reporting
PSS®E Command Line Interface (CLI) Users Guide, Working Case Data Reporting
PSS®E Application Program Interface (API), LIST

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5.38. Listing Components of a Bus

EXAM

Run Activity EXAM - GUI
Power Flow > List Data... *[List Data]* OPTION: Examine Powerflow / sequence data
Run Line Mode Activity EXAM - CLI
ACTIVITY? >>**EXAM** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The data examination activity EXAM produces a tabulation of power flow data organized by bus. For each bus tabulated, its bus data is followed by the data associated with each network element that is connected to the bus.

The EXAM API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".

5.38.1. Operation of Activity EXAM

The user specifies either that the entire working case is to be processed, or that data items for equipment in a designated bus subsystem are to be tabulated.

The report of activity EXAM is directed to the *Report* device.

5.38.2. Listing Format

The blocks of output for buses are ordered as described in Section 4.9, "Subsystem Reporting". The data for each equipment category is tabulated in a format similar to that of activity LIST (refer to Section 5.37.2, "Listing Format").

In the non-transformer branch data tabulation, the column labeled MET contains an F if the bus being reported is the metered end or a T if the bus listed as the to bus is the metered end. Any branch treated as a zero impedance line has a Z printed in the column labeled ZI. The columns labeled ST and OR are the current and original branch status values, respectively. Transformers are omitted from the branch data tabulation.

Two-winding transformer data is printed in three blocks. The column labeled W1 contains an F if the bus being reported is the bus to which Winding 1 of the transformer is connected, or a T if the bus listed as the to bus is the Winding 1 bus. The column labeled MT contains an F if the bus being reported is the metered end, or a T if the bus listed as the to bus is the metered end. The columns labeled ST and OR are the current and original transformer status values, respectively. Transformer impedances, admittances, and winding data are

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printed in the units specified by CZ, CM, and CW, respectively. The column labeled CN is the transformer's automatic adjustment control mode.

Three-winding transformer data is printed in four blocks. The columns labeled ST and OR in the first and second blocks are the current and original three-winding transformer status values (0 through 4; refer to Non-Transformer Branch Data). The columns labeled ST and OR in the third block show the current and original status of the three two-winding transformers comprising the three-winding transformer. The two winding buses in the third block where identifiers are followed by an asterisk (*) are the two metered end buses, and the other bus is the non-metered bus. Except for winding impedances, which are derived from the measured impedances and are tabulated in per unit on system MVA base and winding voltage base, transformer impedances, admittances, and winding data are printed in the units specified by CZ, CM, and CW, respectively. The column labeled CN is the transformer's automatic adjustment control mode.

The adjustable branch reactance device data tabulation includes the to bus and circuit identifier, the branch reactance multiplier magnitude, the nominal branch reactance, and the device status.

Additional Information
PSS®E GUI Users Guide, Listing Buses and their Connected Equipment *PSS®E Command Line Interface (CLI) Users Guide*, Listing Components of a Bus *PSS®E Application Program Interface (API)*, EXAM

5.39. Listing Buses in Alphabetical Order

ALPH

Run Activity ALPH - GUI
Power Flow > List Data... *[List Data]* OPTION: Bus names
Run Line Mode Activity ALPH - CLI
ACTIVITY? >>**ALPH** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The bus alphabetic listing activity ALPH prints an alphabetically sorted table of all buses in a specified subsystem of the working case.

The ALPH API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".

The user specifies either that the entire working case is to be processed, or that buses in a designated bus subsystem are to be tabulated.

The report of activity ALPH is directed to the *Report* device.

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The report of activity ALPH lists the extended bus names and bus numbers of subsystem buses. Buses are listed in alphabetical order by extended bus name, and are grouped as described in Section 4.9, "Subsystem Reporting".

Additional Information
PSS®E GUI Users Guide, Listing Buses in Alphabetical Order
PSS®E Command Line Interface (CLI) Users Guide, Listing Buses in Alphabetical Order
PSS®E Application Program Interface (API), ALPH

5.40. Listing Buses using Partial Bus Name

FIND

Run Line Mode Activity FIND - CLI
ACTIVITY? >>**FIND** ENTER DESIRED BUS NAME:
Interrupt Control Codes
AB

The bus name to number translation activity FIND enables the user to determine the number of a bus with a specified extended bus name. In addition, through the use of a string matching character, activity FIND produces a listing of possible matching names even if the exact spelling of the bus name and/or its base voltage is not known.

The FIND API routine includes the following among its inputs:

- The bus name component of the extended bus name.
- The base voltage component of the extended bus name.

The bus name input string consists of up to twelve alphanumeric characters, plus embedded asterisks (*), which are interpreted as wildcard string matching characters. Each * represents any number of characters, including zero, which may be embedded between other substrings specified in name, and up to seven of them may be included between other alphanumeric characters.

Any buses for which the name and base voltage contain the specified substrings are listed at the *Report* device in ascending bus number or name order, according to the bus output option currently in effect (refer to Saved Case Specific Option Settings and activity OPTN).

The following examples illustrate the response of activity FIND to various input values, with the first column indicating the input for the bus name component, and the second the input for the base voltage component:

AB		All buses of any base voltage whose name begins with "AB".
AB	230	All 230-kV buses whose name begins with "AB".
*AB	115	All 115-kV buses whose name contains the string "AB" anywhere in

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		its twelve-character name. Thus the listing produced would include 115-kV buses "ABCD" and "XYZ-ABC", but not "AXB" or "XAYB".
A*BC		All buses of any base voltage whose name begins with "A" and in addition contains the string "BC". Thus the listing produced would include the buses "ABCXYZ" and "AXYZBC", but not the buses "XABC", "BCA", or "ABXC".
	13.8	All 13.8-kV buses.
	13	All buses with a base voltage equal to or greater than 13-kV and less than 14-kV.
	13.0	All buses with a base voltage of 13.0-kV.
	0	All buses for which a base voltage was not specified.
*		All buses.
		Exit activity FIND.

Activity FIND is not available from the GUI drop-down menus. Rather, the bus filtering capability of the bus subsystem selector and the Find function of the *[Spreadsheet]* provide equivalent capability.

Additional Information
PSS® E GUI Users Guide: Using the Subsystem Menu Menu Options *PSS®E Command Line Interface (CLI) Users Guide*, Listing Buses Matching a Partial Extended Bus Name *PSS®E Application Program Interface (API)*, FIND

5.41. Listing Bus Shunts

SHNT

Run Activity SHNT - GUI
Power Flow > List Data... *[List Data]* OPTION: Bus shunts
Run Line Mode Activity SHNT - CLI
ACTIVITY? >>**SHNT** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

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The bus shunt summary activity SHNT tabulates fixed and/or switched bus shunts contained in the working case.

The SHNT API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".
- A designation of the types of bus shunts to be reported:
- All bus shunts
- Only fixed shunts
- Only switched shunts

The user specifies either that the entire working case is to be processed, or that shunts connected to buses in a designated bus subsystem are to be tabulated.

The report of activity SHNT is directed to the *Report* device.

In the report of activity SHNT, buses are ordered as described in Section 4.9, "Subsystem Reporting". When multiple shunts are present at a bus, fixed shunts at the bus are listed first in ascending shunt identifier order, followed by the switched shunt if one is present.

For each shunt tabulated by activity SHNT, the bus number, name, base voltage, and bus type code are printed, along with the shunt's service status and the nominal values (i.e., at one per unit voltage) of the shunt's admittance. The present bus voltage magnitude is printed along with an identifier indicating whether the shunt is a fixed or switched shunt. For fixed bus shunts, the shunt identifier is also printed. For switched shunts, the switched shunt control mode and switched shunt adjustment method are printed along with the minimum (i.e., all reactors switched on) and maximum (i.e., all capacitors switched on) susceptance values of the switched shunt. If a switched shunt reactor block is out of sequence (i.e., a reactor block is detected after the first capacitor block), the calculated maximum value is followed by an asterisk (*).

Additional Information
PSS®E GUI Users Guide, Listing Bus Shunts *PSS®E Command Line Interface (CLI) Users Guide*, Listing Bus Shunts *PSS®E Application Program Interface (API)*, SHNT *See also*: Fixed Bus Shunt Data Switched Shunt Data

5.42. Listing Outaged Equipment

OUTS

Run Activity OUTS - GUI
Power Flow > List Data... *[List Data]* OPTION: Outaged equipment

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Run Line Mode Activity OUTS - CLI
ACTIVITY? >>**OUTS** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The outaged equipment reporting activity OUTS tabulates those components in the working case that are removed from service.

The OUTS API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".

The user specifies either that the entire working case is to be processed, or that outaged equipment in a designated bus subsystem is to be tabulated.

The report of activity OUTS is directed to the *Report* device.

The following outaged equipment is listed by activity OUTS:

- Out-of-service non-transformer branches and two-winding transformers
- Out-of-service three-winding transformers
- Disconnected and isolated buses
- Out-of-service plants
- Out-of-service synchronous machines at in-service plants
- Out-of-service induction machines at in-service buses.
- Out-of-service loads at in-service buses
- Out-of-service fixed shunts at in-service buses
- Blocked dc transmission lines
- Out-of-service converters at in-service VSC dc lines
- Out-of-service FACTS devices.
- Out-of-service GNE devices.

When a subsystem is being processed and ownership is included among the selection criteria, branch, synchronous machine, load, FACTS device, VSC dc line, and GNE device ownership rather than bus ownership is used in determining which branches, machines, induction machines, loads, FACTS devices, VSC dc lines, and GNE devices are contained in the specified subsystem; any other selection criteria apply to the buses to which the equipment item is connected.

Out-of-service branches are listed in single entry format (refer to Branch Data) and are ordered in ascending numerical (using the *numbers* output option) or alphabetical (using the *names* option) order by from bus,

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and, for each from bus, in ascending order by to bus and circuit identifier. When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), out-of-service multi-section line groupings (from bus, to bus, identifier) rather than their individual line sections are listed in the category of out-of-service ac branches.

Out-of-service three-winding transformers are listed in alphabetical order by transformer name. If only one winding of a three-winding transformer is out-of-service (i.e., the status value of the three-winding transformer is 2 through 4; refer to Transformer Data), the out-of-service winding is noted at the end of the output line.

The classification of isolated buses includes both those buses for which the type code is 4, and those that are not connected to any other bus by an in-service ac branch but where the type code is not 4. Any bus included in the latter group is designated by an asterisk (*) following its bus identifiers, and, unless it is a Type 3 bus connected to a dc line, results in an improperly specified network, which usually causes the power flow solution activities to fail to converge. When the multi-section line reporting option is enabled, Type 4 buses that are dummy buses of a multi-section line grouping (refer to Multi-Section Line Grouping Data) are omitted from the isolated bus report.

The category of disconnected plants includes those buses that have plant data assigned to them but for which their bus type code is 1 or 4.

Buses listed in the isolated bus, disconnected plant, out-of-service synchronous machine, out-of-service induction machine, and out-of-service load sections of the report are in ascending numerical or alphabetical order, as appropriate.

Each out-of-service synchronous machine, out-of-service load, and out-of-service fixed shunt has its identifier printed following the bus identifiers. If more than one synchronous machine, induction machine, load, or fixed shunt is out-of-service at the same bus, they are listed in alphabetical order by synchronous machine, induction machine, load, or fixed shunt identifier, as appropriate.

Two-terminal dc lines, multi-terminal dc lines, and VSC dc lines are listed in dc line name alphabetical order. FACTS devices are listed in FACTS device name alphabetical order, and GNE devices are listed in GNE device name alphabetical order.

When a subsystem is being processed and bus, area, zone, and/or owner are included among the selection criteria, the output of activity OUTS is grouped as described in Section 4.9, "Subsystem Reporting".

Additional Information
PSS®E GUI Users Guide,Outaged Equipment *PSS®E Command Line Interface (CLI) Users Guide*, Listing Outaged Equipment *PSS®E Application Program Interface (API)*, OUTS

5.43. Listing Sequence Data

SQLI

Run Activity SQLI - GUI
Power Flow > List Data... *[List Data]*

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Run Activity SQLI - GUI
OPTION: Sequence data
Run Line Mode Activity SQLI - CLI
ACTIVITY? >>**SQLI** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The sequence data listing activity SQLI tabulates the sequence data arrays in the working case in a form suitable for problem data documentation. The report generated by activity SQLI is separated into several categories of data.

The SQLI API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".
- A designation of the sequence data category to be tabulated.

5.43.1. Operation of Activity SQLI

The user specifies either that the entire working case is to be processed, or that data items for equipment in a designated bus subsystem are to be tabulated.

If output for a bus subsystem is being produced, only equipment in the designated subsystem is included in the report. Any subsystem filtering applied to a data category is described in the data category descriptions given in Section 5.43.2, "Listing Format".

The report of activity SQLI is directed to the *Report* device.

The following data categories are available in activity SQLI:

• All data	• Line shunt data
• Load data	• Two-winding transformer positive sequence data
• Synchronous machine data	• Two-winding transformer zero sequence data
• Induction machine data	• Three-winding transformer general data
• Fixed bus shunt data	• Three-winding transformer positive sequence data
• Switched shunt data	• Three-widing transformer zero sequence data
• Non-transformer branch date	• Zero sequence mutual data

5.43.2. Listing Format

Most of the data items tabulated by activity SQLI correspond to data entered in activities RESQ, TRSQ, READ, TREA, RDCH, and/or MCRE. Refer to Sections Section 5.2.1, "Power Flow Raw Data File Contents", Sec-

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tion 5.4.1, "Machine Impedance Data File Contents", and Section 5.6.1, "Sequence Data File Contents" for detailed descriptions of these quantities.

The following paragraphs describe the data tabulated in the various output categories by activity SQLI. Unless otherwise stated, data in these categories are tabulated in ascending bus number order when the *numbers* output option is in effect and in alphabetical order when the *names* output option is in effect.

Load Data

Data for loads is tabulated in the load data category. Loads are listed in ascending bus order (numeric or alphabetic), and loads at each bus are in ascending load identifier order. The bus type code and load status are tabulated. Loads tabulated are at nominal values (i.e., in MW and Mvar at 1.0 per unit voltage).

In reporting positive sequence loads, the nominal constant MVA load component is not adjusted by any load multiplier associated with an Optimal Power Flow adjustable bus load table; the load multiplier is tabulated in the column labeled PSI. Constant admittance loads do not include any shunt elements modeled at the bus (refer to Fixed Bus Shunt Data and Switched Shunt Data).

Zero sequence and exceptional negative sequence shunt loads, as described in Load Sequence Data, are listed along with the zero sequence grounding flag.

When reporting for a bus subsystem, the bus and/or base voltage selection criteria, if enabled, determine those buses whose connected loads are candidates for reporting; otherwise, all buses are candidate buses. The area, owner, and/or zone selection criterion, if enabled, determine which loads at candidate buses are to be reported; otherwise, all loads at candidate buses are reported.

Generator Data

Data for conventional (synchronous) machines is tabulated in the generator unit data category. Machines are listed in ascending bus order (numeric or alphabetic), and machines in each plant are in ascending machine identifier order.

The data listed includes the machine identifier, the bus type code, the machine status flag, and the generator impedance data described in Generator Sequence Data, along with the values of MBASE, XTRAN, and GENTAP (refer to Generator Data and Section 13.2.3, "Generator Step-Up Transformers"). Recall that the actual status of a machine is determined *both* by its status flag *and* the bus type code (refer to Section 5.10.1, "Equipment Status Changes").

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected machines are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which machines at candidate buses are to be reported; otherwise, all machines at candidate buses are reported.

Induction Machine Data

Data for induction machines is tabulated in the induction machine data listing. Machines are listed in ascending bus order (numeric or alphabetic), and machines at the same bus are in ascending machine identifier order.

The data listed includes the machine identifier, the bus type code, and the machine status flag. Recall that the actual status of a machine is determined *both* by its status flag *and* the bus type code (refer to Section 5.10.1, "Equipment Status Changes"). The machine impedance data described in Induction Machine Sequence Data is listed, along with the vaules of the machine's base power and rated voltage (refer to Induction Machine

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Data). An M (for mechanical) or E (for electrical) indicates the machine base power code and is printed following the machine base power.

When reporting for a bus subsystem, the bus and/or base voltage selection criteria, if enabled, determine those buses whose connected induction machines are candidates for reporting; otherwise, all buses are candidate buses. The area, owner, and/or zone selection criterion, if enabled, determine which induction machines at candidate buses are to be reported; otherwise, all induction machines at candidate buses are reported.

Fixed Shunt Data

Data for fixed bus shunts is tabulated in the fixed shunt data category. Shunts are listed in ascending bus order (numeric or alphabetic), and shunts at each bus are in ascending shunt identifier order.

The data listed includes the shunt identifier, the bus type code, the shunt status flag, the positive sequence fixed shunt admittance, and the zero sequence fixed shunt admittance. Shunt admittances tabulated are at nominal values (i.e., MW and Mvar at 1.0 per unit voltage).

When reporting for a bus subsystem, the fixed bus shunt data tabulation is restricted to subsystem buses.

Switched Shunt Data

Data for switched shunts are tabulated in the switched shunt data listing. The data printed for each switched shunt includes the admittance currently switched on along with the switched shunt blocks present at the bus. Output for each switched shunt consists of two lines of data, with positive sequence values printed on the first line and zero sequence values on the second line. All admittances printed are at nominal values (i.e., at 1.0 per unit voltage.

When reporting for a bus subsystem, the switched shunt data tabulation is restricted to subsystem buses.

Non-Transformer Branch Data

Data for non-transformer ac branches is tabulated in the branch data category. Branches are ordered in ascending numerical or alphabetical order by from bus, and, for each from bus, in ascending order by to bus and circuit identifier.

The non-transformer branch data listing is in single entry format. Each non-transformer ac series branch appears once in the listing: with the lower ordered bus (number or name) listed as the from bus.

For a branch treated as a zero impedance line, a Z is printed in the column labeled ZI. For branches that are not treated as zero impedance lines, this column is left blank.

The column labeled ST indicates the current status of the branch, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the branch whenever the current status of the branch has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for branches which have not been affected by breaker or switch operations.

If a branch is mutually coupled to other branches in the zero sequence network, the mutually coupled branches are listed following the branch data line. Mutuals are listed one per line in single entry format. The from bus, to bus and circuit identifier of the coupled branches are printed along with the mutual impedance. Note that the value printed as the mutual impedance may be the negative of the value initially entered. Careful inspection of the from and to bus ordering for the branches involved will show that the tabulation has re-

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versed the bus ordering for one of the pair of lines, and consequently changed the sign of the complex mutual impedance.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected branches are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which branches at candidate buses are to be reported; otherwise, all non-transformer branches at candidate buses, including branches to non-candidate buses (i.e., ties), are reported.

Line Shunt Data

Data for line connected shunts on non-transformer ac branches is tabulated in the line shunt data category. This report tabulates branches with one or more non-zero line connected shunts in the positive and/or zero sequence networks; branches for which line shunt values are all (0.0 + j 0.0) are omitted from the report.

The line shunt data listing is in single entry format. Each non-transformer ac series branch with a non-zero line shunt admittance appears once in the listing: with the lower ordered bus (number or name) listed as the from bus. Branches are ordered in ascending numerical or alphabetical order by from bus, and, for each from bus, in ascending order by to bus and circuit identifier.

The column labeled ST indicates the current status of the branch, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the branch whenever the current status of the branch has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for branches which have not been affected by breaker or switch operations.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected branches are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which branches at candidate buses are to be reported; otherwise, all non-transformer branches at candidate buses with any non-zero line shunt component, including branches to non-candidate buses (i.e., ties), are reported.

Two-Winding Transformer Data

Data for two-winding transformers is tabulated in the two-winding transformer positive sequence data listing, and in the two-winding transformer zero sequence data listing. Transformer branches are ordered in ascending numerical or alphabetical order by from bus, and, for each from bus, in ascending order by to bus and circuit identifier.

The two-winding transformer data listings are in single entry format. Each two-winding transformer appears once in the listings: with the lower ordered bus (number or name) listed as the from bus.

In the positive sequence data listing, the column labeled ST indicates the current status of the transformer, with 0 indicating out-of-service, and 1 indicating in-service. The column labeled OR is the original status of the transformer whenever the current status of the transformer has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for transformers which have not been affected by breaker or switch operations. If the from bus is the bus to which Winding 1 of the transformer is connected, the column labeled W1 contains an F; otherwise, this column contains a T.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected two-winding transformers are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which transformers at

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candidate buses are to be reported; otherwise, all two-winding transformer branches at candidate buses, including transformers to non-candidate buses (i.e., ties), are reported.

Three-Winding Transformer Data

Data for three-winding transformers is tabulated in the three-winding transformer general data listing, in the three-winding transformer positive sequence data listing, and in the three-winding transformer zero sequence data listing. The three-winding transformer data listing is in single entry format. Each three-winding transformer appears once in the listing, with transformers ordered alphabetically by transformer name.

In the general data listing, the first column labeled ST indicates the current status of the transformer, with values of 0 through 4 as defined in Transformer Data. The column next to it labeled OR is the original status of the transformer whenever the current status of the transformer has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for transformers which have not been affected by breaker or switch operations. The remaining columns labeled ST and OR show the current and original status of the three two-winding transformers used to model the three-winding transformer, as derived from the status flag of the three-winding transformer, with zero indicating out-of-service and 1 indicating in-service. The magnetizing admittance is listed in the units indicted by CM.

In the positive sequence and zero sequence data listings, the column labeled ST indicates the current status of the transformer, with values of 0 through 4 as defined in Transformer Data. The column labeled OR is the original status of the transformer whenever the current status of the transformer has been changed as the result of operations of substation breakers and/or switches; it will differ from the ST value. The OR entry will be blank for transformers which have not been affected by breaker or switch operations.

In the positive sequence sequence data listing, measured impedances are listed in the units indicated by CZ, and tap settings (i.e., WINDVn) are listed in the units indicated by CW.

In the zero sequence data listing, nominal impedances are listed in the units indicated by CZ0, and grounding and neutral impedances are listed in the units indicated by CZG.

When reporting for a bus subsystem, the bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected three-winding transformers are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which transformers at candidate buses are to be reported; otherwise, all three-winding transformer branches at candidate buses, including transformers to non-candidate buses (i.e., ties), are reported.

Mutual Impedance Data

Data for zero sequence mutual couplings is tabulated in the mutual impedance data listing. The data printed for each mutual coupling consists of the from bus, to bus, and circuit identifier of the coupled branches, the mutual impedance and the geographic B factors (refer to Zero Sequence Mutual Impedance Data). As in the case of the mutual portion of the branch data listing described in Non-Transformer Branch Data, the sign of a branch's mutual impedance may be the negative of the value initially entered. Similarly, one or both sets of its B factors may be modified to express the location of the mutual relative to the bus listed as the from bus in the data listing.

Zero sequence mutuals are listed in single entry format.

When reporting for a bus subsystem, a mutual coupling is reported if either of the two branches satisfy the subsystem selection criteria. The bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected branches are candidates for reporting; otherwise, all buses are candidate buses. The owner selection criteria, if enabled, determines which branches involved in mutuals at candidate

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buses satisfy the subsystem selection criteria; otherwise, all branches involved in mutuals at candidate buses, including branches to non-candidate buses (i.e., ties), satisfy the subsystem selection criteria.

Additional Information
PSS®E GUI Users Guide, Listing Sequence Data *PSS®E Command Line Interface (CLI) Users Guide*, Sequence Data *PSS®E Application Program Interface (API)*, SQLI

5.44. Listing Sequence Data for a Bus

SQEX

Run Activity SQEX - GUI
Power Flow > List Data... *[List Data]* OPTION: Examine Powerflow / sequence data > Sequence data
Run Line Mode Activity SQEX - CLI
ACTIVITY? >>**SQEX** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The sequence data examination activity SQEX produces a tabulation of fault analysis data organized by bus. For each bus tabulated, its bus data is followed by the data associated with each network element that is connected to the bus for which sequence data is applicable.

The SQEX API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".

5.44.1. Operation of Activity SQEX

The user specifies either that the entire working case is to be processed, or that data items for equipment in a designated bus subsystem are to be tabulated.

The report of activity SQEX is directed to the *Report* device.

5.44.2. Listing Format

The blocks of output for buses are ordered as described in Section 4.9, "Subsystem Reporting". The output block for each bus tabulated includes bus, load, fixed shunt, switched shunt, synchronous machine, and induction machine quantities along with data for each transmission line, transformer, dc line, and FACTS device connected to the bus. The data for each equipment category is tabulated in a format similar to that of activity SQLI (refer to Section 5.43.2, "Listing Format").

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In the non-transformer branch data tabulation, any branch treated as a zero impedance line has a Z printed in the column labeled ZI at the end of the line; for other branches, this column is left blank.

In the two-winding transformer positive sequence data tabulation, the column labeled W1 contains an F if the bus being reported is the bus to which Winding 1 of the transformer is connected or a T if the bus listed as the to bus is the Winding 1 bus. All transformer impedances, admittances, and tap ratios are printed in the units indicated by the various input/output units codes(i.e., CZ, CM, CW, CZ and CZG).

For dc lines and FACTS devices, the column labeled MDC (MODE for FACTS devices) indicates the current control mode of the dc line or FACTS device. The column labeled OR is the original control mode of the dc line or FACTS device whenever its current control mode has been changed as the result of operations of substation breakers and/or switches; it will differ from the MDC or MODE value. The OR entry will be blank for dc lines and FACTS devices which have not been affected by breaker or switch operations.

For dc lines and FACTS devices, the pu powers as seen by the ac system are printed. For two-terminal dc lines, multi-terminal dc lines, and FACTS devices, the powers are printed as flowing from the ac system into the device at each of the ac buses to which it is connected; for VSC dc lines, the powers are printed as flowing into the ac system.

Additional Information
PSS®E GUI Users Guide, Listing Sequence Data for Buses and their Connected Equipment *PSS®E Command Line Interface (CLI) Users Guide*, Listing Sequence Data for Buses and their Connected Equipment *PSS®E Application Program Interface (API)*, SQEX

5.45. Creating a Saved Case File

SAVE

Run Activity SAVE - GUI
File > Save *[Save Network Data]* Case Data Tab
Run Line Mode Activity SAVE - CLI
ACTIVITY? >>**SAVE** ENTER SAVED CASE FILENAME: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The case saving activity SAVE stores the working case into a user-specified Saved Case File in a compressed format.

The SAVE API routine requires as an input the name of the Saved Case File to be used.

If an asterisk (*) is specified as the name of the Saved Case File, the last Saved Case File accessed by either activity SAVE or activity CASE in the current execution of PSS®E is used.

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Refer to Section 2.4.2, "Specifying Filenames" for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity SAVE.

Along with the working case, several of the PSS®E run-time option settings that are active at the time activity SAVE is selected are preserved in the Saved Case File (refer to Saved Case Specific Option Settings).

Additional Information
PSS®E GUI Users Guide, Creating a Saved Case File *PSS®E Command Line Interface (CLI) Users Guide*, Creating a Saved Case File *PSS®E Application Program Interface (API)*, SAVE *See also:* Section 5.1, "Retrieving a Power Flow Saved Case File"

5.46. Bus Renumbering

BSNM

Run Activity BSNM - GUI
Power Flow > Renumber Buses > By Bus Number... By Bus Name... By Packing... By Subsystem... *[Renumber Buses by Bus Number]* *[Renumber Buses by Bus Name]* *[Renumber Buses by Bus Packing]* *[Renumber Buses by Subsystem]*
Run Line Mode Activity BSNM - CLI
ACTIVITY? >>**BSNM** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
None

The bus renumbering activity BSNM enables the user to change the bus numbers of specified network buses in the working case and retain a tabulation, normally in file form, of the bus number changes made.

The BSNM API routine includes the following among its inputs:

- A flag indicating the method to be used to renumber buses and/or the processing phase to be performed during this call of the BSNM API routine.
- A designation of the subsystem in which bus numbers are to modified if the renumbering method is subsystem renumbering; refer to Section 4.8, "Subsystem Selection".
- Three integer data items; their interpretation and usage is dependent on the value specified for the method flag during this call of the BSNM API routine.
- A character data item that specifies an extended bus name if the renumbering method is Bus Name to Bus Number Translation.

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- A character data item that specifies a translation input file if the renumbering method is a translation method (Bus Number to Bus Number Translation or Bus Name to Bus Number Translation) with input taken from a file.
- A code indicating the disposition of the Output Tabulation of bus number changes made.
- The name of the Bus Renumbering Translation File for recording the bus number changes made; used if a data file is selected for the output tabulation.

5.46.1. Renumbering Methods

Several bus renumbering methods are available in activity BSNM. The appropriate method to use in a given bus renumbering task is usually dictated by the scope and purpose of the renumbering task.

The following paragraphs describe the use of the various bus renumbering methods.

All Buses Without Area Blocking

In this method, activity BSNM renumbers all buses in the working case. The user specifies a starting bus number, and the bus numbers in the working case are resequenced by increments of one, starting at the specified bus number. The new bus number sequence corresponds to the original bus number sequence when the numbers *output* option is in effect, or to bus name alphabetical order when the names *output* option is in effect (see Saved Case Specific Option Settings).

All Buses With Area Blocking

In this method, activity BSNM renumbers all buses in the working case. The user specifies a bus number range for each area that has at least one bus assigned to it. For each area, the bus numbers in the working case are resequenced by increments of one, starting at the area's specified starting bus number. Within each area, the new bus number sequence corresponds to either the original bus number sequence or bus name sequence of area buses, depending on the bus *output* option in effect (see Saved Case Specific Option Settings).

When specifying an area's bus number range, the following conditions must be satisfied:

- The starting bus number must be greater than zero.
- The ending bus number must not exceed 999,997.
- The ending bus number must be greater than the starting bus number.
- The bus number range must be large enough to accommodate all of the buses assigned to the area.

No overlap of bus number ranges is allowed.

Subsystem

In this method, activity BSNM renumbers all buses in a specified bus subsystem. Two subsystem renumbering methods are available: the Bus Number Range Method, and the Bus Number Offset Method.

Bus Number Range Method

In this method, activity BSNM reassigns bus numbers of the specified subsystem so that they all fall within a designated range. The user specifies the starting and ending bus numbers for the range into which the subsystem is to be renumbered.

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When specifying the bus number range, the following conditions must be satisfied:

- The starting bus number must be greater than zero.
- The ending bus number must not exceed 999,997.
- The ending bus number must be greater than the starting bus number.
- The bus number range must be large enough to accommodate all of the buses in the subsystem.

If any violations are detected, an appropriate message is printed and no bus number changes are made for the specified subsystem. Otherwise, the subsystem buses are renumbered. Any buses with numbers already in the specified range are left unchanged.

Bus Number Offset Method

In this method, activity BSNM reassigns bus numbers of the specified subsystem by adding a constant, either positive or negative, to the bus numbers of all buses in the specified subsystem. If, for any bus, the resulting bus number is invalid or already assigned to another bus, an appropriate message is printed and no bus number changes are made for the specified subsystem. Otherwise, the subsystem buses are renumbered.

Bus Number to Bus Number Translation

In this method, activity BSNM changes the number of a specified bus to a designated new bus number. The bus that is to have its number changed is designated by its present bus number.

A bus number pair may be specified as an input to the BSNM API routine. Alternatively, a Bus Renumbering Translation File containing bus number pairs may be specified to the BSNM API routine; each record of the file must be in the form:

```
current bus number   new bus number
```

In either case, when specifying an old number, new number pair, the following conditions must be satisfied:

- The first bus specified must be in the working case.
- The new bus number must be a valid bus number (i.e., between 1 and 999,997).
- The new bus number must not already be assigned to a bus in the working case.

If any violations are detected in processing a bus number pair, an appropriate message is printed and the corresponding bus number change is not implemented. Otherwise, the bus number change is made. Activity BSNM then processes the next bus number pair.

Bus Name to Bus Number Translation

In this method, activity BSNM changes the number of a specified bus to a designated new bus number. The bus that is to have its number changed is designated by its extended bus name.

A bus name, bus number pair may be specified as an input to the BSNM API routine. Alternatively, a translation input file containing bus name, number pairs may be specified to the BSNM API routine; each record of the file must be in the form:

```
'extended bus name'   new bus number
```

In either case, when specifying an bus name, new number pair, the following conditions must be satisfied:

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- The first bus specified must be in the working case.
- The new bus number must be a valid bus number (i.e., between 1 and 999,997).
- The new bus number must not already be assigned to a bus in the working case.

If any violations are detected in processing a bus name, number pair, an appropriate message is printed and the corresponding bus number change is not implemented. Otherwise, the bus number change is made. Activity BSNM then processes the next bus name, number pair.

Bus Number Packing

In this method, activity BSNM changes the numbers of buses in a specified bus number range so that all bus numbers in the range are packed into the low end of the range.

When specifying the bus number range, the following conditions must be satisfied:

- The starting bus number must be greater than zero.
- The ending bus number must not exceed 999,997.
- The ending bus number must be greater than the starting bus number.

If any violations are detected, an appropriate message is printed and no bus number changes are made for the specified range. Otherwise, the bus numbers falling within the specified range are packed sequentially beginning with the starting bus number. Thus, the new bus number sequence corresponds to the original bus number sequence, with gaps in bus number usage eliminated.

5.46.2. Output Tabulation

Activity BSNM provides for a tabulation documenting those buses that have been assigned new bus numbers. This tabulation consists of a series of records in the following format:

```
old bus number   new bus number / bus name and
base voltage
```

The following options are available for the handling of this output tabulation:

- Write these records to a Bus Renumbering Translation File.
- Write these records to the *Report* device.
- Suppress the writing of these records.

Writing these records to a Bus Renumbering Translation File is useful for the situation where several power flow Saved Cases of substantially identical network topology but, for example, different loading levels are to be renumbered. This may be accomplished by directing the output of activity BSNM to a file for the first of the group of cases. That file may then be used as input in subsequent executions of activity BSNM for the remaining cases using the Bus Number to Bus Number Translation renumbering method. This Bus Renumbering Translation File is also directly usable by activity RNFI.

Following the completion of processing, the following message is printed at the *Progress* device by activity BSNM:

```
n BUS NUMBER CHANGES MADE FOR SPECIFIED SUBSYSTEM
```


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If enabled, the tabulation of the bus number changes in Bus Renumbering Translation File format is written just before leaving activity BSNM. Thus, assuming Bus Number to Bus Number Translation, a series of changes such as:

change bus 101 to bus 154 then change bus 154 to bus 101

would be reported as two changes in the *Progress* output, but the output tabulation would show no bus number translations because the effect of the second change was to cancel out the first change.

5.46.3. Examples of Activity BSNM

Suppose that Bus Number Packing is performed on the range starting at 101 and ending at 160 for the *savnw.sav* Saved Case File. The original and resulting new bus numbers are shown in Figure 5.19, "Packing Bus Numbers". It can be seen that the original bus range from 101 to 154 has been packed to a range from 101 to 106.

Bus Number	Bus Name
101	NUC-A
102	NUC-B
151	NUCPANT
152	MID500
153	MID230
154	DOWNTN
201	HYDRO
202	EAST500
203	EAST230

Bus Number	Bus Name
101	NUC-A
102	NUC-B
103	NUCPANT
104	MID500
105	MID230
106	DOWNTN
201	HYDRO
202	EAST500
203	EAST230

Figure 5.19. Packing Bus Numbers

Figure 5.20, "Progress Output Indicating Number of Buses Modified" shows the message at the *Progress* device indicating how many buses have been modified, along with the tabulation of bus number changes at the *Report* device. The latter tabulation could have been sent to a specified output file.

```
 4 BUS NUMBER CHANGES MADE FOR SPECIFIED SUBSYSTEM

 ------------------------------------------------
     151     103   / NUCPANT      500.00
     152     104   / MID500       500.00
     153     105   / MID230       230.00
     154     106   / DOWNTN       230.00
```

Figure 5.20. Progress Output Indicating Number of Buses Modified

Figure 5.21, "Result of Block Renumbering by Areas" shows the *Progress* and *Report* device output resulting from bus renumbering using the All Buses With Area Blocking method on the *savnw.sav* Saved Case File. Each of the three areas in the case is given a new bus number range.

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```
 23 BUS NUMBER CHANGES MADE FOR SPECIFIED SUBSYSTEM

 ------------------------------------------------

      101       1  / NUC-A         21.600
      102       2  / NUC-B         21.600
      151       3  / NUCPANT       500.00
      152       4  / MID500        500.00
      153       5  / MID230        230.00
      154       6  / DOWNTN        230.00
      201     100  / HYDRO         500.00
      202     101  / EAST500       500.00
      203     102  / EAST230       230.00
      204     103  / SUB500        500.00
      205     104  / SUB230        230.00
      206     105  / URBGEN        18.000
      211     106  / HYDRO_G       20.000
     3001     200  / MINE          230.00
     3002     201  / E. MINE       500.00
     3003     202  / S. MINE       230.00
     3004     203  / WEST          500.00
     3005     204  / WEST          230.00
     3006     205  / UPTOWN        230.00
     3007     206  / RURAL         230.00
     3008     207  / CATDOG        230.00
     3011     208  / MINE_G        13.800
     3018     209  / CATDOG_G      13.800
```

Figure 5.21. Result of Block Renumbering by Areas

Additional Information
PSS®E GUI Users Guide, Renumbering Buses *PSS®E Command Line Interface (CLI) Users Guide*, Saving Renumbered Bus Data to a File *PSS®E Application Program Interface (API)*, BSNM

5.47. Saving the Working Case in an IEEE Common Format File

RWCM

Run Activity RWCM - GUI
File > Save... *[Save Network Data]* IEEE Format Power Flow Data Tab
Run Line Mode Activity RWCM - CLI
ACTIVITY? >>**RWCM** ENTER OUTPUT DEVICE CODE:

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Interrupt Control Codes
AB

The IEEE Common Format output activity RWCM writes the working case as a file in IEEE common tape format data records. This format is described in "Common Format for Exchange of Solved Load Flow Data," *IEEE Transactions on Power Apparatus and Systems*, Vol. PAS-92, November/December 1973, pp. 1916-1925.

The RWCM API routine includes the following among its inputs:

- A code indicating the handling of dc lines, FACTS devices, and GNE devices in the IEEE Common Format data records.
- A code indicating the disposition of the IEEE Common Format data records.
- The name of the file to which the IEEE Common Format data records are to be written; used if a data file is selected as the destination for the records written by activity RWCM.

Refer to Section 2.4.2, "Specifying Filenames" for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity RWCM.

If any unblocked dc lines, out-of-service FACTS devices, or out-of-service GNE devices are present in the working case, the user must specify whether to ignore dc lines, FACTS devices, and GNE devices, or add their apparent ac powers to the loads. If dc lines, FACTS devices, and GNE devices are ignored, their representation should be described in the transmittal letter to the party receiving the data file. If power is added to loads, all unblocked dc lines, in-service FACTS devices, and in-service GNE devices are accounted for in the common format data records by treating the powers flowing into them at their ac network buses as complex bus loads.

If more than 9,999 buses are present in the working case, or any buses have bus numbers greater than 9999, an appropriate message is printed at the *Alert* device and activity RWCM is terminated.

Buses with a type code of 4 or greater and out-of-service branches are not included in the output file. Any buses in the working case residing in an area for which the number is greater than 99 are assigned area number 99 in the output file.

For any bus with a switched shunt element, the switched shunt admittance is added to the fixed bus shunt. Line connected shunts are added to the bus shunt of the appropriate buses. In-service constant current load is added to the constant power load and in-service constant admittance load is added to the bus shunt. Powers flowing into in-service induction machines are added to the constant power load.

Branch circuit identifiers may be modified in the output file built by activity RWCM. Circuit numbers are assigned for all circuits between pairs of buses such that they are numbered starting with circuit number 1 and assigned sequential circuit numbers up to 9. If more than 9 parallel circuits exist between any pair of buses, appropriate alarm messages are printed and circuits 10, 11, and so on are assigned and reported. Note that the leading digit is placed in a field that is defined to be blank in the IEEE format, so that other programs reading the data file might have difficulty. It is suggested that parallel circuits be combined so that no more than nine circuits exist between any pair of buses.

For each three-winding transformer contained in the working case, activity RWCM assigns an available four digit bus number to the star point bus, and produces a corresponding bus record. Three two-winding transformer branch records are also produced, one for each winding of the three-winding transformer.

Additional Information
PSS®E GUI Users Guide, Saving the Working Case in an IEEE Common Format File

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Saving the Working Case in an IEEE Common Format File *PSS®E Application Program Interface (API)*, RWCM

5.48. Saving Machine Impedance Data

RWMA

Run Activity RWMA - GUI
File > Save... *[Save Network Data]* Machine Impedance Data Tab
Run Line Mode Activity RWMA - CLI
ACTIVITY? >>**RWMA** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The machine impedance data output activity RWMA writes out machine parametric data from the working case in the form of a Machine Impedance Data File (refer to Section 5.4.1, "Machine Impedance Data File Contents"). That is, when directed to a file, the output of activity RWMA is in a format suitable for input to activity MCRE. A data file of the form written by activity RWMA is also required by the IEEE stability data output program, CMDYRE (refer to *Additional Resources for PSS® E*, CMDYRE).

The RWMA API routine includes the following among its inputs:

- A designation of the subsystem from which the machine parametric data is to be written; refer to Section 4.8, "Subsystem Selection".
- A code indicating whether to include or exclude data records for out-of-service machines.
- A code indicating the option to use in calculating active power split fractions.
- A code indicating the option to use in calculating reactive power split fractions.
- A code indicating the disposition of the machine impedance data records.
- The name of the Machine Impedance Data File to which the machine impedance data records are to be written; used if a data file is selected as the destination for the records written by activity RWMA.

5.48.1. Operation of Activity RWMA

The user specifies either that the entire working case is to be processed, or that machine impedance data in a designated bus subsystem are to be processed.

When a bus subsystem is being processed and ownership is included among the selection criteria, the owners to which machines (rather than buses) are assigned are used in defining the subsystem of machines to

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be processed. The bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected machines are candidates for processing; otherwise, all buses are candidate buses. All machines at candidate buses that are wholly or partly owned by any of the owners specified are processed (refer to Generator Data).

Activity RWMA provides options for calculating the power split fractions FP and FQ (refer to Section 5.4.1, "Machine Impedance Data File Contents"):

- for calculating P fractions:
- PGEN (default)
- MBASE
- PMAX
- for calculating Q fractions:
- QGEN (default)
- MBASE
- QMAX

Details on the calculation of the active and reactive power split fractions are provided in Split Fractions.

Refer to Section 2.4.2, "Specifying Filenames" for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity RWMA.

Data records are produced in ascending bus number order by activity RWMA. The extended bus name (name plus base voltage) is included at the end of each data record. While these are ignored by activity MCRE, they are *required* by the auxiliary program CMDYRE.

5.48.2. Application Notes

The inclusion or omission of out-of-service machines and the proper selection of the quantities used to derive the power split fractions is dependent upon the application at hand.

Transfer Machine Data to Another Case

One use of the Machine Impedance Data File is to introduce generator impedance data, as required for switching studies, fault analysis, and dynamic simulation work, into a power flow case in which the plant modeling terminates at the plant bus. This is handled by activity MCRE, which also provides for the introduction of multiple machine representation at a plant and the splitting of total plant power output and power limits among its machines. The Machine Impedance Data File as built by activity RWMA may be used to transfer this level of plant modeling from one power flow case to another that models essentially the same system but in which the details of the individual machines have not as yet been introduced. Refer to Section 5.4.3, "Application Notes" for additional application notes on the use of the Machine Impedance Data File by activity MCRE.

When transferring machine data from one case to another, it is usually appropriate to include out-of-service machines and to use either generator powers or machine bases to form the active power split fractions. In setting the reactive power split fractions, either the machine base or QMAX should be selected rather than the machine reactive power. This avoids a potential problem if, for example, in a multimachine plant with differently sized units, the lower reactive power limits are all zero and all machines are at their low var limits.

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The reactive power split fractions would then be set equal for these machines, and, when subsequently read by activity MCRE, reactive power limits would be set incorrectly.

Stability Data in IEEE Format

A Machine Impedance Data File, along with its corresponding Dynamics Data File (refer to Section 16.1.1, "Dynamics Model Raw Data File Contents"), is required by the auxiliary program CMDYRE to construct a stability data file in IEEE format. This format is described in "Procedures for the Exchange of Power Plant and Load Data for Synchronous Stability Studies", *IEEE Transactions on Power Apparatus and Systems*, Vol. PAS-100, No. 7 July 1981, pp. 3229-3245.

When the file created by activity RWMA is to be used as input to the auxiliary program CMDYRE, it is usually appropriate to include only in-service machines and to use generator powers to calculate the power split fractions.

Machine Status Considerations

In determining machine status, both the bus type code and the machine status flag are considered. A machine is treated as in-service if its status flag is 1 *and* the bus type code is 2 or 3.

Split Fractions

The split fractions of a machine are calculated as the ratio of the selected machine quantity to the sum of those quantities of all in-service machines being processed at the bus. If this denominator is zero and records for out-of-service machines are being included, the denominator is taken as the sum of the appropriate quantity of all machines being processed at the bus. If it is still zero, the split fraction is set to the reciprocal of the number of in-service machines being processed at the bus; if all machines being processed at the bus are out-of-service, the split fraction is set to the reciprocal of the number of such machines being processed at the bus.

Using this approach, then, the sum of the split fractions of the in-service machines being processed at a bus, exclusive of numerical round off, is normally unity. The only situation in which this is violated is in the case where out-of-service machines are included, both in-service and out-of-service machines are present and being processed at a bus, and the sum of the split fraction quantity is zero for the in-service machines and non-zero for the out-of-service machines.

When generator reactive powers are used in determining the reactive power split fractions, the working case must be solved to an acceptable mismatch level.

Additional Information
PSS®E GUI Users Guide, Saving Machine Impedance Data *PSS®E Command Line Interface (CLI) Users Guide*, Saving Machine Impedance Data *PSS®E Application Program Interface (API)*, RWMA

5.49. Creating a Power Flow Raw Data File

RAWD

Run Activity RAWD - GUI
File > Save...

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Run Activity RAWD - GUI
[Save Network Data] Power Flow Raw Data Tab
Run Line Mode Activity RAWD - CLI
ACTIVITY? >>**RAWD** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The Raw Data File output activity RAWD writes the working case in the form of a Power Flow Raw Data File (refer to Section 5.2.1, "Power Flow Raw Data File Contents"). That is, when directed to a file, the data records written by activity RAWD are in a format suitable for input to activity READ.

The RAWD_2 API routine includes the following among its inputs:

- A designation of the subsystem for which the power flow data records are to be written; refer to Section 4.8, "Subsystem Selection".
- A code indicating whether to include or exclude data records for out-of-service buses (i.e., Type 4 buses).
- A code indicating whether to include or exclude data records for out-of-service branches.
- A code indicating whether to include or exclude data records for equipment within the specified subsystem.
- A code indicating whether to include or exclude data records for tie branches from the specified subsystem.
- A code indicating the option to be used for including load data records.
- A code indicating the use of either bus numbers or extended bus names as bus identifiers.
- A code indicating the intended use of the Power Flow Raw Data file.
- A code indicating the disposition of the power flow data records: either to a file or to the *Progress* device.
- The name of the Power Flow Raw Data File to which the data records are to be written; used if a data file is selected as the destination for the records written by activity RAWD.

5.49.1. Operation of Activity RAWD

The user specifies either that the entire working case is to be processed, or that data records for a designated bus subsystem are to be written.

When a bus subsystem of the working case is specified for processing by activity RAWD, the user has the ability to select as output for that subsystem one of the following:

- All buses in the specified subsystem, all branches having all of its buses in the specified subsystem, and all bus connected equipment connected to buses in the specified subsystem. Branches include non-transformer branches, transformers, dc lines, and FACTS and GNE devices with a series element. Bus connected equipment include fixed shunts, machines, switched shunts, and FACTS and GNE devices without a

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series element. When area, owner and/or zone are among the subsystem selection criteria, options for the handling of loads is described next; otherwise, loads are handled in the same manner as other bus connected equipment.

- All branches having one, but not all, of its buses in the specified subsystem.
- All buses in the specified subsystem, all branches having at least one of its buses in the specified subsystem, and all bus connected equipment connected to buses in the specified subsystem.

When a bus subsystem of the working case is specified for processing by activity RAWD, and area, owner and/or zone are among the subsystem selection criteria, one of the following options is selected for the handling of loads:

- Include load records for all loads at subsystem buses.
- Include load records for subsystem loads at all buses. In this case, the area, owner and/or zone assignments, as appropriate, of a load is used in determining if its load record is output; the area, owner and/or zone assignments of the bus to which the load is connected are not considered. When this option is selected, records for loads that are connected to buses outside of the subsystem may be included in the output of activity RAWD, and records for some loads connected to subsystem buses may be omitted.
- Include load records for all loads at subsystem buses *and* for subsystem loads at non-subsystem buses.

When the entire working case is being processed and the option to include records for subsystem tie branches is specified, activity RAWD produces an output tabulation consisting only of branches between areas. Thus, no bus, load, fixed shunt, generator, switched shunt, area, zone, owner, or interarea transfer data is output, and the only branch, transformer, transformer impedance correction table, dc line, FACTS device, and GNE device data included in the output is for branches with terminals connected to buses in different areas.

The intended use of the Power Flow Raw Data File that is being written governs the appropriate form of the Case Identification Data records at the beginning of the file. One of the following intended uses must be selected:

- By activity READ with initialization of the working case (i.e., with IC = 0 on the first data record).
- By activity READ to add data to the working case (i.e., with IC = 1 on the first data record).
- By activity RDCH (i.e., with the case identification data records omitted), and all other data categories included.
- By activity RDCH (i.e., with the case identification data records omitted), with the substation data category excluded, and all other data categories included.
- By activity RDCH (i.e., with the case identification data records omitted), with only the substation data category included, and all other data categories excluded.

Refer to Section 2.4.2, "Specifying Filenames" for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity RAWD.

5.49.2. Application Notes

Each data category of the output of activity RAWD is tabulated in ascending bus number (or bus name if the NAME bus identifier option is enabled) order, except for:

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- dc line data, which is reported in ascending dc line name order for two-terminal, multi-terminal, and VSC dc lines.
- transformer impedance correction table data, which is reported in table number order.
- area interchange data and area transactions data, which are reported in ascending area number order.
- zone data, which is reported in ascending zone number order.
- owner data, which is reported in ascending owner number order.
- FACTS device data, which is reported in ascending FACTS device name order.
- GNE device data, which is reported in ascending GNE device name order.

While switched shunt data is, for the most part, reported in ascending bus number or name order, those switched shunts that control the admittance setting of other switched shunts are grouped at the end of the switched shunt data records.

If a branch data record is reported for a branch that is a member of a multi-section line grouping, then that multi-section line has a multi-section line grouping data record included in the output tabulation.

A transformer impedance correction table data record is included in the output of activity RAWD for any table referenced in the transformer data record block of at least one transformer where data is included in the output of activity RAWD.

When the data records for the entire working case are being written, an area interchange data record is included in the output of activity RAWD for all areas that satisfy the following:

- at least one bus, load, induction machine, or dc bus record for equipment assigned to the area is written

  and

- the area does not have default area interchange data assigned to it (refer to Area Interchange Data).

When a subsystem of the working case is processed, an area interchange data record is written for all areas that satisfy the following:

- at least one bus or load record for equipment assigned to the area is written

  and

- the area does not have default area interchange data assigned to it.

Similarly, a zone data record is written for any zone for which:

- at least one bus, load, or dc bus record for equipment assigned to the zone is written

  and

- the zone has a non-blank zone name (refer to Zone Data).

When the entire working case is processed, an owner data record is written for any owner for which:

- at least one bus, load, synchronous machine, induction machine, branch, dc bus, FACTS device, VSC dc line, or GNE device record is written

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and

- the owner has a non-blank owner name (refer to Owner Data).

When a subsystem of the working case is processed, an owner data record is written for any owner for which:

- at least one bus, load, machine, branch, dc bus, or VSC dc line record is written

and

- the owner has a non-blank owner name.

When the data records for the entire working case are being written, an interarea transfer data record is included in the output of activity RAWD for any transaction for which at least one bus, load, induction machine, or dc bus record from each of the two areas is written. When a subsystem of the working case is processed, an interarea transfer data record is written for any transaction for which at least one bus, load, or dc bus record from each of the two areas is written.

While a PSS®E Saved Case File is the preferred mechanism for preserving and archiving network data, a PSS®E Power Flow Raw Data File facilitates the merging of network data with another power flow case (see Merging Cases).

Activity RAWD may be used to set up a Power Flow Raw Data File for use by activity READ in the data processing phases of advanced network equivalent construction (refer to Section 5.2.5, "Reading Power Flow Raw Data Files Created by Previous Releases of PSS®E"). When used in this mode, disconnected buses and out-of-service branches should normally not be included in the output file created by activity RAWD.

Additional Information
PSS®E GUI Users Guide, *Creating a Power Flow Raw Data File* *PSS®E Command Line Interface (CLI) Users Guide*, Creating a Power Flow Raw Data File

5.50. Creating a Capability Curve Data File

RWGC

Run Activity RWGC - GUI
File > Save As..... *[Save Network Data]* Machine Capability Curve Data Tab
Interrupt Control Codes
AB

The Machine Capability Curve Data File output activity RWGC writes the capability curve data contained in the working case in the form of a Machine Capability Curve Data File (refer to Section 5.5.1, "Capability Curve Data File Contents"). That is, when directed to a file, the output of activity RWGC is in a format suitable for input to activity REGC.

The WriteCapCurves API routine includes the following among its inputs:

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- A designation of the subsystem for which the capability curve data records are to be written; refer to Section 4.8, "Subsystem Selection".
- A code indicating whether to include or exclude data records for machines at subsystem buses that are outside of the subsystem.
- A code indicating whether to include or exclude data records for out-of-service machines.
- A code indicating the IC value on the first data record (i.e., indicating the intended use of the Machine Capability Curve Data File).
- A code indicating the disposition of the capability curve data records: either to a file or to the *Progress* device.
- The name of the Machine Capability Curve Data File to which the data records are to be written; used if a data file is selected as the destination for the records written by activity RWGC.

5.50.1. Operation of Activity RWGC

The user specifies either that the entire working case is to be processed, or that data records for a designated bus subsystem are to be written.

When a bus subsystem is being processed and ownership is included among the selection criteria, the owners to which machines (rather than buses) are assigned are used in defining the subsystem of machines to be processed. The bus, area, zone, and/or base voltage selection criteria, if enabled, determine those buses whose connected machines are candidates for processing; otherwise, all buses are candidate buses. All machines at candidate buses that are wholly or partly owned by any of the owners specified are processed (refer to Generator Data).

Refer to Section 2.4.2, "Specifying Filenames" for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity RWGC.

5.50.2. Application Notes

The Machine Capability Curve Data File written by activity RWGC always has the IC value on the first data record set to the value specified as an input to the WriteCapCurves API routine. When using such a Machine Capability Curve Data File, it is the responsibility of the user to ensure that this flag is set to the appropriate value for the application at hand. This data value may be changed with a text editor.

The inclusion or omission of out-of-service machines is dependent upon the application at hand.

Additional Information
PSS®E Application Program Interface (API), WriteCapCurves

5.51. Creating a Sequence Data File

RWSQ

Run Activity RWSQ - GUI
File > Save...

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Run Activity RWSQ - GUI
[Save Network Data] Sequence Data Tab
Run Line Mode Activity RWSQ - CLI
ACTIVITY? >>**RWSQ** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The Sequence Data File output activity RWSQ writes the sequence data contained in the working case in the form of a Sequence Data File (refer to Section 5.6.1, "Sequence Data File Contents"). That is, when directed to a file, the output of activity RWSQ is in a format suitable for input to activity RESQ.

The RWSQ_2 API routine includes the following among its inputs:

- A designation of the subsystem for which the sequence data records are to be written; refer to Section 4.8, "Subsystem Selection".
- A code indicating whether to include or exclude data records for out-of-service buses (i.e., Type 4 buses).
- A code indicating whether to include or exclude data records for out-of-service branches.
- A code indicating whether to include or exclude data records for equipment within the specified subsystem.
- A code indicating whether to include or exclude data records for tie branches from the specified subsystem.
- A code indicating the intended use of the Sequence Data File.
- A code indicating the disposition of the sequence data records: either to a file or to the *Progress* device.
- The name of the Sequence Data File to which the data records are to be written; used if a data file is selected as the destination for the records written by activity RWSQ.

5.51.1. Operation of Activity RWSQ

The user specifies either that the entire working case is to be processed, or that data records for a designated bus subsystem are to be written.

When a bus subsystem of the working case is specified for processing by activity RWSQ, the user has the ability to select as output for that subsystem one of the following:

- All buses in the specified subsystem, all branches having all of its buses in the specified subsystem, and all bus connected equipment connected to buses in the specified subsystem.
- All branches having one, but not all, of its buses in the specified subsystem.
- All buses in the specified subsystem, all branches having at least one of its buses in the specified subsystem, and all bus connected equipment connected to buses in the specified subsystem.

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When the entire working case is being processed and the option to include records for subsystem tie branches is specified, activity RWSQ produces an output tabulation consisting only of branches between areas. Thus, no generator, load, switched shunt, fixed shunt, and induction machine data is output, and the only branch, mutual impedance, and transformer data included in the output is for branches with terminals connected to buses in different areas.

Refer to Section 2.4.2, "Specifying Filenames" for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity RWSQ.

5.51.2. Application Notes

The Sequence Data File written by activity RWSQ always has the IC value on the first data record set to the value specified as an input to the RWSQ_2 API routine. When using such a Sequence Data File, it is the responsibility of the user to ensure that this flag is set to the appropriate value for the application at hand. This data value may be changed with a text editor.

Each data category of the output of activity RWSQ is tabulated in ascending bus number order. Transformer records are written with the winding 1 bus as the "from" bus.

Additional Information
PSS®E GUI Users Guide, *Creating a Sequence Data File* *PSS®E Command Line Interface (CLI) Users Guide*, Creating a Sequence Data File *PSS®E Application Program Interface (API)*, RWSQ_2

5.52. Creating a Transactions Raw Data File

RWMM

Run Activity RWMM - GUI
File > Save... *[Save Network Data]* Transaction Data Tab
Run Line Mode Activity RWMM - CLI
ACTIVITY? >>**RWMM** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>
Interrupt Control Codes
AB

The Transactions Raw Data File output activity RWMM writes the transactions data contained in working memory in the form of a Transactions Raw Data File (refer to Section 5.7.1, "Transactions Raw Data File Contents"). That is, when directed to a file, the data records written by activity RWMM are in a format suitable for input to activity REMM.

The RWMM API routine includes the following among its inputs:

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- A code indicating the disposition of the transactions data records: either to a file or to the *Progress* device.
- The name of the Transactions Raw Data File to which the data records are to be written; used if a data file is selected as the destination for the records written by activity RWMM.

5.52.1. Operation of Activity RWMM

Activity RWMM produces data records for all transaction data contained in working memory.

5.52.2. Application Notes

The data records written by activity RWMM include the initial record specifying the current state of the participation factor flag, IPF. In line mode, it is the responsibility of the user to ensure that this record is removed when the file is to be used as input to activity REMM run with the suffix CH.

Additional Information
PSS®E GUI Users Guide, Creating a Transactions Raw Data File
PSS®E Command Line Interface (CLI) Users Guide, Creating a Transactions Raw Data File
PSS®E Application Program Interface (API), RWMM

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Chapter 6
Power Flow Solution Activity Descriptions

6.1. About Power Flow Calculations

The most common power system network simulation is the power flow calculation, described in the following question:

Given the load power consumption at all buses of the electric power system and the generator power production at each power plant, what is the power flow in each line and transformer of the interconnecting network?

Calculated answers to this question are the basic means by which the power system is engineered to serve its load. The power system must operate without overloading transmission lines or transformers, stay within acceptable voltage limits at all buses, and maintain generator reactive power outputs between acceptable limits.

The power flow problem pertains to balanced steady-state operation of the power system. Because it considers balanced operation in which all negative- and zero-sequence voltages are zero, the power flow calculation deals with the *positive-sequence* model of all system components.

The following are the basic known input data for power flow calculations:

- Transmission line impedances and charging admittances.
- Transformer impedances and tap ratios.
- Admittances of shunt-connected devices such as static capacitors and reactors.
- Load-power consumption at each bus of the system.
- Real-power output of each generator or generating plant.
- Either voltage magnitude at each generator bus or reactive power output of each generating plant.
- Maximum and minimum reactive power output capability of each generating plant.

The quantities to be determined are

- The magnitude of the voltage at every bus where this is not specified in the input data.
- The phase of the voltage at every bus, except swing buses.
- The reactive power output of each plant for which it is not specified.
- The real power, reactive power, and current flow in each transmission line and transformer.

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The power flow calculation is a network solution problem. The network of transmission lines and transformers is described by the linear algebraic equation:

$$I_n = Y_{nn} V_n \quad (6.1)$$

where:

$I_{n =}$	Vector of positive-sequence currents flowing into the network at its nodes (buses).
$V_{n =}$	Vector of positive-sequence voltages at the network nodes (buses).
$Y_{nn =}$	Network admittance matrix.

If either I_n or V_n is known, the power flow calculation is straightforward. In practice, neither I_n nor V_n is known and the task of the power flow program is to devise successive trials of both I_n and V_n such that they satisfy both Equation 6.1 and all the load and generation conditions specified in the problem data. After V_n has been determined, all individual transmission line and transformer flows can be obtained directly from the individual component equations.

This chapter contains operational descriptions of the PSS®E power flow solution activities. Those network solution activities used in linear network analysis, equivalent construction, switching studies, and fault analysis are described in subsequent chapters. Interface references are provided, with links to the *PSS®E GUI Users Guide* and the *PSS®E Command Line Interface (CLI) Users Guide*, where appropriate.

6.2. Power Flow Solution Methods

PSS®E includes five power flow solution activities, each of which operates on the bus voltage estimates in the working case to attempt to bring them to a solution of Kirchhoff's laws. Each activity makes successive adjustments to the bus voltages in accordance with a different iterative scheme. The iterative schemes are listed in Table 6.1, "Available Iteration Schemes in PSS®E" along with the activity names familiar to users of previous versions of PSS®E.

Table 6.1. Available Iteration Schemes in PSS®E

Iterative Scheme	Activity Name
Gauss-Seidel	SOLV
Modified Gauss-Seidel handles series capacitors	MSLV
Full Newton-Raphson	FNSL
Decoupled Newton-Raphson	NSOL
Fixed-Slope Decoupled Newton-Raphson	FDNS

Because power flow convergence properties are dependent upon network and load attributes, each of the five iteration methods has its own strengths and weaknesses. The most significant strengths and weaknesses are summarized in Table Power Flow Solution Activities—Selection Guide.

Any of these five methods can fail to converge on the solution of some problems. It is rare, however, to find a problem that has a solution, but the voltage solution vector cannot be found by the application of one or more of the five methods. There are many problems that are difficult or impossible to solve with a single iterative method but which can be solved readily by the successive application of more than one method.

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A general guide to the selection of an iterative method follows:

- The Gauss-Seidel methods are generally tolerant of power system operating conditions involving poor voltage distributions and difficulties with generator reactive power allocation, but do not converge well in situations where real power transfers are close to the limits of the system.
- The Newton-Raphson methods are generally tolerant of power system situations in which there are difficulties in transferring real power, but are prone to failure if there are difficulties in the allocation of generator reactive power output or if the solution has a particularly bad voltage magnitude profile.
- The Gauss-Seidel methods are quite tolerant of poor starting voltage estimates, but converge slowly as the voltage estimate gets close to the true solution.
- The Newton-Raphson methods are prone to failure if given a poor starting voltage estimate, but are usually superior to the Gauss-Seidel methods after the voltage solution has been brought close to the true solution.

Experimentation is needed to determine the optimum combination of iterative methods for each particular power system model.

Experience suggests the following as the most advantageous approach to new power flow cases where the specific characteristics have yet to be learned:

- Initialize all voltages to either unity amplitude, or to scheduled amplitude if given, and initialize all phase angles to zero. (This step is referred to as a *flat* start.)
- Execute Gauss-Seidel iterations until the adjustments to the voltage estimates decrease to, say, 0.01 or 0.005 per unit in both real and imaginary parts.
- Switch to Newton-Raphson iterations until either the problem is converged, or the reactive power output estimates for generators show signs of failure to converge.
- Switch back to Gauss-Seidel iterations if the Newton-Raphson method does not settle down to a smooth convergence within 8 to 10 iterations.

Experience with each specific problem will suggest modifications to this procedure. In particular, the initial Gauss-Seidel iterations and flat-start steps will be bypassed when the result of a previous solution is known to be a close estimate of the expected new solution.

In many power flow cases, the solution can be expedited by using two (or even more) iteration activities in succession in order to take advantage of the complementary strengths of the various iterative schemes. For example, if the starting voltage estimate is known to be poor, it is often advantageous to execute several Gauss-Seidel iterations and to switch over to the Newton-Raphson iterations after the voltage estimate has been brought into rough approximation to the true solution.

A simple approach is to allow a Gauss-Seidel solution to complete its maximum iterations and then select a Newton method and continue the solution from the same convergence point. (The starting voltage estimate for the Newton solution would be the last voltage vector estimated in the preceding execution of the Gauss solution.) At any time the solution can be changed back to a Gauss-Seidel solution or a different Newton method.

If the iterations are slow, it is possible to use the AB interrupt to cancel whichever solution method is in progress and, subsequently, select a different solution method to continue.

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6.2.1. Using Acceleration Factors and Solution Tolerances

The power flow solution activities use the acceleration factors listed in Table 6.2, "Power Flow Iteration Acceleration Factors" and the convergence tolerance values listed in Table 6.3, "Power Flow Convergence Tolerances". The power flow working case contains an iteration limit used by the Gauss-Seidel solutions (ITMX), and another (ITMXN) used by the Newton-Raphson solutions. The default values of these iteration limits are 100 and 20 iterations, respectively.

Table 6.2. Power Flow Iteration Acceleration Factors

Iteration Type	Name	Default Value	Applied To
Gauss-Seidel	ACCP ACCQ	1.6 1.6	Real part of voltage change Imaginary part of voltage change
Modified Gauss-Seidel	ACCM ACCP ACCQ	1.0 1.6 1.6	Complex voltage change at Type 1 buses Real part of voltage change at Type 2 buses Imaginary part of voltage change at Type 2 buses
Newton-Raphson	ACCN	1.0	Magnitude of voltage change at Type 2 buses only

Table 6.3. Power Flow Convergence Tolerances

Name	Used By	Default Value	Applied To
TOL	Gauss-Seidel solutions	0.0001 per unit	Voltage magnitude change convergence tolerance
TOLN	Newton-Raphson solutions	0.1 MW and Mvar	Mismatch convergence tolerance
VCTOLQ	Newton-Raphson solutions	0.1 MW and Mvar	Controlled bus reactive power mismatch convergence tolerance
VCTOLV	Newton-Raphson solutions	0.00001 per unit	Controlled bus voltage error convergence tolerance

Each of these solution parameters is set to a default value based on general experience each time PSS®E is started up. The user should be prepared to tune these solution parameters. The convergence properties of the Gauss-Seidel solutions are particularly dependent upon the acceleration of the voltage adjustment process.

These solution parameters can be changed using the data changing activity CHNG or the *[Solution Parameters]* dialog. They are all saved and retrieved with the power flow case.

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6.2.2. Automatic Power Flow Solution Adjustments

Automatic adjustment options may be used in power flow solutions to allow maneuvering of the system to meet specified voltage, branch flow, and area net interchange schedules. Automatic adjustments can be enabled or disabled when setting program run-time options (see Section 3.3.3, "Program Run-Time Option Settings").

The automatic adjustments available are specified in Table 6.4, "Available Automatic Adjustments". In addition to the options listed in the table, the user has the option to start the power flow solution from a flat start.

When enabled, an adjustment option is applied throughout the entire power flow case. However, it is important to recognize that the adjustment of any individual tap-changing transformer, phase shifter, or area net interchange can be suppressed by editing its adjustment parameters in such a way that no adjustment can occur. Those parameters are accessible in the spreadsheet.

Table 6.4. Available Automatic Adjustments

Type of Adjustment	Normal Default	Available In
Adjustment of transformer turns ratio in steps to hold voltage at a designated bus in specified band.	Disabled	GAUSS NEWTON
Direct adjustment of transformer turns ratio to hold voltage at a designated bus in specified band or to hold reactive power through transformer within a specified band.	Disabled	NEWTON
Adjustment of area swing bus generator powers to hold area net interchange within a specified band.	Disabled	GAUSS NEWTON
Adjustment of transformer phase shift to hold real power through transformer within a specified band.	Disabled	NEWTON
Adjustment of converter transformers of dc transmission terminals to place rectifier delay angle and inverter margin angle within specified ranges.	Enabled	GAUSS NEWTON
Adjustment of switched shunt admittances to hold voltage at the bus as close to specified voltage as permitted by inductance/capacitor module size.	Enabled	GAUSS NEWTON

Adjusting Transformer Off-Nominal Tap Ratio

Depending on the solution technique chosen, transformer tap ratio adjustments can be made to control either voltage at a specified bus, a compensated voltage based on the transformer loading or to control

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Mvar flow at the tapped side of the transformer (see Table 6.5, "Transformer Tap Adjustment Options"). Compensated voltage is often used on radial circuits when the user wishes to remotely regulate a voltage but does not wish to install any communication links. Note that by controlling voltage using the Newton solution activities, the user may choose one of two adjustment techniques: discrete or direct tap adjustment. The Mvar control can be performed using direct tap adjustment only in Newton solutions.

With all solution activities, the request to adjust taps ratio automatically will not be honored until the voltage change is less than the adjustment threshold (ADJTR), a solution parameter with a default value of 0.005 per unit. With discrete tap adjustment, tap ratio will then be adjusted to a maximum of value of tap limit (TAPLIM), a solution parameter with a default value of 0.05 per unit, until tolerance is reached. With direct tap adjustment controlling voltage, tap ratio will be adjusted, as if the taps are continuous, from the iteration when tolerance is first reached through subsequent iterations until the tolerance is again reached. At that time, unless the tap ratio is zero, all taps will be set to the nearest step, locked, and additional iterations made to bring mismatch down to tolerance once again. Similarly, with direct tap adjustment controlling Mvar, adjustment will not be made until tolerance is reached and controlling logic is identical thereafter to that of direct tap adjustment controlling voltage.

The solution parameters ADJTR and TAPLIM can be changed using the data changing activity CHNG or the *[Solution Parameters]* dialog.

Table 6.5. Transformer Tap Adjustment Options

Method	Availability of Voltage Control Adjustment	Availability of Voltage Control Via Direct Tap Adjustment	Availability of Mvar Control Via Direct Tap Adjustment
Gauss-Seidel	Yes	No	No [a]
Modified Gauss-Seidel	Yes	No	No*
Full Newton-Raphson	Yes	Yes	Yes
Decoupled Newton-Raphson	Yes	Yes	Yes
Fixed-slope, decoupled Newton-Raphson	Yes	Yes	Yes

[a]Tap is assumed fixed.

Adjusting Transformer Voltage Control

Voltage control by transformer turns ratio adjustments, as described above, will only be made on transformers having the following conditions: a nonzero regulated bus number, IREG; and a value of 1 for the transformer enable flag, RF. The tap ratio of each transformer is adjusted to hold a voltage magnitude between the limits VMIN to VMAX. The voltage magnitude to be controlled is calculated as follows:

$V_c = |V_{IREG} - I_{transformer} (CR + jCX)|$

where:

CR + jCX	= Compensating impedances entered by the user.
Itransformer	= Current in the transformer calculated on the IREG side of the transformer.

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The recommended method for suppressing ratio adjustment is to set the transformer control flag to zero. Ratio adjustment may also be suppressed by setting IREG to 0.

The tap-step ratio should be properly coordinated with the acceptable voltage band (VMAX-VMIN) because the discrete steps of available tap ratio are recognized. The default tap-step ratio is 0.00625 per unit (0.625%). The band between VMAX and VMIN should normally be twice the transformer tap-step. A band of 2% is recommended for the normal case of 0.625% tap ratio steps. A value of 0 for tap-step should only be used to indicate continuously adjustable taps to the direct tap adjustment algorithm.

Adjusting Transformer Mvar

The tap ratio is adjusted to hold the Mvar on the tap side of the transformer between the limits VMIN-VMAX. As mentioned previously for voltage-controlling transformers, the user should coordinate tap-step and available range for Mvar-controlling transformers.

Adjusting Transformer Phase Shift Angle

The phase-shift angle of each phase shifter is adjusted, as necessary, to keep the real power flow through the phase shifter between the limits VMAX and VMIN.

Phase-shift adjustment is continuous and all adjustable phase shifters are adjusted simultaneously whenever the regulated real power flow of at least one of them falls outside its scheduled band. An unduly narrow band can cause non convergence of the power flow solution. A reasonable band is ± 5 MW of the target flow.

Adjusting Transformer Direct Current (DC) Converter Taps

The control logic adjusts the converter transformer tap positions to attempt to hold the bridge firing angles above minimum values and below maximum. The minimum values of the firing angles, α_{min} and γ_{min}, are firm limits; the bridges will not be operated in power flow solutions with firing angles below these values. The maximum values of the firing angles, α_{max} and γ_{max}, are objectives, but not firm limits; the converters may be operated in power flow solutions with firing angles above these limits if the converter transformer tap positions are at the ends of their ranges or if the desired angle ranges are narrow relative to the tap steps.

Converter taps are adjusted only until the corresponding bridge firing angles are between their specified maximum and minimum values; taps are not adjusted to minimize firing angles when they fall between their corresponding maxima and minima. Accordingly, the ranges of the bridge firing should be treated as the optimum bands for the various converters, with recognition that operation at a firing angle above the specified maximum value may be necessary for some combinations of scheduled dc power, dc voltage, and ac system conditions.

The ac solution options can be modified such that the adjustment of converter transformer ratios are *disabled*. This disabling applies to all dc transmission lines. Selective locking of dc converter transformer taps requires that their adjustment not be disabled via the ac solution options. The tap ratio of any individual converter transformer can be locked by using the program's data editing functions to set the dc converter tap limits and tap ratio equal to the desired value.

Adjusting Net Interchange

Each area of the power system may have one of its generators designated as an area-slack machine. Net interchange adjustment is implemented by manipulating the real power output of all area slack machines except those in areas containing system swing (Type 3) buses. Area slack machine adjustment may be sup-

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pressed for any individual area by setting the area slack bus number in the area interchange data category to 0. Area slack adjustment is continuous, but takes place only if the area net interchange falls outside the band for at least one area: (Scheduled interchange ± Interchange tolerance). An unduly small interchange tolerance can cause non convergence of the power flow solution.

The net interchange tolerance should be related to system interchange capability and system impedances. A typical tolerance for a large electric utility with capability for a 500 MW net interchange might be 10 MW, while an industrial plant with a maximum net interchange capability of 15 MW might be assigned a tolerance of 0.5 MW. The tolerance should be set in relation to interchange capability and remain fixed as the interchange schedule is varied.

Adjusting Switched Shunt Admittance

Automatically switched shunt devices are normally adjusted according to the rules given in Section 6.3.15, "Switched Shunt Devices". When this adjustment option is enabled, either all adjustable switched shunts (i.e., the switched shunt is in-service and its control mode, MODSW, is not 0) are subject to adjustment, or only continuous mode switched shunts (i.e., MODSW is 1) are subject to adjustment. This adjustment is normally enabled in all power flow solutions. It can be suppressed, regardless of the values of MODSW in the Solution Options. Selective locking of automatically switched shunt devices must be handled by leaving the adjustment feature *enabled* in the power flow solution and setting MODSW to 0 at buses where locking is required.

Power Flow Solution Activities—Selection Guide

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Iteration Method/ Old Activity name	Advantages	Disadvantages	Use When	Do Not Use When	Convergence Monitor
Newton-Raphson method with real and reactive power equations decoupled (NSOL)	Rapid convergence on well-conditioned cases. Small bus mismatches can be achieved.	Intolerant of data errors. Cannot start from poor voltage estimates. Cannot handle network with low X/R ratio branches (e.g., equivalents). No indication of cause of problem when failing to converge. Can give problems converging cases where reactive power limits are restrictive.	Poor voltage estimate and network contains negative reactive branch.	Network contains branches with low X/R ratios. Overloading has produced reactive power problems.	As for Full Newton-Raphson.
Newton-Raphson method with real and reactive power equations decoupled using a fixed Jacobian matrix (FDNS)	Rapid convergence on well-conditioned cases. Small bus mismatches can be achieved.	As mismatches are reduced, rate of improvement may be allowed. Intolerant of data errors. No indication of cause of problem when failing to converge. Can give problems converging cases where reactive power limits are restrictive.		Overloading has produced reactive power problems.	As for Full Newton-Raphson.

Iteration Method/ Old Activity name	Advantages	Disadvantages	Use When	Do Not Use When	Convergence Monitor
Newton-Raphson method with real and reactive power equations decoupled (NSOL)	Rapid convergence on well-conditioned cases. Small bus mismatches can be achieved.	Intolerant of data errors. Cannot start from poor voltage estimates. Cannot handle network with low X/R ratio branches (e.g., equivalents). No indication of cause of problem when failing to converge. Can give problems converging cases where reactive power limits are restrictive.	Poor voltage estimate and network contains negative reactive branch.	Network contains branches with low X/R ratios. Overloading has produced reactive power problems.	As for Full Newton-Raphson.
Newton-Raphson method with real and reactive power equations decoupled using a fixed Jacobian matrix (FDNS)	Rapid convergence on well-conditioned cases. Small bus mismatches can be achieved.	As mismatches are reduced, rate of improvement may be allowed. Intolerant of data errors. No indication of cause of problem when failing to converge. Can give problems converging cases where reactive power limits are restrictive.		Overloading has produced reactive power problems.	As for Full Newton-Raphson.

6.3. Applying Gauss-Seidel Power Flow Solution

SOLV

Run Activity SOLV - *GUI*
Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)... *[Loadflow solutions]* Gauss Tab OPTION: Gauss-Seidel
Run Line Mode Activity SOLV - CLI
ACTIVITY? >>**SOLV**
Interrupt Control Codes
AB, NC, NM, DC, FD, MO

6.3.1. About Power Flow Iterative Solution Algorithms

The power flow problem is nonlinear and requires an iterative trial and error process for its solution. One simple but effective iterative scheme follows:

1. Make an initial estimate of the voltage at each bus.
2. Build an estimated current inflow vector, I_n, at each bus from a boundary condition such as:

   $$P_k + jQ_k = v_k i_k^*$$ (6.2)

   where:

   Pk + jQk = Net load and generation demand at bus k.

   vk = Present estimate of voltage at bus k.
3. Use Equation 6.1 to obtain a new estimate of the bus voltage vector, v_n.
4. Return to Step 2 and repeat the cycle until it converges on an unchanging estimate of v_n.

While this scheme is useful in some specific situations (to be discussed later), it does not work well for the general power flow calculation where the terminal voltage magnitude, rather than reactive power output, is specified for many generators.

6.3.2. About Power Flow Boundary Conditions

Steps 2 and 3 of the iterative scheme in Section 6.3.1, "About Power Flow Iterative Solution Algorithms" refer to two aspects of the power flow solution calculation. The solution of the power flow problem is a set of bus (or node) voltages that simultaneously satisfy the network condition, $I_n = Y_{nn}v_n$, derived from Kirchhoff's laws, and the boundary conditions derived from load and generator characteristics, such as $P_k + jQ_k = v_k i_k^*$.

The network condition (Equation 6.1) is linear and can be solved without iteration if either the voltage vector, v_n, or the current vector, I_n, is specified. The solution is a direct calculation if v_n is specified, and requires a standard computer procedure for solving linear simultaneous equations if I_n is given.

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The boundary conditions may be specified quite arbitrarily, depending upon the loads that electricity users choose to connect to the network, and are usually nonlinear. It is the nonlinearity of the boundary conditions that forces the use of an iterative procedure for power flow solution. It must be noted though, that while the network condition *can* be handled in a non iterative, closed-form manner, it is often advantageous to use an iterative method to solve both the network condition and the boundary conditions.

In the commonly used power flow iteration procedures:

- Gauss-Seidel methods solve both network and boundary conditions by iteration.
- Newton-Raphson methods solve the network condition by closed-form calculation, while using iteration to solve the boundary conditions.

6.3.3. Boundary Conditions of Constant MVA Loads

The most common load boundary condition is a specification of load real and/or reactive power consumption

$$\text{Real}\,(v_k\, i_k^{*}) = -P_k \tag{6.3}$$

$$\text{Imag}\,(v_k\, i_k^{*}) = -Q_k \tag{6.4}$$

This characteristic is not realistic for voltages below approximately 0.8 per unit. All PSS®E power flow voltage solutions, in both power flow and dynamic simulation, therefore, modify Equation 6.3 and Equation 6.4 to make P_k and Q_k functions of the magnitude of the bus voltage as shown in Figure 6.1, "Constant MVA Load Characteristic (Top) and Resultant Form of Current/Voltage Curve (Bottom)".

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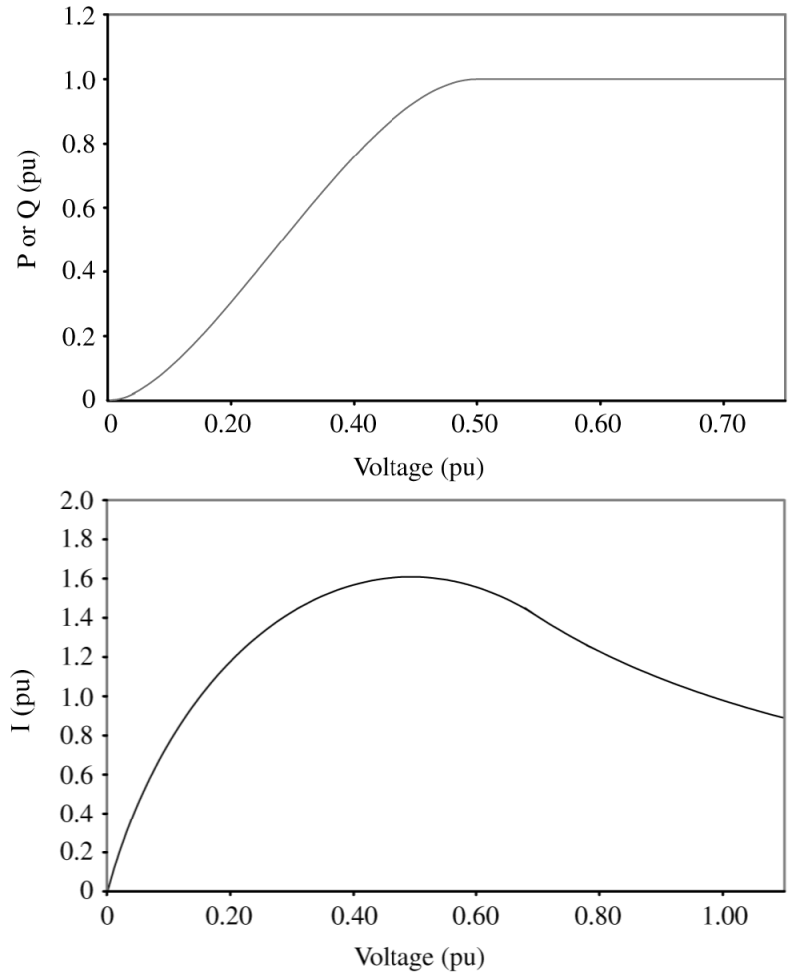

Figure 6.1. Constant MVA Load Characteristic (Top) and Resultant Form of Current/Voltage Curve (Bottom)

The constant power characteristic holds the load power constant as long as the bus voltage exceeds the value specified by the solution parameter PQBRAK, and assumes an elliptical current-voltage characteristic of the corresponding load current for voltages below this threshold.

The user may modify the value of PQBRAK using the data changing activity CHNG or the *[Solution Parameters]* dialog.

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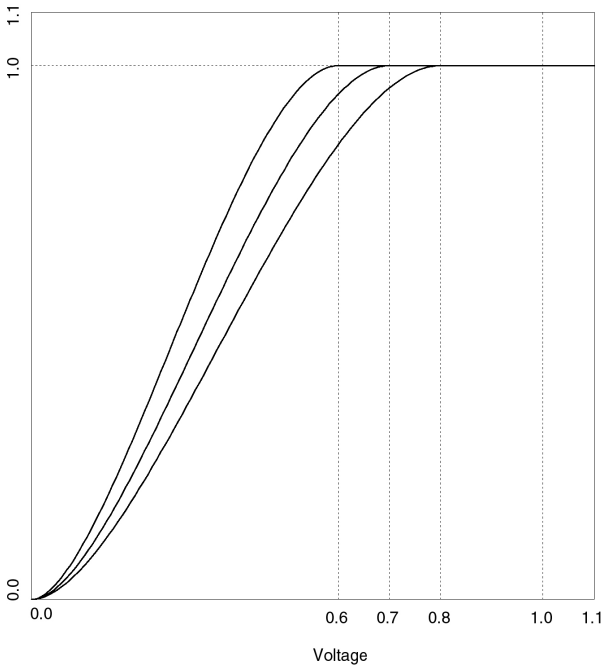

Figure 6.2. yConstant Power Load Characteristic

6.3.4. Boundary Conditions of Constant Current Loads

Load may be specified as a given active or reactive component of current such that:

$$\frac{\text{Real}\,(v_k i_k^*)}{|v_k|} = -I_{pk} \tag{6.5}$$

$$\frac{\text{Imag}\,(v_k i_k^*)}{|v_k|} = -I_{qk} \tag{6.6}$$

Again, because this characteristic is unrealistic for voltages below about 0.5 per unit, all PSS®E power flow solutions modify (Equation 6.5 and Equation 6.6) to make I_{pk} and I_{qk} functions of the magnitude of v_k, as shown in Figure 6.3, “Constant Current Load Characteristic (Top) and Resultant Form of Load MVA/Voltage Curve (Bottom)”.

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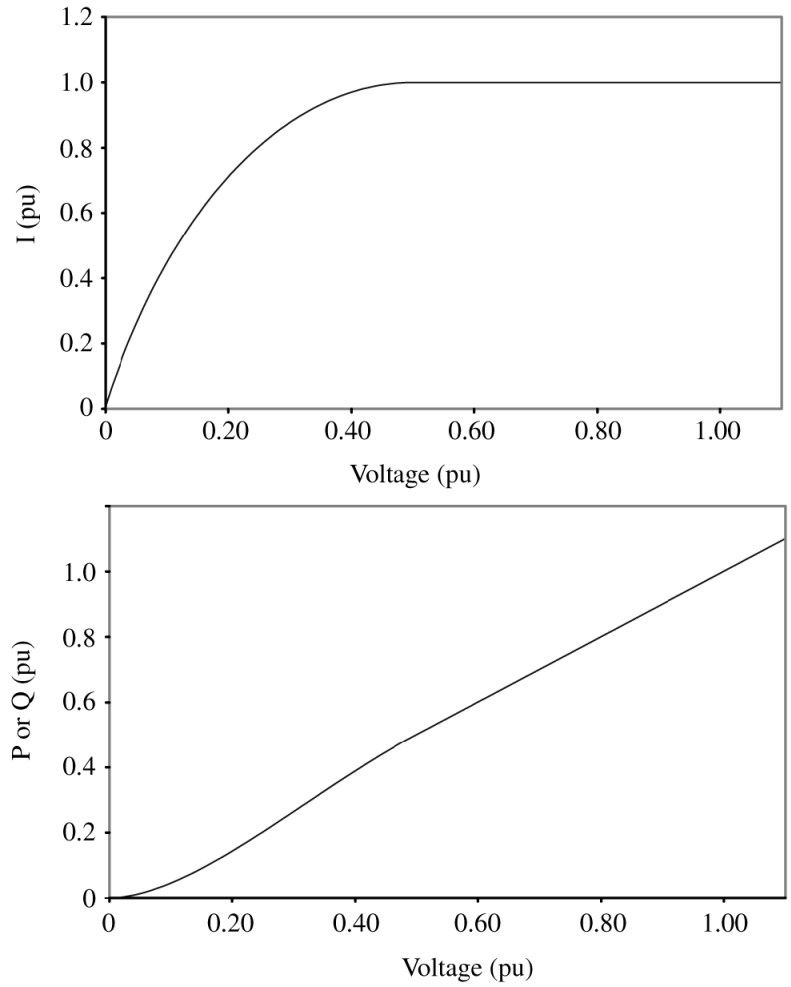

Figure 6.3. Constant Current Load Characteristic (Top) and Resultant Form of Load MVA/Voltage Curve (Bottom)

6.3.5. Boundary Conditions of Constant Impedance Loads

Finally, load may be specified by given real and reactive parts of shunt admittance such that

$$\frac{i_k}{v_k} = G_k + jB_k = \frac{P_k - jQ_k}{v_k v_k^*} = \frac{P_k - jQ_k}{|v_k|^2} = \frac{P_k}{|v_k|^2} - j\frac{Q_k}{|v_k|^2} \quad (6.7)$$

Note that Equation 6.7 is not treated as a boundary condition in the solution process; it can be more convenient to incorporate this type of load as a bus shunt branch unless there is a need, at some time, to convert this load to another type.

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6.3.6. Boundary Conditions of Composite Loads

All PSS®E network solutions allow the load at each bus to be a composite of arbitrary amounts of load with each of the characteristics described in the previous subsections. The composite characteristic becomes the boundary condition used in iterative power flow solutions.

The normal practice is to specify the load at each bus initially as a compendium of constant MVA and constant admittance loads. Subsequently the data can be adjusted to comprise the required amounts of each characteristic. This course gives compatibility with external power flow data formats such as the IEEE Common Format. Conversion and construction of load data is facilitated by activity CONL.

6.3.7. Boundary Conditions of the Swing Bus

Every power flow simulation case must have at least one bus designated as a *swing bus*. The corresponding boundary condition is:

$$v_{k(complex)} = \text{constant} \tag{6.8}$$

The net real and reactive power inflow to a swing bus are free variables and follow from the power flow solution, rather than being boundary conditions imposed upon it.

Power flow solution cases must have at least one swing bus in every separate section (island) of the network. An ac island is defined as the set of all in-service buses such that each bus in the island may be reached from every other bus in the island through the in-service ac network. If a working case contains more than one island, any island is either electrically disjoint from all other islands, or connected to one or more islands by one or more asynchronous ties (e.g., dc lines).

To be a valid power flow case, each ac island in the working case must contain at least one Type 3 (swing) bus. Furthermore, the buses connected by each in-service branch must be in-service (i.e., none of them may be Type 4 buses). No swing bus is needed in fault analysis, switching, and dynamic simulation calculations, although swing buses may be used in these simulations.

When enabled, the solution connectivity checking option checks for the presence of swingless islands at the start of each power flow solution (activity OPTN).

The user can check that every subsystem of a power system power flow case includes a swing bus using activity TREE.

6.3.8. Boundary Conditions of Standard Generators

The standard generator arrangement used throughout PSS®E is shown in Figure 6.4, “Standard PSS®E Generator Configuration”. The generator is assumed to be connected to bus k by a step-up transformer where per unit impedance is Z_t, ($R_t + TX_t$), on generator MVA_{base}. The transformer impedance may be set to zero to model a generator connected directly to the bus such that bus k is the generator terminal bus.

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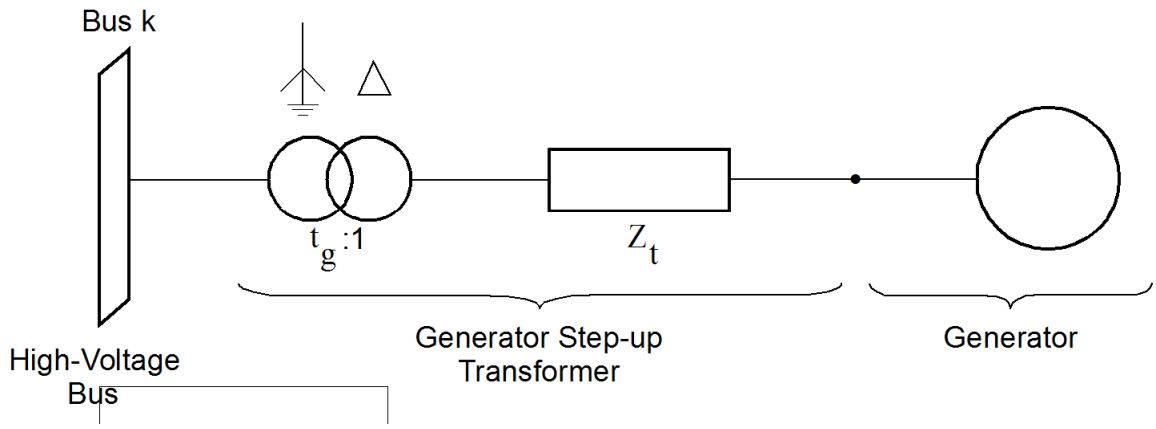

Figure 6.4. Standard PSS®E Generator Configuration

The standard generator boundary condition is a specification of real power output at the high voltage bus, bus k, and of voltage magnitude at some designated bus, not necessarily bus k.

$$\text{Real}\,(v_k\, i_k^*) = P_k \tag{6.9}$$

$$|V_1| = V_{sched} \tag{6.10}$$

This characteristic is subject to the following reactive power output limitations:

$$Q_{mn\,k} \le \text{Imag}(v_k\, i_k^*) \le Q_{mx\,k} \tag{6.11}$$

which overrides the voltage schedule condition (Equation 6.10).

It is important to recognize that the maximum and minimum reactive power limits assigned to bus k apply to generator reactive power output measured at the high-voltage bus and not at the generator terminals. Determination of $Q_{min\,k}$ and $Q_{max\,k}$ must, therefore, recognize the reactive power loss in the step-up transformer reactance. A reasonable assumption for assigning reactive power limits to bus k, in this situation, is to subtract a reactive loss corresponding to full load current (1.0 pu) from the generator terminal reactive power limits.

Because Z_t has a per-unit value with respect to generator MVA base and $Q_{min\,k}$, $Q_{max\,k}$ are in megavars,

$$Q_{limit\,k} = Q_{limit\,g} - X_t \times \text{MBASE} \tag{6.12}$$

for lagging generator terminal power factor, and

$$Q_{limit\,k} = Q_{limit\,g} + X_t \times \text{MBASE} \tag{6.13}$$

for leading generator terminal power factor, where:

X_t	= Step-up transformer reactance in per unit on generator base.
$Q_{limit\,g}$	= Limiting reactive power in Mvar at generator terminals.
$Q_{limit\,k}$	= $Q_{min\,k}$ or $Q_{max\,k}$.

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6.3.9. Boundary Conditions of Multiple Identical Generators

Multiple, *identical* generators and generator step-up transformer units may be represented by the standard model, as shown in Figure 6.4, "Standard PSS®E Generator Configuration", by specifying the generator MVA base to be the total MVA rating of all connected generators and specifying Z_t as the impedance of a single step-up transformer on its own single generator rating. This representation is illustrated in Figure 6.5, "Identical Generators at Bus".

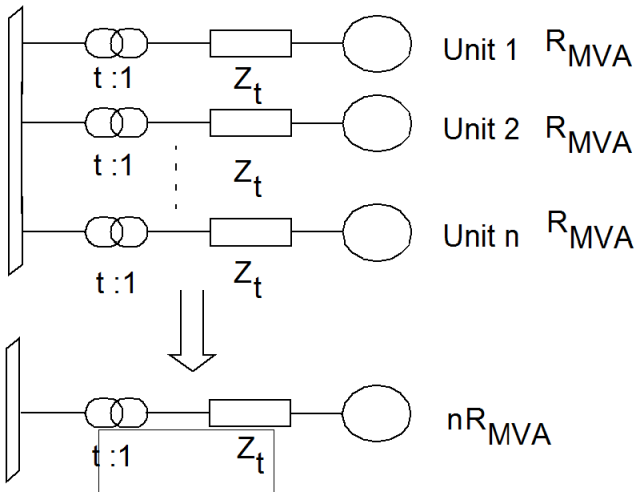

Figure 6.5. Identical Generators at Bus

Lumping several identical generators may, for example, be used where startup and shutdown of individual units in a plant are scheduled to meet increasing and decreasing plant loading. This is achieved by adjusting P_{gen}, $Q_{max\ k}$, $Q_{min\ k}$, and $MBASE_k$. No adjustment of Z_t or of the network branch data is necessary.

Use of this lump approach to handle multiple units implies that real and reactive power output are distributed uniformly between them. If loadings of multiple units are not identical, they must be treated as different generators even though their impedances and other characteristics are identical. This offers the advantage of having to change only unit status flags to account for a change in the number of operating units at the plant.

6.3.10. Boundary Conditions of Multiple Non-Identical Generators

A plant having several different generators connected to its high voltage bus cannot be represented by a single generator model. Correct representation of such plants requires the use of multiple, individual generator models at the plant bus.

6.3.11. Operation of Activity SOLV

The power flow solution activity SOLV uses a Gauss-Seidel iterative algorithm to solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case.

A quick note of value is that the Gauss-Seidel solution cannot handle negative series reactances (series capacitors) but the Newton and Modified Gauss-Seidel can.

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Activity SOLV first does a quick check for valid network topology. In this check, it looks at all series elements (ac branches, dc lines, and FACTS devices with a series element) connected to each bus, and reports the following errors:

- any type 4 (out-of-service) bus with one or more in-service series elements connected to it.
- any in-service bus with no in-service series elements connected to it.

If any topological errors are detected, activity SOLV terminates without attempting its power flow solution calculation.

If the power flow solution network connectivity checking option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings"), activity SOLV checks that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed and activity SOLV is terminated.

When the *Flat start* solution option is selected, all bus voltages are set to a magnitude of 1.0 at a phase angle of 0.0 degrees. Then, at each type 3 (system swing) bus, the voltage magnitude is set to the scheduled voltage specified in the plant data for the bus; any bus which is connected to a type 3 bus by an in-service zero impedance line has its complex voltage set to that of the swing bus to which it is connected.

When the Flat start solution option is not selected, the present voltage profile is used as the starting point for the power flow solution. This is the usual case when changes and/or additions have been imposed on a "solved" power flow case.

However, when buses are added to a solved case with default values applied for voltage magnitude (1.0) and phase angle (0.0), and they are connected to one or more buses whose "solved" voltage values differ substantially from the default values, extremely large mismatches are possible on the first and subsequent iterations. It is possible that these differences are so large that they are "too far away from" the solved voltage profile for the case and the solution attempt can "blow up".

To attempt to overcome such solution failure, activity SOLV looks for "flat" buses (i.e., in-service buses with default voltage and phase angle values), and "boundary" buses (i.e., in-service buses with voltage and/or phase angle values that differ from the default values). Prior to the first iteration, it adjusts complex voltages at "flat" buses as follows:

1. For one "flat" bus radially connected to one "boundary" bus, the complex voltage at the "flat" bus is set to that of the "boundary" bus. If a transformer connects the two buses, the tap settings and phase shift angle of the transformer are considered in setting the voltage at the "flat" bus.

2. For one "flat" bus connected to two or more "boundary" buses, voltages as calculated in 1. are determined for each of the "boundary" buses, and the complex voltage at the "flat" bus is set to the average of these voltages.

3. For an arbitrary sub-network of two through ten "flat" buses connected to a single "boundary" bus, the following procedure is used:

    a. The number of "flat" buses in this sub-network connected directly to the "boundary" bus is determined.

    b. If more than one "flat" bus in this sub-network is connected to the "boundary" bus, the complex voltages of all buses in the "flat" sub-network are set to the complex voltage of the "boundary" bus, and processing of the sub-network is terminated.

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c. If exactly one “flat” bus in this sub-network is connected to the “boundary” bus, the voltage at the “flat” bus is set as in 1. above. This former “flat” bus is removed from the sub-network, and becomes the “boundary” bus of the reduced sub-network. If any “flat” buses remain in the reduced sub-network, go back to step a. and repeat for the reduced sub-network.

4. For all other sub-networks of “flat” buses connected to multiple “boundary” buses, the voltages are left at their “flat” values.

For those in-service machines for which a capability curve has been specified, the machine's reactive power limits are determined from the capability curve based upon the machine's active power setting. Any machine for which its reactive power limits are changed as a result of this determination is reported at the *Progress* device (refer to Section 4.4, “Virtual Output Devices”).

At the completion of each iteration, activity SOLV prints a convergence monitor, which tabulates:

1. The iteration number.
2. The largest voltage magnitude change as a multiple of the convergence tolerance.
3. The number, name and base voltage of the bus experiencing the largest voltage change; alternatively, if the bus is the star point bus of a three-winding transformer, the transformer name followed by * BUS is printed.
4. The real part of the largest voltage change.
5. The imaginary part of the largest voltage change.

At the end of activity SOLV, the largest bus mismatch is tabulated, along with the system total MVA mismatch and a summary of swing (Type 3) bus power outputs. An asterisk (*) following a plant’s active or reactive power output in the swing bus summary indicates that the power output is beyond its limits.

If area interchange adjustment was enabled, each area that has at least one in-service tie element connected to it and which has an area swing assigned to it, but fails to meet is net interchange specification, is logged.

If the case has not converged, the user can modify the solution parameters (refer to Section 6.3.19, “Characteristics of Activity SOLV”) and solution type, if warranted, and re-run the solution. If the solution had diverged (blown up), the *Flat start* control option should be used to establish a feasible starting point for the next solution.

Activity SOLV responds to the following interrupt control codes:

AB	Abandon activity SOLV following completion of the next iteration.
NC	Suppress the convergence monitor.
NM	Suppress any automatic adjustment monitors.

DC	Tabulate the conditions for each in-service dc line after each iteration. The data printed for each non-capacitor commutated two-terminal dc line includes the dc line name followed by two lines including the following quantities:
	
	`IPR IPI ALPHA GAMMA VDCR VDCI`
	`TAPR TAPI DCCUR PACR QACR PACI`

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	The information printed for each capacitor commutated two-terminal dc line includes the dc line name, followed by an iteration report of the Newton solution performed for one or more operating modes, followed by:
	
	`ALPHA_R MU_R VDC_R(kV) TAP_R PAC_R QAC_R`
	`DC_CURRENT GAMA_I MU_I VDC_I(kV) TA`
	`PAC_I QAC_I ALPHA_I`
	
	For multi-terminal lines, the dc line name is followed by conditions at each converter bus:
	
	`bus # ANGLE TAP VDC DCCUR PAC QAC`

	For VSC dc lines with both converters in-service, the dc line name is followed by line and converter bus conditions as follows: dc voltages at the power and voltage controlling converters and dc current if a limit violation is relieved at this iteration, the converter bus at which it occurs, the present ac injections at that bus, and the reduced values are tabulated, followed by the new dc voltages and the dc current dc power, converter losses, and ac power at the power controlling converter dc line losses dc power, converter losses, and ac power at the voltage controlling converter For VSC dc lines with only one converter in-service, the dc line name is followed by the dc power, converter losses, and ac power at the in-service converter.
FD	Tabulate the conditions for each in-service FACTS device after each iteration. The data printed for each FACTS device includes conditions at the sending and terminal buses, series voltage and current in polar coordinates, series voltage with respect to both series current and sending end voltage in rectangular coordinates, and an indication of those quantities currently at a limit.
MO	Tabulate the conditions for each in-service induction machine after each iteration. The data printed for each induction machine includes the bus number and machine identifier, the voltage magnitude at the bus, the voltage error in the induction machine iteration, slip, mechanical torque, mechanical power, the mag-

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	nitude of the voltage across the mutual reactance, the mutual inductance saturation multiplier ALFAXM, the magnitude of armature current, the armature reactance saturation multiplier ALFAXA, the partial derivative of mechanical output power with respect to slip, the partial derivative of mechanical output power with respect to terminal voltage, the partial derivative of mechanical load with respect to slip, and the apparent active and reactive power flowing into the induction machine at the ac bus.

6.3.12. Generation

During the solution, generator buses are treated as follows:

1. Those generators that regulate their own voltage hold their scheduled voltage as long as their reactive power limits are not violated.

2. Those generators that regulate the voltage of a remote type 1 or 2 bus have their reactive power output adjusted as required to hold the desired voltage at the remote bus as long as their reactive power limits are not violated.

3. Those generators that are swing buses (i.e., with a type code of 3) are held at constant voltage and phase angle. Their active and reactive power outputs are set as required prior to leaving activity SOLV. The plant active power is shared among the machines at the plant in proportion to the magnitudes of their active power settings at the time activity SOLV was initiated; if all machines have an active power setting of 0.0 at the time activity SOLV is selected, the plant active power is shared in proportion to the MBASEs of the machines at the plant. For such buses with both generators and synchronous condensers, only the generators share the swing bus active power output.

4. At the end of activity SOLV, generator plant reactive powers (and active powers for Type 3 buses) are apportioned among the machines at the plant; reactive powers are split so as to achieve identical power factors at all machines in a plant, except that machine reactive power limits are honored.

6.3.13. Load

Busbar load data is entered into PSS®E in activities READ, TREA, and RDCH (refer to Load Data) on the basis of the following power flow load characteristics for both the active and reactive components of load: constant MVA, constant current, and constant admittance.

Loads may be apportioned among these three characteristics by activities CONL and RCNL.

The constant power characteristic holds the load power constant as long as the bus voltage exceeds the value specified by the solution parameter PQBRAK, and assumes an elliptical current-voltage characteristic of the corresponding load current for voltages below this threshold. Figure 6.6, “Constant Power Load Characteristic” depicts this characteristic for PQBRAK values of 0.6, 0.7, and 0.8 pu. The user may modify the value of PQBRAK with the data changing activity CHNG or the *[Solution Parameters]* dialog.

The constant current characteristic holds the load current constant as long as the bus voltage exceeds 0.5 pu, and assumes an elliptical current-voltage characteristic as shown in Figure 6.7, “Constant Current Load Characteristic” for voltages below 0.5 pu.

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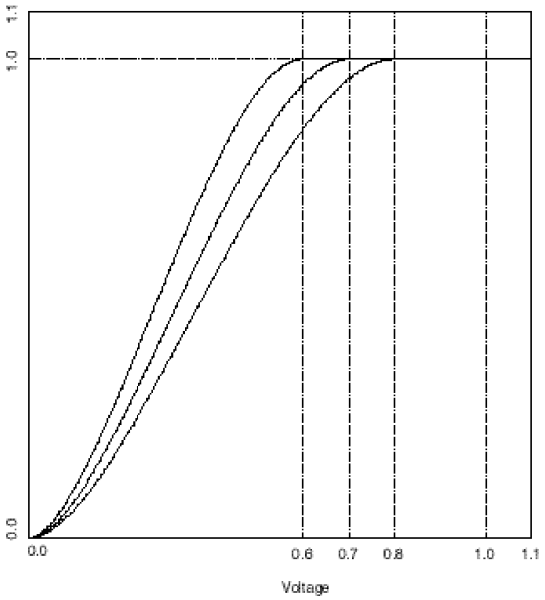

Figure 6.6. Constant Power Load Characteristic

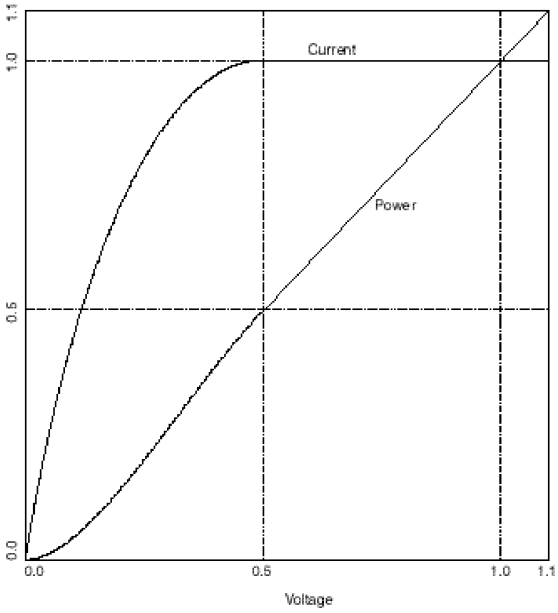

Figure 6.7. Constant Current Load Characteristic

6.3.14. Induction Machines

Buses for which data for one or more induction machines has been specified in activities READ, Section Activity, TREA, or RDCH (refer to Induction Machine Data) are handled according to the following rules:

1. In-service induction machines at type 1, 2 and 3 buses are modeled during power flow solutions.

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2. On an iteration in which a solution for a machine cannot be achieved due to load terminal voltage, the state of the induction machine is changed as follows:

   - If the machine is a motor (i.e., PSET is positive), the machine is treated as "stalled"; it is represented by its locked rotor impedance. In calculating the locked rotor impedance, it is assumed that Xa is fully saturated and that saturation of Xm is negligible
   - If the machine is a generator (i.e., PSET is negative), the machine is treated as "tripped".

3. At each power flow iteration, a complete solution of each in-service machine is performed. Thus, solutions for machines that were placed in the "stalled" and "tripped" states during the previous iteration are again attempted.

4. For each induction machine, the solution keeps count of the number of iterations in which a machine is placed in the "stalled" or "tripped" state. If, during a single power flow solution, a machine experiences more than MXSWIM switchings from the "running" state to the "stalled" or "tripped" state, it remains in the "stalled" or "tripped" state for the remainder of the current power flow solution.

6.3.15. Switched Shunt Devices

Buses for which switched shunt data has been specified in activities READ, TREA, or RDCH (refer to Switched Shunt Data) are handled according to the following rules:

1. Automatically switched shunts are permitted only at type 1 and 2 buses. At type 3 (swing) buses, switched shunts are treated as locked at the value specified as BINIT in activities READ, TREA, or RDCH (refer to Switched Shunt Data), or as subsequently modified with activity CHNG or the *[Spreadsheet]*.

2. For those switched shunts designated as locked (i.e., MODSW = 0 or the bus is a type 3 bus), the specified admittance is held constant during the solution.

3. Those switched shunts controlling local or remote bus voltage to a voltage setpoint (i.e., VSWHI = VSWLO) are handled as follows:

   a. Those switched shunts designated as continuous (i.e., MODSW = 2) hold their scheduled voltage as long as the admittance limits are not violated. The high admittance limit is the admittance when all reactor blocks are switched off and all steps of in-service capacitor blocks are switched on; the low limit represents all steps of in-service reactor blocks switched on and all capacitor blocks switched off. (Note that for reactors, B<0 and for shunt capacitors, B>0).

   b. Switched shunt devices designated as operating in discrete mode (i.e., MODSW = 1) are initially treated as the continuously operating devices described above. When the network solution convergence tolerance is reached, the discretely operating shunts are set to the nearest discrete step. The network solution iteration then continues with the switched shunts held constant.

4. Switched shunts controlling local or remote bus voltage to a voltage band (i.e., VSWHI > VSWLO) *must* be designated as operating in discrete mode. Such devices are moved in single steps between network solution iterations (refer to Switched Shunt Adjustment).

5. Continuous mode control to a voltage band is not allowed. Such devices generate an alarm and are treated as controlling to a voltage setpoint (taken as the midpoint of the specified voltage band) in continuous mode as described in (3a) above.

6. Those switched shunts controlling other voltage controlling devices check the reactive power output of the controlled device against the more restrictive limits described in Switched Shunt Data. If the reactive

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power is outside of these limits, the switched shunt admittance is raised or lowered so as to bring the reactive output of the controlled device toward the nearer of the more restrictive limits. Assuming the input order adjustment method in the example in Switched Shunt Data, if the output of the controlled voltage controlling device exceeds 25 Mvar, the reactive output of the switched shunt will be increased: if the switched shunt has inductors switched on, one step of inductors will be switched off; otherwise one step of capacitors will be switched on.

An adjustment monitor is printed at the *Progress* tab any time a stepwise adjustment as in (4) and (6) above, or the setting to the nearest discrete step as in (3b) above, occurs.

When activity SOLV is run on a system that had previously been solved by one of the PSS®E network solution activities, the presence of discretely operating switched shunts as described in (3b) above may result in more than one network solution iteration being required. Although the network was expected to be in balance, significant voltage changes may initially be imposed as these devices are (temporarily) switched to the continuous mode. This may be overcome by locking the discrete mode switched shunts at their present settings (refer to Section 6.3.20, "Automatic Adjustments" and Switched Shunt Adjustment).

6.3.16. FACTS Devices

PSS®E's FACTS device model (refer to FACTS Device Data and ???) contains a shunt element that is connected between the sending bus and ground, and a series element connected between the sending and terminal buses. A unified power flow controller (UPFC) has both the series and shunt elements active, and allows for the exchange of active power between the two elements (i.e., TRMX is positive). A static series synchronous condenser (SSSC) is modeled by setting both the maximum shunt current limit (SHMX) and the maximum bridge active power transfer limit (TRMX) to zero (i.e., the shunt element is disabled). An interline power flow controller (IPFC) is modeled with a pair of series FACTS devices; the shunt current limits (SHMX) of both devices are set to zero, TRMX of the master device is set to the maximum active power transfer between the two devices, and TRMX of the slave device is set to zero. A static synchronous condenser (STATCON) or static compensator (STATCOM) is modeled by a FACTS device for which the terminal bus is specified as zero (i.e., no series element is present).

For an in-service FACTS device to be modeled during power flow solutions, it must satisfy the following conditions:

1. The sending bus must be either a type 1 or type 2 bus.
2. The sending bus must not be connected by a zero impedance line to type 3 bus.
3. If it is specified, the terminal bus must be a Type 1 bus with exactly one in-service ac branch connected to it; this branch must not be a zero impedance line and it must not be in parallel with the FACTS device.
4. If it is specified, the terminal bus must not have a switched shunt connected to it.
5. If it is specified, the terminal bus must not be a converter bus of a dc line.
6. A bus that is specified as the terminal bus of an in-service FACTS device may have no other in-service FACTS device connected to it. However, multiple FACTS device sending ends on the same bus are permitted.
7. A bus that is specified as the terminal bus of an in-service FACTS device may not have its voltage controlled by any remote generating plant, switched shunt, or VSC dc line converter.

The FACTS device model is called at the start of each iteration to set the boundary conditions to be imposed upon the ac network at the sending and terminal buses during that iteration.

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The shunt element at the sending bus is used to hold the voltage magnitude at either the sending bus or the specified remote bus to VSET, subject to the sending shunt current limit SHMX. This is handled in power flow solutions in a manner similar to that of synchronous condensers and continuous switched shunts. In terms of the boundary conditions they impose at the buses to which they are connected, these three devices differ only in their representation of limits: synchronous condensers have Mvar limits; switched shunts have admittance limits; and the shunt elements of FACTS devices have current limits. Current in the shunt-connected bridge is determined by both the shunt Mvar output and the amount of active power transferred between the shunt and the series elements. If the current limit is violated, sending end voltage control is abandoned and the magnitude of the shunt Mvar output is reduced.

The series element may be set to operate in one of eight modes as described in the following sections.

Normal Mode

The normal operating mode of the series element is enabled by setting the control mode of the FACTS device to 1 (refer to FACTS Device Data). In unconstrained operation, the series element is used to maintain the desired active (PDES) and reactive (QDES) power flow between the sending and terminal buses. If the series current limit (IMX) is violated, the magnitude of the desired reactive power flow is reduced; if reducing the desired reactive power flow to zero still results in a series current limit violation, the magnitude of the desired active power flow is reduced as required. With desired power setpoints established (either as specified or as reduced due to a series current limit violation), the bus boundary conditions, which are to be presented to the power flow solution, are determined.

Limits on three quantities may prohibit the series element from maintaining power flow at the established active and/or reactive power setpoints: the series voltage magnitude (maximum of VSMX); the terminal bus voltage magnitude (maximum of VTMX and minimum of VTMN); and the magnitude of the active power transfer between the shunt and series-connected bridges (maximum of TRMX). These limits result in seven possible states of the model:

1. Series voltage, terminal bus voltage, and bridge active power transfer are all within their limits and are therefore treated as dependent variables. Active and reactive power flows meet their established setpoints.

2. Series voltage is held at VSMX; terminal bus voltage and bridge active power transfer are both dependent variables. If possible, active power flow is maintained at its established setpoint and reactive power flow control is abandoned; otherwise, active power flow control is abandoned and the model attempts to hold reactive power flow at its established setpoint.

3. Terminal bus voltage magnitude is held at either VTMX or VTMN; series voltage and bridge active power transfer are both dependent variables. Reactive power flow control is abandoned and active power flow is maintained at its established setpoint.

4. Bridge active power transfer is held at either TRMX or -TRMX; series voltage and terminal bus voltage are both dependent variables. Reactive power flow control is abandoned and active power flow is maintained at its established setpoint.

5. Series voltage is held at VSMX and terminal bus voltage magnitude is held at either VTMX or VTMN; bridge active power transfer is a dependent variable. Both active and reactive power flow control are abandoned.

6. Series voltage is held at VSMX and bridge active power transfer is held at either TRMX or -TRMX; terminal bus voltage is a dependent variable. Both active and reactive power flow control are abandoned.

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7. Terminal bus voltage magnitude is held at either VTMX or VTMN and bridge active power transfer is held at either TRMX or -TRMX; series voltage is a dependent variable. Both active and reactive power flow control are abandoned.

Generally, the model remains in its current state until network convergence is achieved. Then model quantities are checked to determine if a change in model state is required (i.e., if a limited quantity can be made a dependent variable, or if a dependent variable violates its limit). If a change of model state is applied, the power flow solution continues until convergence is again achieved, and then the model is again tested for a change of state.

The PSS®E FACTS model will not attempt to solve for the state in which all three of the above quantities are simultaneously held at a limit. Such a state defines an over-constrained problem. In such a system condition, the FACTS model will usually cycle among model states (5) through (7) above. It is the user's responsibility to decide which of the three limits is to be relaxed, and then make the appropriate data change.

Bypassed Mode

The bypassed operating mode of the series element is enabled by setting the control mode of the FACTS device to 2 (refer to FACTS Device Data). In this mode, the series element is solved to a series voltage magnitude of zero. No limits are enforced other than SHMX, the maximum shunt current at the sending bus. The solution is identical to that which would be obtained by replacing the series element with a zero impedance line.

Constant Series Impedance Mode

The constant series impedance operating mode of the series element is enabled by setting the control mode of the FACTS device to 3 (refer to FACTS Device Data). In this mode, the series element is modeled with a fixed series impedance specified on system base. No limits are enforced other than SHMX, the maximum shunt current at the sending bus.

Constant Series Voltage Mode

The constant series voltage operating mode of the series element is enabled by setting the control mode of the FACTS device to 4 (refer to FACTS Device Data). In this mode, the series element is solved to a specified complex series voltage expressed relative to the value indicated by VSREF (i.e., relative to either sending end voltage or series current). No limits are enforced other than SHMX, the maximum shunt current at the sending bus.

IPFC Master and Slave Modes

An Interline Power Flow Controller (IPFC) may be modeled using two series FACTS devices. One device of this pair must be assigned as the IPFC master device by setting its FACTS device control mode to 5 or 7; the other must be assigned as its companion IPFC slave device by setting its control mode to 6 or 8 and specifying the name of the master device in its MNAME (refer to FACTS Device Data).

In the IPFC model, both devices have a series element but no shunt element. The master device is solved in the same manner as a series device in Normal Mode when the control mode is 5 or in Constant Series Voltage Mode when the control mode is 7, except that the active power transfer is exchanged with the series element of the slave device. The slave device is always solved with its bridge active power transfer fixed as dictated by the master device; that is, when its control mode is 6, it is always in one of solution states 4, 6, or 7. When its control mode is 8, the specified value of series voltage is modified to reflect the active power exchange dictated by the master.

In control mode 8 when VSREF is 1 (i.e., desired series voltage is specified relative to series current), the specified Vd is replaced with that calculated from the active power exchange determined by the master and

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the series current of the slave. In control mode 8 when VSREF is 0 (i.e., desired series voltage is specified relative to sending bus voltage), the specified Vd and Vq are first transformed to the series current axis, and the resulting Vd is then replaced with that calculated from the active power exchange determined by the master and the series current of the slave.

Therefore, both devices typically have SHMX set to zero, and VSET of both devices is ignored. TRMX of the master device is the maximum active power exchange between the two devices, and TRMX of the slave device is set to zero. QDES of the slave device is ignored.

All Modes

It is possible for the power flow iteration to continue even though its voltage change or mismatch tolerance has been achieved. This can occur in several model states described above when the limited quantity is too far away from the limit being held.

The solution of the series FACTS device in all of the above model states except the constant series impedance mode includes the temporary insertion into the network of ac series and shunt elements and a corresponding Norton current injection at the sending and terminal buses of the FACTS device. This technique can improve the convergence properties of the power flow solution. The insertion of these dummy elements is handled automatically by the FACTS model using the reactance specified as LINX. If the convergence monitor indicates an underaccelerated convergence, increasing LINX (which decreases its effect) may be helpful. When the shunt element is at a current limit, and therefore the sending end voltage magnitude is changing from one iteration to the next, slow network convergence may be observed with several of the model states described above. Manually adjusting LINX can improve the convergence characteristics.

6.3.17. DC Lines

The two-terminal dc line (refer to Two-Terminal DC Transmission Line Data), multi-terminal dc line (refer to Multi-Terminal DC Transmission Line Data), and voltage source converter (VSC) dc line (refer to Voltage Source Converter (VSC) DC Transmission Line Data) models are called at the start of each iteration to set the active and reactive powers flowing from the ac network into the dc line at each converter bus during that iteration. If a dc line is blocked by its model, an appropriate message is printed at the *Progress* tab (refer to Section 4.4, "Virtual Output Devices").

Capacitor Commutated DC Lines

At each power flow solution iteration, capacitor commutated two-terminal dc lines require an iterative Newton solution to compute the active and reactive power boundary conditions. If this Newton solution fails to converge, an appropriate message is printed at the *Progress* tab.

VSC dc Lines

Whenever converter limits are checked by the VSC dc line model, the following procedure is used:

1. Convert the current limit IMAX to MVA using the present ac voltage and the base voltage of the converter bus. If IMAX is positive, but the base voltage at the converter bus is zero, unlimited current loading is permitted.

2. Use IMAX converted to MVA or SMAX, whichever is smaller, as the converter limit.

3. If in power factor mode, limits are checked during each power flow iteration as follows:

    a. Get the MVAR corresponding to the desired MW.

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b. If |P + jQ| exceeds the limit from 2, reduce the magnitude of P and/or Q using PWF.

4. If in ac voltage control mode, limits are checked whenever convergence of the power flow iteration is achieved as follows:

   a. Get the MVAR from the last iteration.

   b. If |P + jQ| exceeds the limit from 2, reduce the magnitude of P and/or the appropriate Q limit using PWF.

   c. If, on any iteration during which limits are checked, both converters are controlling voltage *and* both are loaded beyond a limit, only the converter with the larger overload has its loading reduced so that it is at its limit.

   d. Whenever a limit violation is detected and relieved are described above, the convergence flag is reset and the power flow solution continues.

5. Whenever a converter has its active power order reduced due to a limit violation, the active power order at the other converter is reduced accordingly.

Power orders and/or reactive power limits are reduced such that the following two equations are simultaneously satisfied:

$$(P_i + \Delta P)^2 + (Q_i + \Delta Q)^2 = SMAX^2 \quad (6.14)$$

$$\frac{\Delta P}{\Delta Q} = \frac{P_i}{Q_i} \times \frac{(1.0 - PWF)}{PWF} \quad (6.15)$$

Thus, if PWF is 0.0, only active power is reduced; if PWF is 1.0, only reactive power is reduced; otherwise, the above equations are applied.

One of two solution strategies is used for solving the conditions at a VSC dc line at each iteration. The following approach is used when the following three conditions are satisfied: both converters are in ac voltage control mode; limits are being checked on this iteration; and both converters are overloaded.

1. At the converter where the amount of overload is greater, relieve the overload as described above.

2. Solve for the dc current using the specified dc voltage at the voltage controlling converter, the active power order and the converter loss coefficients at the converter where the overloading was relieved, and the dc line resistance.

3. Calculate the active power order at the other converter.

In all other situations, the following solution approach is used:

1. At the power controlling converter, check limits as described above.

2. Solve for the dc current using the specified dc voltage at the voltage controlling converter, the active power order and the converter loss coefficients at the power controlling converter, and the dc line resistance.

3. Calculate the active power order at the voltage controlling converter.

4. Check limits at the voltage controlling converter as described above.

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5. If a limit is violated at the voltage controlling converter, calculate a new power order for the power controlling converter and repeat steps 1 and 2.

When activity SOLV is run on a system that had previously been solved by one of the PSS®E network solution activities, the presence of VSC converters that control ac voltage and which are at an MVA or current limit usually results in more than one network solution iteration being required. Although the network should have been in balance, significant voltage changes may initially be imposed as these devices are solved with limits ignored until convergence is achieved.

6.3.18. AC Voltage Control

The discussions of the various types of equipment above and in Section 5.2.1, "Power Flow Raw Data File Contents" indicate that the power flow model may include several types of ac voltage controlling equipment. Each such device can be included in one of the following categories:

1. Swing bus.
2. Setpoint mode voltage control device.
3. Band mode voltage control device.

The following paragraphs discuss each of these types of ac voltage control.

Swing Bus

A power flow case may be considered as being composed of one or more ac islands. An ac island is defined as the set of all in-service buses such that each bus in the island may be reached from every other bus in the island through the in-service ac network. If a working case contains more than one island, any island is either electrically disjoint from all other islands, or connected to one or more islands by one or more asynchronous ties (e.g., dc lines).

To be a valid power flow case, each ac island in the working case must contain at least one Type 3 (swing) bus. Furthermore, the buses connected by each in-service branch must be in-service (i.e., none of them may be Type 4 buses). When enabled, the solution connectivity checking option (refer to Saved Case Specific Option Settings) checks for the presence of swingless islands at the start of each power flow solution. Furthermore, activity TREE may be used to check for the presence of swingless islands, as well as to disconnect any or all such islands and to check for the presence of in-service ac branches connected to Type 4 buses.

During power flow solutions, the voltage magnitude *and phase angle* at each Type 3 bus are held constant. The voltage magnitude is held to the value specified by VS in the Type 3 bus generator data (refer to Generator Data), and the phase angle is held to the value specified by VA in the Type 3 bus data (refer to Bus Data) or to zero degrees if a *flat start* is specified. At the completion of each power flow solution, the active and reactive power generation at each Type 3 bus are set such that the mismatch at the bus is zero. Each Type 3 bus *must* have at least one in-service machine connected to it.

While more than one Type 3 bus may be present in an island, this is generally not recommended. It is the user's responsibility to coordinate their phase angle settings when multiple swing buses are present in an island. Furthermore, care must be taken in the use of the *flat start* option of the power flow solution activities, which always sets the phase angles of all buses, including type 3 buses, to zero degrees.

Setpoint Voltage Control

Those devices that provide for voltage control to a setpoint have their reactive power output or consumption set during each power flow iteration such that the voltage magnitude at the controlled bus is held at its

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scheduled value as long as the device's reactive power limits are not violated. These devices provide for either local bus voltage control or voltage control of a remote type 1 or 2 bus.

The following setpoint mode voltage controlling devices may be modeled in PSS®E:

- An in-service generating plant at a type 2 bus may control either local bus voltage or the voltage at a remote bus. Refer to Generator Data and Section 6.3.12, "Generation".
- A switched shunt for which MODSW is 1 or 2 at a type 1 or 2 bus may control either local bus voltage or the voltage at a remote bus to a voltage setpoint (refer to Switched Shunt Data). Such switched shunts for which MODSW = 1 (i.e., discrete control) are initially treated as continuous, then moved to the nearest step and locked, as described in (3b) of Section 6.3.15, "Switched Shunt Devices".
- A voltage controlling converter of a VSC dc line at a type 1 or 2 bus may control either local bus voltage or the voltage at a remote bus. To be a voltage controlling converter, the VSC dc line must be in-service (MDC = 1), the converter must be in-service (TYPE = 1 or 2), and the ac control mode must be set for voltage control (MODE = 1). Refer to Voltage Source Converter (VSC) DC Transmission Line Data and VSC dc Lines.
- The shunt element at the sending end of an in-service FACTS device at a type 1 or 2 bus may control either local bus voltage or the voltage at a remote bus. Refer to FACTS Device Data and Section 6.3.16, "FACTS Devices".

When multiple voltage controlling devices control the same bus voltage, the reactive power requirements are shared among the controlling devices in proportion to their RMPCTs, subject to each device's reactive power, current or admittance limits, as appropriate. The Mvar output of each device controlling the voltage at bus I is set to the product of the total Mvar required at all devices controlling the voltage at bus I times a fraction; the numerator of this fraction is RMPCT of the device and the denominator is the sum of the RMPCTs of all the devices controlling the voltage at bus I. It is the responsibility of the user to ensure that voltage control specifications for all the setpoint mode devices controlling the same bus voltage are identical, and that the RMPCT values are such that the desired reactive power sharing is obtained.

When multiple setpoint mode voltage controlling devices are present at a bus, they should normally all be specified with the same voltage control objective. Similarly, when multiple setpoint mode voltage controlling devices are present among a group of buses connected together by zero impedance lines, the same voltage control objective should be specified for all of them. (Recall that buses connected together by zero impedance lines are treated as the same bus during power flow solutions; refer to Zero Impedance Lines.)

When one or more devices are controlling the voltage at a remote bus, any setpoint mode voltage controlling devices at the controlled bus should be specified so as to control local bus voltage.

Other than generation at the bus itself, setpoint mode voltage controlling devices at type 3 buses and at buses connected to type 3 buses through zero impedance lines may not be used for voltage control purposes. Such devices are handled as follows by the power flow solution activities:

- A zero impedance line connected generating plant is held at its present active and reactive power output.
- A local or zero impedance line connected switched shunt controlling to a voltage setpoint is locked at its present admittance.
- A local or zero impedance line connected VSC dc line is blocked.
- A local or zero impedance line connected FACTS device is blocked.

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Activity CNTB may be used to check for suspect or erroneous voltage control specifications.

Band Mode Voltage Control

Those devices that control voltage at a bus to a voltage band or the reactive power output of another voltage controlling device are termed band mode voltage controlling devices. Their settings are adjusted between (rather than during) power flow iterations.

The following band mode voltage controlling devices may be modeled in PSS®E:

- A switched shunt for which MODSW is 1 at a type 1 or 2 bus may control either local bus voltage or the voltage at a remote bus to a voltage band (refer to Switched Shunt Data and Section 6.3.15, "Switched Shunt Devices").
- A switched shunt for which MODSW is 3 at a type 1 or 2 bus may control the reactive power output of a generating plant to fall within a reduced reactive power limit band (refer to Switched Shunt Data and Section 6.3.15, "Switched Shunt Devices").
- A switched shunt for which MODSW is 4 at a type 1 or 2 bus may control the reactive power output of a voltage controlling VSC dc line converter to fall within a reduced reactive power limit band (refer to Switched Shunt Data and Section 6.3.15, "Switched Shunt Devices").
- A switched shunt for which MODSW is 5 at a type 1 or 2 bus may control the admittance setting of a remote switched shunt to fall within a reduced reactive admittance limit band (refer to Switched Shunt Data and Section 6.3.15, "Switched Shunt Devices").
- A switched shunt for which MODSW is 6 at a type 1 or 2 bus may control the reactive power output of the shunt element of FACTS device to fall within a reduced reactive current limit band (refer to Switched Shunt Data and Section 6.3.15, "Switched Shunt Devices").
- A transformer may have its Winding 1 turns ratio adjusted to hold the voltage at a designated bus within a specified band (refer to Transformer Data and Voltage Control).
- The terminal end of an in-service FACTS device at a type 1 or 2 bus may deviate from its desired power setpoints to keep the voltage at the terminal bus within a specified band. Refer to FACTS Device Data and Section 6.3.16, "FACTS Devices".

6.3.19. Characteristics of Activity SOLV

The Gauss-Seidel and the Modified Gauss-Seidel have much of their operation and characteristics in common. This section will summarize the aspects of the Gauss-Seidel method. Refer to Section 6.4, "Applying Modified Gauss-Seidel Power Flow Solution" to review differences with the Modified method.

The Gauss-Seidel solution method has five available solution control parameters:

- ACCP - Acceleration factor for the real part of the voltage adjustment.
- ACCQ - Acceleration factor for the imaginary part of the voltage adjustment.
- TOL - Convergence tolerance in pu.
- ITMX - Default limit on number of iterations. (= 100)

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- BLOWUP - Largest voltage change threshold.

The acceleration factor ACCM is used only for the Modified Gauss-Seidel method.

The nodal iterative solution method of activity SOLV uses separate acceleration factors (ACCP and ACCQ) for the real and imaginary parts of the voltage adjustment. Both of these factors have default values of 1.6, but ACCP need not be equal to ACCQ. The guide to the tuning of the acceleration factors should be:

- If the voltage change on successive iterations is oscillating in magnitude and/or sign, decrease acceleration.
- If the voltage change on successive iterations is decreasing smoothly, convergence *may* be improved by increasing acceleration (see Figure 6.8, "Dependence of Power Flow Convergence on Acceleration Factors").

The acceleration factors should never be set greater than two, and the optimum seldom exceeds about 1.8. Although the acceleration must be tuned to the system for optimum performance, deviations from the optimum values do not have a dramatic effect on the number of iterations required.

The normal tolerance value for the Gauss-Seidel solution is 0.0001 per unit. This tolerance is applied to the largest voltage change in each iteration and convergence is assumed when | Δ V| is less than this tolerance. The tolerance may be increased as far as 0.0005 per unit if only approximate power flows are needed. Reducing the tolerance as low as 0.00001 per unit is permissible to achieve reduced node mismatches, but such a small tolerance is not recommended because the convergence of the Gauss-Seidel method becomes very slow as the voltage changes fall below about 0.0001 per unit.

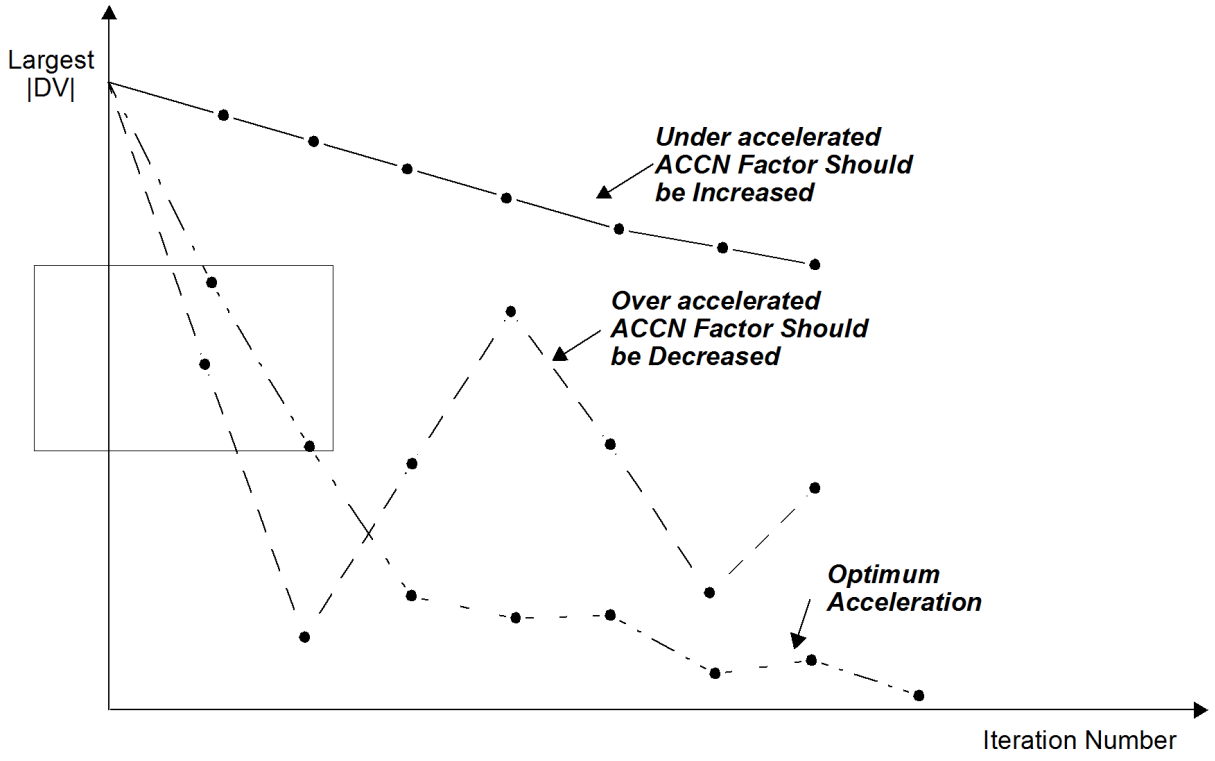

Figure 6.8. Dependence of Power Flow Convergence on Acceleration Factors

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At the completion of each iteration, the largest voltage change is checked against a blowup tolerance, which has a default value of 5.0 pu. If the largest change exceeds this tolerance, an appropriate message is printed and activity SOLV is terminated.

The General solution parameters are independent of the type of iterative solution. In addition to the BLOWUP parameter, previously described, they are:

- *PQBRAK* – This is the voltage level below which the load characteristic will change from constant MVA to an elliptical current-voltage characteristic. (see Section 6.3.3, "Boundary Conditions of Constant MVA Loads").

- *THRSHZ* – This is the threshold reactance for branches below which they will be treated as zero impedance lines. (see Zero Impedance Lines)

- *ADJTHR* – When tap adjustment is enabled for the Gauss-Seidel and the Full Newton-Raphson solutions, those transformers controlling voltage are checked for adjustment whenever the largest voltage magnitude change in per unit on the previous iteration is less than the automatic adjustment threshold tolerance, ADJTHR. When using the decoupled and fixed slope decoupled Newton-Raphson solutions, this adjustment check occurs following any *P-angle half iteration* in which the largest voltage phase angle change in radians is less than ADJTHR.

- *ACCTAP* – This is a tap movement deceleration factor. The voltage error, for a voltage controlling transformer is multiplied by ACCTAP. If ACCTAP is set less than its default value of 1.0, the apparent voltage error is reduced and the number of tap changes occurring during the iterative process is reduced.

- *TAPLIM* – The solution parameter TAPLIM, which has a default value of 0.05, defines the maximum ratio change that may be applied to a transformer during any adjustment calculation. If the tap STEP of a transformer exceeds TAPLIM and the controlled voltage is outside of its voltage band, the ratio is changed by one tap STEP.

- *SWVBND* - The solution parameter SWVBND specifies the percentage of switched shunts controlling to a voltage band, and for which the voltage at the controlled bus is outside of the specified band, that can be adjusted on any single adjustment cycle.

- *MXTPSS* - The solution parameter MXTPSS specifies the maximum number of power flow iterations during which the adjustment of transformer tap ratios and/or the adjustment of switched shunts that control to a voltage band or that control the reactive power output of another voltage controlling device may occur.

- *MXSWIM* - The solution parameter MXSWIM specifies the maximum number of power flow iterations during which an induction machine may be switched from the "running" state to the "stalled" or "tripped" state. Any machine for which this limit is reached remains in the "stalled" or "tripped" state for the remainder of the power flow solution.

Reducing ACCTAP and/or TAPLIM below their default values may be beneficial on systems with a high concentration of Load Tap Changers or where the voltage profile is extremely sensitive to tap settings.

Activity SOLV has a default limit of 100 iterations and a voltage convergence tolerance of 0.0001 pu. This tolerance applies to the largest voltage change each iteration, and convergence is assumed when the magnitude of the largest bus voltage change is less than this tolerance. Reducing the tolerance as low as 0.00001 pu is permissible to achieve reduced node mismatches, but such a small tolerance is not recommended for activity SOLV because the convergence of the Gauss-Seidel method becomes very slow as the voltage changes fall below about 0.0001 pu.

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The user may modify any of these solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

The following rules and convergence characteristics apply to activity SOLV:

1. The presence of negative reactance branches in a network usually causes activity SOLV to diverge.
2. The presence of very low impedance branches (e.g., jumpers with an impedance of j0.0001) that are not being treated as zero impedance lines (refer to Zero Impedance Lines) often results in slow convergence as the voltage adjustments get small, and in mismatches at the buses involved.
3. The number of iterations required to reach the convergence tolerance increases as the system size increases.
4. Activity SOLV may be used if the initial voltage estimate is poor.
5. Activity SOLV is tolerant of reactive power problems.
6. Activity SOLV is tolerant of data errors and insoluble conditions in local areas of the network. It is generally well converged everywhere except in the problem areas.
7. When it diverges, it usually fails gently.

6.3.20. Automatic Adjustments

Activity SOLV provides for the following solution options:

1. Adjustment of transformer off-nominal turns ratio to hold the voltage at a designated bus within a specified band.
2. Adjustment of area swing bus generator powers to hold area net interchange within a specified band.
3. Flat start.
4. Adjustment of the turns ratios of converter transformers at dc transmission terminals and of ac network transformers controlling dc line quantities.
5. Adjustment of switched shunt devices. When this adjustment option is disabled, discretely operating switched shunts are moved to their nearest step, if necessary, and all switched shunts, including continuously operating devices, are held constant. When this adjustment option is enabled, either all adjustable switched shunts may be adjusted, or continuous mode switched shunts are subject to adjustment and discrete mode switched shunts are locked.
6. Ignoring generator reactive power limits at Type 2 buses with unequal var limits.

PSS®E program option settings exist for adjustments (1), (2), (4), and (5) above. Activity OPTN may be used to establish the default setting for each of these adjustment options (refer to Section 3.3.3, "Program Run-Time Option Settings"). Then, on each entry into activity SOLV, these option settings determine which of the adjustment options are activated.

The default responses corresponding to the adjustments for which a PSS®E option setting exists are the current values of these option settings; the default responses for the *flat start* and *var limits* items are zeros.

For transformer ratio adjustment, (#1 above), the adjustment is enabled if the corresponding option setting is either 1 (stepping) or 2 (direct). Refer to Voltage Control.

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For further discussion of solution options, refer to Section 6.5.2, "Automatic Adjustments".

Additional Information
PSS®E GUI Users Guide, Gauss-Seidel Power Flow Solution *PSS®E* Command Line Interface (CLI) Users Guide, Running Gauss-Seidel Power Flow Solution

6.4. Applying Modified Gauss-Seidel Power Flow Solution

MSLV

Run Activity MSLV - GUI
Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)... *[Loadflow solutions]* Gauss Tab OPTION: Modified Gauss-Seidel
Run Line Mode Activity MSLV - CLI
ACTIVITY? >>**MSLV**
Interrupt Control Codes
AB, NC, NM, DC, FD, MO

The power flow solution activity MSLV uses a modified Gauss-Seidel iterative algorithm to solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. A secondary adjustment applied based on each primary voltage change enables negative reactance branches to be represented between Type 1 buses.

Activity MSLV handles the network topology check, the network connectivity checking option, flat start solution, selection and application of automatic adjustments, generator, load, switched shunt, FACTS device, and dc line boundary conditions, the blowup check, and interrupt codes in the same way as activity SOLV. The solution convergence monitor, FACTS device monitor, dc transmission line monitors, induction machine monitor, largest mismatch tabulation, swing bus summary, and area interchange violation summary are identical to those of activity SOLV. Refer to SOLV, particularly the sections regarding Generation, Load, Switched Shunt Devices, FACTS Devices, DC Lines, AC Voltage Control, and Automatic Adjustments.

Unlike the Gauss-Seidel solution method, the Modified Gauss-Seidel iterative algorithm employs a secondary adjustment based on each primary voltage change to enable negative reactance branches to be represented between Type 1 buses. Activity MSLV applies the standard Gauss-Seidel voltage adjustment formula at Type 2 buses and a modified formula at Type 1 buses. This allows it to handle series capacitors connecting Type 1 buses. (MSLV usually *cannot* handle negative reactance branches connected to Type 2 or 3 buses.)

Activity MSLV uses the same acceleration factors, ACCP and ACCQ, at Type 2 buses as does activity SOLV. At Type 1 buses, it uses a separate acceleration factor, ACCM, which has a default value of unity. The principles for setting ACCM are the same as outlined in Section 6.3.19, "Characteristics of Activity SOLV" for setting ACCP and ACCQ. The convergence of activity MSLV is, however, much more sensitive to the value of ACCM than activity SOLV is to the values of ACCP and ACCQ.

Changing ACCP and ACCQ by 0.05 usually has very little effect on the convergence of SOLV and MSLV, but the same change in ACCM may cause a major change in the convergence properties of MSLV. Typical values

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for ACCM range from about 1.2 for well-behaved systems without series capacitors down to slightly below unity in difficult cases with series capacitors.

The parameters designating the maximum number of iterations, the convergence tolerance, and the blowup threshold are shared with activity SOLV. The user may modify any of the solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

The following rules and convergence characteristics apply to activity MSLV:

1. The presence of negative reactance branches connected to Type 2 or 3 buses usually causes activity MSLV to diverge.
2. Series capacitors may be represented between Type 1 buses, as long as the level of compensation does not exceed about 80%.
3. The presence of very low impedance branches (e.g., jumpers with an impedance of j0.0001) which are not being treated as zero impedance lines (refer to Zero Impedance Lines) often results in slow convergence as the voltage adjustments get small, and in mismatches at the buses involved.
4. The number of iterations required to reach the convergence tolerance increases as the system size increases.
5. Activity MSLV may be used if the initial voltage estimate is poor.
6. Activity MSLV is tolerant of reactive power problems.
7. Activity MSLV is tolerant of data errors and insoluble conditions in local areas of the network. It is generally well converged everywhere except in the problem areas.
8. The rate of convergence is very sensitive to the tuning of the acceleration factor ACCM. Modest deviation from the optimum value may result in poor convergence characteristics.
9. Activity MSLV takes somewhat more time per iteration than does activity SOLV, but generally requires fewer iterations.

Additional Information
PSS®E GUI Users Guide, Modified Gauss-Seidel Power Flow Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Modified Gauss-Seidel Power Flow Solution

6.5. Applying Fully-Coupled Newton-Raphson Power Flow Solution

FNSL

Run Activity F*NSL - GUI*
Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)... *[Loadflow solutions]* Newton Tab OPTION: Full Newton-Raphson

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Run Line Mode Activity FNS*L* - *CLI*
ACTIVITY? >>**FNSL** ENTER ITERATION NUMBER FOR VAR LIMITS   0 FOR IMMEDIATELY, -1 TO IGNORE COMPLETELY: >>
Interrupt Control Codes
AB, NC, NM, DC, FD, MO, NV

The power flow solution activity FNSL uses a fully coupled Newton-Raphson iterative algorithm to solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case.

Differences from the Gauss method are as follows:

- Tap and phase adjustment options are broken out to be independent.
- Imposition of VAR limits has more options.
- A non-divergent solution is available only for the full Newton and the fixed slope decoupled Newton solutions.

Activity FNSL requires the prior execution of activity ORDR. If the need for a new bus ordering is detected, activity ORDR is automatically executed before beginning the voltage change calculation.

The user specifies the number of iterations for which generator reactive power limits are applied. In normal (default) mode, reactive power limits are ignored until the largest reactive power mismatch has been reduced to a preset multiple of the mismatch convergence tolerance, TOLN. Set at 0, reactive power limits are recognized on the first mismatch calculation, preceding the first iteration. A positive number < n > causes reactive power limits to be applied either on iteration number < n > or when the largest reactive power mismatch is within a preset multiple of the tolerance, whichever occurs first.

The var limit logic of activity FNSL contains code to prevent the phenomenon of a setpoint mode voltage controlling device oscillating on and off a limit from one iteration to the next, or between high and low limit from one iteration to the next. This is accomplished by going back to check the reactive power requirement and voltage magnitude on the previous iteration. Consequently, because the logic presumes the existence of a previous iteration for the conditions in the working case, the immediate application of generator reactive power limits (i.e., a response of zero) should be specified only when *continuing* the solution of the system in the working case (i.e., following SOLV, MSLV, FNSL, NSOL, or FDNS).

The application of generator reactive power limits on the first iteration following a network change *may* cause setpoint mode voltage controlling devices to be spuriously placed on a limit (e.g., a generator's reactive power set at the low limit but voltage magnitude less than scheduled voltage). Activity FNSL prints a summary of any such voltage controlling devices at the *Progress* tab prior to returning control back to the PSS®E interface (refer to Section 6.5.1, "Characteristics of Activity FNSL").

The reactive power component limits of voltage controlling switched shunts, VSC converters, and shunt elements of FACTS devices are always recognized.

Activity FNSL handles the netowrk topology check, the network connectivity checking option, flat start solution, and load, generator, switched shunt, FACTS device, and dc line boundary conditions in the same way as does activity SOLV. In addition to the interrupt control codes recognized by activity SOLV, the NV interrupt control code may be used to suppress the incorrect var limit tabulation printed at the termination of activity FNSL. The FACTS device monitor, dc transmission line monitors, induction machine monitor, largest

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mismatch tabulation, swing bus summary, and area interchange violation summary are identical to those of activity SOLV. Refer to SOLV, particularly the sections regarding Generation, Load, Switched Shunt Devices, FACTS Devices, DC Lines, and, AC Voltage Control.

During each iteration, activity FNSL prints a convergence monitor, which tabulates:

1. The iteration number.
2. The largest active power mismatch in per unit and the bus at which it occurs (refer to #1 in the convergence criteria described in Section 6.5.1, "Characteristics of Activity FNSL").
3. The largest reactive power mismatch in per unit from among those buses with a PQ boundary condition and the bus at which it occurs (refer to #2 in the convergence criteria described in Section 6.5.1, "Characteristics of Activity FNSL").
4. The largest reactive power mismatch in per unit from among those buses with a PV boundary condition and the bus at which it occurs (refer to #3 in the convergence criteria described in Section 6.5.1, "Characteristics of Activity FNSL"). This line is included only if this value exceeds the reactive power mismatch in 3. above.
5. The largest voltage magnitude change in per unit and the bus at which it occurs.
6. The largest phase angle change in radians and the bus at which it occurs.

Buses are identified in the convergence monitor by their bus numbers, except for the star point buses of three-winding transformers which are identified by their transformer names.

```
 ITER     DELTAP     BUS      DELTAQ     BUS      DELTA/V/     BUS      DELTAANG     BUS
  0       1.6513     152      3.3331     202
                                                  0.02600      202      0.00977      202
  1       0.0479     202      1.4401     211
                                                  0.02646      211      0.00371      211
  2       0.0054     201      0.0713     211
                                                  0.00141      211      0.00020      211
  3       0.0000     201      0.0007     201


REACHED TOLERANCE IN   3 ITERATIONS

LARGEST MISMATCH:     0.00 MW     0.07 MVAR      0.07 MVA AT BUS     211 [HYDRO_G      20.000]
SYSTEM TOTAL ABSOLUTE MISMATCH:                  0.14 MVA

SWING BUS SUMMARY:
   BUS# X-- NAME --X BASKV      PGEN      PMAX     PMIN      QGEN      QMAX     QMIN
   3011 MINE_G       13.800     258.7     900.0      0.0     104.0     600.0  -100.0
```

Figure 6.9. Newton-Raphson Convergence Monitor (Fully Coupled)

6.5.1. Characteristics of Activity FNSL

The Newton-Raphson solution method has seven available solution control parameters. The user may modify any of these solution parameters using the data changing activity CHNG or the *[Solution Parameters]* dialog. The available solution parameters are:

- ACCN - the acceleration factor

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- TOLN - the largest mismatch in MW and Mvar
- ITMXN - the maximum number of iterations
- DVLIM - the largest change in bus voltage
- NDVFCT - the non-divergent improvement factor
- VCTOLQ - the controlled bus Q mismatch convergence tolerance
- VCTOLV - the controlled bus voltage error convergence tolerance

The Newton solutions apply an acceleration factor ACCN to the voltage adjustments made at voltage-controlled buses. This factoring overcomes stability problems that can arise when the Newton method encounters reactive power limits. The normal value of this acceleration factor is unity, but in extremely difficult cases it can be advantageous to set its value as low as 0.1. It will never be advantageous to set this acceleration factor greater than unity.

Activity FNSL scales down the voltage magnitude and voltage angle change vectors if the most negative element of the # *vmag* / *vmag vector* is less than or equal to -1.0. A more restrictive limit on the size of the largest voltage magnitude change that may be applied on any single iteration is provided by the solution parameter DVLIM; the element of the # *vmag* / *vmag vector* with the largest magnitude is checked against DVLIM, and, if it exceeds DVLIM, the change vectors are uniformly scaled down such that the largest voltage magnitude change applied is of magnitude DVLIM. The default value of DVLIM is 0.99; reducing it to as low as 0.05 *may* improve convergence properties in difficult cases.

At the completion of the first iteration, the largest Δ vmag / vmag change is checked against a blowup tolerance, BLOWUP, which has a default value of 5.0 pu; on subsequent iterations, the largest phase angle change as well as the largest voltage magnitude change are checked. The unscaled change vectors are used in the blowup check; i.e., the blown up condition is determined before any scaling down of the change vectors, as described in the preceding paragraph, is applied. If the largest change exceeds the blowup tolerance, an appropriate message is printed and activity FNSL is terminated. The blowup check is bypassed if the non-divergent Newton power flow solution option is enabled (refer to Section 6.5.3, “Non-Divergent Solution Option”).

Convergence is assumed when the following conditions are simultaneously satisfied:

1. The largest active power mismatch at each Type 1 and 2 bus must be less than the mismatch convergence tolerance, TOLN.
2. The largest reactive power mismatch at those buses at which a PQ boundary condition is applied must be less than the mismatch convergence tolerance, TOLN. This test applies to the following categories of Type 1 and 2 buses:
    a. Voltage at the bus is not subject to control to a setpoint.
    b. Voltage at the bus is subject to control to a setpoint (refer to Setpoint Voltage Control), but all of the devices participating in the control of bus voltage to its setpoint are at a reactive power limit.
3. The largest reactive power mismatch at those buses at which a PV boundary condition is applied must be less than the larger of the controlled bus reactive power mismatch convergence tolerance, VCTOLQ, and TOLN. This test applies to each Type 1 and 2 bus where voltage is subject to control to a setpoint, and for which at least one of the devices participating in the control of bus voltage to its setpoint is not at a reactive power limit. While this test is applied to all such buses, the reactive power mismatch will

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normally be zero (or very close to zero) at such buses which have local devices participating in the control of bus voltage to its setpoint, and at least one of these local devices is not at a reactive power limit.

4. The largest difference between actual and scheduled voltage magnitude in per unit at each Type 1 and 2 bus where voltage is subject to control to a setpoint, and for which at least one of the devices participating in the control of bus voltage to its setpoint is not at a reactive power limit, must be less than the controlled bus voltage error convergence tolerance, VCTOLV.

The Newton solution might not be able to reduce mismatch to the 0.001 per-unit value because of the inherent computer precision limit.

Consider a branch with impedance of 0.0001 per unit and voltages close to unity at both ends. The maximum imprecision in calculation of power flow through the branch is approximately:

$$\Delta P = 2\frac{\Delta V}{Z} = 0.0025 \text{ per unit}$$

Because the bus mismatch imprecision is the sum of the imprecisions in flows into connected branches, a power flow case including such low-impedance branches is unlikely to reach the default tolerance of 0.001 per unit.

The default mismatch tolerances (TOLN and VCTOLQ) are 0.1 MW and Mvar (0.001 pu on 100 MVA base), but tolerances as large as 1 MW and Mvar can give acceptable power flow solutions for many purposes. The default Newton power flow solution mismatch tolerance program option setting (refer to User Specified Option Settings) is the value assigned to these tolerances when a new case is read into PSS®E with activity READ. VCTOLQ may be set to a value larger than TOLN for cases with challenging remote voltage control specifications.

Activity FNSL has a default limit of 20 iterations (ITMXN). The default value of VCTOLV is 0.00001.

The user may modify any of these solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

Branch impedances as low as 0.0001 per unit do not occur frequently, but can arise in star-equivalents of three-winding transformers and are sometimes used to represent jumpers between bus sections. When these low-impedance branches exist in a power flow case, it is often advisable to raise the zero-impedance branch threshold to represent these branches as zero-impedance lines and then remove the low-impedance branches by joining the terminal buses of the branch together. (This topological manipulation is described in Section 5.16 Joining Buses). An alternative is to increase the tolerance to 0.0025 or 0.005 per unit.

A tolerance of 0.005 per unit on a 100-MVA system base represents a power flow solution imprecision of 0.5 MW, which is more than acceptable for the great majority of power flow cases.

Activity FNSL continues its iterations until one of the following occurs:

1. The bus mismatch and controlled bus voltage criteria described above are satisfied, and all FACTS devices are satisfactorily solved.

2. The iteration limit is exceeded.

3. The non-divergent solution option is disabled and the blown-up condition is detected.

4. The non-divergent solution option is enabled and applying the current iteration's change vectors does not significantly reduce the system mismatch level.

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5. The user interrupts the solution.

6. A diagonal element of the Jacobian matrix is near zero, indicating a singular matrix.

Prior to terminating, activity FNSL checks for setpoint mode voltage controlled buses with boundary condition inconsistencies.

First, any setpoint mode voltage controlled bus for which at least one of its controlling devices is not at a reactive power limit, but where voltage differs from its designated setpoint by more than the smaller of VCTOLV and 0.00001, is tabulated. For buses where the difference exceeds VCTOLV, the voltage is followed by an asterisk (*). Such buses may be present when ACCN is less than unity and further iterations are needed to step the bus voltage magnitude to within these tolerances of its desired value. Such buses may also be present when, on the last mismatch calculation before the solution is terminated, a voltage controlled bus switches from reactive power limited (a PQ boundary condition) to voltage controlled (a PV boundary condition).

Then, any setpoint mode voltage controlled bus where controlling devices are all at a reactive power limit, but where voltage magnitude is on the wrong side of its voltage setpoint by more than the smaller of VCTOLV and 0.00001, is tabulated (i.e., controlling devices are at their high reactive power limits and the voltage magnitude is greater than the setpoint, or the controlling devices are at their low reactive power limits and the voltage is less than the setpoint). For buses where the difference exceeds VCTOLV, the voltage is followed by an asterisk (*).

Following each report, the number of violations detected is tabulated. The list of buses that are in violation may be suppressed as a user option; the total number of violations is still tabulated.

The following rules and convergence characteristics apply to activity FNSL:

1. Activity FNSL converges in a very few iterations on well-conditioned cases, achieving very small bus mismatches.

2. Negative reactance branches are permitted in the network.

3. The presence of very low impedance branches (e.g., jumpers with an impedance of j0.0001) that are not being treated as zero impedance lines (refer to ORDR) may result in the inability of FNSL to reach the default convergence tolerances. Even with such a network, FNSL is usually capable of reducing the largest bus mismatch to less than 0.2 MVA.

4. The number of iterations required to reach the convergence tolerances is generally insensitive to system size.

5. Activity FNSL may diverge if the initial voltage estimate is poor.

6. Reactive power problems may cause poor convergence characteristics.

7. Applying generator reactive power limits too early may cause FNSL to diverge.

8. Activity FNSL is intolerant of data errors and insoluble conditions in local areas of the network.

9. When it diverges, it often fails catastrophically, giving no indication of where the problems are.

10. Activity FNSL requires the optimal ordering of network nodes. If it detects that a new ordering is required, a message is printed, activity ORDR is automatically executed, and activity FNSL continues.

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11. The time per iteration with FNSL is longer than that required for the other power flow solution activities, but fewer iterations are usually needed.

Convergence Testing Used in Prior Releases

The convergence testing described in the preceding section was implemented in PSS®E-30.3. This section describes the convergence criteria used in earlier releases of PSS ®E, and defines settings of the two solution parameters introduced at PSS®E-30.3 (VCTOLQ and VCTOLV) that will result in matching the convergence criteria used in earlier releases.

For a Newton-Raphson based power flow solution calculation to be considered converged in PSS®E-30.0 through PSS®E-30.2.2, the following conditions must have all been satisfied:

1. The largest active power mismatch at each Type 1 and 2 bus must be less than the mismatch convergence tolerance, TOLN.
2. The largest reactive power mismatch at each Type 1 and 2 bus must be less than the mismatch convergence tolerance, TOLN.
3. The largest difference between actual and scheduled voltage magnitude in per unit at each Type 1 and 2 bus where voltage is subject to control to a setpoint, and for which at least one of the devices participating in the control of bus voltage to its setpoint is not at a reactive power limit, must be less than 0.00001.

To match the convergence criterion of PSS®E-30.0 through PSS®E-30.2.2, set:

```
VCTOLQ = TOLN
VCTOLV = 0.00001
```

For a Newton-Raphson based power flow solution calculation to be considered converged in PSS®E-29.0 through PSS®E-29.5.1, the following conditions must have both been satisfied:

1. The largest active power mismatch at each Type 1 and 2 bus must be less than the mismatch convergence tolerance, TOLN.
2. The largest reactive power mismatch at each Type 1 and 2 bus must be less than the mismatch convergence tolerance, TOLN.

To match the convergence criterion of PSS®E-29.0 through PSS®E-29.5.1, set:

```
VCTOLQ = TOLN
VCTOLV = a large number (e.g., 10.0)
```

For a Newton-Raphson based power flow solution calculation to be considered converged in PSS®E-28 and earlier releases, the following conditions must have all been satisfied:

1. The largest active power mismatch at each Type 1 and 2 bus must be less than the mismatch convergence tolerance, TOLN.
2. The largest reactive power mismatch at the following categories of Type 1 and 2 buses must be less than the mismatch convergence tolerance, TOLN:
    a. Voltage at the bus is not subject to control to a setpoint.
    b. Voltage at the bus is subject to control to a setpoint (refer to Setpoint Voltage Control), but all of the devices participating in the control of bus voltage to its setpoint are at a reactive power limit.

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3. The largest difference between actual and scheduled voltage magnitude in per unit at each Type 1 and 2 bus where voltage is subject to control to a setpoint, and for which at least one of the devices participating in the control of bus voltage to its setpoint is not at a reactive power limit, must be less than TOLN/SBASE.

To match the convergence criterion of PSS®E-28 and earlier releases, set:

```
VCTOLQ = a large number (e.g., 1000.0)
VCTOLV = TOLN/SBASE
```

6.5.2. Automatic Adjustments

Activity FNSL provides for the following solution options:

1. Adjustment of transformer turns ratio to hold the voltage at a designated bus within a specified band, or to hold reactive power flow through the transformer within a specified band.

2. Adjustment of area swing bus generator powers to hold area net interchange within a specified band.

3. Adjustment of transformer phase shift angle to hold the active power flow through the phase shifter within a specified band.

4. Flat start.

5. Adjustment of the turns ratios of converter transformers at dc transmission terminals and of ac network transformers controlling dc line quantities.

6. Adjustment of switched shunt devices. When this adjustment option is disabled, discretely operating switched shunts are moved to their nearest step, if necessary, and all switched shunts, including continuously operating devices, are held constant. When this adjustment option is enabled, either all adjustable switched shunts may be adjusted, or continuous mode switched shunts are subject to adjustment and discrete mode switched shunts are locked.

7. A non-divergent solution option (refer to Section 6.5.3, “Non-Divergent Solution Option”).

PSS®E program option settings exist for adjustments (1), (2), (3), (5), (6) and (7) above. Activity OPTN may be used to establish the default setting for each of these adjustment options (refer to Section 3.3.3, “Program Run-Time Option Settings”). Then, on each entry into activity FNSL, these option settings determine which of the adjustment options are activated.

The following paragraphs discuss the optional automatic adjustments in more detail. The user is referred to Section 5.2.1, “Power Flow Raw Data File Contents” for additional details on the significance of the control parameters mentioned below.

Automatic Transformer Adjustments

Any two-winding ac transformer represented in the PSS®E working case may be treated as either: a fixed tap transformer; a tap changing transformer where Winding 1 turns ratio may be adjusted to control the voltage at a designated bus; a tap changing transformer where Winding 1 turns ratio may be adjusted to control reactive power flow through the transformer; a tap changing transformer where Winding 1 turns ratio may be adjusted to control a dc line quantity; or a regulating phase shifter where Winding 1 phase shift angle may be adjusted to control active power flow through the transformer.

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Similarly, any winding of a three-winding ac transformer represented in the PSS®E working case may be treated as either: a fixed tap transformer winding; a tap changing transformer winding where turns ratio may be adjusted to control the voltage at a designated bus; a tap changing transformer winding where turns ratio may be adjusted to control reactive power flow through the transformer winding; or a regulating phase shifter winding where phase shift angle may be adjusted to control active power flow through the transformer winding.

The data values specified for the transformer automatic adjustment control mode and for the two-terminal dc line ac controlling transformer parameters determine which of the above applies for any transformer (refer to Transformer Data and Two-Terminal DC Transmission Line Data).

An adjustment monitor is printed at the *Progress* tab any time an adjustment occurs.

Voltage Control

To be a candidate for automatic tap ratio adjustment as a voltage controlling transformer, an in-service transformer winding must meet the following conditions:

1. Its automatic adjustment control mode, CODn of Transformer Data, must be set to one.
2. Its specified controlled bus number (CONTn of Transformer Data) must be non-zero, and bus CONTn must have a bus type code of 1 or 2.
3. RMAn and RMIn of Transformer Data define the tap ratio limits of the transformer, and RMAn must be greater than RMIn.
4. VMAn and VMIn of Transformer Data define the desired voltage limits at the controlled bus, and VMAn must be greater than VMIn.
5. NTPn of Transformer Data defines the number of tap positions, and must be greater than one.
6. CRn + jCXn of Transformer Data defines the load drop compensating impedance.

When tap adjustment and reactive power limit application are both enabled during a power flow solution activity, tap adjustment is suppressed until after a voltage magnitude correction with reactive power limits honored has been calculated.

When tap adjustment is enabled in activities SOLV, MSLV, FNSL, INLF, ACCC, PV Analysis, or QV Analysis using FNSL, those transformers controlling voltage are checked for adjustment whenever the largest voltage magnitude change in per unit on the previous iteration is less than the automatic adjustment threshold tolerance, ADJTHR; in activities FDNS, NSOL, ACCC, PV analysis, or QV analysis using FDNS, this adjustment check occurs following any *P-angle half iteration* in which the largest voltage phase angle change in radians is less than ADJTHR.

The controlled voltage is calculated as the voltage at the controlled bus less a compensating voltage that is a function of the transformer's loading:

$$V_C = |V_{CONTn} - I_{transformer} \times (CRn + jCXn)| \quad (6.16)$$

Transformer current is calculated at the controlling winding side bus of the transformer if V_{CONTn} of Transformer Data is either the controlling winding side bus or a negative number; otherwise, current on the Winding 2 side (for a two-winding transformer) or the star bus side (for a three-winding transformer winding) is used. In either case, the current direction reference is always from the uncontrolled side to the controlled side of the transformer. Clearly, load drop compensation by a transformer may be suppressed by setting its CRn and CXn to zero, in which case $V_C = |V_{CONTn}|$.

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The tap adjustment step increment is calculated as:

$$STEP = \frac{RMAn - RMIn}{NTPn - 1} \quad (6.17)$$

A discrete or stepping method of tap adjustment is used by activities SOLV and MSLV; this approach may also be specified in activities FNSL, NSOL, FDNS, INLF, ACCC, PV Analysis, and QV Analysis by specifying one as the tap adjustment code when selecting adjustment options. In addition to the data requirements listed above, the quantity (VMAn - VMIn) should be at least 1.5 * STEP.

In this method, each transformer is checked independently outside of the main power flow iteration. If the controlled voltage is outside of its specified band (i.e., if the relationship shown in the following expression is violated), the tap ratio is moved at least one step.

$$VMIn \le V_C \le VMAn \quad (6.18)$$

The tap ratio may be given a change consisting of multiple steps if the controlled voltage is outside of the voltage band by at least two times STEP. A tap movement deceleration factor, ACCTAP, is applied to the voltage error. Setting ACCTAP below its default value of unity reduces the one-to-one tap movement applied in response to a given voltage error.

The solution parameter TAPLIM, which has a default value of 0.05, defines the maximum ratio change that may be applied to a transformer during any adjustment calculation; if the STEP of a transformer exceeds TAPLIM and the controlled voltage is outside of its voltage band, the ratio is changed by one STEP.

Reducing ACCTAP and/or TAPLIM below their default values may be beneficial on systems with a high concentration of LTCs or where voltage profile is extremely sensitive to tap settings.

Activities FNSL, NSOL, FDNS, INLF, ACCC, PV Analysis, and QV Analysis also provide for a direct Newton-based method of tap adjustment, which is enabled by specifying 2 as the tap adjustment code when selecting adjustment options. In this method, if any tap ratios need to be adjusted, a simultaneous adjustment is made of *all* voltage controlling transformers as well as of all Mvar controlling transformers (refer to Mvar Control) and of all bus voltage magnitudes. Upon convergence of the main power flow iteration, tap ratios of controlling transformers are moved to their nearest step and the solution refined with tap ratios locked at those positions.

The tap movement deceleration factor, ACCTAP, used in the stepping method described above is also used in the direct method. It reduces the elements of the voltage error vector used in the Newton-based direct method. The maximum ratio change solution parameter, TAPLIM, is not used in the direct method.

In the direct method, compensated limits rather than compensated voltages are used to handle load drop compensation. The inequality shown above for the discrete method may be expressed in terms of the voltage magnitude at the controlled bus as follows:

$$VMIn \le (|V_{CONTn}| - \Delta V) \le VMAn \quad (6.19)$$

Adding ΔV to each term above, the relationship may then be written in terms of compensated limits:

$$(VMIn + \Delta V) \le |V_{CONTn}| \le (VMAn + \Delta V) \quad (6.20)$$

It is possible for the direct method to fail as a result of an overconstrained situation. This is most likely to occur in radial or tightly looped portions of the network containing several voltage controlling transformers attempting to control the voltage at multiple buses. Should this occur, tap ratios are moved to their nearest step and the solution continues with subsequent adjustment of voltage controlling transformers handled by the discrete method described above.

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Mvar Control

To be a candidate for automatic tap ratio adjustment as a Mvar flow controlling transformer, an in-service transformer winding must meet the following conditions:

1. It automatic adjustment control mode, CODn of Transformer Data, must be set to 2.
2. RMAn and RMIn of Transformer Data define the tap ratio limits of the transformer, and RMAn must be greater than RMIn.
3. VMAn and VMIn of Transformer Data define the desired Mvar flow limits through the transformer, and VMAn must be greater than VMIn. Flow is calculated at the controlling winding side bus of the transformer and is positive when power is flowing from the controlling winding side bus into the transformer. When specifying VMAn and VMIn, remember that -100 is greater than -110!
4. NTPn of Transformer Data defines the number of tap positions, and must be greater than one.
5. The controlled bus number (CONTn) and compensating impedance (CRn and CXn) of Transformer Data are ignored.

When tap adjustment and reactive power limit application are both enabled during a power flow solution activity, tap adjustment is suppressed until after a voltage magnitude correction with reactive power limits honored has been calculated.

When tap adjustment is enabled in activities FNSL, INLF, ACCC, PV Analysis, or QV Analysis using FNSL, those transformers controlling reactive power flow are checked for adjustment whenever the largest voltage magnitude change in per unit on the previous iteration is less than the automatic adjustment threshold tolerance, ADJTHR; in activities FDNS, NSOL, ACCC, PV analysis, or QV analysis using FDNS, this adjustment check occurs following any *P-angle half iteration* in which the largest voltage phase angle change in radians is less than ADJTHR. Mvar flow control is not available in activities SOLV and MSLV.

Adjustment of Mvar controlling transformers is enabled by entering either 1 or 2 as the tap adjustment code when selecting adjustment options. The direct method described in Voltage Control is used for this adjustment; ACCTAP and TAPLIM are not used in Mvar control.

DC Line Control

To be a candidate for automatic tap ratio adjustment as a dc line quantity controlling transformer, an in-service two-winding transformer must meet the following conditions:

1. Its automatic adjustment control mode, COD1 of Transformer Data, must be set to four.
2. IFR, ITR, and IDR (or IFI, ITI, and IDI, as appropriate) of one (and only one) two-terminal dc line converter must specify the transformer branch (refer to Two-Terminal DC Transmission Line Data).
3. TMXR and TMNR (or TMXI and TMNI, as appropriate) of Two-Terminal DC Transmission Line Data define the tap ratio limits of the transformer.
4. STPR (or STPI as appropriate) of Two-Terminal DC Transmission Line Data defines the tap ratio step increment, and it must be greater than zero.
5. The remaining ac tap adjustment data of Transformer Data is ignored.

Adjustment of ac transformers that control dc line quantities is enabled only when dc tap adjustment is enabled.

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When dc tap adjustment is enabled in activities SOLV, MSLV, FNSL, INLF, ACCC, PV Analysis, or QV Analysis using FNSL, those ac transformers controlling dc line quantities may be adjusted whenever the largest voltage magnitude change in per unit on the previous iteration is less than the automatic threshold tolerance, ADJTHR; in activities FDNS, NSOL, ACCC, PV analysis, or QV analysis using FDNS, adjustment is allowed following any *P-angle half iteration* in which the largest voltage phase angle change in radians is less than ADJTHR.

This adjustment calculation follows the same methodology as is used in the stepping method of tap adjustment for ac voltage control described in Voltage ControlThe tap movement deceleration factor, ACCTAP, and the maximum ratio change solution parameter, TAPLIM, are applied.

Phase Shift Angle Adjustment

To be a candidate for automatic phase shift angle adjustment as a MW flow controlling phase shifter, an in-service transformer winding must meet the following conditions:

1. Its automatic adjustment control mode, CODn of Transformer Data, must be set to 3 (for symmetrical phase shift adjustment) or 5 (for unsymmetrical phase shift adjustment).

2. RMAn and RMIn of Transformer Data define the phase shift angle limits in degrees, and RMAn must be greater than RMIn.

3. VMAn and VMIn of Transformer Data define the desired MW flow limits through the phase shifter, and VMAn must be greater than VMIn. Flow is calculated at the controlling winding side bus of the transformer and is positive when power is flowing from the controlling winding side bus into the transformer. When specifying VMAn and VMIn, remember that -100 MW is greater than -110 MW!

4. CNXAn of Transformer Data defines the winding connection angle in degrees. A non-zero value of the winding connection angle is required when the control mode, CODn, is 5. When the control mode is 3, CNXAn is ignored.

5. The controlled bus number (CONTn), number of tap positions (NTPn), and compensating impedance (CRn and CXn) of Transformer Data are ignored.

When phase shift angle adjustment is enabled in activities FNSL, INLF, ACCC, PV Analysis, or QV Analysis using FNSL, those phase shifters controlling active power flow are checked for adjustment whenever the largest voltage phase angle change in radians on the previous iteration is less than the automatic adjustment threshold tolerance, ADJTHR; in activities FDNS, NSOL, ACCC, PV analysis, or QV analysis using FDNS, this adjustment check occurs following any *QV half iteration* in which the largest voltage magnitude change in per unit is less than ADJTHR. MW flow control is not available in activities SOLV and MSLV. An adjustment monitor is printed on the *Progress* tab any time an adjustment occurs.

The phase shift adjustment utilizes a direct method. If any phase shift angles need to be adjusted, a simultaneous adjustment is made of *all* regulating phase shifters. Normally, all bus voltage phase angles are adjusted as well. The bus voltage phase angle adjustments are suppressed if there are any in-service branches in the working case whose |X/R| ratio exceeds 10.0. ACCTAP and TAPLIM are not used in MW control by phase shifters. The algorithm is normally able to handle the case of parallel phase shifters with similar flow constraints and angle limits, and the presence of phase shifters on area interchange boundaries (see below).

When the phase shift adjustment calculation detects a singular matrix, an alarm is printed and the power flow solution continues with further phase shift adjustment disabled.

In symmetrical phase shift adjustment (i.e., CODn is 3), the adjustable winding's off-nominal turns ratio is not changed as a result of a change in phase shift angle. In the unsymmetrical phase shift adjustment (i.e.,

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CODn is 5), the adjustable winding's off-nominal turns ratio is changed as a function of the phase shift angle and the winding connection angle.

Area Interchange Control

Each bus in the PSS®E working case may be designated as residing in an interchange area, for purposes of both interchange control and selective output and other processing. When the interchange control option is enabled during a power flow solution, each interchange area for which an area slack bus is specified has the active power output of its area slack bus modified such that the desired net interchange for the area falls within a desired band.

The area interchange control code as specified in activity OPTN, or as overridden at the start of the network solution (refer to Section 6.3.20, "Automatic Adjustments" and Section 6.5.2, "Automatic Adjustments"), establishes the net interchange definition to be used by the area interchange control calculation. When a value of 1 is specified, an area's net interchange is defined as the sum of the flows on all of its tie lines; a tie line is a branch (ac line, dc line, series FACTS device or series GNE device) connected to a bus residing in the area where the other end is connected to a bus that is not in the area. Tie flows are calculated at the metered end as power flowing out of the area. When the area interchange control code is 2, a load where area assignment differs from that of the bus to which it is connected is considered a tie branch for net interchange calculation purposes; that is, an area's net interchange includes tie line flows as well as contributions from loads connected to area buses that are assigned to areas other than the bus' area, and from loads assigned to the area that are connected to buses assigned to other areas.

In specifying area interchange control data, the following rules apply:

1. The area slack generator *must* be a Type 2 or 3 bus residing in the area.
2. Any area containing a system swing bus (Type 3) may not have some other generator bus designated as the area slack bus.
3. Any area for which no bus is specified as an area swing is not controlled; it is not mandatory for all areas to be under control.
4. The desired interchange is specified as a positive number for export and a negative number for import.
5. The desired interchanges specified with area interchange data (refer to Area Interchange Data), *not* the sums of each area's interarea transfers (refer to Interarea Transfer Data), are used as targets by the area interchange control algorithm.
6. The specification of PDES of Area Interchange Data must be consistent with the definition of area interchange implied by the area interchange control code (see above).
7. The tolerance band should be at least 2 or 3 MW; 5 or 10 MW are more typical values.

When area interchange control is enabled in activity FNSL, ACCC, PV Analysis, or QV Analysis using FNSL, area interchange is checked whenever the largest voltage phase angle change in radians on the previous iteration is less than the automatic adjustment threshold tolerance, ADJTHR; in activities FDNS, NSOL, ACCC, PV analysis, or QV analysis using FDNS, this adjustment check occurs following any *QV half iteration* in which the voltage magnitude change in per unit is less than ADJTHR. In activities SOLV and MSLV, the check is made when the largest voltage magnitude change in per unit is less than 0.001 and ADJTHR, but no more frequently than every 10 iterations. An adjustment monitor is printed at the *Progress* tab any time an adjustment occurs. An asterisk (*) following a slack bus old or new active power setting indicates that the power output is beyond its limits.

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When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the area assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, a multi-section line grouping is treated as a tie branch if and only if its endpoint buses are in different areas, with tie flow evaluated at the metered end of the multi-section line grouping.

When the multi-section line reporting option is disabled, the area assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

Whenever the area interchange control option is enabled during a power flow solution activity, a warning message is printed at the start of the solution, and again at the completion of the solution, if the sum of the desired interchanges is not zero. At the termination of the solution, an area interchange violation summary tabulates any area that has at least one in-service tie element connected to it and which has an area swing assigned to it, but which fails to meet is net interchange specification.

Switched Shunt Adjustment

The significance of switched shunt data and the treatment of switched shunts during power flow solutions are described in Switched Shunt Data and Section 6.3.15, "Switched Shunt Devices", respectively.

When switched shunt adjustment is enabled, adjustments of switched shunts that control to a voltage setpoint (refer to rule (3) of Section 6.3.15, "Switched Shunt Devices") occur during each iteration of the power flow solution activities as described in Setpoint Voltage Control. Switched shunts that control to a voltage band (refer to rule (4) of Section 6.3.15, "Switched Shunt Devices") are adjusted between network solution iterations. Similarly, switched shunts that control the reactive power output of another voltage controlling device are adjusted between power flow iterations.

In adjusting switched shunts that control to a voltage band, the presence of a large number of such switched shunts among a fairly small number of buses can result in some of them oscillating from one adjustment cycle to the next. The solution parameter SWVBND specifies the percentage of switched shunts controlling to a voltage band and for which the voltage at the controlled bus is outside of the specified band that can be adjusted on any single adjustment cycle. Reducing SWVBND, thereby decreasing the number of switched shunts that are adjusted on any adjustment cycle, may be helpful in overcoming such oscillations.

When adjustment of discrete mode switched shunts is enabled in activities SOLV, MSLV, FNSL, INLF, ACCC, PV Analysis, or QV Analysis using FNSL, such switched shunts are checked for adjustment whenever the largest voltage magnitude change in per unit on the previous iteration is less than the automatic adjustment threshold tolerance, ADJTHR; in activities FDNS, NSOL, ACCC, PV analysis, or QV analysis using FDNS, this adjustment check occurs following any *P-angle half iteration* in which the largest voltage phase angle change in radians is less than ADJTHR.

Oscillating Tap Ratio and/or Switched Shunt Adjustments

Using the solution parameter SWVBND to overcome oscillations in the adjustment of switched shunt settings was discussed in the preceding section. This method addresses the situation where there are a fairly large number of voltage band controlling switched shunts distributed within a relatively small portion of the system.

Another type of oscillation that may occur when either or both of tap ratio adjustment and discrete (i.e., band mode) switched shunt adjustment are enabled involves one or both of:

- very narrow control ranges (i.e., voltage bands) relative to the size of the tap ratio and/or switched shunt steps;

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- interaction among multiple devices controlling the same quantity or "nearby" quantities.

The solution parameter MXTPSS specifies the maximum number of power flow iterations on which tap ratio adjustments and/or discrete switched shunt adjustments may occur. Once that limit is reached, tap ratio and discrete switched shunt adjustments are suppressed for the remainder of the power flow solution. In this case, summaries reporting violations are printed at the end of the solution.

Automatic Adjustment Solution Parameters

The automatic adjustment threshold tolerance, ADJTHR, the tap movement declaration factor, ACCTAP, the maximum ratio change solution parameter, TAPLIM, the voltage band controlling switched shunt percentage, SWVBND, and the maximum number of times tap ratois and/or band mode switched shunts may be adjusted, MXTPSS, may be modified with the data changing activity CHNG or the *[Solution Parameters]* dialog. They are carried along with the working case as it is saved and retrieved with activities SAVE and CASE, respectively.

When a new case is read into PSS®E with activity READ, the solution parameters are set to their default values: ADJTHR to 0.005, ACCTAP to 1.0, TAPLIM to 0.05, SWVBND to 100.0, and MXTPSS to 99.

6.5.3. Non-Divergent Solution Option

When applying the Newton-Raphson algorithm or any of its variants to the power flow problem, one of the following results is usually observed:

- Mismatches for all buses are within tolerance (convergence).
- Increasing mismatches, and usually increasing voltage and angle changes, are observed on successive iterations (divergence). A divergent solution is often characterized by unrealistically small or large voltage magnitudes and by very large bus mismatches; it may be characterized by a Jacobian matrix that is singular or nearly so.
- Mismatch tolerances are not met, but the solution is neither converging nor diverging (non-convergence). This condition could be caused by convergence tolerances that are beyond the precision limits of the combination of host computer and network impedances; it could also result from control ranges that are too narrow relative to the adjustment step (e.g., desired voltage range used by a voltage controlling transformer relative to the tap step), which results in oscillation of the controlling parameter from one adjustment cycle to the next (e.g., tap ratio increasing, then decreasing, then increasing, and so on), or by multiple controlling devices in conflict with each other.

The non-divergent Newton power flow solution option attempts to terminate the iterative procedure before divergent iterations have driven the voltage vector to a state where large mismatches and unrealistic voltages are present. The resulting voltage vector, although not sufficiently accurate to provide a converged power flow solution, often provides a relatively good indication of the state of the network. In particular, voltage collapse situations can be identified by localized areas of bus mismatches and low voltages, with the remainder of the network converged.

The non-divergent solution option may be applied in activities FNSL, FDNS, and the ACCC (refer to Section 6.5.2, “Automatic Adjustments” and Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN).

Non-Divergent Solution Methodology

The application of the Newton method to the power flow problem calculates a pair of change vectors: $\Delta\theta$ for voltage phase angles, and $\Delta vmag / vmag_{old}$ for bus voltage magnitudes. In its usual application, the bus voltages are updated using equations which may be written in the form:

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$$\theta_{new} = \theta_{old} + (ACCFAC * \Delta\theta) \quad (6.21)$$

$$vmag_{new} = vmag_{old} * [1.0 + ACCFAC * (\Delta vmag / vmag_{old})] \quad (6.22)$$

where ACCFAC is equal to 1.0.

In the non-divergent solution scheme, ACCFAC is set to 1.0 at the start of each Newton iteration. If the mismatches that result from the new voltage and angle vectors indicate divergence, the value of ACCFAC is halved, θ_{new} and $vmag_{new}$ recalculated, and mismatches recalculated. This process is repeated until either:

- the indication of divergence is eliminated. In this case, activity FNSL advances to its next iteration.
- ACCFAC has been reduced to a near zero value without eliminating the indication of divergence. In this case, activity FNSL is terminated.

The divergence metric used in this inner loop voltage correction is the sum of squares of the MVA mismatches, expressed in per unit, at all in-service buses in the working case (SUMSQM). The non-divergent algorithm infers the elimination of divergence when the SUMSQM resulting from the application of a set of voltage magnitude and phase angle corrections is less than the SUMSQM at the start of the iteration multiplied by the improvement factor NDVFCT. That is, a new set of voltages is accepted if:

$$SUMSQM_{new} < NDVFCT * SUMSQM_{old}$$

The non-divergent improvement factor, NDVFCT, which has a default value of 0.99, should never be greater than 1.0 nor less than or equal to 0.0. With values very close to 1.0, activity FNSL accepts small reductions in SUMSQM and goes on to the next Newton iteration. As NDVFCT is reduced, activity FNSL requires larger reductions in SUMSQM before inferring the elimination of divergence. The setting of NDVFCT is a trade-off between execution time and a possible incremental improvement in total system mismatch level. The user may modify NDVFCT with the data changing activity CHNG or the *[Solution Parameters]* dialog.

Up to ten inner loop mismatch calculations are performed, and if SUMSQM has not improved sufficiently by the last attempt (with ACCFAC approximately 0.00195), activity FNSL is terminated. The voltage vector is set to either its value at the completion of the prior Newton iteration, or to value corresponding to the last inner loop mismatch calculation, whichever has the smaller value of SUMSQM.

Convergence Monitor

When the non-divergent solution option is enabled, the standard FNSL convergence monitor (refer to FNSL) is augmented by two additional columns containing SUMSQM and ACCFAC. In addition, each inner loop mismatch calculation is reported. Figure 6.10, “Activity FNSL Convergence Monitor” shows the convergence monitors for the attempted FNSL solutions of a difficult case, first with the non-divergent solution option disabled, and then with the non-divergent solution option enabled. DVLIM and NDVFCT are both 0.99.

When the non-divergent solution option is enabled (see Figure 6.10, “Activity FNSL Convergence Monitor”b), the main Newton iteration mismatch lines are identified in the ITER column as whole numbers, just as they are when the non-divergent solution option is disabled (see Figure 6.10, “Activity FNSL Convergence Monitor”a); the inner loop mismatch lines are identified with the corresponding main iteration number, followed by a decimal point, followed by the inner loop iteration number.

In Figure 6.11, “”b, for the first six iterations, the full change vector reduced SUMSQM and hence was applied. The mismatch line labeled 6 indicates the mismatches resulting from the application of the voltage and angle changes listed on the preceding line. At this point, a new pair of change vectors are calculated, and the

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mismatch line labeled 6.1 indicates the resulting mismatch if the full changes are applied (i.e., ACCFAC is 1.0). The resulting SUMSQM exceeds that of line 6, so ACCFAC is halved and the process repeated. Finally, at line 6.4, with ACCFAC at 0.125, a reduction of SUMSQM is achieved. The following line shows the largest voltage and angle changes actually applied (i.e., after being multiplied by ACCFAC).

In Figure 6.10, "Activity FNSL Convergence Monitor"b, for most of the voltage magnitude and phase angle corrections applied, the mismatch lines immediately preceding and following a voltage correction line are identical. While this is often the case, it need not be. Lines 2.1 and 3 differ in the largest reactive power mismatch; this is the iteration on which reactive power limits are first applied. Post-voltage correction mismatch values may also be affected by bus boundary condition modifications imposed for dc lines, FACTS devices, and GNE devices.

Recall that when the solution is terminated by the non-divergent solution option, it is not converged; it has merely been halted in an attempt to capture useful information on the state of the system before the solution blows up catastrophically. Therefore, in addition to mismatch tolerance not being met, specified setpoints and bands of various equipment item parameters may not be satisfied for all controlled equipment in the case.

ITER	DELTAP	BUS	DELTAQ	BUS	DELTA/V/	BUS	DELTAANG	BUS
0	5.6995(	151)	1.2217(	151)	0.07177(	3008)	0.14024(	101)
1	0.5073(	201)	0.6370(	205)	0.00591(	206)	0.00490(	201)
2	0.0028(	152)	0.4046(	206)	0.00379(	206)	0.00034(	206)
3	0.0001(	205)	2.4734(	3008)	0.03639(	3008)	0.00266(	206)
4	0.0065(	154)	1.6332(	205)	0.03136(	205)	0.00779(	101)
5	0.0253(	205)	0.8006(	201)	0.08905(	205)	0.03293(	101)
6	0.2468(	205)	0.0599(	205)	0.17461(	205)	0.06537(	101)
7	0.9226(	205)	0.2194(	205)	0.13294(	154)	0.04131(	101)
8	0.3062(	205)	0.0904(	154)	0.13959(	205)	0.04873(	101)
9	0.5686(	205)	0.1260(	205)	0.08090(	154)	0.03244(	101)
10	0.2733(	205)	0.0589(	205)	0.82297(	205)	0.30216(	101)
11	7.0235(	206)	7.6130(	102)	0.99000(	152)	4.91214(	206)

```
BLOWN UP AFTER  12 ITERATIONS

LARGEST MISMATCH:  495.60 MW 1025.72 MVAR  1139.18 MVA AT BUS    205 [SUB230      230.00]
SYSTEM TOTAL ABSOLUTE MISMATCH:            7280.84 MVA

SWING BUS SUMMARY:
  BUS# X-- NAME --X BASKV      PGEN     PMAX     PMIN      QGEN     QMAX     QMIN
  3001 MINE                -1500.9   9999.0 -9999.0    2959.5*   600.0  -100.0
```

a. Non-Divergent Solution Option Disabled

Figure 6.10. Activity FNSL Convergence Monitor

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```
 ITER      DELTAP      BUS       DELTAQ      BUS       DELTA/V/      BUS       DELTAANG      BUS      SUMSQM       ACCFAC
  0        5.6995(     151   )   1.2217(     151   )                                                   66.093       1.00000
  0.1      0.5073(     201   )   0.6370(     205   )                                                    1.6272      1.00000
                                                       0.07177(     3008   )   0.14024(      101   )
  1        0.5073(     201   )   0.6370(     205   )                                                    1.6272      1.00000
  1.1      0.0028(     152   )   0.4046(     206   )                                                   0.32739      1.00000
                                                       0.00591(      206   )   0.00490(      201   )
  2        0.0028(     152   )   0.4046(     206   )                                                   0.32739      1.00000
  2.1      0.0001(     205   )   0.0388(     206   )                                                  0.30034E-02   1.00000
                                                       0.00379(      206   )   0.00034(      206   )
  3        0.0001(     205   )   2.4734(     3008  )                                                    6.1208      1.00000
  3.1      0.0065(     154   )   1.6332(     205   )                                                    2.7156      1.00000
                                                       0.03639(     3008   )   0.00266(      206   )
  4        0.0065(     154   )   1.6332(     205   )                                                    2.7156      1.00000
  4.1      0.0253(     205   )   0.8006(     201   )                                                   0.64587      1.00000
                                                       0.03136(      205   )   0.00779(      101   )
  5        0.0253(     205   )   0.8006(     201   )                                                   0.64587      1.00000
  5.1      0.2468(     205   )   0.0599(     205   )                                                   0.15622      1.00000
                                                       0.08905(      205   )   0.03293(      101   )
  6        0.2468(     205   )   0.0599(     205   )                                                   0.15622      1.00000
  6.1      0.9226(     205   )   0.2194(     205   )                                                    2.1249      1.00000
  6.2      0.3679(     205   )   0.0859(     205   )                                                   0.33685      0.50000
  6.3      0.2479(     205   )   0.0591(     205   )                                                   0.15538      0.25000
  6.4      0.2319(     205   )   0.0560(     205   )                                                   0.13730      0.12500
                                                       0.02183(      205   )   0.00817(      101   )
  7        0.2319(     205   )   0.0560(     205   )                                                   0.13730      1.00000
  7.1      7.9710(     206   )   9.5520(     102   )                                                    357.47      1.00000
  7.2      3.6765(     206   )   3.8054(     102   )                                                    60.676      0.50000
  7.3      1.5493(     205   )   0.9314(     102   )                                                    8.2484      0.25000
  7.4      0.6542(     205   )   0.1560(     205   )                                                    1.0698      0.12500
  7.5      0.3362(     205   )   0.0800(     205   )                                                   0.28423      0.06250
  7.6      0.2556(     205   )   0.0613(     205   )                                                   0.16584      0.03125
  7.7      0.2364(     205   )   0.0570(     205   )                                                   0.14249      0.01563
  7.8      0.2323(     205   )   0.0560(     205   )                                                   0.13773      0.00781
  7.9      0.2316(     205   )   0.0559(     205   )                                                   0.13698      0.00391
  7.10     0.2316(     205   )   0.0559(     205   )                                                   0.13701      0.00195
                                                       0.00193(      205   )   0.00071(      101   )


TERMINATED AFTER   8 ITERATIONS--NON-DIVERGENT OPTION COULD NOT REDUCE MISMATCH


LARGEST MISMATCH:   23.16 MW    5.59 MVAR    23.83 MVA AT BUS    205 [SUB230      230.00]
SYSTEM TOTAL ABSOLUTE MISMATCH:              93.62 MVA


SWING BUS SUMMARY:
   BUS# X-- NAME --X BASKV       PGEN      PMAX     PMIN       QGEN      QMAX     QMIN
   3001 MINE                   270.5    9999.0 -9999.0      427.0     600.0   -100.0
```

Figure 6.11.

b. Non-Divergent Solution Option Enabled

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Additional Information
PSS®E GUI Users Guide, Fully-Coupled Newton-Raphson Power Flow Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Fully-Coupled Newton-Raphson Power Flow Solution

6.6. Applying Decoupled Newton-Raphson Power Flow Solution

NSOL

Run Activity *NSOL - GUI*
Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)... *[Loadflow solutions]* Newton Tab OPTION: Decoupled Newton-Raphson
Run Line Mode Activity *NSOL - CLI*
ACTIVITY? >>**NSOL** ENTER ITERATION NUMBER FOR VAR LIMITS 0 FOR IMMEDIATELY, -1 TO IGNORE COMPLETELY: >>
Interrupt Control Codes
AB, NC, NM, DC, FD, MO, NV

The power flow solution activity NSOL uses a decoupled Newton-Raphson iterative algorithm to solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case.

Activity NSOL handles the network topology check, the network connectivity checking option, flat start solution, treatment of generator reactive power limits, load, generator, switched shunt, FACTS device, and dc line boundary conditions, the blowup check, scaling of the voltage magnitude change vector, acceleration, convergence criteria, and interrupt control codes in the same way as activity FNSL. The selection and application of automatic adjustments is identical to that of activity FNSL except that the non-divergent solution option is not available in activity NSOL. The FACTS device monitor, dc transmission line monitors, induction machine monitor, largest mismatch tabulation, swing bus summary, and area interchange violation summary are identical to those of activity FNSL.

The solution convergence monitor is similar to that of activity FNSL. Each iteration is identified in the ITER column with its main iteration number, followed by a decimal point, followed by either a zero (for the angle correction calculation) or a five (for the voltage magnitude correction calculation); see Figure 6.12, “Decoupled Newton-Raphson Convergence Monitor”.

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```
ITER      DELTAP      BUS         DELTAQ      BUS        DELTA/V/      BUS       DELTAANG      BUS
  0.0      0.4265(     152   )     1.2845(     202   )
                                                        0.00000(            )   0.01098(     202   )
  0.5      0.0001(     154   )     1.3166(     202   )
                                                        0.02169(     202    )   0.00000(           )
  1.0      0.2135(     205   )     1.0963(     211   )
                                                        0.00000(            )   0.00173(     154   )
  1.5      0.0000(     205   )     1.0909(     211   )
                                                        0.02131(     211    )   0.00000(           )
  2.0      0.1691(     201   )     0.1693(     201   )
                                                        0.00000(            )   0.00295(     211   )
  2.5      0.0000(     201   )     0.1460(     211   )
                                                        0.00268(     211    )   0.00000(           )
  3.0      0.0216(     201   )     0.0401(     201   )
                                                        0.00000(            )   0.00036(     211   )
  3.5      0.0000(     201   )     0.0374(     201   )
                                                        0.00068(     211    )   0.00000(           )
  4.0      0.0055(     201   )     0.0096(     211   )
                                                        0.00000(            )   0.00009(     211   )
  4.5      0.0000(     211   )     0.0089(     201   )
                                                        0.00016(     211    )   0.00000(           )
  5.0      0.0013(     201   )     0.0022(     201   )
                                                        0.00000(            )   0.00002(     211   )
  5.5      0.0000(     211   )     0.0021(     211   )
                                                        0.00004(     211    )   0.00000(           )
  6.0      0.0003(     201   )     0.0005(     211   )

REACHED TOLERANCE IN   6 ITERATIONS

LARGEST MISMATCH:   -0.03 MW    0.05 MVAR     0.06 MVA AT BUS    201 [HYDRO       500.00]
SYSTEM TOTAL ABSOLUTE MISMATCH:               0.19 MVA

SWING BUS SUMMARY:
  BUS# X-- NAME --X BASKV      PGEN     PMAX    PMIN      QGEN     QMAX    QMIN
  3011 MINE_G      13.800     260.8    900.0     0.0     116.3    600.0  -100.0
```

Figure 6.12. Decoupled Newton-Raphson Convergence Monitor

The solution parameters designating the maximum number of iterations, acceleration, convergence tolerances, and maximum voltage magnitude change are shared with activity FNSL, while the blowup threshold and constant power voltage breakpoint (refer to Section 6.3.13, "Load") are shared among all power flow solution activities. The user may modify any of the above solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

Activity NSOL uses an iterative scheme in which the active power-angle calculation is decoupled from the reactive power-voltage adjustment. It is basically a Newton calculation in which each iteration consists of a pair of half iterations; first, with the voltage magnitudes held constant and new voltage phase angles determined, then with the phase angles fixed and new voltage magnitudes calculated.

The rules and characteristics governing the use of activity NSOL are similar to those of activity FNSL. In addition to those rules enumerated in Section 6.5.1, "Characteristics of Activity FNSL", the following apply:

1. Activity NSOL converges well if fairly uniform X/R ratios are present throughout the network.

2. When the network contains branches with resistance close to or greater than the reactance, the iteration usually reaches some mismatch level and then begins to diverge, usually slowly.

3. As the mismatches are reduced, the rate of improvement on successive iterations is slowed.

4. The time per half iteration with activity NSOL is roughly 1/4 of the time per FNSL iteration.

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Figure 6.13, "NSOL Example" is an example of the use of activity NSOL specified to initiate the solution from a flat start voltage profile.

```
ACTIVITY? NSOL,FS

ENTER ITERATION NUMBER FOR VAR LIMITS
 0 FOR IMMEDIATELY, -1 TO IGNORE COMPLETELY:

 ITER       DELTAP      BUS         DELTAQ      BUS        DELTA/V/      BUS       DELTAANG      BUS
  0.0      12.0000(      205   )    7.1856(      152   )    0.00000(            )    0.30269(      101   )
  0.5       0.1299(      152   )    6.9528(      154   )    0.05212(     3002   )    0.00000(            )
  1.0       0.8079(      201   )    0.5285(      206   )    0.00000(            )    0.02701(      101   )
  1.5       0.0011(      204   )    0.5197(      206   )    0.00479(      206   )    0.00000(            )
  2.0       0.0384(      206   )    0.0773(      205   )    0.00000(            )    0.00055(      101   )
  2.5       0.0000(      205   )    2.0215(     3008   )    0.03085(     3008   )    0.00000(            )
  3.0       0.1801(     3008   )    1.7660(      205   )    0.00000(            )    0.00246(      101   )
  3.5       0.0001(      204   )    1.7825(      205   )    0.03084(      205   )    0.00000(            )
  4.0       0.5470(      205   )    0.0599(      205   )    0.00000(            )    0.00752(      101   )
  4.5       0.0001(      152   )    0.1052(      205   )    0.01141(      205   )    0.00000(            )
  5.0       0.2468(      205   )    0.0021(      205   )    0.00000(            )    0.00310(      101   )
  5.5       0.0000(      205   )    0.0306(      154   )    0.00449(      154   )    0.00000(            )
  6.0       0.0987(      205   )    0.0003(      154   )    0.00000(            )    0.00124(      101   )
  6.5       0.0000(      205   )    0.0120(      204   )    0.00179(      205   )    0.00000(            )
  7.0       0.0394(      205   )    0.0000(      154   )    0.00000(            )    0.00049(      101   )
  7.5       0.0000(      152   )    0.0048(      204   )    0.00071(      205   )    0.00000(            )
  8.0       0.0158(      205   )    0.0000(      154   )    0.00000(            )    0.00020(      101   )
  8.5       0.0000(      102   )    0.0019(      204   )    0.00029(      205   )    0.00000(            )
  9.0       0.0063(      205   )    0.0000(      154   )    0.00000(            )    0.00008(      101   )
  9.5       0.0000(      102   )    0.0008(      204   )    0.00011(      205   )    0.00000(            )
 10.0       0.0025(      205   )    0.0000(      154   )    0.00000(            )    0.00003(      101   )
 10.5       0.0000(      102   )    0.0003(      204   )    0.00005(      205   )    0.00000(            )
 11.0       0.0010(      205   )    0.0000(      154   )    0.00000(            )    0.00001(      101   )
 11.5       0.0000(      102   )    0.0001(      204   )    0.00002(      205   )    0.00000(            )
 12.0      0.0004(      205   )    0.0000(      154   )

REACHED TOLERANCE IN  12 ITERATIONS

LARGEST MISMATCH:    0.04 MW    0.00 MVAR     0.04 MVA AT BUS    205 [SUB230      230.00]
SYSTEM TOTAL ABSOLUTE MISMATCH:               0.18 MVA

SWING BUS SUMMARY:
  BUS# X-- NAME --X BASKV      PGEN     PMAX     PMIN      QGEN     QMAX     QMIN
  3001 MINE                  248.9   9999.0 -9999.0     187.2    600.0  -100.0

ACTIVITY?
```

NSOL from flat start

Angle correction #1
Voltage correction #1
Angle correction #2
Voltage correction #2

var limits first applied here; note increased var mismatch

Note slow rate of convergence as we get close

Figure 6.13. NSOL Example

Additional Information
PSS®E GUI Users Guide, Decoupled Newton-Raphson Power Flow Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Decoupled Newton-Raphson Power Flow Solution

6.7. Applying Fixed Slope Decoupled Newton-Raphson Power Flow Solution

FDNS

Run Activity FDNS - GUI
Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)... *[Loadflow solutions]* Newton Tab OPTION: Fixed slope decoupled Newton-Raphson

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Run Line Mode Activity FDNS - CLI
ACTIVITY? >>**FDNS** ENTER ITERATION NUMBER FOR VAR LIMITS 0 FOR IMMEDIATELY, -1 TO IGNORE COMPLETELY: >>
Interrupt Control Codes
AB, NC, NM, DC, FD, MO, NV

The power flow solution activity FDNS uses a fixed-slope decoupled Newton-Raphson iterative algorithm to solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case.

Activity FDNS handles the network topology check, the network connectivity checking option, flat start solution, treatment of generator reactive power limits, load, generator, switched shunt, FACTS device, and dc line boundary conditions, selection and application of automatic adjustments, the blowup check, the scaling of the voltage magnitude change vector, acceleration, convergence criteria, and the non-divergent solution option, and interrupt control codes in the same way as activity FNSL. The FACTS device monitor, dc transmission line monitors, induction machine monitor, largest mismatch tabulation, swing bus summary, and area interchange violation summary are identical to those of activity FNSL. Refer to Section 6.5.1, "Characteristics of Activity FNSL" and Section 6.5.2, "Automatic Adjustments" for details.

Activity FDNS is terminated if the voltage magnitude at a bus is driven to very nearly 0.0.

The solution convergence monitor is similar to that of activity FNSL. Each iteration is identified in the ITER column with its main iteration number, followed by a decimal point, followed by either a zero (for the angle correction calculation) or a five (for the voltage magnitude correction calculation). When the non-divergent solution option is enabled, the inner loop mismatch lines are identified as described above, followed by a second decimal point, followed by the inner loop iteration number.

The solution parameters designating the maximum number of iterations, acceleration, convergence tolerances, and maximum voltage magnitude change are shared with activity FNSL, while the blowup threshold and constant power voltage breakpoint (refer to Section 6.3.13, "Load") are shared among all power flow solution activities. The user may modify any of the above solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

Activity FDNS uses an iterative scheme in which the active power-angle calculation is decoupled from the reactive power-voltage adjustment. It is basically a Newton calculation in which each iteration consists of a pair of half iterations; first with the voltage magnitudes held constant and new voltage phase angles determined, then with the phase angles fixed and new voltage magnitudes calculated.

Activity FDNS uses an approximation of the Jacobian matrix that is insensitive to bus voltages. Thus the matrix used in the active power-angle solution remains fixed throughout the solution, while the matrix used for the reactive power-voltage calculation changes only as voltage controlled buses switch between voltage regulating (PV) and reactive power limited (PQ) boundary conditions.

The rules and characteristics governing the use of activity FDNS are similar to those of activity FNSL. In addition to those rules enumerated in Section 6.5.1, "Characteristics of Activity FNSL", the following apply:

1. Activity FDNS is much less sensitive to a poor initial voltage estimate than is activity FNSL.

2. As the mismatches are reduced, the rate of improvement on successive iterations may be slowed.

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3. The time per half iteration with activity FDNS is roughly 1/5 of the time per FNSL iteration. The start-up time is longer, as the fixed matrices are calculated.

Figure 6.14, “Activity FDNS Convergence Monitor: Non-Divergent Solution Option Enabled” shows the convergence monitor of activity FDNS with the non-divergent solution option enabled. The network condition at the beginning of this example is as it was following the completion of the FNSL solution shown in Figure 6.11, “”b.

```
 ITER       DELTAP      BUS         DELTAQ      BUS        DELTA/V/      BUS       DELTAANG      BUS      SUMSQM       ACCFAC
  0.0       0.2316(      205   )    0.0559(      205   )                                                  0.13701      1.00000
  0.0.1     0.0301(      205   )    0.0794(      154   )                                                  0.19026E-01  1.00000
                                                          0.00000(            )   0.00413(      154   )
  0.5       0.0301(      205   )    0.0794(      154   )                                                  0.19026E-01  1.00000
  0.5.1     0.2790(      205   )    0.0765(      154   )                                                  0.21856      1.00000
  0.5.2     0.1549(      205   )    0.0777(      154   )                                                  0.76493E-01  0.50000
  0.5.3     0.0926(      205   )    0.0785(      154   )                                                  0.37107E-01  0.25000
  0.5.4     0.0614(      205   )    0.0789(      154   )                                                  0.25393E-01  0.12500
  0.5.5     0.0458(      205   )    0.0792(      154   )                                                  0.21540E-01  0.06250
  0.5.6     0.0380(      205   )    0.0793(      154   )                                                  0.20115E-01  0.03125
  0.5.7     0.0341(      205   )    0.0793(      154   )                                                  0.19528E-01  0.01562
  0.5.8     0.0321(      205   )    0.0794(      154   )                                                  0.19267E-01  0.00781
  0.5.9     0.0311(      205   )    0.0794(      154   )                                                  0.19143E-01  0.00391
  0.5.10    0.0306(      205   )    0.0794(      154   )                                                  0.19084E-01  0.00195
  0.5.11    0.0301(      205   )    0.0794(      154   )                                                  0.19026E-01  0.00000
                                                          0.00000(      205   )   0.00000(            )

TERMINATED AFTER   0 ITERATIONS--NON-DIVERGENT OPTION COULD NOT REDUCE MISMATCH

LARGEST MISMATCH:    2.35 MW    7.94 MVAR     8.28 MVA AT BUS    154 [DOWNTN      230.00]
SYSTEM TOTAL ABSOLUTE MISMATCH:              38.45 MVA

SWING BUS SUMMARY:
  BUS# X-- NAME --X BASKV      PGEN     PMAX     PMIN      QGEN     QMAX    QMIN
  3001 MINE                  277.4   9999.0 -9999.0     427.1    600.0  -100.0
```

Figure 6.14. Activity FDNS Convergence Monitor: Non-Divergent Solution Option Enabled

An optimized FDNS solution method is provided in activities that are related to contingency analysis: ac contingency analysis, multi-level contingency analysis, PV analysis and QV analysis. It is a special form of the fixed slope decoupled Newton-Raphson power flow solution method for use in contingency analysis where a set of contingencies is imposed on a base case in turn, and the FDNS solution method is called repeatedly to solve a full ac power flow solution for each contingency. The only difference between the optimized FDNS and the standard FDNS is that, in the former, two techniques are applied to avoid some calculations of the matrices during a power flow solution for a contingency.

- the matrix used in the active power-angle solution (B´ matrix) of the base case is saved and then used in power flow solution for a contingency; the changes in the network due to the contingency are reflected with an additional angle correction vector.
- the matrix used in the reactive power-voltage solution (B¨ matrix) remains fixed when either no bus types switch or bus types switch only from voltage regulating (PV) to reactive power limited (PQ) boundary conditions; the bus type changes are reflected with an additional voltage corrective vector.

The first technique (using the B´ matrix of the base case and the additional angle correction vector reflecting a contingency) produces the identical voltage angles that would be obtained using the B´ matrix of the contingency case at each iteration during the power flow solution for the contingency. If only bus types switches from PV to PQ occur during an iteration, the B¨ matrix from the last iteration is used to calculate voltages at the iteration; then the voltages are modified with the voltage correction vector reflecting bus type switches.

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When the tap setting adjustment is enabled in a contingency analysis, minor discrepancies of voltages between the optimized FDNS and the standard FDNS may be found. That is because there are slight differences in the two methods in the handling of the tap settings in the B¨ matrix.

When the optimized FDNS is enabled in contingency analysis, some matrix calculations are avoided. Therefore, it typically takes less time to run the contingency calculations than it does when the standard FDNS solution is used.

Additional Information
PSS®E GUI Users Guide, Fixed Slope Decoupled Newton-Raphson Power Flow Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Fixed Slope Decoupled Newton-Raphson Power Flow Solution

6.8. Applying Newton-Raphson Power Flow Solution with Inertial or Governor Dispatch

INLF

Run Activity INLF - GUI
Power Flow > Solution > N-R solution with inertial governor dispatch (INLF)... *[N-R Solution with Inertial / Governor Redispatch]*
Run Line Mode Activity INLF - *CLI*
ACTIVITY? >>**INLF** ENTER 0 FOR INERTIAL SOLUTION   1 FOR GOVERNOR RESPONSE SOLUTION: >>
Interrupt Control Codes
AB, NC, NM, DC, FD, MO, NV

The Gauss-Seidel and Newton-Raphson power flow solutions take advantage of the swing bus to absorb all changes in system losses, changes in load and generation dispatch. These assumed steady-state solutions do not account for power flow conditions immediately following a disturbance or system change that modifies the power balance, such as:

- Sudden loss of load, which would be redistributed in the transient period immediately following the disturbance among generators
- Sudden loss of generation, which would be redistributed among all other generators
- System break-up into islands each with its own frequency deviation.

An *Inertial Power Flow* solution represents the effects of governor action and other effects that might stress the system. Following the loss of generation, addition of significant demand or the splitting of a network, conditions can be such that there is a demand resource unbalance. The result is a change in frequency and an adjustment of generator output. Generator output will be controlled initially by machine inertia and fast acting controls, such as excitation and voltage regulators. As time extends beyond the disturbance, the generator governors will have more influence.

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PSS®E provides two power flow solutions with inertia/governor re-dispatch to enable the user to examine system conditions and to obtain a revised generator dispatch for the conditions in which machine inertia plays the major role in modifying generator output and in which governors play a more significant role.

The *Inertial Power Flow* solution gives a quick approximation to system effects in the first 0.5 seconds following a disturbance, during which governor effects are minimal. Generator powers are principally influenced by machine inertias

The *Governor Response* solution represents the system several seconds after an event when governors and exciters have brought the system back to steady-state. The new generator powers are determined by governor droop and load damping characteristics.

The inertial and governor response power flow solution activity INLF uses a Newton-Raphson iterative algorithm to solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. The working case is assumed to have appropriate data changes corresponding to some event imposed upon the solved pre-event power flow case; activity INLF then solves the network for conditions existing either at about half a second following the event (an inertial power flow) or at several seconds after the event (a governor response power flow).

Activity INLF can accommodate switching operations that result in islanding and/or load and generation power unbalances. Swing bus generation changes are redispatched among in-service generator buses, according to either machine inertial (for an inertial power flow) or governor droop and damping (for a governor response power flow) characteristics. Data required for the generation dispatch is specified in an Inertia and Governor Response Data file (refer to Section 6.8.1, "Inertia and Governor Response Data File Contents"). Island average frequencies are estimated and network parameters are made frequency sensitive.

Because INLF permanently changes basic data in the working case to include frequency-dependent effects, it is *strongly* recommended that a Saved Case file of the working case be made with activity SAVE prior to using activity INLF.

6.8.1. Inertia and Governor Response Data File Contents

The input stream to activity INLF consists of a series of records in the following format:

```
I, ID, H, PMAX, PMIN, R, D
```

where:

I	Bus number. Bus I must be present in the working case with a plant sequence number assigned to it (refer to Plant and Machine Sequence Numbers). No default is allowed.
ID	One- or two-character machine identifier used to distinguish among multiple machines at a plant (i.e., at a generator bus). ID = 1 by default.
H	Machine inertia; entered in pu on MBASE base. H = 4.0 by default.
PMAX	Maximum machine active power output; entered in pu on MBASE base. Default PMAX and *PMIN* for this machine are set as described in Section 6.8.4, "Governor Response Power Flow Solution".
PMIN	Minimum machine active power output; entered in pu on MBASE base. If PMAX was set to the default val-

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	ue, the value specified for PMIN is ignored, and PMAX and PMIN for this machine are set as described in Section 6.8.4, "Governor Response Power Flow Solution"; otherwise, PMIN = 0.0 by default.
R	Governor permanent droop; entered in pu on MBASE base. R = 0.05 by default.
D	Turbine damping factor; entered in pu on MBASE base. D = 0.0 by default.

Data records may be entered in any order. Input is terminated with a record specifying an I value of zero.

Any machine for which PMAX = PMIN in the Inertia and Governor Response Data file is treated as non-dispatchable by activity INLF. Its active power remains at its initial value from the working case when its island's swing bus generation change is dispatched among the in-service machines in the island.

In the inertial power flow, except for the use of PMAX and PMIN to categorize machines as either dispatchable or non-dispatchable, only H is used. In the governor response power flow, R, D, PMAX, and PMIN are used.

6.8.2. Operation of Activity INLF

Activity INLF separates the working case into islands. It generates an alarm and terminates if more than ten energized islands are detected. Otherwise, it reports the number of islands detected.

Activity INLF allows the user to select the automatic adjustment options to be enabled. As in the standard power flow solution activities, the default response to the selection request reflects the current automatic adjustment option settings; refer to Sections Section 3.3.3, "Program Run-Time Option Settings" and Section 6.5.2, "Automatic Adjustments" and activity OPTN. Regardless of the selections made, area interchange control is always disabled in activity INLF.

The user must specify the Unit Inertia and Governor Data file containing data records of the form defined in Inertia and Governor Response Data File Contents. If no file is specified, the default data described below is used for all machines. If the file does not exist or some other file system error occurs, an appropriate message is printed. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity INLF.

In reading the input file, gross data errors (e.g., H = 0.0) generate an alarm and the corresponding record is ignored. Data inconsistencies (e.g., PMAX less than initial machine power) generate an alarm and are fixed and used.

Finally, the user must specify the number of the iteration during which generator reactive power limits are to be first applied (refer to activity FNSL). When the inertial power flow calculation had been specified, the default is -1; otherwise, 99 is the default (i.e., the same default used in activities FDNS, FNSL, and NSOL).

Activity INLF checks each island for a swing bus. One (and only one) swing bus must be specified in each island, and, if no swing bus exists in an island, activity INLF assigns the self-regulating generator bus with the largest inertial or governor response, as appropriate, as the island swing. Activity INLF generates an alarm and terminates if:

- Any island contains more than one Type 3 bus
- A Type 3 bus has no dispatchable in-service machines
- An island without a Type 3 bus contains no dispatchable self-regulating generator buses
- An island contains no dispatchable machines.

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Activity INLF functions in a manner similar to activity FNSL. Activity INLF provides for the redispatch of generator powers to allow proper load sharing in a power unbalance condition. This occurs whenever the largest angle change in radians during an iteration is less than the automatic adjustment threshold tolerance, ADJTHR, *and* any island swing bus power has changed by more than the mismatch convergence tolerance since the last dispatch. The largest swing bus power change is tabulated prior to each dispatch calculation.

Island average frequency estimates are updated during the dispatch calculation; they are set to the average of the dispatchable machine frequencies. All bus and line shunts, admittance loads, line charging capacitances and line impedances are modified to reflect their dependency on frequency. Prior to terminating, activity INLF tabulates the frequency estimate and swing bus of each island.

Section 6.8.3, "Inertial Power Flow Solution" and Section 6.8.4, "Governor Response Power Flow Solution" contain details specific to the inertial and governor response power flow solutions respectively.

Where applicable, activity INLF handles the network topology check, treatment of generator reactive power limits, load, generator, switched shunt, FACTS device, VSC dc lines, and dc line boundary conditions, the blowup check, scaling of the voltage magnitude change vector, acceleration, and interrupt control codes in the same way as activity FNSL. The selection and application of automatic adjustments is identical to that of activity FNSL except that the area interchange control and the non-divergent solution options are not available in activity INLF. The basic solution convergence monitor, FACTS device monitor, dc transmission line monitors, induction machine monitor, largest mismatch tabulation, and swing bus summary are identical to those of activity FNSL. Refer to Section 6.5.2, "Automatic Adjustments" for details.

The solution parameters designating the maximum number of iterations, acceleration, convergence tolerances, and maximum voltage magnitude change are shared with activity FNSL, while the blowup threshold and constant power voltage breakpoint (refer to Section 6.3.13, "Load") are shared among all power flow solution activities. The user may modify any of the above solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

6.8.3. Inertial Power Flow Solution

The inertial power flow solution is intended to indicate system conditions that would exist half a second after the initiation of an event on a steady-state system condition. In this time frame, it is assumed that generator overcurrent protection and governor effects are minimal, and that changes in generator powers are influenced principally by machine inertias.

In this solution, generator scheduled voltages at those generator buses that are initially at a reactive power limit are set to their pre-disturbance (i.e., initial working case) voltages. The default to the var limits selection is to ignore reactive power limits. By default, tap adjustment and phase shift angle adjustment are disabled, dc converter taps are locked, and switched shunts are active. These settings may be overridden by the user. Area interchange control and the non-divergent solution option are always disabled.

For machines for which no data record is successfully read (e.g., if no data file is specified, or if it does not contain data records for all in-service machines), a machine inertia constant of 4.0 on machine base (i.e., on MBASE base), is used. Clearly, the proper specification of MBASE for any machine using default data is essential.

Speed deviation and frequency in per unit of each machine are estimated according to the equations:

$$\frac{P_t - P_e}{1 + n} = 2H \frac{dn}{dt} \tag{6.23}$$

$$n = 0.5 \frac{dn}{dt} \tag{6.24}$$

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$$f = 1 + n \quad (6.25)$$

where:

P_t	= Turbine power; assumed equal to the initial machine electrical power.
P_e	= Machine electrical power.
H	= Machine inertia.
n	= Machine speed deviation.
f	= Machine terminal frequency.

Equation 6.24 assumes that the rate of change of frequency is linear over the half second time frame.

6.8.4. Governor Response Power Flow Solution

The governor response power flow solution is intended to indicate system conditions that would exist at least several seconds after the initiation of an event on a steady-state system condition. In this time frame, it is assumed that voltage regulator and turbine governor effects are influential in bringing the system to a new steady-state condition, and that changes in generator powers are determined by governor droop and damping characteristics.

In this solution, generator scheduled voltages are unchanged except as described below, and the default to the var limit selection is to honor generator reactive power limits. By default, tap adjustment by the stepping method (refer to Voltage Control) and phase shift angle adjustment are enabled, dc taps are unlocked, and switched shunts are active. These settings may be overridden by the user. Area interchange control and the non-divergent solution option are always disabled.

A generator bus that is specified as an island swing by activity INLF and that was initially at a reactive power limit generates an alarm, and its scheduled voltage is set to its pre-disturbance (i.e., initial working case) voltage.

For machines for which no data record is successfully read (e.g., if no data file is specified, or if it does not contain data records for all in-service machines), the following default data, on machine base (i.e., on MBASE base), is used: R = 0.05 and D = 0.0. The user may specify one of the following methods of assigning default data for PMAX and PMIN:

- Set them to 1.0 and 0.0 respectively for any machine for which no data record is successfully read, and to the active power limits contained in the working case (refer to Generator Data) for any machine for which a data file record was successfully read but PMAX was set to the default value (Pre Rev-29 approach).
- Set them to 1.0 and 0.0 respectively (0.0 to 1.0 for both types).
- Set them to the working case active power limits (working case values for both types).

Clearly, the proper specification of MBASE for any machine using default data is *essential*.

Speed deviation and frequency in per unit of each machine are estimated according to the equations:

$$n = \frac{P_t - P_e}{D + (1/R)} \quad (6.26)$$

$$f = 1 + n \quad (6.27)$$

where D and R are as defined in Section 6.8.1, “Inertia and Governor Response Data File Contents”, and P_t, P_e, n, and f are as defined in Section 6.8.3, “Inertial Power Flow Solution”.

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If, during the solution, the total generation requirement in an island drives all generators in the island to their high or low power output limits, an appropriate message is printed and activity INLF is terminated.

6.8.5. Example of a Generator Contingency Analysis Solution

An example Inertial solution can be run using the savnw.sav power flow case found in the C:\Program Files \PTI\PSSE33\EXAMPLE directory.

The test will be to trip the machine at bus 101 in the power flow. That unit is generating 750 MW. The inertial data file to be used assumes the following information contained in Table 6.6, "Inertial Power Flow Solution Data File".

Table 6.6. Inertial Power Flow Solution Data File

I	ID	H	PMAX	PMIN	R	D
101	1	4	0.9	0	0.05	0
102	1	4	0.9	0	0.05	0
206	1	4	0.9	0	0.05	0
211	1	5	0.85	0	0.05	0
3011	1	4	0.9	0	0.05	0
3018	1	4	0.9	0	0.05	0

Running the Inertial and Governor power flow solutions will result in a summary of network conditions and a convergence monitor at the *Progress* tab (see Figure 6.15, "Progress Reports for Inertial and Governor Power Flow Solutions").

```
 REACHED TOLERANCE IN   9 ITERATIONS

 LARGEST MISMATCH: 0.00 MW -0.01 MVAR  0.01 MVA AT BUS 211 [HYDRO_G     20.000]
 SYSTEM TOTAL ABSOLUTE MISMATCH:              0.03 MVA

 SWING BUS SUMMARY:
   BUS# X-- NAME --X BASKV      PGEN     PMAX    PMIN      QGEN     QMAX    QMIN
   3011 MINE_G      13.800     446.5    900.0     0.0     109.7    600.0  -100.0

 FREQUENCY FOR ISLAND  1 IS 59.287 HERTZ
   SWING BUS FOR THIS ISLAND IS 3011 [MINE_G  13.800]
```

Inertial Response

```
 REACHED TOLERANCE IN  10 ITERATIONS

 LARGEST MISMATCH: 0.00 MW  -0.04 MVAR   0.04 MVA AT BUS 3008 [CATDOG 230.00]
 SYSTEM TOTAL ABSOLUTE MISMATCH:              0.08 MVA

 SWING BUS SUMMARY:
   BUS X--- NAME ---X      PGEN     PMAX    PMIN      QGEN     QMAX    QMIN
  3011 MINE_G  13.800     809.4    900.0     0.0     77.9    600.0  -100.0

 FREQUENCY FOR ISLAND  1 IS 59.478 HERTZ
   SWING BUS FOR THIS ISLAND IS  3011 [MINE_G  13.8]
```

Governor Response

Figure 6.15. Progress Reports for Inertial and Governor Power Flow Solutions

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It can be seen in Figure 6.15, "Progress Reports for Inertial and Governor Power Flow Solutions" that the frequency in the inertial response, shortly after the disturbance has fallen to 59.287 Hz while following governor action it has recovered to 59.478 Hz. Furthermore, following governor action the swing bus has an increased power generation at 809 MW.

To further examine the solution it is useful to look at the redispatches. In Figure 6.16, "Initial Dispatch Compared to Inertial and Governor Power Flow Redispatch Levels" the dispatches are shown for the Base case, the situation subject to inertial redispatch and the situation subject to governor action.

Inertial redispatch is not controlled by governor action. The amount of power from each unit is a function of its inertia and it can be seen that some units are producing power above PMAX. This will slow down the units. Following governor action, machine powers have been reduced to levels at or within their maximum capabilities.

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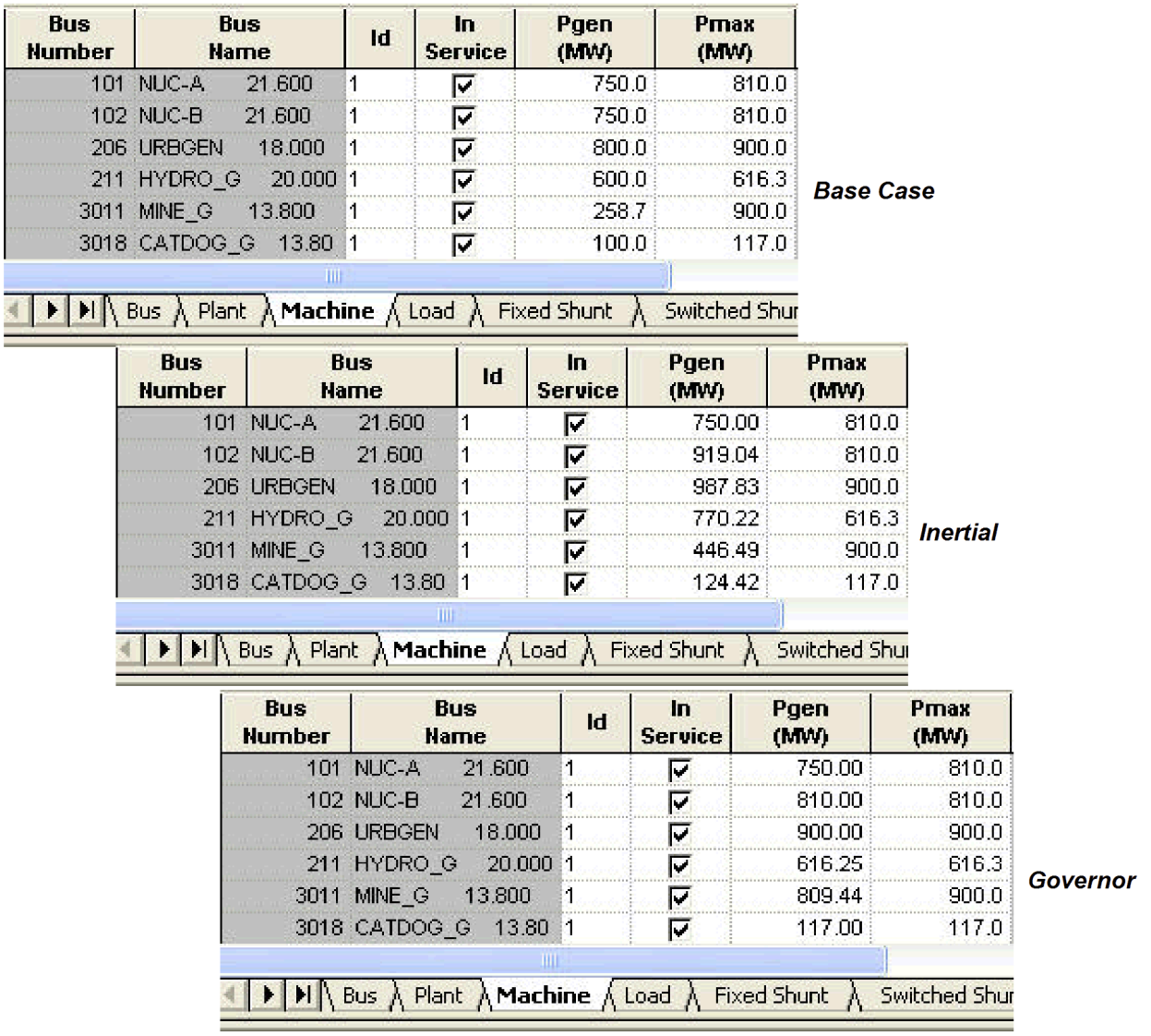

Base Case

Bus Number	Bus Name	Id	In Service	Pgen (MW)	Pmax (MW)
101	NUC-A 21.600	1	☑	750.0	810.0
102	NUC-B 21.600	1	☑	750.0	810.0
206	URBGEN 18.000	1	☑	800.0	900.0
211	HYDRO_G 20.000	1	☑	600.0	616.3
3011	MINE_G 13.800	1	☑	258.7	900.0
3018	CATDOG_G 13.80	1	☑	100.0	117.0

Bus | Plant | **Machine** | Load | Fixed Shunt | Switched Shur

Inertial

Bus Number	Bus Name	Id	In Service	Pgen (MW)	Pmax (MW)
101	NUC-A 21.600	1	☑	750.00	810.0
102	NUC-B 21.600	1	☑	919.04	810.0
206	URBGEN 18.000	1	☑	987.83	900.0
211	HYDRO_G 20.000	1	☑	770.22	616.3
3011	MINE_G 13.800	1	☑	446.49	900.0
3018	CATDOG_G 13.80	1	☑	124.42	117.0

Bus | Plant | **Machine** | Load | Fixed Shunt | Switched Shu

Governor

Bus Number	Bus Name	Id	In Service	Pgen (MW)	Pmax (MW)
101	NUC-A 21.600	1	☑	750.00	810.0
102	NUC-B 21.600	1	☑	810.00	810.0
206	URBGEN 18.000	1	☑	900.00	900.0
211	HYDRO_G 20.000	1	☑	616.25	616.3
3011	MINE_G 13.800	1	☑	809.44	900.0
3018	CATDOG_G 13.80	1	☑	117.00	117.0

Bus | Plant | **Machine** | Load | Fixed Shunt | Switched Shur

Figure 6.16. Initial Dispatch Compared to Inertial and Governor Power Flow Redispatch Levels

6.8.6. Application Notes

The rules and characteristics governing the use of activity INLF are similar to those of activity FNSL (refer to Section 6.5.1, "Characteristics of Activity FNSL").

It may often be appropriate to precede activity INLF with activity CONL to convert the load boundary conditions from the typical constant MVA characteristic used in conventional power flow calculations to a representation suitable for network conditions involving abnormally low or high voltages (refer to Section 6.3.13, "Load" and activity CONL).

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All frequency-sensitive data items (refer to Section 6.8.2, "Operation of Activity INLF") are modified in the working case at the completion of activity INLF. Therefore, the standard power flow reporting activities such as POUT, AREA, and SUBS implicitly calculate results at off-nominal frequencies, even though island frequencies are not preserved. As a result of this, and the possible modification of generator scheduled voltages described above, it is *strongly* recommended that activity SAVE be used with care following any execution of activity INLF. In particular, the user should not overwrite the Saved Case file containing the pre-event network solution, or any milestone Saved Case file preserved prior to selecting activity INLF.

The generator power redispatch and frequency calculation use the machine powers as specified at the time activity INLF is selected. This, coupled with the changes to data in the working case described above, renders the following activity sequence invalid:

- FNSL (to solve the pre-event condition)
- CHNG (to set up the event)
- INLF (solution; not converged)
- INLF (to continue solution)

Activity INLF may require more iterations than activity FNSL, particularly when the system, or an island thereof, is under severe stress. The iteration limit ITMXN may be increased via activity CHNG or the *[Solution Parameters]* dialog.

Should activity INLF fail to converge in an islanding situation, inspection of solution results may indicate that all islands except one are solved. Such a solution may be adequate for the application at hand. Note that in an islanding case, those islands that are not of particular interest may be disconnected with activity TREE. This should be done after implementing the data changes required to model the event being studied, and before selecting activity INLF.

Just as in a conventional power flow solution, selection of a swing bus for each island can have an effect on the solution convergence properties for that island. Proper selection of island swing buses requires an understanding of the system. The user may find it helpful to make several attempts at solution for a particularly troublesome island, each with a different island swing bus.

It is good practice to save the working case in a Saved Case file before selecting activity INLF. This is useful for the situation in which activity INLF fails to converge. The Saved Case may be restored, changes made (e.g., different load characteristics, different island swing buses, or other techniques described above) and a new solution attempted without having to respecify the data changes required to model the event under study.

Note again that the data items described in Section 6.8.1, "Inertia and Governor Response Data File Contents" are specified on machine base.

Additional Information
PSS®E GUI Users Guide, Newton-Raphson Power Flow Solution with Inertial / Governor Dispatch
PSS®E Command Line Interface (CLI) *Users Guide*, Running Newton-Raphson Power Flow Solution with Inertial / Governor Dispatch

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6.9. Voltage droop Control

6.9.1. Introduction to voltage droop control

Generators at a type 2 bus may control either local bus voltage or the voltage at a remote bus in the setpoint mode. These generators that provide for voltage control to a setpoint have their reactive power output or consumption set during each power flow iteration, such that the voltage magnitude at the controlled bus is held at its scheduled value as long as the device's reactive outputs are within their limits.

Renewable generators are often not configured with a traditional "voltage setpoint" control. Instead, they follow a voltage droop control with control centered at the controlled bus, which is the point of interconnection (POI). The voltage droop control scheme is part of the power plant control model used by renewable generators. In voltage droop control, a QV characteristic curve is specified where the reactive power at the point of interconnection is a function of the controlled bus voltage. The purpose of the curve is to enforce that the total reactive flow going into the bus from the generators that belong to the power plant control is equal to this value.

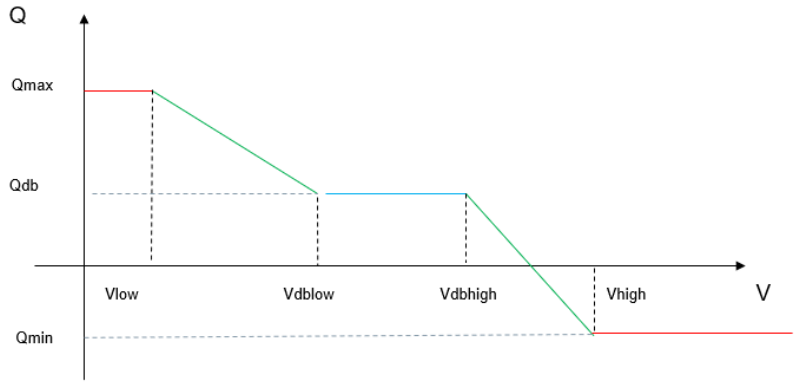

Figure 6.17. QV Characteristic Curve of Voltage Droop Control

A typical QV characteristic curve is shown in the above figure, where V is the voltage at the controlled bus, and Q is the reactive power injection at the controlled bus. The voltage droop control QV characteristics a piece wise linear function defined with 7 input variables and can have up to five segments. There is a deadband region between V_{dblow} and V_{dbhigh} in which the reactive power output remains constant at Q_d. When voltage is less than V_{dblow}, the reactive power linearly increases up to Q_{max} at a voltage of V_{low} and when voltage is above the V_{dbhigh} the reactive power linearly decreases down to Q_{min} at a voltage of V_{high}. The reactive power, Q, at the controlled bus is defined in the form:

$$Q = \begin{cases} Qmax & V \le Vlow \\ Qdb + \left(V - Vdblow\right) * \frac{Qmax - Qdb}{Vlow - Vdblow} & Vlow < V \le Vdblow \\ Qdb & Vdblow < V \le Vdbhigh \\ Qdb + \left(V - Vdbhing\right) * \frac{Qdb - Qmin}{Vdbhigh - Vhigh} & Vdbhigh < V \le Vhigh \\ Qmin & Vhigh < V \end{cases}$$

6.9.2. Voltage Droop Control Data Format

The voltage droop QV characteristic curve is defined with user defined table in Rawx data table format:

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```
"Droop":{
"fields":["DroopName:string","Status:int", "Regbus:int","Qmax:real", "Qdb:real", "Qmin:real",
          "Vlow:real", "Vhigh:real", "Vdblow:real", "Vdbhigh:real"],
"data":[
     ["A", 1, 0, 50.0, 10, -50.0,  0.94, 1.04, 0.97,1.01],
     ["B", 1, 0, 50.0,  0, -50.0,  0.95, 1.05, 0.98,1.02] ]

}
```

Figure 6.18. sample record of QV Characteristic Curve of Voltage Droop Control

DroopName	The non-blank alphanumeric identifier assigned to this voltage droop controller. Each voltage droop control must have a unique NAME.
Status	Status of voltage droop control, one for in service and zero for out of service. If it is set to one, then the power flow solution will attempt to solve to meet the QV characteristic. If set to 0, then the devices that are assigned to this PPC will default back to their original behavior of controlling a regulated bus to a voltage setpoint.
Regbus	Bus number, or extended bus name enclosed in single quotes, of the bus for which voltage is to be regulated by the voltage droop curve. Regbus=0 by default. If Regbus is not specified, the software will automatically determine a Regulated Bus from the generators that belong to the PPC. The regulated bus of one of generators in the power plant control is selected as Regbus for the power plant control.
Q_{max}	Maximum reactive power in Mvar at the Regulated Bus being contributed by the devices in the Power Plant Controller.
Q_{db}	The reactive power in Mvar at Regbus when voltage is within V_{dblow} and V_{dbhigh}.
Q_{min}	Minimum reactive power in Mvar at the Regulated Bus being contributed by the devices in the Power Plant Controller.
V_{low}	Per unit voltage at the Regulated Bus at Q_{max}.
V_{dblow}	Low end of the per unit voltage range at the Regulated Bus when operating at Q_{db}.
V_{dbhigh}	High end of the per unit voltage range at the Regulated Bus when operating at Q_{db}.
V_{high}	Per unit voltage at the Regulated Bus at Q_{min}.

Generators in voltage droop control are defined by extended filed “droopname” of generator table in the rawx data format:

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```
"generator":{
"fields":["ibus", "machid", "droopname:string"],
"data":[
    [9,     "1",   "B"],
    [107,   "1",   "A"] ]
},
```

Figure 6.19. sample record of generators in Voltage Droop Control

DroopName	The non-blank alphanumeric name of voltage droop controller.

6.9.3. Application notes

Currently the voltage droop control mode is only available in Fully-coupled Newton-Raphson power flow solution (FNSL). When the QV characteristic curve and generator voltage droop assignments are defined in the RAWX file, FNSL solution will honor the voltage droop control to regulate the reactive power at the controlled bus in terms of the bus voltage. The difference between the actual reactive power and desired reactive power at the controlled bus must be less than TOLN for power flow solutions to converge.

The program automatically determines the reactive power Q at the regulated bus. Q is the reactive power injection to the regulated bus, it is calculated as the sum of reactive power flows on the branches that are connected to the regulated bus which is remotely controlled by generators, and the reactive power output from the generator at the regulated bus, as defined: $Q=Q_{flow}+Q_{gen}$

Where Q_{flow} is the total reactive power flows on the branches connected to the controlled bus, and Q_{gen} is the total reactive power outputs of the generators at the controlled bus.

Example system is shown as below. The Generators at bus 3018 and 3008 belongs to the same voltage droop control scheme, and control the voltage at the bus 3008. The reactive power Q is the sum of reactive flow on the transformer branch 3008 to bus 3018, measured at the bus 3008 end, and the reactive power output of the generator at bus 3008. It is positive for reactive power Q entering the bus.

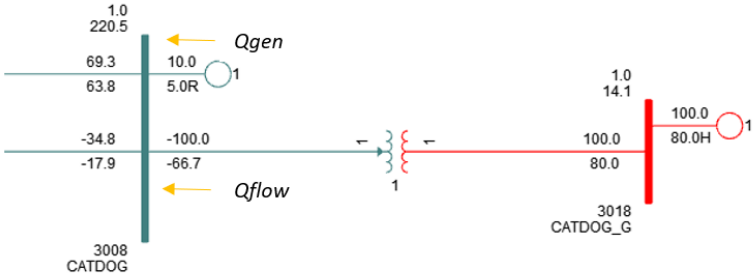

Figure 6.20. Example of Voltage Droop Control

A bus can not be regulated by both generators in voltage droop control mode and devices (generators, switched shunt etc.) in “setpoint” voltage control mode. In this case, the voltage droop control is disabled,

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and generators in voltage droop control are fixed at the current Q outputs. When a bus is remotely regulated by generators in voltage droop control mode, the reactive power flows must be measured on the branches that are connections between the generators and the rest of system, if these branches are turned off, the generators are isolated from the system. An example of invalid voltage droop control scheme is shown in the figure below, where generator at bus 3018 in voltage droop control mode is remotely controlling voltage at the bus 3005. Because there is a branch between 3008 to 154, the reactive power at the bus 3005 is not totally controlled by the generator at bus 3018

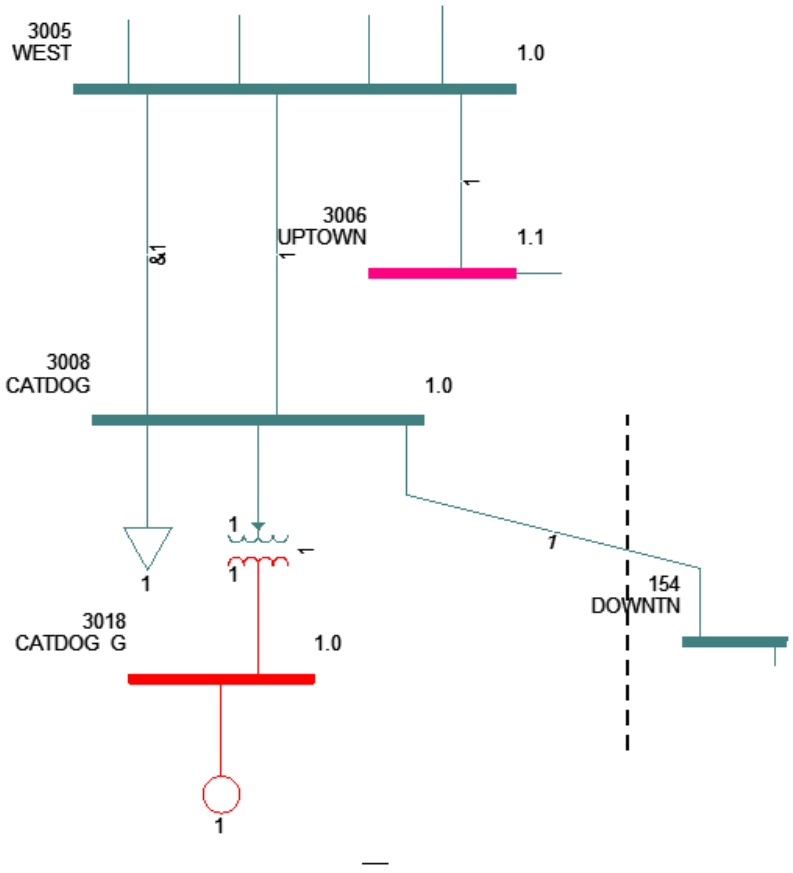

Figure 6.21. Example of Invalid Voltage Droop Control

Generators in power plant control mode will control the reactive power at the regulated bus to follow the QV characteristic curve till the reactive power outputs of generators hit their limits. The figure shows that the reactive power at the regulated bus is limited by the reactive power limits of the generators as shown by the solid purple lines.

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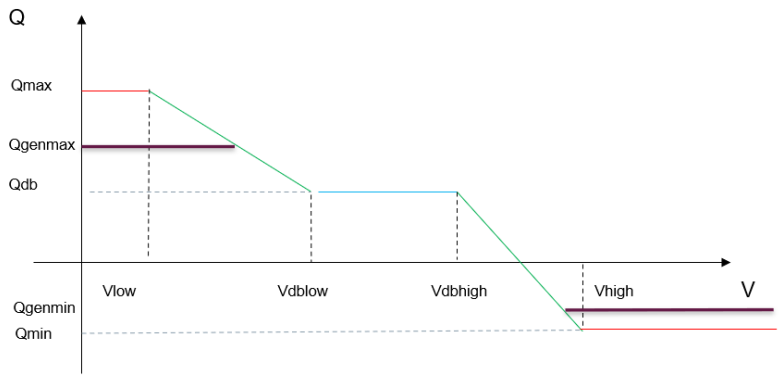

Figure 6.22. QV characteristic curve limited by generators

When there are multiple generators in the same mode, reactive power output of each generator is determined by their RMPCTs.
The Q inputs of voltage droop QV characteristic curve must be monotonically decreasing, and the V inputs of the curve must be monotonically increasing. V_{dblow} must be less than or equal to V_{dbhigh}. If V_{dblow} is less than V_{dbhigh}, to prevent the numerical issue due to the narrow band, the difference of V_{dbhigh} and V_{dblow} must be greater than 0.02pu by default.

6.10. Applying the Robust Solution

6.10.1. Introduction to the Robust Power Flow Solution

It is sometimes challenging to solve a power flow case to find the bus voltage vectors that meet the convergence criteria. The divergence can happen due to one of the causes below: Firstly, a failure in convergence may come from the failure of the power flow solution method, when the solution leaves the feasible space and is unable to return. Secondly, divergence may be due to an infeasible operating point. The power flow may represent a voltage collapse condition, where there is insufficient reactive power to supply load and losses in some portion of the system, or where the load demand exceeds the transfer capability of the network. Thirdly, non-convergence of the power flow can be caused by too many control adjustments in the network models, such as tap-changing or phase-shifting transformers, switchable shunt capacitors or reactors, area interchange control, HVDC lines and FACTS device controls. Incompatible or poorly coordinated control adjustments can result in non-convergence too. If the network models do not represent the physical power system accurately, in terms of the network topology and parameters of all components in the system, it is most likely that the models will result in infeasible solutions. it is recommended that uses apply data checking functionalities such as BRCH and TPCH for checking model parameters. Based on the engineering experiences of solving the 'bad' cases, some strategies to address solution convergence problems are listed as following:

1. Smooth initial values of the bus voltage vector. The Newton-Raphson solution methods, particularly the Full Newton-Raphson algorithm (FNSL), are sensitive to the initial values, sometimes causing a solution to blow up abruptly. Adjusting the initial voltage magnitudes and phase angles at the problematic buses (e.g., by not leaving them at the default values of one per unit voltage and zero degree angle) may aid the solution process. Flat start option should be used if initial values of the bus voltages are 'far away' from the values that meet the convergence criteria.

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2. Switch power flow solution methods. It is rare that a case can not be solved by one of power flow solution methods. The Decoupled Newton-Raphson solution methods (FDNS) may have difficulty converging in cases with branches that have high R/X ratio (transmission lines and transformers typically have low R/X ratios), but the Newton-Raphson solution methods can be used for such cases. On the other hand, FDNS method is not sensitive to the initial values, it can be used to solve the case with 'bad' initial values of bus voltages, or used to estimate the initial values for FNSL solutions.

3. Change the power flow control options. That includes to disable local controls to avoid the effects of small adjustments to the Jacobian that may lead to divergence; For example, a narrow voltage band for a voltage controlling transformer could cause the tap adjustments to oscillate from iteration to iteration; or, ignore the reactive power limits of the generators to increase the feasible solution space; or, prevent the automatic adjustments of switchable shunts and transformer taps after a certain number of iterations to avoid oscillation.

4. Apply control adjustments step by step to prevent from big changes between iterations. For example, apply phase-shifting transformer control first, if the case is converged, apply the tap-changing transformer control, and so on till all required control options are turned on.

5. Change the power flow solution parameters. For example reduce the Newton-Raphson solution acceleration factor to slow down any potential divergence and keep the solution within the feasible region; or, increase the iteration limit in cases where convergence is slow; or, increase the mismatch tolerance in cases with slow convergence or near voltage collapse; or, increase the automatic adjustment threshold tolerance to activate switched shunt adjustments sooner in the iteration process to allow better voltage control before voltage collapse starts to occur.

6. Change the network model. For example, a power flow solution may oscillate due to generators with small reactive power capacity controlling bus voltage, the solution is able to converge if those generators are locked to constant reactive power outputs. Changing the swing bus may help convergence; or, change the remote voltage control to local voltage control for more effective control; or, change the control mode of some switchable shunts from discrete to continuous temporarily.

7. Apply the non-divergent power flow solution method. The non-divergent Newton-Raphson power flow solution is designed to terminate the iterative process before the bus voltage vector is driven to a state where large mismatches and unrealistic voltages are present. The robust power flow solution activity implements these heuristic strategies to solve the 'bad' case. It automatically applies different solution methods (FNSL/FDNS/NSOL) with different combination of power flow options and parameters, as well as technique to smooth initial bus voltages and make model changes, the flow chart of robust power flow solution is shown below briefly.

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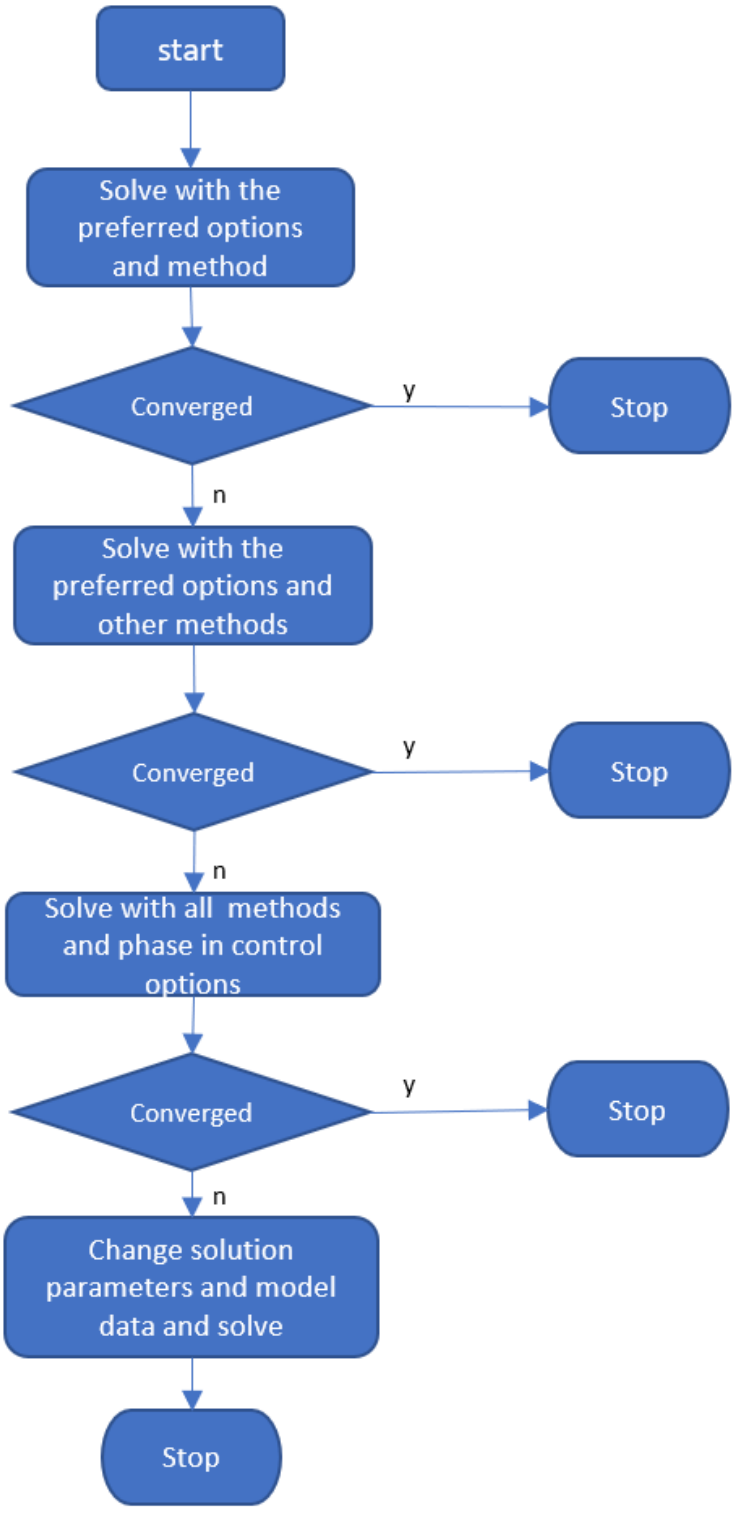

Figure 6.23. Flow Chart of Robust Power Flow Solution Process

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6.10.2. Running the Robust Power Flow Solution

GUI Power Flow ◊ Solution ◊ Solve ◊ Robust Solution

The API used to perform robust solution is ROSL. The following input data is specified for the robust solution.

1. The preferred solution method and power flow options are specified by the users, the RSOL applies the preferred solution method and power flow options first to solve the case; if the case can not be solved, it sequentially applies other methods to solve the case as shown in the flow chart(see section 6.2, “power flow solution methods).

2. Mismatch threshold (MVA) for voltage estimation. If the active power or reactive power mismatch at a bus is greater than the threshold, its bus voltage is adjusted to reduce the mismatch.

3. Minimum Var band (MVar). When reactive power output band of a generator is less than the minimum var band, its reactive power output remains constant at either its Qmax or Qmin during entire solution.

4. On solution failure. If robust power flow solution fails to converge, one of three network conditions can be kept in the working case: restore best found solution, restore original solution and keep final failed solution.

The solution status of robust power flow solution methods is one of successful, unsuccessful, or successful with options being different than given values. In a successful solution, all preferred power flow options are honored and converge criteria are met, though solution parameters and method are determined by the RSOL. When the case is solved successfully with options adjusted by ROSL, the progress message indicates one of three changes made by RSOL to solve the case: mismatch tolerance increase, phase shift locked and conversion of constant PQ load to constant Y load. When a bus voltage is below the PQBRAK (see section, “boundary conditions of constant MVA loads"), RSOL may either use constant power load characteristic to represent the loads at the bus or convert the constant power loads to constant impedance loads. The converted loads are reported in the progress windows. When the solved case is saved, the options used for solution are saved too.

6.11. Overview of Contingency Analysis

Contingency analysis covers a variety of analytical investigations performed by both system planners and operators. The system planner’s objective is to identify the network elements that will be required to maintain system operation within planning criteria. The general requirement is to identify capital investments and operating costs for long term future developments.The system operator’s objective is to identify the manner in which the system must be operated to maintain system security both in the near term (days, weeks, months) with existing network elements or the medium term (one to three years) during which a limited amount of equipment could be installed.

The basic approach to contingency analysis is to:

- Establish generation/demand base case scenarios that are to be tested.

- Identify variations of generation and demand for the base cases for the time period of development or operation.

- Identify the tests (contingencies) to be performed for both steady-state and dynamic analysis and the system conditions that are acceptable or required prior to and during such contingencies. The tests and required post-contingency conditions are generally described by *Reliability Criteria*.

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- Perform the tests.

6.11.1. Reliability Testing Criteria for Contingency Analysis

Criteria describe tests to be performed and required conditions prior to and during the test conditions. While there may be variations in criteria between electric power utilities, regions and countries, there is significant commonality.

Typical steady-state tests can include:

- Base case with all elements in service.
- Single contingencies (N-1). Loss of any transmission line or transformer or generator. These are often termed probable or credible contingencies.
- Double contingencies (N-2). Simultaneous loss of two single-circuit transmission lines, a double-circuit line or dc bipole. Variations on these contingencies exist worldwide specifically with respect to the definition of double circuit and the option of non-simultaneity of loss (N-1-1). These too are credible or probable contingencies.
- Less probable contingences and/or extreme contingencies can include loss of entire substations or multiple generators.

Typical dynamic testing will include the same family of contingencies and are augmented by representation of the severity of the initiating disturbance, which results in the loss of system elements (three-phase and single-phase faults with normal or delayed clearing times for example).

Acceptable system conditions prior to and subsequent to the contingencies depend on the severity of the contingency and include:

- Voltages within defined normal or emergency limits.
- Changes in voltage within defined limits.
- Branch loadings within normal or emergency loading limits.
- Maintenance or loss of limited amounts of load.
- Maintenance of system integrity or breakdown into viable sections.
- Maintenance of transient and dynamic stability.

Such criteria are deterministic in the sense that the scenario being tested must comply with the acceptable system conditions or is considered to have failed the test. A failure implies the need for additional system elements (for planning) or an adjustment of precontingency test conditions (for operations).

An overview of deterministic reliability tests is summarized in Table Deterministic Reliability Tests.

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Table 6-8. Deterministic Reliability Tests

Overview of Deterministic Reliability Tests			
	Test Conditions	**Analysis**	**Acceptable Conditions**
Reliability Test Criteria for Transmission Expansion	Normal Steady-State conditions	Steady-State Power Flow	System within normal loading and voltage limits
	Single Contingencies (Credible - more probable)	Steady-State Power Flow	System within emergency loading and voltage limits immediately after outage and within normal limits after system adjustments
		Dynamic Analysis	Transiently and dynamically stable
	Double Contingencies (Credible - more probable)	Steady-State Power Flow	System within emergency loading and voltage limits after system adjustment.
		Dynamic Analysis	Transiently and dynamically stable
	Less Probable Contingencies Supplementary Tests	Steady-State Analysis	No voltage collapse or overload cascading
		Dynamic Analysis	Transiently and dynamically stable
	Extreme Contingencies Supplementary Tests	Steady-State and Dynamic Analysis	Avoidance of widespread load interruptions, uncontrolled cascading and black-outs

6.11.2. Applying Deterministic Criteria

This type of study comprises the outage of system elements and an examination of voltage and loading conditions both prior to and subsequent to the outage (contingency).

Generally, for the normal condition (base case with all elements in service) the user is establishing acceptable normal conditions of voltage and loading of lines, transformers and generators. Under these conditions the user will normally solve the power flow with control options activated for transformer taps, switched shunts and other devices. Under contingency conditions, depending on the criteria in use, the user will often block control actions as a function of how equipment in the user's system operates.

The basic process is one of:

- Disconnecting a system element.
- Solving the new power flow.
- Examining system conditions using PSS®E reporting facilities.

The user's response to the results of contingency cases will depend on the study objectives.

A limited system analysis can be performed merely by manually performing the steps outlined in the basic process. A more detailed analysis, specifically of a large system can involve the simulation and examination of thousands of contingencies. For this, PSS®E provides procedures for automatic testing and reporting: AC Contingency Analysis and Multi-level AC Contingency Analysis. These procedures are described in this chapter.

Applying Transmission Transfer Limit Analysis

For economic or market requirements, the user will often need to identify the maximum level of power that can be transferred across a network; often from identified sending and receiving areas or systems. Limits are usually identified for both normal and contingency conditions.

As power transfers are increased across a network, limits will be reached when the thermal loading of network elements is exceeded or when voltages fall outside acceptable levels; usually during contingencies. The procedures for identifying transfer limits, available in PSS®E, are described in Chapter 14, *Transmission Pricing and Open Access* and Section 6.19, "Performing PV or QV Analyses".

Applying Voltage Stability Analysis

This type of analysis is similar to transfer limit analysis but is specifically concerned with situations where voltage control becomes the major factor in determining the power transfer capabilities of a network. Voltage stability is a phenomenon concerning the eventual collapse of voltage as system loading is increased and/or power transfers are increased.

Widely accepted measures of a networks vulnerability to voltage instability or collapse are:

- Power - Voltage curves (PV) that identify voltage levels in a system as power transfers increase.
- Reactive - Voltage curves (QV) that identify reactive margins versus system voltage levels.

Both PV and QV curves show limiting operating conditions in a network for both normal and contingency conditions. The procedures for analyzing these conditions with PSS®E are described in Section 6.19, "Performing PV or QV Analyses".

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6.11.3. Applying Probabilistic Reliability Criteria

The behavior of power systems is probabilistic. Loads are always uncertain. The events that cause the outages my vary, but are always in the form of unplanned events such as lightning strikes, falling tree limbs, human errors, and so on; these events are thus of a random nature. The weakness of deterministic reliability analysis is that it does not reflect the probabilistic nature of power system behavior, but treats every contingency tested equally significant. In deterministic reliability analysis, every contingency being tested must comply with the acceptable system conditions or is considered to have failed the test. A failure implies the need for additional system elements (for planning) or an adjustment of precontingency test conditions (for operations). It is impossible to have zero risk of system failures in power systems because contingencies are uncontrollable. tn many cases, a decision must be made to accept a risk as long as it can be technically and financially justified, in other words keeping risk of system failures within a acceptable range. on the other hand, customers have the rights to know how often, for how long these outages cause discontinuity of load supply. Probabilistic reliability assessment (see Section 6.17, “Calculating Probabilistic Reliability”) is provided to answer these questions. The module uses contingency enumeration technology and consists of a calculation part, which evaluates each contingency and models predefined tripping and corrective action sequences, and an analysis part, which conducts a detailed analysis on the basis of the evaluated contingency sequences to calculated following probabilistic indices:

- Probabilistic indices of flow overloads
- Probabilistic indices of voltage violations
- Expected unserved energy
- Interrupted power

The basic procedure to perform probabilistic reliability study is:

- perform a contingency analysis following the procedure for contingency analysis discussed in the above section
- prepare outage statistics data
- perform probabilistic reliability assessment

6.11.4. About Generation Dispatch

Contingency events such as generator outages, the separation of the network into islands and the shedding of load may lead to imbalances in the power resources and demand for interconnected ac systems. An individual dispatch mode can be specified for a contingency causing the imbalance. A DISPATCH keyword should be employed in the contingency specification record that specifies the contingency, and the record be followed by a group of participating bus specification records that defines a dispatch group in the Contingency Description Data file (see *???*), so that the change in the boundary condition is allocated among participating generators selected in the dispatch group.

AC Contingency Analysis also provides an option to perform a common generation dispatch to handle imbalances for all contingency events. When neither individual dispatch mode nor common dispatch mode is specified, these imbalances are totally absorbed by the swing bus in the island; any island without the swing bus is shut down. If both modes are enabled, the common dispatch acts after generation dispatch is done with the individual dispatch mode for a contingency event.

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The common generation dispatch is enabled by selection of one of four dispatch modes and a dispatch subsystem. Imbalances are then apportioned among machines in the dispatch subsystem in specified participation factors. For islands without the swing bus, the bus with largest generation capacity is designated as the island swing bus. When there is insufficient capacity available from the reserves, load is shed. Reactive power loads are shed in proportion to active power loads to retain the power factor at affected buses. The following describes the computation procedures.

1. Identify all islands in the network using a tree searching technique.

2. For each island, assure that a swing bus exists. If there is no existing swing bus, the bus with the largest generation capacity in the island is chosen as the swing bus. If no such bus can be found, the entire island is shut down and all load shed.

3. Sum the present generator output and generation requirement to find if there is a MW imbalance. The generation requirement in the island is given by the total of all bus loads, active power bus shunts, and active power losses, including dc lines.

4. If generation deficiency, DEF, exists, it is met by increasing the present generation of participating machines in the island by an amount proportional to the given participation factors. If DEF still exists after the active power reserves of participating machines are exhausted, all on-line machines in the island are participating in the generation dispatch to meet generation deficiency in the specified dispatch mode.

   $$PGEN_i^1 = PGEN_i^0 + p_i \times DEF \quad (6.28)$$

   where:

   $PGEN_i^0$ is the present dispatch of unit i,

   $PGEN_i^1$ is the new dispatch of unit i, and

   p_i is the participating factor of unit i.

5. Load shedding will take place if the sum of the maximum generation of all on-line machines is insufficient to cover the generation requirements. In such cases, load at all buses is shed proportionately until a load/generation balance is attained. Bus shunts and bus losses will not be shed.

   $$PLOAD_i^1 = PLOAD_i^0 + \frac{PLOAD_i^0}{\sum (PLOAD_i^0)} \times DEF \quad (6.29)$$

   where:

   $PLOAD_i^0$ is the present load at bus i,

   $PLOAD_i^1$ is the new load at bus i.

6. Similarly, if generation excess, EXC, exists, it is corrected by reducing the dispatch of participating machines in this island proportional to their given participation factors. If generation excess still exists after all on-line machines have been reduced to their minimum generation limits, the island will be shut down, and all load shed.

   $$PGEN_i^1 = PGEN_i^0 + p_i \times EXC \quad (6.30)$$

   where p_i is the participating factor of unit i.

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Steps 3 through 6 in the above list are repeated for each island in the network until a load/generation balance is attained.

6.11.5. Tripping Simulation and Corrective Action Analysis

A complete contingency analysis for a contingency event includes a power flow solution on the working base case with the contingency imposed, following sequences including a tripping simulation and a corrective action analysis.

The tripping simulation is discussed in Section 6.15.7, "About Tripping Sequence" and is used to simulate protection relay operations, system adjustments by operators following a contingency event and a system cascading failure.

A power flow solution may have any number of operating limit violations, and these problems can be alleviated by corrective actions. The ac corrective action analysis is formulated as an optimal power flow problem with the objective of minimizing load curtailment, MW generation re-dispatch, transformer phase angle adjustment, off-line generator active power dispatch, tap setting adjustment and switched shunt controls. It includes a standard ac power flow solution with local automatic adjustments, power system network linearization, and a linear programming solution to relieve the overload and voltage limit violations. Contingency analysis may start with corrective action analysis enabled to automatically specify corrective actions for each contingency. Corrective action analysis may be also used to only clean up the current working case. The Corrective Action Analysis is discussed in Section 6.13, "AC Corrective Actions".

Recognizing that thermal ratings and voltage limits in a power system may sometimes be treated as soft limits, the algorithm relaxes the hard limits on the constraints in case of infeasibility, and returns a solution with some minor violations. Hence, incorporation of the soft limits reduces the likelihood of infeasible solutions, which is particularly useful during contingency analysis when a large number of corrective action solutions may need to be performed automatically.

6.11.6. Substation Reliability Assessment Analysis

Substations are junction points of power transfer between generation and transmission systems, or transmission and distribution systems. Substations are a critical part of power systems and play an important role in reliability analysis of power systems. Reliability of a substation is measured in terms of the frequency and duration of substation related outage events leading to system not meeting performance criteria. The reliability results could be used for:

- comparing substation/network configuration alternatives
- evaluating the sensitivity of substation performance to outage statistics
- evaluating the sensitivity of substation performance to equipment rating
- evaluating the sensitivity of substation performance to load level
- determining the impact of equipment maintenance on reliability.

Substation reliability assessment analysis uses state enumeration method, similar to probabilistic reliability assessment, and is discussed in Section 6.18, "Calculating Substation Reliability".

6.11.7. Contingency analysis solution parameters

Contingency analysis solution parameters (power flow>contingency>contingency analysis solution parameters) are used by contingency analysis activities (AC contingency analysis, Multiple level contingency analy-

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sis, N-1-1 and PSCOPF). With the solution options of the activities, they initiate contingency analysis, control the procedure to evaluate a contingency and select results displayed in the spreadsheet table and reports. The parameters include contingency analysis rating sets, AC contingency screening options and contingency analysis solution options.

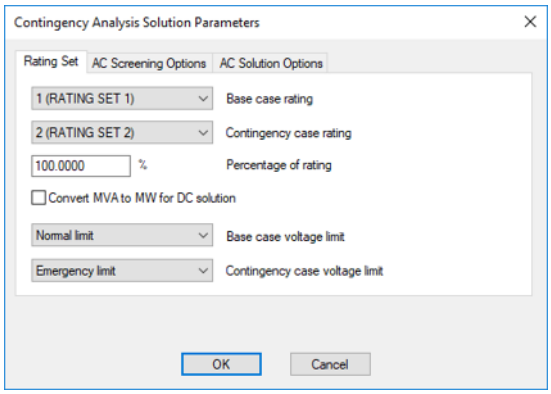

Figure 6.24. Contingency Analysis Rating Set

Rating sets and voltage limits may be required in a contingency analysis run, before 35.1 they are defined as activity inputs. For example, N-1-1 contingency analysis has base case and contingency case rating sets and voltage limits defined prior to the study; the secondary contingencies may be skipped if there are limit violations in their primary contingency solution. Since Ver35.1 the rating sets and voltage limits have been removed from the activity inputs, instead the options defined in contingency analysis solution parameters are used by the activities when it is applicable. The usage of the rating set and voltage limits is discussed in detail in sections for each activity. In summary contingency analysis activities use the rating sets and voltage limits:

- To select and report results in text report tables and spreadsheet tables. When text report tables or spreadsheet tables are enabled in program option, (refer to Section 3.3.3, "Program Run-Time Option Settings"), besides of storing the solutions in contingency solution output file (.acc), the contingency analysis activities (AC contingency analysis, Multiple level contingency analysis, and N-1-1) uses the new reporting capability to display the results in contingency analysis summary table, overload violation table, and voltage violation table, corrective action table if corrective action option is enabled in Multiple level contingency analysis or N-1-1 contingency analysis solution.

  The summary table lists contingencies with one of following issues: Non converged or blown up; Solved by robust power flow solution with one of three conditions: mismatch tolerance increase, phase shift locked and conversion of constant PQ load to constant Y load; Branch flow overload and voltage violations.

  for each solved contingency in the table, the summary table lists the number of branch overload violations and voltage violations in the contingencies.

  The overload violations tables list the monitored elements where loading exceeds the percentage of rating specified in contingency analysis rating set tab, base case rating and contingency rating and loading percentage under base case and contingency case.

  The voltage violation tables list the monitored buses where voltage exceeds the voltage limits. The limits are either specified in the monitored element data file or voltage limits specified in the contingency analysis solution parameters.

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- To calculate the performance index to rank contingencies in Multiple level contingency analysis (refer to Section 6.15, "Performing Multi-Level AC Contingency Solution").
- To identify the limit violation for corrective actions when corrective action option is enabled in Multiple level contingency analysis.
- To skip the secondary contingencies if their primary contingency has overloads and voltage violations in N-1-1 contingency alaysis.

Available Contingency analysis rating set options are listed below.

Table 6.7. Contingency Analysis Solution Parameters

Option	Description
Base case rating	rating set to be used in determining monitored element loading violations in base case or primary contingency solutions in N-1-1 studies
Contingency case rating	rating set to be used in determining monitored element loading violations in contingency solutions and secondary contingency solutions in N-1-1 studies.
Percentage of rating	Specify percent of flow rating to be used in determination of branch and interface flow overloads in primary and secondary contingency solutions.
Convert MVA to MW rating	option to convert MVA rating to MW rating for DC power flow solution-based method.
Base case voltage limit	limit set to be used in determining monitored bus voltage violations in base case or primary contingency solutions in N-1-1 studies.
Contingency case voltage limit	limit set to be used in determining monitored bus voltage violations in contingency solutions and secondary contingency solutions in N-1-1 studies.

When running a contingency analysis, AC contingency screening method can be used to select from the given list the contingencies that need a full AC power flow solution. The screen method is based on the DC power flow network.

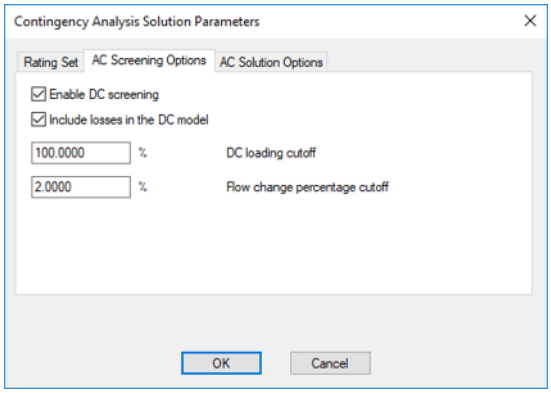

Figure 6.25. AC Contingency Screening Options

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For ACCC, a DC power flow solution is performed for each contingency first, a contingency is selected if it has at least one of monitored branches with MW loading greater than the loading cutoff percentage of rating set. When the flow change percentage cutoff is greater than zero, MW loading increase of the overloaded branch from base case loading must be greater than the cutoff. The selected cases are simulated by AC full power flow solutions, the cases that are not met selection criteria are ignored from the AC contingency analysis. Therefore, a sub-list of contingencies in the given contingency description data file are simulated and their results are stored in result file and displayed in the spreadsheet tables.

For N-1-1 contingency analysis, the secondary N-1 contingencies are screened and selected for AC power flow solutions if the option is enabled. The primary N-1 contingencies are always simulated. For PSCOPF analysis, the screen method is used to select the critical contingency cases.

Available Contingency analysis screening options are listed below.

Table 6.8. Contingency Screening Options

Option	Description
Enable DC screen	the option is to enable or disable screening method in ACCC and N-1-1
Include losses in DC model	the option is to estimate the transmission losses based on the current working condition and include the losses in the DC power flow network model.
DC loading cutoff	the cutoff is the percentage of the rating set for overload checking, the rating set is specified in rating set tab under the contingency analysis solution parameters.
Flow change percentage cutoff	the cutoff is calculated in the form of: $\Delta Flow = (F_{contingency} - F_{base})*100/MW_{rating}$

Example shows that the selected contingency cases based on screening method:

```
3 out of 11 contingency cases are selected by screening tool

Processing contingency "TRIP1NUCLEAR" (#1 of 10):
REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]
```

Figure 6.26. Contingency Screening Example

In contingency analysis solution tab, there are three options: enable robust power flow solution, base loaded machine flag and the total MW generation dispatch limit.

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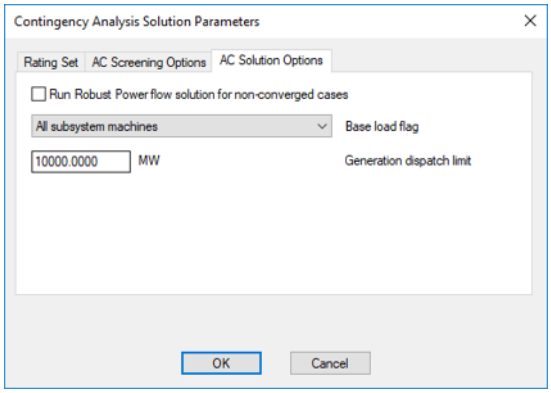

Figure 6.27. AC Contingency Solution Options

Run the robust power flow solution for non-converged contingency cases: robust power flow solution (refer to Section 6.10, “Applying the Robust Solution”) is an effective method to solve the non-converged power flow. It applies either FNSL or FDNS, or bother methods iteratively, adjust power flow solutions parameters to solve the case. During a contingency analysis run, a contingency that fails to converge with pre-defined solution method and options can normally be solved with robust power flow solution. When the option is enabled, if a contingency fails to converge with pre-defined options, the contingency is re-solved by robust power flow solution. The pre-defined method and options become the preferred method and options for robust power flow solution. If a case is re-solved by robust power flow solution, additionally the solution status may be one of three: mismatch tolerance increase, phase shift locked and conversion of constant PQ load to constant Y load. When a bus voltage is below the PQBRAK (see section, “boundary conditions of constant MVA loads"). The option is honored in both AC contingency analysis and N-1-1. The figure shows that the contingencies, “ADDLARGELOAD” and “xfmr204-205” are solved by robust power flow solution, and load changes made by the solution.

	Primary Contingency	Solution Type	Convergence	Worst Bus Mismatch	Number Of Overload	Number of Voltage	Worst Overload% Violation	Worst Undervoltage Violation
	BASE CASE	Contingency	Met convergence tolerances	0.06	0	2	0.00	0.9365
	TRIP1NUCLEAR	Contingency	Met convergence tolerances	0.06	1	0	291.14	0.0000
	TRIP2NUCLEAR	Contingency	Blown up	39.00	0	0	0.00	0.0000
	ADDLARGELOAD	Contingency	RSOL converged with Y load conversion due to low voltage	0.06	1	0	148.08	0.0000
	LOSEWESTGEN	Contingency	Met convergence tolerances	0.04	0	3	0.00	0.9288
	LOSEWESTBIGT	Contingency	Met convergence tolerances	0.06	0	2	0.00	0.9293
	LOSEEASTBIGT	Contingency	Met convergence tolerances	0.05	0	1	0.00	0.9299
	LOSEEASTLOAD	Contingency	Met convergence tolerances	0.10	3	0	677.50	0.0000
	LOSE2LINESWE	Contingency	Met convergence tolerances	0.08	0	2	0.00	0.9219
	LOSE2LINEEA	Contingency	Met convergence tolerances	0.08	0	10	0.00	0.8649
	XFMR204-205	Contingency	RSOL converged with Y load conversion due to low voltage	0.06	1	2	236.11	0.0000

Figure 6.28. AC Contingency Summary Result Table

A machine can be assigned to base load flag. The base loaded machines can operate in one of three modes, up only, down only, and neither up nor down. In contingency analysis, the generation outputs may be changed by the generation dispatch activity (refer to Section 6.11.4, “About Generation Dispatch”) and the corrective action activity (refer to Section 6.15.10, “Corrective Actions Options”). For both activities, the base loaded machines are handled in one of three options:

- All subsystem machines can be adjusted from Pmax to Pmin.
- All subsystem machines and honor base loaded machines: it is enabled to honor base loaded machines, for example, the base loaded machines with neither up nor down are excluded. When system needs to dispatch up generation, non base loaded machines and base loaded machines with up only are available
- All subsystem non-base loaded machines option: it is enabled to exclude all base loaded machines.

The generation dispatch limit is used in corrective action only. It limits either the total generation that is dispatched up or dispatched down within the limit.

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6.12. Performing AC Contingency Analysis

ACCC

(ACCC_PARALLEL_2 and ACCC_WITH_DSP_3)

Run Activity ACCC - *GUI*
Power Flow, Contingency, Reliability, PV/QV Analysis, AC contingency solution (ACCC)... *[AC Contingency Solution]*
Run Line Mode Activity ACCC - CLI
ACTIVITY? >>**ACCC** ENTER AC CONTINGENCY SOLUTION OUTPUT FILENAME: >>
Interrupt Control Codes
AB, CM, NM, DC, FD, MO

The AC Contingency Solution functions calculate full ac power flow solutions for a specified set of contingency cases. Results are stored in a binary file, the Contingency Solution Output file. This file is subsequently processed to produce reports of non-converged contingencies, violations, loadings, and available capacity. An optional ZIP Archive Output File may be produced during the contingency calculation. This file may be subsequently accessed to place any of the contingency case solutions into the working case.

Two APIs can be used to perform the AC Contingency solutions: ACCC_WITH_DSP_3 and ACCC_PARALLEL_2. The former is implemented for single-processor systems where the function evaluates each contingency case in the set sequentially. The latter is implemented for multiple-processor systems. If more than one contingency processor is selected in the Program Settings dialog (see OPTN), ACCC_PARALLEL_2 will be invoked when running ACCC via graphical user interface. The set are divided into as many subsets as number of contingency processor; ACCC_PARALLEL_2 uses each processor to process one subset simultaneously.

This network contingency calculation function calculates full ac power flow solutions for the user's specified set of contingency cases, monitors voltage and loading conditions and stores the results in a binary file. Subsequently, this file can be processed to produce a variety of reports of voltage and loading violations, loadings and available capacity. This feature is a powerful approach for testing large systems with many possible contingencies where the user wishes to monitor specific branches, interfaces or network areas for problems.

One role of the ac contingency calculation is that of a screening tool for which the purpose is to focus attention on those contingency cases that deserve closer study. Such a case may be brought into the working case from an Incremental Saved Case File (an .isv file) contained in the ZIP Archive Output File created during the contingency calculation (see Section 6.12.13, "ZIP Archive Output File").

A useful feature is the ability to produce reports in a format appropriate for import to a spreadsheet program.

To perform ac contingency analysis there are several steps:

1. Produce the Distribution Factor Data file from the **.sub*, **.mon*, **.con* files (activity DFAX).
2. Produce the Contingency Solution Output file using the Distribution Factor Data file.
3. Produce the selected report using the Contingency Solution Output file.

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The process and file path are shown in Figure 6.29, "File Path and Process for AC Contingency Analysis". Remember that the * .*con* file can be partially constructed using the Ranking process (refer to Section 9.3.1, "Contingency Ranking Methodology").

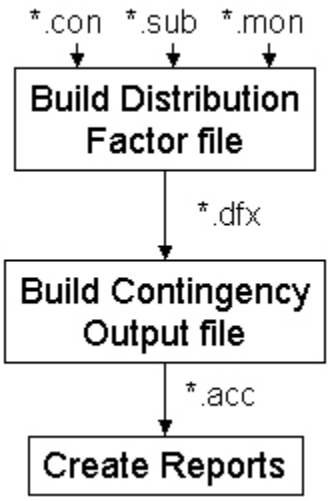

Figure 6.29. File Path and Process for AC Contingency Analysis

6.12.1. AC Contingency Solution Options

The following input data is specified to both of these functions (Table 6.9, "AC Contingency Solution Options").

Table 6.9. AC Contingency Solution Options

Option	Description
Solution options	The desired solution options to be applied during power flow calculations for obtaining post-contingency states (see Section 6.2, "Power Flow Solution Methods") Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads

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Option	Description
	Switched shunt adjustments: • Lock all • Enable all • Enable continuous, disable discrete
Non-divergent solution	Flag to enable or disable option in the power flow solution of each contingency case.
Adjust phase shift	Flag to enable or disable option in the power flow solution of each contingency case.
Adjust dc taps	Flag to enable or disable option in the power flow solution of each contingency case.
Induction motor treatment	When an induction motor fails to solve due to low terminal bus voltage, flag to treat it as, "stalled," or, "tripped".
Induction machine failure	Either: • Treat a contingency as non-coveveraged if any induction machines are placed in the, "stalled," or, "tripped," states. • Treat a contingency case as solved if it converged, even if induction machines were placed in the, "stalled," or, "tripped," states.
Solution Engine	Select the desired power flow solution engine to be used in obtaining postcontingency states (see Section 6.2, “Power Flow Solution Methods”). • Fixed slope decoupled Newton-Raphson (FDNS)* • Full Newton-Raphson (FNSL) * The optimized fixed slope decoupled Newton-Raphson method is used when ACCC is initiated via the graphical user interface. The user may choose either the regular or the optimized fixed slope decoupled Newton-Raphson method when initiating ACCC from an automation file.
Mismatch tolerance	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the process is terminated. This value is also used as the

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Option	Description
	convergence tolerance in the power flow solution of each contingency case (see Section 6.2, "Power Flow Solution Methods").
Dispatch Mode	Dispatch codes for generation dispatch calculations in ACCC analysis.
Subsystem machines (Reserve)	Participating machines are connected to dispatch subsystem buses and have positive active power generation. Each machine's participation factor is its reserve (PT-PG or PG-PB) with positive values. If a Unit Inertia and Governor Data file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used. In generation dispatch analysis, generator active power limits are adjusted by the following rules. If PT had been set to the default value, the larger value of PG (its existing MW output) and machine power base (MVA base) will be used as PT; If PT of a machine is less than its PG, the PT is set equal to the PG. If PB of a machine had been set to the default value, it is set to zero; if PB of a machine is less than its PG, the PB is set equal to the PG.
Subsystem machines (Pmax)	Participating machines are connected to dispatch subsystem buses and have positive active power generation. Each machine's participation factor is its maximum active power generation with positive values. If a Unit Inertia and Governor Data file is specified, maximum machine active power limits are taken from it; otherwise, the maximum machine active power limits in the working case are used. PT is adjusted by the rules discussed in Dispatch Mode of Machine Reserve.
Subsystem machines (Inertia)	Participating machines are connected to dispatch subsystem buses and have positive active power generation. Each machine's participation factor is its inertia constant. For machines for which no data record is successfully read, an inertia constant of 4.0 on machine base is used.
Subsystem machines (Governor droop)	Participating machines are connected to dispatch subsystem buses and have positive active power generation. Each machine's participation factor is its governor permanent droop. For machines for which no data record is successfully read, a droop constant of 0.05 on machine base is used.
Dispatch system	Select the subsystem for the generation dispatch. The subsystem is predefined in a subsystem description file. All subsystem buses with one or more in-service machines where active power generation is positive are participating in the generation dispatch.

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Option	Description
	If several islands exist in the system, participating machines for each island are connected to dispatch subsystem buses in the island.
Distribution Factor Data File	(*.dfx) a required file to be build by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files (refer to ???). The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings"Section and activity OPTN)*must* have the same setting that was in effect when activity DFAX filled the specified Distribution Factor Data file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by the AC Contingency Solution function.
Contingency Solution Output File	(*.acc) a required file designated by the user as the destination for the results of the contingency calculations.
Load Throw-over Data File	(*.thr) an optional file, created by the user, that contains data records of the following form: `IBUS, JBUS` where IBUS and JBUS are bus numbers. If branch outage contingency events isolate bus IBUS, the user can elect to move the load to bus JBUS. Bus IBUS and/or JBUS may be a dummy bus of a multi section line grouping. Data records may be entered in any order. Input is terminated with a record specifying an IBUS value of zero.
Unit Inertia and Governor Data File	(*.inl) a required file if the generation dispatch is enabled and either the inertial or governor droop dispatch method is selected. If the PMAX or reserve method is selected, the specification of this file is optional. If a file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used.
ZIP Archive option	Flag to enable or disable the writing of a ZIP Archive Output File.
ZIP Archive Output File	(*.zip) an optional file designated by the user as the destination for data used to place contingency case solutions into the working case.

6.12.2. Running AC Contingency Analysis

The network contingency calculation function calculates full ac power flow solutions for a specified set of contingency cases. Results are stored in a binary file. This file is subsequently processed to produce reports of violations, loadings and available capacity.

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Invoking the specified AC Contingency Solution function generates an alarm and terminates if generators are converted (refer to CONG). It checks that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network, and an in-service line is connected to an out-of-service bus. If any violations are detected, an appropriate message is printed and the function is terminated. If the largest initial active or reactive power mismatch exceeds the specified mismatch tolerance input data item, the function is terminated. This tolerance is also used as the mismatch convergence tolerance in the power flow solution of each contingency case.

The AC Contingency Solution function sets up PSS®E working memory and temporary scratch files in preparation for the contingency case power flow solutions. This function uses either the same fixed slope decoupled Newton-Raphson iterative power flow solution algorithm as does activity FDNS , or the full Newton-Raphson method of activity FNSL; the same set-up steps are performed here.

The contingency label and contingency events of each contingency case are logged at the *Progress* tab (refer to ORDR) as the solution function encounters it. A message identifies any network conditions that fail to converge or which are skipped.

The AC Contingency Solution function responds to the following interrupt control codes:

AB	Abandon the function following completion of the next iteration.
CM	Print the convergence and automatic adjustment monitors.
NM	Suppress any automatic adjustment monitors (only used with CM interrupt control code).
DC	Tabulate conditions for each dc line after each iteration (see Section 6.3.11, "Operation of Activity SOLV").
FD	Tabulate conditions for each FACTS device after each iteration (see Section 6.3.11, "Operation of Activity SOLV").
MO	Tabulate conditions for induction machines after each iteration (see Section 6.3.11, "Operation of Activity SOLV").

Additional Information
PSS®E GUI Users Guide, Running AC Contingency Analysis *PSS®E* Command Line Interface (CLI) *Users Guide*, Running AC Contingency Analysis

6.12.3. AC Contingency Output

The AC Contingency Solution functions calculate contingency solution results (loadings on monitored branches and interfaces, and bus voltage magnitudes) and place them into the specified Contingency Solution Output file. These results may then be tabulated using any or all of three output processing functions.

- The AC Contingency Single Run Report function (refer to Section 6.12.9, "AC Contingency Single Run Report") may be used to produce a variety of tabular and spreadsheet format output reports of results from a single Contingency Solution Output file.

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- The AC Contingency Multiple Run Report function (refer to Section 6.12.11, "AC Contingency Multiple Run Report") may be used to produce a variety of tabular reports of results from up to nine Contingency Solution Output files.

- The auxiliary program AcccBrwsGrid, which is documented in its Help files, allows spreadsheet-like manipulations of the results from a single Contingency Solution Output file.

For branches, loadings printed are MVA loadings. For each monitored non-transformer branch, the percent loading is either the percent current or the percent MVA loading, according to the non-transformer branch percent loading units program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

For monitored transformers, the percent loading is either the percent current or the percent MVA loading, according to the transformer percent loading units program option setting.

When percent current is used for non-transformer branches or two-winding transformers, both the MVA loading and percent current loading are taken from the end of the branch with the larger current loading. If the current loadings at each end are identical, both the MVA loading and the percent current loading are taken from the end of the branch with the larger MVA loading.

When percent MVA is used for non-transformer branches or two-winding transformers, both the MVA loading and percent MVA loading are taken from the end of the branch with the larger MVA loading. If the MVA loadings at each end are identical, both the MVA loading and the percent MVA loading are taken from the end of the branch listed as the from bus.

For monitored three-winding transformer windings, both the MVA loading and percent loading are calculated at the winding bus.

The flow across an interface is taken as the sum of the MW flows of its members. For three-winding transformer windings, the MW loading is calculated at the winding bus as power flowing into the transformer. For other members, the MW flow is calculated at its metered end in the from bus (i.e., the first bus specified in entering the branch; refer to Monitored Element Data File Contents) toward the to bus direction. The percent loading for each interface is its percent MW loading based on its MW loading and its interface MW rating from the selected rating set.

The user selects those subsystems where bus voltages are to be monitored, the types of voltage check (voltage band or deviation from base case values), and threshold values by specifying records in the Monitored Element Data file (refer to Voltage Monitoring Records). In the AC Contingency Single Run Report function and the AC Contingency Multiple Run Report function, voltage violations reports are presented in groups corresponding to the MONITOR VOLTAGE data records specified in the Monitored Element Data file. One (or two if the second `r` value on the MONITOR VOLTAGE record is specified) reports are generated for each MONITOR VOLTAGE record to be processed. Report blocks are printed in the order in which the MONITOR VOLTAGE data records are contained in the Monitored Element Data file.

Each voltage range record results in a tabulation of those buses from the set of buses defined on the data record where voltage is below the low voltage threshold (i.e., the first `r` value on the data record), followed by the list of buses where voltage is above the high voltage threshold (i.e., the second `r` value on the data record).

Each voltage deviation record results in a tabulation of those buses from the set of buses specified on the data record where voltage drop from their base case values exceeds the voltage drop threshold (i.e., the first `r` value on the data record), followed by the list of buses where voltage rise exceeds the voltage rise threshold (i.e., the second `r` value on the data record).

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6.12.4. AC Contingency Post Processor

The AcccBrwsGrid program is used to perform spreadsheet-like manipulations on the results contained in a Contingency Solution Output file. It executes outside of PSS®E, and is initiated by selecting the ACCC Post Processor (AcccBrwsGrid) entry from the PSS®E Windows® start menu. Refer to *Additional Resources for PSS®E*, ACCC Post Processor.

6.12.5. AC Contingency Result Retrieval Routines

A family of retrieval routines is provided to extract the results in contingency analysis output files and return arrays in FORTRAN format. These return values can then be used to create customized reported. These routines can be called from Python files (see Section 18.1, "Python Modules for Result Retrieval").

6.12.6. Application Notes

The AC Contingency Solution function uses the same network solution calculations as do activities FNSL and FDNS. It uses as its mismatch convergence tolerance the mismatch tolerance data item, which is one of the user inputs to the API, rather than the Newton mismatch tolerance solution parameter contained in the working case. All other network solution parameters are the same as those used by activities FNSL and FDNS.

As in the power flow solution activities, FACTS devices and VSC dc lines that are zero impedance line connected to a Type 3 bus in the base case are blocked (refer to Setpoint Voltage Control). Even if, in a contingency case, zero impedance line outages result in such devices no longer being zero impedance line connected to a Type 3 bus, they remain blocked in the contingency case.

One role of the AC Contingency Solution function is that of a screening tool for which the purpose is to focus attention on those contingency cases that deserve closer study. In this mode, it may be appropriate to use looser convergence tolerances than are used in other power flow solution applications.

If any line outage contingency cases form one or more swingless islands, such islands are disconnected for the solution of that contingency case. Any dc line having at least one converter station bus in such an island is blocked for that contingency case solution. Similarly, any FACTS device or GNE device having at least one of its buses in such an island is removed from service for that contingency case solution. Any such dc line, FACTS device, or GNE device that is a tie branch between a swingless island and the remaining system generates an alarm.

If the DISPATCH option is included on a SET, CHANGE, INCREASE, DECREASE, ADD or REMOVE record, the opposite change in the active power boundary condition is distributed among the Type 2 and 3 buses specified. Note that only Type 2 and 3 buses are allowed to participate in picking up the active power boundary condition change imposed by the contingency event. Any other buses specified as participating generate an alarm and are omitted from the power dispatch calculation. If only non-Type 2 or 3 buses are specified, the system swing bus(es) pick up all of the active power change.

The AC Contingency Solution function detects the specification of duplicate single- and double-line outage contingencies and calculates the contingency case solution only for the first specification of such a contingency. Any contingency cases involving bus boundary condition contingency events or more than two-line outage contingency events are not checked for duplication.

It is required that the working case be solved to an acceptable mismatch tolerance prior to executing the AC Contingency Solution function.

Load Contingencies

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When the SET, CHANGE, INCREASE, DECREASE or MOVE contingency commands operate on LOAD, they operate on the total nominal (i.e., not voltage adjusted) load of the in-service loads connected to the specified bus. For each load at the bus, its pre-contingent nominal load is the sum of the nominal constant power, constant current and constant admittance load components; the effect of any load multiplier associated with an Optimal Power Flow adjustable bus load table is applied to the constant power component in calculating the total nominal load.

When these contingency commands employ the MW keyword, only those in-service loads at the specified bus where pre-contingent nominal active power load is positive are modified in these contingency events.

For the MOVE contingency command, in-service connected load with positive nominal active power must exist at the FROM bus, and at least one in-service load, regardless of its nominal active power specification, must be present at the TO bus. Except for the destination bus in the MOVE contingency command, the total active power change is assigned to individual loads at the bus in proportion to their individual total nominal active power loads. For each participating load at the bus, its reactive power is modified such that the reactive power to active power ratio remains constant; the relative proportions of constant power, constant current and constant admittance load components also remain constant.

When these contingency commands employ the PERCENT keyword, active and reactive power components of all three load characteristics of all in-service loads at the bus are both adjusted by the specified percentage of their pre-contingent nominal load. Thus, the reactive power to active power ratio remains constant, and the relative proportions of constant power, constant current and constant admittance load components remain constant. For the MOVE contingency command, in-service load must be present at both the FROM and TO buses.

In the MOVE contingency command, all nominal active and reactive power load that is removed from the FROM bus is transferred to the TO bus. Each such load component retains its original load characteristic (constant power, constant current or constant admittance).

Shunt Contingencies

When the SET, CHANGE, INCREASE, DECREASE or MOVE contingency commands operate on SHUNT, they operate on the nominal (i.e., at 1.0 pu voltage) shunt admittance of the in-service fixed shunts connected to the bus; any switched shunt connected to the bus is not affected by these commands.

When these contingency commands employ the MW keyword and the initial conductance (i.e., the active power component) of shunt admittance is non-zero, the shunt susceptance (i.e., its reactive power component) is modified such that the susceptance to conductance ratio remains constant.

When these contingency commands employ the MW keyword and the initial conductance of shunt admittance is zero, the specified change is applied to the conductance but the susceptance is unchanged.

When these contingency commands employ the MVAR keyword, the specified change is applied to the susceptance, but the conductance is unchanged.

When these contingency commands employ the PERCENT keyword, the conductance and susceptance components of all in-service fixed shunts at the bus are both adjusted by the specified percentage of their pre-contingent nominal values. Thus, the conductance to susceptance ratio remains constant.

Generation Contingencies

When the SET, CHANGE, INCREASE, DECREASE or MOVE contingency commands operate on GENERATION, the specified buses must have one or more in-service machines connected to them. These contingency commands operate on the total active power output of the in-service machines connected to the specified bus(es); reactive power limits are not modified.

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When these contingency commands employ the MW keyword, the total plant active power is modified accordingly.

When the SET contingency command employs the PERCENT keyword, the total plant active power is set to the specified percentage of its original power. When the CHANGE, INCREASE, DECREASE or MOVE contingency commands employ the PERCENT keyword, the plant active power is modified by the specified percentage of the magnitude of its initial setting.

6.12.7. Implementing Generation Dispatch Algorithm in Contingency Analysis

Run Activity *AC Contingency Solution with Generation Dispatch - GUI*
Power Flow > Solution > Implement generation dispatch... *[Implement Generation Dispatch]*

For each contingency, automatic contingency analysis imposes the contingency, implements generation redispatch if generation dispatch mode is enabled (see Section 6.11.4, "About Generation Dispatch"), and solves ac power flow solution sequentially. To know how system operation condition is changed by a contingency, such as bus disconnection, line open/close and changes in bus boundary condition, and how the imbalances caused by the contingency is apportioned by means of generation dispatch algorithms, the Implement Generation Dispatch feature is used to impose a contingency and/or implement generation dispatch, and activity DIFF can then be used to check the changes made by Implement Generation Dispatch to working case. The modified case can then be solved by power flow solution.

Processing options include:

Study case: Select to impose a contingency from the contingency list specified in Contingency Description Data file (*.con) or to use working case

Dispatch Mode and Dispatch system: Specify generation dispatch mode and dispatch subsystem. If Dispatch mode is disabled, buses that are not connected to a swing bus will be disconnected and imbalances are absorbed by the swing bus in the island.

There are three input files:

Distribution Factor Data File: a required file that contains contingency descriptions and subsystem specifications.

Load Throwover Data File: an optional file that contains load throwover data.

Unit Inertia and Governor Data File: an optional file, that contains generator inertia and governor response data.

6.12.8. Appending Data to a Contingency Solution Output File

Run Append to AC *Contingency Solution Output File - GUI*
Power Flow > Reports > Append to AC Contingency solution output file... *[Append to AC Contingency Solution Output File]*

The Append ACCC function (the APPEND_ACCC API) is used to output system conditions of a contingency case solution, as contained in a designated Saved Case file, in the form of a Contingency Solution Output file. This is useful for the situation in which the specification and/or solution of a contingency case exceeds

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the capability of the AC Contingency Solution function, but the user wishes to have the contingency case reported along with, and in the same format as, the contingency cases calculated by the AC Contingency Solution function. The user specifies the following data items:

1. An append output option flag indicating either that the results for the contingency case specified in 4. below are to be appended to the existing Contingency Solution Output file specified in 5. below, or that a new Contingency Solution Output file is to be written containing results for both the base case specified in 3. below and the contingency case specified in 4. below.

2. The name of the Distribution Factor Data file to be used. This file specifies the network elements to be monitored. When appending results to an existing Contingency Solution Output file from a previous execution of the AC Contingency Solution function, the Distribution Factor Data file *must* be the same one specified to the AC Contingency Solution function.

3. The name of the Saved Case file containing the pre-contingency power flow case. The Saved Case in this Saved Case file *must* be the same case that was present as the PSS®E working case when the Distribution Factor Data file specified in 2. above was constructed with activity DFAX. It *must* also be the Saved Case from which the contingency case was formed and where the solution is contained in the Saved Case file specified in 4. below.

4. The name of the Saved Case file containing the contingency case solution where the results are to be placed in the Contingency Solution Output file specified in 5. below. The contingency case *must* have been formed by imposing status, boundary condition, and/or other data changes in the pre-contingency case specified in 3. above, and then solved using one or more of the power flow solution activities.

5. The name of the Contingency Solution Output file into which the results of the contingency case specified in 4. above are to be placed. When the append output option flag in 1. above indicates that the contingency case results are to be appended to an existing Contingency Solution Output file, the name of that file *must* be specified here, and it *must* be compatible with the Distribution Factor Data file specified in 2. above. This function is able to append to Contingency Solution Output files created by PSS®E-30.0 and later releases of PSS®E.

6. A contingency label of up to 32 characters used to identify the contingency.

7. The number of text lines to be used to describe the contingency. At least one, and no more than 16, such lines may be supplied.

8. Text lines of up to 120 characters each of which are used to describe the contingency. The number of text lines specified *must* be identical to the value specified in 7. above.

In imposing data changes to the base case to define the contingency case, equipment status, boundary condition, and other data changes to the base case may be specified. However, fundamental topological changes are *not* allowed. For example, buses or branches may *not* be added to, or removed from, the case. Furthermore, the use of activities such as SPLT, JOIN, and LTAP is prohibited, as is the use of activity MOVE to re-route a branch, and similar functions.

When results for multiple contingency case solutions are to be appended to the same Contingency Solution Output file, the same Distribution Factor Data file and pre-contingency Saved Case file *must* be specified for each of the Append ACCC functions.

6.12.9. AC Contingency Single Run Report

Run Activity *AC Contingency Report - GUI*
Power Flow > Reports > AC Contingency reports...

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Run Activity *AC Contingency Report - GUI*
[AC Contingency Reports]
Interrupt Control Codes
AB

The AC Contingency Single Run Report function (the ACCC_SINGLE_RUN_REPORT_3 API) is able to process Contingency Solution Output files created by PSS®E-30.0 and later releases of PSS®E.

The following input data is specified to this function:

Table 6.10. AC Contingency Single Run Report Options

Option	Description
Base case rating and voltage limit:	Base case rating defines line loading limits used in determining overloads in base case. Default is set in program options (see activity OPTN). Base case voltage limit defines voltage limits used in determining bus voltage limit violations in base case. The voltage limits apply to buses specified by monitor voltage limit records. When processing a Contingency Solution Output File created by N-1-1 contingency solution (see activity Section 6.16, "N-1-1 Contingency Solution"), the base case rating and voltage limits are used for both base case and primary N-1 contingency case.
Contingency case rating and voltage limit	Contingency case rating define line loading limits used in determining overloads in contingency case. Default is set in program options (see activity OPTN). Contingency case voltage limit defines voltage limits used in determining bus voltage limit violations in base case. When processing a Contingency Solution Output File created by N-1-1 contingency solution, the contingency case rating and voltage limits are used for secondary N-1 contingency case only.
Report format	Reports that can be imported into a spreadsheet program: • Spreadsheet overload report • Spreadsheet loading table • Available capacity table • Non-converged network conditions Reports that are available only for visual inspection: • .Non-spreadsheet overload report

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Option	Description
	• Non-spreadsheet loading table
	• Non-spreadsheet corrective actions report: a report of overloads and voltage violations in each network condition where corrective actions are specified to alleviate the violations, overloads and voltage violations in corresponding post-corrective action network condition, as well as specified corrective actions (see Section 6.15.10, "Corrective Actions Options")
Exclude interfaces from the report	Enable or disable (default) exclusion of interface loadings from all reports.
Perform voltage limit check	The user selects those subsystems where bus voltages are to be monitored, the types of voltage check (voltage band or deviation from base case values), and threshold values by specifying records in the Monitored Element Data file (see ???). In the dialog, however, is the facility to specify the Minimum contingency case voltage change for range violations. If enabled (default), buses where out-of-limits voltages in contingency cases differ from their base case voltages by less than this minimum change parameter are omitted from any voltage range checking reports. Also, a network condition is included in the report if the following conditions are satisfied: • The number of buses in the largest disconnected island is greater than or equal to the threshold. • The number of violations of all three voltage violation categories is greater than or equal to each respective threshold. When voltage limit checking is disabled a network condition is included in the report if the number of buses in the largest disconnected island is greater than or equal to the threshold.
Exclude elements with base case loading violations from contingency reports	Enable or disable (default) exclusion of monitored branches and interfaces that show loading violations in the base case from being checked and reported in overload reports of contingency cases.
Exclude elements with base case voltage violations from contingency reports	Enable or disable (default) exclusion of monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports. This flag applies only to the voltage range violation checks of the overload reports.
Exclude cases with no overloads from non-spreadsheet overload	Enable or disable (default) exclusion of cases with no overloads from the non-spreadsheet overload report or the non-spreadsheet corrective actions re-

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Option	Description
	port. This flag may be used to limit the number of records appearing in these reports. When disabled, reports identify all converged network conditions; it may indicate that some network conditions have no monitored element loading violations. When enabled, reports will identify all network conditions and it may indicate that some network conditions have no overloaded monitored elements.
Report post-tripping simulation solutions	If enabled, post-tripping network conditions are processed in the same way as contingency netowrk conditions. If disabled, post-tripping network conditions are excluded from reports (see *Section 6.15.7, "About Tripping Sequence"*).
Report loss of loads	Enable or disable reporting load. Curtailments in contingency network conditions.
Number of low voltage range violations	Filter limiting the converged network conditions included in a report. (default = 0)
Number of high voltage range violations	Filter limiting the converged network conditions included in a report. (default = 0)
Number of voltage deviation violations	Filter limiting the converged network conditions included in a report. (default = 0)
Number of buses in the largest disconnected island	Filter limiting the converged network conditions included in a report. (default = 0)
Maximum elements in available capacity table	Filter limiting the total number of records (monitored elements) that may appear in the table, and the cutoff threshold for available capacity table parameter excludes any monitored element where available capacity exceeds this threshold. (default = 6000)
Percent of flow rating	Filter used in determining overloads (default = 100.00)
Minimum contingency case flow change for overload report	Overloaded branches in contingency cases where the contingency case flows differ from their base case flows by less than this parameter are omitted from the overload report. (default = 0.000)
Minimum contingency case voltage change for range violations	For voltage violation reports, those buses where out-of-limits voltages in contingency cases differ from their base case voltages by less than this value are omitted from any voltage range checking reports. (default = 0.00)
Minimum contingency case percent loading increase for overload reports	Increase from the base case value. This suppresses the checking and reporting of monitored elements where contingency case loading does not increase above its base case loading by at least a specified percentage of its base case rating. (default = 0.00)
Cutoff threshold for available capacity table	Any monitored element where available capacity exceeds this value is excluded from the table.

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Option	Description
Bus mismatch tolerance (MVA)	Bus mismatch converged tolerance in MW or Mvar used to treat contingency cases that have small mismatches but failed to achieve convergence as converged. (default = 0.50)
System mismatch tolerance (MVA)	Total system MVA mismatch converged tolerance used to treat contingency cases that have small mismatches but failed to achieve convergence as converged. (default = 5.00)
Contingency Solution Output file	Name of the Contingency Solution Output file (*.acc) to be processed.

Results for all system conditions that were successfully solved in the AC Contingency Solution function are included in all but the non-converged network conditions report. In addition, the user may elect to consider as converged those contingency cases that are close to solution; i.e., those contingency cases that satisfy *all* of the following conditions are to be treated as converged:

- the AC Contingency Solution function was terminated either due to the maximum iteration limit being exceeded or by the non-divergent option (refer to activity FNSL)
- the largest bus MW and Mvar mismatch magnitudes are less than the specified bus mismatch converged tolerance data item
- the total system MVA mismatch is less than the specified system mismatch converged tolerance data item.

The non-converged network conditions report identifies those network conditions that failed to achieve convergence and which are not treated as converged based on their termination condition and mismatch level as described above. Thus, the non-converged network conditions report includes those contingency cases for which the solution was terminated under any of the following conditions:

- The BLOWN UP condition.
- The ITERATION LIMIT EXCEEDED or TERMINATED BY NON-DIVERGENT OPTION condition with bus and/or system mismatches exceeding the corresponding mismatch tolerances described above.
- The TERMINATED BY CONSOLE INTERRUPT condition.
- The SINGULAR JACOBIAN OR 0.0 VOLTAGE condition.
- The converged with stalled or tripped induction machines condition.

Such network conditions are identified in the non-Spreadsheet overload report and the non-converged network conditions report, and excluded from all other reports.

The converged network conditions included in a report may be limited by filter criteria. Only those contingencies that satisfy the filter criteria are reported. The following filter criteria are user assigned; the first two apply to all network conditions, while the last two apply only to contingency cases:

- Number of low voltage range violations.
- Number of high voltage range violations.

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- Number of voltage deviation violations.
- Number of buses in the largest disconnected island.

When voltage limit checking is enabled, a network condition is included in the report if the following two conditions are satisfied:

1. For contingency cases, the number of buses in the largest disconnected island is greater than or equal to the threshold.
2. The number of violations of the first two (for bases cases) or all three (for contingency cases) voltage violation categories is greater than or equal to each respective threshold.

When voltage limit checking is disabled, a contingency case is included in the report if the number of buses in the largest disconnected island is greater than or equal to the threshold.

Clearly, specifying a zero for any of the filtering criteria disables filtering according to that criteria.

6.12.10. Single Run Report Formats

In each report format, the first page includes a summary of the linear network analysis data files used in the AC Contingency Solution function along with the percent loading units (MVA or current) for transformers and non-transformer branches, which are based on the percent loading units program option settings at the time the AC Contingency Solution function was executed. If the Contingency Solution Output file was written by PSS®E-30.3 or later, the solution options used during the contingency case power flow solutions is tabulated.

When, during the construction of the Distribution Factor Data file (refer to DFAX), the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), all members of each monitored multi-section line are checked for overloading. For each such line section reported, both the branch itself and the multi-section line of which it is a member are identified in the overload report.

The specified percent of flow rating does not affect the values printed as monitored element ratings in all report formats.

If a line outage contingency forms an island with no Type 3 (swing) bus, the swingless island is disconnected during the solution of the contingency case. In the overload reports, any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, is omitted. In the loading table, such monitored elements have zero reported as their flows. Any bus in a swingless island is omitted from the voltage report.

The reports are directed to the currently selected output destination (refer to Section 4.4, "Virtual Output Devices").

Overload Report

Each converged network condition that satisfies the filter criteria described above (refer to Section 6.12.1, "AC Contingency Solution Options") is included in the overload report. Loadings on monitored branches and interfaces are calculated as described in Section 6.10.3 AC Contingency Output. A branch or interface is reported if the following conditions are satisfied:

- Its rating from the selected rating set is non-zero.

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- Its loading exceeds the specified percentage of the selected rating.
- For contingency cases, the difference in loading between the base and the contingency case exceeds the minimum contingency case flow change tolerance input data item.

For monitored branches in the overload reports, an asterisk (*) is printed between the bus number and name of the bus at the end of the branch at which the loading and percent loading are taken.

In the non-spreadsheet overload report, for each monitored element printed, the pre-contingency and post-contingency loadings, the rating and the post-contingency percent loading are listed. In the spreadsheet overload report, for each monitored element printed, the post-contingency loading, the rating and the post-contingency percent loading are listed.

When the voltage limit checks are enabled in the non-spreadsheet format, the overloaded monitored element report for each system condition is followed by its voltage violation report. The voltage violations report is presented as described in Section 6.12.3, "AC Contingency Output"; the report is limited to only those buses with voltage violations. For each bus printed, both its contingency case and base case values are printed. Any report block for which no violations are detected is omitted from the report.

When the voltage limit checks are enabled in the spreadsheet format, the overloaded monitored element report for all system conditions is followed by the voltage violation report for all system conditions. The voltage violations report is presented in groups corresponding to the voltage monitoring data records specified in the Monitored Element Data file.

For contingency cases, if the voltage change at a bus from its value in the base case is less than the minimum contingency case voltage change input data item, the bus is omitted from the report.

In the spreadsheet format, a final table identifies event descriptions for each network condition for which any flow or voltage was tabulated.

An example spreadsheet overload report is shown, partially in Figure 6.31, "Overload Report from AC Contingency Analysis" and 6-19 overleaf.

The report shows initially the rating set and percentage used for checking overloads, followed by the files used to perform the contingency analysis.

The overload report lists the monitored elements that suffer overloads and the name of the contingency causing the overload. To the right of that listing is the branch Rating, the post-contingency Flow and the post-contingency flow's percentage of rating. It should be noted that the first branch listed, which is from bus 153 to bus 154 has a rating of 350 MVA, an post-contingency flow of 343.2 MVA and a percentage loading of 117%. While the MVA flow is apparently less than the branch rating, it must be remembered that the rating is based on branch current capacity and 100% voltage.

Ratings are assumed to have been entered as:

$$MVA_{rated} = \sqrt{3} \times E_{base} \times I_{rated} \times 10^{-6} \qquad (6.31)$$

where:

E_{base}	= the base voltage in volts of the bus to which the terminal of the branch is connected.

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I_{rated}	= the rated phase current in amps.

For this contingency, the MVA flow shown is concurrent with a voltage of 0.832 pu. This corresponding level of current is 17% higher than the branches current capacity.

The monitored interface listed as suffering an overload is identified in the *jdmsavnw.mon* file, indicating that the interface comprises the three branches shown (see Figure 6.30, "Sample Monitored Interface Description").

```
MONITOR INTERFACE WEST RATING 200 MW
3004 152
3006 153
3008 154
END
```

Figure 6.30. Sample Monitored Interface Description

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Figure 6.31 Overload Report from AC Contingency Analysis

```
ACCC OVERLOAD REPORT: MONITORED ELEMENTS LOADED ABOVE 100.0 % OF RATING SET B
INCLUDES VOLTAGE REPORT

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\ac-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.con

<---------------- MONITORED BRANCH -----------------> CONTINGENCY    RATING     FLOW       %
   153 MID230      230.00    154*DOWNTN      230.00 1  TRIPLINE6       350.0    343.2    117.9
   203 EAST230     230.00    154*DOWNTN      230.00 1  TRIPLINE6       250.0    345.0    165.9
       INTERFACE WEST                                  TRIP1NUCLEAR    200.0    582.3    291.1
   153 MID230      230.00    154*DOWNTN      230.00 1  LOSE2LINEEA     350.0    470.6    162.8

MONITORED VOLTAGE REPORT:
SYSTEM                          CONTINGENCY  <-------- B U S --------->   V-CONT   V-INIT   V-MAX    V-MIN
ALL                       RANGE TRIPLINE2       154 DOWNTN      230.00  0.92754  0.93892  1.50000  0.93000
ALL                       RANGE TRIPLINE3       154 DOWNTN      230.00  0.90864  0.93892  1.50000  0.93000
```

a.)

```
CONTINGENCY LEGEND:
LABEL         EVENTS
TRIPLINE2   : TRIP LINE FROM BUS 152 [MID500      500.00] TO BUS 202 [EAST500     500.00]
TRIPLINE3   : TRIP LINE FROM BUS 153 [MID230      230.00] TO BUS 154 [DOWNTN      230.00]
TRIPLINE6   : TRIP LINE FROM BUS 154 [DOWNTN      230.00] TO BUS 205 [SUB230      230.00]
TRIP1NUCLEAR: REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]
LOSE2LINEEA : TRIP LINE FROM BUS 151 [NUCPANT     500.00] TO BUS 201 [HYDRO       500.00]
              TRIP LINE FROM BUS 152 [MID500      500.00] TO BUS 202 [EAST500     500.00]
```

b.)

Figure 6-18. Overload Report from AC Contingency Analysis

Following the overloaded element list, the report shows buses where voltage falls outside the specified range that has an upper threshold of 1.50 pu (a value of 1.05 is more adequate) and a lower threshold of 0.93 pu. The contingency initiating the voltage reduction is indicated by the bus number. Note that the list of buses outside the range is limited for only two in Figure 6-18. Finally, following the listing of the overloaded elements and buses outside Range, the report presents a legend indicating for each contingency name, the details of the disturbance.

In the non-spreadsheet overload report, for each monitored element printed, the pre-contingency and post-contingency loadings, the rating and the post-contingency percent loading are listed.

Loading Table

Each converged network condition that satisfies the filter criteria described above (refer to Section 6.12.1, "AC Contingency Solution Options") is included in the loading table. Loadings on monitored branches and interfaces are calculated as described in Section 6.12.3, "AC Contingency Output". For each monitored element, the selected rating, loading and percent loading are reported.

In the non-spreadsheet format, results are reported in tabular form with four contingency cases per table. Each group of contingency cases contains a page summarizing the contingency cases being reported. For each contingency, the contingency label and the events defining the contingency are listed. The summary page is followed by the loading table. For each monitored element, this table lists its rating, loading, and percentage loading in the base case network solution, and the loading and percentage loading for each of the contingency cases reported. Any percentage loading above the specified percentage threshold is followed by an asterisk (*). When the multi-section line reporting option is enabled, all members of each monitored multi-section line are reported.

In the spreadsheet format, the monitored flow results are reported in a table that include the following fields: the monitored element description, the network condition label, the element's rating, the element's flow, and the element's percent loading. An asterisk (*) is printed between the bus number and name of the bus at the end of the branch at which the loading and percent loading are taken. A subsequent table identifies event descriptions for each network condition label.

Single branches are listed first, either in the order in which they were specified in the Monitored Element Data file, or in ascending numerical or alphabetical order, according to the option selected during the execution of activity DFAX (refer to Section 9.1.1, "Operation of Activity DFAX"). When interface loadings are included in the report, the branch loadings are followed by the loading on interfaces in the order in which they were specified.

When the voltage limit checks are enabled, each loading table is followed by the voltage violations report.

In the non-spreadsheet format, the voltage violations report is presented as described in Section 6.12.3, "AC Contingency Output"; the report is limited to only those buses with voltage violations. For each bus printed, both its contingency case and base case values are printed. Any report block for which no violations are detected is omitted from the report.

In the spreadsheet format, the voltage violations report includes all buses identified by the monitoring data records.

In the spreadsheet format, a final table identifies event descriptions for the converged network conditions tabulated in the report.

Figure 6.32, "Sample Loading Report" shows a partial sample of a loading report obtained using the *jdmsavnw.sav* power flow case.

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```
ACCC LOADING REPORT: RATING SET B
INCLUDES VOLTAGE REPORT

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\ac-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.con

 <----------------- MONITORED BRANCH ----------------> CONTINGENCY    RATING     FLOW       %
    151*NUCPANT     500.00    152 MID500      500.00 1  BASE CASE      1300.0    495.3     37.7
    153 MID230      230.00    154*DOWNTN      230.00 1  BASE CASE       350.0    259.2     78.9
    201 HYDRO       500.00    151*NUCPANT     500.00 1  BASE CASE      1300.0    623.8     47.4
    202*EAST500     500.00    152 MID500      500.00 1  BASE CASE      1300.0    135.3     10.3
    203 EAST230     230.00    154*DOWNTN      230.00 1  BASE CASE       250.0    133.7     57.0
    205 SUB230      230.00    154*DOWNTN      230.00 1  BASE CASE       660.0    434.4     70.1
   3004 WEST        500.00    152*MID500      500.00 1  BASE CASE         0.0    195.5
   3006 UPTOWN      230.00    153*MID230      230.00 1  BASE CASE         0.0     80.3
   3008 CATDOG      230.00    154*DOWNTN      230.00 1  BASE CASE       440.0    112.1     27.1
        INTERFACE WEST                                 BASE CASE       200.0   -147.9     74.0
        INTERFACE EAST                                 BASE CASE       350.0   -130.8     37.4
    151*NUCPANT     500.00    152 MID500      500.00 1  TRIPLINE2      1300.0    487.2     37.0
    153 MID230      230.00    154*DOWNTN      230.00 1  TRIPLINE2       350.0    272.9     84.0
    201 HYDRO       500.00    151*NUCPANT     500.00 1  TRIPLINE2      1300.0    638.5     48.5
    202*EAST500     500.00    152 MID500      500.00 1  TRIPLINE2      1300.0      0.0      0.0
    203 EAST230     230.00    154*DOWNTN      230.00 1  TRIPLINE2       250.0    118.5     51.1
    205 SUB230      230.00    154*DOWNTN      230.00 1  TRIPLINE2       660.0    412.9     67.4
   3004 WEST        500.00    152*MID500      500.00 1  TRIPLINE2         0.0    195.6
   3006 UPTOWN      230.00    153*MID230      230.00 1  TRIPLINE2         0.0     82.3
   3008 CATDOG      230.00    154*DOWNTN      230.00 1  TRIPLINE2       440.0    126.0     30.9
        INTERFACE WEST                                 TRIPLINE2       200.0   -146.3     73.1
        INTERFACE EAST                                 TRIPLINE2       350.0   -131.9     37.7
```

Figure 6-19. Sample Loading Report

Figure 6.32. Sample Loading Report

It can be seen in the report that there are eleven monitored elements, two of which are defined interfaces. Their ratings, flow and percentage loadings are listed for the base case and subsequently for each contingency for which a converged case was obtained.

Available Capacity

The available capacity report provides a worst case summary over all converged system conditions that satisfy the filter criteria. In this report, each monitored element with a non-zero rating in the specified rating set is printed no more than once: for the system condition (base case or contingency case) in which its available capacity index is smallest.

Two input data items may be specified to control the contents of the available capacity table: the maximum number of elements in the available capacity table data item may be used to restrict the table to the < n > monitored elements with the smallest available capacity indices; the cutoff threshold for the available capacity table data item may be used to restrict the table to those monitored elements where available capacity indices are less than the specified cutoff threshold.

For each monitored element reported, this table lists: its rating; its base case loading; its worst case loading, percentage loading, available capacity index, and the system condition (the contingency case label or BASE CASE) in which the worst case loading occurs. The report indicates the number of other system conditions in which the monitored element has the same available capacity index as in the tabulated worst case, and the impact on the monitored element.

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Base case (the column headed BASE) and worst case (MAXIMUM) loadings are MVA loadings for branches and MW loadings for interfaces. Element loadings in the corresponding system condition are tabulated in the column headed IMPACT. For branches, the IMPACTs are either MVA loadings or current loadings expressed as MVA, according to the transformer and non-transformer branch percent loading units program option settings; for interfaces, the IMPACTS are MW loadings. Worst case percentage loadings are also calculated according to the transformer and non-transformer branch percent loading units program option settings as described in Section 6.12.3, "AC Contingency Output". An available capacity index (the column labeled AVAILABLE) is listed for each tabulated monitored element. The available capacity index is the difference between the RATING and the IMPACT.

In the available capacity report, monitored elements are printed in order of increasing available capacity index. Clearly, elements with negative indices are overloaded in the worst case system condition.

This report is terminated with descriptions of those contingency cases that were identified as worst case system conditions for one or more monitored elements. This contingency case summary lists contingency cases in the order in which they were calculated.

Figure 6.33, "Sample Capacity Report from the savnw.sav Power Flow Case" shows the capacity report for the *savnw.sav* power flow case. It can be seen that the Interface West suffers the greatest capacity deficiency, followed by the branches from bus 154 to buses 153 and 203. Other listed branches and interface show, for the contingencies tested (and converged) an available capacity for increased flow.

```
ACCC AVAILABLE CAPACITY REPORT: RATING SET B

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\ac-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.con

<------- F R O M -------> <--------- T O --------->CKT CONTINGENCY  OTHERS    BASE  MAXIMUM   IMPACT   RATING  PERCENT  AVAILABLE
      INTERFACE WEST                                 TRIP1NUCLEAR     0   147.9    582.3    582.3    200.0    291.1   -382.3
   153 MID230      230.00    154 DOWNTN      230.00 1  LOSE2LINEEA      0   259.2    470.6    570.0    350.0    162.8   -220.0
   203 EAST230     230.00    154 DOWNTN      230.00 1  TRIPLINE6        0   133.7    345.0    414.8    250.0    165.9   -164.8
   205 SUB230      230.00    154 DOWNTN      230.00 1  LOSE2LINEEA      0   434.4    497.5    602.5    660.0     91.3     57.5
  3008 CATDOG      230.00    154 DOWNTN      230.00 1  LOSE2LINEEA      0   112.1    265.3    321.3    440.0     73.0    118.7
      INTERFACE EAST                                 LOSE2LINESWE     0   130.8    138.3    138.3    350.0     39.5    211.7
   151 NUCPANT     500.00    152 MID500      500.00 1  LOSE2LINEEA      0   495.3    734.2    816.9   1300.0     62.8    483.1
   201 HYDRO       500.00    151 NUCPANT     500.00 1  LOSE2LINESWE     0   623.8    676.3    672.4   1300.0     51.7    627.6
   202 EAST500     500.00    152 MID500      500.00 1  LOSEEASTBIGT     0   135.3    448.0    459.8   1300.0     35.4    840.2

CONTINGENCY LEGEND:
LABEL         EVENTS
TRIPLINE6   : TRIP LINE FROM BUS 154 [DOWNTN      230.00] TO BUS 205 [SUB230      230.00]
TRIP1NUCLEAR: REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]
LOSEEASTBIGT: TRIP LINE FROM BUS 151 [NUCPANT     500.00] TO BUS 201 [HYDRO       500.00]
LOSE2LINESWE: TRIP LINE FROM BUS 3004 [WEST        500.00] TO BUS 152 [MID500      500.00]
              TRIP LINE FROM BUS 3006 [UPTOWN      230.00] TO BUS 153 [MID230      230.00]
LOSE2LINEEA : TRIP LINE FROM BUS 151 [NUCPANT     500.00] TO BUS 201 [HYDRO       500.00]
              TRIP LINE FROM BUS 152 [MID500      500.00] TO BUS 202 [EAST500     500.00]
|
```

Figure 6.33. Sample Capacity Report from the savnw.sav Power Flow Case

Non-converged Network Conditions

The non-converged network conditions report identifies each network condition that fails to meet the convergence criteria as described above (refer to Section 6.12.9, "AC Contingency Single Run Report"). The report format is suitable for import into a spreadsheet program.

The user specified values of bus mismatch converged tolerance and system mismatch converged tolerance are identified following the report title. A table of non-converged networks is presented that identifies: the

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contingency label, the largest bus mismatch magnitude, the system mismatch magnitude, and the termination condition from the AC Contingency Solution function. A subsequent table identifies event descriptions for each contingency label.

For the *savnw.sav* power flow case, Figure 6.34, "Non-Converged Network Report Example" shows the Report. It can be seen that five contingencies have failed to converge. The user, at this point has the option of re-testing the ac contingencies using the non-divergent power flow solution with this automated approach or to examine each contingency heuristically using a variety of solution methods and solution controls. Often the problem is a lack of voltage support.

```
ACCC NON-CONVERGED NETWORK REPORT:
BUS MISMATCH TOLERANCE=  0.5000 MVA
SYSTEM MISMATCH TOLERANCE=  5.0000 MVA

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\ac-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.con

CONTINGENCY  X--- MISMATCH (MVA) ----X
X- LABEL --X X-- BUS ---X X- SYSTEM -X TERMINATION CONDITION
TRIPLINE1      10147.       27161.     BLOWN UP
TRIP2NUCLEAR   797.99       3173.8     BLOWN UP
ADDLARGELOAD   795.39       3657.2     BLOWN UP
LOSEEASTLOAD   2496.2       17064.     BLOWN UP
XFMR204-205    797.23       3537.0     BLOWN UP

CONTINGENCY LEGEND:
LABEL         EVENTS
TRIPLINE1   : TRIP LINE FROM BUS 151 [NUCPANT 500.00] TO BUS 152 [MID500  500.00]
TRIP2NUCLEAR: REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]
              REMOVE UNIT 1 FROM BUS 102 [NUC-B       21.600]
ADDLARGELOAD: INCREASE BUS 154 [DOWNTN      230.00] LOAD BY 50 PERCENT
LOSEEASTLOAD: SET BUS 205 [SUB230      230.00] LOAD TO 0 MW
XFMR204-205 : DISCONNECT BRANCH FROM BUS 201 [HYDRO500.00] TO BUS 205 [SUB230 230.00] CKT &1
```

Figure 6.34. Non-Converged Network Report Example

Corrective Action Report

To have corrective action reports, contingency analysis output file must be produced by multiple-level contingency analysis initiated with perform corrective actions (see *Section 6.15, "Performing Multi-Level AC Contingency Solution"*).

Each network condition, which causes violations and for which corrective actions are specified to relieve the violations, is included in the non-spread sheet corrective action report, if it satisfies the filter criteria. After each network condition is solved, flow overloads and bus voltage violations are identified by the operating limits specified prior to multiple level contingency analysis instead of the rating setting and percent of rating specified in single run ACCC report.

The corrective actions results also depend on running mode specified prior to multiple contingency analysis. The running mode is specified according to selections of three options in contingency analysis: perform multiple contingency analysis, perform tripping simulation (see *Section 6.15.7, "About Tripping Sequence"*)

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and perform corrective actions (see *Section 6.15.10, "Corrective Actions Options"*). When contingency analysis is initiated with perform tripping simulation, flows and voltages in post-tripping network conditions are checked against limits, corrective actions are then specified to alleviate flows and voltages violations if any.

To reproduce the complete sequences that had been simulated, and present corrective actions and violations that had been identified and activated corrective actions during contingency analysis, some options in *AC Contingency Single Run report* are not allowed to select, but set by programs when the report option is a corrective action report as follows:

- Enabled:
- *Perform voltage limit check*
- *Report post-tripping simulation solutions*
- Not enabled:
- *Exclude interfaces from report*
- *Exclude elements with base case loading violations from contingency reports*
- *Exclude elements with base case voltage range violations from contingency reports*
- *Exclude cases with no overloads from non-spreadsheet overload report*
- Taken from the rating set specified prior to contingency analysis:
- *Base case rating*
- *Contingency rating*
- *Percent of flow rating*

In the summary page, input files and options used in contingency analysis are presented. For each network condition that satisfies conditions, post-contingency solution is reported first, then followed by post-tripping solution and post-corrective action solution. Each solution is presented in the format similar to non-spread sheet overload report format. For each monitored element printed, the pre-contingency loading, one of post-contingency, post-tripping and post-corrective action loading, the rating and the corresponding percent loading are listed. For each bus printed, voltage in one of post-contingency, post-tripping and post-corrective action solutions, base case voltage are printed.

Violations in solutions are followed by correctives actions specified for the network condition. Corrective actions are grouped by control types: generator active power dispatch, load shedding, phase shift angle adjustment, off-line generator active power dispatch, tap setting adjustment and switched shunt control. In each division, the initial value and the change of a control are reported in its unit. Generator active power dispatch, load shedding and off-line generator active power control are in MW; phase shifter angle adjustment is in degree, tap setting adjustment is in pu, and switched shunt control is in MVAR.

Figure 6.35, "Sample Corrective Action Report" shows a partial sample of a corrective action report. In the sample, contingency 1_5 causes an interface flow overloading and two bus voltage violations. In post-corrective action solution, the loading of the interface is decreased to 100.2 percent of rating A and two bus voltage violations still exist. The corrective actions involved are generation re-dispatch of five generators, load shedding at four buses and one phase shift angle adjustment.

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```
<---------- C O N T I N G E N C Y   E V E N T S ----------><--------- O V E R L O A D E D   L I N E S ----------> <- MVA(MW|FLOW ->
     <---------- MULTI-SECTION LINE GROUPINGS ----------> <------- F R O M -------> <--------- T O --------->CKT  PRE-CNT  POST-CNT  RATING  PERCENT
REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]          ------------------------------------------------------------------------------- CONTINGENCY 1_5
TRIP LINE FROM BUS 151 [NUCPANT     500.00] TO BUS 201 [HYDRO       500.00]
                                                       INTERFACE WEST                                  -147.9    425.0    200.0    212.5

                                                  X-------- B U S --------X  V-CONT  V-INIT  X-------- B U S --------X  V-CONT  V-INIT
        AREA 2 BUSES WITH VOLTAGE GREATER THAN 1.0600:    201 HYDRO       500.00 1.09500 1.04000     211 HYDRO_G     20.000 1.07232 1.04042

<---------------- POST - CORRECTIVE ACTIONS ----------------><--------- O V E R L O A D E D   L I N E S ----------> <- MVA(MW)FLOW ->
     <---------- MULTI-SECTION LINE GROUPINGS ----------> <------- F R O M -------> <--------- T O --------->CKT  PRE-CNT  POST-COR  RATING  PERCENT
-------------------------------------------------------------------------------------------------------------------------------------POST-CORRECTIVE ACTIONS
                                                       INTERFACE WEST                                  -147.9    200.5    200.0    100.2

                                                  X-------- B U S --------X  V-CORR  V-INIT  X-------- B U S --------X  V-CORR  V-INIT
        AREA 2 BUSES WITH VOLTAGE GREATER THAN 1.0600:    201 HYDRO       500.00 1.09923 1.04000     211 HYDRO_G     20.000 1.07668 1.04042

<---------- C O N T R O L   A D J U S T M E N T S --------->
            *** GENERATION  DISPATCH (MW) ***          X-------- B U S --------X  PGEN-INIT  PGEN-DELT
                                                         102 NUC-B       21.600      803.9        6.1
                                                         206 URBGEN      18.000      889.9       10.1
                                                         211 HYDRO_G     20.000      614.6        1.6
                                                        3011 MINE_G      13.800      833.1     -115.1
                                                        3018 CATDOG_G    13.800      115.3     -115.3
            *** LOAD SHEDDING (MW) ***                 X-------- B U S --------X  LOAD-INIT  LOAD-DELT
                                                         153 MID230      230.00      200.0       40.0
                                                         154 DOWNTN      230.00     1000.0       60.0
                                                         203 EAST230     230.00      300.0       40.0
                                                         205 SUB230      230.00     1200.0       66.1
            *** PHASE SHIFTER ADJUSTMENT ***           <------- F R O M -------> <--------- T O --------->CKT  PHAS-INIT  PHAS-DELT
                                                         203 EAST230     230.00   202 EAST500    500.00 1        0.0       -2.6
```

Figure 6.35. Sample Corrective Action Report

6.12.11. AC Contingency Multiple Run Report

Run AC Contingency Multiple Run Report - GUI
Power Flow > Reports > Multiple AC Contingency run report... *[Multiple AC Contingency Run Report]*
Interrupt Control Codes
AB

The AC Contingency Multiple Run Report function (the ACCC_MULTIPLE_RUN_REPORT API) is used to report the results of up to twenty-two executions of the AC Contingency Solution function. This report function is able process Contingency Solution Output files created by PSS®E-30.0 and later releases of PSS®E.

The following input data is specified to this function (Table 6-9 AC Contingency Multiple Run Report Options):

Table 6.11. AC Contingency Multiple Run Report Options

Option	Description
Monitored elements summary	Flag to enable (default) or disable the printing of the Monitored Element Summary Report.
Missing monitored elements	Flag to enable (default) or disable the printing of the Missing Monitored Elements Report.
Missing monitored voltage buses	Flag to enable (default) or disable the printing of the Missing Monitored Voltage Buses Report
Missing contingencies	Flag to enable (default) or disable the printing of the Missing Contingencies Report

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Option	Description
Non-converged contingencies	Flag to enable (default) or disable the printing of the Non-converged Contingencies Report
Contingency legend	Options for the Contingency Summary Report and the Contingency Legend. Control options are: • No: contingency summary and legend reports disabled • Complete master legend report only • Reduced contingency legend with each table in which contingencies are referenced • Both master legend report and a reduced contingency legend with each table in which contingencies are referenced (default)
Loading violations	Options for the Loading Violations Report. Control options are: • No: loading violation reports disabled • Base case and worst case contingency violations only (default) • Base case and all contingency violations • Base case, worst case contingency, and all contingency violations
Voltage violations	Options for the Voltage Violations Report. Control options are: • No: voltage violation reports disabled • Base case and worst case contingency violations only • Base case and all contingency violations • Base case, worst case contingency, and all contingency violations
Column headings	Selection of one of the following to be used as a column heading to identify the run being reported in the column: • AC Contingency report filenames (default) • Saved case filenames as contained in the ac report files
Base case rating set and voltage limit	Rating set to be used in determining base case monitored element loading

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Option	Description
	violations (default set by activity OPTN) • Rate A, B, C Voltage limit to be used in determining bus voltage limit violations in base case Normal, Emergency
Contingency case rating set and voltage limit	Rating set to be used in determining contingency case monitored element loading violations (default set by activity OPTN) • Rate A, B, C Voltage limit to be used in determining bus voltage limit violations in contingency case Normal, Emergency
Include interface loading violations	Flag to enable or disable (default) violation checking and reporting of interface loading violations.
Exclude elements with base case loading violations from contingency reports	Flag to enable or disable (default) exclusion of base case loading violations
Exclude elements with base case voltage range violations from contingency reports	Flag to enable or disable (default) exclusion of voltage range violations
Percent of flow rating for reporting	Percent of flow rating for reporting an element in the loading violation reports (default = 100.00)
Percent of flow rating for counting in worst case contingency violation reports	This value must be at least as large as *Percent of flow rating for reporting* (default = 100.00)
Minimum contingency case flow change for reporting and counting	Loading change from a monitored element's base case loading to enable the checking of its loading in the contingency case loading violation reports. (default = 0.0, to check all monitored elements)
Minimum contingency case % loading increase for overload	Change from a monitored element's base case loading to enable the checking of its loading in the contingency case loading violation reports; the base case rating set is used in calculating these percent loadings. (default = 0.0, to check all monitored elements)
Minimum contingency case voltage for reporting in voltage range violation	Change from a monitored bus' base case voltage magnitude to enable the checking of its voltage in the contingency case voltage range violation reports. (default = 0.0, to check all buses designated for range checking)
Bus mismatch converged tolerance (MVA)	Used to treat contingency cases that have small mismatches but failed to achieve convergence as converged

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Option	Description
System mismatch converged tolerance (MVA)	Used to treat contingency cases that have small mismatches but failed to achieve convergence as converged.
Contingency Solution Output files	Names of the Contingency Solution Output file (*.acc) to be processed; at least one is required; up to 22 may be specified. This report function is able process Contingency Solution Output files created by PSS®E-30.0 through the current release.

The AC Contingency Multiple Run Report function is designed to work best for the situation in which different base cases of essentially the same system (e.g., different seasons, loading levels, and/or years), with the same setting of the transformer percent loading units program option setting, and with the same setting of the non-transformer branch percent loading units program option setting are each presented to the AC Contingency Solution function with its Distribution Factor Data file based on the same set of linear network analysis data files (refer to ???). Typically, each base case is brought into PSS®E, activity DFAX is executed, and the AC Contingency Solution function executed; results of each case may then optionally be individually examined using the Single Run Output Report function (refer to Section 6.12.9, "AC Contingency Single Run Report"). After all base cases have been processed, the AC Contingency Multiple Run Report function may be used to obtain combined reports for up to twenty-two such runs.

The AC Contingency Multiple Run Report function constructs a master monitored element list containing all branches and interfaces that are monitored in at least one of the Contingency Solution Output files being processed. An element's rating corresponding to an Contingency Solution Output file is taken from the appropriate rating set as contained in the file.

The AC Contingency Multiple Run Report function constructs a master monitored voltage bus list containing all buses where voltages are monitored in at least one of the Contingency Solution Output files being processed.

The set of voltage range and deviation checks to be performed is taken from the first Contingency Solution Output file specified. That is, the voltage checks performed by the AC Contingency Multiple Run Report function correspond to the MONITOR VOLTAGE records (refer to Voltage Monitoring Records) contained in the Distribution Factor Data file used in the AC Contingency Solution run corresponding to the first Contingency Solution Output file specified (recall that these MONITOR VOLTAGE records are taken from the Monitored Element Data file specified at the time Distribution Factor Data file is constructed). Each voltage check is performed on the buses specified for it in the first run, as well as for any additional buses specified in other runs containing the *identical* MONITOR VOLTAGE record. Any voltage check not present in the first Contingency Solution Output file but in one of the other files is *omitted* from the voltage violation reports.

The AC Contingency Multiple Run Report function constructs a master contingency list containing all contingencies for which the results are contained in at least one of the Contingency Solution Output files being processed. The process of matching up contingencies from the various runs depends upon the method by which the contingency was specified.

A contingency case where the contingency label contains SINGLE (or DOUBLE) as its first six characters, and which is comprised of one (or two) contingency events, is assumed to have been generated as a result of one of the automatic contingency specification data records (refer to Automatic Contingency Specification) in the Contingency Description Data file. Such contingencies are matched up among the various output files by comparing the text description(s) of the event(s) comprising the contingency. The sixth (or fifth and sixth) character of the contingency label (i.e., E or LE) of such contingencies is replaced by one or two digits(1 through 22) indicating the Contingency Solution Output file in which it is first encountered.

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Similarly, a contingency case where the contingency label contains OVRLOD, VLTAGE, ISLAND, or UNIT as its first six characters, and which is comprised of a single contingency event, is assumed to have been generated by activity RANK. Such contingencies are matched up using the same approach as described above.

All other contingency cases are assumed to have been specified in the Contingency Description Data file using the contingency case block structure (refer to Contingency Case Block Structure). Such contingency cases are matched up among the various output files according to their contingency labels. Clearly, for this process to be effective, unique contingency labels *must* be specified in each Contingency Description Data file; further, if multiple Contingency Description Data files are used, contingency label usage among these files *must* be coordinated.

Results for all system conditions that were successfully solved in the AC Contingency Solution function are included in all but the Non-converged Contingencies report. In addition, the user may elect to consider as converged those contingency cases that are close to solution; i.e., those contingency cases that satisfy *all* of the following conditions are to be treated as converged:

- the AC Contingency Solution function was terminated either due to the maximum iteration limit being exceeded or by the non-divergent option (refer to FNSL)
- the largest bus MW and Mvar mismatch magnitudes are less than the specified bus mismatch converged tolerance data item
- the total system MVA mismatch is less than the specified system mismatch converged tolerance data item.

6.12.12. Multiple Run Output Report Formats

The output of the AC Contingency Multiple Run Report function consists of several reports. The first report is always printed; the remaining reports are included or omitted based upon the values specified for the various report flag input data items.

Many of the reports present results for the runs being tabulated in column format, where columns correspond to the NFILES Contingency Solution Output files being processed. The column heading corresponding to each Contingency Solution Output file is either the name of the Contingency Solution Output file or the name of the Saved Case file as contained in the Contingency Solution Output file, according to the value of the column headings code input data item.

When, during the construction of the Distribution Factor Data file (refer to DFAX), the multi-section line reporting option is enabled (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN), all members of each monitored multi-section line are included as monitored elements. For each such line section reported in the Loading Violations reports, only the branch itself is identified; both the branch itself and the multi-section line of which it is a member are identified in the Monitored Element Summary report.

If a line outage contingency forms an island with no Type 3 (swing) bus, the swingless island is disconnected during the solution of the contingency case. In the loading violations reports, any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, is omitted. Any bus in a swingless island is omitted from the voltage violations reports.

The percent loading values tabulated for each Contingency Solution Output file reflect the settings of the non-transformer branch percent loading units option setting and the transformer percent loading units program setting that existed in the working case at the time the Contingency Solution Output file was constructed. These settings are listed for each Contingency Solution Output file being processed in the AC Contingency Solution Output files report. However, the title boxes of the base case loading violation report, the

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contingency case all loading violations report, and the contingency case worst case loading violations report all include a line indicating the percent loading units for transformers and non-transformer branches as contained in the *last* Contingency Solution Output file specified.

The reports are directed to the currently selected output destination (refer to Section 4.4, "Virtual Output Devices").

Contingency Solution Output Files Report

The Contingency Solution Output files report is always included in the output of the AC Contingency Multiple Run Report function. This report lists the following for each Contingency Solution Output file specified:

- the name of the Contingency Solution Output file.
- the name of the most recently accessed Saved Case file at the time that the AC Contingency Solution function was executed.
- the two line case title of the working case at the time that the AC Contingency Solution function was executed.
- the name of the Distribution Factor Data file specified to the AC Contingency Solution function.
- the name of the Subsystem Description Data file specified to, and used by, activity DFAX in constructing the Distribution Factor Data file.
- the name of the Monitored Element Data file specified to, and used by, activity DFAX in constructing the Distribution Factor Data file.
- the name of the Contingency Description Data file specified to, and used by, activity DFAX in constructing the Distribution Factor Data file.
- the percent loading units (MVA or current) for transformers and non-transformer branches, which are based on the percent loading units program option settings at the time the AC Contingency Solution function was executed.
- if the Contingency Solution Output file was written by PSS®E-30.3 or later, the solution options used during the contingency case power flow solutions.

Monitored Element Summary Report

The Monitored Element Summary report lists the following for each monitored element:

- the description of the branch or interface as obtained from the Contingency Solution Output file in which it is first encountered.
- the element's rating(s) from the appropriate rating set(s) as obtained from the Contingency Solution Output file in which it is present, the element's rating - Contingency Solution Output file pair is blank when the element is not present in the corresponding Contingency Solution Output file; neither of the percent of rating input values affects the value printed as the element's rating(s).
- if a branch is a member of a multi-section line in the Contingency Solution Output file in which it is first encountered, the multi-section line description as obtained from that Contingency Solution Output file.

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Missing Monitored Elements Report

The Missing Monitored Elements report lists members of the master monitored element list that are specified as monitored elements in some, but not all, of the Contingency Solution Output files being processed.

Missing Monitored Voltage Buses Report

The Missing Monitored Voltage Buses report lists members of the master monitored voltage bus list that are specified as monitored voltage buses in some, but not all, of the Contingency Solution Output files being processed. It also tabulates MONITOR VOLTAGE records specified in any of the Contingency Solution Output files being processed that are not present in the first Contingency Solution Output file; the voltage range and deviation checks corresponding to such records are *not* performed by the AC Contingency Multiple Run Report function.

Contingency Summary Report

The Contingency Summary report lists members of the master contingency list. For each contingency, the report tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the contingency legend flag data item is 1 or 3.

Missing Contingencies Report

The Missing Contingencies report lists members of the master contingency list that are specified as contingencies in some, but not all, of the Contingency Solution Output files being processed.

A contingency legend follows this report if the value specified for the contingency legend flag data item is 2 or 3. For each contingency listed in the report, the contingency legend tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency.

Non-converged Contingencies Report

The Non-converged Contingencies report lists members of the master contingency list that failed to meet the convergence criteria as described above (refer to Section 6.12.11, "AC Contingency Multiple Run Report"), along with the network solution termination condition from the AC Contingency Solution function. The user specified values of the bus mismatch converged tolerance and the system mismatch converged tolerance are listed at the beginning of the report.

A contingency legend follows this report if the value specified for the contingency legend flag data item is 2 or 3. For each contingency listed in the report, the contingency legend tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency.

Base Case Loading Violations Report

The Base Case Loading Violations report lists members of the master monitored element list where loading exceeds the specified reporting percentage of the selected rating in the base case of one or more of the Contingency Solution Output files. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the loading violations report flag data item is 1, 2 or 3.

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For each violation, the report tabulates MVA loadings for branches and MW loadings for interfaces. Percentage loadings are calculated as described in Section 6.12.3, "AC Contingency Output", according to the transformer and non-transformer branch percent loading units program option settings. Interface violations are included in the report only if a value of 1 is specified for the interface option flag data item.

Contingency Case All Loading Violations Report

The Contingency Case All Loading Violations report lists members of the master monitored element list where loading exceeds the specified reporting percentage of the selected rating in a contingency case of one or more of the Contingency Solution Output files. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the loading violations report flag data item is 2 or 3.

For each violation, the report tabulates the label of the contingency causing the loading violation. It lists MVA loadings for branches and active power loadings for interfaces. Percentage loadings are calculated as described in Section 6.12.3, "AC Contingency Output", according to the transformer and non-transformer branch percent loading units program option settings.

If a value of 1 is specified for the exclude base case loading violations from contingency case reports flag, a violation is included in the table only if the element was not in violation in the base case. If the minimum contingency case loading change data item is positive, a violation is included in the table only if the element's loading in the contingency case differs from its corresponding base case loading by the specified value. If the minimum contingency case percent loading increase data item is positive, a violation is included in the table only if the element's loading in the contingency case, expressed as its percent loading relative to the its base case rating, exceeds its corresponding base case loading by the specified value. Interface violations are included in the report only if a value of 1 is specified for the interface option flag data item.

A contingency legend follows this report if the value specified for the contingency legend flag data item is 2 or 3. For each contingency listed in the report, the contingency legend tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency.

Contingency Case Worst Case Loading Violations Report

The Contingency Case Worst Case Loading Violations report lists members of the master monitored element list where loading exceeds the specified recording percentage of the selected rating in a contingency case of one or more of the Contingency Solution Output files. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the loading violations report flag data item is 1 or 3.

This report can be considered a compressed form of the Contingency Case All Loading Violations report (refer to Contingency Case All Loading Violations Report). Each row of the report lists a monitored element along with the label of the contingency that causes the most severe contingency case loading violation of that element in one or more of the Contingency Solution Output files. For each monitored element - Contingency Solution Output file pair (i.e., for each cell in the table) having a non-blank entry, this report lists the most severe loading violation of the element from among all contingencies in the Contingency Solution Output file. Thus, while there could be several rows for the same monitored element in the table, each will be paired with a different contingency label and each column of the table will have a non-blank cell in at most one of these rows for the same monitored element.

For each violation reported, the report tabulates MVA loadings for branches and active power loadings for interfaces. Percentage loadings are calculated as described in Section Section 6.12.3, "AC Contingency Output", according to the transformer and non-transformer branch percent loading units program option settings. It also lists a count of the number of contingency cases in this Contingency Solution Output file, in-

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cluding the contingency case listed, in which the loading on the element exceeds the specified percentage of flow rating for counting violations.

If a value of 1 is specified for the exclude base case loading violations from contingency case reports flag, a violation is included in the table only if the element was not in violation in the base case. If the minimum contingency case loading change data item is positive, a violation is included in the table only if the element's loading in the worst case contingency case differs from its corresponding base case loading by the specified value. If the minimum contingency case percent loading increase data item is positive, a violation is included in the table only if the element's loading in the worst case contingency case, expressed as its percent loading relative to the its base case rating, exceeds its corresponding base case loading by the specified value. Interface violations are included in the report only if a value of 1 is specified for the interface option flag data item.

A contingency legend follows this report if the value specified for the contingency legend flag data item is 2 or 3. For each contingency listed in the report, the contingency legend tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency.

Base Case Voltage Violations Reports

The Base Case Voltage Violations reports list members of the master monitored voltage bus list where voltage magnitude fails the corresponding range test in the base case of one or more of the Contingency Solution Output files. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the voltage violations report flag data item is 1, 2 or 3.

The voltage violations report is presented as described in Section 6.12.3, "AC Contingency Output". For each violation, the report tabulates the voltage magnitude in per unit.

Contingency Case All Voltage Violations Reports

The Contingency Case All Voltage Violations reports list members of the master monitored voltage bus list where voltage magnitude fails the corresponding range or deviation test in a contingency case of one or more of the Contingency Solution Output files. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the voltage violations report flag data item is 2 or 3.

The voltage violations report is presented as described in Section 6.12.3, "AC Contingency Output". For each violation, the report tabulates the label of the contingency causing the voltage violation and the voltage magnitude in per unit.

If a value of 1 is specified for the exclude base case voltage range violations from contingency case reports flag, a voltage range violation is included in the table only if bus voltage was within the corresponding range in the base case. If the minimum contingency case voltage change data item is positive, a violation is included in a range violation table only if bus voltage magnitude in the contingency case differs from its base case voltage magnitude by the specified value.

A contingency legend follows this report if the value specified for the contingency legend flag data item is 2 or 3. For each contingency listed in the report, the contingency legend tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency.

Contingency Case Worst Case Voltage Violations Reports

The Contingency Case Worst Case Voltage Violations reports list members of the master monitored voltage bus list where voltage magnitude fails the corresponding range or deviation test in a contingency case of one

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or more of the Contingency Solution Output files. This report is included in the output of the AC Contingency Multiple Run Report function if the value specified for the voltage violations report flag data item is 1 or 3.

These reports can be considered a compressed form of the Contingency Case All Voltage Violations reports (refer to Contingency Case All Voltage Violations Reports). Each row of the report lists a monitored bus along with the label of the contingency that causes the most severe contingency case voltage range or deviation violation of that bus in one or more of the Contingency Solution Output files. For each monitored bus - Contingency Solution Output file pair (i.e., for each cell in the table) having a non-blank entry, this report lists the most severe voltage violation of the bus from among all contingencies in the Contingency Solution Output file. Thus, while there could be several rows for the same monitored bus in the table, each will be paired with a different contingency label and each column of the table will have a non-blank cell in at most one of these rows for the same bus.

The voltage violations report is presented as described in Section 6.12.3, "AC Contingency Output". For each violation reported, the report tabulates the voltage magnitude in per unit. It also lists a count of the number of contingency cases in this Contingency Solution Output file, including the contingency case listed, in which the bus fails the corresponding range or deviation test.

If a value of 1 is specified for the exclude base case voltage range violations from contingency case reports flag, a voltage range violation is included in the table only if bus voltage was within the corresponding range in the base case. If the minimum contingency case voltage change data item is positive, a violation is included in a range violation table only if bus voltage magnitude in the worst case contingency case differs from its base case voltage magnitude by the specified value.

A contingency legend follows this report if the value specified for the contingency legend flag data item is 2 or 3. For each contingency listed in the report, the contingency legend tabulates the contingency label used to identify the contingency in the AC Contingency Multiple Run Report function and the events comprising the contingency.

Additional Information
PSS®E GUI Users Guide, Multiple AC Contingency Run Reports

6.12.13. ZIP Archive Output File

A ZIP Archive Output File may optionally be written during the AC Contingency Analysis, P-V Analysis, and Q-V Analysis functions.

RunDetails.txt

Most of the files that are preserved in the ZIP Archive Output File are in binary form, and are accessible only to PSS®E functions. There is, however, one source file that may be extracted from the ZIP Archive Output File and examined by the user. The source file RunDetails.txt contains information about the execution of the contingency analysis function that created it. This information includes:

- the name of the API routine that created the ZIP Archive Output File.
- the time and date of its writing.
- the names of any files that were used as input to the calculation function.
- the names of any files that were written by the calculation function.

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- the values of other inputs to the API routine which define various solution and other options used in the calculation.

Listing System Conditions Represented in a ZIP Archive Output File

The ListContingencySavedCases API routine is used to produce a tabulation at the Report device of system conditions for which data is stored in the ZIP Archive Output File.

The ListContingencySavedCases API routine includes the following among its inputs:

- the name of the ZIP Archive Output File.
- a label indicating the system condition(s) that are to be tabulated: blank to list all system conditions; a contingency label to list all system conditions for the specified contingency label; or “Base Case” to list all system conditions for the base case (use “Base Case” only for ZIP Archive Output Files written by the P-V and Q-V analysis functions).

For each system condition that is tabulated, the contingency label is listed, along with the name of an Incremental Saved Case File (the .isv file) that contains data pertaining to the solution of the system condition. For ZIP Archive Output Files written by the P-V Analysis function, the incremental transfer level that is applied is also listed; for ZIP Archive Output Files written by the Q-V Analysis function, the voltage at the study bus is also listed.

Bringing Contingency Case Conditions into the Working Case

The GetContingencySavedCase API routine is used to place the working case in the form of a specified system condition as calculated during a previous run of the AC Contingency Analysis, P-V Analysis, or Q-V Analysis function. This function retrieves data from a ZIP Archive Output File that was created during the run of the contingency calculation function.

The GetContingencySavedCase API routine includes the following among its inputs:

- the name of the ZIP Archive Output File.
- the name of an Incremental Saved Case File (.isv file) contained in the ZIP Archive Output File.

6.12.14. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Tables named "CA Summary", "CA Overload", "CA Voltage", "CA Adj" and "CA Cap".

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6.13. AC Corrective Actions

Run AC Corrective Actions - GUI
Power Flow > Solution > AC corrective actions... *[AC Corrective Actions]*
Interrupt Control Codes
AB

Corrective actions are modeled as an optimal power flow problem. The objective function is to minimize the control adjustments needed to remove limit violations in the power system. The constraints include equality and inequality constraints, namely power flow equations and limits of controls and operation conditions.

6.13.1. Constraints and Controls

The corrective action algorithm recognizes several types of constraints and controls. Constraints are operating limits imposed on bus voltages, branch flows, or power transfers over interfaces. The system problems identified in a contingency analysis are the violations of such constraints. Controls include generator active powers, phase shifter angles, bus load curtailments, off-line generator active powers, tap positions and reactive power of switched shunts. The objective of the corrective action algorithm is to observe all constraints while minimizing the weighted sum of control adjustments within their upper and lower bounds. The priorities of control actions are:

1. Phase shifter angle adjustment
2. Tap setting adjustment
3. Switched shunt reactive power control
4. Generator active power dispatch
5. Off-line generator active power dispatch
6. Load curtailment (if required)

This priority order is maintained by assigning a higher cost to the lower priority controls.

Constraints are the power system operating limits that need to be observed. The corrective action algorithm allows three types of constraints that are defined in the Distribution Factor Data file:

Table 6.12. AC Corrective Action Constraints

Constraint	Description
Branch flows	Branch flow limits are obtained from the designated rating set, adjusted by a specified multiplying factor. Percent loadings of monitored elements based on ratings are specified as either Current or MVA percent loadings by percent units program settings (see activity OPTN), and used in determination of over-

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Constraint	Description
	loading. Any branch where the rating in the designated rating set is zero is excluded from the set of constraint; zero impedance lines are also excluded from the set of constraints in the correction actions solution.
Interface MW flows	Interface flow is defined as the sum of MW flow of its members. Each interface is specified in terms of the from bus, to bus and circuit identifier of each of its members. Flow is computed at the from bus end of each circuit in the interface. If an interface consists of one or more zero impedance branches, or its rating in the designated rating set is zero, it will be excluded from constraints.
Bus voltages	High and low voltage limits for each bus are defined in per unit. Voltage limits can be specified on a system-wide basis, for the study area only, or for individual areas, zones and buses. If a bus voltage is monitored by several sets of upper and lower limits, the maximum of upper limits of all sets is selected as the upper limit and the minimum of lower limits of all sets is selected as the lower limit for the bus voltage respectively. Note that voltage deviation limits are not checked in corrective action calculations. For a generator bus, if the generator has sufficient reactive power generation to maintain its scheduled voltage, the generator bus is excluded from the constraint set.

Controls are the actions that can be taken to correct branch loading, bus voltage and interface flow limit violations in a power system. There are six categories of controls modeled in the corrective actions computation:

Table 6.13. AC Corrective Action Controls

Control	Description
Generator active power dispatch	Generators in the generator control subsystem that have a adjustable range are available for corrective action analysis.
Phase shifter angle adjustment	The desired MW flow values in the power flow database for participating phase shifters are ignored in the corrective action analysis. After a corrective action is performed successfully, the upper and lower limits (VMA and VMI) of a phase shifter specified in corrective actions are reset based on its flow in post-corrective solution, while the difference between upper and low limits remains constant.
Load curtailment	Bus active power load can be curtailed as needed to correct system problems. The reactive power load is adjusted accordingly to maintain a constant power factor of load at the bus.

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Control	Description
Off-line generator active power dispatch	Off-line generators in the off-line generator control subsystem that have a adjustable range are available for corrective action analysis. The reactive power outputs of the generators remain zeros. After a corrective action is performed successfully, off-line generators specified in corrective actions will be turn on with the upper and lower limits of reactive power set to zeros.
Tap setting adjustment	The transformers in the tap setting adjustment subsystem that are subject to voltage control mode are available for corrective action analysis. Voltage controls of participating transformers are ignored in the corrective action analysis. After a corrective action is performed successfully, the upper and lower limits (VMA and VMI) of a transformer specified in corrective actions are reset based on bus voltage of the controlled bus in post-corrective solution, while the difference between upper and low limits remains constant.
Switched shunt reactive power control	Switched shunt in the switched shunt control subsystem that subject to voltage control mode are available for corrective action analysis. Voltage controls of participating switched shunts are ignored in the corrective action analysis. After a corrective action is performed successfully, the upper and lower limits (VSWHI and VSWLO) of a switched shunt specified in corrective actions are reset based on bus voltage of the controlled bus in post-corrective solution, while the difference between upper and low limits remains constant.

6.13.2. Weighting and Penalty Functions

The corrective action algorithm finds an optimal solution by minimizing the weighted sum of the individual controls. The weight for each control is given by its weighting function. The corrective actions are influenced by these weighting functions and the relative effectiveness of the respective controls in eliminating the system problems.

Two types of weighting functions for controls are used: a linear function for load shedding and V-curves for other controls such as MW generation and phase shifter angle; Figure 6.36, “Active Power Generation Control Default Weighting Function”, Figure 6.37, “Load Shedding Control Default Weighting Function” and Figure 6.38, “Phase Shifter Angle Control Default Weighting Function” show weighting functions for generator active power dispatch, load curtailment and phase shifter angle adjustment. Controls in one category have the same weighting function.

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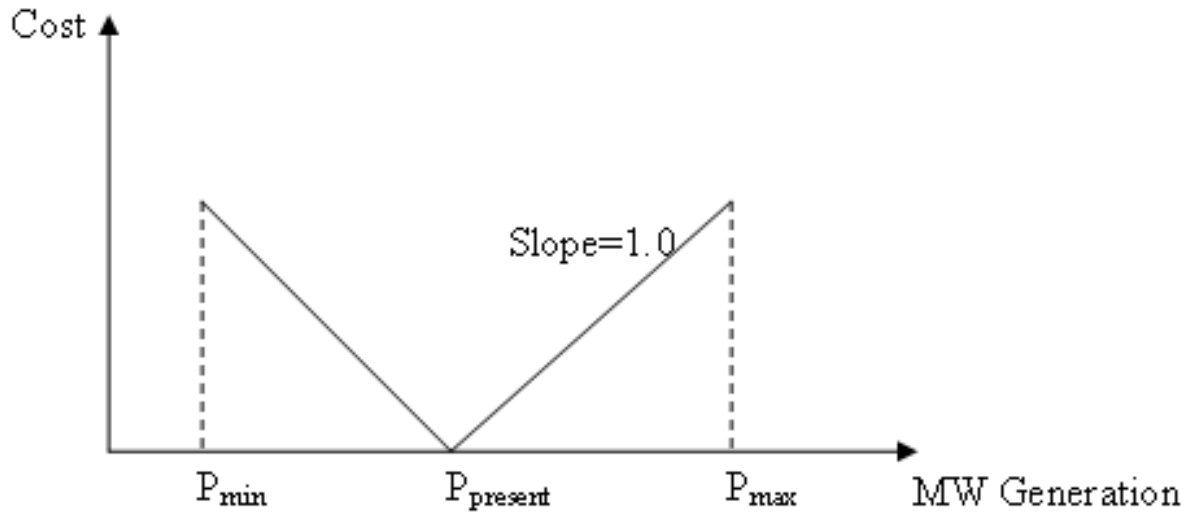

Figure 6.36. Active Power Generation Control Default Weighting Function

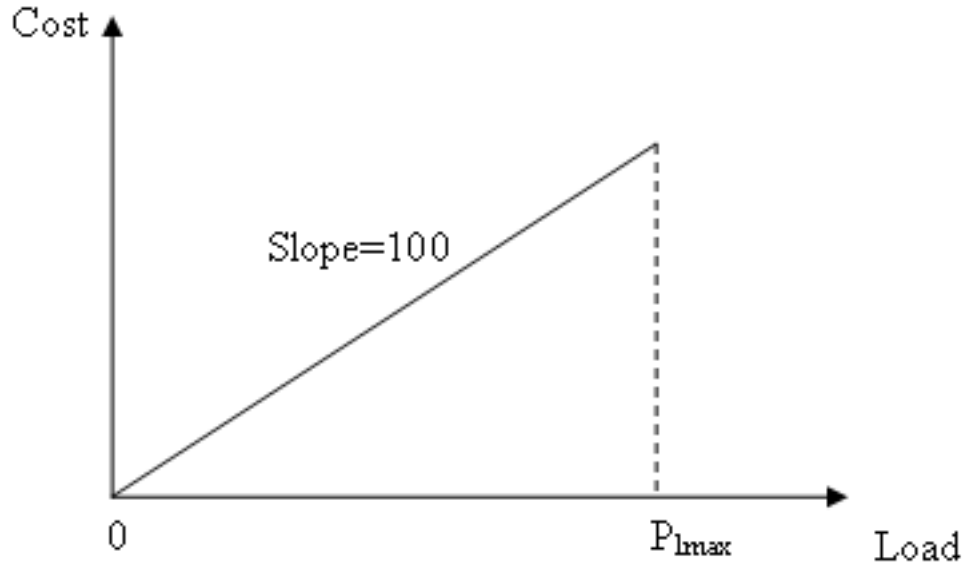

Figure 6.37. Load Shedding Control Default Weighting Function

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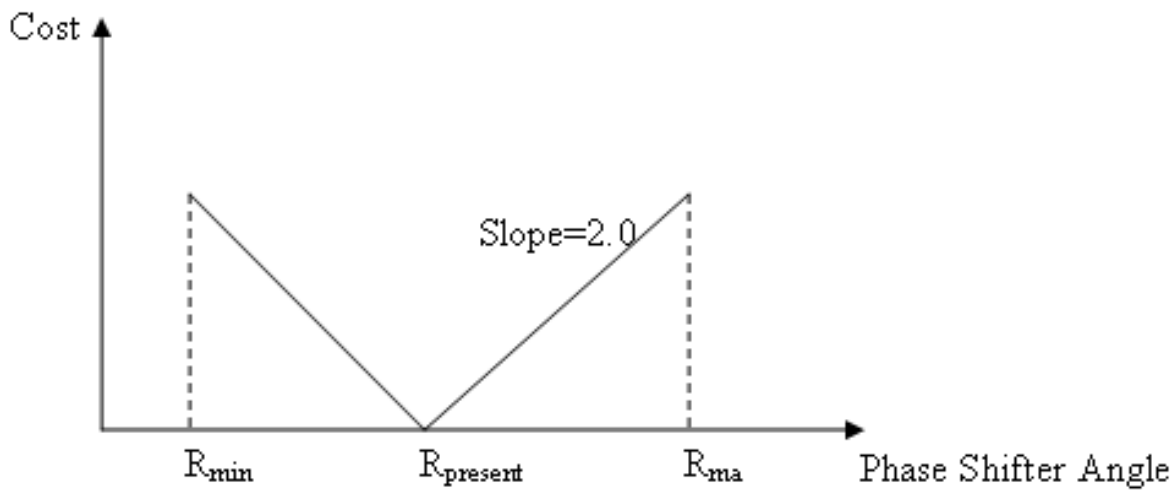

Figure 6.38. Phase Shifter Angle Control Default Weighting Function

Load curtailment should be assigned a much higher weight than generator real power dispatch to prevent unnecessary load curtailment when generation rescheduling is adequate in relieving the system problems. Moderate changes of the weights from the default values generally have little influence on the corrective actions because the sensitivities expressing the effectiveness of individual actions tend to dominate. A large weighting factor discourages the control from being adjusted. Therefore, you can favor certain controls over others by specifying smaller external weights. However, please note that extremely large or small weights may cause solution problems. For this reason, it is recommended that weights be limited to the range from 0.1 to 10.

If the limit violations in the power system cannot all be eliminated using the basic control actions, it may be preferable sometimes to accept certain degree of limit violations instead of applying extreme and ineffective control adjustments. This strategy can be represented by applying soft penalties to the constraints.

The penalties for constraint violations are represented by multi-segment linear functions. The internal penalties for the first two segments for branch loading and bus voltage constraints are 1000 and 10000, respectively, as shown in Figure 6.39, “Branch/Interface Flow Overload Penalty Function” and Figure 6.40, “Bus Voltage Violation Penalty Function”. These penalties cannot be adjusted.

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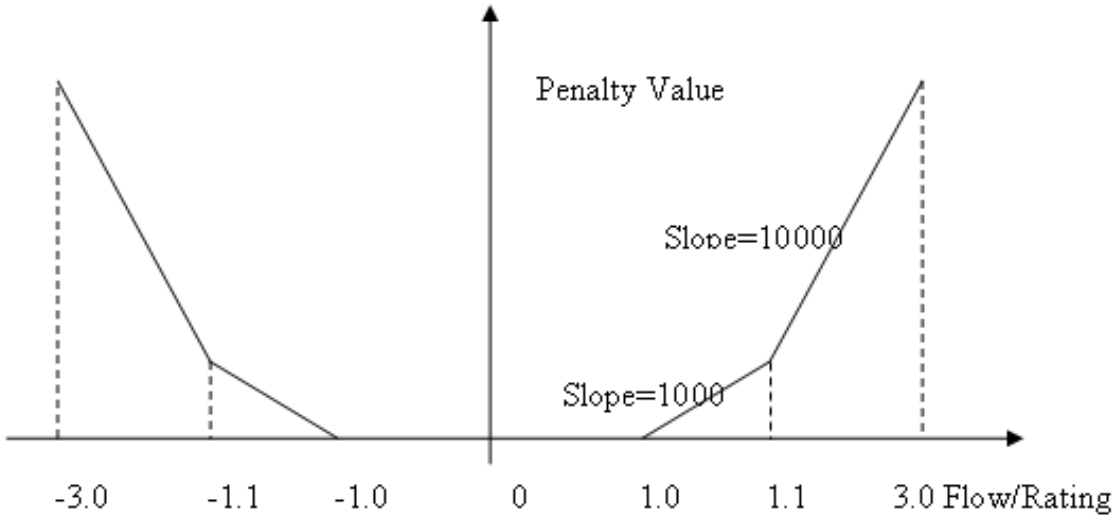

Figure 6.39. Branch/Interface Flow Overload Penalty Function

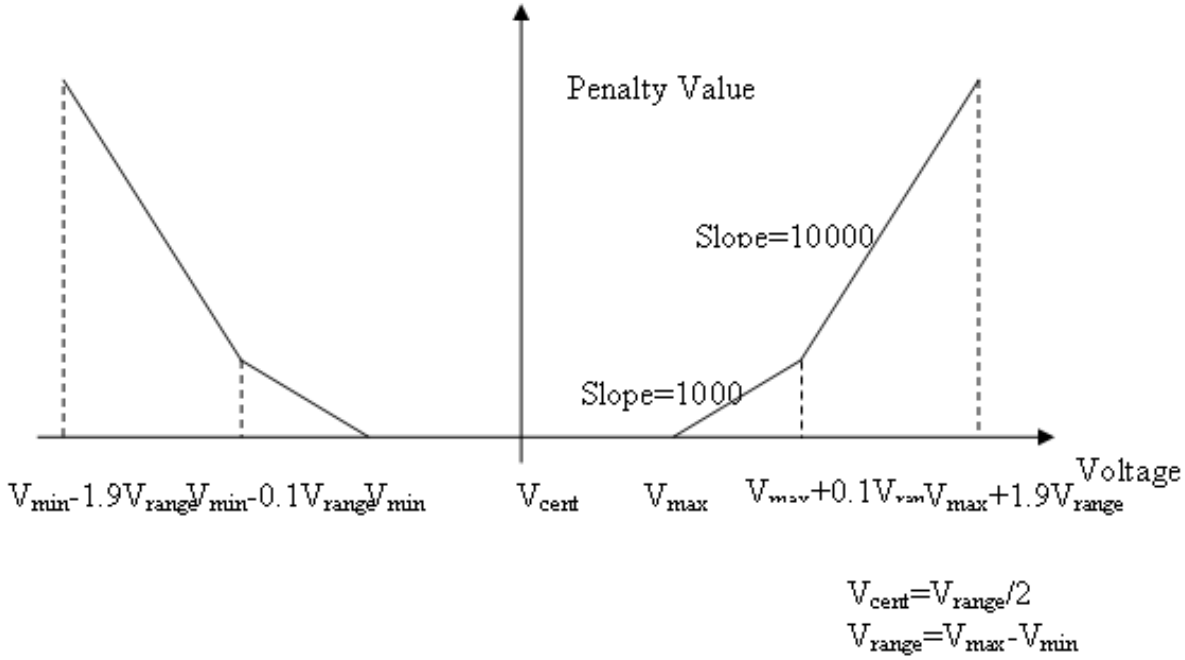

Figure 6.40. Bus Voltage Violation Penalty Function

6.13.3. AC Corrective Action Options

The following options are available in the ac corrective actions menu.

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Table 6.14. AC Corrective Action Options

Option	Description
Solution options	Select the desired solution options to be applied during power flow calculations with corrective actions (see Section 6.12.1, "AC Contingency Solution Options").
Number of ac load flows	Select the desired number of iterations between execution of the linear programming engine to identify corrective actions and subsequent ac power flow solution to verify the feasibility of those actions. The specified number of iterations can be between 1 and 10. Note that, even if a feasible solution appears to have been found, i.e., the identified corrective actions appear to have been successful, the iteration between linear programming computation and ac power flow solution may continue in an attempt to further optimize the solution if not exceeding the iteration limit. The user needs to select this parameter carefully, based on experience. On the other hand, if a solution fails at the nth iteration because the ac power flow diverges, the user may obtain an approximate solution by specifying the number of ac power flows as n-1 and repeating the calculations.
Mismatch tolerance	This tolerance will be used to check for the largest active or reactive power mismatch in the power flow case prior to the start of the ac corrective actions computation. If this tolerance is exceeded, the process is terminated. This value is also used as the convergence tolerance for the ac power flow solution embedded in ac corrective action analysis.
Constraint options	Select branch flow rating set, and limit tolerances for monitored elements.
Rating	Specify rating to be used in determination of branch and interface flow overloads. The default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).
Percent of flow rating	Specify percent of rating to be used in determination of branch and interface flow overloads.
Voltage limit	Defines voltage limits used in determining bus voltage limit violations. The voltage limits apply to buses specified by monitor voltage limit records (see ???).
Branch overload violation tolerance	Specifies branch overload violation tolerance in percent to adjust branch overload limits used in corrective actions. The limits are increased by the tolerance percent of the specified rating when the tol-

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Option	Description
	erance is positive, or reduced when it is negative. For instance, if the tolerance is 0.1% and the overload limit is 100% of rate A, the 100.1% of rating A is used in determining overload violations in corrective actions. In the above case if the tolerance is -0.1%, the 99.9% of rating A is used.
Bus voltage violation tolerance	Specifies bus voltage violation tolerance in pu to adjust the bus voltage limits used in corrective actions. The limits are expanded by the tolerance when it is positive, narrowed when it is negative. For instance, if the tolerance is 0.1 pu and lower and upper limits on a bus voltage are 0.95 and 1.05 in pu respectively. The adjusted voltage limits are 0.949 and 1.051 respectively. In the above case if tolerance is -0.1, the lower and upper limits are 0.951 and 1.049.
Control options	Select types of control adjustments and specify weighting factors for each type.
Enable generator controls	Enable to include active power generation dispatch controls. The adjustable range is from PT to PB of each participating machine. Generator active power limits are taken from a Generator Inertia and Governor Response Data file (see Section 6.8.1, "Inertia and Governor Response Data File Contents") if the file is specified, otherwise the values in base case are used. In corrective action analysis, generator active power limits are adjusted by the following rules. If PT had been set to the default value, the larger value of PG (its existing MW output) and machine power base (MVA base) will be used as PT; If PT of a machine is less than its PG, the PT is set equal to the PG. If PB of a machine had been set to the default value, it is set equal to 0; if PB of a machine is less than its PG, the PB is set equal to the PG. A machine will be excluded from the controls if its PT is equal to its PB.
Generator weighting factors	Specify a weighting factor to adjust the cost factor of active power generation dispatch. The cost of a control adjustment in objective functions is equal to the amount of adjustment times by a adjusted cost factor. The internal cost factor for active power generation dispatch defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls"; the adjusted cost factor is equal to the internal factor multiplied by the user defined weighting factor. e.g., if the weighting factor is set as 1.0, the cost factor for active power generation dispatch of 100 MW will be 1.0.

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Option	Description
Generator control subsystem	Select the subsystem in which all in-service generators with valid range of adjustments will participate in corrective action analysis.
Enable load controls	Enable to include load curtailments. The loads with negative MW values will be excluded from the controls. While performing load curtailment, the algorithm maintains a constant power factor at the buses where load curtailment occurs.
Load weighting factors	Specify a weighting factor to adjust the cost factor of load curtailment. The internal cost factor for load curtailments defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls"; the adjusted cost factor is equal to internal factor multiplied by the the weighting factor. e.g., if the weighting factor of load curtailments is set as 1.0, the cost factor for load shedding of 100 MW will be 100.0.
Load control subsystem	Select the subsystem in which all in-service loads will participate in corrective action analysis. Loads with negative real power will not be included as candidates for load curtailments.
Enable phase shifter controls	Enable to include phase shifter controls. To be adjustable as a phase shifter, transformers must be operated subject to MW control mode. The adjustable range of a participating phase shifter is from Rmax to Rmin.
Phase shifter weighting factors	Specify a weighting factor to adjust the cost factor of phase shifter angle adjustments. The internal cost factor for phase shifter angle adjustments defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls"; the adjusted cost factor is equal to internal factor multiplied by the user-defined weighting factor. e.g., if weighting factor of phase shifter angle adjustment is 1.0, the cost factor of phase shifter angle adjustment of 100 degrees is 3.5.
Phase shifter control subsystem	Select the subsystem in which all in-service phase shifters will participate in corrective actions.
Enable off-line generator controls	Enable to include off-line generator active power shift. The initial active power outputs of participating off-line generators are set to zeros. Their active power ranges are specified in the same way as online generators
Off-line generator weighting factor	Specify a weighting factor to adjust the cost factor of active power generation dispatch of off-line generators. The internal cost factor defined in the pro-

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Option	Description
	grams is shown in Table 6.15, "Weighting Functions and Factors of Controls".
Off-line generator control subsystem	Select the subsystem in which all out-of-service generators with valid range of adjustments will participate in corrective action analysis.
Enable tap setting adjustments	Enable to include tap setting adjustment.
Tap setting weighting factors	Specify a weighting factor to adjust the cost factor of tap setting adjustment. The internal cost factor defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls".
Tap setting adjustment subsystem	Select the subsystem in which all in-service transformers designated in voltage control mode will participate in corrective action analysis.
Enable switched shunt controls	Enable to include switched shunt reactive power controls.
Switched shunt weighting factors	Specify a weighting factor to adjust the cost factor of switched shunt reactive power control. The internal cost factor defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls".
Switched shunt control subsystem	Select the subsystem in which all in-service switched shunts designated in voltage control mode will participate in corrective actions.
Input Data File	The Distribution Factor Data file is used to provide monitored elements, bus voltage limits and control subsystems. When creating a Distribution Factor Data file for corrective action analysis, a Contingency Description Data file where no contingencies are specified is allowed.

Table 6.15. Weighting Functions and Factors of Controls

Control	Weighing Function	Internal Weight
Phase shifter Angle	V-Curve	2 per 57.4 degrees (0.035 per degree)
MW Generation	V-Curve	1 per 100 MW (0.01 per MW)
Load Curtailment	Linear	100 per 100 MW (1.0 per MW)
MW generation of Off-line generator	V-Curve	2 per 100 MW
Tap setting	V-Curve	1 per 0.002 pu
MVAR output of Switched shunt	V-Curve	1 per 100 MVAR

6.13.4. Viewing AC Corrective Actions Results.

After ac corrective actions solution is performed successfully, a corrective action analysis report is produced. In the report, files used in the analysis, and numbers of constraints and controls are presented first. Flow overloads and bus voltage violations prior to and post corrective action analysis are reported. Specified cor-

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rective actions are presented in the order of generation dispatch, load shedding, phase shifter angle adjustment, off-line generator active power dispatch, tap setting adjustment and switched shunt control. For each control adjustment, the initial and new values, as well as the change are reported. Figure 6.41, "Sample Report from Corrective Action Analysis" shows the results of ac corrective actions applied to an example system (savnw case). In this run, percent MVA and percent CURRENT units are specified for monitored transformer and non-transformer branches respectively. 8 branches, 2 interfaces and 6 bus voltages are selected as constraints from monitored elements, 6 generator buses, 7 load buses, as well as 1 phase shifter are selected as controls from components in specified subsystems. In the report, the base case is found to have four flow violations above 60% of rating A (as an illustration) including flow overloading on Interface 'WEST', which is 171% of its normal limit. The report also shows that, after corrective actions have been applied, flows on two branches are slightly higher than their limits.

```
                    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E
 AC CORRECTIVE ACTIONS RESULTS:% LOADING VALUES ARE % MVA FOR TRANSFORMERS AND % CURRENT FOR NON-TRANSFORMER BRANCHES
-------------------------------------------------------------------------------
 Case file:                              C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sav
 Distribution factor file:               C:\Program Files\PTI\PSSE30\EXAMPLE\corrective-actions.dfx
 Subsystem description file:             C:\Program Files\PTI\PSSE30\EXAMPLE\savnwsl.sub
 Monitored element file:                 C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
 Contingency description file:           C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con
 *** Constraints ***
 No. of monitored flows:                 8
 No. of monitored interfaces:            2
 No. of monitored voltages:              6
 *** Controls ***
 No. of Generators:                      6          In Subsystem:  LODSUB
 No. of Load Shedding:                   7          In Subsystem:  LODSUB
 No. of Phase Shifters:                  1          In Subsystem:  LODSUB
-------------------------------------------------------------------------------
 BASE CASE OVERLOAD REPORT: MONITORED BRANCHES LOADED ABOVE  60.0 % OF RATING SET A
 INCLUDES BUS VOLTAGE VIOLATION REPORT
 <---------- O V E R L O A D E D   L I N E S ---------->     FLOW   RATING        %
 <------ F R O M --------> <-------- T O ----------> CKT   MVA/MW      MVA
    203 EAST230      230.00    154 DOWNTN      230.00  1     137.7    200.0    71.22
    205 SUB230       230.00    154 DOWNTN      230.00  1     429.0    600.0    77.11
   3018 CATDOG_G     13.800   3008 CATDOG      230.00  1     122.8    150.0    85.38
 WEST                                                      343.4    200.0   171.70
-------------------------------------------------------------------------------
 POST-CORRECTIVE ACTIONS OVERLOAD REPORT/ INCLUDES BUS VOLTAGE VIOLATION REPORT
 <---------- O V E R L O A D E D   L I N E S ---------->     FLOW   RATING        %
 <------ F R O M --------> <-------- T O ----------> CKT   MVA/MW      MVA
    205 SUB230       230.00    154 DOWNTN      230.00  1     374.3    600.0    66.23
   3018 CATDOG_G     13.800   3008 CATDOG      230.00  1      89.1    150.0    62.53
-------------------------------------------------------------------------------
 BUSES WITH REAL POWER GENERATION ADJUSTMENTS (MW)
 <-------- B U S -------->   GEN-INI   GEN-NEW   GEN-ADJ
    101 NUC-A        21.600     750.0     610.6    -139.4
    102 NUC-B        21.600     750.0     630.0    -120.0
    206 URBGEN       18.000     800.0     840.0      40.0
    211 HYDRO_G      20.000     600.0     616.2      16.2
   3011 MINE_G       13.800     258.7     505.9     247.2
   3018 CATDOG_G     13.800     100.0      48.9     -51.1

 PHASE SHIFTER ADJUSTMENTS (DEGREE)
 <----------- P H A S E   S H I F T E R ----------> CKT  PHASE-INI PHASE-NEW PHASE-ADJ
    203 EAST230      230.00    202 EAST500      500.00  1        0.00      9.55      9.55

 ===== END OF REPORT ============================================================================
```

Figure 6.41. Sample Report from Corrective Action Analysis

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6.13.5. Application Notes

The corrective actions solution algorithm does not place a limit on the numbers of controls and constraints that can be modeled. The computation time of corrective actions solution is approximately proportional to size of the power system and the number of constraints. Because local controls are generally more effective in removing violations, it may be preferable to choose the subsystems in which control actions are allowed to be the same as that in which elements are monitored.

Operating limit violations caused by incorrect limit settings may have a significant impact on the optimal solution found. Hence, suspicious limits should be verified and corrected, if necessary, or even removed from the solution.

Two methods can be used to exclude controls from the corrective actions solution. One is to set the upper and lower limits of a control to be identical. In addition, to exclude an on-line generator and off-line generator from the solution, both the upper and lower limits should be equal to the Pgen. The second method is to specify a control subsystem that does not include undesirable controls in a particular application.

Because the optimization engine is based on a linear power flow, the result from the linear programming solution is iterated with a standard ac power flow solution until one of following conditions occurs:

- all overloads and voltage violations are resolved,
- the adjustable controls are exhausted,
- the number of ac power flow solutions reaches the specified value,
- the difference of the values of objective function between two consecutive iterations is less than a tolerance when both branch overload violation tolerance and voltage violation tolerance are greater than 0.
- the ac power flow diverges during the process

When ac corrective actions computation fails because of ac power flow divergence, warning messages in the progress window indicate at which step the ac power flow diverges. The user may obtain an approximately optimal solution by specifying the number of ac power flow in ac corrective actions input menu as one less than the step in the solution failure occurs. Or, if the user finds that the computation may require more iterations to reach an optimal solution, the desired iteration number may be increased.

Additional Information
PSS®E GUI Users Guide, AC Corrective Actions

6.14. PSCOPF Solution

6.14.1. About PSCOPF analysis

Power flow solution solves a set of non-linear equations, i.e. power balance equations, and adjusts the system to meet branch flows, bus voltages, and area interchange schedules and determines the flows on branches and voltages at buses.

In an optimal power flow, the solution is formulated as an optimization problem with an objective function, a set of constraints to satisfy and a set of controls that can be adjusted (see Section 15.3, “Optimal Power Flow”, although that algorithm is not used in PSCOPF). The settings of controls are determined to achieve the best solution corresponding to a performance measure defined by the objective function. The optimal objective is dependent on the application. Some common objectives are:

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- Economic dispatch
- Maximum power transfer
- MW and MVAr loss minimization
- Minimum deviation from a target schedule
- Minimum control shifts to alleviate violations

These objectives can be set individually or combined to form a composite objective function in an optimal power flow problem. The controls may be applied to existing devices or new installations. Some typical controls are:

- Generator MW dispatch;
- Transformer taps;
- Phase shift angles;
- Shunt capacitor
- or reactor addition;
- Series compensation level;
- Load shed.

The constraints may include:

- Power balance equations
- Limits on controls and operation limits

The security constrained optimal power flow is a special class of OPF problems which takes into consideration the system constraints derived from a base case and a set of predefined contingency cases. System security is the ability to withstand contingencies, in other words, to remain intact even after equipment outages or failures. Security plays a crucial role in the planning and operation of a power system. To ensure a secure system operation condition, system planners and operators can conduct analyses to identify the necessary adjustments such that there are no limit violations in the base case (pre-contingency) condition and following any contingency.

There are two principal classes of methods to enhance the system security: preventive and

corrective methods. The preventive method enables the system to be prepared for certain contingencies beforehand. The preventive method can be considered conservative, since it does not take into account the system's post-contingency control capabilities. All control variables are optimized such that no post–contingency adjustments are necessary in order to avoid violations of base case and post-contingency constraints, and no operator intervention is required following an anticipated contingency. The corrective method helps identify quick and appropriate responses to the contingencies after their occurrence. In recent system security applications, a mixture of the preventive and corrective method has also been used.

The Preventive Security Constrained Optimal Power Flow (PSCOPF) function formulates and solves an optimization problem:

With an objective that minimizes the adjustments of the following types of control:

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- On-line and off-line generator MW generation control
- Phase shifter adjustments
- Load controls
- Tap setting adjustments
- Switched shunt adjustments

Subject to the following types of constraints:

- Power balance equations of base case and contingencies cases
- Limits on controls
- Operation limits under base case and contingency cases

When a huge number of contingency cases are considered, the PSCOPF problem becomes a large-scale optimization problem. Trying to solve this problem directly for a large power system by simultaneously imposing all the post-contingency constraints could lead to prohibitive computer memory requirements and CPU time. Benders decomposition is an appropriate solution method to the large-scale optimization problems. In Benders method, the initial problem is decomposed to several sub-problems that can be solved separately and iteratively. In using the Benders decomposition method in the PSCOPF function, the master problem is set up with the base case condition and the cuts from the contingency cases, and a sub-optimization problem is modeled for each contingency to ensure the feasibility of the solution.

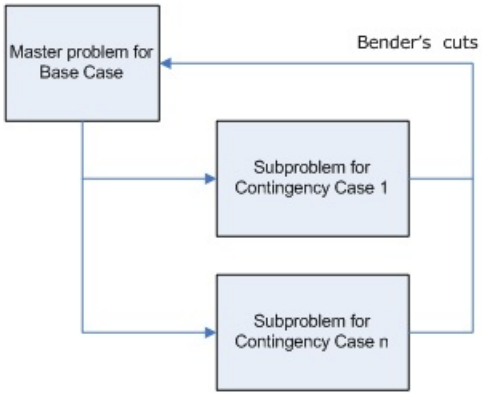

Figure 6.42. Two level structure used in the Bender's decomposition method

A Successive Linear Programming (SLP) method is used to solve the master problem and each sub-problem (see Section 6.13, "AC Corrective Actions"). To keep the number of sub-problems small, a contingency filtering technique that is based on power flow solutions is applied. The critical contingencies are identified and put in a set. Only the contingencies in the set are represented as sub-problems.

The inputs required to perform a PSCOPF consist of a power flow network, the available controls and the constraints. The controls are selected from the control subsystems, which are pre-defined in the subsystem description data file (*.sub), and each type of control is assigned a priority, relative to other control types. The operation constraints are defined in the monitored element data file (*.mon), and the contingencies are specified in the contingency description data file (*.con). Before initiating the PSCOPF solution, a dfx

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file is built using function DFAX to process the three data files and store the information in a dfx file for use by PSCOPF.

6.14.2. PSCOPF solution options

The following options are available in the PCSCOPF function. See the API Manual, PSCOPF.

Table 6.16. PSCOPF Solution Options

Base-case Solution options	Select solution options for power flow calculations performed to obtain the base case solution (see Section 6.2, "Power Flow Solution Methods"). • Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustments: Lock all Enable all Enable continuous, disable discrete Enable or disable phase shift adjustments. Enable or disable DC tap adjustments Induction motor treatment flag applied when an induction motor fails to solve due to low terminal bus voltage • stall • trip
Contingency-case Solution options	Select solution options for power flow calculations performed to obtain the contingency solutions. Tap adjustment: • Lock taps

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	• Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustments: • Lock all • Enable all • Enable continuous, disable discrete Enable or disable phase shift adjustments. Enable or disable DC tap adjustments Induction motor treatment flag applied when an induction motor fails to solve due to low terminal bus voltage • stall • trip
Solution Engine	Select one engine for both base case and contingency solutions (see Section 6.2, "Power Flow Solution Methods"). • Fixed slope decoupled Newton-Raphson (FDNS) (default)* • Full Newton-Raphson * The optimized fixed slope decoupled Newton-Raphson method is used when PSCOPF is initiated via the graphical user interface. The user may choose either the regular or the optimized fixed slope decoupled Newton-Raphson method when initiating PSCOPF from an automation file.
Non-divergent solution	Option to enable or disable non-divergent solution in each power flow solution
Generation dispatch for contingency case	Option for generation dispatch to handle system load/generation imbalance caused by a contingency: • Subsystem machines (Reserve)

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	• Subsystem machines (PMAX) • subsystem machines (Inertia) • subsystem machines (G • overnor droop) Dispatch subsystem specifies the subsystem containing the machines participating in the selected dispatch
Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the PSCOPF solution is terminated. This value is also used as the convergence tolerance in the power flow solution of the base case and contingency cases (see Section 6.2, "Power Flow Solution Methods").
Constraint Options	Ignore violations in base case: flag to ignore the initial violations in the base case, it applies when initial violations are identified, • ignore the violations from PSCOPF process • find the corrective actins to remove the violations; ignore the violations that can not be removed by corrective actions. Rating set: Rating set to be used in determining monitored element loading violations in contingency solutions • A • B • C Specify percent of flow rating to be used in determination of branch and interface flow overloads in contingency solutions. Branch overload violation tolerance (%): Specifies branch overload violation tolerance in percent to adjust branch overload limits used in PSCOPF. The limits are increased by the tolerance percent of the specified rating when the tolerance is positive, or reduced when it is negative. For in-

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Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the PSCOPF solution is terminated. This value is also used as the convergence tolerance in the power flow solution of the base case and contingency cases (see Section 6.2, "Power Flow Solution Methods").
	stance, if the tolerance is 0.1% and the overload limit is 100% of rate A, the 100.1% of rating A is used in determining overload violations in corrective actions. In the above case if the tolerance is -0.1%, the 99.9% of rating A is used. Bus voltage violation tolerance (%): Specifies bus voltage violation tolerance in pu to adjust the bus voltage limits used in PSCOPF. The limits are expanded by the tolerance when it is positive, narrowed when it is negative. For instance, if the tolerance is 0.1 pu and lower and upper limits on a bus voltage are 0.95 and 1.05 in pu respectively. The adjusted voltage limits are 0.949 and 1.051 respectively. In the above case if tolerance is -0.1, the lower and upper limits are 0.951 and 1.049.
Iteration # of PSCOPF	Maximum number of PSCOPF iterations
Cutoff for clean cases	Iteration number n of a contingency to be solved in PSCOPF process. At each iteration of PSCOPF, each contingency that is not in the set will be solved by a power flow solution and its results will be checked to determine flow and voltage limit violations and if it should be included in the set. If a contingency does not cause any limit violations so it is not included in the set in the n consecutive iterations of PSCOPF, it will not be tested any more in subsequent iterations.
Control options	Flag to enable or disable generator active power dispatch (see Section 6.11, AC Corrective Actions) Weighting factor for generator active power dispatch Control subsystem of participating generators Flag to enable or disable load controls Weighting factor for load controls Control subsystem of participating loads

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Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the PSCOPF solution is terminated. This value is also used as the convergence tolerance in the power flow solution of the base case and contingency cases (see Section 6.2, "Power Flow Solution Methods").
	Flag to enable or disable phase shifter controls Weighting factor for phase shifter controls Control subsystem of participating phase shifters Flag to enable or disable off-line generator active power dispatch Weighting factor for off-line generator controls Control subsystem of participating off-line generators Flag to enable or disable tap setting adjustments Weighting factor for tap setting adjustments Control subsystem of participating load tap changing transformer Flag to enable or disable switched shunt controls Weighting factor for switched shunt controls Control subsystem of participating switched shunts
Distribution Factor Data File	(*.dfx) a required file to be build by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files (refer to ???). The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) must have the same setting that was in effect when activity DFAX filled the specified Distribution Factor Datafile.
Load Throwover Data File	(*.thr) an optional file, created by the user, that contains data records of the following form: `IBUS, JBUS` where IBUS and JBUS are bus numbers. If branch outage contingency events isolate bus IBUS, the user can elect to move the load to bus JBUS. Bus

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Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the PSCOPF solution is terminated. This value is also used as the convergence tolerance in the power flow solution of the base case and contingency cases (see Section 6.2, "Power Flow Solution Methods").
	IBUS and/or JBUS may be a dummy bus of a multi section line grouping. Data records may be entered in any order. Input is terminated with a record specifying an IBUS value of zero.
Unit Inertia Data File	(*.inl) a required file if the generation dispatch is enabled and either the inertial or governor droop dispatch method is selected. If the PMAX or reserve method is selected, the specification of this file is optional. If a file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used

6.14.3. Perform PSCOPF Analysis

To launch the PSCOPF analysis, select Power Flow ◊ Contingency, Reliability, PV/QV ◊ Preventive Security Constrained Optimal Power Flow

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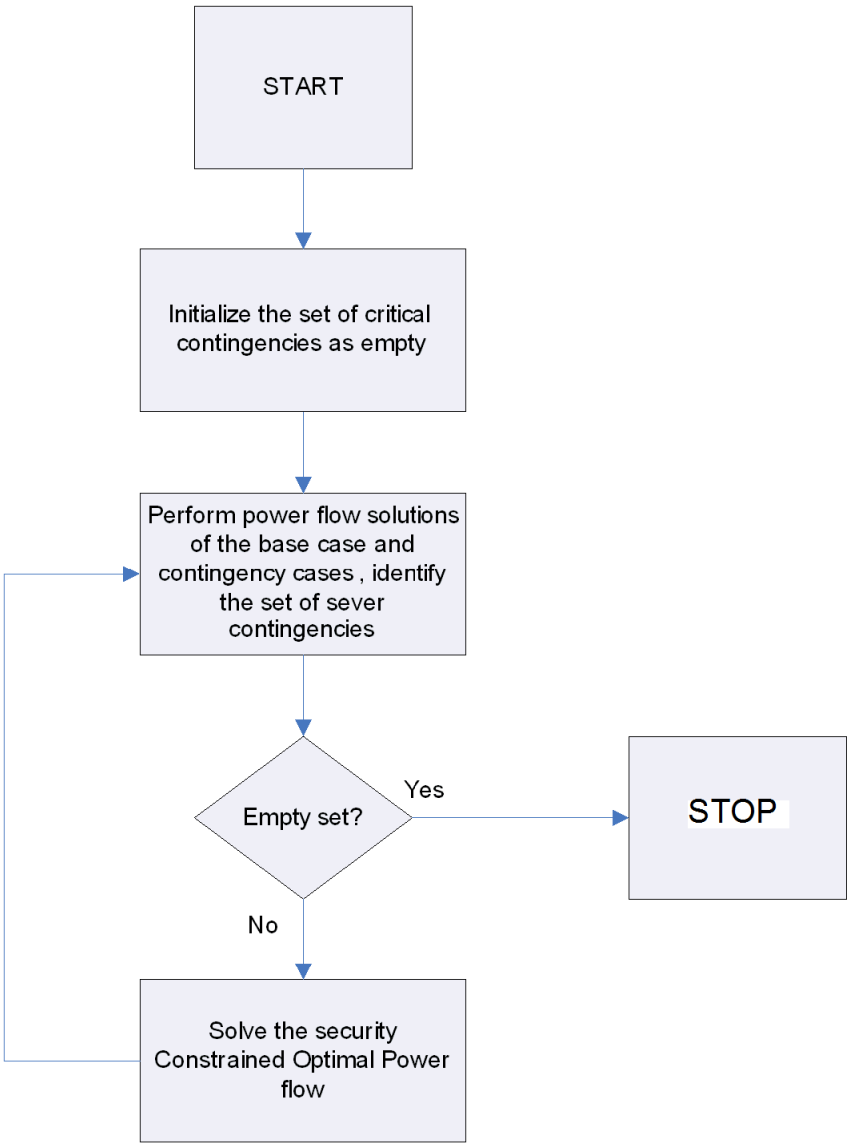

Figure 6.43. Flow chart of PSCOPF process

The PSCOPF is an iterative process, as shown in Figure 6.43, "Flow chart of PSCOPF process", with each iteration consisting of two computational components: contingency testing and solution of the combined master problem and sub-problems. The first part performs power flow solutions of the base case and contingency cases, identifies the set of critical contingencies and builds the Bender's cuts that will be used to formulate the master problem of the security constrained optimal power flow problem. The second part solves the master problem and finds the optimal settings of the controls. At the end of each iteration, the base case system is adjusted according to the identified control settings.

The key function in the testing part is to identify a set of critical contingencies at each iteration of the PSCOPF. The set consists of contingencies that cause flow and/or voltage limit violations, but cannot be represented by any other contingencies in the set. The power flow results from each contingency solution are

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checked against the corresponding performance criteria to identify the limit violations. For example, under contingency A, the line x-y is overloaded by 5 percent of its rating; under contingency B, the same line is overloaded by 10 percent. In theory, if preventive strategies can be found to adjust the system under base case condition so that the line x-y does not violate its rating limit following contingency B, then the line will operate within its limit following contingency A too. If the overload of line x-y is the only violation under both contingencies A and B, B is the dominant contingency and will be added into the set, while A is considered a non-dominant contingency. However, in many situations, multiple violations may be found in a contingency solution. In that case, the solution of the contingency being tested is compared against the solutions of every other contingency already included in the set. If any contingency in the set causes the same violations, more specifically, the number of violations is identical, the same monitored elements are overloaded and the same buses have same low voltage or high voltage problems. The solution of the contingency in the set is further compared to the contingency being tested, if the violations in the contingency in the set are more severe than the ones in the contingency being tested, the contingency being tested is not a dominant contingency; otherwise it is a dominant case and included in the set. This method of selection can reduce the number of cuts significantly and therefore speed up the calculations in the second part.

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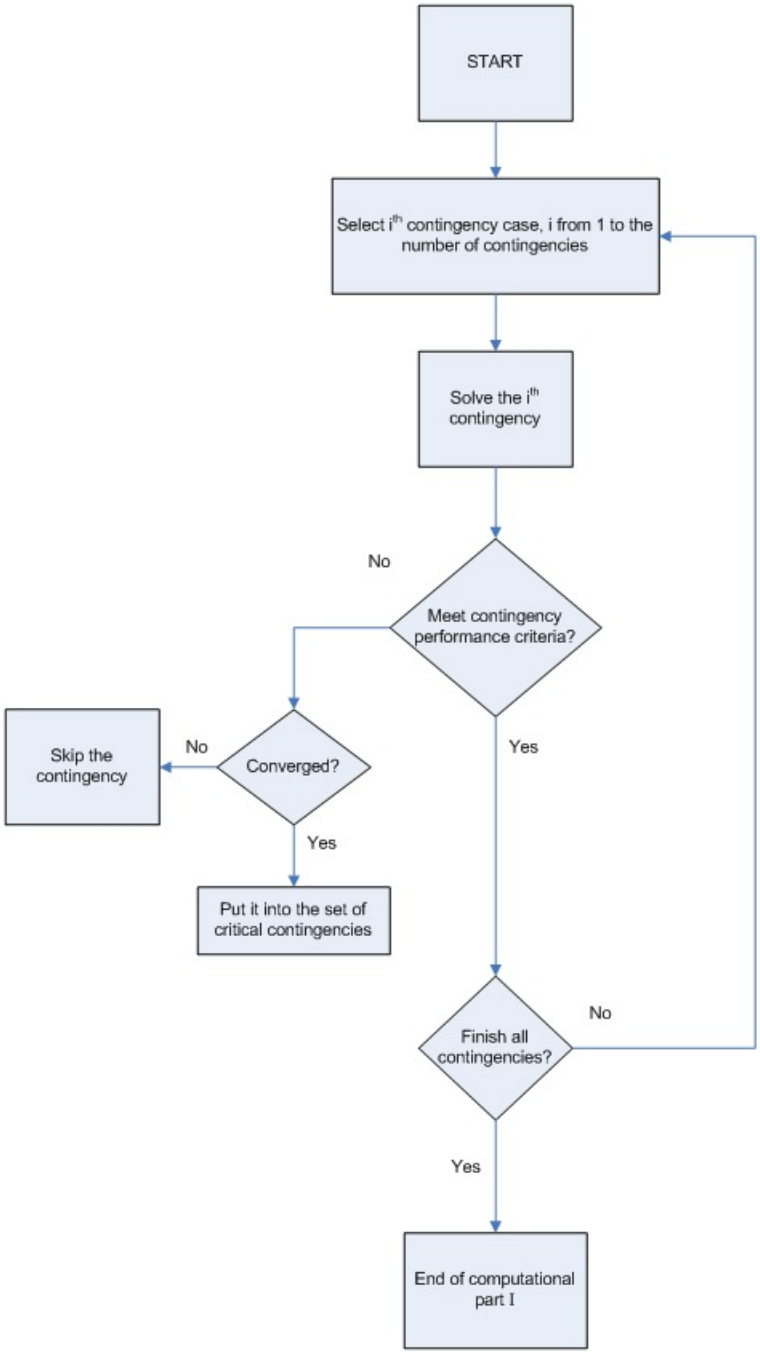

Figure 6.44. Chart for testing of contingencies and selection of cut set

The testing part consists of several steps:

1. Set up the base case, contingency definitions, monitored elements, etc. to prepare for contingency analysis, initialize the set of severe contingencies to be empty.

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2. Perform a power flow solution under base case condition; the solution results are checked against base case performance criteria to identify flow and/or voltage limit violations, if any.

3. Select a contingency, impose it on the base case condition and calculate the power flow solution with contingency-case power flow solution options.

4. Check results from the contingency solution against contingency case performance criteria to identify flow and/or voltage limit violations, if any. If violations are found and the contingency is considered to be dominant, add the contingency to the set of critical contingencies.

5. Return to step 3 to test the next contingency. Stop when all contingences have either been tested or included in the set.

Figure 6.45, "Sample of progress outputs of PSCOPF" shows a sample progress output of contingency testing in an iteration of PSCOPF.

```
PSCOPF is processing the iteration 1 of main loop...


Processing BASE CASE

Processing contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

Processing contingency "N1OVRLOD 2"
OPEN LINE FROM BUS 215 [MIDCOALP    230.00] TO BUS 224 [WESTIE-2    230.00] CKT 1

BUSES WITH LOW/HIGH VOLTAGE LIMIT VIOLATIONS
<-------- B U S --------->  VOLTAGE  V-MAX  V-MIN
   103 WESTIE-1    138.00  0.8982 1.0500 0.9500
   224 WESTIE-2    230.00  0.8658 1.0500 0.9500

Processing contingency "N1OVRLOD 3"
OPEN LINE FROM BUS 213 [BIGOIL      230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

BUSES WITH LOW/HIGH VOLTAGE LIMIT VIOLATIONS
<-------- B U S --------->  VOLTAGE  V-MAX  V-MIN
   224 WESTIE-2    230.00  0.9486 1.0500 0.9500
```

Figure 6.45. Sample of progress outputs of PSCOPF

At each iteration of PSCOPF, every contingency not already in the set will be tested, the results checked to identify possible overloads and/or voltage violations and a decision made whether the contingency should be included in the critical set of not. If no violations are found in the contingency power flow solution, or if the contingency is classified as non-critical in several consecutive iterations of PSCOPF, it is safe to say that any system control adjustments made in subsequent iterations of PSCOPF will have relatively small impact on this contingency. Therefore, this contingency will be excluded from further testing in subsequent iterations. The number of consecutive iterations for making this decision to exclude the contingency is specified in the input option: cutoff for clean cases.

The second calculation part is to solve the master problem and determine the system adjustments, feed the adjustments back to each sub-problem and update the Bender's cuts corresponding to the new system condition. These steps are conducted iteratively until the process arrives at the best solution that satisfies all the constraints. There are several steps in this part:

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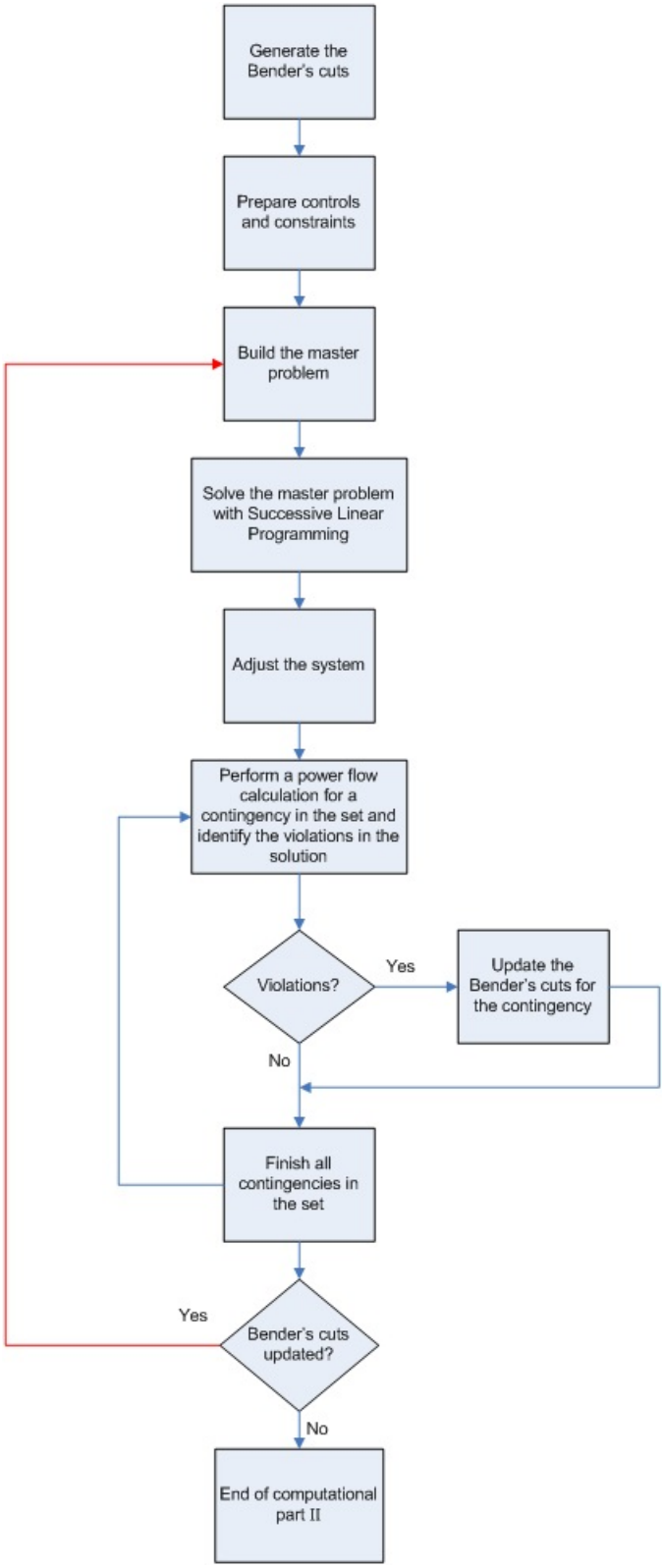

Figure 6.46. The process to determine the control adjustments

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1. Create the Bender cuts for each violation in each contingency in the set.

2. Process controls and constrains, together with Bender cuts to build the master optimization problem.

3. Solve the master optimization problem with Successive Linear Programming.

4. Apply the adjustments determined from the Successive Linear Programming algorithm to the base case system, and perform an AC power flow solution. This solution is used as the initial condition for the following calculations.

5. Impose one contingency case and perform a power flow solution with the contingency case solution options. If automatic system adjustments are enabled, they are determined by the power flow solution to meet their respective specified bus voltage, branch flow or are interchange schedules.

6. Check the results from the contingency solution to identify thermal and/or voltage limit violations, if any. If violations are found, the cuts with respect to the violations are updated

7. If all contingencies in the set have been tested, proceed to the next step; otherwise go to the step 5.

8. If any cuts have been updated in step 6, go to step 1. Otherwise, stop the process.

Figure 6.47, “Sample of progress outputs of PSCOPF” is an example of progress output of solving the master problem to determine the adjustments in one iteration of PSCOPF.

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```
PSCOPF is processing the iteration 1 of inner loop at the iteration 1 of main loop ...


BUSES WITH ACTIVE POWER GENERATION ADJUSTMENTS (MW)
<-------- B U S --------->   GEN-INI   GEN-NEW   GEN-ADJ
   101 COAL-A      138.00      228.0      240.0       12.0
   102 COAL-B      138.00      228.0      240.0       12.0
   107 MIDOIL      138.00      240.0      300.0       60.0
   213 BIGOIL      230.00      346.7      171.9     -174.8
   218 NUKE-A      230.00      400.0      575.0      175.0
   223 BIGCOAL     230.00     1010.0      835.0     -175.0

PHASE SHIFTER ADJUSTMENTS (DEGREE)
<----------------- PHASE SHIFTER ------------------> CKT  PHASE-INI PHASE-NEW PHASE-ADJ
   110 EASTIE-1    138.00    212 EASTIE-2    230.00  1      -24.21    -21.63      2.58
   214 CONDENS     230.00    225 ADDON       230.00  1       22.52     19.95     -2.58

BUSES WITH THE OFF-LINE GENERATORS THAT ARE TURNED ON AND REGULATED BY CORRECTIVE ACTIONS (MW)
<--------- BUS ---------->   GEN-INI   GEN-NEW   GEN-ADJ
   101 COAL-A      138.00        0.0       40.0       40.0
   102 COAL-B      138.00        0.0       40.0       40.0

TAP SETTING ADJUSTMENTS
<----------- TAP CHANGING TRANSFORMERS -----------> CKT   TAP -INI  TAP -NEW  TAP -ADJ
   224 WESTIE-2    230.00    103 WESTIE-1    138.00  1      1.0375    0.9602   -0.0773
   211 MIDTIE-2    230.00    109 MIDTIE-1    138.00  1      1.0469    1.0500    0.0031
   212 EASTIE-2    230.00    109 MIDTIE-1    138.00  1      1.0391    1.0500    0.0109

Processing BASE CASE

Processing contingency "N1OVRLOD 2"
OPEN LINE FROM BUS 215 [MIDCOALP    230.00] TO BUS 224 [WESTIE-2    230.00] CKT 1

BUSES WITH LOW/HIGH VOLTAGE LIMIT VIOLATIONS
<-------- B U S --------->  VOLTAGE  V-MAX  V-MIN
   103 WESTIE-1    138.00  0.9130 1.0500 0.9500

Processing contingency "N1OVRLOD 3"
OPEN LINE FROM BUS 213 [BIGOIL      230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

BUSES WITH LOW/HIGH VOLTAGE LIMIT VIOLATIONS
<-------- B U S --------->  VOLTAGE  V-MAX  V-MIN
   103 WESTIE-1    138.00  0.9428 1.0500 0.9500

PSCOPF is processing the iteration 2 of inner loop at the iteration 1 of main loop ...
```

Figure 6.47. Sample of progress outputs of PSCOPF

6.14.4. Application notes

Invoking the PSCOPF will trigger an alarm and termination of the solution if generators are converted (refer to CONG). The solution then checks to make sure that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any isolated bus is found, an appropriate warning message is displayed and the solution is terminated. If the largest initial active or reactive power mismatch exceeds the specified mismatch tolerance (an input item), the solution is terminated. This tolerance is also used as the bus mismatch convergence tolerance in the power flow solution of each contingency case.

The PSCOPF Solution function responds to the following interrupt control codes:

AB	Abandon the function following completion of the next iteration.
CM	Print the convergence and automatic adjustment monitors.
NM	Suppress any automatic adjustment monitors (only used with CM interrupt control code).

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DC	Tabulate conditions for each dc line after each iteration (see Section 6.3.11, "Operation of Activity SOLV").
FD	Tabulate conditions for each FACTS device after each iteration (see Section 6.3.11, "Operation of Activity SOLV").

When the set of critical contingencies is empty in either one of the two calculation parts, the PSCOPF solution is considered converged and will terminate. If the system condition represented in the working case after a converged PSCOPF solution is then used as the base case for an AC contingency analysis with the identical contingency files (*.sub, *.mon, and *.con) and identical power flow options, there should be no limit violations in the contingency solutions.

PSCOPF may be terminated abnormally under one of following conditions;

- The base case solution does not converge in either one of the calculation parts.
- The Linear Programming solution does not converge.
- The maximum iteration limit is reached.
- Termination initiated by the program user.

Discrete controls of switched shunt and tap setting adjustments are treated as continuous control variables at the beginning of the PSCOPF solution if they are enabled. After the PSCOPF has reasonably converged, the control settings are moved to their respective nearest discrete steps and will remain constant in the remaining iterations of PSCOPF.

PSCOPF solution will update the power flow in the working case with the computed bus voltage magnitudes and angles, and the values of any control variables that have been adjusted, such as generator outputs, loads, phase shifter angels, tap settings, and switched shunt outputs.

Active constraints are limiting constraints which cannot be satisfied in the final solution of PSCOPF. Constraints can be limiting because not enough applicable controls have been specified. For example, if a transmission line along a radial network is overloaded and the loads along the network are not allowed to be adjusted, there is no practical way to eliminate the overload. Another reason is the application of soft penalties to the constraints. Mathematically, because the ineffective controls always involve large adjustments and introduce large penalties to the objective function, the PSCOPF will balance the cost of infeasibility (limit violations) against the costs of the adjustments to determine the optimal solution. Hence, if the limit violations in a power system cannot all be eliminated using the basic control actions, it may be preferable sometimes to accept certain degree of limit violations instead of applying extreme and ineffective control adjustments.

Besides the progress output, PSCOPF also produces a report upon completion of execution. The report summarizes the adjustments made to mitigate the violations in the base case, the list of active constraints, and the adjustments made by PSCOPF process to ensure system security.

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```
 SYSTEM ADJUSTMENTS TO CLEAN THE BASE CASE

------------------------------------------------------------------------------

 BUSES WITH ACTIVE POWER GENERATION ADJUSTMENTS (MW)
 <-------- B U S --------->   GEN-INI   GEN-NEW   GEN-ADJ
    101 COAL-A      138.00      228.0      240.0       12.0
    102 COAL-B      138.00      228.0      240.0       12.0
    107 MIDOIL      138.00      240.0      250.0       10.0
    213 BIGOIL      230.00      346.7      275.8      -70.9

 PHASE SHIFTER ADJUSTMENTS (DEGREE)
 <------------------ PHASE SHIFTER ------------------> CKT  PHASE-INI PHASE-NEW PHASE-ADJ
    110 EASTIE-1    138.00    212 EASTIE-2    230.00  1      -24.21    -13.33      10.88

 TAP SETTING ADJUSTMENTS
 <------------ TAP CHANGING TRANSFORMERS ------------> CKT   TAP -INI  TAP -NEW  TAP -ADJ
    212 EASTIE-2    230.00    109 MIDTIE-1    138.00  1      1.0391    0.9953   -0.0437
    211 MIDTIE-2    230.00    110 EASTIE-1    138.00  1      1.0500    0.9562   -0.0937

 ACTIVE BUS VOLTAGE CONSTRAINT SUMMARY                                  Active constraints
 <-------- B U S --------->  PENALTY     COST CONTINGENCY
    103 WESTIE-1    138.00 0.15E+04 -.10E+06 N1OVRLOD 2

 CONTINGENCY LEGEND:
 <----- CONTINGENCY LABEL ------> EVENTS
 N1OVRLOD 2                       OPEN LINE FROM BUS 215 [MIDCOALP    230.00] TO BUS 224 [WESTIE-2    230.00] CKT 1

------------------------------------------------------------------------------

 BUSES WITH ACTIVE POWER GENERATION ADJUSTMENTS (MW)
 <-------- B U S --------->   GEN-INI   GEN-NEW   GEN-ADJ
    107 MIDOIL      138.00      250.0      300.0       50.0
    216 MIDCOAL     230.00      155.0      125.0      -30.0
    223 BIGCOAL     230.00     1010.0      940.3      -69.7

 BUSES WITH LOAD CURTAILMENS (MW)
 <-------- B U S --------->  LOAD-INI  LOAD-NEW  LOAD-ADJ
    103 WESTIE-1    138.00      225.0      222.4       2.6

 PHASE SHIFTER ADJUSTMENTS (DEGREE)
 <------------------ PHASE SHIFTER ------------------> CKT  PHASE-INI PHASE-NEW PHASE-ADJ
    110 EASTIE-1    138.00    212 EASTIE-2    230.00  1      -13.33     -3.64       9.69
```

6.15. Performing Multi-Level AC Contingency Solution

Run Activity *Multi-Level AC Contingency Solution - GUI*
Power Flow > *Solution* > Multi-Level AC contingency solution... *[Multi-Level AC Contingency Solution]*
Interrupt Control Codes
AB

Multiple level AC contingency analysis provides a comprehensively featured function for testing deterministic reliability as discussed in Section 6.11.2, "Applying Deterministic Criteria". It features easy configuration, built-in contingency ranking function to automatically select contingencies, tripping simulation of post-contingency conditions and corrective action analysis to relieve flow overloads and voltage violations caused by a contingency. The procedure of evaluating a single contingency with multiple level contingency analysis is shown in Figure 6.48, "Outline of Evaluation Procedure Using AC Power Flows for a Single Contingency".

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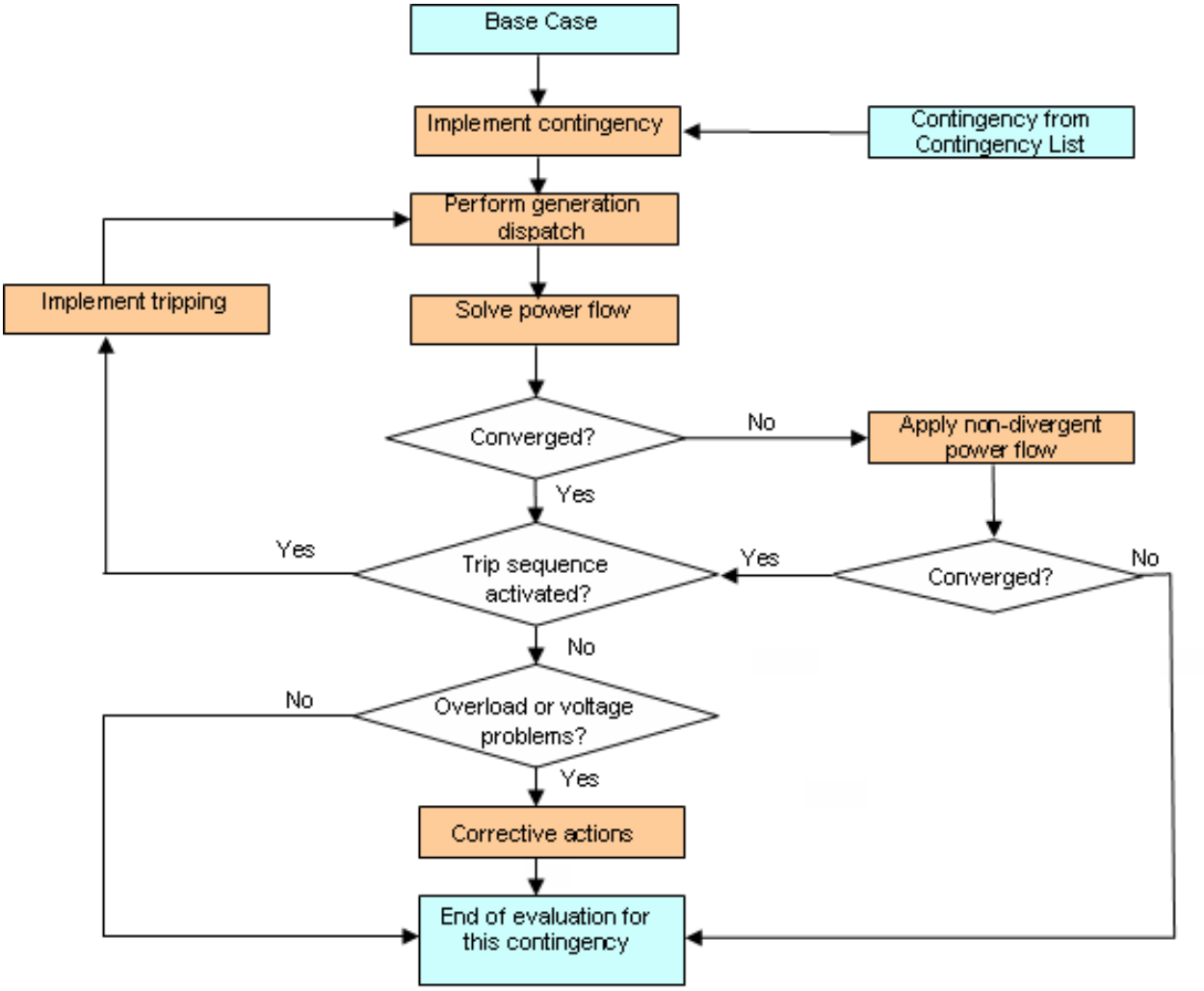

Figure 6.48. Outline of Evaluation Procedure Using AC Power Flows for a Single Contingency

Multiple level AC contingency analysis calculates full AC power flow solutions for the user specified and automatically selected single or multiple contingencies, monitors voltage and loading conditions and stores the results in a binary file. It will accept a user-specified list of contingencies to be included in a contingency study as defined in a Contingency Description Data file (see Contingency Description Data File Contents). Each of these specified contingencies may consist of a combination of generation and transmission outages. In addition, contingencies may be selected by built-in automatic contingency rankers. Two such rankers, for branch overload and voltage collapse contingencies (see Section 9.3.1, "Contingency Ranking Methodology"), are integrated during multiple level contingency analysis. There is full flexibility in defining contingency subsystem; i.e., you can define in which portion of the system you will allow contingencies automatically selected by rankers. A contingency sub-system may comprise of specific equipment, a combination of control areas or zones, or the whole database (see Section 6.12.1, "AC Contingency Solution Options").

The multiple contingency solution function performs deterministic reliability assessment for up to three levels, namely N-1, N-2 and N-3 criterion. User specified and automatically selected contingencies are evaluated individually and in combination with each other as overlapping outages up to the specified evaluation level.

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When used in conjunction with the automatic contingency ranker, a stopping criterion (non-failure cutoff) is applied at each level. This provides the opportunity for enormous savings in computational evaluation of contingencies. Furthermore, PSS®E has built-in logic that identifies if a given set of outages has already been evaluated as a contingency, thus ensuring that an automatic contingency analysis run is comprised only of unique contingencies.

Multiple-level ac contingency analysis models special relay actuation schemes designed to trip or reconnect a generator or a circuit, and shed or transfer load in response to specified low voltage, line flow, interface flow, generator output, or line and generator service status. During contingency analysis, switching will be automatically performed and new power flow solutions obtained whenever trip/reconnect relaying sequences are triggered. The tripping simulation can simulate cascading outages from contingencies.

Multiple-level ac contingency analysis performs corrective action analysis to eliminate flow overloads and voltage violations caused by a contingency with the objective of minimizing control adjustments such as redispatch of generations, curtailment of loads and adjustment of phase-shifting transformers. This can help translate system related reliability measures, such as the location and magnitude of branch overloads and bus voltage violations, to customer-impact indices in terms of the potential amount of service interruptions, which are critical indices in probabilistic reliability assessment.

6.15.1. Terms Used in Multiple-Level Contingency Analysis

Multiple level contingency analysis can perform both simple and complex contingency analysis. Certain terms used in the implementation and documentation are specific to multiple contingency analysis use and may not necessarily reflect those of local electric power analysis practice. Some key definitions of terms applied to multiple contingency analysis are identified as following:

Contingency element: A major component in the network where operation status or set-points can be changed, e.g., one transmission circuit outage, one machine out-of-service, one transmission circuit re-closing, a change in load or generation at a bus.

Contingency event: The contingency of one or more elements originating from a single cause, e.g., each contingency case defined with a block structure in a contingency description file is considered as one contingency event.

Contingency: A contingency condition simulated by PSS®E made up of one or more contingency events.

Level: The number of contingency events within a contingency studied by PSS®E. Level may also be viewed as the number of independent causes studied in a contingency; e.g., N-1, N-2 and N-3.

Contingency events include:

- Single contingency elements, e.g., one transmission circuit outage, and
- Multiple contingency elements, e.g., a combination of circuits and, or machines out-of-service at the same time.

Examples of the latter include outages of multiple circuits terminating at the same bus caused by bus failures and outages of two circuits on the same right-of-way caused by ROW failure.

When multiple events are combined within a contingency, they are assumed to be occurring simultaneously but originating from independent causes, i.e., one event does not trigger the others. Multiple level contingency analysis is deigned to model up to three events per contingency.

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6.15.2. Power Flow Control Options

Table 6.17. Multi-Level AC Contingency Solution Power Flow Control Options

Option	Description
Solution options	The desired solution options to be applied during power flow calculations for obtaining post-contingency states (see Section 6.2, "Power Flow Solution Methods") Tap adjustment: • Lock taps (default) • Stepping • Direct Area interchange control: • Disabled (default) • Tie lines only • Tie lines and loads Switched shunt adjustments: • Lock all • Enable all (default) • Enable continuous, disable discrete
Non-divergent solution	Flag to enable or disable (default) option in the power flow solution of each contingency case.
Adjust phase shift	Flag to enable or disable (default) option in the power flow solution of each contingency case.
Adjust dc taps	Flag to enable (default) or disable option in the power flow solution of each contingency case.
Solution Engine	Select the desired power flow solution engine to be used in obtaining post-contingency states (see Section 6.2, "Power Flow Solution Methods"). • Fixed slope decoupled Newton-Raphson (FDNS) (default) • Full Newton-Raphson (FNSL) • Optimized fixed slope decoupled Newton-Raphson (FDNS)

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Option	Description
Dispatch mode	Dispatch codes for generation dispatch calculations in contingency analysis (see Table 6.10, "AC Contingency Single Run Report Options")
Dispatch system	Select the subsystem for the generation dispatch.
Mismatch tolerance	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the process is terminated. This value is also used as the convergence tolerance in power flow solution (see Section 6.2, "Power Flow Solution Methods").
Distribution Factor Data File	(*.dfx) a required file to be build by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files (refer to ???). The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified Distribution Factor Data file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by the AC Contingency Solution function.
Contingency Solution Output File	(*.acc) a required file designated by the user as the destination for the results of the contingency calculations.
Load Throw-over Data File	(*.thr) an optional file, created by the user, that contains data records of the following form: `IBUS, JBUS` where IBUS and JBUS are bus numbers. If branch outage contingency events isolate bus IBUS, the user can elect to move the load to bus JBUS. Bus IBUS and/or JBUS may be a dummy bus of a multi section line grouping. Data records may be entered in any order. Input is terminated with a record specifying an IBUS value of zero.
Unit Inertia and Governor Data File	(*.inl) a required file if the generation dispatch is enabled and either the inertial or governor droop dispatch method is selected. If the PMAX or reserve method is selected, the specification of this file is optional. If a file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used.

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6.15.3. Multiple Contingency Analysis Options

Table 6.18. Multi-Level AC Contingency Solution Multiple Contingency Analysis Options

Option	Description
Multiple contingency analysis	When disabled, all other options in this dialog are disabled. When enabled, the next-level contingencies (e.g. secondary contingencies of a primary contingency) of each non-failure contingency (see Section 6.15.4, "Classification of Contingency Analysis Results") are to be evaluated. When enabled with failure cutoff disabled, the next-level contingencies of each failure and non-failure contingency will be evaluated.
Max level of contingency analysis	Level of multiple contingency analysis, up to third level. This option enables the contingency combination selections at less or equal levels.
First level of analysis	Contingency groups included in the primary contingency list (see Section 6.15.5, "About Contingency List"). There are three groups of contingencies to choose, namely S, U and B. S represents user specified outages, U and B are machine outages and branch outages respectively. When the ranked contingencies are included in the primary list, the ranking solution is performed on the base power flow case using a linearized network model. The number of ranked contingencies of each group included in the list is determined by the min, max, cutoff values of the ranker.
Second level of analysis	Contingency groups included in the secondary contingency list for each primary contingency. Nine N-2 contingency combinations are represented with a matrix. Each row represents one type of first-level contingencies from which the secondary contingency list is generated based on existing system condition after solving. The name of contingency type is dictated by the letter at the beginning of each row; three elements of a row represent the same categories (S, U, B) in secondary contingency list. When an element is selected, the corresponding N-2 contingency will be evaluated. If one group of contingencies is not specified in primary contingency list, its secondary list is empty and the corresponding row will be disabled.
Third level of analysis	Contingency groups included in the tertiary contingency list for each first-level and second-level con-

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Option	Description
	tingency. There are 27 N-3 contingency combinations for the tertiary contingency; each combination is comprised of one of each of three contingency event types for each level. These combinations are represented with three matrices; each of them has nine elements. Each matrix represents one group of first-level contingency cases and is dictated by the letter at the first column; each row of a matrix represents one type of second-level contingencies and is indicated by the letter at the second column. Three elements of a row of a matrix represent categories S, U and B in tertiary contingency list. The tertiary contingency list is built based on existing system condition after solving first and second level contingencies. Similarly, tertiary contingency list may exist if and only if its corresponding first and second level contingencies have been tested. That is if one group of contingency is not included in primary list, all following lists are empty and the corresponding matrix is disabled; if one group of contingency is not tested at level 2, the corresponding row is disabled and the tertiary list is empty.
Ranking options	Select contingency rankers and specify the number of each group of contingencies to be included in the list (for contingency ranking, see Section 9.3.1, "Contingency Ranking Methodology").
Contingency case rating and percent of rating	These define the loading limits to be used in calculating overload performance indices. By default, contingency case rating is rating set program option setting and percent of rating is 100 percent. They are also used to determine flow overloads in post-contingency solutions and classify the post-contingency solutions into either failure or non-failure groups, and if Perform corrective actions option is enabled, then corrective actions are specified to remove the flow overloads.
Convert to MW rating	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's reactive power loading at the metered end in the base case ac solution.
Contingency subsystem	Subsystem where contingencies will be automatically selected by rankers.
Include subsystem ties	If enabled, include in the contingency list only branches from within the designated subsystem. If disabled, both subsystem branches and ties from the subsystem to other buses in the working case are included. The group of branches that may be

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Option	Description
	outaged is defined according to subsystem selection options.
Small reactance threshold	Reactance threshold; all branches from the contingency list (based on the user's subsystem selection) where magnitude of reactance is less than or equal to this threshold have their performance indices explicitly calculated.
Line overload	Four control parameters. • *Enable:* Include single branch outage that causes overloading in the contingency list. • *Min.:* The minimum of single branch outages from the overload ranking to be tested in the list. • *Max.:* The maximum of single branch outages from the overload ranking to be tested in the list. • *Cutoff:* The number of consecutive single branch outages from the overload ranking that are non-failure
Machine overload	Four control parameters. • *Enable:* Include single machine outages from branch overload ranking in the contingency list. • *Min.:* The minimum of single machine outages from the overload ranking to be tested in the list. • *Max.:* The maximum of single machine outages from the overload ranking to be tested in the list. • *Cutoff:* The number of consecutive single machine outages from the overload ranking that are non-failure.
Voltage depression	Four control parameters. • *Enable:* Include single branch outage that causes voltage collapse in the contingency list. • *Min.:* The minimum of single branch outages from the voltage collapse ranking to be tested in the list. • *Max.:* The maximum of single branch outages from the voltage collapse ranking to be tested in the list.

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Option	Description
	• *Cutoff:* The number of consecutive single branch outages from the voltage collapse ranking that are non-failure.
Islanding ranking	Two control parameters. • *Enable:* Include single branch contingencies that result in the formation of an island of one or more buses, without a swing bus, in the contingency list. • *MW threshold:* The minimum of active power flow on such branches.

6.15.4. Classification of Contingency Analysis Results

There are two cutoffs to control the level of multiple contingency analysis and number of contingencies to be evaluated at each level during analysis: failure cutoff and non-failure cutoff.

The failure cutoff is enabled by selecting *Enable* for multiple contingency analysis option, the next-level contingencies of a failure contingency that results in operating limit violations will be skipped. It is most likely that under a failure contingency condition any next-level contingencies will result in operating limit violations too, so there is no need to apply more contingencies to failure cases. Enabling failure cutoff can reduce the set of tested contingencies and concentrate studies on more probable contingencies. To evaluate complete multiple event contingencies for the specified level, the failure cutoff must be disabled by selecting *Enable with failure cutoff disabled* for **`Perform multiple contingency analysis`** option. After completion of a contingency analysis, only non-failure cases could be used as base cases for next-level contingency analysis. The failure cutoff is applied to both user specified contingencies and ranked contingencies. For example, multiple level contingency analysis is initiated with level 2 and failure cutoff enabled, user specified contingencies for N-1 and combinations SS and SB for N-2 (SS consists of two different user specified contingencies in the user specified contingency list, and SB consists of a user specified contingency and a ranked single branch outage), if all user specified contingencies are failure cases based on the failure criteria in Table 6.20, "Types of Problems Qualifying a Category I Contingency as a Failure", namely all N-1 cases are failure cases, all N-2 contingencies are skipped.

The non-failure cutoff is specified by cutoff numbers within each ranker. If the number of consecutive non-failure contingencies from the ranker is more than the cutoff, it is assumed that the rest contingencies from the same ranker will not lead to violations and be skipped. This section discusses internal the failure criteria to classify a contingency solution.

At the completion of a contingency evaluation, the contingency may be classified according to solution characteristics of the last power flow solution performed on the contingency. This classification is shown in Table 6.19, "Classification of Contingency Evaluation Based on Power Flow Solution".

Table 6.19. Classification of Contingency Evaluation Based on Power Flow Solution

Category I	Converged and maximum bus mismatch within tolerance
Category II	Voltage Collapse solution stopped by non-divergent power flow

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Category III	Not Converged

Category I contingencies are those for which power flow solutions have a maximum bus mismatch less than the mismatch tolerance. Category I contingencies are further classified into: failures and non-failures. A Category I contingency is classified as a failure if the contingency results in problems listed in Table 6.20, "Types of Problems Qualifying a Category I Contingency as a Failure"or if specific events occur during the contingency. The classifications are applied only to internal programs. The failure status of the contingency is used only as stopping criteria for failure and non-failure cutoff algorithms. You can make your failure criteria more stringent in order to obtain a reduced set of tested contingencies; however only monitored flows and voltages specified in monitored element data file of tested contingencies are saved in results files and can be post-processed to generate the different reports. In order to test more contingencies, you have to repeat contingency analysis with new selections of failure criteria (see Section 6.15.6, "Wind Chime Algorithm for Multiple Level Contingency Analysis"). The failure status of a contingency is not stored in result files.

Table 6.20. Types of Problems Qualifying a Category I Contingency as a Failure

Problem	Description	Default Criteria
OVERLOAD	Circuit or interface loading above a specified multiplier of a specified rating (A, B or C).	Specified rating multiplied by a percent factor from GUI, by default, 100 percent of rating set program option setting is used.
HIGH VOLTAGE	Bus voltage above the high voltage limit.	Upper limit defined in monitored element file.
LOW VOLTAGE	Bus voltage below the low voltage limit.	Lower limit defined in monitored element file.
VOLTAGE DEVIATION	Bus voltage change (absolute value) between pre- and post-contingency conditions exceeds the Deviation criterion	Variation defined in monitored element file

The detected overloaded branches are different with those in post-processed report files where the different ratings of loading may be applied.

An islanding event occurs when a contingency causes a physical separation of a bus or group of buses from the base case network. The outage of a radial branch will automatically result in islanding of a radial bus. If the contingency is a multiple element outage, it may cause multiple islands. When dispatch mode and dispatch subsystem are specified, contingency analysis will process islanding events via generation redispatch (see Section 6.11.4, "About Generation Dispatch"), otherwise those islands without a swing bus will be shut down. The islanding problem has been removed from the failure criteria since PSS®E-30.3.

The types of events (as differentiated from troubles) occurring during contingency evaluation that would qualify a contingency as a failure, include:

- TRIPPING – A contingency where at least one trip specification is actuated (see Section 6.15.7, "About Tripping Sequence").

Power flow cases with solutions that are classified as either voltage collapse (Category II) or not converged (Category III) are excluded from tests of failure.

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6.15.5. About Contingency List

Multiple contingency analysis functions work with a contingency event list, also referred to as a contingency list. The list provides a sequence for evaluating contingencies. However, the true power of this list lies in the fact that not all contingencies included in it are evaluated by multiple contingency analyses.

It is useful to think of the contingency list as further divided into contingency groups. Contingency groups are shown in Table 6.21, "Groups within Contingency List". The graphic equivalent of this table is shown in Figure 6.49, "Contingency List".

Table 6.21. Groups within Contingency List

Group	Contents
S	Specified contingencies.
U	Outages of single machines, ranked according to impact on thermal loading of branches.
B1	Outages of single branches that result in separation (islanding) of the network.
B2	Outage of single branches ranked according to impact on thermal loading of branches.
B3	Outage of single branches ranked according to impact on voltage collapse.

In group S, the contingencies are taken directly from a dfax file that is generated with a Contingency Description Data file (see Contingency Description Data File Contents). These can be viewed as must-test contingencies; i.e., no matter how system conditions change, it is important to the reliability assessment to consider these contingencies. The Contingency Description Data file may be:

- Created by the user based on previous operating or planning experience with a system, reliability criteria guidelines or some other arbitrary method. The file is thus one that is created outside of PSS® E, with a text editor.
- Produced by pre-screening contingencies using activity RANK.

Groups U, B1, B2 and B3 are comprised of ranked contingencies from the contingency subsystem, each group is based on different methods of ranking.

Things to note about the contingency list and its groups:

- Within each ranked group, there is internal ordering from most severe to least severe contingencies.
- Contingencies are evaluated according to the sequence of groups.
- If a contingency has been evaluated in an earlier group, the same contingency in a later group will be skipped.

Each contingency evaluation constitutes a Power Flow solution resulting in a post-contingency system state. Conditions in the post-contingency state are then checked against the specified failure criteria.

Ranked contingencies are tested one by one in sequence until an unbroken series of N non-failure contingencies. The stopping criterion (non-failure cutoff) of *N contingencies in a row* is applied here. The evaluation

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of a ranked contingency group is complete when all contingencies are exhausted or the stopping criterion is met.

Groups B1, B2 and B3 may be comprised of the same contingencies, albeit with different ranking. Some of these contingencies may also be included in group S.

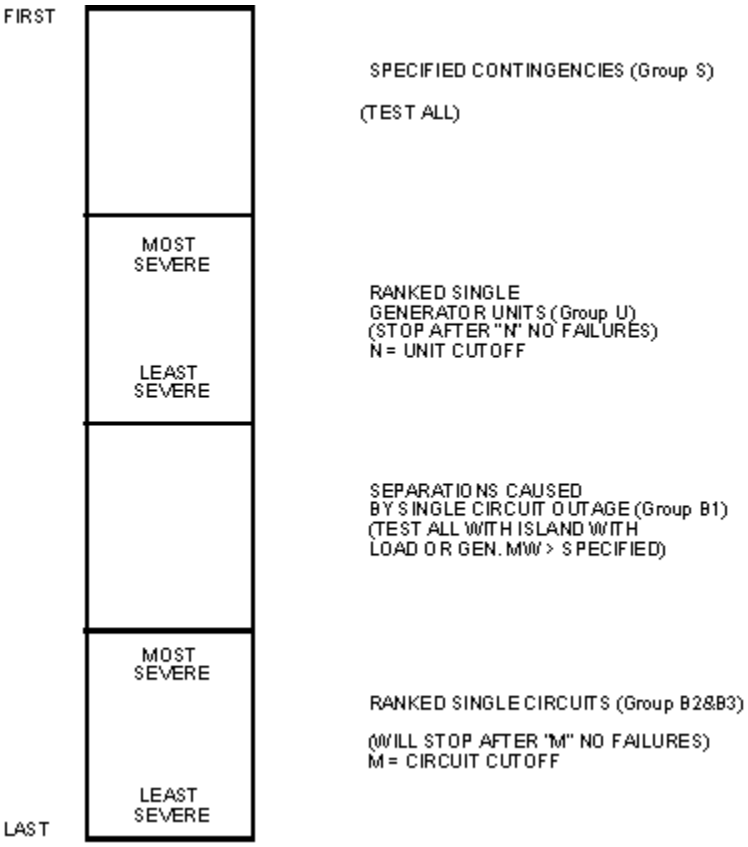

Figure 6.49. Contingency List

6.15.6. Wind Chime Algorithm for Multiple Level Contingency Analysis

If a one-level contingency analysis is specified, all must-test events (group S) are evaluated first. Ranked individual machine outages (group U) follow the group S. These ranked machine outages will be tested one by one until an unbroken sequence of N contingencies without any system problems has been detected or the maximum of contingencies of group U has been tested. When the testing of machine outages is complete, either because the list has been exhausted, or because the stopping criterion has been met, the branch outage events causing system separations (group B1) are evaluated. A simple screening algorithm that searches for continuity can also automatically select these events. The final two groups of events in the list are the ranked branch outages from overload (group B2) and voltage collapse ranking (group B3). These are tested in a similar fashion as the ranked machine outages, in which the individual branch outages are tested one by one and the list is truncated after the corresponding stopping criterion is met. Groups B1, B2 and B3 may be comprised of the same contingencies, albeit with different ranking. Some of these contingencies may also be included in group S.

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The multiple contingency solution procedure is given by the example of a two-level contingency analysis. If a two-level contingency analysis is to be performed, a newly ordered event list is built after solving each first-level contingency Power Flow. These secondary ordered event lists may contain the same categories of outage events as those in the first-level list, with the exclusion of the initial outage event. New rankings are determined based on the system conditions existing after each first-level contingency using a linearized network model. This approach is called Wind Chime approach.

Figure 6.50, "Wind Chime Approach for 2 Level" illustrates the Wind Chime approach for a two-level contingency run using a contingency level of 2; some other options specified for this run are *Enabled with failure cutoff disabled* for **`Perform multiple contingency analysis`** option, B for the first level and BB for the second level analysis, as well as only single branch outages from overload ranking with non-failure cutoff of 2.

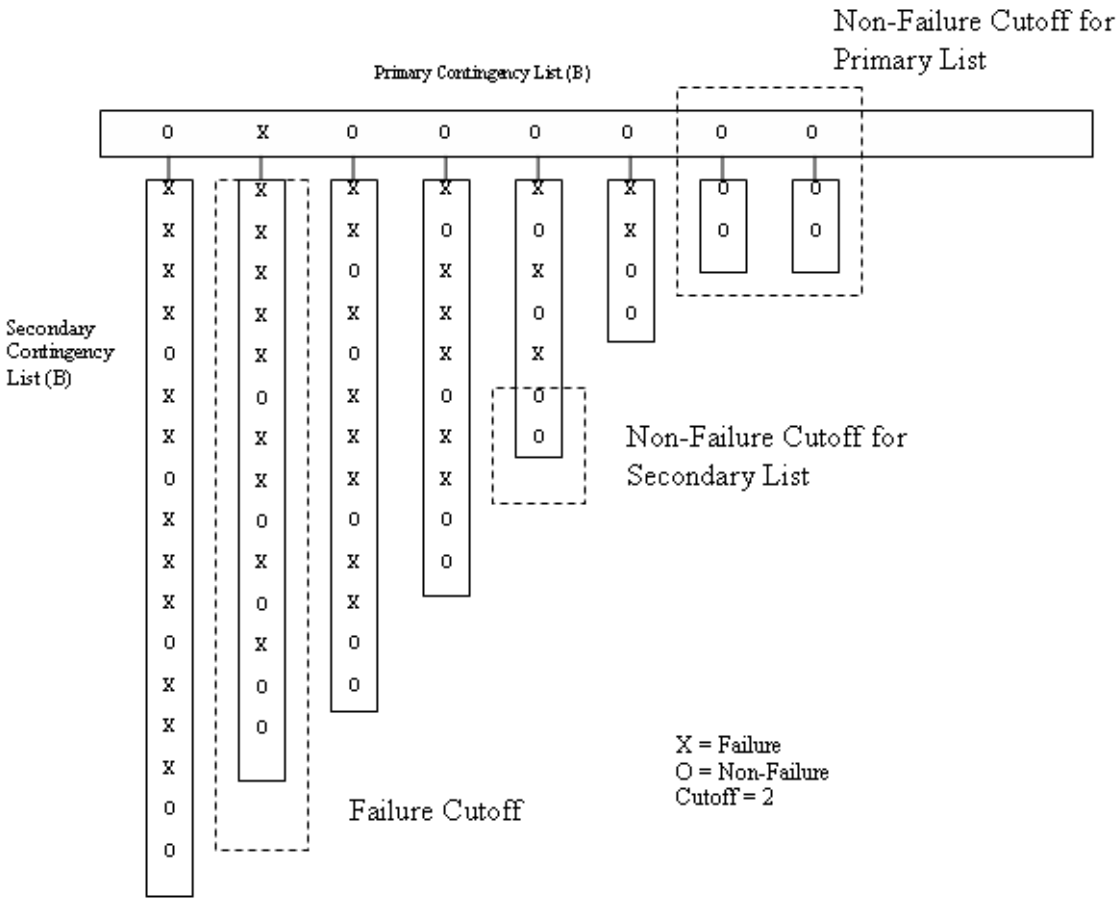

Figure 6.50. Wind Chime Approach for 2 Level

In Figure 6.50, "Wind Chime Approach for 2 Level", the initial first-level contingency is a non-failure. That is, it does not have any criteria violations in Table 6.20, "Types of Problems Qualifying a Category I Contingency as a Failure" and trigger any failure events. A number of second-level contingencies are then run until two consecutive second-level contingencies are non-failures. The next first-level contingency is evaluated and is classified as a failure case; because failure cutoff is disabled, it is followed by second-level contingencies

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until two consecutive second-level contingencies are non-failures. This procedure continues until the top two second-level contingencies are non-failures for two consecutive first level contingencies.

This process can be duplicated for any level of automatic contingency analysis. If third-level contingencies are requested, a ranking is done for each second-level contingency and the tertiary contingencies are run until the maximum is reached or the cutoff criterion is satisfied.

The contingency evaluation functions contain logic to efficiently check whether a contingency consisting of a combination of branches and machines has already been tested. In this way, any particular branch or machine outage combination, whether it originates from one multiple element event or from two or more single element events, will be tested only once. Also, elements that are ranked higher in the primary event list than the present primary contingency will not be included in the present and subsequent secondary ranking lists. This assumes that all two level events associated with those primary events would have been evaluated, either explicitly by power flow solutions or implicitly by ranking list truncation. The ranking algorithm also takes into account the outages of identical machines located at the same bus. Because the impact of these outages on the system will be identical, it is necessary to test only one of such machine outages per contingency level.

The program can still test multiple levels of machines of the same size.

6.15.7. About Tripping Sequence

A contingency analysis may be followed by tripping simulations during multiple level contingency analysis. Tripping sequences are simulations of events ensuing from automatic monitoring equipment such as relays, and automatic circuit breaking and making equipment such as circuit breakers and switches. These sequences are important in order for multiple contingency analysis to obtain the right post-contingency state. Examples of tripping sequences are:

- Transfer of load to another bus on loss of power on a transmission branch.
- Switching of lines following outage or overload of another.
- Tripping of transmission circuits following extreme overloads.

Tripping sequences may also be used to model other automatic actions such as remedial action schemes (RAS), special protection schemes (SPS) and operating guides.

6.15.8. Tripping Simulation Options

Table 6.22. Multi-Level Contingency Solution Tripping Simulation Options

Option	Description
Perform tripping simulation	Enable or disable (default) tripping simulation for post-contingencies states.
Number of trip events per load flow	After a contingency power flow solution, tripping conditions that trigger tripping sequences are checked in the order in which tripping events are specified in Tripping Element Data file (Section 6.15.9, “Tripping Element Data File Contents”). There may be more than one instance of tripping sequences that are activated. This option sets the

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Option	Description
	number of trip events that are activated prior to the next power flow solution.
Number of load flows per contingency	When a new power flow solution is performed following activation of one or more tripping sequences, the trip triggers are reset and all the remaining trip sequences are checked for the new power flow solution. If one or more remaining tipping sequences satisfy conditions, a new power flow is performed with activation of these tripping sequences up to *Number of trip events per load flow*. This process continues till the number of power flows following a contingency power flow reaches the number of power flows per contingency.
Tripping Element Data File	If multiple level contingency analysis is initiated with tripping simulation, this file, created by the user, is required. It contains tripping data.

6.15.9. Tripping Element Data File Contents

The tripping events are defined via the tripping data file. Each specification block structure is comprised of a trip descriptor and one of each of two categories of data records - monitored equipment, trip equipment. The former includes circuit/transformer/interface flows, generator output, bus voltage, the operating status of monitored elements. The latter includes circuit/transformer/unit outages, load shedding, bus outage, and circuit re-closing. The tripping event is defined in a block structure as follows:

TRIPPING label

```
(monitored element)
(trip element)
End
```

The 12-character contingency label is used to identify each tripping event.

Monitored Equipment

The monitored elements include buses, lines, transformers, interfaces and machines. In a block structure of tripping condition definition, there may be several individual conditions up to 10; each condition has its monitored element specification and tripping rating. If the value of logical OR of these conditions is TRUE, tripping actions defined within this block are triggered.

Voltage Monitoring Records

The following data record defines a voltage band with a single bus where voltage is to be checked against the band:

```
[MONITOR]  VOLTAGE RANGE   BUS bsid  TR   r [r]
```

where the first `r` value is the lower bound of the per unit voltage band and the optional second `r` value is the upper bound. If the upper bound is omitted, the upper end of the band is not checked. Token `TR` is key word for trip rating.

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Flow Monitoring Records

The data record format defines a non-transformer branch or two-winding transformer for monitoring. The r value is the tripping rating in specified unit with one of token MW/MVA/ MVAR/AMPS/PERCENT. The monitored flow direction is dictated by FROM bus to TO bus specifications. If the trip rating is a positive number, trip will occur if the monitored flow exceeds this number, if it is a negative number, trip will occur if the monitored flow is below the absolute value of this number. A trip rating of 0 means that tripping should take place if the monitored element is out of service.

```
[MONITOR]|BRANCH| FROM BUS bsid TO BUS bsid [|CIRCUIT| ckid]  TR r |MW     |
         |LINE  |                             |CKT    |             |MVA    |

   |MVAR   |

   |AMPS   |

   |PERCENT| OF |A|

                |B|

                |C|
```

To specify one winding of a three-winding transformer for monitoring, the bus to which the winding is connected must be the first bus specified in the following data record:

```
[MONITOR]|BRANCH| FROM BUS bsid TO BUS bsid TO BUS
bsid [|CIRCUIT | ckid] TR r|MW     |
         |LINE  |                                      |CKT
  |           |MVA    |

              |MVAR   |

              |AMPS   |

              |PERCENT| OF |A|

                           |B|

                           |C|
```

When flow monitoring records employ keyword PERCENT, one of rating sets A/B/C must be specified. Percent loadings of monitored elements based on ratings are specified as either Current or MVA percent loadings by percent units program settings (see activity OPTN).

Unit Monitoring Record

The data record defines a single machine for monitoring:

```
[MONITOR]|UNIT   | mcid AT BUS bsid TR r |MW
|
         |MACHINE|                       |MVA|
                                         |MVAR   |
                                         |AMPS   |
```


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Interface Monitoring Record

```
[MONITOR] INTERFACE label TR r |MW/MVA/ MVAR/AMPS|
(branch specification record, see below)
.
.
(branch specification record, see below)
END
```

where r is trip rating for the monitored interface, if the sum of MW/MVA/ MVAR/AMPS flows at the from bus end of specified branches is greater than r, trip actions occur. A rating of zero will disable the monitored interface flow condition. The branch specification record may specify individual two-winding or non-transformer branches or individual three-winding transformers.

For specifying a two-winding or non-transformer branch the following data specification record is used:

```
|BRANCH| FROM BUS bsid TO BUS bsid [|CIRCUIT|
ckid]
|LINE  |                            |CKT    |
```

For specifying a three-winding transformer the following data specification record is used:

```
|BRANCH| FROM BUS bsid TO BUS bsid TO BUS bsid
[|CIRCUIT| ckid]
|LINE  |                                      |CKT    |
```

Join block structure

A join block structure specifies the monitoring condition through the logical ANDING of two or more of the monitoring records described above. A join group has the following block structure:

```
JOIN
(monitoring record)
.
.
(monitoring record)
END
```

where each monitoring specification data record is one of the simple record types described above (flow monitoring record, unit monitoring record, interface monitoring record, etc) . The logical value of a join block is .TURE. if all monitoring records within the block are .TURE. A tripping block can have one or more join blocks; the join blocks and simple monitoring records within a tripping block are used to determine the value of the block's tripping condition via logical OR.

Trip Equipment

Up to 32 tripping actions may be specified in a tripping event definition block structure. If any one of tripping conditions is satisfied, all given tripping actions might be enforced to post-contingency cases. The tripping actions are defined using following data records.

The outaging of an in-service non-transformer branch or two-winding transformer is specified with the following record:

```
|DISCONNECT||BRANCH| FROM BUS bsid TO BUS bsid
```


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```
[|CIRCUIT| ckid]
|OPEN      ||LINE  |                              |CKT    |
|TRIP      |
```

An out-of-service non-transformer branch or two-winding transformer may be placed in-service with a record of the form:

```
CLOSE |BRANCH| FROM BUS bsid TO BUS bsid  [|CIRCUIT|
ckid]
      |LINE  |                               |CKT    |
```

The outaging of a three-winding transformer is specified with the following record:

```
|DISCONNECT||BRANCH| FROM BUS bsid TO BUS bsid TO
BUS bsid [|CIRCUIT| ] ckid
|OPEN      ||LINE  |                                        |CKT
   |
|TRIP      |
```

An out-of-service three-winding transformer may be placed in-service with a record of the form:

```
CLOSE |BRANCH| FROM BUS bsid TO BUS bsid TO BUS
bsid  [|CIRCUIT| ckid]
      |LINE  |                                          |CKT    |
```

The next four record types allow the user to specify contingency events in which the load and generation boundary conditions may be changed at a selected bus. When changing generation, the bus must have in-service generation connected to it and it may not be a swing bus.

The first data record of this type uses the following data record to set the load or generation at a bus either to a designated value or to a specified percentage of its initial value:

```
SET BUS bsid |GENERATION| TO r |MW     |
             |LOAD      |      |PERCENT|
             |SHUNT     |
```

The number r specified must not be a negative number when the PERCENT keyword is used.

The second data record of this type uses the following data record to change the load or generation at a bus either by a designated amount or by a specified percentage of its initial value:

```
|CHANGE| BUS bsid |GENERATION| BY r |MW     |
|ALTER |          |LOAD      |       |PERCENT|
|MODIFY|          |SHUNT     |
```

The third and fourth data records of this type are similar to the CHANGE record, except the direction of the change is defined by the first keyword, and r must be a positive number:

```
|INCREASE| BUS bsid |GENERATION| BY r |MW
 |
|RAISE   |          |LOAD      |      |PERCENT|
|SHUNT   |

|DECREASE| BUS bsid |GENERATION| BY r |MW     |
|REDUCE  |          |LOAD      |      |PERCENT|
```


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```
|SHUNT    |
```

To transfer load or generation from one bus to another, the following data record is used:

```
MOVE r |MW      | |GENERATION| FROM BUS bsid TO
BUS bsid
       |PERCENT| |LOAD      |
                 |SHUNT     |
```

See Contingency Description Data File Contents for details.

An in-service machine may be removed from service using the following data record:

```
REMOVE |MACHINE| mcid FROM BUS bsid
       |UNIT   |
```

Similarly, an out-of-service machine may be placed in-service with a record of the form:

```
ADD |MACHINE| mcid TO BUS bsid
    |UNIT   |
```

Automatic Single Tripping Record

A series of single branch tripping events is specified with a record of the form:

```
SINGLE LINE IN | AREA i          | TR r PERCENT
OF |A|
               | ZONE i          |                 |B|
               | OWNER i         |                 |C|
               | KV r            |
               | SYSTEM label    |
               | SYBSSYTEM label|
```

In each tripping event, a single branch within a given subsystem is monitored. For non-transformer branches and two-winding transformers, a branch is tripped when flow at either one of its two bus ends exceeds the tripping rating. For three-winding transformers, a three-winding transformer is outaged when the flow at any one of its three winding buses exceeds the tripping rating.

A breaker is a branch where the circuit ID's first character is @, a switch has a * as the first character of its circuit ID. A series of single breaker and switch tripping events is specified with a record of the form:

```
SINGLE BREAKER IN | AREA i          | TR r PERCENT
OF |A|
                  | ZONE i          |                 |B|
                  | OWNER i         |                 |C|
                  | KV r            |
                  | SYSTEM label    |
                  | SYBSSYTEM label|
```

6.15.10. Corrective Actions Options

When multiple level contingency analysis is initiated with corrective action analysis, if there are operating limit violations in post-contingency solutions, or in post-tripping contingency solutions if tripping simulation is enabled and tripping events are activated following a contingency, corrective actions will be specified to relieve these violations.

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Table 6.23. Multi-Level Contingency Solution Corrective Actions Options

Option	Description
Perform Corrective Actions	Enable (default) to perform corrective action analysis; disable to disable all corrective actions options.
Number of AC load flows	The desired number of iterations between execution of the linear programming engine to identify corrective actions and subsequent ac power flow solution to verify the feasibility of those actions. The specified number of iterations can be between 1 and 10. Note that, even if a feasible solution appears to have been found, i.e., the identified corrective actions appear to have been successful, the iteration between linear programming computation and AC power flow solution may continue in an attempt to further optimize the solution if not exceeding the iteration limit.
Branch overload violation tolerance	Specifies branch overload violation tolerance in percent to adjust branch overload limits used in corrective actions. The limits are increased by the tolerance percent of the specified rating when the tolerance is positive, or reduced when it is negative. For instance, if the tolerance is 0.1% and the overload limit is 100% of rate A, the 100.1% of rating A is used in determining overload violations in corrective actions. In the above case if the tolerance is -0.1%, the 99.9% of rating A is used.
Bus voltage violation tolerance	Specifies bus voltage violation tolerance in pu to adjust the bus voltage limits used in corrective actions. The limits are expanded by the tolerance when it is positive, narrowed when it is negative. For instance, if the tolerance is 0.1 pu and lower and upper limits on a bus voltage are 0.95 and 1.05 in pu respectively. The adjusted voltage limits are 0.949 and 1.051 respectively. In the above case if tolerance is -0.1, the lower and upper limits are 0.951 and 1.049.
Control options	Types of control adjustments and weighting factors for each type
Enable generator controls	Enable (default) to include active power generation dispatch controls. The adjustable range is from PT to PB of each participating machine. Generator active power limits are taken from a Unit Inertia and Governor Data file (see Section 6.8.1, “Inertia and Governor Response Data File Contents”) if the file is specified, otherwise the values in base case are used.
	In corrective action analysis, generator active power limits are adjusted by the following rules.

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Option	Description
	• If PT had been set to the default value, the larger value of PG (its existing MW output) and machine power base (MVA base) will be used as PT. • If PT of a machine is less than its PG, the PT is set equal to the PG. • If PB of a machine had been set to the default value, it is set equal to 0. • If PB of a machine is less than its PG, the PB is set equal to the PG. • A machine will be excluded from the controls if its PT is equal to its PB.
Weighting factor of active power generation dispatch	Specify a weighting factor to adjust the cost factor of active power generation dispatch. The cost of a control adjustment in objective functions is equal to the amount of adjustment times by a adjusted cost factor. The internal cost factor for active power generation dispatch defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls"; the adjusted cost factor is equal to the internal factor multiplied by the user defined weighting factor. e.g., if the weighting factor is set as 1.0, the cost factor for active power generation dispatch of 100 MW will be 1.0.
Generator control subsystem	Select the subsystem in which all in-service generators with valid range of adjustments will participate in corrective action analysis.
Enable load controls	Enable to include load shedding controls (default = disabled). The loads with negative MW values will be excluded from the controls. While performing load curtailment, the algorithm maintains a constant power factor at the buses where load curtailment occurs.
Weighting factor of load curtailment	Specify a weighting factor to adjust the cost factor of load curtailment. The internal cost factor for load curtailments defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls"; the adjusted cost factor is equal to internal factor multiplied by the weighting factor. e.g., if the weighting factor of load curtailments is set as 1.0, the cost factor for load shedding of 100 MW will be 100.0.
Load control subsystem	Select the subsystem in which all in-service loads will participate in corrective action analysis. Loads with negative active power will not be included as candidates for load curtailments.

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Option	Description
Enable phase shifter controls	Enable (default) to include phase shifter controls. To be adjustable as a phase shifter, transformers must be operated subject to MW control mode. The adjustable range of a participating phase shifter is from Rmax to Rmin.
Weighting factor of phase shifter angle adjustment	Specify a weighting factor to adjust the cost factor of phase shifter angle adjustments. The internal cost factor for phase shifter angle adjustments defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls"; the adjusted cost factor is equal to internal factor multiplied by the user-defined weighting factor. e.g., if weighting factor of phase shifter angle adjustment is 1.0, the cost factor of phase shifter angle adjustment of 100 degrees is 3.5.
Phase shifter Control subsystem	Select the subsystem in which all in-service phase shifters will participate in corrective actions.
Enable off-line generator controls	Enable to include off-line generator active power shift. The initial active power outputs of participating off-line generators are set to zeros. Their active power ranges are specified in the same way as on-line generators
Weighting factor of off-line generator	Specify a weighting factor to adjust the cost factor of active power generation dispatch of off-line generators. The internal cost factor defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls".
Off-line generator control subsystem	Select the subsystem in which all out-of-service generators with valid range of adjustments will participate in corrective action analysis.
Enable tap setting adjustments	Enable to include tap setting adjustment.
Weighting factor of tap setting	Specify a weighting factor to adjust the cost factor of tap setting adjustment. The internal cost factor defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls".
Tap setting adjustment subsystem	Select the subsystem in which all in-service transformers designated in voltage control mode will participate in corrective action analysis.
Enable switched shunt controls	Enable to include switched shunt reactive power controls.
Weighting factor of switched shunt	Specify a weighting factor to adjust the cost factor of switched shunt reactive power control. The internal cost factor defined in the programs is shown in Table 6.15, "Weighting Functions and Factors of Controls".

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Option	Description
Switched shunt control subsystem	Select the subsystem in which all in-service switched shunts designated in voltage control mode will participate in corrective actions.

6.15.11. Operation of Multiple Contingency Analysis Activity

Table 6.24. Running Modes for Multiple Contingency Analysis

Disabled	Enabled	Action
Multiple Contingency Analysis Tripping Simulation Corrective Actions [a]	Power Flow Control *Options and required filenames have been specified.*	Only user specified contingencies defined in contingency description file are tested.
Multiple Contingency Analysis Corrective Actions*	Power Flow Control Tripping Simulation *Options and required filenames have been specified.*	AC contingency analysis will be performed to all user specified contingencies with tripping simulations. All converged post-contingency states will be checked against the tripping conditions. If tripping actions are activated, the post-tripping condition at the last stage of tripping simulation will be saved in the output file.
Corrective Actions*	Power Flow Control Multiple Contingency Analysis Tripping Simulation *Options and required filenames have been specified.*	Contingency analysis with tripping simulation is performed. Each post-contingency condition will be checked against the tripping conditions. If tripping actions are activated, post-tripping condition at the last stage of tripping simulation will be saved in the output file. Tripping events activated following a contingency will qualify the case as failure, so that its next-level contingencies are skipped if failure cutoff is enabled.

[a]Corrective action analysis can be enabled or disabled for any running mode.

Running mode refers to a combination of selections of three options in multiple-level contingency analysis: perform multiple-level contingency analysis, perform tripping simulation and perform corrective action. Running mode determines which sequences following a contingency are simulated, the solutions with respect to one of the options can be found in ACCC output files only if the option is enabled in a mode.

The contingency label and contingency events of each contingency case are logged at the *Progress* tab as the solution process encounters it. A message identifies any network conditions that fail to converge or which are skipped. The duplicate contingencies at secondary and tertiary levels will not be presented in the *Progress* tab.

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6.15.12. Contingency Analysis Results

The multiple level AC Contingency Solution function calculates post-contingency, post-tripping and post-corrective actions contingency solution results (loadings on monitored branches and interfaces, and bus voltage magnitudes) and places them into the specified Contingency Solution Output file. These results may then be tabulated using any or all of three output processing functions: the AC Contingency Single Run Report function; the AC Contingency Multiple Run Report function; the auxiliary program AcccBrwsGrid (see Section 6.12.3, "AC Contingency Output").

6.15.13. Application Notes

The multiple level AC contingency solution can automatically generate and test single, double, or multiple contingencies. The labels of the single contingencies generated by a contingency ranker in the multiple level AC contingency solution contain the name of the contingency ranker and an identifier. For branch outage the identifier is from bus, to bus and the branch circuit ID, e.g. OVRLOD 219-220(1); for machine outage it is generator bus number and machine ID, e.g. UNIT 221(1). The other rankers for single branch contingencies are VLTAGE and ISLAND.

The label of a double and multiple contingency contains a sequence number for each single contingency which is combined to form the double or multiple contingency; and these numbers are separated by an underscore. As discussed in preceding sections, the function works with a contingency list, the sequence number of a single contingency indicates its position in the corresponding contingency list. The label of a double contingency contains two numbers; the first number indicates the position of the primary contingency in the first level list and the second indicates the position of the secondary contingency at the second level list. These numbers are followed by the labels of the primary and secondary contingencies. A multiple contingency contains three numbers and the last one indicates the position of the tertiary contingency in the third level contingency list.

Example:

A double contingency 4_10(N1OVRLOD 1:N1OVRLOD 3) consists of two contingency events, 'N1OVRLOD 1' and 'N1OVRLOD 3', the number 4 in the label indicates the primary contingency 'N1OVRLOD 1' is the fourth contingency in the first level contingency list, the 'N1OVRLOD 3' is the tenth contingency in the second level contingency list.

Post-tripping solutions at the last stage of tripping simulation for each contingency that triggers tripping events are saved into output files in the same format as contingency cases. The label of a post-tripping solution is generated by the programs, and is comprised of the first ten letters of the label of the contingency activating tripping actions and ended with /T, which identifies the case as a post-tripping case. Normally contingency legends for a tripping solution will include tripping labels, description of the monitored element that triggers the tripping activities, and description of tripping activities, due to limited space to save the legends, when the number of tripping activities and monitored elements in a tripping solution is more than 32, only tripping labels are saved as contingency legends.

The post-corrective action solution and activated corrective actions are stored into output files if the Perform corrective actions option is enabled.

There is no limit on the number of contingencies that can be tested in a contingency analysis, while the size of a contingency analysis output file is limited by Operating system. When the contingency analysis file reaches its size limit, contingency analysis is terminated and the rest of contingencies are ignored.

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6.16. N-1-1 Contingency Solution

6.16.1. About N-1-1 AC contingency analysis

A simultaneous double contingency, e.g. the outage of two single transmission lines, can be simulated with the proceeding functions, ACCC or Multiple-level Contingency Analysis. A double contingency can also occur sequentially, in the form of loss of one system component, possibly followed by some automatic or operator initiated system adjustments, then followed by another loss of a system component. This type of outages is sometimes known as an N-1-1 contingency.

Some definitions of terms applied to the N-1-1 contingency solution documents are identified as follows:

- Base Case: the original base case.
- Primary Contingency: the N-1 Contingency, in the pair of N-1 contingencies, which occurs first.
- Secondary Contingency: the N-1 contingency that occurs after the primary contingency and system adjustments.
- Performance criteria which define system stability, thermal and voltage limits under primary and secondary contingencies
- System security: A system is secure if, following any one of the credible contingencies, all components remain within performance criteria without the need for additional system control adjustments.
- Secure control: A process to determine appropriate system adjustments to ensure system security and is implemented by the Security Constrained Optimal Power Flow (SCOPF).
- System Adjustments: Actions made by regulating the controlling devices, such as phase angle regulators.

The N-1-1 contingencies are credible disturbances that may occur in a power system and are required to be tested, according to many reliability standards. Figure 6.51, “Flow chart to test a pair of N-1 contingencies” shows the evaluation process for a pair of N-1 contingencies. The N-1-1 contingency solution performs an AC power flow to solve the primary contingency, applies the appropriate system adjustments to return the system to an acceptable state, in accordance with the program user's preference, and then runs another AC power flow to solve the secondary contingency.

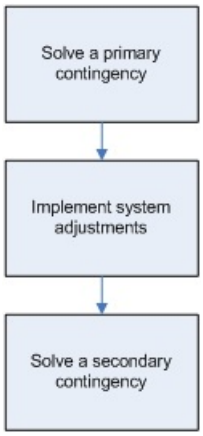

Figure 6.51. Flow chart to test a pair of N-1 contingencies

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The process is repeated until all pairs of N-1 contingencies are tested. The results are stored in a binary file (*.acc), which can subsequently be processed to produce a variety of flow and voltage limit violation reports. To perform an N-1-1 contingency solution, there are several steps:

1. Produce the Distribution Factor Data file from the *.sub, *.mon and *.con files (using activity DFAX). The primary contingencies are specified in the contingency description data file (*.con)

2. Specify another contingency description data file containing definitions of the secondary contingencies.

3. Specify the sets of automatic control options used to conduct power flow solutions for the primary and secondary contingencies, respectively.

4. Specify the control mode for system adjustments between each primary and secondary contingency pair.

5. Perform the N-1-1 Contingency Solution and produce a Contingency Solution Output file.

6. Produce the selected AC Contingency Report(s) using the Contingency Solution Output file.

A challenge in testing N-1-1 contingencies is to determine the appropriate settings of system adjustments following the primary contingency. There are many different types of control adjustments at different locations in the power system. Furthermore, different reliability standards may have different goals and strategies for implementing the system adjustments. The N-1-1 Contingency Solution function in PSSE can be performed with any one of three control modes: (1) Local control mode, (2) Corrective action mode and (3) Preventive Security constrained optimal power flow (PSCOPF) mode. Under each mode, several types of controls are available to adjust system to meet the specified goals.

The following table lists the three control modes to implement the system adjustments.

Table 6.25.

Adjustment mode	Available adjustments	Implementation	Report
Local control mode	• Tap adjustments • Switched shunt adjustments • Phase shift adjustments • DC tap adjustments • Area interchange control • Induction machine response to low voltage	Each adjustment is made to achieve its local objective, i.e., to control a specific bus voltage, power flow or area interchange schedule.s	The adjustments are not available for reporting.
Corrective action mode	• On-line generator MW controls • Load controls • Phase shifter controls	Adjustments are made by a corrective action analysis algorithm to maintain the power system within defined criteria after each contingency	The adjustments made can be reported using the AC Contingency Report function.

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Adjustment mode	Available adjustments	Implementation	Report
	• Off-line generator MW controls • Tap setting adjustments • Switched shunt controls		
Preventive Security constrained optimal power flow mode	• On-line generator MW controls • Load controls • Off-line generator MW controls • Phase shift controls • Tap setting adjustments • Switched shunt controls	Adjustments are made by a preventive security constrained optimal power flow algorithm to ensure a secured system state	The adjustments can be reported with AC contingency single run report.

The primary contingency is solved by a full AC power flow solution. When automatic adjustment options are enabled, the power flow solution will adjust the controls with the aim of meeting pre-defined branch flow, bus voltage and area interchange schedules. For instance, when tap adjustment is enabled, the transformer turns ratio is adjusted to control the voltage at a pre-specified bus. The desired voltage is the control objective of the transformer. The control is local, since the impacts of the turns ratio adjustment on voltages at other buses or on branch flows are not considered. Therefore, adjustments are made by the power flow solution to achieve their respective local objectives.

A solution under a primary contingency is deemed infeasible if the system condition does not comply with steady state performance criteria, because of thermal limit and/or bus voltage violations. If control devices are available, the system problems may be resolved by performing a corrective action analysis, which is formulated as an optimization problem with the objective of removing the violations while minimizing the system adjustments, subject to operational constraints.

Even though the operation strategies determined by the corrective actions analysis can mitigate the system problems after a primary contingency, the system may still not be in a secure state, which means that the system may not be within acceptable criteria if a secondary contingency occurs, unless more corrective actions are taken. To ensure a secure system operation condition, a preventive security constrained optimal power flow can be performed. The control adjustments determined by PSCOPF are meant to ensure a feasible and secure solution, that is, no limit violations after the primary contingency and no limit violations if any one from a list of predefined secondary contingencies occurs.

Therefore, the tool for conducting N-1-1 studies should be powerful and yet flexible enough to allow one to analyze the different scenarios described above. The N-1-1 contingency solution in PSSE has the following features:

- Automatically set up to assess pairs of N-1 contingencies.

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- Determine the system state after each primary contingency under one of three adjustment modes described above: local control, corrective action, and preventive security constrained optimal power flow.
- Produce a variety of reports to document any thermal and voltage limit violations after each primary contingency and each pair of N-1 contingencies, as well as the system adjustments determined by corrective action or by PSCOPF.

N-1-1 contingency solution options

Two N-1-1 contingency solution functions are described:

- N11_ACCC, the function for performing N-1-1 contingency analysis in the local control mode.
- N11_ACCC_PSCOPF, the function for performing N-1-1 contingency analysis in either the corrective action mode or the preventive security constrained optimal power flow mode.

Primary Contingency-case Solution options	Select solution options for power flow calculations performed to obtain the post-primary contingency solutions (see Section 6.2, "Power Flow Solution Methods"). Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustments: • Lock all • Enable all • Enable continuous, disable discrete Enable or disable phase shift adjustments. Enable or disable dc tap adjustments Induction motor treatment flag applied when an induction motor fails to solve due to low terminal bus voltage • stall

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	• trip
Secondary Ccontingency- case Ssolution options	Select solution options for power flow calculations performed to obtain the post-secondary contingency solutions. Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustments: • Lock all • Enable all • Enable continuous, disable discrete Enable or disable phase shift adjustments. Enable or disable dc tap adjustments Induction motor treatment flag applied when an induction motor fails to solve due to low terminal bus voltage: • stall • trip
Solution Engine	Select one solution engine for both primary and secondary contingency solutions (see Section 6.2, "Power Flow Solution Methods"). • Fixed slope decoupled Newton-Raphson (FDNS) (default)* • Full Newton-Raphson * The optimized fixed slope decoupled Newton-Raphhyson method is used when N-1-1 contingency solution is initiated via the graphical user interface. The user may choose either the regu-

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	lar or the optimized fixed slope decoupled Newton-Raphson method when initiating N-1-1 contingency analysis from an automation file.
Non-divergent solution	Option to enable or disable non-divergent solution in each power flow solution
Dispatch mode	Option for generation dispatch to handle system load/generation imbalance caused by a contingency (see Section 6.11.4, "About Generation Dispatch"). • Subsystem machines (Reserve) • Subsystem machines (PMAX) • subsystem machines (Inertia) • subsystem machines (Governor droop) Dispatch system specifies the subsystem containing the machines participating in the selected dispatch
Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the N-1-1 contingnecy solution process is terminated. This value is also used as the convergence tolerance in the power flow solution of each primary and secondary contingency (see Section 6.2, "Power Flow Solution Methods").
Skip the Secondary contingencies if the corresponding Primary contingency has violations	Flag to skip the secondary contingencies. It applies when a primary contingency causes thermal or voltage limit violations, • Skip the analyses of all the secondary contingencies associated with the primary contingency • Continue to assess the secondary contingencies.
System adjustment mode	Select the control mode to implement the system adjustments: • Local mode, automatic adjustments are made during the power flow solution • Corrective action mode, system adjustments are determined by corrective actions analysis. • Security constrained OPF mode, preventive system adjustments are determined by PSCOPF analysis
Constrained Options	Rating set to be used in determining monitored element loading violations in primary contingency solutions

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Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the N-1-1 contingnecy solution process is terminated. This value is also used as the convergence tolerance in the power flow solution of each primary and secondary contingency (see Section 6.2, "Power Flow Solution Methods").
	• A • B • C Rating set to be used in determining monitored element loading violations in secondary contingency solutions • A • B • C Specify percent of flow rating to be used in determination of branch and interface flow overloads in primary and secondary contingency solutions. Branch overload violation tolerance (%) and Bus voltage violation tolerance (%) are used only in the corrective action mode or the PSCOPF mode (see next table).
Distribution Factor Data File	(*.dfx) a required file to be build by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files (refer to ???). The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) must have the same setting that was in effect when activity DFAX filled the specified Distribution Factor Data file.
Contingency Description Data File for secondary contingency	(*.con) an optional file containing the secondary contingency definitions. When not specified, a secondary contingency is selected from the contingency description data file that is used to build the dfax. When specified, a secondary contingency is selected this contingency description data file.
Load Throwover Data File	(*.thr) an optional file, created by the user, that contains data records of the following form:

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Mismatch tolerance (MW)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the N-1-1 contingnecy solution process is terminated. This value is also used as the convergence tolerance in the power flow solution of each primary and secondary contingency (see Section 6.2, "Power Flow Solution Methods").
	IBUS, JBUS where IBUS and JBUS are bus numbers. If branch outage contingency events isolate bus IBUS, the user can elect to move the load to bus JBUS. Bus IBUS and/or JBUS may be a dummy bus of a multi section line grouping. Data records may be entered in any order. Input is terminated with a record specifying an IBUS value of zero.
Unit Inertia Data File	(*.inl) a required file if the generation dispatch is enabled and either the inertial or governor droop dispatch method is selected. If the PMAX or reserve method is selected, the specification of this file is optional. If a file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used
Output file (results)	(*.acc) a required file designated by the user as the destination for the results of the contingency calculations

The following options are used only when the system adjustment mode is set as Corrective action or Security constrained OPF.

Iteration # of PSCOPF	Maximum number of PSCOPF iterations when the system adjustment mode is Security constrained OPF
Cutoff of clean cases	Iteration number of a secondary contingency to be solved in PSCOPF process. At each iteration of PSCOPF, each secondary contingency will be solved by a power flow solution, its results will be checked to determine flow and voltage limit violations. If a secondary contingency does not cause any limit violations in several consecutive iterations of PSCOPF, it will not be tested any more in subsequent iterations.
Control options	Flag to enable or disable generator active power dispatch (see AC Corrective Actions) Weighting factor for generator active power dispatch

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	Control subsystem of participating generators Flag to enable or disable load controls Weighting factor for load controls Control subsystem of participating loads Flag to enable or disable phase shifter controls Weighting factor for phase shifter control Control subsystem of participating phase shifters Flag to enable or disable off-line generator active power dispatch Weighting factor for off-line generator controls Control subsystem of participating off-line generators Flag to enable or disable tap setting adjustments Weighting factor for tap setting adjustments Control subsystem of participating transformers with tap setting adjustment capability Flag to enable or disable switched shunt controls Weighting factor for switched shunt controls Control subsystem of participating switched shunts
Constrained options (for Corrective action mode or Security constrained OPF mode only)	Branch overload violation tolerance (%): The tolerance used in the linear programming algorithm to determine if a flow rating violation has been resolved. E.g., if the tolerance is specified as 0.1% and the percent of flow rating is specified as 100%, the algorithm will ensure that the flow is within 100.1% of rating. Bus voltage violation tolerance (%): The tolerance used in the linear programming algorithm to determine if a bus voltage violation has

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	been resolved. The tolerance is applied to both upper and lower voltage limits. E.g., if the tolerance is specified as 0.1%, and the lower and upper bus voltage limits are 0.95 and 1.05 in pu, respectively, the algorithm will ensure that the bus voltage is between 0.949 and 1.051 pu.

6.16.2. Perform N-1-1 Contingency Analysis

To launch the N-1-1 contingency analysis, select Power Flow ◊ Contingency, Reliability, PV/QV ◊ N-1-1 Analysis

Running in the Local Control Mode

The process to perform N-1-1 contingency analysis is determined by the selection of system adjustment control mode. When the local control mode is specified, the flow chart of the process is shown in Figure 6.52, "Flow chart of evaluation process under local control mode for a pair of N-1 contingencies"

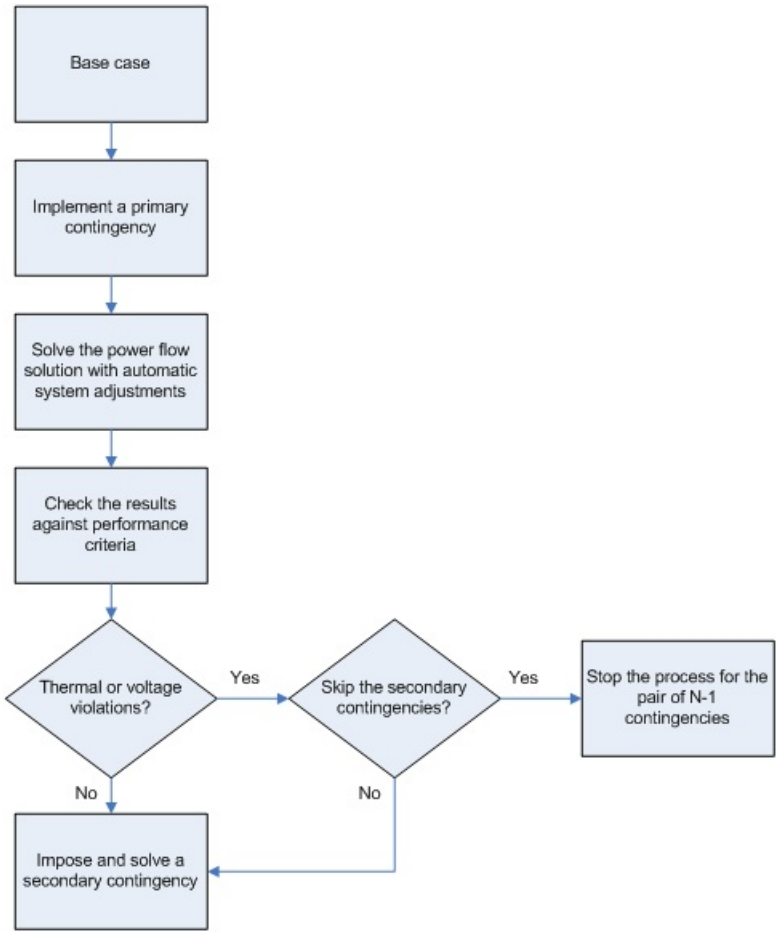

Figure 6.52. Flow chart of evaluation process under local control mode for a pair of N-1 contingencies

The process consists of several steps:

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1. Set up the base case, contingency definitions for primary and secondary contingencies, monitored elements, etc. to prepare for contingency analysis.

2. Impose a primary contingency and calculate the power flow solution with the primary-contingency-case power flow solution options. When automatic system adjustments are enabled, they are implemented by the power flow solution to meet their own respective specified bus voltage, branch flow or area interchange schedules.

3. If the option to skip secondary contingencies if the corresponding Primary contingency has violations is turned on, the results from the primary contingency solution are checked against performance criteria to identify possible flow and voltage limit violations. If violations are detected, the secondary contingencies that are associated with the primary contingency will not be tested. Otherwise proceed to the next step.

4. Impose a secondary contingency and solve it with secondary contingency-case solution options. When automatic system adjustments are enabled, they are implemented by the power flow solution to meet their own respective specified bus voltage, branch flow or area interchange schedules in the post-contingency state.

Figure 6.53, "Progress output from N-1-1 contingency analysis in local control mode" is an example of the progress output from an N-1-1 contingency solution in local control mode.

```
Processing contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1
Diagonals = 24   Off-diagonals = 44   Maximum size = 62

Running N-1-1 contingency analysis for the first N-1 contingency "N1OVRLOD 1"...

*** Contingency "N1OVRLOD 1" following the first N-1 contingency "N1OVRLOD 1" skipped
OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

Processing contingency "N1OVRLOD 2" following the first N-1 contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 215 [MIDCOALP    230.00] TO BUS 224 [WESTIE-2    230.00] CKT 1

Processing contingency "N1OVRLOD 3" following the first N-1 contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 213 [BIGOIL      230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1
*** Successful solution not achieved: Iteration limit exceeded ***
Largest mismatch is       7.30 MW or Mvar at bus 103 [WESTIE-1    138.00]
Total mismatch is        43.53 MVA
3 voltage controlled buses whose voltages differ from their scheduled voltages by
  more than 0.00001--largest error is 0.045511 at bus 223 [BIGCOAL     230.00]
```

Figure 6.53. Progress output from N-1-1 contingency analysis in local control mode

6.16.3. Running in the Corrective Action Mode

When the system adjustment mode is corrective action, the procedure becomes more extensive. Both power flow solution adjustments and corrective actions controls may be involved in the process. Available controls for corrective actions are specified via the control options definitions: the flag to enable or disable a control adjustment type, the weighting factor, and the control subsystem to identify participating control elements. The corrective action has a higher priority in making the control adjustments. For example, if tap setting adjustments is selected as a corrective actions control and tap adjustment is also enabled as a contingency case solution option, the local objective of each transformer is ignored and its tap is adjusted, as necessary, to resolve violations anywhere in the system.

The flow chart of the evaluation process under corrective action mode is shown in Figure 6.54, "Flow chart of evaluation process under corrective action mode for a pair of N-1 contingencies".

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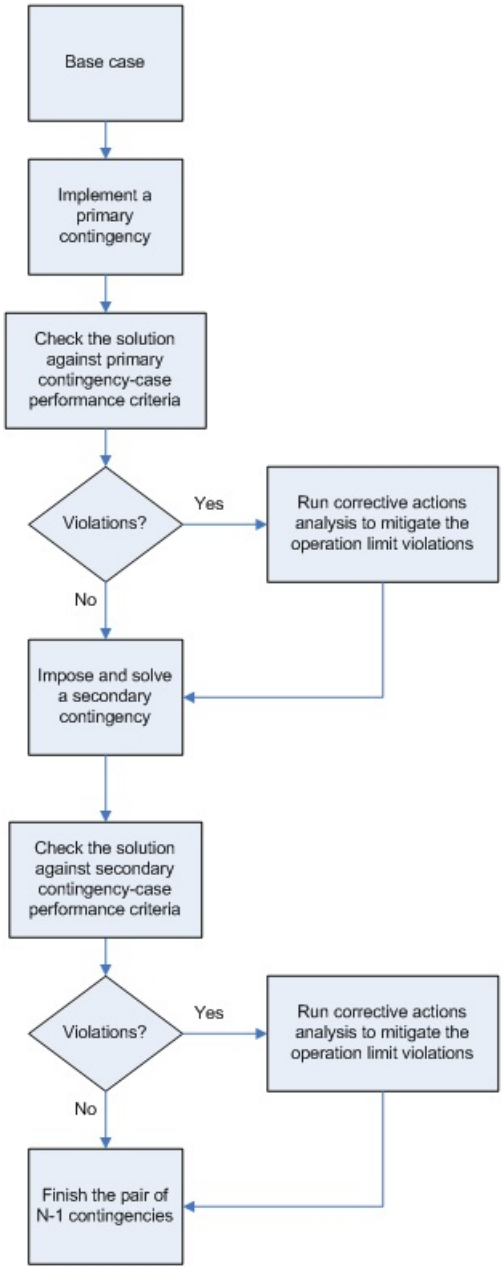

Figure 6.54. Flow chart of evaluation process under corrective action mode for a pair of N-1 contingencies

The process consists of several steps:

1. Set up the base case, contingency definitions for primary and secondary contingencies, monitored elements, etc. to prepare for contingency analysis.

2. Impose a primary contingency and calculate the power flow solution with the primary contingency-case solution options. When automatic system adjustments are enabled, they are implemented by the power

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flow solution to meet their own respective specified bus voltage, branch flow or are interchange schedules.

3. The results from the primary contingency solution are checked against system performance criteria to identify possible thermal and voltage limit violations. If overloads and/or voltage limit violations are found, a corrective action analysis is performed to determine adjustments suitable for eliminating the violations.

4. If the option of skip the secondary contingencies if the corresponding Primary contingency has violations is turned on and the limit violations identified in the previous step cannot be eliminated completely by the corrective actions, the secondary contingencies that are associated with the primary contingency will not be tested and the process of evaluation of the pairs of N-1 contingencies will be terminated. Otherwise proceed to the next step.

5. Apply the control adjustments determined from the corrective action analysis on the system with the primary contingency and perform an AC power flow solution again. This solution is used as the initial condition for the analysis of the secondary contingencies.

6. Impose one secondary contingency and calculate the power flow solution with the secondary contingency-case solution options. When automatic system adjustment options are enabled, they are implemented by the power flow solution to meet their own respective specified bus voltage, branch flow or are interchange schedules.

7. The results from the secondary contingency solution are checked against system performance criteria to identify possible thermal and voltage limit violations. If violations are found, a corrective action analysis is performed to determine the adjustments suitable for eliminating the violations.

The is an example of the progress output from an N-1-1 contingency solution in corrective action control mode. Following the solution of primary contingency "N1VORLOD 1", a corrective action analysis is performed and adjustments are made to remove the violations found. Similarly, following the solution of a secondary contingency, corrective action analysis is performed.

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```
Processing contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

Performing AC corrective action analysis to eliminate the violations in the solution

AC corrective action analysis converges

The following constraints will be ignored from PSCOPF analysis
Diagonals = 24   Off-diagonals = 44   Maximum size = 62

Running N-1-1 contingency analysis for the first N-1 contingency "N1OVRLOD 1"...

*** Contingency "N1OVRLOD 1" following the first N-1 contingency "N1OVRLOD 1" skipped
OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

Processing contingency "N1OVRLOD 2" following the first N-1 contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 215 [MIDCOALP    230.00] TO BUS 224 [WESTIE-2    230.00] CKT 1

Performing AC corrective action analysis to eliminate the violations in the solution

AC corrective action analysis converges

Processing contingency "N1OVRLOD 3" following the first N-1 contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 213 [BIGOIL      230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

Performing AC corrective action analysis to eliminate the violations in the solution

AC corrective action analysis converges
```

Figure 6.55. Exampe of progress output from an N-1-1 contingency solution corrective action control mode

6.16.4. Running in the Security Constrained OPF Mode

When the system adjustment mode is PSCOPF, the power flow solution adjustments, corrective actions as well PSCOPF are all involved in the process. Available controls for corrective actions and PSCOPF are specified via the control options definitions: the flag to enable or disable a control adjustment type, the weighting factor, and the control subsystem to identify participating control elements. The PSCOPF and corrective action have higher priority in making the system adjustments, compared with the local adjustments defined in the power flow solution options. The flow chart of the evaluation process under the security constrained OPF mode is shown in Figure 6.56, “Flow chart of evaluation process with PSCOPF mode for a pair of N-1 contingencies”

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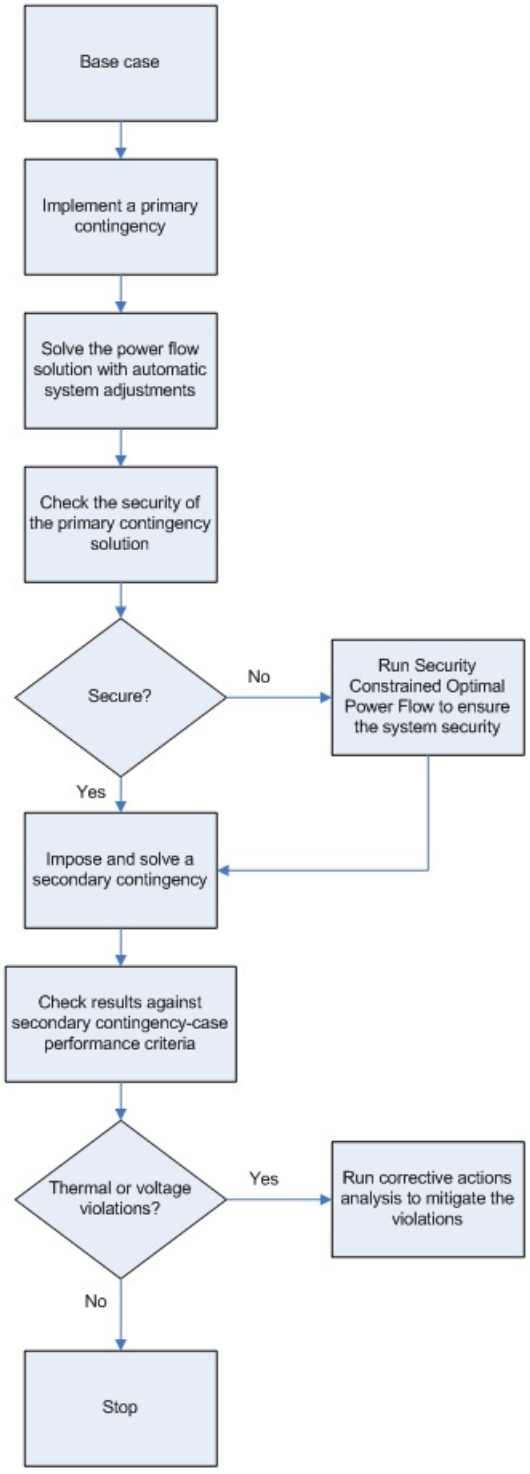

Figure 6.56. Flow chart of evaluation process with PSCOPF mode for a pair of N-1 contingencies

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There are several steps in the N-1-1 contingency solution using the PSCOPF mode:

1. Set up the base case, contingency definitions for primary and secondary contingencies, monitored elements, etc. to prepare for contingency analysis.

2. Impose a primary contingency and calculate the power flow solution with the primary-contingency-case solution options. When automatic system adjustments are enabled, they are implemented by the power flow solution to meet their own respective specified bus voltage, branch flow or area interchange schedules.

3. The results from the primary contingency solution are checked against system performance criteria to identify possible thermal and voltage limit violations. If overloads and/or voltage limit violations are found, a corrective action analysis is performed to determine adjustments suitable for eliminating the violations.

4. If the option of skip the secondary contingencies if the corresponding Primary contingency has violations is is turned on and the limit violations identified in the previous step cannot be eliminated completely by the corrective actions, the secondary contingencies that are associated with the primary contingency will not be tested and the process of evaluation of the pair of N-1 contingencies will be terminated. Otherwise proceed to the next step.

5. Apply the control adjustments determined from the corrective action analysis on the system with the primary contingency and perform an AC power flow solution again. This solution is used as the initial condition for the analysis of the secondary contingencies prior to the preventive security constrained optimal power flow.

6. Perform a preventive security constrained optimal power flow with the solution in step 5 as the base case and the secondary contingencies as contingency constraints. Determine the system adjustments necessary to ensure system security of the post-primary contingency case.

7. Apply the system adjustments determined from the preventive security constrained optimal power flow analysis as well as from the corrective actions analysis in step 3, on the system with the primary contingency and perform an AC power flow solution again. This solution will be used as the initial condition for the analysis of the secondary contingencies.

8. Impose one secondary contingency case and perform a power flow solution with secondary contingency-case solution options. If automatic local system adjustments are enabled, they are implemented by the power flow solution to meet their respective specified bus voltage, branch flow or are interchange schedules.

9. The results in the secondary contingency solution are checked against system performance criteria to determine if there are thermal and voltage limit violations. If any violations are found, a corrective action analysis is automatically performed to determine the system adjustments necessary to eliminate the violations.

This is an example of the progress output from an N-1-1 contingency solution in PSCOPF mode. Following the solution of primary contingency "N1VORLOD 1", the corrective action analysis is performed and the adjustments are made to remove the violations in the solution. Subsequently, PSCOPF starts to perform a secure control.

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```
Processing contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1

Performing AC corrective action analysis to eliminate the violations in the solution

AC corrective action analysis converges

The following constraints will be ignored from PSCOPF analysis
Diagonals = 24   Off-diagonals = 44   Maximum size = 62

Running PSCOPF for the first N-1 contingency "N1OVRLOD 1"...

PSCOPF is processing the iteration 1 of main loop...


Processing BASE CASE

Processing contingency "N1OVRLOD 2" following the first N-1 contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 215 [MIDCOALP    230.00] TO BUS 224 [WESTIE-2    230.00] CKT 1

BUSES WITH LOW/HIGH VOLTAGE LIMIT VIOLATIONS
<-------- B U S --------->  VOLTAGE  V-MAX  V-MIN
   103 WESTIE-1    138.00  0.9170 1.0500 0.9500
   224 WESTIE-2    230.00  0.8839 1.0500 0.9500

<---------- O V E R L O A D E D   L I N E S ---------->     FLOW   RATING        %
<------ F R O M --------> <-------- T O ----------> CKT   MVA/MW      MVA
   110*EASTIE-1    138.00    211 MIDTIE-2    230.00  1     495.3    400.0   123.83
   213 BIGOIL      230.00    223*BIGCOAL     230.00  1     696.4    500.0   132.66

Processing contingency "N1OVRLOD 3" following the first N-1 contingency "N1OVRLOD 1"
OPEN LINE FROM BUS 213 [BIGOIL      230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1
```

Figure 6.57. Progress output from an N-1-1 contingency solution in PSCOPF mode

In each mode, the initial system condition, or base case condition, for secondary contingencies is different. The following table summarizes the base case condition for secondary contingencies under each mode.

Adjustment mode	Initial system condition for the secondary contingencies
Local control	The power flow solution after a primary contingency
Corrective action	The power flow solution after the primary contingency is imposed and the system is adjusted by adjustments made by corrective actins.
PSCOPF	The power flow solution after the primary contingency is imposed and the system is adjusted by adjustments made by PSCOPF

The AC Contingency Solution function responds to the following interrupt control codes:

AB	Abandon the function following completion of the next iteration.
CM	Print the convergence and automatic adjustment monitors.

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NM	Suppress any automatic adjustment monitors (only used with CM interrupt control code).
DC	Tabulate conditions for each dc line after each iteration (see Section 6.3.11, "Operation of Activity SOLV").
FD	Tabulate conditions for each FACTS device after each iteration (see Section 6.3.11, "Operation of Activity SOLV").

The N-1-1 Contingency Solution functions calculate contingency solution results (loadings on monitored branches and interfaces, and bus voltage magnitudes) and place them into the specified Contingency Solution Output file. These results may then be tabulated using the AC Contingency Single Run Report function (Section 6.12.9, "AC Contingency Single Run Report"). In the reports created by AC Contingency Single Run Report function, the contingency labels of the pair of N-1 contingencies are present together. The Figure shows a sample of non-converged network report.

```
<---------------------- N-1-1 CONTINGENCY ----------------------> <--- MISMATCH (MVA) ---->
<---- FIRST N-1 CONTINGENCY----> <--- SECOND N-1 CONTINGENCY ---> <-- BUS ---X X- SYSTEM -> TERMINATION CONDITION
N1OVRLOD 1                      N1OVRLOD 3                        99.090       192.29     Iteration limit exceeded
N1OVRLOD 3                      N1OVRLOD 1                        99.089       192.29     Iteration limit exceeded

CONTINGENCY LEGEND:
<----- CONTINGENCY LABEL ------>  EVENTS
N1OVRLOD 1                      : OPEN LINE FROM BUS 212 [EASTIE-2    230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1
N1OVRLOD 3                      : OPEN LINE FROM BUS 213 [BIGOIL      230.00] TO BUS 223 [BIGCOAL     230.00] CKT 1
```

Figure 6.58. Sample of non-converged network report

In using AC Contingency Single Run Report function to produce one of the report formats: spreadsheet or non-spreadsheet overload report, spreadsheet or non-spreadsheet loading report, available capacity report, Base case rating is used to calculate the loading percent of monitored elements in base case and primary contingency solutions; contingency case rating is used to calculate loading percent of monitored elements in secondary contingency solutions. Figure 6.59, "Sample of spreadsheet overload report" shows a sample of spreadsheet overload report.

Base case rating

```
<----------------- MONITORED BRANCH ------------------> <---------------------- N-1-1 CONTINGENCY ----------------------->
<------- F R O M -------> <--------- T O ---------->CKT <---- FIRST N-1 CONTINGENCY----> <--- SECOND N-1 CONTINGENCY ---> RATING    FLOW      %
   109 MIDTIE-1    138.00    212*EASTIE-2    230.00 1  BASE CASE                                                          450.0    625.7    139.0
   110 EASTIE-1    138.00    211*MIDTIE-2    230.00 1  BASE CASE                                                          400.0    659.6    164.9
   103 WESTIE-1    138.00    224*WESTIE-2    230.00 1  N1OVRLOD 1                                                         400.0    431.8    107.9
   109 MIDTIE-1    138.00    212*EASTIE-2    230.00 1  N1OVRLOD 1                                                         450.0    515.9    114.6
   110 EASTIE-1    138.00    211*MIDTIE-2    230.00 1  N1OVRLOD 1                                                         400.0    721.1    180.3
   110*EASTIE-1    138.00    212 EASTIE-2    230.00 1  N1OVRLOD 1                                                         400.0    416.2    104.1
   213 BIGOIL      230.00    223*BIGCOAL     230.00 1  N1OVRLOD 1                                                         500.0    557.3    106.2
   103*WESTIE-1    138.00    109 MIDTIE-1    138.00 1  N1OVRLOD 1                    N1OVRLOD 2                            208.0    105.5    101.8
   109 MIDTIE-1    138.00    212*EASTIE-2    230.00 1  N1OVRLOD 1                    N1OVRLOD 2                            510.0    594.3    116.5
   110*EASTIE-1    138.00    211 MIDTIE-2    230.00 1  N1OVRLOD 1                    N1OVRLOD 2                            510.0    815.9    160.0
```

Figure 6.59. Sample of spreadsheet overload report

If a N-1-1 contingency solution is started with either corrective action or PSCOPF mode, a corrective action report can be produced to tabulate the flows and voltages in the solutions before and after adjustments, as well as adjustments. If corrective action mode is selected prior to N-1-1 contingency solution, the results of adjustments are corrective actions specified in step 3 and step 7 of N-1-1 analysis procedure with corrective action mode following each primary and secondary contingency solution respectively. In using PSCOPF mode, the results of adjustments following each primary contingency solution are the total adjustments made by corrective action and PSCOPF, step 3 to step 7 in N-1-1 analysis procedure with PSCOPF mode; the

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results following each secondary contingency solution are corrective actions. Figure 6.60, "Sample of corrective action report" shows a sample of corrective action report.

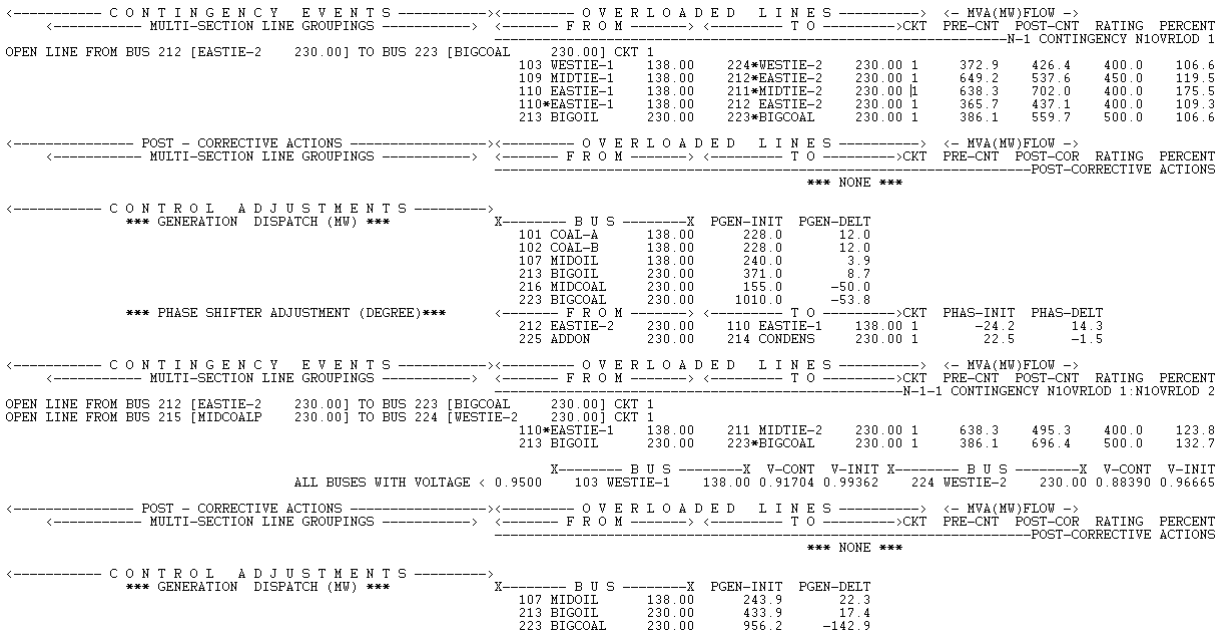

Figure 6.60. Sample of corrective action report

6.17. Calculating Probabilistic Reliability

Run Probabilistic Reliability Assessment - GUI
Power Flow > *Reliability* > Reliability Assessment... *[Probabilistic Reliability Assessment]*
Interrupt Control Codes
AB

The applications of probabilistic methods can provide points of view from transmission planners and operators, focusing on the frequency and duration of system problems, or from customers focusing on the impact of unreliability on load curtailments. Typically probabilistic reliability assessment is applied to:

- Calculation of reliability indices;
- Weak points analysis, i.e. components most affected by outages;
- Comparisons between different operating conditions, network structure as well as planning alternatives.;
- Analysis of effectiveness of corrective actions

The transition from deterministic contingency analysis to probabilistic assessment for system problems is an easy one. Probabilistic reliability analysis is provided via an additional post-processing function to calculate

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probabilistic indices for local and system problems with given outage statistics for each contingency. The reliability assessment package as shown in Figure 6.61, "Process of Probabilistic Reliability Assessment" consists of a calculation part, which evaluates each contingency and models predefined tripping and corrective action sequences, and an analysis part, which conducts a detailed analysis on the basis of the evaluated contingency sequences.

The results are referred to as probabilistic indices. They are composite probabilities of problems given in terms of frequency and duration indices, and determined by probabilities of transitions from 'success' operating conditions to 'failure' operating conditions. Generally, outage statistics are given in terms of frequencies and duration to reflect the probability that a transmission element will be forced out-of-service, and to calculate transition probabilities.

The procedure of performing probabilistic reliability analysis consists of three steps (Figure 6.61, "Process of Probabilistic Reliability Assessment"):

- Perform contingency analysis. The contingency screening results (.acc output files) are required to computed reliability indices.
- Set up options, specify input files, and perform reliability assessment. If deterministic reliability indices are computed, Outage Statistics Data file is not required.
- Analyze the results

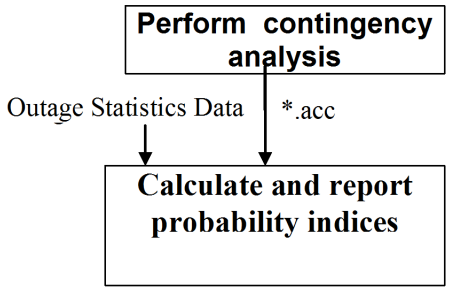

Figure 6.61. Process of Probabilistic Reliability Assessment

6.17.1. Probabilistic Reliability Assessment

Table 6.26. Probabilistic Reliability Assessment Options

Option	Description
Report format	The user selects from among the following reports: • system problem summary • system loss of load • bus loss of load • branch flow overloading

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Option	Description
	• bus voltage violations • contingency summary for • system problem probabilistic indices • system load curtailment probabilistic indices • bus load curtailment probabilistic indices • branch flow overloading probabilistic indices • bus voltage violation probabilistic indices • contingency summary with probabilistic indices When no probabilistic data is available or the studies are limited in deterministic reliability area, the users can produce the first six reports for deterministic reliability analysis, and each contingency will have identical statistic data.
Output mode	Probabilistic reliability analysis can be performed in one of three modes, subject to the corresponding running mode specified during contingency analysis. • post-contingency mode: Select the post-contingency solutions as the states in the determination of system and component problems • post-tripping mode: Select the solutions after tripping events as the states for calculations of probabilistic indices. If contingency analysis has been initiated without tripping simulation, post-tripping solution of a contingency is identical to its post-contingency solution. • post-corrective action mode: Select the solutions after corrective action analysis for probabilistic reliability analysis. If contingency analysis has been initiated without corrective actions; post-corrective action solution of a contingency is identical to its post-contingency solution if tripping simulation enabled in contingency analysis, or to its post-contingency solution if tripping simulation disabled. In this mode a contingency and complete sequences following it are taken account into probabilistic reliability assessment.
Branch rating set	Branch rating set and loading percentage: define branch loading limits used in determining overloads.
Loading percentage	

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Option	Description
Voltage limit	Defines voltage limits used in determining bus voltage limit violations. The voltage limits apply to buses specified by monitor voltage limit records (see ???).
Subsystem	Define the study subsystem where the probability indices are calculated. There is a *Entire system* subsystem set by the programs, when selected, analysis is performed within entire system. Probabilistic Indices can also be obtained for a specific subsystem predefined in Subsystem Description Data file (see ???). When a subsystem is designated, only monitored elements in the subsystem are checked against failure criteria to determine whether a contingency solution has problems. For bus voltages being checked, buses must be part of that subsystem; for line flows being checked, both of ends of a line must be in the subsystem.
Include interfaces	Enable (default) or disable to include interfaces. Interfaces are not part of any subsystems. If enabled, the impacts of interfaces on probability indices of the specified subsystem will be accumulated.
Include ties	Enable (default) or disable to include the impacts of tie lines on probability indices of the specified subsystem.
Probability and frequency cutoffs	Enable (default) or disable to exclude contingencies from probabilistic reliability analysis for which probability and frequency are less than these threshold values. A contingency may be of low probability or frequency that the risk to the system is low, and evaluation is superfluous.
Distribution Factor Data File	(*.dfx) Required file, used when performing contingency analysis (see activity DFAX).
Contingency Solution Output File	(*.acc) Required file, created with the current base case and Distribution Factor Data file. It is recommended that the file been generated by multiple level contingency analysis function (see Section 6.15, “Performing Multi-Level AC Contingency Solution”), When this file is generated by simple ACCC, analysis is limited to deterministic reliability assessment.
Outage Statistics Data File	Required file; the Reliability Outage Statistics file (*.prb) contains outage statistics of user specified contingencies and individual components (see Section 6.17.2, “Outage Statistics Data File Contents”).
Output outage statistics	Enable or disable (default) output of outage statistics to a file.

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Option	Description
Output Outage Statistics Data File	Required when *Output outage statistics* option is selected. The file is used to convert generic statistics to specific statistics for individual components.The resulting file can be edited and modified for future use. It has the same format as Outage Statistics Data file.

6.17.2. Outage Statistics Data File Contents

The Outage Statistics Data file is used to specify frequencies and durations for individual element outages and user specified contingencies. Outage statistics for individual elements, including non-transformer branches, transformers and machines, must be specified. They are also used to calculate outage statistics of multiple independent element contingencies for which statistics are not specified. Statistics for user specified outages can be defined in Outage Statistics Data file or in the Contingency Description Data file; when both files contain statistics for a user specified contingency, values are taken from Outage Statistics Data file. When no statistics data is entered for a multiple element contingency, the contingency is treated as a multiple independent element outage and its statistics are calculated on the basis of outage statistics of each single element within it. If a user specified contingency contains one or more FACTs device outages, closing events, or boundary condition change events, its statistics are set to zeros if not specified. Contingencies generated automatically by the multiple level contingency analysis are treated as independent multiple element contingencies, their outage statistics are thus calculated. The following examples illustrate how to specify outage statistics.

Example 1: There is no need to specify outage statistics of contingencies created by global command 'DOUBLE', each double branch contingency is treated as a multiple independent outage and its outage statistics are calculated from frequency and duration of two branches within it.

Example 2: Contingency A has branch B outage. Statistics for contingency A can be specified in the Outage Statistics Data file or in the Contingency Description Data file. Otherwise its statistics are equal to the statistics for the outage of branch B. If no invalid outage statistics data record is read for branch B in Outage Statistics Data file, outage statistics of contingency A are zero.

Example 3: Contingency A has branches B and C outages. Statistics for contingency A can be specified in the Outage Statistics Data file or in the Contingency Description Data file. If no statistics are specified for contingency A, its statistics are equal to frequency and duration of the multiple independent outaging of branches A and B. If no invalid outage statistics data records are read for either branch B or C, outage statistics of contingency A are zero.

Example 4: Contingency A has branch B and FACTs device C outages. Statistics for contingency A can be specified in the Outage Statistics Data file or in the Contingency Description Data file. If no statistics are specified for contingency A, its statistics are zeros.

Please refer to input data files for AC Contingency Analysis (in Section 6.12.1, “AC Contingency Solution Options”) for notational conventions used in the outage statistics data file. Frequency value is designated in occurrence per year and duration value is in hour. Generally, there are two types of data records: generic outage statistics data record to specify outage statistics for a set of elements, and specific outage data record for a single element. The following data record defines a set of non-transformer branches that have the same outage statistics:

```
|LINES   | |IN   | |AREA  i        | xm bm ft
dt fmt dmt
```


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```
|BRANCHES| |FROM | |ZONE  i        |
                   |OWNER i        |
                   |KV r           |
                   |SYSTEM    label|
                   |SUBSYSTEM label|
```

Where: xm is branch reactance (ohm/mile), bm is branch charging susceptance (S./mile), ft is outage frequency for terminal caused single circuit outages, dt is repair time for terminal caused single circuit outages, fmt is outage frequency for single circuit outages per mile, and dmt is outage durations for single circuit outages. In using system or subsystem keywords, the label must correspond to a subsystem label specified in a previously accessed Subsystem Description Data file. In using IN keyword, both bus ends of a non-transformer branch must be associated with a specified subsystem to be included. In using FROM keyword, either end of a non-transformer branch is associated with a specified subsystem to be included. The frequency F and duration D of a single line outage are calculated as:

F = len*fmt+ft D = (mft*dmt*len+ft*dt)/F (6.32)

Where len is the length of a branch. When length of a branch is not specified in working case, it is estimated by its reactance in the form of len=X/xm. For example, if the branch from bus i to bus j is 10 miles and statistics of the branch are ft=0.01, dt=10, fmt=0.02, dmt=20, then F and D of this single branch outage are:

F = (10*0.02+0.01)=0.21 (Occurrences/year) D = (0.02*20*10+10*0.01)/0.21=19.5 (hour)

The following data record defines outage statistics of breakers or switches within a subsystem:

```
|BREAKERS| |IN   |  |AREA  i        | f d [fu
du fm dm]
|SWITCHES| |FROM|  |ZONE  i        |
                   |OWNER i        |
                   |KV r           |
                   |SYSTEM    label|
                   |SUBSYSTEM label|
```

In using keyword BREAKERS, the record specifies outage statistics of breakers. The breaker is a special branch with the first character of @ in circuit ID. Where f and d are frequency and duration of a breaker fault, optional fu and du are frequency and duration of a breaker unscheduled maintenance outage, optional fm and dm are frequency and duration of a breaker maintenance outage respectively. A record employs SWITCHES keyword to specify outage statistics of switches. A switch is a special branch with the first character of * in circuit ID. The designated values are used as outage statistics with respect to each mode. In Probabilistic Reliability Assessment analysis, a component has only two states: in-service or out-of-service. The probability of a component being out-of-service is the sum of probabilities of three failure modes. When the specification data record is read by Probabilistic Reliability Assessment function, frequency and duration of a breaker or switch being out-of-service are calculated as follows:

F=f+fu+fm (6.33)

D=(f*d+fu*du+fm*dm)/F (6.34)

In Substation Reliability Assessment analysis, each component can have three failure modes: fault, unscheduled maintenance and scheduled maintenance outages. When the record is read by Substation Reliability Assessment function, each pair of frequency and duration values is taken for the corresponding fault or outage mode. By default, the four optional values are set to zeros.

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The following data record defines statistic data of transformers within a subsystem.

```
TRANSFORMERS |IN   | |AREA i          |  f d
             |FROM| |ZONE i          |
                   |OWNER i         |
                   |KV r            |
                   |SYSTEM    label|
                   |SUBSYSTEM label|
```

Where: f is outage frequency for transformer on site outages, d is outage duration for transformer on site outages. In using IN keyword, for a three-winding transformer to be included, all of its in-service windings must be connected to subsystem buses. In using FROM keyword, for a three winding transformer to be included, one of its in-service windings must be connected to a subsystem bus.

The following data record defines generic statistic data for a set of machines.

```
|UNITS    | min_size max_size IN |AREA i
    |  f d
|MACHINES |                      |ZONE i          |
                                 |OWNER i         |
                                 |KV r            |
                                 |SYSTEM    label|
                                 |SUBSYSTEM label|
```

Where: min_size and max_size are minimum and maximum size of generators in the set, the MBASE of generators are checked against the limits. f is frequency of unit outage, d is duration of unit outage.

The following data record defines generic statistic data for buses within a subsystem.

```
|BUSES | IN |AREA i          |  f d [fu du fm dm]
            |ZONE i          |
            |OWNER i         |
            |KV r            |
            |SYSTEM    label|
            |SUBSYSTEM label|
```

The Probabilistic Reliability Assessment and Substation Reliability Assessment functions read the record and match outage statistics of the buses with single bus outage events defined in a Contingency Description Data file. These outages are either single bus outages specified by an automatic single bus outage command SINGLE BUS... or contingency cases labeled as BUS plus the bus number. The probability of the contingencies is the sum of probabilities of three modes. Specially in Substation Reliability Assessment analysis, the record also specifies frequency and duration of three failure modes of single bus outages generated by the function.

The following records define statistic data for a specific user specified contingency:

```
CONTINGENCY label f d
```

Where: f and d are frequency and duration of user specified outages respectively. As discussed before, statistics of user specified contingencies can be specified in either a Contingency Description Data file or an Outage Statistics Data file in above format, if both files have statistics for the same user specified contingencies, the designated values in Outage Statistics Data file will be applied.

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The following data record defines statistic data for a non-transformer branch or two-winding transformer.

```
|LINE  | FROM BUS bsid TO BUS bsid |CKT    |
cktid f d [fu du fm dm]
|BRANCH|                           |CIRCUIT|
```

The following data record defines statistic data for a breaker or switch.

```
|BREAKER| FROM BUS bsid TO BUS bsid |CKT    |
cktid f d [fu du fm dm]
|SWITCH |                           |CIRCUIT|
```

Where: f is frequency of a breaker outage, d is duration of a breaker outage.

Similarly the following data record defines statistic data for a three-winding transformer:

```
|LINE  | FORM BUS bsid TO BUS bsid TO BUS bsid
|CKT    | cktid f d [fu du fm dm]
|BRANCH|                                        |CIRCUIT|
```

The following data record defines statistic data for a machine

```
|UNIT   | macid AT BUS bsid p d
|MACHINE|
```

Where p is probability of machine outage, d is duration of machine outage.

The following data record defines statistic data for a bus.

```
BUS bsid f d [fu du fm dm]
```

When the multi-section reporting option is enabled, the frequency and probability of a multi section line is equal to the sum of frequencies and probabilities of all its members.

6.17.3. Analyzing Probabilistic Assessment Results

For each monitored non-transformer branch, the percent loading is either the percent current or the percent MVA loading, according to the non-transformer branch percent loading units program option setting (see Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN) specified when performing contingency analysis to create Contingency Solution Output files. For monitored transformers, the percent loading is either the percent current or the percent MVA loading, according to the transformer percent loading units program option setting.

Three output modes are available when reporting results: post-contingency, post-tripping simulation, and post-corrective actions. Each output mode corresponds to one stage in the evaluation of a contingency. When performing multiple level contingency analysis (see Section 6.15, “Performing Multi-Level AC Contingency Solution”), a running mode is specified according to selections of options of perform multiple contingency analysis, perform tripping simulations and perform corrective actions. The running mode determines how many stages in a contingency analysis are involved. e.g. if all three options are selected, a contingency and its following sequences: tripping events and corrective actions are simulated and then each contingency may have one power flow solution with respect to each option stored in the Contingency Solution Output file, the solution after a contingency being imposed, the one after simulating tripping events triggered by the contingency, and the last one is after applying corrective actions that are specified to relieve violations in the

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second solution. In other words, a contingency analysis consists of three stages. By specifying output mode option, users have access to power flow solutions at each stage. If power flow solutions for an output mode do not exist in the output files, power flow solutions at previous stage will be used for the output mode. e.g., tripping simulation option is disabled prior to contingency analysis, therefore there are no post-tripping solutions in result files. If output mode is set as post-tripping, reliability results in post-tripping mode are identical to those in post-contingency mode.

When calculating probabilistic load curtailment indices, results may be quite different in different output modes, because load curtailments may be caused by contingencies, tripping events, generation dispatch and corrective actions.

In order to better understand the output, several examples are developed with the savnw.sav power flow case. The .sub, .mon, and .con files for contingency analysis and an example outage statistics data file .prb for reliability assessment are provided in the example directory. Contingency analysis is performed by multiple level contingency analysis function with simulations of tripping and corrective actions sequences.

System Probabilistic Index Summary

Figure 6.62, "System Reliability Indices Summary in Post-Contingency Mode" shows the system reliability indices summary report in post-contingency output mode. The frequency, average duration and probability for each type of problems are calculated. Impact indices of overloads and bus voltage violations with respect to voltage limits 'AREA 2 BUSES WITH VOLTAGE LESS THAN 0.940 (pu)' and 'AREA 2 BUSES WITH VOLTAGE GREATER THAN 1.060 (pu)' are also given. For each type of problems, the number of contingencies causing the problem, the worst violation as well as the contingency causing the worst violation are reported. The study subsystem is 'Entire system', study system total indices are reported at the end. The results in post-corrective actions mode are shown in Figure 6.63, "System Reliability Indices Summary in Post-Corrective Action Mode". The load curtailments are found in the post-corrective actions solutions, while voltage violations and overloads are alleviated by corrective actions. From a system point of view, reliability indices, frequency, average duration and probability, are almost the same in the two stages. Furthermore, the results in post-contingency mode provides reliability indices for system problems, and the results in post-corrective actions mode provide impacts of system problems on electricity customers.

<-- F A I L U R E C R I T E R I A -->	FREQ. (OC/Y)	DURATION (HOURS)	PROB. (H/Y)	IMPACT	NO. OF CONT.	WORST VIOL.	WORST
AREA 2 BUSES WITH VOLTAGE LESS THAN 0.940 (PU)	11.9176	10.0	119.1	1.53	13	0.875	17_17
ZONE 5 BUSES WITH VOLTAGE DROP BEYOND 0.060 (PU)	0.3369	10.0	3.4	0.18	2	0.084	14_20
AREA 2 BUSES WITH VOLTAGE GREATER THAN 1.060 (PU)	8.9740	10.0	89.6	0.27	10	1.106	17_17
OVERLOAD (%)	0.4113	9.1	3.7	3.68	6	211.905	14_20
NOT CONVERGE	0.0511	5.0	0.3		6		
ENTIRE SYSTEM TOTAL	12.3048	10.0	122.7		20		

Figure 6.62. System Reliability Indices Summary in Post-Contingency Mode

<-- F A I L U R E C R I T E R I A -->	FREQ. (OC/Y)	DURATION (HOURS)	PROB. (H/Y)	IMPACT	NO. OF CONT.	WORST VIOL.
AREA 2 BUSES WITH VOLTAGE LESS THAN 0.940 (PU)	1.7018	9.9	16.8	0.29	5	
AREA 2 BUSES WITH VOLTAGE GREATER THAN 1.060 (PU)	1.7198	9.8	16.9	0.07	7	
ZONE 5 BUSES WITH VOLTAGE RISE BEYOND 0.055 (PU)	0.0043	5.0	0.0	0.00	2	
OVERLOAD (%)	0.3361	10.0	3.4	2.49	1	
LOSS OF LOAD (MW)	10.5780	10.0	105.8	14026.67	11	
NOT CONVERGE	0.0511	5.0	0.3		6	
ENTIRE SYSTEM TOTAL	12.3048	10.0	122.7		20	

Figure 6.63. System Reliability Indices Summary in Post-Corrective Action Mode

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System Loss of Load Report

The cumulative system loss of load summary provides detailed indices of loss of load for outages in which one or more buses are separated from the remainder of the network, or load shedding is performed at certain buses. The cumulative loss of load index represents the amount, frequency and energy of load lost as given in increments, and accumulated. Incremental step size is 10 MW; for load curtailment in excess of 1000 MW, it is grouped as one set.

Figure 6.64, "System Load Curtailment Probabilistic Indices" shows system loss of load report in post-corrective actions mode. System load curtailments results are dependent on the options used in the multiple level contingency analysis. The sum of frequency, average duration and probability causing the corresponding amount of load curtailments are reported. The interrupted power I.P. and Expected Unserved Energy E.U.E. are provided. Note for E.U.E, the duration is one year (8760 hours). The study subsystem is specified as 'ENTIRE SYSTEM'

```
<--  LOAD CURTAILMENTS
-->   FREQ.  DURATION  PROB.    I.P.      E.U.E  NO. OF   WORST CONT.
<-----     (MW)     ----->   (OC/Y)   (HOUR)  (H/Y)
 (MW/Y)   (MWH/Y)  CONT.
 80.0 --  90.0
          3.6272    10.0   36.4   290.17   2911.76     2    OVRLOD
 14
160.0 -- 170.0               6.8657    10.0
  68.9  1098.51  11029.22     2    OVRLOD  17
ENTIRE SYSTEM
              10.4928    10.0  105.3  1388.68  13940.98     4
```

Figure 6.64. System Load Curtailment Probabilistic Indices

Branch Flow Overloading Indices

Figure 6.65, "Branch Flow Overloading Deterministic Reliability results" shows the branch flow overload deterministic reliability results, where no outage statistics are required and all contingencies are assumed to have equal statistics. The following items are calculated: average overload and maximum loading of the overloaded circuit, the number of contingencies that result in overload of this circuit, and the contingency causing maximum overload.

Figure 6.66, "Branch Flow Overloading Probabilistic Indices" shows overload probabilistic indices for each circuit and for the study subsystem. Specifically, the sum of frequency, average duration and probability of all contingencies resulting in a circuit overload are reported. The system overload impact index is equal to the sum of overload impact indices of all circuits. As we can see, the total number of contingencies that result in overload problem in Figure 6.65, "Branch Flow Overloading Deterministic Reliability results" is much bigger than that in Figure 6.66, "Branch Flow Overloading Probabilistic Indices"; this is because some contingencies with overload problems had zero statistics or statistics less than cutoff values.

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```
<----- O V E
R L O A D E D  L I N E S ----->  AVG.OVRLOD  MAX.LOADING  NO. OF
  WORST CONT.
<------F R O M ------> <------
T O -----><CKT       (%)          (%)     CONT.
 ENTIRE SYSTEM
                                                          49
  203 EAST230   230.00   154 DOWNTN  230.00      92.55
       244.85       11   5_9
  205 SUB230
 230.00   154 DOWNTN  230.00       9.01        120.68        8
 15_22
       INTERFACE WEST
           193.29        296.76       35   1_12
```

Figure 6.65. Branch Flow Overloading Deterministic Reliability results

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```
<------ O V E R L O A D E D  L I N E S -------->   FREQ.    DURATION   PROB.   IMPACT  MAX VIO.   NO. OF   WORST CONT.<------
------> <------ T O ---- -->CKT   (OC/Y)    (HOUR)    (H/Y)    (PU)     (%)    CONT. ENTIRE SYSTEM
0.4113     9.1     3.7   3.68     0.00         6   203 EAST230  230.00   154 DOWNTN  230.00      0.3518      9.8
3.62   211.90       3   14_20  205 SUB230  230.00   154 DOWNTN  230.00   0.0744      5.0     0.4   0.06   120.68
15_22
```

Figure 6-52. Branch Flow Overloading Probabilistic Indices

Figure 6.66. Branch Flow Overloading Probabilistic Indices

6.17.4. Application Notes

The probabilistic reliability assessment requires a Contingency Solution Output file and a Distribution Factor Data file. It is required that the working case contain the same network representation (i.e., the same Saved Case) on which the contingency analysis calculation was based, and that the same Distribution Factor Data file that was used in contingency analysis be specified; otherwise, the probabilistic assessment function will

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terminate. When the base case remains unchanged, but the Monitored Element Data file and/or the Contingency Description Data file (and therefore the Distribution Factor Data file) are changed, the contingency analysis must be performed again before selecting the reliability assessment function. When adding new subsystems in the Subsystem Description Data file (see ???) without any changes in the Monitored Element Data file and the Contingency Description Data file, only the Distribution Factor Data file needs to be rebuilt, the reliability assessment can be performed with the newly created DFAX file without redoing contingency analysis.

A Contingency Solution Output file created by N-1-1 contingency solution can not be used for reliability assessment studies. A contingency solution output file created by activity ACCC in PSS®E 32 and previous versions can not be used to compute probabilistic reliability indices. When such a file is used, an error message is prompted and the process is terminated.

Contingencies with frequencies and probabilities of zeros have no impact on probability indices. When a user specified contingency does not have statistics specified, it is treated as a multiple independent outage and its statistic data is determined on the basis of the individual outages of which it consists. In case any one among them has unknown or zero frequency or duration, its statistics will be zeros. For contingencies consisting of closing outages, FACTs device outages and boundary condition changes (specified by keywords SET/CHANGE/INCREASE/ DECREASE /MOVE), because statistics of these outages cannot be defined in outage statistic data file, statistics of such contingencies then cannot be calculated must be given explicitly in outage statistics data file or in Contingency Description Data file, otherwise their statistics are zeros.

The multiple level contingency analysis function can simulate a contingency and event sequences following the contingency. For a contingency, there may be three solutions: post-contingency, post-tripping, and post-corrective actions solutions. It is important to properly select the stage at which the reliability assessment will be performed. When analysis is performed on solutions at the stage that does not exist in a Contingency Solution Output file, the solutions at previous stage will be used.

Reliability measures can be obtained as a function of the severity of problems without repeating the contingency analysis. For example, PSS®E can be used to compute failure indices for 10%, 20%, and more severe levels of overloads. Such computations can be done entirely within activity RELIND by adjusting the overload failure criterion and obtaining the reliability index output corresponding to each overload rating. The procedure to change voltage limits is to edit voltage monitor records in Monitored Element Data file (only the values of limits can be changed), and then recreate the Distribution Factor Data file. The reliability assessment can then be performed with newly created DFAX file without redoing contingency analysis. However, when other changes are made to Monitored Element Data file and Contingency Description Data file, the whole procedure must be gone through to perform probabilistic reliability assessment.

Reliability calculations can be obtained for different outage statistics without repeating the contingency analysis. This can be done by preparing two or more sets of outage statistics, and simply repeating the reliability calculations in RELIND after each set of outage data has been read into the program. This feature is useful for studying the impact of severe weather conditions and to determine the sensitivity of results to specific outages and to uncertainties in outage statistics.

If there are violations in the base case with specified failure criteria, all other states will result in system problems on the basis of the second assumption of the algorithm in PSS®E, and then the probability of system problems should be equal to 1; in other words, system problems always exist. In this case, PSS®E will neglect the probability of the base case, and add up probabilities of all outages with system problems to calculate probabilistic indices. However, the double counting problem is more prominent because it is most likely that tested outages have system problems if the base case has problems.

When probabilistic indices are used to compare the relative performance of the system under different operating conditions or with different system configurations, a contingency analysis must be performed for

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each initial system condition (base case). The different operating conditions can be simulated by means of load or dispatch adjustments, or by outages of specific network elements in the pre-disturbance condition. Different system configurations can be represented using appropriate system adjustment activities within PSS®E, such as activity CHNG.

Additional Information
PSS®E GUI Users Guide, Running Probabilistic Reliability Assessment

6.18. Calculating Substation Reliability

Run Substation Reliability Assessment - GUI
Power Flow > Reliability > Substation reliability assessment... *[Substation Reliability Assessment]*
Interrupt Control Codes
AB

6.18.1. Substation Reliability Assessment

Overview

Station reliability evaluations can augment substation design principles to enable quantitative evaluation and comparison of alternative substation configurations. In general, substation reliability studies are organized into seven steps as follows:

1. Define the system extent and list of components to be included. The study system may include substation and incoming and outgoing lines/feeders.
2. Gather failure, repair, and maintenance data for each component within the study system.
3. Specify performance criteria for system failure and select the indices to be calculated.
4. Determine models and modeling assumptions.
5. Develop the reliability models mathematically.
6. Perform component failure effects and system analysis of enumerated states.
7. Compute reliability indices.

Run Options

Table 6.27. Substation Reliability Assessment Options

Option	Description
Report format	Allows user to select from among the following reports: • Substation component report • Contingency report • Bus load curtailment report

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Option	Description
	• Substation load curtailment report
Branch rating set and Flow rating percentage	Specify branch loading limits used in determining the branch MW power transfer capacity. A branch does not have any transfer capacity if its loading limit is 0.
Frequency cutoff	Specify contingency frequency cutoff value in occurrences per year. Contingencies with frequencies of occurrence lower than the cutoff will be excluded from substation reliability assessment.
Peak load percentage	Specify the loads to serve. The desired loads are curtailed when the sum of desired loads is greater than the transfer capacity of a substation. The desired load at a sink bus is the percent of the total MW power load at the bus. If a load consists of constant I and constant Y components, its active power load is calculated at unity voltage. When the bus is connected to a tie, which has one end connected to the bus and another end connected to a bus outside the substation system, the capacity of the tie is included into the total load at the bus.
Switching time:	Specify the elapsed time (in hours) for switching operation(s) to isolate the faulted component(s) and re-close breakers to limit the impact of a fault; the elapsed time is the duration for the fault at post-fault state. The same switching time is assumed for all faults in a study.
Stuck breaker failure rate	Specify the stuck breaker failure rate (in hours per year). The rate will be used to calculate the frequency and probability of a circuit breaker failing to operate following a substation component fault.
Include system ties to source/sink systems	Specify whether the ties connected to the source or sink buses should be considered as part of the substation system (see definitions of source and sink below).
Substation system	Specify the extent of the substation to be considered in the study. Substation reliability assessment processes each failure mode of each component within the substation system only.
Source system	Specify the source system. The power transfer is from the source system to the sink system. A substation can have up to 60 source buses. Each source bus is assumed to be connected to an infinite system with unlimited active power generation. All source buses must reside within the substation system. A source tie (an incoming line) is a branch connecting a source bus with the outside network.

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Option	Description
	When the Include system ties to source/sink systems option is enabled, both ends of the tie are treated as part of the substation, the source bus is replaced by the bus at the other end of the tie and the ties are moved one bus out.
Sink system	Specify the sink system. A sink subsystem includes buses where loads are to be served. A substation can have up to 60 sink buses. All sink buses must reside within the substation system. A sink tie (an outgoing line) is a branch connecting a sink bus with the outside network. When the Include system ties to source/sink systems option is enabled, both ends of the tie are treated as part of the substation, the bus at the other end of the tie is included in the sink system, and the ties are moved one bus out. The equivalent load at a sink bus is the sum of the active power loads at the bus plus the MVA ratings (according to selected rating set and percentage multiplier) of all ties connected to the sink bus.

Input Files

Substation reliability assessment requires a Distribution Factor Data file and an Outage Statistics Data file.

6.18.2. Modeling the Substation

A power flow case may contain one more substations. Substation reliability assessment solution can process only one substation at a time; the extent of the substation is defined by a system in a Subsystem Description Data file. The outside substation in the power flow case is shut down during solutions; therefore the external system has no impacts on the results. Distribution Factor Data file setup activity (DFAX) reads the Subsystem Description Data file, together with a Monitored Element Data file (The file could be empty by having only one END command since none of the contents are used), as well as a Contingency Description Data file which defines common mode faults, and reflects their contents in a Distribution Factor Data file in preparation for substation reliability assessment.

Only in-service buses in the substation system are included in the studies. There are some buses in a substation which are off-line under normal condition and turned on for the emergency. In order to properly simulate switching operations following a fault on those buses, they must be assigned to in-service status. A substation may include several islands, which are isolated by normally-opened breakers; each island must have at lease one swing bus (bus type code 3). Since substation reliability assessment does not perform conventional power flow solutions, the choice of the swing bus has no impact on the results. Besides buses, Components considered in a solution are breakers, switches, branches and three-winding transformers. For a three-winding to be included, all its in-service windings must be connected to substation buses.

For instance, a substation has a main and transfer bus configuration. There is a normally-opened tie breaker between the main and transfer bus, and all branches are connected to the main bus under normal condition; the transfer bus will be isolated under normal condition. when there is a fault at the main bus, all branches will be redirected to the transfer bus and the load supply can be restored. In order to simulate the switching operations in substation reliability assessment solution, the transfer bus must be set in-service and if it is the only bus in an island, it must be set to the swing bus.

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The breakers and switches within the substation system are modeled as special non-transformer branches with unique first characters in the circuit IDs: @ to designate a breaker and * to designate a switch. Though not compulsory, the modeling of breakers and switches as zero impedance lines (each with resistance of zero and reactance less than the zero impedance line threshold) is highly recommended.

Breakers and switches can operate in either normally-open or normally-closed mode. The two modes are represented by the ON and OFF statuses, respectively, of the branch.

Each breaker is assumed to have two switches, one on each side, to isolate the breaker for repair and maintenance. These implicit switches are added automatically during substation reliability assessment. Switches for isolating other substation components must be explicitly modeled by the user.

6.18.3. Failure Modes

Substation reliability assessment analysis performs failure modes and effects analyses of arbitrary bus-breaker configurations and measures the impact of substation equipment failures upon the resulting power transfer capability of the substation. It can evaluate four modes of equipment failures as follows:

- Fault: Requires operation of circuit breakers to clear the fault and isolate the faulted component.
- Common Mode Fault: A common mode fault refers to simultaneous outages of multiple components due to a common cause. It requires operation of circuit breakers to clear the fault and isolate the faulted components. A common mode fault is specified in the Contingency Description Data file.
- Unscheduled Maintenance Outage: Unplanned or uncontrolled removal or transfer to the open circuit state of the affected component. Component is removed immediately, no system fault occurs.
- Scheduled Maintenance Outage: Controlled or planned removal for inspection, overhaul, test, etc. Time of removal is controllable to accommodate maintenance crews, seasonal or weather considerations, load levels, or outage of other components.

Representation of fault events (single component faults and common mode faults) includes the option of substation re-configuration following breaker opening to minimize the impact of the component outages. Faults are represented with a three-state model, as shown in the state transition diagram in Figure 6.67, "Three-State Component Fault Model". λ is the failure rate, μ_{SW} is switching rate, and μ is repair rate.

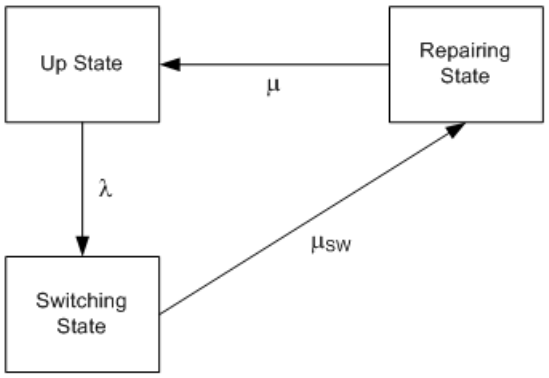

Figure 6.67. Three-State Component Fault Model

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For a fault event, the sequence of switching and assessment is modeled as follows:

1. Identify the fault.

2. Operate breakers to clear the fault.

3. Assess the effect on the substation; this is the post-fault state.

4. Operate switch(es) to isolate the faulted component and, if appropriate, re-close breakers to minimize the extent of the fault effect. The post-fault state persists until switching is completed; the time to switching completion is labeled S.

5. Assess the effect on the substation. This is the post-switching state, which persists until component repair or replacement is completed; the time to repair completion is labeled R.

6. Reconfigure the system with the restored component.

The time line in Figure 6.68, "Time Line Illustrating the Steps in a Fault Simulation" illustrates the steps in a fault simulation:

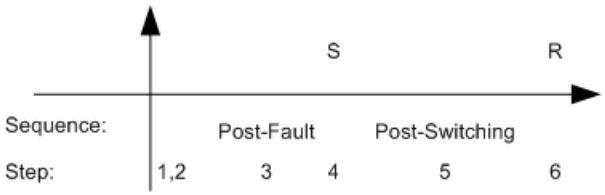

Figure 6.68. Time Line Illustrating the Steps in a Fault Simulation

Any effect of switching operations during system re-configuration in step 6 is not considered in the reliability assessment. Note further that the outage duration is counted from the time of fault or outage initiation. The model for component repair is based on the following premises:

1. Component outage duration is far longer than the switching time, S;

2. Repair time, R, starts upon completion of switching.

For a single-component or common mode fault, the substation capability is evaluated at two stages:

- Post-fault stage consists of steps 1 to 4 in Figure 6.68, "Time Line Illustrating the Steps in a Fault Simulation", corresponding to the period following the fault until re-configuration can be effected by switching, and

- Post-switching stage which consists of steps 5 and 6 corresponding to the period following switching operation until repair is completed.

Each stage is culminated by a check on network connectivity. Equipment that is electrically connected is identified by processing circuit breaker status and disconnect switch positions. Thus, equipment that is isolated and the degree of service lost can be determined. Power transfer capability of the substation is assessed by means of a transportation network flow algorithm (maximum flow-minimum cut). The frequency (in oc-

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currences per year), duration (in hours), and the probability (in hours per year) of finding the system in that condition are also computed.

For an unscheduled maintenance or a scheduled outage event, a model with two states, up and down (repair) states, as shown in Figure 6.69, "Two-State Model for Scheduled and Unscheduled Outage", is used.

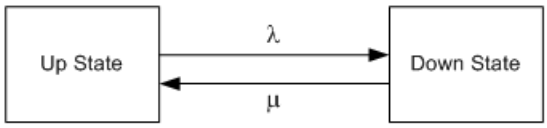

Figure 6.69. Two-State Model for Scheduled and Unscheduled Outage

The sequence of switching and assessment for unscheduled maintenance or scheduled maintenance is modeled as follows:

1. Perform switching to remove the component for maintenance, time to switching completion is S.
2. Assess the effect on the substation. Post-switching state persists until component maintenance is completed, duration is R.

Outage statistics of substation components are specified in the Outage Statistics Data file in the form of frequency of occurrence and duration; by default they are zeros. The statistics data for scheduled and unscheduled maintenance outages is optional and used only in substation reliability study. The duration of an outage mode is repair time only and does not include switching time. In order to include failures of the implicit switches in the substation reliability assessment, the user should specify the outage statistics for the switches using a generic input command (see description in Section 6.17.2, "Outage Statistics Data File Contents"). An example of an Outage Statistics Data file is shown in Figure 6.70, "Example of Outage Statistics Data File". The record to specify the outage statistics data for switches specifies both explicit and implicit switches.

```
lINES IN SYSTEM 'SUBSTATION' 0.1, 0.1, 1, 10, 1, 10
TRANSFORMERS IN SYSTEM 'SUBSTATION' 1, 10
BREAKERS IN SYSTEM 'SUBSTATION' 0.01 96 0.035 24 0.34 24
SWITCHES IN SYSTEM 'SUBSTATION' 0.002 12 0.0 0.0 0.1 0.6
BUSES    IN SYSTEM 'SUBSTATION' 0.14 1.8 0.0 0.0 1 89
```

Figure 6.70. Example of Outage Statistics Data File

Performing the Assessment

The reliability of a substation is assessed by checking the effects of fault and outage of each component within the substation. For each possible failure location and failure mode, the Substation Reliability Assessment (SRA) function in PSS®E simulates the operation of the automatic devices (breakers) to isolate the faulted component and the manual switches to minimize the equipment isolated. The process for SRA evaluation of a component failure is illustrated in Figure 6.71, "Process to Assess a Failure Mode of a Component in SRA". In this manner, SRA methodically looks at each component failure in the substation system and evaluates post-fault and post-switching connectivity and loss of service.

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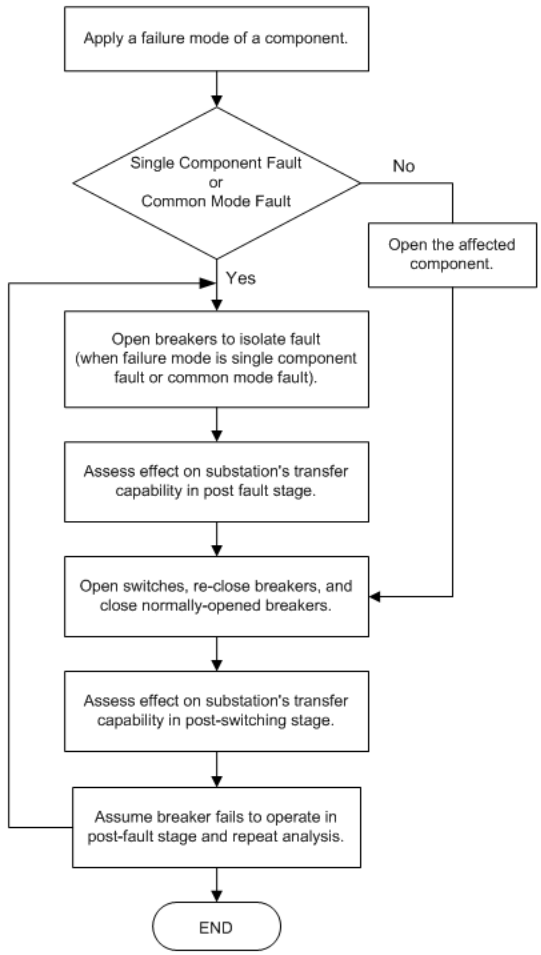

Figure 6.71. Process to Assess a Failure Mode of a Component in SRA

SRA generates a list of outage events; the list contains common mode faults and all in-service substation components. Events in the list are then evaluated with the proceeding process sequentially.

Common mode faults must be defined in the Contingency Description Data file. They are also referred to as user specified outages. For each user specified outage, only the common mode fault is considered in SRA. Unscheduled and scheduled maintenance outages are not considered. SRA honors only these contingencies where a DISCONNECT/OPEN/TRIP keyword is present. In an event description record, for a breaker or switch, a DISCONNECT/OPEN/TRIP keyword indicates that the breaker or switch is taken out-of-service; for other components, it indicates a fault on the component, which should be isolated by breakers.

For each substation component, three failure modes may be considered: fault, unscheduled maintenance, as well as scheduled maintenance. The substation components include transformers, incoming and outgoing lines or feeders, breakers, switches, and bus sections. Switches include implicit ones (on both sides of each breaker) introduced by SRA and those explicitly modeled by the user.

Out-of-service components in the power flow model are excluded from the outage event list, except normally open switching devices (breakers and switches). The normally opened switching devices cannot be

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modeled as a primary failure event (or contingency), however, they may be automatically selected by SRA as a secondary contingency following a single-component fault or a common mode fault, since these devices can be closed in the post-switching stage to restore service interrupted by a primary fault.

SRA can evaluate up to two levels of contingencies. A contingency in SRA refers to a common mode fault or a single component failure in one of three failure modes. A contingency event in the event list is selected as a primary contingency and evaluated. After evaluation of the primary contingency, a two-level contingency is generated by combining the primary contingency with another component failure or common mode fault. Some two-level contingences may not exist in real world, for instance a primary contingency is a fault on a single component followed by a scheduled maintenance on another component. The state transient diagram for some two-level contingencies and the calculations of their frequency and probability are discussed in the PSS®E Program Application Guide, Section 10.1.3, Probabilistic Indices.

A fault is modeled with two stages (post-fault and post-switching). If a primary contingency is a fault or common mode fault, its secondary contingency can occur during either stage of the primary fault. SRA also considers breakers failing to operate. Following each fault, all breakers that have been tripped to clear the fault are cataloged. The possibility of each of these breakers failing to operate correctly is investigated as a new contingency.

Analyzing Results

SRA provides several reports from the substation reliability assessment:

- Substation Component Report
- Contingency Summary Report
- Bus Load Curtailment Report
- Substation Load Curtailment Report

Substation Component Report

This report lists by data category the substation components included in the study and the corresponding frequency and duration of each component failure mode. The column headed ATTRIBUTE in breaker and switch table indicates the status of a breaker or a switch: NORM-CLOSE for normally close and NORM-OPEN for normally open. An * leading the attribute string indicates that the component is an implicit switch. A sample of substation component report is shown in Figure Figure 6.72, “Sample Substation Component Report”.

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```
<----------BUS----------> SYSTEM <-----FAULT-----> <--UNSCHEDULED--> <---SCHEDULED--->
  BUS# X-- NAME --X BASKV          FREQ. DURATION    FREQ. DURATION    FREQ. DURATION
                                   (OC/Y)  (HOUR)    (OC/Y)  (HOUR)    (OC/Y)  (HOUR)
   601 TRANS 1     13.800          0.1400   1.80     0.0000   0.00     1.0000   89.00
   607 LOAD 1      13.800 SINK     0.1400   1.80     0.0000   0.00     1.0000   89.00
  1002 EXTERN 2    115.00 SOURCE   0.1400   1.80     0.0000   0.00     1.0000   89.00

<----NON-TRANSFORMER BRANCH/TWO WINDING TRANSFORMER---> <-----FAULT-----> <--UNSCHEDULED--> <---SCHEDULED
<-------FROM BUS---------> <---------TO BUS----------> CKT   FREQ. DURATION   FREQ. DURATION   FREQ. DURA
  BUS# X-- NAME --X BASKV   BUS# X-- NAME --X BASKV        (OC/Y)  (HOUR)    (OC/Y)  (HOUR)   (OC/Y)  (HO
   601 TRANS 1     13.800   1001 EXTER 1     115.00  1     0.1000   10.00    1.0000   10.00    0.0000
   602 TRANS 2     13.800   1002 EXTERN 2    115.00  1     1.0000   10.00    0.0000    0.00    0.0000

<----------------BREAKER AND SWITCH----------------> ATTRIBUTE  <-----FAULT-----> <--UNSCHEDULED--> <
<-------FROM BUS---------> <---------TO BUS----------> CKT                        FREQ. DURATION   FREQ. DURATION
  BUS# X-- NAME --X BASKV   BUS# X-- NAME --X BASKV                               (OC/Y)  (HOUR)   (OC/Y)  (HOUR)
   601 TRANS 1     13.800   605 BUS SECT 1   13.800  *                *NORM-CLOSE   0.0020   12.00   0.0000    0.00
   605 BUS SECT 1  13.800   609 LOAD 3       13.800  *                *NORM-CLOSE   0.0020   12.00   0.0000    0.00
   612 LOAD 6      13.800   606 BUS SECT 2   13.800  *                *NORM-CLOSE   0.0020   12.00   0.0000    0.00
   601 TRANS 1     13.800   605 BUS SECT 1   13.800  @1               NORM-CLOSE   0.0100   96.00   0.0350   24.00
   606 BUS SECT 2  13.800   612 LOAD 6       13.800  @1               NORM-CLOSE   0.0100   96.00   0.0350   24.00
```

Figure 6-58. Sample Substation Component Report

Figure 6.72. Sample Substation Component Report

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Reactive power is neglected in the substation reliability assessment. The active power load curtailment due to a failure event is equal to the total desired load minus the amount of power that can be transferred from the source system to the sink system. The total load is the sum of the equivalent loads at the sink buses. The equivalent load at a sink bus is the sum of the MW loads at the bus plus the MVA ratings (according to the selected rating set and percentage) of all branches connected between the sink bus and the system outside the substation. Calculation of the maximum transfer capability is based on a transportation network flow algorithm which takes into account only branch capacities (branch MVA ratings) and ignores branch impedances and bus voltages.

The computed interrupted power (I.P.) and expected unserved energy (E.U.E.) are included in the Bus Load Curtailment Report and the Substation Load Curtailment Report shown below. Expected unserved energy indices are normalized for the entire year.

Contingency Summary Report

A sample report shown in Figure 6.73, "Sample Contingency Summary Report" contains the contingency events and the computed reliability results at the post-fault and the post-switching stages. The columns headed by PRIMARY and MD show the primary contingency labels and the corresponding failure modes. Each failure mode is identified by one letter: U for unscheduled outage, F for single-component fault, S for scheduled maintenance and C for common mode fault. Similarly, the columns headed by SECONDARY and MD contain the secondary contingency labels and the corresponding failure modes.

There are two stages for a single-component fault or a common mode fault: post-fault and post-switching. The secondary fault may occur in either stage of the primary fault. A failure mode of "F" indicates that the secondary fault occurs during the post-fault stage of the primary fault, while a failure mode PF indicates that the secondary fault occurs during the post-switching stage of the primary fault. Similarly, failure modes C and PC indicates, respectively, whether the secondary fault occurs during the post-fault or post-switching states of a common mode primary fault.

If a failure event involves a stuck breaker, the label of the breaker that fails to open is shown in the STUCK BREAKER column.

This summary report only displays those contingency events that result in load curtailment during either the post-fault or fault-switching stage. The amount of load curtailed in MW, the frequency in occurrences/year and the probability in hours/year are shown for each stage.

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<-- CONTINGENCY LABEL ----->			STUCK	<------ POST FAULT ------>		<---- POST SWITCHING ---->		
PRIMARY	MD	SECONDARY	MD BREAKER	LOAD CU. (MW)	FREQ. (OC/Y)	PROB. (H/Y)	LOAD CU. (MW)	FREQ. (OC/Y)	PROB. (H/Y)
BUS 4	F	NONE	NONE	60.00	0.1400	0.14	60.00	0.1400	0.39
BUS 4	M	NONE	NONE	0.00	0.0000	0.00	60.00	1.0000	89.00
BUS 4	PF	LINE 9	F NONE	60.00	0.0000	0.00	0.00	0.0000	0.00
SWITCH*5	F	NONE	NONE	60.00	0.0020	0.00	60.00	0.0020	0.03
LINE 6	U	NONE	NONE	0.00	0.0000	0.00	60.00	1.0000	10.00
BUS 7	M	BREAKER 41	F NONE	60.00	0.0001	0.00	25.00	0.0001	0.00
BUS 7	M	BREAKER 44	F NONE	25.00	0.0001	0.00	25.00	0.0001	0.00
SWITCH*8	F	NONE	NONE	25.00	0.0020	0.00	25.00	0.0020	0.03
LINE 9	F	NONE	NONE	25.00	1.0000	1.00	25.00	1.0000	11.00
LINE 9	F	NONE	BREAKER 37	25.00	0.0000	0.00	25.00	0.0000	0.00
LINE 9	F	BREAKER 42	U NONE	0.00	0.0000	0.00	25.00	0.0000	0.00

Figure 6-59. Sample Contingency Summary Report

Figure 6.73: Sample Contingency Summary Report

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As discussed above, SRA adds two implicit switches for each breaker. The faults and outages can occur on both implicit and explicit switches. An * in the contingency event label for a switch, such as SWITCH*5 in the sample report, indicates that the switch is an implicit one.

The bottom of the summary report provides descriptions of the contingencies, as shown in Figure 6.74, "Sample Descriptions of the Contingencies in the Summary Report". For scheduled maintenance and unscheduled maintenance failures, an OPEN keyword in the event description record indicates that the component is taken out of service. For single-component faults, it indicates that a fault is applied on the component which requires operations of breakers and switches to isolate the fault component(s). For an implicit switch fault, the contingency description identifies the breaker branch for which the switch is added, with the from bus corresponding to the side with the faulted switch.

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```
CONTINGENCY LEGEND:
LABEL      EVENTS
BUS 1      : OPEN BUS     601 TRANS 1     13.800
SWITCH 2   : OPEN LINE FROM BUS   601 TRANS 1     13.800 TO BUS    605 BUS SECT 1  13.800 CKT *1
BUS 4      : OPEN BUS     602 TRANS 2     13.800
SWITCH*5   : OPEN LINE FROM BUS   602 TRANS 2     13.800 TO BUS    606 BUS SECT 2  13.800 CKT *
BUS 7      : OPEN BUS     605 BUS SECT 1  13.800
```

Figure 6-60. Sample Descriptions of the Contingencies in the Summary Report

Figure 6.74. Sample Descriptions of the Contingencies in the Summary Report

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Bus Load Curtailment Report

This report lists, for each bus in the sink subsystem, the total load in MW, the total frequency and probability of contingency events that result in load curtailment at the bus, the interrupted power (I.P.) in MW per year and the expected unserved energy (E.U.E.) in MWHR per year in the post-fault and post-switching stages. A sample report is shown in Figure 6.76, "Sample Bus Load Curtailment Report".

Substation Load Curtailment Report

This report is divided into post-fault and post-switching stages. Within each stage, the failure events that resulted in similar amounts of load curtailments are grouped together. For each group, the report lists the corresponding failure event labels, the load curtailment in MW, the frequency in occurrences per year and the probability in hours per year. Also shown for each group are the frequency, the probability, the interrupted power (I.P.) and the expected unserved energy (E.U.E.). Part of a substation load curtailment report is shown in Figure 6.75, "Sample of Substation Load Curtailment Report".

```
CONTINGENCIES CAUSING LOAD CURTAILMENT OF  40.00 (MW)
<----- CONTINGENCY LABEL  ----> STUCK       <---- POST SWITCHING ---->
PRIMARY      MD SCONDARY     MD BREAKER      LOAD CU.    FREQ.     PROB.
                                               (MW)     (OC/Y)    (H/Y)
BREAKER 37    M BUS 13        F NONE          40.00    0.0001     0.00

SUMMARY OF THE GROUP    FREQ.     PROB.      I.P.    E.U.E.
                       (OC/Y)     (H/Y)    (MW/Y)  (MWH/Y)
                       0.0001      0.00      0.01     0.01
```

Figure 6.75. Sample of Substation Load Curtailment Report

```
<-----SINK BUS NAME----->     LOAD <---------- POST FAULT ----------> <---------- POST SWITCHING -
  BUS# X-- NAME --X BASKV      (MW)    FREQ.    PROB.     I.P.   E.U.E.    FREQ.    PROB.     I.P.
                                       (OC/Y)    (H/Y)   (MW/Y)  (MWH/Y)   (OC/Y)    (H/Y)   (MW/Y)
   607 LOAD 1      13.800      10.00   1.6441     0.00    16.34     0.00   6.6766   102.92    38.43
   608 LOAD 2      13.800      15.00   1.6448     0.00    24.52     0.00   6.6774   102.92    58.26
   609 LOAD 3      13.800      10.00   1.6441     0.00    16.34     0.00   6.6766   102.92    38.43
   610 LOAD 4      13.800      10.00   3.8257     0.00    38.16     0.00   6.8260   103.27    39.92
   611 LOAD 5      13.800       8.00   3.8253     0.00    30.52     0.00   6.8256   103.27    31.82
   612 LOAD 6      13.800       7.00   3.8253     0.00    26.71     0.00   6.8256   103.27    27.80
SUBSTATION TOTAL                60.00   3.8263     0.00   152.58     0.00   6.8268   103.27   234.67
```

Figure 6-46. Sample Bus Load Curtailment Report

Figure 6.76. Sample Bus Load Curtailment Report

Additional Information
PSS®E GUI Users Guide, Running Substation Reliability Assessment

6.19. Performing PV or QV Analyses

The PV/QV analyses that are described in this chapter are designed for studies of low voltage stability, which could be analyzed as a steady-state problem. They are power flow based analyses used to assess voltage variations with active and reactive power change. Two methods are used to determine the loading limits imposed by voltage stability under the steady-state conditions.

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The PV/QV analyses do not provide solutions to specific problem but function as tools that can be directed by the user to perform analyses in the solution of problems associated with the steady-state voltage stability of power systems.

6.20. PV Analysis

Run *Activity* PV Analysis - GUI
Power Flow > *Solution* > PV analysis... *[PV Analysis]*
Interrupt Control Codes
AB, CM, NM, DC, FD, MO

Table 6.28. PV Analysis Options

Option	Description
Base-case Solution options	Select solution options for the series of power flow calculations performed to obtain the PV curves for the base case transfer solutions (see Section 6.2, "Power Flow Solution Methods"). Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustment: • Lock all • Enable all • Enable continuous, disable discrete Enable or disable phase shift adjustment. Enable or disable dc tap adjustment. Induction motor treatment flag when it fails to solve due to low terminal voltage: • set to the "stalled" state

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Option	Description
	• set to the "tripped" state
Contingency-case Solution options	The contingency-case solution options allow for a different set of automatic adjustments of the contingency case solutions than those used in the base case transfer increment solutions (see Section 6.2, "Power Flow Solution Methods"). Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustment: • Lock all • Enable all • Enable continuous, disable discrete Enable or disable phase shift adjustment. Enable or disable dc tap adjustment. Induction motor treatment flag when it fails to solve due to low terminal voltage: • set to the "stalled" state • set to the "tripped" state
Apply contingencies to base case transfer solutions (GUI option to select the API routine to use)	If this option is enabled, the PV_ENGINE_6 API routine is used to calculate the analysis; PV_ENGINE_6 *must* be used if the base case and contingency case solution options are not identical, if a negative minimum transfer increment is specified, if a specified power factor is to be used when increasing load using transfer methods 2, 3 and 4 (rather than retaining the original load power factors), or if a re-dispatch is to be performed to satisfy island power unbalances resulting from the application of contingencies.

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Option	Description
	If these options are disabled, the PV_ENGINE_1A API routine is used to calculate the analysis.
Solution Engine	Select the solution engines for the series of power flow calculations and contingency studies to obtain the PV curves (see Section 6.2, "Power Flow Solution Methods"). • Fixed slope decoupled Newton-Raphson (FDNS) • Full Newton-Raphson (FNSL) • Optimized fixed slope decoupled Newton-Raphson (FDNS)
Non-divergent solution	Flag to enable or disable option in each power flow solution.
Var Limit Codes	Select the var limit codes for the following cases: • Base case transfer increment solutions • Contingency case power flow solutions The var limit codes indicate when, during the power flow solutions, plant reactive power limits are to be honored. They can be set as follows: • Apply immediately on the first iteration • Initially ignore, then apply when bus mismatches have been reduced to a pre-set multiple of the convergence tolerance
Transfer Dispatch Methods	Dispatch codes for study and opposing dispatch calculations in PV analysis.
	DFAX generation: Participating buses and their participation factors are taken from the DFAX file. Only subsystem buses with one or more in-service machines with positive active power generation are participating buses. If a PARTICIPATE block is specified for the subsystem in the Subsystem Description Data file, the participation factors specified in the file are used; participating buses for which no participation factor was specified are excluded. If no PARTICIPATE block is specified, it uses the MBASEs of the participating machines.
	DFAX load: Participating buses and their participation factors are taken from the DFAX file. Only subsystem buses with one or more in-service loads with positive constant MVA load active power are par-

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Option	Description
	ticipating buses. Participation factors specified in the subsystem's PARTICIPATE block are used; participating buses for which no participation factor was specified are excluded.
Transfer Dispatch Methods (*continued*)	*DFAX generation or load*: Participating buses and their participation factors are taken from the DFAX file. Only subsystem buses with either one or more in-service machines where active power generation is positive, or one or more in-service loads where constant MVA load active power is positive, are participating buses. Participating buses and participation factors are handled as described in *DFAX generation* and *DFAX load*. If both generation and load meeting the above criteria are present at a bus, the generation is dispatched if the bus is in the study subsystem, and the load is dispatched, if the bus is in the opposing subsystem.
	Subsystem load: Participating buses are subsystem buses with one or more in-service loads where constant MVA load active power is positive; each bus' participation factor is the total of its positive active power constant MVA load.
	Subsystem machines (MW): Participating buses are subsystem buses with one or more in-service machines where active power generation is positive; each bus' participation factor is the total of the active power generation of its machines with positive active power generation.
	Subsystem machines (MBASE): Participating buses are subsystem buses with one or more in-service machines where active power generation is positive; each bus' participation factor is the total of the MBASEs of its machines with positive active power generation.
	Subsystem machines (Reserve): Participating buses are subsystem buses with one or more in-service machines where active power generation is positive; each bus' participation factor is the total of the reserve (PT-PG) of its machines with positive values of both active power generation and reserve.
	Subsystem machines (ECDI w/o Unit Commitment): Participating buses are subsystem buses with one or more in-service machines specified in the Economic Dispatch Data file; machines participate in the transfer based on an equal incremental cost dispatch as implemented by activity ECDI. No machine status changes are implemented.

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Option	Description
	Subsystem machines (ECDI w/ Unit Commitment): Participating buses are subsystem buses with one or more in-service machines specified in the Economic Dispatch Data file; machines participate in the transfer based on an equal incremental cost dispatch as implemented by activity ECDI. If necessary, machines are placed into service or removed from service.
Power factor for load increases	Power factor used in dispatch methods 2, 3 and 4 for load increases. Specify a power factor 0.0 to retain the original power factor at each load that is changed (default).
Generation limits flag for transfer dispatch methods using generation	Select one of the following, if appropriate: • No limits (default) • Honor machine active power limits
Positive load flag for transfer dispatch methods using loads	Select one of the following, if appropriate: • No limits (default) • Enforce non-negative active power constant MVA loads
Mismatch tolerance	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the process is terminated. This value is also used as the convergence tolerance in the power flow solution of each contingency and transfer increment case (see Section 6.2, "Power Flow Solution Methods").
Initial transfer increment (MW)	Starting transfer increment value. This value will be the initial transfer increment step size in MW between the two defined subsystems.
Min monitored bus voltage in pu for terminating analysis for this system condition	Minimum voltage value for termination of calculation. If the low voltage check is enabled, calculations for the current system condition will stop if any monitored voltage is less than this value.
Transfer increment tolerance (MW)	Largest acceptable difference in transfer level between a transfer level for which a solution is found and a higher transfer level which fails to achieve convergence.
Maximum incremental transfer (MW)	Maximum transfer in MW between the two specified subsystems.
Minimum incremental transfer (MW)	Minimum transfer in MW between the two specified subsystems.
Branch overload threshold (%)	Threshold in determining branch overloads.
Rating set	Line loading limit used in determining overloads. If the branch loading check is enabled, calculations

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Option	Description
	will stop if any monitored branches are overloaded beyond this limit.
Low voltage limit check	Flag to enable a low voltage limit check for monitored voltages. If enabled, the calculations for the current system condition will stop if any low voltage violations are detected. This flag may be set as follows: 0 to disable the low voltage check. 1 to enable the low voltage check using the value specified as the minimum monitored bus voltage. 2 to enable the low voltage check using the normal lower limit for base and contingency case solutions. 3 to enable the low voltage check using the normal lower limit for base case solutions, and the emergency lower limit for contingency case solutions. 4 to enable the low voltage check using the emergency lower limit for base and contingency case solutions.
Enable the branch loading check	If enabled, the calculations for the current system condition will stop if any monitored branches are overloaded beyond the limit specified by overload threshold and rating set.
Subsystem selection	Subsystem names of the "source" and "sink" subsystems.
Dispatch mode	Enables the option to re-dispatch to satisfy island power unbalances resulting from the application of contingencies. See Table 6-7 AC Contingency Solution Options.
Dispatch subsystem	Name of the the subsystem for the generation dispatch. All subsystem buses with one or more in-service machines where active power generation is positive are participating in the generation dispatch. If several islands exist in the system, participating machines for each island are connected to dispatch subsystem buses in the island.
Distribution Factor Data File	Name of a Distribution Factor Data File (*.dfx) previously built by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files (refer to ???). The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified Distribu-

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Option	Description
	tion Factor Data file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by the AC Contingency Solution function.
Load Throwover Data File	Name of an optional Load Throwover Data File (*.thr) that contains data records of the following form: `IBUS, JBUS` where IBUS and JBUS are bus numbers. If branch outage contingency events isolate bus IBUS, the user can elect to move the load to bus JBUS. Bus IBUS and/or JBUS may be a dummy bus of a multi section line grouping. Data records may be entered in any order. Input is terminated with a record specifying an IBUS value of zero.
Economic Dispatch File	Name of an Economic Dispatch Data File (*.ecd) that contains machine dispatch data records for use if one of the ECDI-based transfer methods is specified as the transfer dispatch method for the "source" and/or "sink" subsystems.
Unit Inertia and Governor Data File	Name of the Inertia and Governor Response Data File (*.inl). This file is required if the generation dispatch is enabled and either the inertial or governor droop dispatch method is selected. If the PMAX or reserve method is selected, the specification of this file is optional. If a file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used.
Output file (results)	Name of the P-V Results File (*.*pv*) into which the results of the P-V calculation will be stored. The results file is preserved so that you may view the P-V calculation results at a later time by simply selecting the P-V Results File rather than re-running the P-V calculations.
ZIP Archive option	Flag to enable or disable the writing of a ZIP Archive Output File. If enabled, each system condition may be preserved either at its largest solved incremental transfer level, or at all of its solved incremental transfer levels.
ZIP Archive Output File	(*.zip) an optional file designated by the user as the destination for data used to place contingency case solutions into the working case.

Two P_V analysis API routines are available. In the PV_ENGINE_1A API routine, the same set of automatic adjustments is applied in base case and contingency case power flow solutions. The basic solution flow is:

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- Apply the contingency to the base case solution at the initial transfer level (i.e., at 0.0 incremental transfer) and solve the power flow.
- In the solved contingency case at the initial transfer level, increment the transfer to transfer level 1 and solve.
- From the solved contingency case at transfer level 1, increment the transfer to transfer level 2 and solve.
- ...

The PV_ENGINE_6 API routine allows a different set of automatic adjustments to be applied in contingency case power flow solutions. The basic solution flow for contingency cases is:

- Apply the contingency to the base case solution at the initial transfer level (i.e., at 0.0 incremental transfer) and solve the power flow.
- Apply the contingency to the base case solution at transfer level 1 and solve it.
- Apply the contingency to the base case solution at transfer level 2 and solve it.
- ...

PV_ENGINE_6 *must* be used in *any* of the following conditions:

- A different set of automatic adjustments is specified for the base case and contingency case solutions.
- A negative minimum incremental transfer is specified.
- A power unbalance dispatch mode is specified.
- A power factor is specified for use in incrementing the transfer by load increases.
- The ZIP Archive option is specified as either 1 or 2.

Either PV_ENGINE_6 or PV_ENGINE_1A may be used if *all* of the following conditions are satisfied:

- The same set of automatic adjustments is specified for the base case and contingency case solutions.
- 0.0 is specified for the minimum incremental transfer.
- 0 is specified for the dispatch mode.
- 0.0 is specified for the power factor for load increases.
- 0 is specified for the ZIP Archive option.

The results window for PV analysis will be displayed after calculations are initiated.

Select the type of graph you wish to plot and select the items of interest from the following:

Bus voltages: Plots bus voltages against incremental power transfer of selected base case or contingencies.

Generator MW: Plots generator MW output against incremental power transfer of selected base case or contingencies.

Branch MVA Flow: Plots branch MVA against incremental power transfer of selected base case or contingencies.

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Interface MW Flow: Plots interface MW against incremental power transfer of selected base case or contingencies.

Bus voltage against interface flow: Plots the selected bus voltage against interface flow.

Items in the list may be selected individually or as a continuous set. The results are plotted as a graph. Graph axis parameters are fully adjustable. Scale values are adjustable for both vertical and horizontal axes. Additional options are available for printing clarity.

6.20.1. Retrieving Results from a Previous PV Analysis

Run Activity PV Analysis, Previous Results Retrieved - GUI
Power Flow > *Solution* > PV analysis using previous results... *[PV Parameters]*

Select a previously saved PV results file (*.pv). Parameters used for the selected PV analysis will be shown. These values are for reference only and are not editable.

6.20.2. Implementing a Specific PV Transfer

Run Activity PV Transfer -*GUI*
Power Flow > *Solution* > Implement PV transfer... *[Implement PV Transfer]*

It is often important to study the performance of the power system at or near the transfer limit that had been determined by a PV analysis study. The implement PV transfer function may be used to implement a transfer between designated source and sink systems using the same transfer dispatch methods that are available in the PV analysis function.

To impose a designated transfer:

1. Open the Saved Case file containing the power flow case on which the transfer is to be imposed. Normally, the same Saved Case file on which an earlier PV analysis calculation had been performed is specified.
2. Open the *[Implement PV Transfer]* dialog, which will contain default settings.
3. Specify the same Distribution Factor Data file, source and sink systems, and transfer dispatch method data as was specified in the earlier PV analysis calculation.
4. Specify the desired transfer increment.
5. Launch the transfer.

The implement PV transfer function then changes the generation or load, as appropriate, at those subsystem buses participating in the transfer.

Details on the transfer may be examined using activity DIFF.

Additional Information
PSS®E GUI Users Guide, PV Analysis

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6.21. QV Analysis

Run *Activity* QV Analysis - GU*I*
Power Flow > *Solution* > QV analysis... *[QV Analysis]*
Interrupt Control Codes
AB, CM, NM, DC, FD, MO

Table 6.29. QV Analysis Options

Option	Description
Bus number	Bus number for analysis. Filters may be applied to the bus selector for name mask, number range, base voltage, and bus type.
Solution options	Select solution options for the series of power flow calculations performed to obtain the QV curves (see Section 6.2, "Power Flow Solution Methods"). Tap adjustment: • Lock taps • Stepping • Direct Area interchange control: • Disabled • Tie lines only • Tie lines and loads Switched shunt adjustments: • Lock all • Enable all • Enable continuous, disable discrete Enable or disable phase shift adjustment. Enable or disable dc tap adjustment. Induction motor treatment flag when it fails to solve due to low terminal voltage: • set to the "stalled" state

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Option	Description
	• set to the "tripped" state
Non-divergent solution	Flag to enable or disable option in each power flow solution.
Solution Engine	Select the solution engines for the series of power flow calculations and contingency studies to obtain the PV curves (see Section 6.2, "Power Flow Solution Methods"). • Fixed slope decoupled Newton-Raphson (FDNS) • Full Newton-Raphson (FNSL) • Optimized fixed slope decoupled Newton-Raphson (FDNS)
Var Limit Codes	Select the var limit code for the following cases: • VHI power flow solution • Subsequent voltage decrement cases The var limit codes can be set for as follows: • Apply immediately on the first iteration • Initially ignore, then apply when bus mismatches have been reduced to a pre-set multiple of the convergence tolerance
Dispatch mode	Enables the option to re-dispatch to satisfy island power unbalances resulting from the application of contingencies. See Table 6.9, "AC Contingency Solution Options".
Mismatch tolerance (MW and Mvar)	Tolerance used to check for the largest initial active or reactive power mismatch. If exceeded, the process is terminated. This value is also used as the convergence tolerance in the power flow solution of each contingency case (see Section 6.2, "Power Flow Solution Methods").
Initial (maximum) per unit voltage setpoint at study bus (VHI)	Maximum voltage of study bus
Minimum per unit voltage setpoint at study bus (VLO)	Minimum voltage of study bus
Per unit voltage setpoint decrement (positive) at study bus (DLTAV)	Voltage increment size from VLO to VHI.
Dispatch subsystem	Name of the subsystem for the generation dispatch. All subsystem buses with one or more in-service machines where active power generation is positive are participating in the generation dispatch. If several islands exist in the system, participating machines

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Option	Description
	for each island are connected to dispatch subsystem buses in the island.
Distribution Factor Data File	Name of a Distribution Factor Data File (*.dfx) previously built by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files (refer to ???). The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified Distribution Factor Data file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by the AC Contingency Solution function.
Load Throwover Data File	Name of an optional Load Throwover Data File (*.thr) that contains data records of the following form: `IBUS, JBUS` where IBUS and JBUS are bus numbers. If branch outage contingency events isolate bus IBUS, the user can elect to move the load to bus JBUS. Bus IBUS and/or JBUS may be a dummy bus of a multi section line grouping. Data records may be entered in any order. Input is terminated with a record specifying an IBUS value of zero.
Unit Inertia and Governor Data File	Name of the Inertia and Governor Response Data File (*.inl). This file is required if the generation dispatch is enabled and either the inertial or governor droop dispatch method is selected. If the PMAX or reserve method is selected, the specification of this file is optional. If a file is specified, machine active power limits are taken from it; otherwise, the machine active power limits in the working case are used.
Output file (results)	Name of the Q-V Results File (*.*qv*) into which the results of the Q-V calculation will be stored. The results file is preserved so that you may view the Q-V calculation results at a later time by simply selecting the Q-V Results File rather than re-running the Q-V calculations.
ZIP Archive option	Flag to enable or disable the writing of a ZIP Archive Output File. If enabled, each system condition may be preserved either at its lowest solved study bus voltage level, or at all solved study bus voltage levels.

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Option	Description
ZIP Archive Output File	(*.zip) an optional file designated by the user as the destination for data used to place contingency case solutions into the working case.

The results window for QV analysis will be displayed after calculations are initiated. Select the base case and/or contingencies. Items in the list may be selected individually or as a continuous set.

The results are plotted as a graph. Graph axis parameters are fully adjustable. Scale values are adjustable for both vertical and horizontal axes. Additional options are available for printing clarity.

6.21.1. Retrieving Results from a Previous QV Analysis

Run Activity QV Analysis, Previous Results Retrieved - GUI
Power Flow > *Solution* > QV analysis using previous results... *[QV Parameters]*

Select a previously saved QV results file (*.qv). Parameters used for the selected QV analysis will be shown. These values are for reference only and are not editable.

Additional Information
PSS®E GUI Users Guide, QV Analysis

6.22. Basic Engineering Guide to PV and QV Curves Applications

6.22.1. Objective

The objective of a PV and QV curves is to determine the ability of a power system to maintain voltage stability at all the buses in the system under normal and abnormal steady-state operating conditions. They are useful, for example:

- To show the voltage collapse point of the buses in the power system network.
- To study the maximum transfer of power between buses before voltage collapse point.
- To size the reactive power compensation devices required at relevant buses to prevent voltage collapse.
- To study the influence of generator, loads and reactive power compensation devices on the network.

The PV and QV curves are obtained through a series of ac power flow solutions. The PV curve is a representation of voltage change as a result of increased power transfer between two systems, and the QV curve is a representation of reactive power demand by a bus or buses as voltage level changes.

6.22.2. PV Analysis (PV Curves) Applications

PV curves are parametric study involving a series of ac power flows that monitor the changes in one set of power flow variables with respect to another in a systematic fashion. This approach is a powerful method for

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determining transfer limits that account for voltage and reactive flow effects. As power transfer is increased, voltage decreases at some buses on or near the transfer path. The transfer capacity where voltage reaches the low voltage criterion is the low voltage transfer limit. Transfer can continue to increase until the solution identifies a condition of voltage collapse; this is the voltage collapse transfer limit.

This can be demonstrated using a simple two terminals network as in Figure 6.77, "Two Terminals Simple Network". From reference P. Kundur, Power System Stability and Control, McGraw-Hill 1994, the power flow relationship between the source and the load can be summarized by Equation 6.35 and Equation 6.36.

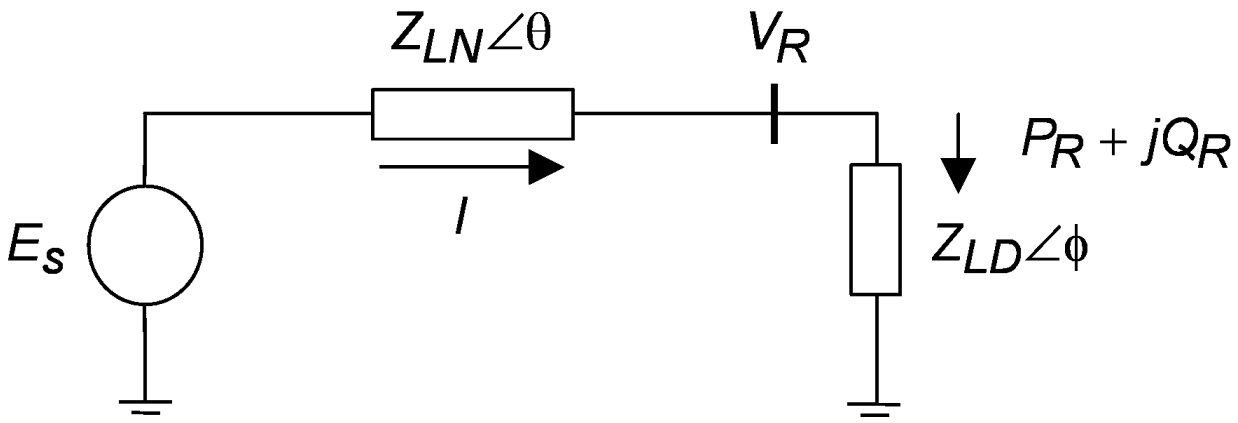

Figure 6.77. Two Terminals Simple Network

$$P_R = V_R I \cos\phi = \frac{Z_{LD}}{F}\left(\frac{E_s}{Z_{LN}}\right)^2 \cos\phi \quad (6.35)$$

$$V_R = \frac{1}{\sqrt{F'}}\frac{Z_{LD}}{Z_{LN}} E_s \quad (6.36)$$

where:

$$F = 1 + \left(\frac{Z_{LD}}{Z_{LN}}\right)^2 + 2\left(\frac{Z_{LD}}{Z_{LN}}\right)\cos(\theta - \phi) \quad (6.37)$$

The loading of the network can be increased by decreasing the value if Z_{LD}. This is done with E_s, load power factor and line parameters fixed. From Equation 6.35, as ZLD is decreased gradually the load power, P_R, increases, hence the power transmitted will increase. As the value of Z_{LD} approaches Z_{LN} the value of P_R starts to decrease gradually due to F. However, from Equation 6.36 as Z_{LD} decreases the receiving voltage V_R decreases gradually.

The plot of the relationship between voltage at the receiving end, V_R, and the load power, P_R, as the power transfer is increased due to increase loading, gives the PV curves similar in characteristic to the curve shown in Figure 6.78, "PV Curves Voltage and Incremental Power Transfer Characteristics".

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PV curves are typically used for the knee curve analysis. It is as named because of its distinctive shape at the point of voltage collapse as the power transfer increases, as shown in Figure 6.78, "PV Curves Voltage and Incremental Power Transfer Characteristics". Depending on the transfer path, different buses have different knee point. The buses closer to the transfer path will normally exhibit a more discernible knee point.

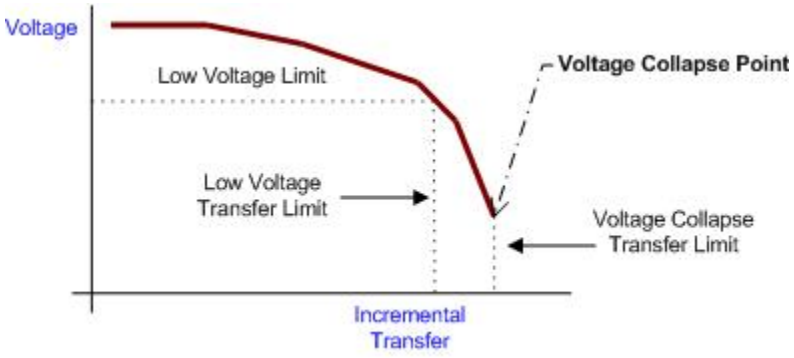

Figure 6.78. PV Curves Voltage and Incremental Power Transfer Characteristics

Voltage instability occurs at the knee point of the PV curve where the voltage drops rapidly with an increase in the transfer power flow. The power flow solution will not converge beyond this limit, indicating voltage instability. Operation at or near the stability limit is impractical and a satisfactory operating condition must be ensured to prevent voltage collapse.

In PSS®E, the PV curves are generated by selecting two subsystems where the power transfer between the subsystems is incremented in a defined step size for a series of ac power flow calculations while the bus voltages, generator outputs and the branch flows of the system are monitored. When the bus voltages are plotted as a function of the incremental power transfer the PV curves are obtained. One of the subsystems in the study must be defined as the study (source) system and another as the opposing (sink) system. The power flows from the study subsystem to the opposing subsystem.

Figure 6.79, "PV Curves Voltage and Incremental Power Transfer Characteristics for Bus 203 under Different Network Conditions" shows the PV curves of a bus 203 in an example network under normal and various contingency conditions. The maximum transfer limit for this bus in base case is approximately 750 MW. The maximum transfer limit decreases under contingency conditions. The response shown is expected because under network contingencies the loading of the line will increase. These curves can be used to set transfers or local generation dispatch so that the system will not fall below the knee point following a disturbance (i.e., loss of lines).

The PV curves for different buses under the same network conditions (base case or contingency) can also be plotted in the results window as shown in Figure 6.80, "PV Curves Voltage and Incremental Power Transfer Characteristics for Different Buses in Base Case". The plots of three different buses in base case indicate that the knee point is below the 90% of the nominal voltage and the transfer limit is approximately 750 MW. However, this transfer limit is not reasonable as voltages below 90% will cause motors in the system to stall. The transfer limit should be set for voltages in the vicinity of 95% of the nominal voltage.

The distribution of generations between the generators in the system as a result of the power transfer can be observed by analyzing their relationship as shown in Figure 6.81, "Generator Output Versus Power Transfer Curves".

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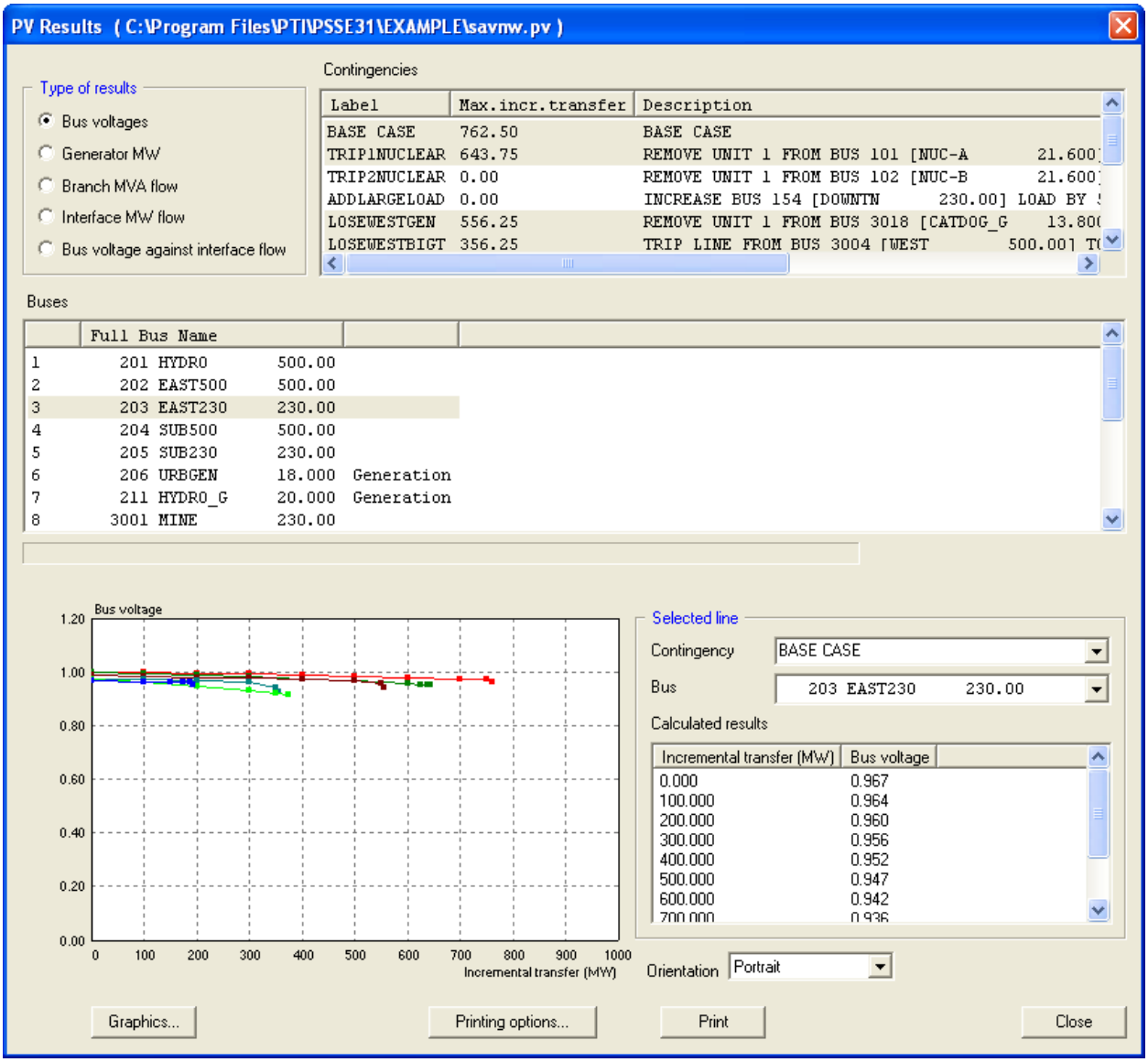

Figure 6.79. PV Curves Voltage and Incremental Power Transfer Characteristics for Bus 203 under Different Network Conditions

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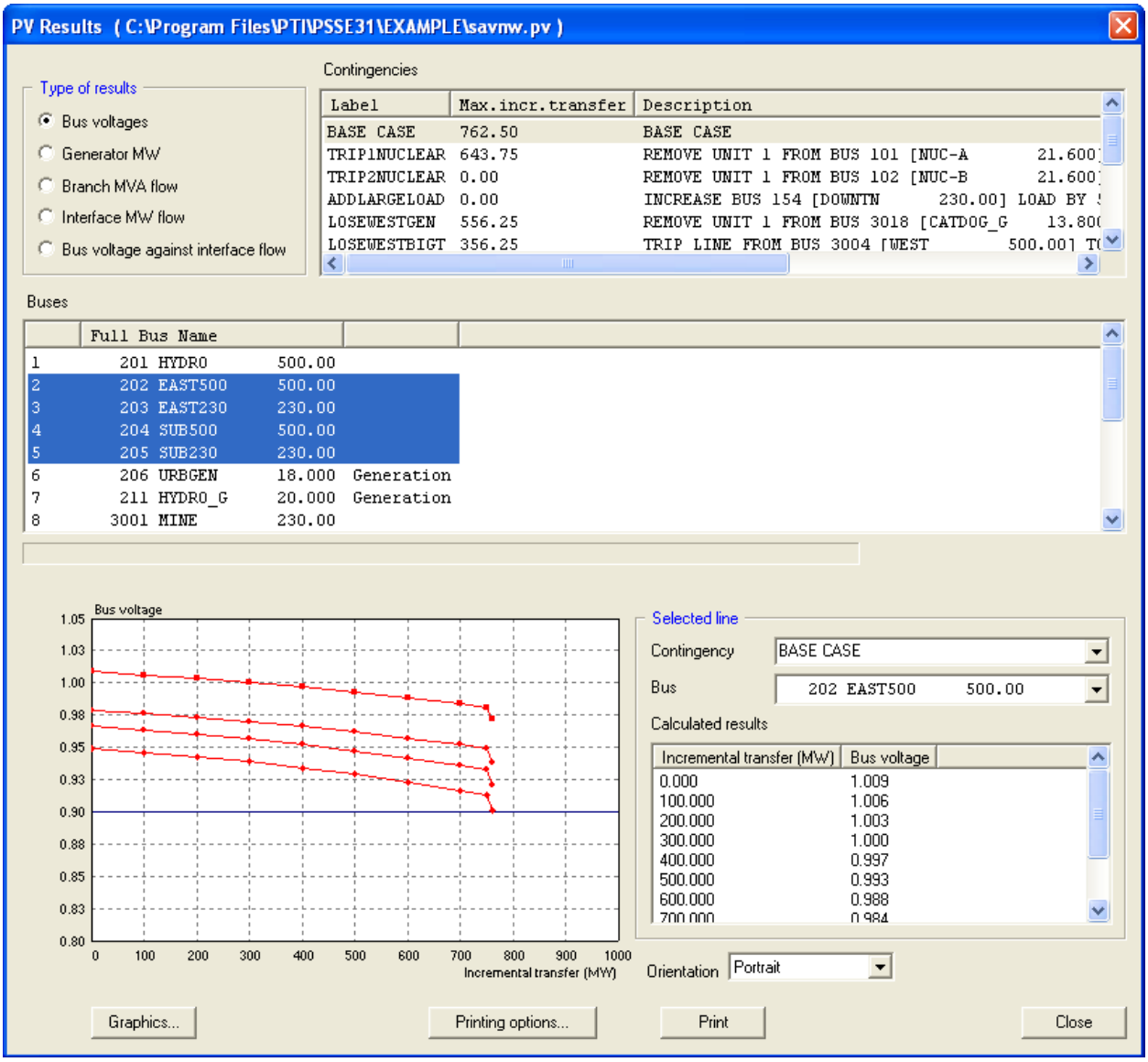

Figure 6.80. PV Curves Voltage and Incremental Power Transfer Characteristics for Different Buses in Base Case

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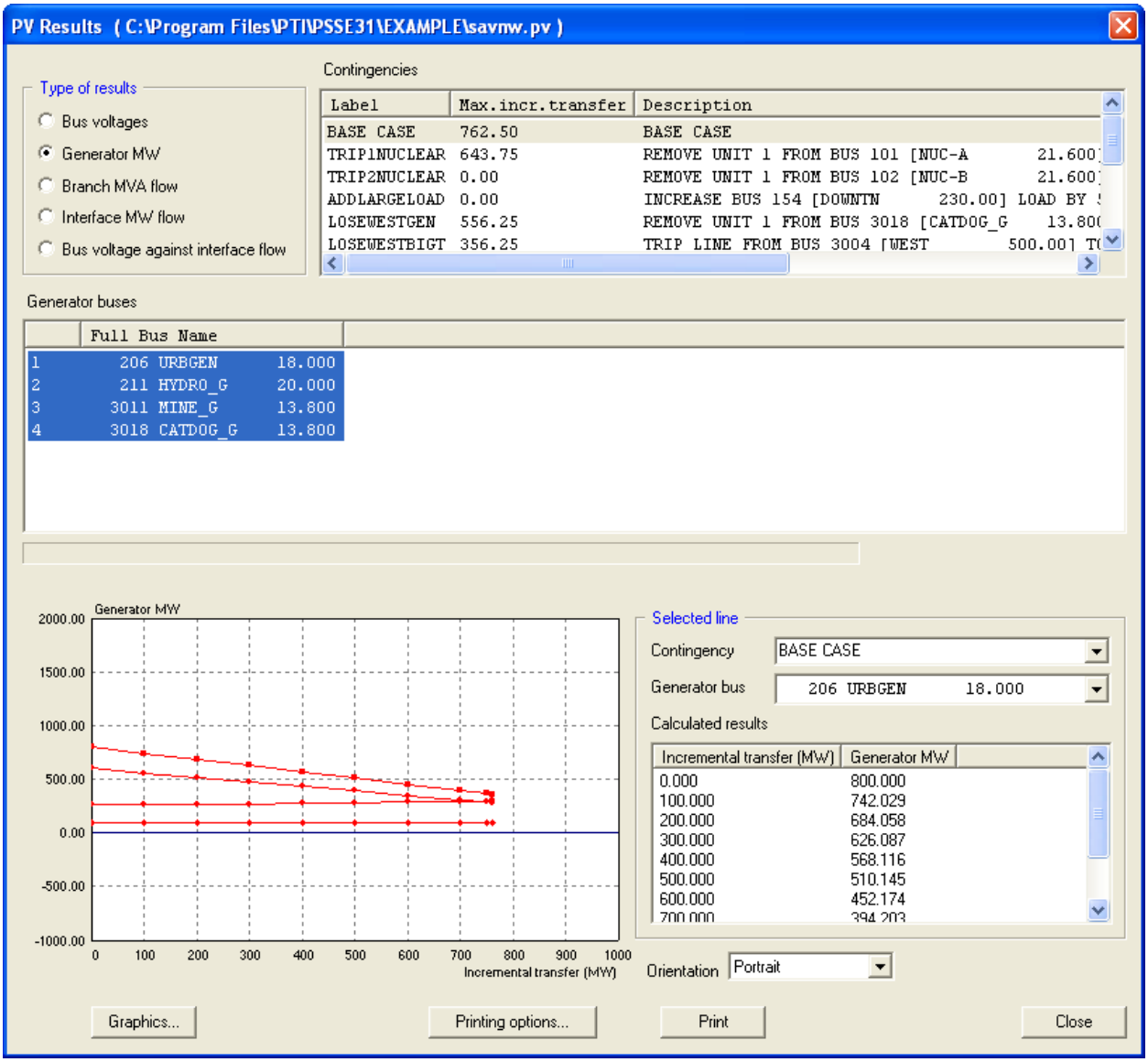

Figure 6.81. Generator Output Versus Power Transfer Curves

6.22.3. QV Analysis (QV Curves) Applications

In the PV curve analysis we have demonstrated the effect of active power flow on voltage instability. However, if we revisit Equation 6.35 and Equation 6.36, we see that the power factor of the load has a significant impact on the overall equations. This is to be expected because the voltage drop in the line is a function of both active and reactive power transfer. Hence, the QV curves may also be used to assess voltage stability of the system.

QV curves are used to determine the reactive power injection required at a bus in order to vary the bus voltage to the required value. The curve is obtained through a series of ac power flow calculations. Starting with the specified maximum per unit voltage setpoint at the study bus, the reactive power injections can

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be computed for a series of power flows as the voltage setpoint is decreased in steps, until the power flow demonstrates convergence difficulties as the system approaches the voltage collapse point.

Figure 6.82, "QV Curves for a Range of System Loading" is a typical of the QV curves that will be generated for a system that is stable at moderate loading and unstable at higher loadings.

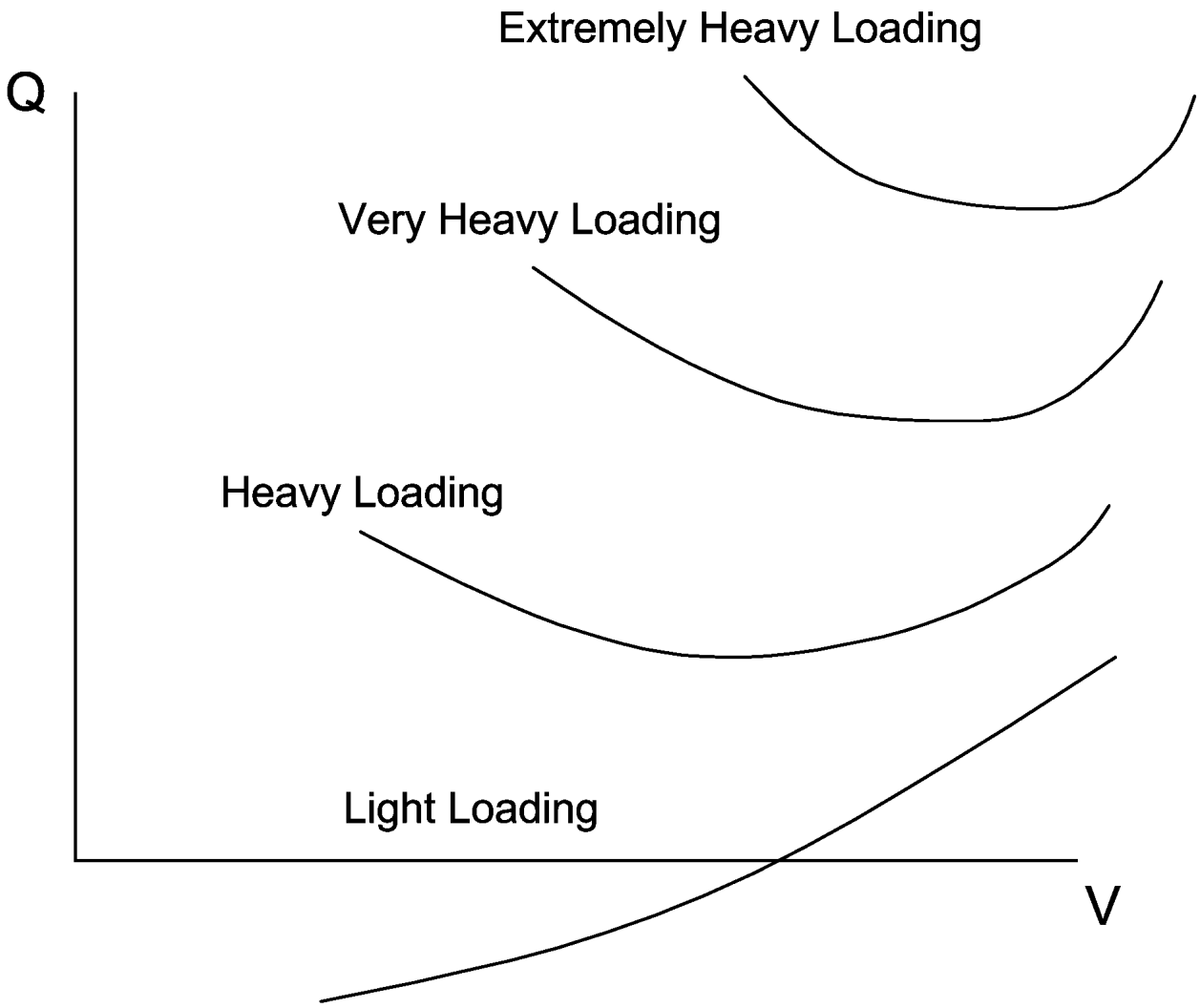

Figure 6.82. QV Curves for a Range of System Loading

The bottom of the QV curve, where the change of reactive power, Q, with respect to voltage, V (or derivative dQ/dV) is equal to zero, represents the voltage stability limit. Because all reactive power compensator devices are designed to operate satisfactorily when an increase in Q is accompanied by an increase in V, the operation on the right side of the QV curve is stable, whereas the operation on the left side is unstable. Also, voltage on the left side may be so low that the protective devices may be activated.

The bottom of the QV curves, in addition to identifying the stability limit, defines the minimum reactive power requirement for the stable operation. Hence, the QV curve can be used to examine the type and size of compensation needed to provide voltage stability. This can be performed by super-imposing the QV charac-

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teristic curves of the compensator devices on that of the system. For instance the capacitor characteristic can be drawn over the system's QV curves as shown in Figure 6.83, "QV Curves and Characteristics of a Capacitor Bank Required at Stable Operating Point".

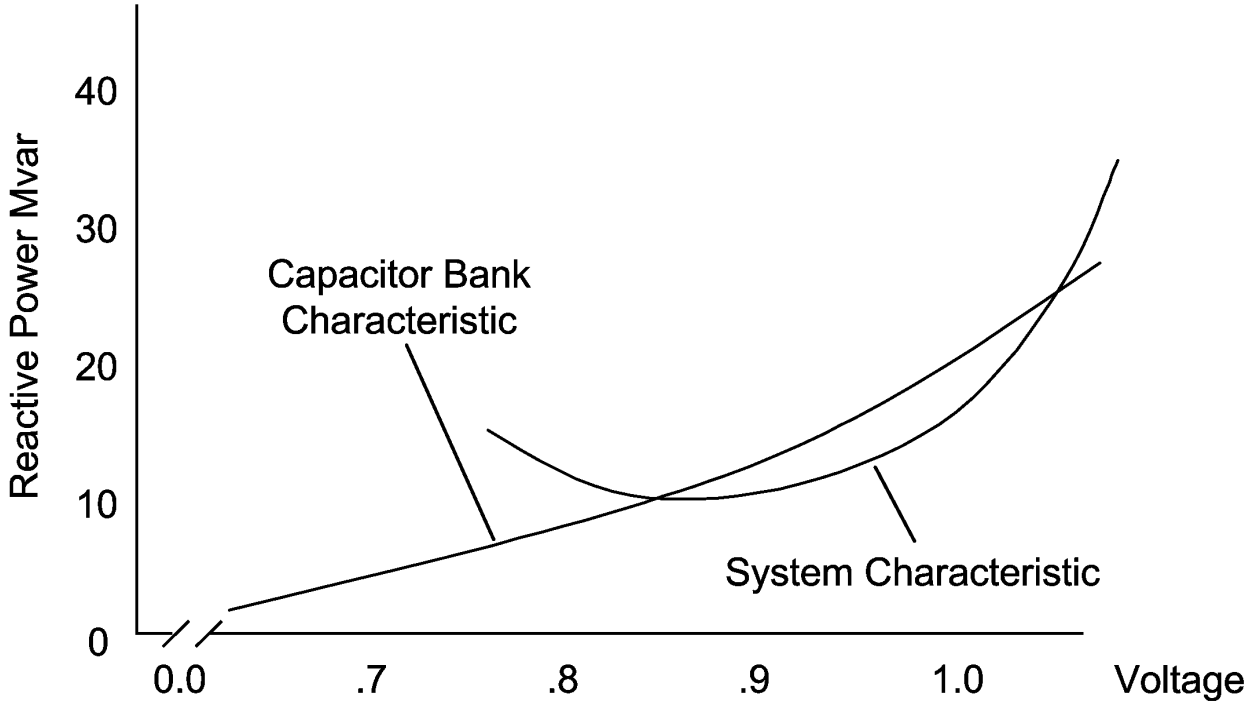

Figure 6.83. QV Curves and Characteristics of a Capacitor Bank Required at Stable Operating Point

Figure 6.84, "Compensator Operations and Size on Voltage Stability using QV Curves" shows the QV curves for a range of capacitor banks with different rating super-imposed on the system's QV curves under different loading conditions. From the plot we can determine that capacitor rating of 300 Mvar is required to maintain 1 pu voltage at loading of 1300 MW, 450 Mvar at 1500 MW and so on.

For the case of very high loading at 1900 MW, even though the capacitor bank rating of 950 Mvar can maintain a voltage of 1 pu, point B is not a stable operating point. If there is a drop in voltage from point B to B', the ability of the capacitor to supply reactive power is decreased more than the drop in requirement of the system. This will result in continuous drop in voltage. Alternatively, if the voltage is increased above point B, the capacitor will supply more reactive power than the increase in requirement of the system. This will result in continuous rise in voltage.

Hence, the criterion for stable operating point when using a reactive power compensator is as follow:

System dQ/dV > Compensator dQcomp/dV

where:

dQ/dV	is the change of the system's reactive power, Q, with respect to voltage, V.
dQcomp/dV	is the change of the compensator's reactive power output, Qcomp, with respect to voltage, V.

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For the case of light loading at 1300 MW with capacitor rating of 300 Mvar, point A is a stable operating point. If the voltage is increased from point A to A', the capacitor will supply less reactive power than the increase in system's demand, hence reducing the voltage to 1 pu. Alternatively, if the voltage is decreased from point A towards the bottom of the QV curve, the capacitor will supply more reactive power than the system's demand, hence returning the voltage to 1 pu.

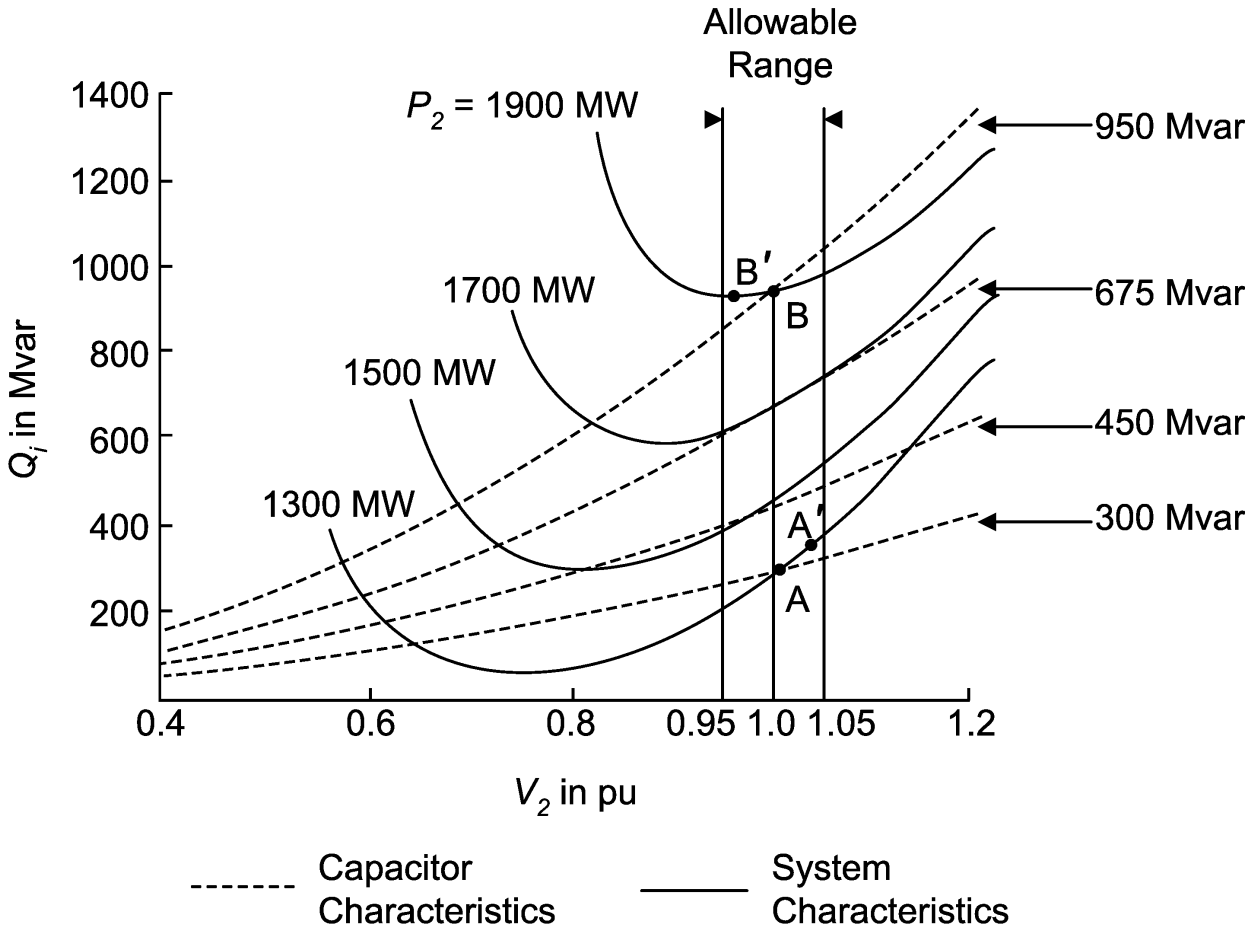

Figure 6.84. Compensator Operations and Size on Voltage Stability using QV Curves

In PSS®E, the QV curves are generated by artificially introducing a synchronous condenser, with high reactive power limits, at a bus to make this a PV bus. As the scheduled voltage set point (bus voltage) of the PV bus is varied in steps for a series of ac power flow calculations, the reactive power output from the condenser is monitored. When the reactive power is plotted as a function of the bus voltage a QV curves are obtained.

QV curves are commonly used to identify voltage stability issues and reactive power margin for specific locations in the power system under various loading and contingency conditions. The QV curves are also used as a method to size shunt reactive compensation at any particular bus to maintain the required scheduled voltage.

Figure 6.85, "QV Curves under Various Contingencies for Bus 103" shows QV curves for bus 203 in an example network. From the curves we can determine the reactive power required to hold the desired scheduled

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voltage in the base case and under various contingencies. In this example, the reactive power required is approximately 110 Mvar to hold the scheduled voltage at bus 203 at 1.0 pu in the base case.

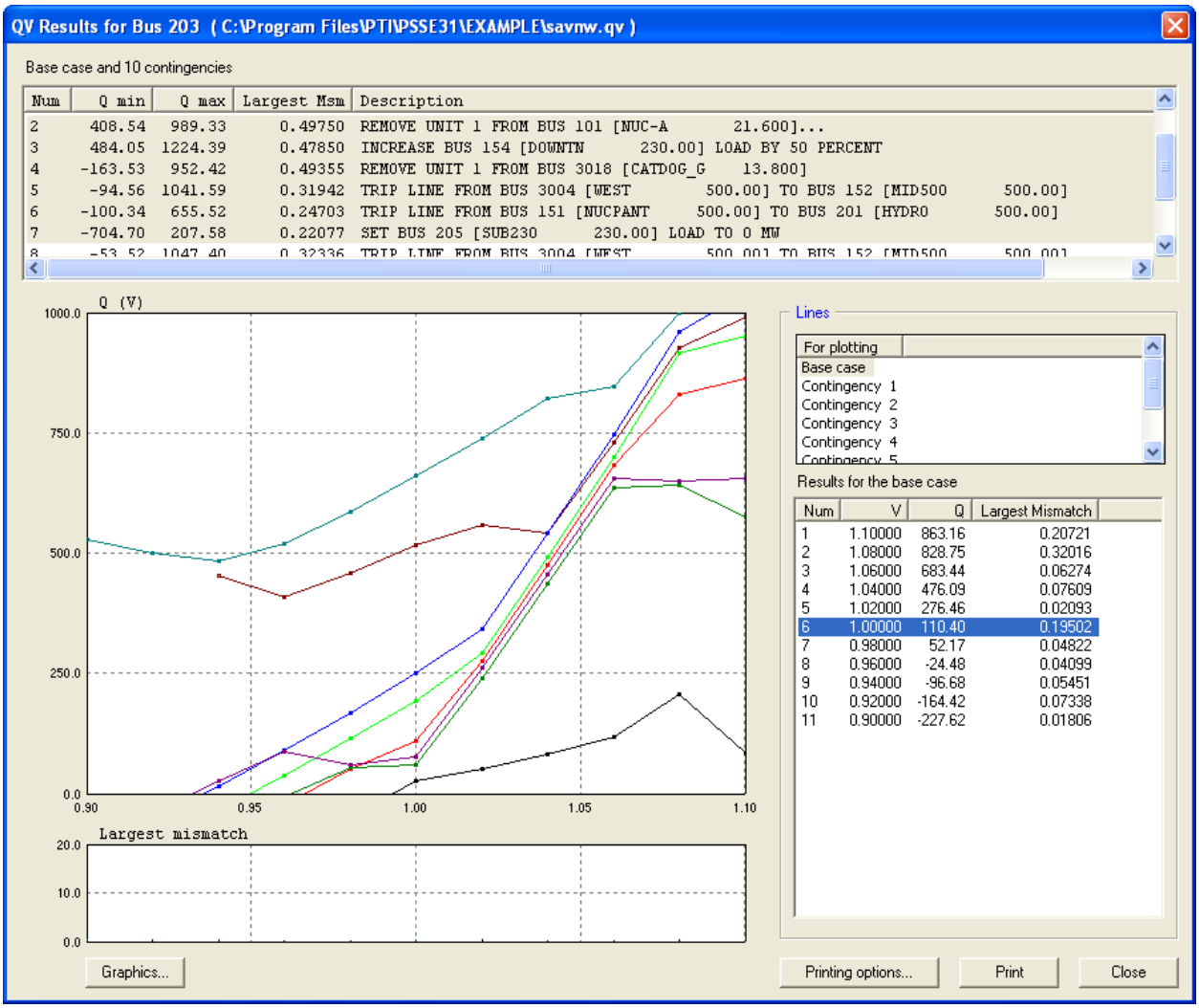

Figure 6.85. QV Curves under Various Contingencies for Bus 103

Figure 6.86, "QV Curve in Base Case with Increase in Load on Bus 103" show the QV curves of bus 103 in base case under higher loading conditions. It is observed that the demand for reactive power to hold the scheduled voltage at 1.0 pu increases with loading.

There are no functions in PSS®E QV Analysis that allow super-imposing of compensator QV characteristics over that of the system. However, this can be performed using IPLAN or Python programming tools.

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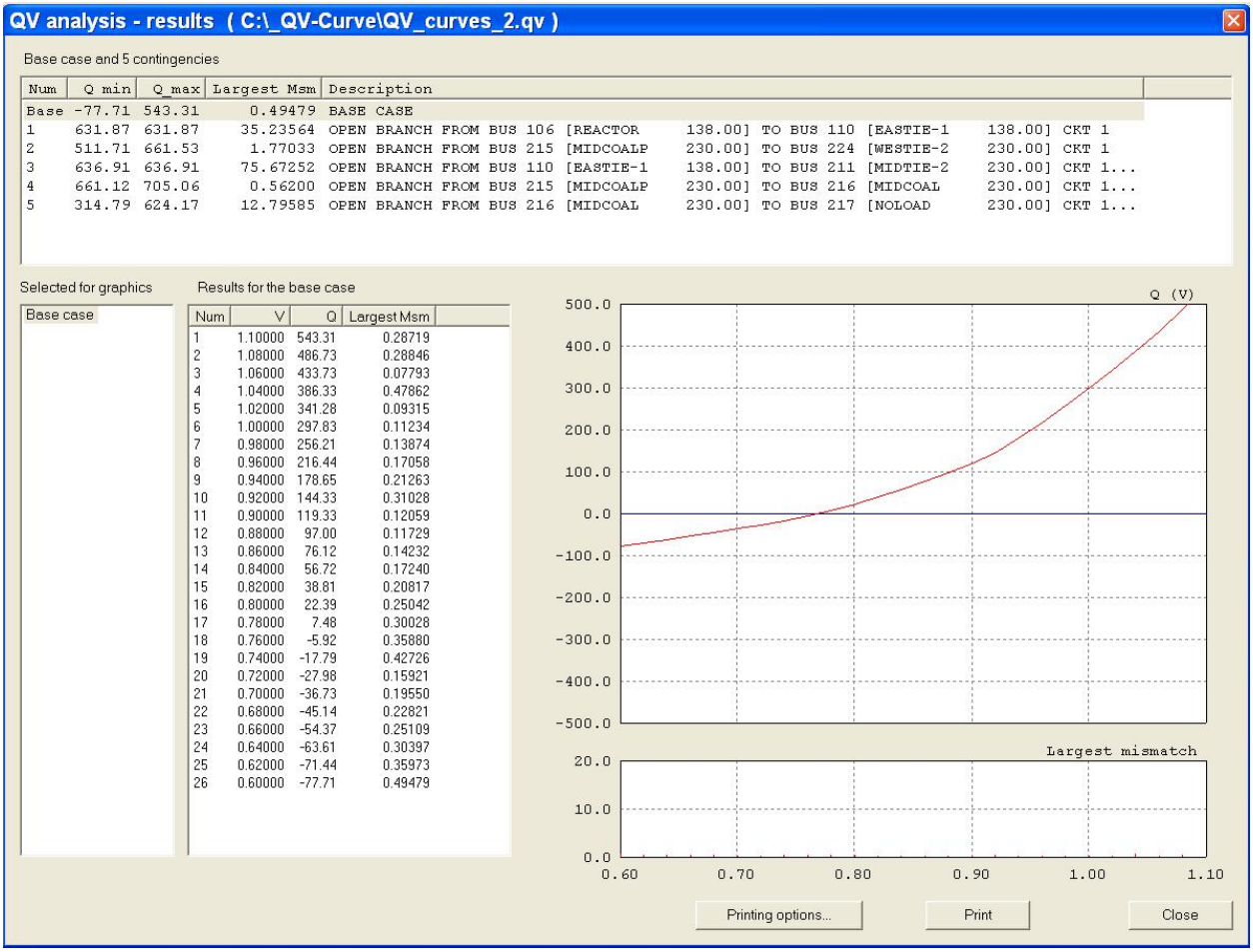

Figure 6.86. QV Curve in Base Case with Increase in Load on Bus 103

The shape of the QV curves can also be used to determine the load characteristic, and study the effect of load tap changer (LTC) transformer on the system. Figure 6.87, "S-Shaped QV Curve on Bus 108" shows the QV curve on bus 108 of the example system. The load on the bus 108 is of constant current type controlled by a LTC transformer. It is observed that the QV curves are slightly shaped like an 'S'. The S-shape characteristics are due to the load type in the system and the action of the LTC transformer as illustrated in Figure 6.88, "QV Curves for Different Load Type with Consideration of LTC".

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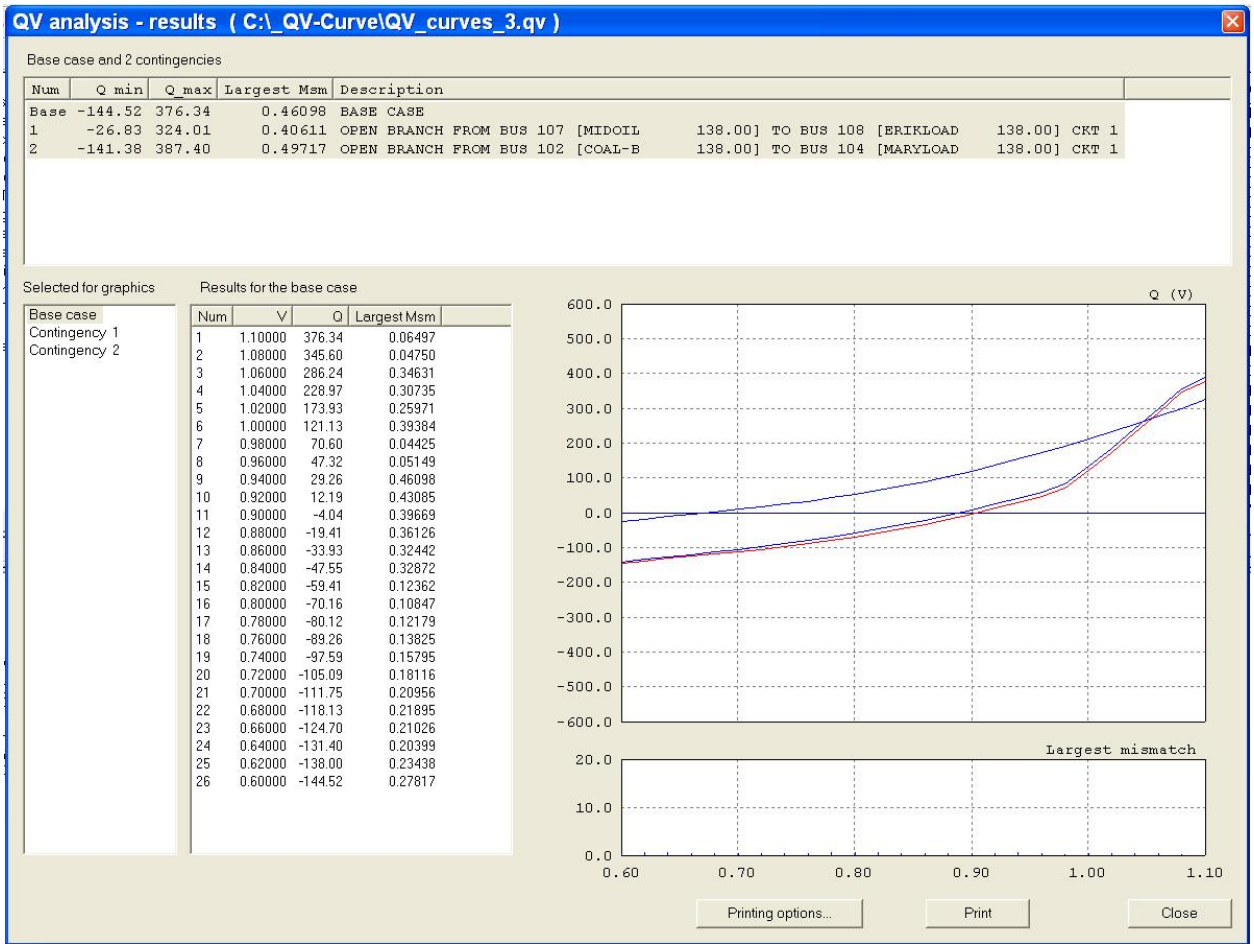

Figure 6.87. S-Shaped QV Curve on Bus 108

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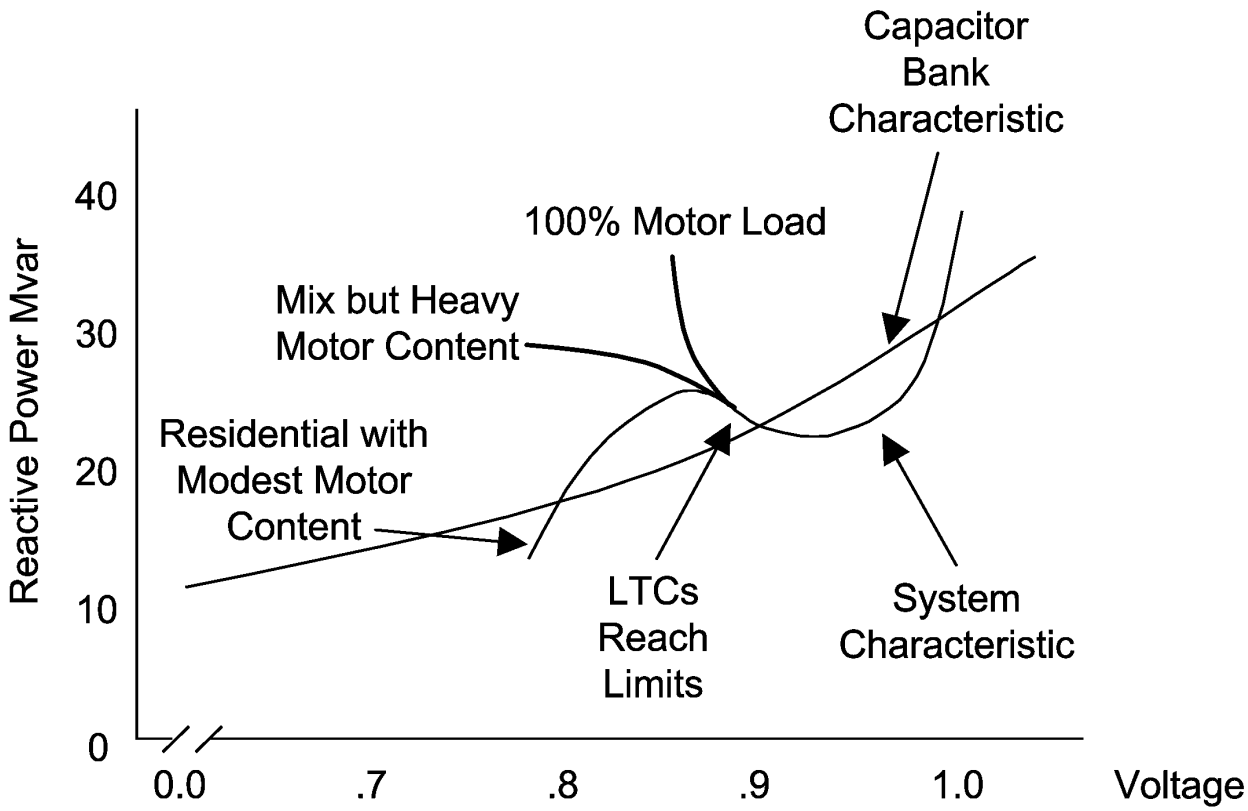

Figure 6.88. QV Curves for Different Load Type with Consideration of LTC

6.23. Sensitivity Analysis

The sensitivity analysis is a method to systematically study the impacts of changes in system operation conditions such as MW and MVar power injections, and phase angles of phase shifters on changes in branch flows and bus voltages. The function calculates sensitivity factors to assess the impacts, and is used in following applications:

- Determine the loading contribution of a branch due to the generation and load injections
- Provide operation strategies to mitigate the thermal or voltage limits violations
- Determine the dispatch mode to maximize power transfer from one area to another

The sensitivity analysis can use a linearized or "DC" network model as used in the dc analogy network solution method of activity DCLF (see Section 9.4, "Applying the DC Linearized Network Solution"). In DC power flow model, the bus voltage magnitudes remain constant, and reactive power flows are neglected. Therefore, the sensitivity analysis for bus voltages and branch flows which include MVar components are not available with DC power flow model. The sensitivity factors are determined by DC power flow network topology.

The sensitivity analysis can also use an AC power flow network model. In the AC network model the sensitivity factors are the first-order partial derivatives of the AC power flow equations at some state which is represented by a set of voltage angles and magnitudes of a converged power flow solution. For example, sensitivity factors of a branch flow to MW power injection at a generator bus is defined as:

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$$\frac{\partial f_{ij}}{\partial pgen_k}$$

which can be re-written as

$$\frac{\partial f_{ij}}{\partial V}\frac{\partial V}{\partial pgen_k}+\frac{\partial f_{ij}}{\partial \theta}\frac{\partial \theta}{\partial pgen_k}$$

with the composite function derivative method.

Where f_{ij} is the branch flow on branch form bus i to bus j, pgenk is the MW power injection at bus k, V is the vector of voltage magnitudes and θ is the vector of bus voltage angles. The AC power flow network model is a full representation of power systems, therefore sensitivity factors of bus voltage magnitudes and branch flows including MVar components can be calculated. The sensitivity factors calculated in this manner are determined by both the network topology and the current system operation condition.

When calculating sensitivity factors to MW change at a bus, an opposing system to which the sensitivity factors are referenced is required. The opposing system determines how the change is to be apportioned among buses with given participation factors to maintain the balance between generation source and demand. The sensitivity factors to MW change at a bus is calculated in the form of: $SF_i = SF'_i - \sum_{j=1}^{n} SF'_i W_j$

Where SF_i is the sensitivity factor at bus i, SF'_j is the sensitivity factor at bus j, W_j is the participation factor of bus j, $SF'_j i$ is the sensitivity factor at bus i which is referenced to the buses from 1 to n. If an opposing system is a bus, the form is simplified to $SF_i = SF'_i - SF'_j$

The sensitivity factors are represented in pu. For generation dispatch, load or switched shunt controls, the factors of flows and voltages in pu are with respect to change of generation MW or Mvar in pu. For tap setting, the factors of flows and voltages in pu are with respect to change of turns ratio in pu. For phase shifters, the factors are with respect to change of phase shift angle in radians.

6.23.1. Options of Sensitivity Analysis

The following options are available in the sensitivity analysis function menu (Table 6.30, "Sensitivity Analysis Options").

Table 6.30. Sensitivity Analysis Options

Network model	Define the power flow network model which sensitivity analysis is based on, select one of the following: • DC network model • AC network model
Initial condition MW mismatch tolerance	With AC network model, the largest MW and MVar mismatches must be less than the mismatch tolerance to continue sensitivity analysis.
Sensitivity factor cutoff	Report sensitivity factors greater than the cutoff value

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Factors for branch flows	Select a single branch via branch type and branch identification selections or branches in a subsystem for sensitivity analysis. Branch type specifies which type of branches the selector is to look for. Branch identification consisting of from bus, to bus and branch circuit ID is specified by the selector.
Flow type	Specify the type of a branch flow for which the sensitivity factors are calculated. Select one of the following (when DC network is used, only the branch MW flow is valid for sensitivity analysis): • MW flow • MVar flow • MVA flow • Amps flow
Factors for bus voltages	Select a single bus or buses in a subsystem for sensitivity analysis if AC network model is enabled.
Factors for interface flow	Select the interface on which sensitivity factors of MW flow are calculated.
Select devices to which sensitivity factors are calculated	Select a subsystem containing devices to which the sensitivity factors are calculated from a list of predefined subsystems. Flag to enable or disable calculation of sensitivity factors to MW injections at buses for branch flows, and MW and MVar injections for bus voltages. Flag to Enable or disable calculation of sensitivity factors to MW injections at generator buses. Flag to Enable or disable calculation of sensitivity factors to MW injections at load buses for branch flows, and to MW and MVar injections at load buses for bus voltages. Flag to enable or disable calculation of sensitivity factors to phase shifters. Flag to enable or disable calculation of sensitivity factors to tap changing transformers in AC network model only. Flag to Enable or disable calculation of sensitivity factors to switched shunts in AC network model only.

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Factors for branch flows	Select a single branch via branch type and branch identification selections or branches in a subsystem for sensitivity analysis. Branch type specifies which type of branches the selector is to look for. Branch identification consisting of from bus, to bus and branch circuit ID is specified by the selector.
Transfer options	Define the direction of the transfer and opposing subsystem. They are required if one of bus, generator and load device flags is selected. The transfer could be either from the device to which sensitivity factors are calculated to the opposing subsystem or the other way around, select one of the following: • From device. The transfer is from a device (bus, generator bus or load bus) to the opposing system. • To device. The transfer is from the opposing system to a device.
Opposing system for MW dispatch	Select to which the sensitivity factors are referenced from one of the following: • Slack bus • Single bus • System
Subsystem and dispatch mode for opposing system	Select a subsystem and a dispatch mode from seven modes if the opposing subsystem type is a subsystem: • Buses & participating factors from DFAX file for buses with positive MW machines • Buses & participating factors from DFAX file for buses with positive MW constant MVA load • Buses & participating factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load • Subsystem buses with positive MW constant MVA load in proportion to their MW load • Subsystem buses with positive MW machines in proportion to their MW output

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Factors for branch flows	Select a single branch via branch type and branch identification selections or branches in a subsystem for sensitivity analysis. Branch type specifies which type of branches the selector is to look for. Branch identification consisting of from bus, to bus and branch circuit ID is specified by the selector.
	• Subsystem buses with positive MW machines in proportion to their MBASE • Subsystem buses with positive MW machines in proportion to their r • eserves (Pmaxmc-Pgenmc)
Distribution factor data file	A required file built by DFAX provides subsystem and interface definitions.

6.23.2. Sensitivity analysis output

The sensitivity analysis will generate a report containing sensitivity factors. Figure 6.89, "Sample of sensitivity analysis report" shows a sample of the report. The network model selection (DC network or AC network) is present at the top of the report, followed by the opposing system selection if any. The sensitivity factors are then listed in different tables, one table for each flow or voltage to each device type.

In the sample, the sensitivity factors of MW flow on the branch from bus 151 to bus 152 to generation injections, loads, as well as the phase shifter from bus 202 to bus 203 are listed respectively. The direction of MW flow is from bus 151 to bus 152. The value of the factor of MW flow to the generation injection at bus 101 is 0.2933, which indicates the line will have an increased MW flow of 0.2933 per unit in the direction from bus 151 to bus 152 when the generation at bus 101 increases by 1 per unit and correspondingly generation at system swing bus decreases by 1 per unit. For the phase shifter, angle is represented in radians. The line will have a decreased MW flow of 0.1912 per unit in the direction from bus 151 to bus 152 when the phase shifter angle increases one radian or 57.3 degrees.

```
                    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E
 SENSITIVITY FACTORS ARE CALCULATED WITH DC NETWORK
 SYSTEM SWING BUS(es) IS USED FOR OPPOSING SYSTEM FOR MW POWER DISPATCH

 ------------------------------------------------------------------------------

 SENSITIVITY FACTORS OF BRANCH FLOW (MW) ON    151 NUCPANT     500.00 TO    152 MID500      500.00 1  GREATER THAN  0.10000

 <---- GENERATOR BUS ----> PGEN(MW) PMAX(MW) PMIN(MW)  MW-SENS
    101 NUC-A       21.600   750.00   810.00     0.00   0.2933
    102 NUC-B       21.600   750.00   810.00     0.00   0.2933
    211 HYDRO_G     20.000   600.00   616.25     0.00   0.1709

 <------ LOAD BUS -------->   LOAD(MW)    MW-SENS
    151 NUCPANT     500.00       0.00     0.2933
    201 HYDRO       500.00       0.00     0.1709

 <----------------- PHASE SHIFTER ----------------->CKT    ANGLE     RMAX     RMIN     SENS
    202 EAST500     500.00   203 EAST230     230.00 1      0.00    30.00   -30.00  -0.1912
```

Figure 6.89. Sample of sensitivity analysis report

The API has been provided to calculate and return sensitivity factors of branch flows to active power injections of generators and loads in a subsystem in Fortran arrays, the arrays can then be exported to a Excel file. The API can be called from Python files (refer to the SENSITIVITY_FLOW_TO_MW in the API Guide).

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6.23.3. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Tables named "SENSITIVITY SENSFLOW" and "SENSITIVITY SENSVOLT".

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Chapter 7
Node-breaker Modeling

7.1. About the Node-breaker Model

Two types of network models are commonly used in power system planning studies: node-breaker and bus-branch. The bus-branch model is typically used for off-line planning studies, where common voltage level elements are consolidated into one logical bus. In contrast, the node-breaker model offers a full-topology model that includes detailed nodes, breakers and other switching devices within substation configuration representations. Real-time system operations typically use the node-breaker model to recognize and operate the individual switching devices when making system operation condition changes.

The recent trend in the power industry is to use node-breaker models for both planning and operation studies for the following reasons:

- There is a lot of redundant work to maintain two models concurrently. It is quite labor intensive and also difficult to identify discrepancies between node-breaker and bus-branch models.
- Mapping bus numbers in the bus-branch model to node numbers in the node-breaker model is very difficult, and validation of the off-line model with real time data is hard to achieve.
- The node-breaker model is ideal for many planning studies: simulating breaker actions and multiple-terminal line contingencies, simulations of splitting and merging buses due to breaker operations.

It should be noted that the full-topology model for planning studies typically does not contain other devices aside from the major switching devices such as breaker disconnect switches; therefore these planning node-breaker models are more a simplified version than the full-topology node-breaker models used by operations. Moving data between a bus-branch model and a node-breaker model can historically prove to be an arduous task, and prone to errors.

Prior to PSS®E version 34, the only way to model node-breaker in PSS®E was to model each node as a bus and each switching device as a regular AC branch or and zero impedance lines. With this method, the simulated node-breaker model is no different than the bus-branch model. However, using this direct method to model the node-breaker topology has some adverse impacts on planning studies, such as:

- Requiring huge numbers of connection nodes (buses) and breakers (lines and zero impedance lines).
- The components are always connected to buses and difficult to identify.
- Transitioning from the bus-branch model to the node-breaker model is not automatic, and is time consuming and difficult.

In contrast, the ideal benefits of an integrated node-breaker modeling system for planning studies are:

- Node-breaker contingency events can be properly simulated and validated.
- Breaker failure events can be simulated.
- The node-breaker model is represented on a substation by substation basis.

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- The ability to display both a bus-branch view and the substation view.
- Power flow results can be reported on bus-branch and node-breaker models.
- The addition of node-breaker representations will not affect computational speed or solution stability.
- Hybrid node-breaker and bus-branch modeling is allowed in the same power flow case.

7.2. Node-breaker Modeling in PSS®E

In order to overcome the disadvantage of having to model node-breaker topology using traditional buses, branches and zero-impedance lines, and to meet the above requirements of an ideal integrated node-breaker modeling system, a dual-level method to model node-breaker is provided in PSS®E by which the full-topology model is represented at both the system and substation levels. This allows users to interact with the network through the traditional bus-branch view, while also having the ability to drill down into the details of breakers and switches as required. By using state-of-the-art topology processing techniques, the design can support tens of thousands of additional nodes without the loss of speed or accuracy in the power flow solution.

The PSS®E node-breaker data model is implemented as an extension to the standard bus-branch data model, providing an interconnection between buses and nodes. The bus-branch and node-breaker models are seamlessly linked with the complex node-breaker detail hidden from view until the user needs to drill down and perform operations at the breaker level.

To illustrate this, the PSS®E node-breaker representation of a bus-branch model is shown in Figure 7.1, "PSS®E Basic Node-breaker Terminology".

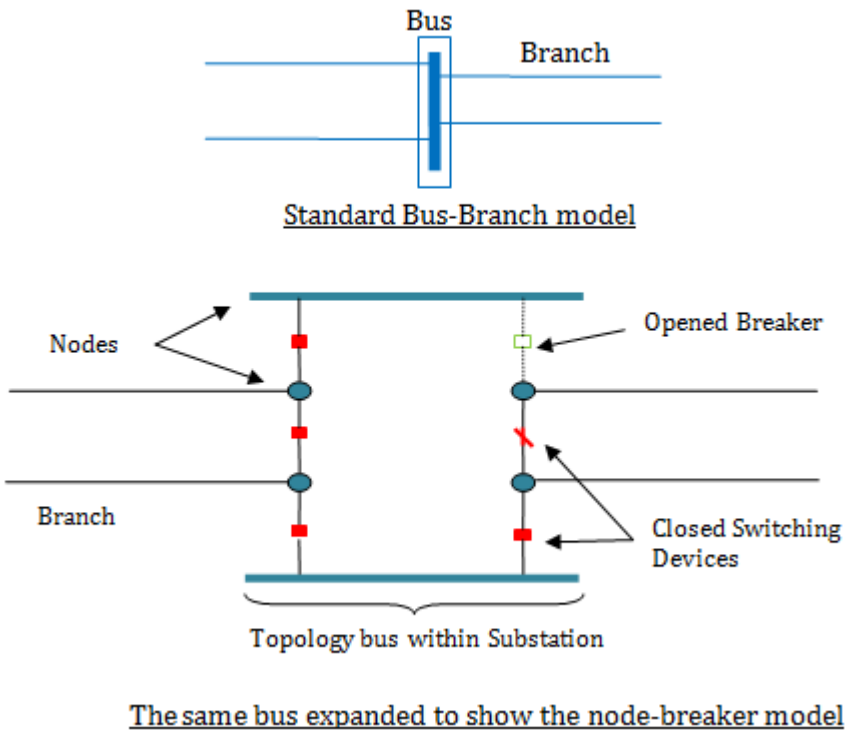

Figure 7.1. PSS®E Basic Node-breaker Terminology

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Each full-topology model consists of the traditional bus-branch model, along with as many substation models as are needed. Substation models are an additional level of detail built on top of bus-branch models. These substation models are optional; if none exist in the case, then the full-topology model is identical to its bus-branch model. When detailed node-breaker substations are present, additional operations and functionalities are available specifically for node-breaker analysis, including calculation of flows on switching devices within a substation, switching device related contingencies, isolation operations and others.

Figure 7.2, "Bus-branch View of a Bus" and Figure 7.3, "Detailed node-breaker view of same bus" show the bus-branch model at bus 203 using GOUT, and a substation diagram of the same bus. Buses with boxes drawn around them indicate that they have an underlying substation model.

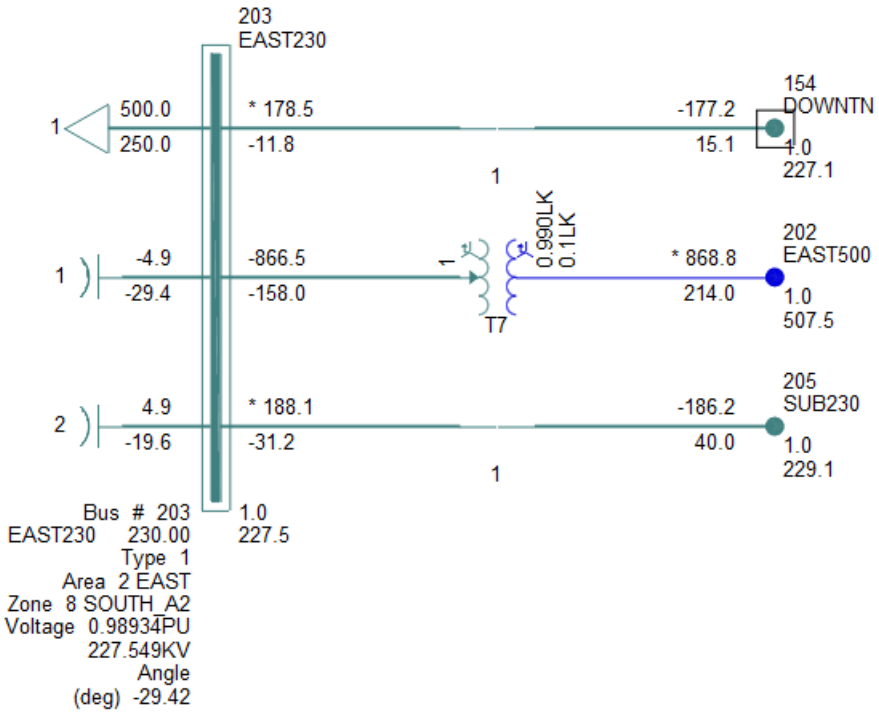

Figure 7.2. Bus-branch View of a Bus

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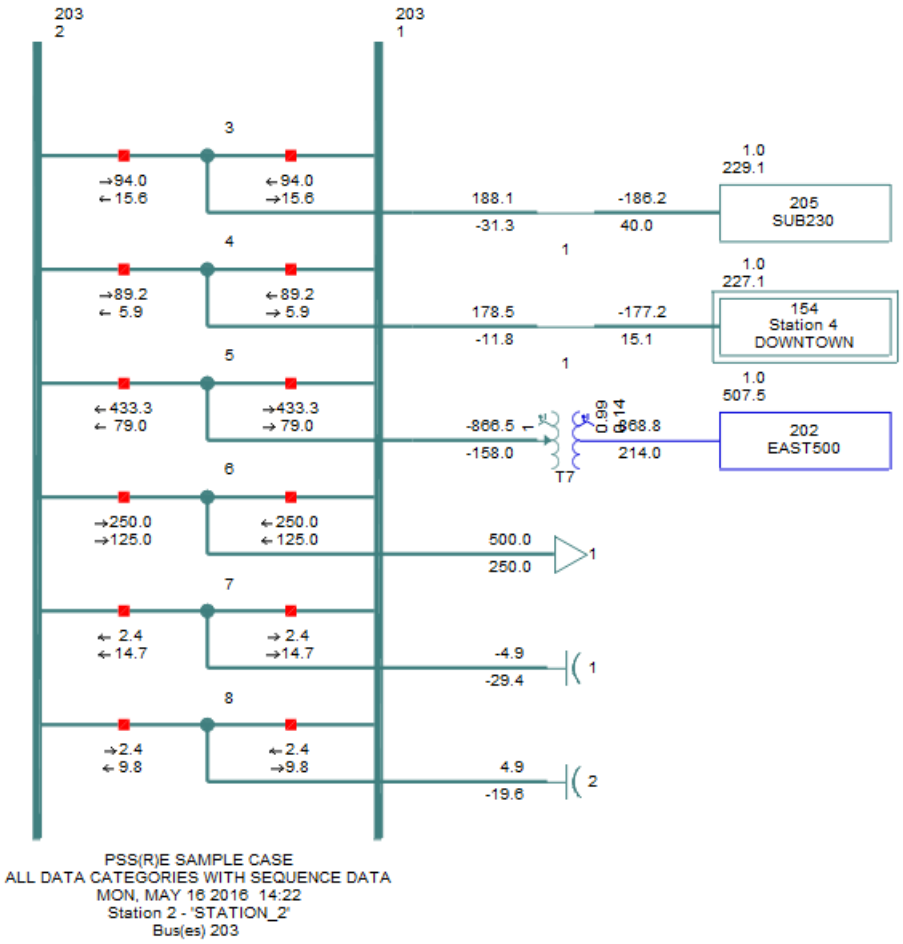

Figure 7.3. Detailed node-breaker view of same bus

In the bus-branch model, buses are the points to which other components such as branches, loads and machines are connected.

In the node-breaker model, terminals describe the connections to the nodes in a substation. The FROM and TO buses of a branch define its connections in the bus-branch model but each branch end may also have a terminal connection to a node.

In Figure 7.3, "Detailed node-breaker view of same bus", above, the branch from Bus 154 to Bus 203 is explicitly modeled in the bus-branch model as being connected between buses 154 and 203. These buses however are also topological buses, each representing a group of nodes within separate substations. In the full-topology model, as shown in Figure 7.4, "Branch 154 to 203 node terminal connections in the Substation Model" below, the branch is more specifically connected between terminal node 4 of Substation 1 containing topological bus 154, and terminal node 4 of Substation 2, containing topological bus 203.

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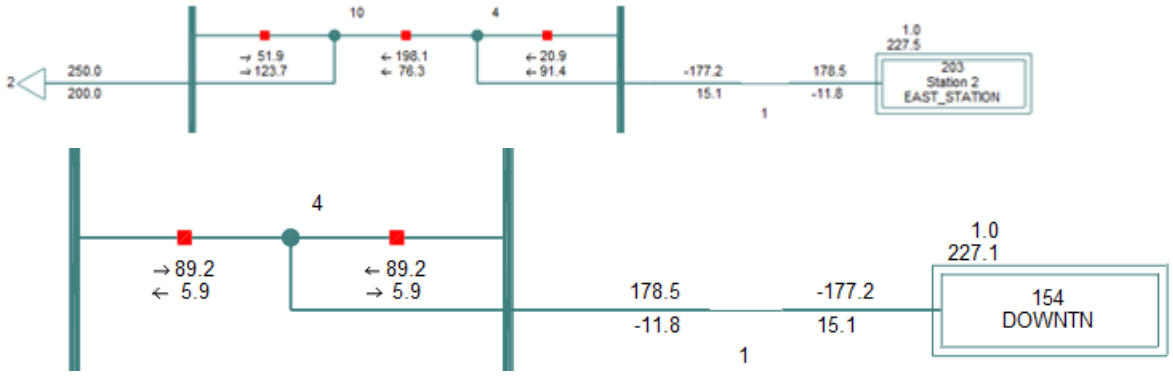

Figure 7.4. Branch 154 to 203 node terminal connections in the Substation Model

Terminal connections to nodes in a substation can be any of the devices supported in the PSS®E power flow raw data file. Details of the format for inputting substation terminal component data are found in Equipment Terminal Data of the Data Formats Reference Manual.

7.3. Topology processor

In the full-topology model, a bus is called a topological bus if it represents a group of nodes in a substation. It is typical to have multiple network buses or topological buses represented in a single substation. A bus without any nodes is a connectivity bus. A full-topology model may consist of both topological buses and connectivity buses.

As shown in Figure 7.3, "Detailed node-breaker view of same bus" above, bus 203 represents nodes 1 through 8 in the substation. When all nodes of a main network bus are connected by closed switching devices, they are represented by one bus in the bus-branch model at the system level. This bus is termed the original, or parent, bus.

As changes are made in bus-branch models, these changes will automatically be propagated to the substation models. For example, if a bus is disconnected (through DSCN), then all its nodes, connected devices and branches are also disconnected. If the original parent bus is simply set out-of-service (type code of 4), then all its nodes are placed out of service, but branches are not affected.

Changes which may cause the automatic running of the node-breaker topology processor include:

- Adding substation models
- Adding/deleting nodes and changing the status of nodes
- Adding/deleting switching devices and changing the status of switching devices
- Changes in terminal nodes of connected equipment
- Changing the status of a branch or transformer connected to a node

The topology processor takes substation switching device connections and statuses as inputs to group the nodes into electrically connected bus sections, assigning a topological bus for each group, and moving components to the right topological bus. Internally, the topology processor will:

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1. Check for connectivity errors in the bus-branch model and substation models

2. Generate node groups by locating all nodes which are connected together by closed circuit breakers and disconnect switches.

3. Assign a topological bus to each group, create a bus-section and re-route components to the bus as needed.

4. Form into electrical "islands" all of the buses connected together by AC lines, including transmission lines, and transformers. An island must have at least one system swing bus (bus type code of 3), otherwise that portion of the system is isolated and shut down.

As substation switching devices are opened, a splitting of the original bus may occur, resulting in two distinct topological bus sections. When a new active node group is created and a topological bus section is needed, the bus data of the original bus is copied to the new bus section, including area/zone/owner numbers, voltage limitations, base kV, voltage magnitudes and angles. The bus number is the same as the original bus, with a section identifier (Bus 154-4 for example).

The section identifier is assigned as the minimum node number of the node group of which the bus section is comprised. Bus sections may come and go as switching devices are opened and closed and, for this reason, bus numbers of bus sections are always referenced and identified by the parent bus number and a bus section number. All equipment attached to the nodes in the group will remain connected to those nodes but will move to the new topology bus section to which those nodes now belong.

For machines that are moved to a new bus section, the new bus section becomes a generator bus with plant data. The parent bus also remains a generator bus. If all machines are connected to the new topological bus, then the original bus still remains a generator bus, although without any machines. Plant data for each will be updated accordingly. If the parent bus is a swing bus (type 3) but does not have any machines, then one of its bus sections will be assigned as the swing bus and the parent will be assigned as a generator bus (type 2).

In Figure 7.5, “Substation bus split into 2 buses sections due to switching operations”, the switching devices between nodes 1 and 4, and between nodes 2 and 10, have been opened, causing nodes 4 and 10 to be electrically separated from the remaining nodes in the substation.

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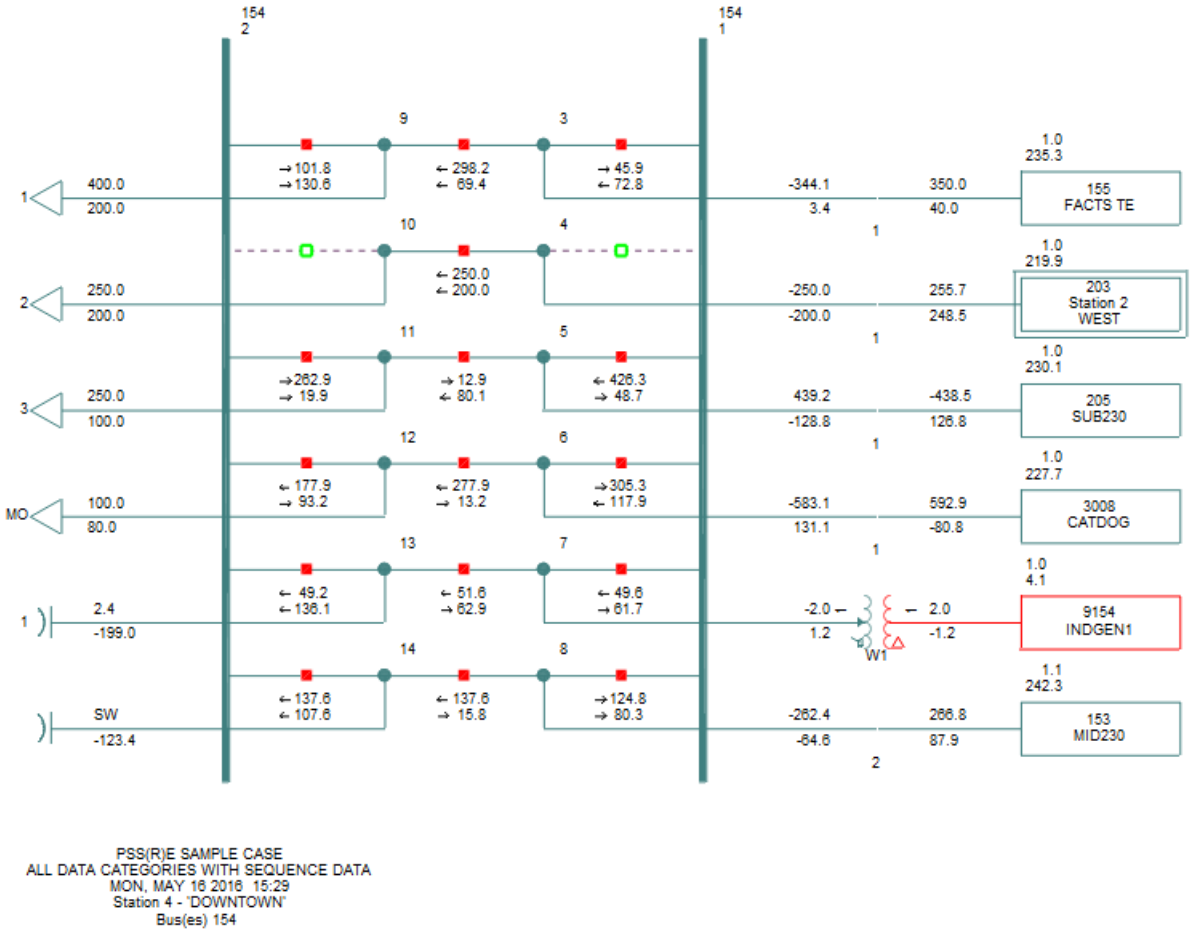

Figure 7.5. Substation bus split into 2 bus sections due to switching operations

Immediately after any switching operation, the node-breaker topology processor automatically updates the underlying bus-branch and node-breaker model connections.

For the above example, an informational message indicates the split, and displays the newly assigned bus section numbers:

```
Bus 154 is split into the following bus sections due to operations of switching devices:
     Bus section    154-  1
     Bus section    154-  4
```

Bus section 154-4 now contains nodes 4 and 10, plus load 2, while bus section 154-1 contains all other remaining nodes and their terminal connections.

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In PSS®E reports and messages, all bus sections will be uniquely identified by the parent bus number and a section number:

```
                                        S H U N T S                                                 X--- NORMAL --X  X- EMERGENCY -X
BUS#-SCT X-- NAME --X BASKV CODE LOADS FIXED SWITCHED  VOLT   ANGLE AREA ZONE OWNER   VMAX     VMIN     VMAX     VMIN
101     NUC-A       21.600   2    0     0      0    1.01000  -11.0   1    1     1    1.10000 0.90000  1.10000 0.90000
102     NUC-B       21.600   2    0     0      0    1.01000  -11.3   1    1     1    1.10000 0.90000  1.10000 0.90000
151     NUCPLNT     500.00   1    0     3      0    1.00071  -14.1   1    1     1    1.10000 0.90000  1.10000 0.90000
152     MID500      500.00   1    1     1      1    1.03913  -24.0   1    2     1    1.10000 0.90000  1.10000 0.90000
153     MID230      230.00   1    1     0      0    1.05336  -25.7   1    3     1    1.10000 0.90000  1.10000 0.90000
154-  1 DOWNTN      230.00   1    3     1      1    0.99751  -33.3   1    3     1    1.10000 0.90000  1.10000 0.90000
154-  4 DOWNTN      230.00   1    1     0      0    0.84512  -36.5   1    3     1    1.10000 0.90000  1.10000 0.90000
```

When switching devices are closed, resulting in node groups being merged back together, an existing bus section will no longer needed. The nodes and their connections are automatically moved back to the bus or bus section to which they are being merged into and the unneeded bus section is deleted.

In Figure 7.5, “Substation bus split into 2 bus sections due to switching operations”, when one of the switching devices between either nodes 1 and 4 or between nodes 2 and 10 is closed, then all nodes are again reconnected, and the following message indicates that the extra bus section is no longer needed:

```
Bus Section 154-4 [DOWNTN      230.00] is not needed due to operations of switching devices
```

At this point, only bus 154 remains, containing all 14 nodes. The additional bus section bus that had been introduced is removed.

When buses become split apart due to the opening of switching devices, some equipment or portions of the substation or even the network may become isolated and taken out of service. When this occurs, the topology processor will automatically preserve an "original status" for all branch devices. This original status is inititalized to the status of the device when the network case is read in or when the device is created or when any explicit changes are made to the status that are not related to node-breaker actions. If, during node-breaker switching actions, a branch device is isolated and taken out of service or blocked, then the original status will be preserved and the present device status be set to out-of-service or blocked. Likewise, if node-breaker switching actions allow for a branch devcie that is currently set as out-of-service or blocked to be placed back in service, then the value of the original status will be used to set the status of the device. If the status of the device was originally out of servcie or blocked, then it will remain as such.

The "original status" value, if it differs from the status of line, is displayed in all List Data reports for branches, line shunts, two-winding and three-winding transformers, FACTS devices and DC lines. It will appear next to the current status (ST) or mode (MODE, MDC) column, under the column heading "OR". If there is no difference between the original status and the present status then the original status (OR) column field for that device will be blank.

7.4. Creating a substation

The procedure to manually build a full-topology model consists of the following steps:

- Build a bus-branch model,
- Expand one or more connectivity buses into topological buses by creating substations.
- Set up node terminals connections for branches and components.
- Place substation switching devices between nodes within the substation.

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The data for the node-breaker model, including the substation definitions, nodes, substation switching devices, and connections to network components is described in the node-breaker raw data file format Substation Data of the Data Formats Reference Manual. This data can be entered or changed through the PSS®E spreadsheet or PSS®E Powerflow data file.

Since the node-breaker model in PSS®E is an extension to the existing bus-branch model, it is possible to add or remove the node-breaker data without any loss of functionality to the bus-branch model. The internal details of the data mapping and topology processing are completely hidden.

To assist with the building the substation models, a substation configuration builder tool provides the means to automatically generate a generic node-breaker topology substation, based on several commonly used substation configurations: ring bus, breaker-and-a-half, double-bus-double-breaker, etc. A graphical representation of the substation is automatically created in the form of a substation single-line diagram (Substation SLD). The layout of the substation SLD can be manually changed for proper placement of terminals to approximate the actual configuration of the substation. New substation components can also be added or existing components deleted. This can be done through either the network data dialog or through the diagram itself.

To run the substation configuration builder, select Node-Breaker>Configuration Builder... from the main PSS®E menu. The Build a Substation Configuration dialog is shown in Figure 7.6, "Automatic Node-Breaker Substation Configuration Builder Dialog".

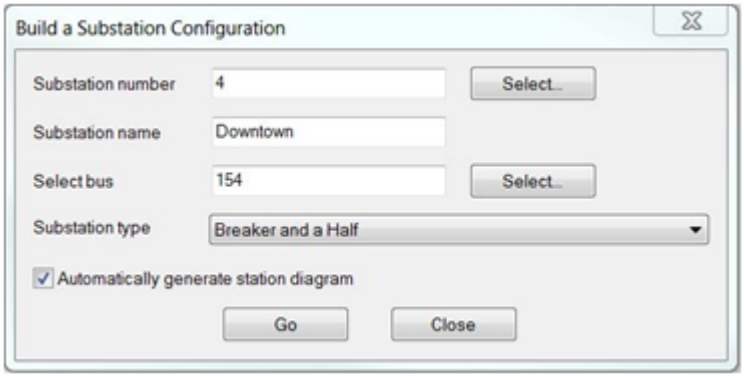

Figure 7.6. Automatic Node-Breaker Substation Configuration Builder Dialog

It is also possible to open the substation builder dialog by doing a right-mouse-click on a bus displayed within the spreadsheet view, or in a diagram, and selecting Build a substation....

Within the dialog:

- The Substation number field will automatically be populated with the lowest available number that has not already been used for other substations. This number can be modified as desired.
- The Substation name can be entered but is optional. It is used in reports and on diagrams.
- The bus number of the bus that is to be expanded into the substation model must be entered. If the Configuration builder was started from a right-mouse-click on a particular bus, then this field will automatically be populated with that bus number.

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More than one bus can belong to the same substation. If a bus is specified and the specfied substation was previously created you will be asked whether the bus should be added to the substation.

- Select the type of substation layout to build. These are just starting configurations for which the diagram builder will use to lay out the substation. Example types include: single bus, ring bus, double bus double breaker, breaker and a half, double bus single breaker, and main bus transfer bus. There are also alternate node number layouts for the breaker and a half, and the double bus single breaker.

- It is possible to have the substation generated without having a new diagram displayed. If this is desired, uncheck the option to automatically generate station diagram.

Once Go is selected and the substation components are created for that bus, the dialog will remain open and another bus can be added for inclusion in the same substation, or a different substation.

7.5. Viewing a substation

To view an existing substation, select Node-Breaker>Generate substation display from the main PSS®E menu. A Select Substation dialog will be displayed where the desired substation number can be entered.

To view a diagram of an existing substation from the bus spreadsheet or a network diagram, right-click on the bus that is within the substation of interest and select Generate substation display PSS®E will either find your custom SLD file for that substation (found by substation name), or it will create one on-the-fly. If a bus is not associated with a substation, then the option to "Build a substation..." will be shown on the menu instead.

7.6. Check Substation Topology (STREE)

The full-topology model is represented by one bus-branch model and additional substation models. It is deemed valid if the neither the bus-branch model nor any of the substation models have topological errors, and the bus-branch model and substation models are consistent. Using Activity TREE, the bus-branch model is topologically correct if:

- All in-service buses are connected back to a Type 3 (swing) bus through the in-service ac network

- There are no in-service branches connected to Type 4 (disconnected) buses

Just as the TREE activity is used to validate the bus-branch model, the STREE activity is used to validate the node-breaker model in the presence of substations. The STREE activity however will rarely, if ever, need to be run manually. Each time a network topology change is detected, either in the bus-branch model or the node-breaker model, that could impact the topology of the node-breaker model, the substation topology processor (STREE) will automatically be run behind the scenes. This could occur when opening or closing substation switching devices, or changing the status of branches connected to a node within a substation or generally changing the topology of the network.

Activity STREE checks for two types of topological errors: continuity of substation networks and consistency between bus-branch and substation models. A substation model is topologically correct if:

- A substation switching device is closed, and its two end nodes have the same service status.

- Nodes connected by closed substation switching devices are in the same node group (bus section).

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A group with only in-service nodes is defined as an active group; each active group within a substation must be represented by an in-service topological bus (or bus section) in the bus-branch model. A node group consisting of out-of-service nodes is defined as an inactive group; an inactive node group may or may not have a unique topological bus in the bus-branch model, and it may share the same topological parent bus as an active node group.

All node statuses are handled internally by the topology processor so it is not likely that topology errors due to differing node statuses will be encountered.

For a branch or two-winding transformer, if one of its end buses is a topological bus, the corresponding terminal node must be one of the nodes in the substation and it must have the same topological bus as the end bus. Furthermore, if the branch is in-service, the terminal must belong to an active node group.

Similarly, any loads, fixed shunts, switched shunts or machines connected to a topological bus must have a terminal node associated with the same topological bus. In the bus-branch model, the equipment is treated as in-service when both its status and the bus status are in-service. Inconsistency may happen if:

- The topological bus and the equipment are in-service, but
- The terminal node belongs to an inactive node group represented by the topological bus

PSS®E will determine when a topological change has occurred within the network and will run automatically run STREE on the impacted substations when required. No explicit running of the activity is required.

If needed though, it is possible run the node-breaker topology processor manually. To launch the STREE activity, select from the main menu bar, Node-Breaker>Check substation topology (STREE)...

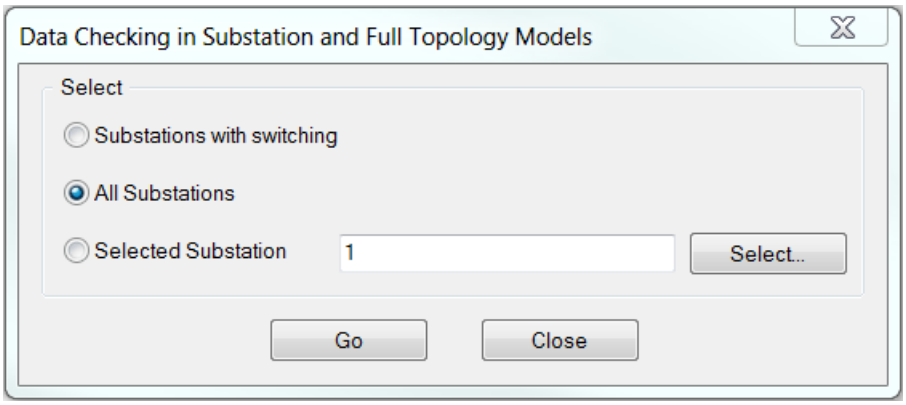

Figure 7.7. Check Substation Topology (STREE) Activity

Within the dialog window shown in Figure 7.7, "Check Substation Topology (STREE) Activity", select which substations to perform the checking on:

- Substations with switching checks only those substations where switching data has been changed
- All substations to checks all substations
- Selected substation to checks only the specified substation

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The activity then checks each selected substation and lists any topology actions that were taken and any errors that exist for each substation.

7.7. List Substation Data (SLIST)

The List Substation data (SLIST) activity lists, for each substation, its number and name, followed by all the components within the substation. The components include nodes, substation switching devices and network system components such as loads, shunts, machines, branches, transformers,dc line and FACTS Device Connections.

To run the activity, select Node-Breaker>List substation data (SLIST)...

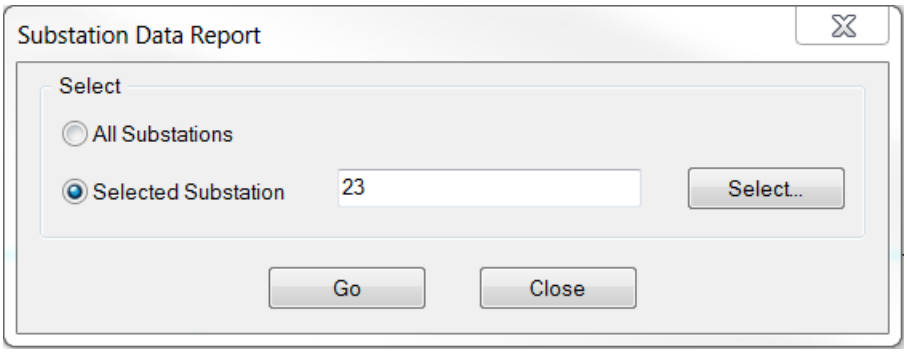

Figure 7.8. List Substation Data Report dialog

Within the dialog shown in Figure 7.8, “List Substation Data Report dialog”, select either:

- All substations to report data for all substations.
- Selected substation to report data for a specific substation.

The resulting report shows each substation number and name, followed by a node report containing node number, name, corresponding bus section, and status. This is followed by the list of substation switching devices, displaying the FROM node, TO node, name of the switching device, its current and normal status, reactance and ratings.

This is followed by a list of all components connected to each node within the substation. For each of the connected loads, bus fixed shunts, switched shunts or machines in the substation, the topological bus section number and name, along with the terminal node number and name are displayed. For each branch and two-winding transformer, it's FROM and TO buses, nodes, and corresponding stations are reported, along with the circuit identifier. For each three winding transformer, it's FROM/TO/TO buses, nodes and stations are displayed, along with the circuit identifier.

A sample substation report is shown in Figure 7.9, “List Substation Data report”.

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```
Substation: 1 [STATION_1]

X----------------- NODE -----------------X X---------- BUS ----------X ST
  1 STATION_1_1_1                          154-  1 DOWNTN 230.00       1
  2 STATION_1_1_2                          154-  1 DOWNTN 230.00       1
  3 STATION_1_1_3                          154-  1 DOWNTN 230.00       1
  4 STATION_1_1_4                          154-  4 DOWNTN 230.00       1
  5 STATION_1_1_5                          154-  1 DOWNTN 230.00       1
  6 STATION_1_1_6                          154-  1 DOWNTN 230.00       1
  7 STATION_1_1_7                          154-  1 DOWNTN 230.00       1
  8 STATION_1_1_8                          154-  1 DOWNTN 230.00       1
  9 STATION_1_1_9                          154-  1 DOWNTN 230.00       1
 10 STATION_1_1_10                         154-  4 DOWNTN 230.00       1
 11 STATION_1_1_11                         154-  1 DOWNTN 230.00       1
 12 STATION_1_1_12                         154-  1 DOWNTN 230.00       1
 13 STATION_1_1_13                         154-  1 DOWNTN 230.00       1
 14 STATION_1_1_14                         154-  1 DOWNTN 230.00       1

Substation switching devices
X-------------- FROM NODE ---------------X X--------------- TO NODE ----------------X CKT X---------------- NAME ----------------X ST NST     X       RATE1    RATE2    RATE3
   1 STATION_1_1_1                          3 STATION_1_1_3                          1                                          1  1  0.00010     0.00     0.00     0.00
   1 STATION_1_1_1                          4 STATION_1_1_4                          1                                          0  1  0.00010     0.00     0.00     0.00
   1 STATION_1_1_1                          5 STATION_1_1_5                          1                                          1  1  0.00010     0.00     0.00     0.00
   1 STATION_1_1_1                          6 STATION_1_1_6                          1                                          1  1  0.00010     0.00     0.00     0.00
   1 STATION_1_1_1                          7 STATION_1_1_7                          1                                          1  1  0.00010     0.00     0.00     0.00
   1 STATION_1_1_1                          8 STATION_1_1_8                          1                                          1  1  0.00010     0.00     0.00     0.00
   2 STATION_1_1_2                          9 STATION_1_1_9                          1                                          1  1  0.00010     0.00     0.00     0.00
   2 STATION_1_1_2                         10 STATION_1_1_10                         1                                          0  1  0.00010     0.00     0.00     0.00
   2 STATION_1_1_2                         11 STATION_1_1_11                         1                                          1  1  0.00010     0.00     0.00     0.00
   2 STATION_1_1_2                         12 STATION_1_1_12                         1                                          1  1  0.00010     0.00     0.00     0.00
   2 STATION_1_1_2                         13 STATION_1_1_13                         1                                          1  1  0.00010     0.00     0.00     0.00
   2 STATION_1_1_2                         14 STATION_1_1_14                         1                                          1  1  0.00010     0.00     0.00     0.00
   3 STATION_1_1_3                          9 STATION_1_1_9                          1                                          1  1  0.00010     0.00     0.00     0.00
   4 STATION_1_1_4                         10 STATION_1_1_10                         1                                          1  1  0.00010     0.00     0.00     0.00
   5 STATION_1_1_5                         11 STATION_1_1_11                         1                                          1  1  0.00010     0.00     0.00     0.00
   6 STATION_1_1_6                         12 STATION_1_1_12                         1                                          1  1  0.00010     0.00     0.00     0.00
   7 STATION_1_1_7                         13 STATION_1_1_13                         1                                          1  1  0.00010     0.00     0.00     0.00
   8 STATION_1_1_8                         14 STATION_1_1_14                         1                                          1  1  0.00010     0.00     0.00     0.00

    X-------- LOAD BUS ---------X ID X------------------ NODE ------------------X
    154-  1 DOWNTN      230.00    1    9 STATION_1_1_9
    154-  1 DOWNTN      230.00    3   11 STATION_1_1_11
    154-  1 DOWNTN      230.00    MO  12 STATION_1_1_12
    154-  4 DOWNTN      230.00    2   10 STATION_1_1_10

X----- FIXED SHUNT BUS -----X ID X------------------ NODE ------------------X
   154-  1 DOWNTN      230.00 1   13 STATION_1_1_13

X--- SWITCHED SHUNT BUS ----X X------------------ NODE ------------------X
   154-  1 DOWNTN      230.00  14 STATION_1_1_14

Non-transformer branches and two winding transformers
     X---------- BUS ----------X X------------------ STATION ------------------X X------------------ NODE ------------------X CKT
FROM    154-  1 DOWNTN      230.00      1 STATION_1                                8 STATION_1_1_8
  TO    153     MID230      230.00      0                                          0                                          2

FROM    154-  1 DOWNTN      230.00      1 STATION_1                                3 STATION_1_1_3
  TO    155     FACTS TE    230.00      0                                          0                                          1

FROM    154-  1 DOWNTN      230.00      1 STATION_1                                5 STATION_1_1_5
  TO    205     SUB230      230.00      0                                          0                                          1

FROM    154-  1 DOWNTN      230.00      1 STATION_1                                6 STATION_1_1_6
  TO   3008     CATDOG      230.00      0                                          0                                          1

FROM    154-  1 DOWNTN      230.00      1 STATION_1                                7 STATION_1_1_7
  TO   9154     INDGEN1     4.1600      0                                          0                                          W1

FROM    154-  4 DOWNTN      230.00      1 STATION_1                                4 STATION_1_1_4
  TO    203     EAST230     230.00      0                                          0                                          1
  TO   3005     WEST        230.00      0                                          0                                          2
```

Figure 7.9. List Substation Data report

7.8. Substation Flow Report (SPOUT)

The Substation Flow Report activity calculates and reports flows on all switching devices within selected substations.

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To launch the activity, select Node-Breaker>Substation flow report (SPOUT)...

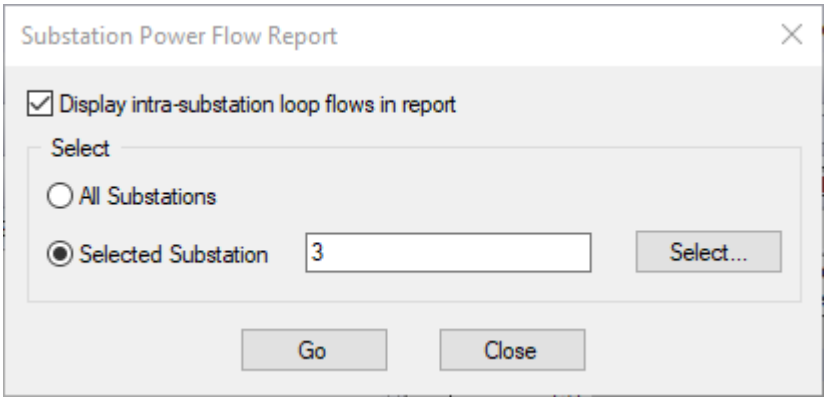

Figure 7.10. Substation Power Flow Report dialog

In the Substation Power Flow Report dialog shown in Figure 7.10, “Substation Power Flow Report dialog”, select which substation to report on:

- All substations to output flow reports for all substation switching devices
- Selected substation to report on only the flows in a specific substation

The "Display intra-substation loop flow in report" toggle indicates whether to include flow values in the report for substation switching devices whose flow has been flagged as being an intra-substation loop flow. When the toggle is selected, all substation flow values will be displayed in the resulting report. When the toggle is not selected, any substation flow value that has been detected as being an intra-substation loop flow will not be printed in the report. The switching device however will still be shown in the report, along with they "!" symbol next to the TYPE column.

For each topological bus section with a substation, a table reports on the FROM and TO node of each substation switching device in the substation, along the type of switching device it is, the MW and MVar flows, and rating and percentage.

A sample report output with intra-substation loop flows displayed looks as shown in Figure 7.11, “Substation Power Flow Report”. Note that "!" is displayed immediately before the MW flow value field for any switching devices that have been flagged as having an intra-substation loop flow.

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```
ALL DATA CATEGORIES WITH SEQUENCE DATA                    RATING SET: 1, %I

SUBSTATION 3 [SS03_ARKANSAS_TYP_4_BH] CONTAINS INTRA-SUBSTATION LOOP FLOWS AS INDICATED BY "!"

SUBSTATION: 3 [SS03_ARKANSAS_TYP_4_BH]  BUS: 154-1 [DOWNTN      230.00]
X--------------- FROM NODE ----------------X X--------------- TO NODE -----------------X CKT X--------------- NAME ----------------X   TYPE         MW     MVAR   RATING       %
  1 SS_ARKANSAS_NODE_1                        3 SS_ARKANSAS_NODE_3                        1  SS_ARKANSAS_SWD_1_3_1                  BREAKER !  -52.17    95.40     0.00
  1 SS_ARKANSAS_NODE_1                        5 SS_ARKANSAS_NODE_5                        1  SS_ARKANSAS_SWD_1_5_1                  BREAKER !  165.50   -25.30     0.00
  1 SS_ARKANSAS_NODE_1                        7 SS_ARKANSAS_NODE_7                        1  SS_ARKANSAS_SWD_1_7_1                  BREAKER !   53.36   -33.97     0.00
  1 SS_ARKANSAS_NODE_1                        8 SS_ARKANSAS_NODE_8                        1  SS_ARKANSAS_SWD_1_8_1                  BREAKER ! -166.68   -36.12     0.00
  2 SS_ARKANSAS_NODE_2                        9 SS_ARKANSAS_NODE_9                        1  SS_ARKANSAS_SWD_2_9_1                  BREAKER !   78.10    93.21     0.00
  2 SS_ARKANSAS_NODE_2                       10 SS_ARKANSAS_NODE_10                       1  SS_ARKANSAS_SWD_2_10_1                 BREAKER     15.80   207.50     0.00
  2 SS_ARKANSAS_NODE_2                       11 SS_ARKANSAS_NODE_11                       1  SS_ARKANSAS_SWD_2_11_1                 BREAKER !  123.18   -11.02     0.00
  2 SS_ARKANSAS_NODE_2                       13 SS_ARKANSAS_NODE_13                       1  SS_ARKANSAS_SWD_2_13_1                 BREAKER !  -52.92  -163.15     0.00
  2 SS_ARKANSAS_NODE_2                       14 SS_ARKANSAS_NODE_14                       1  SS_ARKANSAS_SWD_2_14_1                 BREAKER ! -164.16  -126.54     0.00
  3 SS_ARKANSAS_NODE_3                        9 SS_ARKANSAS_NODE_9                        1  SS_ARKANSAS_SWD_3_9_1                  BREAKER !  291.90    91.79     0.00
  4 SS_ARKANSAS_NODE_4                       10 SS_ARKANSAS_NODE_10                       1  SS_ARKANSAS_SWD_4_10_1                 BREAKER    214.20   -15.50     0.00
  5 SS_ARKANSAS_NODE_5                       11 SS_ARKANSAS_NODE_11                       1  SS_ARKANSAS_SWD_5_11_1                 BREAKER !  119.32   108.27     0.00
  7 SS_ARKANSAS_NODE_7                       13 SS_ARKANSAS_NODE_13                       1  SS_ARKANSAS_SWD_7_13_1                 BREAKER !   55.35   -35.19     0.00
  8 SS_ARKANSAS_NODE_8                       14 SS_ARKANSAS_NODE_14                       1  SS_ARKANSAS_SWD_8_14_1                 BREAKER !  164.16     3.57     0.00

SUBSTATION: 3 [SS03_ARKANSAS_TYP_4_BH]  BUS: 154-6 [DOWNTN      230.00]
X--------------- FROM NODE ----------------X X--------------- TO NODE -----------------X CKT X--------------- NAME ----------------X   TYPE         MW     MVAR   RATING       %
  6 SS_ARKANSAS_NODE_6                       12 SS_ARKANSAS_NODE_12                       1  SS_ARKANSAS_SWD_6_12_1                 BREAKER    100.00    80.00     0.00

SUBSTATION: 3 [SS03_ARKANSAS_TYP_4_BH]  BUS: 9154 [INDGEN1     4.1600]
X--------------- FROM NODE ----------------X X--------------- TO NODE -----------------X CKT X--------------- NAME ----------------X   TYPE         MW     MVAR   RATING       %
 15 SS_ARKANSAS_NODE_15                     17 SS_ARKANSAS_NODE_17                       1  SS_ARKANSAS_SWD_15_17_1                BREAKER      0.00     0.00     0.00
 16 SS_ARKANSAS_NODE_16                     18 SS_ARKANSAS_NODE_18                       1  SS_ARKANSAS_SWD_16_18_1                BREAKER      0.00     0.00     0.00
 17 SS_ARKANSAS_NODE_17                     18 SS_ARKANSAS_NODE_18                       1  SS_ARKANSAS_SWD_17_18_1                BREAKER     -2.00     1.18     0.00
```

Figure 7.11. Substation Power Flow Report

A sample report output without intra-substation loop flows displayed looks as shown in Figure 7.12, "Substation Power Flow Report without Intra-substation Loop Flows Reported".

```
SUBSTATION 3 [SS03_ARKANSAS_TYP_4_BH] CONTAINS INTRA-SUBSTATION LOOP FLOWS AS INDICATED BY "!" AND ARE NOT SHOWN IN THE REPORT

SUBSTATION: 3 [SS03_ARKANSAS_TYP_4_BH]  BUS: 154-1 [DOWNTN      230.00]
X--------------- FROM NODE ----------------X X--------------- TO NODE -----------------X CKT X--------------- NAME ----------------X   TYPE         MW     MVAR   RATING       %
  1 SS_ARKANSAS_NODE_1                        3 SS_ARKANSAS_NODE_3                        1  SS_ARKANSAS_SWD_1_3_1                  BREAKER !
  1 SS_ARKANSAS_NODE_1                        5 SS_ARKANSAS_NODE_5                        1  SS_ARKANSAS_SWD_1_5_1                  BREAKER !
  1 SS_ARKANSAS_NODE_1                        7 SS_ARKANSAS_NODE_7                        1  SS_ARKANSAS_SWD_1_7_1                  BREAKER !
  1 SS_ARKANSAS_NODE_1                        8 SS_ARKANSAS_NODE_8                        1  SS_ARKANSAS_SWD_1_8_1                  BREAKER !
  2 SS_ARKANSAS_NODE_2                        9 SS_ARKANSAS_NODE_9                        1  SS_ARKANSAS_SWD_2_9_1                  BREAKER !
  2 SS_ARKANSAS_NODE_2                       10 SS_ARKANSAS_NODE_10                       1  SS_ARKANSAS_SWD_2_10_1                 BREAKER     15.80   207.50     0.00
  2 SS_ARKANSAS_NODE_2                       11 SS_ARKANSAS_NODE_11                       1  SS_ARKANSAS_SWD_2_11_1                 BREAKER !
  2 SS_ARKANSAS_NODE_2                       13 SS_ARKANSAS_NODE_13                       1  SS_ARKANSAS_SWD_2_13_1                 BREAKER !
  2 SS_ARKANSAS_NODE_2                       14 SS_ARKANSAS_NODE_14                       1  SS_ARKANSAS_SWD_2_14_1                 BREAKER !
  3 SS_ARKANSAS_NODE_3                        9 SS_ARKANSAS_NODE_9                        1  SS_ARKANSAS_SWD_3_9_1                  BREAKER !
  4 SS_ARKANSAS_NODE_4                       10 SS_ARKANSAS_NODE_10                       1  SS_ARKANSAS_SWD_4_10_1                 BREAKER    214.20   -15.50     0.00
  5 SS_ARKANSAS_NODE_5                       11 SS_ARKANSAS_NODE_11                       1  SS_ARKANSAS_SWD_5_11_1                 BREAKER !
  7 SS_ARKANSAS_NODE_7                       13 SS_ARKANSAS_NODE_13                       1  SS_ARKANSAS_SWD_7_13_1                 BREAKER !
  8 SS_ARKANSAS_NODE_8                       14 SS_ARKANSAS_NODE_14                       1  SS_ARKANSAS_SWD_8_14_1                 BREAKER !

SUBSTATION: 3 [SS03_ARKANSAS_TYP_4_BH]  BUS: 154-6 [DOWNTN      230.00]
X--------------- FROM NODE ----------------X X--------------- TO NODE -----------------X CKT X--------------- NAME ----------------X   TYPE         MW     MVAR   RATING       %
  6 SS_ARKANSAS_NODE_6                       12 SS_ARKANSAS_NODE_12                       1  SS_ARKANSAS_SWD_6_12_1                 BREAKER    100.00    80.00     0.00

SUBSTATION: 3 [SS03_ARKANSAS_TYP_4_BH]  BUS: 9154 [INDGEN1     4.1600]
X--------------- FROM NODE ----------------X X--------------- TO NODE -----------------X CKT X--------------- NAME ----------------X   TYPE         MW     MVAR   RATING       %
 15 SS_ARKANSAS_NODE_15                     17 SS_ARKANSAS_NODE_17                       1  SS_ARKANSAS_SWD_15_17_1                BREAKER      0.00     0.00     0.00
 16 SS_ARKANSAS_NODE_16                     18 SS_ARKANSAS_NODE_18                       1  SS_ARKANSAS_SWD_16_18_1                BREAKER      0.00     0.00     0.00
 17 SS_ARKANSAS_NODE_17                     18 SS_ARKANSAS_NODE_18                       1  SS_ARKANSAS_SWD_17_18_1                BREAKER     -2.00     1.18     0.00
```

Figure 7.12. Substation Power Flow Report without Intra-substation Loop Flows Reported

7.9. Isolate Item by Breaker

The Isolate Item by Breaker activity will attempt to isolate a specified component of the network by finding system or substation breakers to open. It can be performed on buses, substation nodes and switching devices, AC branches including system switching devices and transformers, DC lines and FACTS devices, and all bus connected devices such as loads, machines and shunts.

Once invoked, the activity will systematically seek to isolate the component by searching outward from that device until it finds a system or substation breaker that can be opened. Any in service branches, transformers,

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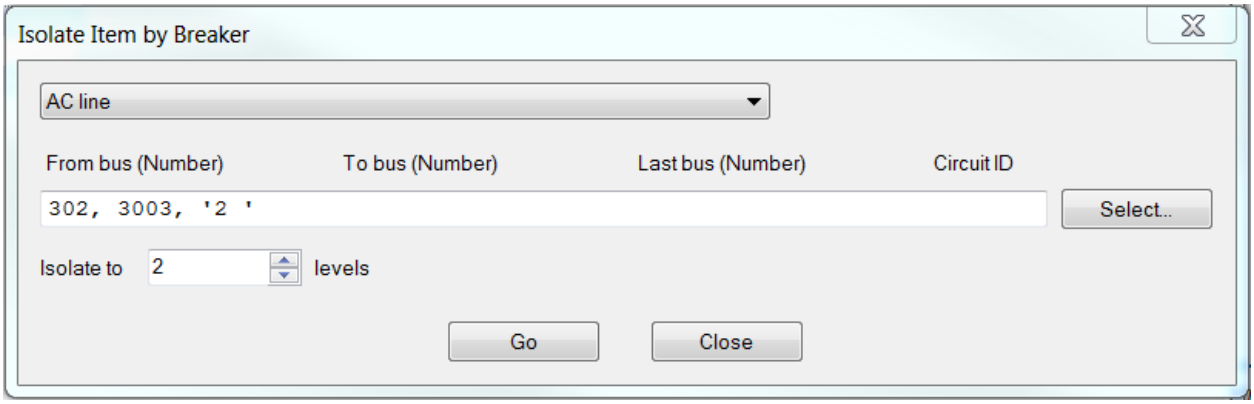
or non-breaker switching devices are skipped over until a valid system or substation breaker is found. If any DC devices are encountered, the process will block these devices and not continue along the search path for breakers. When either a breaker is opened, or a DC device is encountered, the search in that particular direction stops.

The search for breakers will propagate outward for a specified number of bus levels in every direction. For example, starting at a bus and skipping over all connected branches while searching for a breaker, constitutes a step of one level outward in all directions. When breakers are successfully opened in all directions sufficient to isolate the component, the process successfully stops. If the component is not successfully isolated within the given number of levels outward, the process will terminate with a message.

The current maximum number of levels in which the isolate algorithm will search is 50 levels.

To run the Isolate Item by Breaker activity, select Node-Breaker>Isolate item by breaker.



Select the desired type of component to be isolated from the drop-down menu, including lines, transformers, connected equipment, substation switching devices, and nodes. Then select the specific item to be isolated. If a substation switching device is to be isolated, simply enter the 'Substation number, From node, To node, CKT ID'. If a node is to be isolated, enter the 'Substation number, Node number'.

The option to modify the number of levels outward in which to search for breakers can also be modified. The larger the number, the wider and more extensively the network will be taken out if breakers are not located nearby. The default number of levels can also be modified through Misc>Change program settings (OPTN), on the Node-Breaker tab.

It is also possible to invoke the Isolate Item by Breaker operation from a network diagram by doing a right-mouse-click on the component to be isolated and selecting the Isolate item by Breaker option as shown in Figure 7.13, “Performing Isolate by Breaker from a diagram”

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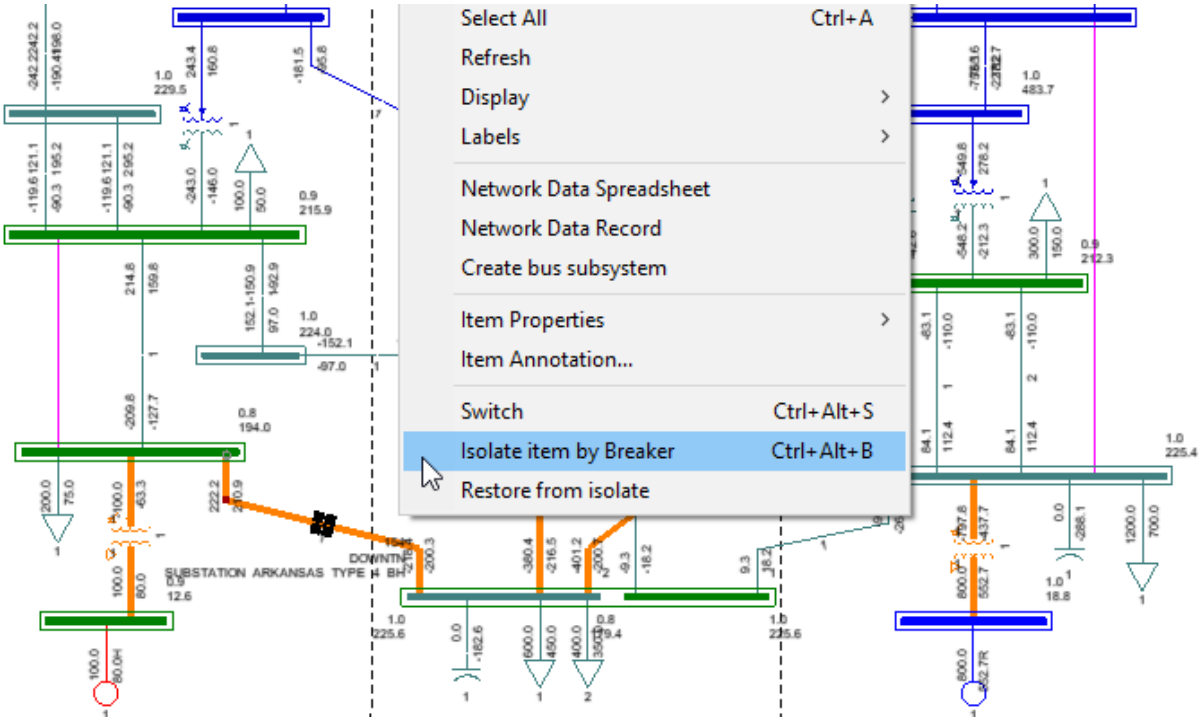

Figure 7.13. Performing Isolate by Breaker from a diagram

During the Isolate operation, progress output will display the breakers that are opened. If successful, this will be followed by a TREE report on any islands created.

```
Opened breaker "1" from node 5 [SUB_STATION_03_5] to node 1 [SUB_STATION_03_1] in substation 3 [SUB_STATION_03]

 Opened breaker "1" from node 5 [SS_ARKANSAS_NODE_5] to node 9 [SS_ARKANSAS_NODE_9] in substation 3 [SUBSTATION_A

 Opened breaker "1" from node 4 [SS_DANUBE_NODE_4] to node 7 [SS_DANUBE_NODE_7] in substation 10 [SUBSTATION_DANU

 Opened breaker "1" from node 4 [SS_DANUBE_NODE_4] to node 1 [SS_DANUBE_NODE_1] in substation 10 [SUBSTATION_DANU

 Bus 154 is split into the following bus sections due to operations of switching devices:
     Bus section    154-  1
     Bus section    154-  2
     Bus section    154-  5

 Bus 3008 is split into the following bus sections due to operations of switching devices:
     Bus section   3008-  1
     Bus section   3008-  4

 Node 5 [SS_ARKANSAS_NODE_5] in substation 3 [SUBSTATION_ARKANSAS_TYPE_4_BH] is isolated

 Status of branch circuit "1" from bus 154-5 [DOWNTN      230.00] to bus 3008-4 [CATDOG      230.00] set to 0

 Bus    154-  5 has been islanded due to operations of switching devices and placed out of service

 Node 4 [SS_DANUBE_NODE_4] in substation 10 [SUBSTATION_DANUBE_TYPE_4_BH] is isolated

 Bus   3008-  4 has been islanded due to operations of switching devices and placed out of service
```


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```
SWING BUSES:
  BUS#-SCT X-- NAME --X BASKV AREA     BUS#-SCT X-- NAME --X BASKV AREA     BUS#-SCT X-- NAME --X BASKV AREA
  3011     MINE_G      13.800    5


ISLAND:
  BUS#-SCT X-- NAME --X BASKV AREA     BUS#-SCT X-- NAME --X BASKV AREA     BUS#-SCT X-- NAME --X BASKV AREA

                                                ** NONE **
```

If the operation is not successful in isolating the component, a message such as the following will be displayed at the end, and the islanding report will not be run.

```
Reached maximum of 4 levels outward; unable to isolate line by opening breakers
```

The impact of the isolate operation will be reflected in the network data and be visible on network diagrams, as shown in Figure 7.14, "Diagram displaying resulting actions of Isolate operation". As compared to the prior diagram, buses 3008 and 154 have both been split due to the opening of breakers at either end of the branch connecting the two buses.

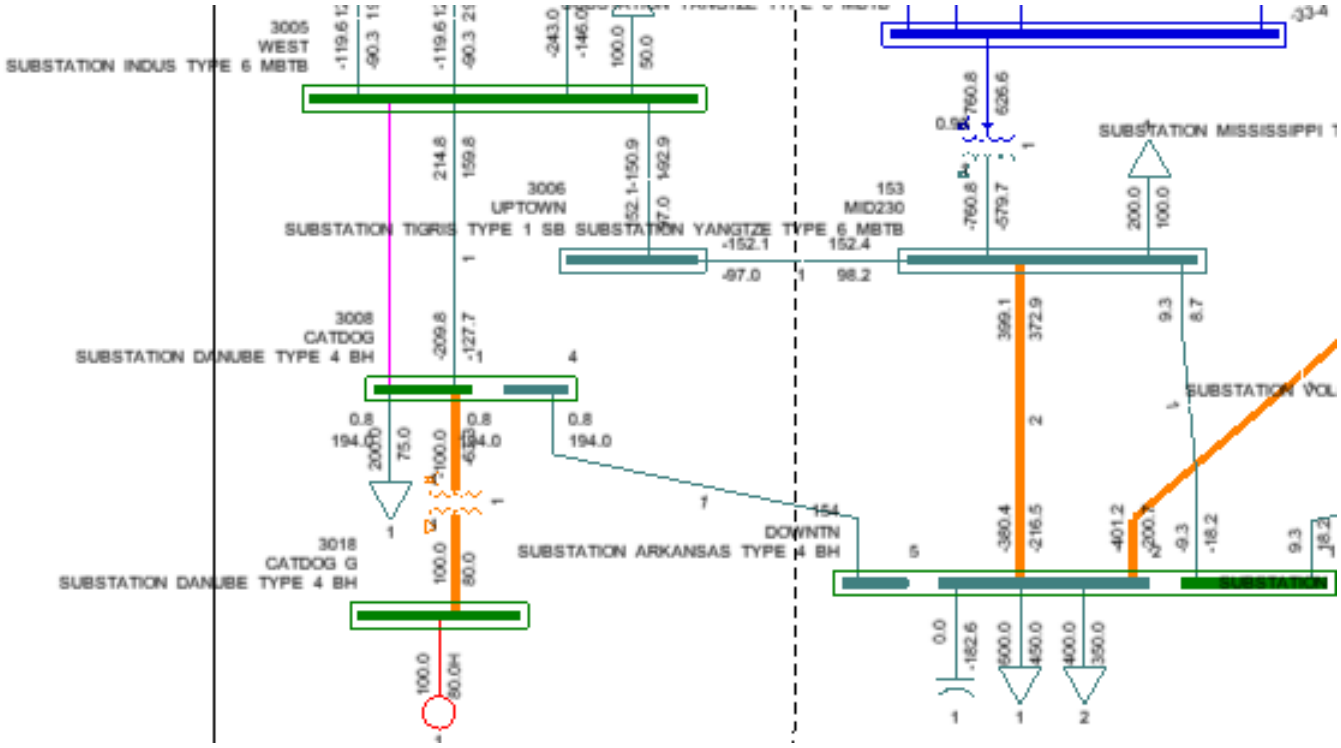

Figure 7.14. Diagram displaying resulting actions of Isolate operation

An Isolate by Breaker report will also be generated displaying the specific network components impacted by the Isolate operation. A sample of the report is shown in Figure 7.15, "Isolate by Breaker Report".

```
The following nodes have been set to out-of-service:

  BUS#-SCT X-- NAME --X BASKV  NODE X--------------- NAME -----------------X
   154-   5 DOWNTN      230.00     5 SS_ARKANSAS_NODE_5
  3008-   4 CATDOG      230.00     4 SS_DANUBE_NODE_4
```


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```
The following system breakers have been opened:

X----------- FROM BUS -----------X X------------ TO BUS ------------X
  BUS#-SCT X-- NAME --X BASKV NODE   BUS#-SCT X-- NAME --X BASKV NODE CKT
   154-  5 DOWNTN      230.00   5    3008-  4 CATDOG      230.00   4   1

The following substation breakers have been opened:

X------------------ STATION ------------------X X----------------- FROM NODE -----------------X X------------------ TO NODE ------------------X CKT
  ST #  NAME                                   NODE #  NAME                                    NODE #  NAME                                    ID
     3 SUBSTATION_ARKANSAS_TYPE_4_BH                 9 SS_ARKANSAS_NODE_9                          5 SS_ARKANSAS_NODE_5                       1
    10 SUBSTATION_DANUBE_TYPE_4_BH                   1 SS_DANUBE_NODE_1                            4 SS_DANUBE_NODE_4                         1
    10 SUBSTATION_DANUBE_TYPE_4_BH                   7 SS_DANUBE_NODE_7                            4 SS_DANUBE_NODE_4                         1
```

Figure 7.15. Isolate by Breaker Report

One or more isolate operations can be sequentially performed, with the network state reflecting the comprehensive outcome. To restore the network to its state prior to the first isolate operation, use the Restore from Isolate activity, described below.

7.10. Restore from Isolate

The Restore from Isolate activity restores the working case to the state is was in just prior to the initial Isolate, or to the state prior to the first Isolate operation after a previous Restore.

To run the Restore from Isolate activity, select Node-Breaker>Restore from Isolate. The operation will be immediate, with no dialog window. When complete, a message will be displayed to the Progress area:

```
Working case restored to state prior to ISOLATE actions
```

The Restore from Isolate operation can also be performed from the diagram by doing a right-mouse-click on any component in the diagram, and selecting Restore from Isolate.

The resetting of all network states back to their pre-isolation values will be reflected in the working case.

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Chapter 8
Geomagnetically Induced Currents

8.1. Activity GIC

- **Run Activity GIC - GUI**

  GIC > GIC Analysis

- ***Requirements / Prerequisites***

  Valid power flow working case

  Valid GIC data (See Data Format Manual, Chapter *GIC Data*)

- **Interrupt Control Codes**

  AB

Activity GIC is to calculate Geomagnetically Induced Currents (GIC) in electric power system network as a result of Geomagnetic Disturbance (GMD). The flow of GIC in transformers causes their half-cycle saturation. This results in increased harmonics, reactive power consumption and increase in transformer hot-spot temperature. After calculating GIC flows, activity GIC calculates transformer reactive power losses.

Activity GIC can be run on entire power flow network or specified sub system.

The user specifies one or all of the following options for the correct execution of activity GIC.

8.1.1. GIC Data

To be able to perform GIC calculations, the additional data like substation locations, dc resistance and grounding state of various network elements is required. This data is not required in power flow studies and hence not part of the case data most of the time. This additional data is provided using "GIC Data Input File (.gic)" file. Refer "Data Formats" manual, Chapter GIC Data File for details. The GIC calculations will not be run without supplying this additional data.

Beginning with PSSE version 35.0, the GIC data is stored in the Saved Case format.

If the Saved Case in PSSE working memory does not have GIC data, use activity GIC_READ to provide additional GIC data.

Application Note:

At PSSE version 35.0 and version 34.8, network equipment location information is consolidated in one source which is base case data. To faciliate that:

- The substation location information is moved from the GIC data file to the Base Case power flow data.
- When GIC data file version 4 is read, "Single Bus" substation configuration is added to each bus-substation data record and non Node-Breaker base case is modified to a case with Node-Breaker configurations.

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- For this to work without any data conflicts, GIC data file version 4 is allowed to be read into base case that has no Node-Breaker configurations.
- Reading GIC data file version 4 adds "Single Bus" substation configuration for each bus-substation record to the working case. Depending on the size the network and number of substations added, this process could take considerable CPU time. So it is recommended that after reading GIC data file version 4 to the working case, save the working case in Saved case format (.sav file) and GIC data to GIC data file version 5 (.gic). Then use this .sav and .gic files in any GIC analysis.

8.1.2. GMD Event

The GMD event is specified in terms of its geoelectric field (E-field) strength and orientation (degrees). Following different GMD events are modeled for calculating GICs flowing in power system network.

a. Uniform Geoelectric Field

   The GMD event is specified by uniform geoelectric field (E) in V/km and its direction in degrees.

   - The branch induced voltages in the entire study subsystem branches are calculated applying uniform geoelectric field.
   - The location factor 'alpha' and earth conductivity factor 'beta' are not applied.

b. Benchmark GMD Event Geoelectric Field

   The benchmark GMD event is specified by geoelectric field (E) of 8 V/km and its direction in degrees.

   The substation location data is provided in terms of geographical coordinates. However, NERC TPL-007 provides alpha scaling factors in terms of geomagnetic coordinates. Therefore, GIC activity "Benchmark Year" option is used to convert substation geographical coordinates to geomagnetic coordinates.

   For a transmission line, higher of 'From End' and 'To End' geomagnetic latitude is considered which gives conservative value of alpha.

   The substation earth model is provided either from activity option argument or in substation record of GIC data file. This substation earth model name is used to get beta scaling factors as per NERC TPL-007.

   For a transmission line, with different 'From End' and 'To End' earth conductivity models, higher value of beta factor is used.

   - The branch induced voltages in the entire study subsystem branches are calculated applying benchmark geoelectric field.
   - The location factor 'alpha' and the benchmark event earth conductivity factor 'beta' are applied.

c. Supplemental GMD Event Geoelectric Event

   The Supplemental GMD event is specified by geoelectric field (E) of 12 V/km and its direction in degrees.

   Similar to the Benchmark event, NERC TPL-007 provides alpha and beta scaling factors for the Supplemental event.

   - The branch induced voltages in the entire study subsystem branches are calculated applying supplemental geoelectric field.

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- The location factor 'alpha' and the supplemental event earth conductivity factor 'beta' are applied.

d. Uniform + Supplemental Geoelectric Field

- The branch induced voltages in the supplemental study subsystem branches are calculated applying supplemental geoelectric field.
- The branch induced voltages in the remaining study subsystem branches are calculated applying uniform geoelectric field.
- The location factor 'alpha' and earth conductivity factor 'beta' are not applied.

e. Benchmark + Supplemental GMD Event

- The branch induced voltages in the supplemental study subsystem branches are calculated applying supplemental geoelectric field, the location factor 'alpha' and the supplemental event earth conductivity factor 'beta'.
- The branch induced voltages in the remaining study subsystem branches are calculated applying benchmark geoelectric field, the location factor 'alpha' and the benchmark event earth conductivity factor 'beta'.

f. Non-Uniform Geoelectric Field from Electrojet

Following data is provided to define the electrojet source current characteristics.

- Electrojet Current Amplitude (million Amps)
- Electrojet Current Density Cauchy Distribution half-width (km)
- Electrojet Period of Variation (minutes)
- Electrojet Height of Current (km)
- Latitude of Electrojet Center (degrees)

g. Non-Uniform Near Real-Time Geoelectric Field

Following types of efield data files can be specfied to calculate the branch induced voltages.

- A comma separated values (csv) file

  Format of data columns: latitude (deg), longitude (deg), Ee (V/km), En (V/km)

  Example of this csv file is as below.

```
latitude(deg),   longitude(deg),     Ee (V/km),     En (V/km)
       32.00,           -80.50,          -2.1,          1.20
       32.00,           -81.00,          1.45,         -1.45
       32.00,           -81.50,          1.65,          2.13
```

- USGS/SWPC json file

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The near real-time E-field mapping project, a joint effort between NOAA/SWPC, the USGS geomagnetism group, and the NASA/CCMC, provides Geoelectric Field Map data.

This data is made available at URLs:

- Fernberg 1D Model

  https://services.swpc.noaa.gov/json/lists/rgeojson/InterMagFB1DLP/

- Empirical 3D Model

  https://services.swpc.noaa.gov/json/lists/rgeojson/InterMagEarthScope/

The format of this json file is as below.

```
   {"type": "FeatureCollection", "features": [
   {"type": "Feature",
   "geometry": {"type": "Point", "coordinates": [lon, lat]},
   "properties": {"Ex": mV, "Ey": mV, "quality_flag": val,
"distance_nearest_station": km}
   },
   {"type": "Feature",
   "geometry": {"type": "Point", "coordinates": [lon, lat]},
   "properties": {"Ex": mV, "Ey": mV, "quality_flag": val,
"distance_nearest_station": km}
   },
   ]
   }
```

8.1.3. Study Subsystems

GIC analysis can be performed either for entire power flow network or for a subsystem.Two subsystem can be defined.

a. Study subsystem

   This subsystem is applied to main event only.

b. Supplemental event subsystem

   Supplemental event subsystem must be subset of the Study subsystem. The Supplemental event subsystem is applied to second (supplemental) event only. It is used only when main event is either uniform or benchmark event. This subsystem is not used when supplemental event is simulated with moving box criterion.

Intertie Levels

In reality during GMD event, the flow of GICs will not be restricted only to subsystem, but GICs will flow in the entire network. So in order to consider the effect of GIC flows in the neighboring network transformers, the "Intertie Levels" can be specified.

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The "Intertie Levels" option is used as below:

- tielevels = 0, consider buses in study subsystem only
- tielevels >0, 1 through 5, consider study subsystem buses + these many levels of inter-tie buses and ignore subsystem definition voltage limits for intertie buses added
- tielevels <0, -5 through -1, consider study subsystem buses + these many levels of inter-tie buses and consider subsystem definition voltage limits for intertie buses added

Examples

- tielevels = 1, consider buses in subsystem + buses 1 level up from subsystem tie buses, ignore the bus voltage of all level 1 buses even when subsystem is defined with voltage limits
- tielevels = -2, consider buses in subsystem + buses 2 level up from subsystem tie buses, consider the bus voltage of all level 1 and level 2 buses when subsystem is defined with voltage limits

Subsystem Boundary Transformers

When study subsystem definition includes base voltage filter this option is used to include/not include of boundary transformer buses in study subsystem. This option is used as below (api option boundary_trn):

- boundary_trn = 0, consider buses in study subsystem only
- boundary_trn = 1, include buses of boundary transformers in study subsystem considering subsystem definition voltage limits
- boundary_trn = 2, include buses of boundary transformers in study subsystem ignoring subsystem definition voltage limits

When using this option, if a substation number is not provided for any bus in the study subsystem, attempt is made to derive the associated substation based on the transformer and zero impedance line connections.

8.1.4. Output Files

a. GIC Case Data Output File (.raw or .sav)

One of the measures of representing GMD effect on power system grid is to add increased reactive power losses in transformers as the constant current loads. This file is to add those loads to the base case as per option selected.

- Option RDCH - The resulting working case can also be saved as read change RDCH (.raw) file. Use this file to apply GMD effect Mvar losses to the base using activity RDCH.
- Option Saved Case - The resulting working case can be saved as a complete case (.sav) file.

b. GIC to Base Case RDCH Output File (.raw)

GMD effect Mvar losses added to the working case can be removed by removing these Mvar losses from the working case. This is a read change RDCH (.raw) file that sets GMD effect constant current loads with "out-of-service" status. Apply this file to the working case using activity RDCH.

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c. GIC Resistive Network Output Raw File (.raw)

The AC power flow base case network is converted to DC resistive network for GIC calculations. This is a complete raw file (.raw) representation of DC resistive network.

The main purpose of providing this file is to check and validate the DC network used to calculate GIC flows. Using activity 'READ', open this file in as raw file and draw a slider diagram of buses of interest to evaluate the DC network used.

This file cannot be used run a power flow. There are no generators, swing bus etc. typical of power flow case. So it cannot be used for any other purpose than to draw DC network single line diagram and verify it.

Note that, there will be many more buses added to the base case when creating the DC network to facilitate transformer winding, shunt elements and substation ground connections.

d. GIC Results Mapping Output File (.pygic)

This file stores the GIC calculation results in Python syntax. This file is used to plot GIC analysis results on network maps drawn to substation geographical locations. This being a Python file of GIC results, it can be used as input to any user developed GIC post processing Python scripts. This file is used by pssarrays.gicmaps or arrbox.gicmaps modules.

e. Transformer Thermal Impact Analysis GIC(t) Output File (.gict)

This file is a comma separated (.csv) output file of E(t) values of GMD events used and GIC(t) values of selected transformers. The GIC(t) data from this file is used as input to the Transformer Thermal Assessment process. This file is used by pssarrays.gicthermal or arrbox.gicthermal modules

Application Note:NERC TPL-007 Benchmark event is 11200 data points of time, eastward and northward geoelectric fields. For this event GIC(t) for each transformer will also contain 11200 data points. So this file tend to a big size file, Use activity report option "top" to limit the size of this file.

Application Notes:

- The "output" files specified here will be created only if their names are provided.
- For moving box calculations, selected output files will be created for each moving box calculation with suffix "_supp_boxN" added to each file name.

8.1.5. Calculation Options

Calculation for specified Storm Strength and Orientation (No scan)

One GIC calculation is done for specified E-field magnitude and degree orientation. This one calculation could include addition of supplemental event to main event.

Degree Scan Calculations

The storm orientation (degree) scan is performed to find a storm orientation that gives maximum Mvar losses when event is ‘uniform’, ‘benchmark’ or ‘supplemental’. The Mvar losses are obtained from converting the effective GIC flow in transformers to corresponding Mvar values. The power flow solution is not needed to run degrees scan. The degree scan is performed in following steps.

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i. The degree scan is performed in steps of a specified degree in an orientation span of 0 through 180 degrees. When step size is not specified, default step size used is 10 degrees. If option to run power flow for each degree scan is selected, power flow solution is performed for each calculation.

ii. Find orientation that gives highest Mvar losses.

iii. If the degree step size is > 1 degree, additionally in steps of ±1 degree from orientation in (ii), Mvar loss calculations are performed to find orientation that gives highest Mvar losses.

NERC TPL-007 requires performing the degree scan. Depending on the report option selected, text report is created either for all orientations simulated or only orientation that gives highest Mvar losses. A summary report of each scanned degree, corresponding Mvar losses and power flow solution state is created as well.

Magnitude Scan Calculations

The E-field magnitude scan is performed to find the highest storm strength (E-field magnitude) that network can withstand. This is determined by solving power flow after applying GMD induced Mvar losses as constant current loads to the base case. The magnitude scan is performed in following steps:

i. Start the magnitude scan only if E-field specified with orientation (specified or obtained from degree scan) yields a converged power flow solution.

ii. The magnitude scan is performed in steps of a specified E-field step size in an E-field span of initial E-field through specified maximum E-field. For example, for a benchmark event, when specified step size is 2 V/km and specified maximum E-field is 50 V/km, magnitude scans are performed from 8 through 50 V/km in steps of 2 V/km. That is 26 magnitude scans will be performed in this run. The total numbers of magnitude scans are limited to 50.

iii. Increase E-field in steps of a specified E-field step size up up to specified maximum E-field or to a magnitude when power flow solution is non-converged.

iv. If E-field in step (iii) is less than specified maximum E-field magnitude and if the step size is > 1 V/km, decrease E-field in steps of 1 V/km till converged power flow solution is obtained.

Depending on the report option selected, text report is created either for all storm strengths simulated or only final storm strength that gives converged power flow solution. A summary report of each scanned E-field magnitude, corresponding Mvar losses and power flow solution state is created as well.

Degree and Magnitude Scan Calculations

First the orientation scan is performed as above to get orientation that results in highest Mvar losses. Using that orientation, magnitude scan is performed as above to determine storm strength that network can withstand.

8.1.6. Supplemental Event Calculations

The supplemental GMD event simulates localized peak geoelectric fields. Per NERC TPL-007 the supplemental GMD vulnerability assessment requirements, the gic flows and GIC(t) profiles can be obtained by defining the subset of study network grid where supplemental event is applied. This subset of study network can be defined by susbsyem definition or moving box (api option supp_evt).

- supp_evt = 0, no supplemental event

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Supplemental Event defined with Subsystem

Define subset of study network grid where supplemental event is applied using bus subsystem.

- supp_evt = 1, consider buses in supplemental event subsystem only

Following criteria is used to calculate the branch induced voltage when supp_evt=1.

1. Both buses of the branch are part of benchmark event subsystem. The branch induced voltage is calculated by applying benchmark (or uniform) efield to the entire branch.

2. Both buses of the branch are part of supplemental event subsystem. The branch induced voltage is calculated by applying supplemental efield to the entire branch.

3. When one bus of the branch is part of benchmark event subsystem and the other bus is part of supplemental event subsystem, following api option determines how induced voltage on such a branch is calculated.

    - brn_seg_efld = 0, the branch induced voltage is calculated by applying benchmark (or uniform) efield to the entire branch.

    - brn_seg_efld = 1, the branch induced voltage is calculated by applying supplemental efield to the entire branch.

Supplemental Event defined with Moving Box

Define subset of study network grid where supplemental event is applied using one box or number of moving boxes.

- supp_evt = 2, Rank substations with maximum GIC flows as center of the moving box from storm orientation scans that results in maximum VAR losses.

- supp_evt = 3, Rank transformers with maximum GIC flows as center of the moving box from storm orientation scans

- supp_evt = 4, Use substation number provided as center of the moving box

- supp_evt = 5, Use location provided as center of the moving box

For supp_evt = 2 or 3, the number of moving boxes to be simulated are specified by api option "supp_box_num".

For supp_evt = 4, the substation number of the center of the moving box is specified by api option "supp_box_num".

For supp_evt = 5, the longitude and latitude co-ordinates of the center of the moving box are specified by api options "supp_box_lon_c" and "supp_box_lat_c".

Size of the Supplemental Event Moving Box

Per NERC TPL-007, the moving box size is specified as 100 km in North-South latitude direction and 500 km in East-West longitude direction. However, any different size can be specified using api options "supp_box_ns_km" and "supp_box_ew_km".

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Branch Intersection with Moving Box and its GMD Induced Voltage

There are three possible scenarios as shown in Figure 8.1, “Branch Intersection with Moving Box”

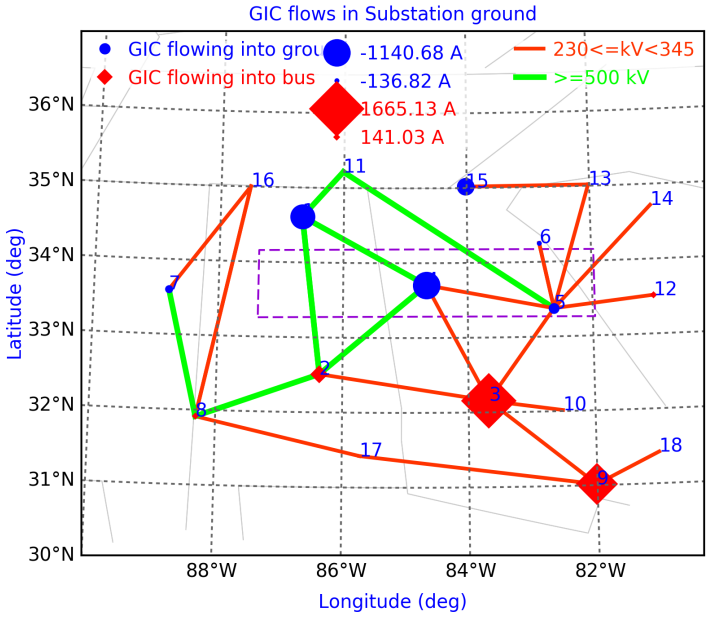

Figure 8.1. Branch Intersection with Moving Box

1. Entire branch is outside of the moving box. The branch induced voltage is calculated by applying main event benchmark (or uniform) Efield.

2. Entire branch is inside of the moving box. The branch induced voltage is calculated by applying supplemental event Efield.

3. Branch segments are inside and outside of the moving box. Following api option determines how induced voltage on such a branch is calculated.

   - brn_seg_efld = 0, apply Supplemental event Efield to line segment inside the box and Benchmark (or uniform) event Efield to line segments outside the box.

   - brn_seg_efld = 1, apply Supplemental event Efield to entire branch.

8.1.7. Non-Uniform Geoelectric Field Event Calculations

Using the non-uniform near real-time geoelectric field data and an adaptive nearest neighbour search algorithm, branch induced voltages are calculated. So for this event, options like degree scan, magnitude scan and moving box, are disabled.

The non-uniform near real-time geoelectric field data is provided either through csv file or json file. The USGS/SWPC json files can be directly downloaded and used. Also if any other entity provides geoelectric field data that conforms the json format of USGS/SWPC file, those can be downloaded and used. The geoelectric field values (mV/km or V/km) provided in these files are calculated considering both location and earth conductivity. Therefore, no other correction factors are applied when using this efield data.

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8.1.8. Worst Case Determination

a. Worst Case for Transformer Effective GIC Flows - NERC TPL-007 requires to determine the maximum effective GIC value for the worst case geoelectric field orientation for the benchmark or Supplemental GMD event. The "Degree Scan" calculations provide the maximum effective GIC and the storm orientation when that happened for each transformer in the study subsystem.

b. Worst Case for GMD Vulnerability Assessment - After "Degree Scan" calculations, the storm orientation that results in highest Mvar losses in the desired/selected portion of the grid is treated as worst case for GMD vulnerability assessment. Following api option determines which transformers to include in worst case determination.

   - worstcase_trn=0, All transformers in study subsystem
   - worstcase_trn=1, All transformers in report subsystem

8.1.9. Default Data

From activity dialog, press button “Defaults” to populate the activity options to default values.

DC Resistance Defaults

- Activity GIC uses the equipment or network element DC resistance data provided in .gic file as is without any conversion or scaling of that data.
- When the AC power flow data of the branches and the transformers do not provide their AC resistance value, GIC activity assumes X/R ratio provided in activity argument to determine DC resistance value from their reactance (X) values.
- The AC to DC conversion factors are used when branch and transformer DC resistance values are default values calculated from AC power flow data.

$$R_{dc} = \text{Conversion Factor} \times R_{ac\text{-}default} \tag{8.1}$$

Benchmark Year

The benchmark and supplemental GMD events in NERC TPL-007 are defined for geomagnetic latitude of 60°. These events need be scaled to account for regional differences based on geomagnetic latitude.

In GIC data file, the substation location is specified in terms of its geographic coordinates. These geographic coordinates need to be converted to its geomagnetic coordinates. The benchmark year value is used in conversion of substation geographic latitude to corresponding geomagnetic latitude to get the alpha (α) scaling factor. Following alpha factors lookup table is implemented.

Table 8.1. E-Field Location Scaling Factors

Geomagnetic Latitude	Scaling Factor
(degrees)	(α)
≤40	0.1
41	0.112
42	0.126
43	0.141

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44	0.158
45	0.178
46	0.2
47	0.224
48	0.251
49	0.282
50	0.316
51	0.355
52	0.398
53	0.447
54	0.501
55	0.562
56	0.631
57	0.708
58	0.794
59	0.891
≥60	1.0

Fernberg 1D Earth Conductivity Models

To run benchmark and supplemental GMD event calculations, each substation must have earth conductivity model specified. The benchmark and supplemental GMD events in NERC TPL-007 are defined for the reference Quebec earth model (SHIELD – QC, ON).

The NERC TPL-007 reference benchmark and supplemental geoelectric field, Epeak, used in a GMD Vulnerability Assessment is scaled using the earth conductivity scaling factor beta (β) of each substation earth conductivity model.

The earth conductivity models of US and Canada defined in TPL-007 are modeled as standard models. Additionally user defined earth conductivity models can be specified in GIC data file.

a. Standard Earth Conductivity Models

Following one dimensional earth conductivity models defined by United States Geological Survey (USGS) are added as standard earth conductivity models of GIC activity.

Table 8.2. 1D Earth Models of United States

Name	Description	Name	Description
AK1A	Adirondack Mountains-1A	IP2	Interior Plains
AK1B	Adirondack Mountains-1B	IP3	Interior Plains (Michigan)
AP1	Appalachian Plateaus	IP4	Interior Plains (Great Plains)
AP2	Northern Appalachian Plateaus	NE1	New England
BR1	Northewest Basin and Range	PB1	Pacific Border (Willamette Valley)
CL1	Colorado Plateau	PB2	Pacific Border (Puget Lowlands)
CO1	Columbia Plateau	PT1	Piedmont
CP1	Coastal Plain (South Carolina)	SL1	St. Lawrence Lowlands

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CP2	Coastal Plain (Georgia)	SU1	Superior Upland
CS1	Cascade-Sierra Mountains	BOU	
FL1	Florida	FBK	
IP1	Interior Plains (North Dakota)	PRU	

Following one dimensional earth conductivity models defined by Natural Resources Canada are added as standard earth conductivity models of GIC activity.

Table 8.3. 1D Earth Model of Canada

Name	Description
BC	British Columbia (BC)
PRARIES	Alberta (AB), Saskatchewan (SK), Manitoba (MB)
SHIELD	Ontario (ON), Quebec (QC)
ATLANTIC	Atlantic Canada

b. User Defined Earth Conductivity Models

User defined earth conductivity models can be specified in the GIC data file.

Refer GIC Data file record for the format and information needed.

Application Notes:

The name of the earth conductivity model should be unique. Activity GIC searches earth conductivity model in "standard" models first then in "user" models. So if any User earth model has same name as "standard" earth model, it will not get used.

The earth conductivity model for any substation is determined in following order.

i. Use earth conductivity model specified in GIC data file

ii. Use earth model specified in GIC activity option when the substation earth conductivity model is not specified in the GIC data file.

GIC flow to Mvar Loss Calculation

The transformer Mvar losses due to GIC flow are calculated as below.

$$\begin{aligned} \text{3-Phase Mvar Losses} &= I_{\text{eff}} \times K_{\text{factor}} \times \frac{V_H}{500} \\ &= I_{\text{eff}} \times K_{\text{factor(2)}} \end{aligned} \tag{8.2}$$

Where:

- I_{eff} = Effective GIC flow in transformer in amperes/phase
- V_H = Transformer Windings highest voltage in kV
- K_{factor} = Mvar/Amp scaling factor defined at 500 kV of V_H
- $K_{factor(2)}$ = Mvar/Amp scaling factor provided in GIC data file

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8.1.10. Output Text Reports

Activity GIC run creates text report. The report subsystem can be defined to limit the content of the report.

- rpt_sid = 0, use study subsystem as report subsystem
- rpt_sid = 1 though 11, use this subsystem as report subsystem.

The text report contain following sections when selected. This report can be created for all elements/equipment or limit to top number of elements/equipment in each section.

1. Transformer Thermal Analysis Effective GIC/phase: Eastward GIC_E and Northward GIC_N (created when Thermal Analysis option selected)

   Eastward GIC_E and Northward GIC_N provided in this report are used to calculate GIC(t) for a transformer.

2. Storm Strength on a Branch considering alpha and beta factors (created when benchmark and/or supplemental GMD events selected) and branch induced voltage option when moving box is simulated.

   For benchmark GMD event this report gives the following for each branch in study subsystem:

   $$E_{field} = \alpha \times \beta \times 8 \ \ \mathrm{V/km}$$

   For supplemental GMD event this report gives the following for each branch in supplemental subsystem:

   $$E_{field} = \alpha \times \beta \times 12 \ \ \mathrm{V/km}$$

3. Branch Induced DC Voltages, positive polarity at To Bus
4. Bus and Substation ground bus DC Voltages
5. GIC flow in Shunt Elements/Eqipment
6. GIC flow in Substation Ground
7. GIC flow in Two Winding Transformers
8. GIC flow in Two Winding Auto Transformer Windings
9. GIC flow in Three Winding Transformers
10. GIC flow in Three Winding Auto Transformer Windings
11. Two Winding Transformer Reactive Power Loss
12. Two Winding Auto Transformer Reactive Power Loss
13. Three Winding Transformer Reactive Power Loss
14. Three Winding Auto Transformer Reactive Power Loss
15. Storm Degree Scan: Transformer Reactive Power Loss Summary
16. Storm Magnitude Scan: Transformer Reactive Power Loss Summary

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17. Maximum GIC flow in Transformer Windings and corresponding Storm Orientation

    This report gives the maximum GIC flow in each transformers among all the degree scan calculations performed and corresponding orientation in degrees.

For moving box calculations following reports are created.

- A report containing above sections for each moving box calculation is created. When report is saved to a file, a report file is created for each moving box calculation with suffix "_supp_boxN" added to each file name.
- A report containing maximum GIC flow in Transformer Windings and corresponding Storm Orientation among all moving box calculation is created. This report also contains transformer reactive power loss summary for all moving box calculations. When this report is saved to a file, a report is created in a file with suffix "_supp_box_Ieff" added to the file name.
- A report containing a) GIC flow in Substation Ground and b) Maximum GIC flow in Transformer Windings and corresponding Storm Orientation for GIC calculation done to arrive at ranking of substations to determine moving box center point. When this report is saved to a file, a report is created in a file with suffix "_rank" added to the file name.

8.2. GIC Data Preparation

The additional data required for GIC calculation is described in Data Format manual Chapter "GIC Data File Contents". A Python module "gicdata" can be used to prepare such a large amount of data needed to run GIC calculations. Following functions from this module can be used at different stages in this data preparation process.

8.2.1. Create Excel Template for GIC Data File

GIC >Create Excel Template for GIC Data File

Use this to create GIC data file template in Excel spreadsheet. This function uses sub system specified to create template of network elements from base power flow case that require GIC data. Using this function as 1st step in GIC data preparation will eliminate many potential errors and provide list of the network elements that must have GIC data.

Important Notes on how to fill this Excel Template:

- Fill data items in these worksheets of only known values for the items. Do not fill with default data item values.
- Many data items have default data (like substation grounding DC resistance R_G=0.1 ohm) which do not need to be filled/specified in substation worksheet.
- The columns with label "No default" must be filled with appropriate value.
- Generally when these worksheets are filled, they are sparsely filled with only known data item values. Activity GIC calculates and uses default data item values when not provided in GIC data file.

8.2.2. Create GIC Data File from Excel File

GIC >Create GIC data file from Excel

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Use this to create GIC data text file (.gic) used by activity GIC from GIC data Excel spreadsheet.

8.2.3. Transfer GIC Data to Excel File

GIC >Transfer GIC data to Excel

Use this to create a new GIC data Excel file by transferring GIC data in .gic file to an empty GIC data Excel template. This function can be used in following situations.

- To transfer .gic file data to a future year (or different) base case
- To transfer .gic file data to a different GIC study subsystem
- To check/validate data in .gic file created by combining GIC data provided by many transmission operators (TOs) and/or Generator Operator (GOs).

It processes and corrects data when possible. It automatically creates bus-substation groups based on branch and transformer connections. It then corrects following data issues for buses found in template data only.

- When these are transformers buses, a bus substation is assigned same as the other transformer bus substation.
- When these are part of substation bus groups in template data, a bus substation is assigned same as the bus group substation.
- When bus is neither a transformer bus nor part of substation bus group, a bus substation needs to be provided.

8.2.4. Merge Multiple GIC Data Excel Files

GIC >Merge multiple GIC Excel files

Use this to create a new GIC data Excel file by merging GIC data from more than one input GIC data Excel files. This function will be useful in combining GIC data provided by many TOs and/or GOs. It does following checks to arrive at valid and unique data for combined data set.

- Check for duplicate/multiple data items. When data item is specified in multiple files, the item found in first input file is used and such progress message is provided.
- Any data item specified in multiple file, data values are checked for difference and such progress message is provided.
- If same substation location is used in multiple files but substation numbers are different, substation number from first input file is used.
- Any bus that appears in multiple substations and if these substation locations are different, the bus is associated with substation from first input file.
- Substation numbers associated with different locations and in different files are preserved as long as they are unique.

8.2.5. Create Excel from GIC Data File

GIC >Create Excel from GIC data file

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Use this to create GIC data Excel spreadsheet from GIC data text file (.gic). This is mainly done to share and update GIC data easily.

8.3. Power Flow with GMD Mvars

When selected, activity GIC also runs power flow solution after adding GMD Mvar losses to the working case. The power flow method and solution options available in working case are used by default or can be specified.

During "Degree Scan" calculations, the power flow solution can be further customized using following options.

a. When to solve the power flow

- degscan_pf_optn=0, Solve power flow only for worst case orientation
- degscan_pf_optn=1, Solve power flow for each degree scan calculation

b. How to add GMD Mvar losses to the working case to solve power.

- pf_qpct_step=0, add 100% GMD Mvar losses in one step
- pf_qpct_step>0, add GMD Mvar looses in specified incremental step until all GMD Mvar losses are added. The power flow is solved after each incremental update of the base case. This is done to achieve power flow solution when local reactive power support is limited.

Application Note - Depending on options degscan_pf_optn, pf_qpct_step, supp_box, and supp_box_num specified there could be large number of power flow solutions. For example:

1. No scan: pf_qpct_step=0, Number of power flows = 1
2. No scan: pf_qpct_step=10, Number of power flows = 10
3. Degree Scan: degscan_pf_optn=1, degscan_step=1 degree, pf_qpct_step=0, Number of power flows = 181
4. Degree Scan: degscan_pf_optn=1, degscan_step=1 degree, pf_qpct_step=1, Number of power flows = 181*100 = 18,100
5. Degree Scan: degscan_pf_optn=1, degscan_step=1 degree, pf_qpct_step=1, supp_box_num=10, Number of power flows = 181*100*10 = 181,000

Additionally, following APIs can be used to specify power flow options and solve power flows.

8.3.1. Activity gic_pf_options

GIC >Set GIC Power Flow Options

Use this activity to specify Newton-Raphson power flow solution options. These options will be used for power flow solutions during activity GIC run. This must be done before running GIC activity.

The sequence of activities to run would be:

- Open power flow case

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- Run GIC Power Flow Options
- Run GIC Analysis

8.3.2. Activity gic_pf

GIC >Run GIC Power Flow

Activity GIC creates "GIC Case Data output RDCH .raw file". Use this activity to run a power flow solution by adding GMD Mvar losses as constant current loads from this RDCH .raw file to the working case. Additionally power flow method, solution options and Mvar step size can be specified to arrived at converged power flow solution.

8.4. Transformer Thermal Assessment GIC(t)

Activity GIC also creates comma separated (.csv) file of the transformer time-series Effective GIC flow GIC(t) data used in performing transformer thermal assessment.

8.4.1. GIC(t) Calculation Method

The method for calculating GIC(t) data is as below.

i. Calculate effective GIC_E due to E_E=1 V/km and E_N=0 (eastward geoelectric field) considering both location factor (α) and earth conductivity factor (β) for the transformer under consideration.

ii. Calculate effective GIC_N due to E_N=1 V/km and E_E=0 (northward geoelectric field) considering both location factor (α) and earth conductivity factor (β) for the transformer under consideration.

iii. Scale the GIC contribution according to the reference (or benchmark) event geoelectric field time series ($E_E(t)$ and $E_N(t)$) to produce the GIC time series, GIC(t), for the transformer under consideration. If the earth conductivity is assumed to be laterally uniform in the transmission system of interest, the transformer GIC (in amperes/phase) for any value of $E_E(t)$ and $E_N(t)$ can be calculated using these equations.

$$
\begin{aligned}
GIC(t) &= |E(t)| \times \{GIC_E \sin\varphi(t) + GIC_N \cos\varphi(t)\} \\
|E(t)| &= \sqrt{E_N^2(t) + E_E^2(t)} \\
\varphi(t) &= \tan^{-1}\frac{E_E(t)}{E_N(t)} \\
GIC(t) &= GIC_E \times E_E(t) + GIC_N \times E_N(t)
\end{aligned}
\tag{8.3}
$$

Note that GIC_E and GIC_N are calculated with the α and β factors considered. Therefore, reference event geoelectric field $E_E(t)$ and $E_N(t)$ are not scaled for the α and β factors.

iv. For the same reference event, the GIC(t) waveshape in every transformer will be different, depending on the location (α and β factors) within the grid and the number and orientation of the circuits connecting to the transformer substation. Therefore, it is necessary to calculate GIC(t) waveshape separately for each individual transformer under consideration.

8.4.2. User E-Field Time Series Input for GIC(t)

GIC >Specify Efiled waveshape for GIC(t)

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The Benchmark and Supplemental Events from TPL-007-2 are used as standard E-field $E_E(t)$ and $E_N(t)$ inputs in transformer GIC(t) calculation.

However, any User specified time series input to calculate GIC(t) can be provided in comma separated (.csv) text file. This must be done before running GIC activity.

The format of the user time series input $E_E(t)$ and $E_N(t)$ is as below.

- It contains three columns of data: time in seconds, $E_E(t)$ in mV/km and $E_N(t)$ in mV/km
- First row data is treated as column labels and ignored.
- Provide as many data rows as desired

The file data would like as below:

```
Time (s),E_EW (mV/km),E_NS (mV/km)

5,19.1912,38.3824

15,15.2473,49.6858

...

...
```

The sequence of activities to run would be:

- Open power flow case
- Specify GIC Efield Waveshape
- Run GIC Analysis

8.4.3. Specific Transformer GIC(t) Calculations

GIC >Calculate transformer thermal analysis GIC(t)

During GIC activity run, GIC(t) is calculated for all selected transformers. However, GIC(t) can be calculated for selected transformer using effective GIC_E and effective GIC_N calculated by GIC activity. When User E-field is specified before performing these calculation that E(t) will be used. Otherwise Benchmark Event E(t) will be used.

8.5. GIC Result Plots

8.5.1. GIC Result on Network Maps

GIC >Plot GIC Results

Using Python Module "arrbox.gicmaps" the GIC calculation results stored in "GIC Results Mapping Output File (.pygic)" are plotted on network maps where substations are drawn per their geographic coordinates. The plots drawn can be saved to one pdf file or separate png, eps or ps format files. All these plots are drawn with pre-configured options. These configuration options can be edited/modified as desired. Following quantities can be plotted.

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1. Bus Voltages - Base case

   These bus voltages are previously solved AC power flow solution bus voltages in the working case.

2. Bus Voltages - GIC case

   These bus voltages are AC power flow solution bus voltages after adding GMD Mvars to the working case.

3. Substation GICs

4. Branch GICs

5. Total Reactive Power Losses

   These will be summary of Orientation Scan and Magnitude Scan results.

6. Transformer Effective GICs

   The GIC flows in a transformer for specified storm E-field and specified orientation or in case of degree scan the orientation that gives highest Mvar losses.

7. Transformer Maximum Effective GICs

   These are maximum GIC flows in a transformer for specified storm E-field and degree scan orientation that gives the highest GIC flow. Only degree scan calculations performed are considered to find this maximum GIC flow.

8.5.2. Transformer GIC(t) Plots

GIC >Plot transformer thermal analysis GIC(t)

Using Python Module "arrbox.gicthermal" the GIC calculation results stored in "Transformer Thermal Impact Analysis GIC(t) Output File (.gict)" are plotted. The plots drawn can be saved to one pdf file or separate png, eps or ps format files. Following quantities can be plotted.

1. Benchmark or User specified GMD event used when .gict file was created

2. Supplemental GMD event used when .gict file was created

3. Selected transformer Effective GIC(t) v/s Time

8.6. Exporting GIC Results to Excel

Using Python Module "arrbox.gic" the GIC calculations are performed and results returned in Python object. The results from the returned object can be used and post-processed in any Python script. However, following methods on this object are available to create pre-configured reports.

Following program listing shows a typical way to do this from Python script.

```
gicobj = arrbox.gic.GIC(savfile, gicfile, efield_mag, efield_deg, **kwds)

gicobj.excel_export(xlfile)

gicobj.text_report(xlfile)
```


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```
gicobj.qtotal_report(xlfile)
```

These are built-in methods that can readily create selected reports. However, the power of this lies in the fact that all GIC results are available in this "gicobj" and can be accessed in any Python script to customize the reports or any further processing.

8.7. GIC Calculations - Python Automation Functions

The GIC calculations and post-processing of results as described in sections above can be run through Python scripts. Following Python functions and modules can be used to accomplish that.

1. gicdata

   This module provides functions to facilitate creating, editing and maintaining GIC data file.

2. psspy.gic_6(..)

   Run GIC calculation using activity GIC.

3. psspy.gic_pf_options(..)

   Set power plow options to use during activity GIC run.

4. psspy.gic_pf(..)

   Run power flow with GMD Mvars added to the working case.

5. arrbox.gic

   This module runs GIC calculations, returns GIC results in Python object and export GIC results to Excel.

6. arrbox.gicmaps

   This module plots GIC results on network maps.

7. arrbox.gicthermal

   This module plots transformer thermal analysis GIC(t).

8. psspy.gic_thermal_impact(..)

   Calculate Transformer Thermal Impact assessment GIC(t) curve using specified effective Eastward GIC_E and Northward GIC_N for one specific transformer.

9. psspy.gic_efield_waveshape(..)

   Set the geoelectric field waveshape for transformer thermal impact assessment.

10. psspy.gic_efield_waveshape_ref()

    Set NERC TPL-007-2 Benchmark event as the geoelectric field waveshape for transformer thermal impact assessment.

11. psspy.gic_efield_waveshape_supp(..)

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Set NERC TPL-007-2 Supplemental event as the geoelectric field supplemental waveshape for transformer thermal impact assessment.

The detailed information on any of these functions and modules can be obtained by executing help(..) statement from any Python interpreter. For example following gives help on gicdata module and gic_6 function.

```
import gicdata

help(gicdata)

import psspy

help(psspy.gic_6)
```

8.8. GIC Calculations - Beginners Steps

8.8.1. How to run GIC Calculations?

- PSSE 35.2 and later OR PSSE 34.8 and later

  In GIC activity API version GIC_8, GIC data file is not part of GIC activity argument. Read GIC data file from:

  - PSSE menu File >Open >Select GIC data file (.gic)
  - psspy.gic_read(..)

  When plan to run GIC "Efield" event calculations, read Efield data file from:

  - PSSE menu File >Open >Select GIC Efield JSON data file (.json or .efldjson)
  - PSSE menu File >Open >Select GIC Efield CSV data file (.csv or .efldjcsv)
  - psspy.gic_read_efield_grid(..)

  The GIC activity gets the substation location co-ordinates and substation-bus association from Power Fow case (.sav or .raw). So it is recommended to use Power Flow case with Node-Breaker data and GIC data file version 5. Please refer Section "GIC Data" in the beginning of this Chapter for more details.

- Versions earlier than PSSE 35.2 and PSSE 34.8

  The GIC data file is part of GIC activity argument. The API to read GIC data file does not exist.

  The GIC "Efield" event calculations do not exist.

8.8.2. Get Familiar

Following GIC calculation example files are installed in "Example" folder of each installed application version.

- ieee_gic_test_case.sav and ieee_gic_test_case.gic

  This is an IEEE test case for the calculation of Geomagnetically Induced Currents.

  This example just calculates GIC flows and does not have required network details to a run power flow.

- sample_nb.sav and sample_fv5.gic OR sample.sav and sample_fv4.gic

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This is a fictitious network and corresponding GIC data file. Use this to calculates GIC flows, GMD Mvar losses and run power flow with GIC Mvar losses added to the working case.

- gic_report.py

  This is a comprehensive Python script that does following.

  - Create GIC data file templates in Excel spreadsheets and create GIC data file from those Excel spreadsheets.
  - Perform GIC analysis, post process GIC results, create customized GIC analysis reports.
  - Perform GIC analysis, post process GIC results, export GIC analysis results to Excel.
  - Map GIC results on network maps

8.8.3. Set GIC Calculation Options

Activity GIC allows running GIC calculations with options for storm orientation scan, storm magnitude scan, and power flow solution with GMD var losses added to the base case. Also one can specify the step size for orientation scan, magnitude scan and GMD var loss. The number of calculations to be done could quickly become a large number.

Example:

Options Selected:

- Run power flow solution for each storm orientation scan
- Storm orientation scan step size = 1 (deg)
- GMD var loss step size = 1 (%)

For these options, number of power flow solutions run will be equal to 181*100 = 18100

As a first step to check and validate the GIC data, have opportunity to correct potential data errors, it is recommended that run GIC calculations without any scans, and may be 100% GMD var loss step size.

8.8.4. Run GIC Calculations

Follow these bare minimum steps to run GIC calculations.

1. Pick power flow base case to run GIC calculations (working case).
2. Define GIC Study Subsystem.
3. Create GIC Data Excel Template.
4. Fill the Excel template with these minimum must have data.
   - substation location coordinates and
   - transformer vector groups

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All the other GIC data items have default data defined.

5. Convert filled GIC data Excel template to text GIC data file (.gic).
6. Run GIC calculations.
7. After the first pass GIC calculations are run successfully, for accurate GIC flows re-visit the GIC data Excel template and provide all available data, re-create GIC data file (.gic) and re-run GIC calculations.

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Chapter 9
Linear Network Analysis

9.1. Building the Distribution Factor Data File

DFAX

Run Activity DFAX - GUI
Power Flow > Linear Network > Build distribution factor data file (DFAX)... *[Build Distribution Factor Data File]*
Run Line Mode Activity DFAX - CLI
ACTIVITY? >>**DFAX** ENTER FILENAME FOR STORING DISTRIBUTION FACTORS
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The Distribution Factor File setup activity DFAX reads a set of Linear Network Analysis Data Files (refer to ???) and reflects their contents in a Distribution Factor Data File in preparation for any of activities OTDF, DCCC, TLTG, POLY, SPIL, IMPC, LLRF, and ACCC. Activity DFAX also calculates line outage distribution factors and preserves them in the Distribution Factor Data File.

Subsystems of the working case are specified in a Subsystem Description Data File, monitored elements are specified in a Monitored Element Data File, and contingencies are specified in a Contingency Definition Data File.

The Distribution Factor File setup activity reads a set of Linear Network Analysis Data Files and reflects their contents in a Distribution Factor Data File in preparation for performing automatic ac contingency analysis as well as a variety of other analyses including dc contingency testing, generator contingency analysis and transfer limit identification.

Input for the process of creating the Distribution Factor file is contained in three data files:

- *Subsystem* Description Data file; relevant subsystems of the working case are specified in this *.SUB file. For more information please see *Subsystem Definition Data File*
- *Monitored Element* Data file; network elements to be monitored for problems are specified in this *.MON file. For more information please see *Monitored Element Data File*
- *Contingency* Definition Data file; contingencies to be tested are specified in this *.CON file. For more information please see *Contingency Definition Data File*

9.1.1. Operation of Activity DFAX

Activity DFAX first checks that generators are not converted (refer to CONG) and that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed.

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The user must specify the filename of the Distribution Factor Data File into which the results of activity DFAX are saved.

The user defines subsystems of the working case from data records taken from either a designated Subsystem Description Data File or the dialog input device. Refer to ???? for data input formats.

Subsystem definitions are required if the Distribution Factor Data File is to be used by activities TLTG, SPIL, and/or POLY, OTDF, DCCC, IMPC, LLRF, and ACCC do not require subsystem definitions *unless* subsystems are referenced in the Monitored Element Data File and/or the Contingency Definition Data File specified below.

The user constructs a monitored element list from data records taken from either a designated Monitored Element Data File or the dialog input device, as appropriate. Refer to Monitored Element Data File Contents for data input formats.

The user may have the branches in the monitored element list sorted in the reports of activities OTDF, DCCC, POLY, IMPC, LLRF, and AC Contingency Solution.

If the sort option is specified, branches are sorted in ascending numerical or alphabetical order according to the bus output option currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). Branches are sorted by from bus, and, for each from bus, by to bus and circuit identifier.

If the sort option is not specified, monitored branches remain in the same order in which they were specified in the Monitored Element Data File. In either case, interfaces are reported after all monitored branches in the order in which they were specified in the Monitored Element Data File.

Contingencies may be specified from either a designated Contingency Definition Data File or the dialog input device. Refer to Contingency Definition Data File Contents for data input formats. Activity DFAX reads the Contingency Definition Data File, updating internal arrays and transferring the contingency definitions into the Distribution Factor Data File. Any contingency case specifications in which errors are detected generate an alarm and are skipped.

Line outage distribution factors are used by activity DFAX only for building Distribution Factor Data Files for specific PSS®E activities, as follows:

- Calculate distribution factors *disabled*: activities ACCC, IMPC, and LLRF
- Calculate distribution factors *enabled*: activities OTDF, DCCC, TLTG, SPIL, and POLY

If the distribution factor calculation is specified (default = enabled), activity DFAX continues its processing. If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed. The base case susceptance matrix is then constructed, followed by the calculation of vectors of line outage distribution factors corresponding to line outage contingency events contained in the Contingency Definition Data File. Whenever a line outage contingency results in the presence of a swingless island, an appropriate message is printed prior to calculating the corresponding distribution factor vector.

A Distribution Factor Data File containing this additional information is required by activities OTDF, DCCC, TLTG, SPIL, and POLY; it may also be used by activities IMPC, LLRF, and ACCC.

Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity DFAX.

9.1.2. Application Notes

Activity DFAX derives line outage distribution factors from the same linearized network model used in the dc analogy network solution method of activity DCLF. While injection methods rather than complete dc power

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flow solutions are used to calculate line outage distribution factors, the comments in Section 9.4.4, "Application Notes" apply here as well. Note, in particular, the approximate nature of the linearized network model.

Activity DFAX needs to be executed before using activities OTDF, DCCC, TLTG, SPIL, POLY, IMPC, LLRF, or ACCC. Because the line outage distribution factors and other information stored in the Distribution Factor Data File are a function of data organization and network topology in the working case, it follows that it must be re-executed before entering these activities any time one or more of the following occurs:

- Change of bus type code
- Change of machine, load or branch status
- Change of branch reactance
- Change in the group of branches modeled as zero impedance lines
- Change of metered end of a multi-section line when the multi-section line reporting option is enabled
- Addition or removal of buses, branches, machines or loads
- Change of any bus attributes by which subsystems, monitored elements, and/or contingencies are specified (e.g., area assignments, base voltages, etc.)
- Change to any of the Linear Network Analysis Data Files

Activity DFAX is sensitive to the multi-section line reporting option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). Its setting at the time activity DFAX is selected is saved in the Distribution Factor Data File and overrides the option setting in subsequent executions of activities OTDF, DCCC, TLTG, SPIL, POLY, IMPC, LLRF, or ACCC.

Activity DFAX is sensitive to the bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). Its setting at the time activity DFAX is selected governs the manner in which buses *must* be identified (i.e., either bus numbers or extended bus names) in the Linear Network Analysis Data Files that are read for activity DFAX. The setting of the bus input option in subsequent executions of those activities that require a Distribution Factor Data File as input (i.e., activities OTDF, DCCC, TLTG, SPIL, POLY, IMPC, LLRF, and AC Contingency Solution) *must* be identical to that which existed when activity DFAX was executed.

In defining subsystems, a bus may be assigned to more than one subsystem. Suppose, for example, that in activity TLTG a study system consisting of all buses in an area is required. Suppose further that the user wishes to include as monitored elements only those branches in the area at 230 kV and above. This can be accommodated by defining one subsystem to be the entire area and a second subsystem (using the join group block structure) to contain those buses in the area at or above 230 kV.

A Distribution Factor Data File specified to one of the analysis functions that utilizes a linear (dc) network model (e.g., activities such as DCCC, TLTG and LLRF) must specify at least one monitored branch or one interface. That is, the Monitored Element Data File specified to activity DFAX to construct such a Distribution Factor Data File must specify at least one monitored branch or one interface. A Distribution Factor Data File that is to be used only for ac contingency calculations in which only voltage violations are of interest need not specify any monitored branches or interfaces.

Care should be taken not to include unnecessary END statements in the Linear Network Analysis Data Files. END statements are used to indicate the termination of block structures as well as the termination of the

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data input stream. Improperly placed END statements are ofteninterpreted by activity DFAX as an end of input data signal. Symptoms of these conditions can occur during the execution of activities that use such a Distribution Factor Data File; they include monitored elements that are omitted from output reports and contingency cases that are skipped.

Additional Information
PSS®E GUI Users Guide, Building the Distribution Factor Data File *PSS®E* Command Line Interface (CLI) *Users Guide*, Building the Distribution Factor Data File

9.2. Calculating Distribution Factors

OTDF

Run Activity OTDF - GUI
Power Flow > Linear Network > Calculate and print distribution factors (OTDF)... *[Select Distribution Factor data file to calculate factors for]*
Run Line Mode Activity OTDF - CLI
ACTIVITY? >>**OTDF** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The distribution factor reporting activity OTDF tabulates distribution factors for single event contingencies using a linearized network model. Distribution factors for line outage contingencies are retrieved from a Distribution Factor Data File constructed by activity DFAX. Distribution factors for other types of contingency events are calculated as needed by activity OTDF.

It is possible for the user to print the distribution factors from the linearized network model Distribution Factor file built as described in Section 9.1, "Building the Distribution Factor Data File" but only for single event line outage contingencies. Distribution factors for other types of single contingency events are calculated as needed.

Because the linearized network model is the basis of the calculation of distribution factors, it is necessary to recognize that their use provides approximate results useful for estimating conditions during contingencies.

In both large and small network models, the factors are useful for the user to make quick estimates of changes in line flows during contingencies that result in:

- Loss of a single line element
- Increase or decrease in generation at a bus
- Increase of decrease of load at a bus

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Line closure distribution factors are generally not very useful and are supported only for compatibility with other linear analyses. They are equal in magnitude to those of the corresponding line outage contingency but with the opposite sign.

9.2.1. Operation of Activity OTDF

Activity OTDF first checks that generators are not converted (refer to CONG) and that each non- Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed and activity OTDF is terminated.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed. Next the largest active power mismatch corresponding to the present ac solution voltage vector in the working case is calculated and printed.

The user must enter the name of a Distribution Factor Data File constructed by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired Linear Network Analysis Data Files (refer to ???). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity OTDF.

The Distribution Factor Data File is read, and the base case dc power flow solution is calculated followed by the processing of the designated contingency cases. Only single event contingencies are allowed in activity OTDF; changing the status of a three-winding transformer is considered a multiple event contingency; disconnecting a bus is often a multiple event contingency. Any multiple event contingencies generate an alarm and are ignored.

Activity OTDF may be terminated by entering the AB interrupt control code.

9.2.2. Output Report Format

Results of activity OTDF are reported in tabular form, with six contingency cases per table. Each group of contingency cases contains a page summarizing the contingency cases performed. For each contingency, the contingency label and the data record defining the contingency are listed, along with the change in power of the affected element (MW shift). For line outage contingencies, the quantity shown is the base case ac active power flow at the metered end of the line in the from bus to to bus direction.

The summary page is followed by the distribution factor table. For each monitored element, this table lists its active power loading in the base case ac network solution as contained in the working case when activity OTDF was entered, its loading in the base case dc analogy network solution, along with the distribution factor for each of the cases reported.

Single branches are listed first, either in the order in which they were specified in the Monitored Element Data File or in ascending numerical or alphabetical order, according to the option selected during the execution of activity DFAX (refer to Section 9.1.1, "Operation of Activity DFAX"). These are followed by the interfaces in the order in which they were specified.

If a line outage contingency forms one or more islands with no Type 3 (swing) bus, such islands are disconnected. Any monitored branch that is located in the swingless island has dashes printed in place of its distribution factor. Similarly, any interface that includes such a member has its distribution factor printed as dashes.

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A partial report is shown in Figure 9.1, "Example Report Listing Distribution Factors", which is curtailed to show only three of the first six contingencies. Note that the reports shows the data files uses and lists each single contingency considered prior to listing the distribution factors for each of the lines and interfaces monitored, for each contingency. The files used are the *savnw.con*/*mon*/*sub* files from the PSS®E EXAMPLE directory.

As an example, It can be seen the if the nuclear plant at bus 101 is tripped, the SHIFT will be 750 MW, (the units output), and the line from Bus 201 to 151 will show a power flow reduction of 310 MW (which is equal to 750 MW x 0.41339). The base case flow is shown as negative value of 564.8 MW at the Bus 201 end. This is the receiving end of the line. Consequently, the 750 MW shift coupled with a positive distribution factor will result in a reduce flow towards Bus 201. An ac solution shows the flow reduces to 255.9 MW. The calculation (-564.8 + 310) gives -254.8 MW (error less than 0.43%).

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```
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE                                    PAGE 1
BASE CASE INCLUDING SEQUENCE DATA
                *** OTDF CONTINGENCY SUMMARY ***

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con

<-CONTINGENCY-> <-MW SHIFT-> <-----------------------------------------CONTINGENCY DESCRIPTION----------

   TRIP1NUCLEAR      750.0     REMOVE UNIT 1 FROM BUS 101 [NUC-A      21.600]

   ADDLARGELOAD      500.0     INCREASE BUS 154 [DOWNTN      230.00] LOAD BY 50 PERCENT

   LOSEWESTGEN       100.0     REMOVE UNIT 1 FROM BUS 3018 [CATDOG_G    13.800]

   LOSEWESTBIGT     -138.2     TRIP LINE FROM BUS 3004 [WEST       500.00] TO BUS 152 [MID500     500.00]

   LOSEEASTBIGT      564.8     TRIP LINE FROM BUS 151 [NUCPANT     500.00] TO BUS 201 [HYDRO      500.00]

   LOSEEASTLOAD    -1200.0     SET BUS 205 [SUB230     230.00] LOAD TO 0 MW

------------------------------------------------------------------------------------------------------

     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E    MON, FEB 16 2004   8:44
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE                                PAGE 2
BASE CASE INCLUDING SEQUENCE DATA
          *** OTDF DISTRIBUTION FACTOR TABLE ***

                                CONTINGENCY LABEL--> BASE CASE MW TRIP1NUCLEAR ADDLARGELOAD  LOSEWESTGEN
                                  POWER SHIFT (MW)-->     AC     DC     750.0      500.0        100.0
<-------- FROM ----------> <---------- TO ----------CKT <----> <----><---------> <---------> <--------->
   201 HYDRO      500.00     151 NUCPANT    500.00 1  -564.8 -558.7   0.41339    -0.07415    -0.04159
   202 EAST500    500.00     152 MID500     500.00 1   -42.6  -50.7  -0.14785    -0.21055    -0.11808
   203 EAST230    230.00     154 DOWNTN     230.00 1   122.4  122.4  -0.04237     0.07663     0.04298
   205 SUB230     230.00     154 DOWNTN     230.00 1   354.2  356.1  -0.22317     0.20808     0.11669
  3004 WEST       500.00     152 MID500     500.00 1  -138.2 -132.2   0.42977     0.34592     0.21485
  3006 UPTOWN     230.00     153 MID230     230.00 1   -78.7  -78.6   0.30190     0.25751     0.11931
  3008 CATDOG     230.00     154 DOWNTN     230.00 1    69.0   62.9   0.26833     0.39657    -0.33416
       INTERFACE WEST                                  -147.9 -147.9   1.00000     1.00000     0.00000
       INTERFACE EAST                                  -130.8 -130.8   0.00000     0.00000     0.00000
```

Figure 8-1. Example Report Listing Distribution Factors

Figure 9.1. Example Report Listing Distribution Factors

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9.2.3. Application Notes

Activity OTDF uses the same linearized network model as is used in the dc analogy network solution method of activity DCLF. Thus, in utilizing the distribution factors reported by activity OTDF, the comments in Section 9.4.4, "Application Notes" apply here as well. Note, in particular, the approximate nature of the linearized network model used by activity OTDF.

Activity OTDF detects the specification of duplicate single line outage contingencies and prints the distribution factor vector only for the first specification of such a contingency. Any contingency cases involving bus boundary contingency events or line closure contingency events are not checked for duplication.

It is recommended that the working case be solved to an acceptable mismatch tolerance prior to launching activity OTDF. Otherwise, while the distribution factors would be valid, the base case line flows printed would not.

Each column of distribution factors is a set of sensitivity coefficients describing the effect of the corresponding contingency on the set of monitored elements. A distribution factor is defined as the ratio of the change in flow on the monitored element in the pre-contingency and post-contingency dc power flow solutions to the dc power change on the element involved in the contingency:

$$\text{D.F.} = \frac{\text{(MW in contingency case) - (MW in base case)}}{\text{Power Shift}} \tag{9.1}$$

Distribution factors are usually used to *estimate* post-contingency line flows when only pre-contingency (base case) conditions are known:

$$P_{new} = P_{base} + (\text{D.F.} \times \Delta P) \tag{9.2}$$

In using distribution factors, the flow directions of P_{base} and ΔP used in the above equation must be consistent with those assumed when the distribution factors were calculated.

For bus boundary condition contingencies, the distribution factors are expressed relative to a change in power withdrawn at the bus. Thus, they are identical for increases in generation, decreases in generation, increases in load and decreases in load. A load increase and a generation decrease both have a positive ΔP, while a load decrease and a generation increase both have a negative ΔP in deriving post-contingency monitored line flows with the above formula.

Line outage distribution factors are applied using the above equation with ΔP equal to the flow on the outaged line before it is opened.

Figure 9.2, "Application of Line Outage Distribution Factor" is an example that shows the outage of a line carrying 100 MW in the base case. If the distribution factor for that specific outage is denoted as FIJ for the line from Bus I to Bus L and its value is equal to 0.05, then the line from Bus I to Bus L will have an increased loading equal to the 100 MW lost from the outaged line multiplied by FIJ, that is 5 MW.

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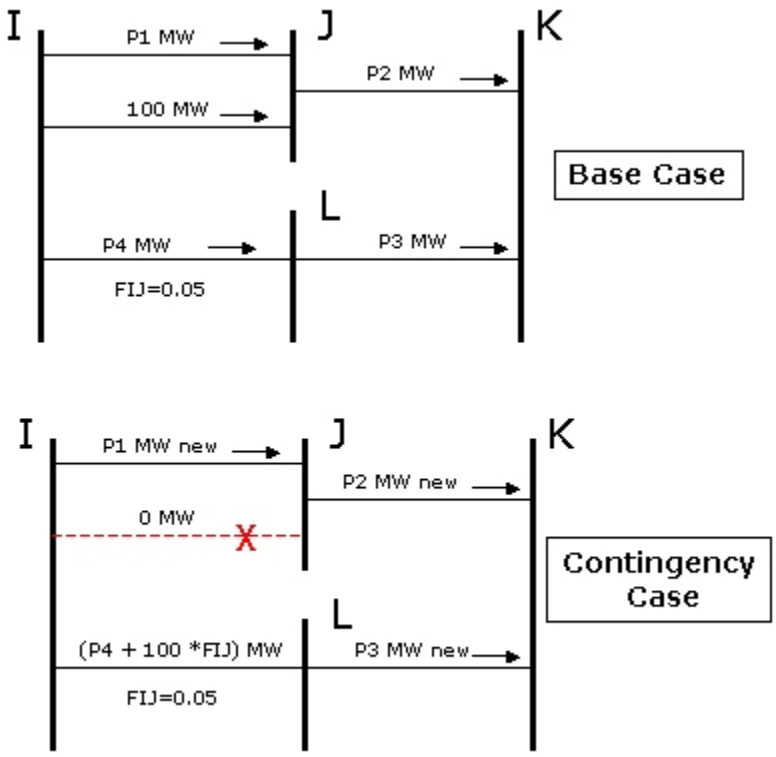

Figure 9.2. Application of Line Outage Distribution Factor

Line closure distribution factors are generally not very useful and are supported only for compatibility with activities DCCC, TLTG, SPIL, POLY, IMPC, and LLRF. They are equal in magnitude to those of the corresponding line outage contingency but with the opposite sign.

In using distribution factors, the flow directions of P_{base} and ΔP used in the above equation must be consistent with those assumed when the distribution factors were calculated.

A line outage distribution factor calculated by activity DFAX and printed by activity OTDF reflects the outage of the series reactance element only and not the removal of the branch's loss estimate and line shunt components. Furthermore, neither line outage nor line closure distribution factors for a multi-section line reflect the removal of any generation or load that may be present at its dummy buses.

Additional Information
PSS®E GUI Users Guide, Calculating and Reporting Distribution Factors
PSS®E Command Line Interface (CLI) *Guide*, Calculating Distribution Factors

9.2.4. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

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Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "OTDF OTDFTab".

9.3. Estimating Severity Rankings for Single Line Outage Contingencies

RANK

Run Activity RANK - GUI
Power Flow > Linear Network > Single contingency ranking (RANK)... *[Single Line Contingency Ranking]*
Run Line Mode Activity RANK - CLI
ACTIVITY? >>**RANK** SOLUTION PARAMETERS ARE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

In large systems with many possible contingencies, especially single branch outages, it is often useful to minimize the computational and subsequent analyses by identifying the most severe contingencies prior to performing the contingency analysis.

The contingency ranking activity RANK estimates the severity of designated single branch outage contingencies and builds a Contingency Description Data File (refer to Contingency Description Data File Contents) with contingencies specified in decreasing order of their estimated severities. Contingency rankings using two different performance criteria may be calculated by activity RANK. Activity RANK may also be instructed to include contingencies that create swingless islands.

9.3.1. Contingency Ranking Methodology

The process of ranking contingencies in order of severity involves the following:

1. Establish the criteria to be considered in formulating the ranking (e.g., overloading, voltage collapse, etc.).

2. For each criteria established in (1), define a scalar mathematical function that has a large value for contingencies that stress the system relative to that criteria, and a small value for those that do not; this function is called a performance index. The performance index should be such that contingencies resulting in system conditions yielding large valued performance indices are considered more severe than system conditions with smaller performance indices.

3. Derive an efficient and accurate procedure for calculating these performance indices for a large number of possible contingencies.

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Activity RANK is able to calculate contingency rankings based on either or both of the following criteria:

1. An overload criteria measuring branch loadings relative to their ratings.
2. A voltage depression criteria that indicates increased reactive power consumption by estimating increases in reactive losses due to increased line loadings.

In the overload ranker, the performance index, PI, is defined as:

$$PI = \sum_{i=1}^{L} \left(\frac{P_i}{PMAX_i} \right)^2 \tag{9.3}$$

where:

P_i	Is the active power flow on branch i.
$PMAX_i$	Is the rating of branch i.
L	Is the set of monitored branches contributing to PI.

Clearly, PI has a small value for system conditions with light branch loadings and a large value when lines exceed their limits. An increase in PI following a contingency indicates that overall loading on the branches contributing to PI has increased.

The set of monitored branches in the overload ranker is as defined in a Monitored Element Data File (refer to Monitored Element Data File Contents). This is normally the same set of monitored elements that are specified to activity DFAX and subsequently, via the Distribution Factor Data File, to activities such as DCCC and ACCC.

In the voltage ranker, the performance index is defined as:

$$PI = \sum_{i=1}^{L} X_i P_i^2$$

where:

X_i	Is the reactance of branch i.
P_i	Is the active power flow on branch i.
L	Is the set of monitored branches contributing to PI.

PI gives an indication of reactive power losses under different system conditions. As line loadings increase, their I^2X losses also increase. This increase in reactive demand generally results in a depression of system voltages.

The set of monitored branches in the voltage ranker is normally all branches in the working case. The user may elect to omit those branches for which no rating is specified in a selected rating set.

Having established ranking criteria and defined performance indices to compare the severity of many contingencies relative to these criteria, we require a procedure for calculating these indices that is substantially less burdensome than explicitly solving each contingency case. Activity RANK employs the ranking algorithm described in *An Advanced Contingency Selection Algorithm* by T.A. Mikolinnas and B.F. Wollenberg, IEEE Transactions on Power Apparatus and Systems, Vol. PAS-100, No. 2, February, 1981, pp. 608-617. This algorithm provides for the evaluation of the performance indices described above without requiring post-contingency branch flows.

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9.3.2. Contingency Ranking Options

When selected, activity RANK first displays its default solution and output control parameters and gives the user the opportunity to change any of these parameters:

The user may specify any of the following options:

Table 9.1. Contingency Ranking Options

Specification	Default	Description
MW mismatch tolerance	0.5	If the largest initial active power mismatch exceeds the specified MW mismatch tolerance, the user may terminate activity RANK.
Small reactance threshold	0.0005	Reactance threshold; all branches from the contingency list (based on the user's subsystem selection, below) where magnitude of reactance is less than or equal to this threshold have their performance indices explicitly calculated. *Note*: The performance index calculation used by activity RANK (refer to Section 9.3.1, "Contingency Ranking Methodology") is inaccurate for the outaging of branches with small reactances.
Allow ties from subsystem as contingency	Disabled	If enabled, include in the contingency list only branches from within the designated subsystem. If disabled, both subsystem branches and ties from the subsystem to other buses in the working case are included. The group of branches that may be outaged are defined according to subsystem selection options.
Number of branch contingencies in file	10	Number < n > of line outages from the overload ranking to be included in the output of activity RANK; a value of zero for this solution parameter causes activity RANK to bypass the overload ranking calculation.
Number of machine contingencies in file	0	Number < m > of machine outages from the overload ranking to be included in the output of the ranking process. The overload ranker is applied to generat-

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Specification	Default	Description
		ing machine outages by assuming that the generation deficiency is redispatched among remaining machines and that the new dispatches cause changes in branch loading. The dispatch is always based on reserve. A value of zero for this solution parameter causes the process to bypass the machine overload ranking calculation.
Contingency case rating	activity OPTN	The contingency case rating set, and the percentage thereof, define the line loading limits used in determining overloads. The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).
Percent of rating	100	
Convert ratings to estimate MW ratings	Disabled	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's reactive power loading at the metered end in the base case ac solution.
Number of voltage contingencies in file	5	Number < k > of voltage contingencies to be included in the output of activity RANK; a value of zero for this solution parameter causes activity RANK to bypass the voltage ranking calculation.
Lines, used in voltage ranker PI	All	Defines the network branches that are to contribute to the voltage ranker performance index: • Only branches with non-zero value in the rating set designated by *contingency case rating* are used • All network branches
Include contingencies creating swingless islands	Disabled	If enabled, the output of contingencies that result in the formation of an island of one or more buses that contains no Type 3 (swing) bus is enabled as follows: if the active power flow on such
MW threshold to include island contingencies	100	

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Specification	Default	Description
		a branch exceeds the MW threshold tolerance specified in the base case dc power flow, the contingency is included in the output of activity RANK. Such contingencies are omitted from the overload and voltage ranking calculations.

9.3.3. Operation of Activity RANK

Activity RANK verifies that at least one of the following parameters:

- number of overload contingencies
- number of voltage contingencies
- contingencies that include swingless islands

is greater than zero, that generators are not converted (refer to CONG), that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network, and that the largest initial active power mismatch is within the specified *MW mismatch tolerance*. If any violations are detected, an appropriate message is printed and activity RANK is terminated.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed.

Activity RANK calculates and prints the largest active power mismatch corresponding to the present ac solution voltage vector in the working case.

Activity RANK constructs a contingency list that defines those single branch outages that are to be considered. The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Unless all buses are specified, activity RANK honors the user specification to either include or omit tie branches from the specified subsystem in the contingency list.

If the determination of an overload ranking is enabled (i.e., if *number of overload contingencies* is greater than zero), the branch ratings to be used are set up according to the specification of *contingency case rating*, *percent of rating*, and *convert ratings*. The appropriate rating set is accessed. If the *convert ratings* setting is enabled, these ratings are converted to estimated MW ratings based upon each branch's reactive power loading at the metered end in the base case ac solution. Finally, the ratings are modified to reflect the percent of rating specified.

The user may define subsystems of the working case with input data records taken from either the designated Subsystem Description Data File or the dialog input device, as appropriate. Refer to ??? for data input formats. Subsystem definitions are required only if subsystems are referenced in the Monitored Element Data File specified below.

The user specifies monitored elements. Activity RANK constructs a monitored branch list, taking input data records from either the designated Monitored Element Data File or the dialog input device, as appropriate.

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Refer to Monitored Element Data File Contents for data input formats. Activity RANK excludes from the monitored branch list any branch with a zero (or negative) rating in the rating set specified by *contingency case rating*; it also ignores any interface definitions.

Note that the above prompts requesting the Subsystem and Monitored Element Description Data Files occur only if overload ranking is enabled.

The contingency ranking algorithm is unable to handle zero impedance lines as contingent or monitored branches. Consequently, if any zero impedance lines are present in either the contingency list, the monitored element list for the overload ranker (if overload ranking is enabled), or the monitored element list for the voltage ranker (if voltage ranking is enabled), zero impedance line modeling is temporarily disabled and a new ordering of network buses is determined with the automatic execution of activity ORDR. The optimal ordering process in PSS®E is used to determine an ordering of the network buses such that sparsity is maintained as the Jacobian matrix is triangularized for ac network solutions or the system admittance matrix is decomposed into its triangular factors in activities for linearized network analyses.

The user specifies either the filename where contingency specification records are to be stored or output of the contingency records to the user's terminal.

Contingency rankings are calculated. If overload ranking is enabled, it is determined and the < n > contingencies with the largest performance indices produce Contingency Description Data File records. Similarly, if voltage depression ranking is enabled, it is determined and the < m > contingencies with the largest performance indices produce Contingency Description Data File records. Finally, if islanding contingency output is enabled, any contingency that results in the formation of a swingless island and for which base case dc power flow loading exceeds the threshold specified by *MW threshold to include island contingencies*, produces Contingency Description Data File records.

The working case is restored to its content at the time activity RANK was selected and activity RANK is terminated.

Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity RANK.

9.3.4. Analyzing AC Contingency Ranking Results

The user supplies the name of an output file (type *.*con*) to which the contingency specification records are to be deposited. If no filename is supplied, the results are written to the *Progress* tab.

If overload ranking is enabled, the < n > contingencies with the largest performance indices produce Contingency Description Data File records. Similarly, if voltage depression ranking is enabled, the < m > contingencies with the largest performance indices produce Contingency Description Data File records. Finally, if islanding contingency output is enabled, any contingency that results in the formation of a swingless island and for which base case dc power flow loading exceeds the threshold specified produces Contingency Description Data File records.

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```
WORKING CASE HAS LARGEST MISMATCH OF      0.02 MW AT    154 [DOWNTN      230.00]

 PROCESSING THE SUBSYSTEM DESCRIPTION FILE...

 PROCESSING THE MONITORED ELEMENT DESCRIPTION FILE...
CONTINGENCY 'OVRLOD 1    '
OPEN LINE FROM BUS    154 TO BUS    205 CKT 1   / PI=    2.3194 'DOWNTN      230.00' TO 'SUB230      230.00'
END
CONTINGENCY 'OVRLOD 2    '
OPEN LINE FROM BUS    151 TO BUS    152 CKT 1   / PI=    1.0727 'NUCPANT     500.00' TO 'MID500      500.00'
END
CONTINGENCY 'OVRLOD 3    '
OPEN LINE FROM BUS    152 TO BUS   3004 CKT 1   / PI=    1.0685 'MID500      500.00' TO 'WEST        500.00'
END
CONTINGENCY 'OVRLOD 4    '
OPEN LINE FROM BUS    154 TO BUS   3008 CKT 1   / PI=    1.0465 'DOWNTN      230.00' TO 'CATDOG      230.00'
END
CONTINGENCY 'VLTAGE 1    '
OPEN LINE FROM BUS    201 TO BUS    204 CKT 1   / PI=   12.9703 'HYDRO       500.00' TO 'SUB500      500.00'
END
CONTINGENCY 'VLTAGE 2    '
OPEN LINE FROM BUS    152 TO BUS    153 CKT 1   / PI=   12.3183 'MID500      500.00' TO 'MID230      230.00'
END
CONTINGENCY 'VLTAGE 3    '
OPEN LINE FROM BUS    151 TO BUS    201 CKT 1   / PI=   12.1484 'NUCPANT     500.00' TO 'HYDRO       500.00'
END
CONTINGENCY 'VLTAGE 4    '
OPEN LINE FROM BUS    202 TO BUS    203 CKT 1   / PI=   12.0872 'EAST500     500.00' TO 'EAST230     230.00'
END
CONTINGENCY 'VLTAGE 5    '
OPEN LINE FROM BUS    201 TO BUS    202 CKT 1   / PI=   11.8464 'HYDRO       500.00' TO 'EAST500     500.00'
END
```

Figure 9.3. Application of Line Outage Distribution Factor

Figure 9.3, "Application of Line Outage Distribution Factor" shows a subset of the Contingency Description Data file produced by the ranking process using the *savnw.sav* power flow case. It can be seen that each contingency included in the ranked set is identified with either an overload or a voltage label and independently numbered.

9.3.5. Application Notes

Activity RANK derives performance indices and establishes severity rankings using the same linearized network model used in the dc analogy network solution of activity DCLF. This level of modeling is well suited for the function performed by activity RANK: the selection of system conditions deserving further study. Contingencies enumerated by the overload ranker may be examined further with activity DCCC and, if the line loadings warrant, with the ACCC and the other ac power flow solution activities. Contingencies produced by the voltage ranker are normally analyzed with the ac power flow solution activities.

The use of a single number to characterize a system condition is convenient for ranking purposes. It does, however, have its limitations. For example, it is possible for a contingency that results in a number of heavily loaded lines but no overloaded lines to be ranked equally with a contingency that produces one or two overloaded lines with other lines being relatively lightly loaded. This is not necessarily a deficiency; it is quite appropriate to conclude that both of these system conditions deserve further study.

The performance indices defined above, and hence the severity rankings that are based on them, are dependent both on (1) network topological and parametric data and (2) the system operating point (i.e., on pre-contingency branch loadings). The same types of modeling changes that require a new execution of activity DFAX (refer to Section 9.1.2, "Application Notes") are also conditions under which new rankings should be determined.

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Furthermore, boundary condition changes (e.g., changes to loads, machine loadings, etc.), while not invalidating a Distribution Factor Data File, may affect the contingency rankings. Boundary condition changes that have only a minor impact on line loadings are not likely to significantly affect the rankings; changes such as increasing an area's load by ten percent or tripping a large unit and dispatching its previous power output among other machines in the system are likely to have a significant effect on line flows that in turn could significantly alter the contingency rankings.

As described in Section 9.3.2, "Contingency Ranking Options", the contingency ranking algorithm is not able to accurately calculate performance indices for branches with small reactances. The *small reactance threshold* parameter, which has a default value of 0.0005, is a reactance threshold; for any branch in the contingency list where reactance magnitude is less than or equal to this threshold, the standard calculation is bypassed and its performance indices are explicitly calculated. The small reactance threshold value should always be at least as large as the zero impedance line threshold tolerance (refer to Section Zero Impedance Lines). Experience to date indicates that the default value is conservative and for most systems it could safely be reduced. For systems with small impedance lines, the user could test the adequacy of this threshold by calculating two sets of overload rankings using different threshold values. If the performance indices for the outaging of small reactance branches are essentially the same, the smaller threshold value may be used.

When the multi-section reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the subsystem assignments of the two endpoint buses of each multi-section line are considered (rather than those of any of its dummy buses) in adding the branch to the contingency list. If a multi-section line is added to the contingency list, the line section at the metered end of the multi-section line is considered to be the outaged element. The performance indices that result are correct *unless* there is any load or generation present at any of the dummy buses of the multi-section line.

For each three-winding transformer in the contingency list, performance indices are calculated for the outaging of each in-service winding in turn, and the largest of these is taken as the performance index for the entire transformer.

In constructing the contingency list, parallel lines between the same pair of buses that have identical impedances, loss estimates, and line shunt and phase shift active power injections (i.e., they are identical in the linear network model used by activity RANK) are not all included in the contingency list. Only one from among such a group of lines is needed. Activity RANK does not check for identical three-winding transformers connecting the same three buses.

If both overload and voltage depression rankings are being determined in the same execution of activity RANK, the overload rankings are calculated first and any branch that is identified by the overload ranker is removed from the contingency list prior to calculating the voltage rankings.

The Contingency Description Data File constructed by activity RANK is a standard source file that may be edited by the user prior to its being submitted to activity DFAX. Bus identifiers are written as bus numbers when the *numbers* input option is in effect, and as extended bus names when the *names* input option is in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). That is, when the Contingency Description Data File built by activity RANK is submitted to activity DFAX along with the same Subsystem Description and Monitored Element Data Files that were used by activity RANK, the files are compatible with each other and with the requirements of activity DFAX (refer to ???).

Additional Information
PSS®E GUI Users Guide, Estimating Severity Rankings for Single Line Outage Contingencies *PSS®E* Command Line Interface (CLI) *Users Guide*, Estimating Severity Rankings for Single Line Outage Contingencies

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9.4. Applying the DC Linearized Network Solution

DCLF

Run Activity DCLF - GUI
Power Flow > Linear Network > DC network solution and report (DCLF)... *[DC Network Solution and Report]*
Run Line Mode Activity DCLF - CLI
ACTIVITY? >>**DCLF** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

9.4.1. DC Linearized Network Solutions

During the early stages of the planning process, when identifying feasible, alternative expansion plans, it may be acceptable to use approximate, computationally more efficient, power flow models. For example, in the daily operation of a power system, the user may be presented, from an online system, a model of the expected load, generation, and transmission system for the day. For this system condition, the user may want to perform a quick analysis of transfer limits for the base case and several contingencies. Because the results may go back to the online system, a computationally more efficient model is needed. One very widely used approximation is the linearized or dc power flow, which converts the nonlinear ac problem into a simple, linear circuit analysis problem. The advantage of this approach is that efficient, non iterative numerical techniques can be used to compute an approximate power flow solution. Many alternatives or contingencies can be investigated with the same computer effort that would be expended to calculate one ac power flow solution.

The dc power flow model is useful for rapid calculation of real power flow. It ignores reactive power flow and changes in voltage magnitudes, and assumes that, for most circuits, $X_{ij} >> r_{ij}$ and the angle between two buses is small. These assumptions result in the power flow from bus i to bus k simplifying to

$$P_{ij} = \frac{\theta_i - \theta_j}{X_{ij}} \tag{9.4}$$

where:

θ_i = Angle at bus i.

θ_j = Angle at bus j.

X_{ij} = Reactance between bus i and bus j.

The power injected into a single bus i is just the sum of the power on all circuits into the bus or

$$P_i = \sum_{j=1}^{n} P_{ij} = \sum_{j=1}^{n} \frac{\theta_i - \theta_j}{X_{ij}} \tag{9.5}$$

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shown in matrix form as

[P] = [B][θ] for a system of n buses.

Note that this equation is linear and that the admittance matrix [B] is sparse because there are only several transmission lines connected to each bus. The solution of this system of equations can be accomplished efficiently by the numerical technique of triangular factorization. The power injections, [P], are known, and the phase angles, [θ], are computed. After the phase-angle solution has been determined, the real power flows can be computed using Equation 9.4.

9.4.2. Operation of Activity DCLF

The linearized network solution activity DCLF applies the dc analogy network solution algorithm to the network modeled in the working case. Optionally, the solution method may also be applied to that network with the status of a specified ac branch changed. The base case and change case solutions are tabulated in a single report.

Activity DCLF checks that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed and activity DCLF is terminated.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed.

The largest active power mismatch corresponding to the present ac solution voltage vector in the working case is calculated and printed. The user may specify that activity DCLF terminates if the largest mismatch is greater than 0.5 MW.

The user specifies which of the 12 branch rating sets are to be used in tabulating line loading percentages in the output report. The default is the rating set established as the default rating set program option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

For a change case solution calculation, the user must designate either of the following options:

- a branch where status is to be changed for the change case solution
- a 3-winding transformer where status is to be changed for the change case solution

Data requirements for a non-transformer branch or a two-winding transformer both buses and the circuit ID. A three-winding transformer is designated by specifying the three buses it connects.

If a branch outage separates the system such that an island with no Type 3 (swing) bus remains, an appropriate message is printed.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), and either an in-service multi-section line grouping or an in-service member of such a grouping is specified, the entire grouping is switched (refer to Multi-Section Line Grouping Data and Section 5.10.1, "Equipment Status Changes"). If the specified multi-section line grouping or member is initially out-of-service, activity DCLF does not allow it to be switched. In outaging a multi-section line grouping, the type codes of the interior dummy buses are automatically changed as required by activity DCLF.

When the multi-section line reporting option is disabled, neither multi-section line groupings nor its members may be switched by activity DCLF.

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If an out-of-service three-winding transformer is specified as the branch where status is to be changed, activity DCLF does not allow it to be switched.

The user may elect to include or exclude a loss estimate. See Section 9.4.4, "Application Notes" for details on the loss estimate used in the dc power flow.

Adjustable phase shifters are those whose control mode is +3 (active power flow control) or +5 (asymmetric active power flow control). The following options are available for their treatment in the dc power flow:

- Fixed phase shift angles in the base case and the change case.
- Fixed MW in the base case, and fixed phase shift angles in the change case.
- Fixed MW in the base case and the change case.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output is generated with buses ordered as described in Section 4.9, "Subsystem Reporting". The user is then given the opportunity to specify an additional group of buses.

The user specifies the voltage vector to incorporate the phase angle vector calculated for either the base case or change case dc power flow solution from the following options, saving:

- Original voltages (default)
- Base case dc power flow voltage angle
- Change case dc power flow voltage angle

If the default is accepted, the voltage vector remains the same as it was prior to performing activity DCLF, and the following message is printed at the *Progress* device:

```
ORIGINAL AC VOLTAGES ARE RETAINED
```

If either the base case or change case voltage angle is specified, activity DCLF sets the voltage at each bus to its voltage magnitude as contained in the working case at a phase angle corresponding to its phase angle in the selected dc solution. An appropriate message is printed:

```
PHASE ANGLES ARE SET TO THOSE OF THE BASE CASE
DC LOAD FLOW SOLUTION
```

or:

```
PHASE ANGLES ARE SET TO THOSE OF THE CHANGE CASE
DC LOAD FLOW SOLUTION
```

If no change case solution is calculated, the first two options incorporate phase angles as determined by the base case dc solution.

Finally, if the change case solution option was enabled, the branch or multi-section line grouping where status was changed for that solution may be either returned to its original status or left with its modified status.

The user may specify that the changed status of the branch or 3-winding transformer be saved. Otherwise, the original status is restored. The following example is printed at the *Progress* tab for activity DCLF with original voltages retained.

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```
The working case has a largest mismatch of 0.02 MW at bus 205 [SUB230      230.00]

Circuit "1" from 151 [NUCPANT    500.00] to 152 [MID500     500.00] is presently in-service

Status of circuit "1" from 151 [NUCPANT     500.00] to 152 [MID500      500.00] is set to
out-of-service

Output completed

Original ac voltages are retained
```

Figure 9.4. Output from Activity DCLF

Activity DCLF may be terminated by entering the AB interrupt control code.

9.4.3. Output Report Format

The general form of the output report of activity DCLF is similar to that of activity LOUT. Figure 9.4, "Output from Activity DCLF" shows an example of the output of activity DCLF.

The bus information presented includes the number of the area in which the bus is located, the bus voltage phase angles for the base case and change case dc power flow solutions, generator MW, and load MW. The value printed as load includes the active power components of all in-service fixed bus shunts, all in-service loads, and all in-service induction machines connected to the bus on the basis of the voltage magnitude at the bus at the time activity DCLF is initiated.

For any induction machine, dc line, or FACTS device that has its ac powers set to zero (e.g., if the case had just been read with activity section / READ and no ac power flow solution attempted), an estimate of its power withdrawn from the network is used based upon its scheduled power or current. Otherwise, regardless of the quality of the ac solution upon entry to activity DCLF, the powers contained in the working case are used.

The star point buses of three-winding transformers (refer to Transformer Data) are not reported by activity DCLF.

Output for FACTS devices and dc lines appears on the right side of the bus output block as the first line or lines of branch data (i.e., before output for ac branches).

For non-transformer branches and for two-winding transformers listed, the number, name, base voltage, and area of the to bus is printed, along with the circuit identifier; for three-winding transformers, the output line contains the string 3WNDTR in the bus number column, the transformer name in the bus name column, and the winding number in the base voltage column, followed by the transformer circuit identifier; if the three-winding transformer to bus reporting option is enabled, two additional lines, containing the bus numbers, names, and base voltages of the two buses connected to the from bus by the three-winding transformer, are printed, followed by the number of the area in which the to bus is located. The active power flowing into the line at the from bus and the corresponding percentage of the selected rating are printed for both the base and change case dc solutions. The selected line rating and the difference in line flow between the base case and change case solutions are also tabulated.

If the current execution of activity DCLF included only a base case dc solution, the change case columns of the report are blank. If a change case solution was calculated, a description of the change case is included in the banner at the top of each page of output. In addition, in the bus output block for the from and to

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buses of the branch where status was changed, the branch output line for this branch is preceded by the character string --->.

If the change case is a line outage that results in the creation of an island without a swing bus, change case phase angles and line flows from within the swingless island are not tabulated; dashes are printed in the output report where these values would normally be shown.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the interior dummy buses of multi-section line groupings (refer to Multi-Section Line Grouping Data) are not reported. In reporting branches, the far end to bus (rather than the closest dummy bus) of each multi-section line connected to the from bus is shown as its to bus. Multi-section lines are identified with an ampersand (&) as the first character of their line identifiers in the branch circuit identifier column (e.g., &1).

9.4.4. Application Notes

Activity DCLF provides only *approximate* power flow solutions. The simplified branch flow equations on which its algorithm is based inherently result in phase angles and branch flows that are *different* from the ac power flow solution, even when the starting point is a fully solved power flow case. Furthermore, they lead to the assumption that bus voltage magnitudes and line losses remain constant as a branch is placed in- or out-of-service. It has the advantage, though, that it is substantially faster than a full ac power flow solution. Thus, its proper role is that of a screening tool to indicate which cases deserve further attention.

The linearized or dc power flow model used by activity DCLF approximates the nonlinear ac power flow with a simplified non-iterative calculation. In this method, the matrix equation:

$$[P] = [B] [\theta] \tag{9.6}$$

expresses net bus power injections as linear functions of bus phase angles.

For any induction machine, dc line, or FACTS device that has its ac powers set to zero (e.g., if the case had just been read with activity READ and no ac power flow solution attempted), an estimate of its power withdrawn from the network is used based upon its scheduled power or current. Otherwise, regardless of the quality of the ac solution upon entry to activity DCLF, the powers contained in the working case are used.

The line loading percentages tabulated in activity DCLF should be used with caution because reactive power flows are neglected in the dc analogy solution. These percentages are calculated based only on the MW loading of the line as determined by the dc analogy solution; activities such as POUT, LOUT and RATE calculate their line loading percentages based upon both the active and reactive power loading of each branch as determined from a full ac power flow solution.

The active power components of any line connected shunts in the working case are treated in the same manner that bus shunts are as far is the dc network solution is concerned (refer to Section 9.4.3, "Output Report Format"). In the output of activity DCLF, they are reported as part of the line flow rather than as load at the bus; that is, they are reported in the same manner as for ac solutions. If the change case is a branch outage, its line shunts are removed from the solution along with the branch. If the change case places a branch in-service, its line shunt components are not added with the branch.

The mathematics of a standard dc analogy network solution neglects line losses. That is, in applying such a method, generation and load exactly balance. Activity DCLF, on the other hand, approximates the effect of line losses on its dc analogy solutions with the following approach for each branch: From the voltage vector in the working case, it calculates the losses on the branch. Then, during its network solutions, activity DCLF injects as load the line losses at the sending end of the branch, as determined from the initial voltage vector.

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In the output report, these loss injections *are not* included in the bus loads tabulated. Rather, their effect is included in the line flows printed for each branch. Thus, instead of having equal flows with opposite signs at the two ends of a branch, these flows as tabulated by activity DCLF differ by the line loss estimate.

The same loss estimate vector is used in both the base case and change case dc analogy solutions. If the change case is a branch outage, its loss estimate is not removed from the solution along with the branch; it remains as an invisible load at the sending end bus described above. If the change case places a branch in-service, no loss estimate is added with the branch.

If the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the losses from the members of each in-service multi-section line are calculated as electrical equivalents at its endpoint buses. Thus, if the change case calculation is the outaging of a multi-section line, all of its losses remain as invisible load.

Similarly, any loss estimate that would normally be injected at the star point bus of a three-winding transformer are calculated as electrical equivalents at its endpoint buses, and remain as invisible load if the change case calculation is the outaging of a three-winding transformer.

From the foregoing discussion, then, it is *strongly* recommended that, when using activity DCLF to check branch outage cases, the working case be solved to an acceptable mismatch tolerance with a full ac solution prior to initiating activity DCLF. While it does not require a solved system condition, the *only* time an unsolved starting point should be provided is in using activity DCLF as a bootstrap to obtain an estimate of phase angles for a full ac power flow solution.

Additional Information
PSS®E GUI Users Guide, Running the DC Linearized Network Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Running the DC Linearized Network Solution

9.5. Calculating Linearized Network Contingency

DCCC

Run Activity DCCC - GUI
Power Flow > Linear Network > DC contingency checking (DCCC)... *[DC Contingency Checking]*
Run Line Mode Activity DCCC - CLI
ACTIVITY? >>**DCCC**
Interrupt Control Codes
AB

The previous section on the dc network solution process described the means by which a linearized base case solution could be obtained along with one change case. Such a change case could be considered a contingency. The linearized network contingency calculation activity DCCC estimates the flows on a set of monitored elements for the base case and for a specified set of contingency cases. Either an overload report or a loading table giving results of each case may be tabulated.

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The user specifies solution and output control parameters for the following:

Table 9.2. Linearized Network Contingency Options

Specification	Default	Description
MW mismatch exceeds tolerance	0.5	If the largest initial active power mismatch exceeds the specified MW mismatch tolerance, the user may terminate activity DCCC. The user can modify tolerance in steps of 0.1.
Contingency case rating	activity OPTN	The contingency case rating set, and the percentage thereof, define the line loading limits used in determining overloads. The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).
Percent of rating	100	
Convert ratings to estimated MW ratings	disabled	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's Mvar loading at the metered end in the base case ac solution.
Line flow code	dc base case	The line flow code defines the base flow to be used in deriving contingency case flow estimates. • dc base case • ac base case
Loss estimate code	include	The loss estimate code allows the user to either include or exclude a line loss estimate calculated from the present voltage vector.
Regulating phase shifter code	Fixed phase shift angles	The regulating phase shifter code defines the treatment of these devices in the base case and contingency case dc power flow solutions. • Fixed angles in the base case and contingency cases. • Fixed MW in the base case, fixed angles in the contingency cases.

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Specification	Default	Description
		• Fixed MW in the base case and contingency cases.
Output code	Summary	The output code allows the user to select either an overload summary report (Summary option) or a loading table (Full option).
Minimum flow change in overload report	1.0	Overloaded branches in contingency cases for which contingency case flows differ from their base case flows by less than the minimum contingency case flow change parameter are omitted from the summary overload report.
Exclude cases with no overloads from overload report	disabled	If enabled, those cases for which no overloads are detected are omitted from the summary overload report.
Distribution Factor Data File	The user must enter the name of a Distribution Factor Data File constructed by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired Linear Network Analysis Data Files (refer to ???). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity DCCC.	

9.5.1. Operation of Activity DCCC

Activity DCCC verifies that generators are not converted (refer to CONG) and that each non Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed.

Activity DCCC calculates and prints the largest active power mismatch corresponding to the present ac solution voltage vector in the working case.

The Distribution Factor Data File is read, and, if the convert ratings option was enabled at the start of activity DCCC, ratings from the selected rating set of those monitored branches having a non-zero rating are modified. For each such branch, the reactive power loading is calculated and a MW rating is determined under the assumption that the reactive power loading is unchanged under contingency conditions.

Finally, the base case dc power flow solution is calculated followed by the processing of the designated contingency cases. If a line outage contingency forms an island with no Type 3 (swing) bus, a singular submatrix is formed. Such islands are omitted from the network solution of activity DCCC.

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Activity DCCC may be terminated by entering the AB interrupt control code.

Two different output reports may be produced by activity DCCC: an overload report and a loading table.

9.5.2. Overload Report

In the overload report, each network condition (i.e., the base case and each contingency case) is reported immediately after its calculation. A branch or interface is reported if the following conditions are satisfied:

- Its rating from the selected rating set is nonzero.
- Its loading estimate exceeds the specified percentage of the selected rating.
- For contingency cases, the difference in loading between the base and contingency cases exceeds the minimum contingency case flow tolerance.

For each monitored element printed, the pre-contingency and post-contingency MW loadings, the rating and the post-contingency percent loading are listed. Any monitored branch that is located in a swingless island, as well as any interface that includes such a branch as a member, is omitted from the overload report.

For three-winding transformer windings, flow estimates are calculated at the winding bus end as power flowing into the transformer. For other ac branches, flow estimates are calculated at the metered end in the from bus to to bus direction. The flow across an interface is taken as the sum of the flows of its members. For three-winding transformer windings, the flow is calculated at the winding bus as power flowing into the transformer. For other members, the flow is calculated at its metered end in the from bus (i.e., the first bus specified in entering the branch; refer to Monitored Element Data File Contents to to bus direction.

Single branches are listed first, either in the order in which they were specified in the Monitored Element Data File, or in ascending numerical or alphabetical order, according to the option selected during the execution of activity DFAX (refer to Section 9.1.1, "Operation of Activity DFAX"). These are followed by the interfaces in the order in which they were specified.

If the convert ratings option was selected at the start of activity DCCC, the values printed as line ratings are the estimated MW ratings; otherwise, the ratings as contained in the working case are printed. In either case, the values printed are not modified by any percentage of rating parameter specified at the start of activity DCCC.

An example dc contingency check run using the *savnw.sav*, *savnw.sub*, *savnw.mon* and *savnw.con* files will produce the Overload/ Summary Report (see Figure 9.5, "Summary Contingency Report from the DC Contingency Checking Process").

It can be seen that four contingencies result in overload conditions. The description of the contingencies are listed at the left hand side of the report. The lines and interfaces that are subjected to overload conditions during each listed contingency are centered. At the right of the table, under the *Name* of the contingency condition, the following are listed:

- Pre-contingency power flow
- Post contingency power flows
- Rating and percent loading based on that rating

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E   WED, FEB 11 2004  12:39
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE                              PAGE 1
BASE CASE INCLUDING SEQUENCE DATA
DCCC OVERLOAD REPORT: MONITORED ELEMENTS LOADED ABOVE 100.0 % OF RATING SET B

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con

                                          <------- O V E R L O A D E D   L I N E S ------>  <--- MW FLOW --->
<---- C O N T I N G E N C Y   E V E N T S---------><------- F R O M -------> <------- T O ----->CKT  PRE-CNT  POST-CNT  RATING  PERCENT
REMOVE UNIT 1 FROM BUS 101 [NUC-A 21.600]          ------------------------------------------------------ CONTINGENCY TRIP1NUCLEAR
                                                          INTERFACE WEST                          -147.9    602.1    200.0    301.0
REMOVE UNIT 1 FROM BUS 101 [NUC-A 21.600]          ------------------------------------------------------ CONTINGENCY TRIP2NUCLEAR
REMOVE UNIT 1 FROM BUS 102 [NUC-B 21.600]
                                                    3008 CATDOG   230.00    154 DOWNTN  230.00 1     62.9    465.4    431.0    108.0
                                                          INTERFACE WEST                          -147.9   1352.1    200.0    676.0
INCREASE BUS 154 [DOWNTN230.00] LOAD BY 50 PERCENT  ------------------------------------------------------ CONTINGENCY ADDLARGELOAD
                                                          INTERFACE WEST                          -147.9    352.1    200.0    176.0
SET BUS 205 [SUB230230.00] LOAD TO 0 MW             ------------------------------------------------------ CONTINGENCY LOSEEASTLOAD
                                                       205 SUB230   230.00    154 DOWNTN 230.00 1    356.1   1198.2    610.2    196.3
                                                          INTERFACE WEST                          -147.9  -1347.9    200.0    674.0
                                                          INTERFACE EAST                          -130.8   1069.2    350.0    305.5
```

Figure 9.5. Summary Contingency Report from the DC Contingency Checking Process

9.5.3. Linear Network DC Loading Report

In the loading table, results are reported in tabular form with four contingency cases per table. Each group of contingency cases contains a page summarizing the contingency cases performed. For each contingency, the contingency label and the events defining the contingency are listed. The contingency case summary page is followed by the loading table. For each monitored element, this table lists its rating, MW loading and percentage loading in the base case network solution, and the MW and percentage loadings for each of the contingency cases reported. Percentage loadings are shown as positive numbers. Any percentage loading above the specified percentage threshold is followed by an asterisk (*). Any monitored branch that is located in a swingless island, as well as any interface that includes such a branch as a member, has dashes printed in place of its flow in the loading table.

An example dc contingency check run using the *savnw.sav*, *savnw.sub*, *savnw.mon* and *savnw.con* files will produce the Loading Report partially shown in Figure 9.6, "Loading Report from the DC Contingency Checking Process".

It should be noted that the cases marked with an asterisk, are summarized on the overload report typical of which is that shown in Figure 8-5.

Any monitored branch that is located in a swingless island, as well as any interface that includes such a branch as a member, has dashes printed in place of its flow in the loading table.

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E   WED, FEB 11 2004  12:38
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE                                PAGE 1
BASE CASE INCLUDING SEQUENCE DATA
                    *** DCCC CONTINGENCY SUMMARY ***

DISTRIBUTION FACTOR FILE:      C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:    C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:        C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con

<-CONTINGENCY-> <-----------------------------------------------------CONTINGENCY DESCRIPTION-----------------------------------

  TRIP1NUCLEAR   REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]

  TRIP2NUCLEAR   REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]
                 REMOVE UNIT 1 FROM BUS 102 [NUC-B       21.600]

  ADDLARGELOAD   INCREASE BUS 154 [DOWNTN      230.00] LOAD BY 50 PERCENT

  LOSEWESTGEN    REMOVE UNIT 1 FROM BUS 3018 [CATDOG_G    13.800]
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E   WED, FEB 11 2004  12:38
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE                                PAGE 2
BASE CASE INCLUDING SEQUENCE DATA
   DCCC LOADING TABLE: * FOR MONITORED ELEMENTS ABOVE 100.0 % OF RATING SET B

                                          CONTINGENCY LABEL-->     BASE CASE     TRIP1NUCLEAR     TRIP2NUCLEAR     ADDLARGELOAD
                                                                     MW     %       MW     %         MW     %         MW     %
<-------- F R O M --------> <---------- T O ---------->CKT RATING  -------------  -------------  -------------  -------------
   201 [HYDRO       500.00]    151 [NUCPANT     500.00] 1  1272.7   -558.7  43.9   -248.6  19.5     61.4   4.8   -595.8  46.8
   202 [EAST500     500.00]    152 [MID500      500.00] 1  1293.6    -50.7   3.9   -161.6  12.5   -272.5  21.1   -156.0  12.1
   203 [EAST230     230.00]    154 [DOWNTN      230.00] 1   244.1    122.4  50.2     90.7  37.1     58.9  24.1    160.8  65.8
   205 [SUB230      230.00]    154 [DOWNTN      230.00] 1   610.2    356.1  58.4    188.8  30.9     21.4   3.5    460.2  75.4
  3004 [WEST        500.00]    152 [MID500      500.00] 1  -NONE-   -132.2   ---    190.1   ---    512.5   ---     40.8   ---
  3006 [UPTOWN      230.00]    153 [MID230      230.00] 1  -NONE-    -78.6   ---    147.8   ---    374.3   ---     50.2   ---
  3008 [CATDOG      230.00]    154 [DOWNTN      230.00] 1   431.0     62.9  14.6    264.1  61.3    465.4 108.0*    261.1  60.6
       INTERFACE WEST                                      200.0   -147.9  74.0    602.1 301.0*  1352.1 676.0*   352.1 176.0*
       INTERFACE EAST                                      350.0   -130.8  37.4   -130.8  37.4   -130.8  37.4   -130.8  37.4
```

Figure 7-7. Loading Report from the DC Contingency Checking Process

Figure 9.6. Loading Report from the DC Contingency Checking Process

9.5.4. Contingency Case Flow Calculations

One of two calculation sequences is used for calculating monitored element flow estimates for each contingency case. For contingency cases involving no line closure contingency events, no three-winding transformer outages, and no more than two line outage contingency events, the following calculations are performed:

1. The contingency case flow estimates are initialized to either the flows given by the base case dc network solution or the base case ac flows from the working case, according to the line flow code solution parameter set at the start of activity DCCC (see above).

2. If any bus boundary condition contingency events have been specified as part of the contingency case, the incremental form of the dc power flow equation

   $$[\Delta P] = [B] [\Delta \theta] \quad (9.7)$$

   is solved for the corresponding phase angle changes, and the resulting incremental line flows are calculated and added to the contingency case flow estimate vector initialized in (1).

3. If one (or two) line outage contingency events have been specified as part of the contingency case, the corresponding vector(s) of line outage distribution factors is (are) retrieved from the Distribution Factor Data File. The contingency case flow estimate vector is updated using the formula in Section 9.2.3, "Application Notes" to reflect the first (or only) line outage.

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4. If two line outage contingency events have been specified as part of the contingency case, the vector of distribution factors for the second outage is updated to reflect the prior outage of the branch processed in (3). The contingency case flow estimate vector is again updated using the formula in Section 9.2.3, "Application Notes" to reflect the second line outage.

For contingency cases involving either line closure contingency event(s), three-winding transformer outages, or more than two line outage contingency events, full dc network solutions are calculated. If the line flow code solution parameter was set to dc base case at the start of activity DCCC (see above), contingency case flows are taken as those from the contingency case dc network solution; if the line flow code was set to ac base case, the changes in flows between the base case and contingency case dc network solutions are added to the base case ac flows to establish the contingency case flow estimates.

In applying the first method described above (the distribution factor method) for contingency cases involving two line outages, if the simultaneous outages create a swingless island that neither of the two outages individually creates, the distribution factor method is abandoned and the second method described above, the full dc network solution, is used.

In either of the two calculation methods, whenever a line outage contingency results in the presence of a swingless island, an appropriate message is printed prior to calculating the contingency case flow estimates.

9.5.5. Application Notes

Activity DCCC uses the same linearized network model as is used in the dc analogy network solution method of activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Note, in particular, the approximate nature of the linearized network model used by activity DCCC. The proper role of activity DCCC, therefore, should be to focus attention on those system conditions that deserve more detailed study.

Activity DCCC detects the specification of duplicate single and double line outage contingencies and calculates the contingency case solution only for the first specification of such a contingency. Any contingency cases involving bus boundary contingency events, line closure contingency events, or more than two line outage contingency events are not checked for duplication.

If the convert ratings code is set to one at the start of activity DCCC, any percentage of rating parameter is applied after the conversion to estimated MW ratings. Interface ratings are not affected by the convert ratings solution parameter.

Line loss estimates and line shunts of outaged lines are handled as in activity DCLF (refer to Section 9.4.4, "Application Notes").

It is required that the working case be solved to an acceptable mismatch tolerance prior to entering activity DCCC.

Additional Information
PSS®E GUI Users Guide, Calculating Linearized Network Contingency *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Linearized Network Contingency

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9.6. Performing DC Corrective Action Analysis

9.6.1. About DC Corrective Action Analysis

The mathematical model is presented in Section 6.13, "AC Corrective Actions". The dc corrective actions algorithm differs from that in the ac corrective actions in that it represents the transmission system by means of a linear model. Control adjustments are computed using a linear programming algorithm, for which this objective is to minimize the cost of generation re-dispatch, phase-shifter angle changes and load shedding. DC corrective actions focus on branch and interface overloads, and MW flows are used to determine circuit loadings. Bus voltage constraints are not taken into account in dc corrective actions. DC corrective actions use the same weighting functions for MW generation, phase angle, and load curtailment as discussed in ac corrective actions (see Section 6.13.2, "Weighting and Penalty Functions"). The penalty functions for soft constraints for branch flows and interface flows are also the same as those discussed in ac corrective actions.

9.6.2. Operation of DC Corrective Actions

DC corrective actions computation represents the transmission system by means of a linear model, and is modeled as an optimal power flow problem solved with a linear programming method. It identifies the necessary adjustments to loads, phase-shifter angles and active power generations to eliminate branch loading violations.

Table 9.3. DC Corrective Actions Options

Specification	Default	Description
Abort if MW mismatch exceeds tolerance	disabled	Enable to terminate dc corrective action analysis if the largest initial active power mismatch exceeds the specified Initial Condition MW Mismatch Tolerance. If disabled, continue to perform dc corrective action analysis with out-of-tolerance mismatch.
Initial condition MW mismatch tolerance	0.50	If the largest initial active power mismatch exceeds the specified MW mismatch tolerance, dc corrective action analysis may be terminated.
Modify network data with corrective action solution	enabled	Enable to modify generation, load and phase shifter angle to incorporate adjustments of control components involved in corrective actions and modify bus voltages to incorporate bus voltage phase angle vectors calculated from corrective actions.
Constraint options		Select branch flow rating set and limit tolerance for monitored branches and interfaces.
Rating	activity OPTN	Define branch rating set and percent of rating. The limits are equal

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Specification	Default	Description
Percent of flow rating	100.0%	to the specified rating multiplied by percent of flow rating.
Convert ratings to estimated MW ratings	disabled	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's MVAR loading at the metered end in the base case ac solution.
Branch overload violation tolerance	0.10%	Specify branch loading limit tolerance for adjustment of load limits to be used in linear programming engine. For example, if the tolerance is 0.1% and the selected flow rating is 100% of Rate A, the linear programming algorithm will consider the solution feasible if the loading of a branch has been reduced within 100.1% of the rating.
Control options		Select types of control adjustments and specify weighting factors for each type (see Section 6.13.3, "AC Corrective Action Options").
Input Data File	The Distribution Factor Data File is used to define branch and interface flow constraints and control subsystems. When creating a Distribution Factor Data File for dc corrective action, a Contingency Description Data File that specifies no contingencies is allowed.	

9.6.3. DC Corrective Action Results

After a dc corrective actions solution is performed successfully, the program lists monitored branches that are overloaded, before and, if any, after the corrective actions solution. In the second part of the report, initial and new values of the controls, as well as changes in controls are reported also. Figure 9.7, "Sample Output of DC Corrective Action Analysis" is the sample result of dc corrective actions applied to an example system. Two branches and an interface flow overloads in the working case are removed by corrective actions, which involve the adjustments of the phase shifter between buses 203 and 202, and generation at two generator buses.

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```
                    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E
                                DC CORRECTIVE ACTIONS RESULTS

------------------------------------------------------------------------------

 Case file:                              C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sav
 Distribution factor file:               C:\Program Files\PTI\PSSE30\EXAMPLE\corrective-actions.dfx
 Subsystem description file:             C:\Program Files\PTI\PSSE30\EXAMPLE\savnwl.sub
 Monitored element file:                 C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
 Contingency description file:           C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con
 *** Constraints ***
 No. of monitored flows:                 8
 No. of monitored interfaces:            2
 *** Controls ***
 No. of Generators:                      6          In Subsystem:  LODSUB
 No. of Load Shedding:                   7          In Subsystem:  LODSUB
 No. of Phase Shifters:                  1          In Subsystem:  LODSUB

------------------------------------------------------------------------------

 BASE CASE OVERLOAD REPORT: MONITORED BRANCHES LOADED ABOVE  60.0 % OF RATING SET A

 <---------- O V E R L O A D E D   L I N E S ---------->     FLOW   RATING        %
 <------ F R O M --------> <-------- T O ----------> CKT       MW      MVA
    203 EAST230     230.00    154 DOWNTN      230.00  1     122.4    200.0    61.22
   3018 CATDOG_G    13.800   3008 CATDOG      230.00  1     100.0    150.0    66.64
 WEST                                                      343.4    200.0   171.70

------------------------------------------------------------------------------

 POST-CORRECTIVE ACTIONS OVERLOAD REPORT

 <---------- O V E R L O A D E D   L I N E S ---------->     FLOW   RATING        %
 WEST                                                      120.0    200.0    60.00

------------------------------------------------------------------------------

 BUSES WITH REAL POWER GENERATION ADJUSTMENTS (MW)
 <-------- B U S -------->   GEN-INI   GEN-NEW   GEN-ADJ
    211 HYDRO_G     20.000     600.0     614.7      14.7
   3018 CATDOG_G    13.800     100.0      83.6     -16.4

 PHASE SHIFTER ADJUSTMENTS (DEGREE)
 <----------- P H A S E   S H I F T E R ----------> CKT  PHASE-INI PHASE-NEW PHASE-ADJ
    203 EAST230     230.00    202 EAST500     500.00  1        0.00     14.45     14.45

 ===== END OF REPORT ==========================================================================
```

Figure 9.7. Sample Output of DC Corrective Action Analysis

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9.6.4. Application Notes

The corrective actions solution algorithm does not place a limit on the number of controls and constraints that can be modeled. The computation time of a corrective actions solution is approximately proportional to the number of constraints. Because local controls are generally more effective in removing violations, it may be preferable to choose the subsystem in which control actions are allowed to be the same as that in which elements are monitored.

Operating limit violations caused by incorrect limit settings may have a significant impact on the optimal solution found. Hence, suspicious limits should be verified and corrected, if necessary, or even removed from the solution.

Two methods can be used to excluded controls from the corrective actions solution. One way is to set the upper and lower limits of a control equal; another is to carefully define the control subsystem to exclude those controls that are not desired in a particular application.

Additional Information
PSS®E GUI Users Guide, Running DC Corrective Action Analysis *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Linearized Network Contingency

9.7. DCPSCOPF Solution

9.7.1. About DCPSCOPF analysis

The preventive security constrained optimal power flow (PSCOPF) is a special class of OPF problems with security constrains. System security is the ability to withstand contingencies, in other words, to remain intact even after equipment outages or failures. Security plays a crucial role in the planning and operation of a power system. To ensure a secure system operation condition, system planners and operators can conduct analyses to identify the necessary adjustments such that there are no limit violations in the base case (pre-contingency) condition and following any contingency.

The preventive security constrained optimal power flow (PSCOPF) can be considered conservative, since it does not take into account the system's post-contingency control capabilities. All control variables are optimized such that no post–contingency adjustments are necessary in order to avoid violations of base case and post-contingency constraints, and no operator intervention is required following an anticipated contingency.

The preventive security constrained optimal power flow (PSCOPF) uses either AC power flow network model (see Section 6.14, "PSCOPF Solution") or DC power flow (linear) network model. The DC based preventive security constrained optimal power flow (DCPSCOPF) features high performance computation, and can eliminate hundreds thermal limit violations in thousands of contingencies within a few of minutes.

The DCPSCOPF function formulates and solves an optimization problem:

With an objective that minimizes the adjustments of the following types of control:

- On-line and off-line generator MW generation control
- Phase shifter adjustments
- Load controls

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Subject to the following types of constraints:

- Power balance equations of base case and contingencies cases
- Limits on controls
- Thermal limits under base case and contingency cases

The inputs required to perform a DCPSCOPF consist of a power flow network, the available controls and the constraints. The controls are selected from the control subsystems, which are pre-defined in the subsystem description data file (*.sub), and each type of control is assigned a priority, relative to other control types. The operation constraints are defined in the monitored element data file (*.mon), and the contingencies are specified in the contingency description data file (*.con). Before initiating the DCPSCOPF solution, a dfx file is built using function DFAX to process the three data files and store the information in a dfx file for use by DCPSCOPF.

9.7.2. DCPSCOPF solution options

The following options are available in the DCSCOPF function. See the API Manual, DCPSCOPF.

Table 9.4. DCPSCOPF Solution Options

Generation dispatch for contingency case	Option for generation dispatch to handle system load/generation imbalance caused by a contingency: • Subsystem machines (Reserve) • Subsystem machines (PMAX) • subsystem machines (Inertia) • subsystem machines (Governor droop) Dispatch subsystem specifies the subsystem containing the machines participating in the selected dispatch
Ignore violations in base case	The option for initial thermal violations in base case. Select the option to exclude thermal violations from the solutions.
Abort if MW Mismatch exceeds tolerance (MW)	The tolerance used to check for the largest initial active power mismatch. If it is exceeded by the largest initial active power mismatch, the DCPSCOPF solution is terminated when the option to abort DCPSCOPF is selected. The transmission losses being treated as MW injections in DC power flow network model are calculated under initial voltage conditions.If the case is not converged or inital active power mismatch is greater than the tolerance, the estimated losses may be inac-

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	curate and have minor impacts on the MW flows on the DC power flow network.
Control options	Flag to enable or disable generator active power dispatch (see Section 6.11, AC Corrective Actions) Weighting factor for generator active power dispatch Control subsystem of participating generators Flag to enable or disable load controls Weighting factor for load controls Control subsystem of participating loads Flag to enable or disable phase shifter controls Weighting factor for phase shifter controls Control subsystem of participating phase shifters Flag to enable or disable off-line generator active power dispatch Weighting factor for off-line generator controls Control subsystem of participating off-line generators
Distribution Factor Data File	(*.dfx) a required file to be build by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired linear network analysis data files. The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) must have the same setting that was in effect when activity DFAX filled the specified Distribution Factor Datafile.

9.7.3. Perform DCPSCOPF Analysis

To launch the DCPSCOPF analysis, select Power Flow ◊ Linear network #DC PSCOPF

The DCPSCOPF uses linear programming to solve the optimal problem constructed with DC power flow network model.

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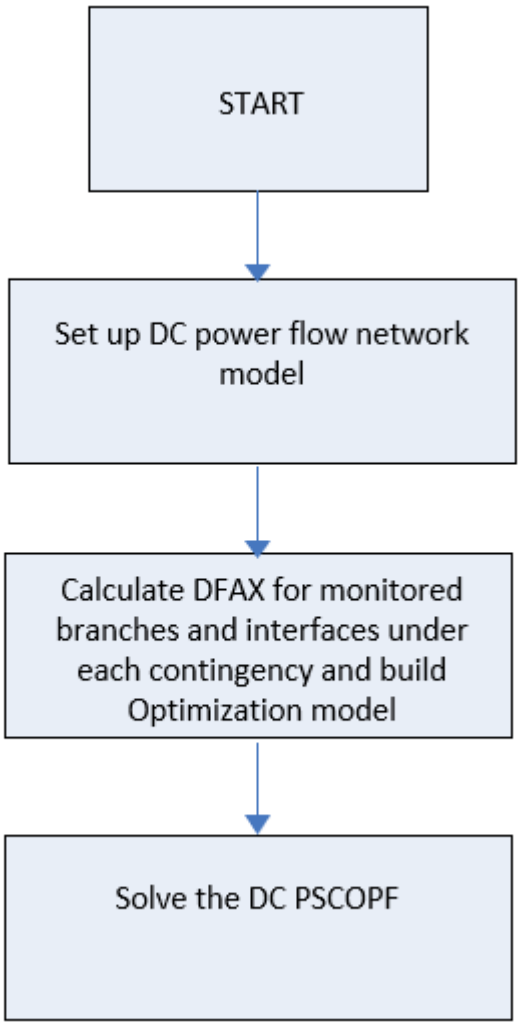

Figure 9.8. Flow Chart of DCPSCOPF

The solution consists of several steps:

1. Read the base case, contingency definitions, monitored elements, etc.
2. Set up the DC power flow netwrok model, estimate transmission losses.
3. calculate DFAX factors for monitored branches and interfaces under each contingency.
4. perform DCPSCOPF solution with linear programming method to find the control settings.

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Network data | Network Results

Contingency Label	Monitored Element	Rating (MVA)	Basecase (%)	Contingency Pre-PSCOPF(%)	Contingency Post-PSCOPF(%)
N1OVRLOD 1	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	109.51	92.96
N1OVRLOD 1	213 BIGOIL 230.00 223 BIGCOAL 230.00 1	500.00	73.01	106.08	94.91
N1OVRLOD 2	103 WESTIE-1 138.00 109 MIDTIE-1 138.00 1	175.00	51.22	101.32	93.32
N1OVRLOD 3	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	109.63	91.63
N1OVRLOD 3	109 MIDTIE-1 138.00 211 MIDTIE-2 230.00 1	400.00	87.85	98.11	65.86
N1OVRLOD 3	212 EASTIE-2 230.00 223 BIGCOAL 230.00 1	500.00	62.26	101.80	94.05
N1OVRLOD 4	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	111.20	94.91
N1OVRLOD 4	109 MIDTIE-1 138.00 211 MIDTIE-2 230.00 1	400.00	87.85	105.58	74.38
N1OVRLOD 5	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	111.17	49.47
N1OVRLOD 5	109 MIDTIE-1 138.00 211 MIDTIE-2 230.00 1	400.00	87.85	105.55	25.33
N1OVRLOD 5	216 MIDCOAL 230.00 225 ADDON 230.00 1	500.00	34.60	0.40	95.10
N1OVRLOD 6	103 WESTIE-1 138.00 109 MIDTIE-1 138.00 1	175.00	51.22	99.96	91.96
N1OVRLOD 7	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	100.75	84.54
N1OVRLOD 8	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	95.92	79.93
N1OVRLOD 8	221 NUKE-B 230.00 222 HYDRO 230.00 1	500.00	55.55	97.58	97.58
N1OVRLOD 10	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	98.80	82.35
N1OVRLOD 10	109 MIDTIE-1 138.00 211 MIDTIE-2 230.00 1	400.00	87.85	130.94	95.10
N1VLTAGE 1	217 NOLOAD 230.00 222 HYDRO 230.00 1	500.00	42.03	97.58	97.58
N1VLTAGE 4	109 MIDTIE-1 138.00 211 MIDTIE-2 230.00 1	400.00	87.85	95.98	64.32
N1VLTAGE 5	103 WESTIE-1 138.00 224 WESTIE-2 230.00 1	400.00	94.78	96.30	80.17

Figure 9.9. Overload outputs of DCPSCOPF

The result table shows thermal overloads in base case and contingencies, and the loadings after the system is adjusted by DCPSCOPF.

9.7.4. Application notes

Invoking the DCPSCOPF will trigger an alarm and termination of the solution if generators are converted (refer to CONG). The solution then checks to make sure that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any isolated bus is found, an appropriate warning message is displayed and the solution is terminated. If the largest initial active power mismatch exceeds the specified mismatch tolerance (an input item) and the option to abort the solution is checked, the solution is terminated.

The DCPSCOPF Solution function responds to the following interrupt control codes:

AB	Abandon the function following completion of the next iteration.

After the DCPSCOPF solution is converged, the working case is updated with the control settings determined by the DCPSCOPF solution, such as generator outputs, loads, phase shifter angels. The working case can be solved by a power flow solution and used by other applications.

DCPSCOPF may be terminated abnormally under one of following conditions;

- The inital MW mismatch is greater than the tolerance.
- The Linear Programming solution does not converge.
- Termination initiated by the program user.

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Network data | Network Results

Adjusted Element	Adjustment Type	Initial Value	Post Adjustment Value
101 COAL-A 138.00	Generator (MW)	228.000000	240.000000
102 COAL-B 138.00	Generator (MW)	228.000000	240.000000
107 MIDOIL 138.00	Generator (MW)	240.000000	275.965600
213 BIGOIL 230.00	Generator (MW)	346.707000	413.435300
215 MIDCOALP 230.00	Generator (MW)	155.000000	26.306230
212 EASTIE-2 230.00 110 EASTIE-1 138.00	Transformer Phase Shift (deg)	-24.205400	-16.898590
225 ADDON 230.00 214 CONDENS 230.00	Transformer Phase Shift (deg)	22.524040	17.10396

Figure 9.10. Control adjustment outputs of DCPSCOPF

DCPSCOPF produces a report/table results upon completion of execution. The report and result table summarize the adjustments made by DCPSCOPF process to ensure system security, and the list of active constraints. Active constraints are limiting constraints which cannot be satisfied in the solution of DCPSCOPF. Constraints can be limiting because not enough applicable controls have been specified. For example, if a transmission line along a radial network is overloaded and the loads along the network are not allowed to be adjusted, there is no practical way to eliminate the overload. Another reason is the application of soft penalties to the constraints. Mathematically, because the ineffective controls always involve large adjustments and introduce large penalties to the objective function, the DCPSCOPF will balance the cost of infeasibility (limit violations) against the costs of the adjustments to determine the optimal solution. Hence, if the limit violations in a power system cannot all be eliminated using the basic control actions, it may be preferable sometimes to accept certain degree of limit violations instead of applying extreme and ineffective control adjustments.

9.8. Calculating Transmission Interchange Limits

TLTG

Run Activity TLTG - GUI
Power Flow > Linear Network > Transmission interchange limits calculation (TLTG)... *[Transmission Interchange Limits Calculation]*
Run Line Mode Activity TLTG - CLI
ACTIVITY? >>**TLTG**
Interrupt Control Codes
AB

9.8.1. Overview: Transmission Transfer Limit Analysis

In the day-to-day operation of interconnected networks, bulk power transfer is often constrained by the ability of transmission elements to withstand thermal effects for various normal and contingency conditions. More recently this bulk power transfer is often constrained by stability limits. If sufficient stability analysis has been performed, megawatt limits can be assigned to elements or groups of elements. Because the transfers of power amount to thousands of megawatts for many hours in every single day, the determination of the ability of the transmission system to support the power transfers is a vital consideration to assure that the interconnected network is operated in a secure and reliable manner.

As shown in Section 9.4.1, "DC Linearized Network Solutions" a dc technique gives a quick solution. A common approach used to find a limiting solution is to start with a base case and calculate the sensitivity of flow

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in monitored elements or groups of elements to a variation in interchange. This technique is often referred to as a distribution factor technique. When the sensitivity of elements is known, linear projections can be used to estimate permissible interchanges based on thermal limits. P_1, P_2 and P_3 represents linear line flow functions of the net import. The horizontal line rating intersects P_1 imposing a limit or net import restriction. Figure 9.11, "Linear Projection Technique Used in Transfer Limit Analysis" graphically shows this technique.

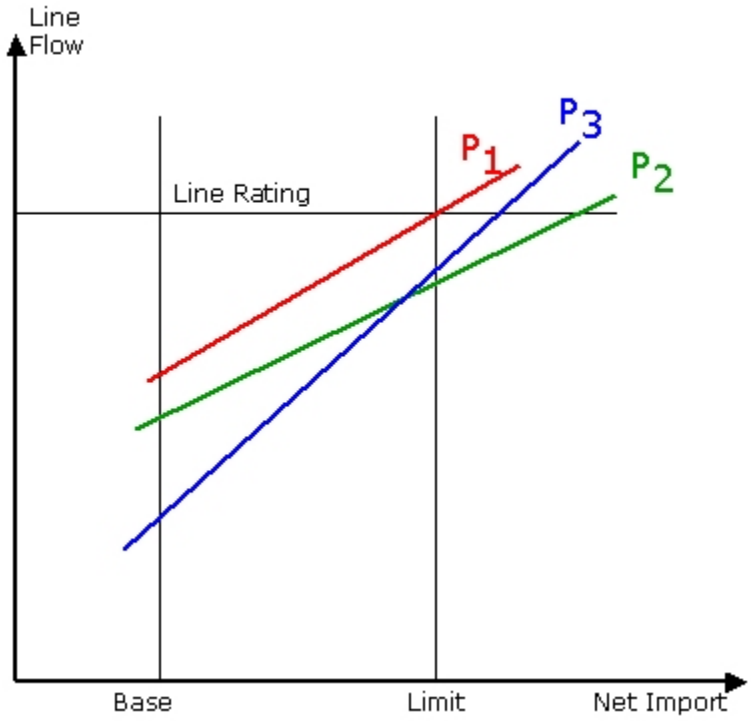

Figure 9.11. Linear Projection Technique Used in Transfer Limit Analysis

9.8.2. Calculating Transmission Transfer Limits

The transmission interchange limit analysis activity TLTG estimates the import or export limits of a specified subsystem of the working case using a linearized network model. The user designates this study system in which the total power injection is to be increased (for export limits) or decreased (for import limits). An opposite change in the total power injection is made in a designated opposing system. The calculation uses the sensitivity type analysis described in Section 9.8.1, "Overview: Transmission Transfer Limit Analysis".

Power transfer distribution factors relating changes in branch and interface flows to a change in study system interchange are determined. The maximum study system export or import is then derived by extrapolation subject to the constraint that no monitored elements exceed a specified percentage of a selected rating.

This process may then be repeated for a designated set of user specified contingency cases.

The process will perform all the contingencies identified in the contingency description file processed when building the distribution factor file. Furthermore, the process uses the monitored list file, which allows the lines to be monitored and interfaced. This feature is important because these interfaces often define stability limits.

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Note that any two subsystems identified in the *.*sub* file can be used for the transfer analysis. Furthermore, however, it should be noted that the limits will be found based on *only* those elements (lines and interfaces) that are monitored, i.e., included in the *.*mon* file.

In Figure 9.12, "Study System (A) and Opposing System (C). Area B Potentially Limiting" It can be seen that Areas A and C form the study and opposing system, respectively. Transfers take place directly and via Area B. It is important therefore to monitor lines within Area B or interfaces between Area B and the other two areas to ensure that the correct limits are identified.

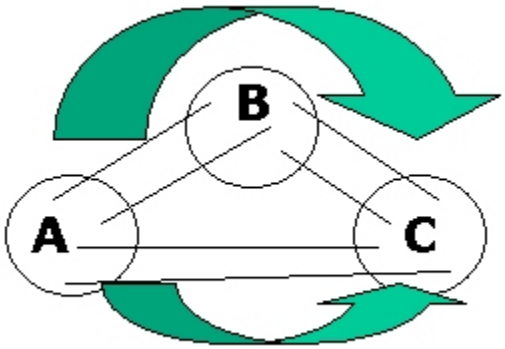

Figure 9.12. Study System (A) and Opposing System (C). Area B Potentially Limiting

9.8.3. Transmission Transfer Limit Analysis Options

The user specifies solution and output control parameters for the following:

Table 9.5. Transmission Transfer Limit Analysis Options

Specification	Default	Description
MW mismatch tolerance	0.5	Largest initial active power mismatch. Note that the user can modify the tolerance in steps of 0.1.
Abort if MW mismatch exceeds tolerance	enabled	Enable or disable termination of activity TLTG if mismatch tolerance exceeded.
Base case rating	activity OPTN	The base case and contingency case rating sets, and the percentage thereof, define the line loading limits used in determining overloads. The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).
Contingency case rating		
Percent of rating	100	

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Specification	Default	Description
Convert ratings to estimated MW ratings	disabled	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's reactive power loading at the metered end in the base case ac solution.
Line flow code	dc base case	The line flow code defines the base flow to be used in deriving contingency case flow estimates. • dc base case • ac base case
Phase shifter in base case	Lock	If set to *Regulate* (in base case), any regulating phase shifter where adjustment control mode CODn (refer to Transformer Data) is set to +3 holds its base case flow in all base case shift solutions. If set to *Lock*, all phase shift angles are locked at their base case settings.
Base case constraints in contingency case	Ignore	If set to *Include*, both base case and contingency case flow constraints are recognized in calculating the pre-contingency interchange limits for each contingency case. If set to *Ignore*, only the contingency case loadings are considered (refer to Section 9.8.8, "Application Notes").
Add study system ties to monitored lines	disabled	If enabled, automatically include study system tie lines as monitored elements.
Minimum contingency case pre-shift flow change	1.0	Provide the means to ignore those monitored elements that, in a contingency case, are not significantly affected by the contingency. If the magnitude change in pre-shift flow from the base case value is less than the *minimum flow change* threshold *and* the magnitude change in the power transfer distribution factor from the base case value is less than
Minimum contingency case distribution factor change	0.0000	

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Specification	Default	Description
		the *distribution factor change* threshold, the monitored element is ignored.
Study system generation shift	100	If the value entered is positive, export limits are to be determined; otherwise, import limits are to be calculated. The generation shift is apportioned among generator and load buses in the study system in proportion to the participation factors specified in the participation block structure of the Subsystem Description Data File (refer to ???). If no participation block was entered for the study system, the generation shift is shared among generator buses in proportion to their plant MBASEs (i.e., the sum of the MBASEs of in-service machines with positive active power output at the bus). In this case, there must be at least one in-service machine with a positive MBASE in the subsystem. The opposite generation shift is applied to buses in the first opposing system, again according to their participation factors or MBASEs. Using the incremental form of the dc power flow equation (refer to activity DCLF) and the power shift vector, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.
Subsystem selection		Study System: network subsystem in which the total power injection is to be increased (for export limits) or decreased (for import limits). The choices available

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Specification	Default	Description
		are those listed in the *.sub type file. Opposing System: network subsystem in which an opposite change from that in the *Study System* is made.
Maximum import / export	99999	Threshold value, number of monitored elements in the summary report.
Minimum distribution factor magnitude	0.0	Threshold value for summary report. For small distribution factors with levels below this threshold there will be no reporting for the related element.
Maximum time report same element	5	Threshold value for summary report limiting the number of times an element is reported in the same report.
Contingency descriptions	Events	Defines the manner in which contingency cases are identified in the summary report as one of the following: • 32-character contingency case label • events comprising the contingency case • both contingency label and the contingency events
List study system buses	disabled	If enabled, list *Study System* buses in output report.
List opposing system buses	disabled	If enabled, list *Opposing System* buses in output report.
List study system tie lines	disabled	If enabled, list *Study System* tie lines in output report.
Output code	Summary	Summary: summary report of all cases considered Full: monitored element flow table for each case considered followed by a summary report of all cases considered (refer to Section 9.9.5, "Output Report Format")

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Specification	Default	Description
Interchange limit output	Incremental	Incremental: incremental transfer capability Total: total transfer capability
Elements in flow tables	60000	Limit the flow table for each case reported to the < n > most restrictive monitored elements.
Apply summary minimum distribution factor to report	disabled	If enabled, apply the minimum distribution factor magnitude threshold for including monitored elements in the summary report to the monitored element flow table of each case reported when full output is selected.
Optional Interface Output	NONE	Summary report is repeated with interface transfer limits and distribution factors for a selected interface listed rather than study system transfer limits and distribution factors. The interface distribution factors listed on the interface summary report are measures of the changes in monitored element flows to a change in base case interface flow.

9.8.4. Operation of Activity TLTG

Activity TLTG checks that generators are not converted (refer to CONG) and that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed and activity TLTG is terminated.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed. Next, activity TLTG calculates and prints the largest active power mismatch corresponding to the present ac solution voltage vector in the working case.

The user may specify that activity TLTG be terminated (default) if this largest mismatch is greater than the MW mismatch tolerance.

The user must enter the name of a Distribution Factor Data File constructed by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, “Application Notes”) and to the desired Linear Network Analysis Data Files (refer to ???). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, “File Usage ” for the file specification conventions used by activity TLTG.

The user must specify the study and opposing systems. If no subsystems are defined, an appropriate error message is printed and activity TLTG is terminated. If only one subsystem was specified, the specified subsys-

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tem is taken as the study system while the remainder of the working case is treated as the opposing system. If two subsystems were specified, the default opposing system is bypassed.

Activity TLTG ensures that there are no buses that are members of both of the selected subsystems. If any such buses are found, they generate an alarm and activity TLTG is terminated.

The largest active power mismatch corresponding to the present ac solution voltage vector in the working case is calculated and printed.

If the convert ratings option was enabled at the start of activity TLTG, ratings from the selected rating set(s) of those monitored branches having non-zero rating(s) are modified. For each such branch, the reactive power loading is calculated and MW rating(s) are determined under the assumption that the reactive power loading is unchanged under power transfer and contingency conditions.

Using the incremental form of the dc power flow equation (see Section 9.4.1, "DC Linearized Network Solutions") and the power shift vector, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.

The base case dc network solution is calculated and the solution summarized as in the following example:

```
STUDY SYSTEM GENERATION IS          1500.0 MW
OPPOSING SYSTEM GENERATION IS       1748.9 MW
STUDY SYSTEM NET INTERCHANGE IS      282.8 MW
```

The user may specify a study system generation shift (default = 100). If the value entered is positive, export limits are to be determined; otherwise, import limits are to be calculated.

The generation shift is apportioned among generator and load buses in the study system in proportion to the participation factors specified in the participation block structure of the Subsystem Description Data File (refer to ???). If no participation block was entered for the study system, the generation shift is shared among generator buses in proportion to their plant MBASEs (i.e., the sum of the MBASEs of in-service machines with positive active power output at the bus). In this case, there must be at least one in-service machine with a positive MBASE in the subsystem. The opposite generation shift is applied to buses in the opposing system, again according to their participation factors or MBASEs. Using the incremental form of the dc power flow equation (refer to Section 9.4.4, "Application Notes" and Section 9.5.4, "Contingency Case Flow Calculations") and the power shift vector, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.

The base case solution is then reported, followed by processing of the designated contingency cases with contingency case flow estimates calculated using the methodology defined in Contingency Case Flow Calculations. Prior to terminating, an ordered summary report from among all the cases calculated is tabulated (refer to Section 9.8.6, "Output Report Format").

The user may specify having the summary report repeated with interface transfer limits and distribution factors for a selected interface listed rather than study system transfer limits and distribution factors. The interface distribution factors listed on the interface summary report are measures of the changes in monitored element flows to a change in base case interface flow.

In listing study system buses, opposing system buses, and study system tie lines, the tabulation may be terminated by entering the AB interrupt control code. In this case, the interrupt is cleared and processing

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continues. After the interchange limit calculations have started, activity TLTG may be terminated by entering the AB interrupt control code.

9.8.5. Analyzing Transfer Limit Results

The results of running an example case will be presented to assist in understanding the report format. The example will be based on the files available in the PSS®E EXAMPLE directory; those being:

- *savnw.sav* – the power flow case
- *savnw.sub* – subsystem definitions
- *savnw.con* – contingency description file
- *savnw.mon* – the monitored element file

To further help clarify the power flow condition, Figure 9.13, "Subsystem file and Power Flow Condition for Transfer Analysis" shows the inter-area flows and the Subsystem Description file, savnw.sub. There it can be seen that the subsystem named STUDY comprises Area 1 of the power flow case and the subsystem named EAST comprises Area 2. Furthermore, Area 1 has a net export of 131 MW + 148 MW; a total of 279 MW

```
SUBSYSTEM STUDY
AREA 1
END
SUBSYSTEM EAST
AREA 2
END
SUBSYSTEM WEST
AREA 5
END
END
```

savnw.sub file

```
          TO AREA:      1       2       5
FROM AREA     *-------------------
          1   *          131     148
FLAPCO        *         -442    -243
              *-------------------
          2   *  -131
LIGHTCO       *    442
              *-------------------
          5   *  -148
WORLD         *    243
              *-------------------
```

Inter-Area Flows in Base Case

Figure 9.13. Subsystem file and Power Flow Condition for Transfer Analysis

9.8.6. Output Report Format

In reporting the base case solution, activity TLTG lists the study system generation, the opposing system generation, and the study system net interchange corresponding to the network solution before and after applying the generation shift. This is followed by a tabulation of the pre-shift and post-shift generation at those study system and opposing system buses participating in the generation shift. Loadings on monitored elements are then reported, sorted such that the most restrictive elements are listed first. Either incremental or total pre-contingency transfer limits, as appropriate, are listed.

For monitored three-winding transformer windings, flow estimates are calculated at the winding bus end as power flowing into the transformer. For other monitored branches, flow estimates are calculated at the metered end in the from bus to to bus direction. The flow across an interface is taken as the sum of the flows of its members. For three-winding transformer windings, the flow is calculated at the winding bus as

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power flowing into the transformer. For other members, the flow is calculated at its metered end in the from bus (i.e., the first bus specified in entering the branch; refer to Monitored Element Data File Contents to to bus direction.

Flow estimates are listed for each of three interchange schedules: with the original generation profile, with the specified generation shift, and at the generation shift corresponding to the transfer level limit required for the most restrictive monitored element (i.e., the one listed first). Any flow at or above the selected percentage of the appropriate rating is followed by an asterisk (*). The power transfer distribution factor relating the change in flow on the monitored element to a change in study system net interchange is also tabulated, along with the rating of the monitored element; for branches, this *is not* the percentage of the rating used in determining the interchange limit, but either the value of rating as entered in activities such as READ, TREA, RDCH, CHNG or the *[Spreadsheet]*, or, if the convert ratings option was selected at the start of activity TLTG, the estimated MW ratings.

When reporting flows for contingency cases, the events comprising the contingency are listed followed by flows on the monitored elements sorted as described above. The flow table includes both the contingency case flows in the same form as the base case report described above, as well as the base case flows at the transfer level required for the most restrictive monitored element. Note that if the *ignore base case constraints in contingency cases* option was specified, only the contingency case constraints are considered in calculating the transfer limit; in this case, base case flows at the transfer limit may be shown as overloaded because the corresponding constraints are ignored in the transfer limit calculation.

A summary report tabulates the number of system condition solutions attempted and the number for which there exists no interchange schedule at which the rating constraint can be satisfied for all monitored elements (refer to Section 9.8.8, "Application Notes"). This is followed by a tabulation of monitored elements for all system conditions that were calculated, in order of increasing incremental or total transfer capability, as appropriate. The limiting element is listed along with its power transfer distribution factor, its pre-shift contingency case flow, its rating, and a description of the system condition.

Only those monitored elements satisfying the cutoff threshold solution parameters set at the start of activity TLTG are included in the summary. After an element has been listed the maximum number of times < n >, its nth occurrence includes an asterisk (*) before its description, and further reporting of the element is suppressed.

This summary report is reprinted for each selected interface with interface transfer limits and distribution factors rather than study system interchange limits and distribution factors tabulated. All distribution factors on the interface transfer limits summary report describe changes in monitored element flow relative to a change in interface flow. The branches that form the interface are listed on the first page of an interface's transfer limit summary report.

The final page of the summary report lists those elements that were reported < n > times, sorted by the number of times they would have been reported.

If a line outage contingency forms one or more islands with no Type 3 (swing) bus, such islands are disconnected. Any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, has dashes printed where its contingency case results would normally be printed.

Figure 9.14, "Base Case Results For Two-Area Transfer Limit Calculation" shows the report obtained using the example files (*savnw.**) from which was built the *dc-contingency.dfx* distribution factor file. The filenames are listed in the report as shown.

The pre- and post-shift generation levels and the net interchange are listed for the Study and Opposing systems. In this example, a shift of 100 MW is assumed. The generators that participate in both systems to

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make the + 100 MW shift are listed. They are the nuclear plants at buses 101 and 102 in the Study system and the generators at buses 206 and 211 in the Opposing system.

For the base case condition, it can be seen that the element that is most restrictive for transfers is the INTERFACE EAST. This interface comprises 4 branches that tie the Study and Opposing systems. Consequently, all transfer takes place across these branches. This is why the Distribution factor has a value of 1.0. This means that 100% of any generation shift will flow on this interface.

```
        *** TLTG EXPORT LIMIT OUTPUT FOR BASE CASE                 ***

DISTRIBUTION FACTOR FILE:     C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:   C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:       C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con

                               PRE-SHIFT    DELTA  POST-SHIFT
STUDY SYSTEM MW GENERATION:       1500.0    100.0     1600.0
OPPOSING SYSTEM MW GENERATION:    1400.0   -100.0     1300.0
STUDY SYSTEM NET INTERCHANGE:      278.7    100.0      378.7

<---------------- STUDY SYSTEM -----------------> <--------------- OPPOSING SYSTEM --------------->
                         <---- GENERATOR MW ----->                          <---- GENERATOR MW ---->
  BUS# X-- NAME --X BASKV    BASE   SHIFT  CHANGE    BUS# X-- NAME --X BASKV    BASE   SHIFT  CHANGE
   101 NUC-A       21.600   750.0   800.0    50.0     206 URBGEN      18.000   800.0   742.0   -58.0
   102 NUC-B       21.600   750.0   800.0    50.0     211 HYDRO_G     20.000   600.0   558.0   -42.0

LOADINGS AT OR ABOVE 100.0 %                                          <------------- BASE CASE ------------->
OF RATING ARE MARKED WITH '*'                                  TOTAL          PRE-    POST-    LIMIT
                                                               TRANS RATING   SHIFT   SHIFT    CASE     DISTR.
<------- F R O M -------> <--------- T O ---------> CKT   CAPAB    A      MW      MW       MW       FACTOR
      INTERFACE EAST                                          497.9   350  -130.8  -230.8   -350.0   -1.00000
   201 HYDRO       500.00    151 NUCPANT     500.00  1    1330.4  1200  -556.7  -619.7   -692.4   -0.60981
   205 SUB230      230.00    154 DOWNTN      230.00  1    3370.1   600   356.1   325.2    288.3   -0.30929
  3008 CATDOG      230.00    154 DOWNTN      230.00  1    4734.9   400    62.9    70.4     79.4    0.07566
   202 EAST500     500.00    152 MID500      500.00  1    9483.9  1200   -50.7   -63.2    -78.1   -0.12486
```

Figure 9.14. Base Case Results For Two-Area Transfer Limit Calculation

The flow in the PRE-SHIFT condition is 130.8 MW on the interface. If a 100 MW shift is imposed, the POST-SHIFT flow on the interface will increase by 100 MW to 230 MW as shown. The shift can be increased until the flow on the interface is 350 MW, its thermal limit. this is the LIMIT case.

It can be seen that the flow on the limiting interface can increase by 220 MW (350-131) before reaching its thermal limit. Consequently, the TOTAL transfer limit will be the pre-shift flow of 278.7 MW plus the possible 220 MW increase. As the report shows, therefore, the transfer limit for the base case condition is 497.9 MW and is caused by the thermal limit on the INTERFACE EAST.

As the listing shows, the next most restrictive element is the branch from Bus 201 to Bus 151. The results are showing that, if the INTERFACE EAST thermal limit were removed (by upgrading or other means), this branch would create the limit; in this case it is 1,330 MW.

How is the 1,330 MW Calculated?

Branch 201 to 151 has a thermal limit of 1,200 MW. The pre-shift flow on the branch is only 558.7 MW. Consequently, the branch has a capacity for an additional flow of 641.3 MW (1200 - 558.3). To increase the flow on this branch by 641.3 the generation shift would have to be 1,051.6 MW because the distribution

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factor for this branch is 0.60981, i.e. only about 60% of the generation shift will flow on this branch. Given a generation shift of 1,051 MW, on top of the pre-shift inter-area flow of 278.7 MW, the total transfer limit is 1,330 MW.

Similar calculations can be done to check the other, less restrictive limits shown in the report listing. Note that the limit imposed by the branch from Bus 205 to Bus 154 is marginally more complicated to check because the generation shift actually decreases flow on this branch. It is necessary to consider the flow direction on the branches (to and from) and the sign of the distribution factors.

9.8.7. Transfer Limit Analysis Flows For Contingency Cases

When reporting flows for contingency cases, the events comprising the contingency are listed followed by flows on the monitored elements sorted as described above. The flow table includes both the contingency case flows in the same form as the base case report described above, as well as the base case flows at the transfer level required for the most restrictive monitored element. Note that if the user so selects (see Section 9.8.3, "Transmission Transfer Limit Analysis Options"), only the contingency case constraints are considered in calculating the transfer limit; in this case, base case flows at the transfer limit may be shown as overloaded because the corresponding constraints are ignored in the transfer limit calculation.

An curtailed example of the report listing for the contingency cases is shown in Figure 9.15, "Partial Listing of Report for Two Area Transfer Limits for Contingency Cases".

A summary report tabulates the number of system condition solutions attempted and the number for which there exists no interchange schedule at which the rating constraint can be satisfied for all monitored elements.

This is followed by a tabulation of monitored elements for all system conditions that were calculated, in order of increasing incremental or total transfer capability, as appropriate. The limiting element is listed along with its power transfer distribution factor, its pre-shift contingency case flow, its rating, and a description of the system condition.

Only those monitored elements satisfying the cutoff threshold solution parameters set by the user are included in the summary. After an element has been listed the maximum number of times < n >, its nth occurrence includes an asterisk (*) before its description, and further reporting of the element is suppressed.

This summary report is reprinted for each selected interface with interface transfer limits and distribution factors rather than study system interchange limits and distribution factors tabulated. All distribution factors on the interface transfer limits summary report describe changes in monitored element flow relative to a change in interface flow. The branches that form the interface are listed on the first page of an interface's transfer limit summary report.

The final page of the summary report lists those elements that were reported < n > times, sorted by the number of times they would have been reported.

If a line outage contingency forms an island with no Type 3 (swing) bus, a singular sub-matrix is formed. Any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, has dashes printed where its contingency case results would normally be printed.

How is the negative transfer limit obtained?

In Figure 9.15, "Partial Listing of Report for Two Area Transfer Limits for Contingency Cases" it can be seen that the most severe contingency identifies a negative transfer limit of -56.5 MW. The contingency is loss of

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the two units at Buses 101 and 102. In the pre-shift condition, in which there is a transfer of 278 MW, the branch from Bus 3008 to Bus 154 has a loading of 465 MW but has a thermal capacity of only 440 MW. This loading has to be reduced by 25.4 MW to keep the branch within its capacity.

The distribution factor is 0.07566. Consequently the transfer has to be reduced by 335 MW (that is 25.4 MW divided by 0.07566) from its current level of 278.7 MW. This, rounding off, is 57 MW less than the base case (pre-shift) transfer.

9.8.8. Application Notes

Activity TLTG uses the same linearized network model as is used in activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Note, in particular, the approximate nature of the linearized network model used by activity TLTG. The proper role of activity TLTG, therefore, should be to focus attention on those system conditions that deserve more detailed study.

Activity TLTG detects the specification of duplicate single and double line outage contingencies and calculates the contingency case solution only for the first specification of such a contingency. Any contingency cases involving bus boundary contingency events, line closure contingency events, or more than two line outage contingency events are not checked for duplication.

The values shown as monitored element flows are set as described in Section 9.5.4, "Contingency Case Flow Calculations". The user selects the base flow value to be used by setting the line flow code solution parameter at the start of activity TLTG (see above).

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```
CONTINGENCY TRIP2NUCLEAR:
     REMOVE UNIT 1 FROM BUS 101 [NUC-A       21.600]
     REMOVE UNIT 1 FROM BUS 102 [NUC-B       21.600]

INSOLUBLE CASE: OVERLOAD IN PRE-SHIFT SOLUTION INSENSITIVE TO INTERCHANGE LEVEL

LOADINGS AT OR ABOVE 100.0 %                                                
```

2. Only post-contingency constraints need be satisfied.

The user selects the method to be used by setting *include/ignore base case constraints in contingency case* at the start of activity TLTG. The setting of this parameter has no effect on the summary report of activity TLTG; for contingency cases, the transfer capability shown is always the one at which the monitored element is at its limit in the contingency case. Thus, the setting of this parameter is meaningful only if the full output option is specified.

When the user has specified *apply summary minimum distribution factor to report* and *minimum distribution factor magnitude* is greater than zero, monitored elements with distribution factor magnitudes below the threshold are ignored in calculating the corresponding monitored element flow table as well as being omitted from the summary report.

If the convert ratings code is set to one at the start of activity TLTG, any percentage of rating parameter is applied after the conversion to estimated MW ratings. Interface ratings are not affected by the convert ratings solution parameter.

Line loss estimates and line shunts of outaged lines are handled as in activity DCLF (refer to Section 9.4.4, "Application Notes").

It is required that the working case be solved to an acceptable mismatch tolerance prior to entering activity TLTG.

It is entirely possible that, for a given system condition, there is no interchange schedule at which the linearized network model results in all monitored elements satisfying their rating constraints. This could occur, for example, if the pre-shift solution had two overloaded branches, and one required an increase in interchange to relieve its overload while the other required a decrease in interchange.

Minimum pre-shift flow change and *minimum distribution factor change* values may be used to exclude from consideration those monitored elements in a contingency case that are not greatly affected by the contingency. This could result in a situation in which none of the remaining monitored elements have non-zero ratings. Such insoluble conditions are described in the output block for that condition when full output is selected; when the summary report is chosen, such cases are identified.

When specifying a rating percentage of other than 100%, note that interface ratings are also scaled by the specified percentage in checking their flows to determine maximum interchange.

Additional Information
PSS®E GUI Users Guide, Running DC Corrective Action Analysis *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Transmission Interchange Limits

9.8.9. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the "Report Tables" section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

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In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "TLTG Limit".

9.9. Calculating Sequential Participation Interchange Limits

SPIL

Run Activity SPIL - GUI
Power Flow > Linear Network > Sequential participation interchange limit (SPIL)... *[Sequential Participation Interchange Limit]*
Run Line Mode Activity SPIL - CLI
ACTIVITY? >>**SPIL**
Interrupt Control Codes
AB

The sequential participation transmission interchange limit analysis activity SPIL estimates the import or export limits of a specified subsystem of the working case using a linearized network model. The user designates this study system in which the total power injection is to be increased (for export limits) or decreased (for import limits). An opposite change in the total power injection is made in a designated opposing system.

It is clear that this analysis is similar to that described in Section 9.8.2, "Calculating Transmission Transfer Limits", the difference here being that the shifts in export or import are defined on the basis of specific participating elements in the study and opposing systems.

Activity SPIL accesses a Distribution Factor Data File to pick up subsystem, monitored element and contingency definitions (refer to activity DFAX). It also accesses a Subsystem Participation Data File in which alternative participation data may be provided for the study and/or opposing systems (refer to Section 9.9.1, "Subsystem Participation Data File Contents").

Power transfer distribution factors relating changes in branch and interface flows to a change in study system interchange are determined. The maximum study system export or import is then derived by interpolation and extrapolation subject to the constraint that no monitored elements exceed a specified percentage of a selected rating.

This process may then be repeated for a designated set of user specified contingency cases.

The process will perform all the contingencies identified in the contingency description file processed when building the distribution factor file. Furthermore, the process uses the monitored list file, which allows the lines to be monitored and interfaced. This feature is important because these interfaces often define stability limits.

Note that any two subsystems identified in the *.sub file can be used for the transfer analysis. Furthermore, however, it should be noted that the limits will be found based on only those elements (lines and interfaces) that are monitored, i.e. included in the *.mon file.

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9.9.1. Subsystem Participation Data File Contents

The Subsystem Participation Data File is used to define participation blocks for one or more of the subsystems that are defined in the Distribution Factor Data File constructed by activity DFAX and specified to activity SPIL. The subsystems were in turn defined in the Subsystem Description Data File (refer to ???) that was read by activity DFAX in the process of constructing the Distribution Factor Data File. The participation data defined in the Subsystem Participation Data File provides for:

- Up to five participation blocks per subsystem.
- For each participation block, the fraction of the maximum permissible power shift (specified in activity SPIL; refer to Section 9.9.3, "Operation of Activity SPIL") to be assigned to the block.
- For each participation block, the list of buses that are to be included in the block and their block participation factors.

The Subsystem Participation Data File consists of one or more sets of subsystem data with a line containing a zero or blanks following the last set of subsystem data. The file format will appear as shown here:

```
set of data for a subsystem

set of data for another subsystem

.
.
.

set of data for another subsystem

0

◀ End of participation data
```

Each set of data for a subsystem contains the subsystem label enclosed in single quotes (refer to ???), followed by data for up to five participation blocks, followed by line containing a zero:

```
'label'

data for participation block 1

.
.
.

data for participation block n

0

◀ End of participation data for subsystem 'label'
```

The format of records for each participation block is as follows:

```
BLOCKF
```


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```
IBUS PF
.
.
.
IBUS PF
0
```

◄ **End of participation block**

where:

BLOCKF	Is a positive value defining the block participation factor. BLOCKF is typically expressed in percent or per unit of the maximum permissible power shift that is to be assigned to the participation block.
IBUS	Is the bus number of a bus assigned to subsystem label. A subsystem bus may be a member of no more than one participation block.
PF	Is a non-zero value defining the bus participation factor. PF is typically expressed in percent or per unit of the power shift assigned to this participation block. While an individual PF value may be negative, the sum of PF factors within each participation block must be positive.

Following is an example of a Subsystem Participation Data File for two subsystems, with each subsystem having two participation blocks.

```
'SUBSYS A'
```

◄ **Start of participation data for subsystem 'SUBSYS A'**

```
.2
```

◄ **Start of participation block 1 for subsystem 'SUBSYS A'**

```
101 1
0
```

◄ **End of participation block 1 for subsystem 'SUBSYS A'**

```
.8
```

◄ **Start of participation block 2 for subsystem 'SUBSYS A'**

```
102 1
0
```

◄ **End of participation block 2 for subsystem 'SUBSYS A'**

```
0
```


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```
 End of participation data for subsystem 'SUBSYS A'
'SUBSYS B'
 Start of participation data for subsystem 'SUBSYS B'
  4
 Start of participation block 1 for subsystem 'SUBSYS B'
  201 .725
  206 1.0
  0
 End of participation block 1 for subsystem 'SUBSYS B'
  6
 Start of participation block 2 for subsystem 'SUBSYS B'
  211 .3
  264 .6
  0
 End of participation block 2 for subsystem 'SUBSYS B'
 0
 End of participation data for subsystem 'SUBSYS B'
0
 End of participation data
```

9.9.2. Interchange Limits Options

Table 9.6. Sequential Participation Interchange Limits Options

Specification	Default	Description
MW mismatch tolerance	0.5	Largest initial active power mismatch. Note that the user can modify the tolerance in steps of 0.1.
Abort if MW mismatch exceeds tolerance	enabled	Enable or disable termination of activity SPIL if mismatch tolerance exceeded.
Base case rating	activity OPTN	The base case and contingency case rating sets, and the percentage thereof, define the line load-
Contingency case rating		
Percent of rating	100	

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Specification	Default	Description
		ing limits used in determining overloads. The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).
Convert ratings to estimated MW ratings	disabled	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's reactive power loading at the metered end in the base case ac solution.
Line flow code	dc base case	The line flow code defines the base flow to be used in deriving branch flow estimates. • dc base case • ac base case
Phase shifter in base case	Lock	If set to *Regulate* (in base case), any regulating phase shifter where adjustment control mode CODn (refer to Transformer Data) is set to +3 holds its base case flow in all base case shift solutions. If set to *Lock*, all phase shift angles are locked at their base case settings.
Add study system ties to monitored lines	disabled	If enabled, automatically include study system tie lines as monitored elements.
Minimum contingency case pre-shift flow change	1.0	Provide the means to ignore those monitored elements that, in a contingency case, are not significantly affected by the contingency. If the magnitude change in pre-shift flow from the base case value is less than the *minimum flow change* threshold *and* the magnitude change in the power transfer distribution factor from the base case value is less than the *distribution factor change* threshold, the monitored element is ignored.
Minimum contingency case distribution factor change	0.0000	

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Specification	Default	Description
Study system generation shift	100	If the value entered is positive, export limits are to be determined; otherwise, import limits are to be calculated. The generation shift is apportioned among generator and load buses in the study system in proportion to the participation factors specified in the participation block structure of the Subsystem Description Data File (refer to ???). If no participation block was entered for the study system, the generation shift is shared among generator buses in proportion to their plant MBASEs (i.e., the sum of the MBASEs of in-service machines with positive active power output at the bus). In this case, there must be at least one in-service machine with a positive MBASE in the subsystem. The opposite generation shift is applied to buses in the first opposing system, again according to their participation factors or MBASEs. Using the incremental form of the dc power flow equation (refer to activity DCLF) and the power shift vector, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.
Subsystem selection		Study System: network subsystem in which the total power injection is to be increased (for export limits) or decreased (for import limits). The choices available are those listed in the *.sub type file.

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Specification	Default	Description
		Opposing System: network subsystem in which an opposite change from that in the *Study System* is made.
Maximum import / export	99999	Threshold value, number of monitored elements in the summary report.
Minimum distribution factor magnitude	0.0	Threshold value for summary report. For small distribution factors with levels below this threshold there will be no reporting for the related element.
Maximum time report same element	5	Threshold value for summary report limiting the number of times an element is reported in the same report.
Summary table contingency descriptions	Events	Defines the manner in which contingency cases are identified in the summary report as one of the following: • 12-character contingency case label • events comprising the contingency case • both contingency label and the contingency events
List study system buses	disabled	If enabled, list *Study System* buses in output report.
List opposing system buses	disabled	If enabled, list *Opposing System* buses in output report.
List study system tie lines	disabled	If enabled, list *Study System* tie lines in output report.
Output code	Summary	Summary: summary report of all cases considered Full: monitored element flow table for each case considered followed by a summary report of all cases considered (refer to Section 9.9.5, "Output Report Format")
Interchange limit output	Incremental	Incremental: incremental transfer capability Total: total transfer capability

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Specification	Default	Description
Elements in flow tables	60000	Limit the flow table for each case reported to the < n > most restrictive monitored elements.
Apply summary minimum distribution factor to report	disabled	If enabled, apply the minimum distribution factor magnitude threshold for including monitored elements in the summary report to the monitored element flow table of each case reported when full output is selected.
Optional Interface Output	NONE	Summary report is repeated with interface transfer limits and distribution factors for a selected interface listed rather than study system transfer limits and distribution factors. The interface distribution factors listed on the interface summary report are measures of the changes in monitored element flows to a change in base case interface flow.

9.9.3. Operation of Activity SPIL

Activity SPIL checks that generators are not converted (refer to CONG) and that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed and activity SPIL is terminated.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed.

The user must enter the name of a Distribution Factor Data File constructed by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, “Application Notes”) and to the desired Linear Network Analysis Data Files (refer to ???). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, “File Usage ” for the file specification conventions used by activity SPIL.

The user must specify the study and opposing systems. If no subsystems are defined, an appropriate error message is printed and activity SPIL is terminated. If only one subsystem was specified, the specified subsystem is taken as the study system while the remainder of the working case is treated as the opposing system. If two subsystems were specified, the default opposing system is bypassed.

Activity SPIL checks to ensure that there are no buses that are members of both of the selected subsystems. If any such buses are found, they generate an alarm and activity SPIL is terminated.

The user may enter the name of a Subsystem Participation Data File that contains subsystem sequential participation data as described in Section 9.9.1, “Subsystem Participation Data File Contents” . Refer to Section 2.4.1, “File Usage ” for the file specification conventions used by activity SPIL.

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Data specified in this file must be consistent with the subsystem definitions contained in the Distribution Factor Data File accessed above. For each of the study and opposing systems, if the designated file contains valid sequential participation data, its data is used rather than the participation data contained in the Distribution Factor Data File; otherwise, the participation data contained in the Distribution Factor Data File is used.

If no Subsystem Participation Data File is specified, activity SPIL uses the participation data contained in the Distribution Factor Data File and produces results identical to those of activity TLTG.

The largest active power mismatch corresponding to the present ac solution voltage vector in the working case is calculated and printed.

If no subsystems are defined an appropriate error message is printed and the interchange calculation process is terminated. If only one subsystem is defined, a second subsystem (WORLD) is assumed containing those buses in the power flow case that are not assigned to the specified subsystem.

If the convert ratings option was enabled at the start of activity SPIL, ratings from the selected rating set(s) of those monitored branches having non-zero rating(s) are modified. For each such branch, the reactive power loading is calculated and MW rating(s) are determined under the assumption that the reactive power loading is unchanged under power transfer and contingency conditions.

Using the incremental form of the dc power flow equation (see Section 9.4.1, "DC Linearized Network Solutions") and the power shift vector, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.

The base case dc network solution is then calculated and the solution summarized as in the following example:

```
STUDY SYSTEM GENERATION IS        1500.0 MW
OPPOSING SYSTEM GENERATION IS     1748.9 MW
STUDY SYSTEM NET INTERCHANGE IS    282.8 MW
```

The generation shift is apportioned among generator and load buses in the study system in proportion to the participation factors specified in the participation block structure of the Subsystem Description Data File (refer to ???) or in a series of sequential shifts as defined in the Subsystem Participation Data File (refer to Section 9.9.1, "Subsystem Participation Data File Contents"). If no participation block was entered for the study system in either file, the generation shift is shared among generator buses in proportion to their plant MBASEs (i.e., the sum of the MBASEs of in-service machines with positive active power output at the bus). In this case, there must be at least one in-service machine with a positive MBASE in the subsystem. Similar opposite generation shifts are applied to buses in the opposing system, again according to their participation factors or MBASEs. Using the incremental form of the dc power flow equation (refer to Section 9.4.4, "Application Notes" and Section 9.5.4, "Contingency Case Flow Calculations") and the generation shift vectors, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into vectors of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.

The generation shift is apportioned among generator and load buses in the study system in a series of sequential shifts as defined in the Subsystem Participation Data File described above.

The base case solution is then reported, followed by processing of the designated contingency cases with contingency case flow estimates calculated using the methodology defined in Contingency Case Flow Cal-

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culations. Prior to terminating, an ordered summary report from among all the cases calculated is tabulated (refer to Section 9.9.5, "Output Report Format").

The user may specify having the summary report repeated with interface transfer limits and distribution factors for a selected interface listed rather than study system transfer limits and distribution factors. The interface distribution factors listed on the interface summary report are measures of the changes in monitored element flows to a change in base case interface flow.

In listing study system buses, opposing system buses, and study system tie lines, the tabulation may be terminated by entering the AB interrupt control code. In this case, the interrupt is cleared and processing continues. After the interchange limit calculations have started, activity SPIL may be terminated by entering the AB interrupt control code.

9.9.4. Analyzing Interchange Limit Results

The significant difference between this Sequential Participation Interchange Limit calculation and the Two-area Transmission Interchange Limit calculation, described in activity TLTG, is that the generation shift here is established by sequential participation of specific buses each with its own participation factor.

When multiple participation blocks are specified in the Subsystem Participation Data File for the study and/or opposing systems, the study system generation shift and the block participation factors specified in the Subsystem Participation Data File are used to allocate portions of the specified shift sequentially among the participation blocks. Within each participation block, its allocation is assigned to the block's participating buses in proportion to their bus participation factors. Thus, the sensitivity of each monitored element's flow to an incremental change in study system interchange is defined by a set of power transfer distribution factors that vary according to the current level of study system interchange.

9.9.5. Output Report Format

In reporting the base case solution, activity SPIL lists the study system generation, the opposing system generation, and the study system net interchange corresponding to the network solution before and after applying the generation shift. This is followed by a tabulation of the pre-shift and post-shift generation at those study system and opposing system buses participating in the generation shift. Loadings on monitored elements are then reported, sorted such that the most restrictive elements are listed first. Either incremental or total pre-contingency transfer limits, as appropriate, are listed.

For monitored three-winding transformer windings, flow estimates are calculated at the winding bus end as power flowing into the transformer. For other monitored branches, flow estimates are calculated at the metered end in the from bus to to bus direction. The flow across an interface is taken as the sum of the flows of its members. For three-winding transformer windings, the flow is calculated at the winding bus as power flowing into the transformer. For other members, the flow is calculated at its metered end in the from bus (i.e., the first bus specified in entering the branch; refer to Monitored Element Data File Contents) to to bus direction.

Flow estimates are listed for each of three interchange schedules: with the original generation profile, with the specified generation shift, and at the generation shift corresponding to the transfer level limit required for the most restrictive monitored element (i.e., the one listed first). Any flow at or above the selected percentage of the appropriate rating is followed by an asterisk (*). The power transfer distribution factor relating the change in flow on the monitored element to a change in study system net interchange is also tabulated. When multiple participation blocks were specified for either the study or opposing system, the power transfer distribution factors printed are those that apply at the transfer level required for the most restrictive monitored element.

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For each monitored element reported, its rating is listed; for branches, this *is not* the percentage of the rating used in determining the interchange limit, but either the value of rating as entered in activities such as READ, TREA, RDCH, CHNG or the *[Spreadsheet]*, or, if the convert ratings option was selected at the start of activity SPIL, the estimated MW ratings.

When reporting flows for contingency cases, the events comprising the contingency are listed followed by flows on the monitored elements sorted as described above. The flow table includes both the contingency case flows in the same form as the base case report described above, as well as the base case flows at the transfer level required for the most restrictive monitored element. Only the contingency case constraints are considered in calculating the transfer limit; base case flows at the transfer limit may be shown as overloaded because the corresponding constraints are ignored in the transfer limit calculation.

A summary report tabulates the number of system condition solutions attempted and the number for which there exists no interchange schedule at which the rating constraint can be satisfied for all monitored elements (refer to Section 9.9.8, "Application Notes"). This is followed by a tabulation of monitored elements for all system conditions that were calculated, in order of increasing incremental or total transfer capability, as appropriate. The limiting element is listed along with its power transfer distribution factor, its pre-shift contingency case flow, its rating, and a description of the system condition.

Only those monitored elements satisfying the cutoff threshold solution parameters set at the start of activity SPIL (refer to Section 9.9.3, "Operation of Activity SPIL") are included in the summary. After an element has been listed the maximum number of times < n >, its nth occurrence includes an asterisk (*) before its description, and further reporting of the element is suppressed.

This summary report is reprinted for each selected interface with interface transfer limits and distribution factors rather than study system interchange limits and distribution factors tabulated. All distribution factors on the interface transfer limits summary report describe changes in monitored element flow relative to a change in interface flow. The branches that form the interface are listed on the first page of an interface's transfer limit summary report.

The final page of the summary report lists those elements that were reported < n > times, sorted by the number of times they would have been reported.

If a line outage contingency forms one or more islands with no Type 3 (swing) bus, such islands are disconnected. Any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, has dashes printed where its contingency case results would normally be printed.

9.9.6. Example Activity SPIL Output

Figure 9.17, "Sequential Participation Interchange Limit Output for Base Case" shows the report obtained using the example files (*savnw.**) from which was built the *dc-contingency.dfx* distribution factor file. The example data files used (**.sub*, **.mon*, **.con*) can be seen listed at the top of the report listing. They are available in the PSS®E EXAMPLE directory.

As an example, a Subsystem Participation Data file was prepared for the two study areas STUDY and EAST identified in the *savnw.sub* file. The file is shown in Figure 9.16, "Participation Factors" where it can be seen that generators at Buses 101 and 102, in the STUDY subsystem are identified as participating at 20% and 80% respectively. In the EAST subsystem, the opposing system, there are two participation blocks, one of which has a participation of 30% and the other with participation of 70%. Each block has a participating generator and load bus.

For the purposes of running and example case, a total shift of 100 MW is assumed. In the Report listing, Figure 9.17, "Sequential Participation Interchange Limit Output for Base Case", it can be seen that this shifts

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in the study and opposing systems are apportioned as defined by the participation block ratios as identified in Figure 9.16, “Participation Factors”.

```
'STUDY'
0.2
101 1
0
0.8
102 1
0
0
'EAST'
0.3
205 0.7
206 1
0
0.7
211 0.3
203 0.6
0
0
0
```

Figure 9.16. Participation Factors

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Figure 8-11 Sequential Participation Interchange Limit Output for Base Case

```
     *** SPIL EXPORT LIMIT OUTPUT FOR BASE CASE                    ***

DISTRIBUTION FACTOR FILE:      C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:    C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:        C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
CONTINGENCY DESCRIPTION FILE:  C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con

                                 PRE-SHIFT    DELTA  POST-SHIFT
STUDY SYSTEM MW GENERATION:        1500.0     100.0    1600.0
OPPOSING SYSTEM MW GENERATION:     1400.0    -100.0    1300.0
STUDY SYSTEM NET INTERCHANGE:       278.7     100.0     378.7

<----------------- STUDY SYSTEM ------------------> <---------------- OPPOSING SYSTEM ---------------->
                         <---- GENERATOR MW ---->                              <---- GENERATOR MW ---->
  BUS# X-- NAME --X BASKV     BASE    SHIFT  CHANGE    BUS# X-- NAME --X BASKV     BASE    SHIFT  CHANGE
   101 NUC-A       21.600   750.0    770.0    20.0     203 EAST230     230.00      0.0    -46.7   -46.7
   102 NUC-B       21.600   750.0    830.0    80.0     205 SUB230      230.00      0.0    -12.4   -12.4
                                                      206 URBGEN      18.000    800.0    782.4   -17.6
                                                      211 HYDRO_G     20.000    600.0    576.7   -23.3

LOADINGS AT OR ABOVE 100.0 %                                          <------------- BASE CASE ------------->
OF RATING ARE MARKED WITH '*'                                 TOTAL           PRE-    POST-    LIMIT
                                                              TRANS RATING    SHIFT    SHIFT     CASE     DISTR.
<------- F R O M -------> <--------- T O ---------> CKT      CAPAB     A       MW       MW       MW      FACTOR
      INTERFACE EAST                                          497.9    350  -130.8   -230.8  -350.0* -1.00000
   201 HYDRO       500.00    151 NUCPANT     500.00  1     1387.8   1200  -558.7   -614.4  -683.6  -0.58028
   205 EAST230     230.00    154 DOWNTN      230.00  1     3241.7    200   122.4    117.0   103.8  -0.11072
   202 EAST500     500.00    152 MID500      500.00  1     5146.4   1200   -50.7    -70.1   -98.4  -0.23699
  3008 CATDOG      230.00    154 DOWNTN      230.00  1     6713.6    400    62.9     69.9    76.2   0.05210
   205 SUB230      230.00    154 DOWNTN      230.00  1    13389.1    600   356.1    336.7   328.2  -0.07200
  3004 WEST        500.00    152 MID500      500.00  1    >99999.   NONE  -132.2   -136.8  -140.9  -0.03407
  3006 UPTOWN      230.00    153 MID230      230.00  1    >99999.   NONE   -78.6    -81.0   -83.2  -0.01803
      INTERFACE WEST                                       >99999.    200  -147.9   -147.9  -147.9   0.00000
```

Figure 8-11. Sequential Participation Interchange Limit Output for Base Case

It can be seen that the transfer capabilities and participation factors are different from those shown in Figure 9.14, "Base Case Results For Two-Area Transfer Limit Calculation" where sequential participation factors were not used. There, the power shift is apportioned relative to the MBASE value for the machines in the two participating systems.

9.9.7. Interchange Limits Contingency Cases

When reporting flows for contingency cases, the events comprising the contingency are listed followed by flows on the monitored elements sorted as described above. The flow table includes both the contingency case flows in the same form as the base case report described above, as well as the base case flows at the transfer level required for the most restrictive monitored element. Only the contingency case constraints are considered in calculating the transfer limit; base case flows at the transfer limit may be shown as overloaded because the corresponding constraints are ignored in the transfer limit calculation.

A summary report tabulates the number of system condition solutions attempted and the number for which there exists no interchange schedule at which the rating constraint can be satisfied for all monitored elements. This is followed by a tabulation of monitored elements for all system conditions that were calculated, in order of increasing incremental or total transfer capability, as appropriate. The limiting element is listed along with its power transfer distribution factor, its pre-shift contingency case flow, its rating, and a description of the system condition.

Only those monitored elements satisfying the cutoff threshold solution parameters set by the user when launching the calculation process are included in the summary. After an element has been listed the maximum number of times < n >, its nth occurrence includes an asterisk (*) before its description, and further reporting of the element is suppressed.

This summary report is reprinted for each selected interface with interface transfer limits and distribution factors rather than study system interchange limits and distribution factors tabulated. All distribution factors on the interface transfer limits summary report describe changes in monitored element flow relative to a change in interface flow. The branches that form the interface are listed on the first page of an interface's transfer limit summary report.

The final page of the summary report lists those elements that were reported < n > times, sorted by the number of times they would have been reported.

If a line outage contingency forms an island with no Type 3 (swing) bus, a singular sub-matrix is formed. Any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, has dashes printed where its contingency case results would normally be printed.

Reference can be made to Figure 9.15, "Partial Listing of Report for Two Area Transfer Limits for Contingency Cases", which shows the report listing for the two-area Transmission Interchange Limit calculation. The format for this two area calculation, using sequential participation factors, is the same. The results, however, will show different interchange limits and distribution factors.

9.9.8. Application Notes

Activity SPIL uses the same linearized network model as is used in activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Note, in particular, the approximate nature of the linearized network model used by activity SPIL. The proper role of activity SPIL, therefore, should be to focus attention on those system conditions that deserve more detailed study.

Activity SPIL detects the specification of duplicate single and double line outage contingencies and calculates the contingency case solution only for the first specification of such a contingency. Any contingency cases

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involving bus boundary contingency events, line closure contingency events, or more than two line outage contingency events are not checked for duplication.

The values shown as monitored element flows are set as described in Section 9.5.4, "Contingency Case Flow Calculations". The user selects the base flow value to be used by setting the line flow code solution parameter at the start of activity SPIL (see above).

When multiple participation blocks are specified in the Subsystem Participation Data File for the study and/or opposing systems (refer to Section 9.9.1, "Subsystem Participation Data File Contents"), the study system generation shift specified in the dialog of activity SPIL and the block participation factors specified in the Subsystem Participation Data File are used to allocate portions of the specified shift sequentially among the participation blocks. Within each participation block, its allocation is assigned to the block's participating buses in proportion to their bus participation factors. Thus, the sensitivity of each monitored element's flow to an incremental change in study system interchange is defined by a set of power transfer distribution factors that vary according to the current level of study system interchange.

Because this transfer limit calculation process uses the linearized network model, note should be taken of the approximate nature of the solution. The proper role of this analysis, therefore, should be to focus attention on those system conditions that deserve more detailed study.

The process detects the specification of duplicate single and double line outage contingencies and calculates the contingency case solution only for the first specification of such a contingency. Any contingency cases involving bus boundary contingency events, line closure contingency events, or more than two line outage contingency events are not checked for duplication.

In the contingency case monitored flow tables, transfer limits are calculated such that only post-contingency constraints need be satisfied.

When *apply summary minimum distribution factor to report* is enabled and *minimum distribution factor magnitude* is greater than zero, monitored elements for which all their distribution factor magnitudes are below the threshold are ignored in calculating the corresponding monitored element flow table as well as being omitted from the summary report.

If *Convert ratings to estimated MW Ratings* is enabled at the start of activity SPIL, any percentage of rating parameter is applied after the conversion to estimated MW ratings. Interface ratings are not affected by the convert ratings solution parameter.

Line loss estimates and line shunts of outaged lines are handled as in activity DCLF (refer to Section 9.4.4, "Application Notes").

It is required that the working case be solved to an acceptable mismatch tolerance prior to entering activity SPIL.

It is entirely possible that, for a given system condition, there is no interchange schedule at which the linearized network model results in all monitored elements satisfying their rating constraints. This could occur, for example, if the pre-shift solution had two overloaded branches, and one required an increase in interchange to relieve its overload while the other required a decrease in interchange.

Options for *Minimum contingency case pre-shift flow change* and *minimum contingency case distribution factor change* may be used to exclude from consideration in a contingency case those monitored elements that are not greatly affected by the contingency. This could result in a situation in which none of the remaining monitored elements have non-zero ratings. Such insoluble conditions are described in the output block for that condition when full output is selected; when the summary report is chosen, such cases are identified.

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Such insoluble conditions are described in the output block for that condition when full output is selected; when the summary report is chosen, such cases are identified.

When specifying a rating percentage of other than 100%, note that interface ratings are also scaled by the specified percentage in checking their flows to determine maximum interchange.

Additional Information
PSS®E GUI Users Guide, Calculating Transmission Interchange Limits *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Sequential Participation Interchange Limits

9.10. Performing Interchange Limit Analysis with Two Opposing Systems

POLY

Run Activity POLY - GUI
Power Flow > Linear Network > Interchange limits with two opposing systems (POLY)... *[Interchange Limits with Two Opposing Systems]*
Interrupt Control Codes
AB

The transmission interchange limit analysis activity POLY estimates the import or export limits of a specified subsystem of the working case using a linearized network model. The user designates this study system in which the total power injection is to be increased (for export limits) or decreased (for import limits). Activity POLY differs from activity SPIL in that it considers simultaneous power injection shifts in two opposing systems in maximizing study system import or export.

Two sets of base case power transfer distribution factors are determined: one set relating changes in branch and interface flows to a change in study system interchange against the first opposing system, and another relating changes in flows to a change in study system interchange against the second opposing system. The maximum study system export or import is then derived by a linear program subject to the constraint that no monitored elements exceed a specified percentage of a selected rating.

This process may then be repeated for a designated set of user specified contingency cases.

Results are calculated and then presented in tabular and, optionally, graphical forms. The graphical output of a previous execution of activity POLY is available via the POLY Results Output File produced during the execution of activity POLY.

9.10.1. Interchange Limit Analysis Options

Table 9.7. Interchange Limits Analysis Options for Two Opposing Systems

Specification	Default	Description
MW mismatch tolerance	0.5	Largest initial active power mismatch. Note that the user can modify the tolerance in steps of 0.1.

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Specification	Default	Description
Abort if MW mismatch exceeds tolerance	enabled	Enable or disable termination of activity SPIL if mismatch tolerance exceeded.
Base case rating	activity OPTN	The base case and contingency case rating sets, and the percentage thereof, define the line loading limits used in determining overloads. The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).
Contingency case rating		
Percent of rating	100	
Convert ratings to estimated MW ratings	disabled	If enabled, ratings of monitored branches are converted to estimated MW ratings based upon each monitored line's reactive power loading at the metered end in the base case ac solution.
Line flow code	dc base case	The line flow code defines the base flow to be used in deriving contingency case flow estimates. • dc base case • ac base case
Phase shifter in base case	Lock	If set to *Regulate* (in base case), any regulating phase shifter where adjustment control mode CODn (refer to Transformer Data) is set to +3 holds its base case flow in all base case shift solutions. If set to *Lock*, all phase shift angles are locked at their base case settings.
Base case constraints in contingency case	Ignore	If set to *Include*, both base case and contingency case flow constraints are recognized in calculating the pre-contingency interchange limits for each contingency case. If set to *Ignore*, only the contingency case loadings are considered (refer to Section 9.8.8, "Application Notes").

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Specification	Default	Description
Add study system ties to monitored lines	Disabled	If enabled, automatically include *Study System* tie lines as monitored elements.
Maximum allowed opposing system generation shift	10000	Places a bound on the permissible generation shift for each of the opposing systems.
Minimum distribution factor magnitude	0.0	Defines the minimum distribution factor magnitude threshold for including monitored elements in the transfer limit calculation and report. Any monitored element for which both of its power transfer distribution factor magnitudes are below this threshold is ignored during the processing and reporting of the corresponding system condition.
Study system generation shift	100	If the value entered is positive, export limits are to be determined; otherwise, import limits are to be calculated. The generation shift is apportioned among generator and load buses in the study system in proportion to the participation factors specified in the participation block structure of the Subsystem Description Data File (refer to ???). If no participation block was entered for the study system, the generation shift is shared among generator buses in proportion to their plant MBASEs (i.e., the sum of the MBASEs of in-service machines with positive active power output at the bus). In this case, there must be at least one in-service machine with a positive MBASE in the subsystem. The opposite generation shift is applied to buses in the first opposing system, again according to their participation factors or MBASEs.
Study system generation shift (continued)		Using the incremental form of the dc power flow equation (refer to activity DCLF) and the power shift vector, phase angle changes are calculated, followed by incre-

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Specification	Default	Description
		mental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.
Subsystem selection		Study System: network subsystem in which the total power injection is to be increased (for export limits) or decreased (for import limits). The choices available are those listed in the *.sub type file. Opposing System 1: network subsystem in which an opposite change from that in the *Study System* is made. Opposing System 2: second network subsystem in which an opposite change from that in the *Study System* is made.
List study system buses	disabled	If enabled, list *Study System* buses in output report.
List opposing system buses	disabled	If enabled, list *Opposing System* buses in output report.
List study system tie lines	disabled	If enabled, list *Study System* tie lines in output report.
Output code	Summary	Summary: summary report of all cases considered Full: monitored element flow table for each case considered followed by a summary report of all cases considered (refer to Section 9.9.5, "Output Report Format")
Maximum opposing system Δ interchange for plotting	1000	Defines the plotting scales if graphical output is specified. Change in interchange in the first opposing system is plotted along the x-axis, and change in interchange in the second opposing system is plotted along the y-axis.

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Specification	Default	Description
Plot code	All system conditions	Defines graphical reports for all system conditions: • POM-Data-ReadingPowerFlow-DataAdditionsFromTheTerminal • Only combined cases • All system conditions A graph is produced for either each optimal solution calculated or for only the *combined cases* solution.
Constraint plot code	Only feasible	Defines constraint line plotting: • All constraints • Only feasible region boundary If *combined cases* specified, all monitored element constraint lines that fall within the plotting limits specified by Δ interchange are plotted. Otherwise, only those constraint lines that form the border of the feasible region are plotted; refer to Section 9.10.3, "Output Report Format".
Graphic output	Standard output	Output display: • Standard output: graphic report reduced to fit on page • Two pages per case: page 1 = plotted results; page 2 = listing of monitored elements corresponding to those constraint lines that form the border of the feasible region; refer to Section 9.10.3, "Output Report Format".
Place	Dialog window	Output location: • Dialog window • Printer
Output file (results)		POLY Results file (*.pol): output file to which the information will be written for current or later plot-

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Specification	Default	Description
		ting of the graphical display of results.

9.10.2. Operation of Activity POLY

Activity POLY checks that generators are not converted (refer to CONG) and that each non-Type 4 bus is connected back to a Type 3 (swing) bus through the in-service ac network. If any violations are detected, an appropriate message is printed and activity POLY is terminated.

If a new ordering of network buses is required, an appropriate message is printed and activity ORDR is automatically executed. Activity POLY calculates and prints the largest active power mismatch corresponding to the present ac solution voltage vector in the working case. The user may specify that activity TLTG be terminated (default) if this largest mismatch is greater than the MW mismatch tolerance.

The user must enter the name of a Distribution Factor Data File constructed by activity DFAX that corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired Linear Network Analysis Data Files (refer to ???). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity POLY.

The user must specify the study and opposing systems. If fewer than two subsystems are defined, an appropriate error message is printed and activity POLY is terminated. If either two or three subsystems were specified, the designated subsystem is taken as the study system while the remaining two subsystems are treated as the opposing systems. If more than three subsystems were defined, the user specifies the two opposing systems. Activity POLY verifies that there are no buses that are members of two or all three of the selected subsystems. If any such buses are found, they generate an alarm and activity POLY is terminated.

Next, the largest active power mismatch corresponding to the present ac solution voltage vector in the working case is calculated. If this largest mismatch is greater than the MW mismatch tolerance, the user is given the option of continuing or terminating: the process.

The program will check to ensure that there are no buses that are members of two or all three of the selected subsystems. If any such buses are found, they generate an alarm and the process is terminated.

If the convert ratings option was enabled at the start of activity POLY, ratings from the selected rating set(s) of those metered branches having non-zero rating(s) are modified. For each such branch, the reactive power loading is calculated and MW rating(s) are determined under the assumption that the reactive power loading is unchanged under power transfer and contingency conditions.

The base case dc network solution is calculated and the solution summarized as in the following example:

```
STUDY SYSTEM GENERATION IS         1500.0 MW
OPPOSING SYSTEM 1 GENERATION IS    1400.0 MW
OPPOSING SYSTEM 2 GENERATION IS     348.9 MW
STUDY SYSTEM NET INTERCHANGE IS     282.8 MW
```

The user may specify a study system generation shift (default = 100). If the value entered is positive, export limits are to be determined; otherwise, import limits are to be calculated.

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The generation shift is apportioned among generator and load buses in the study system in proportion to the participation factors specified in the participation block structure of the Subsystem Description Data File (refer to ???). If no participation block was entered for the study system, the generation shift is shared among generator buses in proportion to their plant MBASEs (i.e., the sum of the MBASEs of in-service machines with positive active power output at the bus). In this case, there must be at least one in-service machine with a positive MBASE in the subsystem. The opposite generation shift is applied to buses in the first opposing system, again according to their participation factors or MBASEs.

Using the incremental form of the dc power flow equation (refer to activity DCLF) and the power shift vector, phase angle changes are calculated, followed by incremental line flows for all monitored lines and interfaces. These are transformed into a vector of base case power transfer distribution factors, and the interchange limit is determined by extrapolation such that all monitored elements satisfy their rating constraints.

A second set of power transfer distribution factors describing the effect on the monitored elements of an interchange shift between the study system and the second opposing system is calculated.

Based on the two sets of power transfer distribution factors, the study system interchange limit is determined by a linear programming technique such that all monitored elements satisfy their rating constraint.

The base case solution is then reported, followed by processing of the designated contingency cases, with contingency case flow estimates calculated using the methodology defined in Contingency Case Flow Calculations.

One final optimal solution is calculated for a combined case. This solution calculation includes constraints for all system conditions (i.e., the base case and all contingency cases) for which an individual optimum solution was found. This calculation therefore determines the maximum study system export (or import, as appropriate) schedule such that no monitored element is overloaded in the base case or under contingency conditions.

The combined case solution is then reported and, prior to terminating, activity POLY summarizes the calculated interchange limits for the base case, for all contingency cases, and for the combined case.

9.10.3. Output Report Format

In reporting the base case solution, activity POLY lists the generation in the study and opposing systems and the study system net interchange corresponding to the network solution before applying any generation shift and in the two generation shift cases. This is followed by a tabulation of the pre-shift and post-shift generation at those study system and opposing system buses participating in the generation shifts.

The generation shifts required to achieve the interchange limit (import or export, as appropriate) are then listed, along with the limiting lines and/or interfaces and their ratings (see Figure 9.18, “Typical Output from Interchange Calculation for Two Opposing System”).

For monitored three-winding transformer windings, flow estimates are calculated at the winding bus end as power flowing into the transformer. For other monitored branches, flow estimates are calculated at the metered end in the from bus to bus direction. The flow across an interface is taken as the sum of the flows of its members. For three-winding transformer windings, the flow is calculated at the winding bus as power flowing into the transformer. For other members, the flow is calculated at its metered end in the from bus (i.e., the first bus specified in entering the branch; refer to Monitored Element Data File Contents) to to bus direction.

Loadings on monitored elements are then reported. The flow estimates are listed for each of four interchange schedules: with the original generation profile; with the specified generation shift in the study system and

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the opposite shift in the first opposing system; with the specified generation shift in the study system and the opposite shift in the second opposing system; and in the case with maximum import or export, as appropriate, in which no monitored elements exceed the specified percentage of the selected rating. Any flow at or above the selected percentage of the appropriate rating is followed by an asterisk (*). The power transfer distribution factors relating the change in flow on the monitored element to a change in study system net interchange with the opposite change in the two opposing systems in turn are also tabulated, along with the rating of the monitored element; for branches, this is *not* the percentage of the rating used in determining the interchange limit, but either the value of rating as entered in activities such as READ, TREA, RDCH, CHNG or the *[Spreadsheet]*, or, if the convert ratings option was selected at the start of activity POLY, the estimated MW ratings (see Figure 9.18, "Typical Output from Interchange Calculation for Two Opposing System").

When reporting results for contingency cases, the events comprising the contingency are listed, followed by the generation shifts required to achieve the transfer limit for the contingency case being reported, along with the limiting elements. The flow table includes both the contingency case flows in the same form as the base case report described above, as well as the base case flows at the transfer limit. Note that if the *ignore base case constraints in contingency cases* option was specified, only the contingency case constraints are considered in calculating the transfer limit; in this case, base case flows at the transfer limit may be shown as overloaded because the corresponding constraints are ignored in the transfer limit calculation.

When reporting results for the combined case, activity POLY tabulates the generation shifts required to achieve the interchange schedule that results in maximum import (or export, as appropriate) while simultaneously avoiding overloading in all system conditions for which individual optima were found. The limiting elements are listed along with the system condition in which they are limiting. The combined case flow table includes only those elements that are part of the border of the feasible region of a successfully solved system condition. Flows are reported in the same form as the base case report described above, and the system condition to which they apply is shown at the right side of each output line.

A summary page tabulates the number of optimal solutions attempted, and the number for which there exists no interchange schedule at which the rating constraint can be satisfied for all monitored elements. Then, for each case processed, the interchange limit is listed along with the corresponding generation shifts required in the two opposing systems.

If a line outage contingency forms one or more islands with no Type 3 (swing) bus, such islands are disconnected. Any monitored branch that is located in the swingless island, as well as any interface that includes such a branch as a member, has dashes printed where its contingency case results would normally be printed.

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```
        *** POLY EXPORT LIMIT OUTPUT FOR BASE CASE                    ***

DISTRIBUTION FACTOR FILE:      C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency3.dfx
SUBSYSTEM DESCRIPTION FILE:    C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:        C:\Program Files\PTI\PSSE30\EXAMPLE\jdmxsmo.mon
CONTINGENCY DESCRIPTION FILE:  C:\Program Files\PTI\PSSE30\EXAMPLE\savnw2.con

                                        X- SHIFT CASE 1 -X  X- SHIFT CASE 2 -X
                                PRE-SHIFT    DELTA  POST-SHIFT   DELTA  POST-SHIFT
STUDY SYSTEM MW GENERATION:        1500.0    100.0     1600.0   100.0     1600.0
OPPOSING SYSTEM 1 MW GENERATION:   1400.0   -100.0     1300.0     0.0     1400.0
OPPOSING SYSTEM 2 MW GENERATION:    358.7      0.0      358.7  -100.0      258.7
STUDY SYSTEM MW NET INTERCHANGE:    278.7    100.0      378.7   100.0      378.7

<---------------- STUDY SYSTEM ----------------> <------------ FIRST OPPOSING SYSTEM ------------> <-------- SECOND OPPOSING SYSTEM -------->
                      <---- GENERATOR MW ---->                      <---- GENERATOR MW ---->                      <---- GENERATOR MW ---->
  BUS# X-- NAME --X BASKV    BASE   SHIFT  CHANGE   BUS# X NAME X BASKV    BASE   SHIFT  CHANGE   BUS# X-NAME -X BASKV   BASE   SHIFT  CHANGE
   101 NUC-A       21.600   750.0   800.0    50.0    205 UBBGEN   18.000   800.0   742.0   -58.0   3011 MINE_G   13.800   258.7   170.2   -88.5
   102 NUC-B       21.600   750.0   800.0    50.0    211 HYDRO_C  20.000   600.0   558.0   -42.0   3018 CATDOC_G 13.800   100.0    88.5   -11.5

                                BASE CASE INCREMENTAL   TOTAL
STUDY SYSTEM GENERATION MW          1500.0    1450.3    2950.3
OPPOSING SYSTEM 1 GENERATION MW     1400.0    -181.9    1218.1
OPPOSING SYSTEM 2 GENERATION MW      358.7   -1268.4    -909.8
STUDY SYSTEM MW NET INTERCHANGE      278.7    1450.3    1729.0

LIMITING MONITORED ELEMENTS LOADED AT 100.0 % OF RATING:
<-------- F R O M --------> <---------- T O ----------> CKT   RATING
   201 [HYDRO       500.00]    151 [NUCPANT     500.00] 1    1200.0
   205 [SUB230      230.00]    154 [DOWNTN      230.00] 1     600.0

LOADINGS AT OR ABOVE 100.0 %                                <---------------------- BASE CASE ---------------------->
OF RATING ARE MARKED WITH '*'                                PRE- CKT SHIFT CASES   LIMIT          DISTRIBUTION
                                                             SHIFT   OPSYS1  OPSYS2   CASE  RATING      FACTORS
<-------- F R O M --------> <---------- T O ----------> CKT    MW      MW      MW      MW      %     OPSYS1  OPSYS2
   151 NUCPANT      500.00    152 MID500       500.00  1    469.0   488.5   498.1   873.5  1200   0.19509  0.29091
   153 MID230       230.00    154 DOWNTN       230.00  1    263.9   264.2   255.1   288.5   300   0.10346  0.01297
   201 HYDRO        500.00    151 NUCPANT      500.00  1   -558.7  -619.7  -600.5 -1200.*  1200  -0.50981 -0.41817
   202 EAST500      500.00    152 MID500       500.00  1    -50.7   -63.2   -37.3    96.9  1200  -0.12486  0.13426
   203 EAST230      230.00    154 DOWNTN       230.00  1    122.4   126.8   127.2   180.4   200   0.04396  0.04931
   205 SUB230       230.00    154 DOWNTN       230.00  1    356.1   325.2   329.8   600.0*  600  -0.30929  0.23660
       INTERFACE WEST                                      -147.9  -147.9  -247.9 -1416.   2800   0.00000 -1.00000
       INTERFACE EAST                                      -130.8  -230.8  -130.8  -312.6  1600  -1.00000  0.00000
```

Figure 9.18. Typical Output from Interchange Calculation for Two Opposing System

9.10.4. Interchange Limit (Two-Opposing Systems) Graphical Output

When the graphical display of results is selected, activity POLY produces a graph for either each optimal solution calculated or for only the combined case solution (see below). Change in interchange in the first opposing system is plotted along the horizontal axis, and change in interchange in the second opposing system is plotted along the vertical axis. The plotting scales are specified at the start of activity POLY using the *maximum opposing system # interchange for plotting* option.

For each monitored element, a pair of straight lines is plotted indicating the set of interchange schedule changes for which that element is exactly at its rating constraint. Moving to any point on one side of any limit line brings the loading on the element below its limit; points on the other side result in a limit violation. On the violation side of the limit line, the background is shaded blue (see Figure 8-16).

When all monitored elements are so plotted, the region of interchange schedule changes at which no monitored element violates its rating constraint is shown as the clean portion of the graph. If the point of maximum transfer (export or import, as appropriate) exists for a given system condition (i.e., a clean portion exists on the graph for the system condition), it occurs at the intersection of a pair of limit lines, and is enclosed in a square. In Figure 9.19, "Graphical Output from Calculation of Interchange Limit with Two Opposing Systems" this is where the interchange with the EAST system is 434 MW and -226 MW with the WEST.

A single-letter code is printed at each end of any limit line that is part of the border of the polygon forming the clean portion of the graph; a key is printed with the graph indicating the elements represented by each such annotated constraint line. The two lines for which the intersection occurs at the transfer limit are assigned the letter codes A and B; continuing along the border of the polygon, successively encountered constraint lines are assigned letter codes in alphabetical order.

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Plot and graphical output options specified at the start of activity POLY provide for the following graphical report options:

1. Producing graphical output only for the combined case or for all cases.

2. Including on each plot for which a transfer limit exists either all constraint lines or only those that form the boundary of the polygon that encloses the feasible region.

3. Including both the graph and the letter code key on the same plot or on separate plots.

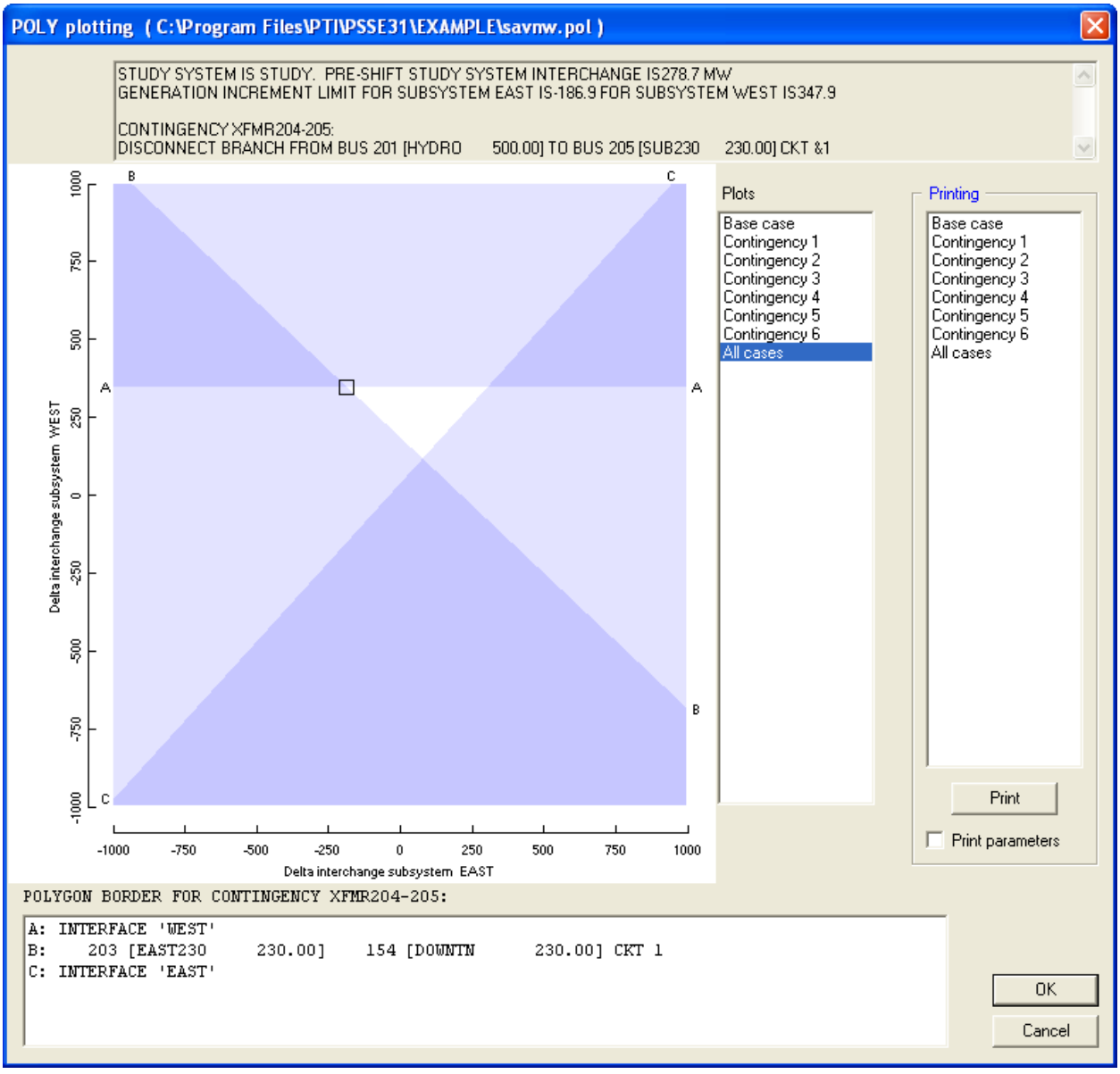

Figure 9.19. Graphical Output from Calculation of Interchange Limit with Two Opposing Systems

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9.10.5. Application Notes

Activity POLY uses the same linearized network model as is used in activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Note, in particular, the approximate nature of the linearized network model used by activity POLY. The proper role of activity POLY, therefore, should be to focus attention on those system and transfer conditions that deserve more detailed study.

Activity POLY detects the specification of duplicate single- and double-line outage contingencies and calculates the contingency case solution only for the first specification of such a contingency. Any contingency cases involving bus boundary contingency events, line closure contingency events, or more than two-line outage contingency events are not checked for duplication.

The values shown as monitored element flows are set as described in Section 9.5.4, "Contingency Case Flow Calculations". The user selects the base flow to be used by setting the *line flow code* option at the start of activity POLY.

Contingency case transfer limits may be calculated using either of the following approaches:

1. Both pre-contingency and post-contingency flows must satisfy their appropriate rating constraints (i.e., both the base case and the contingency case must have no overloads).

2. Only post-contingency constraints need be satisfied.

The user selects the method to be used by setting *include/ignore base case constraints in contingency case* at the start of activity POLY.

When *minimum distribution factor magnitude* is greater than zero, monitored elements with both of the distribution factor magnitudes below the threshold are ignored in calculating the transfer limit for the corresponding system condition.

If the convert ratings code is set to one at the start of activity POLY, any percentage of rating parameter is applied after the conversion to estimated MW ratings. Interface ratings are not affected by the *convert ratings* option.

Line loss estimates and line shunts of outaged lines are handled as in activity DCLF (refer to Section 9.4.4, "Application Notes").

It is required that the working case be solved to an acceptable mismatch tolerance prior to entering activity POLY.

It is entirely possible that, for a given system condition, there is no interchange schedule at which the linearized network model solution results in all monitored elements satisfying their rating constraints. This could occur, for example, if the solution had two overloaded branches, and one required an increase in interchange to relieve its overload while the other required a decrease in interchange. Such insoluble conditions are described in the output block for that condition. When full output is being produced, the flow column corresponding to the maximum import or export for that system condition is left blank. Such cases are identified on the summary page and are omitted from the combined case transfer limit calculation.

The combined case solution is visually equivalent to plotting the graphical output for all successfully calculated system conditions on the same graph and observing the clean area, if it exists, which all of these graphs have in common.

When specifying a rating percentage of other than 100%, note that interface ratings are also scaled by the specified percentage in checking their flows to determine maximum interchange.

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Additional Information
PSS®E GUI Users Guide, Calculating Sequential Participation Interchange Limits

9.11. Midwest MW-Mile Calculation

MWMI

Run Activity MWMI - GUI
Power Flow>Linear Network>Midwest MW-mile calculation (MWMI)... *[Midwest MW-miles calculation]*
Run Line Mode Activity MWMI - CLI
ACTIVITY? >>**MWMI** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

This activity was incorporated into PSS®E for the convenience of customers who use Midwest ISO (http://www.midwestiso.org/home) planning algorithms. Please contact MISO for current updates, as PTI does not provide warranty or support for Midwest Independent Transmission System Operator solutions.

Additional Information
PSS®E GUI Users Guide, Running Midwest MW-mile Calculation

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Chapter 10
Network Reduction

10.1. Overview of Equivalent (Reduced) Networks

PSS®E provides the user with the capability to construct an electrical equivalent of all or a subsystem of the network in the currently open power flow case. The case must be solved to an acceptable mismatch level. The output of the construction process is a case with part or all of the original network model replaced by an equivalent representation.

The primary purpose in constructing equivalents is to represent a portion of a network containing many buses but having only a few boundary buses by a reduced network containing only the boundary buses and, perhaps, a few selected buses from within the original subnetwork. Historically, equivalents have often been used to allow the representation of larger areas of major interconnected systems in studies using computer programs which, due to restrictions imposed by the computing hardware, were of limited dimensional capacity. Advances in computer technology, particularly in the area of memory address capability, have lifted these restrictions to the extent that network modeling capacity is usually no longer an issue. The primary application of equivalents in modern engineering computers is to achieve improved computational speed in simulations by removing buses and branches that are not of specific interest but which do influence system behavior.

10.1.1. Nomenclature of Equivalents

In discussing the application of equivalents, it is useful to adopt the following terminology:

Study System

A group of buses and branches under detailed study. Within the study system, all components are represented explicitly.

External System

A group of buses and branches that is connected to and influences a study system, but which does not need to be represented in detail.

Boundary Buses

Buses from which branches run either into a study system, and one or more external systems, or into more than one external system.

Source System

A power system representation that contains all components of an external system as a subset of its own components. It is used to solve for the base conditions within the external system. The source system does not need to include the study system, but must recognize flows between the external and study system.

Electrical Equivalent

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An artificial group of branches and buses that represents the behavior of the external system as seen from its boundary buses.

Retained Bus

A bus of the external system that is also a bus of the electrical equivalent. A retained bus is not necessarily a boundary bus, but all boundary buses are retained buses.

Deleted Bus

A bus of the external system that is not a bus of the electrical equivalent, but where the effect is represented by the equivalent.

Tie Branch

A branch having one end in one system (study or external) and another end in a different system. Tie branches are connected to boundary buses at each end.

Area

A group of buses designated in power flow input data for interchange control purposes. An area may, but does not necessarily, coincide with a study or external system.

10.2. Methodology of the Electrical Equivalent

An electrical equivalent is constructed by performing a reduction operation on the admittance matrix of the external system that is to be represented by the equivalent. The admittance matrix equation of the external system may be written in the partitioned form

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_1 Y_2 \\ Y_3 Y_4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{10.1}$$

where I_1 and V_1 are node current and voltage at the nodes to be retained and I_2 and V_2 are node current and voltage at the nodes to be deleted.

The desired form of an equivalent is an equation involving only I_1 and V_1 explicitly, with the I_2 and V_2 variables assumed to be linearly dependent upon I_1 and V_1, and recognized implicitly. The equivalent is obtained by rearranging the second row of Equation 10.1 as

$$V_2 = Y_4^{-1}(I_2 - Y_3 V_1) \tag{10.2}$$

and substituting this into the first row of Equation 10.1 to give

$$I_1 = (Y_1 - Y_2 Y_4^{-1} Y_3)V_1 + Y_2 Y_4^{-1} I_2 \tag{10.3}$$

The first term of Equation 10.3 specifies a set of equivalent branches and static shunt elements connecting the retained nodes, while the second term specifies a set of equivalent currents that must be impressed on the

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retained nodes to reproduce the effect of load currents at the deleted nodes. These equivalent currents may be translated into equivalent constant real and reactive power loads at the retained buses. The equivalent obtained by translating the two terms of Equation 10.3(into equivalent branches, shunts, and loads is exact in the base case for which the current vector, I_2, was calculated. When voltage conditions at the boundary buses are changed, the equivalent gives an approximation to the change in power flow into the external system. This approximation is good as long as the changes are small, but may become unreliable when boundary bus voltages and power flows into the external system (or its equivalent) deviate from the base values by large amounts.

An electrical equivalent is, therefore, correctly applied when it represents an external system in which the disturbances or switching operations under study produce only minor effects; but equivalents should not be applied to network segments in the close vicinity of the disturbances or switchings.

10.2.1. Defining Boundaries and Boundary Buses

A boundary cuts a set of tie-lines between areas or otherwise identifiable sections of a network but passes through no buses. A boundary bus is part of only one area (see Figure 10.1, "Separation of Complete Network into Study System and External Systems by Boundaries").

An equivalent makes more efficient use of storage when the ratio of branches to buses in the equivalent is reduced. The relative efficiency of different equivalents of a given system is best determined by trial and error. As a general rule, however, reducing a system into a number of small equivalents is more efficient than reducing of a large system in one step to produce a single equivalent.

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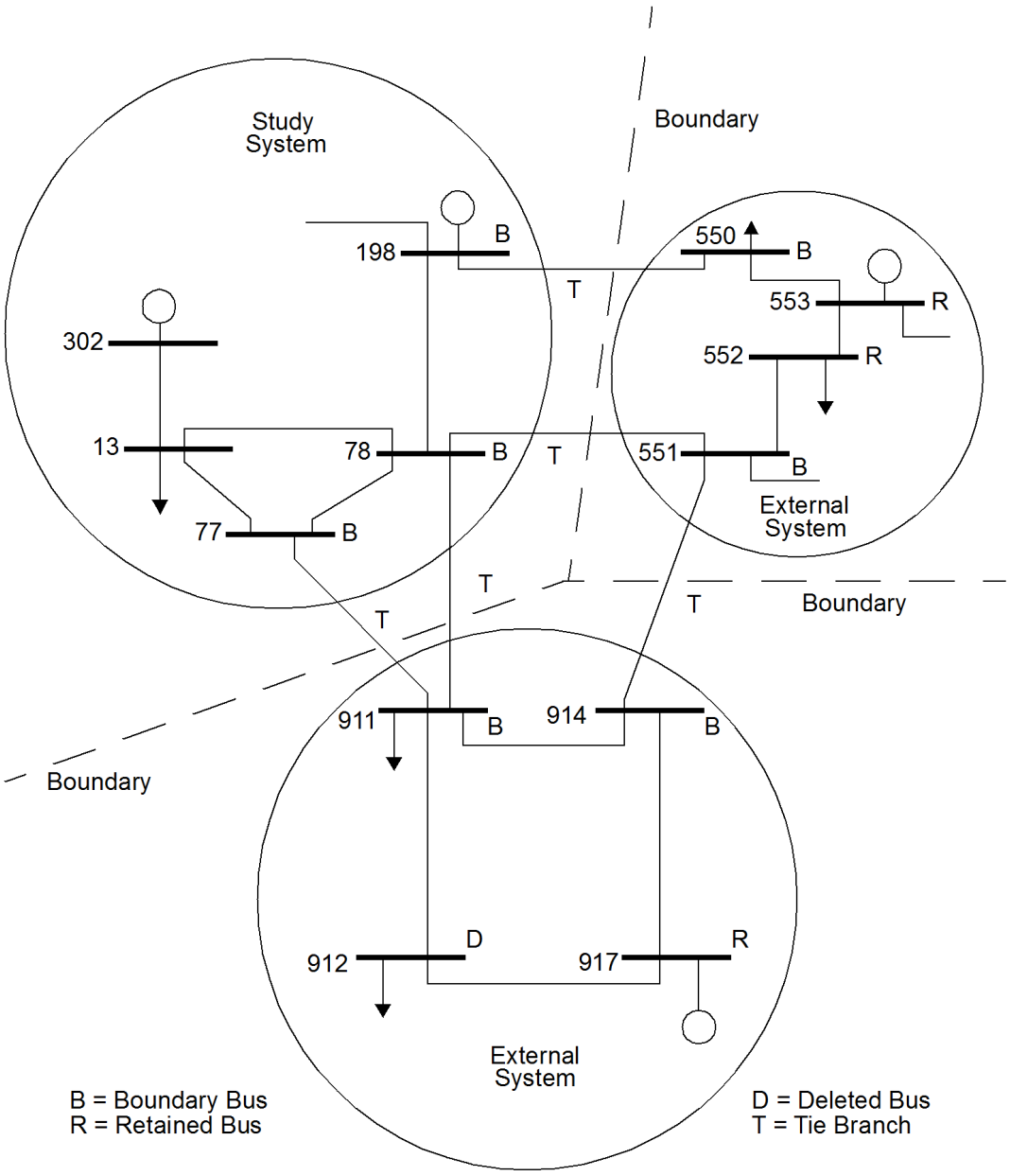

Figure 10.1. Separation of Complete Network into Study System and External Systems by Boundaries

Three special bus type codes are used to designate boundary buses at various stages of the equivalent construction processes:

Type 5	As for Type 1 (load bus); boundary bus or a bus that is not to be deleted by the equivalencing option, retained bus.
Type 6	As for Type 2 (generator bus); boundary bus.

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Type 7	As for Type 3 (swing bus); boundary bus.

It can be seen in Figure 10.1, "Separation of Complete Network into Study System and External Systems by Boundaries" that the area designated as the Study System there are three buses (buses 198,78 and 77) that are the terminals of lines incoming from the other two areas and there are four boundary buses in the external systems. If the external systems bus were to be reduced to an equivalent network, it is possible that boundary 914 could be eliminated. But it would probably be important for the user to retain this bus because of its strategic position in the network. Consequently, it would be important to change the code of this bus, or to retain area boundary buses to avoid having it deleted.

Several analytical processes of PSS®E change the type codes of boundary buses back and forth between Types 1, 2, 3 and 5, 6, 7, respectively. The user may also change type codes manually by editing the data in the spreadsheet view.

It is important to note that type codes 5, 6, 7 are valid only during equivalent construction and system data processing operations. Type codes 5, 6, or 7 must be returned to 1, 2, or 3, respectively, before invoking any optimal ordering, factorization, generator or load conversion, or solution processing.

10.2.2. Handling DC Lines

In PSS®E, the equivalencing process automatically retains converter buses of unblocked dc transmission lines. Hence, all buses that are affected by dc transmission must become part of a study system before commencing the construction of an equivalent.

10.2.3. Approaching the Network Equivalent Process

The equivalencing process in PSS®E is designed to develop an electrical equivalent of a power flow case or a subsystem of it, as described in Section 10.2, "Methodology of the Electrical Equivalent", while allowing the user to control which buses, lines, generators and phase-shifters are retained. This global approach is supported with other reduction processes, which can be useful in making selected equivalents of generators, prior to making the global network equivalent, or radial lines.

In addition, PSS®E provides a process for the construction of network equivalents of the positive and zero sequence networks, and calculates source impedances at equivalent source nodes for all three sequences in preparation for the unbalanced fault analysis activities of PSS®E.

Furthermore, PSS®E provides a boundary bus balance process that converts the mismatch at boundary buses to equivalent load and/or generation. It is intended to be used following the removal of a subsystem from a solved power flow case, with the flows to the deleted subsystem replaced by load and/or generation at those retained buses that were directly connected to the removed subsystem.

10.3. Building an Electrical Equivalent

EEQV

Run Activity EEQV - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Build electrical equivalent (EEQV) Tab

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Run Line Mode Activity EEQV - CLI
ACTIVITY? >>**EEQV** USER SPECIFIES SUBSYSTEM TO BE EQUIVALENCED ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The power flow equivalent construction activity EEQV constructs an electrical equivalent of a subsystem of the network contained *inside* the working case. [1]

The input to activity EEQV (i.e., the working case) must be solved to an acceptable mismatch level. The output of activity EEQV is a working case with part or all of the original network model replaced by an equivalent representation.

10.3.1. Overview

The equivalent constructed by activity EEQV is exact in the base case from which it was calculated. It gives an exact reproduction of the self and transfer impedances of the external system as seen from its boundary buses. The net total of load, generation, and losses in the equivalent matches this total in the complete external system if the bus voltages in the working case were a valid power flow solution. However, the load, generation, and loss totals in the equivalent may not individually match those of the complete external system.

When voltage conditions at the boundary buses of the study system are changed, the equivalent gives an approximation to the change in power flow into the external system. This approximation is good as long as the changes are small, but may become unreliable when boundary bus voltages and power flows into the external system (or its equivalent) deviate from the base values by large amounts.

An electrical equivalent is, therefore, correctly applied when it represents an external system in which disturbances or switching operations under study produce only minor effects. Equivalents should not be applied to network segments in the close vicinity of the disturbances or switchings.

10.3.2. Network Equivalence Options

External systems, following the terminology of Section 10.1.1, "Nomenclature of Equivalents", are defined according to the specified subsystem. The user may specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV.

The user may specify any of the following options:

Table 10.1. Network Equivalence Options

Option	Description
Retain area boundary buses	Enable to keep area boundaries intact.
Retain zone boundary buses	Enable to keep zone boundaries intact.
Suppress equivalencing of phase shifters	If any two-winding transformers with a non-zero phase shift angle are present in the subsystem to be calculated as electrical equivalents, enable to

[1] An electrical equivalent of the portion of the working case *outside of* the specified subsystem may be constructed using the BAT_XEQV command at the PSS®E Command Prompt.

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Option	Description
	suppress building an electrical equivalent of phase shifters. If the user chooses to retain two-winding transformer phase shifters, activity EEQV forces the retention of the buses involved by changing their type codes from 1, 2, or 3 to 5, 6, or 7, respectively. Thus, the phase shifter and the buses it connects are all explicitly retained in the equivalent. If the user chooses to include two-winding transformer phase shifters when building the electrical equivalent, activity EEQV resets the phase shift angle to zero degrees and introduces equivalent MVA load at the buses connected by the phase shifter such that an electrical balance is retained. The phase shifter is then treated like any other transformer.
Retain controlled buses	Enable to suppress building an electrical equivalent of buses where voltage or reactive power output is controlled by remote generation, switched shunt, VSC dc line converter, and/or FACTS device shunt element.
Retain existing branches	Enable to retain the identity of all non-transformer branches and two-winding transformers in the working case connecting buses to be retained in the equivalent. If this option is enabled, the equivalent may contain an additional equivalent branch in parallel with any real branches between any such pair of buses. If this option is disabled, the equivalent branch between any such pair of buses represents the single circuit equivalent of all real branches between the buses and (possibly) an equivalent branch.
Net load and shunt	This option treats the total equivalent load and shunt at retained buses such that occurrences of load and shunt that cancel at the bus (e.g., positive load and negative shunts) are eliminated. The component of smaller magnitude is set to zero with the other component set to the netted quantity. Active and reactive components at each retained bus are processed independently. Disable to allow the equivalent load and shunt components at retained buses to remain as calculated by the equivalent construction matrix reduction process

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Option	Description
Minimum generation for retaining generator buses	Enable to net small plants at Type 2 buses. The user sets the positive threshold below which Type 2 buses will have their generation netted with the bus load and its type code changed to one. This does not apply to boundary Type 2 buses or area swing buses. Both active and reactive power generation magnitudes must be below the threshold. If the threshold specified is zero, all Type 2 generator buses in the subsystem being calculated as electrical equivalents are retained.
Branch threshold tolerance	Any equivalent branch where magnitude of impedance is greater than this tolerance is not retained in the equivalent. The default value is 10 per unit.

10.3.3. Operation of Activity EEQV

First, the portion of the working case to be exempted from the equivalencing activity is (temporarily) removed from the working case, and all radial and two-point Type 1 buses are calculated as electrical equivalents (refer to RDEQ).

The standard output of activity ORDR is tabulated, and the construction of the equivalent is completed.

If the user specified activity EEQV to *net load and shunt* at retained buses, it nets the total equivalent load and shunt at retained buses such that occurrences of load and shunt that cancel at the bus (e.g., positive load and negative shunts) are eliminated. The component of smaller magnitude is set to zero with the other component set to the netted quantity. Active and reactive components at each retained bus are processed independently.

If *net load and shunt* was disabled, the equivalent load and shunt components at retained buses remain as calculated by the equivalent construction matrix reduction process.

The equivalent and detailed system sections are then combined.

10.3.4. Equivalencing Methodology

The activity of constructing and using an electrical equivalent can be separated into three steps:

1. Isolating an external system in the working case. Activity EEQV temporarily discards from the working case all other subsystems (study and external) contained in the source system except the one to be calculated as electrical equivalents. Boundary and retained buses are identified and, if desired, selected generators are replaced by negative load.

2. Constructing the electrical equivalent by performing the required matrix reduction operation on the external system remaining in the working case.

3. Combining system sections together to form a valid system model in the working case. In this step, detailed representations of selected external systems are replaced with their electrical equivalents.

Activity EEQV constructs electrical equivalents on the following basis:

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Table 10.2. Electrical Equivalencing of Buses

Bus Type Code	Treatment By Activity EEQV
1	Bus eliminated
2 and 3	Bus retained
4	Bus ignored in the computation and not included in the equivalent
5, 6, and 7	Bus retained in the equivalent with 4 subtracted from its bus type code

That is, it constructs an equivalent of a subsystem of the working case with its interior Type 1 buses eliminated. Activity EEQV automates many of the data handling functions required in preparation for, and following the execution of, the equivalent construction calculation. It may be used whenever the entire solved source system is able to be brought into the PSS®E working case.

The equivalent constructed by activity EEQV is present in the working case along with the unequivalenced portions of the system in the same form as an unequivalenced power flow case. There are normally no operations required to combine system sections.

Activity EEQV processes only the positive sequence network. If sequence data is included in the working case, the following message is printed upon entering activity EEQV and processing continues:

```
WARNING: SEQUENCE DATA WILL NOT BE EQUIVALENCED
```

Activity EEQV requires that the working case represent a solved system condition.

In constructing the equivalent, all non-boundary Type 1 buses in the designated external system are eliminated, and all Type 5, 6, and 7 buses are retained as Type 1, 2, and 3 buses, respectively. Normally, the boundary buses from within the subsystem being calculated as electrical equivalents are those connected to buses outside of the specified subsystem. However, when the subsystem to be calculated as an electrical equivalent is specified by bus only, Type 1 and netted Type 2 buses specified are calculated as electrical equivalents, with the boundary buses being those buses to which they are connected and that are not among those specified.

When three or more buses are all connected together by zero impedance lines (refer to Section Zero Impedance Lines), either all may be calculated as electrical equivalents, all may be retained, or one may be retained. If the bus type codes are such that more than one but not all the buses in such a group of buses are to be retained, activity EEQV prints an appropriate message and retains all buses in the group.

Three-winding transformers are handled as follows:

- If all three buses connected by an in-service three-winding transformer are to be retained, the transformer is explicitly retained.
- If at least one of the three buses connected by an in-service three-winding transformer is to be calculated as an electrical equivalent, the transformer is calculated as an electrical equivalent.
- Out-of-service three-winding transformers are ignored in constructing the equivalent and are omitted from the equivalent.

Activity EEQV automatically retains the converter buses of unblocked dc lines, the buses to which in-service GNE devices are connected, the sending and terminal buses of in-service FACTS devices, and buses connected to terminal buses of in-service FACTS devices.

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At all type 1 buses being eliminated, their in-service constant admittance load, fixed shunt, and switched shunt are netted before applying the network reduction equations described in Section 10.2, "Methodology of the Electrical Equivalent". Similarly, the powers flowing from the bus to any in-service induction machines are netted with any in-service constant power load and constant current load at the bus.

For any bus where voltage is being controlled by a remote generator, switched shunt, VSC dc line converter, or FACTS device shunt element, an alarm is printed if the controlled bus is calculated as an electrical equivalent while the voltage controlling equipment is retained. The controlling equipment is then set to control local bus voltage and the scheduled voltage setpoint or band is not changed.

For any bus with voltage controlling equipment that is being controlled by a remote switched shunt, an alarm is printed if the controlled bus is calculated as an electrical equivalent while the controlling switched shunt bus is retained. The switched shunt's control mode is *not* changed.

Equivalent branches introduced by activity EEQV are assigned the circuit identifier 99.

Equivalent loads introduced by activity EEQV are assigned the load identifier 99. If a load with the identifier 99 already exists at a retained bus, activity EEQV adds any equivalent load to the load already modeled as load 99; if such a load is associated with an adjustable bus load table for which the load multiplier is zero, a mismatch is introduced at the corresponding retained bus.

Equivalent bus shunts introduced by activity EEQV are assigned the shunt identifier 99. If a shunt with the identifier 99 already exists at a retained bus, activity EEQV adds any equivalent shunt admittance to the fixed shunt already modeled as shunt 99.

10.3.5. Example: Equivalencing a Power Flow Case

In the *savnw.sav* power flow case there are three areas; FLAPCO, LIGHTCO and WORLD. The WORLD area, which will be replaced by its equivalence network has 10 buses, three of which are load buses and three ties to neighboring areas. As seen in Figure 9-2, the eleven buses are from bus 3001 to bus 3018. The load buses are 3005, 3007 and 3008. The ties are between buses 3004, 3006 and 3008 to buses 152, 153 and 154 in adjacent areas respectively.

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```
3001,'MINE    ', 230.0000,1
3002,'E. MINE ', 500.0000,1
3003,'S. MINE ', 230.0000,1
3004,'WEST    ', 500.0000,1
3005,'WEST    ', 230.0000,1
3006,'UPTOWN  ', 230.0000,1
3007,'RURAL   ', 230.0000,1
3008,'CATDOG  ', 230.0000,1
3011,'MINE_G  ',  13.8000,3
3018,'CATDOG_G',  13.8000,2
```

(a) WORLD Buses

```
3005,'1 ',1,   5,   5,   100.000,    50.000
3007,'1 ',1,   5,   5,   200.000,    75.000
3008,'1 ',1,   5,   5,   200.000,    75.000
```

(b) WORLD Load Buses

```
 152,  3004,'1
 153,   154,'1
 153,   154,'2
 153,  3006,'1
 154,   203,'1
 154,   205,'1
 154,  3008,'1
 201,   202,'1
 201,   204,'1
 203,  -205,'1
 203,  -205,'2
3001,  3003,'1
3002,  3004,'1
3003,  3005,'1
3003,  3005,'2
3005,  3006,'1
3005,  3007,'1
3005,  3008,'1
3007,  3008,'1
```

(c) Ties and WORLD Area Lines

Figure 10.2. Components of the WORLD Area in the savnw.sav Power Flow Case

There are also several two-winding transformers in the WORLD area along with generators at buses 3011 and 3018. A plot of the area is shown in Figure 10.3, "Pre-Equivalence Power Flow Diagram for the WORLD Area".

It is to be expected that the result of creating an equivalent of the WORLD area would be to leave the two generator buses intact, (unless the threshold for retaining generation is chosen to be greater that their generation) and leave the boundary buses 3004, 3006 and 3008 intact. This will result in the removal of 5 of the 10 buses. In this example, area boundary buses are retained and the generation threshold is 0.00.

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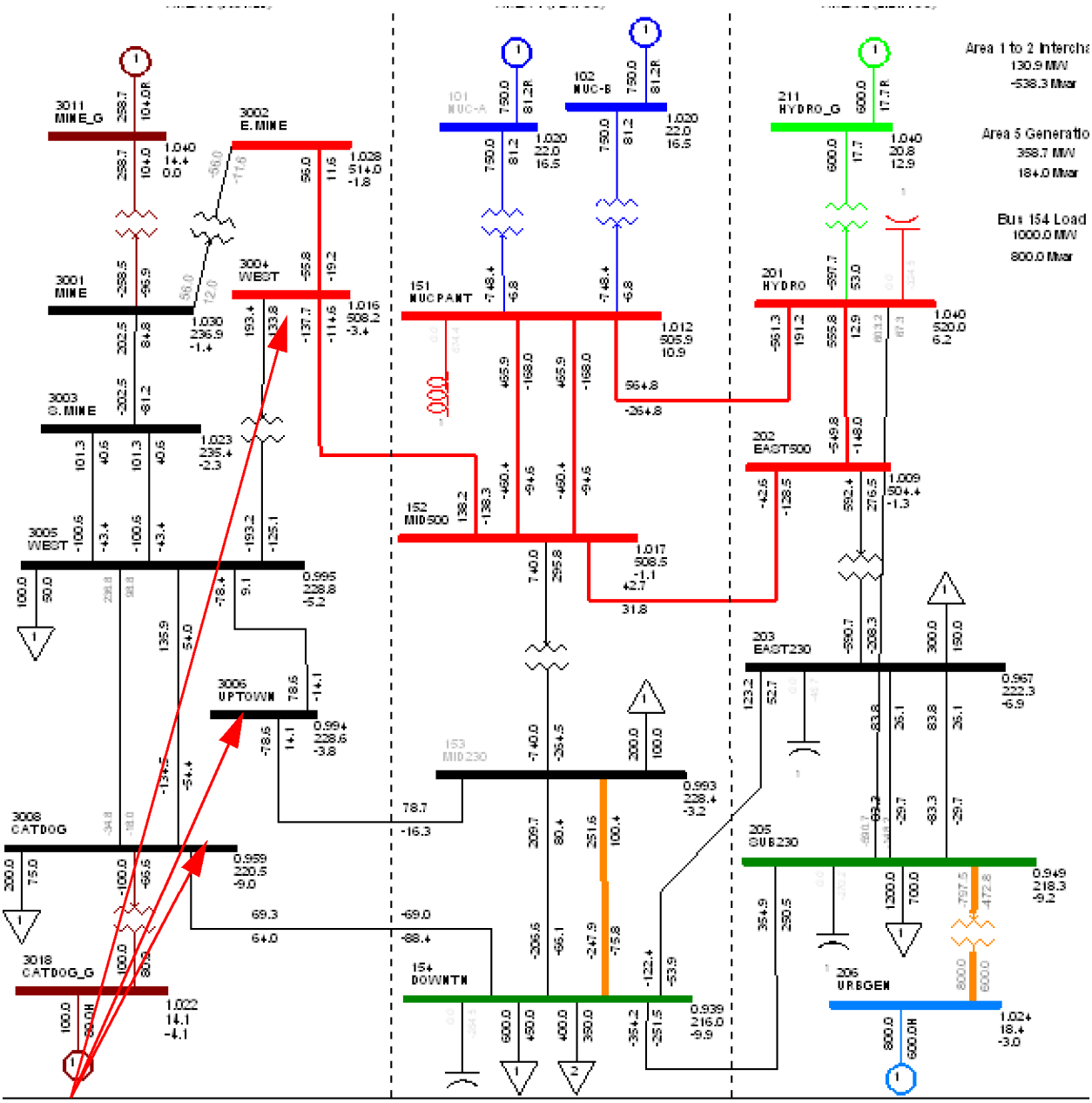

WORLD Area Boundary Buses

Figure 10.3. Pre-Equivalence Power Flow Diagram for the WORLD Area

After building an electrical equivalent of the WORLD area, the raw data appears as shown in Figure 10.4, "Reduced Components of the WORLD Area after Building an Electrical Equivalent".

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It can be seen that the total load of 500 MW in the WORLD area has been retained but allocated to the retained buses 3004, 3006, 3008 and 3011 with an identifier of 99. Note that bus 3008 already had a load of 200 MW with an identifier of 1.

The buses at the boundaries have been maintained but now the number of buses is reduced to five.

There are now 6 new branches with in the WORLD area each with a circuit identifier of 99.

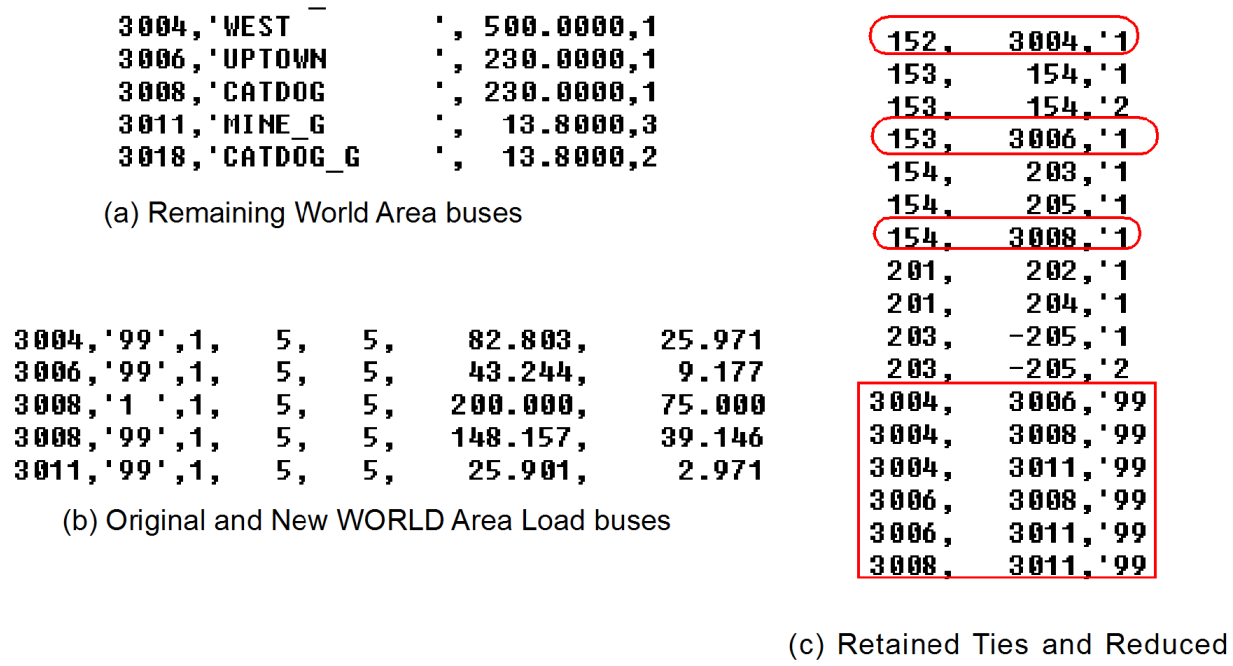

Figure 10.4. Reduced Components of the WORLD Area after Building an Electrical Equivalent

It should be noted that the slider file (*savnw.sld*) that was used to create the plotted results in Figure 10.3, "Pre-Equivalence Power Flow Diagram for the WORLD Area" will not suffice for the new power flow with the WORLD area calculated as an electrical equivalent. Five of the buses have been removed and new branches have been created. The user's selected color coding for the diagram properties will indicate that the buses and branches are unbound, i.e., do not exist in the power flow case. This is because the slider file no longer matches the network. A new or modified slider file will have to be produced.

Figure 10.5, "Redrawn One-Line Diagram to Match Topology after Building an Electrical Equivalent" shows the redrawn slider file presentation with the new branches and loads indicated. Comparing this diagram with that of Figure 10.3, "Pre-Equivalence Power Flow Diagram for the WORLD Area" will clarify the changes in topology due to the changes brought about by the equivalencing process.

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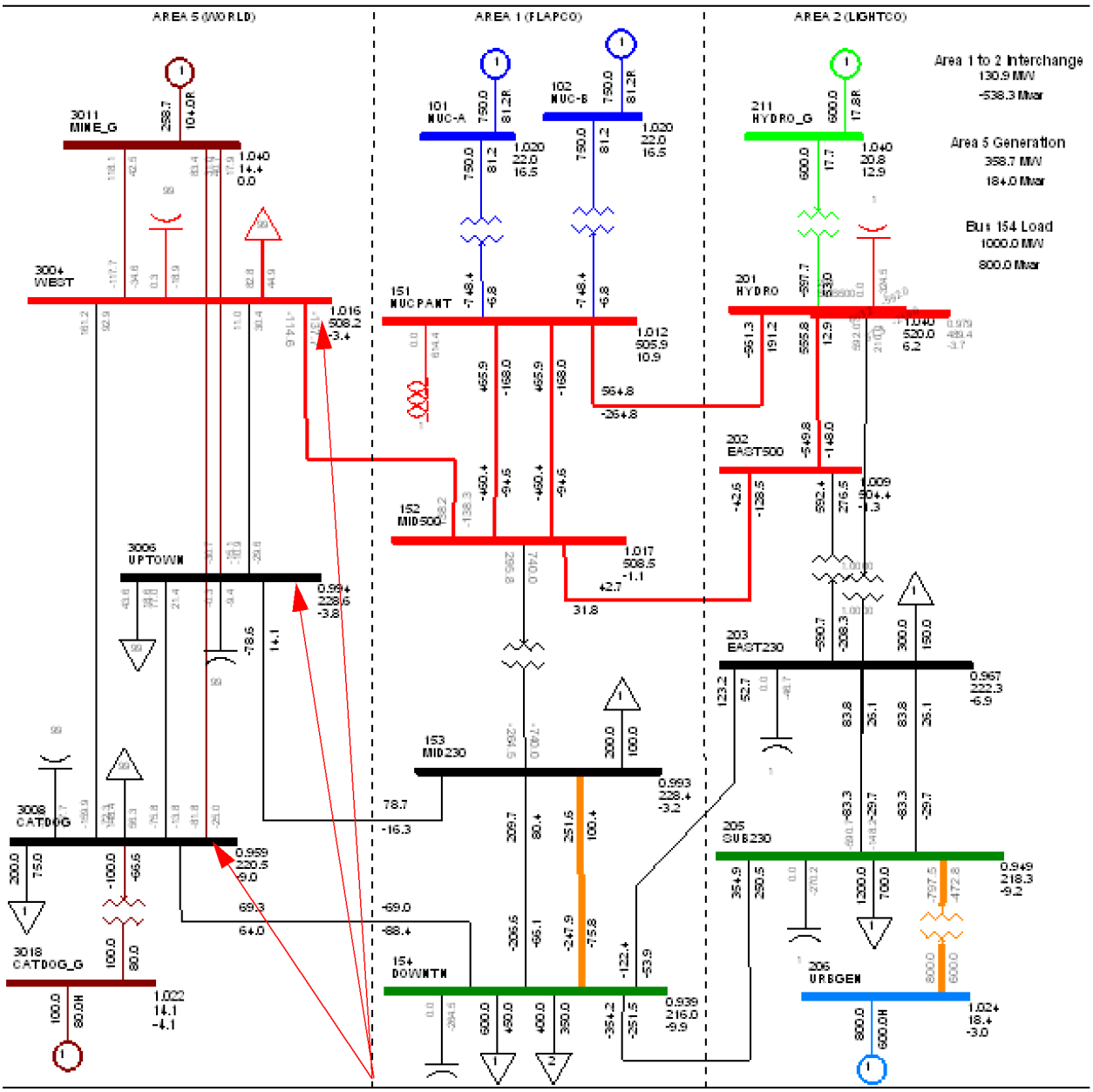

Figure 10.5. Redrawn One-Line Diagram to Match Topology after Building an Electrical Equivalent

10.3.6. Application Notes

This section summarizes some general points regarding the equivalencing process and some potential pitfalls to watch for.

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The goal in equivalent construction is to generate efficient equivalents. An equivalent may be regarded as making more efficient use of storage, and therefore improving computational speed of solution of the network containing the equivalent, when the ratio of branches to buses in the equivalent approaches that of real system (about 1.5). The relative efficiency of different equivalents of a given system is best determined by trial and error. As a general rule, however, the reduction of a group of external systems into a number of small equivalents is more efficient than their reduction in one step to produce a single equivalent.

It is not advisable to net all generation in an external system when constructing an equivalent in which more than a very few buses are to be retained. Type 2 buses have a generally beneficial effect on the convergence of power flow iterations and excessive netting of generation removes this effect from the composite power flow case when the equivalent is finally put to use. It is generally advisable to retain some major generator and synchronous condenser buses and to net the small generators that do not contribute strongly to the regulation of system voltage.

When generator netting has taken place in constructing equivalents, it is advisable to check for the existence of a swing (Type 3) bus in the working case at the completion of the equivalencing calculation. Activity TREE may be used.

Bus type codes of 5, 6, and 7 are valid only during equivalent construction. All such type codes are returned to 1, 2, and 3, respectively, by activity EEQV when the equivalencing process is complete.

The values of load and shunt admittance at each retained bus in the equivalent represent both actual and equivalent quantities at these buses. Changing these values invalidates the equivalent. Similarly, changing the status of any branches in the equivalent invalidates it.

A power flow working case containing elements introduced via an equivalent is identical in form to a power flow case containing only real system elements and may be operated on by all PSS®E activities, including activity EEQV. That is, it is permissible to construct an equivalent of an external system that contains an equivalent.

Additional Information
PSS®E GUI Users Guide, Electrical Equivalent *PSS®E* Command Line Interface (CLI) *Users Guide*, Building a Network Equivalent

10.4. Net Generation with Load In a Subsystem

GNET

Run Activity GNET - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Net generation with load (NETG/GNET/ImNet) Tab
Run Line Mode Activity GNET - CLI
ACTIVITY? >>**GNET** USER SPECIFIES THOSE TO BE NETTED

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Run Line Mode Activity GNET - CLI
ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The generation netting activity GNET changes the in-service generation to negative MVA load at all Type 2 and 3 buses within the subsystem specified by the user.

Prior to running a global network equivalence, or when it is required merely to reduce the number of generators modelled in a power flow case, generators can be removed by netting their output with the load at a bus. Generators at all type 2 and 3 buses will be replaced with equivalent negative load with these exceptions:

- Buses that are designated by the user at the start of the activity.
- Buses that are indicated by type codes 6 or 7 to be boundary or retained buses.

When replacing generation with an equivalent load, the power flow solution will remain the same, i.e. re-solving the case after the netting process will not produce a different solution.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. What is different from most of the other subsystem selections is that, for netting generation, the user can choose to identify those generators that will be netted (see the inside the selected subsystem in the figure) or choose to identify those generators that will not be netted (see the outside option).

Activity GNET processes all Type 2 and 3 buses specified as *inside* the selected subsystem as follows:

1. The type code is set to 1.
2. A new load entry is introduced at the bus.
3. PL of the new load is set to –PG.
4. QL of the new load is set to –QG.

Activity GNET summarizes processing, listing the number of buses netted with their loads.

When activity GNET introduces a load at a bus, the load is assigned the identifier 99. If a load with the identifier 99 already exists at the bus, an unused two-digit identifier is assigned.

Because activity GNET processes only Type 2 and 3 buses, those generator buses that have been designated as boundary buses, and hence have type codes of 6 or 7, do not have their generation netted (refer to Section 5.2.5, "Reading Power Flow Raw Data Files Created by Previous Releases of PSS®E" and activity EXTR).

10.4.1. Example of Net Generation with Load

In the *savnw.sav* power flow case there are three areas each of which has two generation buses. Figure 10.6, "Generation and Load in Power Flow Case savnw.sav" shows the machine and load data from the spreadsheet. Note that the Load tab data has been compressed to show the Load P and Q.

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If the LIGHTCO Area is selected as the subsystem inside of which all generators will be netted, the two generators there, on bus 206 and bus 211 will be netted. In the listing of loads, there will be two new loads each assigned the identifier 99. If a load with the identifier 99 already exists at the bus, an unused two-digit identifier is assigned. The two new loads are shown in the spreadsheet view in Figure 9-7.

Prior to terminating the netting process, PSS®E prints a message at the *Progress* tab, or destination of the user's choice, stating:

```
GENERATION AT nnnnn BUSES NETTED WITH THEIR LOAD
```

Because activity GNET processes only Type 2 and 3 buses, those generator buses that have been designated as boundary buses, and hence have type codes of 6 or 7, do not have their generation netted. Furthermore, the netting process is not sensitive to interruption by the user.

Generation

Bus Number	Bus Name	Id	Code	In Service	Pgen (MW)	Pmax (MW)
101	NUC-A 21.600	1	2	☑	750.0000	810.0001
102	NUC-B 21.600	1	2	☑	750.0000	810.0001
206	URBGEN 18.000	1	2	☑	800.0000	900.0000
211	HYDRO_G 20.000	1	2	☑	600.0000	616.2500
3011	MINE_G 13.800	1	3	☑	258.6570	900.0000
3018	CATDOG_G 13.80	1	2	☑	100.0000	117.0000

Bus | Plant | **Machine** | Load | Fixed Shunt | Switched Shunt | Br

Loads

Bus Number	Bus Name	Id	In Service	Scalable	Pload (MW)	Qload (Mvar)
153	MID230 230.00	1	☑	☑ Yes	200.0	100.0
154	DOWNTN 230.00	1	☑	☑ Yes	600.0	450.0
154	DOWNTN 230.00	2	☑	☑ Yes	400.0	350.0
203	EAST230 230.00	1	☑	☑ Yes	300.0	150.0
205	SUB230 230.00	1	☑	☑ Yes	1200.0	700.0
3005	WEST 230.00	1	☑	☑ Yes	100.0	50.0
3007	RURAL 230.00	1	☑	☑ Yes	200.0	75.0
3008	CATDOG 230.00	1	☑	☑ Yes	200.0	75.0

Bus | Plant | Machine | **Load** | Fixed Shunt | Switched Shunt

Figure 10.6. Generation and Load in Power Flow Case savnw.sav

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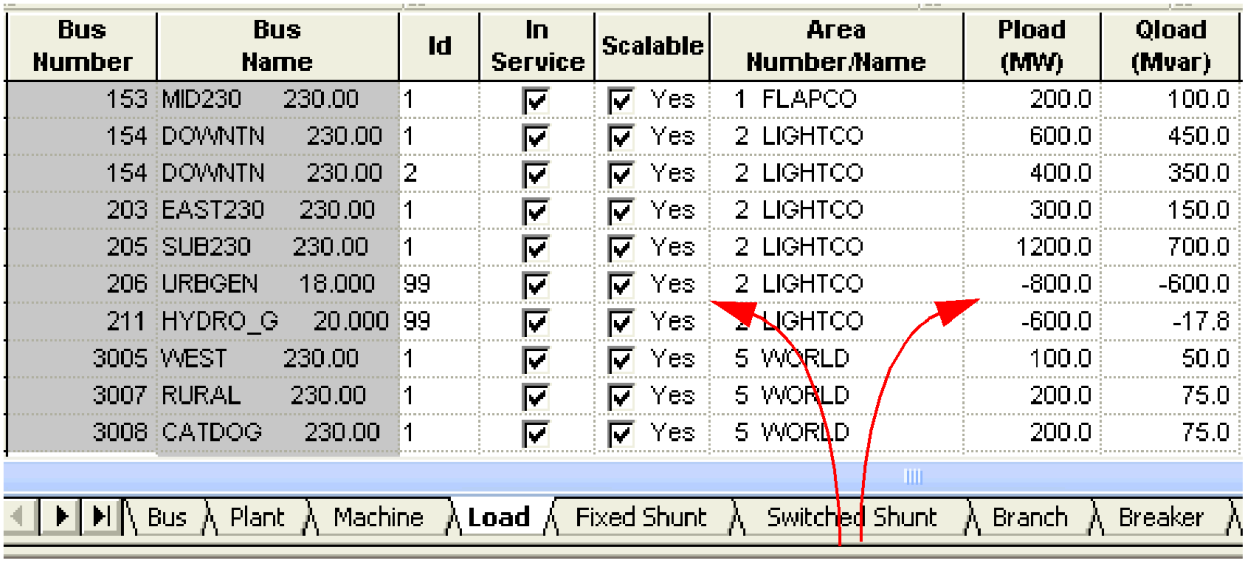

Figure 10.7. Result of Netting Generation in the LIGHTCO Area

Additional Information
PSS®E GUI Users Guide, Net Generation with Load
PSS®E Command Line Interface (CLI) *Users Guide*, Net Generation with Load at All Non-Boundary Buses

10.5. Net Generation with Load Outside of a Subsystem

NETG

Run Activity NETG - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Net generation with load (NETG/GNET/ImNet) Tab
Run Line Mode Activity NETG - CLI
ACTIVITY? >>**NETG** USER SPECIFIES EXCEPTIONS (I.E., THOSE NOT TO BE NETTED) ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The generation netting activity NETG changes the in-service generation to negative MVA load at all Type 2 and 3 buses *except* those within the subsystem specified by the user.

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The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV.

Activity NETG processes all Type 2 and 3 buses except those specified as *outside* the selected subsystem as follows:

1. The type code is set to 1.
2. A new load entry is introduced at the bus.
3. PL of the new load is set to –PG.
4. QL of the new load is set to –QG.

Activity GNET summarizes processing, listing the number of buses netted with their loads.

When activity NETG introduces a load at a bus, the load is assigned the identifier 99. If a load with the identifier 99 already exists at the bus, an unused two-digit identifier is assigned.

Because activity NETG processes only Type 2 and 3 buses, those generator buses that have been designated as boundary buses, and hence have type codes of 6 or 7, do not have their generation netted (refer to Section 5.2.5, "Reading Power Flow Raw Data Files Created by Previous Releases of PSS®E" and activity EXTR).

Additional Information
PSS®E GUI Users Guide, Net Generation with Load *PSS®E* Command Line Interface (CLI) *Users Guide*, Net Generation with Load at All Buses Except Boundary Buses

10.6. Convert Induction Machines to Load

ImNet

NetIm

Run Activity ImNet - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Net generation with load (NETG/GNET/ImNet) Tab
Run Line Mode Activities ImNet, NetIm - CLI
ACTIVITY? >>**IMNET** ACTIVITY? >>**NETIM**
Interrupt Control Codes
None

The induction machine netting API routine, ImNet, converts the boundary conditions of in-service induction machines at Type 1, 2 and 3 buses to MVA load, and changes the status of each machine processed to out-of-service.

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The ImNet API routine includes the following among its inputs:

- A designation of the subsystem in which, or outside of which, induction machines are to be processed; refer to Section 4.8, "Subsystem Selection".
- An option to process induction machines either within, or outside of, the specified subsystem.

Two line mode activities initiate the ImNet API routine: activity IMNET is used to process induction machines *within* a subsystem, and activity NETIM is used to process induction machines *outside of* a subsystem.

The ImNet API routine processes the induction machines at a bus as follows:

1. The sum of the powers flowing from the bus to all of the in-service induction machines at the bus is calculated.
2. A new load is introduced at the bus. The load is assigned the identifier "99". If a load with the identifier "99" already exists at the bus, an unused two-digit identifier is assigned.
3. The constant MVA load component of the new load is set to the sum of powers calculated in 1. The constant current and admittance components of the new load are set to zero.
4. The status of each in-service induction machine at the bus is set to out-of-service.

The ImNet API routine summarizes its processing, listing the number of induction machines that were replaced by load, and the number of buses involved.

When replacing an induction machine with an equivalent load, the power flow solution will remain the same; i.e., re-solving the case after the netting process will not produce a different solution.

Because the ImNet API routine processes only in-service buses, those induction machine buses that have been designated as boundary buses, and hence have type codes of 6 or 7, do not have their induction machines converted to load (refer to Section 5.2.5, "Reading Power Flow Raw Data Files Created by Previous Releases of PSS®E" and activity EXTR).

Additional Information
PSS® E Application Program Interface (API), ImNet

10.7. Equivalencing Radial Buses

EQRD

Run Activity EQRD - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Equivalence radial / 2-point buses (EQRD/RDEQ) Tab
Run Line Mode Activity EQRD - CLI
ACTIVITY? >>**EQRD** ENTER 1 TO EQUIVALENCE RADIAL BUSES ONLY: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

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The radial bus equivalencing activity EQRD build an electrical equivalent for radial and, optionally, two-point Type 1 buses, those buses connected to only two other buses. Activity EQRD uses the same equivalencing algorithm as activity EEQV.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. As in netting generation, the user can choose to identify those buses that will or that will not be calculated as an electrical equivalent. See PSS®E GUI Users Guide, Building a Network Equivalent; buses that *will* be calculated as an electrical equivalent are selected by the Net generation inside the selected subsystem option and buses that will not be calculated as an electrical equivalent are selected by the Net generation outside the selected subsystem option in *[Equivalence Networks]* dialog.

Activity EQRD builds an electrical equivalent for buses *inside* the specified subsystem of the working case.

If sequence data is contained in the working case, the zero sequence network is calculated as an electrical equivalent along with the positive sequence.

The radial equivalent process determines the buses to be calculated as electrical equivalents based upon their current electrical connections. Suppose, for example, that bus I is a Type 1 bus connected to two other buses by in-service branches and that there is an out-of-service branch from bus I to a third bus. Assuming that bus I is not in the subsystem to be exempted from the equivalencing process and that the two-point bus equivalencing option was enabled, bus I will be calculated as an electrical equivalent.

It is neither necessary nor helpful to build an electrical equivalent of radial or two-point buses prior to building a global electrical equivalent of a network or subsystem.

Buses that are always retained by the radial/two-point network reduction function are those:

- Connected by three-winding transformers
- With dc line converters connected to them
- With FACTS devices connected to them
- With GNE devices connected to them.

Whenever a two-point bus and its connected branches are calculated as electrical equivalents, the resulting equivalent branch is assigned the smaller RATEA, RATEB, and RATEC of the connected branches. If load existed at the two-point bus, the ratings of the equivalent branch should be checked.

Whenever a two-point bus and its connected branches are calculated as electrical equivalents, the length assigned to the equivalent branch is the sum of the lengths of the equivalent paths. If parallel branches existed between a retained bus and the equivalenced bus, the length of that path is assumed to be the length of the longest of the parallel branches.

Equivalent branches, introduced by the radial equivalencing process, are assigned the circuit identifier 99. Equivalent branches between the same pair of buses have descending circuit identifiers assigned starting at 99.

Equivalent loads introduced by the radial equivalencing process, are assigned the load identifier 99. If a load with the identifier 99 already exists at a retained bus, the process adds any equivalent load to the load already modeled as load 99. If such a load is associated with an adjustable bus load table for which the load multiplier is zero, a mismatch is introduced at the corresponding retained bus.

The user may specify any of the following options:

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Table 10.3. Radial Bus Equivalencing Options

Specification	Description
Equivalence radial buses only	Limit building an electrical equivalent to only radial buses, otherwise both radial buses and those buses connected to only two other buses are to be eliminated.
Suppress equivalence of transformers	Enable to exclude transformer branches from the equivalencing process.
Suppress equivalence of zero impedance lines	Enable to exempt from a bus electrical equivalent any branches that are treated as zero impedance lines (refer to Section Zero Impedance Lines).
Suppress equivalence of controlled buses	Enable to suppress building an electrical equivalent of buses where voltage or reactive power output is controlled by remote generation, switched shunts, VSC dc line converter, and/or FACTS device shunt element.
Suppress equivalence of mutually coupled branches	If sequence data is contained in the working case, and zero sequence mutual couplings have been specified, any branch (and the buses it connects) involved in a coupling is suppressed, when this option is enabled.
Suppress equivalence of area boundary buses	Keeps area boundaries intact when enabled.
Suppress equivalence of zone boundary buses	Keeps zone boundaries intact when enabled.
Apply transformer impedance correction to zero sequence	If the actual positive sequence impedance of any transformer being calculated as an electrical equivalent differs from its nominal value (refer to Transformer Impedance Correction Table Data), the user may apply transformer impedance correction. If applied, the zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. Otherwise, all zero sequence transformer impedances are left at their nominal values (i.e., the values entered in activities RESQ, TRSQ, or SQCH). The same treatment applies to *all* transformers to be calculated as electrical equivalents that are not at nominal impedance.

10.7.1. Operation of Activity EQRD

Activity EQRD summarizes the number of buses removed from the working case. It rebuilds the data arrays in the working case, compacting them to eliminate gaps created by the deleted equipment. New bus sequence numbers, transformer sequence numbers, and so on are assigned.

At all type 1 buses being eliminated, the powers flowing from the bus to any in-service induction machines are netted with any in-service load at the bus before the equivalent of the bus is calculated.

For any bus where voltage is being controlled by a remote generator, switched shunt, FACTS device shunt element, or VSC dc line converter, an alarm is printed if the controlled bus is deleted while the voltage con-

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trolling equipment is retained. The controlling equipment is then set to control local bus voltage and the scheduled voltage setpoint or band is not changed.

For any bus with voltage controlling equipment that is being controlled by a remote switched shunt, an alarm is printed if the controlled bus is deleted while the controlling switched shunt bus is retained. The switched shunt's control mode is *not* changed.

10.7.2. Example: Equivalencing Radial Buses

Using the *savnw.sav* power flow case, and choosing to build an electrical equivalent for only radial buses, with all other options suppressed, will result in the removal of one radial bus (bus 3007). The load on that bus is moved to the neighboring buses 3005 and 3008 (see Figure 10.8, "Result of Radial Equivalencing Process").

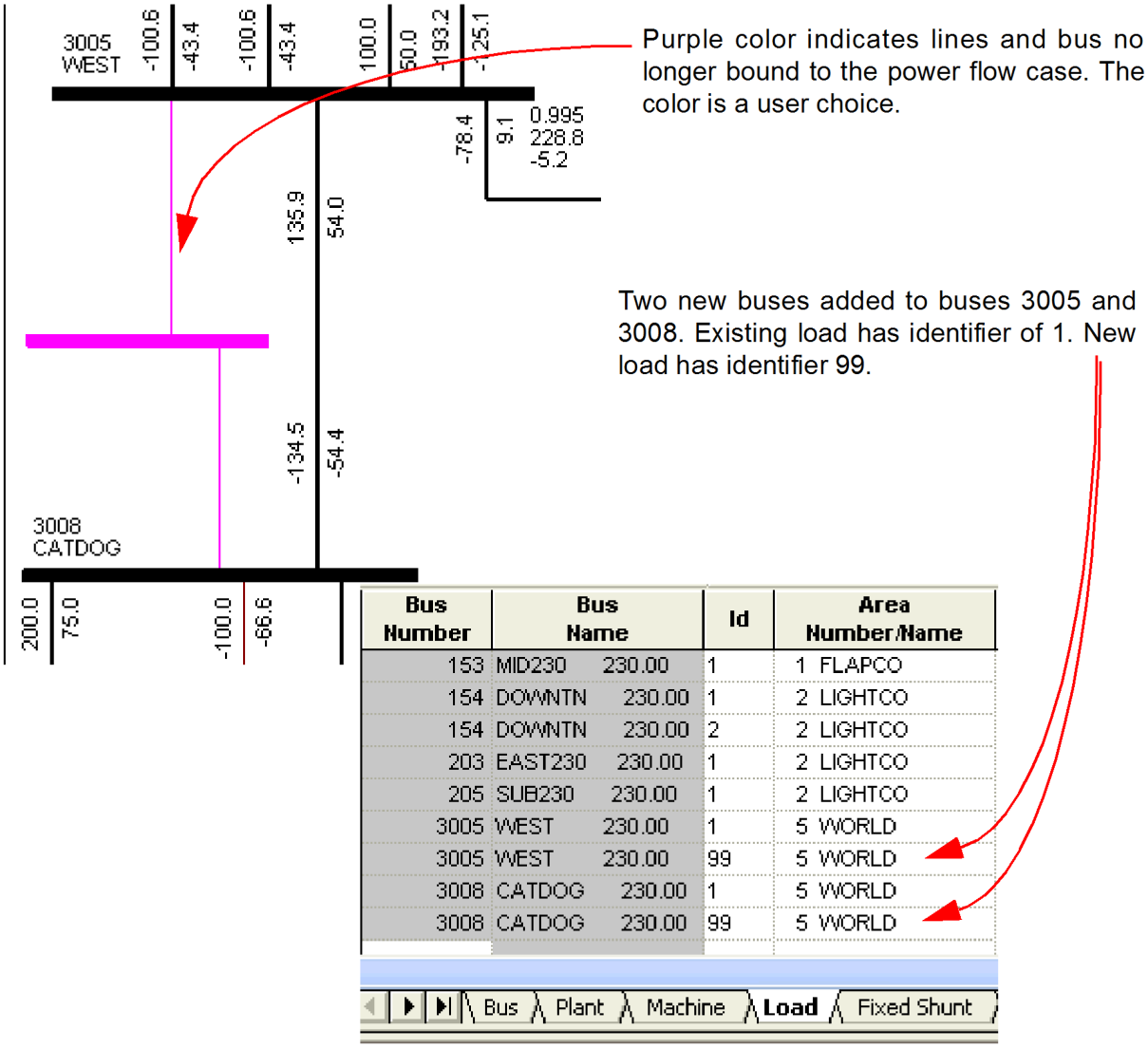

Figure 10.8. Result of Radial Equivalencing Process

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10.7.3. Application Notes

If sequence data is contained in the working case, the zero sequence network is calculated as an electrical equivalent along with the positive sequence.

Activity EQRD determines the buses to be calculated as electrical equivalents based upon their current electrical connections. Suppose, for example, that bus I is a Type 1 bus connected to two other buses by in-service branches and that there is an out-of-service branch from bus I to a third bus. Assuming that bus I is in the subsystem to be calculated as an electrical equivalent and that the two-point bus equivalencing option was enabled, bus I is calculated as an electrical equivalent by activity EQRD.

The use of activity EQRD as a preprocessor to activity EEQV is neither necessary nor helpful.

Buses connected by three-winding transformers are always retained by activity EQRD. The converter station buses of dc lines are always retained by activity EQRD. Similarly, buses with FACTS devices or GNE devices connected to them are always retained.

Whenever a two-point bus and its connected branches are calculated as electrical equivalents, the resulting equivalent branch is assigned the smaller RATEA, RATEB and RATEC (refer to Non-Transformer Branch Data) of the connected branches. If load existed at the two-point bus, the ratings of the equivalent branch should be checked.

Whenever a two-point bus and its connected branches are calculated as electrical equivalents, the length assigned to the equivalent branch is the sum of the lengths of the equivalent paths. If parallel branches existed between a retained bus and the equivalenced bus, the length of that path is assumed to be the length of the longest of the parallel branches.

Equivalent branches introduced by activity EQRD are assigned the circuit identifier 99. Equivalent branches between the same pair of buses have descending circuit identifiers assigned starting at 99.

Equivalent loads introduced by activity EQRD are assigned the load identifier 99. If a load with the identifier 99 already exists at a retained bus, activity EQRD adds any equivalent load to the load already modeled as load 99; if such a load is associated with an adjustable bus load table for which the load multiplier is zero, a mismatch is introduced at the corresponding retained bus.

Equivalent bus shunts introduced by activity EQRD are assigned the shunt identifier 99. If a shunt with the identifier 99 already exists at a retained bus, activity EQRD adds any equivalent shunt admittance to the fixed shunt already modeled as shunt 99.

Additional Information
PSS®E GUI Users Guide, Radial and 2-Point Buses *PSS®E* Command Line Interface (CLI) *Users Guide*, Radial and 2-Point Buses

10.8. Equivalencing Radial Buses, with Exception

RDEQ

Run Activity RDEQ - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Equivalence radial / 2-point buses (EQRD/RDEQ) Tab

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Run Line Mode Activity RDEQ - CLI
ACTIVITY? >>**RDEQ** ENTER 1 TO EQUIVALENCE RADIAL BUSES ONLY: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The radial bus equivalencing activity RDEQ builds an electrical equivalent for radial and, optionally, two-point Type 1 buses *outside* the specified subsystem of the working case. Activity RDEQ uses the same equivalencing algorithm as activity EEQV.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV.

The user may specify any of the following options:

Table 10.4. Radial Bus Equivalencing Options, with Exception

Specification	Description
Equivalence radial buses only	Disable to build an electrical equivalent for both radial buses and those buses connected to only two other buses are to be eliminated
Suppress equivalence of transformers	Enable to suppress building an electrical equivalent of transformers.
Suppress equivalence of zero impedance lines	Enable to suppress building an electrical equivalent of zero impedance lines (refer to Section)
Suppress equivalence of controlled buses	Enable to suppress building an electrical equivalent of buses where voltage or reactive power output is controlled by remote generation, switched shunt, VSC dc line converter, and/or FACTS device shunt element.
Suppress equivalence of mutually coupled branches	If sequence data is contained in the working case, and zero sequence mutual couplings have been specified, any branch (and the buses it connects) involved in a coupling is suppressed, when enabled.
Suppress equivalence of area boundary buses	Keeps area boundaries intact when enabled.
Suppress equivalence of zone boundary buses	Keeps zone boundaries intact when enabled.
Apply transformer impedance correction to zero sequence	If the actual positive sequence impedance of any transformer being calculated as an electrical equivalent differs from its nominal value (refer to Transformer Impedance Correction Table Data), the user may apply transformer impedance correction. If applied, the zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. Otherwise, all zero sequence transformer impedances are left at their nominal values (i.e., the values entered in activities RESQ, TRSQ, or SQCH). The same treatment ap-

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Specification	Description
	plies to *all* transformers to be calculated as electrical equivalents that are not at nominal impedance.

Activity RDEQ summarizes the number of buses removed from the working case. It rebuilds the data arrays in the working case, compacting them to eliminate gaps created by the deleted equipment. New bus sequence numbers, transformer sequence numbers, and so on are assigned.

At all type 1 buses being eliminated, their in-service constant admittance load, fixed shunt, and switched shunt are netted before applying the network reduction equations described in Section 10.2, "Methodology of the Electrical Equivalent". Similarly, the powers flowing from the bus to any in-service induction machines are netted with any in-service constant power load and constant current load at the bus.

For any bus where voltage is being controlled by a remote generator, switched shunt, FACTS device shunt element, or VSC dc line converter, an alarm is printed if the controlled bus is deleted while the voltage controlling equipment is retained. The controlling equipment is then set to control local bus voltage and the scheduled voltage setpoint or band is not changed.

For any bus with voltage controlling equipment that is being controlled by a remote switched shunt, an alarm is printed if the controlled bus is deleted while the controlling switched shunt bus is retained. The switched shunt's control mode is *not* changed.

10.8.1. Application Notes

If sequence data is contained in the working case, the zero sequence network is calculated as an electrical equivalent along with the positive sequence.

Activity RDEQ determines the buses to be calculated as electrical equivalents based upon their current electrical connections. Suppose, for example, that bus I is a Type 1 bus connected to two other buses by in-service branches and that there is an out-of-service branch from bus I to a third bus. Assuming that bus I is not in the subsystem to be exempted from equivalencing and that the two-point bus equivalencing option was enabled, bus I is calculated as an electrical equivalent by activity RDEQ.

The use of activity RDEQ as a preprocessor to activity EEQV is neither necessary nor helpful.

Buses connected by three-winding transformers are always retained by activity RDEQ. The converter station buses of dc lines are always retained by activity RDEQ. Similarly, buses with FACTS devices or GNE devices connected to them are always retained.

Whenever a two-point bus and its connected branches are calculated as electrical equivalents, the resulting equivalent branch is assigned the smaller RATEA, RATEB, and RATEC (refer to Non-Transformer Branch Data) of the connected branches. If load existed at the two-point bus, the ratings of the equivalent branch should be checked.

Whenever a two-point bus and its connected branches are calculated as electrical equivalents, the length assigned to the equivalent branch is the sum of the lengths of the equivalent paths. If parallel branches existed between a retained bus and the equivalenced bus, the length of that path is assumed to be the length of the longest of the parallel branches.

Equivalent branches introduced by activity RDEQ are assigned the circuit identifier 99. Equivalent branches between the same pair of buses have descending circuit identifiers assigned starting at 99.

Equivalent loads introduced by activity RDEQ are assigned the load identifier 99. If a load with the identifier 99 already exists at a retained bus, activity RDEQ adds any equivalent load to the load already modeled as

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load 99; if such a load is associated with an adjustable bus load table for which the load multiplier is zero, a mismatch is introduced at the corresponding retained bus.

Equivalent bus shunts introduced by activity RDEQ are assigned the shunt identifier 99. If a shunt with the identifier 99 already exists at a retained bus, activity RDEQ adds any equivalent shunt admittance to the fixed shunt already modeled as shunt 99.

Additional Information
PSS®E GUI Users Guide, Radial and 2-Point Buses *PSS®E* Command Line Interface (CLI) *Users Guide*, Radial Bus and 2-Point Buses, with Exception

10.9. Converting Net Boundary Bus Mismatch

BGEN

Run Activity BGEN - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Net boundary bus mismatches (BGEN) Tab
Run Line Mode Activity BGEN - CLI
ACTIVITY? >>**BGEN** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

Whenever a subsystem is removed from a solved power flow case, there will be mismatches at those buses that are boundary buses. The boundary bus balance activity BGEN converts the mismatch at boundary buses to equivalent load and/or generation. It is intended to be used following the removal of a subsystem from a solved working case, with the flows to the deleted subsystem replaced by load and/or generation at those retained buses that were directly connected to the removed subsystem.

The user specifies the type of boundary condition modifications to be used from the following options:

Table 10.5. Net Boundary Bus Mismatch Options

Option	Description
Generator for inflow, load for outflow (default)	Model the inflow of power to the retained area as a generator and the outflow of power as a load. A new machine is introduced at any boundary bus at which there was a net inflow of active power from the removed subsystem into the retained subsystem; at boundary buses where there was a net outflow of active power, a new load is introduced.
All equivalent generators	A new machine is introduced at each boundary bus regardless of the direction of power flow at the boundary.

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Option	Description
All equivalent loads	Each flow to the deleted subsystem is replaced by a new load.

The user may process the entire working case or specify the subsystem for mismatch netting (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV.

When a bus has a mismatch in excess of 0.5 MVA, it is assumed to be a boundary bus and a new load or machine, as selected above, is introduced. Boundary buses are reported in ascending bus number order or alphabetical order, as described in Section 4.9, "Subsystem Reporting".

When activity BGEN introduces a machine at a bus, it assigns the machine identifier 99; if a machine 99 already exists at the bus, an unused two-digit identifier is used. Similarly, when activity BGEN introduces a load at a bus, the load is assigned the identifier 99; if a load with the identifier 99 already exists at the bus, an unused two-digit identifier is assigned.

Activity BGEN may be terminated by entering the AB interrupt control code.

Additional Information
PSS®E GUI Users Guide, Net Boundary Bus Mismatch *PSS®E* Command Line Interface (CLI) *Users Guide*, Net Boundary Bus Mismatch

10.10. Building a Three-Sequence Electrical Equivalent

SCEQ

Run Activity SCEQ - GUI
Power Flow > Equivalence Networks... *[Equivalence Networks]* Build three sequence equivalent (SCEQ) Tab
Run Line Mode Activity SCEQ - CLI
ACTIVITY? >>**SCEQ** USER SPECIFIES SUBSYSTEM TO BE EQUIVALENCED ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The three sequence equivalencing activity SCEQ constructs network equivalents of the positive and zero sequence networks, and calculates source impedances at equivalent source nodes for all three sequences in preparation for the unbalanced fault analysis activities of PSS®E. The user may save the equivalents in the form of a Power Flow Raw Data File and a Sequence Data File by specifying filenames.

10.10.1. Creating Short Circuit Equivalents

While network reduction for fault analysis work could be handled with the standard equivalent handling activities covered in the previous sections, PSS®E provides the user with a superior process for creating short-circuit equivalents that is designed specifically for fault analysis applications.

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This process applies all of its operations to all three sequences in parallel, using the same topological boundary definitions for each, as shown in Figure 10.9, "Pictorial Image of Power Flow Case with Equivalenced Sequence Networks". This process handles all steps of the equivalencing process automatically so that its single execution, in effect, accomplishes the following functions:

- Separates the external system in the power flow case.
- Builds an equivalent.
- Replaces the detailed external system model with the equivalent.
- Rejoins the study system onto the equivalent.

A single step in the process replaces the full system model in the power flow case with a reduced model consisting of the detailed study system and an equivalenced external system. The boundary buses of the external system are retained so that tie branches appear as real system elements in the reduced system model.

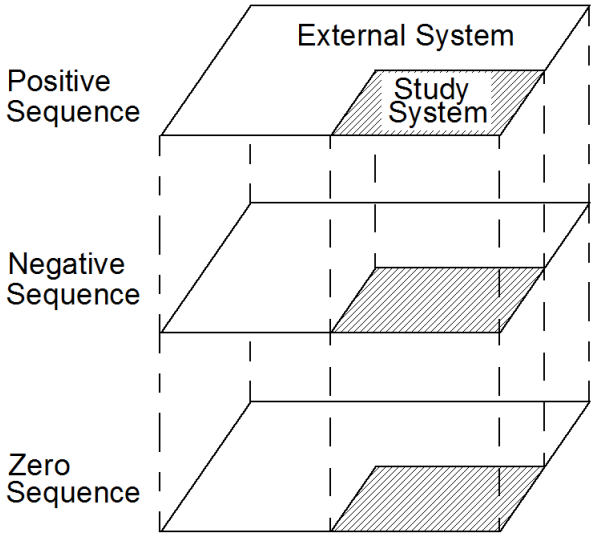

Figure 10.9. Pictorial Image of Power Flow Case with Equivalenced Sequence Networks

The short-circuit equivalencing process performs a simple network reduction without regard to loads, transformer phase shift, or pre-event voltages. All generators in the external system are represented by simple Norton equivalents so that their effective impedances (i.e., Norton shunt admittances) and positive-sequence source currents can be transferred to the boundary buses by standard network mathematics. The process operates on a standard fault analysis model but with FLAT pre-event voltages.

The network reduction is performed by setting up the partitioned admittance matrix of the external system:

$$\begin{bmatrix} I_b \\ I_n \end{bmatrix} = \begin{bmatrix} Y_{bb} Y_{bn} \\ Y_{nb} Y_{nn} \end{bmatrix} \begin{bmatrix} V_b \\ V_n \end{bmatrix} \quad (10.4)$$

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where:

b	Denotes boundary buses.
n	Denotes buses to be deleted.
I_b, I_n	Contains either zero or generator Norton source current as calculated the generator conversion process.

The generator internal impedances, Z_{pos}, Z_{neg}, Z_{zero}, where appropriate, are included in the network as shunt elements and are accounted for by the diagonal terms of the admittance matrix. The positive-sequence source currents are taken as ISORCE = $(1./Z_{pos})$. The admittance matrix and equivalent source currents of reduced network are then obtained from the elimination formula:

$$(I_b - Y_{bn}Y_{nn}^{-1}I_n) = (Y_{bb} - Y_{bn}Y_{nn}^{-1}Y_{nb})V_b \quad (10.5)$$

After calculating these matrix equivalents of the reduced network, the short-circuit equivalencing process automatically deletes the true data for the external system from the working case and replaces it with data for the equivalent network. This equivalent network comprises:

- Branches with series impedance but no charging capacitance or shunts, and with circuit identifier 99.
- Equivalent generators with rating equal to system MVA base and a nonzero positive-sequence source impedance.
- Negative- and zero-sequence source impedance for each equivalent generator.
- Positive-, negative-, and zero-sequence equivalent shunts.

This equivalent can have an equivalent generator at every boundary bus. As indicated by the left-hand side of Equation 10.5, the source currents of these generators are the compendium of source current of any real generators at the boundary bus plus equivalent source currents representing the effect of generators at deleted buses. The short-circuit equivalencing process replaces all real generators at each boundary bus with a single equivalent generator and identifies it as such by assigning it machine number 9. It also assigns the circuit number 99 to all equivalent branches.

All shunt admittances (charging, reactors, capacitors, etc.) arising in the equivalent, except the generator Norton admittances, are collected together and included in the equivalent as a single shunt admittance at each boundary bus. These shunts and the generator Norton admittances are always connected as shunts to ground when the equivalent is used within PSS®E.

The form of the equivalent is illustrated in Figure 10.10, "Form of Sequence Equivalents Built the Short-Circuit Equivalencing Process". Each sequence equivalent is contained in the working case as fault analysis data for equivalent branches and generators. Each equivalent is joined by real tie branches to the study system, which remains in the working case, completely unaltered.

The generator positive- and negative-sequence source impedances in the power flow case need not be identical. If they are different, the positive-sequence and negative-sequence reduced admittance matrices will be different. The standard PSS®E negative-sequence model format recognizes differences between positive and negative sequence only in the generator source impedances, Z_{pos} and Z_{neg}.

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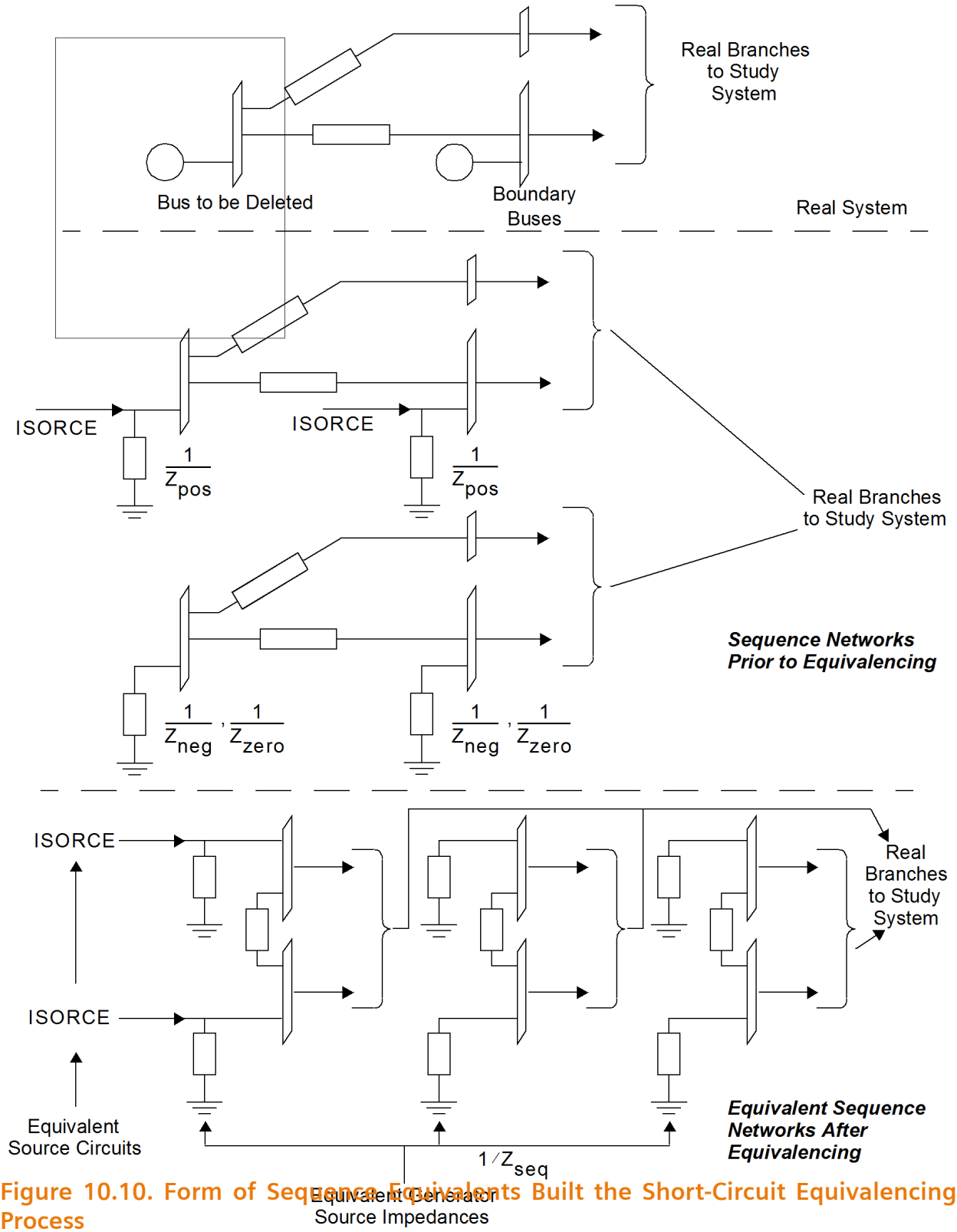

Figure 10.10. Form of Sequence Equivalents Built the Short-Circuit Equivalencing Process

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Hence when the short-circuit equivalencing process encounters a negative-sequence equivalent branch impedance different from the corresponding positive-sequence value, it ignores it and uses the positive-sequence value in both sequences. An alarm message is given whenever the positive and negative-sequence branch impedances differ by more than 5%. This approximation in the reduction and equivalencing of the negative sequence occurs only when generator Z_{pos} and Z_{neg} values are different and is rarely a significant influence on the accuracy of the equivalent.

10.10.2. Constructing a Short Circuit Equivalent

Activity SCEQ works on the basis of classical fault analysis assumptions. External systems, following the terminology of Section 10.1.1, "Nomenclature of Equivalents", are defined according to the specified subsystem; and activity SCEQ handles the data processing functions automatically. Activity SCEQ requires that the working case be in the following format:

1. The network must reflect classical fault analysis assumptions. That is, a uniform voltage profile must be specified, no non-zero in-service loads may be present in any of the sequence networks, and all transformers must be at zero phase shift angle. The set up of these conditions is automated by activity FLAT with classical fault analysis option.

2. The generator source currents must be determined on the basis of the positive sequence generator impedances and the flat generator conditions described in (1) above. This initialization is accomplished with activity CONG using one of the fault analysis positive sequence machine impedances (usually, either transient or subtransient).

If these conditions are not satisfied when activity SCEQ is run, an appropriate alarm message is printed and activity SCEQ is terminated.

10.10.3. Short Circuit Equivalent Options

The user identifies the buses or subsystem to be calculated as electrical equivalents by area, owner, zone, basekV or by individual bus. The equivalencing process generates an alarm and terminates if either none or all buses are to be retained.

Table 10.6. Short Circuit Equivalent Options

Option	Description
Apply impedance correction to zero sequence	If the actual positive sequence impedance of any transformer in the subsystem to be calculated as an electrical equivalent differs from its nominal value, the user is free to specify the treatment of the zero sequence impedance of all such transformers. If enabled, the zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. If disabled (default), all zero sequence transformers are left at their nominal values (i.e., the raw data values entered to populate the power flow case). The same treatment applies to *all* transformers in the subsystem to be calculated as an electrical equivalent that are not at nominal impedance.

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Option	Description
Branch threshold tolerance	Any equivalent branch where magnitude of impedance is greater than this tolerance is not retained in the equivalent. The default value is 10 per unit.
Raw data output file	(*.raw) Optional. The user can save the equivalents in the form of a Power Flow Raw Data File.
Sequence data output file	(*.seq) Optional. The user can save the equivalents in the form of a Sequence Data File.

10.10.4. Operation of Activity SCEQ

If sequence data is not contained in the working case (i.e., activity RESQ has not been executed), an appropriate message is printed, the equivalent is constructed in the normal manner, but only the positive sequence equivalent is valid.

If an equivalent of the entire working case is constructed, Type 5 and 6 buses are retained as Type 1 and 2 buses, respectively. The user may specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Activity SCEQ generates an alarm and terminates if either none or all buses are to be retained.

The portion of the working case to be exempted from equivalencing is (temporarily) removed from the working case, and all radial and two-point Type 1 buses are calculated as electrical equivalents (refer to RDEQ). Then the standard output of activity ORDR is tabulated as the processing of the positive sequence network commences.

The user may specify branch threshold tolerance (default = 10 pu); any equivalent branch where magnitude of impedance is greater than this tolerance is not retained in the equivalent.

If the actual positive sequence impedance of any transformer in the subsystem to be calculated as an electrical equivalent differs from its nominal value (refer to Transformer Impedance Correction Table Data), the user may apply impedance correction to zero sequence. If applied, the zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. Otherwise, all zero sequence transformers are left at their nominal values (i.e., the values entered in activities RESQ, TRSQ, or SQCH). The same treatment applies to *all* transformers in the subsystem to be calculated as electrical equivalents that are not at nominal impedance.

The processing of the positive and negative sequence networks is completed, and the zero sequence calculation commences. The summary output of the optimal ordering function is printed, the zero sequence network equivalent is constructed, and activity SCEQ combines the equivalent and detailed system sections.

When three or more buses are all connected together by zero impedance lines (refer to Zero Impedance Lines), either all may be calculated as electrical equivalents, all may be retained, or one may be retained. If the bus type codes are such that more than one but not all the buses in such a group of buses are to be retained, activity SCEQ prints an appropriate message and retains all buses in the group.

10.10.5. Form of the Equivalent

The equivalent constructed by activity SCEQ is present in the working case along with the unequivalenced portions of the system in the form required for fault analysis work under classical assumptions. There are no operations required to combine system sections.

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In constructing the equivalent, all non-boundary Type 1 and 2 buses in the designated external system are eliminated, and all Type 5 and 6 buses are retained as Type 1 or 2 buses, respectively. Normally, the boundary buses from within the subsystem being calculated as electrical equivalents are those that are connected to buses outside of the specified subsystem. However, when the subsystem to be calculated as an electrical equivalent is specified by bus only, Type 1 and 2 buses specified are calculated as electrical equivalents, with the boundary buses being those buses to which they are connected and which are not among those specified.

All boundary buses that represent equivalent sources are set to Type 2 buses with appropriate Norton source currents and source impedances. Boundary buses with no equivalent source are set to Type 1. Ground paths other than those resulting from equivalent generator impedances are included as positive (and hence, negative) sequence shunts and zero sequence shunts.

The topology of the equivalent network is determined from the positive sequence network. As is the case with the physical system, there may be branches in the positive (and negative) sequence network for which the zero sequence path is open.

If, in the original data, positive and negative sequence generator impedances are equal, the corresponding sequence source and branch impedances in the equivalent will be identical. For generators initially characterized by different positive and negative sequence impedances, the corresponding sequence source and branch impedances will differ in the equivalent. Because the fault analysis activities of PSS®E assume that the branch impedances are identical in the positive and negative sequence networks, activity SCEQ gives an approximation of the negative sequence branch impedances by using those of the equivalent positive sequence network. Activity SCEQ prints a warning message any time the calculated negative sequence branch impedance differs from the positive sequence impedance by more than 5% of the positive sequence value.

10.10.6. Example: Short Circuit Equivalencing of a Power Flow Case

In the *savnw.sav* power flow case, one of the three Areas (FLAPCO) is subject to a short-circuit equivalencing. Figure 10.11, “Pre-Equivalence Power Flow One-Line Diagram Showing the FLAPCO Area Buses” shows the 6 buses in the Area. It can be seen that there are two generators (at buses 101 and 102); one bus (151) with a shunt reactor and one bus (154) with a shunt capacitor. While there are two loads shown on bus 154, the loads themselves are part of Area LIGHTCO.

The short-circuit equivalencing removes the two generator buses and original area branches. An equivalent generator is attached to bus 151. Three new equivalent branches are created linking buses 151, 152 and 153 (see Figure 10.12, “Equivalent FLAPCO Area”). Branch impedances are shown.

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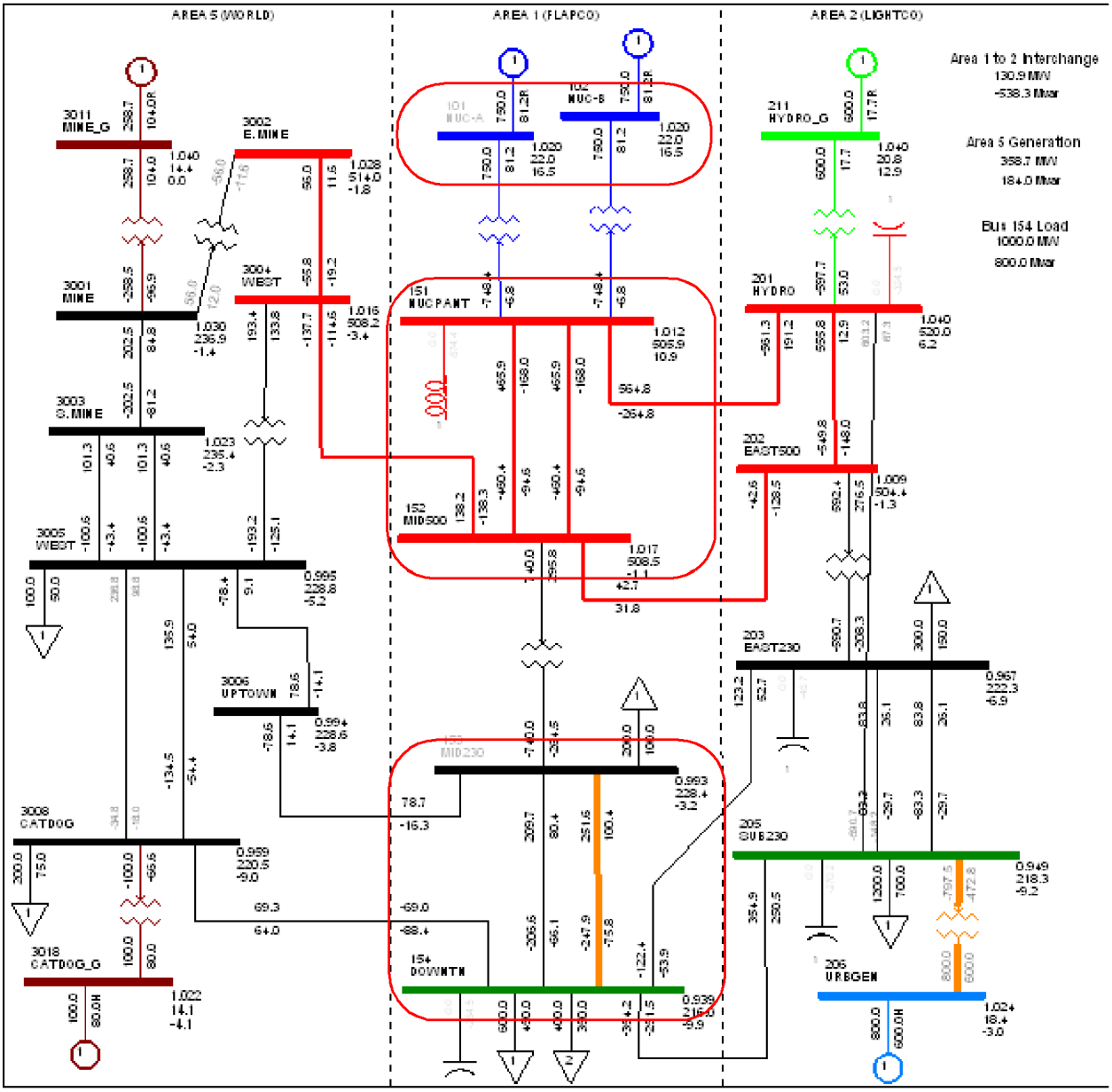

FLAPCO Area Buses

Figure 10.11. Pre-Equivalence Power Flow One-Line Diagram Showing the FLAPCO Area Buses

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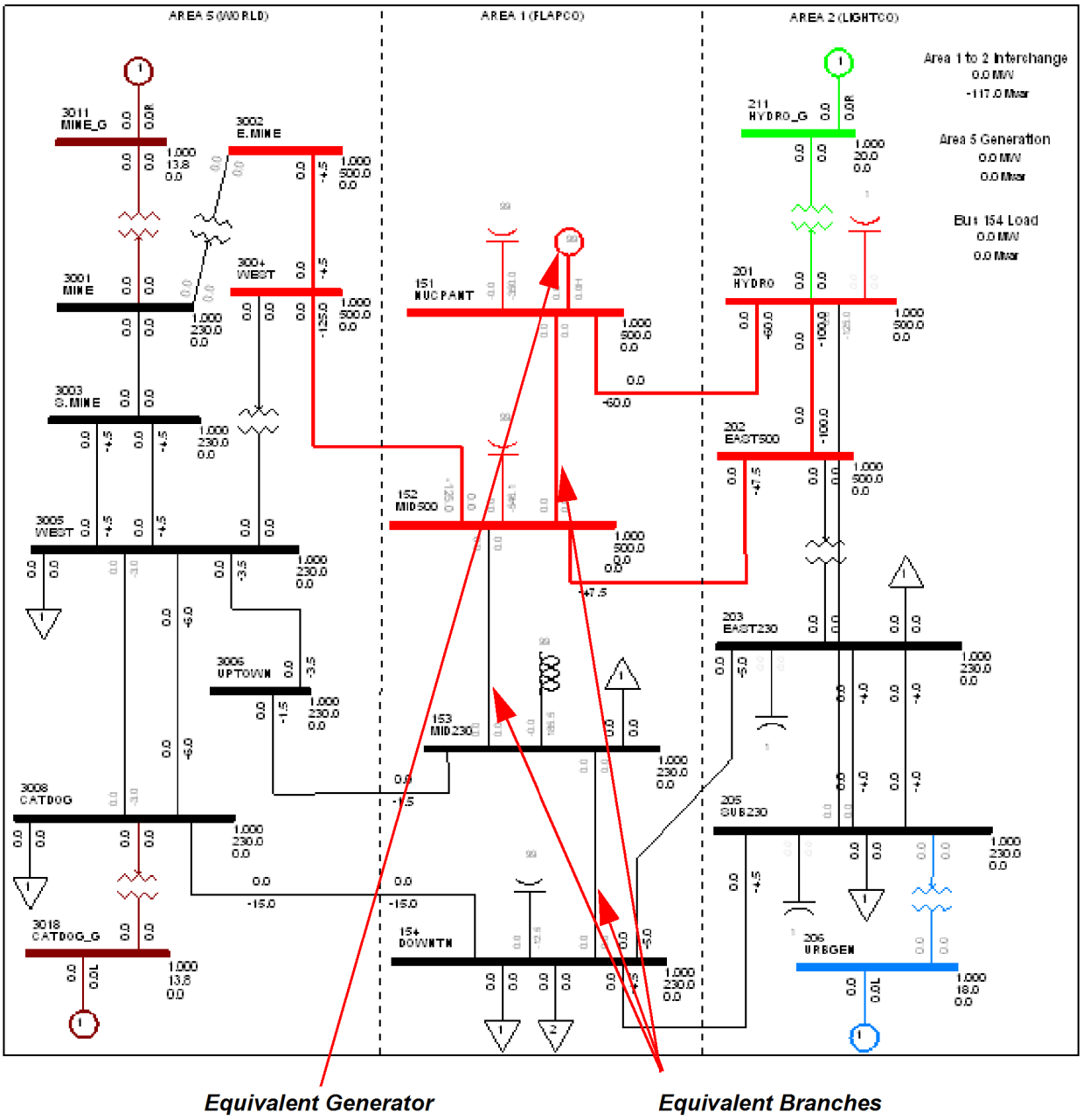

Figure 10.12. Equivalent FLAPCO Area

10.10.7. Application Notes

Activity SCEQ requires that the working case reflect classical assumptions. In setting up this condition, activity CONG using one of the fault analysis machine impedance options must be executed *after* the execution of activity FLAT with classical fault analysis option.

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Activity SCEQ is not capable of handling the case in which zero sequence mutual couplings span the boundary of the external system to be calculated as an electrical equivalent and the portion of the system to be retained in detail. Any such mutual generates an alarm, the mutual is ignored, and processing continues.

Activity SCEQ is not able to build an equivalent of a subsystem in which a dc line, FACTS device, or GNE device is present. If any dc lines, FACTS devices, or GNE devices are encountered, an alarm message is printed but activity SCEQ continues. Upon completion of activity SCEQ, the user should ensure that any such dc lines, FACTS devices, and GNE devices are blocked before using the case in any fault analysis calculation.

Activity SCEQ prints an appropriate error message if the number of equivalent source nodes (i.e., generators), fixed bus shunts, or branches exceeds the capacity limits of PSS®E.

Additional Information
PSS®E GUI Users Guide, Building a Three-Sequence Electrical Equivalent *PSS®E* Command Line Interface (CLI) *Users Guide*, Building a Three-Sequence Electrical Equivalent

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Chapter 11
Unbalanced Fault Analysis

11.1. Overview of Short-Circuit Fault Analysis

Fault analysis in PSS®E is based on a symmetrical component system representation and treated as a direct extension of the power flow activity group. The positive-sequence system model as established in power flow work is used directly in fault analysis. The negative- and zero-sequence system representations for fault analysis work are established simply by appending negative- and zero-sequence parameter values to the parameter lists describing the system for power flow purposes. The power flow file always includes provision for negative- and zero-sequence parameters of *every* system component. This data may be introduced at any time, and after being introduced, it is saved and retrieved as an integral part of a Saved Case.

PSS®E includes several fault analysis procedures.

- Detailed analysis of complicated multiple unbalanced situations presenting a complete system solution and overall system-oriented output comparable to that of a power flow.
- Automatic sequencing calculations intended for the more routine work of examining simple ground faults at a large number of system locations.
- Circuit breaker duty based on ANSI and/or IEC standards
- Circuit breaker Detailed fault analysis
- Positive sequence pi equivalent for representation of transmission line using single pole operation

Each of these will be discussed in this chapter.

The overriding principle of fault analysis data management in PSS®E is that the modeling detail and component status information is dictated by the positive-sequence model. Negative- and zero-sequence data values are held in the power flow file only where their values are different from the corresponding positive-sequence values. In setting up the negative and zero-sequence networks, PSS®E assumes the following:

1. All transmission branches (lines and transformers) have the same impedance, charging, and line-connected shunt characteristics in the negative-sequence as in the positive-sequence.
2. All transformers have phase shift in the negative-sequence equal and opposite to that in the positive-sequence.
3. All zero-sequence branches, both transmission branches, line-connected shunts, and bus connected shunts are assumed to have infinite zero-sequence impedance unless a different value is specified.
4. All constant MVA and constant current load specified in the positive-sequence data are converted automatically to constant shunt admittance in the positive-sequence network.
5. All loads are automatically represented by the same shunt admittance in the negative-sequence as in the positive-sequence unless a different negative-sequence shunt admittance is specified.

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6. Loads are open circuits in the zero-sequence unless represented specifically as grounded with a non-zero by entry of a value of zero sequence shunt admittance.

7. Lines considered zero-impedance branches in the positive-sequence are considered as zero impedance in the negative- and zero-sequence.

It is evident, then, that neglect of the inherent 30° phase shift of wye-delta transformers results in the following:

- Calculated sequence and phase currents that are correct in all branches on the fault side of the transformer.
- Calculated sequence currents that are correct in amplitude but erroneous in phase in all branches that are removed from the fault by a transformer.
- Erroneous values of phase current in all branches separated from the fault by a wye-delta transformer.

11.2. Appending Sequence Data to the Power Flow Case

As described in the previous pages in Section 5.5, Reading Sequence Data, the user is required to prepare a Sequence Data file (type *.seq) in the data record formats shown and in the required order of data categories. After the sequence data has been appended to a power flow case, it accompanies the network as the case is saved and retrieved.

Selecting and opening the file will initiate appending the data. Progress in importing of each data category will show at the *Progress* device (see Figure 11.1, “Output when Appending Sequence Data”).

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```
Reading IC, version number...

Reading machine impedance data...

Reading load data...

Reading non-transformer branch data...

Reading mutual coupling data...

Reading transformer data...

Reading switched shunt data...

Reading fixed bus shunt data...

Reading induction machine data...
```

Figure 11.1. Output when Appending Sequence Data

If the change code parameter IC in the first record of the Sequence Data File is set to one and sequence data had not previously been read for the system in the working case, an alarm is printed and the data is processed as if IC was set to zero. Remember that a code of zero indicates that the data is being appended to the positive sequence model sequence for the first time.

As for positive sequence data, the raw Sequence Data File is read in free format with data items separated by a comma or one or more blanks. Each category of data except the change code is terminated by a record specifying an I value of zero. Specifying a data record with a Q in column one is used to indicate that no more data records are to be supplied.

11.3. Fault Calculation Modeling Assumptions

11.3.1. Detailed Fault Calculation Model

The system under study is modeled by three symmetrical component sequence networks based on the positive-, negative-, and zero-sequence parameters of the three-phase power system elements. The topology of the system, the positive-sequence parameters of all components (except generators in some cases), and the pre disturbance system conditions, are all taken from the power flow Saved Case. Fault analyses may be made with the same level of system modeling as used in a power flow study. Specifically, fault analyses may do the following:

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- Recognize both reactance and resistance and include all actual shunt branches and line charging in the three sequence networks.
- Recognize both the magnitude ratio and phase shift of all transformers, including the inherent shift of delta-wye transformers if it is entered in the power flow data.
- Recognize the actual spread of internal voltage magnitude and phase angle of generators as initialized from a solved power flow case.
- Recognize loads by converting them to equivalent constant shunt admittance.

The level of system modeling detail used in a fault analysis calculation is controlled by manipulating the positive-sequence (or power flow) data in the working case into the required form before commencing fault analysis work. The detailed, unbalanced fault analyses activities, described in Section 11.6, "Performing Fault Analysis with Multiple Unbalances", usually operate on the assumption that the system is modeled in the highest level of detail. In these activities, although results corresponding to a simplified modeling basis are obtained if appropriate elements of data have null values, simplified calculating algorithms are not used. Therefore, no computing time advantage is gained by simplifying the system model. One advantage to simplifying the model, as described in Special Conditions for Fault Calculations is to allow comparison of PSS®E results with those obtained from other software packages which usually use a more simple or classical model.

11.3.2. Detailed Fault Calculation Models for DC Lines and FACTS Devices

If any unblocked dc lines or in-service FACTS devices are present in the working case, the user can specify their treatment in the fault analysis solution. The options are to:

- Block the device: dc lines and FACTS devices are treated as open circuits (i.e., fully blocked bridges) in all three sequences, regardless of their actual prefault loadings as given by the initial condition power flow.
- Represent as load: the apparent ac system complex loads are converted to positive sequence constant admittance load at the buses at which these quantities are injected into the ac system during normal power flow work. In the negative and zero sequence networks, dc lines and series FACTS devices are represented as open circuits. The equivalent positive-sequence shunt admittance is derived from the values of PAC and QAC given by the initial condition power flow at each converter ac bus.

Neither of these two representations should be regarded as exact. The first may be regarded as reasonable for the calculation of fault-current duty on circuit breakers because converter controls are usually designed to limit their fault currents to values equal to or less than normal load current.

Only one of these options may be selected in any execution of the unbalanced network solutions. The selected option applies to *all* dc lines and FACTS devices in the working case. The default handling of these devices is to block.

11.3.3. Simplified Fault Calculation Model

The multiple unbalanced fault calculation described in SCMU and the automatic sequencing fault calculations described in ASCC base their calculations on a fully detailed pre-fault initial condition corresponding to a solved power flow.

Since the use of a simplified model representation is often applicable in the automatic sequencing fault calculations, enabling the flat conditions option of activity ASCC results in the application of the following simplifying assumptions:

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- A uniform voltage profile of unity magnitude at zero phase angle is assumed.
- Generator and induction machine powers are assumed to be zero.
- Loads are neglected in the positive and negative sequence networks.
- Grounded loads are represented in the zero sequence network.
- Fixed bus shunts and switched shunts are neglected in all three sequence networks.
- Line charging capacitances and line connected shunt elements are neglected in all three sequence networks.
- All transformers, including generator step-ups, are assumed to be at nominal ratio, zero phase shift angle.
- Transformer magnetizing admittances are neglected in the positive and negative sequence networks.
- The zero sequence ground ties created by grounded transformer windings are represented.
- Dc lines and FACTS devices are ignored.

When different degrees of simplification than are supplied by the flat conditions option are needed, they may be imposed on the system model before initiating the fault calculations. Such special modeling assumptions are usually implemented with activity FLAT (see Section 11.10, "Setting up Classical Fault Analysis Conditions"). When specifying simplifying assumptions with activity FLAT, note that:

- the simplifying conditions are established in the system model before initiating the fault calculation;
- the flat conditions option of activity ASCC should not be selected;
- the fault calculations assume the detailed calculation model, but they use network parameters that have been set to appropriate flat condition values according to the options specified to activity FLAT.

11.3.4. Special Conditions for Fault Calculations

As mentioned in the previous sections, the detailed analysis (Section 11.4, "Detailed Unbalanced Fault Types") can apply simplifying assumptions for the model. Those conditions are required to be established prior to initiating the calculations.

For the automatic sequencing calculations, similar simplifications can be made to the model but the decision for the user can be made during initiation of the calculations.

Other fault calculations can require special modelling assumptions (refer to activity FLAT).

11.4. Detailed Unbalanced Fault Types

The multiple, unbalanced network solution allows the user to apply simultaneously, at any bus, on any phase, any or all of the following unbalances or faults:

11.4.1. Bus Faults

- One or two single-line-to-ground faults (L-G) with specified fault impedances, Z_F and Z_G (see Figure 11.2, "Using L-G and L-L-G Fault Combinations").

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- One or two line-to-line (L-L) or double-line-to-ground (L-L-G) faults with specified fault impedances, ZF and ZG

Three-phase bus faults are simulated with a combination of one L-G fault and one L-L-G fault. Consequently, if a three-phase bus fault is simulated, there remain one L-G and one L-L-G fault application available. PSS®E assumes that any three-phase fault is simulated using the second single-line-to-ground fault and the second double-line-to-ground fault with all fault impedances set to 0.0+j0.0. Figure 11.2, "Using L-G and L-L-G Fault Combinations" shows the fault combinations together with the impedances available for selection. It shows how a double-line-to-ground fault can be converted to a line-to-line fault by assuming an infinite impedance for the impedance to ground Z_G.

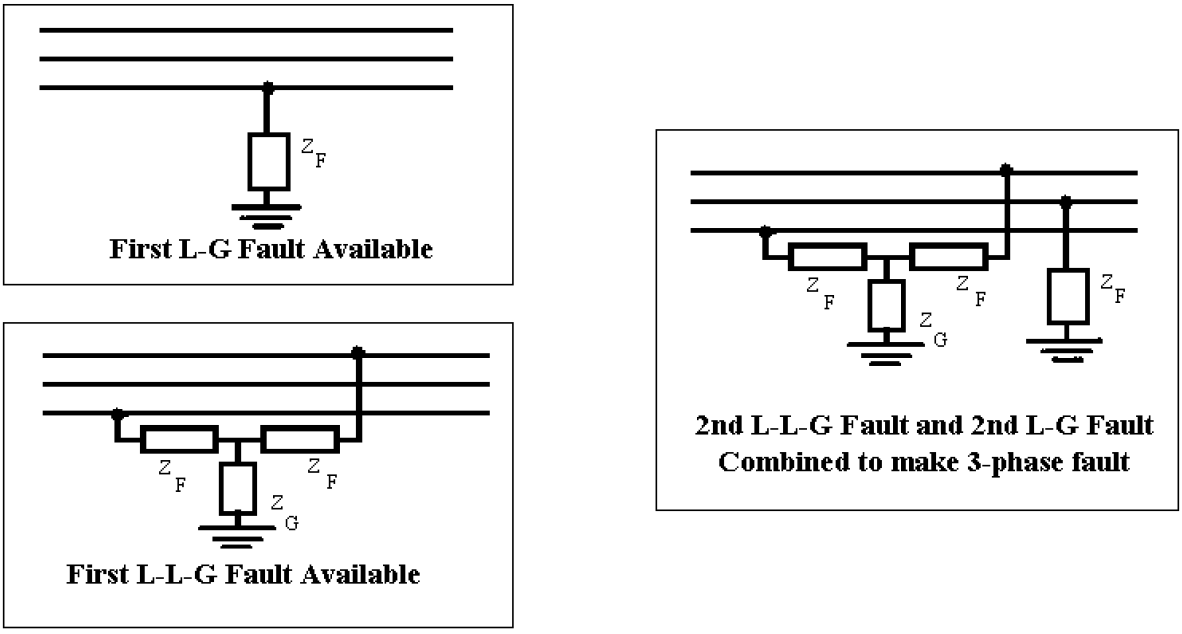

Figure 11.2. Using L-G and L-L-G Fault Combinations

11.4.2. Phase Closed Unbalances

The single and double phase closed unbalances are applicable to represent discrete components such as series capacitors and jumpers, but are not applicable to transmission lines. The unbalance simulation places series elements of the specified *phase* impedance between designated buses in the selected phases (see Figure 11.3, "Phase Closed Series Unbalances"). The unbalances available are:

- Single phase, with specified impedance, closed between a pair of buses.
- Two phases, with equal specified impedance, closed between a pair of buses.

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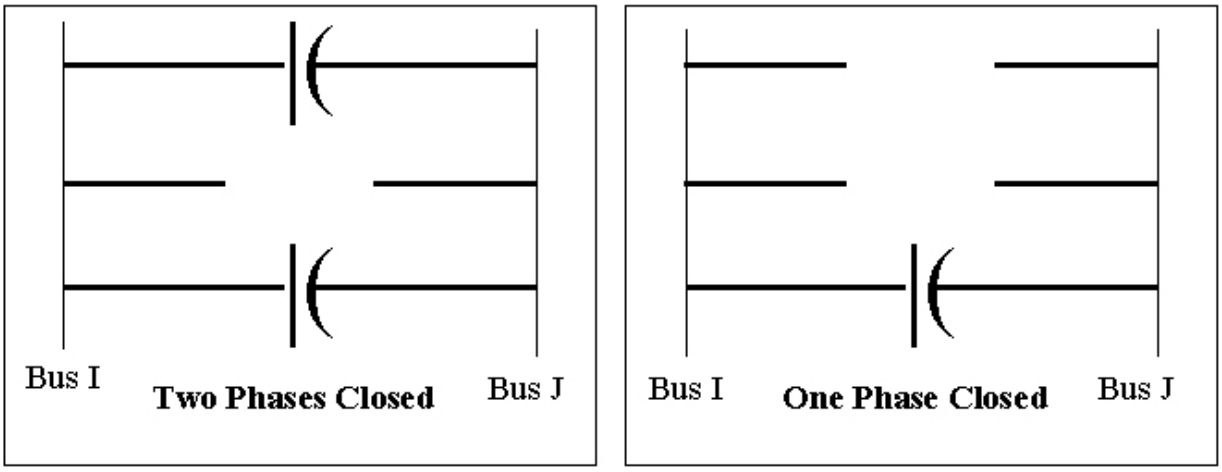

Figure 11.3. Phase Closed Series Unbalances

It should be noted that phase impedances will be placed in parallel with any branch which already exists, in the power flow case, between the selected buses. If the user wishes to use these models to represent an unbalance in an existing branch, then that branch must be switched out of services to avoid duplication (see Section 11.6.5, "Phase Closed Unbalances").

11.4.3. Line Faults

The inter sequence solution method used in PSS®E for unbalanced fault calculations can apply faults only at buses. The application of a fault at some point along a line, therefore, requires the insertion of a dummy bus in the line at the fault location. The fault will be applied at the dummy bus. The user could introduce a dummy bus and re-organize the line topology to accommodate this type of fault. This is not necessary. PSS®E provides for the temporary addition of the required dummy buses and automates the rerouting of branches to the dummy buses and splitting of the zero sequence mutual couplings on either side of the faulted dummy bus.

The simulation of a fault at any point on a line is designated as the fault slider. The line may be represented with both ends closed or with one end opened. In either of these two topologies, one of the following may be applied at any point along the line:

1. The *second* L-G fault.

2. The *second* L-L-G fault.

3. A three phase fault utilizing both the *second* L-G fault and the *second* L-L-G fault.

The dummy bus, introduced at the fault point is automatically numbered 999999. If one end of the faulted line is opened and the fault point is not at the line end position, a second dummy bus, numbered 999998, is introduced at the opened end (see Figure 11.4, "Allocation of Dummy Buses for In-Line Slider Faults").

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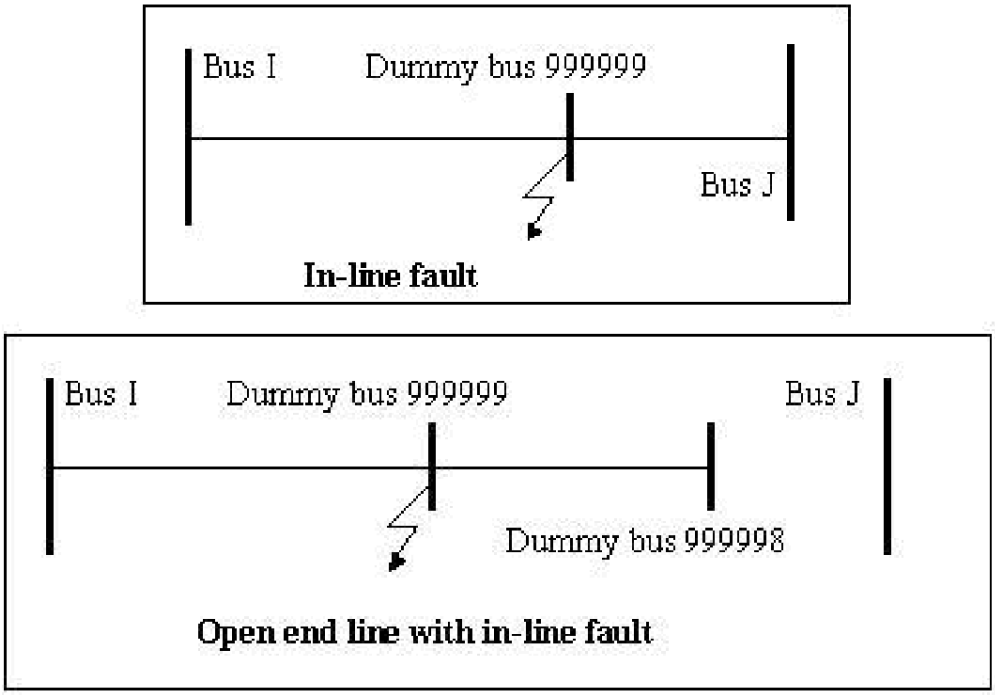

Figure 11.4. Allocation of Dummy Buses for In-Line Slider Faults

11.5. Preparing Sequence Network for Unbalanced Network Solution

SEQD

Run Activity SEQD - GUI
Fault > Setup network for unbalanced solution (SEQD)... *[Setup Network for Unbalanced Solution]*
Run Line Mode Activity SEQD - CLI
ACTIVITY? >>**SEQD**
Interrupt Control Codes
AB

The sequence network set up activity SEQD prepares the working case for the unbalanced network solution activity SCMU or the separate pole circuit breaker duty activity SPCB. This process involves taking the positive sequence network (i.e., the power flow case) and the various sequence data arrays defining the negative and zero sequence networks, and setting up the SEQD temporary file in the form required by activities SCMU and SPCB. Activity SEQD does not modify the contents of the working case itself.

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If sequence data has not been read into the working case via activity RESQ, an appropriate error message is printed and activity SEQD is terminated.

If any unblocked dc lines or in-service FACTS devices are present in the working case, the user must specify their treatment:

- Block and ignore
- Convert to constant admittance load

If dc lines and FACTS devices are to be represented as load, the apparent ac system complex loads are converted to positive sequence constant admittance load at the buses at which these quantities are injected into the ac system during normal power flow work. In the negative and zero sequence networks, dc lines and series FACTS devices are represented as open circuits.

Only one of these options may be selected in any execution of SEQD and the unbalanced network solutions which follow. The selected option applies to *all* dc lines and FACTS devices in the working case. The default handling of these devices is to block.

If the actual positive sequence impedance of any transformer in the working case differs from its nominal value (refer to Transformer Impedance Correction Tables), the user must specify the treatment of the zero sequence impedance of all such transformers.

If transformer correction is applied to the zero sequence, the zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. Otherwise, all zero sequence transformer impedances are left at their nominal values (i.e., the values entered in activities RESQ, TRSQ, or SQCH). The same treatment applies to *all* transformers in the system that are not at nominal impedance.

Activity SEQD performs the following computations on the working case:

1. All positive sequence loads are converted to fixed shunt admittances on the basis of the voltage at each load bus in the working case. This is functionally equivalent to the use of activity CONL in switching study work.

2. For loads for which a non-zero negative sequence load has been specified, it is used to represent the load in the negative sequence. For loads for which the negative sequence load is zero, the positive sequence value calculated in (1) above is used to represent the load in the negative sequence. Refer to Load Sequence Data.

3. All generator and induction machine positive sequence sources are initialized and fixed to correspond to their generator terminal bus conditions in the working case. This is functionally equivalent to the use of activity CONG in switching study work. This step is performed *unless* generators have previously been converted; refer to Application Notes.

4. An ordering for the positive and negative sequence networks is determined and the standard summary of activity ORDR is printed at the *Progress* device. This is functionally equivalent to the use of activity ORDR in switching study work.

5. The positive and negative sequence admittance matrices are constructed and factorized. In switching study work this function is handled by activity FACT.

6. The zero sequence network is ordered and the standard summary is printed.

7. The zero sequence network admittance matrix is constructed and factorized.

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If, in the process of building the sequence network admittance matrices, isolated buses are detected, they generate an alarm. Isolated buses in the positive and negative sequence networks indicate an improperly specified power flow case. Isolated buses in the zero sequence network, although they have generated alarms, are permitted and require no special treatment. When the fault analysis warning option is disabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the isolated bus tabulations are suppressed. In either case, the total number of isolated buses in the sequence networks is tabulated.

Following completion of the processing of the zero sequence network, any three-winding transformers with two windings in-service and one winding out-of-service are tabulated (refer to Zero Sequence Transformer Data). The user should verify that the zero sequence impedances and connection codes of the two in-service windings result in the proper zero sequence modeling of such three-winding transformers.

Activity SEQD is called without being specifically selected during the execution of activity ASCC and during calculation and application of unbalanced bus fault and of branch unbalance.

The isolated bus summaries may be terminated by entering the AB interrupt control code.

11.5.1. Application Notes

The processing of activity SEQD assumes that the working case is in the following condition:

1. The bus boundary conditions and transmission network modeling represent a valid pre-fault condition. The level of detail in network modeling may span the spectrum from classical short circuit study assumptions through full power flow style representation with the initial condition being a solved power flow case.

2. The power flow generator data items defining machine bases and step-up transformer impedances (if the step-up transformer is not explicitly represented as a network branch) must be included in the working case (refer to Generator Data and Section 13.2.3, "Generator Step-Up Transformers").

3. The sequence data arrays must have been filled with the data describing the negative and zero sequence networks, along with the three sequence generator impedances for all in-service machines. Refer to Section 5.6.1, "Sequence Data File Contents".

In unbalanced fault analysis work, generators are represented in the positive sequence in exactly the same way as in power flow cases after execution of activity CONG with generator conversion using fault analysis positive sequence machine impedance ZPOS (see also Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent"). Generators are represented in the negative and zero sequences by the impedances ZNEG and ZZERO connected directly to ground at their terminal buses.

It is often appropriate to model a generator by a different positive sequence impedance in fault analysis than in switching and dynamic simulation studies. The fault analysis activities handle this requirement by using a special positive sequence generator characteristic impedance, ZPOS, in place of the impedance ZSORCE that is associated with the generator in switching and dynamics applications.

It is quite common, and perfectly valid, to have the generator terminal bus isolated in the zero sequence network. This is, in fact, the usual case because the majority of generator step-up transformers are delta connected on the generator side and wye connected on the high side. This is the assumption that is applied in the generator modeling when the step-up transformer is represented as part of the generator data (i.e., XTRAN is non-zero). Activity SEQD prints an alarm message for any generator with non-zero values of both XTRAN and ZZERO and uses only XTRAN in setting up the zero sequence ground tie at the Type 2 (high side) bus.

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The usual network initial condition for unbalanced fault work is a complete solved power flow level model. Through activity FLAT, the user may construct a system model with varying lesser degrees of detail.

The normal sequencing of activities in preparing for fault analysis work is:

1. Set up the positive sequence network at whatever level of detail is required.
2. Append the required sequence data to the working case with activity RESQ.
3. Use activity SEQD to set up the sequence networks on the basis of the boundary conditions contained in the working case.
4. Use activity SCMU to apply unbalances and solve the network.

Activity SEQD is normally entered with generators unconverted; activity SEQD then temporarily converts generators using the generator impedances ZPOS (i.e., it does the equivalent of activity CONG with generator conversion using fault analysis positive sequence machine impedance ZPOS). If generators are converted when activity SEQD is selected, the generator conversion calculation of activity SEQD is bypassed, and the results of the prior execution of activity CONG are used. This is the case, for example, when an equivalent has been constructed (with activity SCEQ), which requires the prior execution of activity CONG with ZPOS generator conversion. Similarly, if the characterization of generators using ZSORCE rather than ZPOS was appropriate for a fault analysis scenario, activity CONG should be executed prior to selecting activity SEQD.

Because activity SEQD builds the sequence network admittance matrices corresponding to the system in the working case, it follows that it must be re-executed before entering activity SCMU any time these matrices change, different bus boundary conditions are imposed, or a new system representation is brought into the working case. Because activity SEQD prepares the SEQD temporary file for unbalanced network solutions, it also needs to be re-executed whenever, in the midst of an unbalanced network study, some other PSS®E activities that utilize these files are executed (e.g., activity ANSI), or when PSS®E is terminated and subsequently reentered.

Activity SEQD is sensitive to the fault analysis modeling option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). This setting, at the time activity SEQD is selected, determines whether subsequent executions of activities SCMU and SCOP use two or three phase network representations. When activity SEQD is used in preparation for activity SPCB, this option setting *must* be set to the normal three phase modeling mode.

Additional Information
PSS®E GUI Users Guide, Preparing Sequence Networks for Unbalanced Network Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Preparing Sequence Network for Unbalanced Network Solution

11.6. Performing Fault Analysis with Multiple Unbalances

SCMU

Run Activity SCMU - GUI
Fault > Solve and report network with unbalances (SCMU/SCOP)... *[Multiple Simultaneous Unbalances]*

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Run Line Mode Activity SCMU - CLI
ACTIVITY? >>**SCMU** ENTER UNBALANCE CODE:
Interrupt Control Codes
AB

The multiply unbalanced network solution activity SCMU allows the user to apply simultaneously, at any bus, on any phase, any or all of the following unbalances or faults:

1. Two single-line-to-ground faults (L-G) with specified fault impedances.
2. Two line-to-line (L-L) or double-line-to-ground (L-L-G) faults with specified fault impedances.
3. Single phase, with specified impedance, closed between a pair of buses.
4. Two phases, with equal specified impedance, closed between a pair of buses.

The network is solved in the presence of the specified unbalances and summary output is printed. Activity SCMU does not modify the contents of the working case or the SEQD temporary file set up by activity SEQD.

11.6.1. Unbalanced Fault Analysis Options

The user specifies the desired unbalances, selecting the types of unbalances (see Simplified Fault Calculation Model and Figure 11.5, "Unbalances Modeled in Activity SCMU") and specifying faulted buses and phases, and fault impedances. Note that all fault impedances are complex numbers and are entered in per unit. Options are available for the following unbalances:

• First L-G	• Second L-L-G	• 3-phase fault
• Second L-G	• 1 phase closed	• 1 end opened
• First L-L-G	• 2 phases closed	• In-line slider

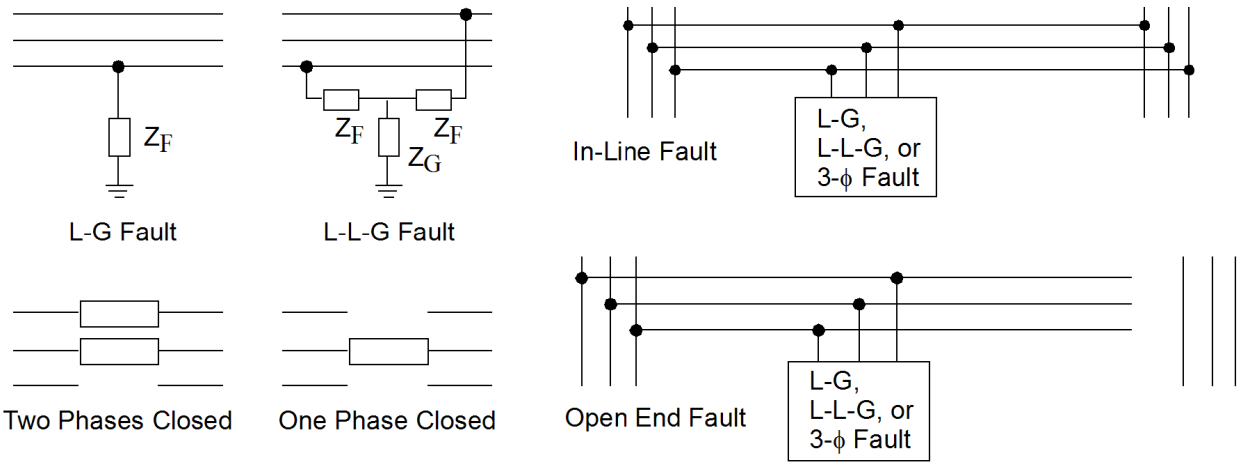

Figure 11.5. Unbalances Modeled in Activity SCMU

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11.6.2. Bus Faults

Bus faults are applied simply by instructing activity SCMU to apply the L-G and L-L-G unbalances, singly or together at the bus, with the appropriate values of the fault impedances Z_F and Z_G (see Figure 11.5, "Unbalances Modeled in Activity SCMU"). For example, a line-to-line fault may be applied by selecting the L-L-G unbalance with infinite impedance specified for Z_G (i.e., the default value). As a second example, a three phase fault may be applied by selecting the L-G unbalance on phase one (i.e., A) with Z_F = 0.0+j0.0 together with the L-L-G unbalance excluding phase one with both Z_F and Z_G specified as 0.0+j0.0. Specifying this condition has been automated by including a special unbalance code for a three phase fault. It automatically applies both the *second* L-G and *second* L-L-G faults at the designated bus, with all fault impedances set to 0.0+j0.0.

The user must specify the faulted bus. In the case of the L-G unbalance, the user specifies the faulted phase and the fault impedance; phases one, two and three designate the A, B, and C phases, respectively. In the case of the L-L-G unbalance, the user specifies the un-faulted phase and the fault impedances Z_F and Z_G. Note again that fault impedances (R + jX) are specified and that *both* resistance and reactance *must* be entered. Fault impedances are specified in per unit of base impedance.

11.6.3. Line Faults

The inter-sequence solution method of activity SCMU can apply faults only at buses. The application of a line fault, therefore, requires the insertion of a dummy bus in the line that is to be subjected to the line fault. The introduction of dummy buses and rerouting of branches may be handled manually in the working case before executing activity SEQD. However, activity SCMU provides for the temporary addition of the required dummy buses and automates the rerouting of branches to the dummy buses and splitting of the zero sequence mutual couplings on either side of the faulted dummy bus.

In the application of the fault slider, the line may be represented in two ways: with both ends closed or with one end opened. In either of these two topologies, one of the following may be applied at any point along the line:

1. The *second* L-G fault.
2. The *second* L-L-G fault.
3. A three phase fault utilizing both the *second* L-G fault and the *second* L-L-G fault.

In modeling line faults, activity SCMU introduces a dummy bus, numbered 999999, at the fault point. When one end of the faulted line is opened and the fault point is not at the line end position, a second dummy bus, numbered 999998, is introduced at the opened end.

To utilize the line faults, the number of buses in the working case must be at least one less (two less for an in-line fault with one end opened) than the maximum number for which PSS®E is dimensioned. In addition, except for the case of a line end fault at the opened end of a line, the following conditions must be met:

1. The number of branches in the working case must be less than the maximum number for which PSS®E is dimensioned.
2. The geographical B factors, BIJ1, BIJ2, BKL1, and BKL2, must have been properly specified for any zero sequence mutual couplings involving the branch (refer to Zero Sequence Mutual Impedance Data).

The user specifies the branch on which the fault slider is to be imposed. For the case of one end of a branch opened, a non-transformer branch, a two-winding transformer, or a three-winding transformer may be des-

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ignated. A non-transformer branch or a two-winding transformer is specified by keeping the default value or entering a zero as the third bus identifier; a three-winding transformer is designated by specifying the three buses it connects.

In the case of one end of the branch opened, the user specifies the opened end and, except for a three-winding transformer, the fault location on the line. The fault location is specified by specifying the fraction of the line between the closed end and the fault point (A factor); the proper designation is a number greater than zero and less than or equal to one (one would indicate a line end fault). For a two-winding transformer, an A factor value of one *must* be specified.

For the case of both ends of the branch closed, the user specifies the fault location by entering the fraction of the line between the from bus as entered above and the fault point; the number entered *must* be greater than zero and less than one.

The user specifies the type of fault (refer to Bus Faults).

Only one line fault may be imposed on the network in any given execution of activity SCMU. Other bus faults and the phase closed unbalances may be simultaneously applied.

For an in-line slider fault with both ends closed, the faulted line may *not* be a transformer, and it may *not* be open in the zero sequence (i.e., its zero sequence impedance *must* be non-zero). In the case of one end opened, the fault *must* be applied at the line end position if the branch is a transformer.

The summary output for either of the line faults is identical in form to that for bus faults. The faulted dummy bus is listed as bus 999999 with the name DUMMYBUS. For the one end opened case, the base voltage of the dummy bus is taken to be the same as the opened end bus. For the case of both ends closed, the base voltage of the dummy bus will be that of one of the two buses involved.

In the case of one end opened with the fault location other than at the opened end, the opened end dummy bus is listed as bus 999998 with the name STUB END and the base voltage of the original bus at the opened end of the line.

11.6.4. Branch with One Open End

For the case of one end of a branch opened, a non transformer branch, a two-winding transformer, or a three-winding transformer may be designated. A non transformer branch or a two-winding transformer is specified by identifying the two associated buses. A three-winding transformer is designated by specifying the three buses it connects.

With one end of the branch opened, the user designates the opened end and, except for a three-winding transformer, the fault location on the branch. For a non-transformer branch, the fault location is specified by specifying the fraction of the line between the closed end and the fault point, as for the slider. The proper designation is a number greater than zero and less than or equal to one (one would indicate a line end fault). For a two-winding transformer, the value of one *must* be specified. For a three-winding transformer, the user can select the open point at any of the three buses to which the transformer is connected.

In the case of one end opened with the fault location other than at the opened end, the opened end dummy bus is listed as bus 999998 with the name STUB END and the base voltage of the original bus at the opened end of the line.

Note that the fault type can be a line-to-ground, a line-to-line-to ground or a 3-phase fault.

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11.6.5. Phase Closed Unbalances

The single and double phase closed unbalances of activity SCMU place series elements of the designated *phase* impedance between designated buses in the selected phases. The user specifies the pair of buses involved, either the phase to be closed (for the single phase closed unbalance) or the opened phase (for the double phase closed unbalance), and the *phase* impedance of the closed phase(s).

If it is intended to open one or two phases of a branch which already exists, that branch must be taken out of service in order to be replaced by the unbalance model.

Following the unbalance specification process, a warning message is printed if there are already any in-service branches between the designated buses. In this case the new series branch, of which only one or two phases are closed, is placed in parallel with those branches that are already present. It is the user's responsibility to ensure that the working case is in a state such that the application of the phase closed unbalance produces the desired overall condition.

The summary output for each of the phase closed unbalances consists of the bus voltage block for each of the two buses and the branch current block for the unbalance only.

11.6.6. Pre-Calculation Network Setup

The normal sequencing of activities in preparing for fault analysis work is:

1. Set up the positive sequence network at whatever level of detail is required.

2. Append the required sequence data to the power flow case (Section 11.2, "Appending Sequence Data to the Power Flow Case").

3. Select the unbalances to be simulated and start the calculation.

When the calculation is initiated, the first step in the process is the setting up of the sequence networks. This involves the preparation of the power flow case for the unbalanced network solution (as well as for the Separate Pole Circuit Breaker simulation discussed in Section 11.13, "Calculating pi-Equivalent, Single Transmission Line Unbalance")

This process involves taking the positive sequence network (i.e., the power flow case) and the various sequence data arrays defining the negative and zero sequence networks, and setting up the SEQD temporary file in the necessary form. The following computations are performed:

- All positive sequence loads are converted to fixed shunt admittances on the basis of the voltage at each load bus in the working case.

- All negative sequence loads, except those for which the user has specified a nonzero negative sequence load are set equal to the positive sequence values calculated in the previous step.

- All generator positive sequence sources are initialized and fixed to correspond to their generator terminal bus conditions in the working case. This is a temporary conversion of the power flow generator model to a Norton equivalent for the fault calculations (see Figure 11.6, "Generator Conversion to Norton Equivalent for Fault Calculations").

- An ordering for the positive and negative sequence networks is determined and summarized at the *Progress* device (see Figure 11.7, "Summary from Network Ordering Prior to Fault Calculations").

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- The positive and negative sequence admittance matrices are constructed and factorized.
- The zero sequence network is ordered and summarized at the *Progress* device (see Figure 11.7, "Summary from Network Ordering Prior to Fault Calculations").
- The zero sequence network admittance matrix is constructed and factorized.

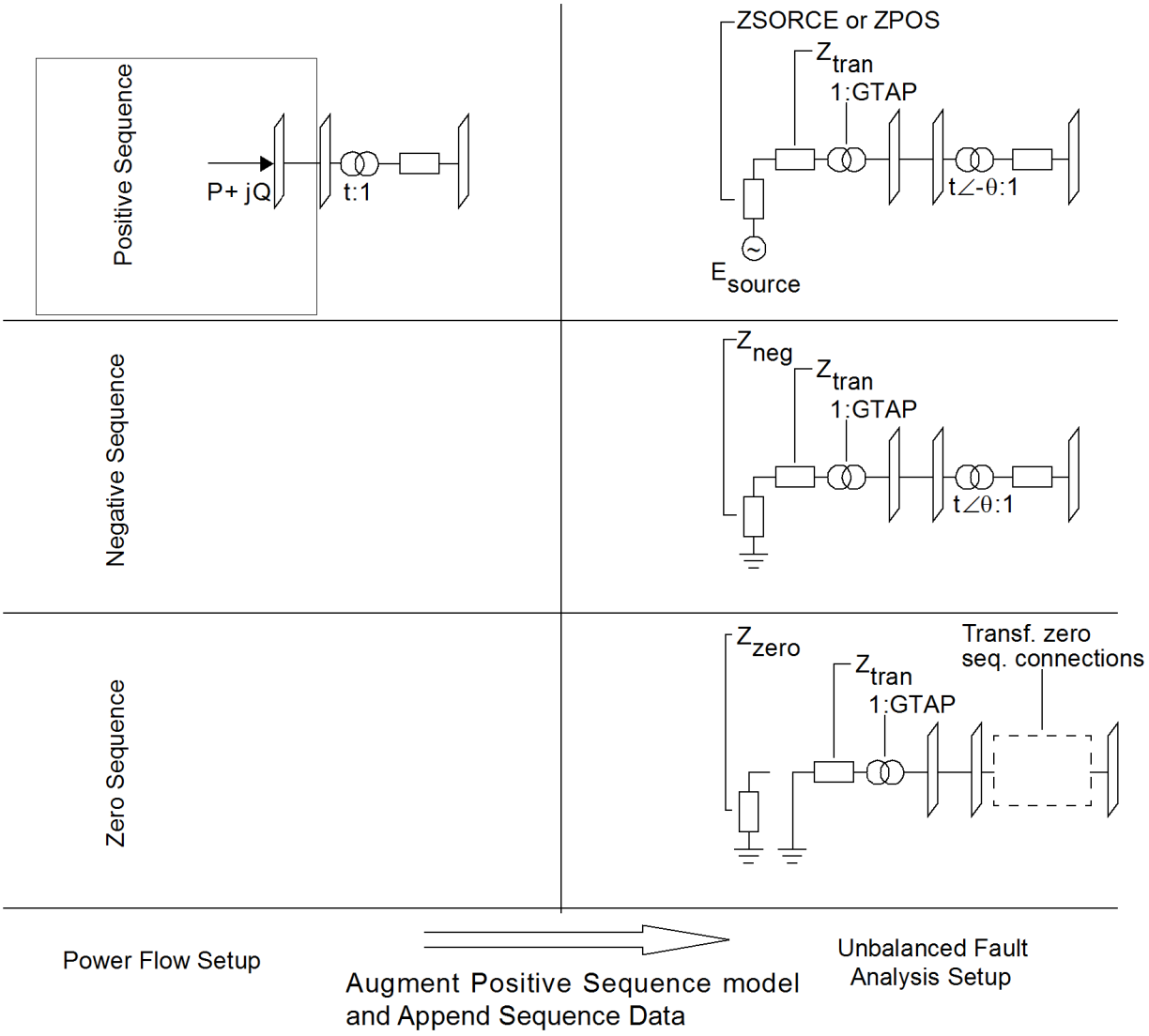

Figure 11.6. Generator Conversion to Norton Equivalent for Fault Calculations

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```
Using machine R and X" from fault analysis data

Diagonals = 23   Off-diagonals = 41   Maximum size = 60

Positive sequence network: Diagonals = 23   Off-diagonals = 41

Diagonals = 23   Off-diagonals = 35   Maximum size = 48

Bus     101 [NUC-A       21.600] is isolated in the zero sequence network

Bus     102 [NUC-B       21.600] is isolated in the zero sequence network

Bus     206 [URBGEN      18.000] is isolated in the zero sequence network

Bus     211 [HYDRO_G     20.000] is isolated in the zero sequence network

Bus    3011 [MINE_G      13.800] is isolated in the zero sequence network

Bus    3018 [CATDOG_G    13.800] is isolated in the zero sequence network

6 buses are isolated in the zero sequence network

Zero sequence network: Diagonals = 23   Off-diagonals = 35
```

Figure 11.7. Summary from Network Ordering Prior to Fault Calculations

With respect to the summary shown in Figure 10-7, it is quite common, and perfectly valid, to have the generator terminal bus isolated in the zero sequence network. This is, in fact, the usual case because the majority of generator step-up transformers are delta connected on the generator side and wye connected on the high side. This is the assumption inherent in the generator modeling when the step-up transformer is represented as part of the generator data (i.e., XTRAN is nonzero). An alarm message will be printed for any generator with nonzero values of both XTRAN and ZZERO. In this case, the value of only XTRAN is used in setting up the zero sequence ground tie at the Type 2 (high side) bus.

11.6.7. Operation of Activity SCMU

If sequence data has not been read into the working case via activity RESQ, an appropriate error message is printed and activity SCMU is terminated.

When the sequence network setup option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings"), activity SEQD is automatically executed. Otherwise, activity SCMU assumes the prior execution of activity SEQD for the network conditions contained in the working case.

After the desired set of unbalances has been specified, the inter-sequence network solution commences. If any line fault has been specified (refer to Section 11.6.3, "Line Faults"), the sequence admittance matrices as constructed by activity SEQD are modified and factorized, and the standard matrix size information (as in activity FACT) is tabulated. Otherwise, the factorized admittance matrices built by activity SEQD are used.

This is followed by a tabulation of the system Thevenin impedances as seen at each bus where an unbalance is to be applied. Each sequence column of this table lists resistive and reactive parts of the Thevenin impedances

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that would be measured by observations made at each bus in turn without the influence of the prescribed set of unbalances.

Activity SCMU interconnects the three sequence networks to represent the unbalanced condition and solves for the sequence voltages. Following solution, a summary report is printed for each unbalance applied. The user is then given the opportunity to get complete output for any bus in the working case.

Activity SCMU requires that the ordered sequence network admittance matrices and their triangular factors are present in the SEQD temporary file; i.e., activity SCMU requires successful execution of activity SEQD. Activity SCMU may be used repeatedly after the successful completion of activity SEQD to apply different unbalances to a given network configuration. However, any time the network configuration is changed (for example, via activities TREA, RESQ, CASE, CHNG, SQCH, or the *[Spreadsheet]*), re-execution of activity SEQD is required.

The summary report for each unbalance may be terminated by entering the AB interrupt control code.

11.6.8. Unbalance Condition Summary Report

Following the solution of the unbalanced network, a summary report is printed at the selected output device for each unbalance that is in effect. Each individual unbalance report consists of a tabulation of bus voltages and branch currents in terms of both symmetrical component (sequence) and phase quantities. Voltages and currents are printed in either physical units (kV L-G and amps) or per unit, and in either rectangular or polar coordinates, according to the fault analysis output options currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). If the physical units option is enabled and a bus has no base voltage specified for it, its output is printed in per unit.

The following blocks of output are included:

1. Fault type and bus(es) involved.

2. Bus voltages for the bus at which the unbalance is applied. The first line of each bus voltage block gives the bus zero, positive and negative sequence voltages as well as three times the zero sequence voltage, and the second line gives the bus phase voltages. When the per unit option is enabled, values are in per unit of rated line to ground voltage. The phase voltages are line to neutral values. For the phase closed unbalances, the bus voltage block is printed for both of the buses involved.

3. For the L-G and L-L-G faults, series branch currents flowing in each branch (including any generator contributions) connected to the faulted node. All currents are tabulated as flowing *into* the faulted node and include the effects of line charging capacitance, line connected shunt admittances, and transformer magnetizing admittance. When the per unit option is enabled, currents are expressed in per unit of base phase current. The first line gives sequence components of current as well as three times the zero sequence current, and the second line gives the phase currents. For non-transformer branches and for two-winding transformers, the number and name of the bus at the other end of the branch is printed, along with the circuit identifier; for three-winding transformers, the output line contains the winding number, the transformer circuit identifier and the transformer name. For the phase closed unbalances, the only branch series currents tabulated are those represented by the unbalance.

4. For the L-G and L-L-G faults, the algebraic sum of contributions of all elements tabulated in (3). The format of these currents is the same as for the series branch currents. This is the total current apparently flowing to ground at the bus, and it includes any load and fixed shunt connected to the bus, fault current, and the ac side current of any unblocked dc line or FACTS device.

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5. For the L-G and L-L-G faults, a value of contributions equivalent positive sequence admittance, expressed in rectangular form, in both per unit relative to system base values and in MVA. This quantity is computed from the sum of contributions in (4) and thus *includes* any load, fixed shunt and unblocked dc line and FACTS device elements connected to the bus. Furthermore, this equivalent admittance is valid *only* in the case of single ground faults. In the case where there are no dc line or FACTS device effects included in this admittance, and only this single unbalance was applied, this shunt admittance in MVA, may be entered as a shunt *replacing* the load and shunt elements at the bus, to give the correct positive sequence equivalent representation of this fault in the dynamic simulation activities.

6. For the L-G and L-L-G faults, the current flowing to ground at the bus exclusive of any fault current. The format of these currents is the same as in (3) and (4) above, and they include any load and shunt current at the bus as well as the ac side current of any unblocked dc line or FACTS device. In the zero sequence, only the load and shunt current is output here; specifically, the zero sequence ground ties created by grounded transformer windings are not shown here but are included in the branch contribution output of (3) above. Load and shunt current output is suppressed if the three sequence shunt and load admittances are all zero.

7. For the L-G and L-L-G faults, when no phase closed unbalances are applied and only one bus is involved in the ground faults, fault current is calculated and tabulated in a format similar to (4) above, and positive sequence equivalent fault admittance is listed in a format similar to (5) above. This admittance in MVA would be *added* to any in-service shunt at the bus to represent this fault in the dynamic simulation activities.

Following the summaries for all active unbalances, the user is able to request output for any bus in the system. Buses are selected one bus at a time and output is written to the selected output device. Output tabulated includes that specified by (2), (3), (4), and (6) above, with branch currents tabulated as *arriving* at the selected bus.

Whenever a single ground fault unbalance is solved by activity SCMU, an entry is appended to a summary file, named SMRYSC, in the user's directory. This file contains fault descriptive information along with the sum of contributions entry described in (4) above. This file is cumulative, and if it does not exist it is created automatically by activity SCMU. It is never automatically deleted; activity SCMU always appends to it. The user may delete this file at any time when it is no longer of use (refer to Section 2.6, "Files Created By PSS®E").

11.6.9. Example: Unbalance Condition Summary Report

Following the sequence network set-up, the system Thevenin impedances, as seen at each bus where an unbalance is to be applied, are reported. Each sequence column of this table lists resistive and reactive parts of the Thevenin impedances that would be measured by observations made at each bus in turn without the influence of the prescribed set of unbalances.

An example is shown in Figure 11.8, "Sequence Thevenin Impedance for Two Line-to-Ground Faults" for a case in which two line-to-ground faults were applied in the savnw.sav power flow case, with a zero fault impedance, on Buses 151 and 3002.

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```
 UNBALANCES APPLIED:

 LINE TO GROUND FAULT AT BUS 151 [NUCPANT     500.00]   PHASE 1

   L-G Z = (0.0,0.0)

 LINE TO GROUND FAULT AT BUS 3002 [E. MINE     500.00]   PHASE 1

   L-G Z = (0.0,0.0)


                    SEQUENCE THEVENIN IMPEDANCES AT FAULTED BUSES


   BUS# X-- NAME --X BASKV          ZERO              POSITIVE            NEGATIVE

    151 NUCPANT     500.00    0.00017  0.00600    0.00409  0.01765    0.00409  0.01765
```

Figure 11.8. Sequence Thevenin Impedance for Two Line-to-Ground Faults

The calculation process interconnects the three sequence networks to represent the unbalanced condition and solves for the sequence voltages. Following solution, a summary report is printed for each unbalance applied. The user is then given the opportunity to get complete output for any bus in the working case.

The summary report is printed at the selected output device for each unbalance that is in effect. Each individual unbalance report consists of a tabulation of bus voltages and branch currents in terms of both symmetrical component (sequence) and phase quantities. Voltages and currents are printed in either physical units (kV L-G and amps) or per unit, and in either rectangular or polar coordinates, according to the fault analysis output options currently in effect (refer to activity OPTN). If the physical units option is enabled and a bus has no base voltage specified for it, its output is printed in per unit.

The following blocks of output are included:

1. Fault type and bus(es) involved.

2. Bus voltages for the bus at which the unbalance is applied. The first line of each bus voltage block gives the bus zero, positive and negative sequence voltages as well as three times the zero sequence voltage, and the second line gives the bus phase voltages. When the per unit option is enabled, values are in per unit of rated line to ground voltage. The phase voltages are line to neutral values. For the phase closed unbalances, the bus voltage block is printed for both of the buses involved.

3. For the L-G and L-L-G faults, series branch currents flowing in each branch (including any generator and induction machine contributions) connected to the faulted node. All currents are tabulated as flowing *into* the faulted node and include the effects of line charging capacitance and line connected shunt admittances. When the per unit option is enabled, currents are expressed in per unit of base phase current. The first line gives sequence components of current as well as three times the zero sequence current, and the second line gives the phase currents. For non transformer branches and for two-winding transformers, the number and name of the to bus is printed, along with the circuit identifier; for three-

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winding transformers, the output line contains the winding number, the transformer circuit identifier and the transformer name. For the phase closed unbalances, the only branch series currents tabulated are those represented by the unbalance.

4. For the L-G and L-L-G faults, the algebraic sum of contributions of all elements tabulated in (3). The format of these currents is the same as for the series branch currents. This is the total current apparently flowing to ground at the bus, and it includes any load connected to the bus, fault current, and the ac side current of any unblocked dc line or FACTS device.

5. For the L-G and L-L-G faults, a value of contributions equivalent positive sequence admittance is expressed in per unit, rectangular form, relative to system base values. This quantity is computed from the sum of contributions in (4) and thus *includes* any load and unblocked dc line and FACTS device elements connected to the bus. Furthermore, this equivalent admittance is valid *only* in the case of single ground faults. In the case where there are no dc line or FACTS device effects included in this admittance, and only this single unbalance was applied, this shunt admittance, multiplied by system base MVA, may be entered as a shunt *replacing* the load and shunt elements at the bus, to give the correct positive sequence equivalent representation of this fault in the dynamic simulation activities.

6. For the L-G and L-L-G faults, the current flowing to ground at the bus exclusive of any fault current. The format of these currents is the same as in (3) and (4) above, and they include any load and shunt current at the bus as well as the ac side current of any unblocked dc line or FACTS device. In the zero sequence, only the shunt load is output here; specifically, the zero sequence ground ties created by grounded transformer windings are not shown here but are included in the branch contribution output of (3) above. Load and shunt current output is suppressed if the three sequence shunt and load admittances are all zero.

7. For the L-G and L-L-G faults, when no phase closed unbalances are applied and only one bus is involved in the ground faults, fault current is calculated and tabulated in a format similar to (4) above, and positive sequence equivalent fault admittance is listed in a format similar to (5) above. This admittance, multiplied by system base MVA, would be *added* to any shunt at the bus to represent this fault in the dynamic simulation activities.

Whenever a single ground fault unbalance is solved an entry is appended to a summary file, named SMRYSC, in the user's directory. This file contains fault descriptive information along with the sum of contributions entry described in (4) above. This file is cumulative, and if it does not exist it is created automatically when the unbalanced fault calculations are performed. It is never automatically deleted; results are always appends to it. The user may delete this file at any time when it is no longer of use.

Figure 10-9 shows the summary report for a single-line-to-ground fault at Bus 151 in the savnw.sav power flow case, with zero fault impedance.

Note that the output options are polar coordinates and physical quantities; i.e., the voltages are in kV and the current is in amperes.

Because there is a 600 Mvar shunt connected to bus 151, the Total Contributions include the current in this shunt.

The total A Phase current is 8420.0 / -67.71 amps. The zero sequence current is 2773.4 /-67.69 amps, These include contributions from the 600 MVAr shunt.

Because the A Phase current in the 600 Mvar shunt is 99.9 /- 69.33, the actual A Phase fault current at the bus is less and is 8320.1 /-67.69 amps.

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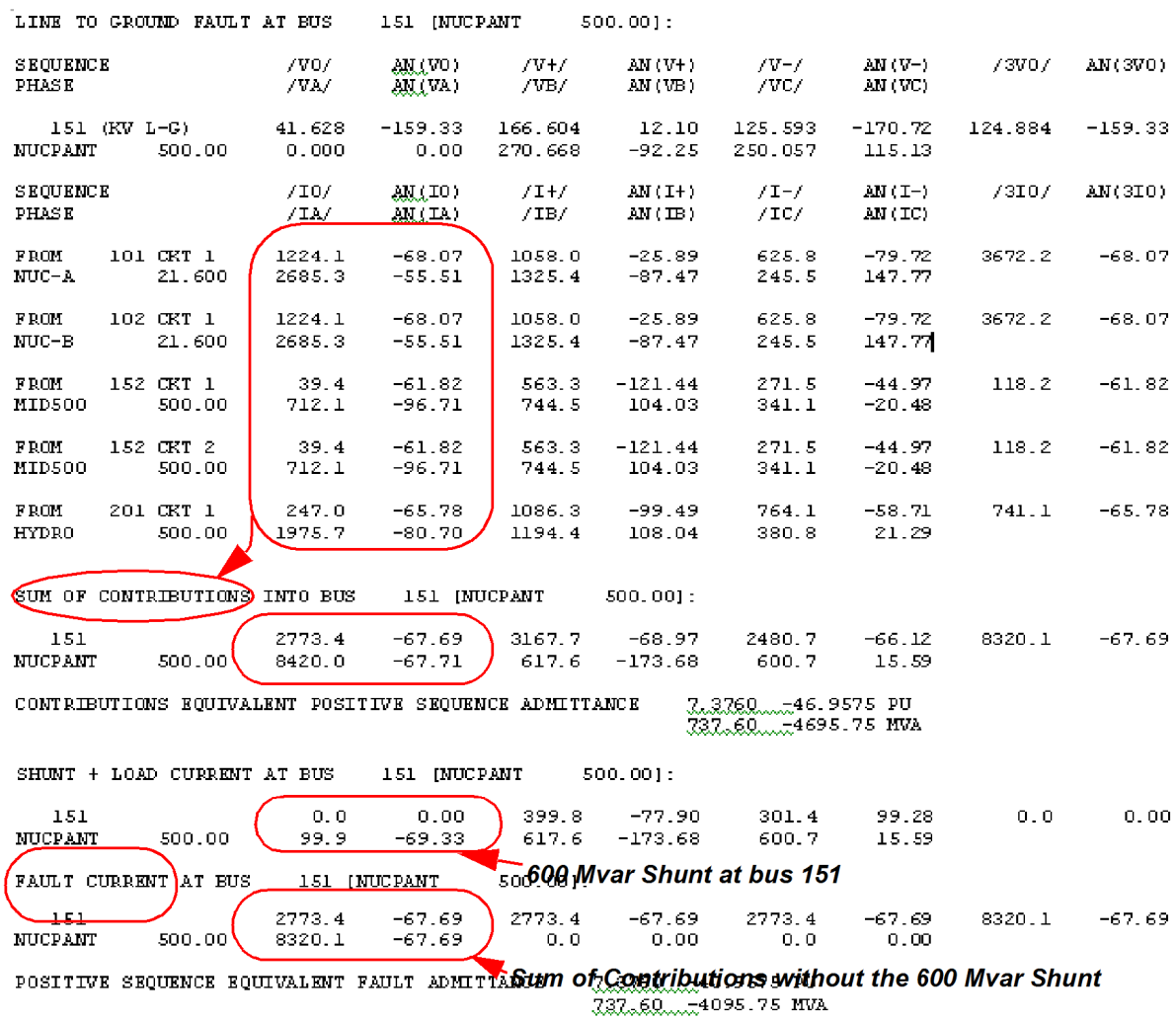

Figure 11.9. Summary Output at Bus 151 with L-G Faults at Buses 151 in Power Flow Case savnw.sav

11.6.10. Unbalanced Fault Analysis Detailed Report

Results summaries are output for each unbalance condition that is in effect. For each fault condition, conditions at other buses in the system can be reported. This is achieved using the Solution Output tab in the unbalance dialog. That tab provides access to selection of buses for which reporting is required and, further, allows the selection of output to be in the following branch quantities:

- Currents
- Apparent impedances
- Apparent admittances

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This reporting facility recognizes the same fault analysis output options, with quantities printed in either physical units or per unit, and in either rectangular or polar coordinates as does the summary report.

The data printed for any selected bus or buses has the same format as for the summary report and consists of:

1. Sequence and phase voltages at the bus.
2. Series branch quantity for each branch and machine connected to the bus, consisting of either:
    a. Branch series currents *leaving* the bus, in either per unit or amps. These are in the reverse direction of the currents shown in the summary report.
    b. Branch apparent impedances looking down each branch from the bus, expressed in either per unit or ohms. These apparent impedances are defined as:

$$Z_{seq\ ij} = \frac{V_{seq\ i}}{I_{seq\ ij}} \qquad Z_{phase\ ij} = \frac{V_{phase\ i}}{I_{phase\ ij}}$$

    c. where the sequence and phase currents are the total currents flowing into the line at the bus, including line charging capacitance and line connected shunt current. Infinite impedance is printed as 9999 per unit or 999999 ohms.
    d. Branch apparent admittances looking down each branch from the bus, expressed in either per unit or mmhos. Apparent admittances are defined as the reciprocal of apparent impedances. Infinite admittance is printed as 9999 per unit or 99999 mmhos.
3. Sum of branch contributions. When branch currents are being tabulated, the sum of all contributions flowing into the bus is the total current apparently flowing to ground at the bus and will be zero unless there is a load or a shunt connected (see Section 11.6.9, "Example: Unbalance Condition Summary Report").
4. When branch currents are being tabulated, the sum of load and shunt current at the bus. Load and shunt current output is suppressed if the three sequence shunt and load contributions are all zero.

Note that the direction of current flow in the series branch output is the *reverse* of that in the summary output. Rather, it follows the power flow output convention of current leaving the bus.

When output is directed to the user's terminal, a summary description of each active unbalance is printed at the top of the first page of output. Otherwise, the unbalance summary is printed on each page of output.

11.6.11. Observation of Transformer Currents

Grounded wye-delta transformers pose a difficulty in unbalanced fault analysis because they behave as a shunt path to ground in the zero-sequence, while being series paths in the positive and negative sequences. There is a distinction between the transformer's symmetrical component equivalent circuits and the actual arrangements of its primary, secondary, and neutral leads.

Consider the currents flowing in the leads of a wye-delta transformer as shown in Figure 11.10, "Lead Current Flowing Into and Out of a Wye-Delta Transformer". The wye-connected winding allows zero-sequence current to flow into the transformer leads and thence to ground via the neutral grounding strap. While the wye-connected winding is certainly a path to ground, observations at the leads that feed this winding see the

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zero-sequence current just as if it were a series current flowing to the bus at the other side. This zero-sequence current in the transformer leads must be recognized in calculations of the phase values of the lead currents.

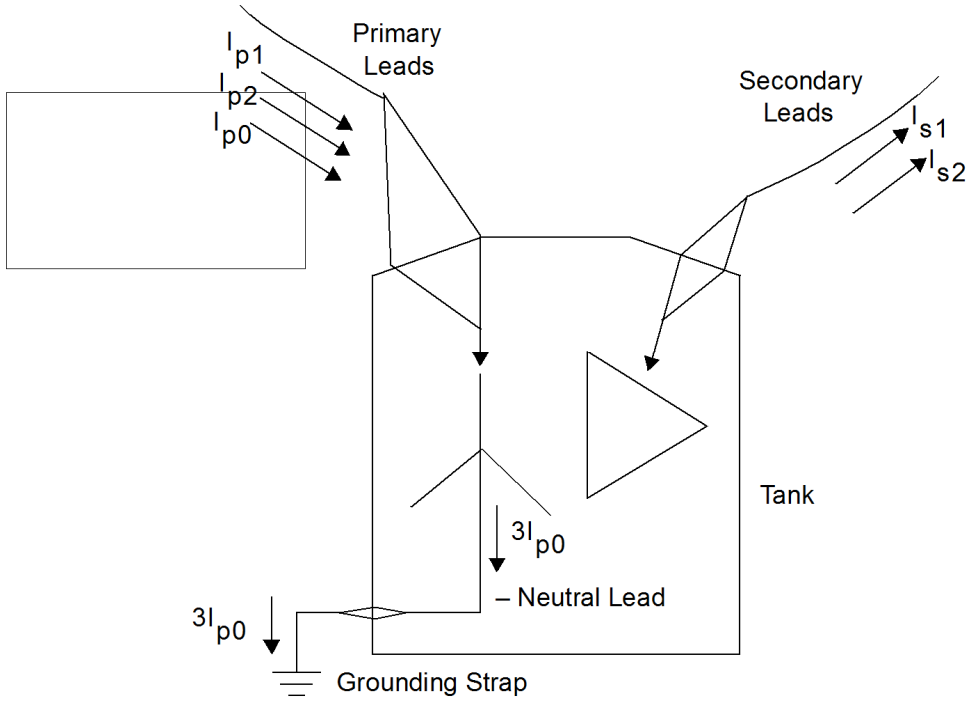

Figure 11.10. Lead Current Flowing Into and Out of a Wye-Delta Transformer

Consider the calculation of conditions at a bus where the following are connected:

- Transmission lines.
- A grounded wye-winding of a wye-delta transformer.
- A grounded wye-connected shunt reactor.
- A phase-to-ground fault somewhere on the bus side of all circuit breakers.

The bus arrangement is shown in Figure 11.11, "Transformer Zero-Sequence Currents Appearing in Alternative Network Representations of the Transformer"a. There are three components of zero-sequence current flowing from the bus to ground: fault current, reactor zero-sequence current, and transformer zero-sequence current. A conventional way of representing the bus is shown in Figure 11.11, "Transformer Zero-Sequence Currents Appearing in Alternative Network Representations of the Transformer"b. Here, the transformer is represented as a ground tie at the bus in the zero-sequence. The bus is assigned a shunt admittance equal to the sum of the reactor and transformer admittances. Use of this representation in system network solutions leads to correct results for bus positive-, negative-, and zero-sequence voltages. The subsequent calculation of transformer lead currents is erroneous, though, because the zero-sequence lead current appears to be zero when it should not be.

The alternative and recommended approach is to represent the transformer via the winding type code option of PSS®E. When the winding type code approach is used, the implied bus representation is as shown in

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Figure 11.11, "Transformer Zero-Sequence Currents Appearing in Alternative Network Representations of the Transformer"c.

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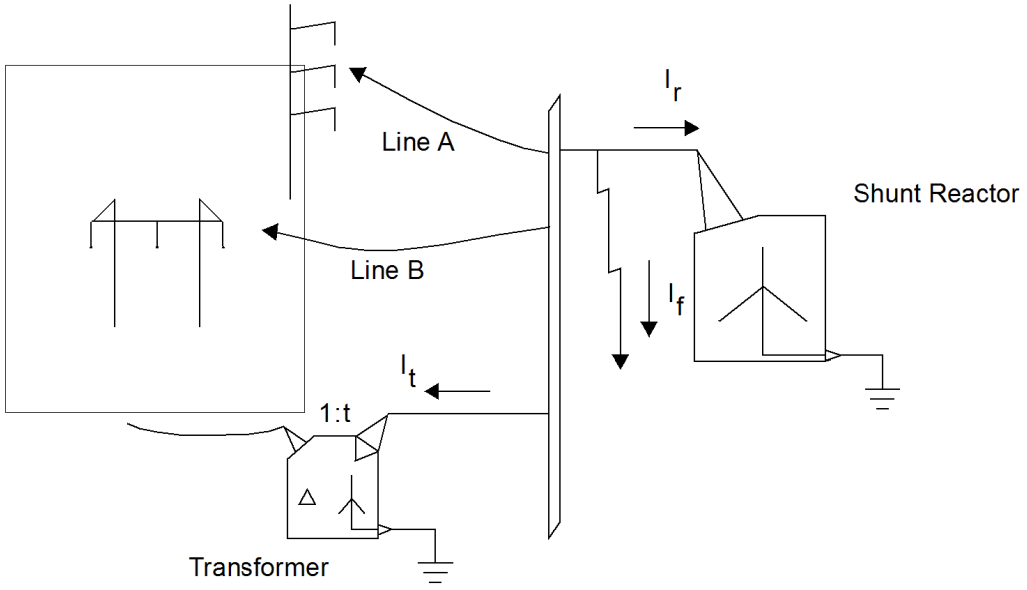
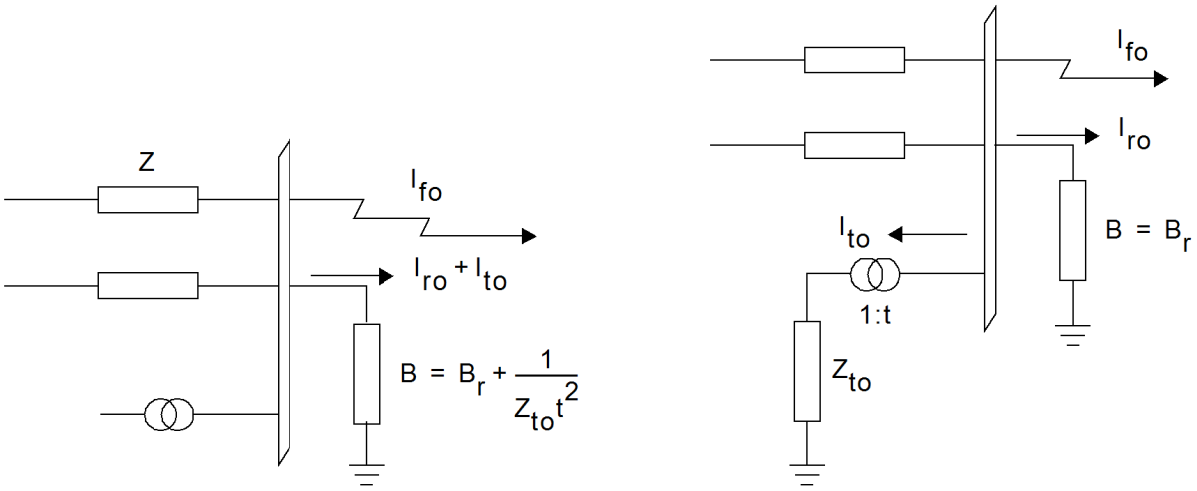


a. Bus with Grounding via Transformer, Shunt Reactor, and Fault



b. Bus Setup with Transformer Treated as Bus Ground Path

Figure 11.11. Transformer Zero-Sequence Currents Appearing in Alternative Network Representations of the Transformer

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This results in the same system solution as obtained with the prior approach, but recognizes the zero-sequence current in the transformer leads. Calculations of transformer lead phase currents are correct, provided that the 30° phase shift of the transformer is handled properly.

While the winding type codes are normally used in the handling of the majority of transformers, use of manually determined shunt paths at buses may still be needed in special situations. When the winding type codes are not used, the following points must be noted:

- If wye windings of wye-delta transformers are to be represented as bus-connected shunt admittances to ground, the shunt admittance should be corrected for off-nominal turns ratio as shown by Figure 11.12, "Assignment of Zero-Sequence Shunt Branch for Typical Tapped Delta-Wye Transformers, Solidly Grounded".

- When wye-delta transformers are represented by bus-connected shunt admittances, the phase currents calculated by PSS®E for the leads to the delta-connected winding are correct, but the phase currents displayed for the wye winding leads are erroneous.

- Because the bus modeling topology shown in Figure 11.11, "Transformer Zero-Sequence Currents Appearing in Alternative Network Representations of the Transformer"b is correct for bus arrangements other than that shown in Figure 11.11, "Transformer Zero-Sequence Currents Appearing in Alternative Network Representations of the Transformer"a, PSS®E must assume that all calculated branch phase currents are valid and leave it for the user to accept or reject the output values.

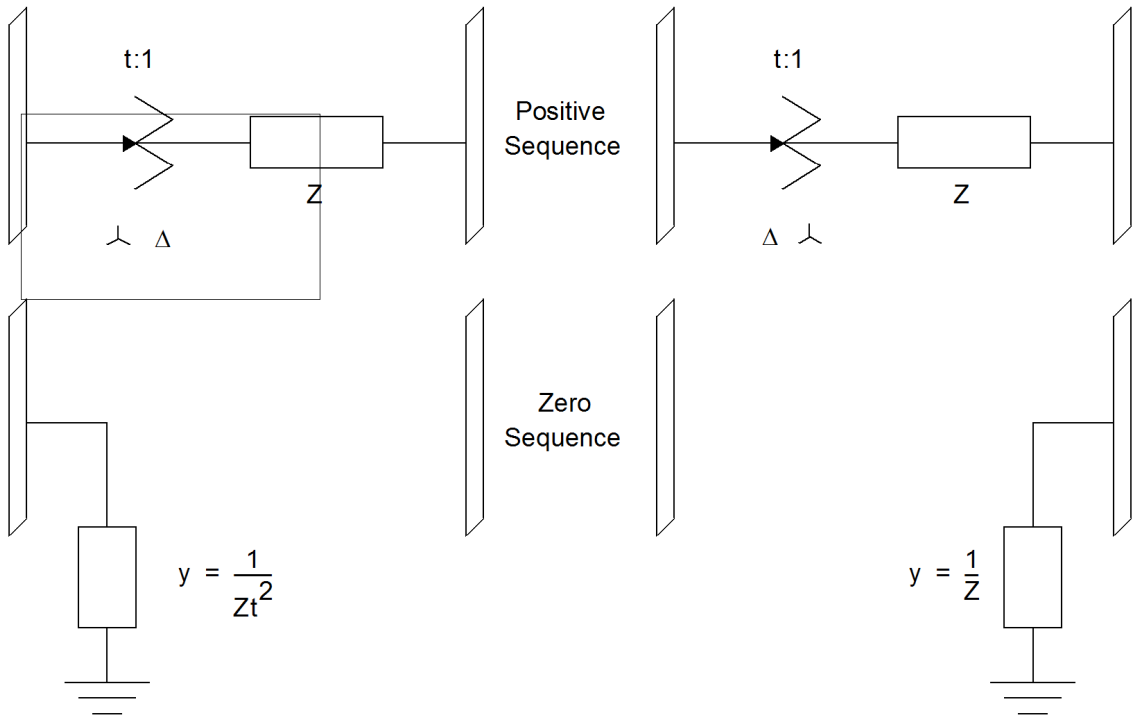

Figure 11.12. Assignment of Zero-Sequence Shunt Branch for Typical Tapped Delta-Wye Transformers, Solidly Grounded

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11.6.12. Working with a Two-Wire System

PSS®E includes an option, in the detailed unbalanced fault analysis and the automatic sequencing fault analysis (Section 11.8, "Calculating Automatic Sequencing Fault"), to handle certain two- and one-phase systems. This option was implemented primarily to handle certain European electric traction systems having a primary transmission system of two phases at 180° displacement feeding a single-phase catenary system. The basic system connections in this system are shown in Figure 11.13, "Two-Phase System Configuration for Railway Application", but other two- and one-phase systems can be handled by the appropriate setup of the sequence network models.

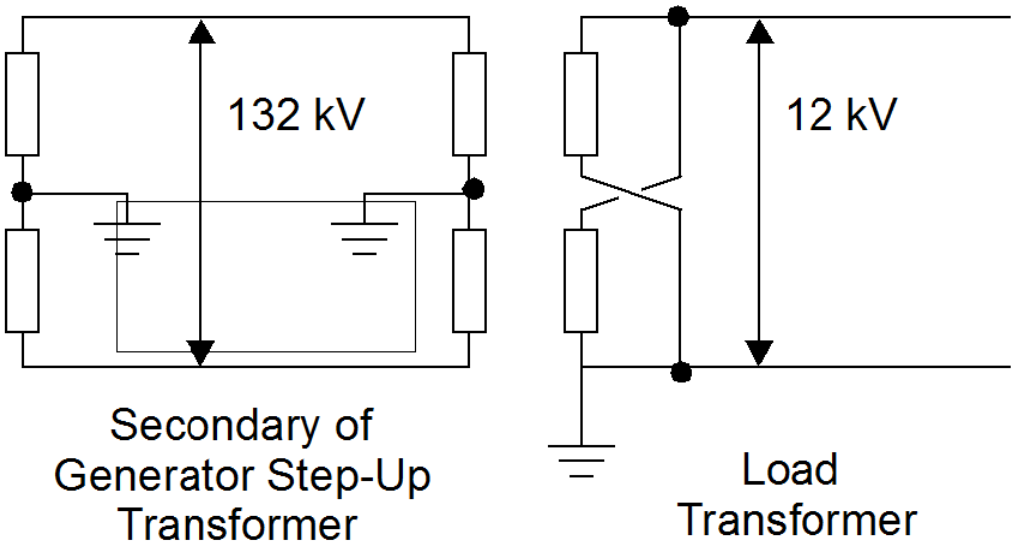

Figure 11.13. Two-Phase System Configuration for Railway Application

The two-phase system modelling assumption is established with the *Short circuit phase modeling* run-time option (Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). This option allows the selection of two-phase instead of the conventional three-phase mode. Selecting the two-phase option has the following effects:

- The symmetrical component < a > operator becomes (−1+j0), and the negative sequence is ignored, giving

$$\begin{bmatrix} i_0 \\ i_1 \end{bmatrix} = 1/2 \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \end{bmatrix}$$

- The base voltage is taken to be the line-to-line voltage at base conditions, and the base current is taken to be the corresponding line current so that

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$$I_{L\ base} = \frac{MVA_{base}}{V_{LL\ base}}$$

$$Z_{LG\ base} = \frac{V^2_{LL\ base}}{2MVA_{base}}$$

- Line-to-ground voltages in kV are calculated as

$V_{LG} = 0.5(V_{LL\ base} \times v_{pu}$

Phase currents in amperes are calculated as

$I_p = I_{L\ base} \times I_{pu}$

where v_{pu} and I_{pu} are the per-unit bus voltage and branch current, respectively.

Using the PSS®E two-phase option is identical to the conventional three-phase mode except for the following:

- Negative-sequence data may be entered for generators, branches, and loads but it is ignored in the fault calculations.
- Unbalanced faults may involve only phases 1 and 3. (The excluded phase in an L-L-G fault must be phase 2.) The three-phase fault selection must not be used.
- The two-phases-closed unbalance should not be used.

11.6.13. Transmission Lines

A two-phase two-conductor transmission line is characterized by positive- and zero-sequence impedances:

$Z_1 = Z_p - Z_m$

$Z_0 = Z_p + Z_m$

where

Z_1 = Positive-sequence impedance.

Z_0 = Zero-sequence impedance.

Z_p = Self impedance of one phase conductor.

Z_m = Mutual impedance between the phase conductors.

Similar expressions hold for the charging capacitances in the positive- and zero-sequences.

11.6.14. Transformers

The positive and zero-sequence characteristics of transformers must be determined from their internal connections. As an example, consider the transformer connection shown in Figure 11.13, “Two-Phase System Configuration for Railway Application”. The behavior of the transformer when positive and zero-sequence currents, respectively, flow in its primary windings are shown in Figure 11.14, “Behavior of Transformer with Secondary Windings Parallel to Single-Phase Load”. When positive-sequence current flows in the primary side, the two secondary windings are effectively in parallel and a positive-sequence current flows in the secondary leads. When a zero-sequence current flows in the primary leads, the direction of current is reversed in

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one primary and secondary winding. The two secondary windings now form a short-circuited loop; a current corresponding to the primary zero-sequence current flows around this loop, but no zero-sequence current flows in the secondary leads. The behavior of the primary and secondary currents in this transformer is not affected by the grounding of one secondary lead; no zero-sequence current can flow into the transformer because the ground is external to it.

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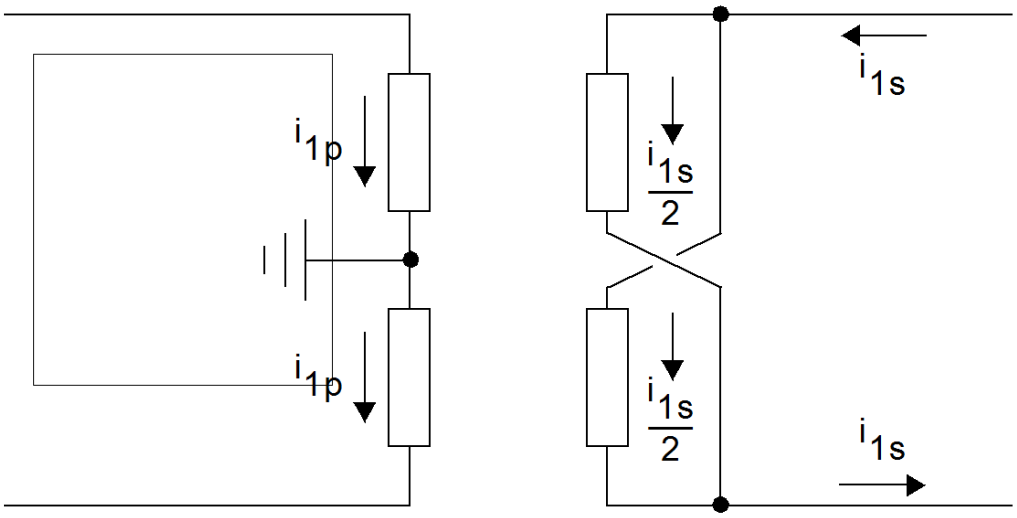
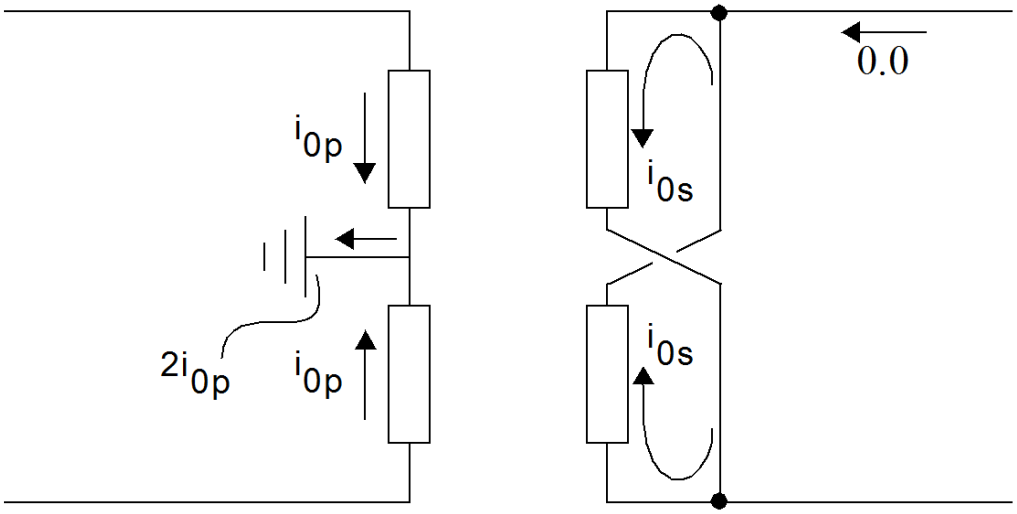


Sequence Behavior



quence Behavior

Figure 11.14. Behavior of Transformer with Secondary Windings Parallel to Single-Phase Load

The modeling of the load transformer in PSS®E can be handled in a completely standard manner by the use of connection codes. The positive- and zero-sequence connections, corresponding to Figure 11.14, "Behavior of Transformer with Secondary Windings Parallel to Single-Phase Load" and Figure 11.13, "Two-Phase System

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Configuration for Railway Application", are shown in Figure 11.15, "Sequence Connections Corresponding to Figure 11.14, "Behavior of Transformer with Secondary Windings Parallel to Single-Phase Load"". They may be specified to PSS®E by connection codes of 2.

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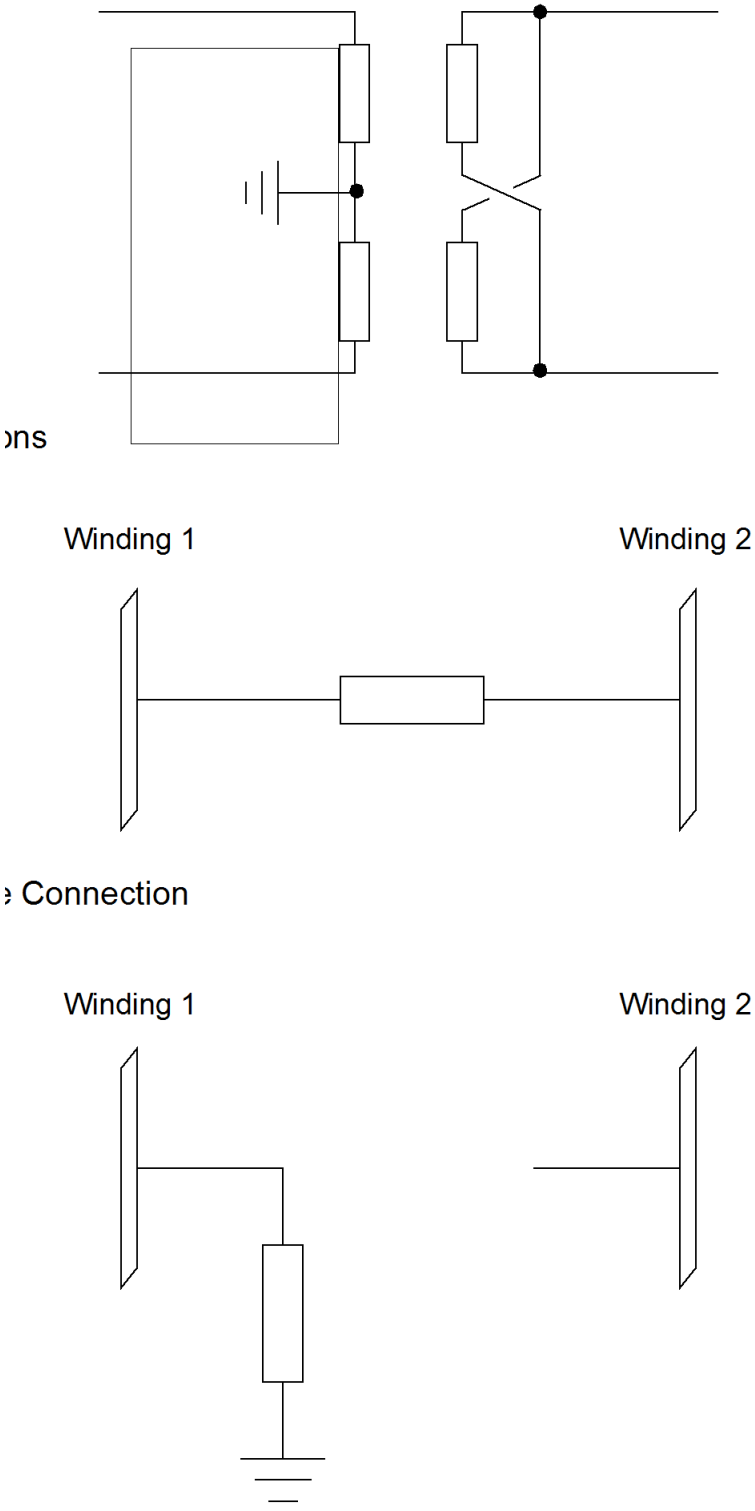

Figure 11.15. Sequence Connections Corresponding to Figure 11.14, "Behavior of Transformer with Secondary Windings Parallel to Single-Phase Load"

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11.6.15. Secondary Circuits

The load on the secondary side of the transformer shown in Figure 11.13, "Two-Phase System Configuration for Railway Application" must be modeled according to its physical connections. Straightforward possibilities are shown in Figure 11.16, "Sequence Circuits for Loads on Two-Phase System". Grounding of one phase of a load does not create a zero-sequence path to ground. A ground must exist inside the load device in order for it to carry a zero-sequence current; such internal ground connections exist in the center-tapped load (Figure 11.16, "Sequence Circuits for Loads on Two-Phase System"a) and in the transmission line section (Figure 11.16, "Sequence Circuits for Loads on Two-Phase System"d) as a result of its shunt charging capacitances.

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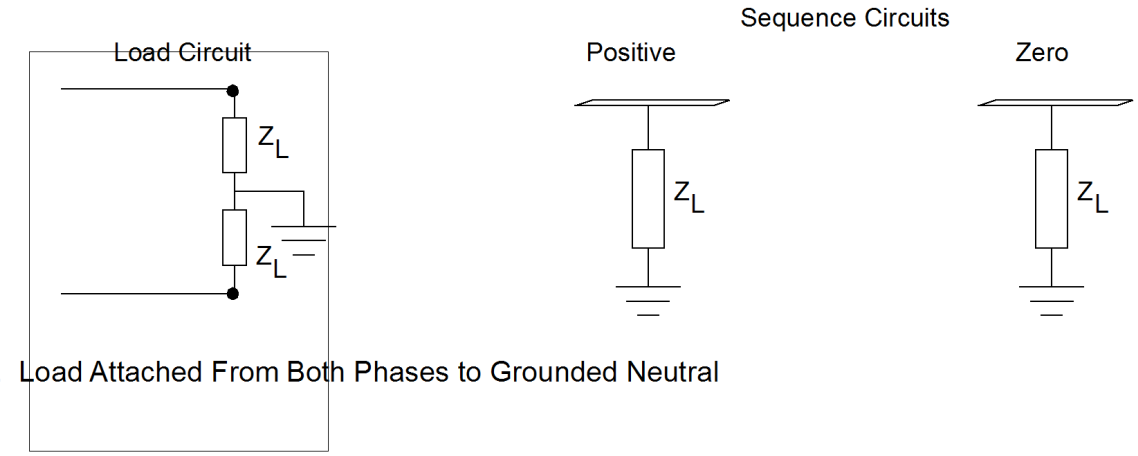

Figure 11.16: Sequence Circuits for Loads on Two-Phase System

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11.6.16. Faults on a Two-Phase System

Faults on the two phase system may be represented by the standard PSS®E repertoire, except that the three-phase and single-phase-open faults must not be used. The two phase system consists of phases a and c. Hence an L-G fault must be on phase a or c, and an L-L-G fault must exclude phase b.

11.6.17. Examples of Two-Wire Systems

Figure 11.17, "Sample System for Two-Phase Example Calculations" shows a small sample system for use in two-phase fault analysis examples. The system data are summarized in Figure 11.18, "Raw Data Files for Two-Phase System" and Figure 11.19, "Data Listings for Two-Phase System".

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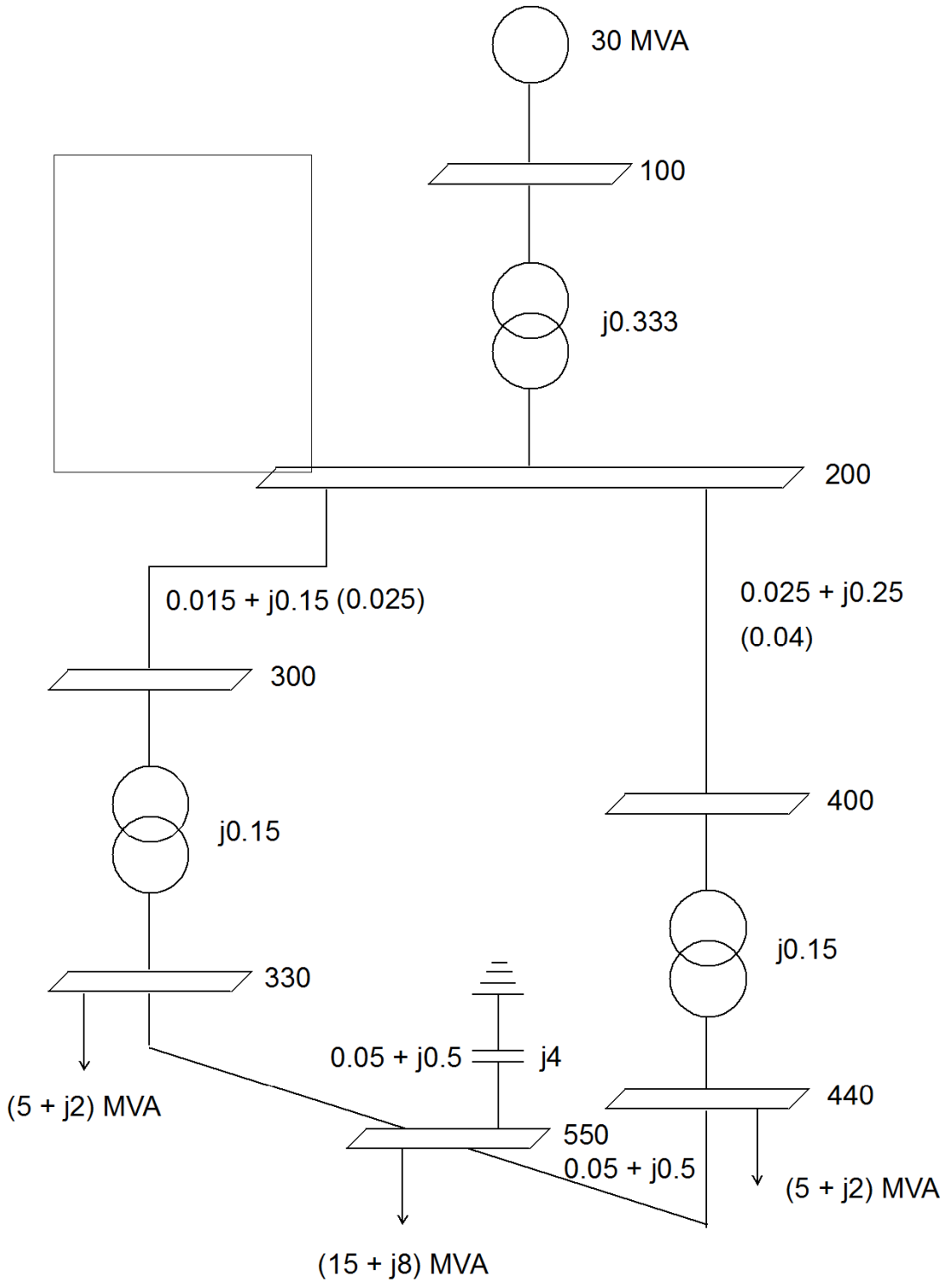

Figure 11.17. Sample System for Two-Phase Example Calculations

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```
0,   100.00          / PSS®E-30.0    THU, AUG 05 2004  11:22
SMALL TWO PHASE EXAMPLE

    100,''GEN-1'     ',  10.0000,3,     0.000,     0.000,   1,   1,1.00000,    0.0000,   1
    200,''HYDRO'     ',  132.0000,1,    0.000,     0.000,   1,   1,0.98277,   -4.8404,   1
    300,''WEST'      ',  132.0000,1,    0.000,     0.000,   1,   1,0.97469,   -5.9957,   1
    330,''EAST-LO    ',  12.0000,1,     0.000,     0.000,   1,   1,0.96723,   -7.2034,   1
    400,''EAST'      ',  132.0000,1,    0.000,     0.000,   1,   1,0.97225,   -6.5672,   1
    440,''WEST-LO    ',  12.0000,1,     0.000,     0.000,   1,   1,0.96527,   -7.6463,   1
    550,''MAIN-LO    ',  12.0000,1,     0.000,     4.000,   1,   1,0.95002,   -9.6922,   1
0 / END OF BUS DATA, BEGIN LOAD DATA
    330,'1 ',1,   1,    5.000,     2.000,     0.000,     0.000,     0.000,    1
    440,'1 ',1,   1,    5.000,     2.000,     0.000,     0.000,     0.000,    1
    550,'1 ',1,   1,   15.000,     8.000,     0.000,     0.000,     0.000,    1
0 / END OF LOAD DATA, BEGIN GENERATOR DATA
    100,'3 ',   25.130,    6.283,    20.000,     0.000,1.00000,   200,    30.000,    0.00000,   0.20000,   0.00000,   0.00000,1.00000,1,    1.0,   9999.000, -9999.000,   1,1.0000
0 / END OF GENERATOR DATA, BEGIN BRANCH DATA
    200,   300,'1 ',  0.01500,   0.15000,   0.02500,   40.00,   40.00,   40.00,   0.00000,   0.00000,   0.00000,   0.00000,1,   0.00,   1,1.0000
    200,   400,'1 ',  0.02500,   0.25000,   0.04000,   40.00,   40.00,   40.00,   0.00000,   0.00000,   0.00000,   0.00000,1,   0.00,   1,1.0000
    330,   550,'1 ',  0.05000,   0.50000,   0.00000,   15.00,   15.00,   15.00,   0.00000,   0.00000,   0.00000,   0.00000,1,   0.00,   1,1.0000
    440,   550,'1 ',  0.05000,   0.50000,   0.00000,   15.00,   15.00,   15.00,   0.00000,   0.00000,   0.00000,   0.00000,1,   0.00,   1,1.0000
0 / END OF BRANCH DATA, BEGIN TRANSFORMER DATA
    200,   100,     0,'1 ',1,1,1,   0.00000,   0.00000,2,'            ',1,   1,1.0000
     0.00000,   0.33000,   100.00
1.00000,   0.000,   0.000,   35.00,   35.00,   35.00,   1,    200, 1.15000, 0.85000, 1.03000, 1.01000,  49,  0, 0.00000, 0.00000
1.00000,   0.000
    330,   300,     0,'1 ',1,1,1,   0.00000,   0.00000,2,'            ',1,   1,1.0000
```

a. Power Flow Raw Data

Figure 11.18. Raw Data Files for Two-Phase System

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```
0           / PSS®E-30.0    THU, AUG 05 2004  11:24
   100,'3 ',     0.00000,     0.20000
0  / END OF POSITIVE SEQ. MACHINE IMPEDANCE DATA, BEGIN NEGATIVE SEQ. MACHINE DATA
   100,'3 ',     0.00000,     0.20000
0  / END OF NEGATIVE SEQ. MACHINE IMPEDANCE DATA, BEGIN ZERO SEQ. MACHINE DATA
   100,'3 ',     0.00000,     0.00000
0  / END OF ZERO SEQ. MACHINE IMPEDANCE DATA, BEGIN NEGATIVE SEQ. SHUNT DATA
0  / END OF NEGATIVE SEQ. SHUNT DATA, BEGIN ZERO SEQ. SHUNT DATA
0  / END OF ZERO SEQ. SHUNT DATA, BEGIN ZERO SEQ. NON-TRANSFORMER BRANCH DATA
   200,   300,'1 ',    0.02000,    0.30000,    0.04000,  0.00000,  0.00000,  0.00000,  0.00000
   200,   400,'1 ',    0.04000,    0.45000,    0.06500,  0.00000,  0.00000,  0.00000,  0.00000
   330,   550,'1 ',    0.04000,    0.80000,    0.00000,  0.00000,  0.00000,  0.00000,  0.00000
   440,   550,'1 ',    0.04000,    0.80000,    0.00000,  0.00000,  0.00000,  0.00000,  0.00000
0  / END OF ZERO SEQ. NON-TRANSFORMER BRANCH DATA, BEGIN ZERO SEQ. MUTUAL DATA
```

Connection codes to represent parallel-secondary transformer connections

b. Sequence Raw Data

Raw Data Files for Two-Phase System

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```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      THU, AUG 05 2004  11:30
 SMALL TWO PHASE EXAMPLE                                    BUS DATA

    BUS# X-- NAME   --X BASKV CODE LOADS  VOLT    ANGLE      S H U N T   AREA ZONE OWNER
     100 'GEN-1'      10.000    3    0  1.0000     0.0        0.0        1    1    1
     200 'HYDRO'      132.00    1    0  0.9828    -4.8        0.0        1    1    1
     300 'WEST'       132.00    1    0  0.9747    -6.0        0.0        1    1    1
     330 'EAST-LO     12.000    1    1  0.9672    -7.2        0.0        1    1    1
     400 'EAST'       132.00    1    0  0.9723    -6.6        0.0        1    1    1
     440 'WEST-LO     12.000    1    1  0.9653    -7.6        0.0        1    1    1
     550 'MAIN-LO     12.000    1    1  0.9500    -9.7        0.0        4.0  1    1

    BUS# X-- NAME   --X BASKV ID CD ST   PSI      MVA-LOAD      CUR-LOAD      Y - LOAD  AREA ZONE OWNER
     330 'EAST-LO     12.000 1   1 1  1.000      5.0    2.0     0.0    0.0     0.0    0.0  1    1    1
     440 'WEST-LO     12.000 1   1 1  1.000      5.0    2.0     0.0    0.0     0.0    0.0  1    1    1
     550 'MAIN-LO     12.000 1   1 1  1.000     15.0    8.0     0.0    0.0     0.0    0.0  1    1    1
                                                                                                X-----  REMOTE BUS  -------X
    BUS# X-- NAME   --X BASKV COD MCNS PGEN     QGEN    QMAX    QMIN VSCHED  VACT.   PCT Q   BUS# X-- NAME  --X BASKV
     100 'GEN-1'      10.000   3    1   25.1      6.3    20.0     0.0 1.0000 0.9828     1.0    200 'HYDRO'       132.00

    BUS# X-- NAME   --X BASKV CD ID ST   PGEN     QGEN    QMAX    QMIN   PMAX   PMIN  OWN FRACT   MBASE  Z S O R C E     X T R A N    GENTAP
     100 'GEN-1'      10.000   3  3  1   25.1      6.3    20.0     0.0 9999.0-9999.0     1 1.000    30.0 0.0000 0.2000

  X------ FROM BUS -------X  X------- TO BUS --------X                                                                Z S
     BUS# X-- NAME   --X BASKV     BUS# X-- NAME   --X BASKV CKT   LINE R   LINE X  CHRGING I T   RATEA   RATEB   RATEC LENGTH OWNR FRACT OWNR FRACT
      200 'HYDRO'      132.00*     300 'WEST'       132.00  1    0.01500  0.15000  0.02500   1    40.0    40.0    40.0     0.0    1 1.000
      200 'HYDRO'      132.00*     400 'EAST'       132.00  1    0.02500  0.25000  0.04000   1    40.0    40.0    40.0     0.0    1 1.000
      330 'EAST-LO     12.000*     550 'MAIN-LO     12.000  1    0.05000  0.50000  0.00000   1    15.0    15.0    15.0     0.0    1 1.000
      440 'WEST-LO     12.000*     550 'MAIN-LO     12.000  1    0.05000  0.50000  0.00000   1    15.0    15.0    15.0     0.0    1 1.000

  X------ FROM BUS -------X X------- TO BUS --------X          XFRMER    S W M C C      A C T U A L                    MAGNETIZING Y          N O M I N A L
     BUS# X-- NAME   --X BASKV     BUS# X-- NAME   --X BASKV CKT X-- NAME   --X T 1 T Z M   R 1-2    X 1-2   W1BASE     MAG1      MAG2  TBL  R 1-2    X 1-2
      100 'GEN-1'      10.000     200 'HYDRO'      132.00  1                    1 T T 1 1  0.00000  0.33000   100.0  0.00000  0.00000    0
      300 'WEST'       132.00     330 'EAST-LO     12.000  1                    1 T T 1 1  0.00000  0.15000   100.0  0.00000  0.00000    0
      400 'EAST'       132.00     440 'WEST-LO     12.000  1                    1 T T 1 1  0.00000  0.15000   100.0  0.00000  0.00000    0

  X------ FROM BUS -------X X------- TO BUS --------X                C
     BUS# X-- NAME   --X BASKV     BUS# X-- NAME   --X BASKV CKT W WINDV1  NOMV1  ANGLE  WINDV2  NOMV2   RATEA   RATEB  RATEC OWNR FRACT OWNR FRACT
      100 'GEN-1'      10.000     200 'HYDRO'      132.00  1   1 1.00000 0.0000    0.0 1.00000 0.0000    35.0    35.0   35.0    1 1.000
      300 'WEST'       132.00     330 'EAST-LO     12.000  1   1 1.00000 0.0000    0.0 1.00000 0.0000    20.0    20.0   20.0    1 1.000
      400 'EAST'       132.00     440 'WEST-LO     12.000  1   1 1.00000 0.0000    0.0 1.00000 0.0000    20.0    20.0   20.0    1 1.000

  X------ FROM BUS -------X X------- TO BUS --------X  W C                                                    X---- CONTROLLED BUS ----X
     BUS# X-- NAME   --X BASKV     BUS# X-- NAME   --X BASKV CKT 1 W CN   RMAX    RMIN    VMAX    VMIN  NTPS     BUS# X-- NAME  --X BASKV    CR    CX
      100 'GEN-1'      10.000     200 'HYDRO'      132.00  1   T 1  1 1.15000 0.85000 1.03000 1.01000   49     -200 'HYDRO'       132.00
      300 'WEST'       132.00     330 'EAST-LO     12.000  1   T 1  1 1.15000 0.85000 1.00000 0.98000   49     -330 'EAST-LO      12.000
      400 'EAST'       132.00     440 'WEST-LO     12.000  1   T 1  1 1.15000 0.85000 1.00000 0.98000   49     -440 'WEST-LO      12.000
```

Figure 11.19. Data Listings for Two-Phase System

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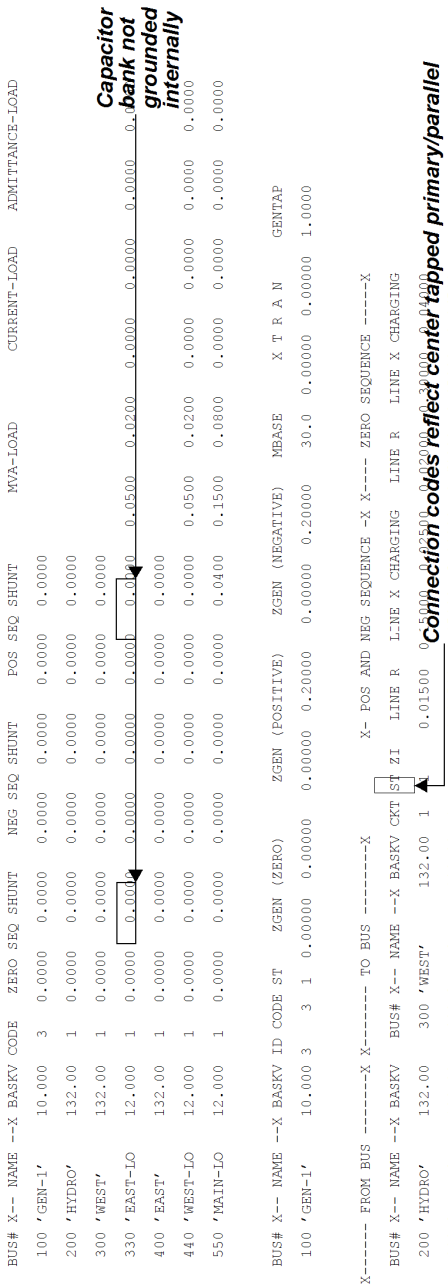

Data Listings for Two-Phase System

The initial condition power flow solution is shown in Figure 11.20, “Initial Condition Power Flow Solution for Two-Phase Sample System”; this is exactly the same as for a three-phase system solution because it involves only balanced operation and the positive sequence. The bus voltages in kilovolts are line-to-line values. This solution was made with the following PSS®E option settings:

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- 50 Hz base frequency.
- Two-phase solution mode.
- Polar output of fault analysis results.

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      TUE, OCT 08 1991  14:58
SMALL TWO PHASE EXAMPLE                                                RATING
SYSTEM FOR PROGRAM APPLICATION GUIDE                                    SET A

BUS    100 GEN-1    10.0 AREA CKT     MW     MVAR     MVA  %I 1.0000PU   0.00  100
 GENERATION                 1         25.1     5.5R   25.7  86 10.000KV
 TO     200 HYDRO     132   1  1      25.1     5.5    25.7  73 1.0313UN

BUS    200 HYDRO     132 AREA CKT     MW     MVAR     MVA  %I 1.0162PU  -4.83  200
                           1                                   134.13KV
 TO     100 GEN-1    10.0   1  1    -25.1    -3.3    25.3  71 1.0313LK
 TO     300 WEST      132   1  1     13.3     2.5    13.5  33
 TO     400 EAST      132   1  1     11.8     0.8    11.9  29

BUS    300 WEST      132 AREA CKT     MW     MVAR     MVA  %I 1.0087PU  -5.91  300
                           1                                   133.15KV
 TO     200 HYDRO     132   1  1    -13.3    -4.8    14.1  35
 TO     330 EAST-LOD12.0   1  1     13.3     4.8    14.1  70 0.9937UN

BUS    330 EAST-LOD12.0 AREA CKT     MW     MVAR     MVA  %I 0.9955PU  -7.04  330
                           1                                   11.946KV
 TO LOAD-PQ                          5.0     2.0     5.4
 TO     300 WEST      132   1  1    -13.3    -4.5    14.0  70 0.9937LK
 TO     550 MAIN-LOD12.0   1  1      8.3     2.5     8.6  58

BUS    400 EAST      132 AREA CKT     MW     MVAR     MVA  %I 1.0066PU  -6.45  400
                           1                                   132.88KV
 TO     200 HYDRO     132   1  1    -11.8    -4.5    12.6  31
 TO     440 WEST-LOD12.0   1  1     11.8     4.5    12.6  63 0.9937UN

BUS    440 WEST-LOD12.0 AREA CKT     MW     MVAR     MVA  %I 0.9938PU  -7.45  440
                           1                                   11.926KV
 TO LOAD-PQ                          5.0     2.0     5.4
 TO     400 EAST      132   1  1    -11.8    -4.3    12.6  63 0.9937LK
 TO     550 MAIN-LOD12.0   1  1      6.8     2.3     7.2  48

BUS    550 MAIN-LOD12.0 AREA CKT     MW     MVAR     MVA  %I 0.9795PU  -9.39  550
                           1                                   11.754KV
 TO LOAD-PQ                         15.0     8.0    17.0
 TO SHUNT                            0.0    -3.8     3.8
 TO     330 EAST-LOD12.0   1  1     -8.2    -2.1     8.5  58
 TO     440 WEST-LOD12.0   1  1     -6.8    -2.0     7.1  48
```

Figure 11.20. Initial Condition Power Flow Solution for Two-Phase Sample System

Figure 11.20, “Initial Condition Power Flow Solution for Two-Phase Sample System” shows output from activity SCOP corresponding directly to Figure 11.19, “Data Listings for Two-Phase System”, with no fault applied. In this report the negative sequence and b-phase fields have no significance. The a and c phase voltage

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fields show sequence and phase voltages on a line-to-ground basis. The current values are in terms of per unit line current.

```
SEQUENCE          /V0/     AN(V0)     /V+/     AN(V+)     /V-/     AN(V-)
PHASE             /VA/     AN(VA)     /VB/     AN(VB)     /VC/     AN(VC)

  550   (P.U.)    0.0000     0.00     0.9795    -9.39     0.0000     0.00
MAIN-LOD12.0      0.9795    -9.39     0.0000     0.00     0.9795   170.61

SEQUENCE          /I0/     AN(I0)     /I+/     AN(I+)     /I-/     AN(I-)
PHASE             /IA/     AN(IA)     /IB/     AN(IB)     /IC/     AN(IC)

TO    330  1      0.0000     0.00     0.0867   155.97     0.0000     0.00
EAST-LOD12.0      0.0867   155.97     0.0000     0.00     0.0867   -24.03

TO    440  1      0.0000     0.00     0.0722   154.05     0.0000     0.00
WEST-LOD12.0      0.0722   154.05     0.0000     0.00     0.0722   -25.95

SUM OF            0.0000     0.00     0.1589   -24.90     0.0000     0.00
CONTRIBUTIONS     0.1589   -24.90     0.0000     0.00     0.1589   155.10


SEQUENCE          /V0/     AN(V0)     /V+/     AN(V+)     /V-/     AN(V-)
PHASE             /VA/     AN(VA)     /VB/     AN(VB)     /VC/     AN(VC)

  330   (P.U.)    0.0000     0.00     0.9955    -7.04     0.0000     0.00
EAST-LOD12.0      0.9955    -7.04     0.0000     0.00     0.9955   172.96

SEQUENCE          /I0/     AN(I0)     /I+/     AN(I+)     /I-/     AN(I-)
PHASE             /IA/     AN(IA)     /IB/     AN(IB)     /IC/     AN(IC)

TO    300  1      0.0000     0.00     0.1407   154.13     0.0000     0.00
WEST     132      0.1407   154.13     0.0000     0.00     0.1407   -25.87
```

Figure 11.21. Output from Short-Circuit Solution Reporting Corresponding to Figure 11.19, "Data Listings for Two-Phase System"

Comparison of the flow into bus 550 from bus 330 in Figure 11.19, "Data Listings for Two-Phase System" and Figure 11.20, "Initial Condition Power Flow Solution for Two-Phase Sample System" and shows:

Received MVA = 8.5

Receiving voltage = 0.9795 per unit = 11.754 kV (L-L) = 5.877 kV (L-G)

Phase current, I_p, from Figure 11.19, "Data Listings for Two-Phase System", =

$$\frac{MVA}{V_{LL}} = \frac{8.5E6}{11.754E3} = 723.2\ A$$

Base phase current =

$$\frac{MVA_{base}}{V_{LL\ base}} = \frac{100E6}{12E3} = 8333.3\ A$$

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Per-unit phase current, from Figure 11.19, "Data Listings for Two-Phase System", =

$$\frac{723.2}{8333.3} = 0.08605 \text{ per unit}$$

Per-unit phase current, from Figure 11.20, "Initial Condition Power Flow Solution for Two-Phase Sample System", = 0.0867 per unit

Rerunning the bus based report with the kVA output option in effect shows that bus 550 receives 8496.5 kVA from bus 330. Redoing the above calculations gives the per-unit current as

$$\frac{8.4965E6}{11.754E3 \times 8333.3} = 0.08674 \text{ per unit}$$

All transformers in the system are connected as shown in Figure 11.13, "Two-Phase System Configuration for Railway Application" with two low-voltage windings in parallel. This is reflected in the connection code data in Figure 11.18, "Raw Data Files for Two-Phase System" and Figure 11.19, "Data Listings for Two-Phase System". The shunt capacitor bank at bus 550 is connected line-to-line and is not grounded internally. Figure 11.22, "Simple L-G Fault at Bus 300" shows the output from the unbalance fault calculation for a single L-G fault applied at bus 300.

As expected, the a-phase bus voltage at the fault is zero, and a current of 1.6417 per unit flows into the fault, and at the fault $I_0 = I_1 = V_{oc} / (Z_0 + Z_1)$.

$$I_0 = I_1 = \frac{1.009}{0.00127 + j0.11696 + 0.28804 + j1.07742}$$

$= 0.1933 - j0.798 = 0.8211 \angle -76.4$

Figure 11.23, "Simple Ground Connection at Bus 330" shows the result for an L-G fault at bus 330. Here, because no zero-sequence ground path exists in the secondary system except for the fault, there is no fault current, and no unbalance appears in the 132 kV primary system.

The ground connection at one point in the system does not affect the balanced load current at bus 330, and the fault current is zero. The a-phase voltage at bus 330 is zero, as expected. The c-phase voltage at bus 330 is listed by SCMU as 1.991 per unit. The phase voltages listed by SCMU are per-unit of phase-to-ground base voltage, and correspond to an a-phase-to-ground voltage of kV.

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```
LINE TO GROUND FAULT AT BUS    300 [WEST      132]    PHASE 1
  L-G Z =    0.0000E+00    0.0000E+00


              SEQUENCE THEVENIN IMPEDANCES AT FAULTED BUSES:

  BUS    NAME   BSKV          ZERO              POSITIVE            NEGATIVE
  300 WEST      132     0.00127  0.11696    0.28804  1.07742    0.00000  0.00000



LINE TO GROUND FAULT AT BUS    300 [WEST      132]:

SEQUENCE            /V0/      AN(V0)      /V+/      AN(V+)      /V-/      AN(V-)
PHASE               /VA/      AN(VA)      /VB/      AN(VB)      /VC/      AN(VC)

  300   (P.U.)      0.0960    -172.91     0.0960       7.09     0.0000       0.00
WEST      132       0.0000       0.00     0.0000       0.00     0.1920    -172.91

SEQUENCE            /I0/      AN(I0)      /I+/      AN(I+)      /I-/      AN(I-)
PHASE               /IA/      AN(IA)      /IB/      AN(IB)      /IC/      AN(IC)

FROM    200 1       0.1810     -80.09     0.7558     -80.88     0.0000       0.00
HYDRO     132       0.9368     -80.73     0.0000       0.00     0.5749      98.87

FROM    330 1       0.6400     -82.91     0.0679     -98.25     0.0000       0.00
EAST-LOD12.0        0.7057     -84.37     0.0000       0.00     0.5749     -81.13


SUM OF CONTRIBUTIONS INTO BUS    300 [WEST      132]:

  300               0.8208     -82.29     0.8208     -82.29     0.0000       0.00
WEST      132       1.6417     -82.29     0.0000       0.00     0.0000       0.00

CONTRIBUTIONS EQUIVALENT POSITIVE SEQUENCE ADMITTANCE     0.0927    -8.5492
```

Load currents flowing in c-phase

a-phase fault current contributions

I0 = I1 for a-phase fault current

Fault current

Figure 11.22. Simple L-G Fault at Bus 300

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```
UNBALANCES APPLIED:

LINE TO GROUND FAULT AT BUS    330 [EAST-LOD12.0]    PHASE 1
  L-G Z =    0.0000E+00    0.0000E+00


              SEQUENCE THEVENIN IMPEDANCES AT FAULTED BUSES:

  BUS    NAME   BSKV          ZERO              POSITIVE             NEGATIVE
  330 EAST-LOD12.0  ******************    0.32217  1.14687    0.00000  0.00000

LINE TO GROUND FAULT AT BUS    330 [EAST-LOD12.0]:

SEQUENCE          /V0/     AN(V0)     /V+/     AN(V+)     /V-/     AN(V-)
PHASE             /VA/     AN(VA)     /VB/     AN(VB)     /VC/     AN(VC)

  330   (P.U.)    0.9955    172.96    0.9955     -7.04    0.0000      0.00
EAST-LOD12.0      0.0000      0.00    0.0000      0.00    1.9910    172.96

SEQUENCE          /I0/     AN(I0)     /I+/     AN(I+)     /I-/     AN(I-)
PHASE             /IA/     AN(IA)     /IB/     AN(IB)     /IC/     AN(IC)

FROM   300 1      0.0000      0.00    0.1407    -25.87    0.0000      0.00
WEST     132      0.1407    -25.87    0.0000      0.00    0.1407    154.13

FROM   550 1      0.0000      0.00    0.0867    155.97    0.0000      0.00
MAIN-LOD12.0      0.0867    155.97    0.0000      0.00    0.0867    -24.03

SUM OF CONTRIBUTIONS INTO BUS    330 [EAST-LOD12.0

  330             0.0000      0.00    0.0541    -28.84    0.0000      0.00
EAST-LOD12.0      0.0541    -28.84    0.0000      0.00    0.0541    151.16

CONTRIBUTIONS EQUIVALENT POSITIVE SEQUENCE ADMITTANCE    0.0505   -0.0202
```

Zero voltage at point of ground on system

Voltage on ungrounded phase is fully offset

Ground does not affect balanced load current; compare with Figure 10-20

Fault current is zero

Figure 11.23. Simple Ground Connection at Bus 330

Figure 11.24, "Secondary System Grounded at Buses 330 and 550" shows the results when the a-phase of the 12-kV system is grounded (by an L-G fault) at both bus 330 and bus 550. In some cases, the ungrounded zero-sequence network cannot be handled by PSS®E because its admittance matrix is singular. This problem is rectified by connecting a low admittance branch to ground in the zero sequence at an appropriate bus; this small shunt admittance is insignificant in relation to the admittances of the system branches, but is adequate to avoid the numerical problems stemming from the singular matrix.

Figure 11.25, "Current Flows (per unit) from Figure 10-24" shows the phase currents from Figure 11.24, "Secondary System Grounded at Buses 330 and 550" on the 2-line diagram of the secondary system. Note the following:

- The SUM OF CONTRIBUTIONS shown by the unbalance fault calculation for the a-phase includes both load and ground current.

Current does flow on the a-phase conductor from bus 330 to 550 even though it is solidly grounded at both ends, as a result of the mutual coupling between the two phases of the line. No current would flow in the a-

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phase conductor if the two phases were of isolated-phase construction and had no phase-to-phase mutual impedance.

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```
LINE TO GROUND FAULT AT BUS    330 [EAST-LOD12.0]    PHASE 1
  L-G Z =    0.0000E+00    0.0000E+00

LINE TO GROUND FAULT AT BUS    550 [MAIN-LOD12.0]    PHASE 1
  L-G Z =    0.0000E+00    0.0000E+00


              SEQUENCE THEVENIN IMPEDANCES AT FAULTED BUSES:

  BUS    NAME   BSKV          ZERO              POSITIVE            NEGATIVE
  330 EAST-LOD12.0    0.00007*********    0.32217  1.14687    0.00000  0.00000
  550 MAIN-LOD12.0    0.04007-99.20067    0.42517  1.25624    0.00000  0.00000


SEQUENCE          /V0/     AN(V0)     /V+/     AN(V+)     /V-/     AN(V-)
PHASE             /VA/     AN(VA)     /VB/     AN(VB)     /VC/     AN(VC)

  330   (P.U.)    1.0066    172.51    1.0066     -7.49    0.0000      0.00
EAST-LOD12.0      0.0000      0.00    0.0000      0.00    2.0132    172.51

SEQUENCE          /I0/     AN(I0)     /I+/     AN(I+)     /I-/     AN(I-)
PHASE             /IA/     AN(IA)     /IB/     AN(IB)     /IC/     AN(IC)

FROM   300 1      0.0000      0.00    0.1517    -23.04    0.0000      0.00
WEST    132       0.1517    -23.04    0.0000      0.00    0.1517    156.96

FROM   550 1      0.0386    -26.93    0.0616    155.92    0.0000      0.00
MAIN-LOD12.0      0.0231    160.69    0.0000      0.00    0.1001    -25.18

SUM OF CONTRIBUTIONS INTO BUS    330 [EAST-LOD12.0]:

  330             0.0386    -26.93    0.0902    -22.33    0.0000      0.00
EAST-LOD12.0      0.1287    -23.71    0.0000      0.00    0.0518    161.10

CONTRIBUTIONS EQUIVALENT POSITIVE SEQUENCE ADMITTANCE    0.0866   -0.0229

LINE TO GROUND FAULT AT BUS    550 [MAIN-LOD12.0]:

SEQUENCE          /V0/     AN(V0)     /V+/     AN(V+)     /V-/     AN(V-)
PHASE             /VA/     AN(VA)     /VB/     AN(VB)     /VC/     AN(VC)

  550   (P.U.)    0.9953    170.86    0.9953     -9.14    0.0000      0.00
MAIN-LOD12.0      0.0000      0.00    0.0000      0.00    1.9906    170.86

SEQUENCE          /I0/     AN(I0)     /I+/     AN(I+)     /I-/     AN(I-)
PHASE             /IA/     AN(IA)     /IB/     AN(IB)     /IC/     AN(IC)

FROM   330 1      0.0386    153.07    0.0616    -24.08    0.0000      0.00
```

Zero voltage at grounded point

Note current flow in a-phase conductor even though it is grounded at both ends

Load current

Zero voltage at grounded point

Algebraic sum of load and ground current

Load current

Figure 11.24. Secondary System Grounded at Buses 330 and 550

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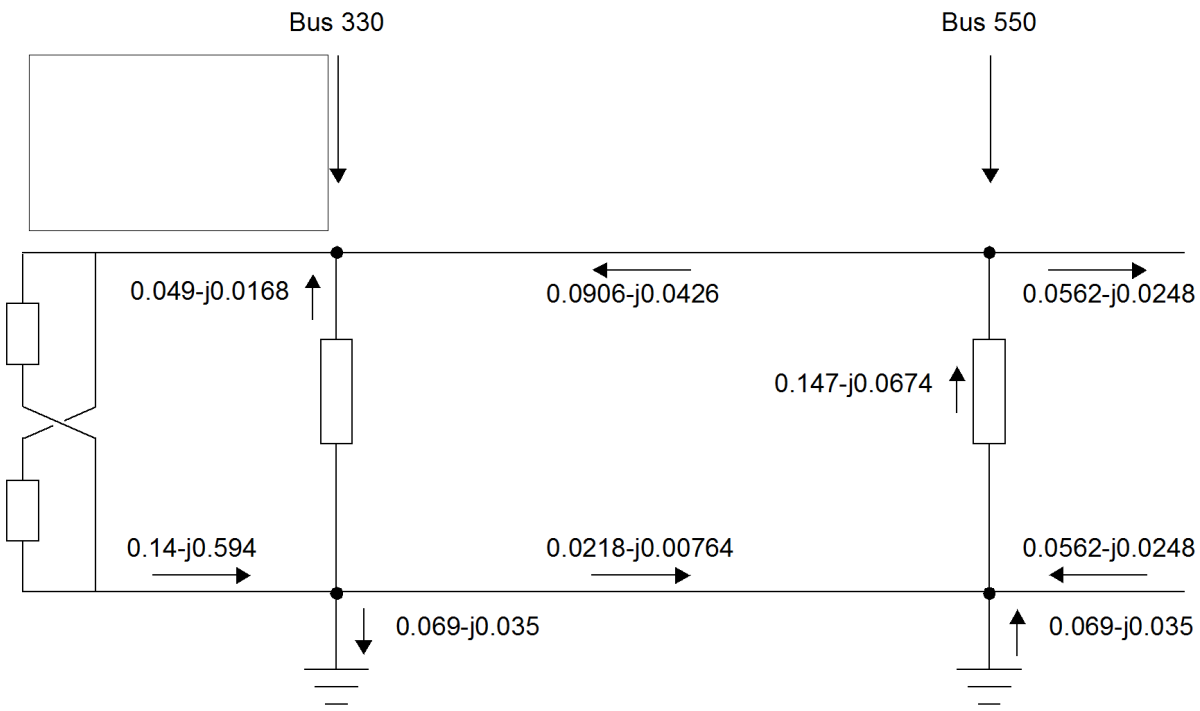

Figure 11.25. Current Flows (per unit) from Figure 10-24

11.6.18. Application Notes

PSS®E treats unbalanced fault analysis as a direct extension of its power flow activities. All that is needed to perform unbalanced fault calculations on a fully detailed, solved power flow case is to augment the positive sequence power flow data with the corresponding negative and zero sequence data, and to use fault analysis activities in place of power flow activities. The working case always includes provision for negative and zero sequence data values corresponding to the positive sequence parameters of the appropriate system components. This data may be introduced into the working case at any time and, after being introduced, it is saved and retrieved by activities SAVE and CASE, respectively, as an integral part of a Saved Case.

The fault analysis activities of PSS®E are capable of working with the transmission network represented in the same degree of detail as in power flow. Specifically, the fault analysis activities recognize:

1. Both resistance and reactance, and include all actual shunt branches and line charging in the three sequence networks.

2. Both the magnitude ratio and phase shift of all transformers, including the inherent shift of delta-wye transformers if it is specified in the positive sequence transformer data.

3. The actual spread of internal voltage magnitude and phase angle of generators as initialized from a solved power flow case.

4. Loads by converting them to equivalent constant shunt admittance.

Each fault calculation is a complete solution for the full sequence and phase specification of conditions at every bus. The user may obtain output of voltage at any bus, or current, apparent impedance or apparent

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admittance at the entry to any branch (refer to activity SCOP). Multiple unbalanced conditions may be applied simultaneously.

After the system positive sequence and fault analysis data have been properly coordinated and saved, the fault analysis process is a direct parallel of the power flow switching study procedure outlined in Section 11.6.7, "Operation of Activity SCMU". The principal steps in a fault analysis calculation are:

1. Set up the power flow case for the pre-switching system condition. When a solved case is to be the initial condition, the working case must be a solved case. Otherwise, the bus boundary conditions must correspond to the pre-fault condition (refer to activity FLAT). This provides the basis for the initialization of all generators and the conversion of all loads to constant admittance.

2. Change branch and generator status as required, or adjust loads and shunts, to produce the desired post-switching system conditions excluding unbalances. This may require the use of activities CHNG and SQCH or the *[Spreadsheet]*.

3. Establish all required sequence network admittance matrices for the post-switching condition with activity SEQD. When the sequence network setup option is disabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), activity SEQD must be executed by the user *before* initiating activity SCMU. When the sequence network setup option is enabled, activity SCMU automatically executes activity SEQD each time it is executed.

4. Apply all unbalances that are to exist in the post-switching condition and solve the resulting interconnected sequence networks for the complete set of system voltages. This step is handled by activity SCMU. Activity SCMU will re-establish the network admittance matrices for the post-switching condition prior to performing the calculations.

5. Use activity SCOP to obtain output as required.

It is important to note the relationship and sequencing of steps 2 through 4. Steps 2 and 3 represent the set-up of the balanced post-switching system condition. In most cases, step 2 is not needed and the procedure is to use activity SEQD followed by activity SCMU. When calculating multiple fault cases from the identical pre-switching system condition when no balanced switchings are required to get to the post-switching conditions, it is not necessary to execute activity SEQD prior to each fault calculation. Activity SEQD needs to be executed following any equipment status changes, or any branch, shunt or load data changes with activities CHNG, SQCH, or the *[Spreadsheet]*.

In applying the unbalances shown in Figure 11.5, "Unbalances Modeled in Activity SCMU", the following points should be noted:

1. Both resistance and reactance must be specified for each impedance shown. Impedances are specified in per unit relative to base impedance.

2. It is *not* necessary to multiply ground fault impedances by three; this is done automatically by activity SCMU.

3. The branch on which an in-line unbalance is to be applied must be in-service in the power flow case before activity SEQD is executed. Activity SCMU opens the branch at one end automatically if the one end opened option is selected and re-establish the network admittance matrices.

4. Only one in-line unbalance may be applied during any execution of activity SCMU. The location of this unbalance is automatically assigned the bus number 999999 by activity SCMU.

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5. For in-line unbalances, the user is required to enter a distance factor, A, specifying the distance of the fault from the from bus or closed end of the line. The A factor is equal to the distance from this bus to the unbalance in per unit of the total line length. The user is cautioned against the use of A factors very close to zero or one because faults very close to the sending or receiving end of the branch can give very low impedances between the dummy bus and the closer terminal bus.

6. The single and double phase closed unbalances place new phase branches in parallel with any that are already present between the specified buses in the working case.

7. The single and double phase closed unbalances are applicable to discrete components such as series capacitors and jumpers, but are not applicable to transmission lines.

8. All unbalances applied in an execution of activity SCMU remain in effect through subsequent executions of activity SCOP. Each execution of activity SEQD or SCMU clears all previously applied unbalances and deletes the dummy buses 999999 and 999998 from the system.

The user may create unbalanced conditions other than those that are automated by activity SCMU by judicious use of dummy buses and low impedance branches. The fault analysis activities are normally able to handle a branch impedance as low as j0.0001 per unit without difficulty. However, when introducing jumper branches which are not being treated as zero impedance lines, the user should examine the impedances of other branches connected to these buses to check for an extremely wide range of impedances, which could result in numerical precision problems.

In the summary output for ground faults (refer to Section 11.6.8, "Unbalance Condition Summary Report"),the sums of the ground currents in (6) and the fault currents in (7) should equal the sum of contributions tabulated in (4). Differences give an indication of the degree of numerical precision error present in the fault case solution.

Activity SCMU is sensitive to the fault analysis modeling option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) in effect at the time activity SEQD is selected (refer to Section 11.5.1, "Application Notes"). When the two phase system modeling option is in effect, L-G faults, either at a bus or along a branch, and the single phase closed unbalance *must not* be applied to the B phase (i.e., *do not* enter a 2 in selecting the phase). Similarly, for the L-L-G faults and the two phase closed unbalance (which are actually balanced conditions), the B phase *must* be designated as the excluded phase. A three phase fault *must not* be applied.

Using the two phase option, negative sequence and B phase quantities are shown in the output reports as zero.

Additional Information
PSS®E GUI Users Guide, Running Fault Analysis under Unbalance Condition *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Fault Analysis under Unbalance Condition

11.7. Multiple Unbalanced Fault Report

SCOP

Run Activity SCOP - GUI
Fault > Solve and report network with unbalances (SCMU/SCOP)...

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Run Activity SCOP - GUI
[Multiple Simultaneous Unbalances] Solution Output Tab
Run Line Mode Activity SCOP - CLI
ACTIVITY? >>**SCOP** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The unbalanced network solution output activity SCOP tabulates the results of activity SCMU. It allows the user to obtain output of all bus voltages and either branch currents, apparent impedances or apparent admittances.

The user may process the entire working case or specify a bus subsystem (refer to Section 4.8, "Subsystem Selection").

The user mus select one of the following branch quantities:

- Currents (default)
- Apparent impedances
- Apparent admittances

Activity SCOP recognizes the fault analysis output options (refer to Section 3.3.3, "Program Run-Time Option Settings"), with quantities printed in either physical units or per unit, and in either rectangular or polar coordinates. These option settings may be modified with activity OPTN.

The data printed for any bus, i, consists of:

1. Sequence and phase voltages at bus i. This is identical to the bus voltage block described in Section 11.6.8, "Unbalance Condition Summary Report".
2. Series branch quantity for each branch and machine connected to bus i consisting of either:
   a. Branch series currents *leaving* bus i, in either per unit or amps. These are in the reverse direction of the currents described in Section 11.6.8, "Unbalance Condition Summary Report".
   b. Branch apparent impedances looking down each branch from bus i, expressed in either per unit or ohms. These apparent impedances are defined as:

$$Z_{seq\ ij} = \frac{V_{seq\ i}}{I_{seq\ ij}} \qquad Z_{phase\ ij} = \frac{V_{phase\ i}}{I_{phase\ ij}}$$

   where the sequence and phase currents are the total currents flowing into the line at bus i, including contributions from line charging capacitance, line connected shunt admittances, and transformer magnetizing admittance. Infinite impedance is printed as 9999 per unit or 999999 ohms.

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c. Branch apparent admittances looking down each branch from bus i, expressed in either per unit or mmhos. Apparent admittances are defined as the reciprocal of apparent impedances. Infinite admittance is printed as 9999 per unit or 99999 mmhos.

3. When branch currents are being tabulated, the sum of all contributions flowing into bus i. This is the total current apparently flowing to ground at bus i and is identical to the sum of contributions block described in Section 11.6.8, "Unbalance Condition Summary Report".

4. When branch currents are being tabulated, the sum of load and shunt current at bus i. This is identical to the load and shunt current block described in Section 11.6.8, "Unbalance Condition Summary Report" Load and shunt current output is suppressed if the three sequence shunt and load contributions are all zero.

Note that the direction of current flow in the series branch output is the *reverse* of that in the summary output of activity SCMU; it follows the power flow output convention of current leaving the bus.

When output is directed to the user's terminal, a summary description of each active unbalance is printed at the top of the first page of output produced by activity SCOP. Otherwise, the unbalance summary is printed on each page of output of activity SCOP.

Activity SCOP may be terminated by entering the AB interrupt control code.

Additional Information
PSS®E GUI Users Guide, Running Detailed Fault Analysis under Unbalance Condition *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Detailed Fault Analysis under Unbalance Condition

11.8. Calculating Automatic Sequencing Fault

ASCC

There are two APIs which run activity ASCC.

ASCC_1A api

- allows only three phase and line-to-ground fault types
- can only be run from BAT commands and automation files
- generates text output report in original format

ASCC_3 api

- in addition to three phase and line-to-ground fault types, allows line-line-to-ground and line-to-line fault types
- allows to display fault currents on slider diagram
- allows to save fault calculation results into results file
- allows contour plots of maximum fault currents
- can be run from GUI, BAT commands and automation files

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- generates text output report in new format

Requirements / Prerequisites, ASCC_1A
Validly specified power flow case with sequence data appended to it. Reading Sequence Data
Run Line Mode Activity ASCC - CLI
ACTIVITY? >>**ASCC** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

Requirements / Prerequisites, ASCC_3
Validly specified power flow case. For unbalanced fault calculations, sequence data appended to saved case. Reading Sequence Data
Run Activity ASCC - GUI
Fault > Automatic sequence fault calculation (ASCC)... *[Automatic Sequence Fault Calculation]*
Interrupt Control Codes
AB

The automatic sequencing short circuit calculation activity ASCC allows the user to apply a series of single faults at various locations in the working case. Activity ASCC provides for:

1. ASCC_1A api: A pre-fault network condition either as specified in the working case or, optionally, corresponding to classical flat fault analysis assumptions.

   ASCC_3 api: A pre-fault network condition can be:

   - as specified in the working case, i.e., from working case power flow solution
   - corresponding to classical flat fault analysis assumptions
   - linear power flow assumptions, i.e, set synchronous and asynchronous machines real and reactive power to zero, and set machine bus voltages to specified voltage.
   - set faulted bus voltage at specified value.

2. The specification of the subsystem to be processed according to the standard selection criteria used throughout PSS®E. Each electrically connected bus in the specified subsystem becomes the home bus for one or more fault analysis network solutions.

3. ASCC_1A api: The application of only three phase faults or both three phase and single-line-to-ground faults at each fault location. ASCC_3 api: If sequence data is not present in the working case, only three phase faults will be calculated.

4. The application of the faults selected in (3) at each home bus selected in (2).

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5. Optionally, (4) again with each branch connected to the home bus in turn removed from service.

6. Optionally, (4) again with the far end of each branch connected to the home bus in turn opened, and the faults applied at the line end position.

7. Optionally, (4) again with specified branches in turn removed from service.

8. A selection of output options for each fault applied, ranging from a tabulation of fault currents through detailed output of conditions at the home bus along with detailed output for all buses up to < n > levels removed from the home bus.

ASCC_1A api: If sequence data has not been read into the working case via activity RESQ, an appropriate error message is printed and activity ASCC is terminated.

ASCC_3 api: If sequence data is not present in the working case, only three phase faults will be calculated.

The user specifies one or all of the following fault calculation options for the correct execution of activity ASCC.

Table 11.1. Automatic Sequencing Short Circuit Calculation Options

Specification	Description
Faults to apply	ASCC_1A api only: Three phase fault: Apply only three phase fault. Line to ground (L-G) fault: If enabled, both a three phase fault and a phase A to ground fault are applied at each fault location. ASCC_3 api only: Apply selected fault types from three phase, line-to-ground, line-line-to-ground, and line-to-line faults.
Line outage cases	If enabled, the selected fault calculations are repeated at each home bus < n > additional times: once with each of its < n > connected branches placed out-of-service. In addition, if any branches have been designated in the Fault Control Data File (refer to Section 11.8.4, "Detailed Bus Output Report") as branches to be outaged for the current home bus, these fault cases are also calculated.
Line end faults	If enabled, the selected fault calculations are repeated for each home bus at least < n > additional times: once (twice for any three-winding transformer which has all its windings in-service) with each of its < n > connected branches opened at the far end, with the fault located at the line end position. As in activity SCMU, a dummy bus, numbered 999999, is introduced at the opened end of the branch for each line end fault (refer to Line Faults).
Impose flat conditions	ASCC_1A api only:

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Specification	Description
	If enabled (default), classical fault analysis conditions are assumed for the network in the working case. • Bus voltages are assumed to be one per unit at zero phase angle. • Constant power, current, and admittance loads are neglected in the positive and negative sequence networks. Grounded loads are represented in the zero sequence network. • Generator and induction machine power outputs are assumed to be zero. • FACTS devices and dc lines are ignored. • Transformer phase shift angles are assumed to be zero. Any transformer impedance, which is a function of phase shift angle, is assumed to be at nominal value. • Transformer turns ratios are assumed to be one. This includes generator step-up transformers, which are modeled as part of the machine representation (i.e., the GENTAP). Any transformer impedance, which is a function of turns ratio, is assumed to be at nominal value. • Line charging is neglected in all three sequence networks. • Fixed bus shunts are neglected in all three sequences. Switched shunts are neglected in all three sequences. Line shunts and magnetizing admittances are neglected in all three sequences. • The zero sequence ground ties created by grounded transformer windings are represented; see ???, and ??? and ???. If disabled, the level of network modeling and the bus boundary conditions used by activity ASCC as the pre-fault network condition are as specified in the working case at the time activity ASCC is selected. If unblocked dc lines or in-service FACTS devices are present in the working case, the user specifies their treatment in the fault analysis solution, either to block or to represent dc lines and FACTS devices as load.

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Specification	Description
Represent dc lines and FACTS devices as load	DC lines and FACTS devices can be blocked or represented as load. If represented as load, the apparent ac system complex loads are converted to positive sequence constant admittance load at the buses at which these quantities are injected into the ac system during normal power flow work. In the negative and zero sequence networks, dc lines and FACTS devices are represented as open circuits. The selected option applies to all dc lines and FACTS devices in the working case.
Apply transformer impedance correction to zero sequence	If the line-to-ground fault option had been selected and the actual positive sequence impedance of any transformer in the working case differs from its nominal value (refer to Transformer Impedance Correction Tables), the user may apply transformer impedance correction to zero sequence impedance. The zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. Otherwise, all zero sequence transformer impedances are left at their nominal values (i.e., the values entered in activities RESQ, TRSQ, or SQCH). The same treatment applies to *all* transformers in the system that are not at nominal impedance.
Synchronous and Asynchronous machines power output option	ASCC_3 api only: When selected synchronous and asynchronous machines real and reactive power poutputs are set to zero.
Transformer tap ratios and phase shift angles option	ASCC_3 api only: Select either of the following options to set transforer off-nominal tap ratios and phase shift angles data. • Leave tap ratios and phase angles unchanged • Set tap ratios = 1.0 pu and phase angles = 0 • Set tap ratios = 1.0 pu and phase angles unchanged • Set tap ratios unchanged and phase angles = 0
Generator Reactance option	ASCC_3 api only: Select either of the following synchronous machine reactances to use in fault analysis.

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Specification	Description
	• Subtransient • Transient • Synchronous
Line charging option	ASCC_3 api only: Select either of the following options to set line charging susceptance data. • Leave line charging unchanged • Set line charging to zero in positive and negative sequence • Set line charging to zero in all sequences
Shunt option	ASCC_3 api only: Select either of the following options to set fixed shunts, loads, and switched shunts and transformer magnetisizing • Leave shunts unchanged • Set shunts to zero is positive and negative sequence • Set shunts to zero is all sequences
Load option	ASCC_3 api only: Select either of the following options to set load data. • Leave loads unchanged • Set loads to zero s positive and negative sequence • Set loads to zero is all sequences
Pre-fault Bus Voltage option	ASCC_3 api only: Select either of the following options to set pre-fault bus voltages. Use the bus voltages as obtained in power flow solution Set all bus voltages to a specified value Set faulted bus voltage to a specified value
Output option	The user specifies one of the following output options:

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Specification	Description
	ASCC_1A api: • Full output at home bus and N levels • Full output at home bus, summary at N levels • Fault current summary table ASCC_3 api: • Fault current summary table • Total fault currents • I"k contributions to N levels away • Total fault currents and I"k contributions to N levels away
Number of levels back for contribution output	If one of the full output options is selected, the user must indicate the portion of the working case for which output is to be generated for each fault calculation. The default entry for the number of levels back for contributions output is 3. If zero is specified, activity ASCC restricts the output to the home bus for each fault. If 1 is specified, output for the home bus is printed followed by the output for all buses connected to it for each fault case solution. Generalizing, a response of a positive number < n > results in output for the home bus, followed by output for all buses connected to it, followed by output for all buses two buses away from it, and so on up to < n > levels removed from the home bus.
Unsymmetrical fault reporting	ASCC_1A api only: This defines the form of output for cases involving L-G faults. If the single-line-to-ground fault and one of the full output at the home bus options have been selected above, the user may specify one of the following options for unsymmetrical fault reporting: • A phase currents / apparent impedances • 3 x zero sequence currents / zero sequence apparent impedances • Both sets of currents / impedances
Fault control input file	Optional Fault Control Data file (*.fcd), a file containing data records of the form defined in Fault Control Data File Contents. Through the use of this

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Specification	Description
	file, various faulting and reporting options may be specified. Refer to Section 11.8.1, "Automatic Sequencing Fault Selection Options" for details and Section 2.4.1, "File Usage " for the file specification conventions used by activity ASCC. This file provides the user with more flexibility in selection of faults to be simulated and in the manner in which output is limited or extended.
Relay file	Optional Relay Output Data file (*.rel) (see Section 11.8.7, "Relay File Output"), a file into which activity ASCC summarizes the faults performed and the short-circuit current levels in pu. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity ASCC. Refer to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity ASCC. The file is generated as a * *.rel* type in ASCII format; readily portable to an Excel type. Bypass the entry to suppress the optional file generation.
Results file	ASCC_3 api only: Optional ASCC Results Output file (*.sc), a file into which ASCC_2 api saves fault calculation results. This file is a binary file, and can be used to reproduce ASCC calculation results in report format and/or on slider diagram without re-running ASCC_2.

11.8.1. Automatic Sequencing Fault Selection Options

The term home bus refers to any bus location selected for application of either three phase faults or both three phase and single-line-to-ground faults. Figure 11.26, "Home Bus" shows a three bus system in which Bus I has been selected for fault application L.

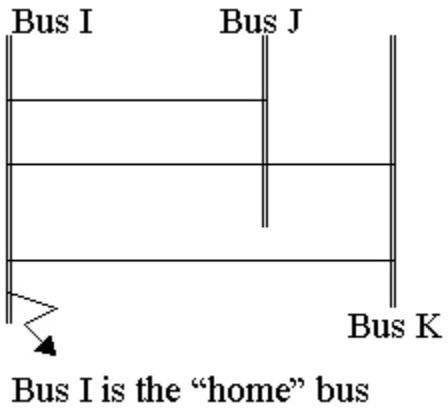

Figure 11.26. Home Bus

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Bus I is the home bus and if faults are to be selected only at home buses there will be a maximum of 2 faults applied at Bus I, if both a three phase and a single-line-to-ground fault are applied. With only a 3-phase fault applied the number of faults will be limited to one.

Figure 11.27, "Faults at Home Bus for Each Outgoing Line" shows the option where a fault is to be applied at the home bus with all circuits in service and with each outgoing line out of service in turn. In this situation, for this home bus, there will be 4 faults if only one fault type is selected. If line outages are performed on a selective basis instead of selecting all outgoing branches, the number of faults applied will clearly be reduced.

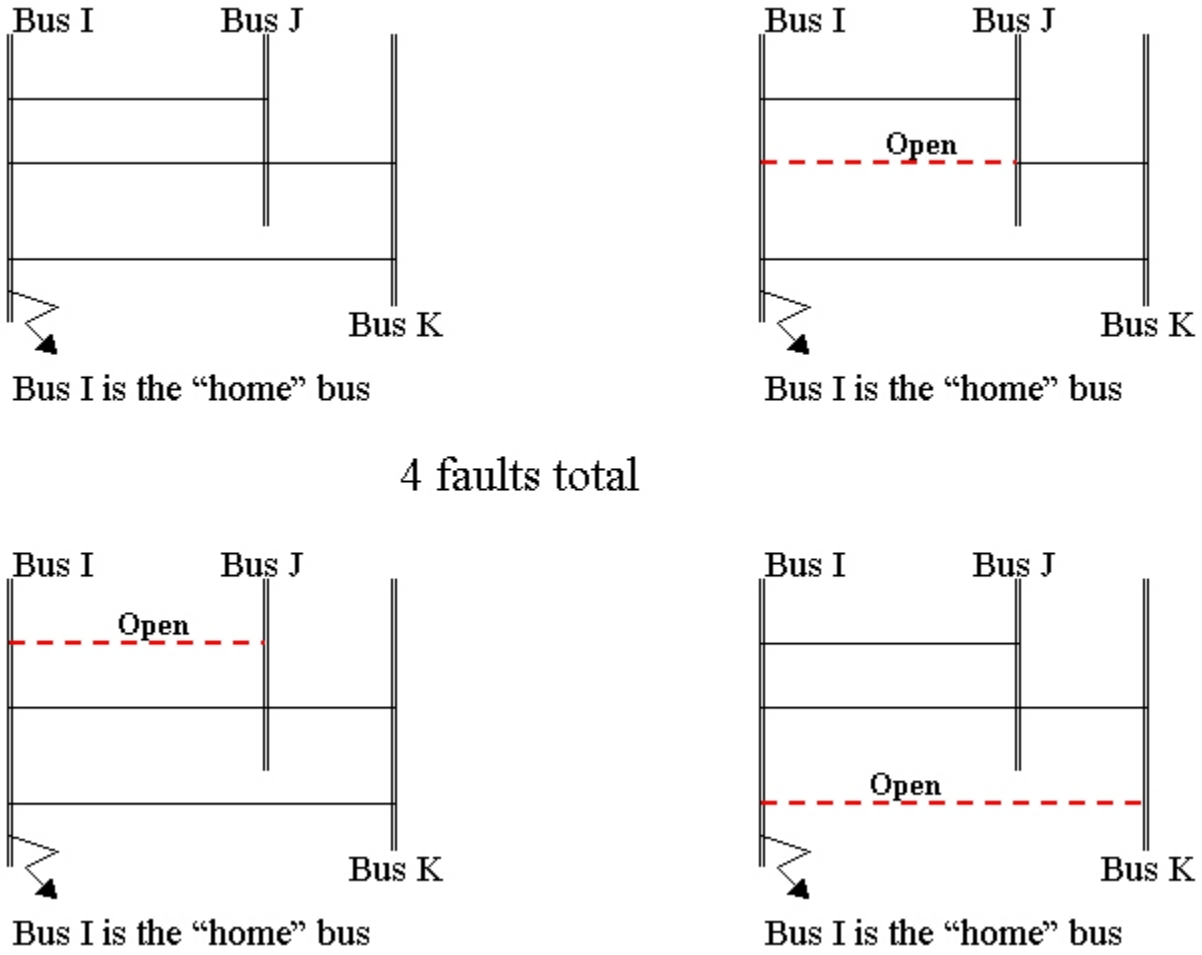

Figure 11.27. Faults at Home Bus for Each Outgoing Line

Figure 10-28 shows the option where a fault is to be applied at the home bus with all circuits in service and faults applied at the open end of each outgoing line. In this situation, for this home bus, there will be 4 faults if only one fault type is selected.

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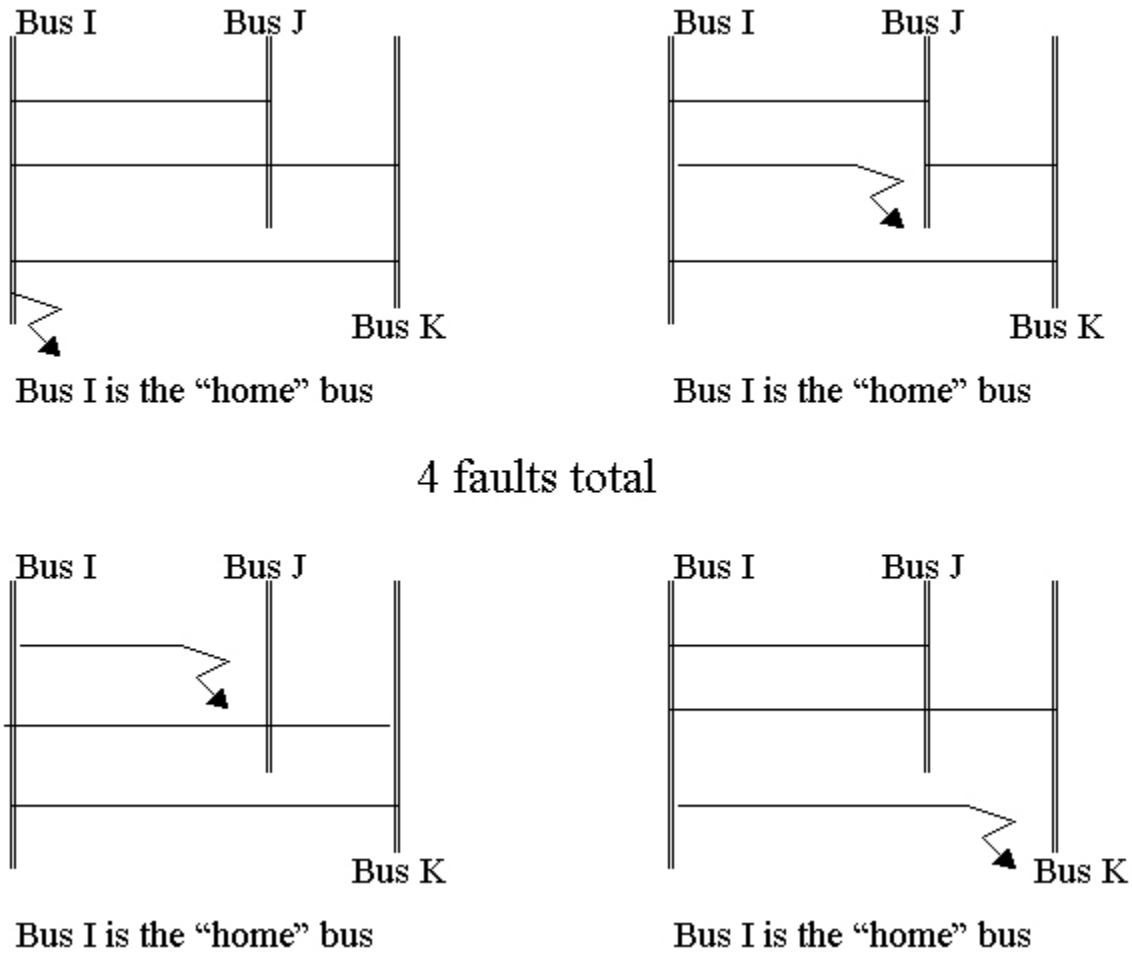

Figure 11.28. Home Bus and Open Line End Faults

The user can select output options for each fault applied, ranging from a tabulation of fault currents through detailed output of conditions at the home bus along with detailed output for all buses up to < n > levels away from the home bus.

Figure 11.29, "Clarification of Home Bus and < n > Levels Away" clarifies the concept of < n > levels away. It can be seen that, given the possibility of a large combination of home buses, line out cases and line open end cases, coupled with a value of < n > greater than 1, the possibility arises for a very large output listing or file.

In the diagram, for the home bus shown, there are 3 buses which are 1 level away and 2 buses which are 2 levels away.

The user should take care in selection of the number of fault and output options.

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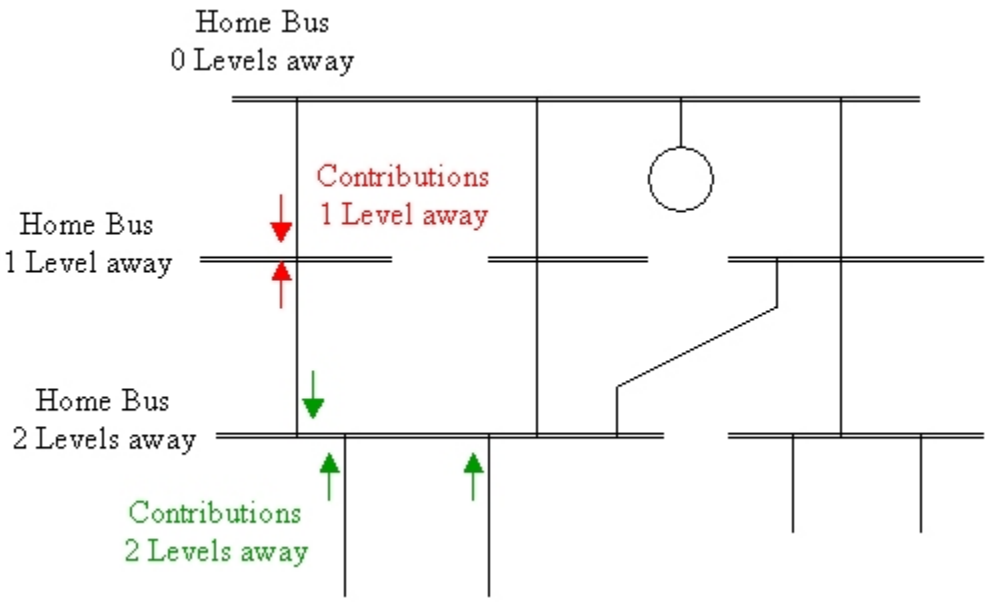

Figure 11.29. Clarification of Home Bus and < n > Levels Away

11.8.2. Fault Control Data File Contents

The Fault Control Data File consists of two groups of records. With the first group of records, buses may be categorized as not to be faulted. This provides, for example, for the suppression of fault calculations at dummy buses. In addition, for the purpose of reporting bus conditions for faults at other buses, each bus may be categorized as being either:

1. Printed and counted in the levels away calculation.
2. Not printed but counted.
3. Not printed and not counted.

This group of records is of the form:

```
IBUS, FCODE, PCODE
```

where:

IBUS	Bus number. Bus IBUS must be present in the working case. No default is allowed.
FCODE	Fault code of either zero (suppress faulting of bus IBUS) or one (allow bus IBUS to be faulted). FCODE = 1 by default.
PCODE	Print control code for conditions at bus IBUS when some other bus is faulted. PCODE may be specified as either zero (no reporting, and not counted in the levels away calculation), one (no reporting, but counted

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	in the levels away calculation), or two (reported and counted). PCODE = 2 by default.

Data records may be entered in any order. This group of records is terminated with a record specifying an IBUS value of zero.

The hidden star point buses of three-winding transformers are always assumed to have FCODE and PCODE values of zero. Any other bus for which no data record is read is assigned the default values given above.

With the second group of records, for any bus to be faulted, the following may be specified:

1. Up to 20 buses where conditions are to be tabulated for all fault cases calculated for this home bus. These buses are in addition to those printed as a result of the levels away selection.

2. Up to eight additional branches to be outaged in turn with the home bus faulted. These fault cases are calculated only if the line out option was selected, and are in addition to the automatic outaging of the branches connected to the home bus. Only non transformer branches and two-winding transformers may be specified.

Data for each home bus specified here is entered on three consecutive data records as follows:

```
IBUS
JBUS1, JBUS2, ... JBUS20
I1, J1, CKT1, ... I8, J8, CKT8
```

where:

IBUS	Bus number. Bus IBUS must be present in the working case. No default is allowed.
$JBUS_i$	Bus number of a bus to be reported for home bus IBUS fault calculations. The first zero value of $JBUS_i$ is interpreted as the end of JBUS values for bus IBUS. $JBUS_i$ = 0 by default.
I_i, J_i, CKT_i	The from bus number, to bus number and branch circuit identifier respectively of a branch to be outaged with bus IBUS faulted. The first zero value of I_i is interpreted as the end of branch specifications for bus IBUS. I_i = 0 and CKT_i = 1 by default; no default is allowed for J_i.

Input is terminated with a record specifying an IBUS value of zero.

If any of the data records in the block for bus IBUS contains an error (e.g., bus not found), an appropriate message is printed and the entire block for bus IBUS is ignored.

All three records must be entered for each bus block specified. If, for example, one or more output buses are to be specified for bus IBUS but no additional branches for outage calculations are required, the third record must still be specified; it may either have a zero as its first bus number or simply be a blank line.

The buses and branches in the block for bus IBUS are used only if bus IBUS is among the buses selected for processing as home buses, and an FCODE value of zero was not specified for bus IBUS.

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Bus JBUSi is not reported if it was assigned a PCODE value of zero or one, or if it is not connected back to the home bus via in-service ac branches.

Outage fault cases are calculated only if the Line outage cases option box is checked in the dialog.

11.8.3. Operation of Activity ASCC

Activity ASCC builds and factorizes the sequence admittance matrices, reporting its progress in the same manner as does activity SEQD. If sequence data has not been appended to the power flow case, an appropriate error message is printed and calculation process is terminated.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Regardless of the method of designating home buses, those buses for which an FCODE value of zero was specified in the Fault Control Input Data File (refer to Section 11.8.2, "Fault Control Data File Contents") are skipped as activity ASCC sequences through fault cases.

The fault calculation of activity ASCC may be terminated by entering the AB interrupt control code.

11.8.4. Detailed Bus Output Report

In order to better understand the output, several examples will be developed each of which assumes a fault on Bus 151 in the *savnw.sav* power flow case. Different fault and output options will be used. The location of Bus 151 relative to neighboring buses is shown in Figure 11.30, "Location of Bus 151 and Buses One Level Away in savnw.sav" where it can be seen that there are 4 buses at one level away. They are buses 101, 102, 152 and 201.

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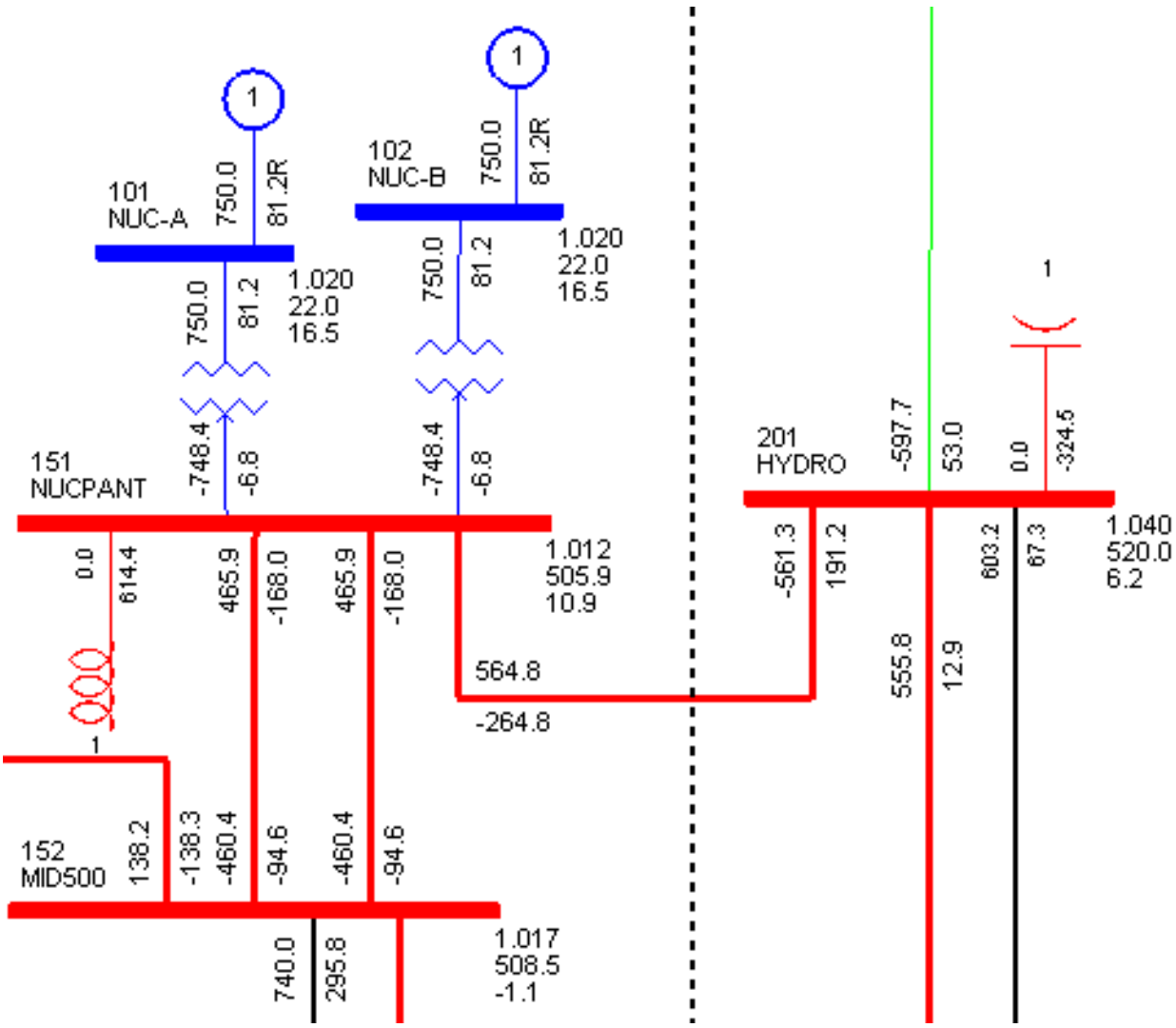

Figure 11.30. Location of Bus 151 and Buses One Level Away in savnw.sav

Each page of the detailed bus output report generated by activity ASCC contains a heading block, which tabulates the current date and time, the two-line case heading, and the home bus number and name. If the *flat conditions* option was specified, this is noted; if the line outage or line end condition is being reported, this is also flagged. If subsystems are being specified by area, zone, and/or owner, the area, zone, and/or owner, as appropriate, currently being processed are also identified.

The home bus, the level number, relative to the home bus, of buses where output is printed on the page, and, if appropriate, a line out or line end identifier are printed on the right side of the banner block. If line-to-ground faults are being calculated and subsystem selection by area, zone, and/or owner is enabled, the current area, zone, and/or owner is also printed here. This facilitates the locating of a desired set of output from within a lengthy report by fanning through the stack of paper.

Voltages, currents and apparent impedances are printed in either physical units (kV L-G, amps and ohms) or per unit, and in either rectangular or polar coordinates, according to the fault analysis output options

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currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). If the physical units option has been enabled and the bus where base voltage is required for the calculation of a quantity has no base voltage specified for it, the quantity is printed in per unit.

For each bus at which conditions are reported (i.e., for each at bus), the following quantities are tabulated:

1. The bus number, name, base voltage, and area number in which it is located. This is followed by the positive sequence bus voltage for the three phase fault case. If single line to ground faults are being calculated, the A phase voltage and the three sequence voltages for the L-G case are printed.

2. If the bus is the home bus, the sequence Thevenin impedances at the fault point in per unit, rectangular coordinates. If only three phase faults are being calculated, only the positive sequence Thevenin impedance is printed.

3. The machine identifier and currents *arriving* at the bus from all in-service machines at the bus. For the L-G fault, either zero sequence, A phase, or both quantities are printed, according to user specification.

4. The from bus number, name, base voltage, area number, and circuit identifier of each non-transformer branch and two-winding transformer connected to the at bus. The branch quantities tabulated include the current *arriving* at the at bus from the from bus, the apparent impedance as seen at the from bus looking down the line toward the at bus (i.e., the V/I ratio at the from bus), and the ratio of apparent X/R. For the L-G fault, either zero sequence, A phase, or both quantities are printed, according to the user specification. Branches are printed in ascending from bus numerical or alphabetical order according to the bus output option in effect.

5. The winding number, transformer name and circuit identifier of each three-winding transformer connected to the at bus. The current *arriving* at the at bus from the transformer is tabulated. For the L-G fault, either zero sequence, A phase, or both currents are printed, according to the user specification. Three-winding transformers are printed in ascending transformer name order.

6. The current flowing to ground at the bus exclusive of any fault current. These currents, annotated TO SHUNT, include any load and shunt current at the bus as well as the ac side current of any unblocked dc line or in-service FACTS device. In the zero sequence, only the shunt load plus fixed and switched shunt is produced here; specifically, the zero sequence ground ties created by grounded transformer windings are not shown here but are included in the branch contribution output of (4) and (5) above. For the L-G fault, either zero sequence, A phase, or both quantities are printed, according to the user specification. Load and shunt current output is suppressed if the corresponding sequence load and shunt admittances are all zero.

7. If the bus is the home bus, the fault current.

All buses reported according to the full output option and all home buses reported according to the full output with summary option are printed as described above.

In reporting remote contributions using the full output with summary option, only machine and branch quantities are tabulated in a form similar to (3), (4) and (5) above. A branch is only reported once: at the end which is closer to (i.e., fewer levels away from) the home bus.

With both full output options printing is suppressed for remote contributions at buses for which a PCODE value of zero or one was specified in the Fault Control Data File. In addition, upon encountering a remote bus with a PCODE value of zero, those buses connected to the bus that would normally be at the next level away are promoted to the current level.

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If the fault current summary table option has been specified, one output line is printed for each fault giving the fault currents. Fault currents are printed in either amps or per unit, and in either rectangular or polar coordinates, according to the fault analysis output options currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). If the physical units option has been enabled and the faulted bus has no base voltage specified for it, fault current is printed in per unit. If line out and/or line end faults are being calculated, for a given home bus, the bus fault is reported first. Then the line out and/or line end faults involving each branch connected to the home bus in turn are processed.

11.8.5. Example: Automatic Sequencing, 3-Phase Faults, ASCC API

This case assumes application of only 3-phase faults at bus 151 with full reporting at the home bus and one level away. Faults are applied with all lines in service, with each outgoing line out of service in turn and with an open line-end fault on each outgoing line in turn. Output is in rectangular coordinates.

In Figure 11.31, "Report Output at the Home Bus (0 level) for a 3-Phase Fault", overleaf, the results for the home bus are listed. It can be seen that although there is a 600 Mvar shunt connected at bus 151, the total shunt current is zero. This is clearly because the voltage at this bus (the faulted home but is zero).

Note that the listing has been compressed slightly from the original output listing for the portrait view.

In Figure 11.32, "Current Flows 1 Level Away from Home Bus 151 for Three-Phase Fault ", the results are listed for the 1 Level away. It can be seen that there is a listing for the four immediate neighbors of bus 151, the home bus. The information on branch flows has the same format as that of the home bus, at the 0 Level because the full report options was selected for these buses.

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```
-           PSS/E
SHORT CIRCUIT OUTPUT   THU, FEB 26 2004  11:35    .HOME BUS IS
  151.
-    PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
              .NUCPANT      500.0.
-    BASE CASE INCLUDING SEQUENCE DATA
                                .
-   *** FAULTED BUS IS:    151 [NUCPANT      500.0] ***
                  0 LEVELS AWAY

AT BUS    151 [NUCPANT     500.00]  AREA   1  (PU) U+:
   0.0000+J 0.0000

THEV. R, X, X/R: POSITIVE  0.00409   0.01765  4.318

                                                    T H
R E E  P H A S E  F A U L T
X--------- FROM --------X AREA  CKT  I/Z     RE(I+)
 IM(I+)  RE(Z+)   IM(Z+)   APP X/R
   101 [NUC-A     21.600]    1  1   PU/PU    9.8734  -11.0030
 0.0003   0.0136   45.333
   102 [NUC-B     21.600]    1  1   PU/PU    9.8734  -11.0030
 0.0003   0.0136   45.333
   152 [MID500    500.00]    1  1   PU/PU   -0.2319
-6.2545  0.0031   0.0500   16.264
   152 [MID500    500.00]    1  2   PU/PU   -0.2319
-6.2545  0.0031   0.0500   16.264
   201 [HYDRO     500.00]    2  1   PU/PU    3.3729  -16.5469
 0.0010   0.0151   14.864
                          TO SHUNT  (P.U.)   0.0000
 0.0000
            TOTAL  FAULT  CURRENT   (P.U.)  22.6560  -51.0619
```

Figure 11.31. Report Output at the Home Bus (0 level) for a 3-Phase Fault

In Figure 11.32, "Current Flows 1 Level Away from Home Bus 151 for Three-Phase Fault " it can be seen that the immediate neighbors to bus 151 are the buses 101, 102, 152 and 201. Those buses, at 1 level away, are now the AT buses. The flows shown in the tabulation are from the from buses towards the respective AT buses.

As an example, it can be seen that the flow shown for bus 101 is the flow from bus 151 to bus 101. The flow is(-9.8734 11.003). This has the opposite sign from that shown in Figure 11.31, "Report Output at the Home Bus (0 level) for a 3-Phase Fault" where the home bus 151 is the AT bus.

Figure 11.33, "Fault Summary Report with 3-Phase Fault on Bus 151 " shows the Fault Current Summary report from the calculation which applies a three phase fault. It shows the fault current for a fault at Bus 151 with all lines in. It can be seen that the result is the same as that shown in Figure 11.31, "Report Output at the Home Bus (0 level) for a 3-Phase Fault". The report then lists the results for a fault at Bus 151 each outgoing line OUT of service and with the fault at the open END.

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```
-           PSS/E
SHORT CIRCUIT OUTPUT   THU, FEB 26 2004  11:35    .HOME BUS IS
  151.
-    PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
              .NUCPANT      500.0.
-    BASE CASE INCLUDING SEQUENCE DATA
                                .
-   *** FAULTED BUS IS:    151 [NUCPANT      500.0] ***
                  1 LEVELS AWAY


AT BUS    101 [NUC-A     21.600]  AREA   1  (PU) U+:
  0.1526+J 0.1310

                                                      T H
R E E  P H A S E  F A U L T
X--------- FROM --------X AREA  CKT  I/Z     RE(I+)
 IM(I+)  RE(Z+)   IM(Z+)   APP X/R
MACHINE 1                            PU/      9.8734  -11.0030
   151 [NUCPANT   500.00]    1  1   PU/PU   -9.8734
11.0030  0.0000   0.0000     0.000


AT BUS    102 [NUC-B     21.600]  AREA   1  (PU) U+:
  0.1526+J 0.1310

                                                      T H
R E E  P H A S E  F A U L T
X--------- FROM --------X AREA  CKT  I/Z     RE(I+)
 IM(I+)  RE(Z+)   IM(Z+)   APP X/R
MACHINE 1                            PU/      9.8734  -11.0030
   151 [NUCPANT   500.00]    1  1   PU/PU   -9.8734
11.0030  0.0000   0.0000     0.000


AT BUS    152 [MID500    500.00]  AREA   1  (PU) U+:
  0.2871+J 0.0269

                                                      T H
R E E  P H A S E  F A U L T
X--------- FROM --------X AREA  CKT  I/Z     RE(I+)
 IM(I+)  RE(Z+)   IM(Z+)   APP X/R
   151 [NUCPANT   500.00]    1  1   PU/PU    0.1847
 5.7520  0.0000   0.0000     0.000
   151 [NUCPANT   500.00]    1  2   PU/PU    0.1847
 5.7520  0.0000   0.0000     0.000
   153 [MID230    230.00]    1  1   PU/PU   -1.2158
-5.5759 -0.0060   0.0542     9.048
   151 [EAST500   500.00]    2  1   PU/PU    0.5901
-1.5481  0.1073   0.1860     1.733
   3004 [WEST     500.00]    3  1   PU/PU    0.2561
-4.3801  0.0193   0.1146     5.929


AT BUS    201 [HYDRO     500.00]  AREA   2  (PU) U+:
  0.2516+J 0.0840

                                                      T H
R E E  P H A S E  F A U L T
X--------- FROM --------X AREA  CKT  I/Z     RE(I+)
```

Figure 11-32: Current Flows 1 Level Away from Home Bus 151 for Three-Phase Fault

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```
    PTI INTERACTIVE
POWER SYSTEM SIMULATOR--PSS/E     THU, FEB 26 2004  11:36
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
         SHORT CIRCUIT
BASE CASE INCLUDING SEQUENCE
DATA                              FAULT CURRENTS


  |   THREE PHASE FAULT
X------ BUS ------X
  X------ TO BUS ------X CKT             RE(I+)    IM(I+)
151 [NUCPANT  500.0                                 P.U.
   22.6560  -51.0619
151 [NUCPANT  500.0   101
[NUC-A     21.600] 1  OUT P.U.    12.7826  -40.0589
151 [NUCPANT  500.0   101 [NUC-A     21.600] 1  END P.U.
    5.5934  -26.1350
151 [NUCPANT  500.0   102
[NUC-B     21.600] 1  OUT P.U.    12.7826  -40.0589
151 [NUCPANT  500.0   102 [NUC-B     21.600] 1  END P.U.
    5.5934  -26.1350
151 [NUCPANT  500.0   152
[MID500    500.00] 1  OUT P.U.    21.9500  -48.2059
151 [NUCPANT  500.0   152 [MID500    500.00] 1  END P.U.
    5.0230  -14.4880
151 [NUCPANT  500.0   201
[HYDRO     500.00] 1  OUT P.U.    18.3905  -41.1311
151 [NUCPANT  500.0   201 [HYDRO     500.00] 1  END P.U.
   10.0377  -24.6822
```

Figure 11.33. Fault Summary Report with 3-Phase Fault on Bus 151

11.8.6. Example: Automatic Sequencing, 3-Phase & Single L-G Faults, ASCC API

This case assumes application of 3-phase faults and single-line-to-ground faults at bus 151 with full reporting at the home bus. Faults are applied with all lines in service and output in polar coordinates. The output will be 3 x zero sequence currents and zero sequence apparent impedances.

With the single-line-to-ground faults included, the report format is the same as for only 3-phase faults. The results now append the zero sequence fault current flows in the branches connected to the home bus. In addition the negative and zero sequence Thevenin impedances are shown, together with the A phase and sequence voltages at the faulted bus.

With polar coordinates used, it can be seen that the total fault current is 55.8625 / -66.07 pu, compared to the rectangular coordinate output which, in Figure 11.31, "Report Output at the Home Bus (0 level) for a 3-Phase Fault", is 22.656 -51.0619 pu. The listing for < n > levels away will use the same format shown in Figure 11.32, "Current Flows 1 Level Away from Home Bus 151 for Three-Phase Fault " with, in this case, the zero sequence (L-G) results appended in the right hand columns.

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```
.                        PSS®E  SHORT  CIRCUIT  OUTPUT                          TUE, APR 19 2011  16:00  .       HOME  BUS  IS  :      .
.                PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                                                .    151 [NUCPANT      500.00] .
.                BASE CASE INCLUDING SEQUENCE DATA                                                        .                            .
.                      ***  FAULTED  BUS  IS :     151 [NUCPANT     500.00]  ***                         .       0  LEVELS  AWAY       .

AT BUS    151 [NUCPANT     500.00] AREA    1      (PU) V+: / 0.0000/    0.00          (PU) VA: / 0.0000/    0.00     V0: / 0.1442/ -159.33
                                                                                              V+: / 0.5771/   12.10     V-: / 0.4351/ -170.72

THEV. R, X, X/R: POSITIVE  0.00409  0.01765    4.318          NEGATIVE  0.00409  0.01765    4.318          ZERO  0.00017  0.00600   34.823

                                                    T H R E E   P H A S E   F A U L T                          O N E   P H A S E   F A U L T
X--------- FROM ----------X AREA CKT  I/Z       /I+/     AN(I+)    /Z+/     AN(Z+)   APP X/R      /3I0/    AN(3I0)    /Z0/     AN(Z0)   APP X/R
   101 [NUC-A       21.600]    1 1   PU/PU    14.7835   -48.10   0.0136    88.74    45.333     31.8020   -68.07   0.0000     0.00     0.000
   102 [NUC-B       21.600]    1 1   PU/PU    14.7835   -48.10   0.0136    88.74    45.333     31.8020   -68.07   0.0000     0.00     0.000
   152 [MID500      500.00]    1 1   PU/PU     6.2588   -92.12   0.0501    86.48    16.264      1.0241   -61.82   0.0915   -94.53    12.626
   152 [MID500      500.00]    1 2   PU/PU     6.2588   -92.12   0.0501    86.48    16.264      1.0241   -61.82   0.0915   -94.53    12.626
   201 [HYDRO       500.00]    2 1   PU/PU    16.8872   -78.48   0.0152    86.15    14.864      6.4181   -65.78   0.0199   -92.57    22.237
                          TO SHUNT  (P.U.)    0.0000     0.00                                    0.0000     0.00
             TOTAL  FAULT  CURRENT  (P.U.)   55.8625   -66.07                                   72.0545   -67.69
```

Figure 11.34. Results for Three-Phase and Single Phase fault at bus 151

11.8.7. Relay File Output

The relay output file optionally generated by the automatic sequence fault calculations activity contains a single-line summary for each fault calculation performed. Each line commences with a code indicating the type of fault:

Code	Type of Fault
3	Three-phase fault at the home bus
1	Line-to-ground fault at the home bus
30	Three-phase line end fault
10	Line-to-ground line end fault
5	Line-line-to-ground fault at the home bus
7	Line-to-line fault at the home bus
50	Line-line-to-ground line end fault
70	Line-to-line line end fault

The following quantities are tabulated in the indicated column positions of each record

2-3:	Fault code
5-10:	Home bus number
12-17:	from bus number (0 if the branch is a three-winding transformer) for line out or line end, or blank
19-24:	to bus number for line out or line end, or blank
26-27:	Circuit identifier for line out or line end, or blank
29-41:	Fault current magnitude in per unit
43-55:	Ratio of imaginary/real components of fault current
58-75:	Extended bus name of home bus
78-95:	Extended bus name of from bus (3WND: followed by the three-winding transformer name if from bus field is 0), or blank
98-115:	Extended bus name of to bus, or blank

11.8.8. Application Notes

Activity ASCC is a totally self-contained fault analysis calculation. All that is required prior to entering activity ASCC is a valid power flow case with sequence data appended to it (i.e., with a activity RESQ having been

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executed). Bus voltages and boundary conditions must correspond to the desired pre-fault network condition, unless the flat conditions option is specified.

Activity ASCC may be preceded by activity FLAT to impose different levels of flattening than is implemented by the *flat conditions* option of activity ASCC. For example, the flat option of activity ASCC retains grounded loads in the zero sequence (refer to Load Sequence Data). To calculate fault cases with these set to zero, the following procedure may be used:

1. Bring the "unconverted" network into the working case.
2. Use activity FLAT specifying the following options:
   - set classical fault analysis conditions.
   - set tap ratios to unity.
   - set charging to zero.
   - set fixed bus shunts to zero in all sequence networks.
   - set switched shunts to zero in all sequence networks.
   - set line shunts to zero in all sequence networks.
   - set magnetizing admittances to zero in the positive sequence network.
   - set loads to zero in all sequence networks.
3. Then run activity ASCC without specifying the flat conditions option.

Although activity ASCC uses the SEQD temporary file (refer to Section 2.2, "The PSS®E Working Case and Temporary Files"), the execution of activity SEQD *is not* required prior to selecting activity ASCC. Activity ASCC sets up this file in the form required for its fault solution calculation. Note that, because activity ASCC uses this file, the following activity sequence would not be valid:

1. SEQD
2. SCMU (perhaps several times)
3. ASCC
4. SCMU

Activity SEQD would need to be re-executed following activity ASCC, prior to the second series of executions of activity SCMU.

Activity ASCC is sensitive to the fault analysis modeling option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). Using the two phase option, negative sequence Thevenin impedances are shown in the output report as zero.

As with all user specified output files used in PSS®E, the relay output summary file designated during an execution of activity ASCC does not have data appended to it if the file already exists. If the contents of a relay output file are to be appended to a previously existing file, a new filename should be specified during activity ASCC and the two files may be subsequently merged with the text editor following termination of activity ASCC.

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Additional Information
PSS®E GUI Users Guide, Calculating Automatic Sequencing Fault *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Automatic Sequencing Fault

11.9. Calculating Fault Currents to ANSI Standards

ANSI

Run Activity ANSI - GUI
Fault > ANSI fault calculation (ANSI)... *[ANSI Fault Current Calculation]*
Interrupt Control Codes
AB

The ANSI fault current calculation activity ANSI calculates fault currents according to the ANSI standard C37.5-1979 *Guide for Calculation of Fault Currents for Application of AC High-Voltage Circuit Breakers Rated on a Total Current Basis*.

The following input data defining solution and reporting options is specified to the ANSI_2 API routine:

The positive sequence generator reactance to be used (refer to Generator Sequence Data) Either subtransient, transient, or synchronous reactance may be used at all in-service synchronous machines.

- The treatment of the zero sequence impedance of all transformers where actual positive sequence impedance differs from its nominal value (refer to Transformer Impedance Correction Tables). The zero sequence impedance of each such transformer may either be scaled by the same factor as is its positive sequence impedance, or all zero sequence transformer impedances may be left at their nominal values (i.e., the values entered in activities RESQ, TRSQ, or SQCH). The same treatment applies to *all* transformers in the system that are not at nominal impedance.

- Decay factors to be applied. Decay factors may be obtained from either Figures 1 and 2 of the ANSI standard, which include the effects of both ac and dc decay, or from Figure 3, which includes only the effects of dc decay.

- Resistance scaling factors. For branches and machines with a zero value of resistance, activity ANSI assigns a non-zero resistance equal to its reactance divided by a scaling factor. Four such factors (for positive sequence branch impedance, positive sequence machine impedance, zero sequence branch impedance, and zero sequence machine impedance) are specified.

- Output option, selected from among the following output report formats:

Summary output in report format using only the ANSI X matrices (output fields are separated by blanks).

Summary output in spreadsheet input format using only the ANSI X matrices (output fields are comma delimited).

Detailed output in report format using only the ANSI X matrices.

Detailed output in report format using both the ANSI R and ANSI X matrices.

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Summary output in report format using both the ANSI R and X matrices (output fields are separated by blanks).

Summary output in spreadsheet input format using both the ANSI R and X matrices (output fields are comma delimited).

- The breaker rating option. All circuit breakers made before 1964 were tested on a, "total current," basis. Those made after 1970 were tested on a, "symmetrical current," basis. Breakers made between 1964 and 1970 could have been made to conform to either test method. Activity ANSI can calculate breaker interrupting currents according to either method.
- Fault specification option. Faulted buses, along with their highest operating voltages and contact parting times, may be specified either along with the other inputs to the ANSI_2 API routine, or separately in an ANSI Fault Specification Data File (see below).

If sequence data is not contained in the working case, an appropriate error message is printed and activity ANSI is terminated.

Activity ANSI performs the following computations on the working case:

1. An ordering for the positive sequence network is determined and the standard summary of activity ORDR is printed at the *Progress* tab.
2. The positive sequence decoupled admittance matrices are constructed and factorized.
3. The zero sequence network is ordered and the standard summary is printed.
4. The zero sequence decoupled admittance matrices are constructed and factorized.
5. Fault cases are calculated.

If, in the process of building the sequence network admittance matrices, isolated buses are detected, they generate an alarm. The presence of isolated buses in the positive sequence network indicates an improperly specified power flow case. Isolated buses in the zero sequence network, although they generated alarms, are permitted and require no special treatment. When the fault analysis warning option is disabled (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN), the isolated bus tabulations are suppressed. The total number of isolated buses in the sequence networks is tabulated.

11.9.1. ANSI Fault Specification Data File Contents

If fault specification data is to be taken from an ANSI Fault Specification Data File, the specified file is opened and processed. Data records must be specified in the following format:

```
IBUS, VMAX, TIME
```

where:

IBUS	Is the bus number of the bus to be faulted. No default is allowed.
VMAX	Is the maximum operating voltage of bus IBUS specified in pu. VMAX = 1.0 by default.
TIME	Is the contact parting time in seconds. TIME = one cycle by default.

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Data records may be entered in any order; fault cases are calculated and reported in the same order in which data records are read. Input is terminated with a record specifying an IBUS value of zero.

If fault specification data is to be taken from the dialog or the BAT_ANSI command, fault cases are calculated and reported in the same order in which the data is specified. No defaults for the maximum operating voltage or the contact parting time are allowed.

11.9.2. Operation of Activity ANSI

If sequence data is not contained in the working case, an appropriate error message is printed and the calculation process is terminated.

The following computations are performed on the power flow case:

1. An ordering for the positive sequence network is determined and the standard summary printed at the progress report output device.
2. The positive sequence decoupled admittance matrices are constructed and factorized.
3. The zero sequence network is ordered and the standard summary is printed.
4. The zero sequence decoupled admittance matrices are constructed and factorized.
5. Fault cases are calculated.

If, in the process of building the sequence network admittance matrices, isolated buses are detected, they generate an alarm. The presence of isolated buses in the positive sequence network indicates an improperly specified power flow case. Isolated buses in the zero sequence network, although they generated alarms, are permitted and require no special treatment.

When the fault analysis warning option is disabled (see Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the isolated bus tabulations are suppressed. In either case, the total number of isolated buses in the sequence networks is tabulated.

As each fault case is calculated, its results are tabulated.

11.9.3. ANSI Fault Calculation Output Report

The summary report for each fault case includes the following:

- The bus number, name and base voltage of the faulted bus, along with the maximum operating voltage and contact parting time input values.
- Three phase fault results, including symmetrical fault MVA, symmetrical fault current in kA, asymmetrical fault current in kA, the ANSI X/R ratio, and the multiplying factor.
- Line-to-ground fault results, including symmetrical fault current in kA, asymmetrical fault current in kA, the ANSI X/R ratio, and the multiplying factor.
- Line-to-line-to-ground fault results, including symmetrical phase current in kA, and three times the zero sequence symmetrical fault current in kA.
- The positive sequence Thevenin impedance as obtained from the decoupled positive sequence admittance matrices.

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- The zero sequence Thevenin impedance as obtained from the decoupled zero sequence admittance matrices.

An example of a summary report, obtained using the *savnw.sav* power flow case, with a fault on Bus 151 (see Figure 11.35, "Summary and Detailed Report of ANSI Fault Calculation"a).

In the detailed output, an output block for each faulted bus is printed. Voltages, currents and apparent impedances are printed in either physical units (kV L-G, amps and ohms) or per unit, and in either rectangular or polar coordinates, according to the fault analysis output options currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). If the physical units option is enabled and the bus where base voltage is required for the calculation of a quantity has no base voltage specified for it, the quantity is printed in per unit.

For each faulted bus (i.e., at each at bus), the following quantities are tabulated:

1. The bus number, name, base voltage, and area number in which it is located, with the specified maximum operating voltage listed below the bus identifiers. This is followed by the positive sequence bus voltage for the three phase fault case, and the A phase voltage and the three sequence voltages for the L-G case.

2. The positive and zero sequence ANSI Thevenin impedances at the fault point in per unit, rectangular coordinates (the positive and negative sequences are identical to the ANSI fault calculation standard).

3. The machine identifier and currents *arriving* at the bus from all in-service machines at the bus. For the L-G fault, three times the zero sequence current (3I0) is printed.

4. The from bus number, name, base voltage, area number, and circuit identifier of each non-transformer branch and two-winding transformer connected to the at bus. The branch quantities tabulated include the current *arriving* at the at bus from the from bus, the apparent impedance as seen at the from bus looking down the line toward the at bus (i.e., the V/I ratio at the from bus), and the ratio of apparent X/R. For the L-G fault, zero sequence quantities are printed. Branches are printed in ascending from bus numerical or alphabetical order according to the bus output option in effect.

5. The winding number, transformer name and circuit identifier of each three-winding transformer connected to the at bus. The current *arriving* at the at bus from the transformer is tabulated. For the L-G fault, zero sequence quantities are printed. Three-winding transformers are printed in ascending transformer name order.

6. The sum of the currents output in (3), (4) and (5) above. These currents are annotated SUM OF CONTRIBUTIONS.

7. The fault current as calculated from the maximum operating voltage and the ANSI Thevenin reactance or impedance, as appropriate.

The quantity shown as the sum of contributions will be essentially identical to the quantity shown as fault current for both three phase and line-to-ground faults when fault currents and contributions are calculated using only the ANSI X matrices. However, these quantities will normally differ when the output is calculated using both the ANSI R and ANSI X matrices. This is due to the error introduced by decoupling the complex sequence admittance matrices into pairs of real matrices, using the real matrices to derive columns of R and X matrices (i.e., the decoupled impedance matrices), and then coupling the columns back together to get approximations of the complex impedance matrix columns.

Figure 11.35, "Summary and Detailed Report of ANSI Fault Calculation"b shows an example of a detailed report obtained for faults on Bus 151 in the *savnw.sav* power flow case.

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      TUE, MAR 02 2004  15:48
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA
                                BRKER X------- THREE PHASE FAULT --------X X-- LINE TO GROUND FAULT --X X- LLG SYMM I -X
  BUS# X-- NAME --X BASKV MAX V  TIME  FLTMVA  SYMM I ASYMM I   X/R FACTOR  SYMM I ASYMM I   X/R FACTOR   PHASE    3IA0    RPOS    XPOS    RZERO
   151 NUCPANT     500.00 1.000 0.017  5148.2   5.945   9.285  41.02 1.562   7.754  12.113  41.06 1.562   7.588  11.149  0.0005  0.0194  0.0001

                                        (a) Summary Report


AT BUS    151 [NUCPANT     500.00] AREA    1      (PU) V+:    0.0000+J 0.0000         (PU) VA:    0.0000+J 0.0000    V0:   -0.1304+J 0.0000
MAXIMUM OPERATING VOLTAGE: 1.0000 PU                                                      V+:    0.5652+J 0.0000    V-:   -0.4348+J 0.0000

POSITIVE SEQUENCE THEVENIN R, X, X/R:  0.00047  0.01942   41.022           ZERO SEQUENCE THEVENIN R, X, X/R:  0.00014  0.00582   41.325

                                                    T H R E E   P H A S E   F A U L T              O N E   P H A S E   F A U L T
X--------- FROM ----------X AREA CKT  I/Z     RE(I+)   IM(I+)   RE(Z+)   IM(Z+)  APP X/R     RE(3I0)  IM(3I0)   RE(Z0)   IM(Z0)  APP X/R
   101 [NUC-A       21.600]    1 1   PU/PU    0.1974 -12.4560  0.0003   0.0136   45.333      0.6385 -28.7427   0.0000   0.0000    0.000
   102 [NUC-B       21.600]    1 1   PU/PU    0.1974 -12.4560  0.0003   0.0136   45.333      0.6385 -28.7427   0.0000   0.0000    0.000
   152 [MID500      500.00]    1 1   PU/PU    0.2483  -5.5877  0.0026   0.0460   17.692      0.1166  -1.7436  -0.0059  -0.0833   14.108
   152 [MID500      500.00]    1 2   PU/PU    0.2483  -5.5877  0.0026   0.0460   17.692      0.1166  -1.7436  -0.0059  -0.0833   14.108
   201 [HYDRO       500.00]    2 1   PU/PU    0.7493 -15.3368  0.0010   0.0150   15.000      0.3883  -6.1349  -0.0010  -0.0185   18.332
             SUM OF CONTRIBUTIONS  (P.U.)    1.6407 -51.4243                               1.8986 -67.1074
            TOTAL  FAULT  CURRENT  (P.U.)    1.2546 -51.4667                               1.6350 -67.1360

                                        (b) Detailed report
```

Figure 11.35. Summary and Detailed Report of ANSI Fault Calculation

11.9.4. Application Notes

While the working case may represent the network at any level of detail from classical fault analysis assumptions through a fully solved power flow model, activity ANSI applies classical fault analysis conditions. Specifically, voltages, loads, machine loadings, bus and line shunts, line charging, phase shift angles, dc lines, and FACTS devices in the working case are ignored. Tap ratios and zero sequence mutual couplings are recognized in constructing the decoupled admittance matrices. It is quite common, and perfectly valid, to have the generator terminal bus isolated in the zero sequence network. This is, in fact, the usual case because the majority of generator step-up transformers are delta connected on the generator side and wye connected on the high side. This is the assumption inherent in the generator modeling when the step-up transformer is represented as part of the generator data (i.e., XTRAN is non-zero). Activity ANSI prints an alarm message for any generator with non-zero values of both XTRAN and ZZERO and uses only XTRAN in setting up the zero sequence ground tie at the Type 2 (high side) bus.

Any bus which is designated as a bus to be faulted, but which has no base voltage specified for it in the working case, is logged and omitted from the fault calculations of activity ANSI.

Any non-zero branch or machine impedance where resistance is zero in the working case has an assumed resistance, equal to its reactance divided by a user-specified scaling factor, used when the corresponding R decoupled admittance matrix is constructed. Such resistances are always positive. Zero sequence mutual impedances (RM + jXM of Zero Sequence Mutual Impedance Data) and zero sequence transformer grounding impedances (RG + jXG of Zero Sequence Transformer Data) are *not* subjected to this treatment.

Any non-zero zero sequence branch impedance where reactance is zero in the working case has an assumed reactance, equal to the zero impedance line threshold tolerance (or to 0.00001 if the zero impedance line threshold tolerance is 0.0), used when the zero sequence X decoupled admittance matrix is constructed. Such reactances are always positive. Machine impedances, zero sequence mutual impedances (RM + jXM of Zero Sequence Mutual Impedance Data), and zero sequence transformer grounding impedances (RG + jXG of Zero Sequence Transformer Data) are *not* subjected to this treatment.

Although activity ANSI uses the SEQD temporary file (refer to Section 2.2, "The PSS®E Working Case and Temporary Files"), the execution of activity SEQD *is not* required prior to selecting activity ANSI. Activity ANSI sets up this file in the form required for its fault solution calculation. Note that, because activity ANSI uses this file, the following activity sequence would not be valid:

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1. SEQD
2. SCMU (perhaps several times)
3. ANSI
4. SCMU

Activity SEQD would need to be re-executed following activity ANSI, prior to the second series of executions of activity SCMU.

Additional Information
PSS®E GUI Users Guide, Calculating Fault Currents to ANSI Standards

11.10. Setting up Classical Fault Analysis Conditions

FLAT

Run Activity FLAT - GUI
Fault > Setup for special fault calculations (FLAT)... *[Setup for Special Fault Calculations]*
Run Line Mode Activity FLAT - CLI
ACTIVITY? >>**FLAT** SELECTED FLAT CHANGES IMPOSED
Interrupt Control Codes
None

The classical conditions network activity FLAT is used primarily to set up network conditions corresponding to classical fault analysis assumptions.

The first input to the FLAT_2 API routine selects the type of model simplification to be imposed on the working case:

- change only voltages.
- set classical fault analysis conditions.
- set conditions in preparation for IEC standard 909 calculations.

These three degrees of simplification are discussed in the following sections.

Note that, for all three types of model simplification, the zero sequence ground ties created by grounded transformer windings are represented; see ???, ???, and ???.

11.10.1. Change Voltages

This option uses none of the other inputs to the FLAT_2 API routine. Under this option, activity FLAT sets *all* bus voltages to one per unit at zero phase angle. This *is not* identical in function to the flat start option that may be specified for the network solution activities of PSS®E, which causes swing (Type 3) bus voltage magnitudes to be set to their scheduled values, and all other bus voltage magnitudes to unity.

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11.10.2. Set Classical Fault Analysis Conditions

Under this option, the following data changes are implemented:

- Bus voltages are set to unity magnitude at zero phase angle.
- Generator and induction machine powers are set to zero.
- Transformer phase shift angles are set to zero. Any transformer impedance that is a function of its phase shift angle (refer to Transformer Impedance Correction Table Data) is set to its nominal value.
- FACTS devices and dc lines are removed from the working case.
- *(Note:* Using load treatment option, set loads to zero in all sequence for classical assumptions.)

Further model simplification may be selected by specifying the following additional inputs to the FLAT_2 API routine:

- Tap ratio treatment option. If this option is enabled, all transformer turns ratios are set to one. This includes generator step-up transformers, which are modeled as part of the machine representation (i.e., the GENTAP; refer to Generator Data and Section 13.2.3, "Generator Step-Up Transformers"). Any transformer impedance which is a function of turns ratio (refer to Transformer Impedance Correction Table Data) is set at nominal value.
- Line charging treatment option. If this option is enabled, line charging is set to zero in all three sequence networks.
- Fixed bus shunt treatment option. One of the following may be specified:
- leave fixed bus shunts unchanged.
- set fixed bus shunts to zero in the positive and negative sequence networks.
- set fixed bus shunts to zero in all three sequence networks.
- Switched shunt treatment option. If this option is enabled, switched shunts are set to zero in all three sequence networks.
- Line shunt treatment option. One of the following may be specified:
- leave line shunts unchanged.
- set line shunts to zero in the positive and negative sequence networks.
- set line shunts to zero in all three sequence networks.
- Transformer magnetizing admittance treatment option. If this option is enabled, transformer magnetizing admittances are set to zero in the positive and negative sequence networks (they are always zero in the zero sequence network).
- Load treatment option. One of the following may be specified:
- leave loads unchanged.
- set the positive sequence constant power, current and admittance loads to zero; model (GNEG + j BNEG) in the negative sequence network; for those loads which are grounded, model (GZERO + j BZERO) in the zero sequence network. Refer to Load Sequence Data.

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- set the positive sequence constant power, current and admittance loads to zero; set the negative sequence load, (GNEG + j BNEG), to zero; for those loads which are grounded, model (GZERO + j BZERO) in the zero sequence network.
- set the positive sequence constant power, current and admittance loads to zero; set the negative sequence load, (GNEG + j BNEG), to zero; set the zero sequence load, (GZERO + j BZERO), to zero. (Use this option for classical assumptions.)

11.10.3. Set IEC 909 Fault Analysis Conditions

Under this option, the following data changes are implemented:

- Bus voltage phase angles are set to zero.
- All synchronous machines with a negative or zero active power have their active and reactive power outputs set to zero.
- All induction machines with a positive or zero active power (i.e., induction motors) have their active and reactive powers drawn from the network set to zero.
- Transformer phase shift angles are set to zero. Any transformer impedance that is a function of its phase shift angle (refer to Transformer Impedance Correction Table Data) is set to its nominal value.
- FACTS devices and dc lines are removed from the working case.

Further model simplification may be selected by specifying the additional inputs to the FLAT_2 API routine. In addition to those that may be specified when setting classical fault analysis conditions, the following additional data inputs to the FLAT_2 API routine are required:

- The voltage magnitude setting. When a positive value is specified, all bus voltage magnitudes are set to the specified value. Otherwise, all bus voltage magnitudes remain at their present values.
- A machine power factor setting. This setting is used at:
- synchronous machines with a positive value of active power.
- induction machines with a negative value of active power (i.e., at induction generators).

  At any such machine, its active power is unchanged, and its reactive power is set as follows:
- if the power factor setting is positive, its reactive power is set such that the specified power factor is maintained.
- otherwise, its reactive power is unchanged.

When the setting of IEC 909 fault analysis conditions is followed by activity BKDY, short circuit currents in conformance with IEC standard 909 are calculated (see Section 11.12, "Calculating Circuit Breaker Interrupting Duty"). Note that loads should normally be set to zero in preparation for IEC 909 fault current calculations.

Additional Information
PSS®E GUI Users Guide, Using Classical Fault Analysis Option *PSS®E* Command Line Interface (CLI) *Users Guide*, Using Classical Fault Analysis Assumptions

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11.11. Calculating Short Circuit Currents According to IEC 60909 Standard

IECS

Requirements / Prerequisites
Validly specified power flow case. For unbalanced fault calculations, sequence data appended to Saved case or Reading Sequence Data
Run Activity IECS - GUI
Fault > IEC 60909 fault calculation (IECS)... *[IEC 60909 Fault Calculation]*
Interrupt Control Codes
None

The activity IECS allows the user to apply shunt faults at various locations in a PSS®E working case to compute short circuit currents according to the IEC 60909 standard.

Activity IECS calculates following currents (refer to the *Program Application Guide, Volume I*, Calculation of Short Circuit Currents for details).

I"k	Initial symmetrical short-circuit current (r.m.s) including phase and sequence components of fault currents
ip(B)	Peak short-circuit current by Method B of IEC standard 60909
ip(C)	Peak short-circuit current by Method C of IEC standard 60909
ib(DC)	DC component of the asymmetrical breaking current using Method C R/X ratio
ib(SYM)	Symmetrical breaking current (r.m.s.)
ib(ASYM)	Asymmetrical breaking current (r.m.s.)

Activity IECS includes the following features.

1. The impedances of synchronous generators, network transformers and power station units are modified with impedance correction factors when calculating short circuit currents with the equivalent voltage source at the short circuit location according to IEC 60909 standard. (The model of a synchronous generator with a step-up transformer (GSU) is called a Power Station Unit in IEC 60909.)

2. The specification of network subsystems to be processed according to PSS®E standard selection criteria. Each electrically connected bus in the specified subsystem becomes the home bus for one or more fault analysis network solutions.

3. The specification of three phase faults (3PH), single-line-to-ground (LG) faults, double-line-to-ground (LLG) faults and line-to-line (LL) faults at each fault location or any combinations of these faults.

4. The application of the faults selected in (3) at each home bus selected in (2).

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5. A selection of output options for each fault applied, ranging from a tabulation of fault currents through detailed output of conditions at the home bus along with detailed output for all buses up to < N > levels away removed from the home bus.

The original values of all data items in the working case are not modified by activity IECS. That is, at the termination of activity IECS, the working case is identical in content to what it was at the time activity IECS was initiated.

The following inputs are required.

Table 11.2. Short Circuit Calculation Options for IEC 60909

Specification	Description
Faults to apply	Fault type selection, select either any combination or all: • Three phase faults • Single line to ground fault • Line-line to ground fault • Line to line fault
Line outage cases	If enabled, the selected fault calculations are repeated at each home bus < n > additional times: once with each of its < n > connected branches placed out-of-service. In addition, if any branches have been designated in the Fault Control Data File (refer to Section 11.8.4, "Detailed Bus Output Report") as branches to be outaged for the current home bus, these fault cases are also calculated.
Line end faults	If enabled, the selected fault calculations are repeated for each home bus at least < n > additional times: once (twice for any three-winding transformer which has all its windings in-service) with each of its < n > connected branches opened at the far end, with the fault located at the line end position. As in activity SCMU, a dummy bus, numbered 999999, is introduced at the opened end of the branch for each line end fault (refer to Line Faults).
Output option	The possible options are: • Fault current summary table • Total fault currents • I"k contributions to < N > levels away • Total fault currents and I"k contributions to < N > levels away
Fault location	The possible options are:

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Specification	Description
	• Fault at Network bus (default) • Fault at LV bus of Power Station Unit • Fault at Auxiliary Transformer (connected to Power Station Unit) LV bus
Shunt option	Select either of the following options to set fixed shunts, loads and switched shunts data. • Set shunts to zero is positive sequence • Set shunts to zero is all sequences • Leave shunts unchanged
Line charging option	Select either of the following options to set line charging susceptance data. • Set line charging to zero in positive sequence • Set line charging to zero in all sequences • Leave line charging unchanged
Transformer taps	When selected transformer off-nominal tap ratios are set to unity, When unselected, transformer off-nominal tap ratios are considered
Represent dc lines and FACTS devices as load	DC lines and FACTS devices can be blocked or represented as load. If represented as load, the apparent ac system complex loads are converted to positive sequence constant admittance load at the buses at which these quantities are injected into the ac system during normal power flow work. In the negative and zero sequence networks, dc lines and FACTS devices are represented as open circuits. The selected option applies to all dc lines and FACTS devices in the working case.
Apply transformer impedance correction to zero sequence	If the line-to-ground fault option had been selected and the actual positive sequence impedance of any transformer in the working case differs from its nominal value (refer to Transformer Impedance Correction Tables), the user may apply transformer impedance correction to zero sequence impedance. The zero sequence impedance of each such transformer is scaled by the same factor as is its positive sequence impedance. Otherwise, all zero sequence transformer impedances are left at their nominal values (i.e., the values entered in activities RESQ,

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Specification	Description
	TRSQ, or SQCH). The same treatment applies to *all* transformers in the system that are not at nominal impedance.
Transformer tap ratios and phase shift angles option	Select either of the following options to set transformer off-nominal tap ratios and phase shift angles data. • Leave tap ratios and phase angles unchanged • Set tap ratios = 1.0 pu and phase angles = 0 • Set tap ratios = 1.0 pu and phase angles unchanged • Set tap ratios unchanged and phase angles = 0
Shunt option	Select either of the following options to set fixed shunts, loads and switched shunts and transfomrer magnetisizing admittance data. • Leave shunts unchanged • Set shunts to zero is positive and negative sequence • Set shunts to zero is all sequences
Line charging option	Select either of the following options to set line charging susceptance data. • Leave line charging unchanged • Set line charging to zero in positive and negative sequence • Set line charging to zero in all sequences
Generator Reactance option	Select either of the following synchronous machine reactances to use in fault analysis. • Subtransient • Transient • Synchronous
Number of levels back for contribution output	If one of the full output options is selected, the user must indicate the portion of the working case for which output is to be generated for each fault calculation. The default entry for the number of levels back for contributions output is 3. If zero is specified, activity ASCC restricts the output to the home bus for each fault. If 1 is specified, output for the home bus is printed followed by the out-

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Specification	Description
	put for all buses connected to it for each fault case solution. Generalizing, a response of a positive number < n > results in output for the home bus, followed by output for all buses connected to it, followed by output for all buses two buses away from it, and so on up to < n > levels removed from the home bus.
Breaker time	Breaker contact parting time in seconds (5.0/Base frequency by default, increased/decreased in seconds per cycle of Base frequency)
Voltage Factor C option	Specify eqivalent voltage source at the fault location as below: • for the calculation of maximum short-circuit currents • for the calculation of minimum short-circuit currents • a particular voltage factor C value
IEC Data file	Optional IEC data input file: IEC data filename (*.iec), a file containing data records of the form defined in IEC Data File Contents.
Fault control input file	Optional Fault Control Data file (*.fcd), a file containing data records of the form defined in Fault Control Data File Contents.
Results file	Optional IEC Results Output file (*.sc), a file into which IECS saves fault calculation results. This file is a binary file, and can be used to reproduce IECS calculation results in report format and/or on slider diagram without re-running IECS.

11.11.1. Fault Control Data File Contents

The fault control data file contents and format is the same as those used in the fault control data file of the activity ASCC. The records in this file are treated same way as they are treated in activity ASCC.

11.11.2. Application Notes

Activity IECS is a totally self-contained fault analysis calculation. All that is required prior to entering activity IECS is a valid working case with sequence data appended to it if unsymmetrical fault currents are to be calculated. If the working case does not have sequence data, only three phase fault currents will be calculated and unsymmetrical faults will be ignored. The fault currents are calculated with positive sequence equivalent voltage sources and network sequence impedances. It is not required to solve the power flow because the power flow solution (i.e., initial state of the network) is not used by the IECS activity when calculating fault currents. When calculating IEC impedance correction factors, default values used are: Generator power factor = 1.0, Generator rated terminal voltage (UrG) = Generator bus voltage (UG), Range of generator voltage regulation (PG) = 0, Tap range of GSU off-load tap-changer transformer (PT) = 0.

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In a PSS®E saved case, if most of the transformer data is specified on system MVA base and transformer winding MVA is not actual winding MVA, it is required to provide nameplate transformer winding MVA as a transformer IEC data record.

Additional Information
PSS®E GUI Users Guide, Calculating Short Circuit Currents According to IEC 60909 Standard

11.12. Calculating Circuit Breaker Interrupting Duty

BKDY

Run Activity BKDY - GUI
Fault > Circuit breaker interrupting duty (BKDY)... *[Circuit Breaker Interrupting Duty]*
Run Line Mode Activity BKDY - CLI
ACTIVITY? >>**BKDY** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB, CM, DC, FD

The circuit breaker duty analysis activity BKDY calculates and reports circuit breaker interrupting duty for three phase faults at all buses in a specified subsystem of the working case. The working case is assumed to represent the pre-fault system condition with generators represented as current sources.

At each bus in the specified subsystem in turn, activity BKDY applies a three phase fault, calculates two network solutions, giving the initial alternating fault current and the decremented alternating fault current at the user specified fault duty time. The decremented dc offset and total fault currents are then derived, and results tabulated.

As an alternative to the standard subsystem selection procedure used throughout PSS®E, fault locations and corresponding breaker operating times may be specified in a Fault Specification Data File (refer to Section 11.12.5, “Operation of Activity BKDY”).

Machine parametric data required by activity BKDY is specified in a Breaker Duty Data File.

The sudden application of a short-circuit fault to a power system produces currents for which transient form is shown in Figure 11.36, “Transient Phase Currents in Suddenly Applied Short Circuit”. In the case of a three-phase fault (simultaneously applied in each phase), the total fault current in each phase consists of the following:

1. An alternating component that decays from an initial subtransient value to a final steady-state value.

2. A decaying unidirectional component for which initial amplitude is equal to the difference between the initial instantaneous value of the alternating component of fault current and the instantaneous current in the phase just prior to fault application.

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Depending upon the standard (e.g., ANSI Standard C37.5 – 1975 or the International Electrotechnical Commission standards) the determination of circuit breaker duty requires the calculation of either of these values:

- The maximum instantaneous value of current in any phase at the instant, a few milliseconds after fault initiation, when the circuit breaker contacts separate
- The root-mean-square (rms) value of current wave consisting of sinusoidal component of constant amplitude equal to the instantaneous amplitude of the decaying alternating component at an instant, superimposed on a constant unidirectional component of amplitude equal to that of the decaying unidirectional component at the same instant.

Figure 10-37 illustrates the two current values of interest. The maximum instantaneous value, as shown in Figure 11.37, "Forms of Expression of Fault Current at Instant of Circuit Breaker Opening"a is of interest in connection with the IEC Circuit Breaker standards, while the rms value shown in Figure 11.37, "Forms of Expression of Fault Current at Instant of Circuit Breaker Opening"b is used by ANSI C37 standards. In this section, we describe the calculations made in PSS®E.

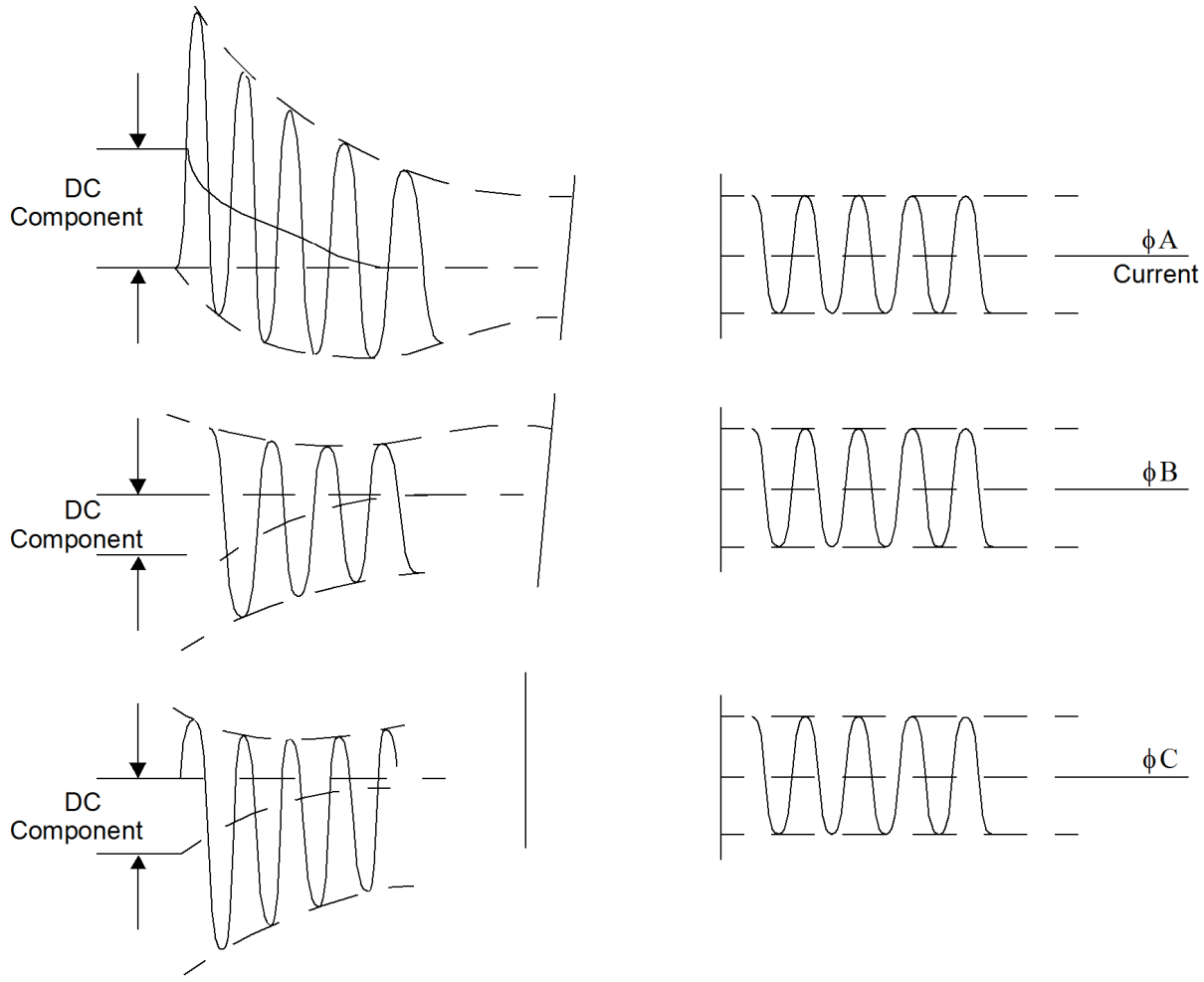

Figure 11.36. Transient Phase Currents in Suddenly Applied Short Circuit

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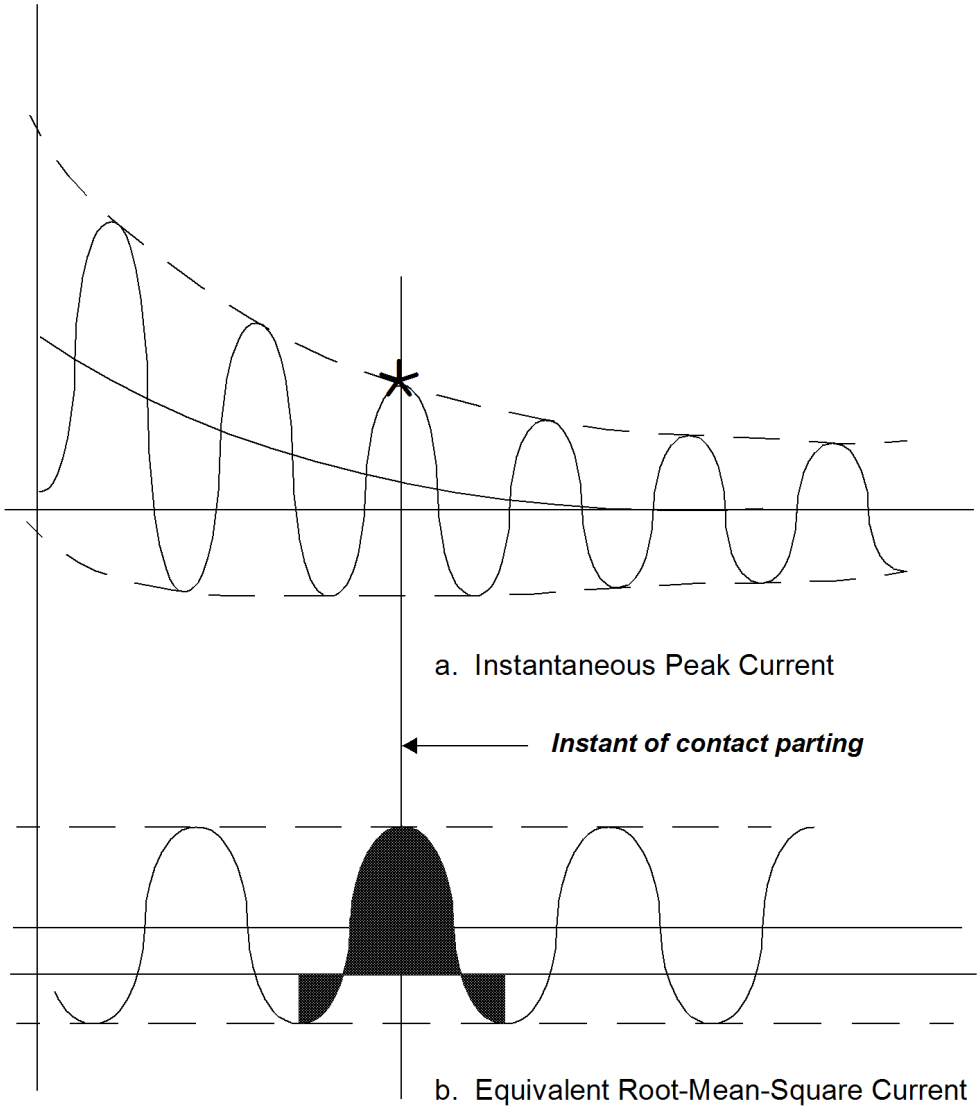

Figure 11.37. Forms of Expression of Fault Current at Instant of Circuit Breaker Opening

To determine these current amplitudes, the following data must be considered:

- The alternating current component results from *decaying* machine internal flux linkages behind *constant* subtransient impedances of the machines.
- The unidirectional component is, at the instant of fault application, determined by the value of the alternating component.

The alternating components can be approximated, with acceptable accuracy for several cycles after fault application, by expressing the generator flux as:

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$$\psi''_d = \psi''_{do}(L''_d + L_e)\left[\left(\frac{1}{L_d + L_e}\right) + \left(\frac{1}{L'_d + L_e} - \frac{1}{L_d + L_e}\right)e^{-t/T'_{dz}} + \left(\frac{1}{L''_d + L_e} - \frac{1}{L'_d + L_e}\right)e^{-t/T''_{dz}}\right] + i_{do}(L''_d + L_e)\left[\left(\frac{L_d - L''_d}{L_d + L_e}(1 - e^{-t/T'_{dz}})\right) + \frac{L'_d - L''_d}{L'_d + L_e}(e^{-t/T'_{dz}} - e^{-t/T''_{dz}})\right]$$

and the equivalent for the q-axis. For induction motors ($L_d = L_q$ and $L'_d = L'_q$), the Ld terms are not in the above equations.

The calculations for initial fluxes, ψ''_{do} and ψ''_{qo}, and initial currents, i_{do} and i_{qo}, are based on the initial conditions in the power flow before fault application. The value of L_e at each generator is calculated by dividing the voltage at each terminal by the current flowing at the terminals at fault initiation. Because an initial loading on the machine, L_e, will never be infinity, this approximation will result in flux decaying even for very remote machines. The user is responsible for not including data for machines where flux decay is not wanted.

The initial value of the unidirectional component of fault current is, in the worst case, equal to the initial amplitude of the alternating component; this corresponds to a fully offset current wave as shown for Phase A in Figure 11.36, "Transient Phase Currents in Suddenly Applied Short Circuit". Using full offset is, of course, conservative. The actual maximum offset depends on the fault point X/R and the point on the wave where the fault occurs. Thus, the peak is reached after some dc decay has occurred. The decay of the unidirectional component of fault current is given by:

$$i_{dc} = I_{ac}(0)\left[a_1 e^{-k_1 t} + a_2 e^{-k_2 t} + a_3 e^{-k_3 t} + \ldots + a_n e^{-k_n t}\right]$$

where $a_1 + a_2 + \ldots + a_n = 1$

and

- The coefficients k_i, characterize the decay of the initial unidirectional components throughout the network.
- The coefficients, a_i, express the contribution of each decaying unidirectional current component to the unidirectional fault current.

In general, an exact expression of the unidirectional fault current would involve a number of k and a coefficients equal to the number of branches in the network. Their determination would require a calculation of the eigenvalues and eigenvectors of the differential equations:

$[L_{net}][sL_{net}] + [R_{net}][i_{net}] = 0$

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where $[R_{net} + sL_{net}]$ is the operational impedance matrix of the complete network. This calculation is impractical for normal system analysis work, and it is usual to approximate the unidirectional fault current by:

$$i_{dc} = I_{ac}(0)e^{-t/T_s}$$

where

$T_s = L_{thev} / R_{thev}$

where L_{thev} and R_{thev} are the Thevenin impedance (inductance, resistance) at the point of the fault.

11.12.1. How PSS®E Calculates Circuit Breaker Duty

PSS®E contains a special short-circuit calculation which reports circuit breaker interrupting duty for three phase faults at all buses in a specified subsystem of the working case. It determines the amplitude of the alternating and unidirectional (dc) components of current flowing in symmetrical faults and in the branches of the network. The alternating and dc components are calculated as indicated in the previous section for the instant of fault application and for a specified time after the fault initiation. These component amplitudes are then used to determine the following:

- The maximum instantaneous current that could flow in any phase at the specified time after fault application, assuming that the fault was initiated at such a time, in relation to the voltage wave that the specified time after application corresponds to a current peak (Figure 11.37, "Forms of Expression of Fault Current at Instant of Circuit Breaker Opening").
- The rms value of the current at the specified time after fault application (Figure 11.37, "Forms of Expression of Fault Current at Instant of Circuit Breaker Opening").

The maximum instantaneous current, $I_{total\ peak}$, and rms currents, $I_{total\ rms}$, are determined by

$I_{total\ peak} = I_{dc} + I_{ac}$

$$I_{total\ rms} = \sqrt{I_{dc}^2 + I_{rms}^2}$$

where:

I_{dc} = Instantaneous amplitude of unidirectional component.

I_{ac} = Peak amplitude of alternating component.

It is convenient to define an instantaneous rms value of the alternating current component, I_{rms}, by

$$I_{rms} = \frac{I_{ac}}{\sqrt{2}}$$

enabling the total rms current to be written as

$$I_{total\ rms} = \sqrt{I_{dc}^2 + I_{rms}^2}$$

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The breaker duty calculation starts with a power flow case corresponding to system conditions just before the application of the fault. The values of the initial internal fluxes, ψ''_{do}, ψ''_{qo}, and initial currents, i_{do} and i_{qo}, are calculated from the prefault machine currents, terminal voltages, and the characteristic machine impedances, ZSORCE.

The calculation recognizes that the time constants T'_{dz}, T''_{dz}, T'_{qz}, T''_{qz} governing the decay of machine internal fluxes and currents are dependent on the relative values of the machine impedances and the impedances of the network between the machines and the fault. The relationships used to calculate these time constants are shown in Figure 11.38, "Relationships Between Machine Time Constants in Radial System". The L_e term is calculated at each machine in the network by dividing terminal voltage by terminal current at the instant the fault is applied.

If the fault location is close to major machines, it will present these machines with conditions approximating a short circuit at their terminals, $L_e = 0$. In this case the short-circuit time constants result. If the fault location is remote from key generators or fault impedances are high, open-circuit time constants result. The calculation process reads the machine reactances and the open-circuit time constants from a Breaker Duty File. The subtransient reactances specified in this file must correspond to the reactive parts of the impedances used in the generator conversion function, CONG, Section 13.1.1, "Objectives of a Balanced Switching Study" which must precede the selection of activity BKDY.

Total dc offset current is calculated in two different manners.

For method one, the instantaneous dc offset for each branch is set equal to the magnitude of the difference between the prefault current on the branch and the instantaneous ac current after the fault. The total dc offset current is the magnitude of the sum of these differences. Decremented dc currents and hence total rms and total peak currents for each path are calculated by decaying each path's initial dc offset current by the Thevenin impedance at the point of the fault looking out each path.

Again the total dc offset current is the magnitude of the sum of the decayed currents, which have been stored and decayed as complex values. The values are listed in the output report on the line headed with FAULT CURRENT.

For method two, the total instantaneous dc offset current is assumed to be equal to the total instantaneous ac current. This instantaneous dc offset current is decremented by the equivalent Thevenin impedance of all paths from the fault location. The total dc offset, total rms, and total peak current are listed following the Thevenin impedance and initial voltage on the row beginning with the word THEVENIN.

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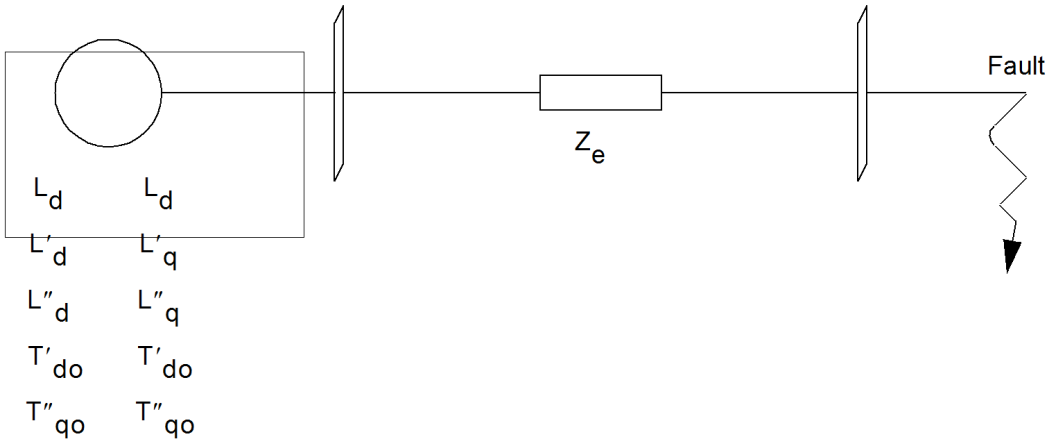

$$T'_{dz} = \frac{L'_d + L_e}{L_d + L_e} T'_{do} \qquad T'_{qz} = \frac{L'_q + L_e}{L_q + L_e} T'_{qo}$$

$$T''_{dz} = \frac{L''_d + L_e}{L'_d + L_e} T''_{do} \qquad T''_{qz} = \frac{L''_q + L_e}{L'_q + L_e} T''_{qo}$$

Figure 11.38. Relationships Between Machine Time Constants in Radial System

11.12.2. Breaker Duty Data File Contents

The input stream to activity BKDY consists of two blocks of records. The first block specifies machine reactances and open-circuit time constants for conventional machines. It consists of a series of records in the following format:

```
I, ID, T'do, T"do, T'qo, T"qo, Xd, Xq, X'd, X'q,
X"
```

where:

I	Bus number. Bus I must be present in the working case with a plant sequence number assigned to it (refer to Plant and Machine Sequence Numbers). No default is allowed.
ID	One- or two-character machine identifier used to distinguish among multiple machines at a plant (i.e., at a generator bus). ID = 1 by default.
T'_{do}	d axis transient open circuit time constant. No default is allowed.
T''_{do}	d axis subtransient open circuit time constant. No default is allowed.

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T'_{qo}	q axis transient open circuit time constant. T'_{qo} = 0. by default.
T''_{qo}	q axis subtransient open circuit time constant. No default is allowed.
X_d	d axis synchronous reactance, entered in pu on MBASE base. No default is allowed.
X_q	q axis synchronous reactance, entered in pu on MBASE base. No default is allowed.
X'_d	d axis transient reactance, entered in pu on MBASE base. No default is allowed.
X'_q	q axis transient reactance, entered in pu on MBASE base. X'_q = 0. by default.
X"	Subtransient reactance, entered in pu on MBASE base. No default is allowed.

The reactances listed should be saturated values.

Data records may be entered in any order. Input is terminated with a record specifying an I value of zero.

The second block of records specifies machine reactances and open-circuit time constants for induction machines. It consists of a series of records in the following format:

```
I, ID, T'o, T"o, X, X', X"
```

where:

I	Bus number. Bus I must be present in the working case with one or more induction machines assigned to it. No default is allowed.
ID	One- or two-character induction machine identifier used to distinguish among multiple induction machines at the bus. ID = 1 by default.
T'_o	Transient open circuit time constant. No default is allowed.
T''_o	Subtransient open circuit time constant. No default is allowed.
X	Synchronous reactance, entered in pu on MBASE base. No default is allowed.
X'	Transient reactance, entered in pu on MBASE base. No default is allowed.
X"	Subtransient reactance, entered in pu on MBASE base. No default is allowed.

Data records may be entered in any order. Induction machine input is terminated with a record specifying an I value of zero.

11.12.3. Fault Specification Data File Contents

This file consists of a series of records in the following format:

```
IBUS, TIME, JBUS, CKT
```


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where:

IBUS	Bus number of the bus to be faulted. No default is allowed.
TIME	Fault duty time. TIME = the value specified during the interactive dialog by default.
JBUS	Branch to bus number. JBUS = 0 by default.
CKT	Branch circuit or multi section line grouping identifier. CKT = '1' by default.

If a nonzero value is specified for JBUS, the branch designated by IBUS, JBUS and CKT must be present in the working case as an in-service branch or multi section line grouping. The specification of a branch is used solely for output identification purposes if several fault calculations are performed at the same bus, each with a different fault duty time.

Data records may be entered in any order; fault cases are calculated and reported in the same order in which data records are read. Input is terminated with a record specifying an IBUS value of zero.

11.12.4. Breaker Duty Calculation Options

Table 11.3. Breaker Duty Calculation Options

Option	Description
Fault selections specified via file	Enable or disable (default) the use of a Fault Specification Data file (see Section 11.12.3, "Fault Specification Data File Contents") as an alternative to interactive subsystem selection. The user enters the name of a file containing information specifically defining fault locations and corresponding breaker operating times to be used.
Default fault duty time	The time after fault application at which the decremented fault current is to be calculated.
Number of levels back for contributions output	The number of levels away from the faulted bus for which contributions will be processed. A level of zero will produce contributions at only the faulted bus.
Breaker duty data file	(*.bkd) This file contains the machine parametric data required for the calculation (see Section 11.12.2, "Breaker Duty Data File Contents").
Fault specification data input file	(*.bkf) This file identifies fault locations and fault duty times required to be investigated (see Section 11.12.2, "Breaker Duty Data File Contents"). Without this file, the user identifies fault locations by Area, Owner, Zone, BasekV or Bus.

11.12.5. Operation of Activity BKDY

Prior to initiating the calculation process, the user should ensure that the power flow case to be used is converged to an acceptable mismatch level using one or more of the conventional Gauss-Seidel or Newton-Raphson solution methods.

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In addition, the user should ensure that:

- All in-service machines are converted from their power flow representation to a constant Norton current source representation (see Section 13.2, "Converting Generators").
- Load boundary conditions are converted from the conventional constant MVA characteristic used in power flow calculations to a representation suitable for network conditions involving abnormally low voltages.

The user has the option to perform a bus ordering. If not, PSS®E will run a new bus ordering of the admittance matrix such that its sparsity characteristics are preserved. A new ordering is required because the generator conversion process introduces a row and column into the admittance matrix for each former Type 3 (swing) bus.

The short-circuit current calculations employ a triangularized Y matrix network solution which is designed for those situations where the internal flux linkages of generators are assumed to remain unchanged as a load or fault is switched onto the system, as a line is opened or closed, or as a load is removed.This activity is used for balanced short circuit, motor starting, voltage dip, and initial load rejection over voltage studies; this class of studies is termed switching studies (see Chapter 13, *Balanced Switching*).

The Y matrix solution handles load boundary conditions and the blowup check in the same way as the Gauss-Seidel solution method. The solution convergence monitor, the dc transmission line monitors for two-terminal and multi-terminal dc lines, and the mismatch summary also are identical to those of the Gauss-Seidel method.

No automatic adjustments are allowed, and switched shunt devices and dc converter transformer tap settings are locked at their pre switching settings. A two-terminal or multi-terminal dc transmission line is blocked during the remainder of the current execution of this breaker duty calculation if, on any iteration, the ac voltage at a rectifier converter station bus falls below 50% or is insufficient to make margin order. Shunt elements of FACTS devices hold their pre switching reactive currents, and series elements are held at their pre switching series voltages.

For each in-service VSC dc line, the solution starts with each converter holding the active and reactive power at the time the generators were converted to the Norton equivalent. If, on any iteration, the corresponding current exceeds the converter's IMAX, the injection is reduced using the power weighting factor fraction (PWF) just as in the conventional power flow solution activities. Each converter is treated independently, so that any reduction at one converter does not affect the injection at the other end of the VSC dc line. The dc transmission line monitor for VSC dc lines includes the dc line name, followed by the ac power injection at each converter bus.

In reading the Breaker Duty Data File, a machine is treated as a salient pole machine if either T'_{qo} or X'_q are specified as zero; a conventional machine is treated as an induction machine if $X_d = X_q$ *and* $X'_d = X'_q$; otherwise, a round rotor machine is assumed. For any in-service machine in the working case for which no data record is read, the same d and q axis currents are used in the two network solutions performed for each fault case.

Note that the user can convert the generators (activity CONG) prior to entering the circuit breaker duty calculation. The power flow case with converted generators can be saved using a different filename for subsequent or frequent calculation of circuit breaker duties. When performing the conversion, the user can elect to use the imaginary part of the ZSOURCE (used in stability analyses) or the ZPOS (used in unbalanced fault calculations) machine impedance.

Regardless of the method of designating buses to be faulted, those buses which are dummy buses of multi section lines are skipped as the calculation sequences through fault cases when the multi section line reporting option is enabled.

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If activity CONG has not been executed, activity BKDY prints an appropriate message and terminates. Otherwise, if activity BKDY detects that a new execution of activity ORDR is required, a message is printed and activity ORDR is automatically executed.

Activity BKDY uses a network solution similar to that of activity TYSL, and it handles load, dc line, and FACTS device boundary conditions as well as the blowup check in the same way as activity TYSL. When enabled, the solution convergence monitor, the dc transmission line monitors, and the FACTS device monitor are identical to those of activity TYSL. Refer to Section 13.7, "Solving the Converted Case" and Section 6.3.11, "Operation of Activity SOLV".

Activity BKDY responds to the following interrupt control codes:

AB	Abandon activity BKDY following completion of the next iteration.
CM	Print network solution convergence monitor.
DC	Tabulate conditions for each dc line after each iteration.
FD	Tabulate the conditions for each in-service FACTS device after each iteration.

11.12.6. Circuit Breaker Interrupting Duty Report

At each faulted bus, the current contributions from all machines and branches connected to the bus are listed, as well as fault current. The Thevenin impedance characterizing the faulted bus as well as its pre-fault voltage are also printed. Five columns of currents are printed:

1. Initial complex alternating current assuming no dc offset.

2. Decremented complex alternating current at the specified fault duty time, again assuming no dc offset.

3. Peak dc offset current magnitude at the fault duty time.

4. Total rms fault current magnitude at the fault duty time.

5. Total peak fault current magnitude at the fault duty time.

The branch decremented peak dc currents in (3) above are calculated assuming that the initial dc offset current follows from the pre-fault to post-fault instantaneous change in current, and that the dc offset current decays at a rate dependent upon the X/R ratio of the Thevenin equivalent of the branch assuming *flat conditions*. Machine currents are similarly calculated, except that the machine (plus step-up transformer if XTRAN is non-zero) impedance is used in setting the X/R ratio. The value printed as FAULT CURRENT is the magnitude of the sum of the decayed change in current of each of the above components. For reference, the final line for each faulted bus' output block shows as peak dc current the total instantaneous fault current decayed according to the faulted bus' Thevenin impedance as determined from the flat conditions impedance matrix. This line also shows the Thevenin impedance (in per unit, rectangular coordinates) and pre-fault bus voltage (in per unit, polar coordinates).

All currents are printed in either amps or per unit, and the complex currents in (1) and (2) above in either rectangular or polar coordinates, according to the fault analysis output options currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). If the physical units option is enabled

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and the bus where base voltage is required for the calculation of the current has no base voltage specified for it, the current is printed in per unit.

In responding to the output option request, the user may suppress the contributions output described above and limit the reporting to fault currents.

Similarly, the output report may be extended to provide contributions up to < n > levels away from the faulted bus. A response of 1 to the output option request causes the output for the faulted bus to be followed by machine and branch contributions output at all buses connected to it. Generalizing, a response of a positive number < n > results in output for the faulted bus, followed by output for all buses connected to it, followed by output for all buses two buses away from it, and so on up to < n > levels removed from the faulted bus. In reporting these remote contributions, only the currents described in (1) and (2) above are tabulated, and a branch is only reported once: at the end which is closer to (i.e., fewer levels away from) the faulted bus.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the far end bus (rather than the closest dummy bus) of each multi-section line connected to the bus being reported (the at bus) is shown as its connected bus (its from bus). Multi-section lines are identified with an ampersand (&) as the first character of their line identifiers in the branch circuit identifier column (e.g., &1; refer to Multi-Section Line Grouping Data). When reporting remote contributions, the dummy buses of multi-section lines are neither reported nor counted in the levels away calculation; i.e., the far end bus is one level removed from the at bus.

Contributions from a three-winding transformer are identified by the winding number, the transformer name and the circuit identifier.

11.12.7. Example: Breaker Duty Results Analysis

Figure 10-39 shows a sample output report using polar coordinates and amperes for clarity. The upper block of output shows decremented values of current based on a contact parting delay time of zero; hence it describes the intersection of the envelope of the fully offset current wave with the t = 0+ axis. The ac component of current is stated in rms terms, the ratio of rms value to amplitude is 1.414, and the initial value of the dc (unidirectional) current component is 1.414 × 18569.7 = 26261.5 A. Also, with no decay taking place, the total peak current is twice the amplitude of the ac component.

The center block of output specifies the situation at 25 milliseconds after fault initiation. Here the ac component of current has decayed to 16239.1 A, while the dc component has decayed more rapidly to 13996.6 A. The final output block, corresponding to 0.5 seconds with the dc component having decayed to essentially zero, shows the expected 1.414 relationship between rms and peak fault current, when expressed in amperes.

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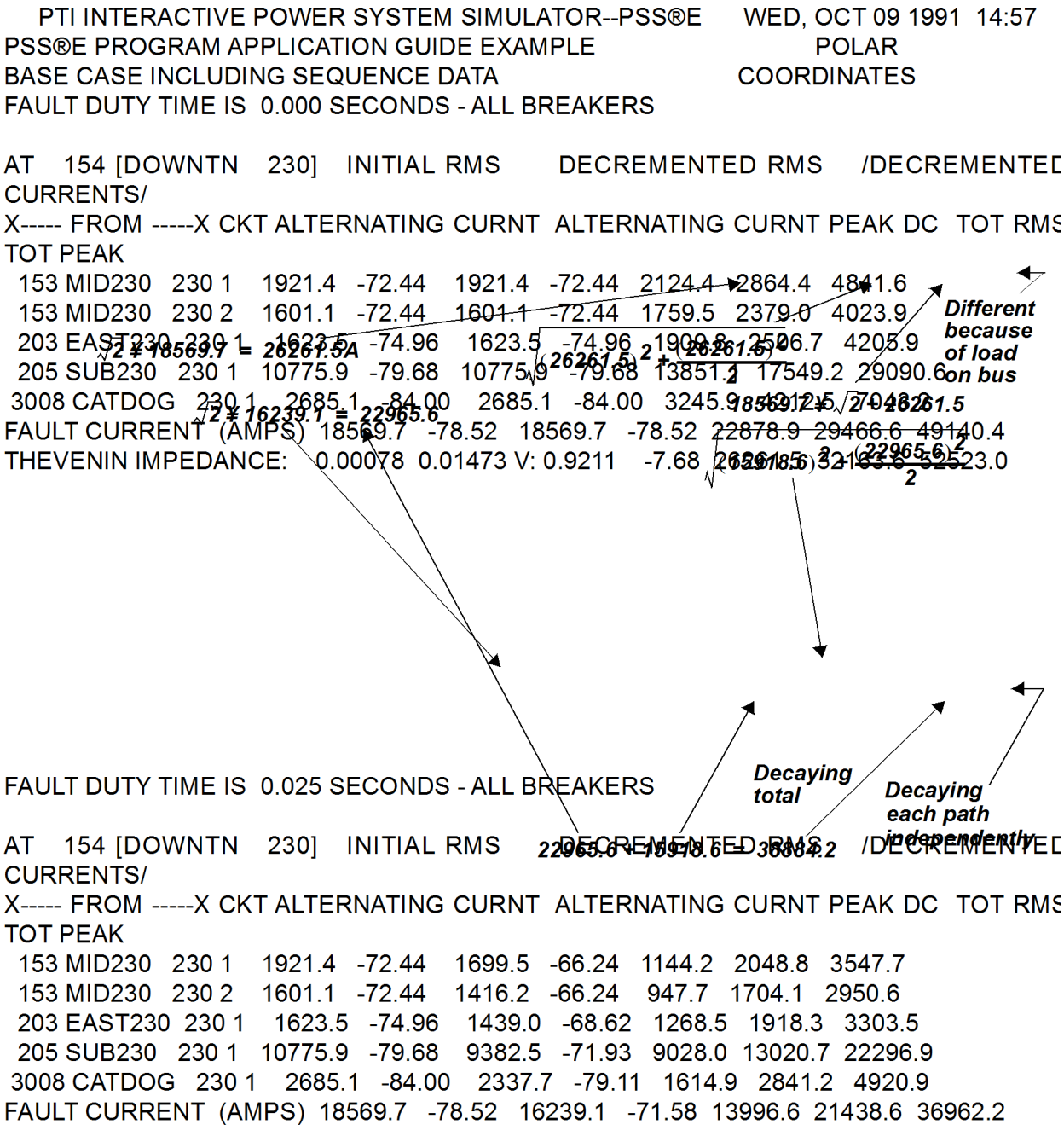

Figure 11.39. Relationship of Outputs to Offset Fault Current Wave (amps)

11.12.8. Application Notes

Activity BKDY uses the same type of network solution as does activity TYSL. A typical sequencing of activities is:

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1. A standard power flow is converged to an acceptable mismatch level using one or more of activities SOLV, MSLV, FNSL,NSOL, or FDNS. It is *strongly* recommended that the solved power flow case be saved in a Saved Case File with activity SAVE.

2. Activity CONG is executed to convert generator buses from their power flow representation to the constant Norton current source representation used in activity BKDY. See also Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent".

3. Activity CONL is executed to convert the load boundary conditions from the conventional constant MVA characteristic used in power flow calculations to a representation suitable for network conditions involving abnormally low voltages (see also Section 6.3.13, "Load").

4. Optionally, activity ORDR is executed to determine a bus ordering of the admittance matrix such that its sparsity characteristics are preserved. A new ordering is required because the execution of activity ,CONG introduces a row and column into the admittance matrix for each former Type 3 (swing) bus. If this step is omitted, activity BKDY will perform the ordering function.

5. Optionally, this converted and ordered case may be saved with activity SAVE if switching solutions are also to be calculated from this base condition.

6. Activity BKDY is entered.

To apply a three phase fault at a bus, activity BKDY temporarily introduces a fixed shunt with a large negative susceptance at the bus. When activity BKDY is initiated, if the fixed bus shunt table is full, an appropriate message is printed and activity BKDY is terminated.

In reading the Breaker Duty Data File, a machine is treated as a salient pole machine if either T'qo or X'q are specified as zero; a conventional machine is treated as an induction machine if Xd = Xq and X'd = X'q; otherwise, a round rotor machine is assumed. For any in-service machine in the working case for which no data record is read, the same d and q axis currents are used in the two network solutions performed for each fault case.

The value specified for subtransient reactance for each conventional machine must be identical to the imaginary part of ZSORCE (refer to Generator Data) when generators have been converted with activity CONG, or to the imaginary part of ZPOS (refer to Generator Sequence Data) when generators have been converted with activity CONG with generator conversion using fault analysis positive sequence machine impedance ZPOS. Similarly, the value specified for subtransient reactance for each induction machine must be identical to the imaginary part of the machine reactance used in activity CONG.

Regardless of the method of designating buses to be faulted, those buses which are dummy buses of multi-section lines are skipped as activity BKDY sequences through fault cases when the multi-section line reporting option is enabled.

Additional Information
PSS®E GUI Users Guide, Calculating Circuit Breaker Interrupting Duty
PSS®E Command Line Interface (CLI) *Users Guide*, Calculating Circuit Breaker Interrupting Duty

11.13. Calculating pi-Equivalent, Single Transmission Line Unbalance

SPCB

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Run Activity SPCB - GUI
Fault > Separate pole circuit breaker (SPCB)... *[Separate Pole Circuit Breaker]*
Run Line Mode Activity SPCB - CLI
ACTIVITY? >>**SPCB** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The separate pole circuit breaker activity SPCB calculates the positive sequence pi-equivalent corresponding to a single *transmission line* unbalance. This equivalent may subsequently be used in dynamic simulations to model the branch unbalance.

11.13.1. Calculating pi-Equivalent for Unbalanced Switching

The short-circuit fault current calculations described in previous sections are based on the assumption that the phase self- and mutual-impedances of each branch are balanced. Several situations exist, especially in dynamics, where the user will want to simulate a situation in which unbalance of the phase impedances and phase-to-phase coupling is a key effect. PSS®E contains an analytical process which is designed to calculate positive-sequence pi-equivalents for these cases. The process constructs positive-sequence equivalents for the following situations:

- One open phase, either grounded or not
- Two open phases
- Mid-line, line-ground, line-line-ground, or three-phase fault
- One breaker open

A series of situations where this activity would be used is shown in Figure 11.40, "Examples of Unbalanced Network Conditions Requiring a PI Equivalent". In this series, a line-ground fault occurs 40% of the way down the line from bus 151 to bus 152. This is the first unbalance condition. Next, the single-pole breaker at one end of the line opens the faulted phase while the breaker at the other end of the line remains closed (stuck) for a period of time. This is the second unbalanced condition. Finally, the second breaker opens to leave an unbalanced condition which is with two phases closed on the line section. The user, in a stability study, may then want to simulate the results of such a sequence of events.

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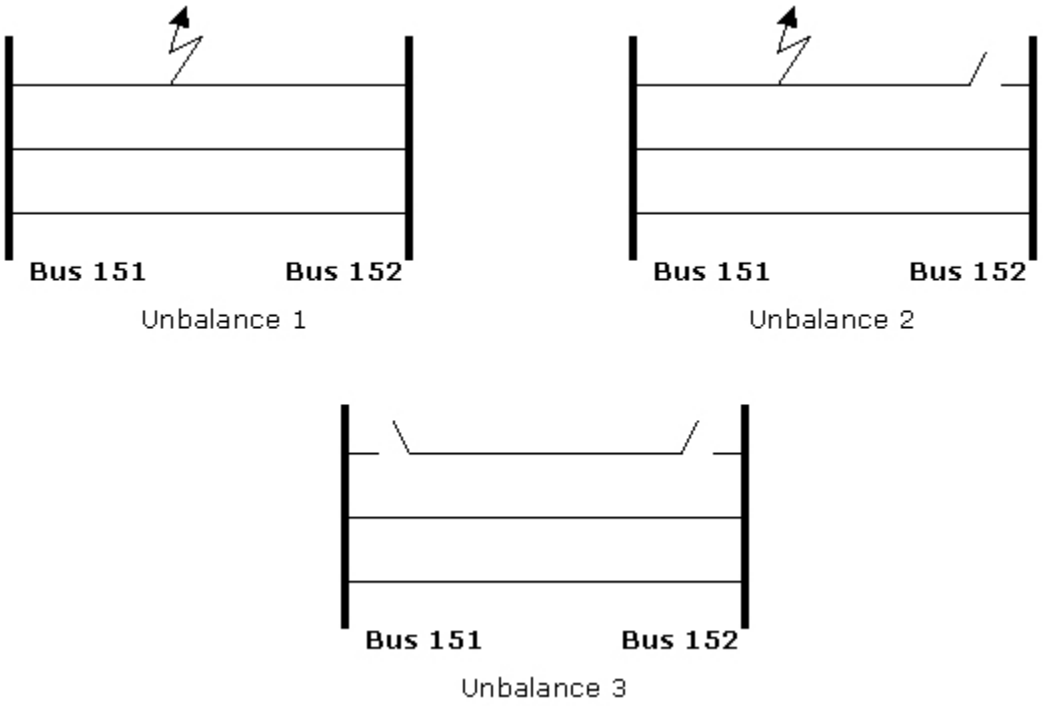

Figure 11.40. Examples of Unbalanced Network Conditions Requiring a PI Equivalent

In order to be able to simulate the three unbalanced conditions, in a stability case, the user would have to replace the positive sequence branch, in the power flow case, with a positive sequence PI equivalent for each of the three conditions, in sequence.

Prior to entering activity SPCB, the PSS®E working case and SEQD temporary file must be set up in the form required by activity SPCB. The procedure is as follows:

1. Solve the working case in the pre-unbalance condition.
2. With activity CHNG or the *[spreadsheet]*, place the transmission line which is to be subjected to the unbalance out-of-service.
3. Ensure that the fault analysis modeling option setting is placed in the normal three phase mode.
4. Run activity SEQD.
5. Then run activity SPCB.

11.13.2. Separate Pole Circuit Breaker Calculation Options

The user identifies the topologies, fault locations and type and grounding impedances to be simulated from among the following options.

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Table 11.4. Separate Pole Circuit Breaker Calculation Options

Option	Description
Out-of-service branch for unbalance	Specification of the circuit on which the unbalance will be placed. Activity SPCB calculates the columns of the three sequence impedance matrices corresponding to the specified buses. These are tabulated in sub-matrix form. Any error conditions (e.g., no such branch in the working case) generate an alarm. • From bus (number) • To bus (number • Circuit ID
Unbalance type	*One phase open*: open at both ends of the branch. Include a path to ground can also be specified, with fault location and line-to-ground impedance. *Two phases open*: this option has no fault applications. *In-line fault*. model assumes both circuit breakers are closed. The fault can be located at any point on the line and can be a single-line-to-ground fault, a line-to-line fault or a 3-phase fault; with selectable fault impedance where appropriate. *One breaker*: option to select the open end. Breaker at from bus (default) or Breaker at to bus Include a path to ground can also be specified, with fault location and line-to-ground impedance.
Set up network before solving with unbalances	PSS®E prepares the power flow case for the unbalanced network solution in a manner similar to that used at the initiation of the detailed unbalanced fault analysis described in Section 11.6.6, "Pre-Calculation Network Setup".

11.13.3. Transmission Line Unbalances

For the one phase open unbalance, the open phase may be grounded (faulted) or ungrounded. The user may choose to include a path to ground. If a ground is specified, the user specifies the grounding (fault) impedance and its location as a fraction of total line length starting at the from bus:

For the two phases open unbalance, no further unbalance specification is necessary.

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For the in-line fault, the user may specify a single-line-to-ground fault, double-line-to-ground (or line-to-line) fault, or three phase fault. For L-G and L-L-G faults, the user specifies the fault impedances in the same manner as in activity SCMU (refer to Bus Faults) and the location of the fault as a fraction of total line length starting at the from bus.

For the unbalance consisting of one breaker on one phase open, the phase with one end opened may include a path to ground. Line to ground impedance and fault location are specified as above.

If the user specifies no unbalance, activity SPCB to go through the calculation process with the data items representing the various unbalances set to default values. This results in no unbalance being imposed and should result in a positive sequence pi-equivalent which is equivalent to the positive sequence branch parameters contained in the working case.

11.13.4. Operation of Activity SPCB

If sequence data has not been read into the working case via activity RESQ, or if activity SEQD had last been executed with the fault analysis option setting at the two phase option, an appropriate error message is printed and activity SPCB is terminated.

Activity SPCB calculates the columns of the three sequence impedance matrices corresponding to the specified buses are calculated in sub-matrix form.

The user describes the unbalance (refer to Transmission Line Unbalances). The positive sequence pi-equivalent of the branch unbalance is calculated, and the results are printed (refer to Section 11.13.5, "Output of Activity SPCB").

11.13.5. Output of Activity SPCB

For each transmission line unbalance calculated by activity SPCB, the following data is tabulated at the selected output device:

1. The admittance matrix terms corresponding to the branch with the selected unbalance present.

2. A summary description of the branch unbalance.

3. The positive sequence pi-equivalent. This is presented in the form of a branch impedance (R + jX) and the equivalent line connected shunts at each end of the branch.

The pi-equivalent calculated by activity SPCB may be used in dynamic simulations to model the branch unbalance. It should *replace* the branch parameters used in modeling the branch without the presence of the unbalance.

Figure 11.41, "Output Report for pi-Equivalent shown in Figure 11.42, "Diagram of pi-Equivalent and the Branch Power Flow Data"" shows the output for a single-line-to-ground fault applied on the branch from bus 151 to bus 152 at a point 40% along the line from bus 151. The fault impedance is zero. Figure 11.42, "Diagram of pi-Equivalent and the Branch Power Flow Data"a and Figure 11.42, "Diagram of pi-Equivalent and the Branch Power Flow Data"b show diagrammatic view of the pi-equivalent.

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```
ZERO SEQUENCE THEVENIN IMPEDANCE SUBMATRIX:
  BUS      151 [NUCPANT  500]       152 [MID500   500]
  151  (   0.00017,   0.00617)  (   0.00014,   0.00235)
  152  (   0.00014,   0.00235)  (   0.00346,   0.03491)

POS. SEQUENCE THEVENIN IMPEDANCE SUBMATRIX:
  BUS      151 [NUCPANT  500]       152 [MID500   500]
  151  (   0.00328,   0.01784)  (   0.00452,   0.01015)
  152  (   0.00452,   0.01015)  (   0.00775,   0.01943)

NEG. SEQUENCE THEVENIN IMPEDANCE SUBMATRIX:
  BUS      151 [NUCPANT  500]       152 [MID500   500]
  151  (   0.00328,   0.01784)  (   0.00452,   0.01015)
  152  (   0.00452,   0.01015)  (   0.00775,   0.01943)

PI EQUIVALENT Y MATRIX IS:
(     1.7790,   -24.6553)   (    -0.8554,    18.5130)
(    -0.8554,    18.5130)   (     1.4711,   -22.0245)

TO SIMULATE:

ONE PHASE GROUNDED WITH IMPEDANCE=   0.0000 +J      0.0000
  40.0 PERCENT OF WAY DOWN LINE FROM   151 [NUCPANT  500]

FOR BRANCH FROM BUS 151 [NUCPANT500] TO BUS 152 [MID500 500] CIRCUIT 1
 USE EQUIVALENT R+JX= (    0.00249,    0.05390)  B= 0.0

AT BUS 151 [NUCPANT500] USE LINE CONNECTED SHUNT= ( 0.92358, -6.14228)

AT BUS 152 [MID500 500] USE LINE CONNECTED SHUNT= ( 0.61572, -3.51151)
```

Figure 11.41. Output Report for pi-Equivalent shown in Figure 11.42, "Diagram of pi-Equivalent and the Branch Power Flow Data"

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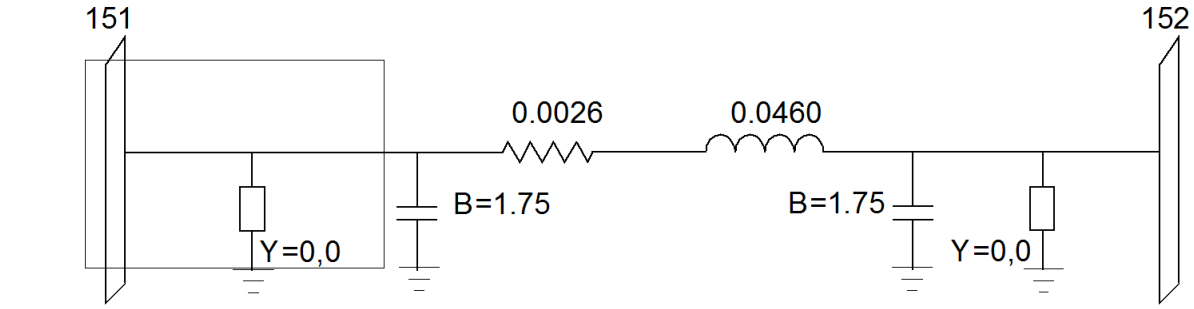
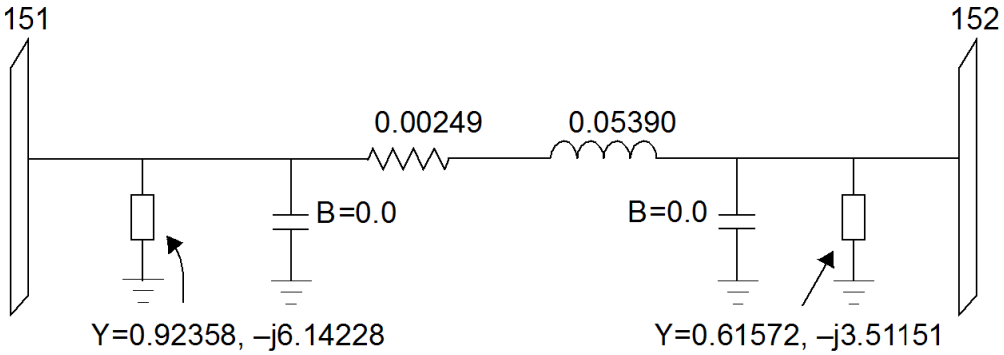


al Line Data



quivalent for Line-Ground Fault 40% Down Line From Bus 151

Figure 11.42. Diagram of pi-Equivalent and the Branch Power Flow Data

11.13.6. Application Notes

Activity SPCB may apply branch unbalances only to transmission lines modeled in full detail in both the positive (and hence, negative) and zero sequences. That is, the proper positive and zero sequence values of branch impedances and line charging (and/or line connected shunts) must be contained in the working case. Transformers, series capacitors, zero impedance lines (refer to Zero impedance Lines) or jumpers may not be subjected to branch unbalances in activity SPCB.

The transmission line to which the unbalance is to be applied must be modeled such that:

1. Y_i^+ or Y_j^+ are non-zero.
2. Y_i^0 and Y_j^0 are non-zero.
3. Y_i^+ and Y_i^0 are different.
4. Y_j^+ and Y_j^0 and different.

where:

Y_i^+	Is the quantity (line shunt + one-half of total line charging) at the bus i end of the branch in the positive sequence.

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Y_i^0	Is the quantity (line shunt + one-half of total line charging) at the bus i end of the branch in the zero sequence.
Y_j^+ and Y_j^0	Are the corresponding quantities at the bus j end of the branch.

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Chapter 12
Power Flow Reports

12.1. Analyzing Power Flow Solution Results

PSS®E can generate a variety of output reports to provide:

- Comprehensive information on the power flow results
- Summaries of conditions in network subsystems
- Checks on system and equipment conditions

This section discusses the available reports, the means by which they are generated and, as appropriate, how to interpret the information presented.

Table 12.1, "Summary of Available PSS®E Power Flow Reports "summarizes the available power flow reports and subsystems (see Section 4.8, "Subsystem Selection") by which they can be limited.

Table 12.1. Summary of Available PSS®E Power Flow Reports

Function	Can be Selected by
Power Flow Results	
Comprehensive power flow output in tabular or graphical format.	Bus subsystem
Multi-terminal dc line solution output	All in-service multi-terminal dc lines are reported
Summary Reports	
Summary of system swing bus and area slack bus conditions, and subsystem totals.	Bus subsystem
Summary of area, zone or owner totals of load, generation, net interchange, shunts and losses.	Area, zone or owner subsystem, as appropriate
Summary of tie-line flows between areas.	Area subsystem
Summary of tie-line flows between zones.	Zone subsystem
Reports on Checking Limit	
Summary of branches exceeding specified percentage of selected rating.	Bus subsystem
Summary of buses with voltage outside specified band.	Bus subsystem
Summary of generator terminal loading conditions on either all, or only overloaded, generators.	Bus subsystem
Summary of generator bus loading conditions on either all, or only var-limited, generators.	Bus subsystem
Summary of controlling transformer's violations	Bus subsystem

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Reports on Checking Limit	
Generator reactive power capability	Bus subsystem
Regulated bus violations and/or conflicts	Bus subsystem
Reports on Comparing Power Flow Cases	
Report on differences between the working case and a specified Saved Case	Bus subsystem
Report area, zone, owner, or complete system totals for the working case and a specified Saved Case	Area, zone or owner subsystem, as appropriate
Report subsystem tie branch differences between the working case and a specified Saved Case	Bus subsystem

Power flow results can be provided either in a tabular format or in a graphical format using the one-line slider format.

In the tabular format, power flow output reports may be organized either numerically by ascending bus number, or alphabetically by bus name. Selection of ordering by numbers or names is made in the Program Settings (see activity OPTN) where either the Number output option, or the Extended Name output option may be chosen. Changing between the Number and Extended Name input options has no effect on the ordering of power flow reports.

All reports, except those selected by the AREA, ZONE and/or OWNER criteria, are made with buses appearing in pure numeric or alphabetic order. Power flow results reports selected by one or more of these criteria order the buses numerically or alphabetically within groupings (see Section 4.9, "Subsystem Reporting"). As an example, if output is selected by AREA, with Areas 2 and 6 selected, the report will list all buses in Area 2 first and then all buses in Area 6.

12.2. Producing a Standard Power Flow Solution Report

POUT

Run Activity POUT - GUI
Power Flow > Reports > Bus based reports... *[Bus Based Reports]* OPTION: Powerflow output / Neither check box
Run Line Mode Activity POUT - CLI
ACTIVITY? >>**POUT** ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
AB

The power flow output activity POUT prints power flow solution output for the working case. The data tabulated for each bus printed includes bus voltage and phase angle, generator power output, constant power, current and admittance loads, bus shunts, and flows into all connected branches, FACTS devices, and GNE devices. Bus mismatch is printed for any bus for which it exceeds 0.5 MVA or kVA, according to the power output option in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). For

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buses where voltage magnitude is less than 10% (such as faulted buses), the quantity FAULT MVA is printed (refer to Fault MVA).

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output is generated with buses ordered as described in Section 4.9, "Subsystem Reporting".

The star point buses of three-winding transformers (refer to Transformer Data) are not reported. When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the interior dummy buses of multi-section line groupings (refer to Multi-Section Line Grouping Data) are not reported.

The output listing of activity POUT may be terminated by entering the AB interrupt control code.

12.2.1. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the "Report Tables" section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

12.2.2. Bus Quantities

The first line of the output block for each bus contains the bus number, name, and base voltage in the left portion of the line, followed by headings for the remaining lines, the bus voltage (in pu), phase angle (in degrees), additional headings for the remaining lines, and either the bus number or the extended bus name again, according to the bus output option currently in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

The second line contains the bus voltage in kV if a base voltage was specified for the bus, followed by headings for the Losses columns, and the numbers and names of the area and zone in which the bus is located. In addition, if the bus is a generator bus, the line contains the identifier GENERATION at the beginning and the plant active power, reactive power, and MVA output under the appropriate headings. This is followed by the percent MVA loading of the plant based on total plant MBASE. The plant reactive power output is followed by a single character flag, where H indicates that the plant reactive power output is at or beyond the total plant upper reactive power limit, L indicates that it is at or below the total plant lower reactive power limit, and R indicates that it is within limits. Note that only plant conditions are tabulated in activity POUT. If more than one machine is modeled at the plant, individual machine conditions may be examined using activities such as LIST, GEOL, and GCAP (see also Generator Unit Data).

Only plant conditions are tabulated. If more than one machine is modeled at the plant, individual machine conditions may be examined using the range checking facilities described in *Section 12.15, "Viewing Network Limit Violations"*.

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These lines are followed by any non-zero constant power load, constant current load, constant admittance load, fixed shunt elements, and switched shunt at the bus, with the appropriate identifier at the left of each line. Note that only totals by load characteristic are tabulated in activity POUT. If more than one load and/or fixed bus shunt is modeled at the bus, individual load and shunt quantities may be examined using activities such as LIST and EXAM (see also Generator Unit Data and Fixed Bus Shunt Data).

These lines are following by output for any single terminal GNE devices at the bus. GNE devices connected to the bus are reported in GNE device name alphabetical order. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service GNE devices are reported. If this option is enabled, out-of-service GNE devices at in-service buses are also reported; X is displayed in the various loading fields.

For each GNE device reported, the following data is printed:

1. An identifier indicating the GNE device type (VAR PQ, VAR I or VAR Y).
2. The identifier GNE; this is printed in the circuit identifier column.
3. The power flowing into the GNE device.
4. The GNE device name.

12.2.3. FACTS Devices

>

Output for FACTS devices appears in the bus output block immediately following the bus quantities output described above, and before output for dc lines and ac branches. FACTS devices connected to the bus are reported in FACTS device name alphabetical order. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service FACTS devices are reported. If this option is enabled, out-of-service FACTS devices at in-service buses are also reported; X is displayed in the various loading and series voltage fields.

For each FACTS device for which the bus being reported is the sending bus of an IPFC master or slave device, the following data pertaining to the device's active power transfer to its companion IPFC device is printed:

1. An identifier indicating the type of the companion IPFC device (IPFC MASTER or IPFC SLAVE).
2. The identifier IPF; this is printed in the circuit identifier column.
3. The active power flowing to the companion IPFC device. If the bus being reported is the IPFC master and the active power transfer is being held at its limit, its value is followed by the tag HI.
4. The FACTS device name.

Except for IPFC devices for which the shunt element is 0.0, for each FACTS device for which the bus being reported is the sending bus, the following data pertaining to the device's shunt element is printed:

1. An identifier indicating the type of FACTS device shunt element and/or active power transfer being reported (STATCON SHUNT, SSSC SHUNT, UPFC SHUNT, or IPFC SHUNT).
2. An identifier indicating the type of FACTS device being reported: STA (for a Statcon or Statcom), SSS (for an SSSC), UPF (for a UPFC) or IPF (for an IPFC); this is printed in the circuit identifier column.

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3. Unless the device being reported is an IPFC, the active power being exchanged between the shunt and series element. If the active power is being held at its limit, its value is followed by the tag HI.

4. The reactive power being absorbed by the shunt element.

5. The MVA corresponding to the active and reactive powers in (3) and (4) above, and the percent loading on the shunt element.

6. The FACTS device name.

If the FACTS device contains a series element, the following data pertaining to it is printed in the output blocks of its sending and terminal buses:

1. The number, name, and base voltage of the other endpoint bus (to bus).

2. An identifier indicating the type of FACTS device being reported: SSS (for an SSSC), UPF (for a UPFC) or IPF (for an IPFC); this is printed in the circuit identifier column.

3. The power flowing into the series element along with its percent loading at the from bus end.

4. The magnitude and phase of the series voltage. If the series voltage magnitude is being held at its limit (control modes 1, 5 and 6) or if the series voltage magnitude exceeds VSMX (control modes 2, 3, 4, 7 or 8), its value is followed by the tag HI. Otherwise, the tag RG is printed.

5. The FACTS device losses calculated as the algebraic sum of the reactive power flowing into the series device from its endpoint buses.

6. The numbers and names of the area and zone in which the other endpoint bus (to bus) is located.

7. The FACTS device name.

12.2.4. Series GNE Devices

Output for multiple terminal GNE devices at the bus appears in the bus output block immediately following the bus and FACTS device output described above, and before output for dc lines. Series GNE devices connected to the bus are reported in GNE device name alphabetical order. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service GNE devices are reported. If this option is enabled, out-of-service GNE devices at in-service buses are also reported; X is displayed in the various loading fields.

For each series GNE device reported, the following data is printed:

1. The number, name, and base voltage of the other endpoint bus (to bus).

2. The identifier GNE; this is printed in the circuit identifier column.

3. The power flowing into the GNE device.

4. GNE device losses calculated as the algebraic sum of the powers flowing into the device from its endpoint buses.

5. The numbers and names of the area and zone in which the other endpoint bus is located.

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6. The GNE device name.

12.2.5. DC Lines

Output for dc transmission lines appears in the bus output block immediately following the bus, FACTS device, and series GNE device output described above, and before output for ac branches. Two-terminal, multi-terminal, and VSC dc lines connected to the bus are reported in dc line name alphabetical order; the order of printing is:

- Two-terminal lines, in ascending dc line number order
- Multi-terminal lines, in ascending dc line number order
- VSC dc line converters connected to the bus, in alphabetical VSC dc line name order.

If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service dc lines are reported. If this option is enabled, out-of-service dc lines at in-service buses are also reported; an X is displayed in the various loading fields and other fields indicating dc line conditions.

The power flowing into a dc line is the apparent ac system complex load as seen at the converter bus. Refer to Sections DC Transmission and Multiterminal DC networks of the *PSS®E Program Application Guide* for additional details on dc line quantities.

The following data is printed in the output block of each two-terminal and multi-terminal dc line converter bus:

1. For two-terminal lines, the number, name, and base voltage of the other converter terminal bus (to bus); for multi-terminal lines, the number and name of the dc bus to which the ac converter bus is connected, with the identifier (DC) in the base voltage column.

2. An identifier indicating the type of dc line being reported: 2D (for two-terminal) or MD (for multi-terminal), and an identifier indicating the type of converter at the from bus: R (for a rectifier) or I (for an inverter); this is printed in the circuit identifier column.

3. The ac power flowing into the dc line.

4. The converter transformer off-nominal turns ratio, TAPR (or TAPI), followed by a two-character tag, which is either HI or LO if the ratio is at its high or low limit, respectively; LK if dc taps were locked during the last power flow solution activity, or if an ac transformer (rather than this converter transformer) is controlling a dc line quantity; or RG if the ratio is in regulating range.

5. The value of converter angle followed by one of the tags RG, HI or LO if the angle is within its limits, at or above its nominal high limit, or at or below its steady-state low limit, respectively.

6. DC line losses calculated as the algebraic sum of the powers flowing into the line from its converter buses.

7. For two-terminal lines, the numbers and names of the area and zone in which the other converter bus (to bus) is located. For multi-terminal lines, the numbers and names of the area and zone in which the dc bus to which the ac converter bus is connected is located.

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8. The dc line name.

The following data is printed in the output block of each VSC dc line converter bus:

1. The number, name, and base voltage of the other converter terminal bus (to bus).
2. The identifier VSC; this is printed in the circuit identifier column.
3. The ac power flowing into the dc converter.
4. The percent of MVA rating, where the rating is taken as the more restrictive of IMAX converted to MVA and SMAX (refer to Voltage Source Converter (VSC) DC Transmission Line Data).
5. DC line losses calculated as the algebraic sum of the powers flowing into the line from its converter buses.
6. The numbers and names of the area and zone in which the other converter bus is located.
7. The VSC dc line name.

Figure 11-1 contains partial output from activity POUT for a system in which bus 201 is the rectifier bus of a two-terminal dc line, and bus 161 is the inverter bus. The output is shown with the numbers output option in effect.

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      WED, JUN 27 2007  16:19
 ADDITION OF DC LINE TO BASE CASE DEMO LOAD FLOW                  RATING   %MVA FOR TRANSFORMERS
 FOR PSS®E MANUAL                                                 SET A   % I  FOR NON-TRANSFORMER BRANCHES

 BUS 201 HYDRO     500.00 CKT     MW    MVAR    MVA   % 0.9913PU    2.55  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X  201
  FROM GENERATION                600.0  400.0H 721.1  99 495.65KV           MW     MVAR     2 LIGHTCO      2
  TO SHUNT                         0.0  294.8  294.8                    1.0000RG   10.94RG   41.76  568.06     1 FLAPCO       1      "1"
  TO 161 INVJCT    18.500 2DR    641.8  286.2  702.7                                          5.35   80.20     1 FLAPCO       1
  TO 151 NUCPANT   500.00  1   -724.8  -51.1  726.6  61                                      2.06   25.76     2 LIGHTCO      2
  TO 202 EAST500   500.00  1    315.1 -142.3  345.8  29                                      4.69   46.92     2 LIGHTCO      2
  TO 204 SUB500    500.00  1    367.9   12.5  368.1  31

 BUS 161 INVJCT    18.500 CKT     MW    MVAR    MVA   % 0.9730PU   -1.60  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X  161
                                                        18.000KV                 MW     MVAR     1 FLAPCO       1
  TO 201 HYDRO     500.00 2DI  -600.0  281.9  662.9      0.9875RG   19.08RG   41.76  568.06     2 LIGHTCO      2      "1"
  TO 154 DOWNTN    230.00  1    600.0   78.4  605.1 101 1.0000UN                  0.00   48.35     1 FLAPCO       1
  TO 162 INVTER    18.500  1      0.0 -360.3  360.3                             0.00   13.71     1 FLAPCO       1
```

Figure 11-1. Example Output From Activity POUT

Figure 12.1. Example Output From Activity POUT

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12.2.6. Branch Quantities

Following the output described above, flows for each ac branch connected to the bus are tabulated. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service branches are reported. If this option setting is enabled, out-of-service branches at in-service buses are also reported; an X is displayed in the various loading and losses fields.

Following the output described above, flows for each in-service ac branch connected to the bus are tabulated.

For non-transformer branches and two-winding transformers, the output line contains the to bus number, name, and base voltage followed by the branch circuit identifier. For three-winding transformers, the output line contains the string 3WNDTR in the bus number column, the transformer name in the bus name column, and the winding number in the base voltage column, followed by the transformer circuit identifier; if the three-winding transformer to bus reporting option is enabled, two additional lines, containing the bus numbers, names, and base voltages of the two buses connected to the from bus by the three-winding transformer, are printed.

The active and reactive power flow on the branch is printed as power leaving the from bus (i.e., positive for outgoing and negative for incoming); MVA flow is also printed.

The percent loading, based on the rating set established as the default rating set program option setting is then printed (refer to Section 3.3.3, "Program Run-Time Option Settings", activity OPTN, Non-Transformer Branch Data, and Transformer Data). For transformers, either percent current or percent MVA loading is reported, according to the transformer percent loading units program option setting. Similarly, for non-transformer branches, either percent current or percent MVA loading is reported, according to the non-transformer branch percent loading units program option setting.

Branch losses which are printed are taken as I^2R and I^2X losses and *exclude* the line charging, line connected shunt, and magnetizing admittance components. For three-winding transformers, if the three-winding transformer to bus reporting option is disabled, total losses on all three windings are printed on the output line for the three-winding transformer; if this option is enabled, losses on each winding are listed separately on the line identifying the bus to which the winding is connected.

The number and name of the area and zone in which the to bus is located are printed for non-transformer branches and for two-winding transformers. For three-winding transformers, if the three-winding transformer to bus reporting option is enabled, the numbers and names of the areas and zones of the two buses connected to the from bus by the three-winding transformer are listed on the lines identifying them.

For transformers, the turns ratio of the winding connected to the from bus is printed following the percent loading. A two-character tag is printed adjacent to the ratio, which has the following significance:

HI	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, and the off-nominal turns ratio is at or beyond its high limit.
LO	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, and the off-nominal turns ratio is at or below its low limit.

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RG	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, its adjustment control mode is set to +1 or +2, the automatic tap ratio adjustment was enabled during the last power flow solution activity, and the off-nominal turns ratio is within its limits.
DC	The from bus is the Winding 1 side of this two-winding transformer which is regulating alpha, gamma, or dc voltage of a dc line; its adjustment control mode is set to +4; dc tap adjustment was enabled during the last power flow solution activity; and the off-nominal turns ratio is within its limits.
LK	The from bus is connected to the Winding 1 side of this two-winding transformer or to any winding of this three-winding transformer, and either this transformer winding is not a tap changing transformer winding, its adjustment control mode is set to zero or negative, or the automatic tap ratio adjustment was disabled during the last power flow solution activity.
UN	The from bus is not the Winding 1 side of this two-winding transformer.

Except for the tags DC and UN, phase shifters are handled in a similar manner. For single section two-winding transformers with non-zero phase shift angle on the Winding 1 side, no indication of phase shift is printed in the output block of the Winding 2 side bus.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the far end to bus (rather than the closest dummy bus) of each multi-section line connected to the from bus is shown as its to bus. Multi-section lines are identified with an ampersand (&) as the first character of their branch identifiers in the circuit identifier column (e.g., &1). The sum of the losses on all of the line sections comprising the multi-section line is shown as losses on the multi-section line. Finally, transformer information is shown only if the line section adjacent to the from bus is a transformer branch, and the data applies to the winding adjacent to the from bus.

If any of the non-transformer branches reported have a non-zero line shunt at the from bus end, or if Winding 1 of any transformer branch connected to the from bus has non-zero magnetizing admittance, and the line shunt reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the branch output lines are followed by lines reporting powers corresponding to each of these admittances. Note that, regardless of the setting of this option, the branch flows reported always include these components.

Figure 12.2, "Power Flow Output for Branches" shows a sample of the tabular output. It covers Buses 152 and 153 which are connected by a 500/230 kV transformer. It can be seen that the tap, at the Bus 152 side is at its high limit (HI) of 1.05 pu. At bus 153 It can be seen that this is not the winding one side of this two-winding transformer (UN).

The figure also shows three non transformer branches from Bus 153. They include circuits #1 and #2 to Bus 154 (DOWNTN), which are overloaded (at 180% and 150% of rating) and a single circuit to Bus 3006 (UPTOWN).

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```
BUS    152 MID500      500.00 CKT     MW     MVAR     MVA  %I 1.0782PU  -15.07
                                                             539.09KV
  TO LOAD-PQ                        1200.0   360.0  1252.8
  TO LOAD-I                          936.2   280.9   977.4
  TO LOAD-Y                          973.9   292.2  1016.8
  TO SHUNT                             0.0  -930.0   930.0
  TO    151 NUCPLNT     500.00  1   -440.3   -46.3   442.7  34
  TO    151 NUCPLNT     500.00  2   -440.3   -46.3   442.7  34
  TO    153 MID230      230.00  1    701.9   255.0   746.8  28 1.0500HI
  TO    202 EAST500     500.00  1    119.0   161.5   200.5  16
  TO   1403 WDUM        18.000  1  -1435.9  -152.6  1444.0  75 1.1000HI
  TO   1404 EDUM        18.000  1  -1435.9  -152.6  1444.0  75 1.1000HI
  TO   3004 WEST        500.00  1   -178.6   -21.8   179.9

BUS    153 MID230      230.00 CKT     MW     MVAR     MVA  %I 1.0150PU  -17.00
                                                             233.45KV
  TO LOAD-PQ                         200.0   100.0   223.6
  TO STATCON         FACTS DEV # 1     0.0    50.7    50.7 100
  TO    152 MID500      500.00  1   -701.9  -228.6   738.2  29 1.0000UN
  TO    154 DOWNTN      230.00  1    362.7    46.9   365.7 180
  TO    154 DOWNTN      230.00  2    302.3    35.7   304.4 150
  TO   3006 UPTOWN      230.00  1   -163.1    -4.7   163.2
```

Figure 12.2. Power Flow Output for Branches

12.2.7. Fault MVA

For buses where voltage magnitude is less than 10% (such as faulted buses), the quantity FAULT MVA is printed. This quantity is intended for use following the solution of a balanced fault with activity TYSL. It is computed using the fixed bus shunt and the bus voltage; as such, it is normally the sum of fault MVA and the MVA consumed by the fixed bus shunt, if any, on the bus.

Under three phase fault conditions, the solved voltage magnitude at the faulted bus, and usually at nearby buses, is very small (e.g., 10^{-6} pu at the faulted bus). Hence, the voltage correction applied during the final TYSL iteration may be a large fraction of the solved voltage. Therefore, to ensure an accurate reporting of FAULT MVA, the acceleration factor applied during the final TYSL iteration should be 1.0. If the solution requires deceleration to achieve convergence, then, after it converges, activity TYSL should be repeated one final time with the acceleration factor reset to unity.

Additional Information
PSS®E GUI Users Guide, Producing a Standard Power Flow Solution Report *PSS®E* Command Line Interface (CLI) *Users Guide*, Standard Power Flow Solution Report

12.2.8. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

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Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "BUS POUT".

12.3. Producing a Wide-Format Power Flow Solution Report

LOUT

Run Activity LOUT - GUI
Power Flow > Reports > Bus based reports... *[Bus Based Reports]* OPTION: Powerflow output / Wide format output checked
Run Line Mode Activity LOUT - CLI
ACTIVITY? >>**LOUT** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The power flow output activity LOUT prints power flow solution output for the working case. The data tabulated for each bus printed includes bus voltage and phase angle, generator power output, loads, bus shunt elements, and flows into all connected branches, FACTS devices, and GNE devices. Bus mismatch is printed for any bus for which it exceeds 0.5 MVA or kVA, according to the power output option in effect (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN). For buses where voltage magnitude is less than 10% (such as faulted buses), the quantity FAULT MVA is printed (refer to Fault MVA

The user may process the entire working case or specify subsystem data (refer to Section 4.8, “Subsystem Selection”) by bus subsystem, area, zone, owner, or basekV. Output is generated with buses ordered as described in Section 4.9, “Subsystem Reporting”.

The star point buses of three-winding transformers (refer to Transformer Data) are not reported. When the multi-section line reporting option is enabled (refer to Section 3.3.3, “Program Run-Time Option Settings”and activity OPTN), the interior dummy buses of multi-section line groupings (refer to Multi-Section Line Grouping Data) are not reported.

The output listing of activity LOUT may be terminated by entering the AB interrupt control code.

Activity LOUT formats its report in traditional power flow output style, with bus quantities on the left side of the page and branch information shown on the right side.

This tabulated output presents the same information about the power flow solution as is seen in the standard output, with the addition of an option to print the branch currents (in amperes). In addition, the rating sets in use are printed.

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The layout of the information is different from the standard layout. Figure 12.3, "Wide Format Power Flow Output including Branch Currents" shows the wide format output for the buses shown in Figure 12.1, "Example Output From Activity POUT" for the standard output format.

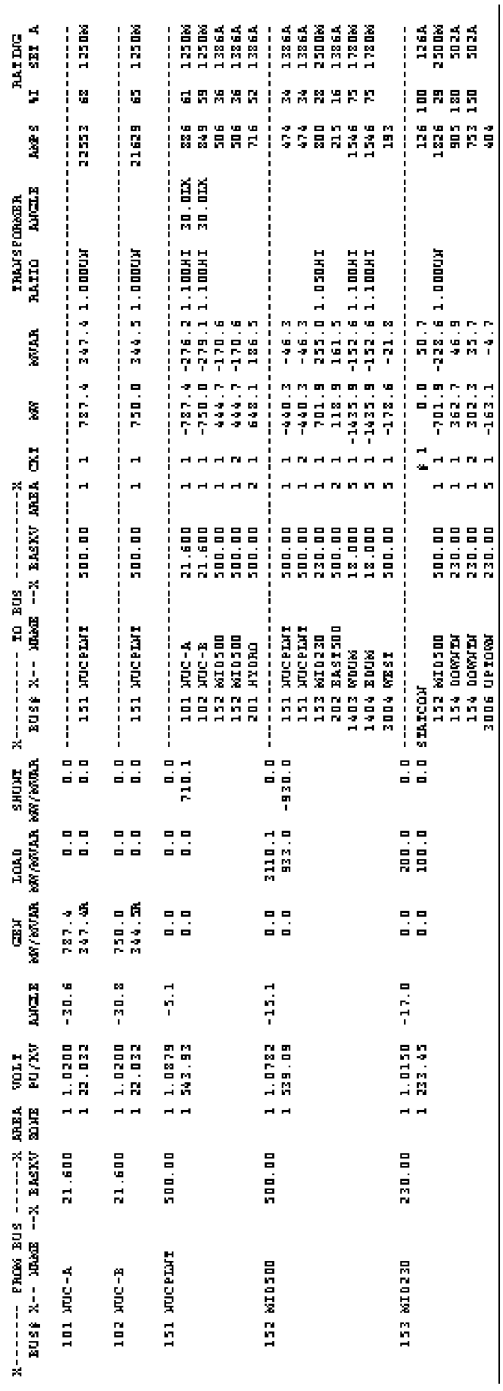

Figure 11-3. Wide Format Power Flow Output including Branch Currents

Figure 12.3. Wide Format Power Flow Output including Branch Currents

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The main difference between the standard and wide format is that, in the former, the data categories are listed in vertical order while in the wide format they are printed from left to right. Those data categories are:

- Bus data
- Generators
- Load
- Shunts
- FACTS devices
- GNE devices
- DC lines
- Ac branches

12.3.1. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "BUS LOUT".

12.3.2. Bus Quantities

The bus identifiers and boundary conditions for each bus reported in activity LOUT are printed on two lines on the left side of the bus’ output block. The bus identifiers of bus number, name, and base voltage are listed followed by the number of the area in which the bus is located and the zone number directly below it on the second line.

The bus voltage (in pu) and phase angle (in degrees) are printed; if the bus base voltage is specified for the bus, the bus voltage in kV is printed directly below the pu voltage on the second line. The generation, load, and shunt at the bus are printed, with active power shown on the first line and reactive power on the second. The plant reactive power output is followed by a single-character flag, where H indicates that the plant reactive power output is at or beyond the total plant upper reactive power limit; L indicates that it is at or below the total plant lower reactive power limit; and R indicates that it is within limits. Note that only plant conditions are tabulated in activity LOUT. If more than one machine is modeled at the plant, individual machine conditions may be examined using activities such as LIST, GEOL, and GCAP (see also Generator Unit Data).

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The quantities printed as load at the bus are the sum of any constant MVA, current, and admittance loads at the bus; the shunt is the sum of any fixed and switched shunts. If more than one load and/or shunt are modeled at the bus, individual load and shunt quantities may be examined using activities such as LIST and EXAM (see also Generator Unit Data and Fixed Bus Shunt Data).

12.3.3. FACTS Devices

Output for FACTS devices appears on the right side of the bus output block as the first line or lines of branch data (i.e., before output for GNE devices, dc lines and ac branches). FACTS devices connected to the bus are reported in FACTS device name alphabetical order. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service FACTS devices are reported. If this option is enabled, out-of-service FACTS devices at in-service buses are also reported; an X is displayed in the various loading and series voltage fields.

For each FACTS device for which the bus being reported is the sending bus of an IPFC master or slave device, the following data pertaining to the device's active power transfer to its companion IPFC device is printed:

1. An identifier indicating the type of the companion IPFC device (IPFC MASTER or IPFC SLAVE).
2. The identifier IPF; this is printed in the circuit identifier column.
3. The active power flowing to the companion IPFC device.
4. The FACTS device name printed in the transformer ratio and angle columns.

Except for IPFC devices for which the shunt element is 0.0, for each FACTS device for which the bus being reported is the sending bus, the following data pertaining to the device's shunt element is printed:

1. An identifier indicating the type of FACTS device shunt element and/or active power transfer being reported (STATCON SHUNT, SSSC SHUNT, UPFC SHUNT, or IPFC SHUNT).
2. An identifier indicating the type of FACTS device being reported: STA (for a Statcon or Statcom), SSS (for an SSSC), UPF (for a UPFC) or IPF (for an IPFC); this is printed in the circuit identifier column.
3. Unless the device being reported is an IPFC, the active power being exchanged between the shunt and series element.
4. The reactive power being absorbed by the shunt element.
5. The FACTS device name printed in the transformer ratio and angle columns.
6. The rating and percent loading of the shunt element.

If the FACTS device contains a series element, the following data pertaining to it is printed in the output blocks of its sending and terminal buses:

1. The number, name, base voltage, and area of the other endpoint bus (to bus).
2. An identifier indicating the type of FACTS device being reported: SSS (for an SSSC), UPF (for a UPFC) or IPF (for an IPFC); this is printed in the circuit identifier column.
3. The power flowing into the series element at the from bus end.

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4. The magnitude and phase of the series voltage. If the series voltage magnitude is being held at its limit (control modes 1, 5 and 6) or if the series voltage magnitude exceeds VSMX (control modes 2, 3, 4, 7 or 8), its value is followed by the tag H. Otherwise, the tag R is printed.

5. The percent current loading and rating.

12.3.4. GNE Devices

Output for GNE devices appears on the right side of the bus output block immediately following any FACTS device output described above, and before output for dc lines and ac branches. GNE devices connected to the bus are reported in GNE device name alphabetical order. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service GNE devices are reported. If this option is enabled, out-of-service GNE devices at in-service buses are also reported; an X is displayed in the various loading fields.

For each single terminal GNE device reported, the following data is printed:

1. An identifier indicating the GNE device type (VAR PQ, VAR I or VAR Y).

2. The identifier GNE; this is printed in the circuit identifier column.

3. The power flowing into the GNE device.

4. The GNE device name.

For each series GNE device reported, the following data is printed:

1. The number, name, base voltage, and area of the other endpoint bus (to bus).

2. The identifier GNE; this is printed in the circuit identifier column.

3. The power flowing into the GNE device.

4. The GNE device name.

12.3.5. DC Lines

Output for dc transmission lines appears on the right side of the bus output block immediately following any GNE device output described above, and before output for ac branches. Two-terminal, multi-terminal, and VSC dc lines connected to the bus are reported in dc line name alphabetical order, with output for two-terminal lines printed first, followed by output for multi-terminal lines, followed by output for VSC dc lines. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service dc lines are reported. If this option is enabled, out-of-service dc lines at in-service buses are also reported; an X is displayed in the various loading fields and other fields indicating dc line conditions.

The power flowing into a dc line is the apparent ac system complex load as seen at the converter bus. Refer to Sections DC Transmission and Multiterminal DC Networks of the *PSS®E Program Application Guide* for additional details on dc line quantities.

The following data is printed in the output block of each two-terminal and multi-terminal dc line converter bus:

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1. For two-terminal lines, the number, name, and base voltage of the other converter terminal bus (to bus), followed by the number of the area in which it is located; for multi-terminal lines, the number and name of the dc bus to which the ac converter bus is connected, with the identifier (DC) in the base voltage column, followed by the number of the area in which the dc bus to which the ac converter bus is connected is located.

2. An identifier indicating the type of dc line being reported: 2D (for two-terminal) or MD (for multi-terminal), and an identifier indicating the type of converter at the from bus: R (for a rectifier) or I (for an inverter); this is printed in the circuit identifier column.

3. The ac power flowing into the dc line.

4. The converter transformer off-nominal turns ratio, followed by a two-character tag, which is either HI or LO if the ratio is at its high or low limit, respectively; LK if dc taps were locked during the last power flow solution activity, or if an ac transformer (rather than this converter transformer) is controlling a dc line quantity; or RG if the ratio is in regulating range.

5. The value of converter angle followed by one of the tags RG, HI, or LO if the angle is within its limits, at or above its nominal high limit, or at or below its steady-state low limit, respectively.

6. The dc line name.

The following data is printed in the output block of each VSC dc line converter bus:

1. The number, name, and base voltage of the other converter terminal bus (to bus), followed by the number of the area in which it is located.

2. The identifier VSC; this is printed in the circuit identifier column.

3. The ac power flowing into the dc converter.

4. The VSC dc line name printed in the transformer ratio and angle columns.

5. The percent of MVA rating and the rating, where the rating is taken as the more restrictive of IMAX converted to MVA and SMAX (refer to Voltage Source Converter (VSC) DC Transmission Line Data).

Figure 12.4, “Example Output From Activity LOUT” contains partial output from activity LOUT for a system in which bus 201 is the rectifier bus of a two-terminal dc line and bus 161 is the inverter bus. The output is shown with the names output option in effect. The system condition in this example is identical to that shown in the example output of activity POUT in Figure 12.1, “Example Output From Activity POUT”.

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```
                    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSSE
     THU, JUN 28 2007   8:31
                        ADDITION OF DC LINE TO BASE CASE
DEMO LOAD FLOW                  %MVA FOR TRANSFORMERS
                        FOR PSSE MANUAL
                                  % I  FOR NON-TRANSFORMER BRANCHES

X------ FROM BUS ------X AREA  VOLT        GEN    LOAD
   SHUNT  X---------TO BUS----------X                        TRANSFORMER
    RATING
X--NAME--X BASKV BUS# ZONE  PU/KV  ANGLE MW/MVAR MW/MVAR
MW/MVAR X--NAME--X BASKV  BUS# AREA CKT   MW     MVAR    RATIO
 ANGLE   %  SET A

HYDRO     500.00  201   2  0.9913    2.5  600.0
0.0     0.0 ------------------------------------------------------------------------
                        2  495.65         400.0H
0.0   294.8 INVJCT    18.500   161    1 2DR  641.8   286.2  1.000RG
 10.9RG "1"

            EAST500   500.00   202    2  1   315.1  -142.3
             29  1200

            NUCPANT   500.00   151    1  1  -724.8   -51.1
             61  1200

            SUB500    500.00   204    2  1   367.9    12.5
             31  1200
INVJCT    18.500  161   1  0.9730   -1.6    0.0
0.0     0.0 ------------------------------------------------------------------------
                        1  18.000           0.0
0.0     0.0 HYDRO     500.00   201    2 2DI -600.0   281.9  0.987RG
 19.1RG "1"

            DOWNTN    230.00   154    1  1   600.0    78.4  1.000UN
        101   600

            INVTER    18.500   162    1  1     0.0  -360.3
```

Figure 12.4. Example Output From Activity LOUT

12.3.6. Branch Quantities

Following the output described above, flows for each ac branch connected to the bus are tabulated on the right side of the bus output block. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service branches are reported. If this option is enabled, out-of-service branches at in-service buses are also reported; an X is displayed in the loading fields.

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Each branch output line for non-transformer branches and for two-winding transformers contains the to bus number, name, and base voltage, followed by the number of the area in which the to bus is located and the branch circuit identifier. For three-winding transformers, the output line contains the string 3WNDTR in the bus number column, the transformer name in the bus name column, and the winding number in the base voltage column, followed by the transformer circuit identifier; if the three-winding transformer to bus reporting option is enabled, two additional lines, containing the bus numbers, names, and base voltages of the two buses connected to the from bus by the three-winding transformer, are printed, followed by the number of the area in which the to bus is located.

The active and reactive power flow on the branch is printed as power leaving the from bus (i.e., positive for outgoing and negative for incoming).

The percent loading, based on the rating set established as the default rating set program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) and the corresponding rating are printed on the far right-hand side of the line. For transformers, either percent current or percent MVA loading is reported, according to the transformer percent loading units program option setting. Similarly, for non-transformer branches, either percent current or percent MVA loading is reported, according to the non-transformer branch percent loading units program option setting.

For transformers, the turns ratio of the winding connected to the from bus is printed. A two-character tag is printed adjacent to the ratio which has the following significance:

HI	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, and the off-nominal turns ratio is at or beyond its high limit.
LO	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, and the off-nominal turns ratio is at or below its low limit.
RG	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, its adjustment control mode is set to +1 or +2, the automatic tap ratio adjustment was enabled during the last power flow solution activity, and the off-nominal turns ratio is within its limits.
DC	The from bus is the Winding 1 side of this two-winding transformer which is regulating alpha, gamma, or dc voltage of a dc line; its adjustment control mode is set to +4; dc tap adjustment was enabled during the last power flow solution activity; and the off-nominal turns ratio is within its limits.
LK	The from bus is connected to the Winding 1 side of this two-winding transformer or to any winding of this three-winding transformer, and either this transformer winding is not a tap changing transformer winding, its adjustment control mode is to zero or negative, or the automatic tap ratio adjustment was disabled during the last power flow solution activity.

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UN	The from bus is not the Winding 1 side of this two-winding transformer.

Except for the tags DC and UN, phase shifters are handled in a similar manner. For single section two-winding transformers with non-zero phase shift angle on the Winding 1 side, no indication of phase shift is printed in the output block of the Winding 2 side bus.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the far end to bus (rather than the closest dummy bus) of each multi-section line connected to the from bus is shown as its to bus. Multi-section lines are identified with an ampersand (&) as the first character of their line identifiers in the branch circuit identifier column (e.g., &1). Transformer information is shown only if the line section adjacent to the from bus is a transformer branch, and the data applies to the winding adjacent to the from bus.

If any of the non-transformer branches reported have a non-zero line shunt at the from bus end, or if Winding 1 of any transformer branch connected to the from bus has non-zero magnetizing admittance, and the line shunt reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the branch output lines are followed by lines reporting powers corresponding to each of these admittances. Note that, regardless of the setting of this option, the branch flows reported always include these components.

12.3.7. Fault MVA

For buses where voltage magnitude is less than 10% (such as faulted buses), the quantity FAULT MVA is printed. This quantity is intended for use following the solution of a balanced fault with activity TYSL. It is computed using the fixed bus shunt and the bus voltage; as such, it is normally the sum of fault MVA and the MVA consumed by the fixed bus shunt, if any, on the bus.

Under three phase fault conditions, the solved voltage magnitude at the faulted bus, and usually at nearby buses, is very small (e.g., 10^{-6} pu at the faulted bus). Hence, the voltage correction applied during the final TYSL iteration may be a large fraction of the solved voltage. Therefore, to ensure an accurate reporting of FAULT MVA, the acceleration factor applied during the final TYSL iteration should be 1.0. If the solution requires deceleration to achieve convergence, then, after it converges, activity TYSL should be repeated one final time with the acceleration factor reset to unity.

Additional Information
PSS®E GUI Users Guide, Wide-Format Power Flow Solution Report *PSS®E* Command Line Interface (CLI) *Users Guide*, Wide-Format Power Flow Solution Report

12.4. Producing a Wide-Format Power Flow Solution Report in Amps

LAMP

Run Activity LAMP - GUI
Power Flow > Reports > Bus based reports... *[Bus Based Reports]* OPTION: Powerflow output / Wide format output and Output with amps both checked

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Run Line Mode Activity LAMP - CLI
ACTIVITY? >>**LAMP** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The power flow output activity LAMP prints power flow solution output for the working case. The data tabulated for each bus printed includes bus voltage and phase angle, generator power output, loads, bus shunt elements, flows into all connected branches, FACTS devices, and GNE devices, and branch current loadings in amps. Bus mismatch is printed for any bus for which it exceeds 0.5 MVA or kVA, according to the power output option in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). For buses where voltage magnitude is less than 10% (such as faulted buses), the quantity FAULT MVA is printed (refer to Fault MVA).

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. The output report is generated with specified buses ordered according to the user option, in ascending bus number order (using the numbers output option) or alphabetical order (using the names output option) (refer to Section 4.9, "Subsystem Reporting").

The star point buses of three-winding transformers (refer to Transformer Data) are not reported. When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the interior dummy buses of multi-section line groupings (refer to Multi-Section Line Grouping Data) are not reported.

The output listing of activity LAMP may be terminated by entering the AB interrupt control code.

Activity LAMP formats its report in traditional power flow output style, with bus quantities on the left side of the page and branch data shown on the right side.

12.4.1. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the "Report Tables" section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "BUS LAMP".

12.4.2. Bus Quantities

The bus identifiers and boundary conditions for each bus reported in activity LAMP are printed on two lines on the left side of the bus' output block. The bus identifiers of bus number, name, and base voltage are listed followed by the number of the area in which the bus is located and the zone number directly below it on the second line.

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The bus voltage (in pu) and phase angle (in degrees) are printed; if the bus base voltage is specified for the bus, the bus voltage in kV is printed directly below the pu voltage on the second line. The generation, load, and shunt at the bus are printed, with active power shown on the first line and reactive power on the second. The plant reactive power output is followed by a single-character flag, where H indicates that the plant reactive power output is at or beyond the total plant upper reactive power limit; L indicates that it is at or below the total plant lower reactive power limit; and R indicates that it is within limits. Note that only plant conditions are tabulated in activity LAMP. If more than one machine is modeled at the plant, individual machine conditions may be examined using activities such as LIST, GEOL, and GCAP (refer to also Generator Unit Data).

The quantities printed as load at the bus are the sum of any constant MVA, current, and admittance loads at the bus; the shunt is the sum of any fixed and switched shunts. If more than one load and/or shunt are modeled at the bus, individual load and shunt quantities may be examined using activities such as LIST and EXAM (refer to Generator Unit Data).

12.4.3. FACTS Devices

Output for FACTS devices appears on the right side of the bus output block as the first line or lines of branch data (i.e., before output for GNE devices, dc lines, and ac branches). FACTS devices connected to the bus are reported in FACTS device name alphabetical order. If the out-of-service branch reporting program option setting is disabled, only in-service FACTS devices are reported. If this option is enabled, out-of-service FACTS devices at in-service buses are also reported; an X is displayed in the various loading and series voltage fields (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

For each FACTS device for which the bus being reported is the sending bus of an IPFC master or slave device, the following data pertaining to the device's active power transfer to its companion IPFC device is printed:

- An identifier indicating the type of the companion IPFC device (IPFC MASTER or IPFC SLAVE).
- The identifier IPF; this is printed in the circuit identifier column.
- The active power flowing to the companion IPFC device.
- The FACTS device name printed in the transformer ratio and angle columns.

Except for IPFC devices for which the shunt element is 0.0, for each FACTS device for which the bus being reported is the sending bus, the following data pertaining to the device's shunt element is printed:

- An identifier indicating the type of FACTS device shunt element and/or active power transfer being reported (STATCON SHUNT, SSSC SHUNT, UPFC SHUNT, or IPFC SHUNT).
- An identifier indicating the type of FACTS device being reported: STA (for a Statcon or Statcom), SSS (for an SSSC), UPF (for a UPFC) or IPF (for an IPFC); this is printed in the circuit identifier column.
- Unless the device being reported is an IPFC, the active power being exchanged between the shunt and series element.
- The reactive power being absorbed by the shunt element.
- The FACTS device name printed in the transformer ratio and angle columns.
- The total shunt current magnitude in amperes if the from bus has its base voltage specified.
- The rating in amperes and percent loading of the shunt element. If no from bus base voltage was specified, these columns are left blank.

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If the FACTS device contains a series element, the following data pertaining to it is printed in the output blocks of its sending and terminal buses:

- The number, name, base voltage, and area of the other endpoint bus (to bus).
- An identifier indicating the type of FACTS device being reported: SSS (for an SSSC), UPF (for a UPFC) or IPF (for an IPFC); this is printed in the circuit identifier column.
- The power flowing into the series element at the from bus end.
- The magnitude and phase of the series voltage. If the series voltage magnitude is being held at its limit (control modes 1, 5 and 6) or if the series voltage magnitude exceeds VSMX (control modes 2, 3, 4, 7 or 8), its value is followed by the tag H. Otherwise, the tag R is printed.
- The total current loading magnitude in amperes if the from bus has its base voltage specified.
- The rating in amperes and percent loading of the series element. If no from bus base voltage or series current limit was specified, these columns are left blank.

12.4.4. GNE Devices

Output for GNE devices appears on the right side of the bus output block immediately following any FACTS device output described above, and before output for dc lines and ac branches. GNE devices connected to the bus are reported in GNE device name alphabetical order. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service GNE devices are reported. If this option is enabled, out-of-service GNE devices at in-service buses are also reported; an X is displayed in the various loading fields.

For each single terminal GNE device reported, the following data is printed:

- An identifier indicating the GNE device type (VAR PQ, VAR I or VAR Y).
- The identifier GNE; this is printed in the circuit identifier column.
- The power flowing into the GNE device.
- The GNE device name.
- The total current loading magnitude in amperes if the from bus has its base voltage specified.

For each series GNE device reported, the following data is printed:

- The number, name, base voltage, and area of the other endpoint bus (to bus).
- The identifier GNE; this is printed in the circuit identifier column.
- The power flowing into the GNE device.
- The GNE device name.
- The total current loading magnitude in amperes if the from bus has its base voltage specified.

12.4.5. DC Lines

Output for dc transmission lines appears on the right side of the bus output block immediately following any GNE device output described above, and before output for ac branches. Two-terminal, multi-terminal,

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and VSC dc lines connected to the bus are reported in dc line name alphabetical order, with output for two-terminal lines printed first, followed by output for multi-terminal lines, followed by output for VSC dc lines. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service dc lines are reported. If this option setting is enabled, out-of-service dc lines at in-service buses are also reported; an X is displayed in the various loading fields and other fields indicating dc line conditions.

The power flowing into a dc line is the apparent ac system complex load as seen at the converter bus. Refer to Sections DC Transmission and Multiterminal DC networks of the *PSS®E Program Application Guide* for additional details on dc line quantities.

The following data is printed in the output block of each two-terminal and multi-terminal dc line converter bus:

- For two-terminal lines, the number, name, and base voltage of the other converter terminal bus (to bus), followed by the number of the area in which it is located; for multi-terminal lines, the number and name of the dc bus to which the ac converter bus is connected, with the identifier (DC) in the base voltage column, followed by the number of the area in which the dc bus to which the ac converter bus is connected is located.
- An identifier indicating the type of dc line being reported: 2D (for two-terminal) or MD (for multi-terminal), and an identifier indicating the type of converter at the from bus: R (for a rectifier) or I (for an inverter); this is printed in the circuit identifier column.
- The ac power flowing into the dc line.
- The converter transformer off-nominal turns ratio, followed by a two-character tag, which is either HI or LO if the ratio is at its high or low limit, respectively; LK if dc taps were locked during the last power flow solution activity, or if an ac transformer (rather than this converter transformer) is controlling a dc line quantity; or RG if the ratio is in regulating range.
- The value of converter angle followed by one of the tags RG, HI or LO if the angle is within its limits, at or above its nominal high limit, or at or below its steady-state low limit, respectively.
- The ac current magnitude in amperes flowing into the dc line, if the from bus has its base voltage specified.
- The dc line name.

The following data is printed in the output block of each VSC dc line converter bus:

- The number, name, and base voltage of the other converter terminal bus (to bus), followed by the number of the area in which it is located.
- The identifier VSC; this is printed in the circuit identifier column.
- The ac power flowing into the dc converter.
- The VSC dc line name printed in the transformer ratio and angle columns.
- The ac current magnitude in amperes flowing into the dc converter, if the from bus has its base voltage specified; if no base voltage is specified, this column is left blank.
- The percent of MVA rating, where the rating is taken as the more restrictive of IMAX converted to MVA and SMAX (refer to Voltage Source Converter (VSC) DC Transmission Line Data). If the from bus has its base

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voltage specified, the rating in amperes is listed followed by an A; if no base voltage is specified, these columns are left blank.

Figure 11-5 contains partial output from activity LAMP for a system in which bus 201 is the rectifier bus of a two-terminal dc line and bus 161 is the inverter bus. The output is shown with the numbers output option in effect. The system condition in this example is identical to that shown in the example output of activity POUT in Figure 12.1, "Example Output From Activity POUT", and of activity LOUT in Figure 12.4, "Example Output From Activity LOUT".

```
PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(tm)E THU, JUN 28 2007 14:20 ADDITION OF DC LINE TO
BASE CASE DEMO LOAD FLOW %MVA FOR TRANSFORMERS FOR PSS®E MANUAL % I FOR NON-TRANSFORMER
BRANCHES X------ FROM BUS ------X AREA VOLT GEN LOAD SHUNT X--------- TO BUS ----------X TRANSFORMER
RATING BUS# X-- NAME --X BASKV ZONE PU/KV ANGLE MW/MVAR MW/MVAR MW/MVAR BUS# X-- NAME --X
BASKV AREA CKT MW MVAR RATIO ANGLE AMPS % SET A 201 HYDRO 500.00 2 0.9913 2.5 600.0 0.0 0.0
---------------------------------------------------------------------------- 2 495.65 400.0H 0.0 294.8 161 INVJCT 18.500
1 2DR 641.8 286.2 1.000RG 10.9RG 818 "1" 151 NUCPANT 500.00 1 1 -724.8 -51.1 846 61 1386A 202
EAST500 500.00 2 1 315.1 -142.3 403 29 1386A 204 SUB500 500.00 2 1 367.9 12.5 429 31 1386A 161
INVJCT 18.500 1 0.9730 -1.6 0.0 0.0 0.0 ---------------------------------------------------------------------------- 1 18.000
0.0 0.0 0.0 201 HYDRO 500.00 2 2DI -600.0 281.9 0.987RG 19.1RG 21263 "1" 154 DOWNTN 230.00 1 1
600.0 78.4 1.000UN 19409 101 600M 162 INVTER 18.500 1 1 0.0 -360.3 11556
```

Figure 12.5. Example Output From Activity LAMP

12.4.6. Branch Quantities

Following the output described above, flows for each ac branch connected to the bus are tabulated on the right side of the bus output block. If the out-of-service branch reporting program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) is disabled, only in-service branches are also reported. If this option setting is enabled, out-of-service branches at in-service buses are reported; an X is displayed in the loading fields.

Each branch output line for non-transformer branches and for two-winding transformers contains the to bus number, name, and base voltage, followed by the number of the area in which the to bus is located and the branch circuit identifier. For three-winding transformers, the output line contains the string 3WNDTR in the bus number column, the transformer name in the bus name column, and the winding number in the base voltage column, followed by the transformer circuit identifier; if the three-winding transformer to bus reporting option is enabled, two additional lines, containing the bus numbers, names, and base voltages of the two buses connected to the from bus by the three-winding transformer, are printed, followed by the number of the area in which the to bus is located.

The active and reactive power flow on the branch is printed as power leaving the from bus (i.e., positive for outgoing and negative for incoming).

If the from bus has its base voltage specified, the current loading at the from bus end of the line is listed in amperes; if no base voltage is specified, this column is left blank.

The percent loading, based on the rating set established as the default rating set program option setting and the corresponding rating are printed on the far right-hand side of the line. For transformers, either percent current or percent MVA loading is reported, according to the transformer percent loading units program option setting. Similarly, for non-transformer branches, either percent current or percent MVA loading is reported, according to the non-transformer branch percent loading units program option setting. For branches with percent current tabulated, the rating is reported in amperes followed by an A as long as the correspond-

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ing branch rating and the from bus base voltage are specified; otherwise, the percent loading and rating columns are blank. For branches with percent MVA tabulated, the rating is printed as is followed by either an M (for MVA) or K (for kVA), depending upon the PSS®E power output option setting. A rating may have been entered as either an MVA rating or a current rating expressed as MVA (refer to Case Identification Data, Generator Data and Non-Transformer Branch Data), but the percent loading calculation always assumes the units consistent with the appropriate percent loading units program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

For transformers, the turns ratio of the winding connected to the from bus is printed. A two-character tag is printed adjacent to the ratio which has the following significance:

HI	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, and the off-nominal turns ratio is at or beyond its high limit.
LO	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, and the off-nominal turns ratio is at or below its low limit.
RG	The from bus is connected to the Winding 1 side of this regulating two-winding transformer or to a regulating winding of this three-winding transformer, its adjustment control mode is set to +1 or +2, the automatic tap ratio adjustment was enabled during the last power flow solution activity, and the off-nominal turns ratio is within its limits.
DC	The from bus is the Winding 1 side of this two-winding transformer which is regulating alpha, gamma, or dc voltage of a dc line; its adjustment control mode is set to +4; dc tap adjustment was enabled during the last power flow solution activity; and the off-nominal turns ratio is within its limits.
LK	The from bus is connected to the Winding 1 side of this two-winding transformer or to any winding of this three-winding transformer, and either this transformer winding is not a tap changing transformer winding, its adjustment control mode is to zero or negative, or the automatic tap ratio adjustment was disabled during the last power flow solution activity.
UN	The from bus is not the Winding 1 side of this two-winding transformer.

Except for the tags DC and UN, phase shifters are handled in a similar manner. For single section two-winding transformers with non-zero phase shift angle on the Winding 1 side, no indication of phase shift is printed in the output block of the Winding 2 side bus.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the far end to bus (rather than the closest dummy bus) of each multi-section line connected to the from bus is shown as its to bus. Multi-section lines are identified with an ampersand

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(&) as the first character of their line identifiers in the branch circuit identifier column (e.g., &1). Transformer information is shown only if the line section adjacent to the from bus is a transformer branch, and the data applies to the winding adjacent to the from bus.

If any of the non-transformer branches reported have a non-zero line shunt at the from bus end, or if Winding 1 of any transformer branch connected to the from bus has non-zero magnetizing admittance, and the line shunt reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the branch output lines are followed by lines reporting powers corresponding to each of these admittances. Current magnitudes in amperes are also printed if the from bus has a non-zero base voltage specified for it. Note that, regardless of the setting of this option, the branch flows reported always include these components.

12.4.7. Fault MVA

For buses where voltage magnitude is less than 10% (such as faulted buses), the quantity FAULT MVA is printed. This quantity is intended for use following the solution of a balanced fault with activity TYSL. It is computed using the fixed bus shunt and the bus voltage; as such, it is normally the sum of fault MVA and the MVA consumed by the fixed bus shunt, if any, on the bus.

Under three phase fault conditions, the solved voltage magnitude at the faulted bus, and usually at nearby buses, is very small (e.g., 10^{-6} pu at the faulted bus). Hence, the voltage correction applied during the final TYSL iteration may be a large fraction of the solved voltage. Therefore, to ensure an accurate reporting of FAULT MVA, the acceleration factor applied during the final TYSL iteration should be 1.0. If the solution requires deceleration to achieve convergence, then, after it converges, activity TYSL should be repeated one final time with the acceleration factor reset to unity.

Additional Information
PSS®E GUI Users Guide, Wide-Format Power Flow Solution Report with Branch Current Loadings *PSS®E* Command Line Interface (CLI) *Users Guide*, Producing a Wide Format Power Flow Solution Report with Branch Current Loadings

12.5. Summarizing Subsystem Conditions

SUBS

Run Activity SUBS - GUI
Power Flow > Reports > Bus based reports... *[Bus Based Reports]* Option: Subsystem summary / totals
Run Line Mode Activity SUBS - CLI
ACTIVITY? >>**SUBS** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The subsystem reporting activity SUBS summarizes conditions in the working case by tabulating the following for a specified subsystem:

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- Conditions at each system swing (Type 3) bus.
- Conditions at each area slack generator bus.
- The number of components.
- Generation, load, shunt and charging totals.
- Loss, line shunt, and charging totals by voltage level.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. The area, zone, and/or owner assignments of buses are used to determine which loads, machines, and branches are to be processed when subsystem selection by area, zone, and/or owner is enabled. The owner assignments of machines, ac branches, VSC dc lines, FACTS devices, and GNE devices, and the area, zone, and owner assignments of loads, are not considered.

When processing a subsystem, if the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the dummy buses of a multi-section line grouping (and hence the members of the grouping) are treated as part of the specified subsystem if and only if the nonmetered endpoint bus of the multi-section line grouping is in the specified subsystem. Therefore, even if such a dummy bus is specifically specified, it is not included in the subsystem unless the nonmetered endpoint bus is also designated as a member of the subsystem. When the multi-section line reporting option is disabled, the multi-section line grouping definitions and their metered end information are ignored.

In calculating branch totals by voltage level, activity SUBS prints an alarm at the dialog output device whenever it encounters a non-transformer branch in the specified subsystem with different bus base voltages at the two ends of the line. Such branches are handled as described in Branch Totals by Voltage Level. If the report of activity SUBS is being directed to the user's terminal, these messages are printed between the area slack bus summary and the subsystem component totals. The tabulation of these error messages may be suppressed by entering the AB interrupt control code.

Following the two-line case heading, the output report of activity SUBS contains a description of the subsystem where conditions are being summarized. This is followed by the five categories of data described in the following paragraphs.

12.5.1. System Swing Bus Summary

Each Type 3 bus in the subsystem being reported is listed, one per line. Swing buses are tabulated in ascending numerical order when the numbers output option is in effect and in alphabetical order using the names output option. The bus number, name, and base voltage are printed, along with the number and name of the interchange area in which it is located and the number and name of the zone to which it is assigned. The total plant active and reactive power output are printed followed by the total plant MBASE (i.e., sum of the MBASEs of the in-service machines at the plant).

12.5.2. Area Slack Bus Summary

Those buses which are area slack buses for area interchange control purposes are tabulated one per line in ascending area number order. The area number and name are printed, followed by the area slack bus number, name, and base voltage and the number and name of the zone in which it is located. Total plant active and reactive power output and total plant MBASE are tabulated. When processing a subsystem, only those area slack buses which are included in the specified subsystem are reported.

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12.5.3. Component Totals

The total number of in-service components in the subsystem is listed in the following categories:

- Buses.
- Plants.
- Machines, including wind machines.
- Fixed shunts.
- Switched shunts.
- Loads.
- AC branches, including transformers.
- Two-winding transformer branches.
- DC lines.
- FACTS devices.
- GNE devices.

The total number of plants is calculated as the number of Type 2 and 3 buses in the subsystem, and the number of machines is the number of in-service machines at these buses. The total number of loads is calculated as the number of in-service loads at Type 1, 2, and 3 buses in the subsystem. The totals of ac branches, transformers, and dc lines include those ties from the specified subsystem to other subsystems in the working case with a nonmetered end bus is in the specified subsystem; the total number of FACTS devices are those where the sending bus is in the specified subsystem; the total number of GNE devices are those where the non-metered bus is in the specified subsystem (refer to Subsystem Totals and Branch Totals by Voltage Level).

The total number of buses does not include the star point bus of any three-winding transformer where the nonmetered end bus is in the specified subsystem. Each three-winding transformer where the nonmetered end bus is in the specified subsystem contributes one to the totals of ac branches and transformers; the three two-winding transformers comprising each such three-winding transformer do not contribute to these totals.

When subsystem specification by area, zone, and/or owner is in effect, both the number of loads and the subsystem load total (refer to Subsystem Totals) indicate the in-service load connected to Type 1, 2, and 3 buses within the subsystem. These totals include loads assigned to other areas, zones, or owners (refer to Load Data), but which are attached to subsystem buses, and exclude loads where area, zone, and/or owner assignments fall within the specified areas, zones, and/or owners, but which are attached to buses outside of the subsystem.

When subsystem specification by owner is in effect, machine, branch, FACTS device, and GNE device totals, as well as VSC dc lines contributing to dc line totals, are handled in a similar manner.

12.5.4. Subsystem Totals

Constant MVA, current and admittance load totals, and bus shunt, FACTS device shunt, single terminal GNE device, line shunt, and charging totals are tabulated, both in terms of nominal values (i.e., at unity voltage)

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and actual values (i.e., at the present voltage profile). Generation totals include swing bus power outputs. Line shunt totals include transformer magnetizing admittance contributions.

12.5.5. Branch Totals by Voltage Level

>

Subsystem losses, line shunt, and line charging are tabulated by voltage level with voltage levels listed in descending order. Line shunt totals include transformer magnetizing admittance contributions. If any series FACTS devices, GNE devices, or dc lines are in the subsystem being summarized, their losses are listed first in the report.

Losses on a series FACTS device (i.e., one with a non-zero terminal bus specified; refer to FACTS Device Data) are defined as the algebraic sum of:

(j QSEND) + (j QTERM)

Similarly, losses on a series GNE device (i.e., one connected to more than one terminal bus) are defined as the algebraic sum of the powers flowing into the device at each of the buses to which it is connected.

Losses on a two-terminal dc line are defined as the algebraic sum of:

(PACR + jQACR) + (PACI + jQACI)

Similarly, losses on a voltage source converter (VSC) dc line are calculated as:

$- (PAC_{cnv1} + jQAC_{cnv1}) - (PAC_{cnv2} + jQAC_{cnv2})$

When a designated subsystem is being tabulated and a two-terminal or VSC dc line is a tie branch to another subsystem, the dc line losses are included in the dc line loss total if the nonmetered end converter bus is in the subsystem being reported.

For multi-terminal dc lines, losses between an ac converter bus and the dc bus(es) to which it is connected are included if the ac bus is in the specified subsystem (i.e., metering at the dc bus is assumed). Losses on a dc link of a multi-terminal line are included in the loss totals when activity SUBS processes all buses in the working case. When a subsystem of the working case is being reported, losses on an individual dc link are included in the dc loss totals unless:

- The subsystem is specified by area and the nonmetered end dc bus is not in one of the specified areas.
- The subsystem is specified by zone and the nonmetered end dc bus is not in one of the specified zones.
- The subsystem is specified by owner and the nonmetered end dc bus is not assigned to one of the specified owners.
- The subsystem is specified by base voltage and the low end of the voltage range is greater than zero.
- The subsystem is specified by bus and at least one converter bus of the multi-terminal line is not in the designated subsystem.

Similarly, losses through the grounding resistance of the second dc bus of the converters of a multi-terminal line are included in the loss totals when activity SUBS processes all buses in the working case. When a subsystem of the working case is being reported, losses through an individual IDC2 ground path are included or omitted using criteria corresponding to that given for dc links above.

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The ac branch losses are taken as I^2R and I^2X losses and *exclude* the line charging and line connected shunt components. Transformer losses *exclude* magnetizing admittance components.

When a designated subsystem is being tabulated, the losses, line shunt, and charging components on any tie line to another subsystem are included in the branch totals if the nonmetered end bus is in the subsystem being reported (i.e., subsystem boundaries are at the metered end of each tie branch). For series FACTS devices, terminal end metering is assumed.

Two- and three-winding transformers have their loss and magnetizing admittance contributions assigned to the voltage level of the Winding 2 side bus. If the second winding of a three-winding transformer is out-of-service with the remaining windings in-service (i.e., its status value is 2), its loss and line shunt contributions are assigned to the voltage level of the Winding 3 side bus.

Non-transformer branches with different base voltages at the two ends of the line generate an alarm, and their contributions are assigned to the lower of the two base voltages. If a base voltage is specified at only one end of such a branch, its contributions are assigned to that voltage level.

Be careful not to confuse the roles of the nonmetered end bus and the Winding 2 side bus of transformers. The nonmetered end bus of all branches, including transformers, is used to determine if the branch is included in the subsystem being reported. If a transformer is in the subsystem, the voltage level of the Winding 2 side bus determines the voltage level totals to which the transformer contributes.

Additional Information
PSS®E GUI Users Guide, Summary of Subsystem Conditions *PSS®E* Command Line Interface (CLI) *Users Guide*, Summary of Subsystem Conditions

12.6. Summarizing Load Reduction

LODR

Run Activity LODR - GUI
Power Flow > Reports > Bus based reports... *[Bus Based Reports]* Option: Load reduction output
Interrupt Control Codes
AB

The load reduction reporting activity LODR summarizes the reduction in:

- Nominal constant MVA load caused when voltage at a load bus falls below PQBRAK.
- Nominal constant current load caused when voltage at a load bus falls below 0.5 pu.

The LOAD_REDUCTION API routine includes the following among its inputs:

- A designation of the subsystem in which reduced loads are to be reported; refer to Section 4.8, "Subsystem Selection".
- A subsystem selection option when area, owner and/or zone are included among the subsystem selection criteria. To determine subsystem loads:

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- Use the area, owner and/or zone assignments of the bus to which the load is connected.
- Use the area, owner and/or zone assignments of the load.

There is no customized question and answer dialog in line mode for activity LODR; in line mode, activity LODR can be accessed through the BAT_LOAD_REDUCTION command.

12.6.1. Operation of Activity LODR

When voltage at a bus falls below the constant power characteristic threshold PQBRAK, any constant power load at the bus is reduced as shown in ??? so that, at a voltage of 0.0, the current drawn by the load is 0.0. Similarly, when voltage at a bus falls below 0.5 pu, any constant current load at the bus is reduced as shown in ??? so that, at a voltage of 0.0, the current drawn by the load is 0.0.

For any subsystem load for which such a reduction is applied, activity LODR tabulates:

- The voltage at the bus.
- If the load has a non-zero constant power component:
- The specified complex constant power load.
- The complex constant power load at the present bus voltage.
- The complex constant power load reduction.
- If the load has a non-zero constant current component:
- The specified complex nominal constant current load.
- The complex nominal constant current load at the present bus voltage.
- The complex nominal constant current load reduction.

Totals of reported quantities are tabulated at the end of each subsystem grouping. An example is provided in Figure 12.6, “Example of Load Reduction Report”.

12.6.2. Application Notes

Note that values listed as constant current load are nominal values (i.e., at unity voltage, as shown in activities such as LIST and in the *[Spreadsheet]* View). Thus, the values shown as actual constant current loads are not identical to those reported in the various power flow solution reporting activities that report loadings in MVA. Clearly, to obtain the MVA corresponding to the reduced current loadings, multiply the complex value shown as actual nominal constant current load by the voltage magnitude at the bus.

PQBRAK may be modified using the data changing activity CHNG or the *[Solution Parameters]* dialog.

Additional Information
PSS®E GUI Users Guide, *Summarizing Load Reduction* *PSS®E Application Program Interface (API)*, LOAD_REDUCTION See also: Constant Power Load Characteristic Constant Current Load Characteristic Section 4.8, “Subsystem Selection” Section 4.9, “Subsystem Reporting”

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```
     PTI
INTERACTIVE POWER SYSTEM SIMULATOR--PSSÆE      TUE, FEB 24 2009
  9:40
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA
LOAD REDUCTION FOR VOLTAGE < PQBRAK=0.700 (MVA
LOAD) & 0.5 (CURRENT LOAD)
                          I         X--------------CONSTANT
MVA LOAD--------------X X--------NOMINAL CONSTANT CURRENT LOAD--------X
  BUS# X-- NAME --X BASKV D VOLTAGE     SCHEDULED
        ACTUAL         REDUCTION       SCHEDULED         ACTUAL
      REDUCTION
   153 MID230       230.00 1  0.3361    60.0    30.0
   30.5    15.2     29.5    14.8    61.1    30.6    56.1    28.1
    5.0     2.5
   154 DOWNTN       230.00 1  0.0000   180.0   135.0
    0.0     0.0   180.0   135.0   195.4   146.6     0.0     0.0
  195.4   146.6
   154 DOWNTN       230.00 2  0.0000   120.0   105.0
    0.0     0.0   120.0   105.0   130.3   114.0     0.0     0.0
  130.3   114.0
   203 EAST230      230.00 1  0.2492    90.0    45.0
   30.0    15.0     60.0    30.0    94.3    47.2    75.6    37.8
   18.7     9.4
   205 SUB230       230.00 1  0.1373   360.0   210.0
   47.2    27.6   312.8   182.4   386.3   225.3   214.8   125.3
  171.5   100.1
  3005 WEST         230.00 1  0.4198    30.0    15.0
   20.2    10.1     9.8     4.9    30.1    15.1    29.5    14.8
    0.6     0.3
  3007 RURAL        230.00 1  0.3116    60.0    22.5
   27.5    10.3     32.5    12.2    62.4    23.4    55.6    20.8
    6.8     2.6
  3008 CATDOG       230.00 1  0.2262    60.0    22.5
   17.3     6.5     42.7    16.0    63.0    23.6    48.0    18.0
   15.1     5.7
TOTALS                                960.0   585.0
  172.7    84.7   787.3   500.3  1023.0   625.7   479.5   244.7
  543.5   381.0
```

Figure 12.6. Example of Load Reduction Report

12.6.3. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

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Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "BUS LORD".

12.7. Summarizing Area Totals

AREA

Run Activity AREA - GUI
Power Flow > Reports > Area / owner / zone totals... *[Area / Owner / Zone Totals]* OPTION: Report on Areas
Run Line Mode Activity AREA - CLI
ACTIVITY? >>**AREA** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The area summary activity AREA provides a tabulation by interchange area of the desired area net interchange, along with area totals of:

• Generation • Load • Induction machine powers • Bus connected shunt elements, including fixed shunts, switched shunts and shunt elements of FACTS devices	• Line connected shunt elements of non-transformer ac branches • Transformer magnetizing powers • Line charging • Net interchange • Losses

The AREA_2 API routine tabulates totals by area. The output block for each area and for system totals consists of two lines: the first contains active power totals and the second the reactive power totals.

The AREA_ZONE API routine tabulates totals by area along with subtotals by zone for each reported area. The output block for each area consists of one line for each zone in the area, followed by one line of area totals.

In both of these reports, the area number and name is printed for each area block reported.

The user specifies the areas to be reported (refer to Section 4.8, “Subsystem Selection”). Totals for the specified areas are printed in ascending area number order, followed by the sums of the totals of those areas.

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Recall that buses, loads and induction machines all have area assignments associated with them (refer to Bus Data, Load Data and Induction Machine Data). In the AREA_2 API routine, one of three load and induction machine reporting options may be selected for calculating each area's total load and induction machine powers:

- Calculate the area totals by summing all in-service loads and induction machines connected to buses that are assigned to the area; the area assignments of the loads and induction machines are not considered. A single set of totals is tabulated in the induction generation, induction motor, and load columns labeled `AT AREA BUSES`.
- Calculate the area totals by summing all in-service loads and induction machines that are assigned to the area; the area assignments of the buses to which they are connected is not considered. A single set of totals is tabulated in the induction generation, induction motor, and load columns labeled `ASSIGNED TO THE AREA`.
- Calculate three subtotals of area load and induction machine powers:
- The sums of all in-service loads and induction machine powers for which both the load or induction machine and the bus to which it is connected are assigned to the area. These are tabulated in the columns labeled `AREA ITEMS AT AREA BUSES`.
- The sums of all in-service loads and induction machine powers for which the load or induction machine is assigned to the area but the bus to which it is connected is assigned to a different area. These are tabulated in the columns labeled `AREA ITEMS AT NON-AREA BUSES`.
- The sums of all in-service loads and induction machines for which the bus to which it is connected is assigned to the area but the load or induction machine is assigned to a different area. These are tabulated in the columns labeled `NON-AREA ITEMS AT AREA BUSES`.

  The output block for `COLUMN TOTALS` is blank for the three sets of load and induction machine columns. Rather, load and induction machine totals for the reported areas are tabulated following this output block. Two complex totals are given for loads, induction generators, and induction motors:

- `TOTAL ASSIGNED TO THESE AREAS`, which are the sums of the columns in the first two sets of totals described above.
- `TOTAL CONNECTED TO BUSES IN THESE AREAS`, which are the sums of the columns in the first and third sets of totals described above.

  Clearly, if all areas are being reported by the AREA_2 API routine, these two sets of totals are identical.

In the AREA_ZONE API routine, an individual in-service bus load is included in the load totals of the area to which the *load* is assigned; this area need not be the same area in which the bus at which the load is connected is located. Similarly, an in-service induction machine is included in the induction machine totals of the area to which it assigned rather than that of the bus to which it is connected.

Line losses are taken as I^2R and I^2X losses and *exclude* the line charging and line connected shunt components; transformer losses *exclude* magnetizing admittance components. Series FACTS device losses are calculated as the algebraic sum of the reactive power flowing into the series device at its endpoint buses. Multiple terminal GNE device losses are calculated as the algebraic sum of the powers flowing into the device at its endpoint buses.

Losses on two-terminal dc transmission lines and on branches within multi-terminal dc lines are assigned to the area of the nonmetered end. For the connections between each multi-terminal dc line ac converter bus

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and the dc bus to which it is connected, metering on the dc bus side is assumed (i.e., losses are assigned to the area of the ac bus). Losses on a voltage source converter (VSC) dc line are calculated as the algebraic sum of the active and reactive powers flowing into the line at its endpoint buses. They are assigned to the area of the voltage controlling converter when both converters are in-service; if only one converter is in-service, losses are assigned to the area in which it is located. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

An area has positive net interchange when it is selling or exporting power. Two net interchange totals are tabulated for each area that is reported:

1. The total power flowing into area tie lines.

2. The total power flowing into area tie lines plus contributions from or to loads where the area assignment of the load differs from that of the bus to which it is connected.

These two values of area net interchange correspond to the two net interchange definitions that may be specified for use by the area interchange control calculation of the power flow solution activities.

Line charging, line shunt, transformer magnetizing, net interchange, and loss totals are evaluated on the basis of the metered end designated for each tie branch; i.e., for each tie branch, these quantities are assigned to the area of the nonmetered end bus. Transformer magnetizing admittance components are excluded from line shunt totals. Tie line flows, and hence interchange, are calculated including the charging, line connected shunt, and transformer magnetizing admittance components.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the area assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, any generation, load, induction machine, and bus shunt present at assigned to the area of the nonmetered endpoint bus of the multi-section line grouping. Similarly, line charging, line shunt, net interchange and loss contributions are evaluated on the basis of the metered end designated for each multi-section line grouping which is a tie branch (i.e., its endpoint buses are located in different areas); for each multi-section tie, these quantities are assigned to the area of the nonmetered endpoint bus.

When the multi-section line reporting option is disabled, the area assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

A three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same area. Tie flows are derived from the flows into the transformer at the two metered ends.

A two-terminal dc transmission line is considered to be a tie line when the converter station buses are located in different areas. The tie flow is taken as the power flowing into the dc converter from the ac system at the metered end. For multi-terminal dc lines, if a converter bus and a dc bus to which it is connected are in different areas, the connection is treated as a tie line and metering on the dc bus side is assumed. DC links within a multi-terminal dc line are handled in the same manner as ac branches. A voltage source converter (VSC) dc transmission line is considered to be a tie line when the converter station buses are located in different areas. The tie flow is taken as the power flowing into the dc converter from the ac system at the power controlling converter end. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

A series FACTS device is considered to be a tie line when the sending and terminal buses are located in different areas. The tie flow is taken as the power flowing into the series element at the terminal bus end.

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A multiple terminal GNE device is considered to be a tie line when its connected buses are located in different areas. The tie flow is taken as the power flowing into the device at the metered end.

Activity AREA may be terminated by entering the AB interrupt control code.

The *savnw.sav* power flow case, which is provided in the Example subdirectory of the PSS®E installation, has three areas. For this power flow case, the Area totals are shown in Figure 11-7. The load reporting option that provides for three subtotals of area load is used in this example.

A brief examination of the results shows:

- The FLAPCO area is transmitting power to the other two areas although it is exporting more (278.7 MW) than scheduled (250 MW). This is because the Area Interchange control was not enabled in the *[Loadflow Solutions]* window.
- LIGHTCO area is receiving 130.8 MW compared to a desired interchange of 100 MW.
- The FLAPCO area is receiving 684.4 Mvar from the other two areas.
- Total Real Power losses are 58.6 MW

Additional Information
PSS®E GUI Users Guide, Area Totals *PSS®E* Command Line Interface (CLI) *Users Guide*, Area Totals

```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, APR 23 2019  13:54
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                          AREA TOTALS
BASE CASE INCLUDING SEQUENCE DATA                                   IN MW/MVAR
                    --AREA ITEMS AT-- --AREA ITEMS AT-- --NON-AREA ITEMS-
                  FROM ----AREA BUSES--- --NON-AREA BUSES- --AT AREA BUSES--              TO            TO XFRMR                    -NET INTERCHANGE-
                 GENE-   TO IND       TO   TO IND       TO   TO IND       TO   TO BUS  GNE BUS  TO LINE  MAGNE-       FROM     TO     TO TIE  TO TIES  DESIRED
X-- AREA --X  RATION   MACHNS     LOAD   MACHNS     LOAD   MACHNS     LOAD    SHUNT  DEVICES    SHUNT   TIZING CHARGING   LOSSES    LINES  + LOADS  NET INT

   1          1500.0      0.0    200.0      0.0      0.0      0.0   1000.0      0.0      0.0      0.0      0.0      0.0     21.3    278.7   1278.7    250.0
FLAPCO         162.6      0.0    100.0      0.0      0.0      0.0    800.0    349.7      0.0     -0.0     -0.0    841.2    438.5   -684.4    115.6

   2          1400.0      0.0   1500.0      0.0   1000.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0     30.8   -130.8  -1130.8   -100.0
LIGHTCO        617.3      0.0    850.0      0.0    800.0      0.0      0.0   -641.5      0.0     -0.0     -0.0    623.0    590.1    441.8   -358.2

   5           358.6      0.0    500.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0      6.6   -147.9   -147.9   -150.0
WORLD          183.6      0.0    200.0      0.0      0.0      0.0      0.0     -0.0      0.0     -0.0     -0.0    345.9     86.8    242.6    242.6

COLUMN        3258.6                                                          0.0      0.0      0.0      0.0      0.0     58.6      0.0      0.0      0.0
TOTALS         963.5                                                       -291.8      0.0      0.0      0.0   1810.1   1115.4      0.0      0.0

                                                                  X------- INDUCTION MACHINES ------X
                                                 X---- LOADS ----X X-- GENERATORS -X X---- MOTORS ---X
                                                      MW      MVAR      MW      MVAR      MW      MVAR
TOTAL ASSIGNED TO THESE AREAS                      3200.0   1950.0       0.0       0.0       0.0       0.0
TOTAL CONNECTED TO BUSES IN THESE AREAS            3200.0   1950.0       0.0       0.0       0.0       0.0
```

Figure 12.7. Area Total Results

12.7.1. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats.

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The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "LF Area".

12.8. Summarizing Owner Totals

OWNR

Run Activity OWNR - GUI
Power Flow > Reports > Area / owner / zone totals... *[Area / Owner / Zone Totals]* OPTION: Report on Owners
Run Line Mode Activity OWNR - CLI
ACTIVITY? >>**OWNR** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The owner summary activity OWNR provides a tabulation by owner of the following owner totals:

- Generation.
- Load.
- Induction machine generation.
- Induction motor powers.
- Bus connected shunt elements, including fixed shunts, switched shunts and shunt elements of FACTS devices.
- Power flow into single terminal GNE devices
- Line connected shunt elements on non-transformer ac branches.
- Transformer magnetizing powers.
- Line charging.
- Losses.

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The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Totals for the selected owners are printed in ascending owner number order, followed by the sums of the totals of those owners.

An individual in-service bus load is included in the load totals of the owner to which the load is assigned; this owner need not be the same owner in which the bus at which the load is connected is located (refer to Bus Data and Load Data). Similarly, generation, induction machine, ac branch, VSC dc line, FACTS device, and GNE device quantities are assigned to the designated conventional machine, induction machine, branch, VSC dc line, FACTS device, and GNE device owners, respectively.

Line losses are taken as I^2R and I^2X losses and *exclude* the line charging and line connected shunt components; transformer losses *exclude* magnetizing admittance components. Series FACTS device losses are calculated as the algebraic sum of the reactive power flowing into the series device at its endpoint buses. Multiple terminal GNE device losses are calculated as the algebraic sum of the powers flowing into the device at its endpoint buses.

Losses on two-terminal dc transmission lines and on branches within multi-terminal dc lines are assigned to the owner of the nonmetered end. For the connections between each multi-terminal dc line ac converter bus and the dc bus to which it is connected, metering on the dc bus side is assumed (i.e., losses are assigned to the owner of the ac bus). Losses on a voltage source converter (VSC) dc line are calculated as the algebraic sum of the active and reactive powers flowing into the line at its endpoint buses. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

Activity OWNR may be terminated by entering the AB interrupt control code.

The *savnw.sav* power flow case, has seven Owners. The Report on Owner totals are shown in Figure 12.8, "Owner Totals Results".

```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, APR 23 2019  14:14
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                          OWNER TOTALS
BASE CASE INCLUDING SEQUENCE DATA                                    IN MW/MVAR
          |X----- FROM -----X     TO                        TO           TO XFRMR
          |X-- GENERATION --X  INDUCTN      TO    TO BUS  GNE BUS  TO LINE  MAGNE-      FROM
X- OWNER --X  SYNCHRN  INDUCTN   MOTORS     LOAD    SHUNT  DEVICES    SHUNT   TIZING CHARGING   LOSSES

   1           499.9      0.0       0.0   800.0      0.0      0.0      0.0      0.0      0.0     26.3
TRAN 1          54.2      0.0       0.0   550.0    349.7      0.0     -0.0     -0.0   1270.7    506.6

   2           560.0      0.0       0.0  1500.0      0.0      0.0      0.0      0.0      0.0      6.2
TRAN 2         246.9      0.0       0.0   850.0   -317.0      0.0     -0.0     -0.0     14.7    264.6

   5           124.0      0.0       0.0   500.0      0.0      0.0      0.0      0.0      0.0      5.4
TRAN 5          67.4      0.0       0.0   200.0     -0.0      0.0     -0.0     -0.0     57.4     53.3

  11          1019.9      0.0       0.0     0.0      0.0      0.0      0.0      0.0      0.0      0.0
GEN 1          116.4      0.0       0.0     0.0     -0.0      0.0     -0.0     -0.0      0.0      0.0

  22           899.7      0.0       0.0     0.0      0.0      0.0      0.0      0.0      0.0     19.5
GEN 2          394.3      0.0       0.0     0.0   -324.5      0.0     -0.0     -0.0    464.9    257.8

  55           155.0      0.0       0.0     0.0      0.0      0.0      0.0      0.0      0.0      0.2
GEN 5           84.3      0.0       0.0     0.0     -0.0      0.0     -0.0     -0.0      0.0     24.6

 100             0.0      0.0       0.0   400.0      0.0      0.0      0.0      0.0      0.0      0.9
NO BUSES         0.0      0.0       0.0   350.0     -0.0      0.0     -0.0     -0.0      2.3      8.5

TOTALS        3258.6      0.0       0.0  3200.0      0.0      0.0      0.0      0.0      0.0     58.6
          |     963.5      0.0       0.0  1950.0   -291.8      0.0      0.0      0.0   1810.1   1115.4
```

Figure 12.8. Owner Totals Results

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A brief examination of the results shows:

- The seven Owners have ownership of all the generators in this network (3258 MW). This can be compared with the Area totals.
- All the shunts in the network, whether they be FACTS devices, switched shunts or bus shunts are assigned to three Owners (TRAN 1, TRAN 2 and GEN 2).
- Three Owners (GEN 1, GEN 2 and GEN 5) supply a total of 2,074.6 MW of generation but have no ownership of demand.
- Note that there is no Owner interchange control available.

Additional Information
PSS®E GUI Users Guide, Owner Totals *PSS®E* Command Line Interface (CLI) *Users Guide*, Owner Totals

12.8.1. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "LF Owner".

12.9. Summarizing Zone Totals

ZONE

Run Activity ZONE - GUI
Power Flow > Reports > Area / owner / zone totals... *[Area / Owner / Zone Totals]* OPTION: Report on Zones
Run Line Mode Activity ZONE - CLI
ACTIVITY? >>**ZONE** ENTER OUTPUT DEVICE CODE:

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Interrupt Control Codes
AB

The zone summary activity ZONE provides a tabulation by zones of the following zone totals:

• Generation • Load • Induction machine powers • Bus connected shunt elements, including fixed shunts, switched shunts and shunt elements of FACTS devices	• Line connected shunt elements of non-transformer ac branches • Transformer magnetizing powers • Line charging • Net interchange • Losses

The ZONE_2 API routine tabulates totals by zone. The output block for each zone and for system totals consists of two lines: the first contains active power totals and the second the reactive power totals.

The ZONE_AREA API routine tabulates totals by zone along with subtotals by area for each reported zone. The output block for each zone consists of one line for each area in the zone, followed by one line of zone totals.

In both of these reports, the zone number and name is printed for each zone block reported.

The user specifies the zones to be reported (refer to Section 4.8, "Subsystem Selection"). Totals for the specified zones are printed in ascending zone number order, followed by the sums of the totals of those zones.

Recall that buses, loads and induction machines all have zone assignments associated with them (refer to Bus Data, Load Data and Induction Machine Data). In the ZONE_2 API routine, one of three load and induction machine reporting options may be selected for calculating each zone's total load and induction machine powers:

- Calculate the zone totals by summing all in-service loads and induction machines connected to buses that are assigned to the zone; the zone assignments of the loads and induction machines are not considered. A single set of totals is tabulated in the induction generation, induction motor, and load columns labeled `AT ZONE BUSES`.

- Calculate the zone totals by summing all in-service loads and induction machines that are assigned to the zone; the zone assignments of the buses to which they are connected is not considered. A single set of totals is tabulated in the induction generation, induction motor, and load columns labeled `ASSIGNED TO THE ZONE`.

- Calculate three subtotals of zone load and induction machine powers:

- The sums of all in-service loads and induction machine powers for which both the load or induction machine and the bus to which it is connected are assigned to the zone. These are tabulated in the columns labeled `ZONE ITEMS AT ZONE BUSES`.

- The sums of all in-service loads and induction machine powers for which the load or induction machine is assigned to the zone but the bus to which it is connected is assigned to a different zone. These are tabulated in the columns labeled ZONE `ITEMS AT NON-ZONE BUSES`.

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- The sums of all in-service loads and induction machines for which the bus to which it is connected is assigned to the zone but the load or induction machine is assigned to a different zone. These are tabulated in the columns labeled `NON-ZONE ITEMS AT ZONE BUSES`.

  The output block for `COLUMN TOTALS` is blank for the three sets of load and induction machine columns. Rather, load and induction machine totals for the reported zones are tabulated following this output block. Two complex totals are given for loads, induction generators, and induction motors:

- `TOTAL ASSIGNED TO THESE ZONES`, which are the sums of the columns in the first two sets of totals described above.

- `TOTAL CONNECTED TO BUSES IN THESE ZONES`, which are the sums of the columns in the first and third sets of totals described above.

  Clearly, if all zones are being reported by the ZONE_2 API routine, these two sets of totals are identical.

In the ZONE_AREA API routine, an individual in-service bus load is included in the load totals of the zone to which the *load* is assigned; this zone need not be the same zone in which the bus at which the load is connected is located. Similarly, an in-service induction machine is included in the induction machine totals of the zone to which it assigned rather than that of the bus to which it is connected.

Line losses are taken as I^2R and I^2X losses and *exclude* the line charging and line connected shunt components; transformer losses *exclude* magnetizing admittance components. Series FACTS device losses are calculated as the algebraic sum of the reactive power flowing into the series device at its endpoint buses. Multiple terminal GNE device losses are calculated as the algebraic sum of the powers flowing into the device at its endpoint buses.

Losses on two-terminal dc transmission lines and on branches within multi-terminal dc lines are assigned to the zone of the nonmetered end. For the connections between each multi-terminal dc line ac converter bus and the dc bus to which it is connected, metering on the dc bus side is assumed (i.e., losses are assigned to the zone of the ac bus). Losses on a voltage source converter (VSC) dc line are calculated as the algebraic sum of the active and reactive powers flowing into the line at its endpoint buses. They are assigned to the zone of the voltage controlling converter when both converters are in-service; if only one converter is in-service, losses are assigned to the zone in which it is located. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

A zone has positive net interchange when it is selling or exporting power. Two net interchange totals are tabulated for each zone that is reported:

1. The total power flowing into zone tie lines.

2. The total power flowing into zone tie lines plus contributions from or to loads where the zone assignment of the load differs from that of the bus to which it is connected.

These two values of zone net interchange correspond to the two net interchange definitions that may be specified for use by the area interchange control calculation of the power flow solution activities.

Line charging, line shunt, transformer magnetizing, net interchange, and loss totals are evaluated on the basis of the metered end designated for each tie branch; i.e., for each tie branch, these quantities are assigned to the zone of the nonmetered end bus. Transformer magnetizing admittance components are excluded from line shunt totals. Tie line flows, and hence interchange, are calculated including the charging, line connected shunt, and transformer magnetizing admittance components.

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When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the zone assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, any generation, load, induction machine, and bus shunt present at assigned to the zone of the nonmetered endpoint bus of the multi-section line grouping. Similarly, line charging, line shunt, net interchange, and loss contributions are evaluated on the basis of the metered end designated for each multi-section line grouping that is a tie branch (i.e., its endpoint buses are located in different zones); for each multi-section tie, these quantities are assigned to the zone of the nonmetered endpoint bus.

When the multi-section line reporting option is disabled, the zone assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

A three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same zone. Tie flows are derived from the flows into the transformer at the two metered ends.

A two-terminal dc transmission line is considered to be a tie line when the converter station buses are located in different zones. The tie flow is taken as the power flowing into the dc converter from the ac system at the metered end. For multi-terminal dc lines, if a converter bus and a dc bus to which it is connected are in different zones, the connection is treated as a tie line and metering on the dc bus side is assumed. DC links within a multi-terminal dc line are handled in the same manner as ac branches. A voltage source converter (VSC) dc transmission line is considered to be a tie line when the converter station buses are located in different zones. The tie flow is taken as the power flowing into the dc converter from the ac system at the power controlling converter end. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

A series FACTS device is considered to be a tie line when the sending and terminal buses are located in different zones. The tie flow is taken as the power flowing into the series element at the terminal bus end.

A multiple terminal GNE device is considered to be a tie line when its connected buses are located in different zones. The tie flow is taken as the power flowing into the device at the metered end.

Activity ZONE may be terminated by entering the AB interrupt control code.

The *savnw.sav* power flow case, has four Zones. The Report on Zone totals is shown in Figure 12.9, "Zone Totals Results". The load reporting option that provides for three subtotals of zone load is used in this example.

A brief examination of the results shows:

- Zone 1 (FIRST) has zero generation but a demand level of 1200 MW. It can be seen that its net interchange is 1218 MW which serves the load demand and the zone's losses of 18 MW. Note that there is no Zone interchange flow control available.

- Similarly, Zone 2 (SECOND) has 1400 MW of generation but 1500 MW of load. Its net interchange therefore is seen to be 130 MW which serves the deficit between generation and demand and the zone's real power losses of 30.8 MW.

Additional Information
PSS®E GUI Users Guide, Zone Totals *PSS®E* Command Line Interface (CLI) *Users Guide*, Zone Totals

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```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, APR 23 2019  14:29
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE                          ZONE TOTALS
BASE CASE INCLUDING SEQUENCE DATA                                   IN MW/MVAR
                  --ZONE ITEMS AT-- --ZONE ITEMS AT-- --NON-ZONE ITEMS-
                FROM ----ZONE BUSES--- --NON-ZONE BUSES- --AT ZONE BUSES--              TO            TO XFRMR                    -NET INTERCHANGE-
               GENE-   TO IND       TO   TO IND       TO   TO IND       TO   TO BUS  GNE BUS  TO LINE  MAGNE-      FROM      TO      TO TIE  TO TIES
X-- ZONE --X  RATION   MACHNS     LOAD   MACHNS     LOAD   MACHNS     LOAD    SHUNT  DEVICES    SHUNT  TIZING CHARGING  LOSSES     LINES  + LOADS

   1             0.0      0.0   1200.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0     0.0      0.0      18.0  -1218.0  -1218.0
FIRST            0.0      0.0    900.0      0.0      0.0      0.0      0.0    349.7      0.0     -0.0    -0.0    841.2     289.7   -698.2   -698.2

   2          1400.0      0.0   1500.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0     0.0      0.0      30.8   -130.8   -130.8
SECOND         617.3      0.0    850.0      0.0      0.0      0.0      0.0   -641.5      0.0     -0.0    -0.0    623.0     590.1    441.8    441.8

   5           358.6      0.0    500.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0     0.0      0.0       6.6   -147.9   -147.9
FIFTH          183.6      0.0    200.0      0.0      0.0      0.0      0.0     -0.0      0.0     -0.0    -0.0    345.9      86.8    242.6    242.6

  77           1500.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0      0.0     0.0      0.0       3.3   1496.7   1496.7
PLANT          162.6      0.0      0.0      0.0      0.0      0.0      0.0     -0.0      0.0     -0.0    -0.0      0.0     148.8     13.8     13.8

COLUMN        3258.6                                                          0.0      0.0      0.0     0.0      0.0      58.6      0.0      0.0
TOTALS         963.5                                                       -291.8      0.0      0.0     0.0   1810.1   1115.4      0.0      0.0

                                                        X------- INDUCTION MACHINES ------X
                                          X---- LOADS ----X X-- GENERATORS -X X---- MOTORS ---X
                                               MW       MVAR       MW       MVAR       MW       MVAR
TOTAL ASSIGNED TO THESE ZONES              3200.0    1950.0      0.0       0.0       0.0       0.0
TOTAL CONNECTED TO BUSES IN THESE ZONES    3200.0    1950.0      0.0       0.0       0.0       0.0
```

Figure 12.9. Zone Totals Results

12.9.1. Report Tables

The results of this function can be output in Report Table format. These tables can be viewed as formatted text reports in an Output window, as a spreadsheet in the Result Table spreadsheet window or in both formats. The selection of the output format for Report Tables can be found at Misc -> Change program settings -> Report Tables. The API is result_table_output.

Report Tables exist in program memory and contain the contents of the last execution of the Activity/API. Existing Report Tables can be accessed from the Network TreeView in the “Report Tables” section.

Report Tables can be exported directly to a CSV file or Excel for further postprocessing and manipulation.

In a future release, a Report Table in program memory will be accessible through Python APIs allowing individual rows, columns and cells to be retrieved.

This Activity/API creates Report Table named "LF Zone".

12.10. Reporting Interchange by Area or Owner or Zone

Table 11-2 Information Provided in Interchange Reports for Areas and Zones shows the information provided in the interchange reports on the bases of area and zone.

Table 12.2. Information Provided in Interchange Reports for Areas and Zones

Information Provided	Area	Zone
Total interchange between Areas (MW, Mvar)	Y	-

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Information Provided	Area	Zone
Interchange flows on tie-lines between Areas (MW, Mvar)	Y	-
Total interchange between Zones (MW, Mvar)	-	Y
Interchange flows on tie-lines between Zones (MW, Mvar)	-	Y

There are two types of summary reports: Inter area/zone flow, which provides total flows between Areas or Zones and Tie line, which provides detailed flows on the tie lines (branches) between the Areas or Zones.

The output is organized by Area or Zone such that all Areas or all Zones are reported or the report may be limited to selected Areas or Zones.

It is useful to note that the inter Area / Zone flow report can be compared to the Totals report described in Section 12.6, "Summarizing Load Reduction".

12.11. Summarizing Area-to-Area Interchange

INTA

Run Activity INTA - GUI
Power Flow > Reports > Area / zone based reports... *[Area / Zone Based Reports]* OPTION: Inter area / zone flow; Output for Areas
Run Line Mode Activity INTA - CLI
ACTIVITY? >>**INTA** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The area interchange summary activity INTA summarizes tie flows between an interchange area and all other areas in the working case.

The user may restrict the report to interchange from specified areas (refer to Section 4.8, "Subsystem Selection"). Tie flow totals from the specified areas to all other areas are printed in ascending from area number order.

The output of activity INTA is in the form of a matrix (Figure 12.10, "Inter-Area Flows by Area"). The two-line row for each from area shows active power interchange on the top line and reactive power interchange on the bottom line. Tie flows, and hence interchange, are evaluated on the basis of the metered end designated for each tie branch, and are calculated including the charging and line connected shunt components. The standard sign convention is used in printing interarea flows: a positive value indicates power leaving the from area.

Tie line flows, and hence interchange, are evaluated on the basis of the metered end designated for each tie branch, and are calculated including the charging, line connected shunt, and transformer magnetizing admittance components.

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When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the area assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, a multi-section line grouping is treated as a tie branch if and only if its endpoint buses are in different areas, with tie flow evaluated at the metered end of the multi-section line grouping.

When the multi-section line reporting option is disabled, the area assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

A three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same area. Tie flows are derived from the flows into the transformer at the two metered ends.

A two-terminal dc transmission line is considered to be a tie line when the converter station buses are located in different areas. The tie flow is taken as the power flowing into the dc converter from the ac system at the metered end. For multi-terminal dc lines, if a converter bus and a dc bus to which it is connected are in different areas, the connection is treated as a tie line and metering on the dc bus side is assumed. DC links within a multi-terminal dc line are handled in the same manner as ac branches. A voltage source converter (VSC) dc transmission line is considered to be a tie line when the converter station buses are located in different areas. The tie flow is taken as the power flowing into the dc converter from the ac system at the power controlling converter end. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

A series FACTS device is considered to be a tie line when the sending and terminal buses are located in different areas. The tie flow is taken as the power flowing into the series element at the terminal bus end.

A multiple terminal GNE device is considered to be a tie line when the buses to which it is connected are located in different areas. The tie flow is taken as the power flowing into the device at the metered bus end.

The net interchange definition follows that of the most recently enabled setting of the area interchange control code via either activity OPTN or a power flow solution activity: tie lines only, or tie lines plus loads for which the load and the bus to which it is connected are in different areas (refer to Area Interchange Control).

The output of activity INTA may be terminated by entering the AB interrupt control code.

The Inter-area summary report for the *savnw.sav* case is seen in Figure 12.10, "Inter-Area Flows by Area".

```
      TO AREA:      1      2      5
FROM AREA     *------------------
        1     *          131    148
FLAPCO        *         -442   -243
              *------------------
        2     *  -131
LIGHTCO       *   442
              *------------------
        5     *  -148
WORLD         *   243
              *------------------|
```

Figure 12.10. Inter-Area Flows by Area

This simple summary on inter-area flow can be compared with the Area Total report. This report confirms the previously documented results which show that Area 1 is exporting 131 MW to Area 2 and exporting

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148 MW to Area 5. Note that these numbers are rounded in comparison to results shown in Figure 12.7, "Area Total Results".

Additional Information
PSS®E GUI Users Guide, Area-to-Area Interchange *PSS®E* Command Line Interface (CLI) *Users Guide*, Area-to-Area Interchange

12.12. Summarizing Zone-to-Zone Interchange

INTZ

Run Activity INTZ - GUI
Power Flow > Reports > Area / zone based reports... *[Area / Zone Based Reports]* OPTION: Inter area / zone flow; Output for Zones
Run Line Mode Activity INTZ - CLI
ACTIVITY? >>**INTZ** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The zone interchange summary activity INTZ summarizes tie flows between each zone and all other zones in the working case.

The user may restrict the report to interchange from specified zones (refer to Section 4.8, "Subsystem Selection"). Tie flow totals from the specified zones to all other zones are printed in ascending from zone number order.

The output of activity INTZ is in the form of a matrix (Figure 12.11, "Inter-Zone Flows by Zone"). The two-line row for each from zone shows active power interchange on the top line and reactive power interchange on the bottom line. Tie flows, and hence interchange, are evaluated on the basis of the metered end designated for each tie branch, and are calculated including the charging and line connected shunt components. The standard sign convention is used in printing interzone flows: a positive value indicates power leaving the from zone.

Tie line flows, and hence interchange, are evaluated on the basis of the metered end designated for each tie branch, and are calculated including the charging, line connected shunt, and transformer magnetizing admittance components.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the zone assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, a multi-section line grouping is treated as a tie branch if and only if its endpoint buses are in different zones, with tie flow evaluated at the metered end of the multi-section line grouping.

When the multi-section line reporting option is disabled, the zone assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

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A three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same zone. Tie flows are derived from the flows into the transformer at the two metered ends.

A two-terminal dc transmission line is considered to be a tie line when the converter station buses are located in different zones. The tie flow is taken as the power flowing into the dc converter from the ac system at the metered end. For multi-terminal dc lines, if a converter bus and a dc bus to which it is connected are in different zones, the connection is treated as a tie line and metering on the dc bus side is assumed. DC links within a multi-terminal dc line are handled in the same manner as ac branches. A voltage source converter (VSC) dc transmission line is considered to be a tie line when the converter station buses are located in different zones. The tie flow is taken as the power flowing into the dc converter from the ac system at the power controlling converter end. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

A series FACTS device is considered to be a tie line when the sending and terminal buses are located in different zones. The tie flow is taken as the power flowing into the series element at the terminal bus end.

A multiple terminal GNE device is considered to be a tie line when the buses to which it is connected are located in different zones. The tie flow is taken as the power flowing into the device at the metered bus end.

The net interchange definition follows that of the most recently enabled setting of the area interchange control code via either activity OPTN or a power flow solution activity: tie lines only, or tie lines plus loads for which the load and the bus to which it is connected are in different zones (refer to Area Interchange Control).

The output of activity INTZ may be terminated by entering the AB interrupt control code.

The Inter-zone summary report for the *savnw.sav* case is seen in Figure 12.11, "Inter-Zone Flows by Zone".

```
      TO ZONE:      1      2      5     77
FROM ZONE     *------------------------
         1    *          131    148 -1497
FIRST         *         -442   -243   -14
              *------------------------
         2    *  -131
SECOND        *   442
              *------------------------
         5    *  -148
FIFTH         *   243
              *------------------------
        77    *  1497
PLANT         *    14
              *-----------------------|
```

Figure 12.11. Inter-Zone Flows by Zone

This simple summary on inter-zone flow can be compared with the Zone Total report (see Figure 12.9, "Zone Totals Results"). This report confirms the previously documented results which show that Zone 1 is importing a total of 1218 MW from the other three zones.

Additional Information
PSS®E GUI Users Guide, Zone-to-Zone Interchange *PSS®E* Command Line Interface (CLI) *Users Guide*, Zone-to-Zone Interchange

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12.13. Summarizing Loadings on Ties from Interchange Areas

TIES

Run Activity TIES - GUI
Power Flow > Reports > Area / zone based reports... *[Area / Zone Based Reports]* OPTION: Tie line; Output for Areas
Run Line Mode Activity TIES - CLI
ACTIVITY? >>**TIES** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The tie line summary activity TIES tabulates the flows on all area ties, with tie flows grouped by area. Interchange between pairs of areas and net interchange from each area are also listed.

The user may restrict the report to tie flows from specified areas (refer to Section 4.8, "Subsystem Selection"). Tie flows from the specified areas to all other areas are printed in ascending from area number order.

Tie line flows, and hence interchange, are evaluated on the basis of the metered end designated for each tie branch, and are calculated including the charging, line connected shunt, and transformer magnetizing admittance components.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the area assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, a multi-section line grouping is treated as a tie branch if and only if its endpoint buses are in different areas, with tie flow evaluated at the metered end of the multi-section line grouping.

When the multi-section line reporting option is disabled, the area assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

A three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same area. Tie flows are derived from the flows into the transformer at the two metered ends.

A two-terminal dc transmission line is considered to be a tie line when the converter station buses are located in different areas. The tie flow is taken as the power flowing into the dc converter from the ac system at the metered end. For multi-terminal dc lines, if a converter bus and a dc bus to which it is connected are in different areas, the connection is treated as a tie line and metering on the dc bus side is assumed. DC links within a multi-terminal dc line are handled in the same manner as ac branches. A voltage source converter (VSC) dc transmission line is considered to be a tie line when the converter station buses are located in different areas. The tie flow is taken as the power flowing into the dc converter from the ac system at the power controlling converter end. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

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A series FACTS device is considered to be a tie line when the sending and terminal buses are located in different areas. The tie flow is taken as the power flowing into the series element at the terminal bus end.

A multiple terminal GNE device is considered to be a tie line when the buses to which it is connected are located in different areas. The tie flow is taken as the power flowing into the device at the metered bus end.

The net interchange definition follows that of the most recently enabled setting of the area interchange control code via either activity OPTN or a power flow solution activity: tie lines only, or tie lines plus loads for which the load and the bus to which it is connected are in different areas (refer to Area Interchange Control).

In the report, the element listed first for each tie is located in the from area, and the second element is in the to area. The metered bus of each tie branch is designated with an asterisk (*). The standard sign convention is used in printing the tie flows: a positive flow indicates power leaving the from area.

Activity TIES may be terminated by entering the AB interrupt control code.

A portion of the inter-area tie-line report for the *savnw.sav* case is seen in Figure 11-12. The report presents information in order of Area. The figure shows the tie-line flows from Area 1 to Areas 2 and 5. The total flows can be compared with the interchange totals shown in Figure 12.10, "Inter-Area Flows by Area"(inter-area flows) and Figure 12.7, "Area Total Results" (area totals report).

```
FROM AREA    1   FLAPCO

  TO AREA    2   LIGHTCO
  X---- FROM AREA BUS ----X   X----- TO AREA BUS -----X
    BUS# X-- NAME --X BASKV     BUS# X-- NAME --X BASKV  CKT      MW      MVAR
     151 NUCPANT     500.00*     201 HYDRO       500.00   1     564.8   -264.8
     152 MID500      500.00      202 EAST500     500.00*  1      42.6    128.5
     154 DOWNTN      230.00*     203 EAST230     230.00   1    -122.4    -53.9
     154 DOWNTN      230.00*     205 SUB230      230.00   1    -354.2   -251.5
  TOTAL FROM AREA    1 TO AREA    2                                          130.8    -441.7

  TO AREA    5   WORLD
  X---- FROM AREA BUS ----X   X----- TO AREA BUS -----X
    BUS# X-- NAME --X BASKV     BUS# X-- NAME --X BASKV  CKT      MW      MVAR
     152 MID500      500.00*    3004 WEST        500.00   1     138.2   -138.3
     153 MID230      230.00*    3006 UPTOWN      230.00   1      78.7    -16.3
     154 DOWNTN      230.00*    3008 CATDOG      230.00   1     -69.0    -88.4
  TOTAL FROM AREA    1 TO AREA    5                                          147.9    -242.9
                                                                          --------  --------
TOTAL FROM AREA    1   FLAPCO                                              278.7    -684.6
```

Figure 12.12. Inter-Area Tie Line Report

Additional Information
PSS®E GUI Users Guide, Loadings on Tie Lines from Interchange Areas *PSS®E* Command Line Interface (CLI) *Users Guide*, Loadings on Ties from Interchange Areas

12.14. Summarizing Loadings on Ties from Zones

TIEZ

Run Activity TIEZ - GUI
Power Flow > Reports > Area / zone based reports... *[Area / Zone Based Reports]*

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Run Activity TIEZ - GUI
OPTION: Tie line; Output for Zones
Run Line Mode Activity TIEZ - CLI
ACTIVITY? >>**TIEZ** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The zone tie line summary activity TIEZ tabulates the flows on all zone ties, with tie flows grouped by zone. Interchange between pairs of zones and net interchange from each zone are also listed.

The user may restrict the report to tie flows from specified zones. The user specifies the zones to be reported (refer to Section 4.8, "Subsystem Selection"). Tie flows from the specified zones to all other zones are printed in ascending from zone number order.

Tie line flows, and hence interchange, are evaluated on the basis of the metered end designated for each tie branch, and are calculated including the charging, line connected shunt, and transformer magnetizing admittance components.

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the zone assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, a multi-section line grouping is treated as a tie branch if and only if its endpoint buses are in different zones, with tie flow evaluated at the metered end of the multi-section line grouping.

When the multi-section line reporting option is disabled, the zone assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

A three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same zone. Tie flows are derived from the flows into the transformer at the two metered ends.

A two-terminal dc transmission line is considered to be a tie line when the converter station buses are located in different zones. The tie flow is taken as the power flowing into the dc converter from the ac system at the metered end. For multi-terminal dc lines, if a converter bus and a dc bus to which it is connected are in different zones, the connection is treated as a tie line and metering on the dc bus side is assumed. DC links within a multi-terminal dc line are handled in the same manner as ac branches. A voltage source converter (VSC) dc transmission line is considered to be a tie line when the converter station buses are located in different zones. The tie flow is taken as the power flowing into the dc converter from the ac system at the power controlling converter end. Refer to the *PSS®E Program Application Guide* for additional details on dc line quantities.

A series FACTS device is considered to be a tie line when the sending and terminal buses are located in different zones. The tie flow is taken as the power flowing into the series element at the terminal bus end.

A multiple terminal GNE device is considered to be a tie line when the buses to which it is connected are located in different zones. The tie flow is taken as the power flowing into the device at the metered bus end.

The net interchange definition follows that of the most recently enabled setting of the area interchange control code via either activity OPTN or a power flow solution activity: tie lines only, or tie lines plus loads for which the load and the bus to which it is connected are in different zones (refer to Area Interchange Control).

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In the report, the element listed first for each tie is located in the from zone, and the second element is in the to zone. The metered bus of each tie branch is designated with an asterisk (*). The standard sign convention is used in printing the tie flows: a positive flow indicates power leaving the from zone.

Activity TIEZ may be terminated by entering the AB interrupt control code.

A portion of the inter-zone tie-line report for the *savnw.sav* case is seen in Figure 12.13, "Inter-Zone Tie Line Report". The report presents information in order of Zone. The figure shows the tie-line flows from Zone 1 to Zones 2, 5 and 77. The total flows can be compared with the interchange totals shown in Figure 12.11, "Inter-Zone Flows by Zone" (inter-zone flows) and Figure 12.9, "Zone Totals Results" (zone totals report).

```
FROM ZONE    1   FIRST

  TO ZONE    2   SECOND
  X---- FROM ZONE BUS ----X   X----- TO ZONE BUS -----X
    BUS# X-- NAME --X BASKV     BUS# X-- NAME --X BASKV  CKT      MW      MVAR
     151 NUCPANT      500.00*     201 HYDRO       500.00   1     564.8   -264.8
     152 MID500       500.00      202 EAST500     500.00*  1      42.6    128.5
     154 DOWNTN       230.00*     203 EAST230     230.00   1    -122.4    -53.9
     154 DOWNTN       230.00*     205 SUB230      230.00   1    -354.2   -251.5
  TOTAL FROM ZONE    1 TO ZONE    2                                          130.8    -441.7

  TO ZONE    5   FIFTH
  X---- FROM ZONE BUS ----X   X----- TO ZONE BUS -----X
    BUS# X-- NAME --X BASKV     BUS# X-- NAME --X BASKV  CKT      MW      MVAR
     152 MID500       500.00*    3004 WEST        500.00   1     138.2   -138.3
     153 MID230       230.00*    3006 UPTOWN      230.00   1      78.7    -16.3
     154 DOWNTN       230.00*    3008 CATDOG      230.00   1     -69.0    -88.4
  TOTAL FROM ZONE    1 TO ZONE    5                                          147.9    -242.9

  TO ZONE   77   PLANT
  X---- FROM ZONE BUS ----X   X----- TO ZONE BUS -----X
    BUS# X-- NAME --X BASKV     BUS# X-- NAME --X BASKV  CKT      MW      MVAR
     151 NUCPANT      500.00*     101 NUC-A       21.600   1    -748.4    -6.8
     151 NUCPANT      500.00*     102 NUC-B       21.600   1    -748.4    -6.8
  TOTAL FROM ZONE    1 TO ZONE   77                                      -1496.7     -13.6
                                                                         --------   --------
TOTAL FROM ZONE    1   FIRST                                            -1218.0    -698.2
```

Figure 12.13. Inter-Zone Tie Line Report

Additional Information
PSS®E GUI Users Guide, Loadings on Tie Lines from Zones *PSS®E* Command Line Interface (CLI) *Users Guide*, Loadings on Ties from Zones

12.15. Viewing Network Limit Violations

PSS®E enables the user to develop seven different types of reports which can assist in rapidly identifying network conditions that are outside of required control limits or to indicate controlling equipment and possible violations and conflicts associated with those controls. The following summary reports are available:

- Summary of branches exceeding specified percentage of selected rating
- Summary of buses with voltage outside specified band
- Summary of generator terminal loading conditions on either all, or only overloaded, generators
- Summary of generator bus loading conditions on either all, or only var-limited, generators

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- Summary of controlling transformers and violations
- Generator reactive power and capability
- Regulated buses, violations and/or conflicts.

12.16. Producing a Branch Overload Checking Report

RAT3

Run Activity RAT3 - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* OPTION: Branches Tab; List overloaded both trans and non-trans branches, all ratings
Run Line Mode Activity RAT3 - CLI
ACTIVITY? >>**RAT3** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The branch overload checking activity RAT3 checks each ac branch current loading against designated branch ratings. For non-transformer branches, loadings are calculated as either current loadings or MVA loadings, according to the non-transformer branch percent loading units program option setting. Similarly, for transformer branches, loadings are calculated as either current loadings or MVA loadings, according to the transformer percent loading units program option setting. Refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN. Branch overloads are reported based on a selected ratings set: RATE A, RATE B or RATE C.

When percent current loadings are calculated, ratings are assumed to have been entered as:

$MVA_{rated} = \sqrt{3} \times E_{base} \times I_{rated} \times 10^{-6}$

where:

E_{base}	Is the branch or transformer winding voltage base in volts.
I_{rated}	Is the rated phase current in amps.

When percent MVA loadings are calculated, ratings are assumed to have been entered as MVA ratings.

The user specifies the percentage loading threshold. Any checked branch where loading, including line charging, line connected shunt, and transformer magnetizing admittance components, exceeds the specified percentage of rating is reported. The default limit is 100%.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output is generated, with all branches having at least one endpoint bus in the specified subsystem and at least one non-zero rating being checked. Then the user is given the opportunity to specify another subsystem.

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When an owner subsystem is specified, branch ownership rather than bus ownership is used in determining which branches are contained in the specified subsystem; any other selection criteria apply to the branch's endpoint buses.

Any branch where the ratings are all zero is not checked for overloading. Any branch where current loading exceeds the designated percentage of at least one non-zero rating generates an alarm. The loading is calculated at each end of the line and the number of the bus at the end of the branch with the higher loading is followed by an asterisk (*). The branch loading, rating and percentage loadings are tabulated for each rating set.

For three-winding transformers, each winding is checked and reported separately. Only those windings connected to buses in the specified subsystem are processed. For each winding reported, the bus to which the winding is connected is shown as the from bus, and the winding number and transformer name are listed as the to bus.

Activity RAT3 may be terminated with the AB interrupt control code.

The output report of activity RAT3 lists branches in single entry list format ordered by from bus (refer to Branch Data) and grouped as described in Section 4.9, "Subsystem Reporting".

Figure 12.14, "Report for Branch Loadings, All Ratings"shows a subset of the report for the *savnw.sav* power flow case. It demonstrates the manner in which the branches are listed in bus number, circuit and Area sequence.

The report shows only one branch is overloaded. It is the 230/18 kV transformer branch from Bus 205 to Bus 206 the loading of which is 108.5% of the RATE A level.

Note that a value for RATE C has been entered with a fictitious value of 1.0 MVA for most branches. The table therefore shows very high percentage loadings for that RATE.

PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E FRI, NOV 28 2003 12:45
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

BRANCH LOADINGS ABOVE 100.0 % OF RATING:

X--------- FROM BUS ----------				X---------- TO BUS ----------					CURRENT (MVA)	RATING SET A		RATING SET B		RATING SET C	
BUS#	X-- NAME	--X BASKV	AREA	BUS#	X-- NAME	--X BASKV	AREA	CKT	LOADING	RATING	PERCENT	RATING	PERCENT	RATING	PERCENT
151	NUCPANT	500.00*	1	152	MID500	500.00	1	1	489.5	1200.0	40.8	1300.0	37.7	1.0	48948.1
151	NUCPANT	500.00*	1	152	MID500	500.00	1	2	489.5	1200.0	40.8	1300.0	37.7	1.0	48948.1
151	NUCPANT	500.00*	1	201	HYDRO	500.00	2	1	616.5	1200.0	51.4	1300.0	47.4	1.0	61647.2
152	MID500	500.00	1	202	EAST500	500.00*	2	1	134.2	1200.0	11.2	1300.0	10.3	1.0	13415.0
152	MID500	500.00*	1	3004	WEST	500.00	5	1	192.2	--	--	--	--	1.0	19223.4
153	MID230	230.00	1	154	DOWNTN	230.00*	1	1	276.0	300.0	92.0	350.0	78.9	1.0	27604.0
153	MID230	230.00	1	154	DOWNTN	230.00*	1	2	231.0	300.0	77.0	350.0	66.0	1.0	23096.7
153	MID230	230.00*	1	3006	UPTOWN	230.00	5	1	80.9	--	--	--	--	1.0	8088.9
154	DOWNTN	230.00*	1	203	EAST230	230.00	2	1	142.4	200.0	71.2	250.0	57.0	1.0	14244.1
154	DOWNTN	230.00*	1	205	SUB230	230.00	2	1	462.6	600.0	77.1	660.0	70.1	1.0	46264.6
154	DOWNTN	230.00*	1	3008	CATDOG	230.00	5	1	119.4	400.0	29.8	440.0	27.1	1.0	11938.8
201	HYDRO	500.00	2	202	EAST500	500.00*	2	1	564.4	1200.0	47.0	1300.0	43.4	1.0	56442.6
201	HYDRO	500.00	2	204	SUB500	500.00*	2	1	641.8	1200.0	53.5	1300.0	49.4	1.0	64175.1
203	EAST230	230.00	2	205	SUB230	230.00*	2	1	93.2	200.0	46.6	250.0	37.3	1.0	9320.5
203	EAST230	230.00	2	205	SUB230	230.00*	2	2	93.2	200.0	46.6	250.0	37.3	1.0	9320.5
205	SUB230	230.00	2	206	URBGEN	18.000*	2	1	976.9	900.0	108.5	1080.0	90.5	1350.0	72.4

Figure 12.14. Report for Branch Loadings, All Ratings

Additional Information
PSS®E GUI Users Guide, Branch Overloads
PSS®E Command Line Interface (CLI) *Users Guide*, Branch Overloads

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12.17. Producing a Transformer Overload Checking Report

OLTR

Run Activity OLTR - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* OPTION: Branches Tab; List overloaded only transformers
Run Line Mode Activity OLTR - CLI
ACTIVITY? >>**OLTR** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The transformer overload checking activity OLTR checks transformer branch loadings against designated branch ratings. Loadings are calculated as either current loadings or MVA loadings, according to the transformer percent loading units program option setting. Refer to Section Saved Case Specific Option Settings and activity OPTN.

When percent current loadings are calculated, ratings are assumed to have been entered as:

$MVA_{rated} = \sqrt{3} \times E_{base} \times I_{rated} \times 10^{-6}$

where:

E_{base}	Is the transformer winding voltage base in volts.
I_{rated}	Is the rated phase current in amps.

When percent MVA loadings are calculated, ratings are assumed to have been entered as MVA ratings.

The user specifies the percentage loading threshold. Any checked transformer branch where loading, including magnetizing admittance, exceeds the specified percentage of rating is reported. The default limit is 100%.

The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). The report is generated with branches having a non-zero value for the specified rating being checked.

The user specifies the subsystem of the working case for which the overload check is to be made (refer to Section 4.8, "Subsystem Selection"). Output is generated with all transformer branches having at least one endpoint bus in the specified subsystem being checked.

If an owner subsystem has been specified, branch ownership rather than bus ownership is used in determining which transformers are contained in the specified subsystem; any other selection criteria apply to the transformer's endpoint buses.

It should be noted, however, that when subsystem selection is by OWNER, it is branch ownership rather than bus ownership which is used in determining which transformer branches are contained in the specified subsystem. Any other selection criteria apply to the transformer's endpoint buses.

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Any transformer where the corresponding rating is zero is not checked for overloading. Any transformer branch where loading exceeds the designated percentage of rating generates an alarm. The loading is calculated at each end of the line and the number of the bus at which the loading is higher is followed by an asterisk (*). The branch loading, rating, and percentage loading are tabulated.

Any transformer where the corresponding rating is zero is not checked for overloading. Any transformer where loading exceeds the designated percentage of rating generates an alarm. The loading is calculated at each end of the transformer, and the number of the bus at the end of the transformer with the higher loading is followed by an asterisk (*). The transformer loading, rating, and percentage loading are tabulated.

For three-winding transformers, each winding is checked and reported separately. Only those windings connected to buses in the specified subsystem are processed. For each winding reported, the bus to which the winding is connected is shown as the from bus, and the winding number and transformer name are listed as the to bus.

The output report of activity OLTR lists transformers in single entry list format ordered by from bus (refer to Branch Data) and grouped as described in Section 4.9, "Subsystem Reporting".

Thus, activity OLTR functions in exactly the same manner as does activity RATE, except that non-transformer branches are omitted from the overload checking.

Activity OLTR may be terminated with the AB interrupt control code.

Figure 12.15, "Report on Transformer Overloads Based on 80% of RATE A" shows a sample output from the *savnw.sav* case with the transformer overload check performed on the basis of 80% of RATE A.

The overload report lists transformer branches in single entry list format ordered by from bus. Transformers are also listed by AREA so that all transformers overloaded in AREA 1 are listed first followed by transformers in AREA 2 and AREA 5. Transformers would also be listed by circuit number if appropriate. This report shows overloads only in AREAS 2 and 5.

In the table, the right most column shows the percentage loading on 100% of the RATE A level. Only one transformer loading exceeds the 100% limit. It is the 230/18 kV transformer branch from Bus 205 to Bus 206.

Two other branches are listed which have loadings less than 100% of RATE A but have loadings exceeding 80%.

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      FRI, NOV 28 2003  12:42
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

TRANSFORMER LOADINGS ABOVE  80.0 % OF RATING SET A:

X---------- FROM BUS ----------X X---------- TO BUS ----------X       MVA       MVA
  BUS# X-- NAME --X BASKV  AREA    BUS# X-- NAME --X BASKV  AREA CKT LOADING  RATING PERCENT
   202 EAST500     500.00*    2     203 EAST230     230.00     2  1    653.7   800.0    81.7
   205 SUB230      230.00     2     206 URBGEN      18.000*    2  1   1000.0   900.0   111.1
  3008 CATDOG      230.00     5    3018 CATDOG_G    13.800*    5  1    128.1   150.0    85.4
```

Figure 12.15. Report on Transformer Overloads Based on 80% of RATE A

Additional Information
PSS®E GUI Users Guide, Transformer Overloads *PSS®E* Command Line Interface (CLI) *Users Guide*, Transformer Overloads

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12.18. Producing a Transmission Line Overload Checking Report

OLTL

Run Activity OLTL - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* OPTION: Branches Tab; List overloaded only non-transformer branches
Run Line Mode Activity OLTL - CLI
ACTIVITY? >>**OLTL** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The transmission line overload checking activity OLTL checks non-transformer branch loadings against designated branch ratings. Loadings are calculated as either current loadings or MVA loadings, according to the non-transformer branch percent loading units program option setting. Refer to Section Saved Case Specific Option Settings and activity OPTN.

When percent current loadings are calculated, ratings are assumed to have been entered as:

$MVA_{rated} = \sqrt{3} \times E_{base} \times I_{rated} \times 10^{-6}$

where:

E_{base}	Is the base voltage in volts of the bus to which the terminal of the branch is connected.
I_{rated}	Is the rated phase current in amps.

When percent MVA loadings are calculated, ratings are assumed to have been entered as MVA ratings.

The user specifies the percentage loading threshold. Any checked branch where current loading, including line charging and line connected shunt components, exceeds the specified percentage of rating is reported. The default limit is 100%.

The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN). The report is generated with branches having a non-zero value for the specified rating being checked.

The user specifies the subsystem of the working case for which the overload check is to be made (refer to Section 4.8, “Subsystem Selection”). Output is generated with all non-transformer branches having at least one endpoint bus in the specified subsystem being checked.

If an owner subsystem has been specified, branch ownership rather than bus ownership is used in determining which branches are contained in the subsystem; any other selection criteria apply to the branch’s endpoint buses.

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Any non-transformer branch where the corresponding rating is zero is not checked for overloading. Any non-transformer branch where loading exceeds the designated percentage of rating generates an alarm. The loading is calculated at each end of the branch, and the number of the bus at the end of the branch with the higher loading is followed by an asterisk (*). The branch loading, rating, and percentage loading are tabulated.

The output report of activity OLTL lists branches in single entry list format ordered by from bus (refer to Branch Data) and grouped as described in Section 4.9, "Subsystem Reporting".

Thus, activity OLTL functions in exactly the same manner as does activity RATE, except that transformer branches are omitted from the overload checking.

Activity OLTL may be terminated with the AB interrupt control code.

Figure 12.16, "Report on Transmission Line Overloads" shows a sample output from the *savnw.sav* case with the transmission line (non-transformer branch) overload check performed on the basis of 75% of RATE A.

The overload report lists branches in single entry list format ordered by from bus. As for Branch and Transformer overloads, branches are also listed by AREA so that all branches overloaded in AREA 1 are listed first followed by branches in AREA 2 and AREA 5. Branches would also be listed by circuit number if appropriate. This report shows overloads only in AREA 1.

In the table, the right most column shows the percentage loading on 100% of the RATE A level. It can be seen that no transmission lines have a loading in excess of 100% of RATE A. Three transmission lines are shown to have loadings in excess of 75% of RATE A. One line from Bus 153 to Bus 154 has two circuits listed in order of circuit number.

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      FRI, NOV 28 2003  12:42
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

TRANSMISSION LINE LOADINGS ABOVE  75.0 % OF RATING SET A:

X--------- FROM BUS ----------X X---------- TO BUS ----------X       CURRENT(MVA)
  BUS# X-- NAME --X BASKV  AREA   BUS# X-- NAME --X BASKV  AREA CKT LOADING  RATING PERCENT
   153 MID230      230.00     1    154 DOWNTN      230.00*    1  1    276.0   300.0    92.0
   153 MID230      230.00     1    154 DOWNTN      230.00*    1  2    231.0   300.0    77.0
   154 DOWNTN      230.00*    1    205 SUB230      230.00     2  1    462.6   600.0    77.1
```

Figure 12.16. Report on Transmission Line Overloads

Additional Information
PSS®E GUI Users Guide, Transmission Line Overloads *PSS®E* Command Line Interface (CLI) *Users Guide*, Transmission Line Overloads

12.19. Producing a Branch Current Ratings Report

RATE

Run Activity RATE - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]*

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Run Activity RATE - GUI
OPTION: Branches Tab; List overloaded both transformer and non-transformer branches
Run Line Mode Activity RATE - CLI
ACTIVITY? >>**RATE** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The branch overload checking activity RATE checks ac branch loadings against designated branch ratings. The user can select from several types of branch loading checks:

- Branch overloads, based on current loading, for transformer and non-transformer branches
- Transformer overloads, based on MVA loading
- Transmission line (non-transformer branches) overloads based on current loading
- Branch current ratings (A, B and C) and percentage loading at each rating

For overload checking purposes, the user can select which of the three ratings is to be used, with the default being the rating set established in the program options (activity OPTN). The default percentage of rating limit is 100%. This limit can be adjusted up or down in 5% steps.

For non-transformer branches, loadings are calculated as either current loadings or MVA loadings, according to the non-transformer branch percent loading units program option setting. Similarly, for transformer branches, loadings are calculated as either current loadings or MVA loadings, according to the transformer percent loading units program option setting. Refer to Section 3.3.3, "Program Run-Time Option Settings", activity OPTN, and Case Identification Data.

When percent current loadings are calculated, ratings are assumed to have been entered as:

$MVA_{rated} = \sqrt{3} \times E_{base} \times I_{rated} \times 10^{-6}$

where:

E_{base}	Is the base voltage in volts of the bus to which the terminal of the branch is connected.
I_{rated}	Is the rated phase current in amps.

When percent MVA loadings are calculated, ratings are assumed to have been entered as MVA ratings.

Any checked branch where current loading, including line charging, line connected shunt, and transformer magnetizing admittance components, exceeds the specified percentage of rating is reported. The default line loading limit is 100%.

The user specifies the rating; the default rating set is established by the program option setting (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). The report is generated with branches having a non-zero value for the specified rating being checked.

The user specifies the subsystem of the working case for which the overload check is to be made (refer to Section 4.8, "Subsystem Selection"). Output is generated with all branches having at least one endpoint bus in the specified subsystem being checked.

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If an owner subsystem has been specified, branch ownership rather than bus ownership is used in determining which branches are contained in the subsystem; any other selection criteria apply to the branch's endpoint buses.

Any branch where the corresponding rating is zero is not checked for overloading. Any branch where loading exceeds the designated percentage of rating generates an alarm. The loading is calculated at each end of the branch, and the number of the bus at the end of the branch with the higher loading is followed by an asterisk (*). The branch loading, rating, and percentage loading are tabulated.

For three-winding transformers, each winding is checked and reported separately. Only those windings connected to buses in the specified subsystem are processed. For each winding reported, the bus to which the winding is connected is shown as the from bus, and the winding number and transformer name are listed as the to bus.

Activity RATE may be terminated with the AB interrupt control code.

The output report of activity RATE lists branches in single entry list format ordered by from bus (refer to Branch Data) and grouped as described in Section 4.9, "Subsystem Reporting".

Figure 12.17, "Output Format for Branch Overloads Based on 80% of Rate A" shows a sample output from the *savnw.sav* case with a branch overload check performed on the basis of 80% of RATE A

The overload report lists branches in single entry list format ordered by from bus. It can be seen that branches are also listed by AREA so that all branches overloaded in AREA 1 are listed first followed by branches in AREA 2 and AREA 5. Branches would also be listed by circuit number if appropriate.

In the table, the right most column shows the percentage loading on 100% of the RATE A level. Only one branch loading exceeds the 100% limit; the 230/18 kV transformer branch from Bus 205 to Bus 206. Four other branches are listed which have loadings less than 100% of RATE A but have loadings exceeding 80%.

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      FRI, NOV 28 2003  12:28
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

BRANCH LOADINGS ABOVE  80.0 % OF RATING SET A:

X---------- FROM BUS ----------X X---------- TO BUS ----------X       CURRENT(MVA)
  BUS# X-- NAME --X BASKV  AREA    BUS# X-- NAME --X BASKV  AREA CKT LOADING  RATING PERCENT
   153 MID230      230.00     1     154 DOWNTN      230.00*    1   1   276.0   300.0    92.0
   202 EAST500     500.00*    2     203 EAST230     230.00     2   1   648.0   800.0    81.0
   204 SUB500      500.00     2     205 SUB230      230.00*    2   1   641.8   800.0    80.2
   205 SUB230      230.00     2     206 URBGEN      18.000*    2   1   976.9   900.0   108.5
  3008 CATDOG      230.00*    5    3018 CATDOG_G    13.800     5   1   125.3   150.0    83.6
```

Figure 12.17. Output Format for Branch Overloads Based on 80% of Rate A

Additional Information
PSS®E GUI Users Guide, Branch Current Ratings *PSS®E* Command Line Interface (CLI) *Users Guide*, Branch Current Ratings

12.20. Producing Out-of-Limits Bus Voltage Reports

Two API routines are available for reporting bus voltage violations:

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1. Activity VCHK checks bus voltages against a user-specified bus voltage magnitude range. The same range is applied at all buses that are to be checked.

2. Activity VLIMS checks bus voltages against either normal or emergency voltage limits. Each bus has its own limits.

12.20.1. Activity VCHK

VCHK

Run Activity VCHK - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Out-of-limit bus voltage Tab
Run Line Mode Activity VCHK - CLI
ACTIVITY? >>**VCHK** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The voltage checking activity VCHK tabulates those buses where voltage magnitude is outside of a specified range.

The VCHK API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".
- A designation of the voltage band to be used. The default settings result in a tabulation of buses where voltage is above 1.05 pu or below 0.95 pu.

The user specifies either that the entire working case is to be processed, or that voltage violations of buses in a designated bus subsystem are to be tabulated.

The report of activity VCHK is directed to the *Report* device.

Activity VCHK produces a listing of those buses where voltage magnitude is greater than VMAX, followed by a listing of buses where voltage is less than VMIN. Both listings are in ascending numerical (using the numbers output option) or alphabetical (using the names output option) bus order; refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN.

The star point buses of three-winding transformers are neither checked nor reported by activity VCHK.

Activity VCHK may be terminated with the AB interrupt control code.

Figure 12.18, "Report for Voltages Out-of-limit" shows the report generated for the *savnw.sav* case using a voltage range from 0.97 pu to 1.01 pu. It can be seen that the bus base voltage is listed together with the actual bus voltages in pu and kV. There are 13 buses with voltages greater than 1.01 pu and 5 buses with voltages less than 0.97 pu.

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      FRI, NOV 28 2003  14:59
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

BUSES WITH VOLTAGE GREATER THAN 1.0100:

   BUS# X-- NAME --X BASKV AREA  V(PU)  V(KV)      BUS# X-- NAME --X BASKV AREA  V(PU)  V(KV)
    101 NUC-A       21.600    1 1.0200 22.032       102 NUC-B       21.600    1 1.0200 22.032
    151 NUCPANT     500.00    1 1.0119 505.95       152 MID500      500.00    1 1.0171 508.54
    201 HYDRO       500.00    2 1.0400 520.00       206 URBGEN      18.000    2 1.0236 18.425
    211 HYDRO_G     20.000    2 1.0404 20.809      3001 MINE        230.00    5 1.0298 236.85
   3002 E. MINE     500.00    5 1.0279 513.96      3003 S. MINE     230.00    5 1.0233 235.36
   3004 WEST        500.00    5 1.0165 508.23      3011 MINE_G      13.800    5 1.0400 14.352
   3018 CATDOG_G    13.800    5 1.0218 14.100

BUSES WITH VOLTAGE LESS THAN 0.9700:

   BUS# X-- NAME --X BASKV AREA  V(PU)  V(KV)      BUS# X-- NAME --X BASKV AREA  V(PU)  V(KV)
    154 DOWNTN      230.00    1 0.9389 215.95       203 EAST230     230.00    2 0.9665 222.30
    205 SUB230      230.00    2 0.9490 218.27      3007 RURAL       230.00    5 0.9637 221.65
   3008 CATDOG      230.00    5 0.9586 220.48
```

Figure 12.18. Report for Voltages Out-of-limit

12.20.2. Activity VLIMS

VLIMS

Run Activity VLIMS - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Out-of-limit bus voltage Tab
Run Line Mode Activity VLIMS - CLI
ACTIVITY? >>**VLIMS** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The voltage checking activity VLIMS tabulates those buses where voltage magnitude is outside of a their normal or emergency voltage limit range.

The VLIMS API routine includes the following among its inputs:

- A designation of the subsystem for which the data tabulation is to be produced; refer to Section 4.8, "Subsystem Selection".
- A designation of the voltage band to be used.
- Normal bus voltage limits
- Emergency bus voltage limits
- A voltage limit threshold that may be used to modify the desitnated voltage limit band. The threshold must be in the range -0.05 through 0.05.

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- 0.0 to report voltages above the upper limit or below the lower limit
- Greater than 0.0 to report voltages beyond, or within the threshold of, a limit
- Less than 0.0 to report voltages outside of a limit by more than the magnitude of the threshold

The user specifies either that the entire working case is to be processed, or that voltage violations of buses in a designated bus subsystem are to be tabulated.

The report of activity VLIMS is directed to the *Report* device.

When checking a bus voltage against the bus' high voltage limit, the appropriate upper limit (i.e., normal or emergency) is modified by subrtacting the specified voltage limit threshold from it. Activity VLIMS produces a listing of those buses where voltage magnitude is greater than the bus' modified limit. The high limit violations report is followed by a listing of buses where voltage is less than a modified lower limit; the modified lower limit is the appropriate lower limit (i.e., normal or emergency), modified by adding the specified voltage limit threshold to it.

Both listings are in ascending numerical (using the numbers output option) or alphabetical (using the names output option) bus order; refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN.

For each bus reported, its bus number and extended bus name are listed, along with its voltage magnitude and the appropriate limit, both in per unit. The bus voltage limits that are tabulated are the unmodified limits (i.e., they are not modified by the voltage limit).

The star point buses of three-winding transformers are neither checked nor reported by activity VLIMS.

Activity VLIMS may be terminated with the AB interrupt control code.

Additional Information
PSS®E GUI Users Guide, Out-of-limit Bus Voltage *PSS®E* Command Line Interface (CLI) *Users Guide*, Out-of-Limits Bus Voltage

12.21. Producing a Machine Reactive Capability Report

GCAP

Run Activity GCAP - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Reactive capability Tab
Run Line Mode Activity GCAP - CLI
`ACTIVITY? >>GCAP ENTER OUTPUT DEVICE CODE:`
Interrupt Control Codes
AB

The machine capability curve checking activity GCAP prints a report of machine loading and limit data. Optionally, machine reactive power limits in the working case may be updated. For each machine that is to be checked, its capability curve must be contained in the working case; for more information please see *Machine Capability Curve Data File*.

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12.21.1. Operation of Activity GCAP

If machine capability curve data had not been read into the working case, an appropriate message is printed and activity GCAP is terminated.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or base voltage. Output is generated with buses ordered as described in Section 4.9, "Subsystem Reporting".

When the owner option is specified, each machine wholly or partly owned by any of the owners specified and for which its capability curve data is contained in the working case is processed (refer to Generator Unit Data). The owner assignment of the bus to which the machine is connected is not considered.

The user may update the reactive power limits of all machines reported by activity GCAP.

The output listing of activity GCAP may be terminated by entering the AB interrupt control code.

The output report of activity GCAP lists the following quantities:

1. Machine active and reactive power loading. If this value is beyond the limits shown in (2), it is followed by an asterisk (*).
2. The reactive power limit settings on the capability curve corresponding to the machine's active power loading.
3. The maximum active power loading from the capability curve (i.e., P_i from the last point entered for this machine).
4. The active and reactive power limits as contained in the working case.

Figure 12-19 shows the report generated for the *savnw.sav* Saved Case. It clearly indicates that the machine at bus 101 has a reactive power output of 81.3 Mvar and a real power output of 750 MW. Based on its capability curve, the reactive power limit should be 243.8 Mvar based on the real power output of 750 MW. This is much less than the assumed limit of 600 Mvar.

```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 24 2019  12:29
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

CAPABILITY CURVE CHECK:
                                                  X-- CAPABILITY CURVE --X X------- WORKING  CASE -------X
  BUS#-SCT X-- NAME --X BASKV ID    PGEN    QGEN     QMAX    QMIN  PLIMIT    QMAX    QMIN    PMAX    PMIN
   101     NUC-A       21.600 1    750.0    81.3    243.8   -37.5   900.0   600.0  -100.0   810.0     0.0
   102     NUC-B       21.600 1    750.0    81.3    243.8   -37.5   900.0   600.0  -100.0   810.0     0.0
   206     URBGEN      18.000 1    800.0   600.0    600.0     0.0  1000.0   600.0     0.0   900.0     0.0
   211     HYDRO_G     20.000 1    600.0    17.3    583.3   -77.8   950.0   400.0  -100.0   616.2     0.0
  3011     MINE_G      13.800 1    258.6   103.6    270.5  -100.0  1000.0   600.0  -100.0   900.0     0.0
  3018     CATDOG_G    13.800 1    100.0    80.0     80.0     0.0   130.0    80.0     0.0   117.0     0.0
```

Figure 12.19. Report Output for Reactive Power Checking with Capability Curve

12.21.2. Application Notes

If the option to update the reactive power limits in the working case was enabled, this updating takes place after all reporting, just prior to leaving activity GCAP. If activity GCAP is terminated by the user, this updating takes place for all machines that were reported.

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Updating reactive power limits by activity GCAP is not reversible. It is recommended that, prior to executing activity GCAP with the option to update reactive power limits enabled, the working case be preserved in a Saved Case File with activity SAVE, and that the new reactive power limit data be examined before permanently overwriting the original data.

Additional Information
PSS®E GUI Users Guide, Machine Reactive Capability *PSS®E* Command Line Interface (CLI) *Users Guide*, Machine Reactive Capability

12.22. Producing a Generator Bus Limits Report

GENS

Run Activity GENS - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Generator bus Tab
Run Line Mode Activity GENS - CLI
ACTIVITY? >>**GENS** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The generating plant summary activity GENS tabulates the loading and voltage conditions at generator buses. The user must choose among the following report options:

- VAR limited plants with unequal VAR limits: only plants at a reactive power limit and where var limits are unequal
- All VAR limited plants: all var-limited plants
- On-line plants: all in-service generator buses
- All plants: both in-service and out-of-service plants.

The user may also specify that voltage output is printed in alternate units rather than in the default units (refer to Section 3.3.3, "Program Run-Time Option Settings"and activity OPTN).

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output is generated with buses ordered as described in Section 4.9, "Subsystem Reporting".

The output listing of activity GENS may be terminated by entering the AB interrupt control code.

The report of activity GENS includes the bus number, name, base voltage, and type code, along with the number of machines modeled at the plant where status flags are 1, the number of machines where status flags are zero, the total plant power outputs and reactive power limits, and scheduled and actual voltages.

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If a plant regulates the voltage of a remote bus, the remote bus identifiers are printed and the actual voltage printed is that at the remote bus; if a plant regulates its own terminal voltage, the remote bus fields are blank. Total plant MBASE of machines at the bus where status flags are 1, along with the numbers of the zone and area in which the bus is located, are also printed. Any plant that is either a system or area swing is identified with an appropriate tag at the end of the output line.

As the last line of each report, subsystem totals of plant power outputs, var limits, and MBASE are tabulated.

Figure 11-21 shows a report for All plants from the *savnw.sav* Saved Case.

From the report it can be seen that:

- Generators at buses 206, 211 and 3018 are regulating voltage at remote buses.
- Generator at buses 206 and 3018 are failing to meet scheduled voltages (VSCHED) at their remote buses. Their actual voltages (VACTUAL) at the remote buses are below scheduled because the machines have reached maximum reactive power limits. Under the TYP column, the machines at buses 206 and 3018 are indicated to be -2; the negative sign indicates that the machines are at a reactive power limit.
- The system swing bus is 3011 while the swing buses for Areas 1 and 2 are Buses 101 and 206 respectively.

```
PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      MON, DEC 01 2003   9:16
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA
GENERATOR SUMMARY:
                          # MACH                                                    X----- REMOTE BUS ------X
  BUS# X-- NAME --X BASKV ON/OFF TYP    MW     MVAR    QMAX    QMIN  VSCHED VACTUAL   BUS# X-- NAME --X BASKV  MVABASE ZONE AREA SWING
   101 NUC-A       21.600  1  0   2   750.0    81.2   600.0  -100.0 1.0200  1.0200                            900.0   77    1  AREA
   102 NUC-B       21.600  1  0   2   750.0    81.2   600.0  -100.0 1.0200  1.0200                            900.0   77    1
   206 URBGEN      18.000  1  0  -2   800.0   600.0   600.0     0.0 0.9800  0.9490    205 SUB230      230.00  1000.0    2    2  AREA
   211 HYDRO_G     20.000  1  0   2   600.0    17.7   400.0  -100.0 1.0400  1.0400    201 HYDRO       500.00   725.0    2    2
  3011 MINE_G      13.800  1  0   3   258.7   104.0   600.0  -100.0 1.0400  1.0400                           1000.0    5    5  SYST
  3018 CATDOG_G    13.800  1  0  -2   100.0    80.0    80.0     0.0 1.0200  0.9586   3008 CATDOG      230.00   130.0    5    5
SUBSYSTEM TOTALS                    3258.7   964.2  2880.0  -400.0                                            4655.0
```

Figure 12.20. Generator Bus Report

Additional Information
PSS®E GUI Users Guide,Generator Bus Limits *PSS®E* Command Line Interface (CLI) *Users Guide*, Generator Bus Limits

12.23. Producing a Machine Terminal Limits Report

GEOL

Run Activity GEOL - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Machine terminal Tab
Run Line Mode Activity GEOL - CLI
ACTIVITY? >>**GEOL** ENTER OUTPUT DEVICE CODE:

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Interrupt Control Codes
AB

The machine terminal conditions summary activity GEOL tabulates the loading and voltage conditions at the generator terminals of in-service machines at Type 2 and 3 buses in the working case. The user specifies whether to restrict the output report to overloaded machines (refer to Section 12.23.1, "Application Notes") or to report conditions for all in-service machines (i.e., at buses with a type code of plus or minus 2, or 3) are listed.)

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV.

If an owner subsystem is specified, each machine wholly or partly owned by any of the owners specified is processed (refer to Non-Transformer Branch Data). The owner assignment of the bus to which the machine is connected is not considered.

The output listing of activity GEOL may be terminated by entering the AB interrupt control code.

Data presented includes the bus number, name, and base voltage, along with the machine identifier and the machine terminal conditions of power output, voltage, current, and power factor. Reactive power limits and the generator base, MBASE, are also printed, along with the step-up transformer impedance and off-nominal turns ratio if the transformer is represented as part of the generator model. The numbers of zone and area in which the bus is located are also printed, and machines at any plant which is either a system or area swing are identified with an appropriate tag at the end of the output line. Generators where loading is outside the assumed capability curve have an asterisk (*) printed following their current loading.

As the last line of each report, subsystem totals of machine power outputs, reactive power limits, and MBASE are tabulated.

Figure 12.21, "Report Tabulation for Machine Terminal Conditions in savnw.sav File"shows the Machine Terminal report selected for All on-line machines in the *savnw.sav* power flow case.

It can be seen that machines at buses 206 and 3018 are overloaded based on the calculations conforming to the assumed capability curve in Figure 12.22, "Assumed Capability Curve of Activity GEOL".

It should be noted that the terminal voltages for those two machines, and the machine at bus 211, are higher than their scheduled voltages (VSCHED) because these machines are attempting to control voltage at remote buses rather than at the machine terminals. The Generator Bus report shows the scheduled voltages and the remote buses (see Figure 12.20, "Generator Bus Report").

```
PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      MON, DEC 01 2003  10:08
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

MACHINE SUMMARY:
  BUS# X-- NAME --X BASKV ID     MW     MVAR    QMAX    QMIN  ETERM CURRENT    PF   MVABASE   X T R A N   GENTAP ZONE AREA SWING
   101 NUC-A       21.600 1    750.0    81.2   600.0  -100.0 1.0200   739.6  0.9942   900.0                        77    1  AREA
   102 NUC-B       21.600 1    750.0    81.2   600.0  -100.0 1.0200   739.6  0.9942   900.0                        77    1
   206 URBGEN      18.000 1    800.0   600.0   600.0     0.0 1.0236   976.9* 0.8000  1000.0                         2    2  AREA
   211 HYDRO_G     20.000 1    600.0    17.7   400.0  -100.0 1.0404   576.9  0.9996   725.0                         2    2
  3011 MINE_G      13.800 1    258.7   104.0   600.0  -100.0 1.0400   268.1  0.9278  1000.0                         5    5  SYST
  3018 CATDOG_G    13.800 1    100.0    80.0    80.0     0.0 1.0218   125.3* 0.7809   130.0                         5    5
SUBSYSTEM TOTALS             3258.7   964.2  2880.0  -400.0                       4655.0
```

Figure 12.21. Report Tabulation for Machine Terminal Conditions in savnw.sav File

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12.23.1. Application Notes

Activity GEOL calculates and reports the generator terminal conditions for in-service machines. When the machine step-up transformer is represented explicitly as a network branch, and hence the Type 2 bus is the generator terminal bus, activity GEOL simply uses the boundary conditions at the bus in generating the report (refer to Section 13.2.3, "Generator Step-Up Transformers" and Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent"b).

When the step-up transformer is modeled with the generator data (i.e., XTRAN is non-zero), the generator bus is the high side bus (see Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent"c). In this case, activity GEOL calculates back through the step-up transformer impedance to determine the conditions that exist at the generator terminals.

In performing its overload check, activity GEOL assumes a machine capability curve as shown in Figure 12.22, "Assumed Capability Curve of Activity GEOL". The radius of the semicircle in the upper half plane, centered at the origin, is taken as:

$$\frac{MBASE}{E_t}$$

When XTRAN is non-zero, the reactive power limits are modified to reflect these limits at the terminals by taking into account losses through the step-up transformer.

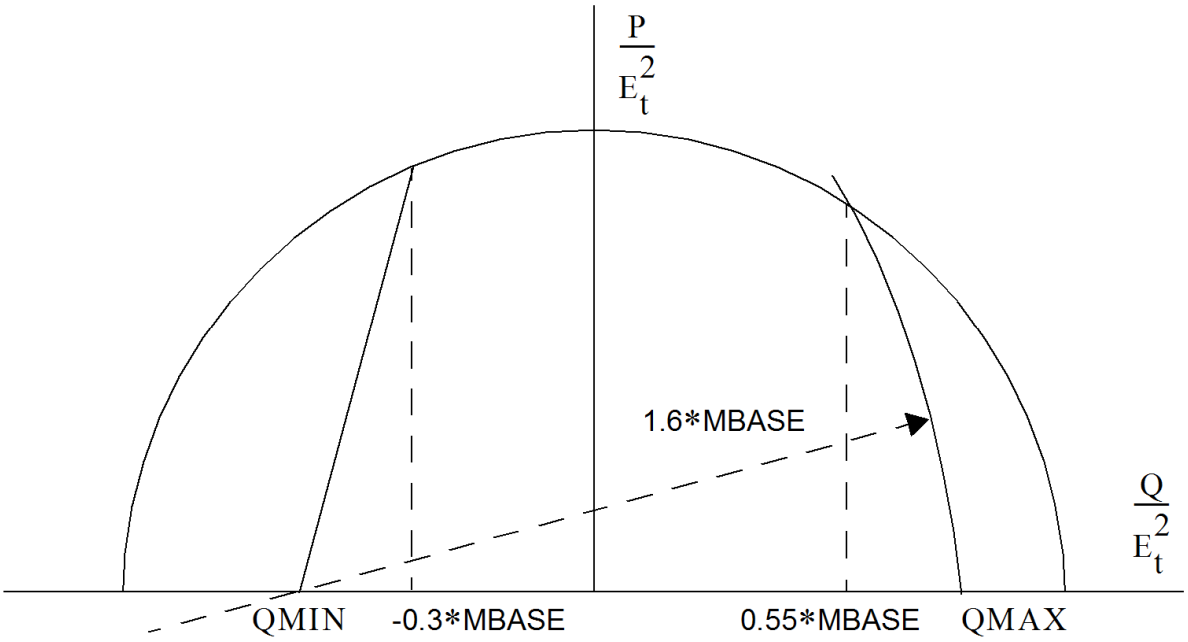

Figure 12.22. Assumed Capability Curve of Activity GEOL

When XTRAN is nonzero, the reactive power limits are modified to reflect these limits at the terminals by taking into account losses through the step-up transformer.

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The assumed operating region in the capability curve of Figure 12.22, "Assumed Capability Curve of Activity GEOL" is bounded as follows:

1. On the bottom, by the Q axis.
2. On the top, by the semicircle in the upper half plane.
3. On the left, by the straight line connecting the point QMIN on the Q axis and the point on the upper half plane circle corresponding to an abscissa of -0.3*MBASE. When QMIN is greater than -0.3*MBASE, this bound is a line parallel to the P axis through the point QMIN on the Q axis.
4. On the right, by a circle of radius 1.6 *MBASE, with center at (0,Qmin) and passing through the point QMAX on the Q axis and the point on the upper half plane circle corresponding to an abscissa of 0.55 *MBASE. When QMAX is less than 0.55*MBASE, this bound is a line parallel to the P axis through the point QMAX on the Q axis.

Any machine where the operating point is such that the quantity:

$$\frac{P_{gen} + jQ_{gen}}{(E_t)^2}$$

is outside this region is considered overloaded. Any machine absorbing active power is checked as if its active power is in the upper half plane.

For determining overloaded conditions, the value specified as MBASE for each machine must be the actual MVA base of the generator. The overload check is invalid for any machine for which the base is something other than this quantity, even if the base conversion has properly been taken into account in specifying ZSORCE and XTRAN.

When the report is restricted to overloaded machines, only those machines considered overloaded under the criteria described above are reported. When all on-line machines in the case (or in the specified subsystem thereof) are being reported, those for which loading is outside the assumed capability curve have an asterisk (*) printed following their current loading.

Additional Information
PSS®E GUI Users Guide, Machine Terminal Limits *PSS®E* Command Line Interface (CLI) *Users Guide*, Machine Terminal Limits

12.24. Producing a Regulated Bus Report

REGB

Run Activity REGB - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Regulated buses Tab
Interrupt Control Codes
AB

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In large databases it is important to have the ability to easily identify which bus voltages are being regulated and which equipment is controlling the regulation. This helps to avoid situations in which there are potential regulation conflicts.

The regulated bus reporting activity REGB tabulates those buses where voltages are controlled by generation, switched shunts, voltage controlling transformers, FACTS devices, and/or VSC dc line converters. Data presented includes the bus number, name, and base voltage of each regulated bus along with its present voltage magnitude. For each controlling equipment item, the desired voltage setpoint or voltage band, as appropriate, is listed along with any deviation between actual and scheduled voltages.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output for the voltage controlled buses contained in the specified subsystem is produced with buses ordered as described in Section 4.9, "Subsystem Reporting".

Only in-service voltage controlled buses (i.e., buses where type codes are less than 4) are reported by activity REGB. The following lists the conditions under which the various voltage controlling equipment items are reported:

Generation	Bus type code is 2 or 3, and at least one in-service machine is present.
Switched shunt	Bus type code is less than 4 and the control mode is 1 or 2.
Transformer	Branch is in-service and its adjustment control mode is set to +1.
FACTS device	Control mode is non-zero for sending bus end; control mode is 1, 5 or 7 for terminal bus end.
VSC dc converter	VSC dc line's control mode (MDC) is 1, the VSC converter's ac control mode (MODE) is 1, and the VSC converter's dc control code (TYPE) is not zero.

For any voltage controlling transformer where load drop compensation impedance (refer to Transformer Data) is non-zero, the controlled bus voltage (or, equivalently, the voltage limits) are compensated as described in Voltage Control. Activity REGB always displays voltage at the controlled bus. Therefore, limits printed for such transformers are the compensated limits and the lower limit is preceded by an asterisk (*). Note that the calculation of these limits requires transformer current; hence, they are valid only if the working case represents a solved system condition.

The output listing of activity REGB may be terminated by entering the AB interrupt control code.

A Regulated Bus report from the *savnw.sav* power flow case is shown in Figure 12.23, "Regulated Bus Report from the Savnw.sav Power Flow Case".

The report shows the following:

- Regulated buses 101, 102 and 3011 are generator terminal buses.
- Bus 154 is a type 1 bus where voltage is regulated by the transformer in the branch from bus 152 to bus 153. Furthermore the voltage control range of the transformer's tap changer is 0.98 to 1.0 pu but the controlled bus voltage is only 0.93892 pu. Consequently, a violation (or error) of -0.04108 pu is indicated.
- Bus 201 is a type 1 bus where voltage is regulated by the remote generator at bus 211. The scheduled voltage of 1.04 pu has been met.

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- Bus 205 is a type 1 bus where voltage is regulated by the remote generator at bus 206 and the transformer tap changer in the transformer branch from the 500 kV bus 204 to the 230 kV bus 205. The generator is scheduled to control voltage at bus 205 to a level of 0.98 pu. The transformer tap changer has a voltage range between 0.98 pu and 1.0 pu and is controlling bus 205. The actual voltage on bus 205 is only 0.949 pu. Consequently, there is a violation indicated of 0.03098 pu.

```
PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E      MON, DEC 01 2003  12:27
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

                          X--VOLTAGE--X X--SCHEDULED--X  X--VIOLATION--X
X---- REGULATED  BUS -----X   P.U.     KV      VOLTAGE        P.U.     KV   X------------------------ CONTROLLING  EQUIPMENT ------------X
   101 [NUC-A       21.600] 1.02000  22.0     1.02000                        LOCAL  GENERATION    101 [NUC-A    21.600]

   102 [NUC-B       21.600] 1.02000  22.0     1.02000                        LOCAL  GENERATION    102 [NUC-B    21.600]

   154 [DOWNTN      230.00] 0.93892 216.0 0.98000 1.00000 -0.04108   -9.45 XFRMER CKT 1  BUS    152 [MID500   500.00] TO 153 [MID230 230.00]

   201 [HYDRO       500.00] 1.04000 520.0     1.04000                        REMOTE GENERATION    211 [HYDRO_G  20.000]

   205 [SUB230      230.00] 0.94902 218.3     0.98000     -0.03098   -7.13 REMOTE GENERATION    206 [URBGEN   18.000]
                                          0.98000 1.00000 -0.03098   -7.13 XFRMER CKT 1  BUS    204 [SUB500   500.00] TO 205 [SUB230 230.00]

  3008 [CATDOG      230.00] 0.95861 220.5     1.02000     -0.06139  -14.12 REMOTE GENERATION   3018 [CATDOG_G 13.800]

  3011 [MINE_G      13.800] 1.04000  14.4     1.04000                        LOCAL  GENERATION   3011 [MINE_G   13.800]
```

Figure 12.23. Regulated Bus Report from the Savnw.sav Power Flow Case

Additional Information
PSS®E GUI Users Guide, Voltage Controlled Buses *PSS®E* Command Line Interface (CLI) *Users Guide*, Voltage-Controlled Buses

12.25. Producing a Controlling Transformer Report

TLST

Run Activity TLST - GUI
Power Flow > Reports > Limit checking reports... *[Limit Checking Reports]* Controlling Transformer Tab
Run Line Mode Activity TLST - CLI
ACTIVITY? >>**TLST** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The controlling transformer summary activity TLST tabulates those transformers in the working case for which off-nominal turns ratio or phase shift angle may be adjusted by the power flow solution activities.

The user may choose to list all adjustable transformers or restrict the report to those where the controlled quantity is outside of its specified band. The user may also specify that voltages and desired voltage bands

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of voltage controlling transformers are printed in alternate units for voltage output rather than in the default units (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output is generated, with all controlling transformers having at least one endpoint bus in the specified subsystem being processed.

If an owner subsystem has been specified, branch ownership rather than bus ownership is used in determining which transformers are contained in the specified subsystem; any other selection criteria apply to the transformer's endpoint buses.

The output listing of activity TLST may be terminated by entering the AB interrupt control code.

Two-winding transformers are listed in the report with their Winding 1 side bus first and with their Winding 2 side bus as the to bus. Three-winding transformer windings are listed in the report with their controlling winding side bus first and with their transformer name and winding number as the to bus; a three-winding transformer winding is processed if it is an adjustable winding and if the bus to which it is connected is in the specified subsystem.

Transformers are ordered in ascending numerical (using the numbers output option) or alphabetical (using the names option) order by controlling winding side bus, and, for each such bus, in ascending order by to bus field and circuit identifier. When subsystems are specified by area, zone, and/or owner, the output of activity TLST is grouped as described in Section 4.9, "Subsystem Reporting".

The output report for each subsystem grouping is printed in up to four sections.

1. Those transformer windings where tap ratio may be adjusted to control the voltage at a designated controlled bus are printed. The present controlled voltage and the desired voltage band are listed, along with the controlled bus, the tap ratio, the ratio limits, and the number of tap positions. When the load drop compensating impedance is non-zero, the compensated voltage is printed preceded by an asterisk (*) (refer to Transformer Data and Voltage Control); otherwise, the voltage at the controlled bus is reported.

2. A list of any transformer winding where phase shift angle may be adjusted to control the active power flow through itself. The data tabulated for these includes the present active power flow through the phase shifter and the desired flow band, the phase shift angle, the angle limits, the transformer control mode, and the winding connection angle.

3. A list of any transformer winding where ratio may be adjusted to control the reactive power flow through itself. The present reactive power flow and the desired flow band, the present tap ratio and its limits, and the number of tap positions are tabulated.

4. A list of those two-winding transformers where ratio may be adjusted to control the firing angle, extinction angle, or voltage of a dc line. The dc line name, the type of quantity controlled, its present value and desired band, the present tap ratio and its limits, and the tap step are tabulated.

Any controlled quantity outside of its desired band is followed by the tag HI or LO as appropriate. Similarly, if the ratio or phase shift angle of a transformer is at or beyond one of its limits, the appropriate identifier is printed.

See Figure 11-25 for an example of a Controlling Transformers report from the savnw.sav power flow case. Controlled buses at 154 and 205 show voltages below their transformers' control range (indicated by LO) and the Phase Shifting transformer controlling power flow shows a flow level above its control range (indicated by HI).

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```
     PTI
INTERACTIVE POWER SYSTEM SIMULATOR--PSSE      TUE, AUG 11 2009
11:46
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
                     TRANSFORMER
BASE CASE INCLUDING SEQUENCE DATA
                        SUMMARY
                    ** TRANSFORMERS CONTROLLING
VOLTAGE **


X--- ADJUSTABLE SIDE ---X X--------- TO ----------X
                                X--- CONTROLLED BUS ----X
  BUS# X-- NAME --X BASKV   BUS# X-- NAME --X
BASKV CKT     VOLT     MAX V   MIN V   BUS# X-- NAME --X BASKV
RATIO    MAX RAT MIN RAT NTAPOS
   152 MID500      500.00    153 MID230      230.00
 1   0.93892LO 1.00000 0.98000    154 DOWNTN      230.00 1.01000
  1.05000 0.95000     33
   204 SUB500      500.00    205 SUB230      230.00
 1   0.94902LO 1.00000 0.98000    205 SUB230      230.00 1.00000
  1.05000 0.95000     33


------------------------------------------------------------------------------


     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSSE
     TUE, AUG 11 2009  11:46
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
                     TRANSFORMER
BASE CASE INCLUDING SEQUENCE DATA
                        SUMMARY
                    ** PHASE SHIFTERS CONTROLLING
MW FLOW **


X--- ADJUSTABLE SIDE ---X X--------- TO ----------X
              MAXIMUM MINIMUM           MAXIMUM MINIMUM CONTROL
WINDING
  BUS# X-- NAME --X BASKV   BUS# X-- NAME --X
BASKV CKT     MW       MW      MW     ANGLE     ANGLE   ANGLE
MODE  CNX ANG
   202 EAST500     500.00    203 EAST230     230.00
 1    592.4HI   555.0   545.0    0.00     30.00  -30.00     3
   0.00
```

Figure 12.24. Controlling Transformers Report for savnw.sav Power Flow Case

Additional Information
PSS®E GUI Users Guide, Controlling Transformers

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Additional Information
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12.26. Comparing Power Flow Case Totals

CMPR

Run Activity CMPR - GUI
File > Compare *[Compare]* Case Totals tab *[Select Powerflow Comparison Options]*
Run Line Mode Activity CMPR - CLI
ACTIVITY? >>**CMPR** ENTER SAVED CASE FILENAME:
Interrupt Control Codes
AB

The case comparison activity CMPR compares the total system values contained in the working case with those of a designated Saved Case. Either system totals, area totals, zone totals, or owner totals may be compared. If the *Substitute working case after comparison* option is selected, then the currently open case will be replaced by the comparison case after the comparison has been completed.

One of the two cases to be compared must be present in working memory. When initiated, activity CMPR requires the user to designate the Saved Case to be compared against the working case. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity CMPR. Any errors encountered in accessing the designated Saved Case File are handled as in activity CASE. The user must specify which of the two cases is to remain as the PSS®E working case.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. When comparing cases by area or zone, the user may specify a difference threshold in either engineering units or percent (default = 0) for the following comparisons:

- generation
- load
- losses
- mismatch
- interchange

The selection of interchange is disabled when totals by owner or the entire case are being compared. The report for the designated category is printed in ascending area, zone, or owner number order, for those areas, zones, or owners selected for which difference meets or exceeds the designated threshold. The report for any category may be terminated by entering the AB interrupt control code, and a new category may then be selected.

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Generally, generation, load, loss, and mismatch subsystem totals are accumulated as shown in Table 12.3, "Subsystem Accumulations in Activity CMPR".

Table 12.3. Subsystem Accumulations in Activity CMPR

	Area Totals	Zone Totals	Owner Totals
Generation	Bus area	Bus zone	Machine owners
Load	Load area	Load zone	Load owner
Losses	Branch nonmetered end bus area	Branch nonmetered end bus zone	Branch owners
Mismatch	Bus area	Bus zone	Bus owner

When the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), generation and mismatch at multi-section line dummy buses are assigned to the area, zone, or owner of the multi-section line nonmetered end bus. When totals by area or zone are being calculated, losses on each multi-section line member are assigned to the area or zone of the multi-section line nonmetered end bus; for totals by owner, losses are always assigned to the branch owners.

In calculating area or zone interchange when the multi-section line reporting option is enabled (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN), the area and zone assignments of the dummy buses of each multi-section line grouping and the metered ends of the individual members of each multi-section line grouping are ignored. Instead, net interchange is evaluated on the basis of the metered end designated for each multi-section line grouping which is a tie branch (i.e., its endpoint buses are located in different areas or zones, as appropriate). When the multi-section line reporting option is disabled, the area or zone assignments of dummy buses as well as the metered end of each member of a multi-section line grouping are recognized; the multi-section line grouping definitions and their metered end information are ignored.

In calculating area or zone interchange, a three-winding transformer is considered to be a tie line when the three buses which is connects are not in the same area or zone, as appropriate. Tie flows are derived from the flows into the transformer at its two metered ends. In calculating area or zone interchange, only tie line flows are considered, even if the interchange definition flag is set to its *tie lines plus loads* setting (refer to Area Interchange Control).

A typical report produced by the comparison is shown in Figure 12.25, "Result of the Compare Case Totals". The report compares the total generation, load, losses and solution mismatches. Differences in the totals for each case as also presented (under the DELTA columns).

```
                IN WORKING CASE    IN C:\Program Files\PTI\PSSE30\EXAMPLE\jdmsavnw.sav
                   MW       MVAR       MW       MVAR  DELTA MW    %  DELTA MVAR    %
GENERATION      3258.7     964.2     3266.3     972.6       7.6   0.2        8.4   0.9
LOAD            3200.0    1950.0     3207.5    1953.9       7.5   0.2        3.9   0.2
LOSS              58.7    1115.7       58.8    1118.0       0.1   0.2        2.3   0.2
MISMATCH             0.018 MVA            0.010 MVA                   -0.008 MVA  43.6
```

Figure 12.25. Result of the Compare Case Totals

Additional Information
PSS®E GUI Users Guide, Comparing Power Flow Case Totals

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Additional Information
PSS®E Command Line Interface (CLI) *Users Guide*, Power Flow Case Totals

12.27. Comparing Power Flow Cases

DIFF

Run Activity DIFF - GUI
File > Compare *[Compare]* Powerflow Cases tab *[Select Powerflow Comparison Options]*
Run Line Mode Activity DIFF - CLI
ACTIVITY? >>**DIFF** ENTER SAVED CASE FILENAME:
Interrupt Control Codes
AB

The case comparison activity DIFF compares certain parametric data and solution results, as contained in the working case, with those of a designated Saved Case. Network elements in the entire case may be subjected to the comparison, or the checks may be restricted to those within a designated subsystem.

The DIFF API routine includes the following among its initial inputs:

- A designation of the subsystem in which data is to be compared; refer to Section 4.8, "Subsystem Selection".
- The name of a Saved Case File whose data is to be compared to that of the working case.
- The method to be used for matching up buses in the two cases. To be considered the same bus, buses must have either:
- the same bus number;
- the same extended bus name; or
- both the same bus number and the same extended bus name.
- The content of the working case at the completion of activity DIFF. The working case may contain either the case that was present at the time DIFF was initiated, or the designated Saved Case which was to be compared against the working case may become the new working case. This is used only if the initialization of activity DIFF fails because there are no buses to compare.

The DIFF API routine is then called once for each quantity that is to be compared. In these calls, the following is included among its inputs:

- the type of comparison to be performed.
- for load and rating comparisons, an additional input that further defines the quantities to be compared.

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- for several comparisons, one or more threshold tolerances.

The DIFF API routine is called one final time to complete the comparision function. In this call, the following input is specified.

- The content of the working case at the completion of activity DIFF. The working case may contain either the case that was present at the time DIFF was initiated, or the designated Saved Case which was compared against the working case may become the new working case.

In the GUI, the DIFF API routine is accessed from the *Powerflow Cases* tab of the *[Compare]* dialog. In line mode, case comparisons are preformed using activity DIFF.

12.27.1. Operation of Activity DIFF

One of the two cases to be compared must be present in working memory. When initiated, activity DIFF requires the user to designate the Saved Case to be compared against the working case. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity DIFF. Any errors encountered in accessing the designated Saved Case File are handled as in activity CASE.

If buses in the two cases are matched using only their extended bus names, each bus in the specified subsystem must have a unique extended bus name. Alarms are printed at the *Progress* device (refer to Section 4.4, "Virtual Output Devices") if there are duplicate extended bus names in the specified subsystem of either the working case or the designated Saved Case. Any buses that have generated alarms are treated as being outside of the specified subsystem, and processing continues using the remaining subsystem buses in the two cases.

A heading page is printed at the selected output device containing the case headings of the working case and the comparison case.

Activity DIFF constructs a bus comparison list containing buses that exist in both cases with the same bus number and/or extended bus name, as appropriate, and are in the specified subsystem in both cases, if subsystem specification was enabled. Star point buses of three-winding transformers are always excluded from the bus comparison list.

Except for star point buses of three winding transformers, any bus from either case that is in the selected subsystem, but not included in the bus comparison list, generates an alarm. This could occur, for example, if a bus is contained in the specified subsystem in one case but not in the other. These messages may be suppressed by entering the AB interrupt control code.

Activity DIFF then constructs a substation comparison list. It contains substations that exist in both cases with the same substation number, and that have at least one bus from the specified subsystem in both of the cases. Then a substation node list is constructed which contains all nodes in the two cases assigned to substations in the substation comparison list. Then a substation switching device comparison list is constructed. It contains all switching devices in the two cases connecting nodes in the node comparison list.

Activity DIFF summarizes the number of buses, branches, substations, nodes, and switching devices in the working case and in the comparison case, the number of buses in the specified subsystem in each case if subsystem specification was enabled, and the number of buses, substations, nodes, and switching devices in the bus, substation, node, and switching device comparison lists, respectively.

It constructs a branch comparison list, which contains all ac branches from either case which connect buses in the bus comparison list. The number of such branches is then tabulated. Three-winding transformer windings are included in the branch comparison list if all three buses are included in the bus comparison list.

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Finally, it constructs a multi-section line comparison list, which contains all multi-section line groupings from either case for which the endpoint buses are contained in the bus comparison list. The number of such multi-section line groupings is then tabulated.

The area, zone, and/or owner assignments of buses are used to determine which buses and branches are to be included in the bus and branch selection lists when subsystem selection by area, zone, and/or owner is enabled. The owner assignments of machines and ac branches, and the area, zone, and owner assignments of loads, are not referenced by activity DIFF.

For several of the data categories that may be compared (see below), the user specifies a threshold tolerance. If the difference between data items in the two cases is within this tolerance, the item is omitted from the output report. The user may specify the tolerance threshold in either percent or engineering units. Comparisons between cases may be specified for the following categories:

Table 12.4. Case Comparison Specifications

Category	Additional Specifications
Bus identifiers	none
Bus type codes	none
Machine status	none
Plant MW	Comparison Threshold in Percent or Engineering Units
Plant MVAR	Comparison Threshold in Percent or Engineering Units
Plant MW or MVAR	Comparison Threshold in Percent or Engineering Units
Loads	Comparison Threshold in Percent or Engineering Units Bus load characteristics • Total nominal load • Constant MVA load • Constant current load • Constant admittance load
Load status	none
Scalable load flags	none
Negative sequence loads	Comparison Threshold in Percent or Engineering Units
Zero sequence loads	Comparison Threshold in Percent or Engineering Units
Fixed bus shunts	Comparison Threshold in Percent or Engineering Units
Fixed bus shunt status	none
Switched shunts	Comparison Threshold in Percent or Engineering Units

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Category	Additional Specifications
Switched shunt status	none
Voltage	Comparison Threshold in Percent or Engineering Units Voltage threshold
Voltage and angle	Comparison Threshold in Percent or Engineering Units Voltage threshold Angle threshold
MBASE & ZSORCE	none
MBASE, RPOS and X"	none
MBASE, RPOS and X'	none
MBASE, RPOS and Xs	none
MBASE & ZNEG	none
MBASE & ZZERO	none
Bus substation assignments	none
Branch status	none
Line R, X, B	none
Line shunts	none
Line ratings	Ratings to compare: • All ratings • Rate A • Rate B • Rate C
Metered end	none
Line lengths	none
Transformers	Ratio threshold Angle threshold
Flows MW or MVAR (from bus)	Comparison Threshold in Percent or Engineering Units
Flows MW (from bus)	Comparison Threshold in Percent or Engineering Units
Flows MVAR (from bus)	Comparison Threshold in Percent or Engineering Units
Flows MW or MVAR (from & to)	Comparison Threshold in Percent or Engineering Units
Flows MW (from & to)	Comparison Threshold in Percent or Engineering Units

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Category	Additional Specifications
Flows MVAR (from & to)	Comparison Threshold in Percent or Engineering Units
Line MW or MVAR losses	Comparison Threshold in Percent or Engineering Units
Line MW losses	Comparison Threshold in Percent or Engineering Units
Line MVAR losses	Comparison Threshold in Percent or Engineering Units
Zero sequence R,X,B	none
Zero sequence line shunts	none
Connection codes	none
Zero sequence mutuals	none
Multi-section lines	none
Multi-section line metered end	none
Substation node status	none
Substation switching device status	none
Substation switching device flow MW or MVAR	Comparison Threshold in Percent or Engineering Units
Substation switching device flow MW	Comparison Threshold in Percent or Engineering Units
Substation switching device flow MVAR	Comparison Threshold in Percent or Engineering Units

The selected check is made and exceptions are listed as described in Section 12.27.2, "Bus Related Data Checks", Section 12.27.3, "Branch Related Data Checks" and Section 12.27.4, "Substation Related Data Checks". The report for any category may be terminated by entering the AB interrupt control code, and a new category may then be selected.

For those categories that allow the user to specify a threshold tolerance, activity DIFF generates an alarm only for those buses or branches for which the designated data item(s) in the two cases differ by more than the specified tolerance, whether the difference threshold is specified in percent or in engineering units. In both cases, the default is zero.

For categories comparing fault analysis data, this data must be present in both cases.

12.27.2. Bus Related Data Checks

Table 12.5. Case Comparison Specifications, Bus Data Checks

Category	Comparison
Bus identifiers, Bus type codes, Machine status, Plant MW, Plant MVAR, Plant MW or MVAR, Loads, Load status, Scalable load flags, Negative sequence loads, Zero sequence loads, Fixed bus shunts, Fixed bus shunt status, Switched shunts, Switched shunt status, Voltage, Voltage and angle, MBASE & ZSORCE, MBASE, RPOS & X", MBASE, RPOS & X',	Check various bus related data items of those buses contained in the bus comparison list. Exceptions are tabulated in ascending bus numerical or alphabetical order, according to the bus output option in effect at the time activity DIFF is selected (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN).

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Category	Comparison
MBASE, RPOS & Xs, MBASE & ZNEG, MBASE & ZZERO, Bus substation assignments	For each bus listed, the appropriate data item(s) from both cases are printed, along with the bus number and extended bus name as contained in the working case.
Plant MW, Plant MVAR, Plant MW or MVAR, Loads, Negative sequence loads, Zero sequence loads, Fixed bus shunts, Switched shunts	Differences between the two cases (Saved Case value minus working case value) are tabulated; differences as a percent of working case values are also printed.
Bus identifiers	Buses in the bus comparison list with different numbers, names, or base voltages are listed.
Bus type codes	Activity DIFF does not report a bus that has a type code of 2 in one case and -2 in the other (refer to Bus Data).
Machine status check	If a machine exists in both cases, it is reported only if its actual status, as implied by the combination of the bus type code and the machine status flag, is different in the two cases (refer to Section 5.10.1, "Equipment Status Changes"). When a machine exists in only one of the cases, it is reported regardless of its implied status. For any machine that is reported, the bus type code and machine status flag from both of the cases are listed.
Plant MW	Reoprts those generator buses for which the total plant active power output differs in the two cases by more than a specified tolerance. If one case contains in-service generation while in the other either no generation is modeled at the bus or it is out-of-service, the bus is reported only if the in-service generation exceeds the threshold tolerance.
Plant MVAR	Similar to *Plant MW* except that a bus is listed if the reactive generator power difference in the two cases exceeds the threshold tolerance.
Plant MW or MVAR	Similar to *Plant MW*, except that a bus is listed if either the active or reactive generator power difference in the two cases exceeds the threshold tolerance.
Loads	Compares either the total in-service constant MVA, constant current, or constant admittance load, or the total in-service nominal load, at each bus in the bus comparison list. If one case contains in-service load while in the other either no load elements are modeled at the bus or they are all out-of-service, the bus is reported only if the in-service load exceeds the threshold tolerance.
Load status	If a load exists in both cases, it is reported only if its actual status, as implied by the combination of the bus type code and the load status flag, is different in

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Category	Comparison
	the two cases. Where a load exists in only one of the cases, it is reported regardless of its implied status. For any load that is reported, the bus type code and the load status flag from both cases are listed.
Scalable load flags	If a load exists in both cases, it is reported only if its scalable load flag is different in the two cases. Where a load exists in only one of the cases, it is reported. For any load that is reported, the bus type code and the scalable load flag from both cases are listed.
Negative sequence loads	Compare negative sequence loads (refer to Load Sequence Data), against a designated threshold tolerance.
Zero sequence loads	Compare zero sequence loads (refer to Load Sequence Data), against a designated threshold tolerance.
Fixed bus shunts	Compares total in-service fixed bus shunts against a designated threshold tolerance. If one case contains in-service fixed shunt while in the other either no fixed shunt elements are modeled at the bus or they are all out-of-service, the bus is reported only if the in-service fixed shunt exceeds the threshold tolerance.
Fixed bus shunt status	If a fixed bus shunt exists in both cases, it is reported only if its actual status, as implied by the combination of the bus type code and the shunt status flag, is different in the two cases. Where a fixed bus shunt exists in only one of the cases, it is reported regardless of its implied status. For any fixed bus shunt that is reported, the bus type code and the shunt status flag from both cases are listed.
Switched shunts	If a bus has a switched shunt in only one of the two cases, it is reported if its magnitude exceeds the tolerance. If the threshold tolerance is specified as zero, the bus is reported even if the value of the switched shunt in the case in which it exists is zero.
Switched shunt status	If a switched shunt exists in both cases, it is reported only if its actual status, as implied by the combination of the bus type code and the shunt status flag, is different in the two cases. Where a switched shunt exists in only one of the cases, it is reported regardless of its implied status. For any switched shunt that is reported, the bus type code and the shunt status flag from both cases are listed.
Voltage	Reports buses where the voltage magnitudes differ by more than a specified threshold. Differences between the two cases (Saved Case value minus working case value) are tabulated. Any bus where the

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Category	Comparison
	type code is 4 in both cases is skipped. If a bus has a type code of 4 in only one of the two cases, it is reported.
Voltage and angle	Reports buses for which either the voltage magnitudes differ by more than a specified tolerance or the phase angle difference exceeds a designated threshold. Differences between the two cases (Saved Case value minus working case value) are tabulated. Any bus where the type code is 4 in both cases is skipped. If a bus has a type code of 4 in only one of the two cases, it is reported.
MBASE & ZSORCE	Compares machines connected to buses in the bus comparison list (refer to Generator Data and Section 13.2.2, "Generator Apparent Impedance"). If a machine exists in both cases, it is reported if either MBASE or ZSORCE is different. Where a machine exists in only one of the cases, it is reported.
MBASE, RPOS & X"	Compare machine MBASE, RPOS and subtransient reactance used in fault analysis work (refer to Generator Sequence Data). This check is handled in the same manner as the ZSORCE comparison described above.
MBASE, RPOS & X'	Compare machine MBASE, RPOS and transient reactance used in fault analysis work (refer to Generator Sequence Data). This check is handled in the same manner as the ZSORCE comparison described above.
MBASE, RPOS & Xs	Compare machine MBASE, RPOS and synchronous reactance used in fault analysis work (refer toGenerator Sequence Data). This check is handled in the same manner as the ZSORCE comparison described above.
MBASE & ZNEG	Compare machine MBASE and the negative sequence impedance used in fault analysis work (refer to Generator Sequence Data). This check is handled in the same manner as the ZSORCE comparison described above.
MBASE & ZZERO	Compare machine MBASE and the zero sequence impedance used in fault analysis work (refer to Generator Sequence Data). This check is handled in the same manner as the ZSORCE comparison described above.
Bus substation assignment	Compare the substation number assigned to the bus in the two cases. Reports buses where the substation assignments differ. The substation number and name in both cases is reported.

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12.27.3. Branch Related Data Checks

Table 12.6. Case Comparison Specifications, Branch Data Checks

Category	Comparison
Branch status, Line R,X,B, Line shunts, Line ratings, Metered end, Line lengths, Transformers, Flows MW or MVAR (from bus), Flows MW (from bus), Flows MVAR (from bus), Flows MW or MVAR (from & to), Flows MW (from & to), Flows MVAR (from & to), Line MW or MVAR losses, Line MW losses, Line MVAR losses, Zero sequence, R,X,B, Zero sequence line shunts, Connection codes, Zero sequence mutuals, Multi-section lines, Multi-section line metered end	Check various branch related data items of branches in the branch and multi-section line comparison lists. Exceptions are tabulated in single entry format (refer to Branch Data), ordered in ascending bus numerical or alphabetical order according to the bus output option in effect at the time activity DIFF is selected (refer to Section 3.3.3, "Program Run-Time Option Settings"). For branches that are present in both cases, bus identifiers and ordering from the working case are used.
Branch status	If a branch appears in both cases, it is reported if the status flags differ. Where a branch exists in only one of the cases, it is reported regardless of its status.
Line R, X, B	Checks for different values of line impedance or charging. If a branch exists in only one of the cases, it is reported. If a branch has the same value of impedance in the two cases but is treated as a zero impedance line in only one of the two (i.e., THRSHZ differs in the two cases; refer to Zero Impedance Lines), it is reported. If a branch is treated as a zero impedance line in both cases, the impedances are considered equal regardless of their values in the two cases. For three-winding transformer windings, winding impedances are compared.
Line shunts	If a branch exists in only one of the cases, it is reported only if the line shunt at either end of the branch is non-zero.
Line ratings	For branches for which the selected rating(s) differ, all three ratings from both cases are tabulated. If a branch exists in only one of the cases, it is excluded from the report.
Metered end	Reports branches with opposite metered ends in the two cases. If a branch exists in only one of the cases, it is excluded from the report.
Line lengths	Reports branches where lengths differ in the two cases. If a branch exists in only one of the cases, it is excluded from the report.
Transformers	Two-winding transformers and individual windings of three-winding transformers are compared. If a branch exists in both of the cases, it is reported under any of the following conditions:

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Category	Comparison
	• It is a two-winding transformer or a three-winding transformer winding in one case, but not in the other. • It is a two-winding transformer in both cases, but the Winding 1 side and Winding 2 side relationship is reversed. • It is a two-winding transformer in both cases and its Winding 1 tap ratio or phase shift angle, or Winding 2 tap ratio, differs.
Transformers (*continued*)	• If a branch exists in one case and not in the other, it is reported if it is a transformer. For two-winding transformers, this report lists the tap ratios of both windings and the Winding 1 phase shift angle from each case, along with an indication of which bus is the Winding 1 side bus; for three-winding transformers, the winding ratio is listed in the RATIO1 column and the RATIO2 column is left blank. Differences in tap ratios and phase shift angles for transformers which are present in both cases are also tabulated.
Flows MW or MVAR (from bus)	Compares line flows against a designated threshold tolerance. A branch in the branch comparison list is reported under any of the following conditions: • It is in-service in both cases and the active or reactive power flow at the from bus end in the two cases differs by more than the threshold tolerance. • It exists in both cases with different service status and its active or reactive power flow at the from bus end in the in-service case exceeds the threshold. • It exists in only one of the cases, is in-service, and its active or reactive power flow at the from bus end exceeds the threshold. This report tabulates the active and reactive power flowing into the from bus end of the line in the two cases. Flow differences between the two cases are printed for each branch reported, both in MW and Mvar and as percentages of the working case flows. For this comparison, both cases should be solved to an acceptable mismatch level.

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Category	Comparison
Flows MW (from bus)	Similar to *Flows MW or MVAR (from bus)* except that a branch is listed if the active power flow difference in the two cases exceeds the threshold tolerance.
Flows MVAR (from bus)	Similar to *Flows MW or MVAR (from bus)* except that a branch is listed if the reactive power flow difference in the two cases exceeds the threshold tolerance.
Flows MW or MVAR (from & to)	similar to *Flows MW or MVAR (from bus)* except the flows at both ends of each line are checked against the threshold. For branches reported, from bus end conditions are shown on the first output line and to bus end conditions on the second.
Flows MW (from & to)	Similar to *Flows MW or MVAR (from & to)* except that a branch is listed if the active power flow difference at either end in the two cases exceeds the threshold tolerance.
Flows MVAR (from & to)	Similar to *Flows MW or MVAR (from & to)* except that a branch is listed if the reactive power flow difference at either end in the two cases exceeds the threshold tolerance.
Line MW or MVAR losses	Compares line losses that are calculated as I^2R and I^2X and *exclude* the line charging, line connected shunt, and transformer magnetizing admittance components. A branch in the branch comparison list is reported under any of the following conditions: • It is in-service in both cases and the active or reactive power losses in the two cases differ by more than the threshold tolerance. • It exists in both cases with different service status and its active or reactive power losses in the in-service case exceeds the threshold. • It exists in only one of the cases, is in-service, and its active or reactive power losses exceeds the threshold. For each branch reported, differences in losses between the two cases are printed, both in MW and Mvar and as percentages of the working case losses.
Line MW losses	Active power losses is similar to *Line MW or MVAR losses* except that a branch is listed if the active power loss difference in the two cases exceeds the threshold tolerance.
Line MVAR losses	Reactive power losses is similar to *Line MW or MVAR losses* except that a branch is listed if the reactive

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Category	Comparison
	power loss difference in the two cases exceeds the threshold tolerance.
Zero sequence R,X,B	Checks for different values of zero sequence line impedance or charging. If a branch exists in only one of the cases, it is included in the report. If a branch has the same value of zero sequence impedance in the two cases but is treated as a zero impedance line in only one of the two (i.e., THRSHZ differs in the two cases; refer to Section Zero Impedance Lines), it is included in the report. If a branch is treated as a zero impedance line in both cases, the zero sequence impedances are considered equal regardless of their values in the two cases. For transformers, the impedances used are those of the "pi" equivalent representing the transformer's zero sequence connections. It is calculated using the transformer's various series and ground impedances and its conection code. For three-winding transformers, winding values are compared.
Zero sequence line shunts	Checks for different values of zero sequence line shunts. If a branch exists in only one of the cases, it is reported only if the zero sequence line shunt at either end of the branch is non-zero. For transformers, the admittances used are those of the "pi" equivalent representing the transformer's zero sequence connections. It is calculated using the transformer's various series and ground impedances and its conection code. For three-winding transformers, winding values are compared.
Connection codes	Checks for different values of the transformer connection code. If a branch exists in both of the cases, it is reported under any of the following conditions: • It is a two-winding transformer or a three-winding transformer winding in one case, but not in the other. • It is a two-winding transformer in both cases, but the Winding 1 side and Winding 2 side relationship is reversed. • It is a two-winding transformer in both cases, and its winding connection codes differ. • It is a winding of a three-winding transformer in both cases, and the connection codes applied at the winding differ.

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Category	Comparison
	If a branch exists in one case and not in the other, it is reported if it is a transformer. This report lists the connection code from each case, along with an indication of which bus is the Winding 1 side bus.
Zero sequence mutuals	For mutuals to be checked, the branches coupled must both be members of the branch comparison list. If a mutual coupling exists in one case but not in the other, it is reported. If the mutual is in both cases, it is reported if its mutual impedance, its geographic B factors, or its polarity is not identical in the two cases (refer to Zero Sequence Mutual Impedance Data). The report lists one of the branches along with the mutual data on one line, followed by a line listing the other branch.
Multi-section lines	Compares multi-section groupings for multi-section lines in the multi-section line comparison list (refer to Multi-Section Line Grouping Data). If a multi-section line grouping exists in one case but not in the other, it is reported. If a multi-section line grouping exists in both cases, it is reported under either of the following conditions: • Any of its dummy buses in either case is not contained in the bus comparison list. • It contains different members in the two cases. The report lists the endpoint buses and multi-section line identifier, with the bus identifiers taken from the working case. It also lists the members of the multi-section line as contained in the working case (using bus identifiers from the working case) and in the comparison case (using bus identifiers from the comparison case).
Multi-section line metered end	Reports multi-section line groupings with opposite metered ends in the two cases. If a multi-section line exists in only one of the cases, it is excluded from the report.

12.27.4. Substation Related Data Checks

Table 12.7. Case Comparison Specifications, Substation Data Checks

Category	Comparison
Substation node status	Compare the status of nodes in the node compare list. If a substation node appears in both cases, it is reported if the status settings differ. Where a substation node exists in only one of the cases, it is reported regardless of its status. Exceptions are tablu-

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Category	Comparison
	ated in ascending substation number order, and, for each substation, in ascending node number order. The substation and node identifiers from the working case are used.
Substation switching device status, Substation switching device flow MW or MVAR, Substation switching device flow MW, Substation switching device flow MVAR	Check various data items of switching devices in the switching device compare list. Exceptions are tabulated in ascending substation number order, and, for each substation, in single entry format in ascending node number order by from node, and, for each from node, in ascending order by to node. Parallel devices between any pair of nodes are tabulated in ascending switching device identifier order. Any reported switching device appears once in the listing: with the lower node number listed as the from node. For switching devices that are present in both cases, the substation, node and switching device identifiers from the working case are used.
Substation switching device status	Compare the status of switching devices in the switching device compare list. If a substation switching device appears in both cases, it is reported if the status settings differ. Where a substation switching device exists in only one of the cases, it is reported regardless of its status.
Substation switching device flow MW or MVAR	Compares substation switching device flows against a designated threshold tolerance. A device in the substation switching device comparison list is reported under any of the following conditions: • It is in-service in both cases and the active or reactive power flow at the from node end in the two cases differs by more than the threshold tolerance. • It exists in both cases with different service status and its active or reactive power flow at the from node end in the in-service case exceeds the threshold. • It exists in only one of the cases, is in-service, and its active or reactive power flow at the from node end exceeds the threshold. This report tabulates the active and reactive power flowing into the from node end of the switching device in the two cases. Flow differences between the two cases are printed for each switching device reported, both in MW and Mvar and as percentages of the working case flows. For this comparison, both

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Category	Comparison
	cases should be solved to an acceptable mismatch level.
Substation switching device flow MW	Similar to *Substation switching device flow MW or MVAR* except that a switching device is listed if the active power flow difference in the two cases exceeds the threshold tolerance.
Substation switching device flow MVAR	Similar to *Substation switching device flow MW or MVAR* except that a switching device is listed if the reactive power flow difference in the two cases exceeds the threshold tolerance.

Additional Information
PSS®E GUI Users Guide, Comparing Power Flow Cases *PSS®E* Command Line Interface (CLI) *Users Guide*, Power Flow Cases

12.28. Displaying Power Flow Solution Differences on a Diagram

GDIF

Run Activity GDIF - GUI
Diagram > Results > Graphical difference data...
Run Line Mode Activity GDIF - CLI
ACTIVITY? >>**GDIF** ENTER COORDINATE FILE NAME, BINARY OPTION (-1 TO EXIT): >>
Interrupt Control Codes
AB

The graphical case comparison activity GDIF calculates differences in power flow solution results and certain power flow boundary condition data contained in the working case with those of a designated Saved Case. Results are produced in the Diagram View in a form similar to the display of power flow output.

Prior to initiating activity GDIF, one of the two cases to be compared must be present in the working case.

If buses in the two cases are matched using only their extended bus names, each bus must have a unique extended bus name. Alarms are printed at the dialog output device (refer to Section 4.4, "Virtual Output Devices") if there are duplicate extended bus names in either the working case or the designated Saved Case.

A heading page is printed at the *Progress* tab containing the case headings of the working case and the comparison case.

GDIF builds a diagram bus list of buses where difference quantities are to be calculated. A bus comparison list is then constructed. It contains a list of buses for which the following are all true:

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1. They are contained in both cases with the same bus number and/or extended bus name, as appropriate.
2. They are in the diagram bus list.
3. They did not generate an alarm because they had duplicate extended bus names (see above) in either the working case or the comparison case.

Any bus from the working case which is in the diagram bus list but which is not included in the comparison case generates an alarm.

Activity GDIF summarizes the number of buses and branches in the working case and in the comparison case, the number of buses from each case in the diagram bus list, and the number of buses in the bus comparison list.

It constructs a branch comparison list that contains all ac branches from either case that connect buses in the bus comparison list. The number of such branches, three-winding transformers, and multi-section lines is then tabulated.

Activity GDIF calculates differences in solution results for equipment at all buses in the bus comparison list and for all branches in the branch comparison list. Activity GDIF generates the diagram showing the case comparison results.

The output of activity GDIF consists of a single one-line diagram showing the differences in solution results and bus boundary conditions between the working case and the designated Saved Case. Differences are always calculated as comparison case values minus working case values. At each bus in the bus comparison list, voltage difference in per unit and phase angle difference in degrees are shown. All other difference values are shown in MW and Mvar.

The difference fields are left blank for equipment items present in the working case, but not in the comparison case. Those items present in the comparison case that are not included in the working case are omitted from the diagram because it is the working case and its diagram definition data which determine the network elements drawn on the one-line diagram.

Load and shunt differences include voltage sensitivity effects.

Differences in flow into a converter bus of dc line <name> are shown if all of the following conditions apply:

1. DC line <name> is present in both cases.
2. The converter bus is in the bus compare list.
3. The same converter bus is specified in both cases.

Differences in the sending bus shunt element of FACTS device <name> are shown if all of the following conditions apply:

1. FACTS device <name> is present in both cases.
2. The sending bus is in the bus compare list.
3. The same sending bus is specified in both cases.

If the above conditions are satisfied and FACTS device <name> has a series element in the working case, differences in series flow at the sending bus are also shown.

Differences in series flow at the terminal bus are shown if all of the following conditions apply:

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1. FACTS device <name> is present and has a series element in both cases.
2. The terminal bus is in the bus compare list.
3. The same terminal bus is specified in both cases.

Activity GDIF lists the name of the Saved Case File against which the working case is being compared.

Additional Information
PSS®E Command Line Interface (CLI) *Users Guide*, Displaying Power Flow Solution on the Diagram

12.29. Comparing AC Tie Branches

DFTI

Run Activity DFTI - GUI
File > Compare... *[Compare]* Tie Lines Tab
Interrupt Control Codes
AB

The tie line case comparison activity DFTI compares power flow branch data and solution results contained in the working case with those of a designated Saved Case. Only ac tie branches from the designated subsystem are subjected to processing and reporting.

Prior to selecting activity DFTI, one of the two cases to be compared must be present in the working case. The report of activity DFTI is directed to the Progress tab unless an output destination had been preselected via activity OPEN.

The following input data defining comparison and reporting options is specified to DFTI:

- The subsystem where tie branches are to be compared.
- The name of the Saved Case File for which tie branches from the specified subsystem are to be compared to those of the working case.
- The method to be used in matching up buses in the two cases.
- Data defining the quantities to be compared and any appropriate comparison options.
- The case to be left in PSS®E working memory at the termination of activity DFTI (i.e., the original working case or the specified Saved Case).

Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity DFTI. Any errors encountered in accessing the designated Saved Case File are handled as in activity CASE. Following the successful pickup of the comparison case, a heading page containing the case headings of the working case and the comparison case is printed at the output device.

Activity DFTI constructs a table of tie point buses in the two cases. Then, for both the working case and the comparison case, it summarizes the number of buses and branches in the case, the number of tie point buses in the specified subsystem, and the number of tie point buses outside of the specified subsystem.

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It then matches up tie point buses in the two cases using the method specified by the user (same bus number, same extended bus name, or both the same bus number and same extended bus name). If buses in the two cases are matched using only their extended bus names, each tie point bus must have a unique extended bus name. Alarms are printed at the dialog output device (refer to Section 4.4, "Virtual Output Devices") if there are duplicate extended bus names of tie point buses in either the working case or the designated Saved Case. Any tie point buses that generated an alarm are ignored in the subsequent processing and reporting of activity DFTI.

Tie point bus discrepancies between the two cases are then reported at the output device.

It then constructs a tie branch comparison list, which contains any ac branch from either case which connects a subsystem tie point bus to a non-subsystem tie point bus. Three-winding transformers which have at least one winding connected to a subsystem tie point bus and at least one winding connected to a non-subsystem tie point bus have those windings which are connected to subsystem tie point buses included in the tie branch comparison list.

The area, zone, and/or owner assignments of buses are used to determine which buses are tie point buses when subsystem selection by area, zone, and/or owner is enabled. The owner assignments of ac branches are not referenced by activity DFTI.

The user specifies one of the following data types for comparison:

Table 12.8. Case Comparison Specifications, AC Tie Branch Data Checks

ref #	Specification	Description
1	Branch status	If a tie branch appears in both cases, it generates an alarm if the status flags differ. Where a tie branch exists in only one of the cases, it generates an alarm regardless of its status.
2	Line R,X,B	Check for different values of line impedance or charging. If a tie branch exists in only one of the cases, it is included in the report. If a tie branch has the same value of impedance in the two cases but is treated as a zero impedance line in only one of the two (i.e., THRSHZ differs in the two cases; refer to Section Zero Impedance Lines), it is included in the report. If a tie branch is treated as a zero impedance line in both cases, the impedances are considered equal regardless of their values in the two cases.

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ref #	Specification	Description
		For three-winding transformer windings, winding impedances are compared.
3	Line shunts	If a tie branch exists in only one of the cases, it is included in the tabulation only if the line shunt at either end of the branch is non-zero.
4	Line ratings	The user specifies the rating sets to be compared (all ratings, RATEA only, RATEB only, or RATEC only). For tie branches for which the selected rating(s) differ, all three ratings from both cases are tabulated. If a tie branch exists in only one of the cases, it is excluded from the report.
5	Metered end	Report tie branches with opposite metered ends in the two cases. If a tie branch exists in only one of the cases, it is excluded from the report.
6	Line length	Report tie branches where lengths differ in the two cases. If a tie branch exists in only one of the cases, it is excluded from the report.
7	Transformers	Two-winding transformers and individual windings of three-winding transformers are compared. If a tie branch exists in both of the cases, it generates an alarm under any of the following conditions: • It is a two-winding transformer or a three-winding transformer winding in one case, but not in the other. • It is a two-winding transformer in both cases, but the Winding 1 side and Winding 2 side relationship is reversed. • It is a two-winding transformer in both cases, and its Winding 1 tap ratio or phase shift angle, or Winding 2 tap ratio, differs.

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ref #	Specification	Description
		• It is a winding of a three-winding transformer in both cases, and its winding tap ratio or phase shift angle differs. If a tie branch exists in one case and not in the other, it generates an alarm if it is a transformer. For two-winding transformers, this report lists the tap ratios of both windings and the Winding 1 phase shift angle from each case, along with an indication of which bus is the Winding 1 side bus; for three-winding transformers, the winding ratio is listed in the RATIO1 column and the RATIO2 column has a value of 1.0.
8	Flows MW or MVAR (from bus)	Compare line flows against a designated threshold tolerance. A branch in the tie branch comparison list is included in this report under any of the following conditions: • It is in-service in both cases, and the active or reactive power flow at the from bus end in the two cases differs by more than the threshold tolerance. • It exists in both cases with different service status, and its active or reactive power flow at the from bus end in the in-service case exceeds the threshold. • It exists in only one of the cases, is in-service, and its active or reactive power flow at the from bus end exceeds the threshold. This report tabulates the active and reactive power flowing into the from bus end of the line in the two cases. Flow differences between the two cases are printed for each branch reported, both in MW and Mvar and as percentages of the working case flows. For this

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ref #	Specification	Description
		comparison, both cases should be solved to an acceptable mismatch level.
9	Flows MW or MVAR (from and to buses)	Similar to previous specification, except the flows at both ends of each tie line are checked against the threshold. For branches reported, from bus end conditions are shown on the first output line and to bus end conditions on the second.
10	Line MW or MVAR losses	Compare line losses that are calculated as I^2R and I^2X and *exclude* the line charging, line connected shunt, and transformer magnetizing admittance components. A branch in the tie branch comparison list is included in this report under any of the following conditions: • It is in-service in both cases, and the active or reactive power losses in the two cases differ by more than the threshold tolerance. • It exists in both cases with different service status, and its active or reactive power losses in the in-service case exceeds the threshold. • It exists in only one of the cases, is in-service, and its active or reactive power losses exceeds the threshold. For each branch reported, differences in losses between the two cases are printed, both in MW and Mvar and as percentages of the working case losses.
11	Zero sequence R,X,B	Check for different values of zero sequence line impedance or charging.

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ref #	Specification	Description
		If a tie branch exists in only one of the cases, it is included in the report. If a tie branch has the same value of zero sequence impedance in the two cases but is treated as a zero impedance line in only one of the two (i.e., THRSHZ differs in the two cases; refer to Section Zero Impedance Lines), it is included in the report. If a tie branch is treated as a zero impedance line in both cases, the zero sequence impedances are considered equal regardless of their values in the two cases. For transformers, the impedances used are those of the "pi" equivalent representing the transformer's zero sequence connections. It is calculated using the transformer's various series and ground impedances and its conection code. For three-winding transformers, winding values are compared. Both the working case and the comparison case must contain sequence data.
12	Zero sequence line shunts	If a tie branch exists in only one of the cases, it is included in the tabulation only if the zero sequence line shunt at either end of the branch is non-zero. For transformers, the admittances used are those of the "pi" equivalent representing the transformer's zero sequence connections. It is calculated using the transformer's various series and ground impedances and its conection code. For three-winding transformers, winding values are compared.

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ref #	Specification	Description
		Both the working case and the comparison case must contain sequence data.
13	Connection codes	In the transformer connection code check, if a tie branch exists in both of the cases, it generates an alarm under any of the following conditions: • It is a two-winding transformer or a three-winding transformer winding in one case, but not in the other. • It is a two-winding transformer in both cases, but the Winding 1 side and Winding 2 side relationship is reversed. • It is a two-winding transformer in both cases, and its winding connection codes differ. • It is a winding of a three-winding transformer in both cases, and the connection codes applied at the winding differ. If a tie branch exists in one case and not in the other, it generates an alarm if it is a transformer. This report lists the connection code from each case, along with an indication of which bus is the Winding 1 side bus.
14	Flow MW (from bus)	Similar to *flows MW or MVAR (from bus)* except that a branch is listed if the active power flow difference in the two cases exceeds the threshold tolerance.
15	Flow MVAR (from bus)	Similar to *flows MW or MVAR (from bus)* except that a branch is listed if the reactive power flow difference in the two cases exceeds the threshold tolerance.
16	Flow MW (from and to buses)	Similar to *flows MW or MVAR (from and to buses)* except that a branch is listed if the active power flow difference at either end in

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ref #	Specification	Description
		the two cases exceeds the threshold tolerance.
17	Flow MVAR (from and to buses)	Similar to *flows MW or MVAR (from and to buses)* except that a branch is listed if the reactive power flow difference at either end in the two cases exceeds the threshold tolerance.
18	Line MW losses	Similar to *line MW or MVAR losses* except that a branch is listed if the active power loss difference in the two cases exceeds the threshold tolerance.
19	Line MVAR losses	Similar to *line MW or MVAR losses* except that a branch is listed if the reactive power loss difference in the two cases exceeds the threshold tolerance.

In comparing line flows and losses (*Flows MW or MVAR (from bus)*, *Flows MW or MVAR (from and to buses)*, and *Line MW or MVAR losses*), a threshold tolerance is used. If the difference between data items in the two cases is within this tolerance, the item is omitted from the output report. Tolerances are allowed to be specified either in engineering units (e.g., MW and Mvar), or as a percentage of the data item values in the working case.

The selected check is made and exceptions are listed as described.

For categories comparing fault analysis data, this data must be present in both cases. At the completion of activity DFTI, the user designation of which of the two cases is to be left as the PSS®E working case is honored.

Exceptions are tabulated in up to three groups, each of which is in single entry format (refer to Branch Data), ordered in ascending bus numerical or alphabetical order according to the bus output option in effect at the time activity DFTI is selected (refer to Section 3.3.3, "Program Run-Time Option Settings"). The first group contains tie branches connecting tie point buses which are present in both cases. The second group contains those tie branches in the working case with at least one of the buses it connects not present in the designated Saved Case. The third group contains those tie branches in the designated Saved Case with at least one of the buses it connects not present in the working case.

Bus identifiers and ordering from the working case are used in tabulating the first two groups; bus identifiers and ordering from the designated Saved Case are used in tabulating the third group.

When reporting individual windings of three winding transformers, winding data is tabulated.

Additional Information
PSS®E GUI Users Guide, Comparing AC Tie Branches

12.30. Generating a Graphical Report

GRPG

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Run Activity GRPG - GUI
Diagram > Generate graphical report... Graphical Report Data file (*.grp) GRED Library File (*.sgf) Graphical Report Definition Data File (*.grb)
Run Line Mode Activity GRPG - CLI
ACTIVITY? >>**GRPG** ENTER INPUT FILE NAME, FILE TYPE OPTION (0=EXIT): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The graphic report generating activity GRPG allows the user to generate arbitrary diagrams on a customer-supported graphics output device from a file.

A dialog requesting selection from among the customer-supported graphics output devices pops up.

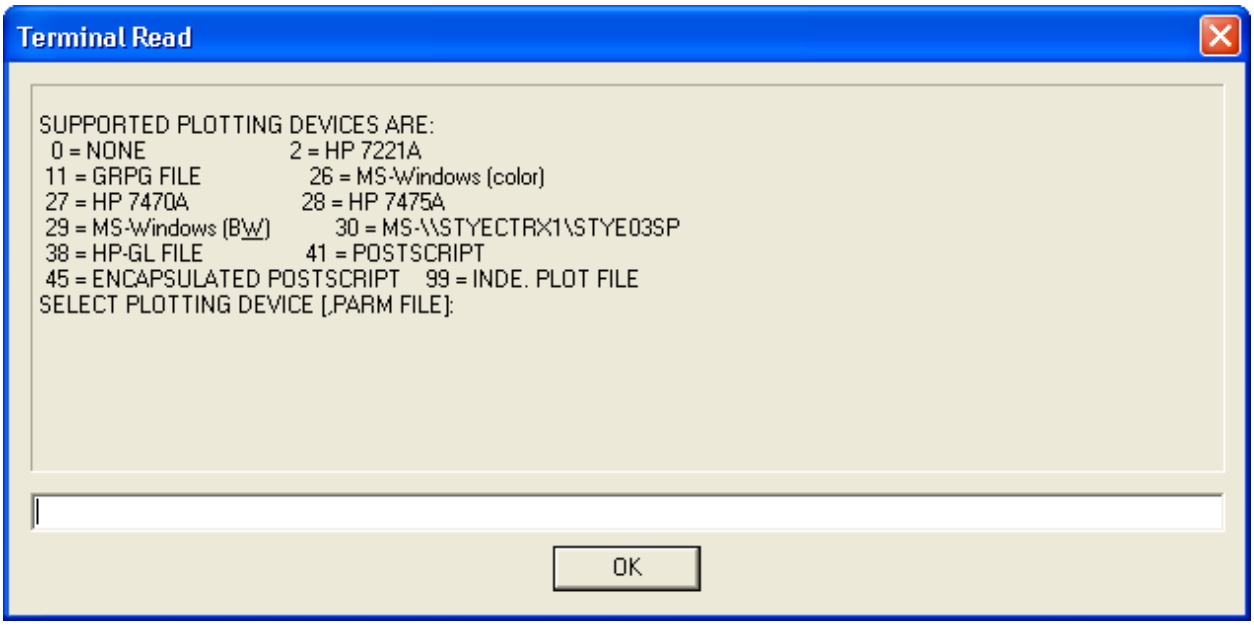

Figure 12.26. Plotting Device Popup Dialog

For terminal display, enter the device number for MS-Windows (color) or MS-Windows (BW). To close the graphic display and return to the *[Terminal Read]* dialog, use *[Enter]*.

12.30.1. GRPG Commands

GRPG commands are comprised of a verb, followed by zero or more keywords, variables, and/or comments. In describing the GRPG command syntax, the following notational conventions are used:

CAPITALS	Keyword that must be specified exactly as shown. No keyword abbreviations are allowed.

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lowercase	Items shown in lowercase are GRPG variables. The variable < c > is always enclosed in square brackets and denotes optional comments; other variables are described along with the appropriate command.
[...]	Items enclosed in square brackets are optional keywords and/or variables.
|A| or A|B |B|	Specify one from the list separated by, or enclosed in, the vertical bars.

The GRPG commands are organized into seven categories: setup, miscellaneous, drawing, text, assignment, conditional, and termination. These are described in the following sections.

Setup Commands

The function of these commands is to set the size and orientation of the picture produced by activity GRPG. Any setup commands *must* be issued prior to any use of the commands in the drawing, text, and terminate categories.

```
SET X|Y COORDINATE PAPER LIMITS TO min AND max
```

The SET PAPER LIMITS command is used to change the size of the drawing that is produced. The default limits are 0 to 8 in. for the x-coordinate, and 0 to 10.5 in. for the y-coordinate. Separate commands must be issued for X and Y if the paper limits of both are to be changed. The limits *must* be specified in inches.

```
SET PLOT ORIENTATION TO PORTRAIT|LANDSCAPE
```

The SET PLOT ORIENTATION command is used to rotate the diagram 90 degrees. The default orientation is PORTRAIT; if LANDSCAPE orientation is specified, the picture is rotated prior to output.

Use of the following command sets the drawing limits and orientation for the drawing produced by activity GRPG to those specified in a one-line diagram Drawing Coordinate Data File. Therefore, the above SET commands *should not* be used in conjunction with the INCLUDE ONE-LINE command.

```
INCLUDE [BINARY] ONE-LINE [c] FROM filename
```

The INCLUDE ONE-LINE command is used to include one-line diagrams of the form produced by activity DRAW in the drawing created by activity GRPG. A Graphical Report Definition Data File along with all its INCLUDEd files may contain only one INCLUDE ONE-LINE command.

Miscellaneous Commands

The miscellaneous commands may be entered at any time during the execution of activity GRPG.

```
blank or *
```

Blank lines and lines that contain an asterisk (*) in column one are ignored when processing a GRPG input data file. They may be used to enhance the readability of the input file.

```
COMMENT string
```

The COMMENT command simply echoes the command to the user's terminal. It is useful in monitoring the progress during the execution of longer GRPG data files.

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```
HELP [command]
```

The HELP command is used to display the syntax of a GRPG command. If a GRPG command is not entered, a list of GRPG commands is displayed.

```
INCLUDE filename [c] [OFFSET x,y] [c] [SCALE scale]
```

The INCLUDE command is used to insert commonly used subdrawings into a Graphical Report Definition Data File. The optional parameters OFFSET and SCALE may be used to modify the location and size of the subdrawing being INCLUDEd. By default, the subdrawing is not offset or scaled (i.e., x,y = 0.0,0.0 and scale = 1.0).

The file <filename> must be in the form of Graphical Report Definition Data File; specifically, it may not be a GRED Library File.

```
ROTATE PLOT angle [c] AROUND x,y
```

The ROTATE PLOT command causes subsequent plotting to be rotated the specified angle around the point <x,y>. The most common use of this command is in conjunction with the INCLUDE command described above. This command must be included *after* any INCLUDE ONE-LINE command.

```
SET ANGLE TO DEGREES|RADIANS
```

The SET ANGLE command is used to change the units by which angles are specified in subsequent GRPG commands. By default, angles are specified in degrees.

```
SET UNITS TO INCHES|CENTIMETERS
```

The SET UNITS command is used to change the units by which distances are specified in subsequent GRPG commands. The default is inches.

```
DECLARE REAL|STRING &label_l [&label_2
... &label_10]
```

The DECLARE command is used to define the names of variables to be used in subsequent GRPG commands. All variables (labels) must be declared exactly once before they are used in other GRPG commands. The form of each label is a 12-character string, the first of which must be an ampersand (&); commas and spaces are not allowed in label names. Up to 10 variables may be defined in a single DECLARE command. A Graphical Report Definition Data File along with all its INCLUDEd files may have defined up to 2000 REAL variables and 500 STRING variables of up to 60 characters each. REAL variables are initialized to zero and STRING variables to a single blank at the time they are DECLAREd.

In describing the remainder of GRPG commands, GRPG variables are often referred to as ampersand labels.

```
DEFINE MACRO macroname
```

The DEFINE MACRO command is used to define and name a series of GRPG commands, which may later be executed with the INVOKE command. A Graphical Report Definition Data File along with all its INCLUDEd files may define up to 50 macros. Any GRPG command (except another DEFINE MACRO command) may be used between the DEFINE MACRO command and its corresponding ENDMACRO command. However, DECLARE commands should not be contained within any macro that will be INVOKEd more than once. Macros may be contained in INCLUDE files.

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Within a macro, the special strings %n% (where < n > is a single-digit 1 through 9) are used to represent the values of up to nine arguments specified in the INVOKE command and evaluated at the time the macro is INVOKEd. An %n% string may be used in any keyword or variable position except for the command name in any GRPG command within a macro.

ENDMACRO

The ENDMACRO command terminates the definition of a GRPG macro.

Drawing Commands

The drawing commands, along with the text commands, are used to define the desired drawing. Four of the drawing commands have several forms. The general description of those commands are as follows.

MOVE	Causes an invisible line to be drawn to the specified point.
DRAW	Causes one or more lines to be drawn, depending on the command. See the SET LINE commands in this section for a description of the attributes of the line.
SHADE	Causes the figure defined by the command to be filled or toned with the pattern described by the last SET SHADING PATTERN command.
APPEND	Causes one or more lines (or a figure) to be appended to a list of polygons to be shaded. If the APPEND command causes a move, the current figure is closed (but not printed) and a new figure is started. All APPEND commands except APPEND LINE TO, APPEND LINE FOR, and APPEND VECTOR FOR result in a move. The SHADE ACCUMULATED POLYGONS command causes any appended figures to be filled.

The following commands are described using only one verb, but the description applies to all verbs, with the action taken defined above.

```
|DRAW   |
|MOVE   | LINE [FROM x1,y1] TO x2,y2
|APPEND|
```

This form of the DRAW LINE command causes a line to be drawn from < x1,y1 > to < x2,y2 >. If the FROM coordinate < x1,y1 > is omitted, the line is drawn from the current position to < x2,y2 >.

```
|DRAW   |
|MOVE   | LINE FOR x,y
|APPEND|
```

This form of the DRAW LINE command causes a line to be drawn from the current position < x0,y0 > to < x0+x,y0+y >.

```
|DRAW   |
|MOVE   | VECTOR [FROM x,y] FOR len [c] ANGLE ang [c]
|APPEND|
```


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The DRAW VECTOR command causes a line of length < len > and at the angle < ang > to be drawn from the coordinate < x,y >. If the from coordinate < x,y > is omitted, the line is drawn from the current position.

```
|DRAW   |
|APPEND| BOX FROM x1,y1 TO x2,y2 [c] [RADIUS rad]
|SHADE  |
```

This form of the DRAW BOX command results in a rectangle being drawn. The rectangle is defined by the intersections of horizontal and vertical lines through the points < x1,y1 > and < x2,y2 >. If the optional radius is included, the corners of the box are rounded with the specified radius < rad >.

```
|DRAW   |
|APPEND| BOX CENTERED AT x,y WITH SIDES xl BY yl [c] [RADIUS rad]
|SHADE  |
```

This form of the DRAW BOX command results in a box being drawn with the center at the coordinate < x,y >. The lengths of the sides of the box are specified by < xl > and < yl >. If the optional radius is included, the corners of the box are rounded with the specified radius < rad >.

```
|DRAW   |
|APPEND| CIRCLE [c] AT x,y [c] RADIUS rad
|SHADE  |
```

The DRAW CIRCLE command results in a circle with the specified radius being drawn with the center at the coordinate < x,y >.

```
|DRAW   |
|APPEND| ARC [c] AT x,y [c] RADIUS rad [c] FROM a1 [c] TO a2 [c]
|SHADE  |
```

The DRAW ARC command results in a counterclockwise arc being drawn with the focus specified by < x,y >, the radius by < rad >, the beginning angle by < a1 > and the ending angle by < a2 >.

```
|DRAW   | |CLOCKWISE        |
|APPEND| |CW               | ELLIPSE FROM x1,y1 TO x2,y2
|SHADE  | |COUNTERCLOCKWISE|
          |CCW              |
```

The DRAW ELLIPSE command results in an ellipse being drawn from the coordinate < x1,y1 > to the coordinate < x2,y2 >. The direction (or curvature) of the ellipse is determined by the specification of CLOCKWISE (CW) or COUNTERCLOCKWISE (CCW).

```
|DRAW   |
|APPEND| POLYGON[S] FROM file [c] [OFFSET x,y] [c] [SCALE scale]
|SHADE  |
```

The DRAW POLYGON command causes the contents of the file < file > to be drawn on the picture. Each line in the file must be blank or must have an x-y pair with the numbers separated by a comma or one or more blanks. An implied move is performed for the first x-y coordinate in the file. In addition, a blank line in the file causes a move to the next x-y coordinate pair.

```
DRAW GRID FROM x1,y1 TO x2,y2 [c] DELTA xd,yd
```


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The DRAW GRID command results in a grid being drawn with the lower left corner specified by < x1,y1 > and the upper right corner specified by < x2,y2 >. In the x direction, lines are drawn spaced at intervals specified by < xd >; in the y direction, lines are drawn spaced at intervals specified by < yd >.

```
DRAW LOGO AT x,y WITH SIZE siz [c] [ANGLE ang]
[c]
```

The DRAW LOGO command results in the PTI logo being drawn with the lower left corner specified by < x,y > at a height of < siz > and at the angle < ang >.

```
     |SCATTER PLOT  |
PLOT |STRAIGHT LINES| FROM filename
     |SPLINE CURVES |
```

The PLOT command is used to plot a series of x-y coordinate pairs stored in the file < filename > on the drawing. Unlike the DRAW POLYGON command, the PLOT command scales the data before it is included on the diagram. If the SCATTER PLOT option is selected, the data points are plotted using the symbol designated by the SET PLOT SYMBOL command at the size and angle designated by the SET SYMBOL command. If the STRAIGHT LINES option is selected, the data points are connected by straight lines. If the SPLINE CURVES option is selected, the data points are connected using a curve approximation algorithm to generate smooth lines between the data points. At least three data points must be included to approximate a curve. If only two data points are included, a straight line is drawn between the two points. See the following SET commands for further information on scaling.

```
SET X|Y COORDINATE PLOTTING LIMITS TO min AND max
```

The SET PLOTTING LIMITS command sets the limits for the portion of the drawing to which scaled data is mapped. This allows the user to map plotted data into, say, a gridded area, which was generated by use of the DRAW GRID command. By default, the plotting limits are the same as the paper limits. This command is normally used in conjunction with the following command.

```
SET X|Y COORDINATE DATA LIMITS TO min AND max
```

The SET DATA LIMITS command sets the limits for data to be plotted via the PLOT command. As x-y pairs are read from the file, the values are scaled according to where they fall in the data limit range. Data points outside the data limit range receive no special treatment, but are plotted outside of the limits set in the SET COORDINATE PLOTTING LIMITS command. By default, the data limits are the same as the paper limits.

```
SET PLOT SYMBOL TO number
```

The SET PLOT SYMBOL command may be used to change the symbol to be drawn by the PLOT SCATTER PLOT command, where < number > is the number of a symbol from the font specified in the last SET FONT TYPE command (see Figure 12.27, "STANDARD Font" and Figure 12.28, "LOWERCASE Font"). A < number > of zero may be specified to use a circle centered around each plotted point; this is the default if no SET PLOT SYMBOL command is specified. It is recommended that one of the centered symbols (0 through 14) be specified.

```
SET SYMBOL [HEIGHT TO size] [c] [ANGLE TO ang]
[c]
```

The SET SYMBOL command may be used to change the height and/or rotational angle of symbols drawn via the PLOT SCATTER PLOT command. By default, symbols are drawn with a height of 0.1 in. and rotated zero degrees.

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`SET LINE TYPE TO type`

The SET LINE TYPE command specifies the line pattern to be used when drawing lines to be of type < type > where < type > is an integer with a range of one to six. A line type of one is a solid line. Other line patterns are plotting device specific. By default, < type > is one.

`SET LINE COLOR TO color`

The SET LINE COLOR command specifies the color to be used for subsequent lines and text to be of color < color >, where < color > is an integer with a range of 1 to 8. On some devices, however, not all eight colors can be supported. In this case, the value of < color > is adjusted to a valid color number. The resulting color for each value of < color > is device dependent. For monochrome plotting devices, the SET LINE COLOR command is ignored. By default, < color > is one.

`SET LINE WIDTH TO width`

For monochromatic plotting devices, the SET LINE WIDTH command specifies the line thickness to be used for subsequent lines. The value of < width > can be 1 through 5. This command is ignored for plotting devices which support color. By default, < width > is one.

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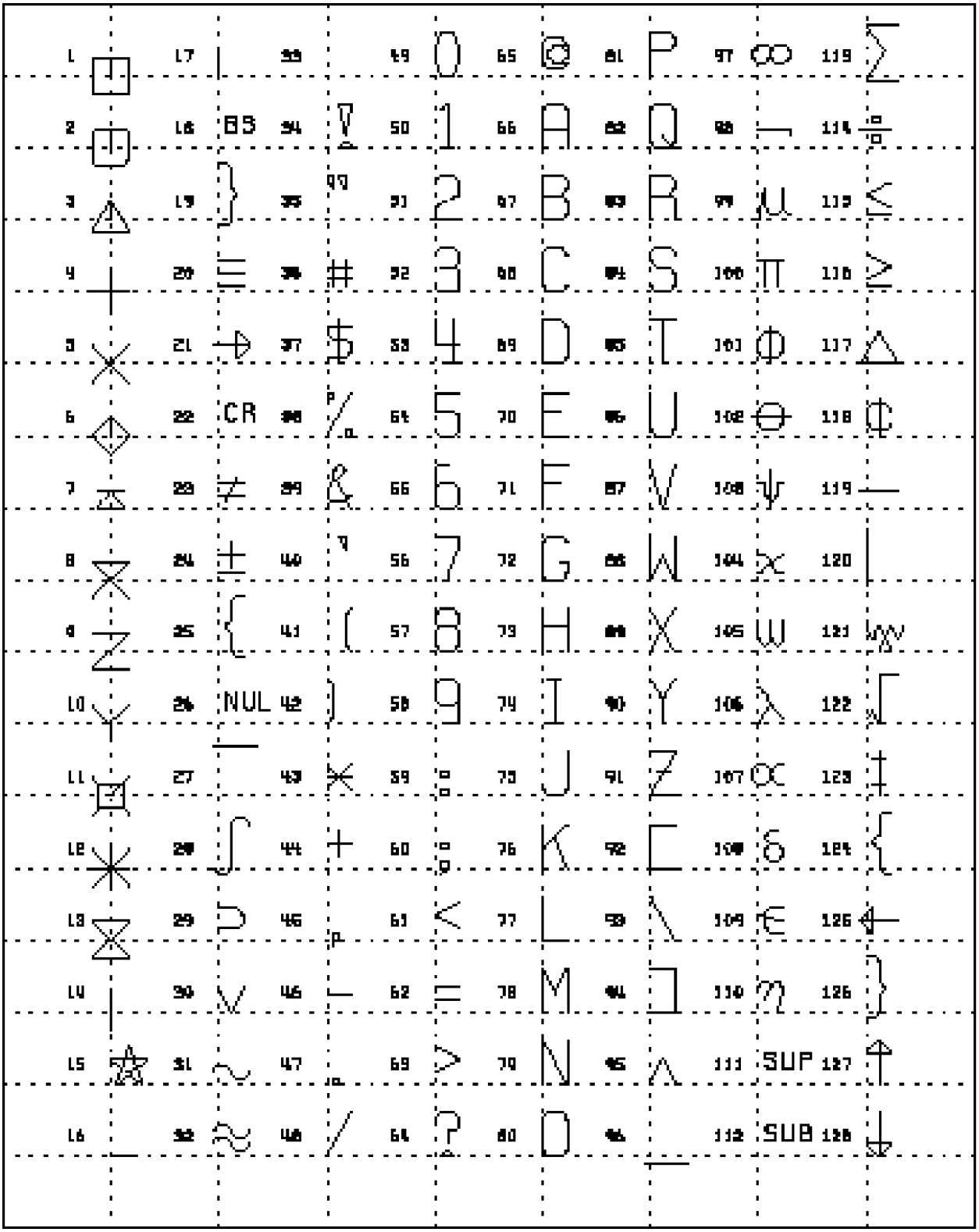

Figure 12.27. STANDARD Font

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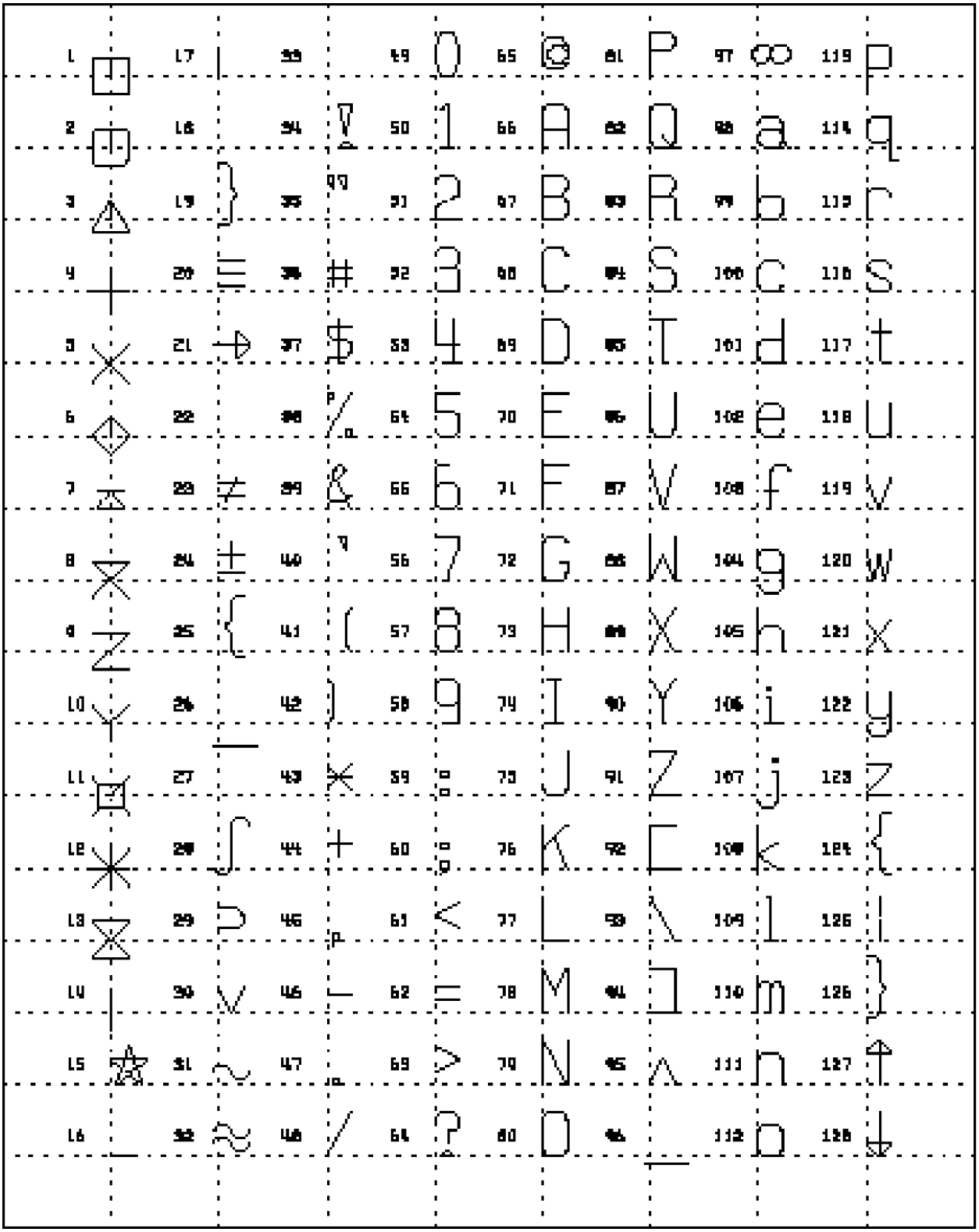

Figure 12.28. LOWERCASE Font

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```
SET SHADING PATTERN TO pattern
```

The SET SHADING PATTERN command specifies the shading color or pattern to be used when filling or toning polygon areas. The value of < pattern > is 1 through 16. The resulting pattern or color is device dependent. For plotting devices that do not support polygon filling, shading is simulated by drawing closely spaced lines in the polygonal area.

```
SET SHADING OUTLINE VISIBLE|INVISIBLE
```

The SET SHADING OUTLINE command specifies whether the boundary lines of polygons to be shaded are to be drawn. By default, the boundary lines are not drawn.

```
CLOSE POLYGON
```

The CLOSE POLYGON command may be used after an APPEND command to force the end of a polygon area. This may be needed when it is desired to tone several overlapping polygon areas with a single shade command.

```
SHADE ACCUMULATED POLYGON[S]
```

This form of the SHADE command closes an open polygon, if necessary, and causes one or more polygons that have been accumulated by the APPEND commands to be toned or filled. The accumulated polygons are then removed from the list of polygons to be shaded.

```
FLUSH ACCUMULATED POLYGON[S]
```

The FLUSH command causes any polygons that have been generated by the APPEND commands to be removed from the list of polygons to be shaded.

```
INVOKE MACRO macroname USING P1 [P2 ... P9]
```

The INVOKE command executes the macro < macroname >, passing up to nine arguments to the macro. The macro < macroname > must have been previously defined using the DEFINE MACRO and ENDMACRO commands.

If fewer arguments are specified in the INVOKE command than are used within the macro, the unspecified arguments are evaluated as the string .NULL.. If more arguments are specified in the INVOKE command than are used within the macro, the extra arguments are ignored.

Text Commands

The text commands are used to control output of graphic text on the drawing.

```
JUSTIFY TEXT LEFT|RIGHT|CENTERED
```

The JUSTIFY TEXT command can be used to modify the reference point for placement of text specified by the WRITE command. For left justified text, the location specified by the WRITE command is at the lower left corner of the text string. If the text is to be centered, the WRITE command specifies the center point of the text. If the text is to be right justified, then the WRITE command specifies the lower right corner of the text block. The default reference for text is left justified.

```
SET FONT TYPE TO STANDARD|LOWERCASE|fontnumber
```

The SET FONT TYPE command is used to select the character font to be used in subsequent WRITE commands. The STANDARD font, or font one (see Figure 11-28), which does not contain lowercase characters, is used by

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default. The LOWERCASE font, or font two (see Figure 12.28, "LOWERCASE Font"), contains both uppercase and lowercase characters.

The variable < fontnumber > is a positive integer through which the user may select one of the hardware fonts available on certain PostScript printers. Refer to *Guide to Printing and Plotting* for details on setting up the graphics device parameter file.

```
SET TEXT [HEIGHT TO size] [c] [ANGLE TO ang] [c]
```

The SET TEXT command may be used to change the height and/or rotational angle of text drawn via the WRITE command. By default, text is drawn with a height of 0.1 in. and rotated zero degrees.

```
SET PRINT FORMAT TO width,decimal
```

The SET PRINT FORMAT command specifies the format to be used for printing numeric data in subsequent WRITE commands. The variable < width > specifies the total width of the numeric field, including the decimal point and any leading spaces. < decimal > specifies the number of decimal places to be printed following the decimal point. If the value is too large to fit in < width > digits, < decimal > is reduced to accommodate it. By default, < width > is 8 and < decimal > is 2.

```
WRITE item [item2 ... item10] [AT x,y]
```

The WRITE command builds an output line of up to 132 characters formed by the concatenation of the < items > specified. Items may be quoted or unquoted strings of up to 60 characters, reserved keywords enclosed in percent symbols (%), symbol codes enclosed in angled brackets (< >), or ampersand labels. Reserved keywords and symbol codes must be specified as uppercase characters.

If a quoted string is specified as an item, the quotes are removed before it is appended to the output line. If the string contains any blanks, commas (,), or slashes (/), it must be enclosed in single quotes. If a quoted string is to contain quotes, each quote within the string must be specified as two consecutive quotes.

If a reserved keyword of the form %name% is specified as an item, a value designated by the reserved name is appended to the output line. If the reserved name %DATE% is specified, a 17-character string containing the current date is used. If the reserved name %TIME% is specified, a 5-character string containing the current time is used.

If a string of the form <SYMnnn> is specified as an item, the symbol for which the symbol number is nnn is appended to the output line. Symbol numbers are dependent upon the character font specified in the last SET FONT TYPE command (see Figure 12.27, "STANDARD Font" and Figure 12.28, "LOWERCASE Font").

If an item is an ampersand label (&label), the value of the corresponding GRPG variable is appended to the output line. If the variable is of type STRING, the output includes any trailing blanks that are a part of the variable's value. If the variable is of type REAL, the value is formatted according to the specifications in the last SET PRINT FORMAT command.

If < AT x,y > is specified, that coordinate value is used as a reference point for the output line. See the JUSTIFY TEXT command above for more information on text reference points. If < AT x,y > is not specified, the output line is positioned according to the following:

- If text is left justified, the output line is placed at the end of the previous output.
- If text is centered, the output line is centered 1.3 times text height below the previous output.
- If text is right justified, the output line is right justified 1.3 times text height below the previous output.

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When in the left justify mode, the string <CR> embedded in the output line causes a carriage return and line feed to be simulated. The resulting position is 1.3 times the text height relative to the last position specified by explicitly entering < x,y >. Additional <CR> strings cause simulation of additional carriage returns and line feeds. The string <CR> should not be used in the centered or right justify modes.

Assignment Commands

The assignment commands are used to place values into previously DECLAREd GRPG ampersand labels (i.e., variables). They provide for access to the PSS®E working case and for arithmetic operations involving ampersand labels.

```
                |constant|
LET &label_1 = |string  |
                |&label_2|
```

This form of the LET command assigns the value from the right side of the equal sign to the ampersand label on the left side. If &label_1 is of type REAL, the right side must contain either a numeric constant or another REAL ampersand label. If &label_1 is of type STRING, the right side must contain either a character string or another STRING ampersand label. If type conflicts occur, an error message is printed.

```
LET &label_1 = |constant| oper |constant|
                |&label_2|      |&label_3|
```

This form of the LET command assigns the value resulting from the arithmetic computation on the right side of the equal sign to the ampersand label on the left side. All values must be of type REAL or an error is reported. The specification of < oper > must be one of the arithmetic operators +, -, *, / or their corresponding keywords PLUS, MINUS, TIMES, or OVER.

```
LET &label_1 = THE CABS OF |constant| AND |constant|
                           |&label_2|     |&label_3|
```

This form of the LET command assigns the magnitude of a complex number to the ampersand label on the left side of the equal sign. The complex number operated upon is the one for which real and imaginary components respectively consist of the two values specified on the right side of the equal sign. All values must be of type REAL or an error is reported.

The following forms of the LET command are used to access data from the PSS®E working case. Results are stored in ampersand labels, which must have been previously DECLAREd. When a form of this command returns more than one value (e.g., a complex power), the values are stored in consecutively DECLAREd REAL ampersand labels, starting at the one specified in the LET command. In the following forms of the LET command, if the variable on the left side is < &label >, the command returns a single value; the variable < &label2 > on the left side indicates that two values are returned by the command. Whenever a bus identifier is required in the LET command (busid), either the extended bus name enclosed in single quotes or the bus number may be specified.

```
LET &label2 = THE CASE TITLE
```

Returns the two-line case title in the STRING variables &label2 and the next DECLAREd STRING ampersand label.

```
LET &label = THE NAME FOR AREA|ZONE n
```


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Returns the name of the specified area or zone in the STRING variable &label.

```
LET &label2 = THE INTERCHANGE FROM AREA n TO
AREA n
LET &label2 = THE INTERCHANGE FROM ZONE n TO ZONE n
```

Returns the complex interchange between a pair of areas or zones.

```
LET &label = THE MW |TRANSACTION| FROM AREA
n TO AREA n [IDENTIFIER id]
                    |TRANSFER   |
```

Returns the specified interarea transfer between a pair of areas in MW. If omitted, a transfer identifier of 1 is assumed.

```
LET &label2 = THE NET INTERCHANGE FROM AREA|ZONE
n
```

Returns the complex total interchange from a specified area or zone.

```
LET &label2 = THE GENERATION|LOAD|LOSSES FOR
AREA|ZONE n
```

Returns the complex total generation, load or losses for a specified area or zone.

```
LET &label2 = THE SYSTEM GENERATION|LOAD|LOSSES
```

Returns the complex system total generation, load or losses.

```
LET &label = THE [EXTENDED] NAME FOR BUS busid
```

Returns the 8-character bus name or the 12-character extended bus name in the STRING variable &label.

```
LET &label2 = THE GENERATION AT BUS busid
```

Returns the complex generation at the specified bus.

```
LET &label2 = THE LOAD AT BUS busid [LOADID
id]
```

Returns the complex load at the specified bus. If the optional load identifier is omitted, total bus load is returned; otherwise, the specified load is returned.

```
LET &label2 = THE ACTUAL|NOMINAL FIXED|SWITCHED
SHUNT AT BUS busid
```

Returns the complex fixed or switched bus shunt either in MW and Mvar or in admittance.

```
LET &label = THE BASE VOLTAGE FOR BUS busid
```

Returns the base voltage of a specified bus.

```
LET &label = THE VOLTAGE AT BUS busid [ IN
PU|KV ]
```


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Returns the voltage magnitude at the specified bus in either per unit or kV. If the units specification is omitted, per unit voltage is returned.

```
LET &label = THE ANGLE AT BUS busid
```

Returns the voltage phase angle at the specified bus in degrees.

```
LET &label = THE STATUS FOR |BRANCH| FROM BUS
busid TO BUS busid [CIRCUIT id]
                           |LINE  |
```

Returns the service status of the specified branch as either zero for out-of-service or one for in-service into a REAL ampersand label. If omitted, a circuit identifier of 1 is assumed.

```
LET &label2 = THE FLOW FOR |BRANCH| FROM BUS
busid TO BUS busid [CIRCUIT id]
                          |LINE  |
```

Returns the complex power flow on the specified branch. If omitted, a circuit identifier of 1 is assumed.

```
LET &label2 = THE |RECTIFIER| FLOW FOR 2-TERMINAL
DC LINE n
                  |INVERTER |
```

Returns the complex power flowing into the two-terminal dc line as seen from the appropriate ac converter bus.

```
LET &label2 = THE CONVERTER FLOW AT BUS busid
FOR N-TERMINAL DC LINE n
```

Returns the complex power flowing into the multi-terminal dc line as seen from the appropriate ac converter bus.

```
LET &label = THE CONTROL MODE FOR |2-TERMINAL|
DC LINE n
                                  |N-TERMINAL|
```

Returns the control mode value MDC for the specified dc line.

Conditional Commands

The conditional commands are used to control output based upon the results of assignment commands. These commands are used to form IF command blocks, which consist of:

- One IF command followed by a series of commands.
- Optionally, one or more ELSEIF commands, each followed by a series of commands.
- Optionally, one ELSE command followed by a series of commands.
- One ENDIF command.

```
   |constant|      |constant|
IF |string  | oper |string  |
   |&label_1|      |&label_2|
```


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The IF command compares a pair of operands to determine if the next series of commands is to be executed. Both operands must be of the same type. The specification of < oper > must be one of the logical operators =, <>, >, >=, <, <=, or their corresponding keywords EQ, NE, GT, GE, LT, or LE.

If the specified expression evaluates to a true condition, the commands in the range of the IF command are executed. The range consists of those commands between the IF command and the next ELSEIF, ELSE, or ENDIF command.

```
       |constant|      |constant|
ELSEIF |string  | oper |string  |
       |&label_1|      |&label_2|
```

The ELSEIF command must follow an IF or another ELSEIF command. The logical expression is evaluated only if the IF and all other preceding ELSEIF commands in the IF command block had evaluated to a false condition. Otherwise, the ELSEIF command behaves in the same manner as the IF command.

```
ELSE
```

The ELSE command must follow an IF or ELSEIF command. The series of commands following an ELSE command are executed only if the IF and all preceding ELSEIF commands in the IF command block had evaluated to a false condition.

```
ENDIF
```

The ENDIF command terminates each IF command block.

There is an unresolved problem with the ELSEIF command. It is recommended that, until the problem is corrected, instead of using ELSEIF commands, nested IF command blocks be used as in the following structural example:

```
IF ...                  IF ...
   a...                    a...
   ELSEIF                  ELSE
   b...                    IF ...
   ELSE                       b...
   c...                       ELSE
ENDIF                         c...
                           ENDIF
                        ENDIF
```

Termination Commands

The termination commands are used to terminate the current plot.

```
ABORT
```

The ABORT command is used to terminate the execution of GRPG without the generation of plotting output.

```
END [c]
```

The END command is used to terminate all GRPG data files. If the data file currently being processed is an INCLUDE file, processing of the INCLUDE file terminates and processing of the original file resumes. Otherwise, the plotting page is terminated and the prompt for selection of the desired plotting device is displayed.

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Only one drawing in such a file may contain SET PAPER LIMITS and SET PLOT ORIENTATION commands. It is strongly recommended that it be the first one.

```
START NEW PLOT
```

The START NEW PLOT command is used to terminate a plotting page. Additional plotting commands may follow and results in another plot being produced. This command, then, is used between groups of data records specifying multiple drawings in a single data input file.

Additional Information
PSS®E GUI Users Guide, Generating a Graphical Report *PSS®E* Command Line Interface (CLI) *Users Guide*, Generating a Graphical Report

12.31. Viewing Graphical Output of Fault Calculations

The fault currents calculated by activity ASCC (ASCC_2 api) and activity IECS (IECS_2 api) can be displayed on slider diagrams.

12.31.1. Display Options

Fault Calculation Results Displayed

The following types of fault currents can be displayed on one slider diagram:

- Total fault currents for all faulted buses and for all fault types considered
- Total fault currents and fault current contributions at home bus
- Total fault currents and fault current contributions across entire network
- Contour diagram of maximum fault currents at each faulted bus considered

Units of Voltage and Currents Displayed

Engineering units are established for a slider diagram using the [Program Settings] dialog. Refer to Section 22.10, "Changing PSS®E Program Settings" for the following options:

- Fault currents are displayed in either per unit on a system apparent power and voltage base or in AMP using the Short circuit output option.
- Fault currents are displayed in either polar or rectangular coordinates using the Short circuit coordinates option.
- Post-fault bus voltages are displayed in either per unit on a system voltage base or kV using the Short circuit output option.

Voltage and Currents Displayed

Voltage and current values are displayed on the slider diagram as follows:

- The post-fault bus voltage and total fault current are displayed as bus quantities on the faulted bus.
- The post-fault bus voltages on all buses except the faulted bus are displayed as bus quantities.

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- The fault currents on each branch, shunts and machines show their contributions to the total fault current.

Slider Diagram Title

The titles on the slider diagram display the activity name and fault description.

Graphical user interface dialog options

The various options in the graphical user interface dialog (Figure 11-30) are:

Table 12.9. ASCC Fault Analysis Annotation Options

Option	Description
Select fault to display	Fault type • Fault currents (only bus faults): display total fault currents for all calculated fault types and all faulted buses This selection does not display total fault currents for the Line Out and Line End fault options, even when they are calculated. • Three phase (3PH) faults: display three phase fault currents • Line to ground (LG) faults: display line-to-ground fault currents • Line Line to Ground (LLG) faults: display double line-to-ground fault currents • Line to Line (LL) faults: display line-to-line fault currents Contributions at home bus for bus faults To display fault contributions at home buses for all faulted buses select this option in addition to the 3PH, LG, LLG or LL fault type selection. This selection does not display fault contributions at home buses for the Line Out and Line End fault options, even when they are calculated. Faulted bus To display fault currents for a selected faulted bus specify the faulted bus in addition to 3PH, LG, LLG or LL fault type selection.
Display quantity	Sequence components of fault current: positive sequence (Ia1), negative sequence (Ia2) and Zero sequence (Ia0) or 3Ia0

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Option	Description
	Phase components of fault current: Phase A (IA), Phase B (IB), Phase C (IC)
Line Out Faults	For Line Out (LOUT) fault cases, in addition to selection of fault to display and display quantity, specify the branch or three winding transformer that is intentionally outaged.
Line End Faults	For Line End (LEND) fault cases, in addition to selection of fault to display and display quantity, specify the branch connected to faulted bus whose far end (from faulted bus) is being faulted. When a three winding transformer is connected to the faulted bus, two Line End faults are calculated at two buses other than the faulted bus. For these two Line End faults, specify: From bus: Line End bus To bus: Faulted bus Last bus: Last bus Note: To Bus is always the faulted bus, but the From Bus is selected depending on which Line End is faulted.

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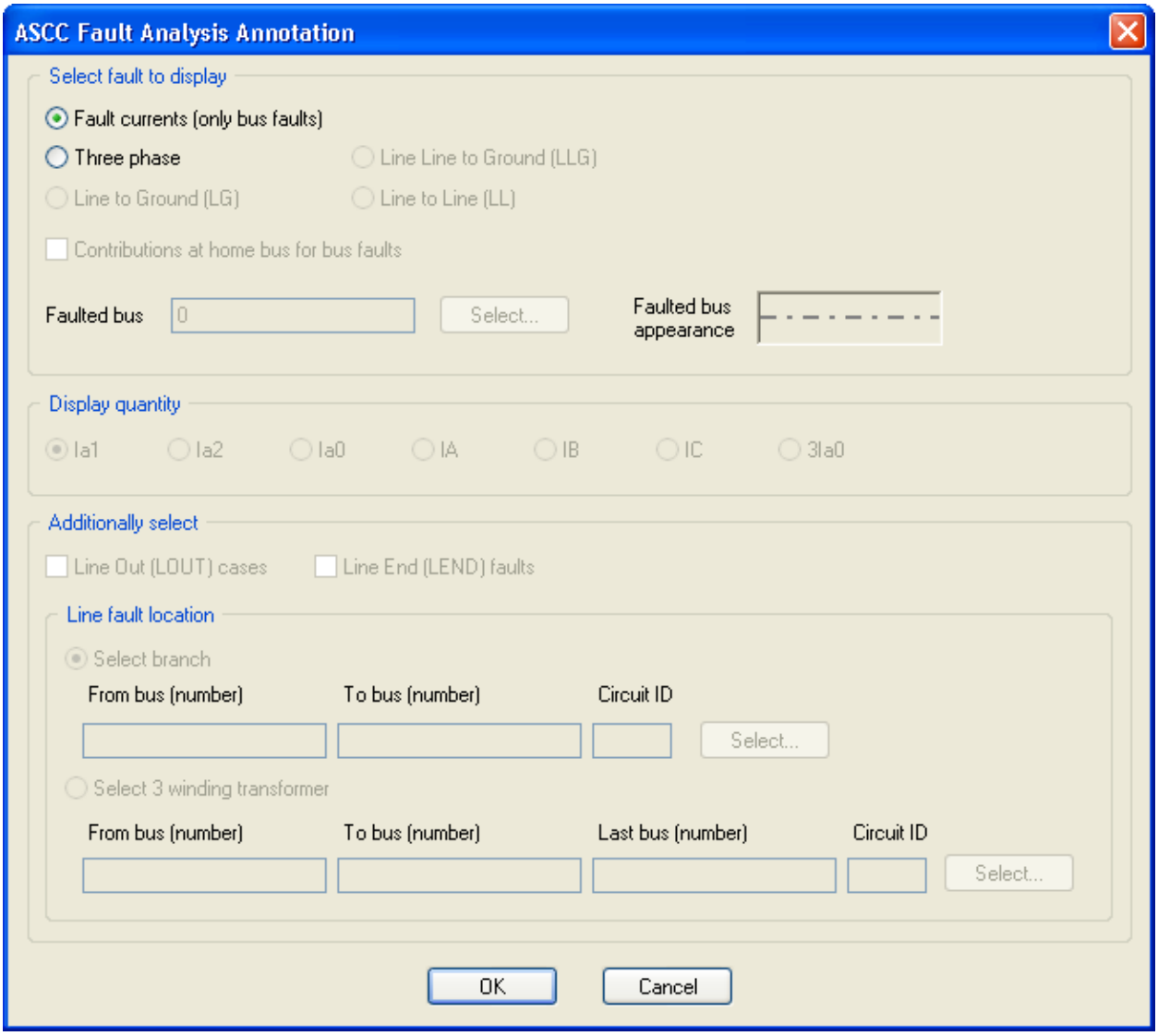

Figure 12.29. ASCC Fault Analysis Annotation Dialog

12.31.2. Example

The slider diagrams in this section show the ASCC_2 fault currents for following faults:

- Case: sample.sav
- 3PH, LG, LLG and LL faults are calculated for faults on buses 152, 3001, 3002, 3004, 3011
- Faults with Line Out as well as Line End

Similarly, activity IECS_2 fault currents can be displayed in the slider diagram.

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Total Fault Currents, All Faulted Buses, All Selected Fault Types

The slider diagram below shows total fault currents at faulted buses for 3PH, LG, LLG and LL faults on buses 152, 3001, 3002, 3004 and 3011.

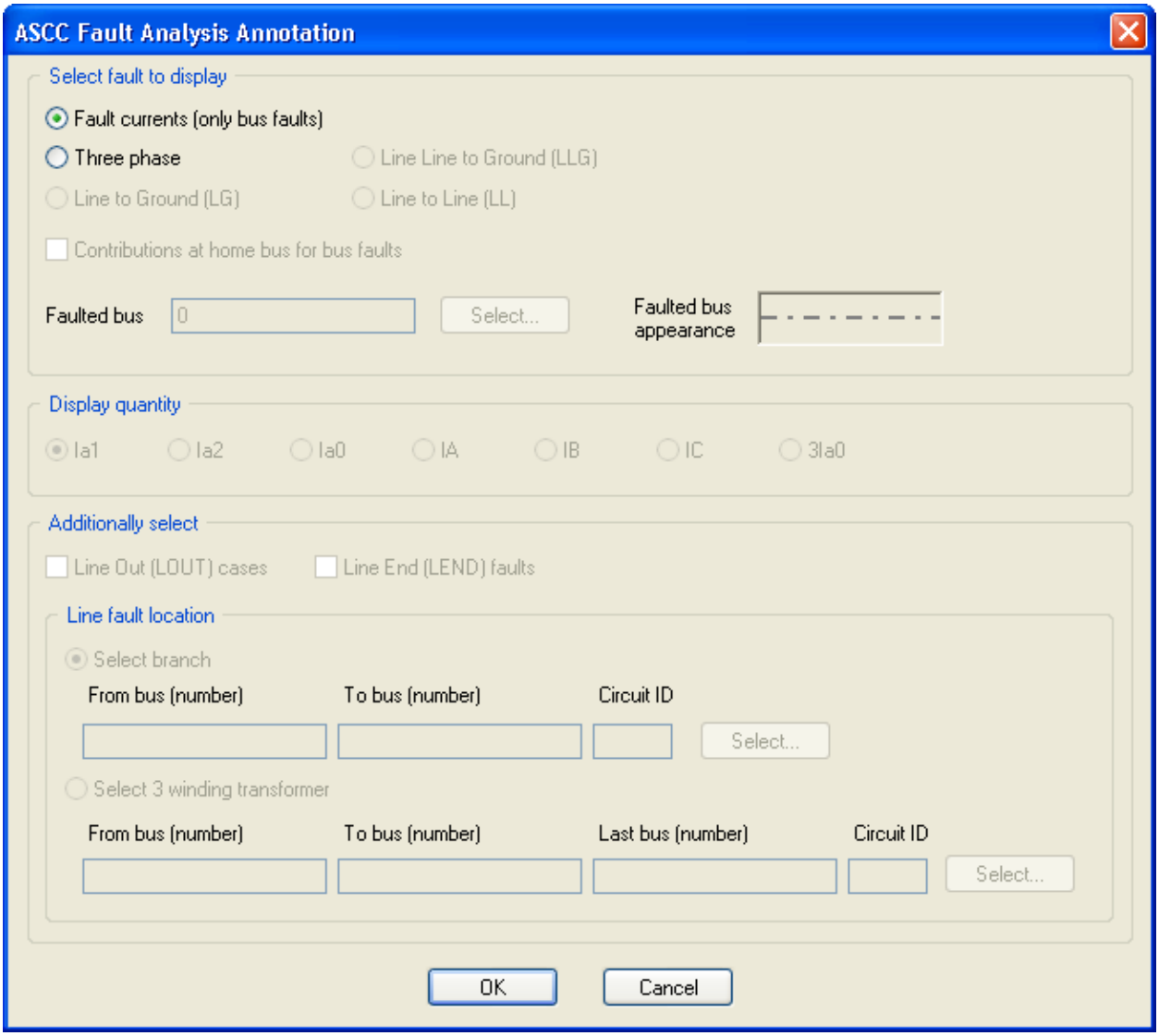

Figure 12.30. Fault Analysis Annotation Dialog, Total Fault Currents

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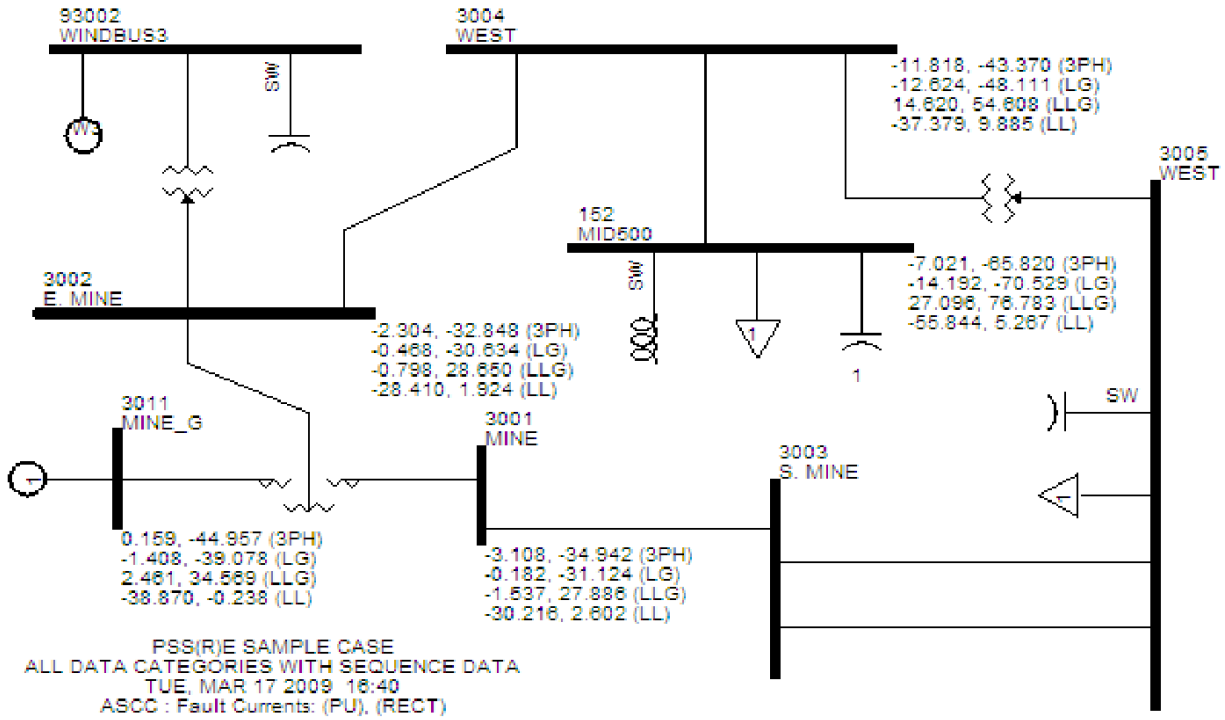

Figure 12.31. Slider Diagram, Total Fault Currents

Total Fault Currents and Fault Current Contributions, Home Bus

The slider diagram below shows total fault currents and fault current contributions at home bus for a three phase fault on buses 152, 3001, 3002, 3004 and 3011.

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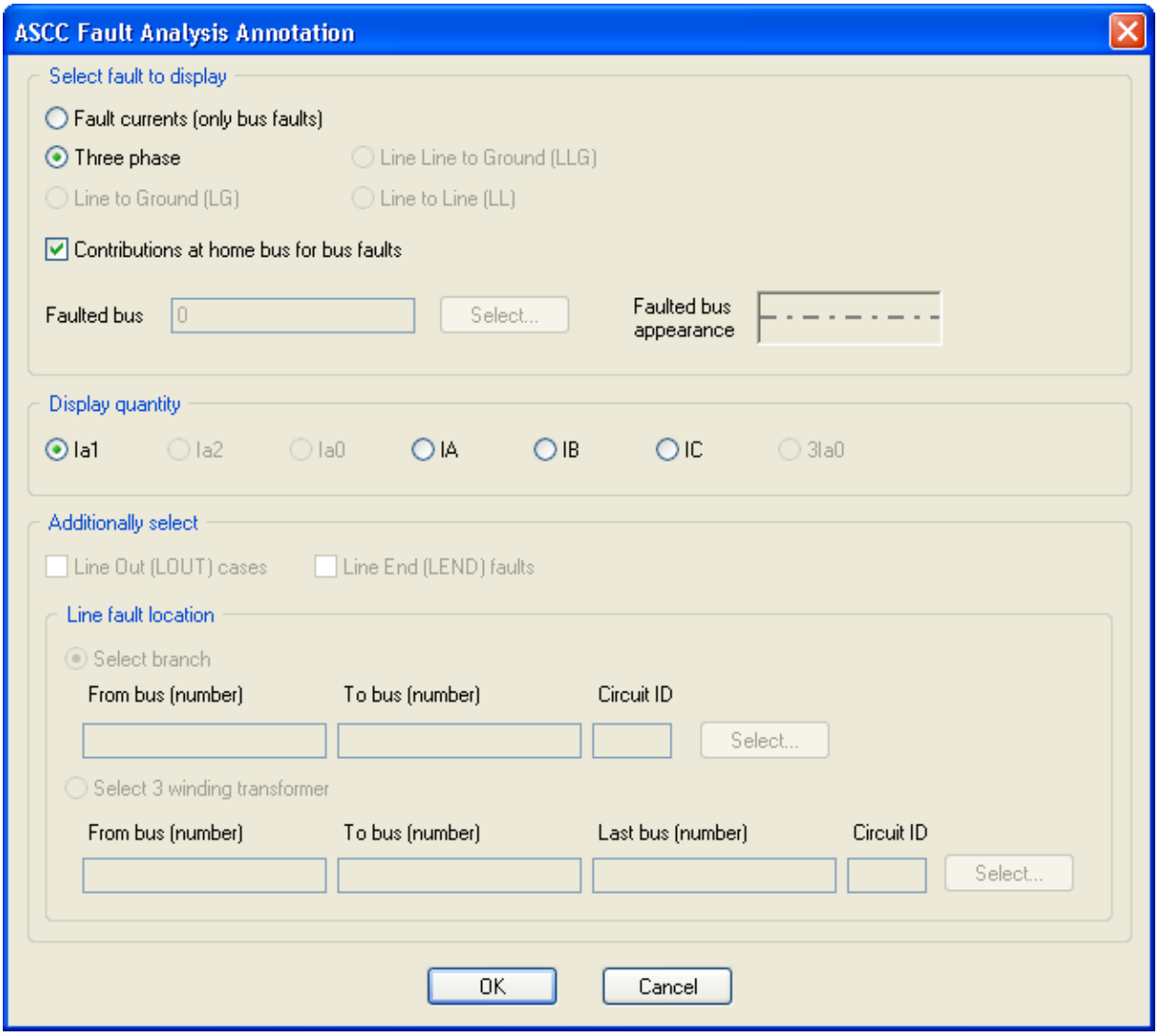

Figure 12.32. Fault Analysis Annotation Dialog, Contributions at Home Bus

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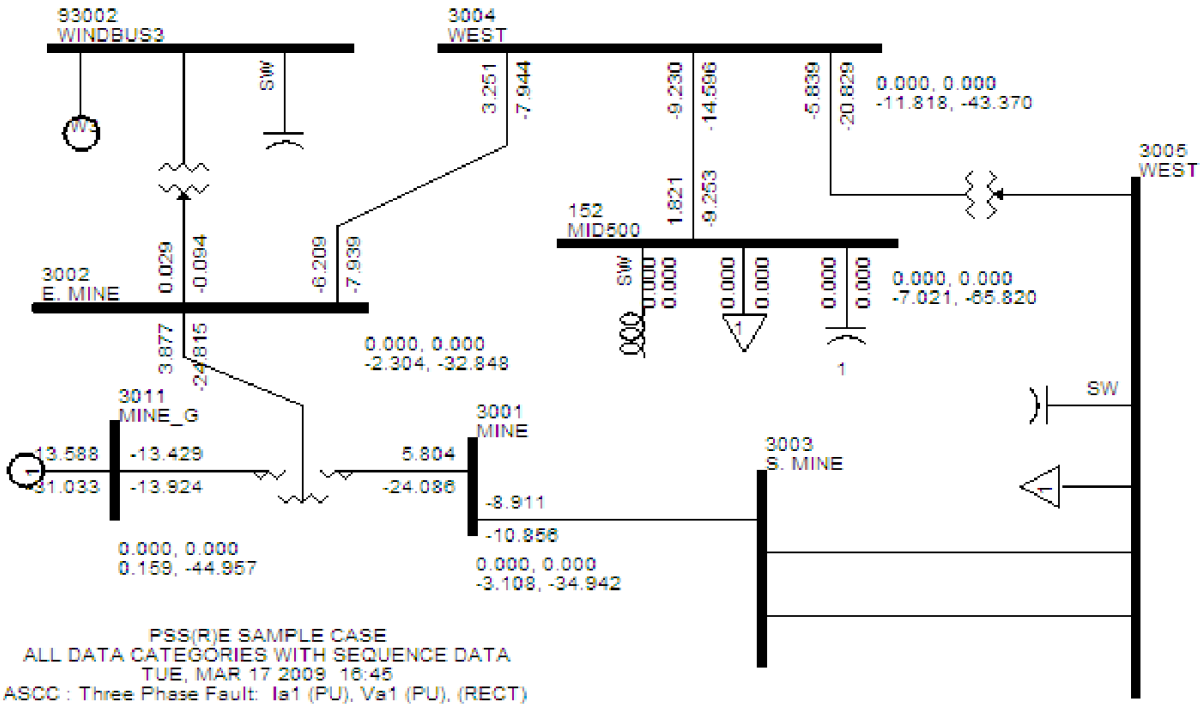

Figure 12.33. Slider Diagram, Fault Current Contributions at Home Bus

Total Fault Currents and Fault Current Contributions, Entire Network

The slider diagram below shows fault currents and post-fault voltages for a three phase fault at bus 3001.

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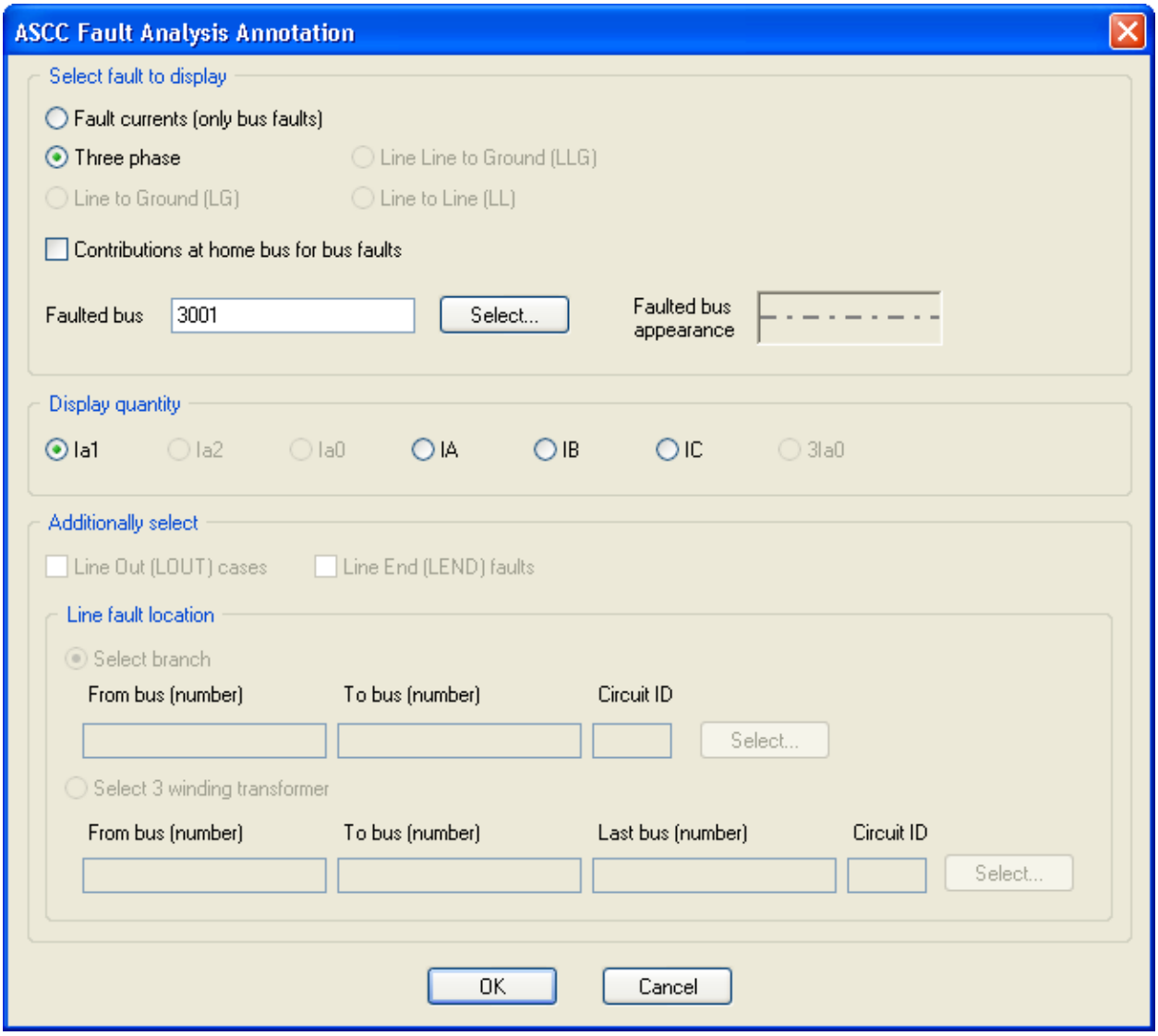

Figure 12.34. Fault Analysis Annotation Dialog, Selected Fault

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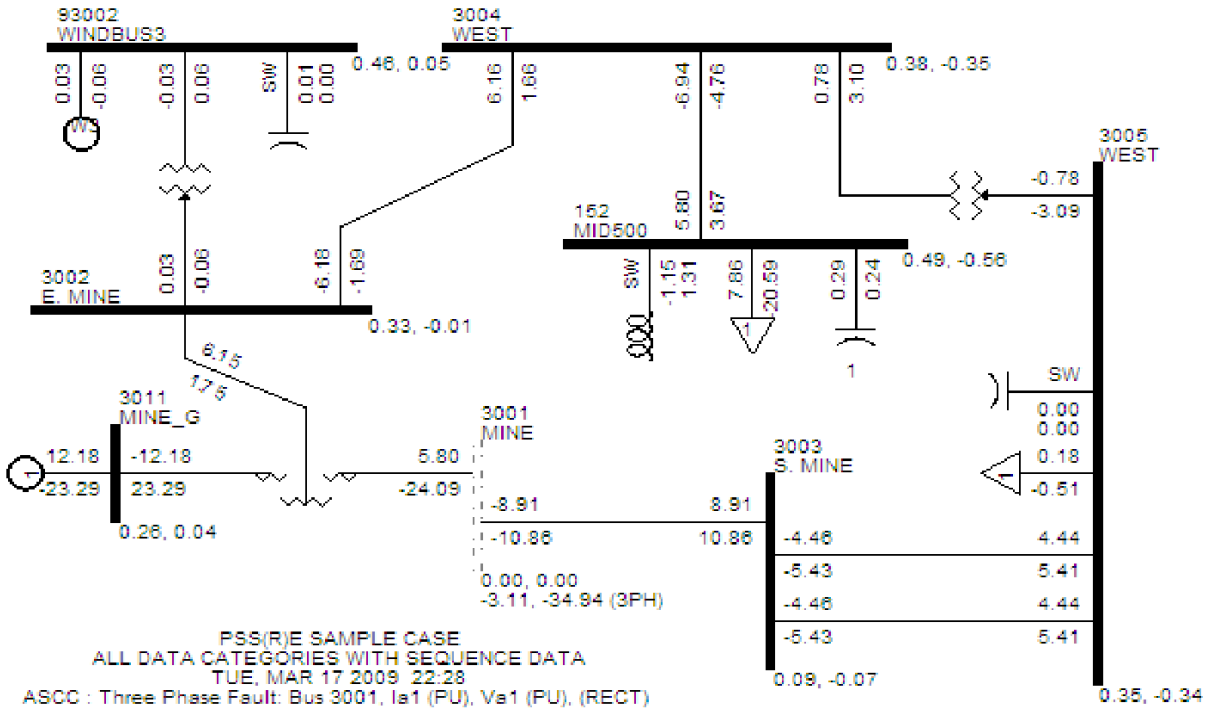

Figure 12.35. Slider Diagram, Fault Currents for a Selected Fault

Total Fault Currents and Fault Current Contributions, Line Outage

The slider diagram below shows fault currents and post-fault voltages for a three phase fault at bus 3001 with branch 3001-3003-1 out.

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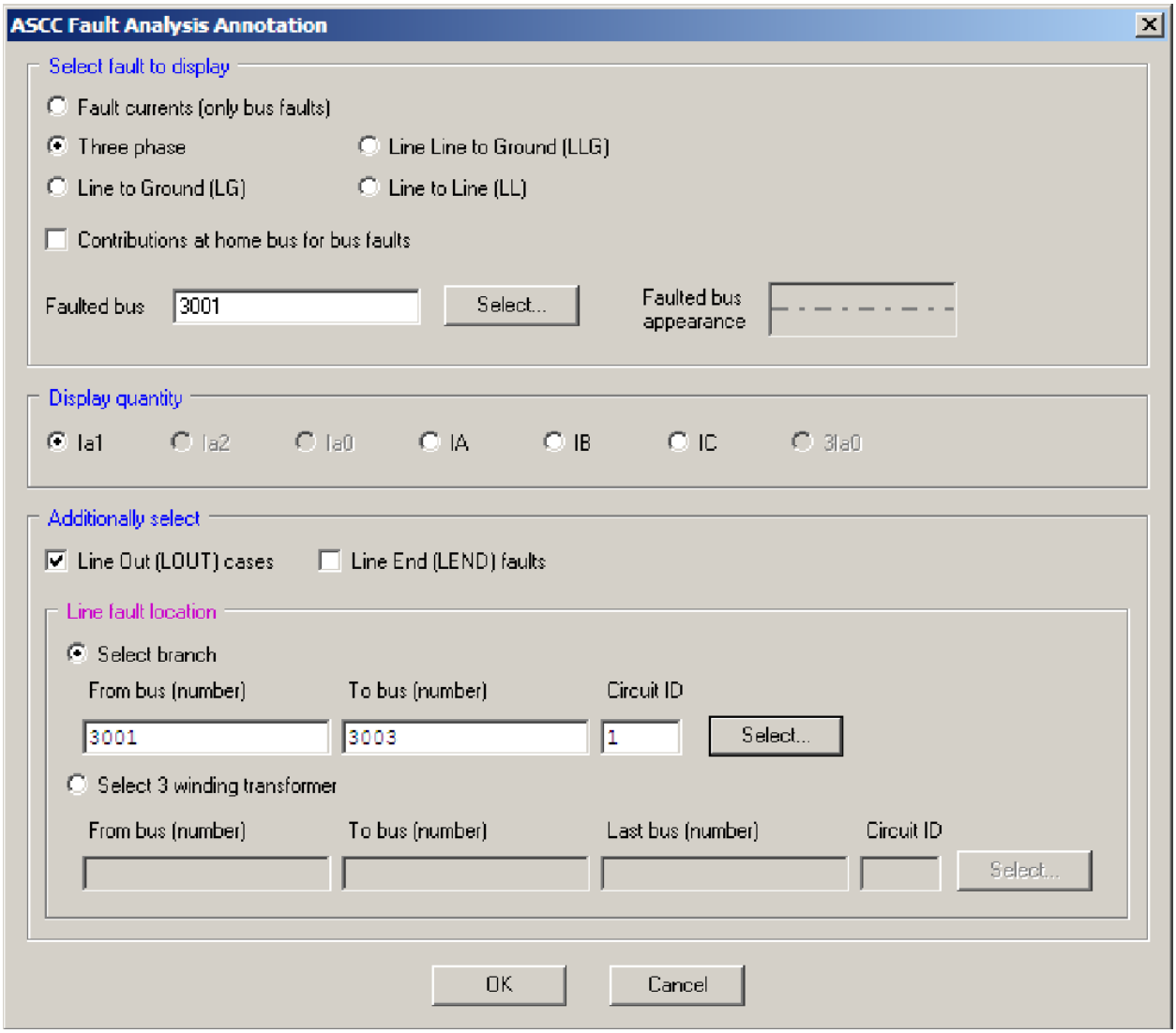

Figure 12.36. Fault Analysis Annotation Dialog, Line Outage Fault

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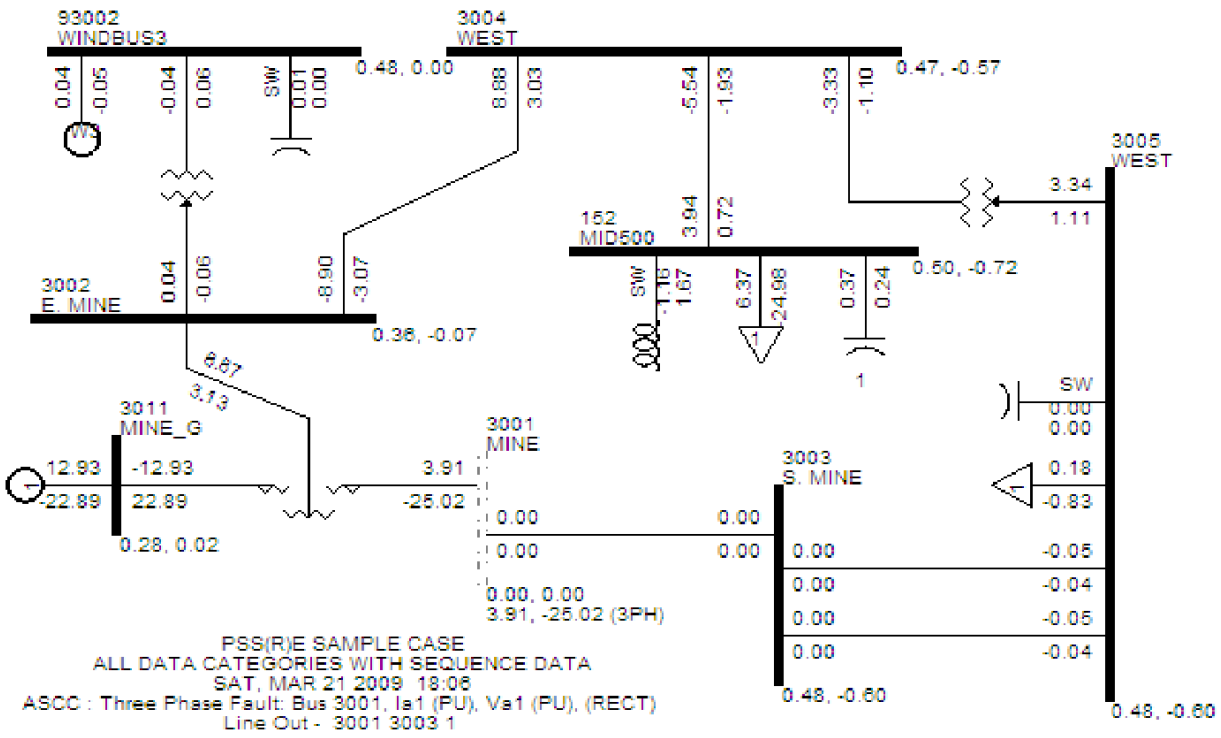

Figure 12.37. Slider Diagram, Line Outage Fault Currents

Total Fault Currents and Fault Current Contributions, Three Winding Transformer Winding Outage

The slider diagram below shows fault currents and post-fault voltages for a three phase fault at bus 3001 with three winding transformer 3001-3002-3011-1, winding connected to faulted bus 3001 out.

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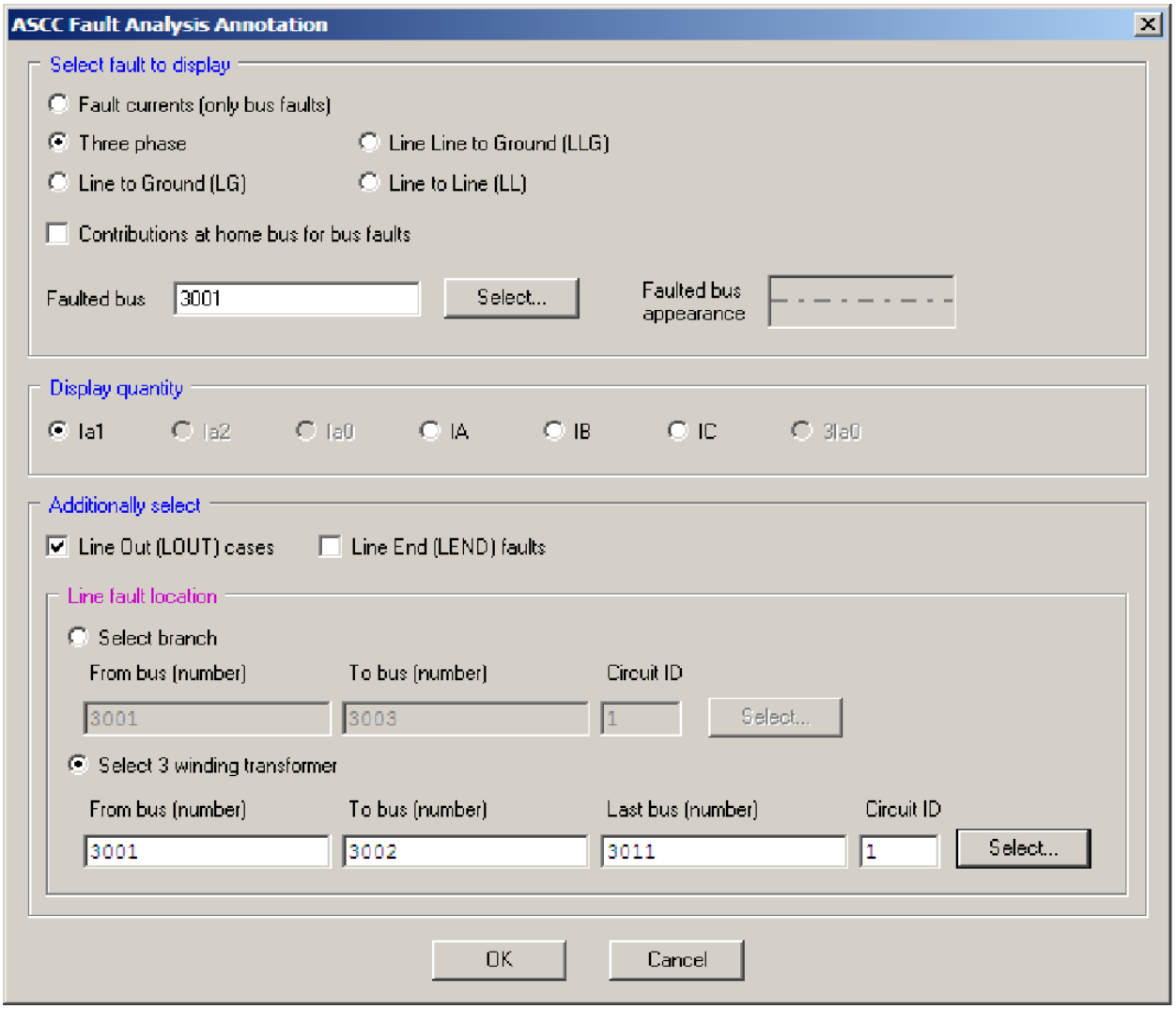

Figure 12.38. Fault Analysis Annotation Dialog, Three Winding Transformer Winding Outage

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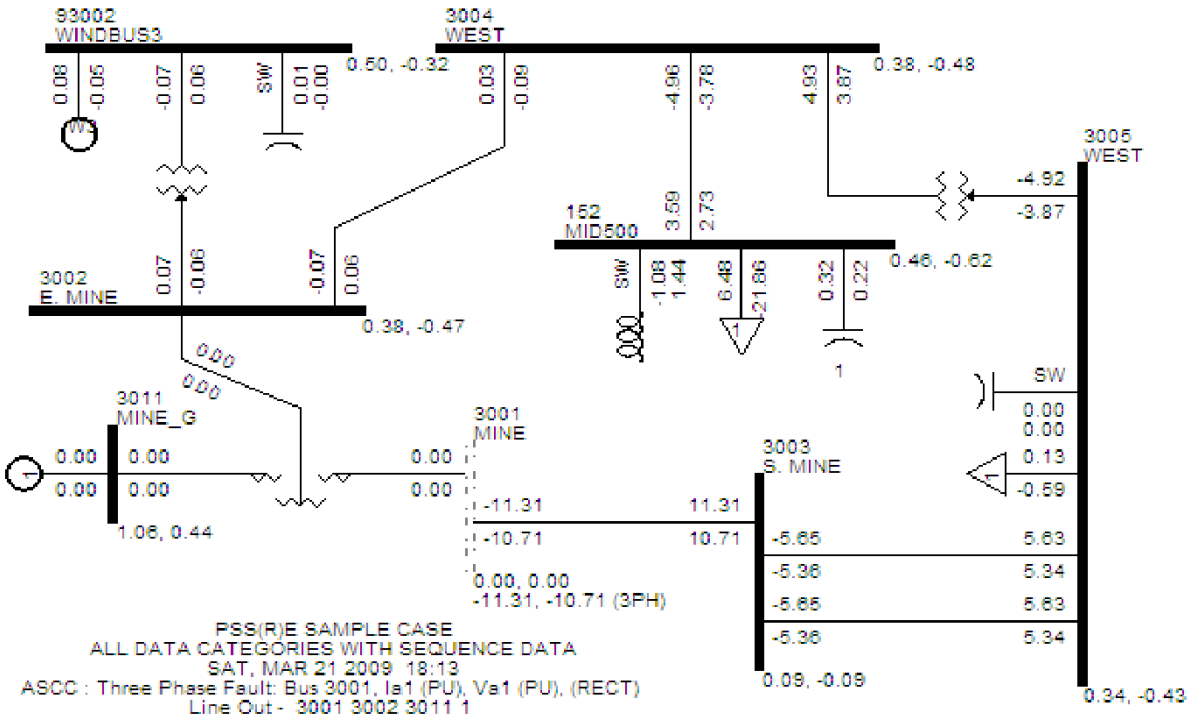

Figure 12.39. Slider Diagram, Three Winding Transformer Winding Outage Fault Currents

Total Fault Currents and Fault Current Contributions, Three Winding Transformer Winding Line End Fault – 1st Line End Fault

The slider diagram below shows fault currents and post-fault voltages for a three phase fault at the end of three winding transformer winding: faulted bus 3001, three winding transformer 3001-3002-3011-1, and fault at the end of winding connected to bus 3002.

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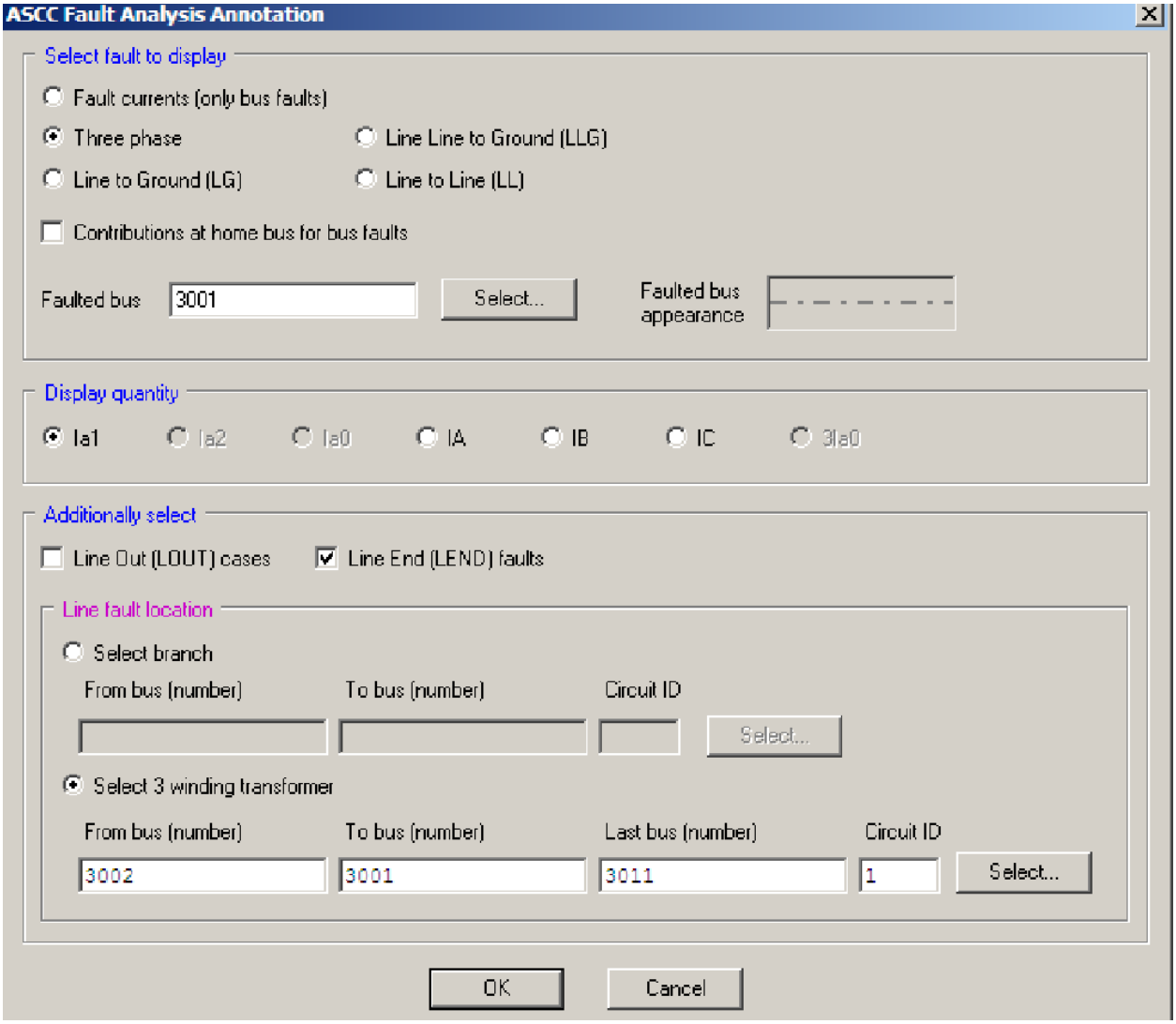

Figure 12.40. Fault Analysis Annotation Dialog, Three Winding Transformer, 1st Line End Fault

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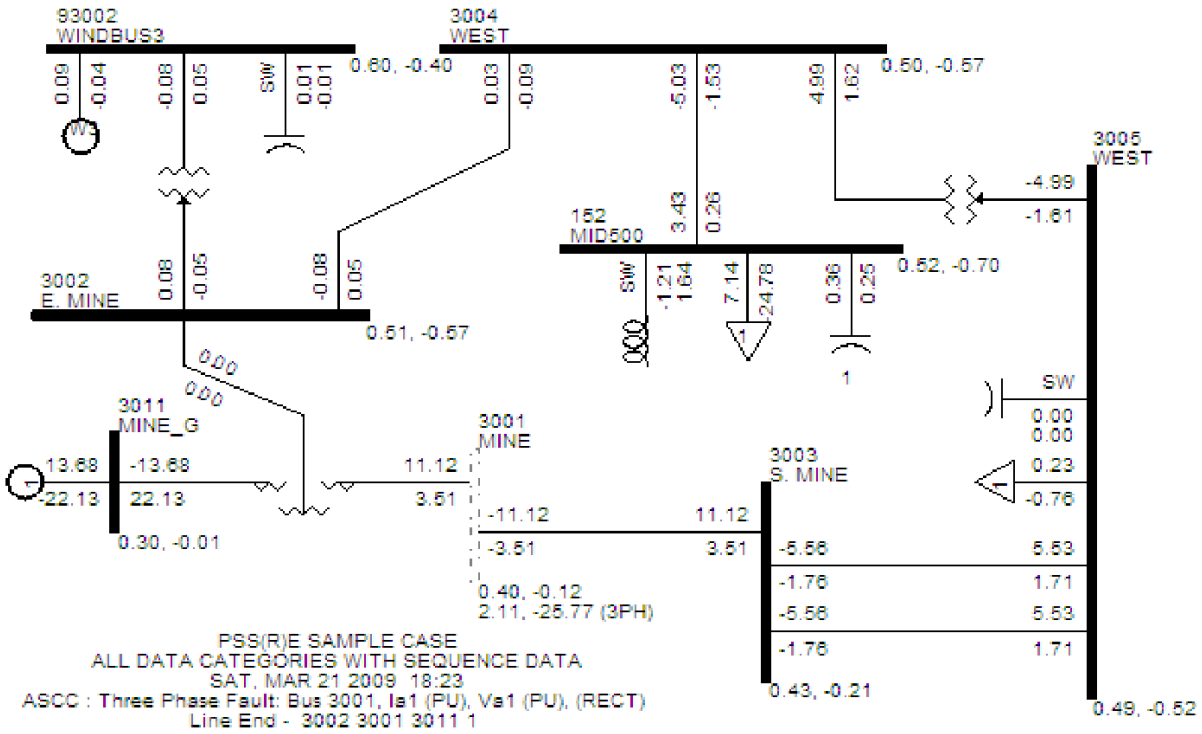

Figure 12.41. Slider Diagram, Three Winding Transformer, 1st Line End Fault Currents

Total Fault Currents and Fault Current Contributions, Three Winding Transformer Winding Line End Fault – 2nd Line End Fault

The slider diagram below shows fault currents and post-fault voltages for a three phase fault at the end of three winding transformer winding: faulted bus 3001, three winding transformer 3001-3002-3011-1, and fault at the end of winding connected to bus 3011.

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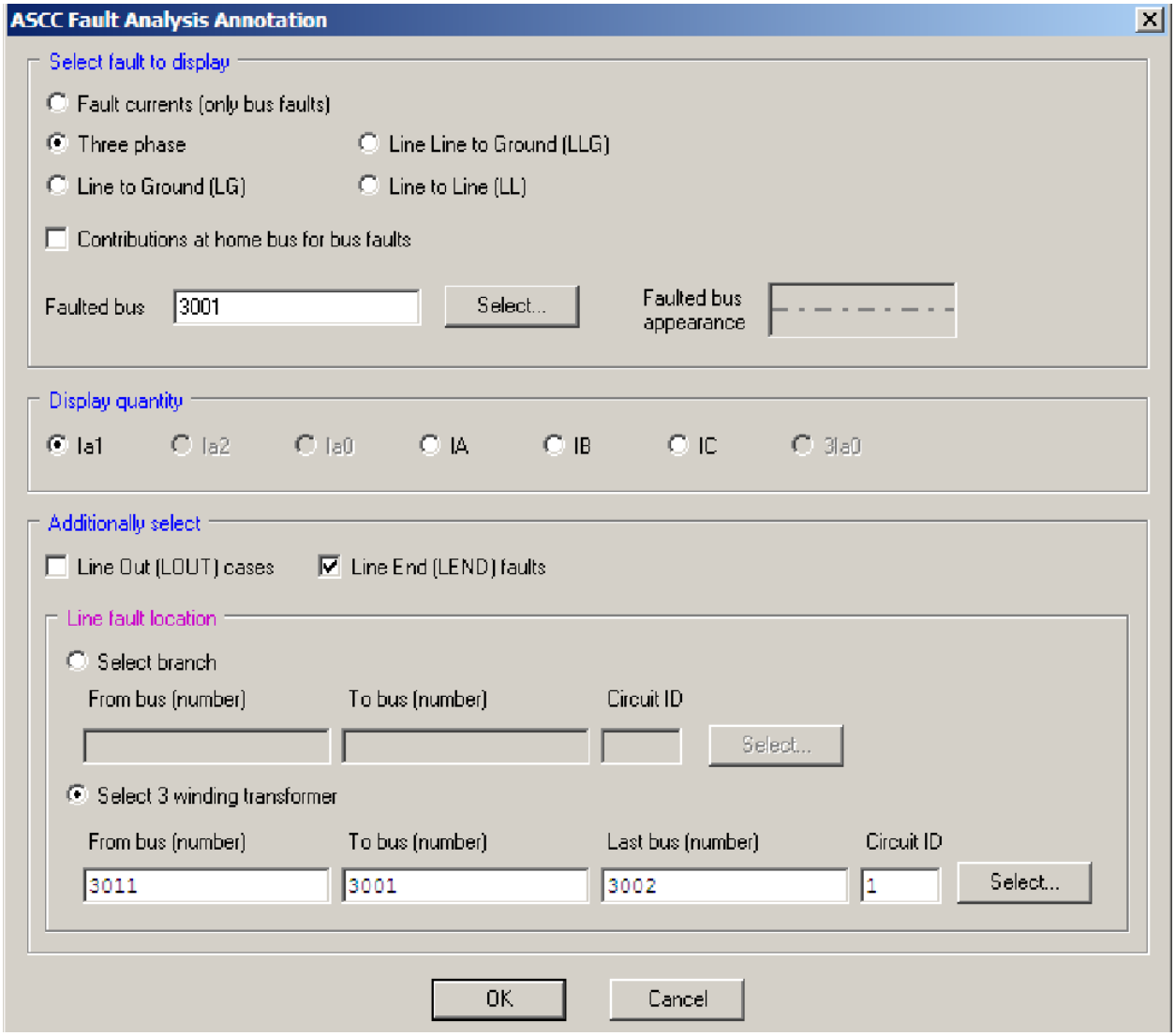

Figure 12.42. Fault Analysis Annotation Dialog, Three Winding Transformer, 2nd Line End Fault

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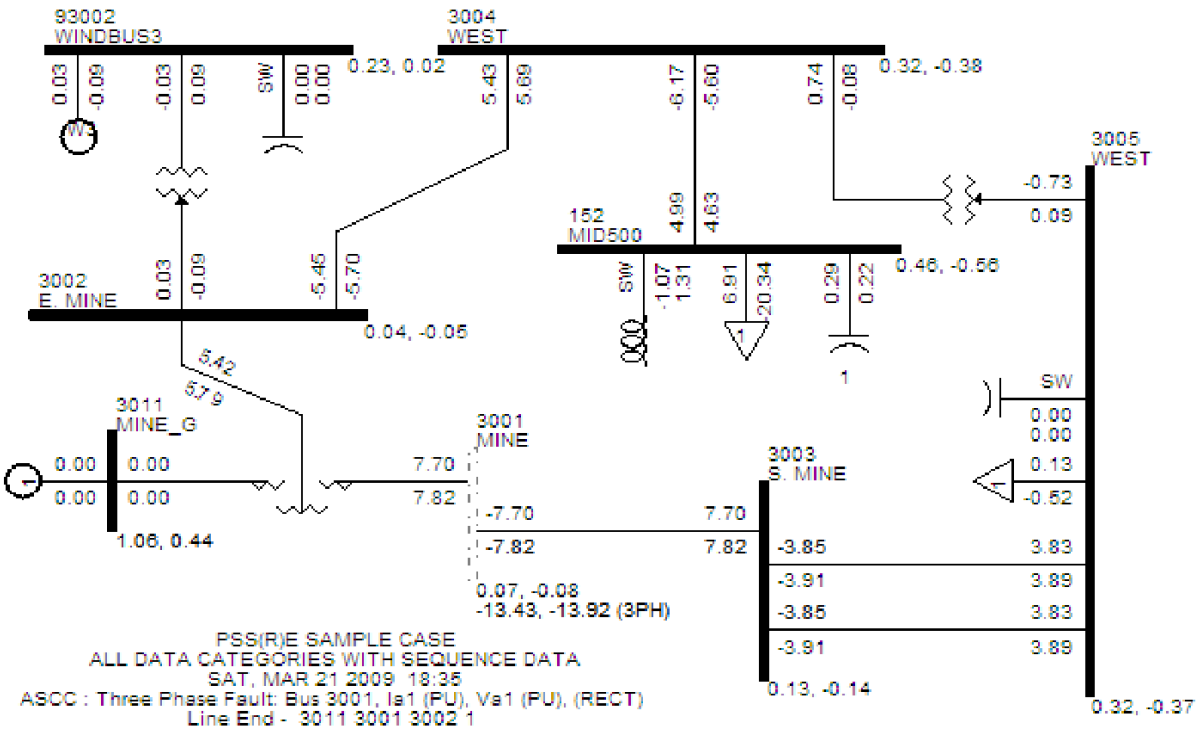

Figure 12.43. Slider Diagram, Three Winding Transformer Winding, 2nd Line End Fault Currents

Unbalanced Fault Currents

The slider diagram below shows fault currents and post-fault voltages for a line-to-ground (LG) fault at bus 3004.

The sequence data for shunts (loads, fixed shunts and switched shunts) are entered as negative and zero sequence shunt data records in sequence data file. The shunt current contributions from these shunt elements for unbalanced faults are displayed as combined shunt current of all shunts on one of these shunt elements. For example, in the slider diagram below, combined shunt current is displayed on loads at bus 152 and at bus 3005.

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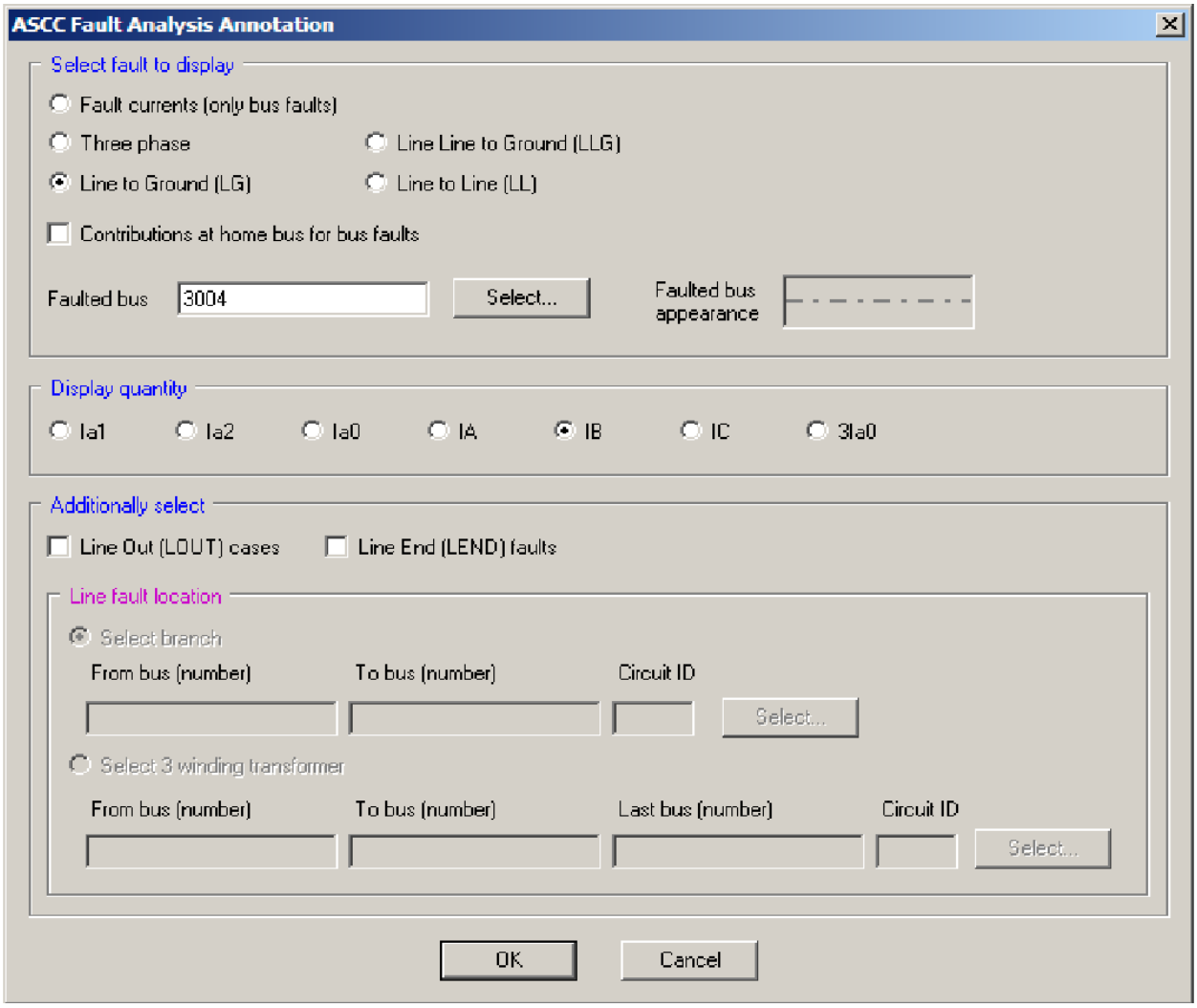

Figure 12.44. Fault Analysis Annotation Dialog, Unbalanced Fault Currents

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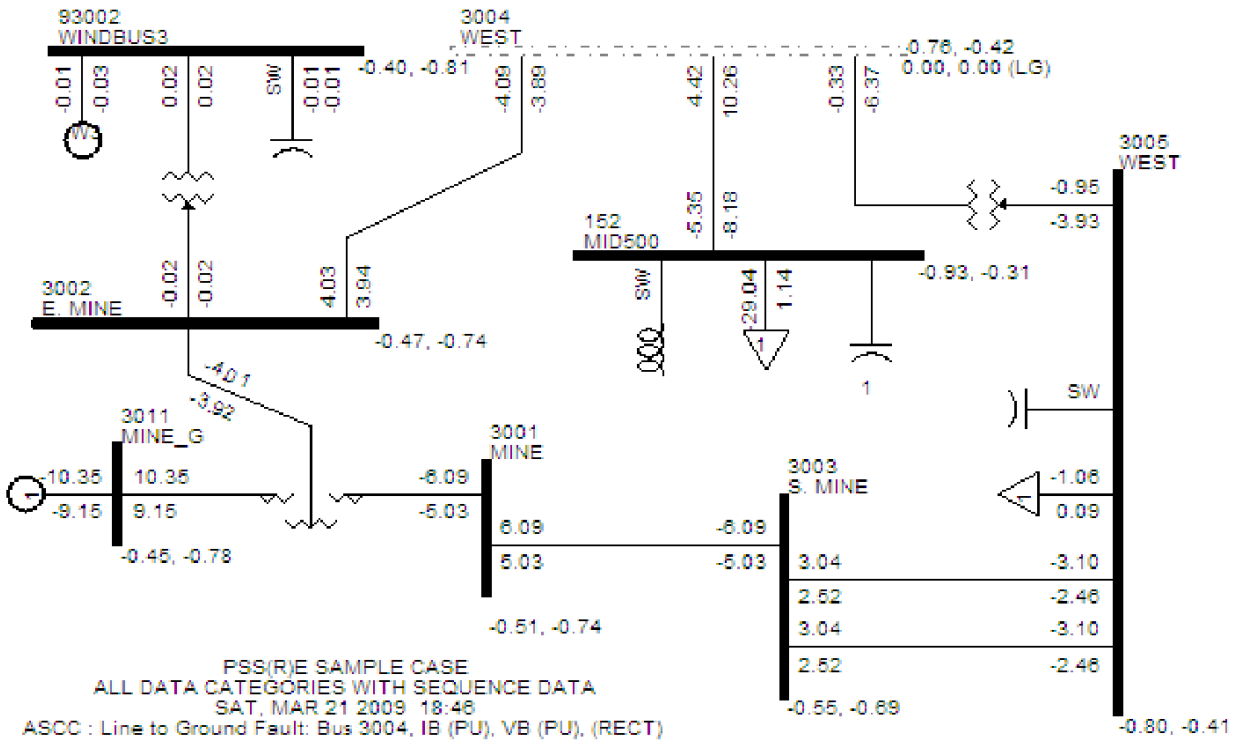

Figure 12.45. Slider Diagram, Selected Fault, Unbalanced Fault Currents

12.32. Exporting Power Flow Results to Excel

Run Activity Export Results - GUI
Power Flow > Reports > Export ACCC, PV/QV results to Excel... *Start > Programs > PSSExx > Export results to Excel* *[Export PSS®E Data/Result to Excel]*

Table 12.10. PSS®E to Excel Export Options

Option	Description
Data/Results to Export	Report type to export, with data Options. • ACCC Analysis Results • PV Solution Results • QV Solution Results
ACCC Analysis Results Options	*[Specify pssexcel.accc Options]* dialog, with the following options: • Solution Type

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Option	Description
	- Contingency - Corrective Action - Tripping • Rating - Rating A, Rating B, or Rating C • Bus Mismatch (default = 0.5) • System Mismatch (default = 5.0)
Data/Result File	ACCC analysis output file (*.acc) to be exported
Excel Workbook File	(*.xls) User-specified filename or Excel default, if no name is specified.
Quantities to Export	One worksheet for each selected item in this field is created in a specified workbook. The name of the worksheet is the name of the quantity to export.
Excel Sheet Prefix	The worksheet name will be prefixed by the specified text.
Overwrite Worksheets	Enable (default) to overwrite existing worksheets. Disable to copy existing worksheets; filenames are automatically appended with (#), where # is the next sequence number.
Show Workbook	Enable (default) to open and display workbook as it is being written to. Disable to populate workbook without displaying it.
Split Extended Bus Names	Enable (default) to split extended bus names into three columns - Bus Number, Bus Name and Bus Voltage.

Additional Information
PSS®E GUI Users Guide, Exporting Power Flow Results to Microsoft Excel®

12.33. Displaying Power Flow Solution on the Diagram

GOUT

Run Line Mode Activity GOUT - CLI

```
ACTIVITY?
>>GOUT
Enter  0 to exit
       26 for interactive display
       30 for default printer
>>
```


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Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The graphical power flow output activity GOUT has been replaced by a similar function in the Diagram View of the Graphical User Interface (GUI). The *PSS®E GUI Users Guide*, Diagram View describes the Diagram View.

The old line mode dialog of activity GOUT is implemented at the command line input and results in the execution of the Diagram View's GOUT function.

The user specifies the bus or branch to be displayed, and the drawing is generated. While the dialog allows the selection of either a bus display or an expanded branch display, only the bus display is supported. If the branch display is selected, a bus display of the first bus specified is produced.

Additional Information
PSS®E GUI Users Guide,Displaying Power Flow Results *PSS®E Command Line Interface (CLI) Users Guide*, Displaying Power Flow Solution on the Diagram

12.34. Displaying Power Flow Data on the Diagram

GEXM

Run Line Mode Activity GEXM - CLI
ACTIVITY? >>**GEXM** Enter 0 to exit 26 for interactive display 30 for default printer >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The graphical power flow data examination activity GEXM has been replaced by a similar function in the Diagram View of the Graphical User Interface (GUI). The *PSS®E GUI Users Guide*, Diagram View describes the Diagram View.

The old line mode dialog of activity GEXM is implemented at the command line input and results in the execution of the Diagram View's GEXM function.

The user specifies the bus or branch to be displayed, and the drawing is generated. While the dialog allows the selection of either a bus display or an expanded branch display, only the bus display is supported. If the branch display is selected, a bus display of the first bus specified is produced.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Displaying Power Flow Data on the Diagram

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Chapter 13
Balanced Switching

13.1. Overview of Balanced Switching

The voltage versus current characteristics of synchronous generators are different in different time regimes. Power system network solutions must, therefore, be categorized according to the conditions just before and at the instant for which the solution is to apply. The time regimes of significance in network solutions are illustrated in Figure 13.1, "Time Regimes Considered in Power System Simulations".

All power system simulations assume that the system is in the steady state for an extended period prior to the time, t. It is assumed that the first incident of interest, such as a switching operation, fault, load change, or control setpoint change, occurs at, t. Power flow calculations apply to the instant, t^- (t minus). The power flow generator boundary conditions, therefore, assume that the system is in the steady state, or more practically, experiencing the gentle motions of normal undisturbed operation. Every power flow calculation establishes the condition of the entire transmission network, outward from generator terminals to load terminals, for an instant, t^-.

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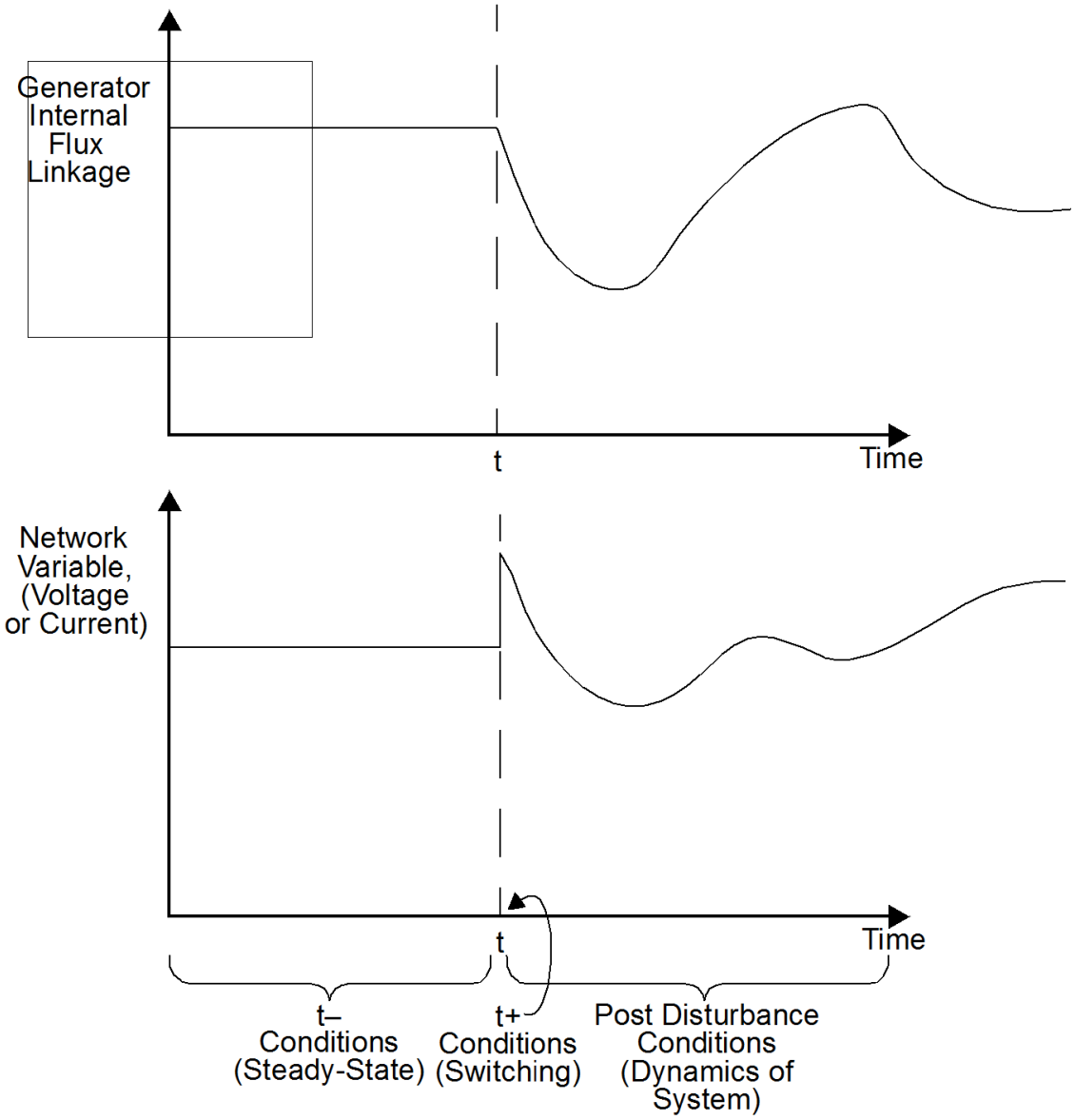

Figure 13.1. Time Regimes Considered in Power System Simulations

Any switching operation, fault, significant load change or change of control inputs (governor or excitation system reference) starts the system moving. PSS®E simulations assume that high frequency switching transients in the electric network die away very rapidly in relation to the time durations of the transients that are of interest. The power frequency phasors representing currents and voltages throughout the network are assumed to change instantaneously at time t.

The transients of the flux linkages in the rotors of electrical machines are of prime interest and must be accurately accounted for. Simulations of conditions at, t^+ (t plus), and later must, therefore, use boundary conditions that recognize dynamic, rather than steady-state, behavior of equipment. Both generator and load characteristics applicable to t^+ are different from those applying at t^-. Loads are commonly assumed to

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have a constant MVA steady-state characteristic in steady-state power flow solutions applying to t^-, but to be better modeled by a mixture of constant current and constant impedance characteristics at t^+ and later.

The instantaneous change caused by a switching is followed by a period when all generator flux linkages, rotor angular positions, and other power plant quantities vary, as dictated by the differential equations governing their dynamic behavior.

The power flow database of PSS®E allows the entry of generator and load data pertaining to pre-disturbance conditions at t^-, to switching conditions at t^+, and to system dynamic behavior over an arbitrarily long period after the initiation of the disturbance. The user of PSS®E may then obtain a solution for any of the three time regimes, *steady-state*, *switching*, or *dynamic behavior*, by executing appropriately selected sequences of PSS®E analytical processes.

Switching, or t^+, solutions and dynamic simulations require the generator boundary conditions to be set in accordance with the electromagnetic laws governing rotor flux linkages. Therefore, the power flow boundary condition in which power output and bus voltage are specified, must be replaced by a specification of a Thevenin or Norton source where instantaneous value is determined by instantaneous values of flux linkages. The generator boundary conditions applying at instants, t^- and t^+, are illustrated by Figure 13.2, "Standard Power Flow Model and Norton Equivalent Used for Switching and Dynamic Studies". The t^+ boundary condition recognizes that generator rotor flux linkages must obey Lenz's and Maxwell's laws.

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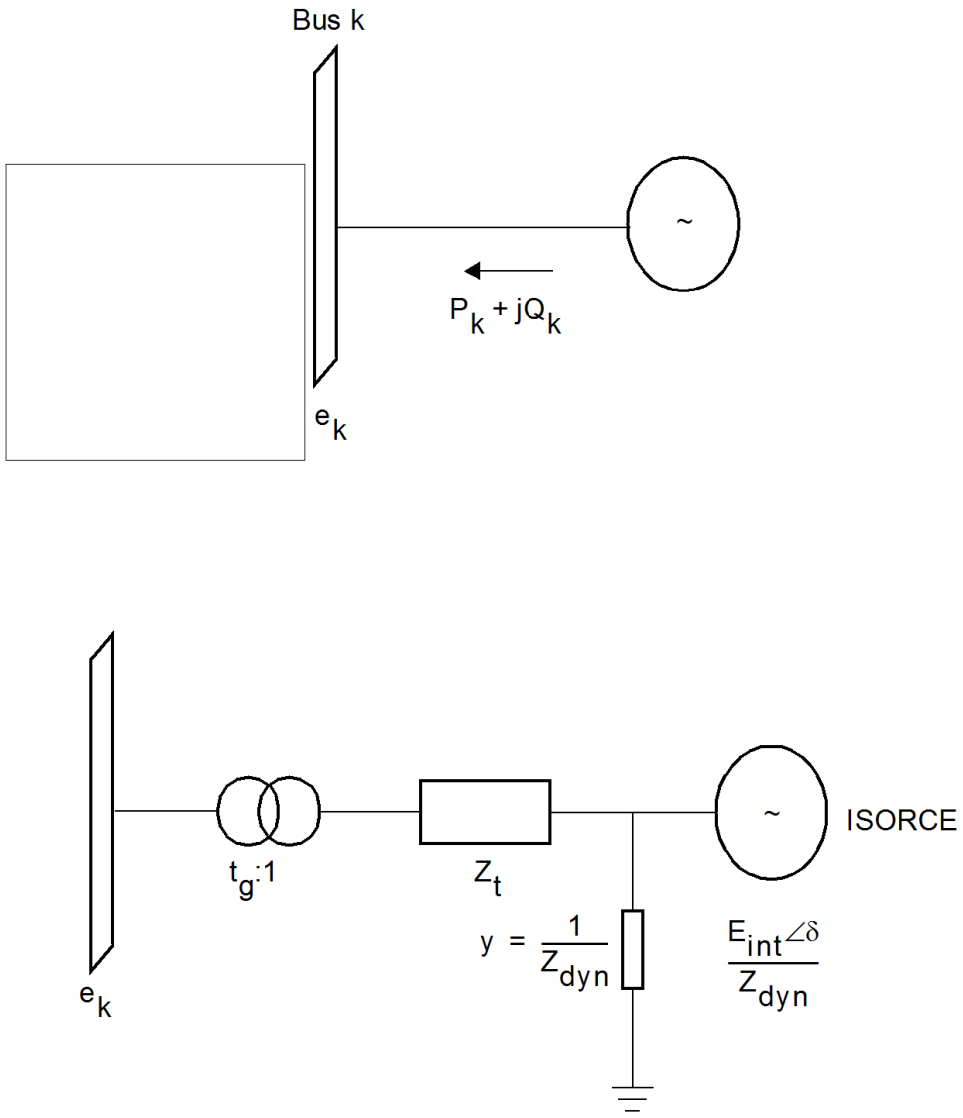

rton Generator Equivalent for Switching and Dynamic Simulation Calculations

Figure 13.2. Standard Power Flow Model and Norton Equivalent Used for Switching and Dynamic Studies

13.1.1. Objectives of a Balanced Switching Study

The objective of a switching study is to calculate the conditions that will exist in the power system just after a sudden change such as the opening of a transmission line, switching on of a large load, application of a fault, or tripping of a generator. These calculations are useful, for example:

1. To show the immediate voltage dip caused by switching on of a large motor.

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2. To give symmetrical fault duty at a bus.

3. To show the immediate change of flow that will be seen on each tie-line into an area (before the inertial swing of rotor angles) when one such tie or a generator within the area is tripped.

4. To show the voltage rise which occurs when a line is opened at one end. This could separate a generation plant from the network and result in what is commonly known as load rejection.

A switching study is a calculation of conditions at time, t^+. A switching study is, in effect, the calculation of the transmission system conditions at the first instant, $t = t^+$, of a dynamic simulation, separated from the subsequent calculation of conditions at later instants, $t > t^+$. Switching study results are presented and examined with the same output and limit checking activities as are used in power flow work studies.

13.1.2. Preparing a Power Flow Case for Balanced Switching

Prior to performing the switching operations, the user must prepare the power flow case. The three steps are to:

- Ensure the power flow case is solved and contains the required generator dynamic data
- Convert the generators to a Norton Equivalent (generator boundary condition, activity CONG)
- Convert the loads to a voltage dependent model (load boundary condition, activity CONL)

These operations result in a converted power flow case which can be saved in its converted form for subsequent switching analyses. Note that this converted case should be saved using a different name because the conversion process is not reversible.

13.1.3. Establishing the Power Flow Base Case for Balanced Switching

The starting point for a switching study is a conventional solved power flow case giving data and solution corresponding to the pre switching (t^-) condition. Conventional procedures are used with the provision that the dynamic impedance, ZSORCE, must be included in the database for *every* generator, and step-up transformer data, XTRAN and GENTAP, must be present for all generators where the implicit step-up transformer treatment is used.

The generator dynamic impedance and step-up transformer parameters are not needed during power flow solutions; they may be added to the power flow but a solved power flow case need not be resolved after any item of generator dynamic impedance or step-up transformer is added or changed because the dynamic impedance and step-up transformer parameters affect only the Norton equivalent current sources, ISORCE, that will represent the generators in the post switching calculation.

13.1.4. Performing a Balanced Switching Study

The sequence of steps required to perform a balanced switching study is shown in Figure 13.3, “Switching Analysis – Sequence of Processes”. It can be seen that the basic preparation is as described as discussed in the previous sections. After that preparation is complete, the user can perform a variety of switching studies on the converted power flow case. The method involves:

- Performing the switching action by changing the network as required

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- Obtaining a solution and producing the results

The switching of network elements is performed using the processes used by the conventional power flow facilities. Furthermore, the output of results and network conditions can be performed using the same output and reporting facilities used for the conventional power flow and getting a solution.

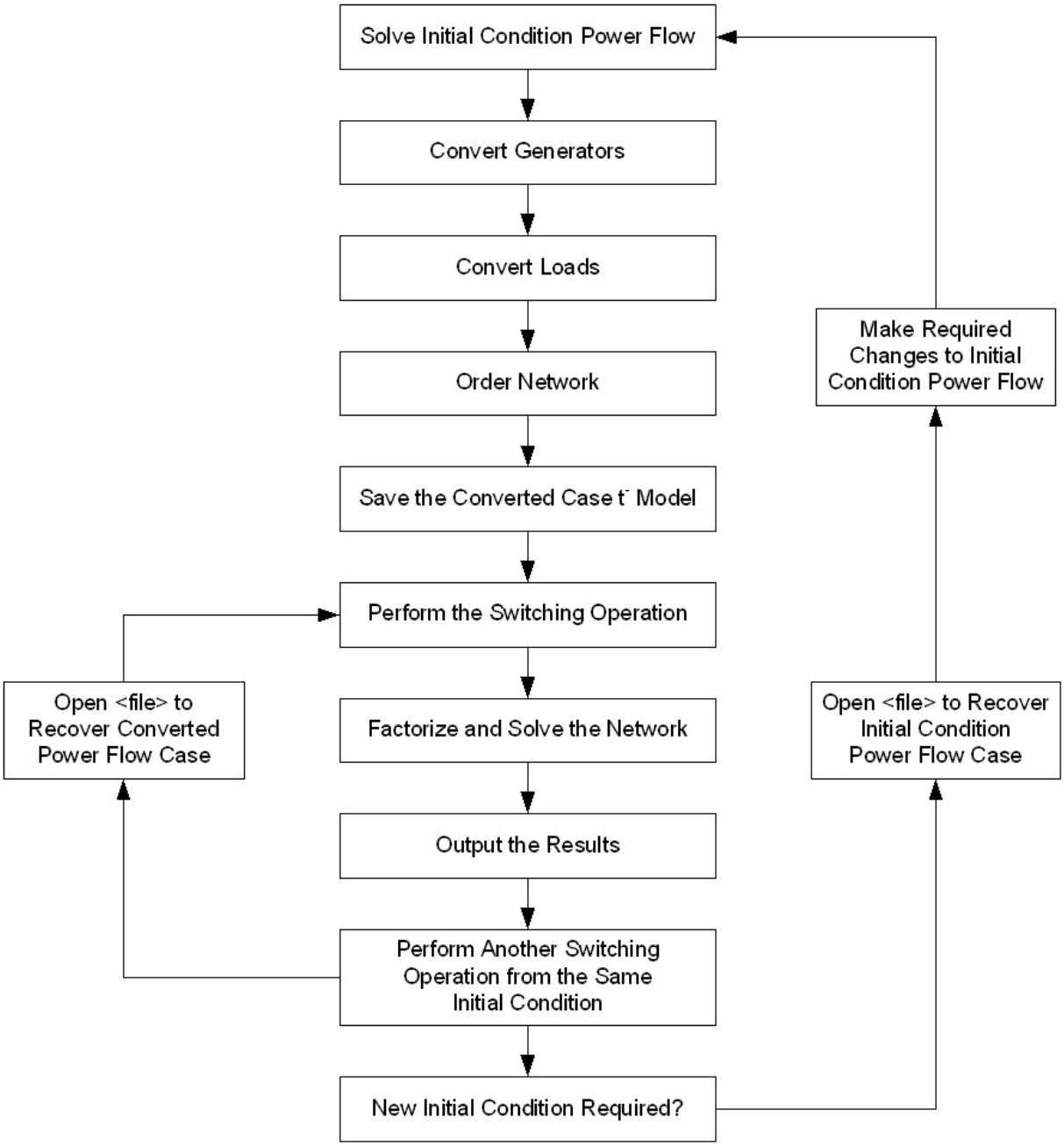

Figure 13.3. Switching Analysis – Sequence of Processes

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13.2. Converting Generators

CONG

Run Activity CONG - GUI
Power Flow > Convert Loads and Generators... *[Convert / Reconstruct Loads and Generators]* OPTION: Convert Generators
Run Line Mode Activity CONG - CLI
ACTIVITY? >>**CONG**
Interrupt Control Codes
None

The generator and induction machine conversion activity CONG initializes the in-service machines in the working case in preparation for dynamic simulation calculations, network solutions used in switching studies (activity TYSL), and the three sequence equivalencing activity SCEQ. It also retains the conditions at each FACTS device and VSC dc line for use in their modeling in switching studies.

13.2.1. Converting Machines for Balanced Switching

PSS®E uses the same power flow working case for power flow, fault analysis, and dynamic simulation calculations. The representation of a machine as a voltage behind an apparent impedance, as required for fault analysis and dynamic simulation, is handled by a Norton equivalent for each machine. The Norton source current is determined from the machine boundary conditions as determined from the power flow (t^-) solution.

For conventional machines, the generator dynamic impedance may be taken to be either the generator power flow attribute, ZSORCE, or the fault analysis generator attribute, ZPOS. Normal practice is to set ZSORCE equal to subtransient impedance. In setting ZPOS, either subtransient, transient or synchronous reactance may be selected, depending upon the requirements of fault analysis. Those switching studies taking the dynamic impedance to be subtransient machine impedance, Z´´, give an *accurate* calculation of conditions immediately after a switching, while those switching studies taking the dynamic impedance to be transient machine impedance, Z´, give an *approximate* calculation of conditions roughly three to five cycles after a switching.

When the power flow impedance ZSORCE is selected as the impedance characterizing conventional machines, IMZPSN, the impedance characterizing induction machines, has a resistance of Ra, the machine armature resistance, and a reactance X" that is calculated from the equivalent circuit parameters as

$$X'' = X_a + \frac{X_m\left[X_3 + \frac{X_1 X_2}{X_1 + X_2}\right]}{X_m + X_3 + \frac{X_1 X_2}{X_1 + X_2}}$$

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When the fault analysis impedance ZPOS is selected as the impedance characterizing conventional machines, IMZPSN is set using the induction machine locked rotor to rated current ratio, ILR2IR, if it is specified as a non-zero value with the induction machine sequence data. The magnitude of the impedance is set to the reciprocal if ILR2IR, and the resistance and reactance of IMZPSN are set according to this impedance magnitude and the resistance to reactance ratio, RM2XM, specified with the induction machine sequence data. If ILR2IR is zero, IMZPSN is set from the equivalent circuit parameters as described above for the case of ZSORCE being used to characterize conventional machines.

The initialization of the Norton sources and changeover of boundary conditions is illustrated in Figure 13.5, "Change of Generator Boundary Condition for Switching". The conversion process, which initializes all source currents to correspond to the flux linkages behind the dynamic impedance (ZSORCE or ZPOS), sets flags to cause ISORCE to be held constant in all subsequent solutions, and changes all Type 3 buses to Type 2. The choice of dynamic impedance is controlled during the conversion process and selected by the user.

13.2.2. Generator Apparent Impedance

The data characterizing the Norton equivalent for each generator consists of the machine MVA base (MBASE) and the machine impedance (ZSORCE). Values for these data items are entered as generator data via the PSS®E activities READ, TREA, RDCH, or MCRE, and they may be modified with the data changing activity CHNG or the *[Spreadsheet]*. The value specified for ZSORCE may be the generator transient impedance, subtransient impedance, or any other value of impedance that may be a meaningful characterization of the generator for the study at hand. In addition, the generator step-up transformer may be represented as part of the generator model itself or explicitly as a network branch (refer to Section 13.2.3, "Generator Step-Up Transformers"). Refer to Generator Data for additional details on these data parameters. Note that the value of ZSORCE must be specified on the generator MVA base (i.e., on MBASE base) and that it is a complex number (resistance and reactance).

The correspondence between the commonly recognized Thevenin equivalent generator representation and the Norton equivalent used in PSS®E is summarized in Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent". Note that the Thevenin internal bus *must not* be included in the PSS®E power flow case. A generator is modeled throughout PSS®E by its terminal and/or high side buses only. Furthermore, the fictitious shunt branches representing generators are automatically separated from real shunt admittance at the generator buses so that their presence does not affect power flow output reports from PSS®E. Only the shunt elements representing actual reactors and capacitors are shown under the heading SHUNT in PSS®E reports.

ZSORCE and MBASE data need not be entered if the working case being constructed is to be used only for standard power flow solutions. When no data is entered, MBASE and ZSORCE take on default values equal to the system base MVA and j1.0 pu, respectively.

The report generated by activity GEOL will not be valid for any generators for which MBASE and ZSORCE are not specified.

13.2.3. Generator Step-Up Transformers

The user may select one of two methods of representing the generator step-up transformer for each machine.

The traditional method involves representing the step-up transformer as a standard power flow transformer branch. Both the generator terminal bus and the high side bus are contained in the case along with a transformer branch between them representing the step-up transformer. The terminal bus is the Type 2 (or 3)

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bus and the high side bus has a Type code of 1. This representation is shown by Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent"b.

The alternative approach is to represent the step-up transformer as part of the generator modeling. In this method, the high side bus is represented in the power flow case as a Type 2 bus and the terminal bus is not present. The step-up transformer impedance and off-nominal ratio are represented by the generator data items XTRAN (i.e., RT + jXT) and GTAP, respectively (refer to Generator Data). XTRAN is entered on *generator* base (i.e., on MBASE base) and is a complex number (resistance and reactance). This representation is shown by Figure 12-4c.

When XTRAN is non-zero (i.e., the generator bus is the high side bus), the generator boundary conditions (i.e., voltage setpoint, active power output, and reactive power limits) are as seen at the high side bus. Generator terminal conditions may be examined with activity GEOL.

When XTRAN is zero, PSS®E assumes that the step-up transformer is represented by an ac branch and the Type 2 bus is the terminal bus.

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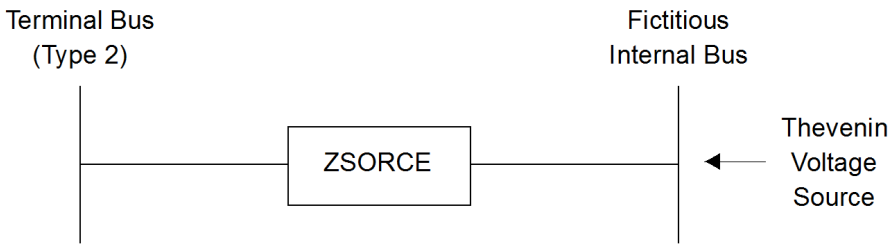
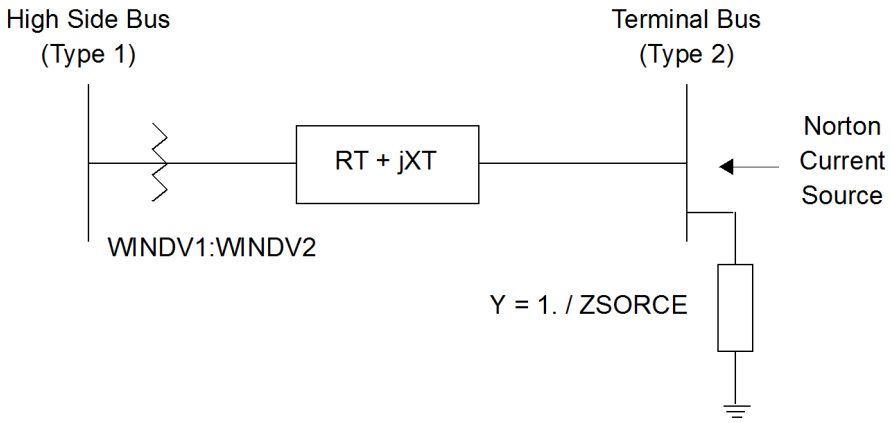
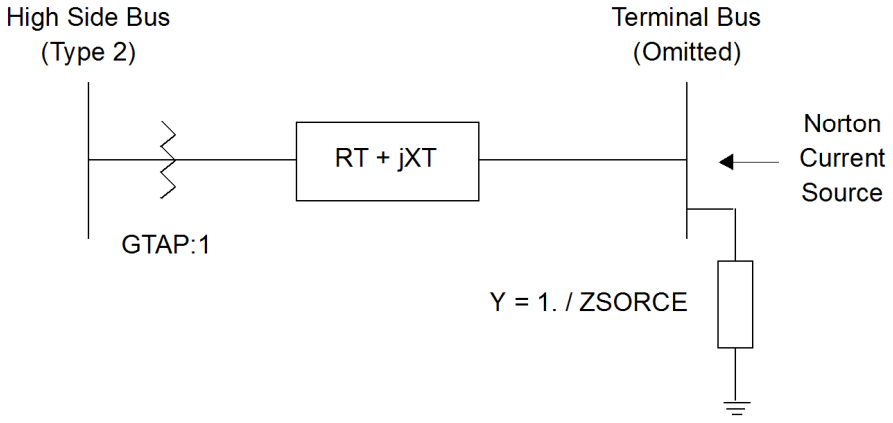


a. Thevenin Generator Equivalent



b. Norton Generator Equivalent Used in PSS®E (XTRAN = 0.0)



c. Norton Generator Equivalent Used in PSS®E (XTRAN > 0.0)

Figure 13.4. Relationship Between PSS E Generator Equivalent and Thevenin Equivalent

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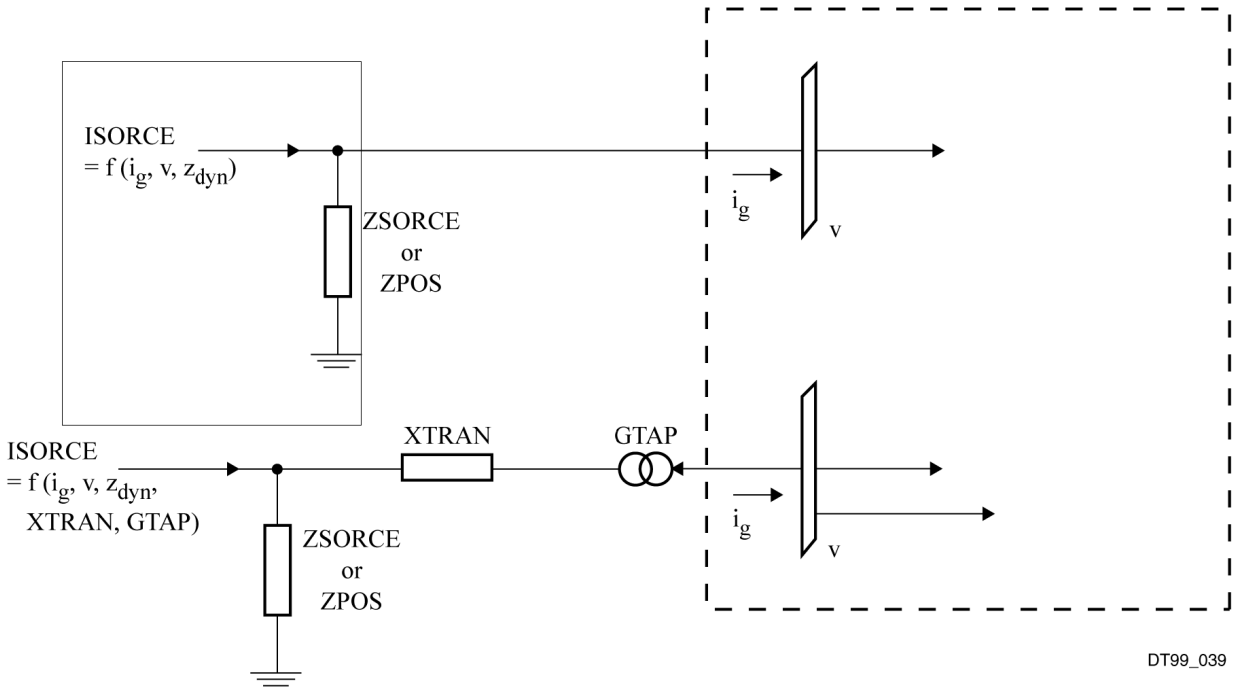
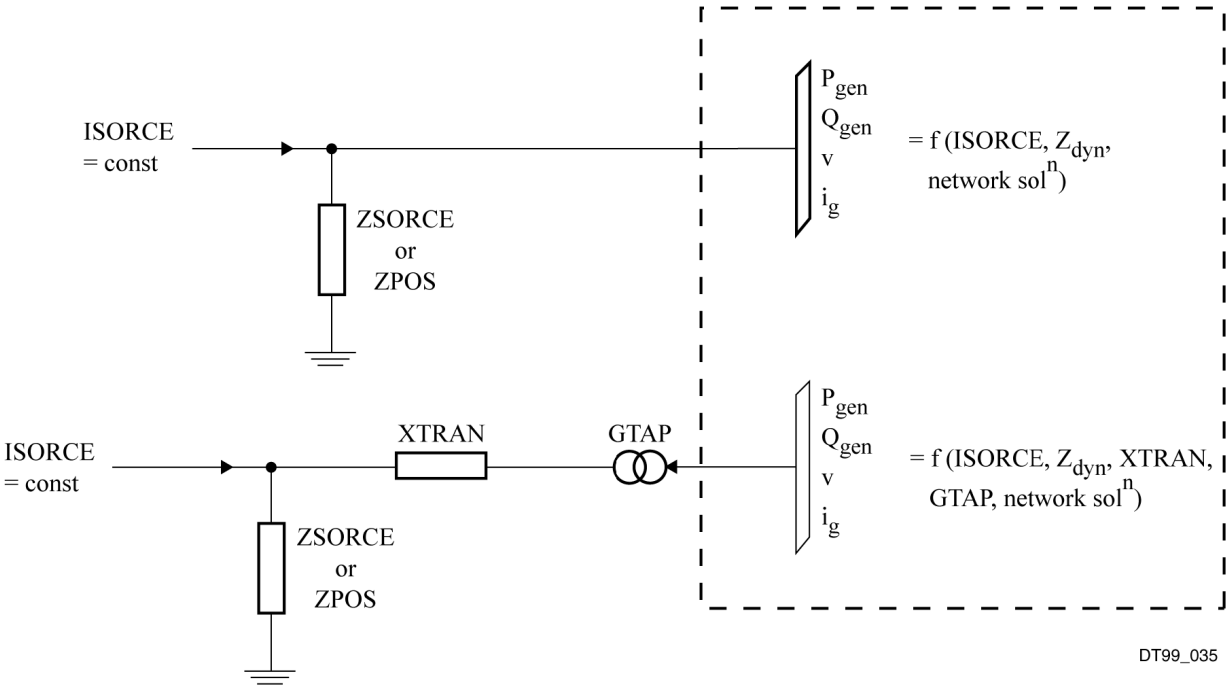


a. Power Flow - ISORCE Follows Network Solution



b. Switching - Network Solution Follows ISORCE

Figure 13.5. Change of Generator Boundary Condition for Switching

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13.2.4. Operation of Activity CONG

Activity CONG calculates the Norton source currents for in-service conventional machines at all Type 2 and 3 buses on the basis of the network conditions at the bus (voltage and power output) and the values specified for MBASE, ZSORCE, XTRAN, and GTAP (see Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent"). In addition, the Type codes of system swing buses are changed from 3 to 2, and of var-limited generator buses from -2 to 2. Similarly, activity CONG calculates the Norton source currents for in-service induction machines at all type 1, 2 and 3 buses on the basis of the network conditions at the bus and the data items describing the equivalent circuit of the machine.

The working case must be solved to an acceptable mismatch level and appropriate values of MBASE, ZSORCE, XTRAN, and GTAP must have been specified for all in-service generators prior to executing activity CONG.

When the *Use ZPOS* option is specified, the positive sequence generator impedances as specified in sequence data input (ZPOS; refer to Generator Sequence Data) are used in place of ZSORCE in determining the Norton source currents. If sequence data had not previously been read into the working case, activity CONG proceeds with the generator conversion using ZSORCE.

Activity CONG is neither reversible nor repeatable. Therefore, it is *strongly* recommended that the power flow case be saved prior to executing activity CONG. Furthermore, the only *network* solution permitted following activity CONG is activity TYSL.

Activity CONG generates an alarm for any bus with a machine for which an impedance of zero is detected during the Norton current calculation. When XTRAN is zero, the generator impedance ZSORCE or ZPOS, as appropriate, must be non-zero; when XTRAN is non-zero, the effective impedance

$GTAP^2$ (ZG + XTRAN)

must be non-zero, where ZG is ZSORCE or ZPOS, as appropriate. If this restriction is violated, an appropriate message is printed at the conclusion of activity CONG and the working case is unchanged. Appropriate data corrections must then be made before executing activity CONG again.

Activity CONG generates an alarm for any Type 2 or 3 bus that does not have a plant sequence number assigned to it (i.e., any Type 2 or 3 bus for which no generator data record was entered via activities READ, TREA, or RDCH). In this case, the bus type code is set to 1, the conversion is not valid, and the unconverted case must be picked up and the appropriate corrections made before proceeding with CONG. (Refer to Generator Data and Plant and Machine Sequence Numbers.)

Activity CONG preserves the current magnitude of the shunt element of each in-service FACTS device, and the series voltage of each in-service series FACTS device for use in activities TYSL and BKDY. Similarly, it preserves the ac active and reactive power injection at each VSC dc line converter bus for switching study and circuit breaker duty solutions. Refer to Section 13.7, "Solving the Converted Case" for details on the handling of these devices in activities TYSL and BKDY.

Additional Information
PSS®E GUI Users Guide, Generators *PSS®E* Command Line Interface (CLI) *Users Guide*, Converting Generators

13.3. Converting Load Characteristics

CONL

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Run Activity CONL - GUI
Power Flow > Convert Loads and Generators... *[Convert / Reconstruct Loads and Generators]* OPTION: Convert / Reconstruct Loads; Convert constant MVA loads
Run Line Mode Activity CONL - CLI
ACTIVITY? >>**CONL** ENTER UP TO 20 BUS NUMBERS >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The load conversion activity CONL converts the constant MVA load for a specified grouping of network loads to a specified percentage for constant current or constant admittance load characteristics of the existing constant MVA load (refer to Load Data and Section 6.3.13, "Load").

13.3.1. Modeling Load Characteristics

Normal practice in power flow work is to assume that distribution system tap changers and voltage regulators have brought customer voltages to nominal values and, hence, that load at the buses represented in the power flow case may be treated as a constant real and reactive power demand. It is generally recommended, therefore, that load is treated as constant MVA from one power flow case to the next.

This is particularly true of conventional planning studies where demand level is a given quantity which must be maintained under a variety of system normal and abnormal conditions. The Planner's job is to identify the lines and equipment required to sustain the defined demand levels; it is axiomatic that the demand level is a given (constant MVA).

In steady-state and quasi steady-state operation studies, it is reasonable to recognize the loads' dependence on voltage. Such studies include balanced switching, short-circuit analysis (Chapter 11, *Unbalanced Fault Analysis*) and examination of potential voltage collapse or voltage instability. These are conditions under which voltage excursions can be significant.

Similarly, under dynamic conditions, which would be simulated with the PSS®E dynamic simulator, treating loads as purely constant MVA at the reference load value is not acceptable because time delays in distribution voltage-regulating devices prevent them from adjusting customer voltages in the period of interest. Furthermore, loads have a frequency dependence. For dynamic studies, therefore, a variety of load models of varying complexity are available. These models recognize, in more detail, the specific characteristics of loads such as lighting, heating and motor loads. The PSS®E models available for dynamic simulations are discussed in detail in *PSS® E Model Library*.

13.3.2. Basic Load Characteristics

The network solutions of PSS®E recognize load components with constant MVA, constant current, and constant admittance characteristics as discussed in Load Data.

The input functions of PSS®E identify the load at each bus and allow the user to indicate the following components of each load.

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PL	Real power, constant MVA load.
QL	Reactive power, constant MVA load.
IP	Real power, constant current load.
IQ	Reactive power, constant current load.
YP	Real power, constant admittance load.
YQ	Reactive power, constant admittance load.

All of these components are specified in MW and Mvar. The MW and Mvar values for the constant current and constant admittance components are the values that would be consumed by these loads when the bus voltage is unity per unit. The values of these components can be changed, using the editing facility in the Spreadsheet View or the load can be taken out of service by changing its status in the Spreadsheet View.

While it is possible to enter bus loads as combinations of constant MVA, constant current and constant admittance, it is preferable to enter the total load as constant MVA mainly because the majority of steady-state analysis will be performed at or close to nominal voltage and the total demand level is critical to the study at hand. It is a simple exercise to convert system loads from the constant MVA characteristic to either of the voltage dependent characteristics or a combination of all three depending on the type of analysis being carried out. It is equally simple to reconvert the loads to have their original characteristic or any other combination.

13.3.3. Converting Load Characteristics

The conversion process in PSS®E takes constant MVA load and reassigns it all or in part as constant current or constant admittance load. A fraction of the constant MVA load at each bus are transferred to each of the other two load characteristics, according to the following rules

$$S_I = S_i + \frac{aS_p}{V}$$

$$S_Y = S_y + \frac{bS_p}{V^2}$$

$$S_P = S_p \times (1 - a - b)$$

where:

S_p = Original constant MVA load.

S_i = Original constant current load.

S_y = Original constant shunt admittance load.

S_P = Final constant MVA load on bus (real, MW, or reactive, Mvar).

S_I = Final nominal constant current load on bus (real, MW, or reactive, Mvar).

S_Y = Final nominal constant shunt admittance load on bus (real or reactive).

a,b = Load transfer fractions, (a + b) < 1 (Real and reactive load may be allocated by different a and b values.)

v = Magnitude of bus voltage when load conversion is made.

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The conversion and re-allocation process is diagrammed in Figure 12-6.

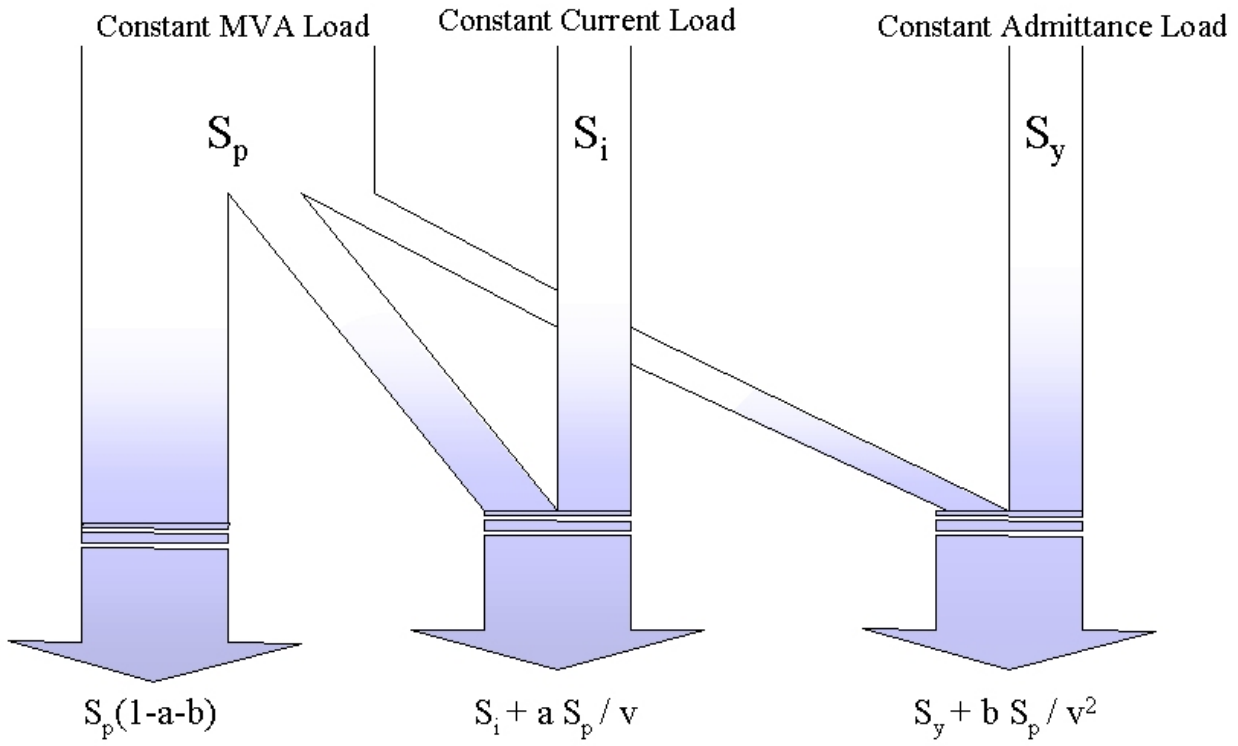

Figure 13.6. Re-allocation of Constant MVA Bus Load

It is useful to note the following points regarding the load conversions:

1. The conversion is performed only on the constant MVA load.
2. Load converted to constant current and constant admittance is added to any existing load represented by those characteristics.
3. The load is converted on the basis of the actual voltage at the network buses. Prior to conversion the working case should be solved to an acceptable mismatch level. Following conversion, the network is still in balance.
4. The split of load among the three characteristics may be different for the active and reactive components of load.
5. Admittance load is stored separately from the bus shunt.

Furthermore, following conversion, only the new components are accessible for editing.

13.3.4. Converting Loads for Balanced Switching

The load boundary condition for switching studies should reflect the relationship between load voltage and current during sudden changes of voltage. The constant (P,Q) characteristic used in power flow studies is generally regarded as unsuitable for switching studies. A mixture of constant current and constant imped-

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ance is usually regarded as a more accurate treatment of loads when the supply voltage is changing rapidly. Load characteristics should be adapted for switching studies as described in Section 13.1.2, "Preparing a Power Flow Case for Balanced Switching".

13.3.5. Operation of Activity CONL

The user may specify the manner of selecting loads to be processed and/or the buses where loads are to be converted (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. When processed by basekV, activity CONL processes all loads at each bus where base voltage falls within the specified base voltage band. When processed by area, zone, or owner, each load assigned to one of the areas, zones, or owners specified is converted (refer to Load Data). The area, zone, or owner assignments of buses (refer to Bus Data) are not considered in these subsystem selection modes.

The user specifies the manner in which the existing constant MVA load is to be apportioned for each grouping of loads to be processed by entering the percentages of the constant current and constant admittance load characteristics.

Activity CONL prints the number of loads converted out of the total number of loads in the working case.

13.3.6. Example of Load Conversion

This example describes a case in which 30% of the real part of the constant MVA load is to be converted to constant current and 20% to constant admittance. For the imaginary component, the conversion is to 20% constant current and 10% constant admittance.

It can be seen further that the loads to be converted can be selected by the Area, Zone or Owner number to which the loads are assigned (NOT to the bus assignment to which the loads are connected). When selected by kV, all loads connected to buses with the selected voltage(s) will be converted in the loads' selected Areas, Zones and/or Owners.

Figure 13.7, "Reference Load Values in Savnw.sav Power Flow Case" shows the MVA loads in the *savnw.sav* power flow case, as initially defined. If the conversion selections above were chosen, the result would be to re-allocate the loads as shown in Figure 13.8, "Converted Loads Re-allocations as per Figure 12-7". The conversion has been done, as can be seen, on all loads.

Note that conversion is done on the basis of the current load bus voltages in the power flow case. The user should be sure to solve the case to an acceptable tolerance before converting the loads.

Pload (MW)	Qload (Mvar)	IPload (MW)	IQload (Mvar)	YPload (MW)	YQload (Mvar)
200.0	100.0	0.0	0.0	0.0	0.0
600.0	450.0	0.0	0.0	0.0	0.0
400.0	350.0	0.0	0.0	0.0	0.0
300.0	150.0	0.0	0.0	0.0	0.0
1200.0	700.0	0.0	0.0	0.0	0.0
100.0	50.0	0.0	0.0	0.0	0.0
200.0	75.0	0.0	0.0	0.0	0.0
200.0	75.0	0.0	0.0	0.0	0.0

Figure 13.7. Reference Load Values in Savnw.sav Power Flow Case

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Pload (MW)	Qload (Mvar)	IPload (MW)	IQload (Mvar)	YPload (MW)	YQload (Mvar)
100.0	70.0	60.4	20.1	40.6	-10.1
300.0	315.0	191.7	95.9	136.1	-51.0
200.0	245.0	127.8	74.6	90.7	-39.7
150.0	105.0	93.1	31.0	64.2	-16.1
600.0	490.0	379.3	147.5	266.5	-77.7
50.0	35.0	30.2	10.1	20.2	-5.1
100.0	52.5	62.3	15.6	43.1	-8.1
100.0	52.5	62.6	15.6	43.5	-8.2

Figure 13.8. Converted Loads Re-allocations as per Figure 12-7

13.3.7. Application Notes

The portion of the load which is represented as constant MVA load is converted as specified by activity CONL. Any of that load to be converted to the constant current and admittance characteristics is added to the load already represented by those characteristics.

It is useful to note the following points regarding the load conversions:

1. Activity CONL converts only the constant MVA load.
2. Load converted to constant current and constant admittance is added to any existing load represented by those characteristics.
3. The load is converted on the basis of the actual voltage at the network buses. Thus, activity CONL requires that the working case be solved to an acceptable mismatch level. Following completion of activity CONL, the network is still in balance.
4. The split of load among the three characteristics may be different for the active and reactive components of load.
5. The presence of constant MVA and/or constant current reactive power load may require slight deceleration in activity TYSL and dynamic simulations for optimum performance.
6. Admittance load is stored separately from the bus shunt.

Additional Information
PSS®E GUI Users Guide, Load Characteristics *PSS®E* Command Line Interface (CLI) *Users Guide*, Converting Load Characteristics

13.4. Reconverting Load Characteristics

RCNL

Run Activity RCNL - GUI
Power Flow > Convert Loads and Generators...

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Run Activity RCNL - GUI
[Convert / Reconstruct Loads and Generators] OPTION: Convert / Reconstruct Loads; Reconstruct loads...
Run Line Mode Activity RCNL - CLI
ACTIVITY? >>**RCNL** ENTER LOAD RECONSTRUCTION OPTION: 1 FOR PRESENT VOLTAGE 2 OR 3 FOR UNITY VOLTAGE: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The load reconstruction and conversion activity RCNL reconstructs the constant MVA load for specified network loads. It then converts the reconstructed MVA load to a specified mixture of the constant MVA, constant current, and constant admittance load characteristics (refer to Load Data and Sections Section 6.3.13, “Load” and Section 13.3, “Converting Load Characteristics”).

13.4.1. Reconverting Load Characteristics

It is possible to reconstruct the constant MVA load from a mixture of the three load characteristics for selected network loads. Subsequently, it is possible to then convert the reconstructed constant MVA load to a different specified mixture of the constant MVA, constant current, and constant admittance load characteristics. The reconstruction process as illustrated in Figure 13.9, “Load Collection and Re-allocation”. The process first collects all load back into a single constant MVA load and then reallocates it into constant MVA, constant current, and constant admittance components in the same manner as the conversion process.

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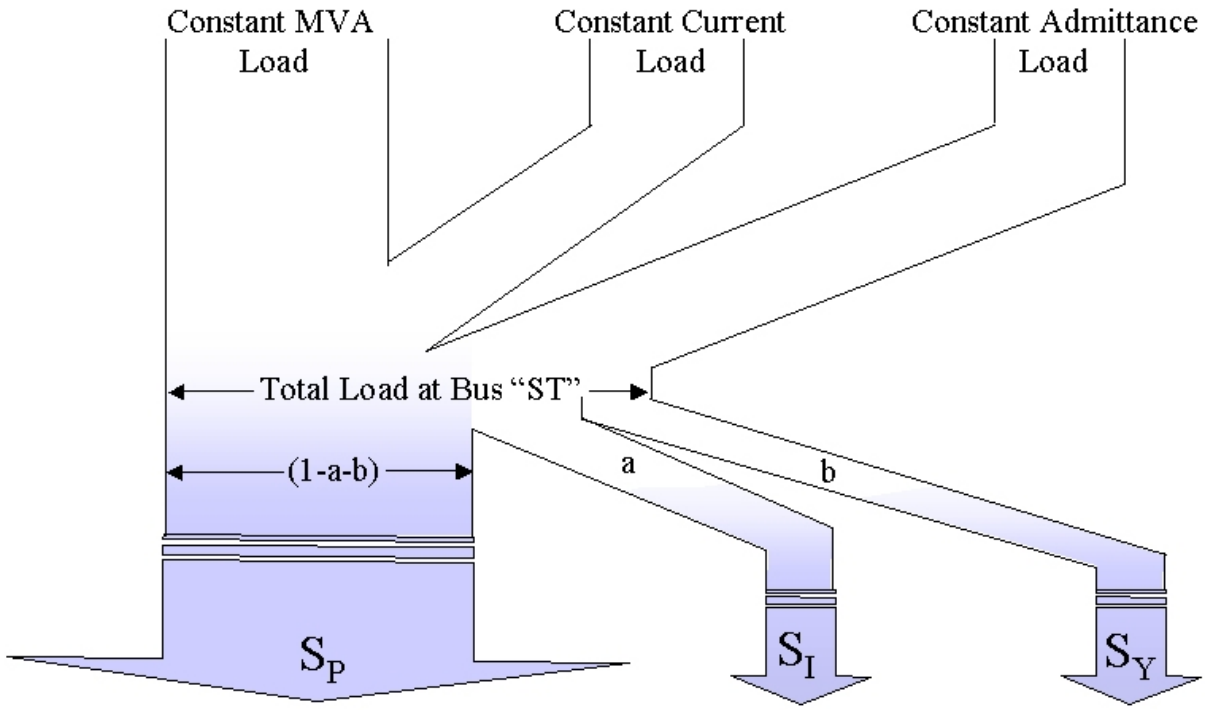

Figure 13.9. Load Collection and Re-allocation

There are two approaches to reconstruction and re-allocation of loads:

- Reconstruct MVA loads based on present bus voltage.
- Reconstruct MVA loads based on unity voltage.

When reconstructing MVA loads based upon present bus voltages, for each bus processed, the individual MVA loads represented by the three load characteristics are summed. This total load is then split among the three load characteristics as specified. Using this load reconstruction option, the network should be solved to an acceptable mismatch level before initiating the reconstruction process. The network will remain in balance at completion of this process.

This process first collects all load into the constant MVA component by

$$ST = S_p + vS_i + v^2S_y$$

where ST is the total of the three load components and then reallocates it by the formula

$$S_I = \frac{aST}{v}$$

$$S_Y = \frac{bST}{v^2}$$

$$S_P = ST(1 - a - b)$$

The symbols used are defined in Section 13.3.3, "Converting Load Characteristics".

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When reconstructing loads based on unity voltage, the three load components are summed assuming one per unit voltage, (v = 1.0) and this total load is split among the three load characteristics as specified. The network will normally not be in balance following this process with this load reconstruction option.

13.4.2. Operation of Activity RCNL

The user specifies the option to be used in reconstructing the MVA load: present voltage or unity voltage. Refer to Section 13.4.4, "Application Notes"for details on the load reconstruction options. The user may specify the buses where loads are to be processed (refer to Section 4.8, "Subsystem Selection") by area, zone, owner, or basekV.

When processed by basekV, activity RCNL processes all loads at each bus where base voltage falls within the specified base voltage band. When processed by area, zone, or owner, each load assigned to one of the areas, zones, or owners specified is converted (refer to Load Data). The area, zone, or owner assignments of buses (refer to Bus Data) are not considered in these subsystem specifications.

Activity RCNL prints the number of loads reconstructed out of the total number of loads in the working case.

13.4.3. Example of Load Reconstruction

In this example the option to reconstruct using present voltage has been selected and the re-allocation is to 10% of total load for both real and reactive power for constant current and constant impedance loads.

Section 13.3.6, "Example of Load Conversion" shows an example conversion process in which Figure 13.7, "Reference Load Values in Savnw.sav Power Flow Case" shows the MVA loads in the *savnw.sav* power flow case, as initially defined. That conversion example re-allocates the loads with the result shown in Figure 13.8, "Converted Loads Re-allocations as per Figure 12-7". The conversion was done on all loads.

With that converted load as a starting point, the reconstruction can be done based on the reallocations shown in Figure 13.9, "Load Collection and Re-allocation" above. The result is shown in Figure 13.10, "Reconstruction of Loads Based on Global 10% Reallocation".

Pload (MW)	Qload (Mvar)	IPload (MW)	IQload (Mvar)	YPload (MW)	YQload (Mvar)
160.0	80.0	20.1	10.1	20.3	-10.1
480.0	360.0	63.9	47.9	68.1	-51.0
320.0	280.0	42.6	37.3	45.4	-39.7
240.0	120.0	31.0	15.5	32.1	-16.1
960.0	560.0	126.4	73.8	133.2	-77.7
80.0	40.0	10.1	5.0	10.1	-5.1
160.0	60.0	20.8	7.8	21.5	-8.1
160.0	60.0	20.9	7.8	21.8	-8.2

Machine | Load | Fixed Shunt | Switched Shunt | Branch | Breaker

Figure 13.10. Reconstruction of Loads Based on Global 10% Reallocation

13.4.4. Application Notes

When reconstructing MVA loads based upon present bus voltages, for each bus processed, the individual MVA loads represented by the three load characteristics are summed. This total load is then split among the

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three load characteristics as specified. Using this load reconstruction option, the network should be solved to an acceptable mismatch level when activity RCNL is specified, and it remains in balance following the execution of activity RCNL.

When reconstructing loads based on unity voltage, the three load components are summed assuming one per unit voltage, and this total load is split among the three load characteristics as specified. The network will normally not be in balance following the execution of activity RCNL with this load reconstruction option.

Additional Information
PSS®E GUI Users Guide, Reconverting Load Characteristics *PSS®E* Command Line Interface (CLI) *Users Guide*, Reconverting Load Characteristics

13.5. Ordering Network Buses for Matrix Manipulation

ORDR

Run Activity ORDR - GUI
Power Flow > Solution > Order network for matrix operations (ORDR)... *[Order Network]*
Run Line Mode Activity ORDR - CLI
ACTIVITY? >>**ORDR**
Interrupt Control Codes
None

The optimal ordering activity ORDR determines an ordering of the network buses such that sparsity is maintained as the Jacobian matrix is triangularized in activities FNSL, NSOL, FDNS, INLF, and ACCC, or the system admittance matrix is decomposed into its triangular factors in activities FACT and BKDY and the linearized network analysis activities DCLF, DFAX, DCCC, OTDF, TLTG, SPIL, and POLY.

Because it changes at least one bus type code, the generator conversion process (activity CONG) must be followed by an ordering of the network buses such that sparsity is maintained as the Jacobian matrix is triangularized in a variety of PSS®E analytical processes. The network bus ordering generally takes place in the background. In the case, however, where the user is specifically setting up conditions for the switching analysis, it is necessary to convert generators and order the network as separate activities.

When the *Assume all branches are in-service* option is specified, activity ORDR ignores the service status of all branches and assumes all branches connected to Type 1, 2, or 3 buses are in-service. Thus, following the ordering, simple branch status changes do not require an additional execution of activity ORDR.

When the *Ignore out-of-service branches* option is specified, activity ORDR recognizes the status of network branches and ignores out-of-service branches in determining the bus ordering. Thus, when a branch previously out-of-service is returned to service, activity ORDR should be executed again. Branch status changes that do not isolate a bus do not require repeated executions of activity ORDR. However, when a large number of branches have been taken out-of-service, this option *may* result in a more efficient matrix storage, thereby decreasing both the time and memory required to perform calculations using the network matrices.

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Activity ORDR must be re-executed whenever any of the following takes place:

1. The addition of a bus or branch to the case.

2. A change to the Type code of a bus from 1 or 2 to 3 or 4, or vice versa. Note that this includes the execution of those PSS®E activities such as CONG, DSCN, and TREE which have the ability to modify bus type codes.

3. The removal of a bus from the case (e.g., via activities EXTR or EEQV).

4. If *Ignore out-of-service branches* was specified for the last execution of ORDR, the returning to service of a branch which was out-of-service at the time of the previous ordering.

5. A change of an in-service branch from treatment as a zero impedance line to a non-zero impedance line, or vice versa (refer to Section).

Activities FNSL, FDNS, NSOL, INLF, ACCC, FACT, BKDY, and the linearized network activities automatically run activity ORDR if the user has not explicitly done so prior to their selection.

The optimal ordering function is also performed, without being explicitly invoked, during activities EEQV, SCEQ, SEQD, ASCC, ANSI, when network switching by dynamics models requires it, and optionally following network data changes in activity ALTR.

Activity ORDR summarizes the results of the ordering by tabulating the following:

1. The number of DIAGONALS (rows) in the matrix.

2. The number of OFF-DIAGONALS corresponding to the number of non-zero super-diagonal elements in the triangularized or factorized matrix during activities FACT and TYSL, the linearized network analysis activities, the fault analysis activities, and dynamic simulations.

3. The number MAX SIZE, which represents the largest number of non-zero super-diagonal elements present in the Jacobian matrix at some stage of the triangularization during activities FNSL, NSOL, FDNS, INLF, and ACCC.

Additional Information
PSS®E GUI Users Guide, Ordering Network Buses for Matrix Manipulation *PSS®E* Command Line Interface (CLI) *Users Guide*, Ordering Network Buses for Matrix Manipulation

13.6. Factorizing the Network Admittance Matrix

FACT

Run Activity FACT - GUI
Power Flow > Solution > Factorize admittance matrix (FACT)
Run Line Mode Activity FACT - CLI
ACTIVITY? >>**FACT**

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Interrupt Control Codes
None

13.6.1. About the Power Flow Network Admittance Matrix

Consider a bus, i, of a power network which connects two transmission lines, two transformers, and a shunt element (see Figure 13.11, "Equipment Connected at Bus, i").

Bus i is connected only to buses j, k, m, and n. Also note the designation of the tap-changing side of each transformer.

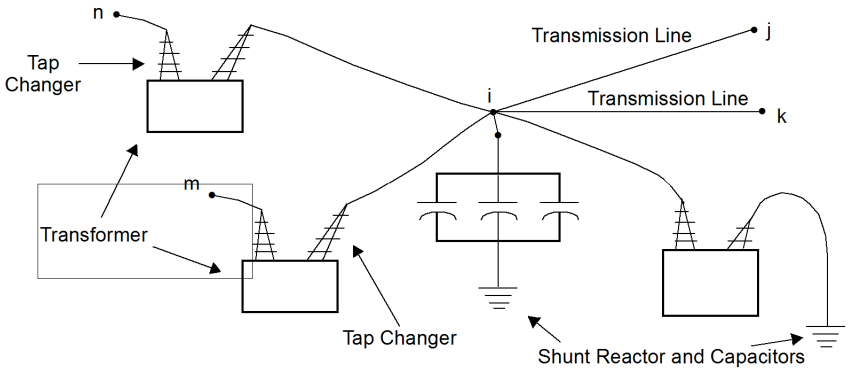

Figure 13.11. Equipment Connected at Bus, i

Each line and transformer may be represented by a per-unit equivalent circuit, with per-unit parameters being calculated from spacing, length, and nameplate data with respect to a common system MVA base. The resulting equivalent circuit surrounding network node, i, which represents bus i in the equivalent circuit of the whole system, is shown in Figure 13.12, "Equivalent Circuit for Node i of Transmission Network Model".

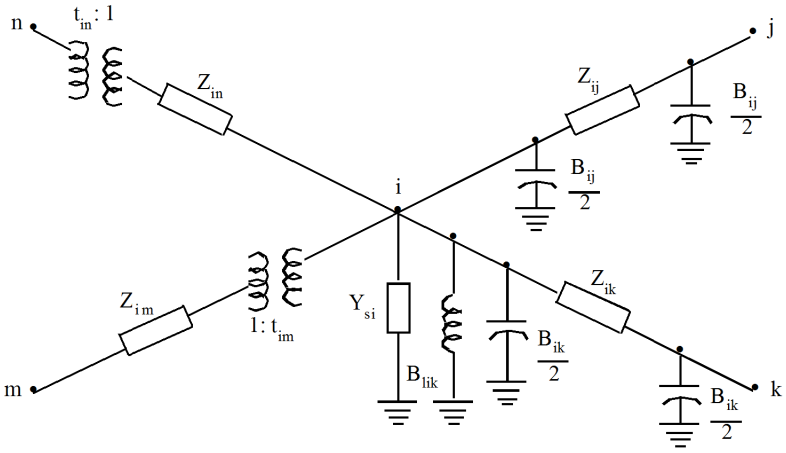

Figure 13.12. Equivalent Circuit for Node i of Transmission Network Model

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The total current, i_i, flowing into node i when voltages, v_i, v_j, v_k, v_m, and v_n are applied to the network are determined by adding up the flow into each leg of Figure 13.12, "Equivalent Circuit for Node i of Transmission Network Model" at its connection to node i. In the case shown above, the total current, i_i, is zero because there is no source or load connected there. Hence:

$$\begin{aligned} i_i &= v_i Y_{si} \\ &+ (v_i - v_j) / z_{ij} + \frac{v_i B_{ij}}{2} \\ &+ (v_i - v_k) / z_{ik} + \frac{v_i B_{ik}}{2} + v_i B_{ik} \\ &+ \left(\frac{v_i}{t_{im}} - v_m \right) / z_{im} t_{im} \\ &+ \left(v_i - \frac{v_n}{t_{in}} \right) / z_{in} \end{aligned} \tag{13.1}$$

Expansion of Equation 13.1 for the i-th element of I_n, which is i_i, gives:

$$i_i = \sum_{n=1}^{n} y_{ih} v_h \tag{13.2}$$

where

y_{ih} are the elements of Y_{nn}

and

v_h are the elements of the node voltage vector, V_n

The expressions for the elements, y_{ih}, in terms of transmission line and transformer parameters can be found by collecting terms in Equation 13.1and comparing the result with Equation 13.2. The construction of the individual elements, y_{ii}, of the network admittance matrix from the line and transformer data is a key section of all power flow solution procedures of PSS®E.

Examination of Equation 13.1 shows that only the diagonal element, y_{ii}, and four off-diagonal elements, y_{ih}, of the i-th row of the admittance matrix are nonzero. That is, a line or transformer from bus i to bus j contributes nonzero elements only to the i-th and j-th rows of Y_{nn}. Realistic power systems have between 1.5 and 2 transmission lines or transformers for each node. A transmission network of 2000 buses might, therefore, be expected to have approximately 4000 branches and, correspondingly, 8000 nonzero off-diagonal elements in its admittance matrix. This typical bus-to-branch ratio results in very sparse admittance matrices. In the above example only 8000 out of (2000) (2000-1), or 0.2% of the off-diagonal elements of Y_{nn} are nonzero.

The great majority of modern power flow calculation programs, including those of PSS®E, take advantage of this sparsity in their management of computer storage. They also take advantage of procedures that allow the admittance matrix to be manipulated into triangular factor and partial inverse forms which have similar sparsity properties, but where the number of nonzero off-diagonal elements is typically two to three times the number of such elements in the original Y_{nn} matrix.

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Presenting a power flow program with a power system network model, for which original Y_{nn} matrix (or derivatives therefrom) has a number of nonzero elements that exceed the program's allocated capacity, results in an error condition that prevents the use of some, but not all, of the available power flow iteration algorithms.

13.6.2. Operation of Activity FACT

The triangular factorization activity FACT decomposes the network admittance matrix (Y matrix) into its upper and lower triangular factors for use in the triangularized Y matrix network solution (activity TYSL) or in the network balance of dynamic simulations. This computation also takes place, without being specifically invoked, by the following functions:

ASCC	ASTR	BKDY	EEQV	IECS
MSTR	MRUN	SEQD	SCMU	SCEQ
STRT	RUN		DYNAMICSMODE	

Because activity FACT performs a computation involving the admittance matrix, it follows that it must be re-executed any time this matrix changes and switching studies or dynamic simulations are to be run. Thus, in these applications, activity FACT must be rerun any time one or more of the following occurs:

- Change of load characteristic modeling via activities CONL or RCNL.
- Change of bus type code.
- Change of machine, branch or load status.
- Change of machine impedance parameters.
- Change of branch impedance or charging.
- Change of which branches are modeled as zero impedance lines.
- Change of transformer ratio or phase shift angle.
- Change of bus or line connected shunt.
- Change of constant admittance load.

Activity FACT tabulates the number of non-zero diagonal and off-diagonal terms in each factor of the matrix at the *Progress* tab.

Activity FACT requires the prior execution of activity CONG, usually activity CONL, and activity ORDR. Refer also to Section 13.7.5, "Application Notes". It prints an appropriate error message and terminates if generators have not been converted (i.e., activity CONG has not been executed). If activity FACT detects that a new execution of activity ORDR is required, a message is printed and activity ORDR is automatically executed prior to the factorization.

13.6.3. Exporting the Power Flow Network Admittance Matrix

All or part of the network admittance (Y) matrix, as used in the PSS®E power flow solutions, may be exported in the form of a source file. To initiate the output Y matrix function, select *File > Export > Network*

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admittance matrix... from the GUI. The user selects the portion of the admittance matrix to be produced from the following options:

- The entire admittance matrix.
- Admittance matrix rows corresponding to buses in the current bus subsystem, either as currently defined, or as redefined in the [*Bus Subsystem Selector]*.
- A specified set of buses.

Unless the *all buses* option is selected, all non-zero off-diagonal terms in the rows being processed are produced by default. If this option is disabled, only off-diagonal terms corresponding to other rows being processed are produced. The user directs the output to either a named file or to the *Progress* tab.

The admittance matrix terms produced are extracted from the admittance constructed by, and used during, the PSS®E power flow solution functions. As such, they include contributuins for constant admittance loads, off-nominal tap ratios, phase shifts, and GNE-Y devices, but exclude constant MVA load, constant current load, generator, switched shunt, dc line, FACTS device, GNE-PQ device, and GNE-I device contributions. Furthermore, the admittance matrix used during the power flow solutions excludes rows and columns for swing (i.e., Type 3) buses; this function also *excludes* terms from these rows and columns *unless* generators are converted (i.e., activity CONG is executed) prior to initiating this function.

Because the admittance matrix is extremely sparse, its terms are produced as a set of matrix terms, using external bus numbers. Unfamiliar large bus numbers (greater than six digits) indicate terms in admittance matrix rows and columns corresponding to the star point buses of three-winding transformers. The following example contains the non-zero admittance matrix output for the row corresponding to bus 154:

```
154,   154,  41.9126    , -404.786
154,  3008, -5.49574    ,  44.7801
154,   153, -4.47154    ,  40.2439
154,   205, -29.4701    ,  297.380
154,   203, -2.47525    ,  24.7525
```

Additional Information
PSS®E GUI Users Guide, Factorizing the Network Admittance Matrix *PSS®E* Command Line Interface (CLI) *Users Guide*, Factorizing the Network Admittance Matrix

13.7. Solving the Converted Case

TYSL

Run Activity TYSL - GUI
Power Flow > Solution > Solution for switching studies (TYSL)... *[Solution for Switching Studies]*
Run Line Mode Activity TYSL - CLI
ACTIVITY? >>**TYSL**

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Interrupt Control Codes
AB, NC, DC, FD

The triangularized Y matrix network solution activity TYSL is designed for those situations where the internal flux linkages of generators are assumed to remain unchanged as a load or fault is switched onto the system, as a line is opened or closed, or as a load is removed. It determines the instantaneous change in network voltages as the switching operation takes place. This activity is used for balanced short circuit, motor starting, voltage dip, and initial load rejection overvoltage studies; this class of studies is called switching studies.

When the *flat start* option is specified, all bus voltages are reset to unity magnitude at zero phase angle. Otherwise, the existing voltage vector in the working case is used as the initial voltage estimate. During the solution, generator buses are not treated as they are in power flow solution calculations. Rather, the Type 2 bus voltages become free variables just as the voltages at Type 1 buses, and a fixed source current is injected into the network at each generator bus (refer to Figure 13.4, "Relationship Between PSS®E Generator Equivalent and Thevenin Equivalent").

Activity TYSL handles load boundary conditions and the blowup check in the same way as activity SOLV. The solution convergence monitor, the dc transmission line monitors for two-terminal and multi-terminal dc lines, and the mismatch summary are identical to those of activity SOLV. Refer to Section 6.3.11, "Operation of Activity SOLV" and Section 6.3.13, "Load".

No automatic adjustments are allowed, and switched shunt devices and dc converter transformer tap settings are locked at their pre-switching settings. A two-terminal or multi-terminal dc transmission line is blocked for the remainder of the current execution of activity TYSL if, on any iteration, the ac voltage at a rectifier converter station bus falls below 50% or is insufficient to make margin order. Shunt elements of FACTS devices hold their pre-switching reactive currents, and series elements are held at their pre-switching series voltages.

For each in-service VSC dc line, the solution starts with each converter holding the active and reactive power at the time activity CONG was executed. If, on any iteration, the corresponding current exceeds the converter's IMAX, the injection is reduced using the power weighting factor fraction (PWF) just as in the conventional power flow solution activities. Each converter is treated independently, so that any reduction at one converter does not affect the injection at the other end of the VSC dc line. The dc transmission line monitor for VSC dc lines includes the dc line name, followed by the ac power injection at each converter bus.

By default, activity TYSL has a convergence tolerance of 0.00001 on voltage change and a limit of 20 iterations. A deceleration factor, which has a default value of unity, is applied to the voltage change at each bus. For optimum convergence characteristics, it may be advantageous to reduce the deceleration factor, especially when a large fraction of the reactive load is represented by the constant MVA and/or current characteristics. The presence of dc lines may also require some deceleration of the solution. The guideline of Section 6.3.19 Characteristics of Activity SOLV should be used in tuning this parameter. The user may modify any of these solution parameters with the data changing activity CHNG or the *[Solution Parameters]* dialog.

Activity TYSL requires that the factors of the network admittance matrix to be present in the admittance matrix temporary file (i.e., it requires the prior execution of activity FACT).

Activity TYSL responds to the following interrupt control codes:

AB	Abandon activity TYSL following completion of the next iteration.
NC	Suppress the convergence monitor.
DC	Tabulate conditions for each dc line after each iteration.

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FD	Tabulate the conditions for each in-service FACTS device after each iteration.

The following examples show switching cases and provide additional information for the process:

- Voltage rise on a line end
- Motor starting
- Fault application.

13.7.1. Example of Voltage Rise on Open Line End

The calculation of voltage on the open end of a line is often required to examine the conditions which could be encountered by line connected reactors; by surge arresters and by breakers. The latter have limits of voltage across the open break within the equipment. Voltage on the line side could be high while voltage on the system side could have fallen due to loss of the line's incoming power.

This example will open the line from bus 151 to bus 152, circuit 1, at the end terminating on bus 152, in the *savnw.sav* power flow case. Figure 12-13 shows the power flow diagram for these buses.

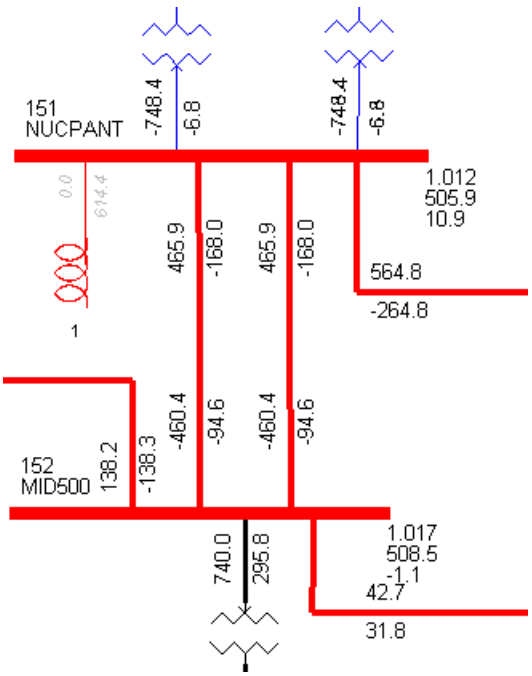

Figure 13.13. Example – Circuits from Bus 151 to Bus 152

One way of performing the breaker operation, is to establish a dummy bus on the circuit from bus 151 to bus 152 which is connected to Bus 152 via a zero impedance line. To open the breaker it would require taking the zero impedance line out of service (see Figure 13.14, "Opening a Line at One End"). The circuit from bus 151 to the dummy bus 999 will have the same line parameters as the actual circuit from bus 151 to bus 152 circuit 1. When the zero impedance line between bus 999 and bus 152 is opened, circuit 1 from bus 151 will

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be connected at bus 151 and hanging open. The post-switching voltage at bus 999 is the line-end voltage of interest together with the voltage at bus 152.

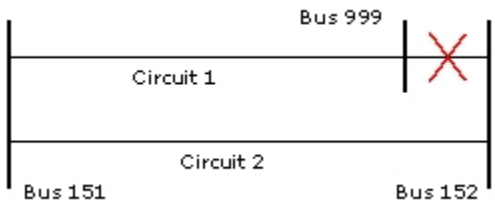

Figure 13.14. Opening a Line at One End

To avoid having to pre-view the need for dummy buses, activity SPLT can be used. Here the user would split bus 152 and move circuit 1 to bus 151 to a new bus automatically created by the program and provided with a unique new number.

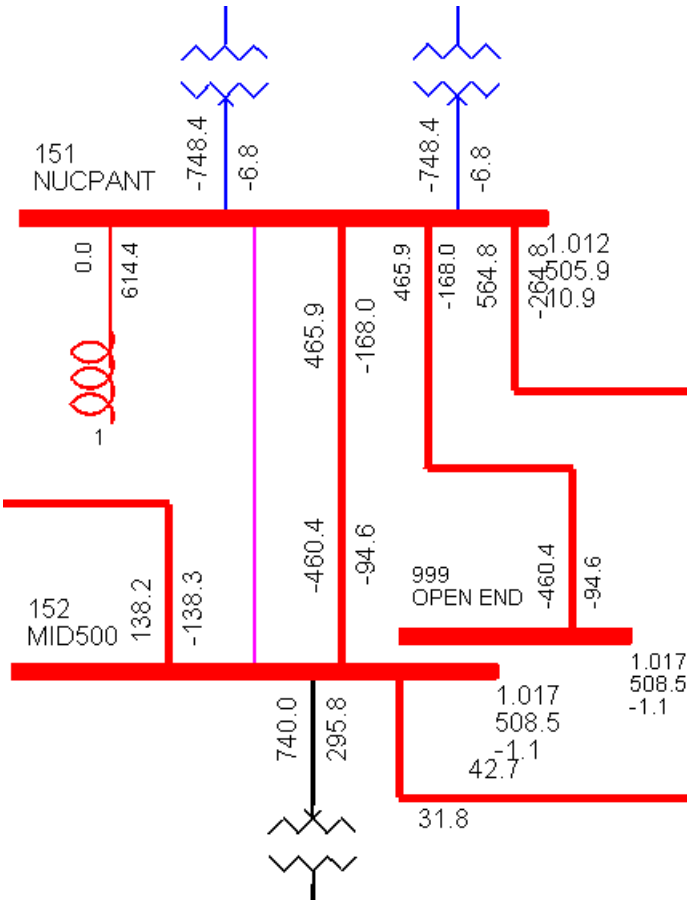

Figure 13.15. Producing an Open Line End with Activity SPLT

The diagram will indicate that circuit 1 is open (purple line) but the user has to grow the diagram to show the new bus and line to 151.

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The bus split will introduce a new zero impedance branch between bus 999 and bus 152. The user will open this zero impedance line to perform the line opening. Furthermore, the new bus; the new line to bus 151 and the zero impedance branch will be seen in the Tree View and the Spreadsheet View.

The sequence of events for this model is:

1. Convert the power flow case (activity CONG) and order the network buses (activity ORDR).
2. Split the bus (activity SPLT) and open the new zero impedance line to create an open ended line.
3. Factorize the matrix (activity FACT).
4. Perform the Y matrix solution (activity TYSL).

Now the voltages can be examined with a bus based report or by examining the solution shown in the diagram.

13.7.2. Example of Open Line End Results

It is interesting to look at the bus voltages, in this example, on buses 151, 152 and 9999 for the power flow condition, for the post switching condition and for a condition obtained by opening the line in the unconverted power flow case and solving with a conventional solution (e.g., Gauss-Seidel). This latter case will show conditions after all the voltage regulation devices have had time to operate to adjust back to set-point voltages (V long-term).

Table 13.1. Results of Line Open End Balanced Switching Calculation

Results of Line Open Operation			
Bus Number	V pre-switch	V post-switch	V long-term
151	1.012	1.023	1.015
152	1.017	0.996	0.981
9999	--	1.112	1.104

It can be seen that immediately after switching the voltage at bus 152 drops because of the sudden loss of incoming power and the need for greater voltage support. The voltage at bus 151 rises slightly because, in part, of unconsumed reactive power from the open line. The line end voltage has increased from its original level of 1.017 pu to 1.112 pu (see Table 13.1, "Results of Line Open End Balanced Switching Calculation").

Following voltage regulator action, the voltage at all buses has been reduced to a new steady-state level. The voltage at the open end is elevated but reasonable for sustained operation.

13.7.3. Example of Motor Starting

A fairly common switching study requirement is illustrated by Figure 13.16, "Motor Starting Example". One of a pair of large motors is running under load, and the second one is to be started by switching it directly onto the bus. The extent of the voltage dip produced by switching on the second motor is needed for the coordination of overcurrent and undervoltage protection.

The example will assume a load at bus 3006, in the *savnw.sav* power flow case, to consist of two large motors which, at full load and unity voltage, draw (7.5 + j3.95) MVA each. The starting current for one motor is given in specification data as 2.72 times rated current at 0.275 power factor. The calculation in Figure 13.16,

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"Motor Starting Example" shows that this starting duty corresponds to an initial shunt admittance of (0.0625 j0.218) per unit, relative to a 100-MVA base (I = YV), and nominal voltage.

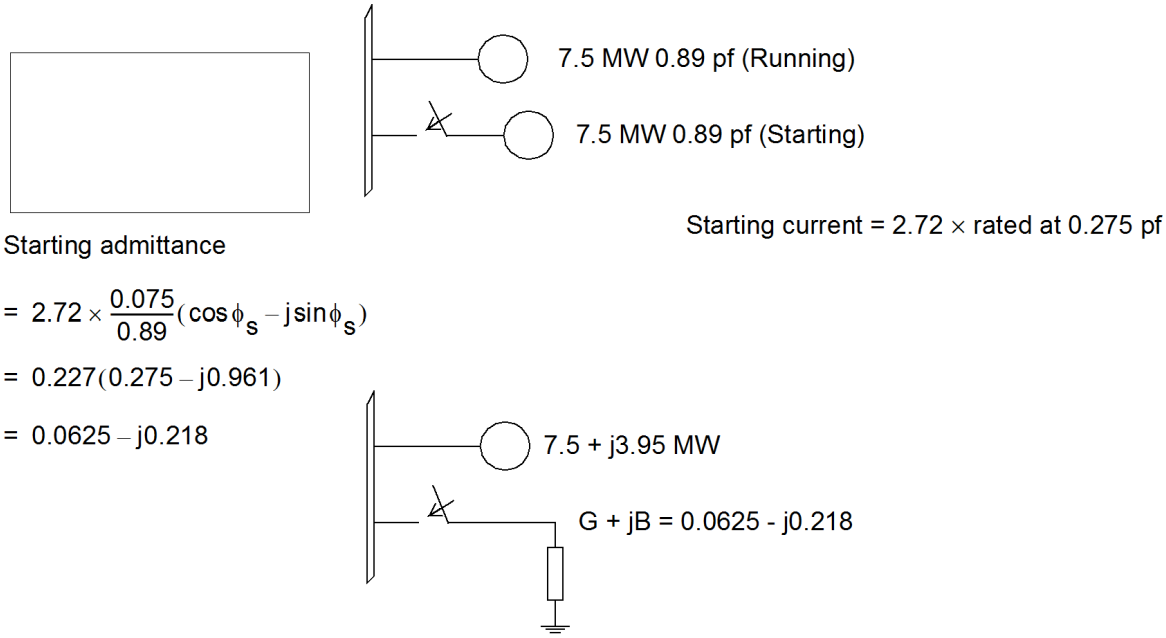

Figure 13.16. Motor Starting Example

The switching operation comprises the connection of the starting admittance to the case. As previously discussed, the user will convert and order the power flow case; then connect the admittance to bus 3006 then factorize and perform the Y matrix solution. Note that the starting admittance is on a 100 MVA base. This means that the user should add a shunt of 6.25 -j21.8 MVA which is the MVA at 1.0 pu voltage. Figure 12-17 shows the bus based report for bus 3006.

BUS	3006 UPTOWN	230.00	CKT	MW	MVAR	MVA	%I	0.9871PU 227.04KV
TO	LOAD-I			7.5	0.0	7.5		
TO	LOAD-Y			0.0	3.9	3.9		
TO	SHUNT			6.1	21.2	22.1		
TO	153 MID230	230.00	1	-85.5	-3.2	85.6		
TO	3005 WEST	230.00	1	72.0	-21.9	75.3		

Figure 13.17. Bus Based Report Following Motor Starting

It can be seen that the inrush is 85.6 MVA at 0.987 pu voltage. The initial voltage was 0.993 pu, indicating a voltage dip of only less than 1.0%.

Notice that the load on the bus (the other motor) was converted to 100% constant current for real power and 100% constant admittance for reactive power.

13.7.4. Example of Fault Current

Because these examples lie within the realm of balanced switching, the only faults which can be analyzed are three-phase faults.

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In the case of a three-phase fault, the switching operation which is performed before factorization and solution is the connection of a very large reactive admittance to ground at the fault point. A value of -j1E9 is adequate. After a solution has been obtained for the instant, t^+ output of results is handled by the standard power flow output formats such as the bus based reports. Such a bus based report will provide the fault point and branch information (see Figure 13.18, "Calculation of Branch Currents Flowing into a Bus Faulted through Zero-Impedance").

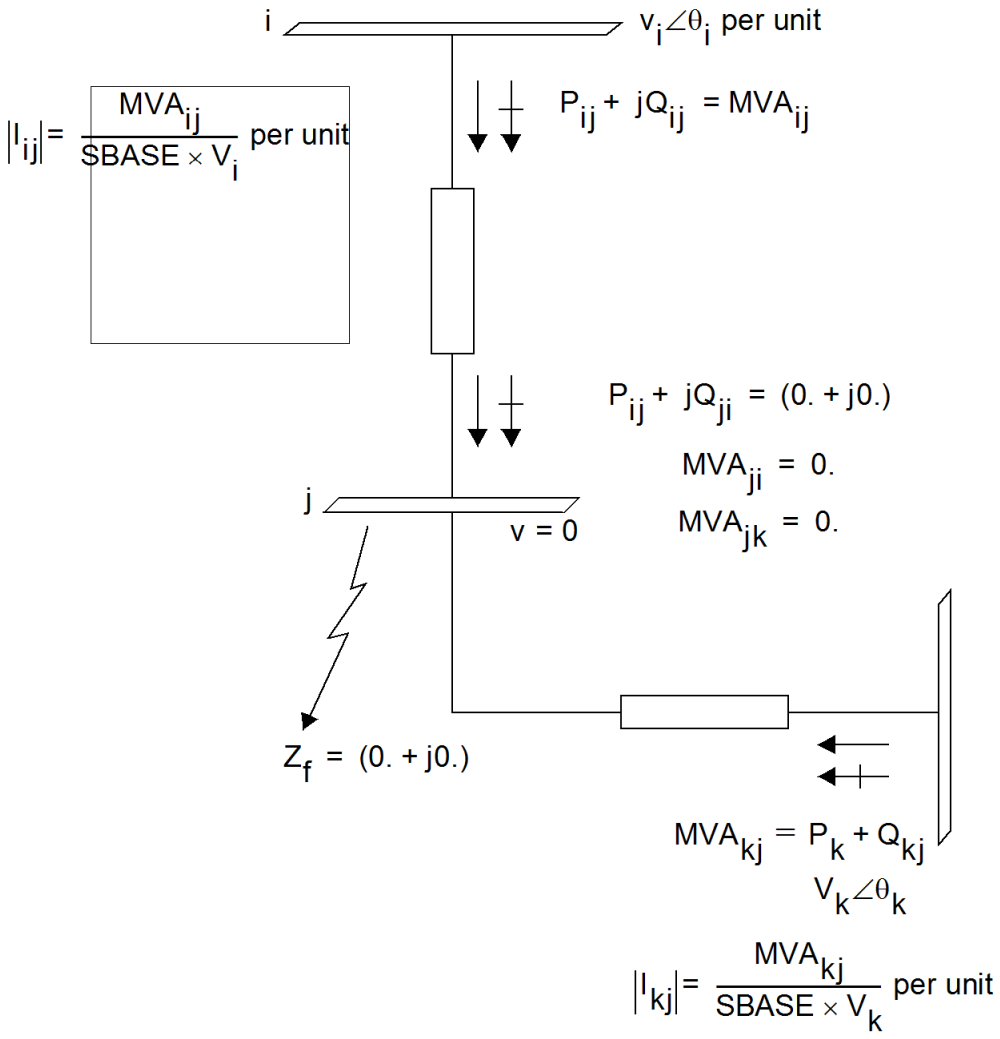

Figure 13.18. Calculation of Branch Currents Flowing into a Bus Faulted through Zero-Impedance

The format of the bus based report will be unchanged except that it will display the net current flowing to ground at any bus where voltage is below 0.1 per unit. This current value is expressed in terms of fault MVA where

FAULT MVA = $\sqrt{3} \times$ (Base Voltage, kV) $\times$ (Current, A) $\times 10^{-3}$

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Because the output reports display power flow results, the current flowing in faults and branches must be calculated from the displays of voltage and complex power or MVA flow. Current in amperes is obtained from

$$|\text{Current}| = \frac{\text{MVA flow}}{\sqrt{3} \times (\text{Voltage, kV}) \times (1 \times 10^{-3})} \text{ A}$$

Per-unit current is obtained from

$$|\text{Current}| = \frac{\text{MVA flow}}{(\text{System Base MVA}) \times (\text{Voltage per unit})} \text{ per unit}$$

The calculation of currents flowing into a bus faulted through a zero impedance is illustrated by Figure 13.18, "Calculation of Branch Currents Flowing into a Bus Faulted through Zero-Impedance". Because the voltage at the faulted bus is zero, the MVA flows into it are zero even though the currents are nonzero. The currents in the branches feeding the fault must be determined from the MVA flows and voltages at the ends that are away from the fault.

Figure 12-19 shows the output report for a fault on bus 3006, in the *savnw.sav* power flow case. Output is for the faulted and two adjacent buses.

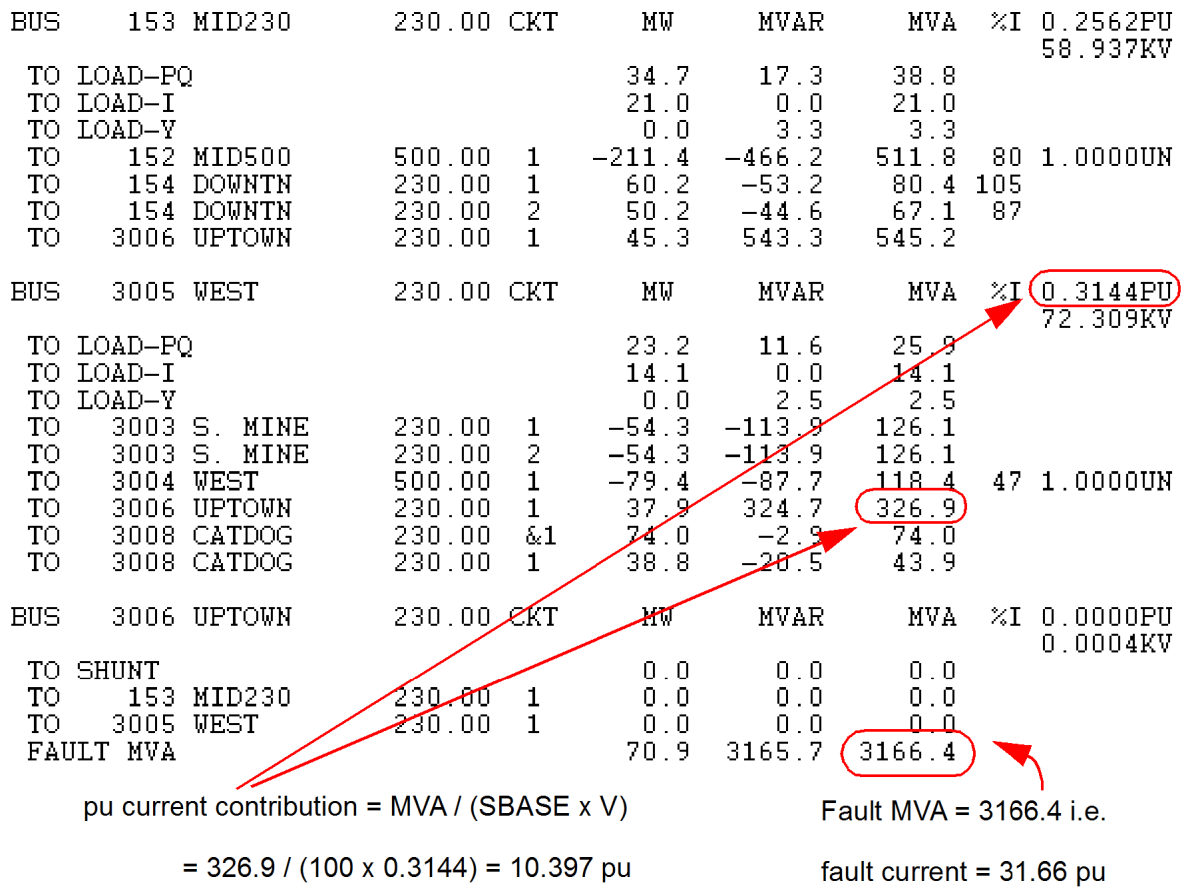

Figure 13.19. Bus Based Output for Three-phase Balanced Fault on Bus 3006

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13.7.5. Application Notes

Activity TYSL *is not* a power flow solution; it is unable to handle the power flow style representation of generator buses (i.e., fixed power and voltage). But, for networks in which generators are represented by constant internal flux linkages (i.e., following activity CONG), activity TYSL is capable of producing very small mismatches in only a few iterations. (Note that its default convergence tolerance is an order of magnitude smaller than that of activities SOLV and MSLV!). It is much less sensitive to the values of branch impedances than the other network solution activities.

The use of activity TYSL for switching study network solutions involves operations upon the factorized system admittance matrix. Thus, a typical sequencing of activities is:

1. A standard power flow is converged to an acceptable mismatch level using one or more of activities SOLV, MSLV, FNSL, NSOL, or FDNS. It is *strongly* recommended that the solved power flow case be saved in a Saved Case File with activity SAVE.

2. Activity CONG is executed to convert generator buses from their power flow representation to the constant Norton current source representation used in activity TYSL. See also Figure 12-4.

3. Activity CONL is executed to convert the load boundary conditions from the conventional constant MVA characteristic used in power flow calculations to a representation suitable for network conditions involving abnormally low or high voltages (refer to Load Data and Section 6.3.13, "Load").

4. Activity ORDR is executed to determine a bus ordering of the admittance matrix such that its sparsity characteristics are preserved. A new bus ordering is required because the execution of activity CONG introduces a row and column into the admittance matrix for each former Type 3 (swing) bus.

5. Optionally, this converted and ordered case may be saved with activity SAVE if several switching solutions are to be calculated from this base condition.

6. Apply the switching via activity CHNG or the *[Solution Parameters]* dialog, factorize the corresponding admittance matrix with activity FACT, and balance the network with activity TYSL.

7. The standard power flow output reporting activities (e.g., POUT, VCHK, etc.) are used to examine the results of the post-switching network solution.

8. Additional switchings may be imposed on the same base condition by retrieving the converted case saved in step 5 and proceeding to step 6.

Note again that activity TYSL requires that the factorized admittance matrix, representing the switching operation imposed upon the pre-switching network condition, be present in the admittance matrix temporary file.

Additional Information
PSS®E GUI Users Guide, Solving the Converted Case *PSS®E* Command Line Interface (CLI) *Users Guide*, Solving the Converted Case

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Chapter 14
Transmission Pricing and Open Access

14.1. Overview of Open Access and Pricing

The electric utility industry restructuring, prevalent in the United States and many other countries in the close of the 20th century has generated interest in analytic tools for transaction processing. Open Access and Pricing Activities (OPA) have been introduced to the power flow processor to aid in assessing transaction feasibility, and cost allocation. In PSS®E, the transaction feasibility assessment is supported by a transaction impact calculator and a line loading relief calculator. The transmission cost allocation technique mandated by the Public Utility Commission of Texas for computing the impact fee component of transmission usage fee is provided in the PSS®E allocation function. Transmission loss allocation techniques also are provided.

OPA requires a power flow model, which can be employed to produce a valid dc power flow solution, be imported into PSS®E. This model is then augmented with transaction data. Transaction data includes transaction event definitions.

A transaction event consists of the following attributes:

- A unique number used for reporting, for defining sequence order, and for referring to transactions in program dialog.
- An alpha numeric label.
- Service status.
- Priority for grouping and ranking.
- A magnitude value in MW.
- A curtailment value in MW.
- A list of participating network buses.

Buses participate in a transaction as points of power injection into or demand from the power system network. An in-service transaction of nonzero magnitude will result in power flows on transmission branch elements. Load and generation input values are associated with each participating bus. A bus participates as a power injection point when its generation value is greater than zero or its load value is less than zero. A bus participates as a power demand point when its load value is greater than zero or its generation value is less than zero. The injection or demand magnitude is a function of the transaction magnitude, the load and generation values, and for some functions, the transaction curtailment value.

Transmission cost and loss allocation methods are concerned with branch ownership and control area, respectively. The flow or loss impact of a branch is computed and accumulated by owner or control area.

14.2. Managing Transaction Data

Each transaction event the user first assigns a set of transaction attributes and then identifies which buses will be participating in the transaction event. The transaction attributes are the following:

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ID	A Transaction event number. ID = 0 by default.
DESCRIPTION	Alphanumeric label assigned to the transaction event. The label may be up to twenty four characters and may contain any combination of blanks, upper-case letters, numbers and special characters. LABEL is twenty four blanks by default.
STATUS	Transaction event status: 0 for out-of-service, 1 for in-service. STATUS = 0 by default.
PRIORITY	Transaction event priority. This integer value priority number is used to group transactions into sets. IPR = 0 by default.
MAGNITUDE	Transaction event magnitude in MW. MAG = 0.0 by default.
CURTAILMENT	Transaction event curtailment magnitude in MW. CURT = 0.0 by default.

For each transaction event the user identifies which will be the participating buses. For each participating bus, the following information is required:

I	Bus number. I = 0 by default.
LOAD	Participating bus load value. LV = 0.0 by default.Can be negative.
GENERATION	Participating bus generation value. GV = 0.0 by default.Can be negative.

Table 13-1 Example of Transaction Data for Two Transactions shows an example transaction data consisting of two transaction events.

Table 14.1. Example of Transaction Data for Two Transactions

Typical Transaction Data	
First Transaction	
`6 UPSTART 1 1 200.0 0.0`	Transaction Attributes
`3008  1.0000   .0000`	Participating Buses
`102    .0000   .2667`	
Second Transaction	
`5 WORLD 1 1  354.0  0.0`	Transaction Attributes
`3008   .0000  1.0000`	Participating Buses
`3007  1.0000  1.0000`	
`3006  1.0000  1.0000`	
`3005  1.0000  1.0000`	
`3004   .0000  1.0000`	
`3003   .0000  1.0000`	
`3002   .0000  1.0000`	

Data is facilitated in the *[Transaction Data]* dialog. Both attribute and bus data can be modified. Note that the data is neither saved nor retrieved with the power flow case. It is only temporarily available for the Transmission Access analyses.

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14.2.1. Transmission Access Calculators

PSS®E provides two calculators. They are:

- Calculation of incremental monitored element MW flow impacts due to transaction events (activity IMPC)
- Calculation of the active power flow on monitored elements as a function of transaction events. This Line Loading Relief calculation (activity LLRF) is used to: determine the transfer distribution factors on all monitored elements due to all transaction events; determine the incremental curtailment of transaction events required to achieve an incremental flow change on a target monitored element; and determine the incremental restoration of previously curtailed transaction events needed to achieve an incremental flow change on a target monitored element.

14.3. Calculating Transaction Event Impact on Monitored Elements

IMPC

Run Activity IMPC - GUI
Trans Access > Calculators > Impact on monitored elements... *[Impact on Monitored Elements]*
Run Line Mode Activity IMPC - CLI
ACTIVITY? >>**IMPC** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

Activity IMPC computes and reports the incremental MW flow impact on a set of monitored elements due to a single transaction event.

The user must enter the name of a Distribution Factor Data File constructed by activity DFAX which corresponds to the network condition contained in the working case (refer to Section 9.1.2, "Application Notes") and to the desired Linear Network Analysis Data Files (refer to Section 8.1.1 Linear Network Analysis Data Files). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity IMPC.

The Distribution Factor Data File defines the monitored element set. It employs the transaction magnitude, without regard to the curtailment value, to determine the power injections and demands at participating buses. It employs the linearized solution technique to compute the incremental element flows. Regulating in-service phase shifting transformers have zero incremental MW flow.

The user may provide the name of an ATC Updates File, which is optional. This file may modify the monitored interface ratings employed in the report as available transfer capability (ATC). The ATC Updates File consists

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of one or more records, each consisting of two fields, a text field which identifies an interface by twelve character label and a numeric value field which identifies the interface ATC in MW.

Activity IMPC may be terminated by entering the AB interrupt control code.

The impact calculator produces a report which illustrates the incremental MW flow impacts on each of the monitored elements, along with the initial available transfer capability (ATC) for the element and a final ATC which is adjusted for the flow impact due to the transaction. For monitored interfaces, a value is provided which illustrates the maximum transaction magnitude that can be supported by the element's ATC.

14.3.1. Output Report Format

Both narrow and wide report formats are employed. The column width employed for the output report depends on the selection of output device. The report begins with an initial page header followed by four records which identify the source of Distribution Factor and Linear Network Analysis Data Files.

The remainder of the report is divided into two sections. The first section reports impacts for all monitored branches contained in the Distribution Factor Data File. The second section reports impacts for all monitored interfaces. The selected transaction event is identified and followed by a table of monitored elements. The transaction event identification includes: transaction event number, transaction event label and transaction magnitude. Each record of the monitored element table includes: a monitored element description, an initial ATC value in MW, the incremental MW flow impact on the monitored element due to the transaction event, and a final (adjusted for incremental flow) ATC value in MW. Branch RATEA values are employed as ATC values for monitored branches. Monitored interface ATC values default to the interface RATEA and may be adjusted by entries in the ATC Updates File.

For monitored branches the monitored element description consists of: from bus number and name, to bus number and name, and circuit id. For monitored interfaces the twelve character monitored element label is employed as the monitored element description.

The monitored interface table includes an additional column of numeric values. These values illustrate the transaction magnitude, in MW, which would result in a final ATC value of zero for that element. This maximum transaction field will take a value only when the initial ATC and incremental flow values are of the same sign and when the incremental flow value magnitude significantly exceeds zero.

14.3.2. Example: Transaction Event Impact on Monitored Elements

Using the *savnw.sav* power flow case and the associated data files (*savnw.sub/.mon/.con*), all of which are available in the PSSE/EXAMPLE directory in the user's installation, an example case is run using Transaction #6.

The report generated by the impact calculation is shown in Figure 14.1, "Results of Impact Calculation".

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E   THU, MAR 11 2004  11:35
PSS/E PROGRAM APPLICATION GUIDE EXAMPLE                              PAGE 1
BASE CASE INCLUDING SEQUENCE DATA
                        ** TRANSACTION IMPACT SUMMARY **

DISTRIBUTION FACTOR FILE:      C:\Program Files\PTI\PSSE30\EXAMPLE\dc-contingency.dfx
SUBSYSTEM DESCRIPTION FILE:    C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.sub
MONITORED ELEMENT FILE:        C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.mon
CONTINGENCY DESCRIPTION FILE: C:\Program Files\PTI\PSSE30\EXAMPLE\savnw.con

TRANSACTION:             6 Transaction #1
<------- F R O M -------> <--------- T O ---------->CKT  <INT ATC MW> <IMPACT MW-> <FIN ATC MW>
   201 HYDRO       500.00    151 NUCPANT     500.00 1       1200.00       -91.75      1291.75
   202 EAST500     500.00    152 MID500      500.00 1       1200.00        31.82      1168.18
   203 EAST230     230.00    154 DOWNTN      230.00 1        200.00         3.03       196.97
   205 SUB230      230.00    154 DOWNTN      230.00 1        600.00        56.90       543.10
  3004 WEST        500.00    152 MID500      500.00 1          0.00       -47.66        47.66
  3006 UPTOWN      230.00    153 MID230      230.00 1          0.00       -39.00        39.00
  3008 CATDOG      230.00    154 DOWNTN      230.00 1        400.00      -113.34       513.34

TRANSACTION:             6 Transaction #1
<-INTERFACE-> <INT ATC MW> <IMPACT  MW> <FIN ATC MW> <MAX TRN MW>
WEST                200.00      -200.00       400.00
EAST                350.00         0.00       350.00
```

Figure 14.1. Results of Impact Calculation

14.3.3. Application Notes

Activity IMPC uses the same linearized network model as activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Regulating in-service phase shifting transformers hold constant MW flow.

Additional Information
PSS®E GUI Users Guide, Calculating Transaction Event Impact on Monitored Elements *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Transaction Event Impact on Monitored Elements

14.4. Calculating Line Loading Relief

LLRF

Run Activity LLRF - GUI
Trans Access > Calculators > Line loading relief by transaction adjustment... *[Line Loading Relief by Transaction Adjustment]*
Run Line Mode Activity LLRF - CLI
ACTIVITY? >>**LLRF** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

Activity LLRF provides a menu of functions related to computing transaction magnitude adjustments which result in feasible monitored element branch MW flows. It performs various analyses related to identifying the active power flow on monitored elements as a function of transaction events. LLRF is used to:

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- Determine the transfer distribution factors on all monitored elements due to all transaction events
- Determine the incremental curtailment of transaction events required to achieve an incremental flow change on a target monitored element
- Determine the incremental restoration of previously curtailed transaction events needed to achieve an incremental flow change on a target monitored element.

Activity LLRF uses a Distribution Factor Data File to define the monitored element set. It employs the transaction magnitude an the curtailment value to determine the power injections and demands at participating buses. It employs the linearized solution technique to compute monitored element distribution factors and the incremental element flows. Regulating in-service phase shifting transformers have zero incremental MW flow. It reflects adjustments in the transaction curtailment value.

The user may specify the desired MW flow increment on a monitored element. This incremental flow value may, for example, represent the amount by which the element is overloaded. The line loading relief calculator will report adjustments to transaction magnitudes required to satisfy this incremental flow. These transaction adjustments may be limited to transaction curtailment (i.e., increases in transaction curtailment value magnitude not to exceed the transaction magnitude value) or transaction restoration (i.e., decreases in transaction curtailment value magnitude).

Transactions participate in adjustment as a function of their transaction priority and the applied adjustment method. Four adjustment methods are available: first in last out, decreasing order by distribution factor magnitude, pro rata by distribution factor magnitude, and pro rata by the product of distribution factor with transaction schedule (transaction magnitude less curtailment). Transactions are excluded from adjustment when their distribution factor magnitude is less than a user-specified tolerance. Individual events are grouped by priority. The curtail transactions function proceeds to investigate the priorities from lowest to highest priority number until the incremental flow target is satisfied or all transaction priority groups are exhausted. The restore transactions function proceeds to investigate the priorities from highest to lowest priority number until the incremental flow target is satisfied or all transaction priority groups are exhausted.

The line loading relief calculator produces reports which illustrate:

- Transaction adjustments which result in a specified incremental flow on a selected monitored element.
- Distribution factors (i.e., sensitivity of monitored element MW flow to transaction magnitude) for all monitored elements and transaction events).

14.4.1. Operation of Activity LLRF

The user must enter the name of a Distribution Factor Data File constructed by activity DFAX which corresponds to the network condition contained in the working case (refer to Section 9.1.2, “Application Notes”) and to the desired Linear Network Analysis Data Files (refer to ???). The Distribution Factor Data File must specify at least one monitored branch or interface. The bus input option (refer to Section 3.3.3, “Program Run-Time Option Settings” and activity OPTN) *must* have the same setting that was in effect when activity DFAX filled the specified file. Refer to Section 2.4.1, “File Usage ” for the file specification conventions used by activity LLRF.

When the distribution factor file is identified, the monitored element list (from a *.*mon* file) is tabulated in the window.

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Table 14.2. Line Loading Relief Calculator Functions

Line Loading Relief Function	Calculation
Curtail or Restore	Specify the target element from the list of monitored elements and define the incremental flow target element adjustment value in MW. The conventions for positive and negative element flow are defined when presenting the monitored element list to activity DFAX.
DF Report	Activity LLRF produces a report of transfer distribution factors for all monitored elements due to all transactions events. Out-of-service transaction events will have distribution factors of zero.
Distribution factor tolerance	If the reset distribution factor tolerance is specified, activity LLRF requires a new value of distribution factor tolerance. This tolerance is employed to preclude those transaction events with negligible effect on the target monitored element from the transaction adjustment functions

The user specifies the association method for adjustment. Four transaction event adjustment methods are available:

- First in last out (`FILO`)
- Decreasing order of distribution factor magnitude (`DF-ORDER`)
- Distribution factor pro rata (`DF-PRO RATA`)
- Pro rata base on the product of distribution factor with transaction schedule (`DF-SCHED-PRO RATA`)

A transaction's schedule is the transaction magnitude less curtailment. The user may alter the adjustment method associated with each transaction event priority.

Individual events, where distribution factors exceed a tolerance, are grouped by priority. The curtail transactions function investigates the priorities from lowest to highest priority number until the incremental flow target is satisfied or all transaction priority groups are exhausted. The restore transactions function investigates the priorities from highest to lowest priority number until the incremental flow target is satisfied or all transaction priority groups are exhausted.

Having updated the adjustment method associations activity LLRF produces a report of transaction adjustments and the effect of each on the target monitored element. The transaction event curtailment values are updated by the reported adjustments.

Activity LLRF may be terminated by entering the AB interrupt control code.

14.4.2. Output Report Format

Both narrow and wide report formats are employed. The column width employed for the output reports depends on the selection of output device. All reports begin with an initial page header followed by four records which identify the source of the Distribution Factor Data File and the Linear Network Analysis Data Files.

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Curtailing and Restoring Transactions

Following the Distribution Factor Data File and the Linear Network Analysis Data Files summary, the target element and target adjustment MW value are reported. This is followed by a table which illustrates for each affected transaction event: the transaction event number, the transaction event priority, the transaction event label, the initial transaction schedule (transaction event magnitude less curtailment) in MW, the schedule adjustment in MW (increment used to update the transaction event curtailment), the final transaction schedule in MW, and the impact of this adjustment on the target element in MW. A statement which indicates if the adjustment target MW value is satisfied follows the table.

Distribution Factor Matrix

The distribution factor matrix is reported after the Distribution Factor Data File and the Linear Network Analysis Data Files summary. Rows of this matrix are the transaction events. These are identified by transaction event number, transaction priority, and transaction label. Columns of the matrix are monitored elements. The column headers are limited to twelve characters. For monitored branches these headers are references to a monitored branch legend. For monitored interfaces the headers are the interface labels. The matrix presentation is wrapped across multiple records when the number of columns to be presented exceeds the presentation width.

The distribution factor values of the matrix represent the sensitivity of monitored element MW flow to transaction event magnitude. A positive value indicates that the monitored element flow will increase with increasing transaction event magnitude.

The distribution factor matrix is followed by a monitored branch legend which cross references the column labels employed in the matrix with monitored branches. The branch descriptions include from bus number and extended name, to bus number and extended name and circuit id.

14.4.3. Application Notes

Activity LLRF uses the same linearized network model as is used in activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Regulating in-service phase shifting transformers hold constant MW flow.

Transaction events may be adjusted by the curtailment or restoration functions when their distribution factor magnitudes exceed the distribution factor tolerance and are of the correct sign with respect to the sign of the adjustment MW value (e.g., having opposite sign for curtailment and the same sign for restoration).

When performing curtailment, the adjustment magnitude limit for each transaction event is equal to the transaction event magnitude less the initial curtailment magnitude. When performing restoration the adjustment magnitude limit for each transaction event is equal to the initial transaction event curtailment magnitude. Adjustments resulting from the curtailment and restoration functions update the transaction event curtailment magnitude.

Additional Information
PSS®E GUI Users Guide, Calculating Line Loading Relief *PSS®E* Command Line Interface (CLI) *Users Guide*, Calculating Line Loading Relief

14.5. Making Allocations

ALOC

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Run Activity ALOC - GUI
Trans Access > Allocations > Megawatt shift factors... *[Megawatt Shift Factors]* *Trans Access > Allocations > Vector Absolute MW-mile* *[Vector Absolute MW-mile]* *Trans Access > Allocations > Vector Absolute MW-ohm* *[Vector Absolute MW-ohm]* *Trans Access > Allocations > Vector Sum MW-ohm* *[Vector Sum MW-ohm]*
Run Line Mode Activity ALOC - CLI
ACTIVITY? >>**ALOC** SELECT THE ALLOCATION TYPE,
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Activity ALOC performs various analyses related to identifying the MW-mile impact of transaction events on transmission facility owners. ALOC is used to:

- Determine the MW-mile impacts of all transaction events on all transmission facility owners.
- Determine the active power generation on MW-mile shift factors of an individual event on all transmission facility owners.
- Determine the MW-ohm impacts of all transaction events on all control areas.
- Determine the active power generation on MW-ohm shift factors of a individual event on all control areas.
- Report the total branch mileage of each transmission facility owner.

Activity ALOC employs the Vector Absolute MW-Mile method as required by the Public Utility Commission of Texas for computing the impact fee (refer to PUCT Substantive Rule 23.67, *Open-Access Comparable Transmission Service*, and PUCT Substantive Rule 23.70, *Terms and Conditions of Open-Access Comparable Transmission Service*).

The user must specify one of the following reports;

- Vector Absolute Megawatt-mile
- Generation on Megawatt-mile shift factors
- Vector Absolute Megawatt-ohm
- Generation on MW-ohm shift factors (employing the Vector Absolute method)
- Vector Sum Megawatt-ohm

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- Branch mileage summary by owner

If specifying Vector Absolute MW-mile, Vector Absolute MW-ohm, Vector MW-ohm, or mileage summary, the report is produced at the specified output destination.

If specifying Generation on MW-mile shift factors or Generation on MW-ohm shift factors the user must specify a transaction event. The report for that event is produced at the specified output destination (refer to Section 4.4, "Virtual Output Devices").

14.5.1. Output Report Format

Transmission allocation reports and related transaction event worksheets are generated to provide an accounting basis which illustrates the impact on the transmission system due to transaction events. The accounting basis may be used to allocate embedded transmission costs or controls area losses among the various transaction events making use of the transmission networks.

All other reports produced by activity ALOC are formatted for import to a spreadsheet program. Selections from the Graphical User Interface menu bar result in spreadsheet formatted reports.

Reports formatted for import to a spreadsheet program include an initial record of text fields for use as column headings in a spreadsheet table. The first field of each subsequent record provides a row label. All other fields are numeric data.

The Branch Mileage Report contains only the branch mileage summary by owner and is displayed at the *Report* tab.

A Vector Absolute MW-Mile (VAMM) method is used to produce the transmission embedded cost allocation. MW-mile methods are techniques for ascribing the use of the electric power transmission system among the various beneficiaries. These are accounting practices which rely on engineering analysis to determine the basis. This basic accounting unit is the product of branch MW flow with branch length.

The VAMM method determines the impact for each individual transaction event by perturbing each bus generation associated with the event against the associated event load. For each perturbation, the absolute value of change in branch flow (i.e., change from the initial condition power flow model) is multiplied by the branch length in miles, and this result is accumulated in the event's MW-Mile vector, one entry for each branch owner of the power flow model (i.e., each facility owner) as illustrated in Equation 14.1.

$$MWM_{ij} = \sum_{k}^{NAG_j} \sum_{l}^{NAL_i} |\Delta P_l| L_l \qquad (14.1)$$

where:

MWM_{ij}	= MW-Mile impact for the i^{th} owner and j^{th} transaction event.
NAG_j	= Number of j^{th} transaction event associated generators.
NAL_i	= Number of branches owned by the i^{th} owner (transmission facility owner).
ΔP	= Incremental MW branch flow due to perturbing MW generation.

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L	= Branch length in miles.

The union of these vectors for each transaction event forms an aggregate MW-Mile allocation matrix. This matrix is then employed in a spreadsheet program to allocate each facilities owner's cost of service among the transaction events, each transaction event being responsible for costs in proportion to the ratio of its impact to the total impact on that facilities owner.

The aggregate MW-Mile allocation matrix may be constructed directly by using a procedure which repeatedly perturbs the participating generation buses for each of the various transaction events, computing the flow impact and accumulating the impacts into the appropriate matrix positions. The aggregate matrix may also be constructed as the union of impact vectors, one for each transaction, where these impact vectors are computed by a transaction event worksheet.

Transaction event worksheets compute an impact vector as a product of generation vector with a coefficient matrix; a generation on MW-Mile shift factor matrix. An element of this coefficient matrix is the sensitivity of the MW-mile impact on a facility owner to the generation output of a participating bus. The matrix elements have units of MW-Mile per MW. The inner summation of Equation 14.1 is used to compute the elements of the generation on MW-Mile shift factor matrix, where the generation perturbation magnitude is 1.0 MW. Computing the aggregate MW-Mile allocation matrix in this manner while more involved, provides for analyzing the impact of various generation dispatch scenarios with the spreadsheet program. A generation vector can be selected which satisfies the demand and minimizes the transmission cost.

The reports generated can produce information for allocating control area transmission losses among the various transaction events. Two loss allocation methods are available, the Vector Absolute MW-Ohm (VAMO) and the Vector Sum MW-Ohm (VSMO). As with the MW-Mile methods, these MW-Ohm methods are accounting practices which rely on engineering analysis to determine the basis. This basic accounting unit is the product of branch MW flow with branch per unit resistance.

The computation required to produce both the VAMO and the VSMO aggregate allocation matrices is that described in Equation 14.1 modified as follows:

- Branch length is replaced by branch per unit resistance.
- The inner summation is over branches as part of the control area as opposed to branch owners.
- For VSMO, only the signed value as opposed to the absolute value of incremental branch flow is employed.

The generated reports may be directed to produce the aggregate allocation matrix for any of the VAMM, VAMO, or VSMO methods. It may also be directed to report the generation vector and the shift factor matrix for both the VAMM and VAMO methods. Furthermore the generated reports can include the summation of branch mileage for each facility owning area. This mileage report is useful for checking the integrity of the mileage and joint branch ownership data.

14.5.2. Application Notes

Activity ALOC uses the same linearized network model as is used in activity DCLF. Thus, the comments in Section 9.4.4, "Application Notes" apply here as well. Regulating in-service phase shifting transformers hold constant angle.

Additional Information
PSS®E GUI Users Guide, Making Allocations *PSS®E* Command Line Interface (CLI) *Users Guide*, Making Allocations

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Chapter 15
Optimal Power Flow

15.1. Overview

The PSS®E OPF application complements the main PSS®E power flow program. A valid power flow data model must be present in the current PSS®E power flow working case before any OPF activities can be successfully executed. The power flow case does not necessarily need to be solved.

To put the power of the OPF in perspective, it may be easiest to contrast it with the traditional power flow. With the traditional power flow, a significant amount of time is often spent trying to achieve an good solution. Multiple iterations are required in which solution executions are performed, results are analyzed and new estimates of the control values are determined for use in the next solution. Much time may be spent trying to parametrically determine what values of the controls will provide a feasible solution. If estimates prove to be unacceptable, as may result in a violation in the operating criteria, new estimates must be provided and the procedure started again. This can often prove to be a very costly cycle.

The optimal power flow, on the other hand, provides a completely analytic model, one which automatically changes certain control variables to arrive at the best solution with respect to a stated quantitative performance measure (i.e. objective function). It efficiently achieves this result by formulating and solving an optimization problem, defining the goal as a combination of objective functions and a set of variable constraints to satisfy. After the problem statement is set up, the OPF often requires much less user intervention than the power flow.

PSS®E OPF formulates and solves the optimization problem within constraints, and automatically updates any power flow data values affected by the solution. Upon completion of the solution, results are presented in one or more formatted reports and output files.

Optimal power flow activities are not sensitive to any interrupt control code options.

The following sections provide more detailed information for:

- the optimal power flow analytical model
- how it relates to the standard power flow
- a description for each power flow data model used within the PSS®E OPF
- models employed specifically within the OPF
- OPF algorithm and the solution process.

15.2. Conventional Power Flow

The OPF solves an optimization problem with side conditions. To place this problem in context, the conventional power flow problem is presented first and then contrasted with the optimal power flow problem.

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The conventional power flow problem solves a series of simultaneous nonlinear equations which ensure that the net complex power injection at every bus is equivalent to the sum of complex power flows on each connected branch.

$$S_{Gi} - S_{Li} = \sum_{j=1}^{N} S_{ij} \quad (15.1)$$

where:

N = Total number of buses,

S_{Gi} = Net complex power generation at bus i,

S_{Li} = Net complex load at bus i,

S_{ij} = Complex power flow on branch connecting buses i and j.

Bus power injections and branch power flows are expressed as functions of complex bus voltage. Through the iterative solution process, a bus voltage vector is determined which satisfies these equality constraints to within a mismatch tolerance.

One criteria for a desired solution might require that a dependent variable set, such as a group of bus voltage magnitudes and branch flows, must satisfy maximum and minimum limits. The power flow model may offer a set of automatic controls for which the values vary independently as functions of local objectives. For example, generator reactive power varies continuously to control the machine's terminal bus or some remote bus voltage magnitude. Other controlling equipment may include transformer tap ratios, transformer phase shift angles, susceptive bus shunts, dc converter control angles, etc.

It is unlikely that sufficient controlling equipment exists such that an individual control may be assigned to each dependent variable. In addition, the control variable range is limited and may be insufficient to satisfy a local objective. This latter condition is especially apparent when local objectives conflict. It is necessary therefore, for the analyst to manually vary either the control values or their local objectives to attempt to achieve a good solution. By observing the impact of parametric variations, the analyst develops an intuition of the power system model. This is likely to be a long and tedious process however.

15.3. Optimal Power Flow

NOPF

The optimal power flow is distinguished from the traditional power flow in that it solves an optimization problem consisting of an objective function augmented by equality and inequality constraints. The optimal power flow algorithms solve a nonlinear problem of the following form:

minimize	f(x,y)
subject to:	equality constraints, and inequality constraints

where:

- The minimization objective may consist of one or more functions such as fuel cost, transmission losses, etc.
- The equality constraints constitute such items as the power flow equations presented in Equation 15.1 which ensure that the net power injection at each bus equals the sum of the power flows on the connected branches.

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- The inequality constraints consist of variables such as the bus voltage magnitudes and angles or active power output which can be varied between a minimum and maximum value.

Power system performance quality is implied by satisfying both the global objective and the constraints. By adjusting control variables, the solution process determines conditions that simultaneously satisfy the constraint equations and minimize the objective function. A complete optimal power flow problem statement thus requires specification of the objective function, the controls, and the constraints.

15.3.1. Objective Functions

Objective functions are expressions of cost in terms of the power system variables. For example, the fuel cost incurred to produce power is a function of the active power generation among participating machines. The OPF automatically adjusts the participating machines' active power generation, within capability limits, to reduce the total fuel cost.

Objective functions have both explicit and implicit components. Explicit objective components are identified by auxiliary modeling data and are generally functions expressed in terms of the optimized subsystem variable set. Implicit objective functions may be introduced as approximations for discrete or discontinuous control actions. These implicit objectives are weighted quadratic penalties applied to the excursion of a variable from an offset: $\rho\ (x_j - x_{j0})^2$. The scalar weight or cost, ρ , is assignable.

15.3.2. Constraints and Controls

Reducing participating active power generation to zero will certainly minimize fuel cost, but it does not provide for the customer power demand. It is necessary therefore, to augment the objective function with constraint equations, including both equality and inequality constraints. Inequality constraints define upper and lower bounds on a variable. For the fuel cost dispatch example, these constraint equations will ensure that the active power generation is sufficient to provide for customer demand and power system transmission losses.

The complex power flow equations of the conventional power flow problem are always included. Controls, such as active power generation, generator voltage magnitude, transformer tap ratio, transformer phase shift angle, etc., may be assigned a fixed value or assigned upper and lower limits. Certain dependent variables, such as load bus voltage magnitudes and branch flows, are assigned upper and lower bounds. It is possible however, for the entire constraint set to not be simultaneously satisfied. This situation results in an infeasible termination.

Unlike the conventional power flow there is no independent local objective associated with each control. The solution process adjusts each control to find the setting that satisfies all stipulated constraints, and further minimizes the objective function. The constraints are applied to, and the objective is written in terms of a variable set. Two variable types exist in the model: control variables (also referred to as independent or decision variables) and dependent variables (also referred to as state variables). These are identified through both the conventional power flow model and the auxiliary optimal power flow modeling data.

One primary difference between the optimal power flow solution and a standard power flow solution when applied to a basic power flow case is that the OPF may introduce barrier terms into the objective function. These barrier terms come from constraints that have hard limits enforced. Bus voltage constraints for example have, by default, hard limits automatically introduced for every bus in the system.

15.3.3. Sensitivities

Each variable, both independent and dependent, has a sensitivity associated with it. Sensitivity values quantify the expected change in the objective in response to a change in the variable. A negative sensitivity indi-

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cates that an increase in the variable's value will decrease the objective's value. The optimal setting for any variable is one which results in a sensitivity of zero. For some variables, the optimal setting exists outside of the variable's limits. In this case, the OPF will drive the variable to the limit and report the sensitivity value. The relative magnitude of the sensitivity values direct attention to the constraints or fixed controls that have the most influence on the objective.

Additional Information
PSS®E GUI Users Guide, Running the Optimal Power Flow Solution *PSS®E* Command Line Interface (CLI) *Users Guide*, Running the Optimal Power Flow Solution

15.4. Conventional Power Flow Models

The OPF model is primarily a conventional power flow model. Complex power flows across transmission branches that connect buses at points of power generation and consumption. The conventional power flow data is prepared and maintained through the PSS®E power flow data functions and stored and utilized within the PSS ®E working case.

The following sections identify the conventional power flow models and details how each one is utilized and modeled within the optimal power flow. The categories are presented in the same order as they would appear in the PSS®E Power Flow Raw Data File.

15.4.1. Bus Data

Only in-service buses (not type 4) defined within the power flow working case are recognized in the OPF model. Any conductive and susceptive shunts that may exist at an in-service bus are recognized.

Voltage magnitude and bus angle are initialized from the conventional model. Swing (Type 3) bus angles are fixed at their initial values while all other bus angles are allowed to vary without limit. Swing bus voltage magnitudes and reactive power generations may vary between the stated upper and lower limits. For problem statements in which generation is dispatched to minimize fuel cost, the swing bus active powers vary only when a fuel cost model is applied. For all other problem statements, the swing bus active power may vary without limit.

By default, all bus voltage limits are set to 0.2 per unit Vmin and 2.0 per unit Vmax, with hard lim-its imposed. These generally wide voltage limits have the potential of contributing to either an infeasible solution or an extraneous solution. Setting the limit mode to *Reporting* may have a similar impact because it essentially causes the voltage limits to be ignored. It is highly recommended that the voltage limit ranges on all buses in the OPF problem statement be set to desired values before commencing the solution.

15.4.2. Load Data

The PSS®E load model is fully recognized by the PSS®E OPF. An auxiliary OPF Adjustable Bus Load data model may be applied to existing loads. This model may be used to determine an optimal load multiplier value that is reported, but not directly applied to the current load values.

The data record for defining the auxiliary data is described in Section Bus Load Data and the adjustable bus load data model itself is described in Section 15.6.3, "Load Adjustment".

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15.4.3. Fixed Shunt Data

Fixed Shunt G-Shunt (MW) and B-Shunt (Mvar) values are included directly into the corresponding bus power injection equations of the OPF solution. Multiple fixed shunts at a bus are treated as a cumulative value.

Although the OPF adjustable bus shunt data records are independent of the fixed shunt data records, they do impact the fixed shunt data after an OPF solution in which the objective to minimize adjustable bus shunts is specified. If a corresponding bus and bus shunt identifier is found within the fixed shunt data, then the B-Shunt value will be updated with the new OPF solution value added in; otherwise a new bus shunt will be added to the power flow network data with same identifiers as given for the adjustable bus shunt record, and the new OPF solution Mvar value specified for B-Shunt.

15.4.4. Generator Data

One or more machines may be associated with a generator bus (type 2) or a swing bus (type 3). Active power and reactive power generation and reactive power generation limits are initialized from the conventional power flow data model. Should the lower limit total equal or exceed the upper limit total at any bus, the bus is implicitly altered to a load bus (type 1). Bus numbers of remotely regulated voltage buses are retained for reporting purposes and optionally for local voltage control of nonoptimized equipment, although the remote regulation concept is invalid for optimized controls. The OPF exercises all optimized controls to satisfy all constraints and beyond that to minimize the objective function. Generator base MVA, source impedance, implicit step-up transformer impedance, and tap ratio data are ignored by PSS®E OPF.

In the conventional power flow, generator control action is discontinuous. Generators regulate a single bus voltage magnitude. This voltage is considered fixed and the reactive generation varies until a reactive generation limit is reached, at which point the regulated voltage becomes free to vary in one direction and the reactive generation is fixed at the limit. If the regulated voltage should return to the set point, the voltage once again becomes fixed and the reactive generation is free.

Generator bus voltage magnitudes are control variables that are varied by the OPF to optimally settle on a value that respect limits. The preferred treatment for nonoptimized generators is the conventional generator model, but this model's discontinuity severely impedes the reliability of the OPF solution. The PSS®E OPF does not directly accommodate the discontinuous model, but approximates it through one of two methods:

1. Through the application of a quadratic penalty on any excursion of the generator voltage magnitude from its initial value. By default a quadratic penalty with a penalty weight of 100.0 is employed. The penalty weight may be changed through the OPF solution option settings and a scheduled voltage may be assigned as the penalty offset for locally regulating units; or

2. Through the introduction of an equality constraint (clamp) which restricts the total error in nonoptimized generator voltage versus reactive generation performance.

These two treatments are mutually exclusive of each other and are globally applied to all nonoptimized generators. Both options are further discussed in Section 15.5, "Modeling Power Flow Controls".

15.4.5. Non-Transformer Branch Data

Resistance, reactance, charging, from bus conductive and susceptive shunt, and to bus conductive and susceptive shunt values are all obtained from the branch data of the standard power flow branch model.

The PSS®E OPF adheres to the PSS ®E convention for zero impedance line modeling. Any branch where resistance is zero and where reactance is smaller than the threshold value THRSHZ (as set in the power flow solution parameters), is treated as a zero impedance line.

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Branch flow constraints for existing branches in the working case may be introduced through the OPF branch flow data model, as presented in Section Branch Flow Constraint Data.

15.4.6. Transformer Adjustment Data

A control variable allocation is made for every transformer defined in the power flow data model, as identified by a branch with a nonzero tap ratio. The optimization status of the transformer is based upon whether the branch is located in an area or zone selected for optimization. Branches are owned by the area and zone associated with the non-metered bus.

Upper and lower limits in the tap ratio or phase shift angle, and tap ratio step size are established as part of the transformer adjustment data in the power flow model. The regulated bus number and adjustment enable flag are used by the OPF, but the controlled value limits (voltage, MW or Mvar flow) are only used to discern the transformer type, tap ratio control or phase angle control, and then discarded. Regulation status is implied by the data as with PSS®E, but as discussed in Section 15.2, "Conventional Power Flow" and Section 15.3, "Optimal Power Flow", optimized controls are adjusted to accommodate the entire optimization problem, not simply a local objective.

The PSS®E program accommodates a transformer model for which tap is a complex number in polar form: the tap ratio is the magnitude and the phase shift is the angle. Only one of these two components on each transformer is selected as the available control. When the transformer adjustment range (i.e. RMAX - RMIN) exceeds 1.0 then phase shift angle is the control variable, otherwise the tap ratio is the control variable. In both situations, the alternate component is fixed. Sensitivities are calculated only for the control variable. Therefore, a transformer that has a fixed phase angle shift to represent a delta-wye connection is reported within the tap ratio transformer summary report (Section 15.14.9, "Tap Changing Transformers"), as opposed to the phase shift angle transformer summary report (Section 15.14.10, "Phase Shifting Transformers").

15.4.7. Area Interchange Data

Area interchange data is only used if the option to Regulate Area Interchange is enabled. In this situation, a constraint equation is introduced for every area interchange record defined as part of the power flow data within the working case. Refer to Section 15.6.7, "Regulated Area Interchange" for further information on the area interchange constraint.

15.4.8. Two Terminal DC Data

Two terminal dc systems are modeled as equivalent active and reactive power demand. All in-service two terminal dc systems of the working case are automatically replaced by equivalent bus loads during the OPF solution. The equivalent load is obtained from the dc bus injection as calculated by the PSS®E power flow solution activities and stored within the working case. It is therefore best to perform a power flow solution prior to the OPF solution in order to start with a condition that has been solved for the dc boundary conditions of interest.

15.4.9. Voltage Source Converter Data

VSC devices are modeled as equivalent load injections, obtained directly from the PSS®E power flow solution as stored in the working case. It is best to start the OPF solution from a power flow condition that has already been solved for the boundary conditions of interest.

15.4.10. Transformer Impedance Correction Data

Transformer impedance adjustment as a function of tap position is not accommodated in the PSS®E OPF. Transformer impedance is initialized to the adjusted impedance and held fixed.

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15.4.11. Multi-Terminal DC Line Data

Multi-terminal dc systems are modeled as equivalent active and reactive power demand. All in-service multi-terminal dc systems within the working case are automatically replaced by equivalent bus loads. The equivalent dc bus injection is calculated by the PSS®E power flow solution activities and stored in the working case. It is therefore best to perform a power flow solution prior to the OPF solution in order to start with a condition that has been solved for the dc boundary conditions of interest.

15.4.12. FACTS Device Data

FACTS devices are modeled as equivalent active and reactive power demand. During the OPF solution, all in-service FACTS devices are automatically replaced by equivalent bus loads. The loads are obtained directly from the equivalent bus injection as calculated by the PSS®E power flow solution and stored in the working case. It is therefore best to start the OPF solution from a power flow condition that has already been solved for the boundary conditions of interest.

15.4.13. Induction Generators and Motors

Induction generators and motors are modeled as equivalent active and reactive power demand at their corresponding buses. During the OPF solution, all induction generators and motors that are in-service and not tripped are automatically replaced by the equivalent active and reactive bus loads. The load injections are obtained directly from the equivalent bus injection as calculated by the PSS®E power flow solution and stored in the working case. It is therefore best to start the OPF solution from a power flow condition that has already been solved for the boundary conditions of interest.

15.4.14. Switched Shunt Data

Bus shunt control variables are identified by both the auxiliary OPF adjustable bus shunt data model and the switched shunt model in the conventional power flow data. The adjustable bus shunt data model is presented in Section 15.6.2, “Adjustable Bus Shunt”.

For switched shunts defined within the power flow, the bus number, control mode, initial value and switching information for the eight blocks are retained by the program.

By default the switched shunt susceptance is adjusted continuously. Non-regulating switched shunts and var sites in nonoptimized subsystems are fixed at their initial value (BINIT).

The control mode values of switched shunts that are not fixed are treated as follows:

- Modes 1 and 2: The desired voltage limits and the remote bus number are discarded in lieu of explicit bus voltage limits applied to buses via the OPF Bus Voltage Attribute Data Record.
- Mode 3: Reactive generation limits of regulated generators are also restricted by the specified band.
- Mode 4: Handled the same as modes 1 and 2. Because VSC are handled as equivalent bus load injections in the OPF solution, no adjustments are made to VSC limits.
- Mode 5: Admittance limits of switched shunts (local or remote) are also restricted by the specified band.

15.5. Modeling Power Flow Controls

The conventional power flow models pertaining to generator bus voltage magnitudes, transformer tap ratios, transformer phase shift angles and switched shunts, as presented in Section 15.4, “Conventional Power Flow

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Models", introduce a number of controls into the OPF problem statement. Within the OPF however, unlike in the standard power flow, it is possible to influence the motion of these controls.

Controls within both optimized and nonoptimized subsystems (selected areas or zones) of the power flow may be represented. Most of the control variables introduced through the power flow model residing in the nonoptimized subsystems are fixed at their initial value, but special treatment is given to nonoptimized generator bus voltage magnitudes. This is described in further detail below.

15.5.1. Scalar Quadratic Penalties

A weighted quadratic penalty is inherently applied to four power flow controls: generator bus voltage magnitude, transformer tap ratio, transformer phase shift angle and switched shunt admittance. The OPF will adjust these controls to achieve a feasible solution and to balance the penalty against the other selected objective components. This quadratic penalty is all or none; it may not be selectively applied to certain nonoptimized generator units. Global quadratic penalties take the form,

$$\rho \sum_{i=1}^{N} (x_i - x_{io})^2$$

where:

N = number of penalized control variables,

ρ = scalar quadratic penalty weight,

x = control variable current value (per unit),

x_o = control variable penalty offset (per unit).

By default, the nonoptimized generator bus voltage magnitudes are penalized with a default weight of 100.0 and the quadratic penalty weight for the transformer and switched shunt controls is 0.0. These values are globally applied to all nonoptimized controls and may be altered by modifying the *Penalty for fixed voltage* value and the *Quadratic penalty coefficient* value, respectively, in the OPF change parameters interface.

15.5.2. Soft Limit Penalties

Unlike the global quadratic penalty, soft limit penalties, which include a penalty weight (cost) and offset value, may be selectively applied to specific constraint variables. This includes power flow bus voltage constraints as well as branch flows and interface flows, both of which are introduced as part of the OPF models discussed in Section 15.6, "OPF Modeling Extensions" By default, soft limit penalty weights of 1.0 are assigned. These values may be modified and applied to specific variables, or a common penalty weight may be applied to an entire subsystem by employing the corresponding OPF subsystem spreadsheet (refer to Section 15.8.4, "Optimal Power Flow Spreadsheet View").

Details of the quadratic and linear soft penalty treatments are further discussed in Soft Limit Treatment.

15.5.3. Treatment of Transformers and Switched Shunts

For transformers and switched shunts residing in an area or zone selected for optimization, a zero cost quadratic penalty is automatically applied. This penalty weight value may be altered through the OPF solution tolerance for the Quadratic Penalty Coefficient.

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Transformers and switched shunts not residing within an optimized subsystem, or those where the setting has been fixed (either in the power flow model or through the *Fix* setting), will be treated as nonoptimized. In this situation, these controls are fixed at their current working case values. There is the possibility however, for bus voltage penalties to also be imposed on the regulated buses, although this is done independent of whether the transformer or switched shunt is being treated as a nonoptimized control.

Discrete transformer tap ratio adjustments or switched shunt adjustments are approximated by initially treating these controls as continuous. If the option to Round Transformer Tap Ratios and/or Round Switched Shunt Vars is imposed, then after an optimal solution has been achieved with continuous treatment, the control is fixed to the nearest discrete value and the problem is resolved from this condition.

15.5.4. Treatment of Local Generator Voltage Controls

Similar penalty treatments exist for generator voltage controls, but vary depending upon whether the treatment is applied to an optimized or nonoptimized generator. Nonoptimized generators are those that are not located in an optimized subsystem. If the Treat all Generators as Non-optimized option is enabled, then all generators in the system are treated as nonoptimized, independent of whether they are located within a nonoptimized area or zone.

Using the generator and area designations defined in the one-line diagram of Figure 15.1, "Generator and Area Designations from the savnw.sav Case", the far right-hand column of Table 15.1, "Local Generator Voltage Controls" indicates which generators would result in being treated as nonoptimized based upon the setting of the *Treat all generators as non-optimized* option and the subsystem of areas selected for optimization.

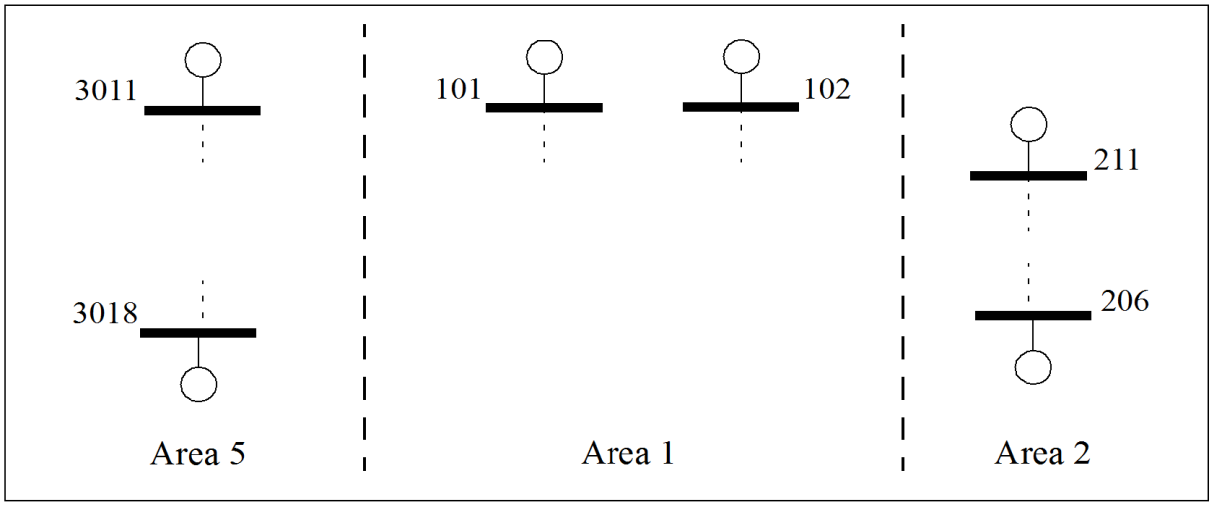

Figure 15.1. Generator and Area Designations from the savnw.sav Case

Table 15.1. Local Generator Voltage Controls

Treat all generators as non-optimized?	Areas selected for optimization	Resulting nonoptimized generators
Yes	1	All generators
Yes	1, 2 and 5	All generators
No	1	211, 206, 3011, 3018

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Treat all generators as non-optimized?	Areas selected for optimization	Resulting nonoptimized generators
No	1, 2 and 5	No generators

Treatment of Optimized Generators

Generators that are located in an optimized subsystem are treated according to the voltage magnitude penalty limits defined at the voltage regulated bus. One of four limits may be applied: reporting only (no penalties), hard limits, or soft limits with either a linear or quadratic penalty. The hard limits introduce a barrier term into the objective function. This treatment is discussed in Hard Limit Barrier Treatment. The soft limits utilize a soft limit penalty weight in conjunction with either a linear or quadratic curve to penalize the motion of the controls. The higher the cost, the less likely the variable is to violate its limits. Employing the quadratic penalty also imposes a more severe penalty than will the linear penalty.

Treatment of Non-optimized Generators

The preferred treatment for nonoptimized generators is the local voltage control model employed by the conventional power flow solution. This however poses problems when modeled in the optimal power flow. Figure 15.2, "Conventional Power Flow Generator Voltage Regulation" illustrates the relationship between the voltage magnitude at the regulated bus and the regulating reactive power generation employed by the conventional power flow solution.

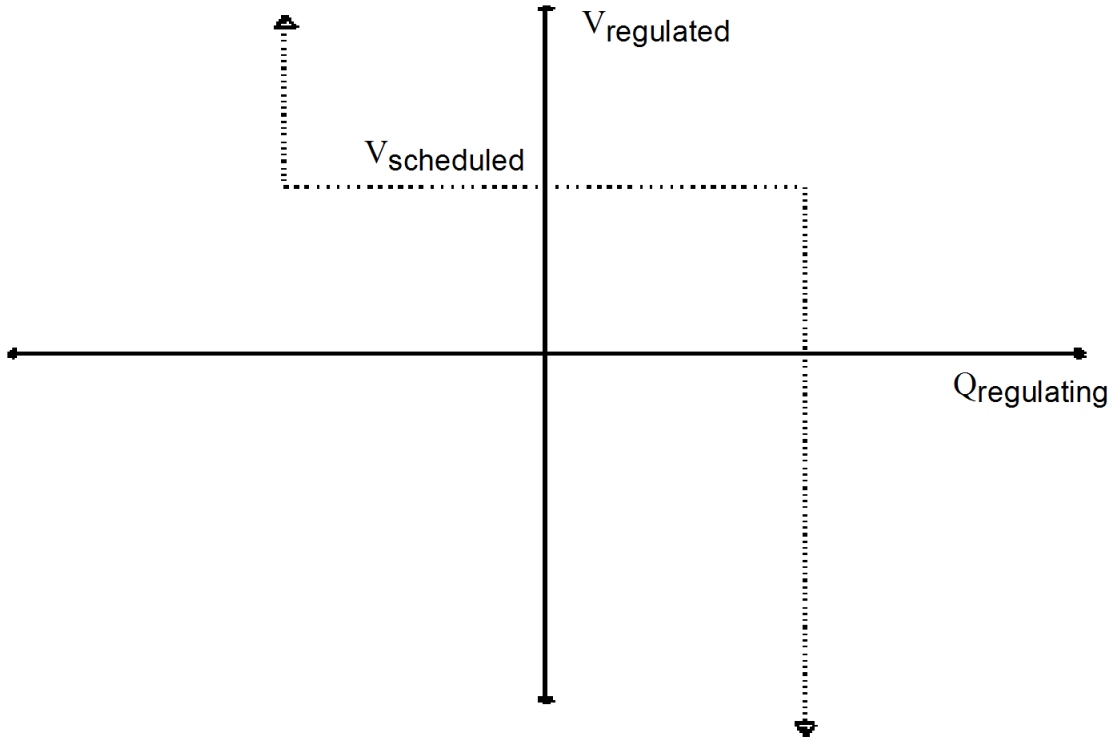

Figure 15.2. Conventional Power Flow Generator Voltage Regulation

The regulated voltage may be that of a generator bus or a remote non generator bus. The regulating reactive power supply is the sum of the reactive generation of all the units that regulate the bus. Note the discontinuity

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at the upper and lower reactive power generation limits. It is this discontinuity which confounds modeling this relationship in the OPF, and thus approximations for this treatment must be employed.

Several different penalty techniques are available for approximating the local control objective at nonoptimized generators:

- bus voltage magnitude soft limit penalties (linear or quadratic),
- scalar quadratic penalties
- constraint (clamp) equation.

Soft Limit Penalty Techniques

Soft limit penalty techniques are applied on a per bus basis through the bus voltage magnitude constraints defined at corresponding voltage regulated buses (see Section Bus Voltage Constraint Data). The soft limit penalty introduces an objective function component that penalizes the excursion of the regulated voltage magnitude from an offset.

There are two soft limit penalty treatments available: linear and quadratic. Both are discussed in further detail in Soft Limit Treatment. When applied to nonoptimized generator units, the soft limit penalty treatment supersedes either the scalar quadratic penalty or clamp treatments discussed in the following text.

Scalar Quadratic Penalty

A second technique for approximating the local control objective of nonoptimized generators is by means of a scalar quadratic penalty. This quadratic penalty is all or none; it may not be selectively applied to certain nonoptimized generator units. Unless the clamp equation is employed (as discussed below), or a penalty limit is applied to the individual bus voltage constraint, a quadratic penalty is inherently applied to all nonoptimized generator units. The default penalty weight of 100.0 may be modified prior to the OPF solution. Unlike the soft limit penalty weight which may be uniquely defined for individual bus voltage constraints, this value is applied as a scalar to the entire quadratic penalty equation. Refer to Section 15.5.1, "Scalar Quadratic Penalties" for further details.

Clamp Treatment

The final penalty technique available for approximating the local control objective for nonoptimized generators is by means of a constraint, or clamp, equation. This equation limits the total deviation from the voltage and reactive power relationship presented in Figure 15.2, "Conventional Power Flow Generator Voltage Regulation". Consider the equation,

$$\sum_{i=1}^{N} \frac{(Q_{max,i} - Q_i)\max(0, (V_{s,i} - V_i)) + (Q_i - Q_{min,i})\max(0, (V_i - V_{s,i}))}{(Q_{max,i} - Q_{min,i})} \leq \tau$$

where:

Q = reactive power generation

V = regulated (local or remote) bus voltage magnitude

V_s = regulated voltage set-point

N = number of nonoptimized generators

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τ = clamp tolerance,

and note:

$(Q_{max} - Q) \max (0, (V_s - V)) \geq 0$

$(Q - Q_{min}) \max (0, (V - V_s)) \geq 0$

Armature reaction (E_{fd}) is employed in place of reactive generation (Q) when reactive capability models are introduced for nonoptimized generators. With $\tau = 0.0$, the equation is satisfied only when $V = V_s$ or Q is limited, and the discontinuous performance is accommodated exactly.

The PSS®E OPF interior point solution technique described in Section 15.7.2, "Accommodating Inequality Constraints" prevents a solution when τ equals 0.0. The consequence of having a nonzero clamp tolerance is illustrated in Figure 15.3, "Feasible Clamp Region for Non-zero τ ". Clamp constraint limits are applied in an attempt to reduce the total error in nonoptimized generator performance. The strategy employed in the PSS®E OPF is to initialize the problem with a relatively large clamp tolerance, and then systematically and gradually reduce this tolerance to a final tolerance as the solution progresses.

If the voltage set point falls outside of the minimum or maximum voltage range specified for the regulated voltage bus, the set point voltage will be set to the value of the violated limit.

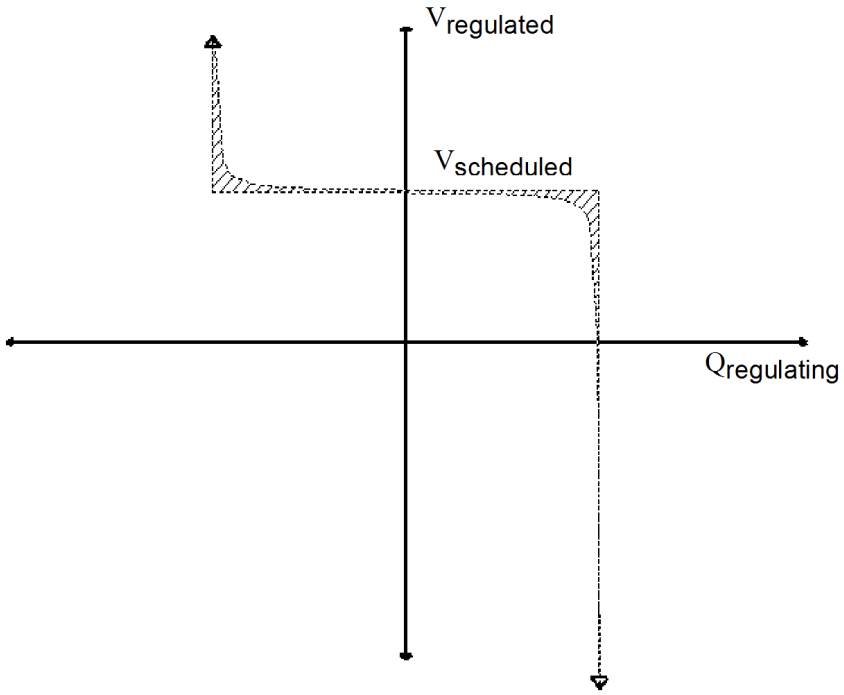

Figure 15.3. Feasible Clamp Region for Non-zero #

15.5.5. Reactive Generation Reserve

A reactive generation reserve objective function component may be included in the opf problem statement. This objective component will influence the magnitude of reactive power generation held in reserve. All in-service generators that are members of the optimized subsystem contribute to this objective. The OPF will

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adjust the reactive power generation of these generators to achieve a feasible solution and to balance the cost of this objective component against the other selected objective components. The reactive generation reserve objective takes the form:

$$\rho \sum_{i=1}^{NGEN} (Q_{MAX,i} - Q_{GEN,i})$$

where:

NGEN	= Number of in-service optimized generators.
ρ	= Objective cost coefficient in cost units/Mvar.
Q_{MAX}	= Maximum reactive power generation limit in Mvar.
Q_{GEN}	= Reactive power generation in Mvar.

The reactive generation reserve will tend to be minimized by applying a positive objective cost coefficient. The reactive generation reserve will tend to be maximized by applying a negative objective cost coefficient. The objective component may be applied by enabling the Minimize Reactive Generation Reserve option, and the cost coefficient may be assigned using the input field labeled *Reactive generation reserve cost ($/MVAR)*.

15.6. OPF Modeling Extensions

The following subsections introduce the power system models recognized by the PSS®E OPF beyond the conventional power flow models introduced in Section 15.4, "Conventional Power Flow Models".

15.6.1. Active Power Generation

Active power generation control variables may be introduced by presenting cost as a function of active power generation. Individual generator units may be associated with certain active power controls. The amount of dispatch is adjusted to satisfy the constraints and minimize the total generation cost component of the objective function. Generators in nonoptimized subsystems that are associated with an active power dispatch control will participate in the dispatch.

One of three cost curves may be employed to model the control: piece-wise linear, piece-wise quadratic and polynomial.

15.6.2. Adjustable Bus Shunt

A subset of buses may be identified as susceptive bus shunt adjustment candidates for shunt compensation. When defined, adjustable bus shunt controls are employed to minimize the total adjustable bus shunt component of the objective function (refer to Minimize Adjustable Bus Shunts). The candidate buses for shunt compensation are introduced through the adjustable bus shunt data records described in Section Adjustable Bus Shunt Data.

This objective function employs a linear penalty of the following form:

$$F(B) = \sum_{i=1}^{NVARS} |B_i| C_i$$

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and

$$B_{IND,\,MAXi} \le B_i \le B_{CAP,\,MAXi}$$

where:

B	= adjustable bus shunt susceptance (per unit)
$B_{CAP,MAX}$	= capacitive limit (per unit)
$B_{IND,MAX}$	= inductive limit (per unit)
C	= shunt cost / per unit Mvar
NVARS	= number of adjustable shunt candidate buses

The cost, C_i, is assigned to individual adjustable bus shunt controls, in cost units / per unit Mvar. Multiple shunt device controls may be assigned to the same bus, with each one distinguished by a different cost for deployment. For example, a relatively low cost may be assigned to a control representing an existing installation and a high cost to a control representing a potentially new installation. In this situation the existing installation is ensured of being deployed to its limit before any portion of the new capacity is applied.

The shunt control is assigned a minimum and maximum var limit and may operate in both an inductive (-B) and capacitive (+B) range. The resulting shunt value is added to the corresponding fixed shunt in the power flow model. If no corresponding fixed shunt is found, then a new fixed shunt with the same identifiers as the adjustable bus shunt will be added. If an initial offset (no cost, fixed shunt component) is desired, it may be included in the power flow model as a fixed shunt prior to the OPF solution.

15.6.3. Load Adjustment

Any number of predefined loads within the power flow case may be identified as adjustable load candidates. The constant MVA component of these loads is adjusted by a multiplier to minimize the total load adjustment component of the objective function. The multiplier may vary continuously between predefined minimum and maximum limits.

The load adjustment objective function is of the following form:

$$F(\Psi) = \sum_{i=1}^{NSHED} C_i \bullet MW_{0_i} \, (1 - \Psi_i)$$

$$\Psi_{MIN_i} \le \Psi_i \le \Psi_{MAX_i}$$

$$MVA_{LOAD_i} = \Psi_i \bullet MVA_{0_i}$$

where:

MVA_{LOAD}	= MVA bus load (per unit)
MVA_0	= initial MVA bus load (per unit)
MW_0	= initial active power bus load (per unit)
Ψ	= load adjustment factor
Ψ_{MIN}	= minimum load adjustment factor
Ψ_{MAX}	= maximum load adjustment factor

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C	= load shed cost / per unit MW
NSHED	= number of load shed candidate buses

Bus load is shed at constant power factor. The cost, C_i, is assigned per candidate load adjustment group, in cost units / per unit MW. The control is limited between minimum and maximum values specified as a fraction of initial total load.

15.6.4. Adjustable Branch Reactance

The adjustable branch reactance model is generally employed to identify the best transmission lines of a candidate set for which to apply optimal series compensation. The control modifies the branch reactance by adjusting a reactance multiplier between a minimum and maximum limit in order to meet the minimal adjustable branch reactance component of the objective. Only non transformer branches may be considered as candidates for reactance adjustment.

The form of the adjustable branch reactance objective function is as follows:

$$f(X_c) = \sum_{i=1}^{NCOMP} C_i \bullet X_{0_i} (1 - \Psi_i)$$

and

$$\Psi_{MIN_i} \leq \Psi_i \leq \Psi_{MAX_i}$$

$$X_{COMP_i} = \Psi_i \bullet X_{0_i}$$

where:

XCOMP	= Branch reactance value (per unit)
X0	= initial branch reactance (per unit)
Ψ	= reactance adjustment factor
Ψ_{MIN}	= minimum reactance multiplier
Ψ_{MAX}	= maximum reactance multiplier
C	= compensation cost / per unit ohms
NCOMP	= number of compensation candidate branches

The cost, C_i, is assigned per candidate branch, in cost units / per unit ohms. The control may vary between the minimum and maximum values specified as a fraction of initial compensation.

15.6.5. Generator Reactive Capability

Synchronous generator performance is limited primarily by stator heat transfer, rotor heat transfer, and a minimum sustainable exciter voltage. These limits are often abstracted into other related variables depending upon the application. For example, plant operating personnel may be provided with limits depicted by the reactive power generation (Q_{gen}) versus active power generation (P_{gen}) plane illustrated in Figure 15.4, “Reactive Capability, Q_{gen} vs. P_{gen}”. When abstracted in this way, certain assumptions may apply. Any one set of limiting point locus from the Q_{gen} vs. P_{gen} representation assumes constant (often unity) terminal voltage and constant cooling medium consideration (e.g. cold gas inlet temperature, or hydrogen pressure).

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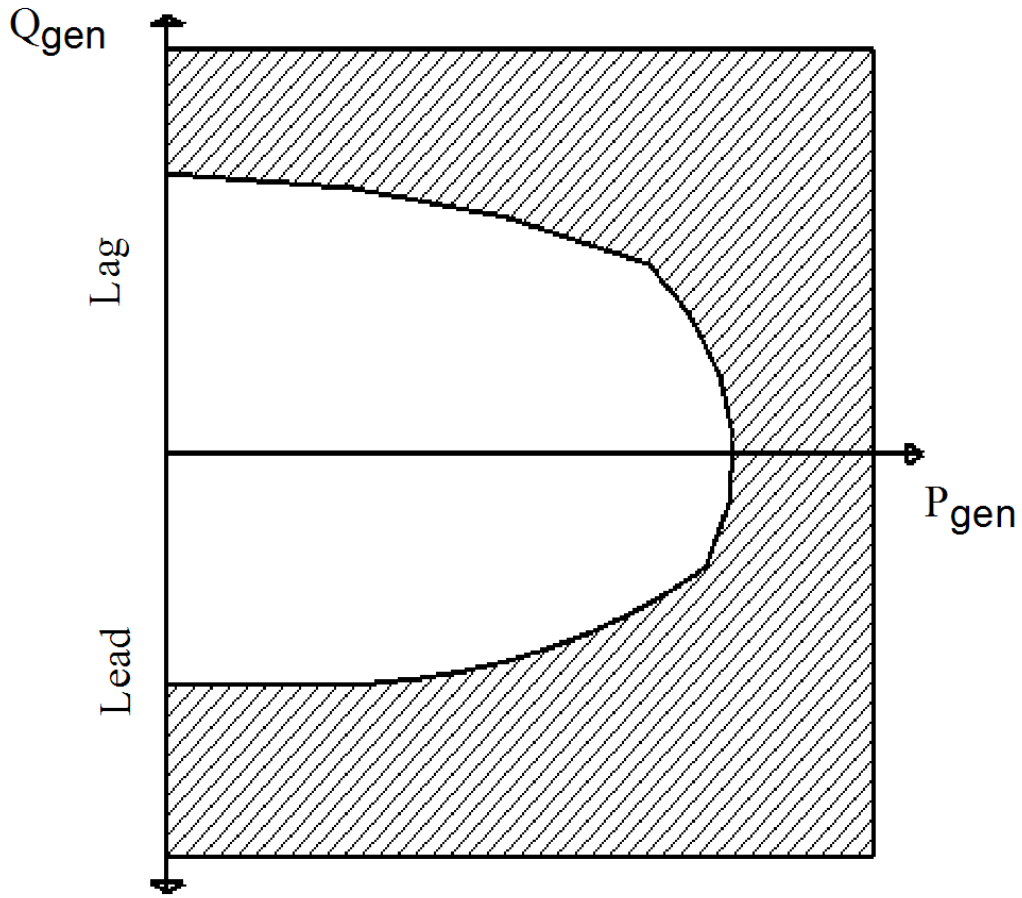

Figure 15.4. Reactive Capability, Q_{gen} vs. P_{gen}

The Q_{gen} vs. P_{gen} limits description is amenable to the conventional power flow problem because of the simple inequality treatment of reactive power limits, $Q_{min} \leq Q_{gen} \leq Q_{max}$, fixed voltage schedule, and fixed active power generation. Because the OPF is able to dispatch active power generation and voltage schedule, a more elaborate model that describes the machine's capability is appropriate. This is achieved by replacing the generator representation of P_{gen} and Q_{gen} injections at the terminal bus, with one of P_{gen} and Q'_{gen} injections at a point removed from the terminal bus through direct axis synchronous reactance, as illustrated in Figure 15.5, "Alternate Generator Model (Injection Behind Synchronous Reactance, X_d)".

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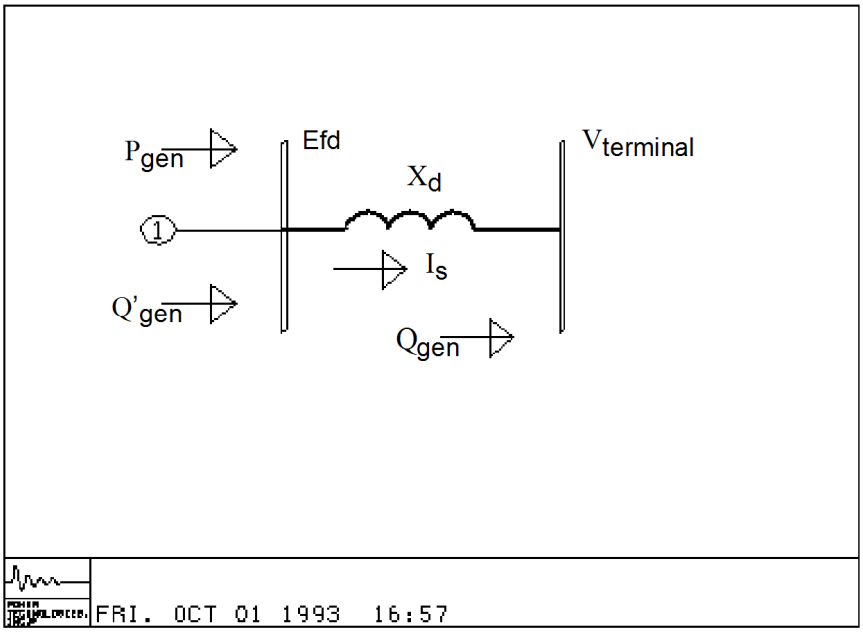

Figure 15.5. Alternate Generator Model (Injection Behind Synchronous Reactance, X_d)

By ignoring machine saliency and saturation, this model defines both armature reaction (E_{fd}) and stator current (I_s). The Q_{gen} vs. P_{gen} depiction of generation capability is constructed as a mapping, at constant terminal voltage magnitude, of voltage invariant limits in the I_s vs. E_{fd} plane, as illustrated in Figure 15.6, "Is vs. Efd Plane".

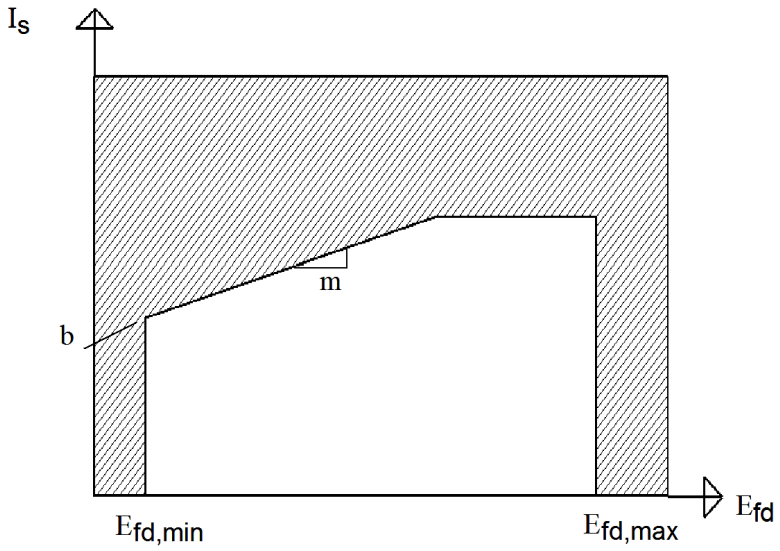

Figure 15.6. Is vs. Efd Plane

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The machine capability is then fully described by the following inequality limits applied to these parameters,

$E_{fd,min} \le E_{fd} \le E_{fd,max}$

$0 \le I_s \le I_{s,max}$

$0 \le I_s \le mE_{fd} + b$

In addition, all of the limit coordinates: $E_{fd,min}$, $E_{fd,max}$, m and b, can be derived from machine parameters generally provided with the Q_{gen} vs. P_{gen} capability description, including: machine MVA rating (MVA_r), direct axis synchronous reactance (X_d), rated lagging power factor (pf_{lag}), rated leading power factor (pf_{lead}), maximum reactive absorption at zero power factor (Q_{0pf}), and a choice of maximum stator current ($I_{s,max}$) reflecting a particular cooling condition (e.g 1.0 per unit current for H_2 pressure of 30 psig). The vector arithmetic required to derive the limits is illustrated in Figure 15.7, "Vector Arithmetic for Deriving Limits".

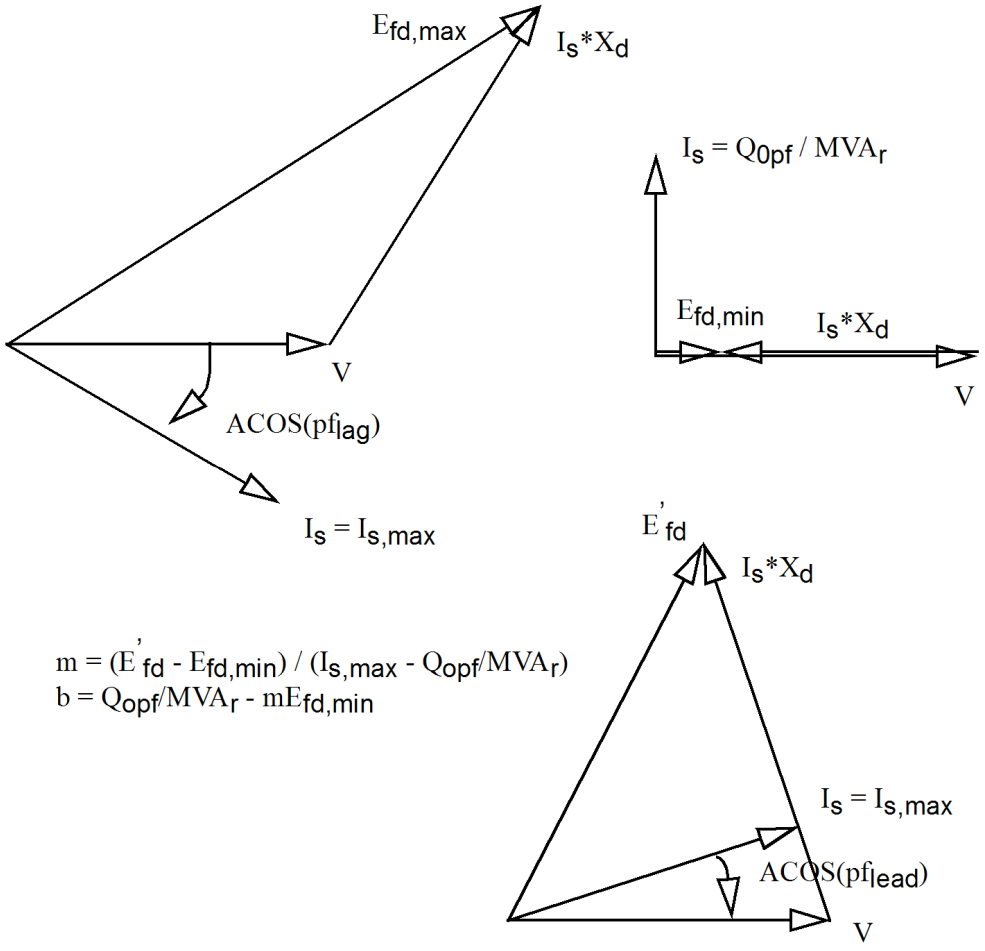

Figure 15.7. Vector Arithmetic for Deriving Limits

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15.6.6. Generator Period Reserve

A common practice in power system operation is to commit and dispatch generation such that sufficient excess capacity exists to accommodate some reserve constraint. This may be compounded with a time dimension, such that the reserve capacity must be fully available within a stated number of minutes, and the units that participate in supplying the reserve are limited by their active power generation ramp capability and emergency maximum capacity. A minimum generation cost dispatch may be significantly impacted by these generation period reserve constraints.

The PSS®E OPF generation period reserve model provides for a reserve in MW, and a time period in minutes, to be stated. Participating generators associated with each of these constraints are also defined. A maximum unit capability in MW and an active generation ramp rate in MW per minute is identified for each participating generator. The sum of participating unit reserves must exceed the reserve constraint. The amount of reserve contributed by each of the participating units is the lesser of the unit capability minus its active power dispatch, or its ramp rate times the period. This is exemplified in the following equation.

$$\sum_{i=1}^{N} \min((P_{max,i} - P_{gen,i}), T \bullet Ramp_i) \geq RES$$

where:

$P_{max,i}$	= maximum unit capacity for the reserve contribution
$P_{gen,i}$	= unit active power dispatch
$Ramp_i$	= unit ramp rate
T	= reserve period
N	= number of reserve constraint participating generators
RES	= reserve magnitude

Multiple coincident period reserve constraints are permitted. For example, a problem statement may include both a 200 MW reserve requirement in 10 minutes and a 700 MW reserve requirement in 30 minutes. Individual units may participate in multiple constraints. Those units with fixed active power generation merely act to offset the reserve requirement, otherwise the period reserve constraints may impact the dispatch of participating units.

15.6.7. Regulated Area Interchange

The regulated area interchange model provides a convenient means of defining a closed MW interface constraint. Area interchange is measured as the sum of MW flows on tie branches emanating from the stated area. As such, the tie branches should consist entirely of impedance branches.

This model introduces an interface flow constraint defined by the components described below. It does not introduce any MW associated controls. While not necessary, it is appropriate to include in the problem statement those controls for which area tie branch flows are likely to be sufficiently sensitive (e.g. dispatchable active power generation spanning the controlled area).

The area interchange data model is introduced as part of the standard power flow data model as defined by the following information:

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- Area number
- Bus number of the area slack bus
- Desired net interchange leaving the area (export), in MW
- Interchange tolerance band width, in MW

With the *Regulate area interchange* option enabled, the optimal power flow transforms the above information into a constraint equation with the following area interchange limits:

AIntmin = (DesInt - DesTol)

AIntmax = (DesInt + DesTol)

where:

AInt	= the minimum and maximum constraint limits placed on the OPF area interchange constraint equation.
DesInt	= the desired interchange.
DesTol	= the interchange tolerance band width.

Each new area interchange constraint model implicitly defines an interface consisting of the following components:

- area tie branches, excluding zero impedance lines;
- flows associated with dc lines, multi-terminal dc lines, FACTs and VSC devices;
- loads, provided that the option to include loads in area interchange calculations has been enabled (refer to *PSS®E GUI Users Guide*, Changing Program Settings).

15.6.8. Linear Constraint Dependency Equation

The linear constraint dependency equations provide the capability to introduce customized constraint equations into the OPF problem statement. The form of the constraint equation is as follows:

$$\left(\sum_{i=1}^{NTERM} C_i \bullet X_i \right) - s = 0$$

$$s_{min} \leq s \leq s_{max}$$

where:

C	variable term coefficient
x	power system variable (voltage, angle, flow, etc.)
s	the slack value

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smin	minimum slack variable value
smax	maximum slack variable value
NTERM	number of terms within a single equation.

Any number of equations may be introduced into the problem statement. The equations themselves may consist of one or more variable terms selectable from up to ten different power system variable types, including:

- voltage magnitude, in pu
- voltage angle, in radians (degrees/57.29578)
- active power generation, in per unit of reactive power based on system base(i.e. 400 MW limit base on a system base of 100 MVA is entered as 4.0)
- reactive power generation, in per unit of reactive power based on system base(i.e. 400 Mvar limit base on a system base of 100 MVA is entered as 4.0)
- transformer tap ratio, entered as the inverse of the tap ratio
- transformer phase shift angle, in radians
- branch flow, in per unit based on system base
- interface flow, in per unit flow value based on system base
- adjustable bus shunt, in per unit Mvar value based on system base
- switched shunt, in per unit Mvar value based on system base
- load adjustment, entered in terms of the load multiplier (i.e. 0.8 for 80% of load or 1.8 for 180% of load

The units of the slack variable are dependent upon the type of variables used within the equation.

Example

Suppose the voltage angles between two generator buses need to be held to within three degrees of each other. A constraint equation of the following form can be introduced:

(1)*Ang1 + (-1)*Ang2 - slk = 0

and

-3° < slk < 0°, or

-0.052 < slk < 0 radians

15.6.9. Other Dependent Variable Limits

Several power system variables may have limits selectively applied, including those for bus voltage magnitudes, branch flows and interface flows.

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Branch flow limits may be applied to any of MW, Mvar, MVA or current values. The first three limit values are stated in physical units. Current limits are stated in MVA units at nominal voltage so as to preclude the need for a valid base kV in the power flow model.

Interface flow limits may be applied to either MW or Mvar constraints, and are stated in physical units. Interfaces are defined as a collection of branches. The limited flow quantity is the scalar sum of the individual branch MW or Mvar flows. All flow quantities are measured at the branch metered point, as identified in the power flow data.

Zero impedance branches may not be branch flow constraint candidates, and may not be included in interface definitions.

15.7. Solution Process

15.7.1. The Nonlinear Problem

PSS®E OPF solves a nonlinear problem consisting of an objective function and a linear combination of equality constraints to form a scalar Lagrangian function of the following form.

$$L(x, \lambda) = f(x) + [\lambda]^t [h(x)] \quad (15.2)$$

and

$$h(x) = 0$$

$$x_{MIN} \leq x \leq x_{MAX}$$

where:

x = power system variables (control and dependent, column vector),

f(x) = objective function (e.g. active power dispatch cost),

h(x) = equality constraints including bus-power mismatch equations (column vector),

λ = Lagrange multiplier variable (column vector).

The Lagrangian function is constructed in terms of both the power system variables and the Lagrange multipliers, thereby resulting in a dual variable problem. The equality constrained optimal solution is a Lagrangian stationary point, and is determined by equating the function's gradient to zero and solving for x and λ. This is the Kuhn-Tucker optimality condition [1], and the resulting set of simultaneous equations is referred to as the KT formulation. Solving the KT problem produces a set of optimally defined power system variables (x), along with the objective function sensitivities to changes in the constraints (λ).

The OPF formulation is always nonlinear in x because the constraints include the power flow mismatch equations. The objective function is also likely to be nonlinear in x either through explicitly selected components, like active power loss, or the implicit introduction of quadratic penalties. Therefore, an iterative solution technique which will recognize the limits in x is required. A solution to the nonlinear problem results from solving a series of simpler subproblems, wherein the inequality constraints are satisfied at every subproblem.

[1] Kuhn, H.W. and Tucker, A.W., "Nonlinear Programing", *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability* (J. Neyman, ed.), Berkeley, University of California Press, 1951.

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15.7.2. Accommodating Inequality Constraints

Inequality constraints are imposed through implicit objective terms expressed as a function of the constrained variable. These limits have two types, hard and soft. Hard limits introduce objective terms that are asymptotic to the variables limits (i.e. barrier terms), while soft limits introduce objective terms which are defined in the infeasible region and grow in magnitude as the variable value departs further from its violated limit (i.e. penalty terms).

Limits imposed on the power system control variables, such as transformer tap settings, are considered extremes in the equipment's physical range, and thus are always treated as hard limits. This treatment is in addition to the global quadratic penalty that may optionally be imposed on power flow control values as they move away from their initial values. As discussed in Section 15.5, "Modeling Power Flow Controls", this quadratic penalty is imposed on all optimized transformers and switched shunts provided that the quadratic penalty coefficient is greater than zero.

Limits applied to power system dependent variables, including voltage magnitude, branch flows, and interface flows, represent operating criteria and may receive either hard or soft limit treatment.

Soft Limit Treatment

Two forms of the soft limit penalty are available: quadratic and (two point) linear. These are discussed in further detail in the following two sections.

Quadratic Penalties

The soft limit quadratic penalty introduces a penalty term of the following form into the objective function.

$$\sum_{i=1}^{N} \rho_i \bullet \left(x_i - \frac{(x_{i,min} + x_{i,max})}{2} \right)^2$$

where:

N	= number of quadratically penalized control variables
x	= control variable current value (per unit)
x_{min}	= control variable minimum (per unit)
x_{max}	= control variable maximum (per unit)
ρ	= vector penalty weight. This value is unique to each control.

A penalty term is introduced any time a variable takes an excursion away from the midpoint of its two limits. The quadratic growth of the penalty tends to discourage large and presumably infeasible excursions.

Linear Penalties

The soft limit linear penalty introduces a penalty term of the following form into the objective function.

$$\sum_{i=1}^{N} \rho_i \bullet (\max(0, x_i - x_{MAX,i}) - \min(0, x_i - x_{MIN,i}))$$

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where:

N	number of linearly penalized control variables
x	control variable current value (per unit)
x_{MAX}	control variable maximum value
x_{MIN}	control variable minimum value
ρ	vector linear penalty weight

A penalty term is introduced into the objective that is directly proportional to the variable's excursion from a violated limit. The penalty is zero for feasible variables. The linear form provides a relatively larger objective contribution for fairly small excursions in the control value as compared to the quadratic penalty.

Hard Limit Barrier Treatment

The hard inequality constraints are rigorously enforced at each sub-problem by augmenting the Lagrangian with objective-like equations in terms of x, x_{MAX}, and x_{MIN}. These equations, called barrier functions (B(x)), are continuous and asymptotic to the limits, x_{MAX} and x_{MIN}, as depicted in Figure 15.8, "Logarithmic Barrier Function".

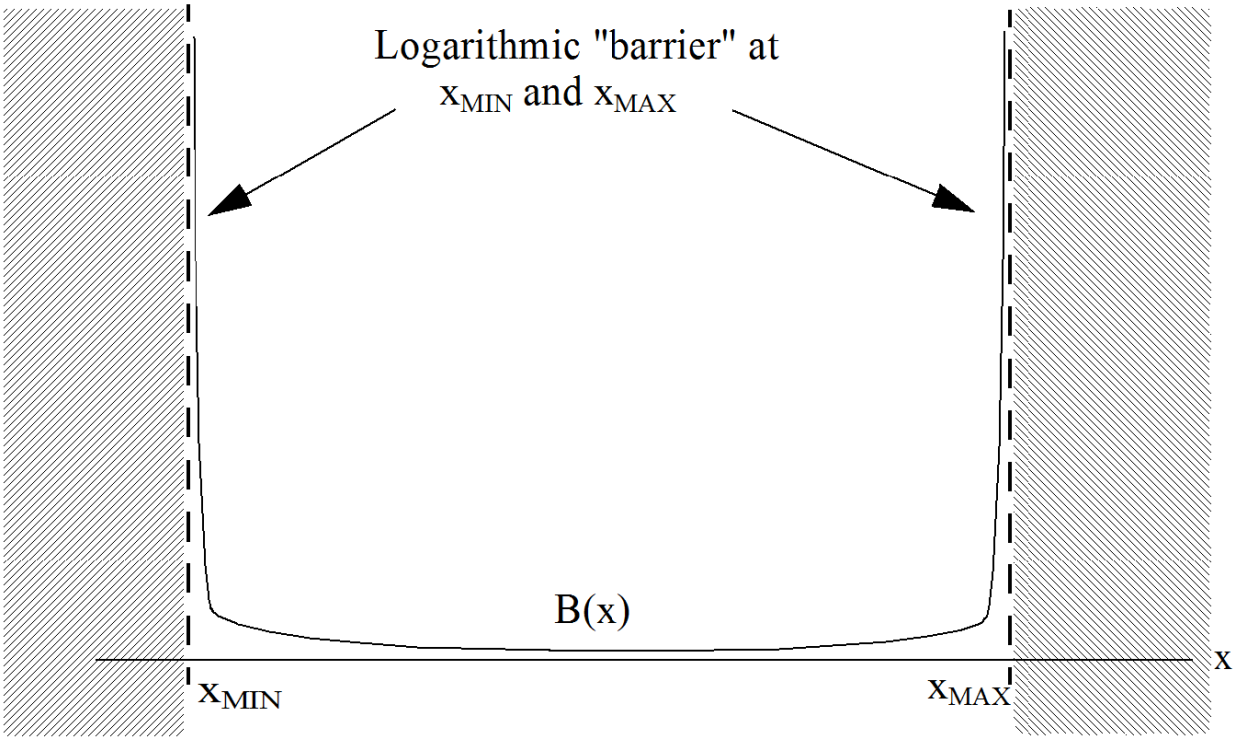

Figure 15.8. Logarithmic Barrier Function

As shown, the function is not defined for values of x outside of the specified range; thus barriers are formed in which all of x is corralled within a feasible region. The optimization problem itself is infeasible if any equality constraint (e.g. bus-power mismatch) requires that x violate this feasible region.

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PSS®E OPF employs a logarithmic barrier function of the form,

$$B(x) = -(10.^{\mu}) \sum_{i=1}^{N} \{\log(x_i - x_{MIN,i}) + \log(x_{MAX,i} - x_i)\} \quad (15.3)$$

The barrier functions aid the solution process by transforming inequality constraints into objective terms, eliminating an additional iterative process that would otherwise be required to prevent violations when solving a subproblem.

The addition of the barrier function transforms the constrained problem into a more easily handled unconstrained problem. The unconstrained minimum to the combined objective function always occurs within the feasible region of x, or interior to the limits imposed by x_{MAX} and x_{MIN}. For this reason, this technique is referred to as the Interior Point Method.

When included in the nonlinear problem formulation of Equation 15.2, the resulting augmented Lagrangian, as denoted by L´, is as follows.

$$L'(x, \lambda) = f(x) + B(x) + [\lambda]^t [h(x)] \quad (15.4)$$

and

$h(x) = 0$,

where:

B(x) = the barrier equations to satisfy the inequality constraints.

Whereas the original problem statement sought to find the constrained minimum of f(x), the new formulation ks to find the constrained minimum of the combined objective function, f(x) + B(x). Not only must the original objective function be decreased but the barrier objective contribution must also be decreased. This is accomplished by systematically reducing the barrier parameter #; , as contained in the equation of Equation 15.3.

Example

An illustration of the effect of augmenting the objective function with the barrier terms is shown in Figure 15.9, “Objective Function with Barrier Term”. Only the barrier term associated with the upper limit, x_{MAX} = 1.0, is illustrated for simplicity. The constrained optimal solution to the simple objective f(x), occurs at x_{MAX}. The constrained optimal solution to the combined objective, f(x) + B(x), occurs at x_{OPT}= 0.8. This value may be confirmed by setting the combined objective’s first derivative in x equal to zero and solving for x, the stationary point.

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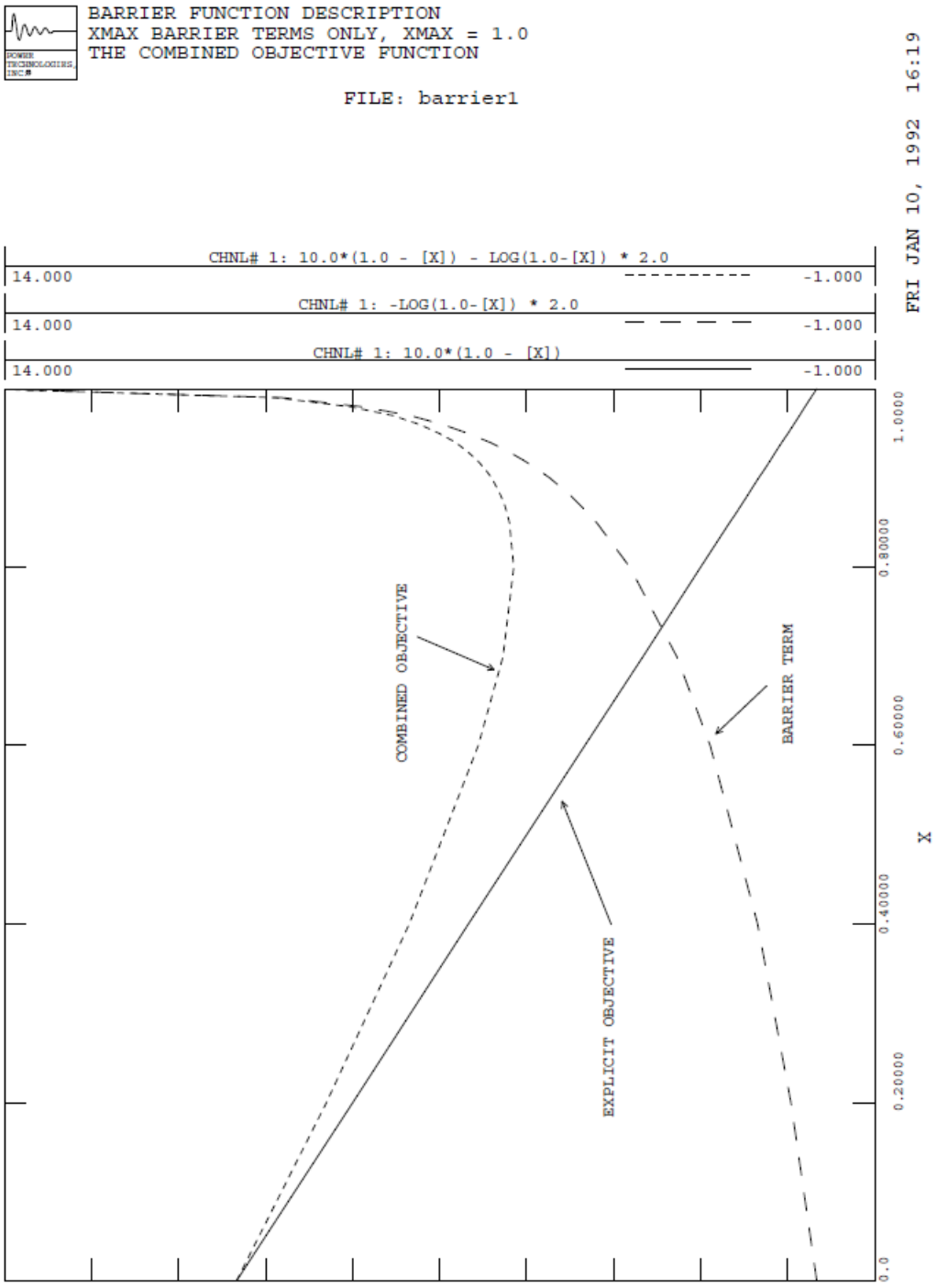

Figure 15.9. Objective Function with Barrier Term

Note that $f(x_{MAX}) < |f(x_{OPT}) + B(x_{OPT})|$, and $x_{OPT} < x_{MAX}$. The interior point solution technique will decrement the barrier objective contribution as it proceeds in order to reduce the objective discrepancy. Figure 15.10, "Effects of Reducing the Barrier Coefficient" illustrates the effects of diminishing the barrier.

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As the barrier contribution decreases, the combined objective minimum approaches the explicit objective minimum. In Figure 15.10, "Effects of Reducing the Barrier Coefficient", the minimum of the combined objective with a barrier coefficient of 2.0 is at $x_{OPT} = 0.8$, whereas a barrier coefficient of 1.0 yields an x_{OPT} minimum of 0.9. Because the combined objective optimum cannot converge exactly to the x_{MAX} limit, some discrepancy between it and the explicit objective optimum must remain. A final barrier coefficient value that is defined to be rather small should drive the discrepancy to a fairly insignificant value.

15.7.3. Formulating the Equation

Based upon the augmented Lagrangian ($L'(x, \lambda)$) formulated in Equation 15.4, the optimum solution to this nonlinear problem is sought by employing Newton's second ordered solution method. The standard Kuhn-Tucker conditions formulate the following equation.

$$\begin{bmatrix} \nabla_x \nabla_x L'(x, \lambda) & \nabla_x \nabla_\lambda L'(x, \lambda) \\ \nabla_\lambda \nabla_x L'(x, \lambda) & \nabla_\lambda \nabla_\lambda L'(x, \lambda) \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} -\nabla_x L'(x, \lambda) \\ -\nabla_\lambda L'(x, \lambda) \end{bmatrix} \quad (15.5)$$

The solution to this equation provides corrections to the dual variables, x and λ, and solves a quadratic approximation to the nonlinear system. These variable corrections are driven by the right-hand side (RHS) of the system of equations.

The RHS is of length N+M where N is the number of power system variables (x) and M is the number of equality constraint equations ($h(x) = 0$). The RHS matrix is composed of a lambda gradient term, $-\nabla_\lambda L'(x, \lambda)$, encompassing the equality constraint residuals, -h(x), as observed in Equation 15.4. This term encourages the value of x to move in a direction that reduces the mismatch. The x gradient term of the RHS, $-\nabla_x L'(x, \lambda)$, encourages x to move in a direction that minimizes the combined objective function.

The four terms of the square KT matrix on the left side of the equation are themselves matrices. Because the Lagrangian is linear in λ, the lower diagonal submatrix is uniquely zero. The off-diagonal sub-matricies are merely derivatives of equality constraint equations in terms of the power system variables, and are the Jacobian (MxN) and its transpose (NxM). The short-hand reference to the upper diagonal submatrix is the Hessian.

Newton's second ordered solution method approximates the combined objective function with a quadratic equation in x. The quadratic equation is constructed for some point x´ such that the tangent to both the quadratic and the nonlinear expressions are equivalent at x´. Figure 15.11, "Quadratic Approximation to Combined Objective" illustrates the quadratic approximation to the combined objective function for a single barrier coefficient, with a linear explicit objective, and a number of evaluated conditions x´. The values of x´ determined for each curve are shown in the table provided within the graph. As an example, the value of x for which the tangent to curve C and the tangent to the combined objective is identical occurs at x´ = 0.7. Additional analysis of this graph is provided by an example in Iterative Solution Strategy.

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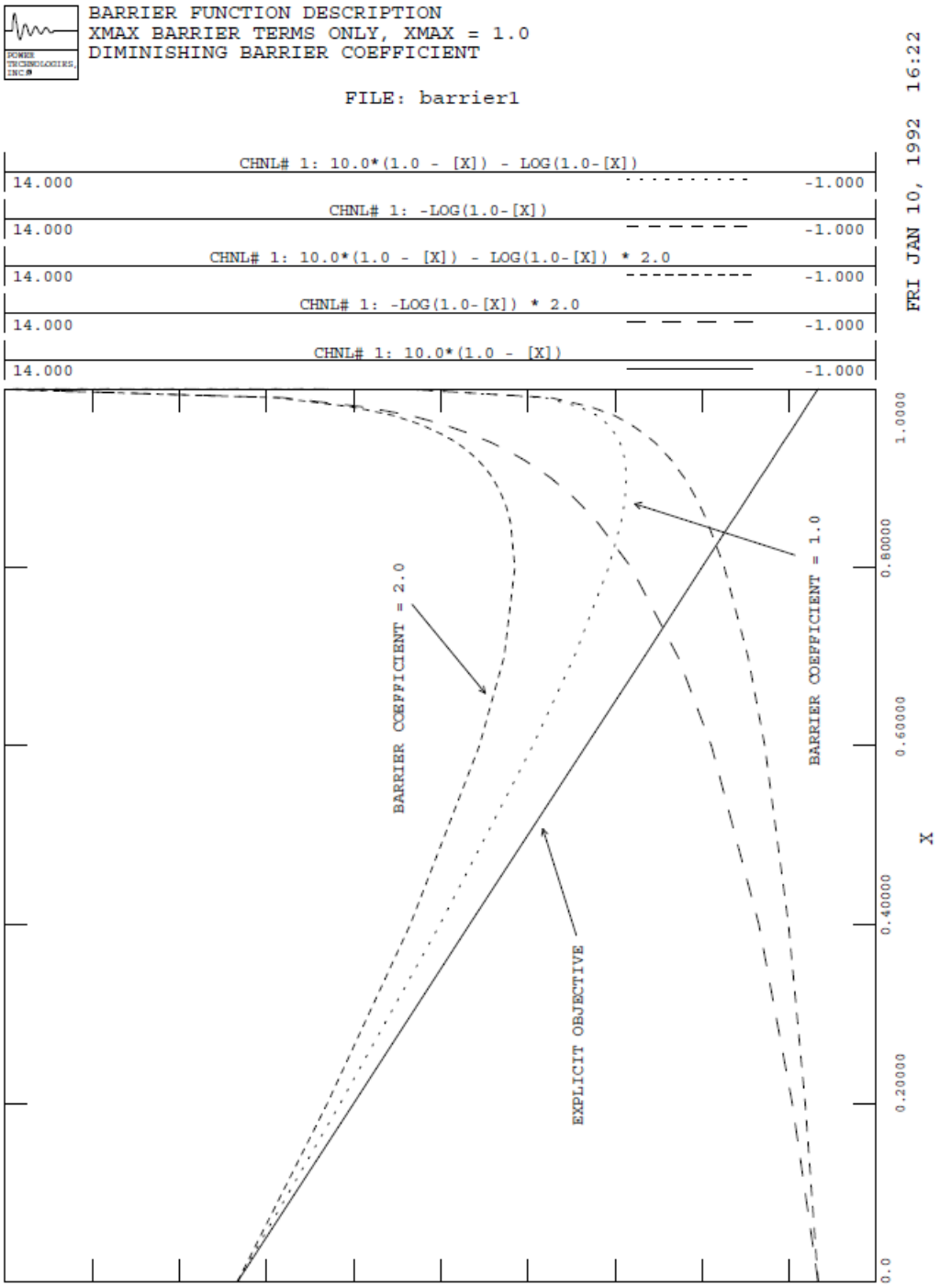

Figure 15.10. Effects of Reducing the Barrier Coefficient

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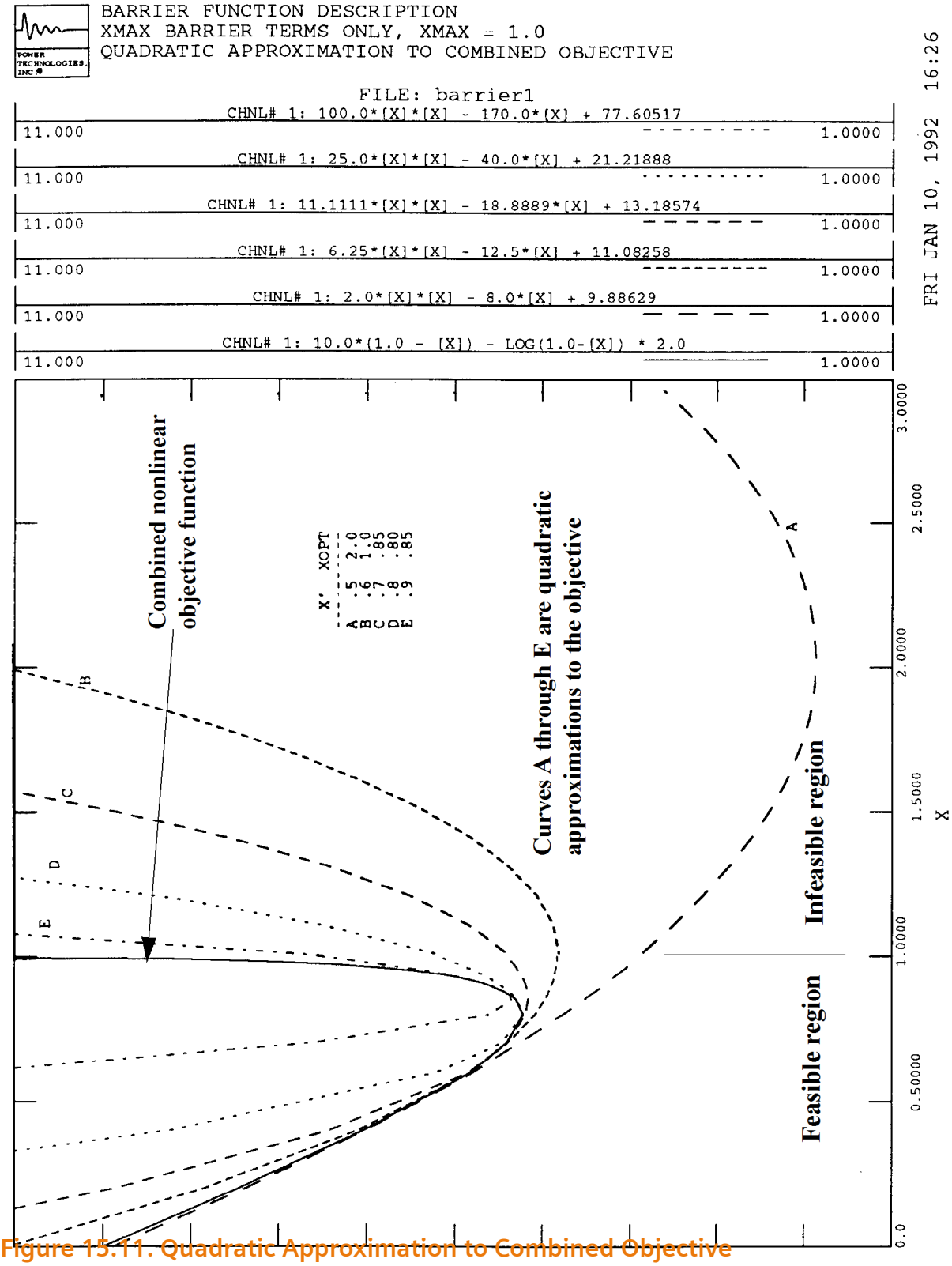

Figure 15.11. Quadratic Approximation to Combined Objective

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Iterative Solution Strategy

The solution to the equation formulated in Equation 15.5 provides corrections to the dual variables, x and λ , as driven by the right-hand side (RHS) of the system of equations. The solution algorithm is iterative. It constructs a KT problem formulation about some estimated x and λ , calculates corrections and updates the variables.The KT problem is formulated using initial estimate of x and λ , and iteratively solved to produce corrections in x and λ .

$x' = x_{old} + \alpha \Delta x$

$\lambda' = \lambda_{old} + \alpha \Delta x$

The computed correction, or step (Δ x), when added to x´, determines the new minimum of the quadratic equation. From this point a new quadratic equation is developed and a new step is computed. As x´ approaches the nonlinear optimum, the minimum of the quadratic equation converges to the minimum of the nonlinear combined objective. When the quadratic solution estimate is developed for an x´ that is far from the optimum, the computed step may force some elements of x to violate their limits. Because the combined objective is not defined for infeasible values of x, it becomes necessary to reduce, or decelerate the correction to x and take a smaller Δ x step.

Within the OPF iteration log, the step is reported in per unit. A step of 1.0 implies that 100% of the correction can be added to x' without violating any limits. Otherwise, the reported step is less than 1.0 and the new value indicates by how much the most limiting element of x reduces the correction.

Example

In Figure 15.11, "Quadratic Approximation to Combined Objective", the combined nonlinear objective function for one particular barrier coefficient (in this case 2.0) is shown by the solid, unlabeled line. Each of the other five lines are different quadratic approximations to the objective function curve. The values x´ displayed in the table provided within the graph indicate the value at which the tangent to the objective function and the tangent to the quadratic approximation is the same. The x_{OPT} values reflect the optimal minimum of x and in some situations the value of x may need to be corrected so that it achieves a feasible value as close to x_{OPT} as possible. For this particular example, all values greater than 1.0 are infeasible.

The quadratic curves in Figure 15.11, "Quadratic Approximation to Combined Objective" exhibit a number of situations that may be encountered by the program during the solution process. The following scenarios illustrate the necessary corrections in x and the resulting step size.

Scenario A	For the quadratic approximation exhibited by curve A, the value at which the tangent to the objective function and the tangent to curve A is identical, occurs at x' =.5. The optimal value of x for this approximation occurs at x_{OPT} = 2.0. The optimal correction is therefore (2.0 - 0.5), resulting in a Δ x of +1.5. Because an x_{OPT} value larger than 1.0 falls in the infeasible region, the correction must be truncated to Δ x< +.5 to ensure a feasible result. This results in a reported step of 0.5/1.5, or 0.33333. All of Δ x is scaled by the most limiting element.
Scenario B	For curve B, the condition where the tangents are identical occurs at x´=.6 with an optimal value of x

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	occurring at x_{OPT} = 1.0. In this situation, the optimal value of x lies right on the border of the feasible region and a full correction of Δ x = 1.0 - 0.6 = 0.4 can be made. Because 100% of the step can be made, a value of 1.0 is reported for the step.
Scenario C	Likewise to Scenario B, except that the optimal x is at x_{OPT} = 0.85. This again is present in the feasible region resulting in a full optimal correction of 0.15 and a recorded step of 1.0 (100%).
Scenario D	For this particular approximation the values at for the condition x' and the optimal minimum x_{OPT} both occur at 0.8. In this situation, there is no correction required and a step size of 1.0 is recorded.
Scenario E	This situation is similar to Scenario C except that the corrected step must be taken backwards, thereby resulting in a Δ x of -0.15.

In these examples all calculations are based on the actual values presented in each of the scenarios. Within the program, the corrections in x and the step size ratio also take into consideration a tolerance value to ensure feasibility. This implies that in situations such as the one encountered in Scenario B, a slightly smaller step would likely be taken to take into account the small safety margin around the point of infeasibility. The step size would therefore be a little bit less than 1.0.

Reduction of the Barrier Terms

The barrier terms contribute to the KT formulation and their impact is diminished as the solution progresses. It is not necessary to solve to the optimal condition every time the value of the barrier coefficient is modified. In the PSS®E OPF, the barrier coefficient is reduced by a factor of 5 at the start of the solution for every iteration which results in a favorable step larger than 0.5, This allows for the solver to take larger steps towards the optimal solution. After the solution has made some progress the barrier coefficient is reduced by a factor of 5 at every iteration. This allows the solver to progress more quickly towards the optimal solution. The PSS®E OPF starts with a default of 1.0 for the initial barrier coefficient value and progressively reduces it until the final value of the barrier co-efficient is reached, as defined by the power flow Convergence Tolerance value. The exponential decay of the barrier coefficient is illustrated in Figure 15.12, "Diminishing the Barrier Coefficient".

In some instances, particularly for large or difficult cases, it is suggested to use a larger positive initial value for the barrier parameter. Values that are too small may prevent the solver from converging to an optimal solution fast enough, or in some cases, may cause a failure to solve. Although there is no universal initial value that works for every problem, it has generally been found that a value on the order of 100 can improve the efficiency and robustness of the OPF, especially for the large-scale optimization problems. If the optimization problem you have presented to the PSS®E OPF does not solve with the initial default value of 1.0, then try values such as 10, 20, 100, or others. As always, there may be some problems that do better with a smaller initial barrier coefficient value, such as 0.1 or 0.01.

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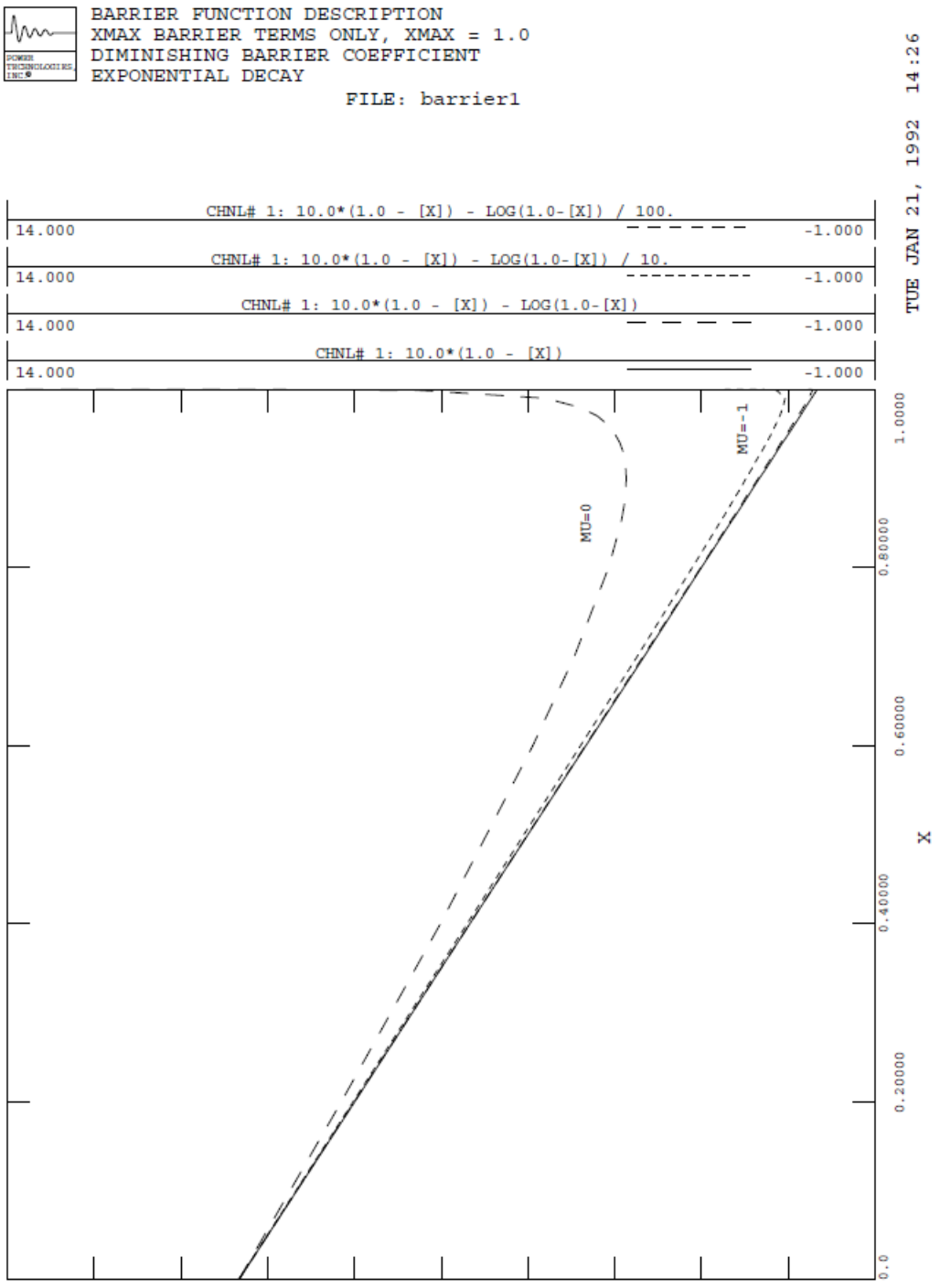

Figure 15.12. Diminishing the Barrier Coefficient

The Right-hand Side (RHS)

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The entries in the RHS of the KT problem are the mismatch terms. These values drive the variable corrections. When the RHS becomes small, the correction magnitude will also be small. Therefore, the mismatch magnitude is a good solution convergence indicator.

The mismatch criteria may be applied to the entire RHS or, by default, to only the lambda gradient terms (i.e. the bus-power mismatch equations). The latter is a single variable mismatch criteria that ensures the power flow equations are solved, but does not ensure convergence in λ. Generally, iterations beyond those required to meet the single variable criteria do not appreciably decrease the objective function. In some applications however, the objective sensitivity information may be a primary concern. For example, the value of λ may be used to obtain cost signals for transmission system access in wheeling power under third party agreements. In situations such as this, where accurate sensitivities are desired, the dual variable mismatch criteria should be employed (see Use Dual Variable Convergence Criteria).

When viewing the OPF iteration log, the single variable power flow mismatch value is reported under the column labeled *Mismatch*, whereas the dual variable mismatch is reported under the column labeled *Norm RHS*. The solution tolerance is obtained from the value specified for the Newton tolerance, TOLN, as defined within the standard power flow solution parameters.

Solution Convergence Criteria

The iterative solution process is considered converged when the following criteria are met:

- the final value of the barrier parameter is obtained,
- the final value of the clamp tolerance is obtained (this value is examined only if the clamp equation has been employed for nonoptimized generator controls),
- a unit step (Step = 1.0) has resulted in the final iteration, and
- the mismatch is below a user-controlled tolerance.

When all convergence criteria are satisfied, the solution process terminates with an `Optimal Solution Found` message. An abnormal termination of the OPF solution will result in an error message.

Terminated After XX Iterations

If the solution process fails to converge within the maximum iteration limit, the following error message appears at the bottom of the solution iteration log:

```
Error: Terminated after XX iterations--reached
iteration limit
```

This error indicates that the maximum iteration limit value (ITMXN), as displayed to the far right of the message, has been exceeded. Review the progress of the solution iteration log and examine the formatted report for any violations or inordinately large sensitivities to determine whether to proceed with the same problem statement at an increased iteration limit, or to reexamine and modify the original problem statement.

If it appears that the solution is making some progress, and not diverging, then it is likely that the maximum iteration limit (ITMXN) simply needs to be increased. This value can be altered (refer to *PSS®E GUI Users Guide*, OPF Parameters or Specifying Solution Parameters).

Blown Up after XX Iterations

If the solution is progressing poorly, then the iterative process will be terminated with the following error message:

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```
Error: Blown up after XX iterations.
```

This could occur for a number of reasons, of which the most common are:

- There are many constraints and not enough available controls.
- Constraints overlap and do not provide a feasible solution space.
- The solution started at a point which makes it difficult to find the optimal solution.

  In the first two instances, try opening up some of the constraint limits or setting the limit type to *Reporting Only* to try to get a sense of which constraints are most critical. It may be necessary to also introduce more controls.

  In the latter situation, changing the Initial Barrier Coefficient value may help to attain a solution. By default the initial value is 1.0, but for some large-scale problems a larger value improves the robustness of the solution. Try values such as 10, 20, 100 or 120. There may also be problems that do better with a small initial value (0.1 or 0.01) as well. Refer to the Reduction of the Barrier Terms and Section 15.10.3, "Tolerance Options" for more information.

  In general, identifying an infeasible problem is often heuristic and takes some experience before becoming proficient at recognizing which actions need to be taken. There are two places to look at for additional information: the iteration log and the OPF Report File.

  The formatted OPF output report illustrates the objective sensitivities of the constrained variables. By comparing the relative magnitudes of these sensitivities, the most penalizing constraints can be identified. Sensitivities are useful even for infeasible terminations in that they help identify the most effective places where additional control variables may be needed and/or limits may need to be relaxed.

  If, upon examination of the OPF report, nothing appears to be inherently wrong with the problem statement, the iteration log may provide additional insight into why the solution may have terminated as infeasible.

  An iterate of the solution process is considered to be *good* when the following criteria is met:

- The difference between the values of the nonlinear objective function (*NL Objective*) in the prior and present iterations is greater than, or equal to, zero. A value less than zero indicates that the objective function was not significantly reduced in the prior iteration.
- The ratio of largest mismatch between the present and prior iteration is less than 0.95. A value larger than, or equal to, 0.95 indicates that the mismatch was not significantly reduced in the prior iteration.

  When one, or both, or the above criteria are not met, the iteration is considered *bad* and the iteration number displayed in the solution log is marked with an x. Under certain conditions, the solution process will terminate as infeasible:

- If the value of the barrier is fairly large (greater than 0.011) and the number of successive bad iterations equals the *Bad iteration coarse limit* as specified (refer to *PSS®E GUI Users Guide*, OPF Parameters) OR
- the value of the barrier is small (less than, or equal to, 0.011) and the number of successive bad iterations equals the *Bad iteration fine limit*, as specified (refer to *PSS®E GUI Users Guide*, OPF Parameters) OR
- the step size (as displayed under the *Step* column in the solution log) is below the minimum barrier step size tolerance.

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If the solution is making progress, albeit slow, then it is possible that an increase in the bad iteration limits will give the solution more time to proceed and an optimal solution will be found (refer to Bad Iteration Coarse Limit and Bad Iteration Fine Limit).

15.8. Data Input and Storage

The constraint, control and solution options data associated with a given optimal power flow problem statement can be introduced, stored and retrieved through the use of data files and interactive spreadsheets. The principal files and tools used by the optimal power flow activities for data manipulation and interrogation are as follows.

PSS® E Saved Case File

A binary file that efficiently stores a complete image of everything within the PSS®E working case, including optimal power flow data and solution settings.

Optimal Power Flow Raw Data File

An ASCII text file containing the optimal power flow control and constraint data records.

Optimal Power Flow Options File, PSSOPF.OPT

A binary file containing the optimal power solution parameter settings. This file may be used to override default options established at the time the program is initiated.

Optimal Power Flow Spreadsheet View and Data Tables

A set of interactive spreadsheets and tables for viewing and editing optimal power flow data.

The following sections discuss, in detail, how each of the above files and functions are utilized by the optimal power flow in the preservation, modification and restoration of optimal power flow data.

15.8.1. PSS®E Saved Case

All data pertaining to the optimal power flow problem statement, including controls, constraints, objectives and solution parameters are preserved in the PSS®E binary Saved Case File. Each saved case represents a complete power flow study, including all power flow data and optionally, any optimal power flow data and solution parameter settings. After all of the information for a study has been brought into the PSS®E working case and preserved within a PSS®E Saved Case, it becomes almost unnecessary to refer to any other files.

Existing saved cases may be brought into the working case at any time. When a PSS®E Saved Case is brought into PSS®E, the complete power flow description, including that of the optimal power flow problem, will automatically be loaded into the working case. If a saved case is read in that does *NOT* contain any OPF data, then the only OPF related data that will be initialized within the working case will be the default settings of the OPF solution parameters. These values are either initialized to default values, or to the values stored within the PSSOPF.OPT file, if such a file exists. Refer to Section 15.8.3, "Optimal Power Flow Options File: PSSOPF.OPT" for details on the use of the PSSOPF.OPT file.

Optimal power flow data can easily be added to the working case by reading in data through an OPF Raw Data File or by entering data via the OPF spreadsheets and data tables. It is not possible to append data by reading in a second saved case file.

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15.8.2. Optimal Power Flow Raw Data File

ROPF

The OPF Raw Data File populates the working case with new or revised optimal power flow data and stores only optimal power flow data for use in future sessions. It specifically contains data records for the controls and constraints of an OPF problem statement. It does not contain any information related to stated objectives, tolerances or other solution parameters.

The file itself is in ASCII format and contains a series of data records group together by OPF data category. The data categories must be entered in a particular order, as defined by Figure 15.13, "Optimal Power Flow Raw Data File Structure". A single data record containing only a zero indicates that all entries for a particular data category have been entered. If no data records are to be entered for a particular category, then only a single zero needs to be placed in the data file. If a record with a single q is encountered during processing, then the file is closed and the program does not attempt to read any further data records.

Additional Information
PSS®E GUI Users Guide, Reading Optimal Power Flow Data *PSS®E* Command Line Interface (CLI) *Users Guide*, Reading Optimal Power Flow Data

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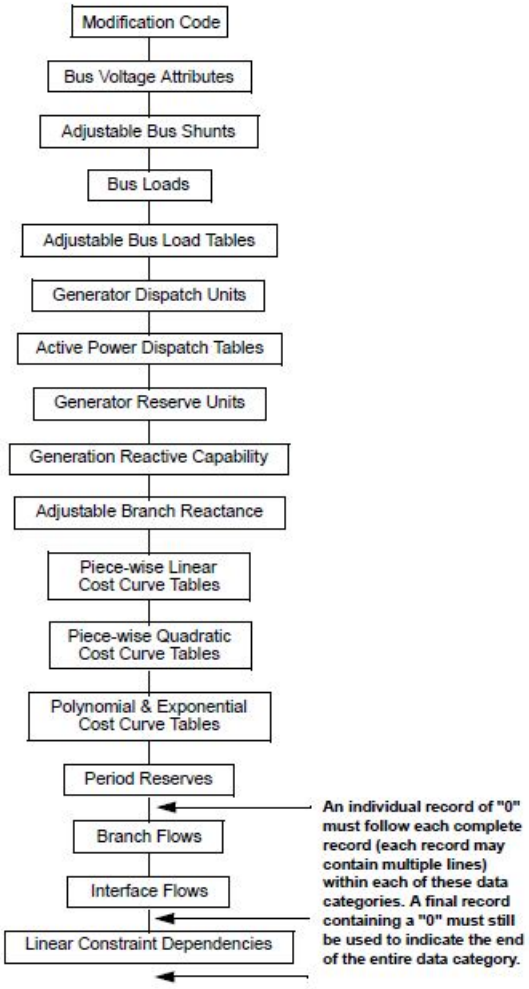

Figure 15.13. Optimal Power Flow Raw Data File Structure

15.8.3. Optimal Power Flow Options File: PSSOPF.OPT

The Optimal Power Flow Options File provides a means for saving and restoring optimal power flow solution parameter settings between program sessions. When PSS®E is installed, a set of default run-time options is established. Upon initiation of the PSS®E program, the default optimal power flow options may be overridden by the values contained within the PSSOPF.OPT file. Likewise, by bringing in a PSS®E Saved Case file, the values of the option settings stored within it may override all previous settings.

In addition to the options described in Section 3.3.3, "Program Run-Time Option Settings", there are a number of option settings specific to the optimal power flow program module. Details of each option can be found in Section 15.10, "Optimal Power Flow Solution Options". The OPF Parameters specified may be saved to the PSSOPF.OPT options file. The resulting PSSOPF.OPT file is located in the current working directory. The only information that is not currently stored is that of a subsystem.

Each time PSS®E is initiated, it will look for the existence of the PSSOPF.OPT file using a particular directory search sequence, as outlined below.

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1. *The current working directory*. If the file is found, the settings within the file will automatically be applied and further searches in the alternate locations defined in steps 2 and 3 will not be pursued.

2. *The user's home directory*. If the file is located here, the settings within the file will automatically be applied and further searches in the alternate location defined in step 3 will not be pursued.

3. *The program default directory*. If the file is located in the PSS®E subdirectory PSSPRM, then the settings within the file will automatically be applied. If the file is not found, then no further searches are ensued and the original default values remain unchanged.

15.8.4. Optimal Power Flow Spreadsheet View

General use of the Spreadsheet View is covered in the *PSS®E GUI Users Guide,* Chapter 2.

The optimal power flow spreadsheet view allows viewing and editing all bus oriented optimal power flow data records. If optimal power flow data is not present in the working case prior to invoking the OPF data spreadsheet, then the option to initialize the OPF data is provided.

After being initiated, the OPF spreadsheet view is populated with the default and defined data values for the optimal power flow data present in the working case. Clicking on the tabs at the bottom of the spreadsheet allow for navigation between the various bus oriented OPF data records. By hovering the mouse cursor over a particular tab, the full name of the tab will pop up.

For information on adding records using the OPF spreadsheet and modifying default values refer to *PSS®E GUI Users Guide,* Data Initialization for OPF.

15.8.5. Optimal Power Flow Data Tables

OPF Data Tables allow access to view and edit all table oriented optimal power flow data records. This includes cost curve tables, period reserve constraints, interface flow constraints, and linear dependency constraint equations.

15.9. Constraint and Control Data

For more information please see *Power Flow Raw Data File Contents*.

15.10. Optimal Power Flow Solution Options

Optimal power flow solution parameters may be applied prior to initiation of the OPF solution. These solution parameters are used to further define the optimal power flow problem statement and may direct the program to perform additional functions after the solution process has completed.

The five solution parameter groups are as follows:

- *Objectives:* A list of all available solution objectives. Cost coefficient values are also listed for those objectives that may have an associated cost applied.

- *General:* A list of general optimal power flow solution options and additional constraint equation directives that may be activated during the solution process.

- *Tolerances:* A list of solution tolerance values and advanced level solution parameters that may be altered to influence the progress of the solution algorithm.

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- *Control:* A list of options pertaining to the treatment of nonoptimized controls within the optimal power flow problem statement.
- *Reporting:* A list of output reporting options. These must be defined prior to solution initiation.

15.10.1. Objectives

More than one objective may be applied simultaneously to formulate a composite objective function. The exception is the *Minimize fuel cost* and *Minimize active power loss* objectives, of which neither can solve for their respective goals if the other objective is active. This is covered in more detail in Minimize Fuel Cost and Minimize Active Power Loss.

Table 15.2. Optimal Power Flow (OPF) Objectives

OPF Objective	Description
Minimize Fuel Cost	Directs the OPF to find the minimal amount of active power required to serve losses and loads. The system swing bus is automatically considered into the problem statement although its active power generation is fixed at the initial value and not considered an available control. In addition to specifying the *Minimize fuel cost* objective, three key data elements are essential to the fuel cost model: • *Generators for dispatch*: Each eligible participating generator in the active power dispatch model must have a Generator Dispatch record defined. The information within the Generator Dispatch record includes the percentage of total generation capacity available for dispatch. • *Active power dispatch constraints*: Each Generator Dispatch record references a single Active Power Dispatch Table that defines the minimum and maximum active power available for dispatch, and references a Cost Curve Table. • *Generator fuel cost curves*: Each Active Power Dispatch Table references one of three available cost models: piece-wise linear, piece-wise quadratic and polynomial.
When the *Minimize fuel cost* objective is enabled, the *Minimize active power loss* objective is automatically disabled. The combination of these two objectives does not formulate a feasible problem statement. The minimization of active power losses is achieved by holding active power generation fixed and varying the available reactive power. This is in direct conflict with the fuel cost minimization objective which relies on the ability to vary active power generation to make up for losses or loads.	
Minimize Active Power Slack Generation	Historically employed for minimization of active power losses. In contrast to the separate minimize

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OPF Objective	Description
	active power loss objective, the active power slack objective has the disadvantage of not being able to distinguish between optimized and non-optimized subsystems. This objective function adds the change in the active power injection of the retained slack buses (type 3) to the composite objective. When the active power dispatch is fixed at every generator except the slack bus, then the minimization of active power losses is mathematically equivalent to the minimization of slack bus active power. There are no additional data models required to complement this objective function.
Minimize Reactive Power Slack Generation	Adds the reactive power generation of all in-service and regulating generators to the composite objective. There are no additional data models required to complement this objective function.
Minimize Active Power Loss	Attempts to reduce active power losses in the system. It is frequently employed, either individually or in combination with other objectives. A small reduction in losses can often produce substantial cost savings. A global loss objective coefficient may be assigned by entering a cost value directly into the input field labeled *Active power loss cost coefficient ($ / pu MW)*. A coefficient value of 1.0 has no effect on the active power loss objective. If multiple objectives are employed, the relative costs of each objective to one another determines which objective is treated with a higher priority. There are no additional data models required to complement this objective function.
The *Minimize active power loss* objective and *Minimize fuel cost* objective cannot be simultaneously employed due to the competing goals. The minimization of active power loss is achieved by holding active power generation fixed and varying the available reactive power which is in direct conflict with the minimization of fuel cost which relies on the ability to vary active power generation to make up for losses and serve loads.	
Minimize Reactive Power Loss	Attempts to minimize vars or other voltage related objectives to fine tune the results and help to eliminate circulating reactive flows and other undesirable phenomena which may result in the transmission network.

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OPF Objective	Description
	A global loss objective coefficient may be assigned by entering a value directly into the input field labeled *Reactive power loss cost coefficient ($ / pu MVAR)*. A coefficient value of 1.0 has no effect on the reactive power loss objective. If multiple objectives are employed, the relative costs of each objective to one another determines which objective will be treated with the higher priority. There are no additional data models required to complement this objective function.
Minimize Adjustable Branch Reactances	Determines the minimal amount of series compensation required to relieve flow constraints. This objective is often used in conjunction with branch flow (see section Branch Flow Constraint Data) and interface flow constraints (Section Interface Flow Constraint Data). Candidate branches for series compensation must be defined prior to initiation of the OPF solution (Section Adjustable Bus Shunt Data).
Minimize Adjustable Bus Shunts	Determines the minimal amount of shunt reactors and capacitors that need be placed at candidate buses. It may be employed in voltage stability studies or to assist in the minimization of a composite objective function. Candidate buses for var adjustment must be defined through Adjustable Bus Shunt records prior to the OPF solution (Section Adjustable Bus Shunt Data). To encourage the deployment of vars, particularly for voltage stability studies, it may be desirable to fix transformer taps, phase shifters and switched shunts (see Fix Transformer Tap Ratios). During the OPF solution, var values may optionally be rounded to the nearest discrete block step by choosing to *Round switched shunt vars* (see Round Switched Shunt Vars). After the solution process has completed, the new shunt additions will automatically be reflected in the power flow bus data of the working case as bus shunts.
Minimize Adjustable Bus Loads	Determines the minimal amount of load adjustment to make at candidate buses. It is most commonly used to study the effects of load shedding to ease voltage constraint violations due to a severe disturbance on the system. It is possible however to for-

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OPF Objective	Description
	mulate a problem statement in which the minimization of load additions, as opposed to load shedding, is the objective. Candidate buses for load adjustment must be defined through OPF Adjustable Bus Load records (Section Bus Load Data) and Adjustable Bus Load Table records (Section Adjustable Bus Load Table Data) prior to the solution. After the OPF solution process, new load adjustment factors will be reflected in the value Psi, and displayed with the load values in PSS®E output reports (for example, refer to Section 5.37, "Displaying Power Flow Data").
Minimize Interface Flows	Used in inter-area studies to encourage or curtail flows across a particular interface. Interface flow limits and participating branches must be defined through the Interface Flow records described in Section Interface Flow Constraint Data prior to the solution. In conjunction, the Constrain Interface Flows option, must be enabled for the interface flow objective and the interface flow constraints to be recognized during the OPF solution process. A global interface flow cost coefficient may be assigned by entering a value directly into the input field labeled *Interface flow cost coefficient ($ / pu Flow)*. A negative value will attempt to *maximize* interface flows. A coefficient value of 1.0 has no effect on the interface flow objective. If multiple objectives are being simultaneously employed, the relative cost of the objectives to one another determines which objectives are be treated with a higher priority during the solution process.
Minimize Reactive Generation Reserve	Will influence the reactive generation reserves. It introduces to the combined objective function a component term proportional to the excursion of reactive power generation from the maximum reactive power generation limit for each in-service generator that is a member of the optimized subsystem. A scalar objective coefficient may be assigned by entering a cost value directly into the input field labeled *Reactive generation reserve cost ($ / MVAR)*. When the objective is employed with a negative cost coefficient, the reactive generation reserves

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OPF Objective	Description
	will tend to be increased. When a positive cost coefficient is employed, the reactive generation reserves will tend to be decreased.

15.10.2. General Options

Table 15.3. Optimal Power Flow (OPF) Parameters - General Options

Option	Description
Regulate Area Interchange	Introduces area interchange constraints into the OPF problem statement for each area interchange record defined in the standard power flow model. By default, area interchange constraints are not introduced. For a complete description of the model, refer to Section 15.6.7, "Regulated Area Interchange".
Constrain Interface Flows	Introduces defined interface flow constraints (Section Interface Flow Constraint Data) and the *Minimize interface flows* objective, if enabled, into the OPF problem statement. If the *Constrain interface flows* option is not enabled, the *Minimize interface flows* objective and any interface flow constraints will be ignored and not used.
Use Automatic Scaling	Automatically scales the objective coefficients and derivatives toward a value better suited to the solution calculation. The variables are scaled to a value that is numerically closer to unity and then rescaled back after the solution process has completed. In some situations this may improve convergence, without significantly altering the solution.
Use Dual Variable Convergence Criteria	Directs the solution process to continue until the magnitudes of the entire KT problem right-hand side (both x and λ) fall below a specified tolerance. When the option is disabled, only the bus power mismatch equations (x) are examined. The use of the dual variable convergence criteria is most useful when accurate sensitivities are desired. Refer to Section 15.7.3, "Formulating the Equation" and more specifically The Right-hand Side (RHS) for further information. The convergence tolerance is obtained directly from the PSS®E Newton Solution tolerance, TOLN, and may be modified (refer to *PSS®E GUI Users Guide*, Specifying Solution Parameters or OPF Parameters).
Fix Transformer Tap Ratios	Fixes all transformer tap ratios to their current values, thus preventing them from being employed as

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Option	Description
	controls during the OPF solution process. It is often used to encourage the deployment of other controls, such as var adjustment, during the OPF solution.
Fix Transformer Phase Shift Angles	Fixes all transformer phase shift angles at their current values, thus preventing them from being active during the optimal power flow solution process. This forces the possible employment of alternate controls in order to meet the desired objective.
Fix Switched Shunts	Fixes all switched shunts at their current values, preventing them from being altered during the optimal power flow solution process. This action may allow alternate controls to be deployed during the solution.
Round Transformer Tap Ratios	Directs the solution process to round transformer tap ratios to their nearest discrete tap setting after an initial solution is obtained. It then attempts to pursue a feasible solution with the tap settings fixed. Rounding to the nearest tap setting attempts to mimic the discrete nature of transformer taps. During the OPF solution, transformer taps are treated as continuous controls. In some situations a final solution may not result after rounding to a discrete tap position, although this is unlikely. When the rounding takes place, upper and lower bus voltage limits may need to be slightly relaxed to attain a solution. If this should happen, an integer value will appear to the far right-hand side of the OPF solution iteration log under an unlabeled column. This value indicates the number of variables for which limits needed to be relaxed as a result of rounding.
Round Switched Shunt Vars	Directs the OPF solution process to round switched shunt values to their nearest block setting after an initial solution is obtained and then attempts to continue to a feasible solution with the switched shunt settings fixed. This option only applies to switched shunts within the general power flow model. It does not affect any OPF adjustable bus shunts. Rounding to the nearest var block attempts to mimic the discrete nature of switched shunts. During the main OPF solution switched shunt var adjustments

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Option	Description
	are treated as continuous functions. In some situations the rounding may lead to serious voltage infeasibilities, particularly if the var block size is rather large. If this should occur it may be prudent to manually round var settings instead. When the rounding takes place, upper and lower bus voltage limits may need to be relaxed to attain a solution. If this should happen, an integer value will appear to the far right-hand side of the OPF solution iteration log under an unlabeled column. This value indicates the total number of variables for which limits needed to be relaxed as a result of rounding.
Automatically Adjust Bus Voltages for Feasibility	Temporarily adjusts minimum or maximum bus voltage limits to ensure that the current voltage is within the specified range. For example, if the lower and upper voltage limits at a particular bus are set at 0.95 and 1.02 pu respectively, and the initial voltage in the power flow bus data is 0.94, then the voltage bounds at the bus would temporarily be set to 0.94 and 1.02. Only the infeasible bound is modified; the actual voltage limits as defined in the corresponding OPF Bus Voltage Attribute data record are not affected. To permanently change the constraint limits for feasibility, use the *Adjust limits* option in the Bus Voltage Attribute Spreadsheet. This option may be selectively applied to individual OPF bus voltage constraint records or applied to all buses within a selected subsystem.
Impose Emergency Bus Voltage Limits	Directs the OPF solution process to employ the maximum and minimum *emergency* bus voltage limits defined in the OPF Bus Attribute data records instead of the normal bus voltage limits. Refer to Bus Voltage Attribute Data Record for further information.
Impose Emergency Branch Flow Limits	Directs the OPF solution process to employ the maximum and minimum *emergency* branch flow limits defined in the Branch Flow Constraint Data Record instead of the normal branch flow limits.

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15.10.3. Tolerance Options

Table 15.4. Optimal Power Flow (OPF) Parameters - Tolerance Options

Option	Description
Quadratic Penalty Coefficient	Indicates, in cost units/pu, the severity for which a variable should be penalized for an excursion away from its initial value. This penalty is only employed as part of the quadratic penalty equation applied to transformers and switched shunt controls residing in optimized subsystems. Nonoptimized transformers and switched shunts are strictly fixed at their initial working case values. A separate and unique penalty coefficient may be applied to nonoptimized generators (Section 15.10.4, "Control Options"). A full explanation on the treatment of all power flow controls and the use of the quadratic penalty coefficients is presented in Section 15.5, "Modeling Power Flow Controls". A hard or soft (linear or quadratic) limit may also be applied to specific constraints on bus voltages, interface flows or branch flows. When in effect, these treatments will override the general quadratic penalty treatments for optimized transformers or switched shunt controls. By default, the quadratic penalty coefficient is zero for transformer and switched shunt controls. The quadratic penalty will not be applied unless the quadratic penalty coefficient is assigned a value greater than zero.
Initial Barrier Coefficient	The initial value of the barrier function coefficient. At each of the first few iterations of the solution algorithm, the barrier coefficient value is reduced by a factor of 5 provided the solution algorithm takes a step of at least 0.5. This allows the solver to take larger steps towards the optimal solution. The progress of the minimization of the barrier function can be gauged by examining the values of #; (mu) printed in the OPF solution iteration log. After some solution progress has been made, the barrier coefficient is reduced by a factor of 5 at each iteration, regardless of whether the step-size is greater than 0.5 or not. This allows the solver to make faster progress towards the optimal solution. By default, the initial barrier coefficient is assigned a value of 1.0, which corresponds to a value of 0 for (1.0 = 100). This may be too small for some opti-

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Option	Description
	mization problems and may prevent the solver from converging to an optimal solution fast enough, or to fail. Initial values on the order of 10 or 100 can improve the efficiency and robustness of the solution for some optimization studies..
Final Barrier Coefficient	Final coefficient value that the barrier function is attempting to attain. At each iteration of the solution algorithm, the value of the barrier coefficient is systematically reduced. As long as no infeasible conditions occur, the process will continue until all of the convergence criteria are met. One of the criteria (as outlined in Solution Convergence Criteria) is that the final value of the barrier parameter be less than, or equal to, the final barrier coefficient. By default, the final barrier coefficient is assigned a value of 0.0001, which corresponds to a value of -4 for #; (mu). As the iteration log progresses, you should see the value of #; steadily descend toward this final value.
Minimum Barrier Step Length Tolerance	At each iteration of the nonlinear barrier function, the step length taken by the barrier function algorithm is compared against this value. If the required step length is less than the minimum barrier step length tolerance value, then the solution terminates with an infeasible condition. The default value for the minimum barrier step length tolerance is 0.00001. It is unlikely that you will encounter a situation where you need to modify this value.
Bad Iteration Coarse Limit	Specifies the number of bad iterations allowed when the barrier coefficient is greater than 0.011 (mu greater than -2) before the solution progress is terminated as being infeasible. At each iteration of the solution process several criteria are evaluated to assess whether the solution should continue or be terminated due to infeasibility. In general, the solution will terminate on an infeasible condition if at least one of the following conditions are encountered. • the number of successive bad iterations hits a predefined limit, or

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Option	Description
	• the solution process is forced to take an algorithmic step that is smaller than a predefined minimum tolerance. As used in the context above, a bad iteration is defined as one in which: • the difference between the nonlinear objective function in the present and prior iterations is less than zero, or • the ratio of the largest mismatch value between the current and prior iterations is greater than, or equal to, 0.95. By default, the coarse limit on the number of bad iterations is set at 3. If the solution process appears to be making very small progress, it is possible for this limit to be exceeded. In situations where the mismatch value is making slow, but steady, progress, and the reduction of the objective is also making some progress, it is reasonable to try to increase the bad iteration limit.
Bad Iteration Fine Limit	Specifies the number of bad iterations allowed when the barrier coefficient is less than, or equal to, 0.011 (mu less than or equal to -2) before the solution progress is terminated as being infeasible. By default, the fine limit on the number of bad iterations is set to 10. If the solution process appears to be making very small progress it is possible for this limit to be exceeded prematurely. In situations where the mismatch value is making slow, but steady, progress, and the reduction of the objective is also making some progress, it is reasonable to try to increase the iteration limit.
Maximum Iteration Limit	Indicates the maximum number of allowable iterations before the OPF solution process will end. This value is identical to the Iteration limit (ITMXN) value used for the Newton power flow solution activities. It may also be modified (refer to *PSS®E GUI Users Guide*, Specifying Solution Parameters).
Convergence Tolerance	Defines the largest allowable mismatch tolerance in MW and Mvar. For the OPF solution process to terminate successfully, the largest mismatch must be at, or below, the convergence tolerance specified. This value is identical to the largest mismatch in MW and Mvar (TOLN) value used in the main power flow Newton solutions. It may also be modified (refer to

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Option	Description
	PSS®E GUI Users Guide, Specifying Solution Parameters).
Advanced Algorithmic Decision Tolerances	The following options affect the decision process of the solution algorithm and are strictly provided for highly unique situations. There are no particular instances which would warrant modifications to any of these values. Short descriptions of each option are offered below.
	Apply Step Size Limit indicates to the solution algorithm that an alternate lambda mismatch step size is to be taken. In conjunction with this option, the Step Length for Barrier Function specified determines the step size to be taken. By default, the algorithm automatically determines the appropriate step size to be taken along the search direction and a full step is always used for the mismatch.
	LaGrange Multiplier Blow-up Tolerance After the first iteration of the solution process, a check is performed to ensure that the values of the right hand side of the equation are not diverging. If the largest lambda value exceeds the *Lagrange multiplier blow-up tolerance*, then an asterisk (*) is placed to the right of the *Normal RHS* value in the iteration log output report. By default the Lagrange multiplier blow-up tolerance is set at 100000.0.
	Interior Shift Parameter indicates by how much a variable should be moved more interior of the upper and lower limits. By default, the interior shift value is set at 0.1.
	Step Length for Barrier Function By default, this value is set to 0.99. This implies that 99% of the largest possible step is taken. If an exact Newton step can be taken without hitting a bound, then the step size will not be proportionately reduced.
	Minimum Tap Ratio Step allows the minimum limit of all fixed transformer taps to be opened up. Any adjustable transformers with a valid tap range are not affected.

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Option	Description
	The value specified for the *Minimum tap ratio step* establishes a minimum boundary on the tap setting. For example, if the step value is set at 0.02, then each fixed tap will have a minimum value of (t-0.02), where t is the per unit value of the fixed tap. By default, the tap ratios of all fixed transformers are not altered. This option is only provided for highly unique situations. Often it is much better to explicitly define transformer tap limits if tap settings are to be altered.
	Maximum Tap Ratio Step allows the maximum limit of all fixed transformer taps to be opened up. Any adjustable transformers with a valid tap range are not affected. The value specified for the *Maximum tap ratio step* establishes a maximum boundary on the tap setting. For example, if the step value is set at 0.04, then each fixed tap will have a maximum value of (t+0.04), where t is the per unit value of the fixed tap. By default, the tap ratios of all fixed transformers are not altered. This option is only provided for highly unique situations. Often it is much better to explicitly define transformer tap limits if tap settings are to be altered.

15.10.4. Control Options

Table 15.5. Optimal Power Flow (OPF) Parameters - Control Options

Option	Description
Treat all Generators as Non-optimized	Forces all generators in the network to be treated as nonoptimized. This overrides whether the generator is in an optimized subsystem or not.
Clamp Non-optimized Generators	Applies the clamp equation logic to all nonoptimized generators. This overrides the quadratic penalty treatment, but does not supersede bus voltage constraints that may be applied at generator buses. Values for *Final clamp tolerance*, *Initial clamp tolerance* and *Clamp slack variable decay factor* may be entered directly into the fields provided. As described in Clamp Treatment, the OPF solution initializes the problem with the relatively large *Ini-*

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Option	Description
	tial clamp tolerance, and then systematically and gradually reduces this tolerance by the *Clamp slack variable decay factor* until a value less than or equal to the *Final clamp tolerance* is attained. Reductions in both the clamp equation slack variable and the clamp tolerance are taken when a solution step size of at least 0.5 is achieved. For example, with a default initial clamp tolerance of 1.0 and a decay factor of 0.1, the clamp tolerance will gradually be reduced by 10 (multiplied by 0.1) at each healthy step of the solution process, until a final tolerance value of 0.0001 is attained. The *Scale clamp constraint by reactive generator limits* option scales the clamp equation by the maximum of either 1.0 or the difference between the upper and lower generator reactive limits. This is done internal to the solution to potentially improve mathematical calculations and should be transparent.
Use Generator Scheduled Voltage	Directs the solution to hold voltage to the generator scheduled voltage instead of the current voltage.
Penalty for Fixed Voltage	The Quadratic penalty treatment of nonoptimized generators is the default if the *Clamp non-optimized generators* option is not enabled, and bus voltage constraints on generator buses are not applied. Quadratic terms are introduced into the objective function to penalize the excursion of regulated voltage magnitudes from their offsets. The value specified in the *Penalty for fixed voltage* field indicates the extent to which excursions are penalized.

15.10.5. Reporting Options

Table 15.6. Optimal Power Flow (OPF) Parameters - Reporting Options

Option	Description
Produce an Optimization Log File	Preserves all information on the OPF solution progress to a text file. The contents of the Optimization Log File is presented in Section 15.11.3, "OPF Optimization Log".
Optimization Log File	By default, the Optimization log filename will be given the name DETAIL if an alternate name is not specified. The filename extension is specific to the computer system PSS®E is being run on.

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Option	Description
	If the option to *Produce an optimization log file* is enabled, then the name of the log file will be preserved between subsequent OPF solution runs within the same PSS®E session. The specified filename will not however be preserved within an external data storage file such as a PSS®E Saved Case or the Raw Data File. Each time a saved case is brought into PSS®E for which the *Produce an optimization log file* option is enabled, the log filename will always revert to the default name of DETAIL.
Print a Detailed Listing	appends the optimization log file with detailed information on each variable and a complete list of the Lagrange multipliers for each constraint equation. This is the only place for which this information can be obtained; it is not printed as part of the normal solution process. By default, a detailed listing will not be produced as part of the specified optimization log file.
OPF Output Report Information	The OPF solution report automatically generated after the OPF solution includes information on the control variables affected by the solution process. The *OPF output report information* allows a subsystem to be defined so that the records contained within the report will be limited to just the equipment residing within the defined subsystem. This includes bounded and infeasible reactive power generation, regulated buses, transformer tap ratios, transformer phase shift angles and bounded switched shunts. By default, information for all equipment is presented in the output report. A full description of the contents of the OPF Report File can be found in Section 15.14, "Optimal Power Flow Solution Report".

15.11. Solution Results and Reports

Several output reports may optionally and automatically be generated after the OPF solution:

OPF Progress Summary Report

All information pertinent to the progress of the optimal power flow solution is contained within this summary. By default, the output is directed to the progress device but may optionally be directed to an alternate device such as a file or printer.

OPF Solution Report File

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A detailed report containing a summary of the optimal power flow problem statement, and the resulting control variable settings and sensitivities is produced. By default, this report is directed to the report device.

OPF Optimization Log File

An OPF Optimization Log File may optionally be generated. In addition to the information included in the OPF solution report, the log file also contains problem statement statistics, post solution results, and optionally, a detailed listing of Lagrange multipliers. The option to produce the OPF Optimization Log File file is set by enabling the *Produce optimization log file* option on the *Reporting* tab of *[OPF - Change Parameters]*.

OPF Diagram Results

OPF sensitivity results for bus voltage magnitude, bus active and reactive power in-jection, loads, fixed bus shunts and switched shunts can be selectively displayed on the diagram. A contour plot of values may also be produced.

The following subsections describe the contents of each of the above output reports and files in more detail.

15.11.1. OPF Progress Summary Report

During the optimal power flow solution information pertaining to its progress will automatically be displayed to the *Progress* tab of the PSS®E interface. The output may also be directed to another device such as a printer or file using *I/O Control>Direct Progress output (PDEV)*....

The contents of the OPF Progress Summary Report are defined in the following sections.

Problem Size Statistics

A table describing the size of the optimal power flow problem statement is the first item printed in the OPF Progress Summary Report.

```
   Matrix Size                 Problem      Size
   -----------                 -----------------
CM Rows:                 73    Angles         28
CM Columns:             118    Voltages       28
Jacobian elements:      559    Xformers       11
KTM elements:           962    MW gens         4
                               MV gens         6
                               Bus Shunts      0
                               Sw. Shunts      0
                               Load Shed       0
                               Flows           3
                               Interfaces      1
                               S. Compen.      1
                               R-cap.          5
                               Res. Units      4
                               Reserves        1
                               Area Int.       0
                               Lin. Dep.       0
                               Segments       21
```

The left-hand column, labeled Matrix Size, provides statistics about the formulated KT problem discussed in Section 15.7, “Solution Process”. The information is defined by the following:

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Rows: The number of equality constraints.

Columns: The number of power system variables.

Jacobian elements: The number of nonzero entries within the Jacobian matrix.

KTM elements: The number of nonzero entries within the KT matrix.

The right-hand column, labeled Problem Size, contains information on the total number of power system variables that formulate the columns of the Jacobian matrix. The labels identifying each variable are defined in Table 15.7, "Problem Size Label Descriptions".

Table 15.7. Problem Size Label Descriptions

Label	Description
Angles/ Voltages	The number of power flow bus nodes processed for voltage magnitude and angle information. Every bus in the network will have a voltage magnitude column and angle column introduced in the matrix. Buses connected by zero impedance branches are treated as a single node.
Xformers	The number of transformers processed from the power flow data model. Each transformer branch has one column introduced for either the tap ratio or phase shift angle.
MW gens	The number of generator units participating in the active power dispatch model. When minimizing fuel cost, piece-wise segmented cost curves will introduce a column for each in-service segment.
MV gens	The number of generator units participating in the reactive power dispatch
Bus Shunts	The number of participating added var devices. Each added var introduces two columns, one for the capacitive segments and one for the inductive segments.
Sw. Shunts	The number of participating switched shunts.
Load Shed	The number of participating load adjustment variables.
Flows	The number of participating branch flow constraints.
Interfaces	The number of defined interfaces.
S. Compen.	The number of participating branch reactance adjustment variables.
R-cap.	The number of generator units that have a reactive capability model defined. Each reactive capability model introduces four columns.
Res. Units	The number of generating units participating in the generation period reserve constraints. Each participating unit introduces two columns.

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Label	Description
Reserves	The number of generation period reserve constraints.
Area Int.	The number of interchange regulated areas.
Lin. Dep.	The number of linear constraint dependency equations.
Segments	The number of segments used to fully represent the variables of certain models (e.g., added vars, linear penalties, etc.). Additional columns are employed to accommodate all but the first segment. The number indicated is the total number of additional columns required for these models.

The problem size information is immediately followed by a qualitative statement indicating the initial conditions of the optimal power flow problem. An example follows:

```
 Note:  There were    3 infeasible variables found.
        The sum of the violations is                0.2055,
        The  largest   violation  is Efd       206   0.1934.
```

The first line specifies the number of variables for which initial values violate their stated limits.

The second line specifies the sum of the absolute values of the per unit values for each constraint violation.

The third line specifies the variable having the largest constraint limit violation, along with the value in pu. The nomenclature used to identify the variables (i.e., Efd), is defined in Table 15.8, "Jacobian Column Label Descriptions".

Solution Iteration Log

A summary of the nonlinear solution progress is displayed to the progress device and, optionally, to the OPF Optimization Log File. As discussed in Section 15.7.1, "The Nonlinear Problem", the nonlinear nature of the OPF problem requires that a series of simpler subproblems be solved. At each iterate of the solution process, one of the subproblems is solved and a row of information is displayed describing the progress made. This iteration log is the primary means of monitoring the progress of the solution.

The output is of the following form:

```
-------------------------------------------------------------------------------------------
Iter  Mu NL Objective  Norm RHS  Mismatch(PU)  Row Equation       Nearest Variable    Step
----  -- ------------ ---------- ------------------------------  ------------------------
1    0  2.88120E+01   2.42E+02     6.34E+00    RC-Q       206      Flow      153    u   4.
2x   0  2.95010E+01   1.18E+02     3.39E+00    RC-Q       206      Trns      204    u   7.
3   -1  3.03835E+01   2.23E+02     9.16E-01    F-eq       153                          1.
4   -2  3.04181E+01   7.35E+01     8.11E-02    Peqn       204      Volt     3018    u   1.
5   -2  3.02021E+01   2.23E+02     7.37E-02    Peqn       204      Segm      +70        3.
6   -2  2.98706E+01   1.70E+02     4.72E-02    Peqn       204                          1.
7   -3  2.97907E+01   5.33E+00     6.03E-03    Qeqn       153      Segm      +70        1.
. . .
```

The information contained within each row is defined as follows:

Iter

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The current subproblem, or iteration number. An x following the number indicates that the prior iteration was determined to be bad according to the definition given in Blown Up after XX Iterations.

Mu

The barrier coefficient exponent, also referred to as the barrier parameter. As discussed in Initial Barrier Coefficient, the number #; is the exponent of 10. A value of -3 signifies a barrier parameter value of 10-3, or 0.001.

The barrier parameter is the coefficient in the barrier objective function. The smaller the barrier parameter, the closer the variables (voltages, taps, and others) can approach their bounds, if that is the direction that they are driven towards. A large barrier parameter of about 0.01 will keep just about all variables off their bounds and towards the middle. A small barrier parameter of 0.00001 (#; = -5) or less will allow those variables to get very close to their limits.

NL Objective

The sum of all the objective components. This value includes all explicitly defined objectives, as well as any implicit objective terms that may have been introduced (i.e., terms of the quadratic penalty equation introduced for certain power flow controls (Section 15.5, "Modeling Power Flow Controls") or the barrier terms required for constrained variables at a hard limit).

Norm RHS

The infinity norm (i.e., the largest element absolute value) of the right-hand side (RHS) vector in the optimization solution. An asterisk (*) to the right of the value indicates that the value exceeded a predefined lambda tolerance. By default the tolerance is set to 105 and should not need to be modified (Advanced Algorithmic Decision Tolerances). If the values are becoming increasing larger, the solution is likely to terminate due to infeasibility problems. This indicates that the problem is most likely over-constrained.

When the dual variable mismatch criteria (Use Dual Variable Convergence Criteria) is imposed, the values reported for the *Norm RHS* and the *Mismatch* must be less than the tolerance specified by TOLN in the power flow solution parameters.

Mismatch

The infinity norm of the equality constraints or the lambda gradient portion of the RHS, better known as the familiar power flow mismatch. To arrive at this value, the absolute values of the MW and Mvar mismatches at each bus are added together. The resulting value must be less than the tolerance specified by TOLN in the Power Flow Solution Parameters option. The mismatch value is in per unit on the system MVA base.

Row Equation

Description of the Jacobian row having the largest mismatch value as displayed under the Mismatch column. The nomenclature employed to identify the entry is presented in Table 15.9, "Jacobian Row Label Descriptions".

Nearest Variable

Description of the most limiting variable, if the value reported under *Step Size* is less than 1.0 per unit. The nomenclature used to identify the variables (i.e., Efd), is defined in Table 15.8, "Jacobian Column Label Descriptions". A u placed after the description indicates that the corresponding variable is constrained by an upper limit.

Step Size

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The size of the correction step taken towards the optimal solution point. A 1.0 per unit step implies that 100% of the correction was added to the power system variables.

NE

The number of negative eigenvalues in the factorization of the KT matrix. It reflects the number of power system variables that make the KT problem indefinite. The problem becomes indefinite if the quadratic representation of the problem has negative curvature.

This value may change at each solution iteration, generally increasing for subproblems that result in a small step size. It may grow quite large, but for problems making good progress, it should shrink again. If the execution terminates as optimal, but with a nonzero NE value, then the power balance will be satisfied but the solution may not be a true optimal point. (You can try increasing the final value of the barrier parameter by an order of 10.)

Unlabeled column

An integer value may appear in an unlabeled column immediately to the right of the *NE* column. This number reflects the number of variables for which limits have been relaxed. This is likely to occur when the discrete treatment of transformer tap ratios or switched shunts is enabled.

During solution, transformer settings and switched shunts are treated as continuous variables, provided that they have not been fixed (Fix Transformer Tap Ratios, Fix Transformer Phase Shift Angles, or Fix Switched Shunts) at their current settings. If the option to Round Transformer Tap Ratios or Round Switched Shunt Vars is specified, then the resulting values as obtain after an initial optimal solution is found, are rounded to the nearest discrete tap or var setting. When this occurs, the program may find that some limits need to be relaxed. The total number of relaxed limits is reflected in the value displayed in this column.

Solution Termination Message

A termination message will be displayed immediately following the solution progress log. An itemized list of the objective functions and their values follows, along with a solution timing report. The following is one such example:

```
 Optimal Solution Found.
     Minimum Fuel cost  objective:  29115.463109
     Minimum series var objective:     23.121500

 Elapsed time:   0 minutes,  0.2 seconds.
     CPU time:   0 minutes,  0.1 seconds.
```

In the above example, the optimal power flow problem minimized fuel cost and adjustable branch reactances (series compensation).

Labeling Nomenclature

The following two tables define the labeling conventions used in a number of the solution results tables.

Jacobian Columns

Within the OPF output reports, the power system variables placed in the columns of the Jacobian matrix are described with an abbreviated label. The labeling nomenclature consists of a character string followed by an integer number. The character labels are defined in Table 15.8, "Jacobian Column Label Descriptions". The integer number appended to the label corresponds to one of two values:

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- If the number is prefixed with a +, the number that follows is a sequence number. For example, the label Intf +3 refers to the third interface defined in the working case.

- If a number is specified without a + prefix, the value simply refers to the identification number describing the power system element. For example, the label Angl 101 refers to the angle at bus 101.

Jacobian Rows

The labeling nomenclature used within the OPF reports to describe the constraints that make up the rows of the Jacobian matrix consists of a character label followed by a number. The labels are described in Table 15.9, "Jacobian Row Label Descriptions". The numbers refer to one of two values:

- If the number is prefixed with a +, it refers to a sequence number. For example, the label RTEQ +1 refers to the first generation period reserve constraint defined within the working case. Sequence numbers are used when the constraint cannot be referenced back to a power flow bus.

- If the number is not prefixed with a +, the value given is the power flow bus number for which the constraint equation is defined at. For example, Peqn 101 refers to the active power constraint equation at bus number 101 in the working case.

Table 15.8. Jacobian Column Label Descriptions

Label	Variable Description
Angl	Voltage angle at an explicit network bus
Del	Voltage angle at an implicit reactive capability bus
Volt	Voltage magnitude at an explicit network bus
Efd	Voltage magnitude at an implicit reactive capability bus
Trns	Transformer control element: tap ratio or phase angle
Pgen	Active power slack variable associated with a fuel cost segment
Pslk	Active power slack variable associated with a system slack bus
Qgen	Reactive power generator
Cap	Capacitive added var component
Shnt	Switched shunt
Shed	Bus load adjustment component
Flow	Constrained branch flow slack variable
Intf	Interface flow slack variable
Comp	Adjustable branch reactance variable
RC-I	Generator stator current slack variable, to limit reactive capability model I_{max}
RC-Y	Generator stator current slack variable, to limit reactive capability model $mE_{fd}+b$
RTMW	Generation period reserve participating unit slack variable
RT	Generation period reserve slack variable

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Label	Variable Description
LnCs	Linear dependency constraint equation variable
AI	Regulated area interchange slack variable
Clmp	Local generator voltage control clamp equation slack variable
Segm	Auxiliary variable segments (e.g., employed for linear type penalties). The number which follows the *Segm* label indicates the column number of the variable.

Table 15.9. Jacobian Row Label Descriptions

Label	Constraint Equation Description
Peqn	Active power balance at an explicit network node
RC-P	Active power balance at an implicit reactive capability node
Qeqn	Reactive power balance at an explicit network node
RC-Q	Reactive power balance at an implicit reactive capability node
F-eq	Branch flow constraint
I-eq	Interface flow constraint (or report)
RC-I	Generator stator current, to limit reactive capability model I_{max}
RC-Y	Generator stator current, to limit reactive capability model $mE_{fd}+b$
RTMW	Generation period reserve participating unit
RTEQ	Generation period reserve
LNCS	Customized linear dependency
AI	Regulated area interchange
CGEN	Local generator voltage control clamp

15.11.2. OPF Solution Results Report

An OPF Solution Results Report is automatically produced at the completion of the optimal power flow solution. It is sent to the report device by default but may be redirected to an alternate device or file through the *I/O Control > Direct report output (OPEN)...* menu option.

The OPF Solution Report consists of tables containing summaries of the problem statement control and constraint variable settings, and their sensitivities to the objective. The report itself is printed in a format suitable for printing and is necessary for performing post processing and analysis.

It is not possible to send the OPF Report to the Progress View! To avoid potential confusion that would arise from having output from both the OPF Solution Report and the Solution Iteration Log simultaneously sent to the Progress View and getting mixed together, the OPF Solution Report information is prevented from being sent to the Progress View. If the device specified by the *I/O Control > Direct report output (OPEN)...* menu option is set to the Current progress device, then the report will automatically be redirected to the report device. This action does not cause the output of any other activity to be redirected.

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15.11.3. OPF Optimization Log

The OPF Optimization Log File is optional and, by default, will not be produced. To produce an Optimization Log File, the *Produce optimization log file* option must be enabled prior to initiating the solution. By default the resulting file will be called DETAIL.DAT. An alternate name may be specified by entering a name directly in the *Optimization log* input field or selecting from a list of existing files by clicking *[...]* to the right of the input field. Existing files will be overwritten! Refer to Produce an Optimization Log File for further information on the use of this dialog.

In addition to the information printed within the standard OPF Progress Summary Report discussed in Section 15.11.1, "OPF Progress Summary Report", the Optimization Log File also includes information on problem size statistics and optionally, a detailed listing of the power system variables and Lagrange multipliers. The following section describe items that are unique to the Optimization Log File.

Immediately following the problem size statistics and a report of the initial conditions, a table of KKT Matrix Factorize statistics is printed. An example of this table is shown below.

```
 KKT Matrix Factorize statistics...
 ---------------------------------
 Symbolic Factors. Opts:     203   Elements:    203
Compres:       0    % Fill-in:     28
 Symbolic Factors. Opts:    6441   Elements:   1373
Compres:       1    % Fill-in:     70
 Unfactored.    Columns:     169   Elements:    806
Density:    2.82
 Numeric  Factors. 2x2s:       8   Elements:   1413
Compres:       1    % Fill-in:     75
       Neg. Eigenvalues:      67  Max front:     17  N-Steps
:      82
```

Immediately following this table is the solution iteration log and termination message, the same ones as displayed in the Progress View during the solution process. Refer to Solution Iteration Log and Solution Termination Message for details.

Infeasible Variable Report

Details of any infeasible variables discovered in the problem statement will be printed in a table. This will take place independent of whether the *Print a detailed listing* option is enabled or not.

Post Solution Results

If the option to *Print a detailed listing* is enabled (refer to Produce an Optimization Log File), then a complete table detailing the final values for *all* power system variables is produced. The detailed listing is the only way to obtain this information as it is not displayed in the Progress View during the solution process.

The table takes the following form:

```
       Column        Name            F S    Value
    Lower          Upper          Cost
-------- ------------------  ----  ---------  ------------
 ------------    ----------
           1    Angl      101         3   0.19434
  -6.2832       6.2832
```


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```
          2    Angl      102          3   0.19434
-6.2832       6.2832
          3    Angl      151          3   0.10051
-6.2832       6.2832
            . . .
         29    Volt      101        U 1   1.04992
 0.9500       1.0500     -1206.06170
         30    Volt      102        U 1   1.04992
 0.9500       1.0500     -1208.82517
```

The information presented under each column heading is defined as follows:

Column

The sequence number of the column entry within the Jacobian matrix where the associated variable is present.

Name

The identifier of the variable associated with the column entry of the Jacobian matrix. The description of each Name entry is defined in Table 15.8, "Jacobian Column Label Descriptions".

F

The final status of the variable. This flag may take on one of the following values:

U:	The variable is limited by an upper bound;
L:	The variable is limited by a lower bound;
F:	The variable is fixed at a given value;
-:	The lower bound of the variable has been relaxed;
+:	The upper bound of the variable has been relaxed;
I:	The variable is infeasible. Variables that are infeasible are always displayed in the Optimization Log, independent of whether the full detailed listing is produced, or not.

S

The final state of the variable. The state is described by an integer according to the following table:

Table 15.10. Variable State Values

Variable State, S	Variable (x) Condition
0	x = xmin
1	x = xmax
2	xmin< x < xmax
3	-∞ < x < +∞
4	x = x0 (x0 ≡ initial condition)
5	xmin < x < +∞
6	-∞ < x < xmax
7	-∞ < x < +∞

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Value

The final value of the variable. The units associated with the values correspond to the following:

- Voltage magnitudes are in per unit;
- Voltage angles are in radians;
- Branch flow constraints (MVA and Ampere) are in terms of their magnitudes squared;
- Transformer tap ratios are inverted.

Lower

The value of the variable's lower bound. The units of the values are as stated within the definition for *Value*.

Upper

The value of the variable's upper bound. The units of the values are as stated within the definition for *Value*.

Cost

The sensitivity of the variable. Whenever a variable reaches an upper or lower bound, or is fixed, there will be a nonzero entry in this column. The sensitivity values indicate how much impact it can have on the objective with a small move beyond its boundary limits. All values are in terms of Δ cost/ Δ variable. The number may not be identical to those produced in the output report due to unit conversions.

Lagrange Multipliers

When the *Print a detailed listing* option is selected, a table of Lagrange multiplier values is produced.

The Lagrange multiplier table takes the following form:

```
  Row        Name             Lagrange Mult.
 -------- ------------------ ---------------
       1    Peqn      101        -2456.88
       2    Peqn      102        -2456.89
       3    Peqn      151        -2469.08
       4    Peqn      152        -2542.66
       5    Peqn      153        -2545.58
 . . .
```

where:

Row

The sequence number of the row entry in the Jacobian matrix for the associated constraint equation.

Name

The identifier of the constraint equation associated with the row of the Jacobian matrix. The description of each Name heading is defined in Table 15.9, "Jacobian Row Label Descriptions".

Lagrange Mult

The final value of λ as used in the equation described in Section 15.7.1, "The Nonlinear Problem". These values indicate the sensitivity of the objective function to a change in the constraint.

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15.12. Examining OPF Data and Results

Optimal power flow data values may be examined through several different functions, including:

Power Flow > List Data... , *OPF* button

Produces a tabular report of any or all of the OPF constraint and control data within the working case.

The OPF Solution Report

A tabular report automatically generated subsequent to the OPF solution process. The report contains a solution summary and sensitivity tables pertaining to the power system variables of the OPF problem statement.

OPF > Data... and *OPF > Data tables...*

Presents the power flow and optimal power flow data in spreadsheets and data tables suitable for viewing and editing.

Some of these functions provide additional capabilities for editing the data values displayed.

Each subsection of this chapter covers the above program functions in some detail.

15.13. Listing Control and Constraint Data

LSTO

OPF data output lists control and constraint data, providing options for a number of customized reports in addition to the full listing.

Table 15.11. Optimal Power Flow (OPF) Data Output Options

Report Type	Description
Case Summary	Produces a tabular report of the total number of records defined within each OPF data category, along with the maximum number of allowable entries for each category. Counts of Bus loads and Generator dispatch units correspond directly to the load and generator power flow data models and may be obtained through activity SIZE. Immediately following the tabular listing of data item counts will be a list of the OPF solution objectives enabled in the working case. If no objectives have been explicitly specified, the word POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal will appear after the *Desired objectives* heading.
All Data	A complete report consisting of all the OPF data categories is produced.
Active Power Dispatch Tables	All active power dispatch tables within the working case are listed in the active power dispatch table report, independent of the subsystem selected for

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Report Type	Description
	output. The data values displayed in the report are defined in Active Power Dispatch Data Record.
Adjustable Bus Loads	All adjustable bus load records referencing an adjustable bus load table number greater than zero are listed. The values listed under the *Tbl* heading should correspond to an existing adjustable bus load table record. Data values listed in each adjustable bus load record are defined in Bus Load Data Record.
Adjustable Bus Load Tables	All adjustable bus load tables within the working case are listed in the report, independent of whether a subsystem has been specified. The values listed under the *Load ratio* headings, as well as the *Cost type* and the *Cost table* headings show default values. They are displayed for completeness but are not utilized by the current release of the program. The data values displayed in the table correspond to the definitions provided in Adjustable Bus Load Table Data Record.
Adjustable Branch Reactance Data	All adjustable branch reactance data records within the selected subsystem are listed in the output report. Branches are listed in ascending bus order (numeric or alphabetic) with the lower ordered bus listed as the from bus. The data values listed within each adjustable branch reactance record correspond directly to the definitions provided in Adjustable Branch Reactance Data Record.
Branch Flow Constraint Data	All branch flow constraint records within the defined subsystem are listed in the output report. Branches are listed in ascending from bus order (numeric or alphabetic) and then ascending to bus order, with the lower ordered bus listed as the from bus. Parallel circuits between any two buses are tabulated in ascending circuit identifier order and identical circuits are listed in ascending branch flow identifier order. The data values listed within each branch flow constraint record correspond directly to the definitions provided in Branch Flow Constraint Data Record.
Bus Data	All bus voltage constraint records for buses within the defined subsystem are displayed in the output report. Default data is displayed for those buses voltage constraint records that have not been explicitly modified. Data values displayed within the

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Report Type	Description
	bus attribute records are defined in Bus Voltage Attribute Data Record.
Bus Shunt	All adjustable bus shunt records within the defined subsystem are displayed in the output report. Multiple bus shunts at the same bus are differentiated by unique shunt identifiers. The data values displayed in the table correspond directly to definitions provided in Adjustable Bus Shunt Data Record. Values listed under *Cost type* and *Cost table* are displayed for completeness but are not utilized by the current release of the program.
Generator Dispatch Data	All generator dispatch records referencing an active power dispatch table number greater than zero and residing within the selected subsystem for printing are listed in the generator dispatch report. The data values for each generator dispatch record are defined in Generator Dispatch Data Record.
Generation Reactive Capability Data	All generation reactive capability data records of machines residing within the defined subsystem are listed in the output report. Machines are listed in ascending bus order (numeric or alphabetic) and multiple machines at the same bus are listed in ascending machine identifier order. The data values listed under each heading of the record correspond directly to the definitions provided in Generation Reactive Capability Data Record.
Generation Reserve Data	All generation reserve data records for machines residing in the defined subsystem are listed in the output report. Machines are listed in ascending bus order (numeric or alphabetic) and multiple machines at the same plant bus are listed in ascending machine identifier order. The values listed under *Ramp rate* and *Capability* correspond directly to the definitions provided in Generation Reserve Data Record.
Interface Flow Constraint Data	All interface flow constraint records contained in the working case are listed in the report. Interfaces are listed in ascending order based upon the interface flow identifier. Each interface flow record may contain one or more participating branches that define the interface. The participating branches are listed to the far right of the report on separate lines, in ascending from bus order (numeric or alphabetic) and then ascending to

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Report Type	Description
	bus order. Parallel circuits between any pair of buses are tabulated in ascending circuit identifier order. The data values listed within each interface flow constraint data record are defined in Interface Flow Data Input Values.
Linear Constraint Dependency Data	All linear constraint dependency records having one or more participating variable records are listed in ascending linear constraint identifier order within the report. Participating variable records are displayed to the far right of the output report with each participating record listed on a separate line. Each participating variable record is listed in the order in which it is stored in the working case. The *Type code* value indicates the type of variable being listed. The entries listed under the *Participating entry description* label identify the variable record being included in the linear constraint dependency equation. These may include identifiers such as table numbers, branches (from bus, to bus and circuit identifier) and machines (bus and generator identifier). The data values listed within each linear constraint dependency data record, including a description of the variable type codes and their corresponding identifiers, are defined in Linear Constraint Dependency Data Record.
Period Reserve Data	All period reserve records having one or more participating generator reserve units defined are listed in ascending numerical order in the report. There can be at most fifteen period reserve records listed. If the period reserve record contains more than one participating generator reserve unit, then subsequent reserve units will be listed on separate lines immediately following the main period reserve record, in ascending order (numerical or alphabetical). The data values in each record correspond directly to the generator reserve values defined in Period Reserve Data Input Values.
Piece-wise Linear Cost Table Data	All piece-wise linear cost tables within the working case are listed in ascending numerical order in the output report. A main header line is displayed for each linear cost table record, along with up to three coordinate pairs of data. If a cost table contains more than three coordinate data points, further data is displayed on subsequent lines, with-

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Report Type	Description
	out the main header. The data values are defined in Piece-wise Linear Cost Table Data Record.
Piece-wise Quadratic Cost Table Data	All piece-wise quadratic cost tables within the working case are listed in ascending numerical order in the report. A main header line is displayed for each quadratic cost table record, along with up to three coordinate pairs of data. If a cost table contains more than three coordinate data points, further data are displayed on subsequent lines, without the main header. The data values are defined in Piece-wise Quadratic Cost Data Record.
Polynomial and Exponential Cost Table Data	All polynomial cost tables within the working case are listed in ascending numerical order in the report. Each record is displayed on one line of output. The data values listed within each polynomial cost table record are defined in Polynomial and Exponential Cost Curve Data Record.

Additional Information
PSS®E GUI Users Guide, Displaying OPF Data *PSS®E* Command Line Interface (CLI) *Users Guide*, Listing Optimal Power Flow Data

15.14. Optimal Power Flow Solution Report

The OPF Solution Report is automatically produced after an OPF solution. By default, output is sent to the report device but may optionally be redirected to an alternate device through the use of activity OPEN.

The following sections describe each portion of the OPF Solution Report, in the order in which they appear in the report.

15.14.1. Solution Termination Summary

A qualitative summary of the power flow problem is output at the top of the report. This information indicates the initial conditions of the optimal power flow problem statement, as shown below:

```
Note:   There were  4 infeasible variables found.
        The sum of the violations is  0.4437,
        The largest violation is Efd   206   0.2377.
```

A complete description of the information presented in this summary can be found in the Solution Termination Message.

Immediately following the above summary a statement is printed indicating whether an optimal solution was found or not. Along with this will be an itemized list of each resulting objective function value.

```
Optimal Solution Found.
     Minimum Fuel cost  objective:  29208.031976
```


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```
     Minimum series var objective:     29.271418
```

If the message states something other than `Optimal Solution Found`, then the results presented in the rest of the report will not represent a solved problem! Messages such as `Terminated after XX iterations--reached iteration limit" or "Blown up after XX iterations" are analyzed in Maximum Iteration Limit Exceeded and Problem Unable to Solve.`

15.14.2. Interface Flows

Each interface flow constraint record in the working case is displayed in the Interface Flow report. One table is produced for each interface flow constraint specified.

```
  -----  Interface Flow  -----: AREA 5 TO AREA
1                 Continued.
  Minimum and Maximum:    -160.000    -155.000,
Sensitivity:       -513.5541 Status: Constrained

  From Bus    Name       kV       Area  Zone
To Bus      Name       kV       Area  Zone    MW Flow       Mvar Flow
      Mva Flow
  ---------------------------------------
----------------------------------------   ---------    -----------
    ----------
    3004  WEST          500.00     5     5
   152m MID500        500.00     1     1     -79.770         64.503
       102.586
    3006  UPTOWN        230.00     5     5
   153m MID230        230.00     1     1     -49.223        -27.433
        56.351
    3008  CATDOG        230.00     5     5
   154m DOWNTN        230.00     1     1     -26.026         65.054
        70.067

                                          ----------     ----------
    ----------

                              Totals:       -155.019        102.124
      185.635
```

The top line of the report specifies the name of the interface flow constraint being reported.

Immediately following the *Minimum and Maximum:* field are two values reflecting the minimum and maximum interface flow limits defined across the interface. Values are expressed in either MW or Mvar, depending upon the flow condition defined in the data record.

The *Sensitivity:* field has a value immediately following it if all following conditions are met: a flow limit has been reached, the OPF Solution Parameter option to Constrain Interface Flows is enabled, and the interface flow constraint being reported upon does not have a Limit type of *Reporting*. The Sensitivity field is left blank if the interface flow constraint limit type is set to *Reporting* or the *Constrain interface flows* option is not selected prior to the solution.

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The sensitivity value is expressed in units of Δ cost/ Δ per unit MW (on system MVA base) or Δ cost/ Δ per unit Mvar. For every per unit value in which the constrained limit can be relaxed, the cost will proportionally be reduced by one cost unit.

The *Status:* field reflects a value of *Constrained* if the OPF Solution Parameter option to Constrain Interface Flows is enabled prior to the optimal power flow solution; otherwise the word *Unconstrained* is displayed.

Within each Interface Flow report is a list of participating branches, identified by From Bus and To Bus number, Name, base kV, Area and Zone.

The MW flow, Mvar flow and MVA flow is reported for each participating branch of the interface. The total MW, Mvar and MVA flows across all branches in the interface are summed together and given at the bottom of the report.

15.14.3. Bounded, Infeasible and Opened Voltage Magnitudes

The Bounded, Infeasible and Opened Voltage Magnitudes report displays buses within the selected subsystem where voltage magnitudes have either violated or reached their voltage limits, or where limits may have been temporarily adjusted.

```
 Bounded, Infeasible and Opened Voltage Magnitudes:
 -------------------------------------------------

   Bus # Name          kV      V-old     V-change
    V-new     Vmin       Vmax       Sensitivity Violation
 ------- ------------ ------  -------- ------------
--------- ---------- ---------- ---------- ----------
     101 NUC-A        21.600  1.0200    -.2175E-03
  1.0198    0.9500     1.0200(o)  -458.4
     102 NUC-B        21.600  1.0200    0.2955E-01
  1.0496    0.9500     1.0500     -222.8
     154 DOWNTN       230.00  0.9389    0.4382E-02
  0.9433    0.9389(o)  1.0500
     201 HYDRO        500.00  1.0400    0.9920E-02
  1.0499    0.9500     1.0500     -1246.
    3018 CATDOG_G     13.800  1.0218    0.2818E-01
  1.0499    0.9500     1.0500     -1980.
```

Violations may occur in situations where discrete controls (transformer taps or switched shunts) have been employed, soft limits (linear or quadratic) have been assigned to the OPF bus voltage constraint records, or the OPF solution was not able to attain a feasible state.

Bounded or infeasible buses are identified by bus number (*Bus #*), *Name* and base *kV*.

Under the *V-old*, *V-change* and *V-new* headings are the initial bus voltage, the change in voltage and the final bus voltage, all in per unit.

The *Vmin* and *Vmax* columns list the imposed lower and upper voltage bounds at the bus. These limits reflect either the Normal or Emergency voltage limits according to the option selected prior to the OPF solution (Impose Emergency Bus Voltage Limits). All voltage values are presented in per unit.

A value is displayed in the *Sensitivity* column if the corresponding bus voltage magnitude has reached or exceeded a voltage limit, and where sensitivity value is larger than the greater of 0.1 or 10% of the largest voltage magnitude sensitivity reported. The sensitivity value is specified in Δ cost/ Δ per unit voltage.

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Values shown in the *Violation* column indicate the amount by which the final voltage magnitude value at the bus exceeds the violated lower or upper voltage limit. This may occur if the Limit Type of the bus within the OPF Bus Voltage constraint record was set to *Reporting*.

15.14.4. Bounded and Infeasible Reactive Generation

The Bounded and Infeasible Reactive Generation report displays generators within the selected subsystem where reactive generation values have reached or violated the upper or lower bound.

```
  Bounded and Infeasible Reactive Generation:
  -------------------------------------------

     Bus # Name         kV     ID    Variable Description
       Sensitivity Violation
  -------- ------------ ------ -- ------------------------------
----------  ----------
       206 URBGEN       18.000 1  Efd  (PU)          =
 2.2795    -762.
      3018 CATDOG_G     13.800 1  Qgen (Mvar)        =
 109.66                 29.7
      3018 CATDOG_G     13.800 1  Efd  (PU)          =
 2.4677     741.       0.188
```

Reactive generation limits (Qmax and Qmin) are introduced through the main power flow generator model. Additional limits on armature reaction and stator current can be applied using the OPF generator reactive capability model.

The bus number (*Bus #*), *Name*, *base kV* and machine identifier (*ID*) describe the reported generator unit.

The *Variable Description* column displays the reactive generation value being reported. The labels may be one of the following:

- Qgen (Mvar) - The reactive power injection in Mvars.
- Efd (pu) - The armature reaction in per unit.
- Is (pu on MVARC) - The stator current value in per unit on generator MVA rating.

The *Sensitivity* column displays a sensitivity value for generators where reactive generation value has reached or exceeded a limit, and where sensitivity value is larger than the greater of 0.1 or 10% of the largest reactive generation related variable sensitivity. Values are shown in Δ cost/ Δ per unit based upon the system MVA.

The *Violation* column displays the amount by which the reported value has exceeded its limit. Violations may result in situations where soft limits are applied or reactive capability models have been employed. In the above report, two violations have occurred at bus 3018, machine identifier 1. The first violation indicates that the reactive power injection has exceeded the specified limit by 29.7 Mvars. The second violation indicates that the armature reaction limit has been exceeded by 0.188 per unit.

15.14.5. Bounded or Infeasible Line Flows

The Line Flows that are Infeasible or at a Bound report lists branches within the specified subsystem where flows have reached or violated defined flow limits.

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```
Line Flows that are Infeasible or at a Bound:
-----------------------------------------

 From Bus Name      kV       To Bus   Name
  kV     Ckt  FlowID      Flow          Minimum   Maximum Violation
Sensitivity
 ------------------------------------------------------------------------------------------
     153 MID230     230.00      154 DOWNTN
  230.00  1      1       199.905 MVA      0.000   200.000     0.000
   -104.858
```

The *From Bus*, *To Bus*, *Ckt Id*, *base kV* and *Flow Id* identify the reported branch.

The *Flow*, *Minimum* and *Maximum* columns reflect the final flow value, the type and units of the reported flow value and the imposed lower and upper flow limits. Flow limit values reflect either Normal or Emergency flow limits according to the option selected for the OPF solution (Impose Emergency Branch Flow Limits). Physical units are used for the MW, Mvar and MVA flow values and are indicated by the label next to the Flow value. Current flow values (I) are presented in per unit on the system MVA base.

The *Violation* column displays the amount by which the flow value has exceeded its limit.

The *Sensitivity* column displays the sensitivity value for all flow values that are at a limit. Sensitivities are reported in the following units for the flow type indicated.

- MW, in Δ cost/ Δ per unit MW (on system MVA base)
- Mvar, in Δ cost/ Δ per unit Mvar (on system MVA base)
- MVA, in Δ cost/ Δ per unit MVA (on system MVA base)
- Current, in Δ cost/ Δ per unit amperes (on system MVA base)

15.14.6. Generator Fuel Cost Summary

The Generator Fuel Cost Summary is displayed if the Minimize Fuel Cost objective had been specified prior to solution in association with defined generator dispatch records and corresponding cost curves.

```
Summary Table of Linear Dependency Constraints:
-----------------------------------------------
                          Constraint Slack
Equation # Name          Value    Max      Min
---------------------     -------- -------- --------
2  NEW ONE              6.00     6.00    -2.00
  Participating Variables
  Coeff  Value       Description
  Element
  ------ ----------  -------------------------
------------
     1.0       0.88  Bus Volt Mag (PU)
 152 [MID500      500.00]
     2.0      19.08  Bus Volt Ang (Degrees)
 101 [NUC-A       21.600]
     3.0     773.92  Active Power Dispatch (MW)
```


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```
Dispatch table:      2
     4.0     -99.96  Reactive Power Generation
 102 [NUC-B       21.600]  1
     5.0       1.05  Tap Changing Transformer
 152 [MID500      500.00]  to 153 [MID230      230.00]  Ckt 1
      6.0       222.78  Branch Flow Constraint
     (MVA )  153 [MID230      230.00]  to 154 [DOWNTN      230.00]
 Ckt 1
     7.0    -159.78  Interface Flow Constraint
 (MW  )  Flow ID   1:  AREA 5 TO AREA 1
     8.0       0.04  Adjustable Bus Shunt(MVar)
204 [SUB500      500.00]  1
     9.0       0.08  Switched Shunt (MVar)
 154 [DOWNTN      230.00]
    10.0       0.60  Load Adjust Multiplier
 Load Table:      2
```

The *id* and *Type* fields identify the active power dispatch constraint table and fuel cost curve type being employed for the generators identified under the *Bus #* and *ID* headings. The Type may be either Lin, Quad or Poly to respectively indicate a linear, quadratic or polynomial cost curve.

The *Fuel $* value indicates the cost of the fuel dispatched, as calculated from the corresponding cost curve table.

The *MW Output* value reflects the total active power output from all units participating in the defined constraint. This includes all the units listed under the *Bus #* and *ID* headings for the active power dispatch constraint Id.

The *MW Minimum* and *MW Maximum* columns indicate the minimum and maximum active power limits defined for the active power dispatch constraint.

Totals for the Fuel and MW value columns are provided at the bottom of the table.

The *Bus #*, *Name*, *kV*, *ID*, *Pgen* and *Pfrac* columns provide information on each generator participating in the active power dispatch constraint defined. If only one generator participates in a given active power dispatch constraint, as is shown in the summary report above, then the *Pgen* value is the same as the *MW Output* value, and *Pfrac* should be 1.0.

15.14.7. Generation Reserve

A Generation Reserve summary report is displayed for each generator period reserve constraint in the working case.

```
  Generation Reserve,  200.00 MW in   5.000 minutes.

      Bus # Name          kV     ID   MW Reserve   MW Maximum
     ------ ------------------- -- ------------ ------------
        206 URBGEN       18.000 1         50.000
50.000
        211 HYDRO_G      20.000 1        150.019       500.000
                                    ============ ============
        Period MW reserve totals:        200.019       550.000,
```


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```
Sensitivity =    0.161E+04
```

The header line of each Generation Reserve report indicates the active power reserve, in MW, and the amount of time, in minutes, by which the reserve constraint must be satisfied.

Within each Generation Reserve report is a list of contributing generator reserve units, identified by the bus number (*Bus #*), *Name*, *base kV* and machine identifier (*ID*).

The *MW Reserve* and *MW Maximum* columns reflects the final reserve and maximum reserve capabilities, both in units of MW. The final unit reserves and the maximum values for all participating reserve units are summed together and given at the bottom of each report.

A *Sensitivity* value is calculated and displayed to the right of the reserve totals if the final reserve value has reached the reserve limit specified. The sensitivity value reflects the impact that a one per unit MW change has on the objective (Δ cost/ Δ per unit MW, on system MVA base).

15.14.8. Regulated Voltage Buses

A summary record is produced for every voltage regulated bus located within the selected subsystem. By default, all regulated buses in all areas are printed.

```
 Summary Table for Regulated Voltage Buses:
 ------------------------------------------

   Bus # Name          kV       Area     MW          Pmin
   Pmax    Pchange              Voltage    Vmin       V-Sens
                                 Zone     Mvar        Qmin
   Qmax    Qchange   Lambda     Vchange    Vmax       Q-Sens
  ------------------------------------------------------------------------------------------
     101 NUC-A         21.600      1     750.00
     810.00          -2456.884    1.0499 U 0.9500   -1206.0617
                                  77     154.74     -100.00
 600.00    73.541    -0.027    0.0299    1.0500      -0.0270

     102 NUC-B         21.600      1     750.00
     810.00          -2456.887    1.0499 U 0.9500   -1208.8252
                                  77     154.74     -100.00
 600.00    73.543     0.017    0.0299    1.0500       0.0168

     206 URBGEN        18.000      2     836.40
     900.00    36.401-2609.142    1.0136    0.9500
                                   2     428.08        0.00
 600.00 -171.919 -102.021   -0.0101    1.0500    -102.0211
```

The report for each regulated bus includes two lines of output.

The bus number (*Bus #*), *Name* and base *kV* identify the voltage regulated bus. The *Area* and *Zone* numbers of the bus are shown in the upper and lower lines under the *Area* and *Zone* heading.

Under the *MW* and *Mvar* heading is the final value of the active power, in MW, and the reactive power, in Mvar. They are shown in the upper and lower lines of the record, respectively.

An additional character may appear to the right of the reactive power (Mvar) value:

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- L indicates that the reactive power is limited by a lower bound (Qmin).
- U indicates that the reactive power is limited by an upper bound (Qmax).
- F indicates that the reactive power is fixed at a particular value.
- + indicates that the reactive power has violated the upper bound (Qmax).
- - indicates that the reactive power has violated the lower bound (Qmin).

Under the *Pmin* and *Qmin* heading is the minimum active power limit, in MW, and the minimum reactive power limit, in Mvar. The values are obtained directly from the power flow generator data model. A blank field indicates that the magnitude of the defined limit is significantly high.

The Pmin value shown may not be the actual active power limit employed by the OPF. If the associated bus has a generator with a corresponding OPF generator fuel cost model, then the Pmin and Pmax limits defined by the fuel cost model will be employed. These limits are shown in the Generator Fuel Cost Summary (Section 15.14.6, "Generator Fuel Cost Summary").

The *Pmax* and *Qmax* column displays the maximum active power limit, in MW, and the maximum reactive power limit, in Mvar.

Under the *Pchange* and *Qchange* heading, the value shown in the upper line is the difference between the final active power value and the initial active power value, in MW. The value shown in the lower line is the difference between the final reactive power and the initial reactive power value, in Mvar.

The *Lambda* column displays the Lagrange multipliers for the active and reactive power injections at the bus, shown in the upper and lower lines respectively.

The Lambda value for the *active power injection* is the partial derivative of the objective function with respect to the active power injection at the regulated bus. It indicates the impact that a one per unit adjustment in the active power has on the objective function (Δ cost/ Δ per unit MW on system MVA base). It is most useful for determining the incremental effects of load or generation changes on the losses or production costs, depending upon the stated objective.

The Lambda value for the *reactive power injection* is the partial derivative of the objective function with respect to the reactive power injection at the regulated bus and indicates the impact that a one per unit adjustment in the reactive power has on the objective function (Δ cost/ Δ per unit Mvar on system MVA base).

The *Voltage* and *Vchange* column displays the final bus voltage magnitude, in per unit, and the difference between the final voltage value and the initial bus voltage value, also in per unit. Values are shown in the upper and lower line respectively.

If the final bus voltage magnitude (Voltage) is limited by either an upper or lower bound, or is fixed, then one of the following characters will appear after the voltage value:

- L indicates that the voltage value is limited by a lower bound (Vmin).
- U indicates that the voltage value is limited by an upper bound (Vmax).
- F indicates that the voltage value is fixed at a particular value.

The *Vmin* and *Vmax* column displays the minimum and maximum bus voltage magnitude limits, in per unit, shown in the upper and lower line respectively. They reflect either the Normal or Emergency voltage limits

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defined in the OPF bus voltage attribute data record depending upon the option selected prior to the OPF solution (Impose Emergency Bus Voltage Limits).

Under the *V-Sens* heading, a sensitivity value of the voltage to the objective function is displayed in the upper line if the voltage is at a limit (as indicated by a U or L after the Voltage value), or is fixed at the current value (as indicated by an F after the Voltage value). The voltage sensitivity value reflects the impact that a one per unit adjustment in the voltage has on the comprehensive objective function (Δ cost/ Δ per unit voltage magnitude). This is the same value as shown in the Bounded and Infeasible Voltage Magnitude report (see Section 15.14.3, "Bounded, Infeasible and Opened Voltage Magnitudes").

Under the *Q-Sens* heading, a sensitivity value of the reactive generation to the object is displayed in the lower line if the reactive power of the bus is at a limit (as indicated by a U or L after the Mvar value), or fixed (as indicated by an F after the Mvar value). The reactive sensitivity value reflects the impact that a one per unit adjustment in the reactive power has on the objective (Δ cost/ Δ per unit Mvar on system MVA base). This is the same information as shown in the Bounded and Infeasible Reactive Generation report (see Section 15.14.4, "Bounded and Infeasible Reactive Generation").

15.14.9. Tap Changing Transformers

The Summary Table for Tap Changing Transformers reports on all tap ratio varying transformers within the selected subsystem that have a significant sensitivity value or a change in tap ratio.

```
Summary Table for Tap Changing Transformers:
--------------------------------------------

  From Bus            kV      Area  Zone    Voltage
Ratio         Minimum   MW Flow        Sensitivity
  To Bus              kV      Area  Zone    Voltage
Change        Maximum   MV Flow     Tap          Volt
 ------------------------------------------------------------------------------------------

    151 NUCPANT     500.00      1     1    1.03227
1.00000 F     0.90000  -748.404  -1133.62749
    101 NUC-A       21.600      1    77    1.04992u
             1.10000   -82.382                -1206.06170

    151 NUCPANT     500.00      1     1    1.03227
1.00000 F     0.90000  -748.404  -1137.12650
    102 NUC-B       21.600      1    77    1.04992u
             1.10000   -82.384                -1208.82517

    152 MID500      500.00      1     1    1.02750
1.01506       0.95000   578.878
    153 MID230      230.00      1     1    0.99759
0.00506 ( 2 ) 1.05000   305.148
```

Each two line transformer record identifies the *From Bus*, *To Bus*, base *kV*, *Area* and *Zone* number of the end buses of the transformer.

The *Voltage* value reflects the from bus and to bus voltages corresponding to each side of the tap. A single character displayed to the right of either value indicates whether the voltage is limited by a lower bound (L), upper bound (u), or has been fixed (f).

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The *Ratio* value reflects the tap ratio, in per unit, and is shown on the upper line of the transformer record. A single character displayed immediately to the right of this value indicates whether the tap ratio is limited by a lower bound (L), upper bound (U), or is fixed (F).

A *Change* value may be displayed under the *Ratio* value if the final tap ratio, as provided in the top number, has changed from the initial tap ratio.

To the right of the Change value, in parentheses ( ), is the number of discrete tap steps taken. A zero per unit change will result in a 0 being printed. If the option to Round Transformer Tap Ratios had been specified, then the change in tap ratio ought to be an exact multiple of the tap step. The tap ratio will be annotated with a fixed flag (F) to indicate that tap settings were fixed at their nearest step during the last phase of the solution process. Tap ratio changes may have occurred during the first phase of the solution process however, prior to the discretization and display of the fixed flag!

The *Minimum* and *Maximum* tap ratio limits are displayed in the upper and lower line of the transformer record respectively. These values are obtained from the transformer information in the power flow data model of the working case. The minimum and maximum tap ratios of a fixed transformer are shown as 1.0.

The *MW Flow* and *MV Flow* column displays the final MW and Mvar flows on the branch as seen at the from bus. The MW flow value is shown on the upper line and the Mvar flow value below that.

The *Sensitivity Tap* column displays a tap ratio sensitivity value if the tap ratio value is either fixed or at a limit. The value is expressed in units of Δ cost/ Δ per unit tap ratio and is only displayed in the upper line of the transformer record.

The *Sensitivity Voltage* column displays a voltage sensitivity value for the from bus and/or to bus if the voltage magnitude of the corresponding bus is limited by either its upper or lower constraint. The values are reported in units of Δ cost/ Δ per unit voltage magnitude and are the same values as displayed in the Bounded and Infeasible Voltage Magnitude report (see Section 15.14.3, "Bounded, Infeasible and Opened Voltage Magnitudes").

Transformer tap records *will not be reported* if the tap ratio has not changed by more than 0.001 per unit, or if the objective sensitivity of the tap ratio is less than 1.0!

15.14.10. Phase Shifting Transformers

The Summary Table for Phase Shifting Transformers reports on all phase angle varying transformers within the selected subsystem.

```
 Summary Table for Phase Shifting Transformers:
 ----------------------------------------------

   From Bus          kV         Area  Zone    Angle
  Minimum     MW Flow  Sensitivity
   To Bus            kV         Area  Zone    Change
  Maximum     MV Flow
  ------------------------------------------------------------------------------------------
     202 EAST500     500.00      2     2   -3.40702
-30.00000     600.000
     203 EAST230     230.00      2     2   -3.40702
 30.00000     271.184
```


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Each two line phase shifting transformer record identifies the *From Bus*, *To Bus*, base *kV*, *Area* and *Zone* number of each end bus of the transformer.

The *Angle* value is shown in the upper line of the displayed transformer record and reflects the final phase shift value, in degrees. A single character displayed immediately to the right of the value indicates whether its limited by a lower bound (L), upper bound (U) or is fixed (F).

A *Change* value may be displayed under the *Angle* value if the final phase shift value, as provided in the top number, has changed from the initial phase angle, in degrees. A Change value will not be displayed if the difference between the initial and final phase angle is less than or equal to 0.01 degrees.

The *Minimum* and *Maximum* phase angle limits, in degrees, are displayed in the upper and lower line of the phase shifting transformer record respectively. These values are obtained directly from the transformer information in the power flow data model of the working case. Minimum and maximum phase angle limit values will not be displayed if the phase shifting transformer is fixed or if the phase angle is less than or equal to 0.01 degrees

A *MW Flow* and *MV Flow* column displays the final MW and Mvar flows on the transformer as seen at the from bus.

A *Sensitivity* value is displayed in the upper line of the phase shifting transformer record if the phase shifter is fixed or constrained by either its upper or lower limit. The value is expressed in units of Δ cost/ Δ degrees.

15.14.11. Added Shunts

When the Minimize Adjustable Bus Shunts objective is enabled, all adjustable bus shunts adjusted during the solution and within the specified subsystem for printing are included in the Summary Table for Added Shunts.

```
Summary Table for Added Shunt (Mvar):
 ------------------------------------
    Bus # Name         kV     ID  Area  Zone
 Initial     Final     Change    Minimum    Maximum    Cost ($/Mvar)
 Sensitivity
 ------- ------------------- -- ----- -----
--------- ---------- ---------- ---------- ---------- -------------
------------
    153 MID230     230.00     1    1     1
    0.000      4.217     4.217    -40.000     68.00      1.000
    154 DOWNTN     230.00     1    1     1
    0.000     68.000    68.000    -40.000     68.00      1.000
      -151.0977

    -------    -------
                                 Total:
     0.000     72.217
```

The bus number (*Bus #*), *Name*, base *kV*, *ID*, *Area* and *Zone* columns provide identification information for each adjustable bus shunt control. Multiple adjustable bus shunts defined at a common bus are listed on separate lines and are uniquely identified by the shunt ID.

The *Initial*, *Final* and *Change* values reflect the initial additional shunt installation (in Mvar), the final additional shunt value (in Mvar) and the difference between the initial and final values, in Mvars. The *Final* value

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is added to the B-Shunt value of the corresponding fixed shunt within the power flow data. If no corresponding fixed shunt is found, then a new fixed shunt is added with the given ID.

The *Minimum* and *Maximum* columns display the upper and lower Mvar limits of the adjustable bus shunt devices.

The *Cost* value is the cost for each Mvar deployed. The values are obtained directly from the working case.

A *Sensitivity* value is reported for each adjustable bus shunt record where the final Mvar value is at a minimum or maximum Mvar limit. The value is expressed in units of Δ cost/ Δ per unit susceptance (Mvar).

The *Total* number of initial Mvars and final Mvars is displayed at the end of the report.

Only adjustable bus shunt controls for which the final shunt value exceeds 0.1 Mvar or the nominal shunt value has changed by more than 0.1 Mvar will be included in the solution report!

15.14.12. Switched Shunts

All switched shunt devices that have a significant sensitivity value and are located within the specified subsystem for printing are included in the Bounded Switched Shunt report.

```
  Bounded Switched Shunts:
  ------------------------
    Bus # Name         kV      Area  Zone    Final Mvar
  Minimum    Maximum   Sensitivity
  -------------------------- ----- -----   ------------
---------- ---------- ------------
    154   DOWNTN       230.00     1     1       45.000
U   -45.000     45.00    -100.00
```

The bus number (*Bus #*), *Name*, base *kV*, *Area* and *Zone* columns provide identification information for each switched shunt.

The *Final Mvar* column lists the final resulting susceptive value, in Mvar.

The *Minimum* and *Maximum* columns display the susceptive limits, as obtained directly from the power flow switched shunt data model. The values are expressed in units of Mvar at nominal voltage.

A *Sensitivity* value is reported for those switched shunts where the final Mvar values are at a minimum or maximum limit. The value is expressed in units of Δ cost/ Δ per unit susceptance (Mvar).

Only switched shunts where the sensitivity value is larger than the greater of 0.1 or 10% of the largest switched shunt sensitivity are included in the report!

15.14.13. Load Adjustment

When the Minimize Adjustable Bus Loads objective is enabled, all adjustable bus loads adjusted during the solution and within the specified subsystem for printing are included in the Summary Table for MW Load Adjustment.

```
Summary Table for MW Load Adjustments:
```


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```
-------------------------------------

Bus # Name         kV     ID   Initial MW Load Mult.
Effective MW  Maximum MW  Minimum MW  Sensitivity
------- ------------------ --   ---------- ----------
 ------------  ----------  ----------  ----------
154 DOWNTN 230.00 1  600.00 0.400  240.05  600.00 240.00 1.0823
154 DOWNTN 230.00 2  400.00 0.400  160.04  400.00 160.00 1.0823
205 SUB230 230.00 1 1200.00 1.328 1593.57 2400.00 600.00
```

The bus number (*Bus #*), *Name*, base *kV* level and load identifier (*ID*) describe the reported bus load. The ID differentiates between multiple loads at the same bus.

The *Initial MW* value is the real power load prior to the load adjustment.

The *Load Mult.* value reflects the resulting multiplier of the original load applied at the conclusion of the OPF solution.

The *Effective MW* value reflects the resultant real power load with the final load multiplier applied to the initial load value.

The *Maximum* and *Minimum* active power load limits, in MW, indicate the valid range for MW load adjustments during the OPF solution as calculated by applying the corresponding maximum and minimum load multipliers to the current MW load.

A *Sensitivity* value is reported for those load adjustment controls for which the final load multiplier is at a minimum or maximum limit. The value is expressed in units of Δ cost/ Δ per unit MW (on system MVA base).

15.14.14. Series Compensation

When the Minimize Adjustable Branch Reactances objective is enabled, all adjustable branch reactance records within the specified subsystem for printing are processed for inclusion in the Summary Table for Series Compensation.

```
  Summary Table for Series Compensation:
  -------------------------------------

      From Bus Name    kV      Area  Zone    Psi
      Psi Max   Voltage   MW Flow  Sensitivity
        To Bus Name    kV      Area  Zone    X
(PU)    Psi Min   Voltage   MV Flow  Cost
      ------------------------------------------------------------------------------------
      201  HYDRO      500.00     2     2    0.305
      1.000     1.043   794.387  -2067.50921
      204  SUB500     500.00     2     2    0.00916
    0.300     0.989   244.003     30.01278
```

Each adjustable branch reactance record is identified by the *From bus*, *To Bus*, base *kV* level, *Area* and *Zone* of each end bus of the branch.

The *Psi* value is shown in the upper line of the displayed branch record and reflects the final value of the reactance multiplier, ψ .

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The *X (pu)* value, displayed below the Psi value, is the resulting branch reactance value, in per unit.

The *Psi Max* and *Psi Min* column displays the maximum and minimum reactance multipliers as defined in the adjustable branch reactance record.

The *MW Flow* and *MV Flow* column displays the final MW and Mvar flow on the branch at the from bus.

A *Sensitivity* value is displayed in the upper line of the last column if the final reactance multiplier is at a minimum or maximum limit. The sensitivity value is expressed in units of Δ cost/ Δ per unit ψ .

A *Cost* value is reported in the lower line of the last column for situations in which ψ is not equal to 1.0. The cost is in units of cost/per unit ψ .

15.14.15. Regulated Area Interchange

When the option to Regulate Area Interchange is enabled during the OPF solution, a Summary Table of Area Interchanges is printed for all area interchange constraints, independent of whether the areas correspond to areas within a printed subsystem or not.

```
  Summary Table of Area Interchange:
  ----------------------------------

      Area # Name            MW Value     Minimum     Maximum
Sensitivity
    -------- ------------ ---------- ---------- ----------
-----------
           1 FLAPCO          284.341    210.000    290.000
           2 LIGHTCO        -139.979   -140.000    -60.000
   492.
           5 WORLD          -144.362   -160.000   -140.000
```

The area number (*Area #*) and *Name* identify the reported area.

The *MW Value* indicates the final area interchange value, in MW.

The *Minimum* and *Maximum* values are the defined interchange limits, in MW.

A *Sensitivity* value is reported if the final interchange value is at a minimum or maximum limit. This value is expressed in units of Δ cost/ Δ per unit MW (on system MVA base).

15.14.16. Linear Dependency Constraints

The Summary Table of Linear Dependency Constraints reports on all linear dependency constraints that have been defined, along with each of their corresponding variables.

```
Summary Table of Linear Dependency Constraints:
 ------------------------------------------------
                             Constraint Slack
   Equation # Name          Value    Max      Min
  ---------------------    -------- -------- --------
        2  NEW ONE             6.00     6.00
  -2.00
             Participating Variables
```


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```
             Coeff  Value       Description
              Element
             ------ ----------  -------------------------
------------
                1.0       0.88  Bus Volt Mag (PU)
         152 [MID500      500.00]
                2.0      19.08  Bus Volt Ang (Degrees)
    101 [NUC-A       21.600]
                3.0     773.92  Active Power Dispatch
(MW) Dispatch table:  2
                4.0     -99.96  Reactive Power
Generation  102 [NUC-B       21.600]  1
                5.0       1.05  Tap Changing Transformer
  152 [MID500      500.00]  to 153 [MID230      230.00]  Ckt 1
                6.0     222.78  Branch Flow Constraint
  MVA )  153 [MID230 230.00]  to 154 [DOWNTN      230.00]  Ckt 1
                7.0    -159.78  Interface Flow
Constraint  MW  )  Flow ID   1:  AREA 5 TO AREA 1
                8.0       0.04  Adjustable Bus
Shunt(MVar) 204 [SUB500      500.00]  1
                9.0       0.08  Switched Shunt
(MVar)      154 [DOWNTN      230.00]
               10.0       0.60  Load Adjust Multiplier
    Load Table:      2
```

The first line of the linear dependency constraint record identifies the linear constraint equation number and name. The final calculated constraint slack value and the defined maximum and minimum constraint slack values are also displayed.

The remainder of the multiple line record lists each of the participating variables within the linear dependency constraint equation. Only those variables that are active and in-service at the time of the solution are reported. The Adjustable Bus MVar and Load Multiplier variables will only be active if the corresponding objective functions to Minimize adjustable bus shunts and Minimize adjustable bus loads, respectively, are enabled.

Each of the participating variables display the Coeff data value followed by the calculated value of the individual variable and the description of the variable. At the end of the each participating variable record is the element corresponding to the description.

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Chapter 16
Dynamic Simulation Activity Descriptions

This chapter contains operational descriptions of the PSS®E dynamic simulation activities. Each section of this chapter documents a single PSS®E activity. Interface references are provided, with links to the *PSS®E GUI Users Guide* and the *PSS®E Command Line Interface (CLI) Users Guide*, where appropriate.

16.1. Reading Dynamics Model Data

DYRE

Run Activity DYRE - GUI
File>Open... *[Open]* Dynamics Model Raw Data File (*.dyr)
Run Line Mode Activity DYRE - CLI
ACTIVITY? >>**DYRE,ADD** ENTER DYNAMICS DATA SOURCE FILENAME : >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The dynamics data input activity DYRE reads simulation model data from a Dynamics Model Raw Data File into the dynamics data arrays in PSS®E working memory.

16.1.1. Dynamics Model Raw Data File Contents

The input stream to activity DYRE consists of a Dynamics Model Raw Data File. This file contains a group of logical records, each of which defines the location in the network (by bus, machine, load, dc line, etc.) of a dynamic equipment model from the PSS®E model library, along with the constant parameters of the model. The *PSS®E Model Library* contains data sheets for each of the equipment models in the PSS®E model library.

PSS®E Dynamic models can be broadly classified into the following categories:

Device models. These are attached to specific equipments in PSS®E power flow (e.g., models of generators, loads etc.). For any given equipment, there can be only one device model attached to it.

Protection models and models of type “other” attached to specific equipment in PSS®E power flow. For this group, for any given equipment although there can be more than one model attached, the restriction is that there can be only one instance of any model attached to that equipment. In other words for any equipment there can not be more than one instance of the same model attached to it.

Miscellaneous model of type “other”. These are unattached models in the sense that these may or may not be attached to any specific equipment in PSS®E power flow. For miscellaneous “other” type, the same model

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can applied more than once. In other words there can be more than one instance of the same model. The different models instances are identified by a unique model instance (MINS)value assigned to each model. The MINS values are assigned by users while reading the model DYRE record (when the model is being added interactively the MINS value can be generated by PSS®E or can be input by the user). The rule in applying models of this category is that no two miscellaneous "other" models can have the same MINS value. If there are two models with the same MINS value in the same DYR file, then this condition is treated as a duplicate model and the second model instance is ignored. However if a second model with the same MINS value is read in via activity "DYR,ADD" (activity to add dynamics model) then the model being added replaces the one that is already present in the case.

For model categories (a) and (b), each logical record is of the following general format:

```
BUSID 'model name' data list /
```

where:

BUSID	Is the bus identifier (number or extended bus name) of the bus at which this equipment model is to be placed (from bus for a line relay model), a dc line or FACTS device name (in single quotes), area, zone, or owner number, or zero.
'model name'	Is the name by which the model is referenced. The model name is limited to sixteen characters, and *must* be enclosed in single quotes.
data list	Specifies the constant parameters associated with the model. Generally, these parameters are specified in the order in which constants are listed on the data sheets in the blocks labeled ICONs and CONs.

For model category (c), each logical record is of the following general format:

```
MINS   'model name'  data list /
```

where:

MINS	Model instance value (a whole number greater than zero).
'model name'	Is the name by which the model is referenced. The model name is limited to sixteen characters, and *must* be enclosed in single quotes.
data list	Specifies the constant parameters associated with the model. Generally, these parameters are specified in the order in which constants are listed on the data sheets in the blocks labeled ICONs and CONs.

Data records for device models, protection models, and models of type "other" that are associated with plant, wind machine, and induction machine are specified in data records of the form:

```
BUSID 'model name' ID  data list /
```

where ID specifies the machine identifier and the other data items are as described above.

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Data records for device models, load relay (a.k.a protection) models, and models of type “other” that are associated with loads are specified in data records of the form:

```
BUSID 'model name' ID  data list /
```

where ID specifies the load identifier or a wildcard (*) designating all loads for BUSID. BUSID may be a bus, area, zone, or owner number.

Data records for relay models associated with branches and 2-winding transformers are specified in data records of the form:

```
BUSID 'model name' BUSID2 ID RS data list /
```

where:

BUSID2	Is the bus identifier (number or extended bus name) of the to bus.
ID	Is the circuit identifier.
RS	Is a relay slot, either 1 or 2. Two relays are allowed at each from bus location on a branch.

Data records for device models and models of “other” type associated with branches and 2-winding transformers are specified in data records of the form;

```
BUSID 'model name' BUSID2 ID data list /
```

Where, BUSID, BUSID2, and ID are as defined above for line relay models.

Data records for device models, protection models, and models of “other” type associated with 3-winding transformers are specified in data records of the form;

```
BUSID 'model name' BUSID2 BUSID3 ID data list /
```

Where, BUSID, BUSID2, and ID are as defined above for line relay models, and BUSID3 is the bus identifier (number or extended bus name) of the third bus.

Auxiliary-signal models can be associated with dc line models (two-terminal dc, multi-terminal dc, VSC dc) and FACTS device models. Data records for auxiliary-signal models are specified in the form:

```
BUSID   'model name'   DVTYP     SGNDX     data list  /
```

where:

DVTYP	Indicates the device type with which the auxiliary-signal is associated, and could be any of the following: 1 : for two-terminal dc line 2 : for multi-terminal dc line 3 : for VSC dc line 4 : for FACTS device
SGNDX	Is the signal injection point number. This is any number 1 through maximum number of auxiliary-signals that can be associated with the device type under consideration (refer to Table 3.1, “Standard Maximum PSS®E Program Capacities” for the maximum number of auxiliary signal injection points, i.e., the

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	maximum number of auxiliary-signals that can be attached to each device type).

Data records for device models, protection models, and "other" models associated with dc line (two-terminal dc, multi-terminal dc, and VSC dc), FACTS models, and switched shunt models are specified in the form:

```
BUSID   'model name'   data list  /
```

In the case of dc lines and FACTS models, BUSID refers to dc line and FACTS device name specified in single quotes. In the case of switched shunt models, BUSID refers to bus identifier.

The "Miscellaneous Other" category can be used for those categories of models that are not associated with any device or equipment in the power flow. Data records for such models are specified in the form:

```
MINS   'model name'  data list  /
```

where, MINS is the model instance, and is used to distinguish one "Miscellaneous Other" model from another "Miscellaneous other" model. The model instance can be any whole number greater than zero. There cannot be more than one "Miscellaneous Other" model with the same MINS value.

When specifying a bus, BUSID, BUSID2, and BUSID3 must specify a bus number when the *numbers* input option is in effect (refer to Section 3.3.3, "Program Run-Time Option Settings" and activity OPTN). Using the *names* input option specify either a bus number or the extended bus name enclosed in single quotes. When BUSID, BUSID2 and BUSID3 specify a number, it must be entered without the quotes.

The model name is limited to sixteen characters, and must *always* be enclosed in single quotes.

As with the majority of data input records in PSS®E, data is read by activity DYRE in free format with data items separated by a comma or one or more blanks. Each logical record may occupy several sequential lines in the data file and *must* be terminated with a slash. These logical records may be in any order within the data file.

The following paragraphs discuss details specific to each model type.

Device, Protection, and Other type models associated with machines

Generator Models

Generator models are classified as device models. Each in-service machine in the working case must have a generator device model data record specified for it. The list of generator device models, generator protection models, and generator that activity DYRE recognizes is provided in Generator Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the generator models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

It is to be noted that reactances, H and D are specified on MBASE base as contained in the working case (refer to Generator Data and Section 13.2.2, "Generator Apparent Impedance"). In the GENROE, GENROU, GENSAE, GENSAL, GENDCO and FRECHG models, the value specified for X'' must be identical to the imaginary part of ZSORCE as contained in the working case. For the GENCLS and GENTRA models, the imaginary component of ZSORCE must be identical to the transient reactance, X'. In the CIMTR1, CIMTR2, CIMTR3 and CIMTR4 models, a single cage induction machine has T'' and X'' set to zero and X' must be identical to the imaginary part of ZSORCE; a double cage machine has nonzero values for the subtransient data items and X'' must be identical to the imaginary component of ZSORCE.

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Current Compensating Models

Compensating models are classified as device models. Any machine may have a voltage regulator current compensating model specified for it. These models allow the voltage regulator to sense the voltage either at a remote bus or at a point separated from the machine terminals by a specified impedance which must be entered on MBASE base.

Activity DYRE recognizes the compensating models that are listed in Compensator Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the compensating models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Stabilizer and Excitation Limiter Models

Stabilizer and Excitation Limiter models are classified as device models. Any machine for which an appropriate generator/exciter model pair has been specified may have a power system stabilizer model, a minimum excitation limiter model, and/or a maximum excitation limiter model specified for it. These units are devices that inject supplementary signals into the voltage regulator units. Stabilizer signals are usually derived from either generator speed or generator power in such a way as to improve the damping of generator rotor angle swings. Minimum and maximum excitation limiters are relatively slow-acting protective devices which limit field voltage, EFD. The stabilizer model output, and the minimum and maximum excitation limiter model outputs are accessed as VOTHSG, VUEL and VOEL inputs respectively by the excitation system model.

The stabilizer and the excitation limiter models that are recognized by activity DYRE are provided in Stabilizer Model Data Sheets; Minimum Excitation Limiter Model Data Sheets; and Maximum Excitation Limiter Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the stabilizer and excitation limiter models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

In stabilizer models that require a remote bus ICON to be specified, the remote bus ICON must be specified as zero if a local signal is used.

Excitation System Models

Excitation system models are classified as device models. Any machine which is modeled by one of the generator models GENROE, GENROU, GENSAE, GENSAL, GENDCO, GENTRA, CGEN1 or FRECHG may have an excitation system model specified for it.

Activity DYRE recognizes the exciter models that are listed in Excitation System Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the excitation system models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Turbine Governor Models

Turbine governor models are classified as device models. Any machine which is modeled by one of the generator models GENROE, GENROU, GENSAE, GENSAL, GENDCO, GENTRA, CIMTR1, CIMTR2, CIMTR3, CIMTR4 or CGEN1 may have a turbine governor model specified for it.

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The list of governor models that are recognized by activity DYRE is provided in Turbine-Governor Model Data Sheets of the *PSS®E Model Library*.

For those governor models for which the turbine rating cannot be specified, the turbine governor data would have to be specified on the base of its generating unit (i.e., on MBASE base).

Data records specifying most of the turbine governor models should have, following the model name, the identifier of the machine to which the model is being assigned, the ICON data (if applicable), and CON data in the same order as they appear in the respective model data sheets.

For the cross compound models, BUSID and ID specify the high pressure unit. ID is followed by the bus and machine identifiers of the low pressure unit. The remaining data items are the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

For the IEEE Type 1 models, IEEEG1, TGOV5 and WSIEG1, BUSID and ID specify the machine at which the model is to be called. If a cross compound is to be modeled, these specify the high pressure unit, and ID is followed by the bus and machine identifiers of the low pressure unit; otherwise two data items of zero must be specified following ID.

Turbine Load Controller Models

Turbine load controller models are classified as device models. Any machine which has a turbine governor model may have a turbine load controller model specified for it. The function of these models is to maintain turbine power at a set value by adjusting the turbine-governor reference.

The list of turbine load controller models that are recognized by activity DYRE is provided in Turbine Load Controller Model Data Sheets of the *PSS®E Model Library*.

For those governor models for which the turbine rating cannot be specified, the turbine load controller data would have to be specified on the base of its generating unit (i.e., on MBASE base).

Data records specifying the turbine load controller models should have, following the model name, the identifier of the machine to which the model is being assigned, the ICON data (if applicable), and CON data in the same order as they appear in the respective model data sheets.

Machine and Wind machine Protection Models

Any machine (including wind machines) may have a protection model specified for it. Activity DYRE recognizes the machine protection models that are listed in the Machine protection Model chapter of the *PSS®E Model Library*.

Data records specifying the machine protection models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Load Characteristic Models

Load characteristic models are classified as device models. Any load in the working case may have a load characteristic model data record specified for it.

The list of load characteristic models that are recognized by activity DYRE is provided in Load Characteristic Model Data Sheets of the *PSS®E Model Library*.

Data records specifying each of the above models should have the form:

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```
BUSID  'model name'  ID  data list /
```

where:

BUSID	Bus number or name for type BL models. Owner number for type OW models. Zone number for type ZN models. Area number for type AR models. 0 for type AL models.
ID	Load identifier or * to specify the model for all loads at BUSID.
data list	As specified on the model data sheets.

Load Relay (a.k.a, load protection) Models

Any load in the working case may have a load relay model data record specified for it.

The list of load relay models that are recognized by activity DYRE is provided in Load Relay Model Data Sheets of the *PSS®E Model Library*.

Data records specifying each of the above models should have the form:

```
BUSID  'model name'  ID  data list /
```

where:

BUSID	Bus number or name for type BL models. Owner number for type OW models. Zone number for type ZN models. Area number for type AR models. 0 for type AL models.
ID	Load identifier or * to specify the model for all loads at BUSID.
data list	As specified on the model data sheets.

Branch and 2-winding transformer device models

Any branch or a 2-winding transformer in the network may have a device model attached to it. The device models for branches that are recognized by activity DYRE are provided in the Branch Device Model chapter of the *PSS®E Model Library*, while the device models for 2-winding transformer that are recognized by activity DYRE are provided in the Branch Device Model chapter of the *PSS®E Model Library*.

Data records specifying each of the above models should have the form:

```
BUSID 'model name' BUSID2  ID  data list /
```

where:

BUSID	Is the branch from bus number or name; this is the end of the line at which the relay model is placed.
BUSID2	Is the branch to bus number or name.
ID	Is the branch circuit identifier.
data list	As specified on the model data sheets.

The relay slot number (RS) must be either 1 or 2. Up to two models per branch per from bus, or four models per branch, can be accommodated.

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Branch Relay and 2-winding transformer protection Models

Any branch in the network may have up to two line relay models at each end. A relay models is placed at the from bus end of the line, as specified on the model data record. The line relay models that are recognized by activity DYRE are provided in Line Relay Model Data Sheets of the *PSS®E Model Library*.

Data records specifying each of the above models should have the form:

```
BUSID 'model name' BUSID2  ID  RS  data list /
```

where:

BUSID	Is the branch from bus number or name; this is the end of the line at which the relay model is placed.
BUSID2	Is the branch to bus number or name.
ID	Is the branch circuit identifier.
RS	Is the relay slot number.
data list	As specified on the model data sheets.

The relay slot number (RS) must be either 1 or 2. Up to two models per branch per from bus, or four models per branch, can be accommodated.

3-winding transformer device models

Any 3-winding transformer in the network may have a device model attached to it. The device models for branches that are recognized by activity DYRE are provided in the Model Library.

Data records specifying each of the above models should have the form:

```
BUSID   'model name'   BUSID2     BUSID3   ID data list  /
```

where:

BUSID	Is the branch from bus number or name; this is the end of the line at which the relay model is placed.
BUSID2	Is the branch to bus number or name.
BUSID3	Is the 3-winding transformer third bus
ID	Is the branch circuit identifier.
data list	As specified on the model data sheets.

Auxiliary-Signal Models

Activity DYRE recognizes the auxiliary-signal models that are provided in Auxiliary-Signal Model Data Sheets of the *PSS®E Model Library*:

With the exception of models RBKELT and SQBAUT, all other auxiliary-signal models can be associated with dc line models (2-terminal dc, multi-terminal dc, VSC dc) and FACTS device models. RBKELT and SQBAUT can be associated only with 2-terminal dc line models.

Data records for auxiliary-signal models are specified in the form:

```
BUSID   'model name'   DVTYP     SGNDX     data list  /
```

where:

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DVTYP	Indicates the device type with which the auxiliary-signal is associated, and could be any of the following: 1 : for two-terminal dc line 2 : for multi-terminal dc line 3 : for VSC dc line 4 : for FACTS device
SGNDX	Is the signal injection point number. This is any number 1 through maximum number of auxiliary-signals that can be associated with the device type under consideration (refer to Table 3.1, "Standard Maximum PSS®E Program Capacities" for the maximum number of auxiliary signal injection points, i.e., the maximum number of auxiliary-signals that can be attached to each device type).

Each of the above "tableized" auxiliary-signal models calculates the model output and stores it in the array VAR(L), where L is the starting VAR index for that auxiliary-signal model. Subsequently, depending on the type of device with which the auxiliary-signal model is associated with, the model output contained in VAR(L) is then transferred by PSS®E into one of the PSS®E standard arrays, DC2SIG or DCMSIG or VSCSIG or FCTSIG. The array locations in DC2SIG, DCMSIG, VSCSIG, and FCTSIG into which VAR(L) is transferred, corresponds to the internal device index, and the signal index for which the auxiliary-signal model is called.

DC Line Device Models

DC lines could be one of the following: two-terminal, multi-terminal or VSC dc.

The list of dc two-terminal, multi-terminal, and VSC dc line models recognized by activity DYRE is provided in Two-Terminal dc Line Model Data Sheets; Multi-Terminal dc Line Model Data Sheets; and VSC dc Line Model Data Sheets of the *PSS®E Model Library*.

Data records for dc line models are specified in the form:

```
BUSID   'model name'   data list  /
```

where:

BUSID	Is the dc line name (specified in single quotes).

2-Terminal DC Line other Models

The list of dc two-terminal "other" type models recognized by activity DYRE is provided in the section on 2-terminal dc found in the Other Models chapter of the *PSS®E Model Library*.

Data records for dc line models are specified in the form:

```
BUSID   'model name'   data list  /
```

where:

BUSID	Is the dc line name (specified in single quotes).

FACTS Device Model

The list of FACTS device models recognized by activity DYRE is provided in FACTS Device Model Data Sheets of the *PSS®E Model Library*.

Data records for FACTS device models are specified in the form:

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```
BUSID   'model name'   data list  /
```

where:

BUSID	Is the FACTS device name (specified in single quotes).

Wind Generator Models

Each in-service wind-machine in the working case must have a wind generator model data record specified for it. The list of wind generator models that activity DYRE recognizes is provided in Generic Wind Generator Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the wind-generator models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Wind Electrical Control Models

Any wind machine may have an electrical control model specified for it. These -models control the active and reactive power to be delivered to the system. The list of wind electrical control models recognized by activity DYRE is listed in Generic Wind Electrical Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the wind electrical control models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Wind Turbine Models

Any wind machine may have a turbine model specified for it. These -models calculate the mechanical power and speed of the wind turbine. The list of wind turbine models recognized by activity DYRE is listed in Generic Wind Mechanical Model Data Sheets of the *PSS®E Model Library*.

Data records specifying the wind turbine models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Wind Pitch Control Models

Any wind machine may have a pitch control model specified for it. These models control the blade pitch angle using the rotor speed and torque command input signals.

Activity DYRE recognizes the pitch control models that are listed in Generic Wind Pitch Control Model Data Sheetsof the *PSS®E Model Library*.

Data records specifying the pitch control models should have, following the model name, the identifier of the machine to which the model is being assigned, followed by the ICON data (if applicable) and CON data in the same order as they appear in the respective model data sheets.

Switched Shunt Models

Any device that is modeled as a switched shunt in PSS ®E power flow can have a switched shunt dynamic model specified for it.

The list of switched shunt models that are recognized by activity DYRE is provided in Switched Shunt Model Data Sheets of the *PSS®E Model Library*.

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Data records specifying switched models should have, following the model name, the ID of the switched shunt followed by the ICON data (if applicable), and CON data in the same order as they appear in the respective model data sheets.

Miscellaneous “other” Models

The list of miscellaneous “other” models that are recognized by activity DYRE is provided in Miscellaneous and other Models chapter of the *PSS®E Model Library*.

Being unattached models, in principle these models are similar to the CONEC and CONET type models. However the miscellaneous “other” models unlike the CONEC and CONET type models will not generate any calls in the CONEC and CONET subroutines.

Data records specifying switched models should have, following the model instance value “MINS” (which is any unique whole number greater than zero), followed by the model name, the model the ICON data (if applicable), and CON data in the same order as they appear in the respective model data sheets.

CONEC and CONET Models

The PSS®E model library contains a group of models that are presently called in CONEC and/or CONET. Some of these models are used to model the dynamic characteristics of equipment, while others pertain to electric network rather than the dynamic characteristics of the equipment. These models (called CONEC and CONET models) are listed in Chapter 23, CONEC and CONET Model Data Sheets of the *PSS®E Model Library*.

The ICON and CON lists contain the constant data values and are specified in the same order as they appear in the ICON and CON blocks on the model data sheets.

User-Written Models

In addition to processing data records for most of the models contained in the PSS®E model library, activity DYRE provides for the processing of records specifying data for user-written models (refer to Chapter 25, *Model Writing* for model writing details and Chapter 23, *Dynamic Simulation Setup and Procedures* for information on the PSS®E dynamic simulation structure). For such data records to be handled correctly by activity DYRE:

1. The special formats described below must be used.
2. The models must be written so as to be consistent with the calling sequences assumed (refer to Section 25.4, “Model Calling Sequence Rules”).

Data records for load characteristic or load relay user-written models must be in the following format:

```
BUSID ’USRLOD’ ID ’model name’ IC IT NI NC NS NV NRI data list /
```

Data records for load “other” type user-written models must be in the following format:

```
BUSID ’USRLOD’ ID ’model name’ IC IT NI NC NS NV data list /
```

Data records for branch device and branch “other” user-written models must be in the following format:

```
BUSID ’USRBRN’ BUSID2 ID ’model name’ IC IT NI NC NS NV data list /
```

Data records for 2-winding transformer device and 2-winding transformer “other” user-written models must be in the following format:

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```
BUSID 'USR2WT' BUSID2 ID 'model name' IC IT NI NC NS NV data list /
```

Data records for line (branch and 2-winding transformer) relay user-written models must be in the following format:

```
BUSID 'USRREL' BUSID2 ID RS 'model name' IC IT NI NC NS NV NRI data list /
```

Data records for 3-winding transformer device, and 3-winding transformer "other", and 3-winding transformer protection type user-written models must be in the following format:

```
BUSID 'USR3WT' BUSID2 BUSID3 ID 'model name' IC IT NI NC NS NV data list /
```

Data records for bus connected protection and "other" type user-written models must be in the following format:

```
BUSID  'USRBUS'  'model name' IC IT NI NC NS NV data list /
```

Data records for user-written auxiliary-signal models must be specified in the following format:

```
BUSID 'USRAUX' DVTYP SGNDX 'model name' IC IT NI NC NS NV data list /
```

Data records for user-written dc line (two-terminal dc, multi-terminal dc and VSC dc) device models, protection models, and model of type "other" are specified in the form:

```
BUSID 'USRDCL' 'model name' IC IT NI NC NS NV data list /
```

Data records for user-written FACTS device models, protection models, and model of type "other" are specified in the form:

```
BUSID 'USRFCT' 'model name' IC IT NI NC NS NV data list /
```

Data records for user-written switched shunt device models, protection models, and model of type "other" are specified in the form:

```
BUSID 'USRSWS2'  ID 'model name' IC IT NI NC NS NV data list /
```

User-written switched shunt related models with the keyword 'USRSWS' instead of 'USRSWS2' will still be read, but the model in this case will be applied to the first switched shunt at that bus

Data records for user-written induction machine device, protection and "other" type models must be in the following format:

```
BUSID  'USRIMC'  ID  'model name' IC IT NI NC NS NV data list /
```

Data records for user-written machine (including wind machine) device, protection and "other" type models must be in the following format:

```
BUSID  'USRMDL'  ID  'model name' IC IT NI NC NS NV data list /
```

Data records for user-written miscellaneous "other" type models (these are unattached models) must be in the following format:

```
MINS  'USRMSC'  'model name' IC IT NI NC NS NV  data list /
```

Data records for all other types of user-written models must be in the following format:

```
BUSID 'USRMDL' ID 'model name' IC IT NI NC NS NV data list /
```


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where BUSID, BUSID2,ID, RS, MINS, and < model name > are as described at the beginning of Section 16.1.1, "Dynamics Model Raw Data File Contents".The dynamics data input activity DYRE reads simulation model data from a Dynamics Model Raw Data File into the dynamics data arrays in PSS®E working memory., and:

IC	Is the user model type code: 0 = Model called from subroutine CONET and not called from subroutine CONEC. 1 = Generator model. 2 = Current compensating model. 3 = Stabilizer model. 4 = Excitation system model. 5 = Turbine-governor device model. 7 = Two-terminal dc line model in PSS®E that is not yet in table-driven form (this is obsolete and should not be used). 8 = Other subroutine CONEC model; place before RETURN statement in CONEC. 9 = Minimum excitation limiter model. 10 = Maximum excitation limiter model. 11 = Cross-compound turbine-governor model. 12 = Load characteristic model. 13 = Load relay (protection) model. 14 = Line relay (protection) model. 17 = Auxiliary signal model. 18 = Two-terminal dc line model. 19 = Multi-terminal dc line model. 20 = VSC (Voltage Source Converter) dc line model.
	21 = FACTS device model. 23 = Turbine load controller model. 24 = Switched Shunt model. 25 = Induction machine model 101 = Renewable generator model.

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	102 = Renewable electrical control model. 103 = Renewable turbine (mechanical) model. 104 = Renewable pitch control model. 105 = Renewable aerodynamic model. 106 = Renewable gust model. 107 = Renewable auxiliary control model. 108 = Renewable torque control model. 301 = Branch device model 302 = 2-Winding transformer device model 303 = 3-Winding transformer device model 403 = 3-Winding transformer protection model 404 = Bus protection model (i.e., protection model attached to bus) 405 = Machine (including wind machine) protection model 406 = 2-Terminal dc line protection model 407 = N-Terminal dc line protection model 408 = VSC dc line protection model 409 = FACTS device protection model 410 = Switched shunt protection model 413 = Induction machine protection model 501 = Branch “other” model 502 = 2-Winding transformer “other” model 303 = 3-Winding transformer device model 503 = 3-Winding transformer “other” model 504 = Bus “other” model (i.e., “other” type model attached to bus) 505 = Machine (including wind machine) “other” type model 506 = 2-Terminal dc line “other” model

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	507 = N-Terminal dc line “other” model 508 = VSC dc line “other” model 509 = FACTS device “other” model 510 = Switched shunt “other” model
	511 = Load “other” model 512 = Miscellaneous “other” model (the model instance, MINS value, is required only for this category of models) 513 = Induction Machine “other” model
IT	For user models that are called from CONEC and/or CONET (i.e., when IC is 0, 7, or 8), IT could be 0, 1 or 2: 0 = not a current injection or metering model 1 = current injection model, place before IFLAG test in subroutine CONET 2 = metering model, place after IFLAG test in subroutine CONET For user models with IC values of 1 through 5, or 9 through 14, 23, 17 through 21, or 101 through 107, the IT value depends on the value of IC. When IC is 1 through 5, 9 through 11, 23, 17 through 21, or 101 through 107, IT could be 0 (not a current injection model) or 1 (current injection model). When IC is either 12 or 13 (i.e., user written load characteristic or load relay model), IT represents the subsystem type for the load-related model and could be any value 1 through 5: 1 = Bus-type 2 = Owner-type 3 = Zone-type 4 = Area-type 5 = All-type When IC is 14 (line relay user model), then IT indicates whether the relay is a supervisory relay: 0 = Not supervisory 1 = Supervisory
	When IC is 24, IT could be 1 or 2:

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	1 = current injection model 2 = metering model When IC is 25, IT should be 1 (indicating it is a current injecting type model) When IC is 301 through 303, IT could be any value 0 through 2 When IC is 403 through 410, or when IC is 413, IT should be 2 When IC is 501 through 513, IT could be any value 0 through 2
NI	Is the number of ICONs used by the model (NI ≤ 500).
NC	Is the number of CONs used by the model (NC ≤ 1000).
NS	Is the number of STATEs used by the model.
NV	Is the number of VARs used by the model.
NRI	Is the number of additional (reserved) ICONs used by the model.
data list	Specifies NI ICONs, followed by NC CONs.

In specifying ICONs, character ICONs (e.g., circuit identifiers) may be specified; they *must* be enclosed in single quotes.

The 'model name' is limited to sixteen characters. This string is employed as a subroutine name within the connection subroutines generated by activity DYRE (refer to Section 16.1.2, "Operation of Activity DYRE".For CONET models which are not coordinated call models (i.e., IC is zero and IT is either one or two), BUSID and IM are both specified as zero.) so it must conform to FORTRAN conventions for a subroutine name.

The following paragraphs discuss details specific to various model types. The CALL statements for these models are generated as defined in Model Calling Sequence Rules.

Machine Related Models

For plant-related model records (i.e., IC values of one through five and nine through eleven), BUSID and IM designate the bus and machine identifiers respectively. Except as described below, all machine related models must have a zero specified for IT.

For user-written cross-compound governor model records (i.e., IC values of eleven), BUSID and IM specify the high-pressure unit. NV is followed by the bus and machine identifiers of the low pressure unit. The remaining data items are the model constants (NI ICONs followed by NC CONs).

Machine-related user-written models are called at those phases of the dynamic simulation calculation concerned with the handling of differential equations. Certain generator models such as static var systems also involve a voltage sensitive current injection during the network balance. For such models:

1. IC and IT must be specified as one.
2. The model must contain a supplementary ENTRY point with the first character of the model (SUBROUTINE) name changed to a T.

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3. The model is called at its principal entry point for simulation calculations involving the differential equations, and at its supplementary entry point during the current injection calculation of the network solution.

The implementation of this type of modeling requirement is referred to as a coordinated call model.

Load-Related Models

Load-related model records have an IC value of 12 or 13 and must have an IT value of 1 through 5.

The value of IT determines the type of subsystem for which the model applies, determines the nature of BUSID and imposes constraints on the user 'model name' for the data record:

Table 16.1. Load-Related Model Subsystem Types

IT	TYPE	BUSID	Model Name	IL
1	Bus	Bus Number	'xxxxBL'	Load ID or *
2	Owner	Owner Number	'xxxxOW'	Load ID or *
3	Zone	Zone Number	'xxxxZN'	Load ID or *
4	Area	Area Number	'xxxxAR'	Load ID or *
5	All		'xxxxAL'	Load ID or *

The model name must be sixteen characters with the last two characters specifying the model type (e.g., BL for Bus type model). IL may be specified as a specific load identifier for which the model is to be applied, or a wildcard (*) indicating that the model is to be applied to all loads in BUSID.

All load-characteristic-user-written models must be set up as coordinated call models. Each load-characteristic model will result in a subroutine call to 'model name' and a call to a supplementary entry point with the first character of 'model name' changed to a T. For example, the model which employs a 'model name' of MYLDBL will result in a calls to MYLDBL and TYLDBL. The 'model name' should preclude an initial character of T to ensure that each subroutine reference is unique.

The NRI field of the user-written model data record identifies the number of reserved ICONs employed by the model. A single data record for a load-related model may generate multiple instances of the model's application. The ICON and CON data presented on the data record will be common to all application instances and will be stored in the data arrays only once. Any VAR, STATE and reserved ICON requirements will apply to each instance. An allocation of the required VARs, STATEs and reserved ICONs will be made in the data arrays for each application instance.

Line Relay Models

Line relay model records have an IC value of 14. The IT value must be 1 to indicate that it is a relay acting in a supervisory role, or 0 if it is not a supervisory relay.

The BUSID value indicates the branch from bus where the relay is located and BUSID2 indicates the branch to bus. ID is the branch circuit identifier. Two relays are allowed at each from bus. The user must enter a 1 or 2 for the relay slot (RS).

Line relay user-written models are called from subroutine USRREL.

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Auxiliary Signal Models

Auxiliary-signal user-model records have an IC value of 17. The IT value must be zero to indicate that it is not a current injection model.

Auxiliary-signal models can be associated with dc line models (two-terminal dc, multi-terminal dc, VSC dc) and FACTS device models. The BUSID in the dynamic data record is the device name in single quotes with which the auxiliary-signal is associated. DVTYP indicates the type of the device with which the auxiliary-signal is associated (1 for 2-terminal dc line, 2 for multi-terminal dc line, 3 for VSC dc line, 4 for FACTS device). SGNDX is the auxiliary-signal index. This is any number 1 through maximum number of auxiliary-signals that can be associated with the device type under consideration (refer to Table 3-1 for the maximum number of auxiliary-signals that can be attached to each device type).

The user-written auxiliary-signal model should calculate the model output and store it in VAR(L), where L is the starting VAR index for that auxiliary-signal model. Subsequently, depending on the type of device with which the auxiliary-signal model is associated, the model output contained in VAR(L) is then transferred by PSS®E into one of the PSS®E standard arrays, DC2SIG or DCMSIG or VSCSIG or FCTSIG. The array locations in DC2SIG, DCMSIG, VSCSIG, and FCTSIG into which VAR(L) is transferred, corresponds to the internal device index and the signal index for which the auxiliary-signal model is called.

DC Line Models

DC line user-model records have an IC value of 18 for two-terminal dc lines, 19 for multi-terminal dc lines, and 20 for VSC dc lines. The IT value must be 1 to indicate that it is a current injection model. The BUSID is the dc-line line name in single quotes.

All dc line user-written models must be set up as coordinated call models. Each dc line model will result in a subroutine call to 'model name' and a call to a supplementary entry point with the first character of the 'model name' changed to a T. The model is called at its principal entry point for simulation calculations involving the differential equations, and its supplementary entry point during the current injection calculation of the network.

FACTS Device Models

FACTS device user-model records have an IC value of 21. The IT value must be 1 to indicate that it is a current injection model. The BUSID indicates the FACTS device name in single quotes.

All FACTS device user-written models must be set up as coordinated call models. Each FACTS device model will result in a subroutine call to 'model name' and a call to a supplementary entry point with the first character of the 'model name' changed to a T. The model is called at its principal entry point for simulation calculations involving the differential equations, and its supplementary entry point during the current injection calculation of the network.

Wind Machine Related Models

For wind machine model records (i.e., IC values of 101 through 107), BUSID and IM designate the bus and machine identifiers respectively. Except for the wind generator models, all other wind machine related models must have a zero specified for IT.

Wind machine related user-written models are called at those phases of the dynamic simulation calculation concerned with the handling of differential equations. Wind generator models also involve a voltage sensitive current injection during the network balance. For such models:

1. IC and IT must be specified as one.

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2. The model must contain a supplementary ENTRY point with the first character of the model (SUBROUTINE) name changed to a T.

3. The model is called at its principal entry point for simulation calculations involving the differential equations, and at its supplementary entry point during the current injection calculation of the network solution.

Switched Shunt Models

Switched shunt user-model records have an IC value of 24. The IT value could be 1 (for current injecting models), or could be 2 (metering model that turns shunts ON/OFF).

The BUSID indicates the bus at which the switched shunt is connected.

User-written switched shunt model with IT=1 (current injecting type), will be set up as a coordinated call model. Such switched shunt models will result in a subroutine call to 'model name' and a call to a supplementary entry point with the first character of the 'model name' changed to a T. The model is called at its principal entry point for simulation calculations involving the differential equations, and its supplementary entry point during the current injection calculation of the network.

User written switched shunt models with IT=2 are called after the dynamic simulation network solution is converged.

CONEC Models

Those CONEC models with an IC value of eight may be implemented either as simple CONEC models (IT is specified as zero) or as coordinated call model pairs (IT is specified as one). In the latter case, model calls are generated in both subroutines CONEC and CONET, with the first character of the model name changed to a T in the CONET model call.

CONET Models

For CONET models which are not coordinated call models (i.e., IC is zero and IT is either one or two), BUSID and IM are both specified as zero.

16.1.2. Operation of Activity DYRE

In order to conduct the time domain simulation it is necessary to associate (connect) dynamic equipment and other models with power flow network equipment (i.e. buses, machines, loads, etc.). The dynamic models are resolved either in the PSS®E simulation model library or by user supplied user-written model code. A programmatic association is made for those models which are resolved in the PSS®E simulation model library and which represent generator plant related equipment, load characteristics, load relays, branch relays, auxiliary signals, dc lines, FACTS device, switched shunt, and some monitoring functions, and for user supplied models of plant related equipment, load characteristics, load relays, branch relays, auxiliary signals, dc lines, FACTS device, switched shunt, and induction machine models.

Activity DYRE performs these major functions as it reads through the Dynamics Model Raw Data File:

1. It assigns storage locations in the various dynamics data storage arrays for each of the models referenced (refer to Section 23.1.2, “Data Structure”).

2. It initializes the constant parameters in these data arrays for each model referenced in the data file.

3. It creates tables of pointers for programmatic associations of dynamic models with network equipment where appropriate.

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4. If required, it constructs the connection subroutines (refer to Section 23.4, "The Dynamic Model").

Activity DYRE requires that the network corresponding to the dynamics data being read is present in the working case. It further requires that the in-service machines in the working case contain the proper specification of the machine attributes MBASE, ZSORCE, XTRAN and GENTAP (refer to Generator Data), and that they be represented as current sources (i.e., with activity section / CONG having been executed.

The user specifies the Dynamics Model Raw Data File which contains data records in the form given in Section 16.1.1, "Dynamics Model Raw Data File Contents".The dynamics data input activity DYRE reads simulation model data from a Dynamics Model Raw Data File into the dynamics data arrays in PSS®E working memory.

The user also specifies the files into which activity DYRE is to place the connection subroutines. Refer to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity DYRE. If the user does not provide filenames for either CONEC or CONET files, an image of the corresponding subroutine's CALL statements is written to the user's terminal rather than to a file.

After the required files are read into the working case, activity DYRE provides the following information:

```
NEXT AVAILABLE ADDRESSES ARE:
   CON  STATE    VAR   ICON
   nnn    nnn    nnn    nnn
```

The values < nnn > indicate the next available locations in the corresponding dynamics data arrays. When PSS®E is started up, these indices are set to one and are subsequently updated by activities DYRE, CHAN, and CHSB. They may be changed by the user via activity ALTR using the category of solution parameter changes. These indices are preserved with dynamics working memory as it is saved and retrieved with activities SNAP and RSTR respectively.

When constructing a new dynamics setup, the normal entry is 1 for each of the starting indices. When activity DYRE is used in its *add models* mode (refer to Section 16.1.3, "Adding Models"), the usual procedure is to let activity DYRE assign locations in the dynamics data arrays for the models being added beginning at the next unused slots.

Activity DYRE generates an alarm for any model data record for which either too many, or not enough, data items have been specified. Upon completing its pass through the Dynamics Model Raw Data File, activity DYRE builds the connection subroutines and deposits values in locations in the dynamics data constant arrays.

As it completes each of the model types described above, a summary of the storage locations allocated in the various data storage arrays is printed. A single similar summary is printed for the model types described in CONEC and CONET Models. These are followed by a summary of the number of model references read from the Dynamics Model Raw Data File. These tabulations are printed at the *Progress* tab.

Activity DYRE then prints a final summary of the storage locations used. This is printed at the *Progress* tab; it is also printed at the dialog output device if these two output destinations are different. The next available indices are updated to reflect these dynamics data array allocations.

Activity DYRE summarizes the status of the connection subroutines generated by printing at the dialog output device, one of the messages as explained below.

If the case has user models for which compilation is required, and if no model calls were generated in the connection routines CONEC and CONET, the message generated at the dialog output device would be:

```
COMPILE USER MODELS, & CREATE USER DLL
```


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If there are user models in the case, and if model calls were generated only in CONEC, the message generated at the dialog output device would be:

```
COMPILE CONEC, USER MODELS, & CREATE USER
DLL
```

If there are user models in the case, and if model calls were generated only in CONET, the message generated at the dialog output device would be:

```
COMPILE CONET, USER MODELS, & CREATE USER
DLL
```

If there are user models in the case, and if model calls were generated both in CONEC and in CONET, the message generated at the dialog output device would be:

```
COMPILE CONEC, CONET, AND USER MODELS, & CREATE
USER DLL
```

If the case does not have any user models for which compilation is required, the following message is generated at the dialog output device:

```
NO USER WRITTEN MODELS IN CASE. COMPILATION
NOT REQUIRED
```

If the user specifies filename for a Compile File, unnecessary if CONEC and CONET filenames were not read into the working case, activity DYRE creates a file which, when executed at operating system level, compiles the connection subroutines just constructed. The use of this feature is strongly encouraged because it guarantees the correct specification of compiler option settings in compiling these subroutines. A response of zero to the above instruction suppresses the generation of the compiling file.

Those data items identified as BUSID in the data record formats described in Section 16.1.1, "Dynamics Model Raw Data File Contents" must specify bus numbers when the *numbers* input option is in effect. Using the *names* input option specify either a bus number or an extended bus name enclosed in single quotes. In specifying ICONs in the data list portion of data records, any ICON in which a bus is to be specified must contain a bus number regardless of the bus input option setting.

Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity DYRE.

16.1.3. Adding Models

Activity DYRE may be operated in an *add models* mode. The usual procedure in adding models to an existing (or partially completed) system model is as follows:

1. Initiate PSS®E.
2. Use activity
3. to restore RSTR the Snapshot containing the dynamics data of the existing system model.
4. Pick up the network model corresponding to the Snapshot accessed in (2).
5. Run activity DYRE,ADD or navigate *File > Open > Files of type: Add Dynamics Model Data (*.dyr)* from the GUI menu.
6. Take a Snapshot with activity

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7. Exit PSS®E with activity section / STOP.

8. If there are model calls in the connection routines CONEC and/or CONET, merge the model calls generated in (4) with the original versions of these connection subroutines.

Compile the resulting subroutines and link the connection subroutines into PSS®E in the standard manner (refer to Sections Section 4.10, "Loading the Connection Subroutines" and Section 23.4.4, "Program Set-Up").

If no additional model calls are generated in the connection subroutine additions files, steps 6 through 8 are not required (see below).

After specifying the appropriate Dynamics Model Raw Data File to activity DYRE in (4) above, the user enters the names of the files into which activity DYRE is to place any new connection subroutine model calls.

The user *should not* specify the pair of files containing the versions of these subroutines corresponding to the existing system model; activity DYRE does not append to the specified files but *overwrites* the previous contents of existing files. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity DYRE.

The normal response to the request for starting data array indices (refer to Section 16.1.2, "Operation of Activity DYRE") is the default, after which activity DYRE uses the next available locations as contained in the Snapshot restored in (2) above.

In the *add models* mode, activity DYRE summarizes the status of the connection subroutines by printing one of the following messages as explained below.

If the DYRE addition has user models for which compilation is required, and if no model calls were generated in the connection routines CONEC and CONET, the message generated at the dialog output device would be:

```
CONEC & CONET (IF PRESENT) ARE UNCHANGED.
COMPILE USER MODELS, AND CONEC & CONET (IF PRESENT), & CREATE
DLL
```

If the DYRE addition generates user model calls only in CONEC, the message generated at the dialog output device would be:

```
EDIT NEW CALLS INTO CONEC. CONET (IF PRESENT)
IS UNCHANGED. THEN COMPILE CONEC, USER MODELS, AND CONET (IF PRESENT), & CREATE
USER DLL
```

If the DYRE addition generates user model calls only in CONET, the message generated at the dialog output device would be:

```
EDIT NEW CALLS INTO CONET. CONEC (IF PRESENT)
IS UNCHANGED. THEN COMPILE CONET, USER MODELS, AND CONEC (IF PRESENT), & CREATE
USER DLL
```

If the DYRE addition generates user model calls in CONEC as well as CONET, the message generated at the dialog output device would be:

```
EDIT NEW CALLS INTO CONEC & CONET. THEN
COMPILE CONEC, CONET, AND USER MODELS, & CREATE USER DLL
```

If the DYRE addition does not have any user model for which compilation is required, the following message is generated at the dialog output device:

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```
CONEC & CONET (IF PRESENT) ARE UNCHANGED
```

The generation of a new compiling file is suppressed in the *add models* mode. In (7) above, the usual approach is to store the merged versions of the connection subroutines with their original filenames rather than with the filenames specified to activity DYRE in (4) above. In this case, the original compiling file may be used. If filenames other than the original filenames are used in (7) above, the original compiling file may be modified as appropriate with the text editor.

In the *add models* mode, the CONEC and CONET subroutine additions files constructed are not in FORTRAN subroutine form. Rather, they consist of only the CALL statements and are set up to be conveniently merged, via the text editor, with the original versions of these two subroutines.

When replacing a model called from internal tables (e.g., changing the excitation system model from a SEXS to an IEEET1), activity DYRE removes references to the old model from the dynamics model connection tables before adding the new model reference to these data tables. A message is printed at the *Progress* tab each time a model reference is deleted. When replacing a model called from subroutine CONEC or CONET, the user must remove the old model call from the connection subroutines when merging the original versions of these subroutines with the newly generated model subroutine calls.

16.1.4. Subsystem Models

Load characteristic and load relay models may be introduced by subsystem. Subsystems can be designated as: bus, owner, zone, area, or all. The presentation of a single data record for a subsystem model can generate multiple instances of the model's application. For example, presenting a bus specific model with a wildcard element identifier (e.g., a load-related model with IL='*') generates an instance of the model for each load connected to the bus. Similarly, presenting an area specific subsystem model with a specific identifier (e.g., a load-related model with IL='1') generates as many instances of the model as there are loads within the area which are assigned that identifier.

Precedence is an issue unique to subsystem models. The data presentation for subsystem models may involve intersection among the subsystems. The data may contain a presentation for both a bus-specific model and an area-specific model, and the bus associated with the first model may be part of the area associated with the second. Therefore a precedence order is provided to accommodate subsystem models. The decreasing order of precedence for resolving conflicts is from the most specific to most general subsystem: bus, owner, zone, area, and all. A model applied to a bus takes precedence over all conflicting models applied to any other subsystem in which the bus is located. A model applied to an owner takes precedence over any conflicting model applied to a zone, area, or all buses.

A precedence order is provided to accommodate conflicts in element identifier as well. A data record which introduces a subsystem model with a specific element identifier will take precedence over a subsystem model with a wildcard ('*') identifier.

Models employing programmatic association with network elements may be bypassed. This includes all models except those connected to the network by subroutine calls within the CONEC and/or CONET subroutines. When the model is bypassed, its function is excluded from the time domain simulation without extracting its reference or data from the simulation model. The model may be restored to an active state through program dialog.

A subsystem model can be bypassed at both the subsystem instance and the element instance. When a model is bypassed at the subsystem instance the model's function is excluded for each element to which the subsystem model applies. When the subsystem model is bypassed at the element instance, the model's function is excluded for only that element, and the function will be invoked for all other active elements to which the subsystem model applies.

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A bypassed model remains connected to the element. The bypassed model is not superseded by another active model which could potentially replace it due to precedence. However, if the model is removed, the next subsystem model in the precedence order automatically connects to the element. For example, if a Bus Subsystem model with a specific identifier (e.g.'1') were to be bypassed, another Bus Subsystem model at the same bus with a wildcard (*) identifier does not replace it. If the element specific model were removed, then the wildcard model applies.

16.1.5. Application Notes

The processing of activity DYRE results in three related but separate sets of data needed in formulating the dynamic system model:

1. Dynamics working memory with its constant data arrays initialized to the model constants obtained from the data file.

2. A data file containing the FORTRAN subroutine CONEC.

3. A data file containing the FORTRAN subroutine CONET.

After an error-free execution of activity DYRE is completed, and prior to terminating PSS®E, it is strongly recommended that dynamics working memory be preserved in the form of a Snapshot by using activity SNAP.

When specifying data for the compensating models (refer to Current Compensating Models), the user should carefully note the significance of the sign of their data values (refer to the data sheets in Compensator Model Data Sheets of the *PSS®E Model Library*). With a negative reactance, the COMP model queries from the terminal bus into the machine to determine the compensated voltage, while with a positive reactance, it queries into the network. The IEEEVC model uses the opposite sign convention.

The 1968 IEEE Type 1 and 2 exciter models, IEEET1 and IEEET2, use the same model subroutine which distinguishes between these two excitation systems on the basis of the value specified for the second feedback time constant. A zero for this constant is used to designate the Type 1 model, and a nonzero value is interpreted as the second feedback time constant of the Type 2 model. Activity DYRE examines the value specified for this data item, and if appropriate, converts a Type 1 reference to a Type 2 or vice versa. The 1979 IEEE Type 1 and 2 models, IEEEX1 and IEEEX2, are handled in the same manner.

In the summary tabulating the number of references of each PSS®E model, the IEEE Type 1 governor models, IEEEG1, TGOV5 and WSIEG1, may each be shown twice. The first listing shows its number of uses as a cross compound turbine governor model, and its second gives the number of occurrences with a single machine.

Models CIROS1 and SLNOS1 have an ICON that must be modified if these models are used in a supervisory mode. These should be set to the number of the appropriate permissive flag ICON of the supervised relay model. This can be done through the model edit feature in the Dynamics Models Spreadsheet; refer to *PSS®E GUI Users Guide*, Dynamics Spreadsheet.

16.1.6. Error Messages

As it processes the Dynamics Model Raw Data File, activity DYRE generates an alarm for errors resulting from the conditions listed below. In alarming a data record, the error message block often includes the first physical line of the offending data record.

1. A model name is encountered which is not recognized. The following error message is generated:

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```
UNDEFINED MODEL aaaaaa
```

Activity DYRE ignores the data record and processing continues. In reading Dynamics Model Raw Data Files prepared for, or generated by, release nineteen or earlier of PSS®E, data records for the models STAB2 and GAST2 generate an alarm as undefined. These models were replaced by the models STAB2A and GAST2A at PSS®E-20. For additional details refer to the PSS®E-20 release notes in *PSS®E Compatibility Reference*.

2. Too many models of a given type (e.g., two generator models specified for a machine). One of the following error messages is generated:

```
DUPLICATE MODEL TYPE: aaaaaa AT BUS nnnn MACHINE
n
DUPLICATE MODEL TYPE: aaaaaa AT BUS nnnn LOAD n
DUPLICATE MODEL TYPE: aaaaaa AT AREA nnnn LOAD n
DUPLICATE MODEL TYPE: aaaaaa AT ZONE nnnn LOAD n
DUPLICATE MODEL TYPE: aaaaaa AT OWNER nnnn LOAD n
DUPLICATE MODEL TYPE: aaaaaa FROM BUS nnnn TO BUS nnnn SLOT i
DUPLICATE MODEL TYPE: aaaaaa FOR ALL-TYPE LOAD aa
DUPLICATE MODEL TYPE: aaaaaa WITH SAME SIGNAL INDEX nn
                            ASSOCIATED WITH 2-TERM DC LINE name
DUPLICATE MODEL TYPE: aaaaaa WITH SAME SIGNAL
INDEX nn
                          ASSOCIATED WITH MULTI-TERM DC LINE name
DUPLICATE MODEL TYPE: aaaaaa WITH SAME SIGNAL INDEX nn
                               ASSOCIATED WITH VSC DC LINE name
DUPLICATE MODEL TYPE: aaaaaa WITH SAME SIGNAL INDEX nn
                              ASSOCIATED WITH FACTS DEVICE name
DUPLICATE MODEL TYPE: aaaaaa FOR 2-TERMINAL DC LINE   nn
DUPLICATE MODEL TYPE: aaaaaa FOR MULTI-TERMINAL DC LINE name
DUPLICATE MODEL TYPE: aaaaaa FOR VSC  DC LINE name
DUPLICATE MODEL TYPE: aaaaaa FOR FACTS DEVICE name
```

Activity DYRE ignores the data record that generated an alarm (i.e., it uses the data record previously read) and processing continues.

3. Error in decoding a data record. The standard data input error message is printed followed by:

```
RECORD IGNORED
```

This message is the first physical line of the offending record. In addition to typographical errors, this error is encountered when the *numbers* input option is in effect and a BUSID field on a data record specifies an extended bus name.

4. Error in the specification of the low pressure unit on a data record specifying the cross compound governor model CRCMGV, IEEEG1, TGOV5 or WSIEG1, the second machine of a FRECHG or COMPCC model, or the low pressure unit of a user-written cross-compound governor model. The following error message is printed, the record is ignored, and processing continues:

```
aaaaaa BUS nnnn MACHINE n NOT FOUND
```

5. Either the bus or designated machine is not in the working case.

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6. Too many bus and network oriented models. The following error message is printed, the record is ignored, and processing continues:

   ```
   TOO MANY MISCELLANEOUS MODELS
   ```

7. Activity DYRE can process a total number of the models listed in

8. of up to the maximum number of machines for which PSS®E is dimensioned. Too few data items on a record. The following error message is printed followed by a line indicating that either the record is ignored or that zeros are assumed for the missing data items:

   ```
   WARNING: INCOMPLETE DATA RECORD FOR MODEL aaaaaa
   ```

9. Too many data items on a record. The following error message is printed followed by the first physical line of the data record, and processing continues:

   ```
   EXCESS DATA IGNORED ON RECORD
   ```

10. The bus specified on a bus oriented model data record is not in the working case. The following error message is printed, the record is ignored, and processing continues:

    ```
    BUS nnnn NOT FOUND
    ```

11. Auxiliary-signal models SQBAUT and RBKELT can be used only with two-terminal dc line models. One of the following error message is printed, the record is ignored, and processing continues:

    ```
    AUX. SIGNAL MODEL SQBAUT CAN BE USED ONLY WITH
    2-TERM DC LINES
    AUX. SIGNAL MODEL RBKELT CAN BE USED ONLY WITH 2-TERM DC LINES
    ```

12. The device type (DVTYP) in the auxiliary-signal data record is invalid. This error message could be due to the following reason:

    - DVTYP is a value other than 1 through 4.

      The following error message is printed, the record is ignored, and processing continues:

      ```
      INVALID DEVICE TYPE
      ```

13. The signal index (SGNDX) in the auxiliary-signal data record is invalid. This error occurs if the SGNDX value specified is less than or equal to zero, or if it is greater than the maximum number of auxiliary signals that can be associated with the device type.

    The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID AUX. SIGNAL INDEX FOR THE DEVICE
    ```

14. An invalid two-terminal dc line name is specified either in the data record for a two-terminal dc line model, or in the auxiliary-signal data record that is associated with a two-terminal dc line. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID 2-TERMINAL DC LINE
    ```


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15. An invalid multi-terminal dc line name is specified either in the data record for a multi-terminal dc line model, or in the auxiliary-signal data record that is associated with a multi-terminal dc line. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID MULTI-TERMINAL DC LINE
    ```

16. An invalid VSC dc line name is specified either in the data record for a VSC dc line model, or in the auxiliary-signal data record that is associated with a VSC dc line. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID VSC DC LINE NAME
    ```

17. An invalid FACTS device is specified either in the data record for a FACTS device model, or in the auxiliary-signal data record that is associated with a FACTS device. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID FACTS DEVICE
    ```

18. An invalid user model type code (IC) is specified for a user-written model. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID USER MODEL TYPE CODE: nnn
    ```

19. An invalid user model network code (IT) is specified for a user-written model. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID USER MODEL NETWORK CODE: nnn
    ```

20. An invalid combination of model type and network codes is specified for a user-written model. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID USER MODEL TYPE/NETWORK CODE COMBINATION
    ```

21. Less than zero or more than five hundred array locations in the CON, STATE or VAR array, or less than zero or more than two hundred array locations in the ICON array, specified for a user-written model. The following error message is printed, the record is ignored, and processing continues:

    ```
    INVALID NUMBER OF ARRAY LOCATIONS: nnn
    ```

22. The same user-written model is given different characteristics (IC through NV or NRI) on two user model data records. The following error message is printed, the record is ignored, and processing continues:

    ```
    INCONSISTENT MODEL DEFINITIONS FOR SAME USER
    MODEL
    ```

23. More than 100 user-written model names have been defined for plant-related, load characteristic, load relay, branch relay, auxiliary signal, dc line, and FACTS device -models. The following error message is printed, the record is ignored, and processing continues:

    ```
    TOO MANY USER MODEL DEFINITIONS
    ```

    The *Remove unused user models* function available from the *Dynamics>Model maintenance...* menu entry for Plant model, Load model, Line relay model, Aux Signal model, DC line model, FACTS model, switched, and wind model may be used to eliminate unused user model definitions. *Refer to PSS®E GUI Users Guide*, Modifying Dynamics Model Pointer Tables.

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24. In reading the ICONs for a user-written model, a number with a decimal point was entered. The following error message is printed, the record is ignored, and processing continues:

    ```
    REAL ENTERED WHERE INTEGER OR CHARACTER EXPECTED:
    X.XXXX
    ```

25. In the *add models* mode, the number of entries needed in the table driven machine model connection tables exceeds the number of machines for which PSS®E is dimensioned. The following error message is printed and activity DYRE is terminated:

    ```
    MACHINE MODEL CONNECTION TABLES FULL--USE DYCH
    TO PACK TABLES
    ```

    The *Remove unconnected models* and *Pack models* functions of the *Dynamics>Model maintenance...* menu entry for Plant model may be used to eliminate slots in these tables for machines not in the working case. Refer to *PSS®E GUI Users Guide*, Modifying Dynamics Model Pointer Tables.

26. In the *add models* mode, the number of entries needed in the table driven load model connection tables exceeds the number of table entries for which PSS®E is dimensioned. The following error message is printed and activity DYRE is terminated:

    ```
    LOAD MODEL CONNECTION TABLES FULL--USE PACK
    LOAD MODELS FUNCTION
    ```

    The *Remove unconnected models* and *Pack models* functions of the *Dynamics>Model maintenance...* menu entry for Load model may be used to eliminate slots in these tables for loads not in the working case. Refer to *PSS®E GUI Users Guide*, Modifying Dynamics Model Pointer Tables.

27. In the *add models* mode, the number of entries needed in the table driven line relay model connection tables exceeds the number of table entries for which PSS®E is dimensioned. The following error message is printed and activity DYRE is terminated:

    ```
    LINE RELAY CONNECTION TABLES FULL--USE PACK LINE
    RELAY MODELS FUNCTION
    ```

    The *Remove unconnected models* and *Pack models* functions of the *Dynamics>Model maintenance...* menu entry for Line relay model may be used to eliminate slots in these tables for branches not in the working case. Refer to *PSS®E GUI Users Guide*, Modifying Dynamics Model Pointer Tables.

28. In the *add models* mode, the number of entries in the auxiliary-signal connection tables exceeds the number of table entries for which PSS®E is dimensioned. The following error message is printed and activity DYRE is terminated:

    ```
    AUX. SIGNAL CONNECTION TABLES FULL-USE PACK
    AUX. SIGNAL MODEL
    ```

29. In the *add models* mode, the number of entries in the dc line connection tables exceeds the number of table entries for which PSS®E is dimensioned. Depending on the dc line type (two-terminal or multi-terminal or VSC), one of the following error messages is printed and activity DYRE is terminated:

    ```
    TWO-TERMINAL DC LINE MODEL CONNECTION TABLES
    FULL-USE PACK DC LINE MODEL
    MULTI-TERMINAL DC LINE MODEL CONNECTION TABLES FULL-USE PACK DC
    ```


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```
LINE MODEL
VSC DC LINE MODEL CONNECTION TABLES FULL-USE PACK DC LINE MODEL
```

30. In the *add models* mode, the number of entries in the FACTS device connection tables exceeds the number of table entries for which PSS®E is dimensioned. The following error message is printed and activity DYRE is terminated:

    ```
    FACTS DEVICE MODEL CONNECTION TABLES FULL-USE
    PACK FACTS DEVICE MODEL
    ```

31. One of the data storage arrays has no more available locations. The following message is printed and activity DYRE is terminated:

    ```
    DYRE TERMINATED--SYSTEM MODEL REQUIRES TOO MANY
    aaaaaa
    ```

32. The same machine was specified for both machines in a two machine model. The following -message is printed, the record is ignored, and processing continues:

    ```
    DIFFERENT MACHINES MUST BE SPECIFIED IN TWO
    MACHINE MODEL
    ```

33. Too many load models of a given subsystem type (BUS, OWNER, ZONE, AREA or ALL-TYPE) were specified. The following -message is printed, the record is ignored, and processing continues:

    ```
    xxxxx LOAD MODEL TABLE OVERFLOW
    ```

34. Mismatch between the network code (IT) and the two character suffix of a user load model name:

    ```
    LOAD MODEL SUBSYSTEM CODE AND MODEL NAME INCONSISTENT
    ```

35. An invalid relay slot (RS) was specified. The following error message is printed, followed by the first physical line of the data record, and processing continues:

    ```
    INVALID RELAY SLOT: aaaaa CKT id FROM BUS nnnn
    TO BUS nnnn SLOT i
    ```

36. An obsolete system-wide model, for which the function is now handled via solution options, is specified. The following error message is printed and processing continues:

    ```
    MODEL aaaaa IGNORED--USE SIMULATION OPTIONS
    TO INVOKE
    ```

37. An obsolete monitoring model, for which the function is now enabled via activity

    , is specified. One of the following error messages is printed and processing coCHSBntinues:

    ```
    MODEL aaaaa IGNORED--USE ACTIVITY CHSB
    MODEL TOTA FOR AREA NUMBER nnn IGNORED--USE ACTIVITY CHSB
    MODEL TOTZ FOR ZONE NUMBER nnn IGNORED--USE ACTIVITY CHSB
    ```

38. In converting an old branch relay model to its new table-driven replacement, two branch relay models are already present on the line in the same direction. The following error message is printed and processing continues:

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```
MODEL aaaaa FROM BUS nnn TO BUS nnn CKT aa IGNORED--BOTH
RELAY SLOTS TAKEN
```

The user should reconcile any error messages before completing the setup of the dynamic model of the system under study.

Additional Information
PSS®E GUI Users Guide, Reading Dynamics Model Data *PSS®E* Command Line Interface (CLI) *Users Guide*, Reading Dynamics Model Data

16.2. Assigning Simulation Variables to Output Channels

CHAN

Run Activity CHAN - GUI
Dynamics>Define simulation output (CHAN) *[Assign Channels for Bus Quantities]* *[Assign Channels for Line Quantities]* *[Assign Channels for Load Quantities]* *[Assign Channels for Machine Quantities]* *[Assign Channels for Misc. Quantities]*
Run Line Mode Activity CHAN - CLI
ACTIVITY? >>**CHAN** NEXT AVAILABLE ADDRESSES ARE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The output channel selection activity CHAN enables the user to select simulation variables which are to be monitored during dynamic simulation runs. These quantities may be tabulated and/or placed into a Channel Output File at regular intervals during a simulation run (refer to activity RUN). The simulation results, as contained in the Channel Output File, are able to be processed by the PSS®E plot package following completion of the simulation. Up to six channels can be defined for interactive plotting during the simulation run.

16.2.1. Operation of Activity CHAN

The user identifies quantities to be placed into output channels and activity CHAN places their memory addresses and alphanumeric identifiers into the PSS®E dynamics data arrays IPRINT and IDENT respectively.

Activity CHAN displays the next available location pointers for Channel, VAR, and ICON for the corresponding dynamics data arrays (refer to Section 16.1.2, “Operation of Activity DYRE” and activity SNAP). In a simulation,

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output channels are assigned sequentially starting with channel one. If there are < n > quantities being monitored, they normally occupy output channels one through < n >. Activity CHAN assigns locations in these arrays at their next available positions.

The user specifies the type of data to be placed in output channels, the element to be monitored, and an optional thirty-two character identifier to be assigned to the output channel. The user also specifies the bus number, machine identifier, and an alphanumeric channel identifier which, if supplied, *must* be enclosed in single quotes. If no identifier is specified, activity CHAN supplies a default identifier. Table 16.2, "Activity CHAN Summary" summarizes each category of data with the default identifier assigned by activity CHAN.

Table 16.2. Activity CHAN Summary

Quantity	Units	Selected By	Default Identifier*
ANGLE: rotor angle	degrees	Bus, machine id	ANGL xx
PELEC: electrical power	pu on system base	Bus, machine id	POWR xx
QELEC: reactive power	pu on system base	Bus, machine id	VARS xx
ETERM: terminal voltage	pu	Bus, machine id	ETRM xx
EFD: field voltage	pu	Bus, machine id	EFD xx
PMECH: mechanical power	pu on machine base	Bus, machine id	PMEC xx
SPEED: speed deviation	pu	Bus, machine id	SPD xx
XADIFD: field current	pu	Bus, machine id	IFD xx
ECOMP: compensated voltage	pu	Bus, machine id	ECMP xx
VOTHSG: stabilizer signal	pu	Bus, machine id	AUX xx
VREF: regulator reference	pu	Bus, machine id	VREF xx
VUEL: minimum excitation limiter signal	pu	Bus, machine id	VUEL xx
VOEL: maximum excitation limiter signal	pu	Bus, machine id	VOEL xx
BSFREQ: frequency deviation	pu	Bus	FREQ yy
Bus voltage	pu	Bus	VOLT yy
Bus voltage and angle	pu and degrees	Bus	VOLT yy and ANGL yy
Branch flow (P)	MW	from bus, to bus, circuit identifier, third bus	POWR zz
Branch flow (P and Q)	MW and Mvar	from bus, to bus, circuit identifier, third bus	POWR zz and VARS zz
Branch flow (MVA)	MVA	from bus, to bus, circuit identifier, third bus	MVA zz
Apparent R and X	pu and pu	from bus, to bus, circuit identifier, third bus	APPR zz and APPX zz
VAR	Varies	VAR number	VAR nn
STATE	Varies	STATE number	STATE nn
Machine terminal current (1)	pu	Bus, machine id	ITRM xx

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Machine apparent impedance [1]	pu and pu	Bus, machine id	APPR xx and APPX xx
PLOAD: active power load	pu	Bus, load id	PLOD vv
QLOAD: reactive power load	pu	Bus, load id	QLOD vv
Governor reference	pu	Bus, machine id	GREF xx
Turbine Load Controller reference	pu	Bus, machine id	LCRF xx
Wind velocity	pu	Bus, wind machine id	WVEL xx
Wind turbine speed	pu	Bus, wind machine id	WTSP xx
Wind Pitch	degree	Bus, wind machine id	WPCH xx
Wind aerodynamic torque	pu	Bus, wind machine id	WAET xx
Wind rotor voltage	pu	Bus, wind machine id	WRTV xx
Wind rotor current	pu	Bus, wind machine id	WRTI xx
Wind real power command from electrical control	pu	Bus, wind machine id	WPCM xx
WInd reactive power command from electrical control	pu	Bus, wind machine id	WQCM xx
Wind Auxiliary control output	pu	Bus, wind machine id	WAUX xx
SUBSYSTEM TOTALS[2]	MVA	Subsystem number	PMECH subsystem PELEC subsystem PACCL subsystem PLOAD subsystem PE-PL subsystem

*Notes:

xx Contains the bus number, the extended bus name, and the machine identifier

yy Contains the bus number and the extended bus name.

zz Contains the from bus number, to bus number or, for a three-winding transformer, the string 3W: followed by the transformer name, and the circuit identifier.

nn Is the VAR or STATE index.

vv Contains the bus number, load identifier, and the extended bus name

[1] Valid for synchronous machine models only.

[2] From activity CHSB, SUBS.

When selecting branches for monitoring, up to three buses may be specified. Nontransformer branches and two-winding transformers are specified by making no entry in the third bus field. For three-winding transformers, all three buses must be specified. Branch quantities are calculated at the end of the branch corresponding to the first bus specified. Note that the following categories generate two output channel assignments for each element selected for monitoring:

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- Voltage & Angle
- Flow (P&Q)
- Relay2 (R & X)
- Machine apparent impedance

Activity CHAN summarizes the next available positions in the pertinent arrays as a result of its current execution. The value of the solution parameter NCHAN, which indicates the highest numbered channel being monitored during simulations (refer to activity ALTR and Section 23.6, "Simulation Outputs"), is set to NCHAN-1.

When activity CHAN is being used to change the channel assignments assigned to a subset of previously generated assignments, the value printed for < NCHAN > is the larger of the original next available channel and that which would be set as a result of the current execution of activity CHAN. Consider, for example, the situation in which fifty output channels have been assigned and are being monitored and the user wishes to change the quantity associated with channel ten. Activity CHAN would be entered, a starting channel number of ten would be specified rather than the default of fifty-one, and the appropriate channel assignment made. < NCHAN > would be fifty-one, and hence NCHAN would retain its previous value of fifty.

16.2.2. Application Notes

As shown in Table 16.2, "Activity CHAN Summary" the following data of activity CHAN's output category selector designate quantities associated with plant equipment models.

• Angle • Pelec • Qelec • Eterm • EFD • Pmech • Speed • Xadifd	• Ecomp • Vothsg • Vref • Vuel • Voel • Gref • Lcref • Wind velocity	• Wind turbine speed • Wind pitch • Wind aerodynamic torque • Wind rotor voltage • Wind rotor current • Wind real power command from electrical control • Wind reactive power command from electrical control • Wind auxiliary control output

Pload and *Qload* designate the active and reactive load quantities associated with each load in the network. In the PSS®E dynamics data structure, there are dedicated arrays containing each of these quantities for each machine and load in the working case. These data items are automatically calculated by the various equipment models used to model each machine and its controls and by the various load characteristic models applied to loads in the network (refer to Section 16.1.2, "Operation of Activity DYRE" and 21.4.1). Activity CHAN simply sets up the appropriate address pointers in the dynamics data array IPRINT so that the channel output routine picks up the proper values.

Similarly, *bus frequency deviation* is stored in a dedicated array for each bus in the working case and activity CHAN sets up the appropriate pointer. VARs and STATEs (categories nineteen and twenty) are handled in the same way.

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However, the following categories specify data items that are not readily accessible from the standard dynamics data arrays.

• Voltage	• Flow (MVA)
• Voltage and Angle	• Relay2 (R&X)
• Flow (P)	• Machine terminal current
• Flow (P&Q)	• Machine apparent impedance

For these quantities to be available as output channel variables, they must be calculated and placed into a data array which can be accessed by the channel output routine. This function is performed by a group of network monitoring models which calculate certain network quantities and place them into specified locations in the VAR array.

Thus, when activity CHAN is programmed to place bus voltages, branch flows, branch apparent impedances, machine currents and machine apparent impedances into output channels, it takes the following steps:

1. It allocates positions in the ICON array and sets these elements to the values required to designate the bus, branch or machine to be monitored.
2. It allocates positions in the VAR array which are to contain the monitored quantities.
3. It generates entries in the CHAN monitoring model tables for the appropriate model call (VOLMAG, FLOW1, FLOW3, RELAY2, RELAY3, GENTMC or GENTMZ).
4. As with the other data categories, it establishes the appropriate IPRINT and IDENT array entries.

During dynamic simulations, model calls corresponding to the model references contained in the CHAN monitoring model tables are issued each time step at the completion of the network balance. Thus, after being selected by activity CHAN, these network quantities are automatically calculated during dynamic simulation runs and placed into output channels.

As activity CHAN assigns each selected simulation quantity to an output channel, it makes a corresponding entry in a set of channel descriptor arrays. These arrays describe the type of quantity assigned to the channel, and its location in the network in terms of external bus numbers and machine, load and circuit identifiers (or in a dynamics data array in the case of categories nineteen and twenty). These arrays are also used in describing the content of each channel in activities ALTR and DLST.

This information is also used to reconcile differences in the PSS®E internal numbering scheme which may exist when a Saved Case is retrieved for which internal storage of network quantities differs from that which was contained in the working case during the execution of activity CHAN. This reconciliation occurs at the execution of activities RSTR, SRRS, and section / RTRN whenever both network and dynamics data have been read into PSS®E working memory. It also occurs following network data changes via activity ALTR, and at the completion of the power flow data edits.

When the value specified as the starting channel index exceeds the next available channel value contained in dynamics working memory, the channels in this gap have their IPRINT, IDENT and channel descriptor array entries set so as to leave the channel unused. Dynamic simulation quantities may be assigned to such channels in subsequent executions of activity CHAN as described earlier.

As in activity DYRE (refer to Section 15.1.3, Adding Models), the user should take care to take a Snapshot following successful execution of activity CHAN and prior to terminating PSS®E. If a Snapshot is not taken,

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the newly established IPRINT, IDENT, ICON and channel descriptor array entries are not preserved and activity CHAN will need to be re-executed.

The user must exercise caution in specifying the starting VAR and ICON indices in response to the input request issued by activity CHAN. In particular, the location indices specified and successive locations in the respective arrays *must not* be used by other models. For simulation execution efficiency, it is desirable that the locations specified be the next available slots in the respective data arrays. Thus, the normal specification of these indices is to enter the default response.

As can be seen in the use of activity CHAN, as well as in the requirements of activities SNAP and DYRE, it is strongly recommended that the next available position indices maintained within each Snapshot be used whenever possible. PSS®E is able to perform the bookkeeping required for the storage allocations assigned by activities DYRE, CHAN and CHSB. It is the user's responsibility to update these indices, with activity ALTR, whenever models or user written code utilizing additional dynamics data array elements are manually added to the connection subroutines.

16.2.3. Error Messages

During its implementation, activity CHAN generates an alarm for errors resulting from the following conditions:

1. A bus, machine, load or branch specified by the user is not in the working case. An appropriate error message is printed and the user re-specifies the equipment identifier(s).

2. In allocating an output channel, the preceding channel selection used the highest numbered output channel for which PSS®E is currently dimensioned. The following error message is printed and activity CHAN is terminated:

   ```
   ALL CHANNELS USED
   ```

3. In selecting for output a quantity requiring a network monitoring model call, there are not enough ICONs remaining to accommodate those required by the model call. The following message is printed:

   ```
   nnnn IS END OF ICON ARRAY
   ```

4. In selecting for output a quantity requiring a network monitoring model call, there are not enough VARs remaining to accommodate those required by the model call. The following message is printed:

   ```
   nnnn IS END OF VAR ARRAY
   ```

5. In selecting for output a quantity requiring a network monitoring model call, the table accommodating the CHAN monitoring model references is full. The following message is printed:

   ```
   nnnn MONITORING MODELS SPECIFIED--POINTER TABLE
   FULL
   ```

6. In placing a VAR or STATE in an output channel, a VAR or STATE number greater than the largest for which PSS®E is currently dimensioned is specified. The following message is printed:

   ```
   nnnn IS END OF aaaaa ARRAY
   ```

Additional Information
PSS®E GUI Users Guide, Simulation Variables

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Simulation Variables

16.3. Saving Dynamics Working Memory in a Binary File

SNAP

Run Activity SNAP - GUI
File>Save... *[Save Dynamics Data]* Snapshot Data Tab
Run Line Mode Activity SNAP - CLI
ACTIVITY? >>**SNAP** ENTER SNAPSHOT FILENAME: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The snapshot activity SNAP preserves the contents of dynamics working memory in a user specified Snapshot File. PSS®E optionally runs activity SNAP on completion of activity STRT.

The user specifies the name of the file in which the Snapshot is to be saved. If an asterisk (*) is specified as the name of the Snapshot file, the last Snapshot File accessed by either activity SNAP or activity RSTR during the current session of PSS®E is used. Refer to Section 2.4.1, "File Usage "for the file specification conventions and to Section 2.6, "Files Created By PSS®E" for the file overwriting conventions used by activity SNAP.

When activity SNAP is run, output quantities are displayed for CONs, STATEs, VARs, ICONs, and Channels. The user specifies the number of elements in the data arrays and the number of output channels (refer to activity CHAN and Section 23.6, "Simulation Outputs") to be written to the Snapshot File. The displayed values reflect the next available position indices as maintained by activities DYRE, CHAN, and CHSB (refer to Section 16.1.2, "Operation of Activity DYRE" and activities CHAN and CHSB). When specifying other than the default values (i.e., the indices displayed), the number specified for each of the above quantities should be at least as large as the number of elements in use in the corresponding data arrays.

The contents of dynamics working memory are then written to the file, along with the current date and time. The following message is printed at the user's terminal and activity SNAP is terminated:

```
SNAPSHOT STORED IN FILE <filename> AT TIME
= X.XXX
```

16.3.1. Application Notes

Activity SNAP records only the data arrays associated with the modeling of dynamic equipment; it *does not* record any other data arrays describing the electric network as contained in the working case. The user must, therefore, take careful note of the network model as contained in the working case at the time the Snapshot is taken. It is generally advisable to use activity section / SAVE to create a power flow Saved Case File corresponding exactly to each Snapshot.

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A Snapshot in conjunction with the corresponding power flow Saved Case completely specifies the instantaneous condition of the system at the time of their creation. It is strongly recommended that the user always preserve and retrieve Snapshots and Saved Cases in coordinated pairs, making careful note of which power flow Saved Case is associated with which Snapshot. The utilization of some meaningful file naming convention can be very useful in this regard.

The most common application of Snapshots is to save a valid setup so that future runs from the same network condition can be started without the need of re-executing activities DYRE, CHAN, and so on (refer to Section 23.4, "The Dynamic Model"). The network condition associated with these starting condition Snapshots is normally a converted base case power flow which is already saved. Thus, all that is needed to be able to reestablish the network data arrays is to note the name of the appropriate power flow Saved Case File. A Snapshot may also be taken within activity STRT for this purpose of saving valid checked out initial conditions.

Another possible application is to preserve a dynamic simulation run for future running to an extended value of TIME (refer to Restarting From a Snapshot). In this case, the network model is likely to be significantly different from the base case, and it is necessary that the present condition of the working case be saved immediately following execution of activity SNAP. When taking a mid-run Snapshot, it is recommended that the user preserve the initial condition Snapshot by specifying a different filename for the mid-run Snapshot.

Taking a Snapshot has no effect on the simulation and the user may proceed to any activity immediately after completion of activity SNAP.

In specifying the number of elements to be saved in the Snapshot, the usual procedure is to respond with the default (i.e., a carriage return). The next available position indices are maintained by activities DYRE, CHAN, and CHSB to record the utilization of storage for the model references generated by these activities. These pointers are preserved with the Snapshot and are used to set the default responses for activities SNAP, DYRE, CHAN, and CHSB.

For those applications in which models are manually added to a dynamics setup or user written code in the connection subroutines utilizes additional locations in the dynamics data arrays, it is strongly recommended that the user manually update these next available pointers via activity ALTR. Subsequent executions of activities SNAP, CHAN, CHSB, and DYRE may then use, via the default response, the pointers maintained with the Snapshot without the user having to remember and enter these values.

The user is permitted to record portions of the dynamics data arrays beyond those currently in use by overriding the default response. However, the practice of preserving large portions of these arrays which are unused by the simulation is discouraged. Large amounts of disk space is unnecessarily used and the time required to write the Snapshot and subsequently restore it with activity RSTR is increased.

Additional Information
PSS®E GUI Users Guide,Saving Dynamics Working Memory in a Binary File *PSS®E* Command Line Interface (CLI) *Users Guide*, Saving Dynamics Working Memory in a Binary File

16.4. Restoring Dynamics Working Memory from a Binary Snapshot File

RSTR

Run Activity RSTR - GUI
File>Open...

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Run Activity RSTR - GUI
[Open] Dynamics Snapshot Data File (*.snp)
Run Line Mode Activity RSTR - CLI
ACTIVITY? >>**RSTR** ENTER SNAPSHOT FILENAME: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The snapshot retrieval activity RSTR restores the dynamics data arrays contained in the specified Snapshot File into dynamics working memory. The contents of working memory are overwritten while the contents of the specified Snapshot File are unchanged.

The user specifies the name of the file in which the Snapshot is located. If an asterisk (*) is specified as the name of the Snapshot file, the last Snapshot File accessed by either activity RSTR or activity SNAP during the current session of PSS®E is read into the working case. Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity RSTR.

If the specified file is not in a Snapshot File format recognized by this release of PSS®E, an error message is printed.

If the Snapshot exceeds the capacity limits of the size level of PSS®E currently installed in the system, an error message is printed.

Generally, activity RSTR is able to access Snapshot Files filled by activity SNAP of the current and previous releases of PSS®E. However, a Snapshot File written by the current version of PSS®E is normally not able to be read by earlier releases of the program.

In restoring the output channel specification data, activity RSTR automatically updates the channel addresses if the Snapshot was taken in a version of PSS®E of different dimensional capacity than the one currently installed on the system.

In retrieving Snapshots created by PSS®E-16 through PSS®E-19, activity RSTR checks for the use of the models STAB2 and GAST2; at PSS®E-20, these models were replaced by the models STAB2A and GAST2A respectively. Activity RSTR deletes all references to the old models. The replacement models, which have different storage requirements than their obsolete counterparts, must be introduced into the simulation setup with activity DYRE. Upon encountering a reference to either of these models, activity RSTR allows the user to specify a file into which it will place data records for the new models; this file may subsequently be used as input (refer to Section 16.1.3, "Adding Models").

Following the reading of the dynamics data from the Snapshot File, if network data is contained in the working case, the dynamics model table array pointers and output channel addresses are updated to reflect the internal bus, machine, load, branch, and VSC dc line numbering of the working case. The dynamic model table array pointers are also updated for two-terminal dc line, multi-terminal dc line, and FACTS device to reflect if the corresponding device is present in the working case.

Finally, the case heading read from the file is printed at the user's terminal, along with the time and date at which the Snapshot was written, and the number of elements in the dynamics data arrays that were restored.

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16.4.1. Restarting From a Snapshot

Because a Snapshot in conjunction with the corresponding power flow Saved Case completely specifies the instantaneous condition of the system at the time they were written, a simulation run may be restarted from a Snapshot and continued just as if it had never been halted.

As pointed out in Section 16.3.1, "Application Notes" network data is not saved with the dynamics data arrays in the Snapshot File. Thus, to continue a simulation from mid-run Snapshot and Saved Case Files, it is necessary to retrieve *both* the appropriate Snapshot *and* its corresponding power flow Saved Case. The procedure is illustrated by example.

Consider a simulation run that has been stopped for intermediate examination, with the mid-run Snapshot in the file temp.snp, the corresponding network in the Saved Case File temp.cnv, and with the output of the simulation up to the snapshot TIME in the file case1. *out*. To continue the run, the restart sequence of PSS®E activities is:

1. Enter PSS®E.

2. Restore the dynamics data with the activity command `rstr,temp.snp`.

3. Pick up the electrical network using the command `case`:

   ```
   case,temp.cnv
   ```

4. Proceed with the simulation as if it were not interrupted (activities

or if, before continuing with the simulation, changes are to be made to ALTR the system condition as restored into memory).

The name of the simulation output file, case1.*out* in this example, is retained in the Snapshot along with a file position pointer for the output file. Thus, when restarting from an intermediate point Snapshot, the output from the continuation of the run is appended onto the output contained in the output file, and so the output file should not be changed. It should not have been used for any other purpose (except for examination with the channel output file processing program) after the simulation was halted and the Snapshot taken.

When the Snapshot contains an initial condition (i.e., it was written in activity STRT), the user does not need to re-execute activity STRT. However, the user should check, with activity ALTR, that the simulation output file, as specified in the Snapshot, is the desired output file. If not, an output file may be established by specifying the appropriate filename in activity ALTR. In this situation, it is still advisable to execute activity STRT before proceeding with the simulation because this allows for:

1. A cross check of dynamics and network data to ensure proper correspondence of the Snapshot and Saved Case Files.

2. Specifying a new simulation output file and having it initialized with the output channel headings without having to go through activity ALTR.

16.4.2. Application Notes

If implementing the supplemental file for STAB2A / GAST2A models optionally constructed by activity RSTR, note that GAST2A requires one more data item than did GAST2. The user should edit the data file to add the value of T_C to each GAST2A data record. The STAB2A data records may be used without change.

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Additional Information
PSS®E GUI Users Guide, Restoring Dynamics Working Memory from a Binary Snapshot File *PSS®E* Command Line Interface (CLI) *Users Guide*, Restoring Dynamics Working Memory from a Binary Snapshot File

16.5. Restoring Dynamics Working Memory from a Snapshot File Created in PSS® E-26 or Earlier

SRRS

Run Activity SRRS - GUI
File>Open... *[Open]* Dynamics Snapshot Raw Data File (*.srs)
Run Line Mode Activity SRRS - CLI
ACTIVITY? >>**SRRS** ENTER SOURCE FORM SNAPSHOT FILENAME: >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The Dynamics Snapshot Raw Data File retrieval activity SRRS restores the dynamics data arrays contained in the specified Dynamics Snapshot Raw Data File into dynamics working memory. The contents of working memory are overwritten while the contents of the specified Snapshot File are unchanged. Source snapshots could only be created prior to PSS®E-27.

The user specifies the name of the file in which the Snapshot is located. If no file is specified, or a file system error condition is encountered in opening the designated file, or the specified file is not in Source Form Snapshot File format, an appropriate error message is displayed.

If the number of CONs, ICONs or output channels contained in the Snapshot File exceeds the capacity limits of the size level of PSS®E currently installed in the system, an appropriate error message is displayed along with sizing information, and activity SRRS is terminated. Similarly, if any of the next available position pointers specified in the Snapshot File exceed the capacity limits of the size level of PSS®E currently installed, an appropriate error message is displayed.

In retrieving Source Form Snapshots created by PSS®E-16 through PSS®E-19, activity SRRS checks for the use of the models STAB2 and GAST2; at PSS®E-20, these models were replaced by the models STAB2A and GAST2A respectively. Activity SRRS deletes all references to the old models. The replacement models, which have different storage requirements than their obsolete counterparts, must be introduced into the simulation setup with activity DYRE. Upon encountering a reference to either of these models, activity SRRS allows the user to specify a file into which it will place data records for the new models; this file may subsequently be used as input (refer to Adding Models).

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Following the reading of the dynamics data from the Snapshot File, the case heading as read in from the file is printed at the user's terminal, along with the time and date at which the Snapshot was written, and the number of elements in the dynamics data arrays which were restored.

Prior to terminating, activity SRRS gives the user the option of generating a file for the compiling of the connection subroutines, CONEC and CONET. (These were presumably obtained from the same party as the Source Form Snapshot which was just read by activity SRRS.)

When a filename is specified, activity SRRS creates a file which, when executed at operating system level, compiles the CONEC and/or CONET subroutines which are contained in the files specified. The use of this feature is strongly encouraged because it guarantees the correct specification of compiler option settings in compiling these subroutines.

Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity SRRS.

16.5.1. Application Notes

The file specified in invoking activity SRRS must be in the format of a Source Form Snapshot File; that is, it must have been written by activity SRSN, which was available prior to PSS®E-27.

Activity SRRS requires that the network corresponding to the Source Form Snapshot File to be read be present in the working case at the time activity SRRS is selected.

In restoring the output channel specification data, activity SRRS automatically updates the channel addresses if the Snapshot was taken in a version of PSS®E of different dimensional capacity than the one currently installed on the system.

If implementing the supplemental file for STAB2A / GAST2A models optionally constructed by activity RSTR, note that GAST2A requires one more data item than did GAST2. The user should edit the data file to add the value of T_C to each GAST2A data record. The STAB2A data records may be used without change.

A change in the Source Form Snapshot File data defining the system variables assigned to output channels occurred between releases eight and nine of PSS®E. Therefore, when one of the parties is using release eight of PSS®E while the other is running release nine or above, it is *strongly* recommended that the output of activity DMPC be included in the transmittal of dynamic simulation setups. This Response File may then be used to override the output channels specified in the Source Form Snapshot File after it has been restored.

Additional Information
PSS®E GUI Users Guide, Restoring Dynamics Working Memory from a Binary Snapshot File Created in PSS®E-26 or Earlier *PSS®E* Command Line Interface (CLI) *Users Guide*, Restoring Dynamics Working Memory from a Snapshot File Created in PSS®E-26 or Earlier

16.6. Modifying Data in Dynamics Working Memory

ALTR

Run Activity ALTR - GUI
Dynamics *[spreadsheet]*

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Run Line Mode Activity ALTR - CLI
ACTIVITY? >>**ALTR** ENTER CHANGE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The dynamics data change activity ALTR allows the user to examine and make changes to the values of data items contained in dynamics data working memory. When using line mode, it also provides for the specification of network data changes through an automatic link to the power flow activities section / CASE, section / CHNG, section / ORDR, and section / FACT; using the GUI, a new Saved Case may be picked up, but access to activity CHNG is not provided.

Additional Information
PSS®E GUI Users Guide, Modifying Data in Dynamics Working Memory *PSS®E* Command Line Interface (CLI) *Users Guide*, Modifying Data in Dynamics Working Memory

16.7. Modifying Dynamics Model Pointer Tables

DYCH

Run Activity DYCH - GUI
Dynamics>Model maintenance... *[Model Maintenance]*
Run Line Mode Activity DYCH - CLI
ACTIVITY? >>**DYCH**
Interrupt Control Codes
None

The table driven data table maintenance activity DYCH allows the user to interrogate and modify the data array entries used in associating plant related equipment models with machines represented in the working case.

Activity DYCH first displays table utilization, the number of current entries and the maximum allowable number of entries, for the following dynamic tables:

Table 16.3. Dynamic Tables

Table	Description
Model Connection Table	Each entry is associated with a specific machine. Normally, there are as many entries in this table as there are machines in the working case. However, when the working case differs topologically from that which was present when the simulation model data was read into PSS®E, this may not be the case. For dynamic simulation calculations, each in-service

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Table	Description
	machine in the power flow *must* have a model connection table entry. The model connection table entry for a machine contains pointers to the *array allocation table* entries assigned to that machine.
Array Allocation Table	Each entry is associated with a single model reference. It defines the specific model being used along with the starting indices in the dynamics data storage arrays assigned to that model call (refer to Section 23.1.2, "Data Structure"). Normally, there are as many entries in this table as there are plant-related model references in the simulation setup.
User Model Definition Table	One entry appears for each user-written model 'model name' that was defined during the construction of the dynamic simulation setup (refer to User-Written Models), and which is called from internal PSS®E tables. User-written models which are called from CONEC or CONET do not have entries in the user model definition table.
Activity CHAN Monitoring Model Table	Contains an entry for each network monitoring model reference generated through activities CHAN andCHSB .

A function selector provides access to the tables. Some selections may be limited to specific models: load, plant, wind, relay, auxiliary signal, FACTS device, dc line, or switched shunt models.

The user may select from the following options, one at a time:

- List unconnected models
- Remove unconnected models
- Pack models
- List user models
- Remove unused user models
- Consistency check (plant and wind models only)

The following sections describe each of the above functions.

16.7.1. List Unconnected Models

The unconnected model listing function lists those model connection table entries for which the corresponding machine is not contained in the working case as in the following example:

```
PLANT RELATED MODELS WITH NO MACHINE IN LOAD
FLOW:
```


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```
MACH     GEN.     COMP.   STAB.   MINEXL   MAXEXL    EXC.     GOV.     TLC
 1      GENROU     --      --       --        --     IEEET1   *TGOV1    ---
```

16.7.2. Remove Unconnected Models

The unconnected model removal function marks as unused those model connection table entries for which the corresponding machine is not contained in the working case. The corresponding array allocation table entries are also marked as unused. Each model reference removed from the model tables is reported at the *Progress* tab.

16.7.3. Pack Tables

The table packing function removes gaps (i.e., entries which are marked as unused) from the model connection table and the array allocation table. If either of these tables are packed, messages are printed at the *Progress* tab as in the following example:

```
PLANT MODEL TABLES UPDATED:
                                          OLD       NEW      MAXIMUM
CONNECTIONS FROM MACHINES TO MODELS:        6         5        33050
POINTERS FROM MODELS TO DATA:              17        13        66100
```

16.7.4. List User Models

The list user model definitions function prints the user model definition table which reflects the definition information specified to DYRE on the USRMDL data records of plant-related user-written models (refer to User-Written Models). The data printed for each model definition includes its name, the type of model, the number of locations required in the data storage arrays for each reference of the model, and the number of references of the model (i.e., the number of machines at which it is called). The report also indicates if a model is a coordinated call model (e.g., an SVS) or a two machine model (i.e., a cross-compound governor model).

16.7.5. Remove Unused User Models

The unused model definition removal function deletes from the user model definition table any model definitions which are no longer referenced from the array allocation table. Vacated slots are available to activity DYRE, ADD for new model definitions.

16.7.6. Consistency Check

The consistency checking function loops through all machines in the working case and generates an alarm for any invalid model combinations. This report is in ascending bus number order when the *numbers* output option is in effect, and in alphabetical bus name order using the *names* option. For any alarm messages printed by activity DYCH for a given machine, a heading line identifies the bus and machine being processed. The following conditions generate an alarm:

1. No generator model specified for a machine. The following error message is printed:

   ```
   NO GENERATOR MODEL. MACHINE STATUS IS IN (or
   OUT)
   ```

   The remaining model combination checks in (2) through (12) below are bypassed. In executing dynamic simulation calculations, *all* in-service machines must have an active generator model assigned to them.

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2. The generator model reference is bypassed (refer to Section 16.7.8, "Change Model Status"). The following message is printed:

   ```
   GENERATOR MODEL BYPASSED. MACHINE STATUS IS
   IN (or OUT)
   ```

   The remaining model combination checks in (3) through (12) below are bypassed. In executing dynamic simulation calculations, *all* in-service machines must have an active generator model assigned to them.

3. At a machine with active generator and exciter models (i.e., neither model is bypassed; refer to Change Model Status), the generator model does not allow an excitation system model. The following error message is printed:

   ```
   EXCITER MODEL NOT ALLOWED WITH GENERATOR MODEL
   name
   ```

   An excitation system is not permitted in combination with the classical generator model GENCLS, with the static shunt compensator models CSVGN1, CSVGN3, CSVGN4, CSVGN5 and CSVGN6, with the induction machine models CIMTR1, CIMTR2, CIMTR3 and CIMTR4, with the DSMES model CDSMS1, or with CBEST, CSMEST and CSTATT models.

4. At a machine with active generator and stabilizer models, the generator model does not allow a stabilizer model. The following error message is printed:

   ```
   STABILIZER MODEL NOT ALLOWED WITH GENERATOR
   MODEL name
   ```

   A stabilizer model is not permitted in conjunction with the classical generator model GENCLS, with the induction machine models CIMTR1, CIMTR2, CIMTR3 and CIMTR4, with the DSMES model CDSMS1, or with CBEST, CSMEST and CSTATT models.

5. At a machine with active generator and stabilizer models, a power sensitive stabilizer model is used in combination with a static var system model. The following error -message is printed:

   ```
   STABILIZER MODEL name NOT ALLOWED WITH SVS MODEL
   name
   ```

   The stabilizer models STAB2A, STAB3, STAB4, STABNI, IVOST, PTIST1 and PTIST3 are not permitted as supplementary signal models for the static shunt compensator models CSVGN1, CSVGN3, CSVGN4, CSVGN5 and CSVGN6; the STAB1, IEEEST, IEE2ST, PSS2A, ST2CUT and STBSVC models or user-written stabilizer models may be used.

6. At a machine with active generator and stabilizer models, the generator model is not a static var system model and either there is no excitation system referenced or the exciter model is bypassed (refer to). The following error message is printed:

   ```
   STABILIZER MODEL REQUIRES EXCITATION SYSTEM
   MODEL
   ```

7. At a machine with active generator, stabilizer and exciter models, the generator model is not a static var system model and the excitation system model does not allow a supplementary signal. The following error message is printed:

   ```
   STABILIZER MODEL NOT ALLOWED WITH EXCITER MODEL
   ```


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```
name
```

A stabilizer model is not permitted in conjunction with the IEEET4, IEEET5, IEET5A, IEEEX4 and EXELI excitation system models.

8. At a machine with active generator and excitation limiter models, the generator model does not allow an excitation limiter model. The following error message is printed:

   ```
   EXCITATION LIMITER MODEL NOT ALLOWED WITH GENERATOR
   MODEL name
   ```

   An excitation limiter model is not permitted in conjunction with the classical generator model GENCLS, with the static shunt compensator models CSVGN1, CSVGN3, CSVGN4, CS-VGN5 and CSVGN6, with the induction machine models CIMTR1, CIMTR2, CIMTR3 and CIMTR4, with the DSMES model CDSMS1, or with CBEST, CSMEST and CSTATT models.

9. At a machine with active generator and excitation limiter models, either there is no excitation system referenced or the exciter model is bypassed (refer to Section 16.7.8, "Change Model Status"). The following error message is printed:

   ```
   EXCITATION LIMITER MODEL REQUIRES EXCITATION
   SYSTEM MODEL
   ```

10. At a machine with active generator, excitation limiter and exciter models, the excitation system model does not allow a supplementary signal. The following error message is printed:

    ```
    EXCITATION LIMITER MODEL NOT ALLOWED WITH EXCITER
    MODEL name
    ```

    An excitation limiter model is not permitted in conjunction with the IEEET4, IEEET5, IEET5A, IEEEX4 and EXELI excitation system models.

11. At a machine with active generator and turbine-governor models, the generator model does not allow a governor model. The following error message is printed:

    ```
    GOVERNOR MODEL NOT ALLOWED WITH GENERATOR MODEL
    name
    ```

    A turbine governor model is not permitted in conjunction with the classical generator model GENCLS, the static shunt compensator models CSVGN1, CSVGN3, CSVGN4, CSVGN5 and CSVGN6, the salient pole frequency changer model FRECHG, with the DSMES model CDSMS1, or with CBEST, CSMEST and CSTATT models.

12. At a machine with active generator and turbine-governor models, the torsional shaft model SHAF25 is used in combination with a generator model other than GENDCO. The following error message is printed:

    ```
    SHAF25 MODEL REQUIRES GENDCO GENERATOR MODEL
    ```

13. One machine of an active two-machine PSS®E Library model is not in the working case. The following error message is printed:

    ```
    MODEL name--OTHER MACHINE NOT IN CASE: MACHINE
    i BUS nnn
    ```


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The following sections describe functions available in line mode only.

16.7.7. List Plant Models

The list model references function tabulates the plant related model references for all machines at selected buses. Activity DYCH instructs the user to select the bus for which output is desired. Buses are entered one at a time and the report is printed following the user specification of the bus. Then the request for a bus selection is repeated. A response of zero to the bus selection request terminates the listing function of activity DYCH.

The report of the listing function is as shown in the following example:

```
PLANT RELATED MODELS AT BUS    101 [NUC-A
    21.600]:
MACH    GEN.    COMP.  STAB.  MINEXL  MAXEXL   EXC.    GOV.    TLC
 1     GENROU    --     --     --       --     IEEET1  *TGOV1   ---
```

For any model reference which is bypassed (refer to Section 15.7.8, Change Model Status), the model name is preceded by an asterisk (*).

16.7.8. Change Model Status

The model status modification function allows individual models to be characterized as either active or bypassed. Activity DYCH instructs the user to select the bus where plant related models are to be listed and, possibly, have their status changed. Buses are entered one at a time and activity DYCH cycles through the machines connected to the bus in ascending machine identifier order.

First, the model names are listed as described in Section 16.7.7, “List Plant Models”. Then the consistency checks given in Section 16.7.6, “Consistency Check” are performed and any error or warning messages are printed. Finally, the user may toggle the status of any of the models at the machine as in the following example:

```
MODEL CODES FOR MACHINE 1  AT BUS    101 [NUC-A
      21.600] ARE:
 1 FOR  GENROU
 6 FOR  IEEET1
 7 FOR *TGOV1
ENTER CODE OF MODEL TO BE SWITCHED (0 FOR NO MORE):
```

Following the entry of a valid model code, the model status is changed followed by the printing of a new model name listing and code menu. This process is repeated until a zero or carriage return is entered in response to the above request. The consistency checks are then again performed, and the next machine at the selected bus is processed.

After all machines at the bus have been processed, the request for a bus selection is repeated. A response of zero to the bus selection request terminates the status changing function of activity DYCH.

16.7.9. Remove Plant Models

The model removal function allows individual models at a machine to be removed from the model connection table entry for the machine, and the corresponding array allocation table entry to be marked as unused. Activity DYCH instructs the user to select the bus where plant related models are to be listed and, possibly,

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removed. Buses are entered one at a time and activity DYCH cycles through the machines connected to the bus in ascending machine identifier order.

First, the model names are listed as described in Section 16.7.7, "List Plant Models". Then the consistency checks given in Section 16.7.6, "Consistency Check" are performed and any error or warning messages are printed. Finally, the user may remove the pointer to any of the models at the machine as in the following example:

```
MODEL CODES FOR MACHINE 1  AT BUS    101 [NUC-A
      21.600] ARE:
 1 FOR  GENROU
 6 FOR  IEEET1
 7 FOR *TGOV1
ENTER CODE OF MODEL TO BE REMOVED (0 FOR NO MORE):
```

Following the entry of a valid model code, the appropriate model pointer is removed from the model connection table and a confirming message is printed at the *Progress* tab. This is followed by the printing of an updated model name listing and code menu. This process is repeated until a zero or carriage return is entered in response to the above request. The consistency checks are then again performed, and the next machine at the selected bus is processed.

After all machines at the bus have been processed, the request for a bus selection is repeated. A response of zero to the bus selection request terminates the model removal function of activity DYCH.

Note that the blocks of storage locations in the data storage arrays that were allocated to a model which is removed are available to the user. They are not, however, automatically reused by activities DYRE, ADD, CHAN, or CHSB.

16.7.10. List CHAN Models

The activity CHAN model listing function lists those monitoring model references which were generated as a result of the execution of activities CHAN and CHSB (refer to Section 16.2.2, "Application Notes" When activity CHAN is being used to change the channel assignments assigned to a subset of previously generated assignments, the value printed for < NCHAN > is the larger of the original next available channel and that which would be set as a result of the current execution of activity CHAN. Consider, for example, the situation in which fifty output channels have been assigned and are being monitored and the user wishes to change the quantity associated with channel ten. Activity CHAN would be entered, a starting channel number of ten would be specified rather than the default of fifty-one, and the appropriate channel assignment made. < NCHAN > would be fifty-one, and hence NCHAN would retain its previous value of fifty. and Section 16.25.2, "Application Notes"). This tabulation includes the model name, the type of quantity calculated by the model, the bus, machine, branch or subsystem at which it is called, and the starting ICON and VAR indices used.

When this function is selected, the user is instructed to:

```
SELECT CHAN MODELS TO LIST:
  0 = ALL
  1 = THOSE MONITORING EQUIPMENT IN POWER FLOW
  2 = THOSE AT EQUIPMENT NOT FOUND IN POWER FLOW
ENTER SELECTION CODE:
```

If a two is entered, only models for which the corresponding bus, branch or machine is not contained in the working case are listed. If a one is entered, only models for which the monitored element exists in the working case are listed. If a zero is entered, all CHAN model calls are tabulated.

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16.7.11. Remove Unused CHAN Models

The unused activity CHAN model removal function deletes network monitoring references for which the monitored element is not contained in the working case. Each deleted model reference is listed using the same format as in the CHAN model listing function (refer to List CHAN Models). The CHAN monitoring model table is then packed so as to remove gaps in the numeric sequence.

Note that, while deleted CHAN model table entries are available to subsequent executions of activities CHAN and CHSB, the ICON, VAR and output channel entries which are released are not automatically made available for reuse.

Additional Information
PSS®E GUI Users Guide, Modifying Dynamics Model Pointer Tables *PSS®E* Command Line Interface (CLI) *Users Guide*, Modifying Dynamics Model Pointer Tables

16.8. Linking to Power Flow Activities from Dynamic Simulation

LOFL

Run Line Mode Activity LOFL - CLI
ACTIVITY? >>**LOFL**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Note: Starting with PSS®E-31, mode switching is not apparent to the user and is retained only for backwards compatibility.

The power flow link activity LOFL allows the user to execute any power flow activity after having been operating in dynamics mode.

Activity LOFL is used primarily to:

1. Set up the working case to correspond to the starting condition for a dynamic simulation run. The activity sequence to be used to set up the initial condition power flow case is:

   a. Link to Power Flow Activities from Dynamic Simulation (LOFL)

   b. Retrieve a Power Flow Saved Case File (CASE) <file>

   c. Return to Dynamic Simulation (RTRN)

   where <file> is the name of the Saved Case File containing the initial condition converted power flow case (refer to Section 23.3, “The Network Model”).

2. Allow examination of the network conditions existing at any instant during the system disturbance being simulated. When run during a state-space or extended term dynamic simulation calculation, activ-

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ity LOFL sets the plant and machine power output arrays in the working case to correspond to those contained in the dynamics data arrays PELEC and QELEC. Thus, the machine powers at that point in the simulation will be recognized in any power flow reporting activities which are executed while the power flow linkage is in effect.

When examining network conditions during a dynamics run, any of the power flow reporting activities may be used. However, these activities, in particular activities such as section / POUT and section / LOUT, do not have a knowledge of any load-related models or other non-plant boundary condition models that might be active in the simulation; they only recognize the standard power flow load characteristics (refer to Section 6.3.13, "Load"). Thus, if -network frequency dependence, algebraic load models such as IEELAR, or equipment such as saturable reactors are being modeled, mismatches will occur at the buses involved. These mismatches represent the power variation due to these models at the present instant of the simulation.

When examining network conditions during a dynamics run, the user should exercise caution in executing any power flow activity which modifies the working case unless a system disturbance or re-initialization of the dynamic simulation is intended. Reporting activities may be executed without risk of disturbing the dynamic simulation; all other activities, particularly activities section / CASE, section / CHNG, and all other activities which affect the content of the working case, should be used with great care during a simulation run.

While the power flow linkage is in effect, dynamics working memory is preserved.

Activity LOFL itself is not sensitive to any interrupt control code options. However, any power flow activity run while the power flow linkage is in effect is sensitive to the same interrupt control code to which it responds in normal power flow work.

Additional Information
PSS®E Command Line Interface (CLI) *Users Guide*, Linking to Power Flow Activities from Dynamic Simulation

16.9. Initializing Models for State-Space Simulation

STRT

Run Activity STRT - GUI
Dynamics>Simulation>Perform simulation (STRT/RUN)... *[Perform Dynamic Simulation]*
Run Line Mode Activity STRT - CLI
ACTIVITY? >>**STRT** ENTER CHANNEL OUTPUT FILENAME: >>
Interrupt Control Codes
AB, CM, NC, MO

The state-space dynamic simulation initialization activity STRT calculates the initial values of all variables for each equipment model as a function of the model's constant data and the boundary condition at the bus in

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the working case at which it is referenced. Thus, for a valid execution of activity STRT, dynamics data must be present in dynamics data working memory and the converted network must be represented in the working case. The normal activity sequence which precedes activity STRT is:

1. Restore Dynamics Working Memory from a Binary Snapshot File (RSTR) <file1>

2. Retrieve a Power Flow Saved Case File (CASE) <file2>

where <file1> is the Snapshot File containing the appropriate dynamics model data and output channel specifications, and < file2 > is the Saved Case File containing the converted power flow case.

When run, activity STRT generates an alarm and terminates if generators are not converted. It then ensures that the machine power arrays in the power flow working case are set to their original initial condition values (they may have been changed by a prior execution of activity STRT or MSTR). Activity STRT completes its setup phase by checking for the presence of the factorized network admittance matrix. If necessary, activity FACT (or activities section / ORDR and section / FACT) is executed.

Activity STRT initializes the dynamic simulation by first doing a network solution and then, based on the bus boundary conditions, initializing the state variables (STATEs) and algebraic variables (VARs) of each equipment model. If, in setting up for the initial network solution, any CIMTR5, CIMWSC, CIM5xx, CIMWxx, CLODxx, or IEELxx model removes constant admittance load from the load at which it is called, an appropriate message is printed and the network admittance matrix is automatically refactorized prior to commencing the network solution.

Any machine which is online in the working case but which does not have a generator model assigned to it generates an alarm with a message of the form:

```
NO ACTIVE GENERATOR MODEL CALL FOR MACHINE i
AT BUS nnnn
```

The source current (ISORCE) of any such machine is set to zero. The presence of any such machines renders the initial condition invalid, and activity RUN will not allow itself to be executed following such an initialization.

Similarly, any two-terminal, multi-terminal, or VSC dc line, any FACTS device, or any switched shunt device that is under continuous control which is in-service in the working case but which does not have a dynamic table model assigned to it generates an alarm with one of the following error messages:

```
NO ACTIVE TABLE MODEL CALL FOR TWO-TERMINAL
DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR MULTI-TERMINAL
DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR VSC DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR FACTS DEVICE
"name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR SWITCHED SHUNT
AT BUS "bus name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
```


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During the model initialization process, any model variable which is initialized beyond its prescribed limits (e.g., the voltage regulator output, STATE(K+1), of the IEEE type one excitation system model, IEEET1) is alarmed with a message of the form:

```
name AT BUS nnnn MACHINE i INITIALIZED OUT OF
LIMITS
```

where < name > is the model name (e.g., IEEET1), < nnnn > is the bus number, and < i > is the machine identifier of the model containing the out-of-limits variable. The user should resolve any such error conditions before running a simulation.

Activity STRT then prints, at the progress report output device, the case heading followed by the message:

```
INITIAL CONDITION LOAD FLOW USED n ITERATIONS
```

A value of < n > greater than one usually (but not always) indicates some sort of modeling error.

Similarly, any two-terminal, multi-terminal, or VSC dc line, any FACTS device, or any switched shunt device that is under continuous control which is in-service in the working case but which does not have a dynamic table model assigned to it generates an alarm. The presence of any such dc line, FACTS or switched shunt device could indicate one of the following:

- A table model was assigned to the dc line, FACTS device or the switched shunt, but the model was later removed. When this condition is detected during activity STRT, activity RUN will not allow itself to be executed for dc lines and FACTS devices. In the case of switched shunts, RUN will still be allowed, because the simulation will use the power flow value of shunt in the admittance matrix.
- No table model was ever assigned to the dc line, FACTS, or the switched shunt device, or that the model assigned is a CONEC and CONET called model. When this condition is detected during activity STRT, activity RUN

will allow itself to be executed.

During the model initialization process, any model variable which is initialized beyond its prescribed limits (e.g., the voltage regulator output, STATE(K+1), of the IEEE Type 1 excitation system model, IEEET1) generates an alarm. The user should resolve any such error conditions before running a simulation. The presence of such error alarms renders the initial condition invalid in the sense that the system is not in a steady state; that is, if activity STRT were to be followed by activity RUN with no disturbances imposed, there would be movement as the out-of-limits quantities are brought back to their appropriate limits. The exception is the condition where the bus boundary conditions are such that the variable in question is initialized *exactly* at its limit. Such a situation may occur, for example, when a generator is fully loaded and its turbine mechanical power initializes precisely at its ceiling.

Activity STRT then prints, at the *Progress* tab, the case heading followed by the number of iterations used to calculate the initial power flow. A value greater than one usually (but not always) indicates some sort of modeling error. This is especially true if the converted power flow network was solved with activity TYSL prior to the execution of activity STRT. (The network solution method used in the dynamic simulation is identical to that used by activity TYSL, and the default convergence tolerance is an order of magnitude smaller in section / TYSL.)

The most common causes of this type of error are:

1. No dynamics model corresponding to an in-service dc line or FACTS device in the working case.

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2. Forgetting to compile the connection subroutines following a change to these subroutines, or forgetting to relink following a recompilation of one of these subroutines.

The types of errors described above and all other initialization errors and warnings should be resolved before executing simulations.

Activity STRT summarizes the conditions at each online machine by tabulating the following quantities:

- Terminal voltage (ETERM).
- Exciter output voltage (EFD).
- Active and reactive power output (PELEC and QELEC).
- Power factor.
- Machine angle in degrees (ANGLE).
- d and q axis currents on machine base.

This report is in ascending bus number order when the *numbers* output option is in effect, and in alphabetical bus name order using the *names* option. Note that for wind machines, because the quantity EFD is not of any relevance, this is set to zero.

Following the calculation of the system initial condition, a dummy step calculation is made and the time derivatives (DSTATEs) of all STATE variables are checked for a steady state. Any STATE for which the time derivative is nonzero generates an alarm (nonzero as used here is interpreted as significantly different from zero in relation to the value of the STATE). For any such STATE variable, its index is printed along with the values of its time derivative and the STATE variable itself. For STATEs that generated an alarm that are assigned to plant-related or wind-machine related models, the model and machine are also identified. For STATEs that generated an alarm that are assigned to load-related models, the model and load are also identified. For STATEs that generated an alarm that are assigned to auxiliary-signal models, the model and the name of the device (the two-terminal, multi-terminal or VSC dc line or the FACTS device) with which the auxiliary-signal is associated are identified. For STATEs that generated an alarm that are assigned to dc line models, the model and dc line are identified. For STATEs that generated an alarm that are assigned to FACTS device models, the model and FACTS device are identified. For STATEs that generated an alarm that are assigned to dc line models, the model and dc line are identified. For STATEs that generated an alarm that are assigned to switched shunt models, the model and switched shunt device are identified.

In each case, the ac bus is also identified. In the case of machines, loads, and switched shunts, this will be the ac bus at which the machine, load or the switched shunt are connected; for two-terminal dc lines, VSC dc lines, as well as for auxiliary signals attached to these devices, this is the rectifier ac bus; for multi-terminal dc lines and for auxiliary signals attached to multi-terminal dc lines, this will be the ac bus associated with converter number one; for FACTS devices and for auxiliary signals attached to FACTS devices this will be the sending end ac bus.

If no nonzero STATEs are found, activity STRT outputs a message. The absence of suspect initial conditions generally (though not always) indicates a valid steady state. Conversely, the presence of initial condition alarms usually indicates some error in setup.

The same types of procedural errors which cause the initial network solution to use more than one iteration may also result in nonsteady STATEs. In addition, the following situations may cause these errors:

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1. Model variables initialized out of limits. When this occurs, the STATE variable downstream of the limit violation may also show some perturbation.

2. STATE variables characterized by a high gain and a short time constant. This often occurs in the voltage regulator transfer function (e.g., a gain of four hundred and a time constant of 0.05 seconds). This can usually be accepted if the value of the derivative is small relative to the value of the STATE (say, two or three percent).

3. Improperly specified data (e.g., zero time constants where they are not allowed, unrealistic limits, etc.).

The user specifies the name of the Channel Output File to be used in the subsequent dynamic simulation run. If no filename is specified in response to the above instruction, the writing of the output channel variable values to a file is suppressed during the simulation run.

The user must specify a Dynamics Snapshot Data File to preserve the initial system condition, as contained in the dynamics data arrays. If no filename is specified, activity STRT is terminated. If a Snapshot File name is specified, a Snapshot is taken prior to terminating activity STRT.

Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity STRT.

The default time step in PSS®E is half a cycle (i.e., 0.008333 of a second when base frequency is sixty cycles and 0.01 of a second for fifty cycle systems). Activity STRT sets the value of simulation TIME to minus two time steps. This provides for a period of simulation under steady-state conditions prior to initiating a disturbance.

The user may specify that the network solution convergence monitor be automatically printed. This is particularly useful when PSS®E is being executed as a batch job.

Activity STRT responds to the following interrupt control codes:

AB	(1) Suppress machine initial conditions summary. (2) Suppress suspect initial conditions summary.
CM	Print the network solution convergence monitor.
NC	Suppress network convergence monitor when the CM suffix was specified in selecting activity STRT.
MO	Print induction motor local iteration convergence monitor.

The format of the network convergence monitor is the same as that of activities section / TYSL and section / SOLV.

In the induction motor model CMOTOR and the complex load models CLOAD and CLODxx, a local iteration is used to solve for the equivalent load drawn by the model at each iteration of the network solution.

The CMOTOR convergence monitor tabulates the bus number at which the model is being called, the local iteration number, and the ratio of the change in motor slip to its convergence tolerance of 0.00001.

The CLOAD and CLODxx convergence monitor tabulates the bus number at which the model is being called, the local iteration number, the voltage magnitude at the low side bus, the change in voltage magnitude, and the complex voltage at the low side bus. Preceding each such local iteration monitor line, the inner loop convergence monitors of the large and small motors (listed as motors one and two respectively) are printed. These contain the motor number, the inner loop iteration number, the motor slip, and the ratio of the change in slip to its convergence tolerance of one-tenth of the network solution convergence tolerance.

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Additional Information
PSS®E GUI Users Guide, Running State-Space Simulation *PSS®E* Command Line Interface (CLI) *Users Guide*, Initializing Models for State-Space Simulation

16.10. Performing State-Space Simulation in Time Steps

RUN

Run Activity RUN - GUI
Dynamics>Simulation>Perform simulation (STRT/RUN)... *[Perform Dynamic Simulation]*
Run Line Mode Activity RUN - CLI
ACTIVITY? >>**RUN** AT TIME = 0.017 ENTER TPAUSE, NPRT, NPLT, CRTPLT: >>
Interrupt Control Codes
AB, CM, NC, MO, CH, TI

The state-space time simulation activity RUN sequences through time solving the system's differential equations and the electrical network at each time step. When run, activity RUN generates an alarm and terminates if activity STRT had not previously been successfully executed or if generators are not converted.

The user specifies the following simulation options:

TPAUSE	Run to n.nnnn secs : the value of simulated TIME at which the simulation activity is to be terminated. Activity RUN steps through time simulating the system, starting at the present value of simulation TIME, until TIME > TPAUSE.
NPRT	Print every <n> time steps: integer value defining the interval, in units of the number of simulation time steps, between the recording of output channel values. This setting is recorded and becomes the initial value the next time RUN is run. The values of the first NCHAN output channel variables are written to the *Progress* tab every NPRT time steps. They are printed for the present value of TIME, every NPRT time steps thereafter, and for TIME = TPAUSE. If NPRT is entered as zero, the tabulation of output channel values is bypassed.
NPLT	Write every < n > time steps: integer value defining the interval, in units of the number of simulation time steps, between the recording of output chan-

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	nel values. This setting is recorded and becomes the initial value the next time RUN is run. The values of the first NCHAN output channel variables are written to the Channel Output File every NPLT time steps. They are written to the file for the present value of TIME, every NPLT time steps thereafter, and for TIME = TPAUSE. If NPLT is entered as zero (or one), channel values are written to the output file at the completion of each time step. NPLT has significance only if a channel output file has been specified either in activities STRT or ALTR.
CRTPLT	Plot every <n> time steps: integer value defining the interval, in units of the number of simulation time steps, between the recording of output channel values. This setting is recorded and becomes the initial value the next time RUN is run.

If the value specified for TPAUSE is less than the current value of simulation TIME, activity RUN is terminated.

If a Channel Output File has been opened (in the most recent execution of activity STRT or subsequently via ALTR), the filename is displayed in a message.

Activity RUN then checks for islands which contain no in-service machines, generates an alarm for buses in such islands, and disconnects them. Activity RUN completes its setup phase by checking for the presence of the factorized network admittance matrix. If necessary, activity FACT (or activities ORDR and FACT) is executed.

The simulation starts at the present value of simulation TIME.

The user may specify that the network solution convergence monitor be automatically printed. This is particularly useful when PSS®E is being executed as a batch job.

Activity RUN responds to the following interrupt control codes:

AB	Force a pause by setting TPAUSE to the current value of simulation TIME. This overrides the value of TPAUSE which was specified at the time activity RUN was run. In addition, if PSS®E is operating in its Response File mode, all active Response Files are closed and subsequent user commands are taken from the user's terminal.
CM	Print the network solution convergence monitor.
NC	Suppress network convergence monitor when the CM suffix was specified in selecting activity RUN.
MO	Print the induction motor local iteration convergence monitor.
CH	Print the output channel values at each time step. The value of NPRT which was entered at the time activity RUN was run is overridden while this option

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	is in effect. When this option is cleared, the original value of NPRT is restored and the next printing of channels occurs NPRT time steps from the TIME of the clearing of the option.
TI	Print the value of simulation TIME at the user's terminal.

The form of the network convergence monitor is identical to that of activities STRT, TYSL, and SOLV. The induction motor convergence monitor is identical to that of activity STRT.

16.10.1. Application Notes

When setting up a new system model, it is good practice, after a clean run of activity STRT is obtained, to execute activity RUN for some period of simulation TIME without applying any disturbances. This can often reveal time constants which are too small in relation to the simulation time step, DELT, or other modeling errors. Another useful test is to initiate some system movement with a minor disturbance and execute the simulation until a new steady state is reached. This might be, for example, the shedding of a small amount of load, the tripping of a small unit, or the tripping of a lightly loaded branch.

Activity STRT sets the output control variables NPRT and NPLT to one, and CRTPLT to zero. On the first subsequent execution of activity RUN, these values are the defaults for these variables. On following executions of activity RUN, these variables default to the values specified on the previous execution of activity RUN.

Whenever the value of simulation TIME reaches the time specified for TPAUSE, the time derivative and network solution calculations are performed with a flag set to inform the simulation that this is a t^- calculation. For any of the channel output options which are active, the appropriate output is generated regardless of whether the present time step is a print or plot time step as determined by the variables NPRT, and NPLT. When activity RUN is reentered, with perhaps some switching operation having been specified, the time step is repeated as a t^+ calculation. Appropriate channel output is performed at the completion of the time step as determined by the new specification of the output control variables. This applies both for the case in which activity RUN has advanced TIME to the value specified for TPAUSE and when interrupted by the user.

There are several PSS®E models which activate automatic switchings (e.g., blocking or unblocking of a dc line by a dc line model, branch tripping by a relay model, load shedding, generator tripping, and so on). When one of these switchings takes place, a dual time step calculation is performed as described above. If a Channel Output File is active, the output channel values are written to it for both the pre- and post-switching time step calculations made at the switching TIME; similarly, CRT plotting takes place for both the pre- and post-switching time step calculations.

Both upon entering activity RUN and before performing the t^+ time step calculation following automatic switchings, activity RUN checks for islands which contain no in-service machines, generates an alarm for buses in such islands, and disconnects them. It then calculates a new sparsity-preserving ordering (activity section / ORDR) and rebuilds the factorized admittance matrix (activity section / FACT).

It is not permissible to change the simulation time step, DELT, during a state-space simulation run. Output channels may however be added during the course of a simulation run.

If the user forces a pause in the simulation, care must be taken when PSS®E is being executed from a command file with PSS®E commands and responses taken from a Response File. As described above, activity RUN closes any active PSS®E Response Files. The source of subsequent PSS®E dialog varies among the different PSS®E host computers.

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If the network solution fails to reach its tolerance in the prescribed number of iterations, an alarm is printed and the simulation continues as though convergence had been achieved. Refer to Section 23.7, "Running a Simulation"for additional details.

Additional Information
PSS®E GUI Users Guide, Running State-Space Simulation *PSS®E* Command Line Interface (CLI) *Users Guide*, Running State-Space Simulation in Time Steps

16.11. Initializing Excitation System Models

ESTR

Run Activity ESTR - GUI
Dynamics>Simulation>Perform exciter simulation (ESTR/ERUN)... *[Perform Exciter Simulation Test]*
Run Line Mode Activity ESTR - CLI
ACTIVITY? >>**ESTR** ENTER 0 FOR RESPONSE RATIO TEST 1 FOR OPEN CIRCUIT SETPOINT STEP TEST: >>
Interrupt Control Codes
AB

The excitation system performance initialization activity ESTR initializes the dynamic models in preparation for excitation system performance checking. Activity ESTR is used in conjunction with activity ERUN to verify excitation system data by simulating the step response of excitation systems in isolation. (Section 23.5, "Model Verification" contains details on the use of this pair of activities.)

Activities ESTR and ERUN, then, are special versions of activity STRT and RUN respectively which are used in verifying excitation system data. As such, their operating characteristics are similar to those of activities STRT and RUN.

The user may specify a single bus for processing. Activity ESTR (and the subsequent execution of activity ERUN) then tests the excitation system response of all machines at the specified bus having a connected excitation system model. Otherwise, all machines with connected exciter models are tested.

Following the initialization calculation, activity ESTR prints the case heading followed by the number of iterations used to calculate the initial power flow. This is followed by the machine initial conditions summary for all machines being tested in a form identical to that of activity STRT.

The user must specify the name of the Channel Output File to be used in the subsequent exciter test simulation. If no filename is specified in response to the above instruction, the writing of the output channel variable values to a file is suppressed during the simulation.

Activity ESTR overrides the user specified set of main output channel assignments (refer to Section 23.6.1, "Output Channel Selection") with a set of channels suitable for excitation system testing and sets all CRT plot channel selections (refer to Section 23.6.3, "CRT Plot Channels") to zero. Therefore, activity ESTR does not enable the user to take an optional Snapshot as does activity STRT. In fact, the user should *never* overwrite

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the main simulation Snapshot File following execution of activity ESTR because its execution destroys the user specified output channel assignments.

When using line mode, two separate step response tests may be simulated: a response ratio test, and an open circuit test with a step of the voltage regulator setpoint.

Activity ESTR may be terminated by entering the AB interrupt control code.

16.11.1. Response Ratio Test

In this test, activity ESTR initializes each generator to rated MVA (i.e., to MBASE as contained in the power flow generator data) at a user specified power factor (default = 0.95, lagging). The user may also specify those machines which are to be initialized at another power factor (e.g., synchronous condensers).

Activity ESTR initializes all machines to rated MVA at their specified power factor. Because the machine initialization to rated MVA is based on the value specified as MBASE for each machine, the initialization, and subsequent simulation with activity ERUN, is not valid for any machines where data is specified on a base other than its actual MVA base. As pointed out several times throughout this manual, it is therefore strongly recommended that the PSS®E user adopt the practice of specifying machine data parameters on actual machine MVA base.

The user selected output channels are overridden, and the exciter output voltage, EFD, for each of the < n > online machines being tested are placed in channels one through < n >. Channel assignments are made in ascending bus number order and identifiers are assigned as listed in Table 16.2, "Activity CHAN Summary".

Following activity ESTR, the user should check the value of EFD for each machine before proceeding to activity ERUN. This value of full load excitation should be compared with that obtained from generator test data for rated load or from the generator data verification program, VCV (see *Additional Resources for PSS® E*, VCV). An unacceptable value of full load EFD most often indicates an erroneous specification of the saturation data for the main generator. All generator data, in particular SE(1.2), should be checked before proceeding.

16.11.2. Open Circuit Step Response Test

In this test, activity ESTR initializes each generator to unity terminal voltage on open circuit. The user selected output channels are overridden, and the exciter output voltage, EFD, and machine terminal voltage, ETERM, for each of the < n > online machines being tested are placed in channels 1 through 2n. Channel assignments are made in ascending bus number order with EFDs in odd numbered channels and ETERMs in even numbered channels. The identifiers assigned by activity ESTR are as listed in Table 16.2, "Activity CHAN Summary".

The initial value of EFD for each unit will be slightly greater than the air gap line value of unity due to saturation in the main generator. An initial open circuit value of EFD exceeding about 1.2 pu is a fair indication that the saturation curve specified for the generator is erroneous.

Additional Information
PSS®E GUI Users Guide,Running Exciter Simulation *PSS®E* Command Line Interface (CLI) *Users Guide*, Initializing Excitation System Models

16.12. Performing Exciter Simulation in Time Steps

ERUN

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Run Activity ERUN - GUI
Dynamics>Simulation>Perform exciter simulation (ESTR/ERUN)... *[Perform Exciter Simulation Test]*
Run Line Mode Activity ERUN - CLI
ACTIVITY? >>**ERUN** AT TIME = 0.000 ENTER TPAUSE, NPRT, NPLT, CRTPLT >>
Interrupt Control Codes
AB, CH, TI

The excitation system performance checking activity ERUN simulates the step response of excitation systems in isolation. Activity ERUN performs the test selected in the previous execution of activity ESTR; the selection of activity ERUN must, therefore, be preceded by activity ESTR.

The user specifies the simulation options TPAUSE, NPRT, NPLT, CRTPLT as in activity RUN. Re-entering these parameters is necessary because the required activity ESTR has overridden channel assignments. Recall that activity ESTR overrides the user selected output channel assignments with channel values appropriate for the excitation system response tests and that it sets CRT plot channel selections to zero. Thus, if the terminal plot option is to be selected in activity ERUN (i.e., CRTPLT is to be specified as nonzero), the user must reestablish the CRT plot channels and scales to correspond to the desired quantities (refer to Section 23.6, "Simulation Outputs"). This must be done *after* execution of activity ESTR.

If the value specified for TPAUSE is less than the current value of simulation TIME, activity ERUN is terminated.

If a Channel Output File has been opened (in the most recent execution of activity ESTR or subsequently via activity ALTR), the filename is displayed in a message.

The simulation starts at the present value of simulation TIME.

Activity ERUN responds to the following interrupt control codes:

AB	Force a pause by setting TPAUSE to the current value of simulation TIME. This overrides the value of TPAUSE which was specified at the time activity ERUN was run. In addition, if PSS®E is operating in its Response File mode, all active Response Files are closed and subsequent user commands are taken from the user's terminal.
CH	Print the output channel values at each time step. The value of NPRT which was entered at the time activity ERUN was run is overridden while this option is in effect. When this option is cleared, the original value of NPRT is restored and the next printing of channels occurs NPRT time steps from the TIME of the clearing of the option.
TI	Print the value of simulation TIME at the user's terminal.

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16.12.1. Response Ratio Test

Activity ERUN automatically raises the voltage regulator reference settings by a large amount at TIME equals zero seconds. This will drive all excitation systems being tested to their ceilings as -rapidly as possible.

At TIME = 0.5 seconds, activity ERUN tabulates at the *Progress* tab the values of full load excitation and of Response Ratio for all machines being tested. The tabulation is in ascending bus number or alphabetical order, depending on the output option currently in effect. Although response ratio is printed for all machines, its main significance is for rotating machine exciters. SCR bridge type exciters, such as that modeled by SCRX, will generally exhibit very high response ratios because of their ability to reach ceiling output practically instantaneously.

This test should be carried out until at least two seconds to allow all rotating exciters to reach their ceilings.

Using activities ESTR and ERUN to perform this simulation yields three major items of information on each generator-exciter combination:

1. Full load excitation; note that this quantity is determined exclusively as a function of the generator model data parameters, particularly its saturation curve.

2. The ceiling value of EFD; note that EFDs are expressed in per unit of excitation voltage for rated open circuit generator voltage on the air gap line, not in per unit of rated load excitation or any other value of excitation voltage.

3. The excitation system response ratio.

16.12.2. Open Circuit Step Response Test

The response ratio test discussed in Response Ratio Test provides information on the modeling of rotating exciters but gives no information on the correctness of the voltage regulator gains and time constants. These data are checked for correctness by ensuring that the excitation system gives stable and effective control of generator terminal voltage when the generator is operating at rated speed on open circuit (e.g., just before synchronizing). The open circuit response test is intended to check this aspect of excitation system performance.

In this test, a simple step change of about five percent is applied to the voltage regulator references and the resulting responses of field voltage and generator terminal voltage are observed. At TIME equals zero seconds, the activity ERUN applies the user-specified Vref pu step change (typically 0.02 to 0.1) to all voltage regulator setpoints. The step magnitude should not exceed about ten percent (0.1 pu) because the object of this test is to reveal small disturbance behavior.

This simulation should normally be carried out until at least five seconds; it may have to be extended to as long as ten seconds if any slow responding exciters have not reached their steady state.

The values of excitation system parameters should be such that a well damped but not overly sluggish response is obtained. The response to a setpoint step of about 0.05 pu should normally show a slight overshoot but be well damped with no persistent oscillations.

CRT plotting must not be enabled until *after* the voltage reference step has been specified at TIME equals zero seconds.

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16.12.3. Application Notes

The procedure in simulating the excitation system response tests follows that used in running conventional state-space dynamic simulations. Instead of using activities STRT and RUN after the data has been brought into memory, activities ESTR and ERUN are used.

The response tests calculate the excitation system response of either all generating units in the system with connected excitation system models, or all machines at a bus (designated during activity ESTR initialization of a single bus) with connected exciter models. Because machines are operating in isolation, no network solution is required in the simulation. Furthermore, only the generator and excitation system models are active in these simulations; all other models perform no calculations. Therefore, these simulations require substantially less computer time than complete system simulations.

The typical sequencing of PSS®E activities in preparing for these simulations is as follows:

1. Restore Dynamics Working Memory from a Binary Snapshot File () <file1>

2. Retrieve a Power Flow Saved Case File (section / CASE) <file2>

where <file1> is the name of the appropriate dynamics data Snapshot File and <file2> is the name of the corresponding converted power flow Saved Case File. Generators in the working case must be converted (i.e., activity section / CONG must have been executed). Because the network is not involved in these simulations, the load characteristic is immaterial and the execution of activity section / FACT is not required.

Following execution of activity ESTR, activity ERUN should have its first TPAUSE specified as zero, because the perturbation of the voltage regulator references occurs at that TIME. Activity ERUN should then be reentered, specifying the desired duration of the simulation.

Note again that because activity ESTR replaces the user's output channel assignments with other quantities, activity SNAP *should not* be executed following activity ESTR.

The comments given in Section 16.10.1, "Application Notes" for activity RUN regarding the defaults for the output control variables, the repeat time step calculation at each simulation pause, changing of the simulation time step, adding output channels, and interrupting the simulation when running from a command file also apply to activity ERUN.

Additional Information
PSS®E GUI Users Guide, Running Exciter Simulation *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Exciter Simulation in Time Steps

16.13. Initializing Turbine Governor Models

GSTR

Run Activity GSTR - GUI
Dynamics>Simulation>Perform governor response simulation (GSTR/GRUN)... *[Perform Governor Response Simulation]*

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Run Line Mode Activity GSTR - CLI
ACTIVITY? >>**GSTR** ENTER INITIAL LOADING, STEP (P.U.): >>
Interrupt Control Codes
AB

The governor performance initialization activity GSTR initializes the dynamic models in preparation for governor performance checking. Activity GSTR is used in conjunction with activity GRUN to verify turbine governor data by simulating the response of individual units in isolation. (Section 23.5, "Model Verification" contains details on the use of this pair of activities.)

The principal purpose of the governor response test is to ensure that the governor gain and time constant parameters correspond to a correctly tuned, well damped response. This test simulates the response of the governing loops of units in isolation to a step change in load. The load active power is held constant and the reactive load is adjusted to maintain unity terminal voltage. This test, then, indicates the damping due to the turbine and governor loop only.

Activities GSTR and GRUN are special versions of activities STRT and RUN, respectively which are used in verifying governor data. As such, their dialog and operating characteristics are similar to those of activities STRT and RUN.

If the user has specified processing a single bus, activity GSTR (and the subsequent execution of activity GRUN) then tests the governor response of all machines at the specified bus having a connected governor model. Otherwise, all machines with connected governor models are tested.

The user specifies initial machine loading in per unit of machine base, MBASE, and the load step change to be applied at TIME equals zero by activity GRUN. Because the machine initialization to a specified fraction of rated MVA is based on the value specified as MBASE for each machine, this test assumes that generator parameters are entered on actual machine base. The governor droop is to be specified on turbine base if the turbine governor model has the provision to specify the turbine rating; otherwise the governor droop has to be specified on the machine base.

Activity GSTR prints the case heading followed by the number of iterations used to calculate the initial power flow. This is followed by the machine initial conditions summary for all machines being tested in a form identical to that of activity STRT.

The user specifies the Channel Output File to be used in the subsequent governor test simulation. If no filename is specified in response to the above instruction, the writing of the output channel variable values to a file is suppressed during the simulation.

Activity GSTR overrides the user specified set of output channel assignments (refer to Section 23.6.1, "Output Channel Selection"), and the machine speed deviation, SPEED, and turbine mechanical power, PMECH, for each of the < n > online machines being tested are placed in channels 1 through 2n. Channel assignments are made in ascending bus number order with SPEEDs in odd numbered channels and PMECHs in even numbered channels; the identifiers assigned by activity GSTR are as listed in Table 14-2. Activity GSTR also sets all CRT channel selections (refer to Section 18.6.3 CRT Plot Channels) to zero. Therefore, activity GSTR does not enable the user to take an optional snapshot as does activity STRT. In fact, the user should *never* overwrite the main simulation Snapshot File following execution of activity GSTR because its execution destroys the user specified output channel assignments.

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The machine initial conditions summary of activity GSTR may be terminated with the AB interrupt control code.

16.13.1. Application Notes

The testing of cross-compound governor models requires that both generator models be active.

Additional Information
PSS®E GUI Users Guide, Running Governor Response Simulation Test *PSS®E* Command Line Interface (CLI) *Users Guide*, Initializing Turbine Governor Models

16.14. Performing Governor Response Simulation in Time Steps

GRUN

Run Activity GRUN - GUI
Dynamics>Simulation>Perform governor response simulation (GSTR/GRUN)... *[Perform Governor Response Simulation]*
Run Line Mode Activity GRUN - CLI
ACTIVITY? >>**GRUN** AT TIME = 0.000 ENTER TPAUSE, NPRT, NPLT, CRTPLT: >>
Interrupt Control Codes
AB, CH, TI

The governor performance checking activity GRUN simulates the response of the governing loops of units in isolation to a step change in load. Activity GRUN uses the step specified in the previous execution of activity GSTR; the selection of activity GRUN must, therefore, be preceded by activity GSTR.

The user specifies the simulation options TPAUSE, NPRT, NPLT, CRTPLT as in activity RUN. Re-entering these parameters is necessary because the required activity GSTR has overridden channel assignments. Recall that activity GSTR overrides the user selected output channel assignments with channel values appropriate for the governor response test and that it sets CRT plot channel selections to zero. Thus, if the terminal plot option is to be selected in activity GRUN (i.e., CRTPLT is to be specified as nonzero), the user must reestablish the CRT plot channels and scales to correspond to the desired quantities (refer to Section 23.6, "Simulation Outputs"). This must be done *after* execution of activity GSTR.

If the value specified for TPAUSE is less than the current value of simulation TIME, activity GRUN is terminated.

If a Channel Output File has been opened (in the most recent execution of activity GSTR or subsequently via activity ALTR), the filename is displayed in a message.

The simulation starts at the present value of simulation TIME.

Activity GRUN responds to the following interrupt control codes:

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AB	Force a pause by setting TPAUSE to the current value of simulation TIME. This overrides the value of TPAUSE which was specified at the time activity GRUN was run. In addition, if PSS®E is operating in its Response File mode, all active Response Files are closed and subsequent user commands are taken from the user's terminal.
CH	Print the output channel values at each time step. The value of NPRT which was entered at the time activity GRUN was run is overridden while this option is in effect. When this option is cleared, the original value of NPRT is restored and the next printing of channels occurs NPRT time steps from the TIME of the clearing of the option.
TI	Print the value of simulation TIME at the user's terminal.

Activity GRUN automatically applies the load increment (typically 0.05 to 0.1) specified in activity GSTR to each unit at TIME equals zero seconds.

This simulation should normally be carried out until at least ten seconds; the simulation may have to be extended to as long as twenty seconds if hydro units are present.

16.14.1. Application Notes

Using activities GSTR and GRUN to perform this simulation reveals the transient variation of turbine power and machine speed. All governing units should have a well damped response. Hydro governors will generally exhibit a somewhat greater overshoot than steam turbine governors, but should still be well damped, showing no persistent oscillations. A small negative change in hydro turbine power before it follows a positive change in load power is normal and does not indicate incorrect governor tuning.

The procedure in simulating the governor response test follows that used in running conventional state-space dynamic simulations. Instead of using activities STRT and RUN after the data has been brought into memory, activity GSTR and GRUN are used.

The response test calculates the governor response of either all generating units in the system with connected governor models, or all machines at a bus (designated during activity GSTR initialization of a single bus) with connected governor models. Because machines are operating in isolation, no network solution is required in the simulation. Furthermore, only the generator and turbine-governor models are active in these simulations; all other models perform no calculations. Therefore, these simulations require substantially less computer time than complete system simulations.

The typical sequencing of PSS®E activities in preparing for these simulations is as follows:

1. Restore Dynamics Working Memory from a Binary Snapshot File (RSTR) <file1>

2. Retrieve a Power Flow Saved Case File (CASE) <file2>

where <file1> is the name of the appropriate dynamics data Snapshot File and <file2> is the name of the corresponding converted power flow Saved Case File. Generators in the working case must be converted (i.e.,

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activity section / CONG must have been executed). Because the network is not involved in these simulations, the load characteristic is immaterial and the execution of activity section / FACT is not required.

Following execution of activity GSTR, activity GRUN should have its first TPAUSE specified as zero, because the load increment is applied at that TIME. Activity GRUN should then be reentered, specifying the desired duration of the simulation.

Note again that because activity GSTR replaces the user's output channel assignments with other quantities, activity SNAP *should not* be executed following activity GSTR.

The comments given in Section 16.10.1, "Application Notes" for activity RUN regarding the defaults for the output control variables, the repeat time step calculation at each simulation pause, changing of the simulation time step, adding output channels, and interrupting the simulation when running from a command file also apply to activity GRUN.

Additional Information
PSS®E GUI Users Guide, Running Governor Response Simulation Test *PSS®E* Command Line Interface (CLI) *Users Guide*, Running Governor Response Simulation in Time Steps

16.15. Building a State Variable Matrix for Linear Dynamic Analysis (LSYSAN)

ASTR

Run Activity ASTR - GUI
Dynamics>Build matrices for LSYSAN program (ASTR)... *[Build Matrices for LSYSAN]*
Run Line Mode Activity ASTR - CLI
ACTIVITY? >>**ASTR** ENTER LARGEST DERIVATIVE CHANGE ALLOWED (DEFAULT IS 0.01): >>
Interrupt Control Codes
AB, CM, NC, MO

The state variable matrix construction activity ASTR calculates the initial values of all variables for each equipment model as a function of the model's constant data and the boundary condition at the bus in the working case at which it is referenced. It then infers the state variable system matrices and processes them into a form suitable for input into the Linear Dynamic Analysis program, LSYSAN.

For a valid execution of activity ASTR, dynamics data must be present in dynamics data working memory and the converted network must be represented in the working case. The normal activity sequence which precedes activity ASTR is:

1. Restore Dynamics Working Memory from a Binary Snapshot File (RSTR) <file1>

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2. Retrieve a Power Flow Saved Case File (CASE) <file2>

where <file1> is the Snapshot File containing the appropriate dynamics model data and output channel specifications, and <file2> is the Saved Case File containing the converted power flow case.

When run, activity ASTR generates an alarm and terminates if generators are not converted. It then ensures that the machine power arrays in the power flow working case are set to their original initial condition values (they may have been changed by a prior execution of activity STRT or MSTR). Activity ASTR completes its setup phase by checking for the presence of the factorized network admittance matrix. If necessary, activity FACT (or activities section / ORDR and section / FACT) is executed.

Activity ASTR does a network solution and then, based on the bus boundary conditions, initializes the state variables (STATEs) and algebraic variables (VARs) of each equipment model. If, in setting up for the initial network solution, any CIMTR5, CIMWSC, CIM5xx, CLODxx, or IEELxx model removes constant admittance load from the load at which it is called, an appropriate message is printed and the network admittance matrix is automatically refactorized prior to commencing the network solution.

Any machine which is online in the working case but which does not have a generator model assigned to it generates an alarm with a message of the form:

```
NO ACTIVE GENERATOR MODEL CALL FOR MACHINE i
AT BUS nnnn
```

Similarly, any two-terminal, multi-terminal, VSC dc line, FACTS device, or switched shunt device in continuous control mode, which is in-service in the working case but which does not have a dynamic table model assigned to it generates an alarm with one of the following error messages:

```
NO ACTIVE TABLE MODEL CALL FOR TWO-TERMINAL
DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR MULTI-TERMINAL
DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR VSC DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR FACTS DEVICE
"name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR SWITCHED SHUNT
AT BUS "bus name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
```

The presence of any such dc line, FACTS device or switched shunt could indicate one of the following:

- a table model was assigned to the dc line, FACTS device, or the switched shunt, but the model was later removed by selecting Remove Model form the popup menu that appears by right-clicking on model name in the Dynamics Models Spreadsheet; refer to *PSS®E GUI Users Guide*, Models Spreadsheet.
- no table model was ever assigned to the dc line, FACTS device or the switched shunt, or that the model assigned is a CONEC and CONET called model.

The presence of any dc line or FACTS device for which there is no dynamic model renders the initial condition invalid.

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During the model initialization process, any model variable which is initialized beyond its prescribed limits (e.g., the voltage regulator output, STATE(K+1), of the IEEE Type 1 excitation system model, IEEET1) generates an alarm with a message of the form:

```
name AT BUS nnnn MACHINE i INITIALIZED OUT OF
LIMITS
```

where < name > is the model name (e.g., IEEET1), < nnnn > is the bus number, and < i > is the machine identifier of the model containing the out-of-limits variable. The user should resolve any such error conditions before using the matrices constructed by activity ASTR. The presence of such error alarms renders the initial condition invalid in the sense that the system is not in a steady state. The exception is the condition where the bus boundary conditions are such that the variable in question is initialized *exactly* at its limit. Such a situation may occur, for example, when a generator is fully loaded and its turbine mechanical power initializes precisely at its ceiling.

Activity ASTR then prints case heading followed by the number of iterations used to calculate the initial power flow. A value of < n > greater than one usually (but not always) indicates some sort of modeling error. This is especially true if the converted power flow network was solved with activity TYSL prior to the execution of activity ASTR. (The network solution method used here is identical to that used by activity TYSL, and the default convergence tolerance is an order of magnitude smaller in section / TYSL.)

The most common causes of this type of error are the same as for activity STRT.

Activity ASTR then summarizes the conditions at each online machine by tabulating the following quantities:

- Terminal voltage (ETERM).
- Exciter output voltage (EFD).
- Active and reactive power output (PELEC and QELEC).
- Power factor.
- Machine angle in degrees (ANGLE).
- d and q axis currents on machine base.

This report is in ascending bus number order when the *numbers* output option is in effect, and in alphabetical bus name order using the *names* option. Note that for wind machines, because the quantity EFD is not of any relevance, this is set to zero.

Following the calculation of the system initial condition, the time derivatives (DSTATEs) of all STATE variables are calculated and then checked for a steady state. The user specifies the largest derivative change to be allowed. Any STATE for which the time derivative exceeds the specified threshold generates an alarm; its index is printed along with the values of its time derivative and the STATE variable itself. For STATEs that have generated an alarm that are assigned to plant-related models, the model and machine are also identified. For STATEs that have generated an alarm that are assigned to load-related models, the model and load are also identified. For STATEs that have generated an alarm that are assigned to auxiliary-signal models, the model and the name of the device (the two-terminal, multi-terminal or VSC dc line or the FACTS device) with which the auxiliary-signal is associated are identified. For STATEs that have generated an alarm that are assigned to dc line models, the model and dc line are identified. For STATEs that have generated an alarm that are assigned to FACTS device models, the model and FACTS device are identified. For STATEs that have generated an alarm that are assigned to switched shunt models, the model and switched shunt device are identified.

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In each case, the ac bus is also identified. In each case, the ac bus is also identified. In the case of machines, loads, and switched shunts, this will be the ac bus at which the machine, load or the switched shunt.

In the case of machines and loads, this will be the ac bus at which the machine and load are connected; for two-terminal dc lines, VSC dc lines, as well as for auxiliary signals attached to these devices, this is the rectifier ac bus; for multi-terminal dc lines and for auxiliary signals attached to multi-terminal dc lines, this will be the ac bus associated with converter number one; for FACTS devices and for auxiliary signals attached to FACTS devices this will be the sending end ac bus. If any such STATE variables are detected, activity ASTR is terminated. As in activity STRT, the presence of nonsteady STATEs may be caused by a variety of factors (e.g., procedural errors, incorrect model data, numerical precision).

The user must specify the name of the Matrix Output File into which the system matrices and related information are to be written. If no filename is specified in response to the above instruction, activity ASTR is terminated. Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity ASTR.

The user specifies those STATE variables to be included in the linear system matrices, either as single STATEs or range of STATEs to be included in the matrices.

The user specifies the perturbation factor (default = 0.01).

The LSYSAN program provides for up to fifty elements in the system output vector and related matrices. Activity ASTR treats the output channel selections in dynamics working memory (refer to activity CHAN and Section 23.6, "Simulation Outputs") as potential output quantities. If no output channel selections have been introduced into the simulation setup, an appropriate message is printed and the H and F matrices are not constructed. Otherwise, the user may specify those output channels to be included as linear system outputs, either as single channels or range of channels to be included as linear system outputs. If no specification is made, the first < nn > (or fifty, whichever is smaller) output channels are used.

Activity ASTR then perturbs each of the selected STATE variables in turn, calculates time derivatives, constructs the corresponding columns of the A and H matrices (refer to Application Notes), and writes them to the matrix output file.

The user specifies input variables to be perturbed. The LSYSAN program provides for up to twenty input variables. Input quantities for the designated machine and a perturbation factor by which the initial condition value of the selected quantity is modified may be any of the following:

- EFD
- Pmech
- Vothsg
- Vref
- VAR (L)

The VAR index may be specified.

The user may specify a thirty-two character identifier to be included in the matrix output file. If no identifier is specified, activity ASTR assigns an identifier as listed in Table 16.2, "Activity CHAN Summary". The corresponding columns of the B and F matrices are calculated and written to the matrix output file.

The user may specify that the network solution convergence monitor be automatically printed. This is particularly useful when PSS®E is being executed as a batch job.

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Activity ASTR responds to the following interrupt control codes:

AB	(1) Suppress machine initial conditions summary. (2) Suppress suspect initial conditions summary.
CM	Print the network solution convergence monitor.
NC	Suppress network convergence monitor when the CM suffix was specified in selecting activity ASTR.
MO	Print induction motor local iteration convergence monitor.

The format of the network convergence monitor is the same as that of activity STRT.

16.15.1. Application Notes

Activity ASTR converts the system model into its state-space form represented by the following equations:

$x = Ax + Bu$

$v = Hx + Fu$

where x is the state variable vector, u is the vector of input variables, v is the output variable vector, and A, B, H and F are the real matrices describing the small perturbation behavior of the system.

While zero time constants for certain transfer function blocks such as voltage transducers may be appropriate for dynamic simulation runs, any such state variable included in the system matrices described above will result in its row in the A matrix having every term zero. The analysis in the LSYSAN program prohibits this condition; LSYSAN automatically removes the corresponding row and column from the A matrix, the corresponding column from the H matrix, and the corresponding row from the B matrix. Alternatively, the user may exclude such STATEs from the specification of those STATEs to be included in the linear system. Note that several PSS®E models treat small (relative to the simulation time step, DELT) time constants as if they were zero; therefore, the comments above apply to such STATEs as well.

If the user wishes to include such STATEs in the linear system model, then, prior to using activity ASTR, those time constants which are zero must be set to appropriate small values and the simulation time step DELT must be set such that all time constants included in the linear system are at least five times DELT. DELT may be set to as small a value as necessary with no execution time penalty or change in results because no actual time simulation takes place during or following activity ASTR. DELT is only used by the equipment models to determine which transfer function blocks are to be treated algebraically; the models perform the same calculations that they do in normal time simulations.

In designating those inputs to be perturbed, only variables which are external inputs into the system model may be selected. For example, an EFD may be perturbed only if there is no excitation system model for the machine; similarly, a machine's PMECH may not be perturbed if its governor is -modeled.

The mathematical basis of the calculations performed in activity ASTR and by the LSYSAN program require that the matrices constructed be based on a system model initialized in equilibrium. Any out-of-limits or non steady-state variable errors should be checked before using these matrices in LSYSAN; care should be exercised in increasing the derivative threshold tolerance simply to force activity ASTR to run to completion.

The matrix output file constructed by activity ASTR is a binary file in the form required by the LSYSAN activity BCAS.

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Additional Information
PSS®E GUI Users Guide, Building a State Variable Matrix for Linear Dynamic Analysis (LSYSAN) *PSS®E* Command Line Interface (CLI) *Users Guide*, Building a State Variable Matrix for Linear Dynamic Analysis (LSYSAN)

16.16. Listing Dynamics Model Data

DOCU

Run Activity DOCU - GUI
Dynamics>List>Models and data (DOCU)... *[List Dynamics Model Data]*
Run Line Mode Activity DOCU - CLI
ACTIVITY? >>**DOCU** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
AB

The model documenting activity DOCU produces a tabulation of the data associated with equipment models referenced in the user's simulation setup. The tabulation for each model includes a listing of the locations in the various dynamics data arrays used by the model, and a listing of the values of the constant data parameters used by the model.

The user must specify one of the following modes of operation:

- Reporting mode
- Data checking mode

In *reporting mode*, all model references being processed are tabulated. In *data checking mode*, activity DOCU compares the data of each model reference being processed to typical parameter ranges and performs certain relational checks (e.g., generator subtransient reactance greater than leakage reactance). For any model for which the data is found to be suspect, the data item(s) in question are tabulated followed by the standard model data tabulation for that model.

The user specifies the model type to be listed or processed:

- CONEC models: restricted to models called from subroutine CONEC
- CONET models: restricted to models called from subroutine CONET
- CHAN models: restricted to activity

and monitoring models (refer to CHAN and CHSB). The monitoring models are listed in the order in which the output channel assignments were made.

- Models by subsystem

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- Miscellaneous other models

If all buses in a *models by subsystem* list are processed, activity DOCU displays all models referenced in the simulation setup in the following order:

1. plant-related equipment models
2. load-related models
3. line relay models
4. auxiliary signal models in the following order:
    a. two-terminal dc line auxiliary signal models
    b. multi-terminal dc line auxiliary signal models
    c. VSC dc line auxiliary signal models
    d. FACTS device auxiliary signal models
5. dc line models in the following order:
    a. two-terminal dc line models
    b. multi-terminal dc line models
    c. VSC dc line models
6. FACTS device models
7. all miscellaneous other models and models called from subroutines CONEC and CONET

The various dynamic simulation options are also displayed.

Plant-related and load-related models are tabulated in ascending bus numerical (using the *numbers* output option) or alphabetical (using the *names* option) order; multiple elements (machines or loads) at the same bus are printed in ascending identifier order. Plant- and load-related models for a given network element are listed together in the report in the same order in which they are called during a dynamic simulation calculation. For machines, this order is: generator, compensator, stabilizer, minimum excitation limiter, maximum excitation limiter, excitation system, and governor. For loads, the model order is load characteristic followed by load relay.

Line relay models are also tabulated in ascending bus numerical or alphabetical order. Models on branches connected to a given bus are listed in ascending order of the other bus; line relay models on parallel circuits are printed in ascending circuit identifier order.

Auxiliary-signal models are tabulated in ascending bus numerical or alphabetical order. At any bus, the auxiliary-signal models are listed in the following order: auxiliary models associated with two-terminal dc lines followed by auxiliary models associated with multi-terminal dc lines followed by auxiliary models associated with VSC dc lines followed by auxiliary models associated with FACTS devices.

DC line models are tabulated in ascending bus numerical or alphabetical order. At any bus, dc line models are listed in the following order: two-terminal dc line models followed by multi-terminal dc line models followed by VSC dc line models.

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FACTS devices, and switched shunt models are also tabulated in ascending bus numerical or alphabetical order.

Wind machine models (i.e., wind related models attached to machines designated as wind machines in power flow) are tabulated in ascending bus numerical (using the *numbers* output option) or alphabetical (using the *names* option) order; multiple machines at the same bus are printed in ascending identifier order. Wind machine related models for a given network element are listed in the following order: generator, electrical control, mechanical model, pitch control, aerodynamic model, gust model, and auxiliary control.

The CONEC models are tabulated in the order in which they are called; then the CONET models are tabulated in the order they are called.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. In any of the subsystem selection modes the user may specify any of the following types of models:

• All models	• VSC dc line models
• All plant models	• FACTS device models
• Generator models	• CONEC models
• Compensator models	• CONET models
• Stabilizer models	• CHAN models
• Minimum excitation limiter models	• Turbine load controller models
• Maximum excitation limiter models	• Switched shunt models
• Excitation system models	• Wind models
• Turbine governor models	• Wind generator models
• Load characteristic models	• Wind electrical control models
• Load relay models	• Wind mechanical models
• Line relay models	• Wind pitch models
• Auxiliary signal models	• Wind aerodynamic models
• 2-terminal dc line models	• Wind gust models
• N-terminal dc line models	• Wind auxiliary control models

Output for the appropriate grouping of buses is produced with buses ordered as described in Section 4.9, "Subsystem Reporting".

16.16.1. Application Notes

The data checking mode of activity DOCU is intended to identify gross data errors such as misplaced decimal points, improper relationships between parameters and so on. It provides an initial editing of the complete

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set of model data, restricting the output to models for which the values of parameters are outside of a representative range of values.

Being reported in the data checking mode of activity DOCU does not indicate that parameter values are necessarily wrong; it simply indicates that they deserve to be checked. Conversely, the absence of warning messages from this screening by activity DOCU *does not* indicate that all data is valid. Most power system parameters can fall within reasonable normal value ranges and still be totally invalid. For this reason, the checking of parameter ranges is not an adequate form of data verification; it is only a rough initial check for typing and stenographic errors. The checking of parameter ranges must be followed by performance checks to verify that the parameter values specified for the dynamic equipment models actually represent realistic system performance. Model performance verification is described in Section 23.5, "Model Verification".

When generator data is expressed on a base other than the actual machine base (i.e., when MBASE is not actual machine base), the data checking mode of activity DOCU may generate alarms for data items which are perfectly reasonable. This is due to the assumption in the models that data is expressed on actual machine base. Quantities such as machine inertias and turbine governor limits are most likely to generate an alarm for such machines. It is therefore strongly recommended that generator data be expressed on actual generator base.

Additional Information
PSS®E GUI Users Guide, Model Data *PSS®E* Command Line Interface (CLI) *Users Guide*, Listing Dynamics Model Data

16.17. Listing Dynamics Data Arrays

DLST

Run Activity DLST - GUI
Dynamics>List>Dynamics data (DLST)... *[List Dynamics Data Common]*
Run Line Mode Activity DLST - CLI
ACTIVITY? >>**DLST** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The dynamics data array listing activity DLST provides a tabulation of user selected portions of any or all of the following categories of data:

- CON array
- VAR array
- STATE array
- ICON array

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- Main output channel selections
- CRT plot channel selections

The user specifies the desired starting and ending indices for each of these arrays. A null response for the range of data values of any category of data causes that category to be skipped.

STATEs and VARs are generally meaningful only during and following dynamic simulation calculations. These arrays are initialized during the initialization of the simulation (i.e., during activities STRT, MSTR, ESTR, or GSTR), and not during activity DYRE. When performing extended term simulations with activities MSTR and MRUN, the STATE array is used for different purposes than it is during state-space simulations. The value displayed for a STATE by activity DLST is unrelated to the STATE index displayed.

For output channel data, the listing consists of the output channel number, the channel address (IPRINT), the current value of the variable assigned to the channel, the channel identifier, a description of the type of quantity assigned to the channel (refer to activity CHAN and Table 16.2, "Activity CHAN Summary"), and its location in the network or dynamics data arrays.

For CRT plot channel data, the listing consists of the CRT plot channel number, the number of the main output channel assigned to this CRT plot channel, the current value of the variable assigned to the channel, the plotting limits (CMIN and CMAX), the channel identifier, a description of the type of quantity assigned to the channel (refer to activity CHAN and Table 16.2, "Activity CHAN Summary"), and its location in the network or dynamics data arrays.

Additional Information
PSS®E GUI Users Guide, Data Arrays *PSS®E* Command Line Interface (CLI) *Users Guide*, Listing Dynamics Data Arrays

16.18. Creating a Dynamics Model Raw Data File

DYDA

Run Activity DYDA - GUI
File>Save... *[Save / Show Dynamics Data]* Dynamics Model Data Tab
Run Line Mode Activity DYDA - CLI
ACTIVITY? >>**DYDA** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The dynamics data output activity DYDA processes dynamics model data into the form of a Dynamics Model Raw Data File (refer to Dynamics Model Raw Data File Contents)suitable for input to activity DYRE. It may be viewed as the logical inverse of activity DYRE.The dynamics data input activity DYRE reads simulation model data from a Dynamics Model Raw Data File into the dynamics data arrays in PSS®E working memory.

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The user may process the entire dynamics simulation setup or specify model specification records for those models called from subroutines CONEC or CONET. Data is written in the following order:

1. Data records for plant-related models contained in the model connection tables are written in ascending numerical or alphabetical bus order, according to the bus output option currently in effect.

2. Data records for load-related models contained in the model connection tables are written in order of decreasing precedence. Bus oriented load-related models are presented in ascending numerical or alphabetical bus order, according to the bus output option currently in effect. Load-related models pertaining to other subsystems are presented in subsystem numeric order.

3. Line relay models are written in ascending numerical or alphabetical from bus order according to the bus output option in effect; when multiple line relay models are present at the same from bus, they are written in ascending to bus and circuit identifier order.

4. Auxiliary signal models are written in ascending numerical or alphabetical bus order, according to the bus output option currently in effect. At any bus, the auxiliary-signal models are written in the following order: auxiliary models associated with two-terminal dc lines followed by auxiliary models associated with multi-terminal dc lines followed by auxiliary models associated with VSC dc lines followed by auxiliary models associated with FACTS devices.

5. The dc line models are written in ascending numerical or alphabetical bus order, according to the bus output option currently in effect. At any bus, dc line models are written in the following order: two-terminal dc line models followed by multi-terminal dc line models followed by VSC dc line models.

6. FACTS devices, and switched shunt models are written in ascending numerical or alphabetical bus order, according to the bus output option currently in effect.

7. Wind machine model (i.e., wind related models attached to machines designated as wind machines in power flow) data records contained in the model connection tables are written out in ascending numerical or alphabetical bus order, according to the bus output option currently in effect.

8. Data records for the remaining simulation models are written in the same order in which the equipment models are referenced in subroutines CONEC and CONET.

In any of the subsystem selection modes the user may specify any of the following types of models:

• All models	• VSC dc line models
• All plant models	• FACTS device models
• Generator models	• CONEC models
• Compensator models	• CONET models
• Stabilizer models	• Turbine load controller models
• Minimum excitation limiter models	• Switched shunt models
• Maximum excitation limiter models	• Wind models
• Excitation system models	• Wind generator models
• Turbine governor models	• Wind electrical control models

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• Load characteristic models	• Wind mechanical models
• Load relay models	• Wind pitch models
• Line relay models	• Wind aerodynamic models
• Auxiliary signal models	• Wind gust models
• 2-terminal dc line models	• Wind auxiliary control models
• N-terminal dc line models	

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output for the appropriate grouping of buses is produced with buses ordered as described in Section 4.9, "Subsystem Reporting".

Activity DYDA produces model specification records for any subsystem load-related model connected to a load which is a member of selected subsystem. This means that, for example, an Owner type subsystem model is produced when processing a subsystem consisting only of zones if the Owner type subsystem model is connected to a load which is a member of the selected zones.

16.18.1. Application Notes

Prior to initiating activity DYDA, the following requirements must be met:

1. Dynamics data must be present in working memory (i.e., the appropriate Snapshot File must be restored into memory with activity).
2. The corresponding electrical network must be present in the power flow working case as a converted case.
3. The associated connection subroutines (if present) must be linked into the PSS®E program structure (i.e., the createusrdll command to create the user dll must have been executed; refer to *Additional Resources for PSS® E*, CreateUsrDll).

Thus, assuming that the connection subroutines have been linked as described in Section 4.10, "Loading the Connection Subroutines", the following activity sequence normally precedes activity DYDA:

1. Restore Dynamics Working Memory from a Binary Snapshot File (RSTR) <file1>
2. Retrieve a Power Flow Saved Case File (CASE) <file2>

where <file1> is the name of the appropriate dynamics data Snapshot File and <file2> is the name of the corresponding converted power flow Saved Case File.

Data records for plant-related models, load-related models, line relay models, dc line models, FACTS device models, and auxiliary signal models which are bypassed (i.e., off) are omitted from the output of activity DYDA. Similarly, data records for such models which are unconnected (i.e., the network element to which they are connected is not present in the power flow working case) are excluded.

In subsystem selection mode in the graphical user interface, when specifying subsystems and model types, the user should be careful not to write data records for any model more than once.

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When writing two machine models (e.g., FRECHG, COMPCC, CRCMGV), a special convention is used by activity DYDA. The model is assumed to be associated with the first machine of the two for which it is applied.

When writing auxiliary-signal models, dc line models, and FACTS device models, the following convention is used: for two-terminal dc lines, and for auxiliary-signal models associated with two-terminal dc lines, DYDA assumes that the model is associated with the rectifier bus; for multi-terminal dc lines and auxiliary-signal models combined with multi-terminal dc lines, DYDA assumes that the model is associated with the ac bus of the first converter; for VSC dc lines, and for auxiliary-signed models combined with VSC dc lines, DYDA assumes that the model is associated with the first converter; for FACTS devices and auxiliary-signal models combined with FACTS devices, DYDA assumes that the model is associated with the sending end ac bus.

In subsystem selection mode, a similar convention is used in processing line relay, CONEC, and/or CONET models following specification of *all models*, *line relay models*, *CONEC models*, or *CONET models*. For line relay models and for those CONEC and CONET models that refer to more than one bus, activity DYDA assumes that the model is associated with the first bus that defines the equipment being modeled (e.g., for DISTR1, the from bus of the branch).

GUI only: The *[Save / Show Dynamics Data]* dialog available from the *File > Save* menu entry when dynamics data is present in the working case provides an additional control beyond those available using line mode. This control is the *Load model output* toggle which includes two conditions: *All subsystem models* and *Selected subsystem models*. The toggle is available when bus selection is limited to one bus subsystem or individually specified buses.

The *All subsystem models* selection is equivalent to the line mode behavior of activity DYDA.

Selecting the *Selected subsystem models* condition may further limit the reported model specification records of subsystem load-related models.

Under either condition the output includes model specification records for Bus type subsystem models connected to any element which is a member of the selected subsystem. When subject to the *Selected subsystem models* condition, the output excludes all non-Bus type subsystem models for which the subsystem type is not part of the subsystem selection criteria. For example, if the selected subsystem consists only of zones, then the reported model specification records excludes all Owner, Area and All type subsystem models, and includes Zone type subsystem models which are connected to elements in the selected zones.

Additional Information
PSS®E GUI Users Guide, Creating a Dynamics Model Raw Data File *PSS®E* Command Line Interface (CLI) *Users Guide*, Creating a Dynamics Model Raw Data File

16.19. Dumping Dynamic Simulation Output Channels into a Response File

DMPC

Run Activity DMPC - GUI
File>Save... *[Save / Show Dynamics Data]* Dump Output Channels Tab

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Run Line Mode Activity DMPC - CLI

```
ACTIVITY?
>>DMPC
ENTER 0 TO SKIP CHANNELS CONTAINING STATES
AND VARS
       1 TO INCLUDE ALL CHANNELS:
>>
```

Interrupt Control Codes

POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The output channel dumping activity DMPC writes the simulation output channel selections as contained in dynamics working memory in the form of a PSS®E Response File suitable for directing activities CHAN and/or CHSB. Thus, activity DMPC may be viewed as the logical inverse of the channel selection activities. The normal application of this file is to automate the transfer of output channel specifications from an existing simulation setup to a new Snapshot which models essentially the same system.

The user specifies either:

- A filename. If an error occurs in opening the file, an appropriate error message is printed and the request for the output device is repeated. Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity DMPC.
- Output of records to the user's terminal.

The user specifies one of the following conditions:

- Skip channels containing VARs and STATEs: all channels containing STATEs, as well as those containing VARs not associated with activity CHAN and CHSB monitoring models are omitted from the set of responses produced by activity DMPC.
- Include all channels: responses for all channels are generated.

Activity DMPC writes to the selected output destination the dialog required to specify channels one through NCHAN (refer to ALTR and RUN) via activities CHAN and/or CHSB.

Activity DMPC generates an alarm at the *Progress* tab any channel for which the channel generation dialog is not written. Output channels are skipped under the following conditions:

1. An invalid channel address is assigned to the channel (refer to Section 23.6.1, "Output Channel Selection").
2. The channel quantity is a VAR not associated with a CHAN monitoring model or a STATE, and the option to include these was not enabled.

In the Response File built by activity DMPC, a blank line is included as the response for the starting channel, VAR and ICON request by activities CHAN and CHSB. Thus, the next available locations in these arrays at the time the Response File is executed will be used.

Additional Information

PSS®E GUI Users Guide, Dumping Dynamic Simulation Output Channels into a Response File

PSS®E Command Line Interface (CLI) *Users Guide*, Dumping Dynamic Simulation Output Channels into a Response File

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16.20. Creating Dynamic Data Records for Use by Other Activities

RWDY

Run Activity RWDY - GUI
File>Save... *[Save / Show Dynamics Data]* Machine Parametric Source Data Tab
Run Line Mode Activity RWDY - CLI
ACTIVITY? >>**RWDY** ENTER OUTPUT DEVICE CODE: >>
Interrupt Control Codes
AB

The auxiliary dynamics data output activity RWDY writes machine parametric data from dynamics working memory in the form of either an Inertia and Governor Response Data File (refer to Section 6.8.1, "Inertia and Governor Response Data File Contents") for input to activity section / INLF, or a Breaker Duty Data File (refer to Section 11.12.2, "Breaker Duty Data File Contents") for input to activity section / BKDY. Activity RWDY may also be used to produce line relay data and branch impedances in the form required by the PSSPLT activity RELY.

The user must specify one of the following file types to be generated:

- Breaker Duty Data File
- Unit Inertia and Governor Data File
- Relay Characteristic Data File

If the user specifies the Unit Inertia and Governor Data File, the following options are also available:

- Suppress machine power limits: When output is used by activity INLF the machine active power limits from the working case are used rather than those from the turbine governor models.
- Specify treatment of machines without governor models by activity INLF:
- Use INLF defaults (default = working case limits) OR
- Treat machine as non-dispatchable

If working case limits are specified, PMAX and PMIN are set to default values for such machines so that activity INLF will assign active power limits as described in Section 6.8.4, "Governor Response Power Flow Solution". If working case limits are to be treated as non-dispatchable, PMAX and PMIN are both reported as zero so that activity section / INLF will treat such machines as fixed MW output machines and will exclude them from participating in any swing bus power change dispatches.

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The appropriate output tabulation is then produced and activity RWDY is terminated.

16.20.1. Application Notes

When a Breaker Duty Data File is being written by activity RWDY, a data record is produced for each machine in the working case which is modeled by one of the following generator models: GENROU, GENROE, GENSAL, GENSAE, GENDCO, FRECHG,CIMTR1,CIMTR2, CIMTR3 or CIMTR4. Machines modeled by other machine models are omitted from the Breaker Duty Data File.

When an Inertia and Governor Response Data File is being written by activity RWDY, a data record is produced for each machine in the working case which is modeled by one of the following generator models: GENROU, GENROE, GENSAL, GENSAE, GENCLS, GENDCO, GENTRA, CGEN1, FRECHG, CIMTR1, CIMTR2, CIMTR3 or CIMTR4. Machines modeled as a static var system, by the CDSMS1, CSMEST, CBEST and CSTATT model, or by a user-written model are omitted from the Inertia and Governor Response Data File. If the machine has a governor model other than the SHAF25 model or a user-written model, governor response data is included in its data record; otherwise, only the machine inertia is written and the remaining data items will take on default values when the data file is read by activity section / INLF.

When a Relay Characteristic Data File is being written by activity RWDY, a data record is produced for each occurrence in the simulation setup of any of the following line relay models for which the branch is in the working case: CIROS1, SLNOS1, DISTR1, RXR1,SLLP1, and SLYPN1. A data record is also generated for each RELAY2 and RELAY3 monitoring model reference generated by activities CHAN and CHSB for which the monitored branch is in the working case. The 16-character identifier fields are set by activity RWDY so as to identify the branch at which the model is applied. For the line relay models, each identifier is set to the two bus numbers, the circuit identifier, and the relay slot number, all separated by dashes (e.g., 7533-248-1-1). For the RELAY2 monitoring model, each identifier is set to the two bus numbers and the circuit identifier, all separated by dashes (e.g., 7533-248-1). For the RELAY3 model, each identifier is set to the number of the bus to which the monitored winding is attached, the three-winding transformer name, and the circuit identifier, all separated by dashes (e.g., 7533-TRNSFRMR-1); the relay type field is set to RELAY2. These identifiers may be changed by the user to conform with the naming conventions and to assure uniqueness.

Additional Information
PSS®E GUI Users Guide, Creating Dynamic Data Records for Use by Other Activities *PSS®E* Command Line Interface (CLI) *Users Guide*, Creating Dynamic Data Records for Use by Other Activities

16.21. Listing Dynamics Model Storage Locations

MLST

Run Activity MLST - GUI
Dynamics>List>Model storage locations... *[List Model Storage Locations]*
Run Line Mode Activity MLST - CLI
ACTIVITY? >>**MLST** ENTER OUTPUT DEVICE CODE: >>

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Interrupt Control Codes
AB

The model listing activity MLST produces a tabulation of plant-related equipment models referenced in the user's simulation setup. The tabulation for each model includes a listing of the locations in the various dynamics data arrays used by the model, and a flag for models which are bypassed (refer to Section 16.7.8, "Change Model Status").

The user may restrict the tabulation to active models, bypassed models, or both (refer to Section 16.7.8, "Change Model Status").

In any of the subsystem selection modes the user may specify any of the following types of models:

• All machine models	• Maximum excitation limiter models
• Generator models	• Excitation system models
• Compensator models	• Turbine governor models
• Stabilizer models	• Turbine load controller models
• Minimum excitation limiter models	

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV. Output for the appropriate grouping of buses is produced with buses ordered as described in Section 4.9, "Subsystem Reporting". Then the user is given the opportunity to specify another subsystem.

If an owner subsystem is specified, the owners to which machines (rather than buses) are assigned are used in defining the subsystem of machines to be processed. If a machine has multiple owners (refer to Generator Data), its output is included in the block for each of its owners which is contained in the specified subsystem.

For each model listed by activity MLST, the locations in the CON, STATE, VAR and ICON arrays allocated to that model reference are printed. For any model which is bypassed (refer to Section 16.7.8, "Change Model Status"), the model name is preceded by an asterisk (*).

Additional Information
PSS®E GUI Users Guide, Model Storage Locations *PSS®E* Command Line Interface (CLI) *Users Guide*, Listing Dynamics Model Storage Locations

16.22. Initializing Models for Extended Term Simulation

MSTR

Run Activity MSTR - GUI
Dynamics>Simulation>Perform extended term simulation (MSTR/MRUN)... *[Perform Extended Term Dynamic Simulation]*
Run Line Mode Activity MSTR - CLI
`ACTIVITY?`

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Run Line Mode Activity MSTR - CLI
>>**MSTR** ENTER CHANNEL OUTPUT FILENAME: >>
Interrupt Control Codes
AB, CM, NC, MO

The extended term dynamic simulation initialization activity MSTR initializes the dynamic models in preparation for extended term simulation calculations. Activity MSTR is used in conjunction with activity MRUN in performing these simulations. Refer to Section 23.10, "Extended Term Simulations" of this manual, and the *PSS®E Program Application Guide* for additional details on extended term simulations.

Activities MSTR and MRUN are special versions of the state-space simulation activities STRT and RUN respectively which are used to perform extended term simulations. As such, their dialog and operating characteristics are similar to those of activities STRT and RUN.

Working memory requirements for activity MSTR are identical to those of activity STRT. The normal activity sequence which precedes activity MSTR is:

1. Restore Dynamics Working Memory from a Binary Snapshot File (RSTR) < file1 >
2. Retrieve a Power Flow Saved Case File (CASE) < file2 >

where < file1 > is the Snapshot File containing the appropriate dynamics model data and output channel specifications, and < file2 > is the Saved Case File containing the ordered converted power flow case.

When initiated, activity MSTR first checks that:

1. The simulation time step is less than the *intermediate time step* mode threshold.
2. The *long time step* mode threshold is greater than the *intermediate time step* mode threshold.

If either of these requirements is violated, an appropriate error message is printed and activity MSTR is terminated.

Activity MSTR generates an alarm and terminates if generators are not converted. It then ensures that the machine power arrays in the power flow working case are set to their original initial condition values (they may have been changed by a prior execution of activity STRT or MSTR). Activity MSTR completes its setup phase by checking for the presence of the factorized network admittance matrix. If necessary, activity FACT (or activities section / ORDR and section / FACT) is executed.

Activity MSTR initializes the dynamic simulation by first doing a network solution and then, based on the bus boundary conditions, initializing the Z-form integration cells (STOREs) and algebraic variables (VARs) of each equipment model. If, in setting up for the initial network solution, any CIMTR5, CIMWSC, CIM5xx, CIMWxx, or IEELxx model removes constant admittance load from the load at which it is called, an appropriate message is printed and the network admittance matrix is automatically refactorized prior to commencing the network solution.

Any machine which is online in the working case but which does not have a generator model assigned to it generates an alarm with a message of the form:

```
NO ACTIVE GENERATOR MODEL CALL FOR MACHINE i
AT BUS nnnn
```


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The source current (ISORCE) of any such machine is set to zero. The presence of any such machines renders the initial condition invalid, and activity MRUN will not allow itself to be executed following such an initialization.

Similarly, any two-terminal, multi-terminal, VSC dc line, FACTS device, or any switched shunt device that is in the continuous control mode, which is in-service in the working case but which does not have a dynamic table model assigned to it generates an alarm with one of the following error messages:

```
NO ACTIVE TABLE MODEL CALL FOR TWO-TERMINAL
DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR MULTI-TERMINAL
DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR VSC DC LINE "name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR FACTS DEVICE
"name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
NO ACTIVE TABLE MODEL CALL FOR SWITCHED SHUNT
AT BUS "bus name"
PLEASE ENSURE THAT THIS EQUIPMENT HAS A MODEL CALL IN CONEC
```

The presence of any such dc line, FACTS or switched shunt device could indicate one of the following:

- A table model waswas ever assigned to the dc line, FACTS, or the switched shunt device, or the model assigned is a CONEC and CONET called model. When this condition is detected during activity MSTR, activity MRUN will allow itself to be executed.

During the model initialization process, any model variable which is initialized beyond its prescribed limits (e.g., the voltage regulator output of the IEEE Type 1 excitation system model, IEEET1) generates an alarm with a message of the form:

```
name AT BUS nnnn MACHINE i INITIALIZED OUT OF
LIMITS
```

where < name > is the model name (e.g., IEEET1), < nnnn > is the bus number, and < i > is the machine identifier of the model containing the out-of-limits variable. The user should resolve any such error conditions before running a simulation. The presence of such error alarms renders the initial condition invalid in the sense that the system is not in a steady state; that is, if activity MSTR were to be followed by activity MRUN with no disturbances imposed, there would be movement as the out-of-limits quantities are brought back to their appropriate limits. The exception is the condition where the bus boundary conditions are such that the variable in question is initialized *exactly* at its limit. Such a situation may occur, for example, when a generator is fully loaded and its turbine mechanical power initializes precisely at its ceiling.

Activity MSTR then prints, at the *Progress* tab, the case heading followed by the message:

```
INITIAL CONDITION LOAD FLOW USED n ITERATIONS
```

A value of < n > greater than one usually (but not always) indicates some sort of modeling error. This is especially true if the converted power flow network was solved with activity TYSL prior to the execution of activity MSTR. (The network solution method used in the dynamic simulation is identical to that used by activity TYSL, and the default convergence tolerance is an order of magnitude smaller in section / TYSL.) The

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most common causes of this type of error are the same as for activity STRT. The types of errors described above and all other initialization errors and warnings should be resolved before executing simulations (refer to Application Notes).

Activity MSTR then summarizes the conditions at each online machine by tabulating the following quantities:

- Terminal voltage (ETERM).
- Exciter output voltage (EFD).
- Active and reactive power output (PELEC and QELEC).
- Power factor.
- Machine angle in degrees (ANGLE).
- d and q axis currents on machine base.

This report is in ascending bus number order when the *numbers* output option is in effect, and in alphabetical bus name order using the *names* option. Note that for wind machines, because the quantity EFD is not of any relevance, this is set to zero.

The user specifies the Channel Output File to be used in the subsequent dynamic simulation run. If no filename is specified in response to the above instruction, the writing of the output channel variable values to a file is suppressed during the simulation run. Refer to Section 2.4.1, "File Usage "for the file specification conventions used by activity MSTR.

The default time step in PSS®E is half a cycle (i.e., 0.008333 of a second when base frequency is sixty cycles and 0.01 of a second for fifty cycle systems). Activity MSTR sets the value of simulation TIME to minus two time steps. This provides for a period of simulation under steady-state conditions prior to initiating a disturbance.

The user may specify that the network solution convergence monitor be automatically printed. This is particularly useful when PSS®E is being executed as a batch job.

Activity MSTR responds to the following interrupt control codes:

AB	(1) Suppress machine initial conditions summary. (2) Suppress suspect initial conditions summary.
CM	Print the network solution convergence monitor.
NC	Suppress network convergence monitor when the CM suffix was specified in selecting activity MSTR.
MO	Print induction motor local iteration convergence monitor.

The format of the network convergence monitor is the same as that of activities section / TYSL, section / SOLV, and STRT.

In the induction machine models CIMTR5, CIMWSC, CIM5xx, CIMWxx, CIMTR3, and CIMTR4, a local iteration is used in extended term simulations to solve for the equivalent load drawn by the model at each iteration of the network solution. The induction machine convergence monitor tabulates the load or machine to which the model is connected, the local iteration number, electrical torque (VAR(L+2)), and the real and imaginary components of the complex machine internal voltage.

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16.22.1. Application Notes

While a faithful system model is critical to all simulations, it is especially so in extended term simulations because of their duration. As with state-space simulations, it is strongly recommended that the system model data be validated using the techniques given in Sections Section 23.5.1, "Parameter Range Checking " and Section 23.5.2, "Performance Verification ".

Furthermore, because a steady-state check such as that performed during activity STRT is *not* performed by activity MSTR, it is recommended that the initial condition operating point be checked using techniques similar to those described in Section 23.5.3, "Simulation Initialization". This involves the following steps:

1. Using activity STRT to achieve a clean initialization.
2. Running a no disturbance state-space simulation of several seconds duration with activity RUN.
3. Running a standard state-space simulation of at least ten seconds.

When these test simulation runs look reasonable, they should be repeated using the extended term simulation activities MSTR and MRUN in place of activities STRT and RUN. The simulation time step should not be changed during these first runs, and results should be essentially identical to their state-space counterparts.

In any simulation setups containing user-written models which utilize STATEs, such models *must* be modified to allow them to be used in extended term simulations. Refer to the *PSS®E Program Application Guide* for details.

Most models supplied with PSS®E have been enhanced to allow their use in extended term simulations. Several models, however, have *not* been so modified and any simulation setups containing any of them may not be used in extended term simulations. Such models are listed here, grouped as their data sheets appear in the *PSS®E Model Library*.

Table 16.4. PSS®E Models Unavailable for Extended Term Simulation POM_DynSim.fm

Generator and Compensator Models				
CBEST	CGEN1	CIMTR1	CIMTR2	CSMEST
CSTATT	FRECHG	GENDCO	GENTRA	
Stabilizer and Excitation Limiter Models				
IVOST	STABNI	PSS3B	UEL2	UEL1
PSS4B				
Excitation System Models				
BUDCZT	EMAC1T	EXNEBB	EXNI	IVOEX
REXSYS	REXSY1	URHIDT	AC7B	AC8B
DC4B	ST6B	ST5B	ST7B	DC3A
Governor and Turbine Load Controller Models				
HYGOV2	HYGOVM	HYGOVT	IVOGO	SHAF25
TGOV4	TURCZT			
Load Characteristic and Relay Models				
CLODxx				
Auxuliary Signal Models				
RBKELT				

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DC Line Models				
CDC1T	CDCABT	MTDC2T	VSCDCT	CDC7T
Generic Wind Models				
WT3G1	WT3E1	WT3T1	WT3P1	WT1G1
WT2G1	WT3G2	WT4G1	WT2E1	WT4E1
WT12T1	WT12A1	WT4G2	WT4E2	
Switched Shunt Models				
CHSVCT	ABBSVC1			
CONEC and CONET Models				
CASEA1	CDC1	CDCAB1	CDCRL	CDCVUP
CEELRI	CHESVC	CMDWAS	DCPOW	MTDC02
RBKELR				

Activity MSTR generates an alarm for the use of any of the models listed above and sets a flag such that activity MRUN will not allow itself to be executed. However, it is *not* able to detect the presence of user-written models for which the code has not been modified for use in extended term simulations. Moreover, it is likely that such models will result in other models behaving incorrectly. The test simulation runs recommended at the beginning of this section are likely to reveal this situation.

Additional Information
PSS®E GUI Users Guide, Running Extended Term Dynamic Simulation *PSS®E* Command Line Interface (CLI) *Users Guide*, Initializing Models for Extended Term Simulation

16.23. Performing Extended Term Simulation in Time Steps

MRUN

Run Activity MRUN - GUI
Dynamics>Simulation>Perform extended term simulation (MSTR/MRUN)... *[Perform Extended Term Dynamic Simulation]*
Run Line Mode Activity MRUN - CLI
ACTIVITY? >>**MRUN** AT TIME = 0.000 ENTER TPAUSE, NPRT, NPLT, CRTPLT: >>
Interrupt Control Codes
AB, CM, NC, MO, CH, TI

The extended term time simulation activity MRUN sequences through time solving the system's differential equations and the electrical network at each time step.

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Activity MRUN requires the prior successful execution of activity MSTR. It also requires that the *long time step* mode threshold be greater than the *intermediate time step* mode threshold, and that generators be converted. If any of these requirements is violated, an appropriate error message is printed and activity MRUN is terminated.

The user specifies the four simulation options described in activity section / RUN. If the value specified for TPAUSE is less than the current value of simulation TIME, activity MRUN is terminated.

If a Channel Output File has been opened (in the most recent execution of activity MSTR or subsequently via activity ALTR), its filename is printed.

If the simulation time step has changed since the last execution of MRUN, both the old and new time steps are listed along with simulation mode to be used (refer to Section 23.10, "Extended Term Simulations").

Activity MRUN checks for islands which contain no in-service machines, generates an alarm for buses in such islands, and disconnects them.

If activity MRUN is operating in the *uniform island frequency* mode, the number of energized islands in the system is determined and each non-Type 4 bus is assigned to an island. Up to ten such islands are permitted.

Activity MRUN completes its setup phase by checking for the presence of the factorized network admittance matrix. If necessary, activity FACT (or activities section / ORDR and section / FACT) is executed.

The simulation starts at the present value of simulation TIME.

The user may specify that the network solution convergence monitor be automatically printed. This is particularly useful when PSS®E is being executed as a batch job.

Activity MRUN responds to the following interrupt control codes:

AB	Force a pause by setting TPAUSE to the current value of simulation TIME. This overrides the value of TPAUSE which was specified at the time activity MRUN was run. In addition, if PSS®E is operating in its Response File mode, all active Response Files are closed and subsequent user commands are taken from the user's terminal.
CM	Print the network solution convergence monitor.
NC	Suppress network convergence monitor when the CM suffix was specified in selecting activity MRUN.
MO	Print the induction motor local iteration convergence monitor.
CH	Print the output channel values at each time step. The value of NPRT which was entered at the time activity MRUN was run is overridden while this option is in effect. When this option is cleared, the original value of NPRT is restored and the next printing of channels occurs NPRT time steps from the TIME of the clearing of the option.
TI	Print the value of simulation TIME at the user's terminal.

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The form of the network convergence monitor is identical to that of activities STRT, RUN, MSTR, section / TYSL, and section / SOLV in the *small* and *intermediate time step* modes. In the *large time step* (*uniform island frequency*) mode, each convergence monitor line is extended to include:

- The magnitude of the largest machine angle change as a multiple of the island frequency convergence tolerance.
- The number of the bus with the machine experiencing the largest angle change.
- The largest machine angle change.

The form of the induction machine convergence monitor is identical to that of activity MSTR.

16.23.1. Application Notes

As discussed in Application Notes, before executing extended term simulations with a system model, state-space simulations should be performed. A steady-state run of several seconds and a reference standard simulation of at least ten seconds are recommend. After any suspect modeling has been resolved, this pair of simulations should be repeated in extended term mode using the same time step as was used in the state-space simulations.

Activity MSTR sets the output control variables NPRT and NPLT to one, and CRTPLT to zero. On the first subsequent execution of activity MRUN, these values are the defaults for these variables. On following executions of activity MRUN, these variables reset to its default values, those values specified on the previous execution of activity MRUN.

Whenever the value of simulation TIME reaches the time specified for TPAUSE, the transfer function output and network solution calculations are performed with a flag set to inform the simulation that this is a t^- calculation; the Z-form integration array elements (STOREs) are not updated.For any of the channel output options which are active, the appropriate output is generated regardless of whether the present time step is a print or plot time step as determined by the variables NPRT, NPLT and CRTPLT. When activity MRUN is reentered, with perhaps some switching operation having been specified, the time step is repeated as a t^+ calculation. Appropriate channel output is performed at the completion of the time step as determined by the new specification of the output control variables. This applies both for the case in which activity MRUN has advanced TIME to the value specified for TPAUSE and when interrupted by the user.

It is permissible to change the simulation time step, DELT, during an extended term simulation run (refer to Section 23.10, “Extended Term Simulations”). Time step increases should be implemented only after the higher frequency oscillations have been reduced; decreasing the time step is permitted at any point in the simulation. It is strongly recommended that all disturbances and major switchings be modeled only in the *small time step* mode.

Care must be taken in modifying the time step thresholds during extended term simulations. It is not valid to change the time step thresholds such that a new simulation mode results for the simulation time step in use before the threshold change. For example, t^- values of:

- DELT = 0.07.
- DLTBKW = 0.05.
- DLTEXT = 0.1.

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would have resulted in machines being modeled in the *intermediate time step* mode. Changing DLTBKW to 0.06 or DLTEXT to 0.09 would be valid; changing DLTBKW to 0.08 or DLTEXT to 0.07 would not.

A typical extended term simulation might start with the default time step of half a cycle and would initially be performed just as a state-space simulation would, except that activity MSTR and MRUN are used in place of activities STRT and RUN. After the higher frequency effects have more or less subsided, the simulation time step is increased so that running time is reduced. If another disturbance or switching is to be applied, it is prudent to reduce the simulation time step so that the higher frequency effects may again be modeled and observed; otherwise, dynamic instabilities may be completely masked out.

When changing the simulation time step, particularly into and out of the *long time step* (*uniform island frequency*) mode, bumps may occur on plotting simulation results at time coordinates where modeling assumptions were changed. Usually, these are minor and any oscillations introduced die out relatively quickly.

Longer time steps may require more network solution iterations than the default maximum allowed and/or some deceleration of the network solution. These parameters may be changed with activity ALTR. In the *long time step* mode, note that two acceleration factors and tolerances are involved (refer to Section 23.10, "Extended Term Simulations"). Acceleration factors should be tuned using the guide given in Section 6.3.19, "Characteristics of Activity SOLV".

There are several PSS®E models which activate automatic switchings (e.g., blocking or unblocking of a dc line by a dc line model, branch tripping by a relay model, load shedding, generator tripping, and so on). When one of these switchings takes place, a dual time step calculation is performed as described above. If a Channel Output File is active, the output channel values are written to it for both the pre and postswitching time step calculations made at the switching TIME; similarly, CRT plotting takes place for both the pre- and post-switching time step calculations.

As described above, it is appropriate that switchings be implemented in *small time step* mode; this is not so critical for load shedding and capacitor or reactor switching. In simulations where relays are being modeled, it is recommended that Snapshots and Saved Cases be preserved periodically during the simulation. Because relay action cannot always be anticipated, this allows the user to back up and repeat part of the simulation with a smaller simulation time step without having to rerun the entire simulation.

Both upon entering activity MRUN and before performing the t^+ time step calculation following automatic switchings, activity MRUN checks for islands which contain no in-service machines, generates an alarm for buses in such islands, and disconnects them. It then calculates a new sparsity-preserving ordering (activity section / ORDR) and rebuilds the factorized admittance matrix (activity section / FACT).

If an automatic line switching occurs in *uniform island frequency* mode which causes changes in the island assignments of buses (e.g., if a new island is defined), results are unpredictable. Establishing the relationship of buses to islands occurs only upon entry into activity MRUN and not following automatic switchings.

If the network solution fails to reach its tolerance in the prescribed number of iterations, an alarm is printed and the simulation continues as though convergence had been achieved. Refer to Section 23.7, "Running a Simulation" for additional details.

The computer time per simulation time step is much longer in extended term simulations than it is in state-space simulations. However, extended term simulations can achieve significant execution time advantages over state-space simulations when there are significant periods of simulated time where a much larger simulation time step is appropriate.

Additional Information
PSS®E GUI Users Guide, Running Extended Term Dynamic Simulation

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Additional Information
PSS®E Command Line Interface (CLI) *Users Guide*, Running Extended Term Simulation in Time Steps

16.24. Changing Dynamics Model Constants

CCON

Run Line Mode Activity CCON - CLI
ACTIVITY? >>**READ** ENTER BUS NUMBER, MACHINE ID (0 TO EXIT): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The dynamics data modification activity CCON allows the user to display and modify the constant data values being used by plant-related models contained in the simulation setup.

Activity CCON uses the CRT terminal lines per page option setting to group data items into pages. When a model is selected, any CONs which it uses are processed first, followed by its ICONs. If the model uses both CONs and ICONs, the ICON display always starts on a new page.

The user specifies the bus and machine identifier of the machine where plant-related model data is to be examined. The user specifies a model for any equipment being modeled at the machine. The constant data associated with the selected model is displayed.

The user may change the value of any of the displayed data items by entering its code as shown on the display along with the new value. The data display is repeated with the new data value.

The user continues with data changes as prompted by the dialog. Except for a response of -1 when the -1 FOR ICONS response is valid, any negative response is treated the same as a response of zero.

16.24.1. Application Notes

When changing the ICON values of a model, the user may specify either integer or character constants. Character ICON values must be entered as one or two alphanumeric characters enclosed in single quotes (e.g., 'H').

When activity CCON builds the display for a selected model, information on the number of CONs and ICONs it utilizes is taken from the internal tables describing the standard PSS®E models or from the *user model definition table*. Similarly, the starting CON and ICON array indices used by the model are obtained from the *array allocation table* (refer to activity DYCH).

The descriptive information on each data item is read from a binary file named MODELS (on most systems, this file has the extension RWF). When initiated, activity CCON searches the current directory for this file. If the file is not found locally, activity CCON follows the directory search sequence described in Section 2.1, "Directories and Files Overview".

As supplied with PSS®E, the MODELS file is located in the PSS®E master directory and contains descriptive data for plant-related models distributed with PSS®E; it does not, of course, contain information on any other

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user-written models which may be included in your simulation setup. When using this MODELS file, CONs of user-written models for which the descriptive information is not contained in the file are described as J, J+1, J+2, and so on, and ICONs are described as M, M+1, and so on.

With the auxiliary program DBUILD, users may construct customized MODELS files which include descriptive information on their user-written models. With such a customized MODELS file in the user's current or home directory, activity CCON automatically accesses the user's copy rather than the MODELS file as supplied on the PSS®E installation tape. Details on the operation of DBUILD may be found in *Additional Resources for PSS® E*, section / DBUILD.

Additional Information
PSS®E Command Line Interface (CLI) *Users Guide*, Changing Dynamics Model Constants

16.25. Assigning Subsystem Simulation Data to Output Channels

CHSB

Run Activity CHSB - GUI
Dynamics>Define simulation output by subsystem (CHSB) *[Select Channels by Subsystem]*
Run Line Mode Activity CHSB - CLI
ACTIVITY? >>**CHSB** NEXT AVAILABLE ADDRESSES ARE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The output channel selection activity CHSB enables the user to select either simulation variables which are to be monitored during dynamic simulation runs on a subsystem basis, or certain machine angle-related quantities. These quantities may be tabulated, placed into a Channel Output File and/or plotted at the terminal at regular intervals during a simulation run (refer to activity RUN). The simulation results, as contained in the Channel Output File, are able to be processed by the plotting package provided with PSS®E.

16.25.1. Operation of Activity CHSB

The user identifies quantities to be placed into output channels and activity CHSB places their memory addresses and alphanumeric identifiers into the PSS®E dynamics data arrays IPRINT and IDENT respectively.

The user may specify starting indices for the output channel and associated VAR and ICON or keep the default settings, which are the next available location pointers for the corresponding dynamics data arrays (refer to Section 16.1.2, "Operation of Activity DYRE" and activity SNAP). In a simulation, output channels are assigned sequentially starting with channel one. By default, activity CHSB assigns locations in these arrays at their next available positions.

Machine, Bus, Load, and Branch Quantities		
• Angle	• Bus frequency deviation	• Active power command from wind control

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Machine, Bus, Load, and Branch Quantities		
• Pelec • Qelec • Eterm • EFD • Pmech • Speed • Xadifd • Ecomp • Vothsg • Vref	• Voltage • Voltage & Angle • Mach Iterm • Machine Apparent Impedance • Vuel • Voel • Pload • Qload • Gref • Lcref	• Wind turbine speed • Wind pitch • Wind aerodynamic torque • Wind rotor voltage • Wind rotor current • Wind real power command from electrical control • Wind reactive power command from electrical control • Wind auxiliary control output
Machine, Bus, Load, and Branch Quantities Tie Line Quantities		
• Flow (P) • Flow (P&Q)	• Flow (MVA)	• Relay2 (R&X)

The user specifies any of the following quantities for monitoring:

In addition, the user may specify the inclusion of out-of-service equipment. Otherwise, output channels are generated only for in-service machines, loads, buses or branches, as appropriate. The default is in-service equipment only.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection").

For load and generating-plant quantities, activity CHSB processes all elements in the selected subsystem. For bus quantities, activity CHSB processes all buses in the specified subsystem. For branch quantities, activity CHSB processes all branches where endpoint buses are in the specified subsystem. Output channels are generated, and then the user may select another type of quantity, service status option, and, if appropriate, subsystem.

Note that categories *Voltage & Angle*, *Flow (P&Q)*, *Relay2 (R&X),* and *Machine Apparent Impedance* generate two output channel assignments for each element selected for monitoring.

When processing branches between areas or zones, the user must specify ties from either areas or zones, as appropriate, using the methodology given in Section 4.8, "Subsystem Selection". The selected branch quantity or quantities for all tie branches from each specified area or zone are placed into output channels.

Subsystem power totals may be specified by area, zone, owner, or all buses. This produces summations by subsystem of: machine electrical power, machine mechanical power, accelerating power, and load demand.

The user may process the entire working case or specify subsystem data (refer to Section 4.8, "Subsystem Selection") by bus subsystem, area, zone, owner, or basekV.

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If machine angle statistics are specified, the following quantities are selected for monitoring:

• Average angle	• Smallest angle
• Largest angle	• Bus with smallest angle
• Bus with largest angle	• Angle spread

Activity CHSB supplies a channel identifier for each output channel that is generated. These identifiers are the same format as the default identifiers generated by activity CHAN (refer to Table 15-2 Activity CHAN Summary).

Following the completion of the channel selection process, activity CHSB summarizes the next available positions in the pertinent arrays as a result of its current execution.

The value of the solution parameter NCHAN indicates the highest numbered channel being monitored during simulations (refer to activity ALTR and Section 23.6, "Simulation Outputs"), is set to NCHAN-1.

When activity CHSB is being used to change the channel assignments of a subset of previously generated output channels, the value printed for < NCHAN > is the larger of the original next available channel and that which would be set as a result of the current execution of activity CHSB.

16.25.2. Application Notes

As shown in Table 16.2, "Activity CHAN Summary", values for Angle, Pelec, Qelec, Eterm, EFD, Pmech, Speed, Xadifd, Ecomp, Vothsg, Vref, Vuel, and Voel of activity CHSB's output category selector designate quantities associated with generating plant equipment models.

Values for *Pload* and *Qload* designate the active and reactive load quantities associated with each load in the network. In the PSS®E dynamics data structure, there are dedicated arrays containing each of these quantities for each machine and load in the working case. These data items are automatically calculated by the various equipment models used to model each machine and its controls and by the various load characteristic models applied to loads in the network (refer to Section 16.1.2, "Operation of Activity DYRE" and Section 23.4.1, "Subroutines TBLCNC and CONEC "). Activity CHSB simply sets up the appropriate address pointers in the dynamics data array IPRINT so that the channel output routine picks up the proper values.

Similarly, *bus frequency deviation* is stored in a dedicated array for each bus in the working case, and activity CHSB sets up the appropriate pointer.

However, values for *Voltage*, *Voltage & Angle*, *Flow (P)*, *Flow (P&Q)*, *Flow (MVA)*, *Relay2 (R&X)*, *Mach Iterm*, and *Machine Apparent Impedance*, as well as totals by subsystem, specify data items that are not readily accessible from the standard dynamics data arrays. For these quantities to be available as output channel variables, they must be calculated and placed into a data array which can be accessed by the channel output routine. This function is performed by a group of network monitoring models which calculate certain network quantities and place them into specified locations in the VAR array.

Thus, when activity CHSB is programmed to place bus voltages, branch flows, branch apparent impedances, machine currents, machine apparent impedances and subsystem powers into output channels, it takes the following steps:

1. It allocates positions in the ICON array and sets these elements to the values required to designate the bus, branch, machine, or subsystem to be monitored.

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2. It allocates positions in the VAR array which are to contain the monitored -quantities.

3. It generates entries in the CHAN monitoring model tables for the appropriate model call (VOLMAG, FLOW1, FLOW3, RELAY2, RELAY3, GENTMC, GENTMZ, PTOTOW, PTOTZN, PTOTAR, or PTOTAL).

4. As with the other data categories, it establishes the appropriate IPRINT and IDENT array entries.

During dynamic simulations, model calls corresponding to the model references contained in the CHAN monitoring model tables are issued each time step at the completion of the network balance. Thus, after being selected by activity CHSB, these network quantities are automatically calculated during dynamic simulation runs and placed into output channels.

VARs and STATEs are not valid options in activity CHSB. In order to assign VARs and STATEs to output channels activity CHAN must be used.

As activity CHSB assigns each selected simulation quantity to an output channel, it makes a corresponding entry in a set of channel descriptor arrays. These arrays describe the type of quantity assigned to the channel, and its location in the network in terms of external bus numbers, machine and circuit identifiers, and so on. These arrays are also used in describing the content of each channel in activities ALTR and DLST.

This information is also used to reconcile differences in the PSS®E internal numbering scheme which may exist when a Saved Case is retrieved for which the internal storage of network quantities differs from that which was contained in the working case during the execution of activity CHSB. This reconciliation occurs at the execution of activities RSTR SRRS, and RTRN (see also Application Notes) whenever both network and dynamics data have been read into PSS®E working memory.

When the value specified as the starting channel index exceeds the next available channel value contained in dynamics working memory, the channels in this gap have their IPRINT, IDENT and channel descriptor array entries set so as to leave the channel unused. Dynamic simulation quantities may be assigned to such channels in subsequent executions of activity CHAN or CHSB using the approach given at the end of Section 16.2.1, “Operation of Activity CHAN” and Section 16.25.1, “Operation of Activity CHSB”.

As in activity DYRE (refer to Section 16.1.3, “Adding Models”), the user should take care to take a Snapshot following successful execution of activity CHSB and prior to terminating PSS®E. If a Snapshot is not taken, the newly established IPRINT, IDENT, ICON and channel descriptor array entries are not preserved and activity CHSB will need to be re-executed.

The user must exercise caution in specifying the starting VAR and ICON indices in response to the input request issued by activity CHSB. In particular, the location indices specified and successive locations in the respective arrays must not be used by other models. For simulation execution efficiency, it is desirable that the locations specified be the next available slots in the respective data arrays. Thus, the normal specification of these indices is to enter the default response.

As can be seen in the use of activity CHSB, as well as in the requirements of activity CHAN, SNAP, and DYRE, it is strongly recommended that the next available position indices maintained within each Snapshot be used whenever possible. PSS®E is able to perform the bookkeeping required for the storage allocations assigned by activities DYRE, CHAN, and CHSB. It is the user’s responsibility to update these indices, with activity ALTR, whenever models or user written code utilizing additional dynamics data array elements are manually added to the connection subroutines.

16.25.3. Error Messages

During its dialog, activity CHSB generates an alarm for errors resulting from the following conditions:

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1. In allocating an output channel, the preceding channel selection used the highest numbered output channel for which PSS®E is currently dimensioned. The following error message is printed and activity CHSB is terminated:

   ```
   ALL CHANNELS USED
   ```

2. In selecting for output a quantity requiring a network monitoring model call, there are not enough ICONs remaining to accommodate those required by the model call. The following message is printed:

   ```
   nnnn IS END OF ICON ARRAY
   ```

3. In selecting for output a quantity requiring a network monitoring model call, there are not enough VARs remaining to accommodate those required by the model call. The following message is printed:

   ```
   nnnn IS END OF VAR ARRAY
   ```

4. In selecting for output a quantity requiring a network monitoring model call, the table accommodating the CHAN

monitoring model references is full. The following message is printed:

```
nnnn MONITORING MODELS SPECIFIED--POINTER TABLE
FULL
```

5. In selecting angle statistics, the following message is printed if the function was previously selected:

   ```
   ANGLE STATISTICS STARTING AT VAR nnnn ALREADY
   IN CHANNELS
   ```

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Subsystem Simulation Data

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Chapter 17
Program Automation

17.1. What is Program Automation

Program automation in PSS®E provides the mechanism to control PSS®E execution other than by direct user interaction. This is the ability to define a set of operations for PSS®E to perform in a file of some kind (explained in the following sections) and to tell PSS®E to use the instructions in that file as a single command. We will call these files Automation Files. There are two primary ways to use Automation Files:

- As a program extension. In this case the user, in the course of using the program interactively, can decide to select an existing Automation File as the next instruction to the program.
- For un-attended execution. This is commonly called batch execution. In this case the entirety of the user's interaction would be the specification of the source of the instructions for it to execute (although there may be other options specified at the same time).

Although there are a few minor exceptions, for the most part the specification of the instructions, i.e. the contents of the Automation File, is the same in both cases.

The following sections will concentrate on the first alternative. Near the end of the chapter is a section on batch execution. There is also a section on entering single commands; while not a program automation method it uses the commands defined here and is best understood in this context.

17.2. Controlling PSSE Execution Using the API

An Application Program Interface (API) is a set of callable routines for building software applications. PSS®E has a supported, documented API available for use with PSS®E. The API is the primary and recommended method for accessing PSS®E functionality for automation. The PSS®E API is described in *PSS®E Application Program Interface (API)*. That manual includes descriptions of the syntax for using each API routine in each automation method in which it is supported. Not every API routine is supported in every automation method.

In most cases the API manual will also describe the syntax for calling an equivalent Fortran subroutine.

Siemens PTI will support users who make use of the API through the automation methods described here, such as batch commands and Python programs, only. No support is provided for Fortran or, say, C applications developed by users of PSS®E.

17.3. Automation Methods in PSSE

There are six automation processors in PSS®E:

- An embedded Python interpreter (refer to Section 17.4, "Python Programs")
- The Batch (or BAT_) Command interpreter (Section 17.5, "Batch Commands")

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- The Line Mode Interpreter (LMI) (Section 17.11, "Line Mode Commands")
- The PSS®E Engineering Basic (PSEB) macro processor (Section 17.14, "Building a Response File,Power Flow Calculation")
- The PSS®E Simulation Run Assembler (PSAS) macro processor (Section 17.15, "Building a Response File,Dynamic Simulation")
- The IPLAN simulator (Section 17.16, "Launching an IPLAN Program File")

Only five file types are available in the GUI, because line mode commands and batch commands can be intermingled in the same file, so the *Response File* file type (*.idv) is used for either (or both).

Some general observations are appropriate at this point:

- Python and IPLAN are both programming languages, i.e. they can be used to write any kind of a program, not just control PSS®E (although IPLAN's features are fairly limited, relatively speaking).
- IPLAN programs must be compiled (they do not need to be linked). The compiler (called IPLAN) is a Siemens PTI product and supplied with PSS®E. The source files are of file type *.ipl. The result (*.irf) are not object code files in the normal sense, but a unique format.
- Python programs are interpreted (one command at a time is executed - no compilation is necessary); although Python does support optional byte-compilation.
- PSEB and PSAS commands can also be embedded in Response Files, albeit in a particular context.
- Python programs can use existing IPLAN programs and Response Files. The other automation methods can all interact with each other.
- Response Files, PSAS, and PSEB Command Files are free-format data files.

17.4. Python Programs

Python is an OSI certified, freely available, object-oriented, interpreted programming language. Its syntax and usage are well-documented on the web and in numerous publications, so we will concentrate here solely on its interaction with PSS®E.

A Python installer is included in the PSS®E program installation package for your convenience. The language can also be downloaded from the official Python site at http://www.wxpython.org/. Python language documentation, extension modules, book recommendations, and user forums are also available directly from this site. Siemens PTI recommends the following textbooks for reference:

- *Learning Python* - Lutz and Ascher
- *Python Essential Reference* - Beazley
- *Programming Python* - Lutz
- *Python in a Nutshell* - Martelli
- *Python Cookbook* - Martelli and Ascher
- *Python Programming on Win32* - Hammond and Robinson

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Siemens PTI supports the use of Python programs within PSS®E, however, the language itself is not supported nor maintained by Siemens PTI staff.

The PSS®E installation package will also install the following freely available third-party extension modules:

- *wxPython* (http://www.wxpython.org/l)
- *pywin32* (Python Extensions for Windows) (https://github.com/mhammond/pywin32)
- Other 3rd-party modules needed to support those above are also installed.

Python files can be in source form (*.py, *.pyw) or byte-compiled (*.pyc). Python Files can be used in PSS®E in any of the following ways:

- From *Run Program Automation File...* (see Section 17.3, "Automation Methods in PSSE")
- At program startup by using the *-pyfile* option (see Section 17.9, "Unattended Execution of PSSE")

17.4.1. PSS®E Extension Modules For Python

One of Python's greatest strengths is its ability to be easily extended. These extensions can be written in Python or in system languages (C/C++, Fortran, etc.). Libraries and legacy code can be therefore be made accessible in Python. It is also an excellent transition language in that it can communicate with other programs and processes (e.g. Microsoft® Excel). Extensions are accomplished through the creation and use of packages that Python calls modules.

Many extension modules are currently available on the Internet for download including: numeric processing and optimization, graphic and data visualization (plotting), database connectivity, graphical user interface programming, XML and text processing, network programming and connections to Microsoft® Excel and Microsoft® Access.

Your PSS®E installation will include over 20 Python extension modules, documented in *PSS®E Application Program Interface*. They can be imported into user programs inside or outside of the PSS®E GUI or console interface.

The following modules provide access to the core API for PSS®E (i.e. PSS®E operations and primary data retrieval):

- *psspy* - provides access to the PSS®E API
- *pssplot* - provides access to the PSS®E plotting API
- *mustpy* - provides access to the PSS®MUST API
- *sliderPy* - provides functions to directly manipulate PSS®E diagram elements
- *redirect* - some tools to connect I/O streams between PSS®E and Python

The following is a partial list of other modules supplied with PSS®E. (i.e. PSS®E. Again, see the *PSS®E Application Program Interface* for more information on these and other modules:

- *excelpy* - provides Python functions to interface with Excel; these functions can be used to create, populate and format Excel workbooks from Python.

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- *pssarrays* - provides Python functions to retrieve PSS®E solution results
- *pssexcel* - provides Python functions to export PSS®E data or solution results to Microsoft® Excel spreadsheets.
- *bsntools - a set of simple bus name manipulation tools*
- caspy - provides access to the Saved Case Data Extraction routines (USRCAS)
- *dyntools* - tools for processing channel output files
- excelexportw - provides a graphical user interface to PSSEXCEL for exporting PSS®E results into Excel® spreadsheets.
- *gicdata* - help to create GIC data file applicable to a specific PSSE Power Flow Network Case for an entire network or just desired study subsystem.
- plot2wordw - provides a graphical user interface to WORDPY for inserting plot/picture files into a Microsoft® Word document.
- pssgrpg - provides functions intended to be used to supply valves to report nodes on I-line diagrams
- wordpy - provides Python functions to interface with Microsoft® Word; these functions can be used to create, populate, and format a Microsoft® Word document from Python. This is especially useful to insert plot files (*.eps, *.wmf, *.png, etc.) created by PSSPLT/PSSPLOT or any picture file into a Word document.

These modules are fully supported as part of the PSS®E program product. In addition to the documented modules, there are other Python modules and programs that are supplied with PSS®E that either support the above modules or are used by other program features. Modules that are designed as standalone programs are also supplied. You will also find several *.py files supplied in the Example directory for you to use and modify as you see fit.

Each of the modules above provides built-in documentation which can be accessed via the Python help command. In the case of psspy, however, the result will be too voluminous to be useful. A summary is available by entering:

```
help(psspy.psspyc)
```

Help is also available for individual API routines, e.g:

```
help(psspy.pout)
```

Some features of the psspy module that are not part of the API itself are documented in the doc string for psspy. Enter:

```
print psspy.__doc__
```

17.4.2. The Embedded Interpreter Environment

PSS®E contains an embedded Python interpreter. This is what allows PSS®E to execute Python files and commands. When the PSS®E GUI is started, it initializes the Python interpreter and executes a few initial commands. Those commands are:

```
import sys
```


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```
sys.path.append(pathname)
import os
import psspy
_i=psspy.getdefaultint()
_f=psspy.getdefaultreal()
_s=psspy.getdefaultchar()
import redirect
redirect.py2psse()
import pssplot
import mustpy
import sliderPy
```

(where pathname is a variable containing a value retrieved from the registry; you will generally not need to be concerned with this)

When the PSS®E Command Prompt program is started, it initializes Python in a similar way, but does not include the pssplot and sliderPy modules as they require the GUI execute correctly.

The os and sys modules are supplied with Python and contain commonly used functions for accessing operating system features. The other modules are defined above. The use of the _i, _f, and _s variables is explained in the section about default values in Python Files, below, and in the *PSS® E Application Program Interface (API)*. The redirect.py2psse() function causes Python's stdin (standard input) to use the function userin from the psspy module (psspy.userin) and Python's stdout (standard output) and stderr (standard error) to be redirected to PSS®E's progress device.

Note in particular that none of the modules are imported into the local name space. That means, for example, that to use a PSS®E API routine defined in the API manual as, say:

```
ierr = abc()
```

You would need to reference it as:

```
ierr = psspy.abc()
```

This is in keeping with standard practice for the Python community.

A Python program is shown below that performs the following:

1. Read a PSS®E Saved Case file, *savnw.sav*, into working memory.
2. Solve the power flow.
3. Loop through the buses and print the bus number and area to the *Report* tab.

```
psspy.case('savnw.sav')
psspy.fnsl((0,0,0,1,1,0,99,0))
psspy.inibus(0)

while 1:

    ierr,busnum,busname = psspy.nxtbus()
    if ierr!=0: break
    ierr,busarea = psspy.busint(busnum,'AREA')
```


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```
psspy.report('%s %s' % (busnum,busarea))
```

One idiosyncrasy of the current embedded interpreter is that results are not automatically echoed to the terminal as they are in the stand-alone Python interpreter. To view the values of expressions you must use the print function, as in the following:

```
print(psspy.size())
```

Or assign the result of the psse API call to a variable that you can print or otherwise manipulate later, as in the following:

```
result=psspy.size()
```

17.4.3. The External Interpreter Environment

Python programs that access PSS®E functions can be used without starting the PSS®E GUI. Python is installed with its own command line interpreter, its own integrated Development Environment (IDE) - called IDLE -, and programs can be written to create their own GUIs. Any of these can use the PSS®E supplied extension modules shown above.

In order to use the PSS®E supplied extension modules in this way, you must first make sure that the Python import path can find all the modules by name, and that the windows load path can find any required DLLs. The following example initializes PSS®E in the Python command line interpreter:

>>> import psse3500

>>> import psspy

>>> psspy.psseinit(80000)

>>> _i=psspy.getdefaultint()

>>> _f=psspy.getdefaultreal()

>>> _s=psspy.getdefaultchar()

This example assumes a starting bus size of 80000. Other values could have been chosen. It omits python built-in modules os and sys, which you can certainly use if you need to. It also omits other PSSE supplied modules. The module psse3500 is specific to PSS®E release 35.0. Since multiple minor versions of PSS®E (i.e. say, release 35.0 and 35.1) can be installed on the same machine, directories involved in resolving references to PSSE modules are specific to that particular minor release, so the version of the initial import is also specific to the particular minor release that you wish to work with.

There is also a module psse35. It will default to the lastest minor release of PSS®E installed. You can specify a different choice if you wish, for example:

>>> import psse35

>>> psse35.set_minor(n) # here n is minor version number, like 0 in 35.0.x

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17.4.4. The Embedded Interpreter Version

PSS®E version 35 uses the standard interpreter as installed by the standard Python installer, for those versions of Python that are currently supported.

PSS®E release 35.0.0 supports the following:

Python 2.7

Python 3.4

Python 3.7

"Supports" means that those versions of the Python interpreter can be used as the embedded Python interpreter in PSS®E (the GUI or the console interface). Also PSS®E modules for each of those Python versions are installed with the product.

To specify which version of Python is used by PSS®E specify the start-up command -pyver. For example:

PSSE35.EXE -pyver 37

The Python version is specified without the period delimiting the major and minor versions, i.e. 27 for Python 2.7. Python 3.7 is the default.

17.4.5. Finding user-created modules

If you wish to write your own modules for import into PSS®E or your scripts, you have a the following choices.

1. The startup directory should be on the import path. Files located there should be found.
2. The PYTHONPATH environment variable is added to the front of the import path. PSS®E will add it's paths in the PSSE install tree ahead of those in PYTHONPATH. You may place multiple paths here - separate them with a semi-colon.
3. When you are running a script, an empty string (indicating the current directory) should be temporarily added to the front of the install path.
4. The import path list is accessible and can be modified. It is sys.path. Care will need to be exercised when changing this.
5. You can add your module in the site-packages directory of your Python installation.
6. You can add a path configuration file to the site-packages directory of your Python installation.

There are actually several other methods, but they require changes either to the registry or the Python installation itself.

17.5. Batch Commands

Batch commands are free-format forms of inputs to the PSS®E API. What this means is best illustrated by an example.

Let's say there is a PSS®E API routine called `abcd`. It takes an integer argument (IVAL), a two-element integer array (IDATA) and returns a floating-point value and an integer return code. The batch command syntax for this hypothetical API routine would be:

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```
BAT_ABCD IVAL IDATA(1) IDATA(2)
```

In other words, three integers after the command name that correspond to the arguments IVAL, the first element of IDATA and the second element of IDATA, respectively. The returned values are not represented. Furthermore, if the function of the API routine was only to return those values, then there will not be a batch command syntax defined for it. The reason for this is that the batch commands are not a language; there are no variables or logic controls. A batch command is really data to the batch command interpreter, which then calls the indicated API routine with the input values defined after the command name.

The format of the batch commands is the same as the standard free-format rules used for all PSS®E data files. The following additional rules apply to batch commands, though:

- The command name itself is case insensitive, i.e. it doesn't matter whether you use upper case or lower case letters, or mix them (the documentation will always define them as all upper case).
- The commands may be continued on multiple lines. The interpreter keeps reading data until all the arguments that it needs for a given API routine are satisfied.
- There is a special token, a semi-colon, that can be used to tell the interpreter that it should return all remaining inputs to default settings for that particular API routine. It must be a separate token, not part of the previous value. In our example above,

  ```
  BAT_ABCD 1 34 ;
  ```

  will reset the third value to the default value, but

  ```
  BAT_ABCD 1 34;
  ```

  will return a syntax error because 34; is not a valid integer.

Batch commands can be used in Response Files or entered as Command Line input.

17.6. Recording

The PSS®E interface is capable of recording PSS®E API commands. Recorded API commands can be in the form of either Python statements or batch commands (Response File) depending on a specified user option. The easiest way to record a Python script (*.py, for Python statements) or a Response File (*.idv, for batch commands) for later execution is to use the *I/O Control>Start Recording...* option. Selecting this option will start the recorder. The entry of subsequent selections of PSS®E commands will record the API routines used to an automation file for later use until the *I/O Control>Stop Recording* option is selected.

As an example, suppose you wanted to print power flow results to the Report View for buses 101, 201, 205, 154, and 3001. To record these interactions:

1. Select *I/O Control>Start Recording...* and select the type of file you wish to record: Python File (*.*py*) or Response File (*.*idv*).
2. Specify, say, *busout.idv* or *busout.py* as the filename depending on your selection of file type in Step 1, and open it.
3. Select *File>Open*, select *savnw.sav*, and open it.
4. Select *Power Flow>Solution>Solve*. The power flow solutions dialog is displayed. Solve it and then close it.

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5. Select *Power Flow>Reports>Bus based reports...*. The bus based reports dialog is displayed. Enter buses 101, 201, 205, 143, 3001 in the text box provided. Select **Go** and then close it.

6. Select *I/O Control>Stop Recording*.

Using a text editor, open the file *busout.idv* (???) or *busout.py* (???) and you will see the following:

```
@! File:"D:\dev\psse\psse_next\all\base\example\busout.idv", generated on FRI, JUL 19 2019
BAT_CASE,'D:\dev\psse\psse_next\all\base\example\savnw.sav'
BAT_FDNS,0,0,0,1,1,0,99,0
BAT_BSYS,1,0,0.0,0.0,0,4,101,201,205,3001,0,0
BAT_POUT,1,0
```

```
# File:"D:\dev\psse\psse_next\all\base\example\busout.py", generated on FRI, JUL 19 2019
psspy.case(r"""D:\dev\psse\psse_next\all\base\example\savnw.sav""")
psspy.fdns([0,0,0,1,1,0,99,0])
psspy.bsys(1,0,[0.0,0.0],0,[],4,[101,201,205,3001],0,[],0,[])
psspy.pout(1,0)
```

17.7. Argument Passing

Arguments can be passed to a response file by using the **Arguments** button located on the *[Select Program Automation File to Run]* dialog.

Arguments can be passed to the following types of automation files:

- Python files
- Response files
- IPLAN files

but how they are handled by each varies. Note that the dialog permits specifying arguments for PSAS and PSEB files, but those languages have no facility for retrieving the argument values.

17.7.1. Arguments in Python Files

The list argv in the module sys can be used to access the program arguments. Consider the following Python program named arg0.py:

```
import sys
print(sys.argv)
```

When run using the Python interpreter from a Command Prompt the input is assumed to be space delimited and is parsed into separate values, e.g.:

Enter:

```
python arg0.py a list of words
```

and the following is displayed:

```
['arg0.py', 'a', 'list', 'of', 'words']
```


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From within PSS®E the input string is entered as a single argument. If the file arg0.py is selected (say, in the root directory of the C drive) and the string *a list of words* is entered in the argument string dialog box, the following will be displayed at the *Progress* tab:

```
Executing Python file:C:\LocalDocs\dev\testing\arg0.py
['C:\\arg0.py', 'a list of words']
```

17.7.2. Arguments in Response Files

The argument string is parsed according to the standard PSS®E free-format input rules for strings. Arguments in the Response File are then replaced by simple string substitution. If an argument contains a blank, comma, or slash, the argument must be enclosed in single quotes; otherwise the quotes are optional.

A Response File that is receiving arguments has the appropriate argument, less any surrounding quotes, substituted by the input processor before the dialog input line is passed to PSS®E itself. Arguments replace argument designator strings, which are of the form %n%, where n > is one of the numbers 1 through 15. Any recording of input lines contains the argument values in place of the argument designator strings. A Response File may pass an argument that it receives to another Response File, which it, in turn, initiates. It is usually good practice to place such argument designators in quotes so that a single argument that contains blanks and/or commas not be interpreted by the designated Response File as multiple arguments.

An example of an argument would be %1%, which could appear anywhere in the file. In this case %1% would be replaced by the first value parsed from the argument string. Likewise %5% would be replaced by the fifth value parsed from the argument string.

Response Files require that all arguments receive values. When a Response File references an argument that was not passed to it at the time it was initiated, an error condition occurs A useful technique is to document all arguments referenced in a Response File using the TEXT command at the beginning of the Response File. This approach will help detect missing argument errors before any lengthy computations specified in the Response File are performed.

The following are examples of specifying arguments to a response file that can be entered in the command line:

```
@INPUT , filename , 'argument 1' , 'argument
2' ... 'argument 15'
```

or:

```
@CHAIN , filename , 'argument 1' , 'argument
2' ... 'argument 15'
```

17.7.3. Arguments in IPLAN Files

The entire argument string is placed in a buffer. The IPLAN command ARGUMENT can then be used to parse the string in any way - and repeatedly - using the standard PSS®E free-format input rules. Refer to Program Arguments of the *IPLAN Program Manual*.

17.8. Default Values

Many of the statements in automation files specify data for which defaults exist. The idea of a default value is one that is simply not specified. Typically this means that either some function or action associated with

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that value is ignored, or a specific documented value is used; that specific value might be a fixed value - say, zero - or the current value of some quantity in the network case or a program setting.

While it is possible in some programming languages for a value to be missing, the general case is that in order to indicate this situation a special value is used. When input values are omitted in automation files such special values are substituted for them. The various API routines check for the special values and take appropriate default actions when they are encountered.

Currently spaces and false are used for filenames and logical values when they are omitted. The special values for integers, real numbers (floats), and characters (strings) are returned from the API routines GETDEFAULTINT, GETDEFAULTREAL, and GETDEFAULTCHAR. No assumptions should be made that these precise values will be identical from one release of the PSS®E to another. If it is necessary to acquire these values, always run the API routine to get the current values.

Default values can be specified in references to API routines in Python Files. Data items can be omitted in Response Files. The concept of default values does not apply to IPLAN programs in general, but there are some functions that take optional arguments, and because PUSH statements can contain anything that could be in a Response File, any comments about Response Files will apply there. All data values are required in PSEB and PSAS files, unless the syntax specifically indicates otherwise.

17.8.1. Defaults in Python Functions

The Python language provides an easy mechanism for permitting optional arguments to functions. However it requires that each function be coded in such a way as to permit this. Not every one is. All the functions in the psspy module are coded in this way. For other modules delivered with PSS®E please check the help facility.

17.8.2. Defaults and Keywords in Module psspy

All arguments to functions defined in module psspy may be omitted and default values will be supplied. All arguments to functions defined in module psspy may be specified by keyword.

In addition psspy contains a unique feature for specifying values for specific elements in an input list. This is best illustrated by example. Let's say a function `mysub` takes a single argument, opt, which is a 2-element list of integers, then the following are equivalent:

```
mysub([_i,1])
mysub(opt=[_i,1])
mysub(opt2=1)
mysub(opt=POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal,opt2=1)
mysub(opt=[_i,42],opt2=1)
mysub(opt2=1,opt=[_i,352])
mysub(opt1=_i,opt2=1)
```

where _i is a variable equal to the default integer value.

What this means is that all the functions in psspy recognize special keywords formed by concatenating any input list name and the string representation all the possible index values (starting at 1) and will use them to override just that specific element of the list.

17.8.3. Defaults in Recorded Python Files

When recording is enabled and the Python format is chosen, values for all arguments are specified. Where default values are recorded, actual values are used except for the following:

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- in place of default integer values, the variable _i is used.
- in place of default float values, the variable _f is used.
- in place of default string values (not filenames) the variable _s is used.

This will work directly inside of the PSS®E GUI because those variable are automatically assigned the appropriate values at program startup. This is why the statements

```
_i=psspy.getdefaultint() _f=psspy.getdefaultreal() _s=psspy.getdefaultchar()
```

are recommended in Section 17.4.3, "The External Interpreter Environment".

17.8.4. Defaults in Response Files

Because Response Files are simply free-format data files, an input value can be omitted by two commas either together or separated by only spaces.

For batch commands there is a special token, a semi-colon, that can be used to tell the interpreter that it should return all remaining inputs to the default settings (see Section 17.5, "Batch Commands").

Files containing recorded batch commands will use consecutive commas for default values specified to the API routine.

17.9. Unattended Execution of PSSE

In some cases, it may be desirable to run PSS®E in batch mode, that is, without any user interface. PSS®E provides command line options that can be used for this purpose. Section 17.9.1, " Start-up Commands" provides a list of all the available command line options; these four specifically pertain to this purpose:

	-rspfile	-pyfile	-argstr	-embed

- note that the initial - is part of the command
- -rspfile, -pyfile, and -argstr all expect to be followed by a string
- -pyfile will be ignored if -rspfile is specified
- -argstr and -embed will be ignored if neither -rspfile nor -pyfile are specified
- the command line is blank delimited; if the string specified to -rspfile, -pyfile, or -argstr contains a blank then the string must be quoted
- if the same option is specified more than once, only the last one will be used
- notwithstanding the previous point, the order of the options is unimportant
- the string value for the -argstr option is the argument string (see Section 17.7, "Argument Passing")

Provision is made for specifying only Response Files or Python Files. Both of those file types can use other types of Automation Files so, indirectly at least, any type can be used.

The -embed option has the effect of suppressing the display of the PSS®E GUI. If the Automation File terminates without stopping the execution of PSS®E the GUI will be displayed at that time.

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The following are a few points to be aware of when preparing to run PSS®E in an unattended mode:

- PSS®E does not read input from stdin or write to stdout, so pipes cannot be used for batch execution of PSS®E.
- Be aware of the nature of the operating system tool you are using to start PSS®E. In Microsoft® Windows, for example, non-console application programs, such as PSS®E, are executed asynchronously. This means that should you have a batch file that starts PSS®E it will not wait for the program to stop before continuing on the next statement in the batch file.
- When running in batch mode (i.e. with a startup automation file specified that stops the program) the *Progress* tab and *Report* tab output should be re-directed to a printer or file, otherwise output will be sent to windows that would be closed when the program stops.

Here are some examples of starting PSS®E from a Command Prompt:

```
psse35 -rspfile busout.idv
psse35 -pyfile busout.py
psse35 -pyfile busout.py -embed
psse53 -embed -pyfile "study number 2.py" -argstr
101
```

Appendix D - Command Line Options contains a complete list of all the command line options that are available.

17.9.1. Start-up Commands

The following arguments can be specified on the command line when starting the program:

-buses Used to specify startup size of PSS®E, 1K->150k.

-rspfile Immediately run a Response file.

-pyfile Immediately run a Python file.

-argstr Specify arguments for automation files.

-embed Run without display (until automation file terminates).

-inifile Run program with alternative ini file.

17.10. Running a Python Program

Run Python Program - GUI
I/O Control>Start recording... *[Select Program Automation File to Record]* Python file (*.py)
Interrupt Control Codes
None

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Python programs run in this manner execute in the embedded interpreter environment as described in Section 17.4.2, "The Embedded Interpreter Environment". All code runs in that local namespace and can therefore be impacted by previous Python programs, or affect subsequent Python programs. Specifically, local variables retain their bindings (values) for the entirety of the execution of PSS®E. Use Modules, Classes, or functional programming techniques to create independent namespaces.

Python's interaction with PSS®E is strictly through the API. PSS®E functions can be executed, and data can be retrieved from PSS®E's working memory. API routines are provided to run Response Files and IPLAN programs from within a Python program, so the other automation methods are integrated with Python in this way, and functionality that was developed using this other methods, which all predate Python's introduction into PSS®E, is preserved and can be used with minimal rework.

17.11. Line Mode Commands

The line mode was the primary method of controlling PSS®E through rev 29 (and the only method through rev 18). It was highly efficient, very thorough, but completely procedural. It was essentially a question-and-answer session with a program that began with the question:

```
Activity?
```

The appropriate response to this question was any of about 170-180 (in the last few releases) activities that comprised the primary functions that PSS®E could perform. Each activity name was four characters long; most could be abbreviated. Some activities could accept optional suffixes which modified their behavior somewhat. Almost all would then ask a series of questions, unique to each activity, to which the user would respond to operate the program. Many of those activity names are preserved today in API routines that perform similar functions.

Line mode commands can be used to automate PSS®E through Response Files (*.idv), so called because they contain responses to the questions that the command line of PSS®E asks.

The modern GUI has replaced this previously common method of operation, but there is a wealth of existing Response Files containing line mode commands used to automate PSS®E. Therefore PSS®E currently contains a feature called the Line Mode Interpreter (LMI) - a new program written to imitate the traditional line mode of PSS®E, but that actually uses the PSS®E API. In a way it is a translator from line mode to batch commands. It is not a perfect imitation, and those differences are captured in PSS®E *Command Line Interface (CLI) Users Guide*,Appendix A.

It is worth noting that, although the line mode now comprises a subset of the full functionality of PSS®E, for those functions it can perform, and for those users who are thoroughly familiar with the command line (and are good typists), the program can still be operated more quickly from the keyboard than is physically possible with a point-and-click interface (i.e., a mouse). So while command line capabilities are not likely to be enhanced in the future, its current operation is considered to be an important feature of the program.

17.11.1. Mixing Line Mode and Batch Commands

Response files can contain both line mode and batch commands. The batch commands can be thought of as additional activities of the line mode. They are not, and they are handled very differently; but for purposes of understanding how they may be combined in the same file, that image provides the guidelines that need to be followed.

The format of the line mode commands also conforms to the PSS®E standard free-format rules

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17.11.2. Immediate Commands

The line mode has been augmented over the years with a set of Immediate (@) Commands, so called because their actions occur without the involvement of any of the API routines in PSS®E. They are functions of a utility library (a Siemens PTI product) that is used by PSS®E but separately maintained. Immediate Commands can also be mixed with line mode and batch commands. Unlike batch commands, however, they can be used anywhere, even in the middle of a series of responses to a particular activity using the command line. The following list is provided for convenience:

@CHAIN filename - Open the specified file as a chained response file.

@INPUT filename - Open the specified file as a nested response file.

@PAUSE - Temporarily suspend response file operation.

@CONTINUE - Resume PAUSEd response file operation.

@END - Close the current response file.

@SYSTEM command - Issue a designated operating system specific command

@RELEASE - display the PSS®E release number and date, along with similar information about selected sub-systems.

@DONGLE - display time used statistics for time lock users

@! - Ignore the entire line.

The last one, @!, will work from any input file. The others are only recognized from terminal input, such as a Response File.

17.11.3. Version

The LMI provides an additional feature that the original line mode did not - the ability to change versions. As releases of PSS®E come out, sometimes changes are necessary in the sequence of line mode commands. The creates maintenance work for existing Response Files. The LMI provides a command, VERSION, which displays or sets the version of the line mode language being processed. Introduced at rev 31 of PSS®E, it recognizes versions from rev 30 and newer, for example:

`VERSION 30` - sets the version to rev 30

`VERSION 31` - sets the version to rev 31

etc.

`VERSION` - displays the current version

17.12. Running a Response File

Run Response File - GUI
I/O Control>Run Program Automation file... *[Select Program Automation File to Run]* Response file (*.idv)

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Run Line Mode Activity IDEV - CLI
ACTIVITY? >>**IDEV <filename>**
Immediate Commands - CLI
@ input filename OR @ chain filename
Python Command
`psspy.runrspnsfile <filename>`
Interrupt Control Codes
None

IDEV

Executing activity IDEV in Line Mode should be considered an obsolete capability which is retained for backward compatibility purposes. Activity IDEV itself is not sensitive to any interrupt control code options. While one or more Response Files are active, the CL interrupt control code may be used to close all open Response Files.

17.12.1. Application Notes

The Response File must contain PSS®E Batch Commands and/or an exact image of the Line Mode user responses which would normally be entered by the user from the Command Line and/or Immediate Commands. Therefore, the manual creation of a Response File, which is done with the text editor, requires an intimate familiarity with these commands, particularly for dialog intensive Line Mode activities such as CHNG and ALTR.

Additional Information
PSS®E GUI Users Guide, Response File *PSS®E Command Line Interface (CLI) Users Guide*, Assigning Interactive Input to a Response File

17.13. Recording User Actions

ECHO

Run Line Mode Activity ECHO - CLI
ACTIVITY? >>**ECHO**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The dialog echoing activity ECHO enables the writing of all subsequent user dialog input to a designated file, which may subsequently be used as a Operation of activity ECHO.

Users of PSS®E's traditional command line interface will be familiar with the activity ECHO. It performs a function similar in effect to the recording function in the current program. It is a very different thing, however, and a remnant of it remains.

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ECHO copies terminal input to a file. Because all PSS®E input in a console application either came from a terminal or a device that imitated the terminal, that meant all (interactive) input. This capability is retained in the PSSECMD console application.

The current recording function is a feature of the API routines. When they are called, if recording is enabled, they record a copy of their input in the designated format in the designated file; it doesn't matter how the call occurred - by menu selection, through line mode, from an automation file, etc.

The ECHO function still exists in the GUI, and it still does the same thing, i.e. copies all terminal input to a file. The difference is that there is very little terminal input. The API routine *userin*, the IPLAN function *INPUT* (and *INPUTLN*), and the *@pause* immediate command all attempt to read from the terminal, which is handled in via a pop-up window. So data entered there will show up in an ECHO file.

Related to this is the traditional command line activity IDEV, meaning Input DEVice. IDEV was used to replace the terminal with a file so that subsequent READs from the terminal would find the responses in the file (hence the name Response File). IDEV never executed the commands in a Response File, although that appeared to be the situation. The implementation of IDEV in the GUI does what most people would expect, therefore; it causes the commands in the Response File to be interpreted by the LMI. This means, however, that IDEV no longer runs the way it used to, i.e. replaces terminal input with a file. That function is now performed by the API routine SET_INPUT_DEV. The outcome of that API is opening a file to respond to input requests from the terminal (except @pause which will still force an actual terminal read, handled via a pop-up window).

When activity ECHO is initiated, if a file for capturing responses had previously been opened with an earlier execution of activity ECHO, it is closed.

17.13.1. Application Notes

With a console application like PSSECMD the file built as a result of activity ECHO can be used as a PSS®E Response File with @INPUT or @CHAIN or IDEV command to exactly reproduce a sequence of PSS®E activity commands and responses to instructions. Files written by activity ECHO are source files which may be modified with the text editor (e.g., to specify different faulted buses, switched branches, Saved Case and Snapshot File names, and so on) to tailor the original set of user responses to the application at hand.

Activity ECHO copies to the designated file those user inputs that are entered into PSS®E as part of the interactive dialog. In addition, whenever a PSS®E bulk data input activity is reading data from the dialog input device (rather than from a bulk data input file), these data records are also recorded in the file. This applies to activities such as READ, TREA, RESQ, TRSQ, MCRE, and PSAS.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Recording User Actions

17.14. Building a Response File,Power Flow Calculation

PSEB

Run Activity PSEB - GUI
I/O Control>Run program Automation file... *[Select Program Automation File to Run]* PSEB Command file (*.pse)

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Run Line Mode Activity PSEB - CLI
ACTIVITY? >>**PSEB** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The power flow run assembler activity PSEB allows the user to specify routine power flow runs in an English-like language. Input to activity PSEB is taken either from a PSEB Command File or from the dialog input device. This can be in-stream in a Response File, typed into the command line in the GUI, or entered at the console in PSSECMD. The output from activity PSEB is in the form of a PSS®E Response File.

17.14.1. PSS®E PSEB Macro Language

PSEB (PSS®E Engineering Basic) is a built-in macro language that supports PSS®E Power Flow operations through the use of English-like command macros. PSEB commands are created within a text file using a text editor. Each record is in the form of a command starting with a verb from a previously defined vocabulary. The PSEB commands are documented below and in the *PSEB User's Ready Reference*.

PSEB commands do not interact with PSS®E directly. The PSEB macro processor is really a translator from the PSEB language to a Response File, which may contain a mixture of command line and batch commands. The PSEB command line command normally automatically performs the commands in the Response File after the translation. When used with the suffix CHECK (i.e. PSEB, CHECK) only the translation is performed. When using the PSEB API routine only the translation is performed.

Each PSEB command begins with a verb from the vocabulary recognized by activity PSEB. Some commands require additional keywords (e.g., BUS, BRANCH), numeric quantities (e.g., bus number, load MW), and/or character identifiers (e.g., bus name, circuit identifier) in specific fields of the command format. Some commands provide for the optional specification of keywords, numeric, and/or character quantities; these are shown in the command syntax enclosed in braces ([ ]).

Where one of several keywords may be specified, they are shown in the command format enclosed in vertical bars (| |). One of the keywords listed must be entered in the field.

Each keyword or data value must be followed by one or more blanks. For clarity, a comma or equal sign may be used in place of, or in addition to, a blank space. For example, the following two PSEB commands are equivalent:

```
OPEN TIE FROM BUS 153 TO BUS 154 CKT 2
OPEN TIE, FROM BUS = 153, TO BUS = 154, CKT = 2
```

Any field labeled bus id designates that a bus identifier is to be entered. When the *numbers* input option is in effect, the bus number must be specified; using the *names* input option, the extended bus name enclosed in single quotes is required.

Any field labeled dc id designates that a two-terminal dc line name is to be entered. If the name contains any blanks or special characters, it must be enclosed in single quotes.

I and R indicate that integer and real (i.e., it may include a decimal point) numeric values respectively are to be specified.

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Many PSEB commands provide for the insertion of optional descriptive text in designated fields, indicated by [n] in the command syntax. For example, the following two PSEB commands translate into the same PSS®E dialog:

```
RECOVER FROM SAVNW.SAV
RECOVER saved case FROM SAVNW.SAV
```

A dollar sign ($) at the end of a PSEB command line is used to continue the command on the following line. At least one space must precede the dollar sign. As many physical lines as required may be used to specify a single PSEB command as long as all lines except the last one are terminated with a dollar sign. This multiline capability may be used with all PSEB commands *except* the USE command (see below). In addition, each PSS®E activity command and response to an instruction entered in *passthru* mode (see below) must be entered on a single line.

17.14.2. PSEB Commands

The detailed syntax of each PSEB command is given below and also on the *PSS®E Engineering Basic PSEB User's Ready Reference* sheet. Keywords shown in uppercase letters must be entered as shown in either upper or lowercase characters.

```
HOLD [n] IN (saved-case-filename) [NOW]
```

The HOLD command is used to preserve the power flow working case in the designated Saved Case File; refer to Section 5.45, "Creating a Saved Case File". When the HOLD command includes the optional keyword NOW, activity SAVE is immediately executed. In this case, the PSS®E responses for activity SAVE are *not* written to the Response File being constructed by activity PSEB.

When the HOLD command does not include the optional keyword NOW, the appropriate entries in the Response File are made to save the working case during the subsequent execution of the Response File.

```
RECOVER [n] FROM (saved-case-filename)
```

The RECOVER command is used to access the designated Saved Case File; refer to Section 5.1, "Retrieving a Power Flow Saved Case File". Activity CASE is executed immediately and the PSS®E responses for activity CASE are written to the Response File being constructed by activity PSEB. Note that the RECOVER command overwrites the working case. The HOLD command with the NOW option may be used before the RECOVER command to save the working case.

```
          |MW    |
          |MWP   |
          |MWI   |
|ALTER |  |MWY   |                                                  [ |MW
 | ]
|CHANGE|  |MVAR | LOAD [n] BUS (bus id) [LOAD (id)] TO (R) [ |MVAR|
]
          |MVARQ|
          |MVARI|
          |MVARY|
```

This form of the ALTER command is used to change a specified component of a designated load at a designated bus to a specified value. If the optional load identifier tokens are omitted, a load identifier of 1 is assumed. The second token specified indicates the component of load as follows:

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	MW, MWP	Active power component, constant power characteristic.
	MWI	Active power component, constant current characteristic.
	MWY	Active power component, constant admittance characteristic.
	MVAR, MVARQ	Reactive power component, constant power characteristic.
	MVARI	Reactive power component, constant current characteristic.
	MVARY	Reactive power component, constant admittance characteristic; R is negative for an inductive load.

The value specified is always in MW or Mvar at unity voltage. Refer to Load Data.

```
SHED LOAD [n] BUS (bus id)
```

The SHED command is used to set all loads at the specified bus out-of-service. Refer to Load Data.

```
          |MW   |
|ALTER |  |MWG  |                                                  [|MW
 |]
|CHANGE|  |MVAR | SHUNT [n] BUS (bus id) [SHUNT (id)] TO (R) [|MVAR|]
          |MVARB|
```

This form of the ALTER command is used to change a specified component of a specified fixed shunt at a designated bus to a specified value. If the optional shunt identifier tokens are omitted, a shunt identifier of 1 is assumed. The second token specified selects either the active (MW or MWG) or reactive (MVAR or MVARB) component of bus shunt. Values entered are in MW or Mvar at unity voltage. The reactive component is specified as a negative quantity for an inductive load. Refer to Fixed Bus Shunt Data.

```
|ALTER |
|CHANGE| BUS (bus id) CODE TO (I)
```

This form of the ALTER command is used to change the bus type code of a designated bus. Refer to Bus Data.

```
|ALTER |  |R|
                   [|CKT    |     ]
|CHANGE|  |X| TO (R) [n] FROM [n] BUS (bus id) TO [n] BUS (bus id)
[|CIRCUIT| (id)]
          |B|
```

This form of the ALTER command is used to change the per unit resistance, reactance or charging of a designated non-transformer branch or two-winding transformer to a specified value. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. Refer to Non-Transformer Branch Data.

```
DROP PLANT [n] BUS (bus id)
```

This form of the DROP command is used to disconnect all generation at a specified bus. The bus type code is set to one; refer to Bus Data and Section 5.10.1, "Equipment Status Changes".

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```
RECONNECT PLANT [n] BUS (bus id)
```

This form of the RECONNECT command is used to reconnect generation at a specified bus. The bus type code is set to two; machine status flags are not changed. Refer to Bus Data, Generator Data, and Section 5.10.1, "Equipment Status Changes".

```
     |UNIT     |
DROP |GENERATOR| (id) [n] BUS (bus id)
     |MACHINE  |
     |LOAD     |
     |SHUNT    |
```

This form of the DROP command is used to disconnect a specified machine, load, or fixed shunt at a designated bus. The machine, load, or shunt status flag is set to zero; refer to Load Data, Fixed Bus Shunt Data, Generator Data, and Section 5.10.1, "Equipment Status Changes".

```
          |UNIT     |
RECONNECT |GENERATOR| (id) [n] BUS (bus id)
          |MACHINE  |
          |LOAD     |
          |SHUNT    |
```

This form of the RECONNECT command is used to reconnect a specified machine, load, or fixed shunt at a designated bus. The machine, load, or shunt status flag is set to 1; the bus type code is not changed. Refer to Bus Data, Load Data, Fixed Bus Shunt Data, Generator Data, and Section 5.10.1, "Equipment Status Changes".

```
DISCONNECT BUS (bus id)
```

This form of the DISCONNECT command is used to electrically disconnect a bus and all equipment connected to it. This command results in the use of activities DSCN and ORDR.

```
|DISCONNECT| |LINE  |
             [|CKT    |     ]
|TRIP      | |TIE   |[n] FROM [n] BUS (bus id) TO [n] BUS (bus id)
[|CIRCUIT| (id)]
|OPEN      | |BRANCH|
```

This form of the DISCONNECT command is used to remove a specified branch from service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The branch status flag is set to zero; refer to Non-Transformer Branch Data and Section 5.10.1, "Equipment Status Changes".

```
|RECONNECT| |LINE  |
             [|CKT    |    ]
|CLOSE    | |TIE   | [n] FROM [n] BUS (bus id) TO [n] BUS (bus id)
[|CIRCUIT|(id)]
|RECLOSE  | |BRANCH|
```

This form of the RECONNECT command is used to return a specified branch to service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The branch status flag is set to 1; refer to Non-Transformer Branch Data and Section 5.10.1, "Equipment Status Changes".

```
|DISCONNECT|
                                          [|CKT    |     ]
```


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```
|TRIP      |THREEWINDING [n] FROM [n] BUS (bus id) TO [n] BUS (bus
id) TO [n] BUS (bus id) [|CIRCUIT| (id)]
|OPEN      |
```

This form of the DISCONNECT command is used to remove all windings of a specified three-winding transformer from service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The transformer status flag is set to zero; refer to Transformer Data and Section 5.10.1, "Equipment Status Changes".

```
|DISCONNECT|
                                          [|CKT    |     ]
|TRIP      |THREEWINDING [n] AT [n] BUS (bus id) TO [n] BUS (bus
id) TO [n] BUS (bus id) [|CIRCUIT| (id)]
|OPEN      |
```

This form of the DISCONNECT command is used to remove the winding connected to the bus specified as the AT bus of a specified three-winding transformer from service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The transformer status flag is set to 2, 3 or 4; refer to Transformer Data and Section 5.10.1, "Equipment Status Changes".

```
|RECONNECT|
                                            [|CKT    |     ]
|CLOSE    |THREEWINDING [n] FROM [n] BUS (bus id) TO [n] BUS (bus
id) TO [n] BUS (bus id) [|CIRCUIT| (id)]
|RECLOSE  |
```

This form of the RECONNECT command is used to return all windings of a specified three-winding transformer to service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The transformer status flag is set to 1; refer to Transformer Data and Section 5.10.1, "Equipment Status Changes".

```
BLOCK DCLINE (dc id)
```

The BLOCK command is used to block a specified two-terminal dc line. The control mode is set to zero; refer to Two-Terminal DC Transmission Line Data.

```
UNBLOCK DCLINE (dc id)
```

The UNBLOCK command is used to unblock a specified two-terminal dc line. The control mode is set to 1 (i.e., the UNBLOCK command assumes power control mode). Refer to Two-Terminal DC Transmission Line Data.

```
                    |SETPOINT|
SET DCLINE (dc id) |SCHEDULE| TO (R)
```

This form of the SET command is used to change the current or power demand (SETPOINT) or scheduled dc voltage (SCHEDULE) on a specified two-terminal dc line to a designated value; refer to Two-Terminal DC Transmission Line Data.

```
SET LOADFLOW TITLE LINE |1| TO '(title)'
                        |2|
```

This form of the SET command is used to change the designated line of the two-line case heading to a specified string. The title may be up to 60 characters and must be enclosed in single quotes; refer to Case Identification Data.

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```
                    |BUS   |
NET GENERATION [n] |BUSES| (bus id) [(bus id)...[(bus id)]]
```

This form of the NET command is used to net the generation with the load at specified buses; refer to activity GNET. Up to 25 buses may be specified in each use of this command.

```
NET GENERATION [n] FOR BUSES (I) THRU (I)
```

This form of the NET command is used to net the generation with the load at all buses within a specified bus number range; refer to activity GNET. This command is valid only when the bus input option is in its numbers setting.

```
SET SOLUTION OPTION[S] TO DEFAULT
```

This form of the SET command is used to instruct activity PSEB that subsequent SOLVE commands (see below) are to execute power flow solutions with the present power flow solution adjustment option settings honored. Refer to Sections Section 3.3.3, "Program Run-Time Option Settings" and Section 6.3.20, "Automatic Adjustments" and activity OPTN.

```
SET SOLUTION OPTION[S] TO [|DISCRETE-TAP|],[|AREA-LOCKED|],[FLAT-START],[|DC-LOCKED|],[|SH
|]
                          [|DIRECT-TAP  |] [|AREA-INTCHG|]
        [|DC-ADJUST|] [|SHUNT-ADJUST|] [|PSHFT-LOCKED|]
                          [|TAP-LOCKED  |] [|TIES+LOADS |]
                      [|SHN-ADJ-CONT|]
```

This form of the SET command is used to enable and/or disable the power flow adjustments to be used on subsequent power flow solutions initiated with the SOLVE command (see below). For any adjustment option not specified, the present program option setting is honored. Refer to Sections Section 3.3.3, "Program Run-Time Option Settings" and Section 6.3.20, "Automatic Adjustments" and activity OPTN.

```
                |FNSL|
                |FDNS| [                    |AFTER (I) [ITERATION[S]]|]
SOLVE [n] USING |NSOL| [WITH VAR LIMITS [n] |IMMEDIATE[LY]
     |]
                |SOLV| [                    |IGNORED
     |]
                |MSLV|
                |INLF|
```

The SOLVE command is used to execute one of the power flow solution activities. Solution options as established by the last SET SOLUTION OPTIONS command are honored; if no SET SOLUTION OPTIONS command was specified, the current solution adjustment option settings are honored. Refer to activities SOLV, MSLV, FNSL, NSOL, FDNS, INLF, and OPTN and Section 3.3.3, "Program Run-Time Option Settings".

```
|PSS     |
|PASS    |
|PASSTHRU|
```

The PASSTHRU command is used to inform activity PSEB that subsequent command input is in the form of BAT_ records (refer to *PSS®E Application Program Interface (API)*) and/or PSS®E activity commands and responses to questions rather than in the form of PSEB commands. While *passthru* mode is in effect, each input to activity PSEB is simply appended without modification to the Response File being built. If input to

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activity PSEB is being taken from the user's terminal, the prompt PASSTHRU: is issued while this mode is in effect. *Passthru* mode is terminated and PSEB command mode reinstated by entering the command FIN (see below). @INPUT and @CHAIN commands must not be specified while *passthru* mode is in effect; the IDEV activity command may be specified.

```
FIN
```

Terminate *passthru* mode (see above) and reinstate PSEB command mode.

```
|USE |
|COPY| (response-file-filename) [arg1 ... [arg9]]
|IDEV|
```

The USE command allows the user to include the BAT_ records (refer to *PSS®E Application Program Interface (API)*) and/or PSS®E line mode commands and responses to instructions which are contained in a Response File within the PSEB commands which define a PSS®E calculation sequence. The USE filename command is converted to an @INPUT filename command and inserted into the Response File being constructed by activity PSEB. Note that entries in the designated file must be in the form of BAT_ records and/or activity commands and responses to questions rather than in the form of PSEB commands.

The designated file may contain @INPUT commands but must not contain any @CHAIN commands or IDEV activity commands. The designated file must be terminated with an @END command; it *must not* be terminated with an IDEV activity command or a FIN PSEB command.

```
EXECUTE (iplan-program-filename)
```

The EXECUTE command allows the user to include the execution of an IPLAN program within the PSEB commands which define a calculation sequence. The EXECUTE filename PSEB command is converted to an EXEC filename activity command (refer to activity EXEC) and inserted into the Response File being constructed by activity PSEB.

```
WATCH [OFF]
```

The WATCH command is used to enable or disable the echoing of responses generated from PSEB commands to the dialog output device (normally the user's terminal). Response echoing is disabled upon entering activity PSEB; the PSEB command WATCH enables response echoing and WATCH,OFF disables it.

```
CHECK
```

The CHECK command instructs activity PSEB to bypass the automatic execution of the Response File that it constructs.

```
END
```

The END command terminates PSEB command input. Unless it is otherwise suppressed, the Response File constructed by activity PSEB is automatically executed.

```
ABORT
```

The ABORT command terminates activity PSEB and bypasses the automatic execution of the Response File that it has constructed.

```
HELP
```


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This form of the HELP command displays the list of PSEB commands at the dialog output device (refer to Section 4.4, "Virtual Output Devices").

`HELP,command`

This form of the HELP command displays the syntax format of the specified PSEB command at the dialog output device (refer to Section 4.4, "Virtual Output Devices").

Additional Information
PSS®E GUI Users Guide, PSEB Command File, Power Flow Calculation *PSS®E Command Line Interface (CLI) Users Guide*, Building a Response File, Power Flow Calculation

17.15. Building a Response File,Dynamic Simulation

PSAS

Run Activity PSAS - GUI
I/O Control>Run program Automation file... *[Select Program Automation File to Run]* PSAS Command file (*.psa)
Run Line Mode Activity PSAS - CLI
ACTIVITY? >>**PSAS** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The simulation run assembler activity PSAS allows the user to specify dynamic simulation runs in an English-like language. Input to activity PSAS is taken either from a PSAS Command File or from the dialog input device. This can be in-stream in a Response File, typed into the command line in the GUI, or entered at the console in PSSECMD. The output from activity PSAS is in the form of a PSS®E Response File.

17.15.1. PSS®E PSAS Macro Language

PSAS (Simulation Run Assembler) is a built-in macro language that supports PSS®E Dynamics simulations through the use of English-like command macros. PSAS commands are created within a text file using a text editor. Each record is in the form of a command starting with a verb from a previously defined vocabulary. The PSAS commands are documented below and in the *PSAS User's Ready Reference*.

PSAS commands do not interact with PSS®E directly. The PSAS macro processor is really a translator from the PSAS language to a Response File, which may contain a mixture of command line and batch commands. The PSAS command line command normally automatically performs the commands in the Response File after the translation. When used with the suffix CHECK (i.e. PSAS, CHECK) only the translation is performed. When using the PSAS API routine only the translation is performed.

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17.15.2. PSAS Commands

Each PSAS command begins with a verb from the vocabulary recognized by activity PSAS. Some commands require additional keywords (e.g., BUS, VAR), numeric quantities (e.g., bus number, VAR index), and/or character identifiers (e.g., dc line name, circuit identifier) in specific fields of the command format. Some commands provide for the optional specification of keywords, numeric and/or character quantities; these are shown in the command syntax enclosed in braces ([ ]).

Where one of several keywords may be specified, they are shown in the command format enclosed in bars (| |). One of the keywords listed must be entered in the field.

Each keyword or data value must be followed by one or more blanks. For clarity, a comma or equal sign may be used in place of a blank space. For example, the following two PSAS commands are equivalent:

```
START OUTPUT SAVNW.PLT
START, OUTPUT=SAVNW.PLT
```

Any field labeled bus id designates that a bus identifier is to be entered. When the *numbers* input option is in effect, the bus number must be specified; using the *names* input option, the extended bus name enclosed in single quotes is required.

Any field labeled dc id designates that a two-terminal dc line name is to be entered. If the name contains any blanks or special characters, it must be enclosed in single quotes.

I and R indicate that integer and real (i.e., it may include a decimal point) numeric values respectively are to be specified. C indicates that a one or two character alphanumeric value is to be specified.

Many PSAS commands provide for the insertion of optional descriptive text in designated fields, indicated by [n] in the command syntax. For example, the following two PSAS commands translate into the same PSS®E dialog:

```
RECOVER FROM SAVNW.SNP AND SAVNW.SAV
RECOVER snapshot and saved case FROM SAVNW.SNP AND SAVNW.SAV
```

A dollar sign ($) at the end of a PSAS command line is used to continue the command on the following line. At least one space must precede the dollar sign. As many physical lines as required may be used to specify a single PSAS command as long as all lines except the last one are terminated with a dollar sign. This multiline capability may be used with all PSAS commands *except* the USE command (see below). In addition, each PSS®E activity command and response to an instruction entered in *passthru* mode (see below) must be entered on a single line.

The detailed syntax of each PSAS command is given below and also on the *PSS®E Simulation Run Assembler PSAS User's Ready Reference* sheet. Keywords shown in uppercase letters must be entered as shown in either upper or lowercase characters.

```
     [|NOW                          |]
HOLD[n]IN(snapshot-filename)AND(saved-case-filename) [|SIZE[S](I1)(I2)(I3)(I4)(I5)|]
```

The HOLD command is used to preserve dynamics working memory and the power flow working case in Snapshot and Saved Case Files respectively (refer to activities SNAP and SAVE). When the HOLD command includes the optional keyword NOW, activity SNAP and SAVE are immediately executed and the user speci-

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fies the number of dynamic simulation array elements to save in the same manner as when activity SNAP is specifically invoked. In this case, the PSS®E responses for activity SNAP and SAVE are *not* written to the Response File being constructed by activity PSAS.

When the HOLD command does not include the optional keyword NOW, the appropriate entries in the Response File are made to take a Snapshot and save the working case during the subsequent execution of the Response File. In this case, the user may optionally specify the keyword SIZES followed by the number of CONs, STATEs, VARs, ICONs and output channels to be preserved. If this keyword and its data are omitted, a blank response is generated in the Response File and, when the Response File is subsequently executed, activity SNAP preserves up to the next available positions in the respective dynamics data arrays.

```
RECOVER[n] FROM (snapshot-filename) AND (saved-case-filename)
[NOFACT] [NOTYSL] [NORETURN]
```

The RECOVER command is used to access the designated Snapshot and Saved Case Files (refer to activities RSTR and CASE). Activities RSTR and CASE are executed immediately *and* the PSS®E responses for activities RSTR, LOFL, and CASE are written to the Response File being constructed by activity PSAS. The activity commands FACT, TYSL, and RTRN are included in the Response File unless they are suppressed with one or more of the optional tokens NOFACT, NOTYSL and NORETURN. If NOFACT is specified, NOTYSL is assumed. If NORETURN is specified, NOFACT and NOTYSL are also assumed, and the next PSAS command must be either CONVERT, USE or PASS; the CONVERT command (see below) generates the RTRN activity command but in *passthru* mode (see below), the RTRN activity command must be specified before terminating *passthru* mode.

The RECOVER command overwrites dynamics working memory and the power flow working case. The HOLD command with the NOW option may be used before the RECOVER command to preserve these two sets of data.

```
|INITIALIZE| OUTPUT |(filename)| [SNAPSHOT |(filename)|]
|START     |        |NONE      | [         |NONE      |]
```

This form of the START command is used to initialize the dynamic model in preparation for state-space simulations. The STRT activity command is generated with user specified channel output and Snapshot filename responses. The reserved token NONE may be used to indicate that the corresponding filename is not to be specified.

```
|INITIALIZE| EXTENDED [n] OUTPUT |(filename)| [n]
|START     |                     |NONE      |
```

This form of the START command is used to initialize the dynamic model in preparation for extended term simulations. The MSTR activity command is generated with a user specified channel output filename response. The reserved token NONE may be used to indicate no channel output file is to be specified.

```
        |TO |     |CYCLE[S] |
RUN [n] |FOR| (R) |SECOND[S]| [PRINT (I)] [PLOT (I)] [CRTPLT (I)]
```

The RUN command is used to perform dynamic simulation calculations with activity RUN or MRUN. If the RUN command is preceded by the START command (see above), the form of the START command determines whether activities RUN or MRUN is to be used. Otherwise, state-space simulations are assumed and activity RUN is used. The real value indicates the duration of the simulation calculation and may be specified in either seconds or cycles. It is the value of simulation time (i.e., TPAUSE) when the token TO precedes it and an incremental value of time when the token FOR precedes it. The optional integer tokens and their corresponding keywords allow the user to specify channel printing and plotting intervals (refer to Section 16.10, "Performing State-Space Simulation in Time Steps"). By default, PRINT=20, PLOT=5 and CRTPLT=0.

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```
                    |OFF|
CONVERGENCE MONITOR |ON |
```

The CONVERGENCE MONITOR command is used to enable or disable the printing of the network solution convergence monitor during the execution of the PSS®E activity calls generated by subsequent START and RUN commands (see above). By default, the convergence monitor is suppressed.

```
APPLY FAULT [n] BUS (bus id) [n]
```

This form of the APPLY FAULT command is used to apply a solidly grounded three-phase fault at the designated bus. The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the fixed shunt at the bus to j(-2. *10^9); refer to Bus Data, activity ALTR and Section 21.7.2 Applying Disturbances.

```
                                  |Y         |
       |MVA    |
APPLY FAULT [n] BUS (bus id) [n] |ADMITTANCE| (R1) (R2) |MHO[S]|
[BASEKV (R)]
```

This form of the APPLY FAULT command is used to apply a fault with a specified fault admittance, (R1) + j(R2), at a designated bus. The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the fixed shunt at the bus. (R2) should be entered as a negative number. Refer to Bus Data, activity ALTR and Section 23.7.2, “Applying Disturbances”.

If the MVA token is specified, the active and reactive components of fixed shunt at the bus are set to the values specified as (R1) and (R2) respectively; i.e., the fault admittance is specified in the same MVA units as is bus shunt.

If the MHO or MHOS token is specified, (R1) and (R2) are entered in mhos and activity PSAS converts them to the MVA units used for specifying bus shunts. If the optional base voltage tokens are specified, the base voltage specified on the PSAS command is used in the conversion calculation; otherwise, the base voltage contained in the working case for the designated bus is used. In either case, a positive base voltage value is required.

```
                                 |Z         |
APPLY FAULT [n] BUS (bus id) [n] |IMPEDANCE| (R1) (R2) OHM[S] [BASEKV
(R)]
```

This form of the APPLY FAULT command is used to apply a fault with a specified fault impedance, (R1) + j(R2), at a designated bus. The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the fixed shunt at the bus. (R2) should be entered as a positive number and (R1) and (R2) are specified in ohms. Activity PSAS converts them to the MVA units used for specifying bus shunts. If the optional base voltage tokens are specified, the base voltage specified on the PSAS command is used in the conversion calculation; otherwise, the base voltage contained in the working case for the designated bus is used. In either case, a positive base voltage value is

```
     |L-G   |
SCMU |L-L   | FAULT [n] BUS (bus id) [n] [ZL-G (R1) (R2)] [n]
     |L-L-G|
                     [ZL-L (R3) (R4)] [n] [CONVERTDC] [n] [APPLYZCOREC]
```

The SCMU FAULT command is used to calculate and apply a single line-to-ground, line-to-line, or line-to-line-to-ground fault at a designated bus. For the L-G and L-L-G faults, (R1) and (R2) may optionally specify the

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complex line-to-ground fault impedance to be used; similarly, for the L-L and L-L-G faults, (R3) and (R4) may optionally specify the complex line-to-line fault impedance to be used. These impedances are specified in per unit. For any of these impedance values that are not specified, 0.0 is assumed.

If the optional token CONVERTDC is specified, dc lines and FACTS device boundary conditions are converted to constant admittance load for the fault calculation. Otherwise, they are neglected.

If the optional token APPLYZCOREC is specified, for those transformers which include a transformer impedance adjustment table, the same correction factor is applied to the zero sequence impedance as is applied to the positive sequence impedance. Otherwise, transformer zero sequence nominal impedances are used.

```
     |L-G   |
     |L-L   |
     |L-L-G |
       [|CKT    |     ]
SPCB |3PHASE| FAULT (R1) [n] FROM [n] BUS (bsid) TO [n] BUS (bsid)
[n] [|CIRCUIT| (id)]
                        [n] [ZL-G (R2) (R3)] [n]
[ZL-L (R4) (R5)] [n] [CONVERTDC] [n] [APPLYZCOREC]
```

This form of the SPCB FAULT command is used to calculate and apply a ground fault at a designated point along a transmission line. (R1) specifies the fault location as the fraction of the line from the FROM bus; the default value of 0.5 applies the fault at the midpoint of the line.

For the L-G and L-L-G faults, (R2) and (R3) may optionally specify the complex line-to-ground fault impedance to be used; similarly, for the L-L and L-L-G faults, (R4) and (R5) may optionally specify the complex line-to-line fault impedance to be used. These impedances are specified in per unit. For any of these impedance values that are not specified, 0.0 is assumed.

If the optional token CONVERTDC is specified, dc lines and FACTS device boundary conditions are converted to constant admittance load for the fault calculation. Otherwise, they are neglected.

If the optional token APPLYZCOREC is specified, for those transformers which include a transformer impedance adjustment table, the same correction factor is applied to the zero sequence impedance as is applied to the positive sequence impedance. Otherwise, transformer zero sequence nominal impedances are used.

```
                [|CKT    |     ]
SPCB 2PHASE OPEN [n] FROM [n] BUS (bsid) TO [n] BUS (bsid) [n] [|CIRCUIT| (id)]

[n] [CONVERTDC] [n] [APPLYZCOREC]
```

This form of the SPCB FAULT command is used to calculate and apply a transmission line unbalance with two of its three phases open.

If the optional token CONVERTDC is specified, dc lines and FACTS device boundary conditions are converted to constant admittance load for the fault calculation. Otherwise, they are neglected.

If the optional token APPLYZCOREC is specified, for those transformers which include a transformer impedance adjustment table, the same correction factor is applied to the zero sequence impedance as is applied to the positive sequence impedance. Otherwise, transformer zero sequence nominal impedances are used.

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```
                [|CKT    |     ]
SPCB 1PHASE OPEN [n] FROM [n] BUS (bsid) TO [n] BUS (bsid) [n] [|CIRCUIT| (id)]

          [n] [PATH (R1) [ [n] [ZL-G (R2) (R3)]
] [n] [CONVERTDC] [n] [APPLYZCOREC]
```

This form of the SPCB FAULT command is used to calculate and apply a transmission line unbalance with one of its three phases open.

This branch unbalance may optionally include a ground fault at a designated point of the open phase by specifying the optional token PATH, along with (R1) which specifies the fault location as the fraction of the line from the FROM bus; the default value of 0.5 applies the fault at the midpoint of the open phase. (R2) and(R3) may optionally specify the complex line-to-ground fault impedance in per unit to be used. For any of these impedance values that are not specified, 0.0 is assumed.

If the optional token CONVERTDC is specified, dc lines and FACTS device boundary conditions are converted to constant admittance load for the fault calculation. Otherwise, they are neglected.

If the optional token APPLYZCOREC is specified, for those transformers which include a transformer impedance adjustment table, the same correction factor is applied to the zero sequence impedance as is applied to the positive sequence impedance. Otherwise, transformer zero sequence nominal impedances are used.

```
               [|CKT    |     ]
SPCB BREAKER OPEN [n] AT [n] BUS (bsid) TO [n] BUS (bsid) [n] [|CIRCUIT| (id)]
          [n] [PATH (R1) [ [n] [ZL-G (R2) (R3)]
] [n] [CONVERTDC] [n] [APPLYZCOREC]
```

This form of the SPCB FAULT command is used to calculate and apply a transmission line unbalance with the breaker of one phase open. The breaker at the AT bus is assumed to be the open breaker.

This branch unbalance may optionally include a ground fault at a designated point of the open phase by specifying the optional token PATH, along with (R1) which specifies the fault location as the fraction of the line from the FROM bus; the default value of 0.5 applies the fault at the midpoint of the open phase. (R2) and(R3) may optionally specify the complex line-to-ground fault impedance in per unit to be used. For any of these impedance values that are not specified, 0.0 is assumed.

If the optional token CONVERTDC is specified, dc lines and FACTS device boundary conditions are converted to constant admittance load for the fault calculation. Otherwise, they are neglected.

If the optional token APPLYZCOREC is specified, for those transformers which include a transformer impedance adjustment table, the same correction factor is applied to the zero sequence impedance as is applied to the positive sequence impedance. Otherwise, transformer zero sequence nominal impedances are used.

```
CLEAR FAULT [n] [BUS (bus id)]

CLEAR BUS [FAULT] [n] [BUS (bus id)]
```

These forms of the CLEAR command are used to clear bus faults which were previously applied with the APPLY FAULT command (see above). The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the fixed bus shunt at the bus back to its original value.

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Activity PSAS remembers the faulted bus and the fixed bus shunt which was replaced with the fault admittance for up to ten bus faults. If the optional bus identification tokens are omitted, activity PSAS clears the last bus fault which was applied and which has not already been cleared.

```
                |LINE  |
                       [|CKT    |    ]
APPLY FAULT [n]|TIE   |[n] FROM [n] BUS (bus id) TO [n] BUS (bus
id) [n][|CIRCUIT|(id)] [n]
                |BRANCH|
```

This form of the APPLY FAULT command is used to apply a solidly grounded three-phase fault at the from bus end of a designated non-transformer branch. The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the line shunt at the from bus end of the branch to j(-2.*10 ^{7}). If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. Refer to Bus Data, activity ALTR and Section 23.7.2, "Applying Disturbances".

```
                 |LINE  |
                          [|CKT    |     ]
APPLY FAULT [n] |TIE   | [n] FROM [n] BUS (bus id) TO [n] BUS (bus
id) [n] [|CIRCUIT| (id)] [n]
                 |BRANCH|
                                                    |Y         |
           |MVA   |
                                                    |ADMITTANCE|
(R1) (R2) |MHO[S]| [BASEKV (R)]
```

This form of the APPLY FAULT command is used to apply a fault with a specified fault admittance, (R1) + j(R2), at the from bus end of a designated non-transformer branch. The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the line shunt at the from bus end of the branch. (R2) should be entered as a negative number. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. Refer to Non-Transformer Branch Data, activity ALTR and Section 23.7.2, "Applying Disturbances".

If the MVA token is specified, the values specified as (R1) and (R2) are divided by the system base MVA and the active and reactive components respectively of the line shunt at the from bus end of the branch are set to these values. The fault admittance is therefore specified on the record is in the same MVA units as is bus shunt (refer to Bus Data).

If the MHO or MHOS token is specified, (R1) and (R2) are entered in mhos and activity PSAS converts them to the per unit values used for specifying line shunts. If the optional base voltage tokens are specified, the base voltage specified on the PSAS command is used in the conversion calculation; otherwise, the base voltage contained in the working case for the designated from bus is used. In either case, a positive base voltage value is required.

```
                 |LINE  |
                          [|CKT    |     ]
APPLY FAULT [n] |TIE   | [n] FROM [n] BUS (bus id) TO [n] BUS (bus
id) [n] [|CIRCUIT| (id)] [n]
                 |BRANCH|
                                                       |Z
 |
                                                       |IMPEDANCE|
```


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```
(R1) (R2) OHM[S] [BASEKV (R)]
```

This form of the APPLY FAULT command is used to apply a fault with a specified fault impedance, (R1) + j(R2), at the from bus end of a designated non-transformer branch. The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the line shunt at the from bus end of the branch. (R2) should be entered as a positive number and (R1) and (R2) are specified in ohms. Activity PSAS converts them to the per unit values used for specifying line shunts. If the optional base voltage tokens are specified, the base voltage specified on the PSAS command is used in the conversion calculation; otherwise, the base voltage contained in the working case for the designated from bus is used. In either case, a positive base voltage value is required. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. Refer to Non-Transformer Branch Data, activity ALTR and Section 23.7.2, "Applying Disturbances".

```
       |LINE   |
CLEAR  |TIE    | FAULT [n]
       |BRANCH|
```

This form of the CLEAR command is used to clear the last line fault which was previously applied with the APPLY FAULT command (see above). The Response File being constructed by activity PSAS includes responses to use activity ALTR to change the appropriate line shunt on the faulted branch back to its original value.

Activity PSAS remembers the faulted branch and the from and to bus orientation specified on the last APPLY FAULT command. Note that only the last line fault command is remembered by activity PSAS.

```
CONVERT [n] MW TO (R) (R) [MVAR TO (R) (R)]
```

The CONVERT command is used to convert the constant MVA load at all buses in the working case to a specified mix of the constant MVA, constant current and constant admittance load characteristics. The (R) values are specified as percents where the first value of each pair is the percentage of load to be converted to the constant current characteristic and the second is the percentage to be converted to constant admittance. If the optional Mvar tokens are omitted, the same split of load is used for the active and reactive power.

The Response File being constructed by activity PSAS includes responses to use activity CONL to convert the load as specified in the CONVERT command, followed by the activity commands CONG, ORDR, FACT, TYSL, and RTRN.

The CONVERT command must be used immediately following a RECOVER command on which the optional token NORETURN is specified.

```
          |MWP   |
          |MWI   |
|ALTER |  |MWY   |                                                  [ |MW
 | ]
|CHANGE|  |MVARQ| LOAD [n] BUS (bus id) [LOAD (id)] TO (R) [ |MVAR|
]
          |MVARI|
          |MVARY|
```

This form of the ALTER command is used to change a specified component of a designated load at a designated bus to a specified value. If the optional load identifier tokens are omitted, a load identifier of 1 is assumed. The second token specified indicates the component of load as follows:

	MWP	Active power component, constant power characteristic.

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	MWI	Active power component, constant current characteristic.
	MWY	Active power component, constant admittance characteristic.
	MVARQ	Reactive power component, constant power characteristic.
	MVARI	Reactive power component, constant current characteristic.
	MVARY	Reactive power component, constant admittance characteristic; R is negative for an inductive load.

The value specified is always in MW or Mvar at unity voltage. Refer to Load Data.

```
SHED LOAD [n] BUS (bus id)
```

The SHED command is used to set all loads to out-of-service at the specified bus. Refer to Load Data.

```
|ALTER | |MWG  | |LOAD |
               [|MW  |]
|CHANGE| |MVARB| |SHUNT| [n] BUS (bus id) [SHUNT (id)] TO (R) [|MVAR|]
```

This form of the ALTER command is used to change a specified component of a specified fixed shunt at a designated bus to a specified value. If the optional shunt identifier tokens are omitted, a shunt identifier of 1 is assumed. The second token specified selects either the active (MWG) or reactive (MVARB) component of bus shunt. Values entered are in MW or Mvar at unity voltage. The reactive component is specified as a negative quantity for an inductive load. Refer to Fixed Bus Shunt Data.

```
|ALTER |
|CHANGE| BUS (bus id) CODE TO (I)
```

This form of the ALTER command is used to change the bus type code of a designated bus. Refer to Bus Data.

```
|ALTER | |R|
                  [|CKT    |     ]
|CHANGE| |X| TO (R) [n] FROM [n] BUS (bus id) TO [n] BUS (bus id)
[|CIRCUIT| (id)]
         |B|
```

This form of the ALTER command is used to change the per unit resistance, reactance or charging of a designated non-transformer branch or two-winding transformer to a specified value. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. Refer to Bus Data.

```
DROP PLANT [n] BUS (bus id)
```

This form of the DROP command is used to disconnect all generation at a specified bus. The bus type code is set to one; refer to Bus Data and Section 5.10.1, "Equipment Status Changes".

```
RECONNECT PLANT [n] BUS (bus id)
```

This form of the RECONNECT command is used to reconnect generation at a specified bus. The bus type code is set to two; machine status flags are not changed. Refer to Bus Data, Generator Data, and Section 5.10.1, "Equipment Status Changes".

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```
     |UNIT     |
DROP |GENERATOR| (id) [n] BUS (bus id)
     |MACHINE  |
     |LOAD     |
     |SHUNT    |
```

This form of the DROP command is used to disconnect a specified machine, load, or fixed shunt at a designated bus. The machine, load, or shunt status flag is set to zero; refer to Bus Data, Fixed Bus Shunt Data, and Section 5.10.1, "Equipment Status Changes".

```
          |UNIT     |
RECONNECT |GENERATOR| (id) [n] BUS (bus id)
          |MACHINE  |
          |LOAD     |
          |SHUNT    |
```

This form of the RECONNECT command is used to reconnect a specified machine, load, or fixed shunt at a designated bus. The machine, load, or shunt status flag is set to one; the bus type code is not changed. Refer to Bus Data, Load Data, Fixed Bus Shunt Data, Generator Data, and Section 5.10.1, "Equipment Status Changes".

```
DISCONNECT BUS (bus id)
```

This form of the DISCONNECT command is used to electrically disconnect a bus and all equipment connected to it. This command results in the use of activities DSCN and ORDR.

```
|DISCONNECT| |LINE  |
                    [|CKT    |     ]
|TRIP      | |TIE   | [n] FROM [n] BUS (bus id) TO [n] BUS (bus
id) [|CIRCUIT| (id)]
|OPEN      | |BRANCH|
```

This form of the DISCONNECT command is used to remove a specified branch from service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The branch status flag is set to zero; refer to Non-Transformer Branch Data and Section 5.10.1, "Equipment Status Changes".

```
|RECONNECT| |LINE  |
                   [|CKT    |    ]
|CLOSE    | |TIE   | [n] FROM [n] BUS (bus id) TO [n] BUS (bus id)
[|CIRCUIT|(id)]
|RECLOSE  | |BRANCH|
```

This form of the RECONNECT command is used to return a specified branch to service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The branch status flag is set to one; refer to Non-Transformer Branch Data and Section 5.10.1, "Equipment Status Changes".

```
|DISCONNECT|
                                      [|CKT    |     ]
|TRIP      |THREEWINDING [n] FROM [n] BUS (bsid) TO [n] BUS (bsid)
TO [n] BUS (bsid) [|CIRCUIT| (id)]
|OPEN      |
```

This form of the DISCONNECT command is used to remove all windings of a specified three-winding transformer from service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed.

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The transformer status flag is set to zero; refer to Transformer Data and Section 5.10.1, "Equipment Status Changes".

```
|DISCONNECT|
                                        [|CKT    |     ]
|TRIP      |THREEWINDING [n] AT [n] BUS (bsid) TO [n] BUS (bsid)
TO [n] BUS (bsid) [|CIRCUIT| (id)]
|OPEN      |
```

This form of the DISCONNECT command is used to remove the winding connected to the bus specified as the AT bus of a specified three-winding transformer from service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The transformer status flag is set to 2, 3 or 4; refer to Transformer Data and Section 5.10.1, "Equipment Status Changes".

```
|RECONNECT|
                                      [|CKT    |     ]
|CLOSE    |THREEWINDING [n] FROM [n] BUS (bsid) TO [n] BUS (bsid)
TO [n] BUS (bsid) [|CIRCUIT| (id)]
|RECLOSE  |
```

This form of the RECONNECT command is used to return all windings of a specified three-winding transformer to service. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The transformer status flag is set to one; refer to Transformer Data and Section 5.10.1, "Equipment Status Changes".

```
BLOCK DCLINE (dc id)
```

The BLOCK command is used to block a specified two-terminal dc line. The control mode is set to zero; refer to Two-Terminal DC Transmission Line Data.

```
UNBLOCK DCLINE (dc id)
```

The UNBLOCK command is used to unblock a specified two-terminal dc line. The control mode is set to one (i.e., the UNBLOCK command assumes power control mode). Refer to Two-Terminal DC Transmission Line Data.

```
                         |POWER   |
                         |CURRENT |
SET [n] DCLINE (dc id) |SETPOINT| TO (R)
                         |SCHEDULE|
                         |VOLTAGE |
```

This form of the SET command is used to change the current or power demand (POWER, CURRENT, SETPOINT) or scheduled dc voltage (SCHEDULE, VOLTAGE) on a specified two-terminal dc line to a designated value; refer to Two-Terminal DC Transmission Line Data.

```
        |VAR|
SET [n] |CON| (I) TO (R)
```

This form of the SET command is used to change a specified VAR or CON to a designated value. Refer to activity ALTR and Section 23.1.2, "Data Structure".

```
                      |(C)|
SET [n] ICON (I) TO |(I)|
```


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This form of the SET command is used to change a specified ICON to a designated integer or character value. Refer to activity ALTR and Section 23.1.2, "Data Structure".

```
        |VAR[S]|
SET [n] |CON[S]| (I1) THRU (I2) TO (R1) ... (Rm)
```

This form of the SET command is used to change up to twenty-five consecutive VARs or CONs to designated values. Refer to activity ALTR and Section 23.1.2, "Data Structure".

```
                                    |(C1)|     |(Cm)|
SET [n] ICON[S] (I1) THRU (I2) TO |(I1)| ... |(Im)|
```

This form of the SET command is used to change up to twenty-five consecutive ICONs to designated integer and/or character values. Refer to activity ALTR and Section 23.1.2, "Data Structure".

```
                          [|CYCLE[S] |]
SET [n] STEP TO (R) (n) [|SECOND[S]|] [n]
```

This form of the SET command is used to change the simulation time step, DELT, to a designated value. The new time step may be specified in either seconds or cycles. When neither of the optional tokens is specified, SECONDS is assumed.

```
NEXT [CHANNEL[S] (I)] [VAR[S] (I)] [ICON[S] (I)]
```

The NEXT command is used in conjunction with PLACE commands to form the dialog required by activity CHAN to assign simulation variables to PSS®E output channels. It is used in setting the responses for the starting channel, VAR and ICON indices to be used by activity CHAN. If a NEXT command is not entered, or if the CHANNEL, VAR and/or ICON keywords and values are omitted from the NEXT command, a corresponding default response is generated in the Response File. When the Response File is subsequently executed, activity CHAN will use the next available location pointers as contained in dynamics data memory.

```
PLACE quantity [n] BUS (bus id) [MACHINE (id)]
IN CHANNEL[S]
                                [WITH IDENTIFIER[S] (ident) [AND (ident)]]
```

This form of the PLACE command is used to assign a single machine quantity (two quantities in the case of MACHAPPIMP) to an output channel. < quantity > may be specified as one of the following machine related tokens:

ANGLE	PELEC	QELEC	ETERM	ECOMP	EFD	PMECH
VREF	SPEED	XADIFD	VOTHSG	VUEL	VOEL	
MACHAPPIMP		MACHITERM				

If the optional machine identifier tokens are omitted, channels for all machines at the bus are generated. The user may either specify the channel identifier(s) to be assigned to the output channel(s) or let activity CHAN assign them. Refer to Table 16.2, "Activity CHAN Summary" and Table 23.1, "Dynamic Simulation Arrays", activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
      |BSFREQ  |
      |VOLTAGE |
PLACE |VOLT&ANG| [n] BUS (bus id) IN CHANNEL[S]
                                [WITH IDENTIFIER[S] (ident) [AND (ident)]]
```


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This form of the PLACE command is used to assign a single bus quantity (two quantities in the case of VOLT&ANG) to an output channel. The user may either specify the channel identifier(s) to be assigned to the output channel(s) or let activity CHAN assign them. Refer to Table 16.2, "Activity CHAN Summary" and Table 23.1, "Dynamic Simulation Arrays", activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
      |PLOAD|
PLACE |QLOAD| [n] BUS (bus id) [LOAD (id)] IN CHANNEL[S]
                                                  [WITH IDENTIFIER
(ident)]
```

This form of the PLACE command is used to assign a single load quantity to an output channel. If the optional load identifier token is omitted, channels for all loads at the bus are generated. The user may either specify the channel identifier to be assigned to the output channel(s) or let activity CHAN assign it. Refer to Table 16.2, "Activity CHAN Summary" and 21-1, activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
PLACE quantity [n] BUSES (I1) THRU (I2) IN CHANNEL[S]
```

This form of the PLACE command is used to assign a specified machine, bus, or load related quantity to output channels for all machines, buses, or loads within a designated bus number range. quantity > is specified as in the three BUS forms of the PLACE command given above. Refer to Table 15-2 Activity CHAN Summary and Table 23.1, "Dynamic Simulation Arrays", activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
      |FLOWP  |                                        [|CKT
   |     ]
PLACE |FLOWPQ | [n] BUS (bus id) TO [n] BUS (bus id) [|CIRCUIT|
(id)]
      |FLOWMVA|
      |RELAY2 |
                        IN CHANNEL[S] [WITH IDENTIFIER[S] (ident)
[AND (ident)]]
```

This form of the PLACE command is used to assign one or two branch related quantities to output channels for a designated branch. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The second token specified indicates quantities to be monitored as follows:

	FLOWP	One channel containing the active power flow on the branch.
	FLOWPQ	Two channels containing the active and reactive power flow on the branch.
	FLOWMVA	One channel containing the MVA flow on the branch.
	RELAY2	Two channels containing the apparent impedance on the branch as n by the RELAY2 model.

The user may either specify the channel identifier(s) to be assigned to the output channel(s) or let activity CHAN assign them. Refer to Table 16.2, "Activity CHAN Summary" and Table 23.1, "Dynamic Simulation Arrays", activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
      |3WNDFLOWP  |
```


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```
      |3WNDFLOWPQ |
      |3WNDFLOWMVA|
             [|CKT    |     ]
PLACE |3WNDRELAY3 | [n] BUS (bus id) TO [n] BUS (bus id) TO [n]
BUS (bus id) [|CIRCUIT| (id)]

                                    IN CHANNEL[S] [WITH IDENTIFIER[S]
(ident) [AND (ident)]]
```

This form of the PLACE command is used to assign one or two branch related quantities to output channels for the three-winding transformer winding connected to the first bus specified. If the optional circuit identifier tokens are omitted, a circuit identifier of 1 is assumed. The second token specified indicates quantities to be monitored as follows:

	3WNDFLOWP	One channel containing the active power flow into the winding.
	3WNDFLOWPQ	Two channels containing the active and reactive power flow into the winding.
	3WNDFLOWMVA	One channel containing the MVA flow on the winding.
	3WNDRELAY3	Two channels containing the apparent impedance on the winding as n by the RELAY3 model.

The user may either specify the channel identifier(s) to be assigned to the output channel(s) or let activity CHAN assign them. Refer to Table 16.2, "Activity CHAN Summary" and Table 23.1, "Dynamic Simulation Arrays", activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
      |VAR[S]   |
PLACE |STATE[S]| (I1) [THRU (I2)] IN CHANNEL[S]
```

This form of the PLACE command is used to assign one or more consecutive VARs or STATEs to output channels. Refer to Table 16.2, "Activity CHAN Summary" and Table 23.1, "Dynamic Simulation Arrays", activity ALTR, and Section 23.6.1, "Output Channel Selection".

```
|PSS      |
|PASS     |
|PASSTHRU|
```

The PASSTHRU command is used to inform activity PSAS that subsequent command input is in the form of BAT_ records (refer to *PSS®E Application Program Interface (API)*) and/or PSS®E activity commands and responses to questions rather than in the form of PSAS commands. While *passthru* mode is in effect, each input to activity PSAS is simply appended to the Response File being built without modification. If input to activity PSAS is being taken from the user's terminal, the prompt PASSTHRU: is issued while this mode is in effect. *Passthru* mode is terminated and PSAS command mode reinstated by entering the command FIN (see below). @INPUT and @CHAIN commands must not be specified while *passthru* mode is in effect; the IDEV activity command may be specified.

```
FIN
```

Terminate *passthru* mode (see above) and reinstate PSAS command mode.

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```
|USE |
|COPY| (response-file-filename) [arg1 ... [arg9]]
|IDEV|
```

The USE command allows the user to include the BAT_ records (refer to *PSS®E Application Program Interface (API)*) and/or PSS®E line mode commands and responses to instructions which are contained in a Response File within the PSAS commands which define a PSS®E calculation sequence. The USE filename command is converted to an @INPUT filename command and inserted into the Response File being constructed by activity PSAS. Note that entries in the designated file must be in the form of BAT_ records and/or activity commands and responses to questions rather than in the form of PSAS commands.

The designated file may contain @INPUT commands but must not contain any @CHAIN commands or IDEV activity commands. The designated file must be terminated with an @END command; it *must not* be terminated with an IDEV activity command or a FIN PSEB command.

```
EXECUTE (iplan-program-filename)
```

The EXECUTE command allows the user to include the execution of an IPLAN program within the PSAS commands which define a calculation sequence. The EXECUTE filename PSAS command is converted to an EXEC filename activity command (refer to activity EXEC) and inserted into the Response File being constructed by activity PSAS.

```
WATCH [OFF]
```

The WATCH command is used to enable or disable the echoing of responses generated from PSAS commands to the dialog output device (normally the user's terminal). Response echoing is disabled upon entering activity PSAS; the PSAS command WATCH enables response echoing and WATCH,OFF disables it.

```
CHECK
```

The CHECK command instructs activity PSAS to bypass the automatic execution of the Response File which it constructs (refer to Section 17.15.3, "Operation of Activity PSAS").

```
END
```

The END command terminates PSAS command input. Unless it is otherwise suppressed (refer to Section 17.15.3, "Operation of Activity PSAS"), the Response File constructed by activity PSAS is automatically executed.

```
ABORT
```

The ABORT command terminates activity PSAS and bypasses the automatic execution of the Response File which it has constructed.

```
HELP
```

This form of the HELP command displays the list of PSAS commands at the dialog output device (refer to Section 4.4, "Virtual Output Devices").

```
HELP,command
```

This form of the HELP command displays the syntax format of the specified PSAS command at the dialog output device (refer to Section 4.4, "Virtual Output Devices").

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17.15.3. Operation of Activity PSAS

The user specifies either:

- The name of the appropriate PSAS Command File (refer to Section 2.4.1, "File Usage "). If the specified file does not exist or some other file system related error occurs, an appropriate error message is printed and the request for the input device is repeated.
- Data input from the user's terminal. This applies to both interactive and Response File operation of PSS®E.

The user specifies an output filename. Refer to Section 2.4.1, "File Usage " for the file specification conventions used by activity PSAS. If a file system error occurs while opening the specified file, an appropriate error message is printed. If no filename is specified, activity PSAS writes its output into a file for which the name is of the form PSASnnn.IDV, where < nnn > is a number which is set such that the resulting filename is a new file in the user's directory.

When terminal input had been specified, activity PSAS issues the prompt PSAS: each time it is ready to accept a new PSAS command. PSAS command input is terminated by entering either the END or ABORT command.

When input is taken from a PSAS Command File, no prompts are issued. If the input file is not terminated with either the END or ABORT command, the following message is printed:

```
OUT OF FILE DATA--SWITCH TO TERMINAL INPUT MODE
```

Additional PSAS command input records may then be entered from the user's terminal.

If an error occurs in processing an input record, an appropriate error message is printed, the offending record is ignored, and processing continues.

Upon entry of either the END or ABORT command, activity PSAS is terminated and, except as described below, an @INPUT command is automatically issued to execute the Response File constructed by activity PSAS. If PSS®E is already being operated in Response File mode (i.e., activity PSAS was initiated from a Response File), any remaining commands contained in the original Response File are executed following the execution of the Response File created by activity PSAS.

The execution of the PSAS Response File is bypassed if any one of the following has occurred:

1. Command line only: Activity PSAS was invoked with the suffix CHECK.
2. The PSAS command CHECK was entered during PSAS command input.
3. Command line only: PSAS command input was terminated with the command ABORT rather than END.
4. *Any* errors were encountered during the processing of PSAS command input. In this case, if PSS®E is being operated in its Response File mode, all active Response Files are closed and further user inputs to PSS®E are taken from the user's terminal.

When automatic Response File execution is bypassed, activity PSAS prints the message:

```
YOUR RESPONSE FILE IS filename
```

where < filename > is the name of the PSS®E Response File corresponding to the run described by the PSAS commands which were specified.

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While activity PSAS may be used with an ECHO file enable, care must be exercised if the automatic execution of the Response File built by activity PSAS occurs. In this case, the ECHO file will contain the commands used to execute activity PSAS as well as the Response File commands generated by activity PSAS. That is, if subsequently executed as a PSS®E Response File, the ECHO file will perform the sequence defined by the PSAS commands twice.

17.15.4. Application Notes

When using the SET command to assign values to VARs, CONs and/or ICONs, the corresponding Response File dialog is not generated until either:

1. A START, RUN, HOLD, PASS, USE, EXECUTE, END, or ABORT command, or one of their synonyms, is entered; or

2. Two hundred data items have been specified in SET commands.

This minimizes the number of entries into activity ALTR.

In a similar manner, the Response File dialog for PLACE commands is not generated until either:

1. A NEXT, START, RUN, HOLD, PASS, USE, EXECUTE, END, or ABORT command, or one of their synonyms, is entered; or

2. Two hundred simulation variables have been specified in PLACE commands.

This minimizes the number of entries into activity CHAN.

The generation of the Response File entries corresponding to previously specified PLACE commands cancels the parameters established by any previous NEXT command.

Additional Information
PSS®E GUI Users Guide, PSAS Command File, Dynamic Simulation *PSS®E Command Line Interface (CLI) Users Guide*, Building a Response File, Dynamic Simulation

17.16. Launching an IPLAN Program File

EXEC

Run an IPLAN Program - GUI
I/O Control>Run program Automation file... *[Select Program Automation File to Run]* IPLAN File (*.irf)
Run Line Mode Activity EXEC - CLI
ACTIVITY? >>**EXEC,filename** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR

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Run Line Mode Activity EXEC - CLI
TERMINAL): >>
Python Command
psspy.runiplanfile <filename>
Interrupt Control Codes
None

The programmable PSS®E execution activity EXEC retrieves an IPLAN executable program and starts its execution. The IPLAN program must have been compiled by the IPLAN compiler from its source form into an IPLAN Executable Program File, and the name of the IPLAN Executable Program File must be specified.

Activity EXEC itself is not sensitive to any interrupt control code options. However, PSS®E activities executed under control of an IPLAN program respond to interrupt control codes in their normal fashion. In addition, the IP interrupt control code may be used to terminate execution of an active IPLAN program.

17.16.1. IPLAN Programs

IPLAN is high-level programming language designed to be utilized as an enhancement to existing application programs such as PSS®E, but which can also be used as a stand-alone product. IPLAN is a Siemens PTI proprietary programming language that requires compilation in order to be executed in PSS®E. The compiled IPLAN program in binary format is used as input to the IPLAN simulator which is built into PSS®E. Programs written in IPLAN are created by the user with a standard text editor and compiled using the IPLAN compiler program supplied with PSS®E.

Compiled IPLAN Files can be used in PSS®E in any of the following ways:

- from *I/O Control >Run Program Automation File...* (see Section 17.3, “Automation Methods in PSSE”)
- via line mode activity EXEC (see Section 17.11, “Line Mode Commands”)
- via the EXECUTE command in PSAS and PSEB (see Section 17.14.1, “PSS®E PSEB Macro Language” and Section 17.15.1, “ PSS®E PSAS Macro Language”)
- via the runiplanfile function in the Python extension module psspy (see Section 17.4.1, “PSS®E Extension Modules For Python”)

For specific detailed information and instructions on how to use the IPLAN language, please refer to the *IPLAN Program Manual*.

17.16.2. Interaction With PSS®E

The IPLAN compiler and simulator recognize the Single Element Data Retrieval routines from the PSS®E API. They are used in IPLAN CALL statements. This means that IPLAN programs can assign various values from the PSS®E network case to variables and act on them.

IPLAN cannot modify PSS®E directly, or call other routines in the PSS®E API directly. However, via the PUSH (and PUSHX, QPUSH, QPUSHX) command(s) it can do anything that can be done in a Response File. That includes, say, @input, which will open a Response File; in that case, the commands in that response file will not take effect immediately unless other steps are taken. This is explained in detail in the *IPLAN Program Manual*.

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One of the commands in line mode is EXEC, which is used to load an IPLAN program. If an IPLAN program invokes EXEC through a PUSH statement or by causing a Response File containing EXEC to run, that program will replace the current one in the IPLAN simulator. IPLAN programs do not nest.

17.16.3. The IPLAN Stand-Alone Simulator

The IPLAN compiler can call the IPLAN simulator directly. When used in this mode PUSH commands are just displayed at the terminal, and fetch routines return default values (non-zero return codes where appropriate, else zeros and blanks).

17.16.4. Operation of Activity EXEC

If no file is specified, or if a file system error condition is encountered in opening the designated file, an error message is printed and activity EXEC is terminated.

Activity EXEC first searches the current directory for the IPLAN program file. If it is not found there, activity EXEC follows the directory search sequence described in Section 2.1, "Directories and Files Overview" and looks to the user's home directory and the PSS®E master directory PROGRAMS for the file. If the file is not found, an appropriate message is printed and activity EXEC is terminated. Refer to Section 2.4.1, "File Usage " for the file specification conventions used in specifying filenames to activity EXEC.

If the IPLAN program in the above example requires run-time arguments, it may be initiated with either of the activity commands:

```
exec,job1, argument_list
exec,job1, -arg
```

When the token -arg is specified, activity EXEC instructs the user to specify the argument list which is to be passed to the IPLAN program.

After the IPLAN program is loaded into PSS®E's working memory, activity EXEC is terminated and execution of PSS ®E continues. PSS®E's next request for input (i.e., for the user's specification of the next function or activity to be performed) triggers the execution of the IPLAN program. When the PSS®E GUI is being used, a read from the dialog input device is forced after the IPLAN program is loaded. While an IPLAN program is controlling the execution of PSS®E, input to PSS ®E is normally taken from the IPLAN program rather than from the PSS®E dialog input device. On the other hand, input to the IPLAN program to satisfy any IPLAN INPUT and INPUTLN statements is always taken from the dialog input device. When the PSS®E GUI is being used, a General Input Window is created containing whatever prompts are awaiting display and taking the request input.

PSS®E continues to accept input from the IPLAN program until either an IPLAN PAUSE, PAUSE UNTIL, or PAUSE READ statement is executed; or the IPLAN program is terminated. At this time, PSS®E looks to the dialog input device for its input, or control returns to the GUI.

The input conventions described above apply regardless of whether interactive operation is active, the execution of the IPLAN program is initiated from a Response File, or the IPLAN program PUSHes an @INPUT or @CHAIN command or an IDEV,filename activity command. *IPLAN Program Manual* gives some examples describing the interaction of Response Files and IPLAN programs when both are active simultaneously.

While the redirection of input to PSS®E provided by activity EXEC is similar in concept to the use of Response Files, an IPLAN program is much more flexible than a Response File; it may receive information from PSS®E and make decisions based upon PSS®E execution results.

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For details on the IPLAN programming language, the IPLAN language interface to the PSS®E working case and dynamics data, and the IPLAN compiler, refer to the *IPLAN Program Manual*.

Additional Information
PSS®E GUI Users Guide, IPLAN Program File *PSS®E Command Line Interface (CLI) Users Guide*, Launching an IPLAN Program File

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Chapter 18
Result Retrieval

18.1. Python Modules for Result Retrieval

PSS®E result retrieval APIs (Chapter 9, *PSS®E Application Program Interface (API)*) are used to retrieve PSS®E activity results into Python lists. Such lists then can be used to make customized reports or for further processing using Python scripts. This chapter describes how to invoke these APIs from PSS®E Python automation files or the PSS®E CLI (command line interface), and write simple Python scripts using these APIs to further process the results. The Python modules available for such a use are:

- PSSARRAYS module returns data and results from PSS®E activities to Python lists.
- PSSEXCEL module creates and populates Excel spreadsheets with data and results from PSS®E activities. This module is an example of how to use Python lists returned by PSSARRAYS for further processing. This modules uses the PSSARRAYS module to retrieve PSS®E activity data and results and the EXCELPY module to write the data and results to Excel spreadsheets.
- EXCELPY is an auxiliary module that provides Python functions to create and populate Excel spreadsheets.

Following sections describe just a few functions and give sample Python commands/code to illustrate how to use Python functions available in these modules. These Python commands can be executed from either PSS®E program automation files or the PSS®E CLI.

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18.2. pssarrays.accc_summary

18.2.1. CLI

```
import pssarrays (1)

smry = pssarrays.accc_summary(r"savnw.acc") (2)
or
smry = pssarrays.accc_summary(accfile=r"savnw.acc")
or
smry = pssarrays.accc_summary(accfile=r"c:\filepath\savnw.acc")

print smry.ierr (3)
0

print smry.acccsize.nmline
11

print smry.file.acc
C:\Program Files\PTI\PSSE31 Alpha 2\EXAMPLE\savnw.acc

print smry.melement
('   201 HYDRO       500.00    151 NUCPANT     500.00 1', '   202
EAST500     500.00    152 MID500      500.00 1', '   203 EAST230
230.00    154 DOWNTN      230.00 1', '   205 SUB230      230.00    15
DOWNTN      230.00 1', '  3001 MINE        230.00   3002 E. MINE
500.00 1', '  3004 WEST        500.00    152 MID500      500.00 1', '
3004 WEST        500.00   3005 WEST        230.00 1', '  3005 WEST
230.00   3008 CATDOG      230.00 1', '  3006 UPTOWN      230.00    15
MID230      230.00 1', '  3008 CATDOG      230.00    154 DOWNTN
230.00 1', '  3008 CATDOG      230.00   3018 CATDOG_G    13.800 1', '
INTERFACE WEST', '       INTERFACE EAST')

print smry.mvbuslabel
('  3001 MINE        230.00', '   201 HYDRO       500.00', '   202
EAST500     500.00', '   203 EAST230     230.00', '   204 SUB500
500.00', '   205 SUB230      230.00', '   206 URBGEN      18.000', '
211 HYDRO_G     20.000', '  3001 MINE        230.00', '  3002 E. MINE
500.00', '  3003 S. MINE     230.00', '  3004 WEST        500.00', '
3005 WEST        230.00', '  3006 UPTOWN      230.00', '  3007 RURAL
230.00', '  3008 CATDOG      230.00', '  3011 MINE_G      13.800', '
3018 CATDOG_G    13.800')
```

(1) By importing the pssarrays module, the functions defined inside this module are made available outside of the module.

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2 The accc_summary function can be called in any of the following ways:

- Contingency Solution Output file is available in the working folder.

```
smry = pssarrays.accc_summary(r"savnw.acc")
```

- Contingency Solution Output file is available in the working folder, accc_summary function called with keyword argument `accfile`.

```
smry = pssarrays.accc_summary(accfile=r"savnw.acc")
```

- Contingency Solution Output file is not available in the working folder, then its location needs to be specified.

```
smry = pssarrays.accc_summary(accfile=r"c:\filepath\savnw.acc")
```

Note: The accfile name is specified in r"..." format, to make the filename a Python raw string. It suppresses escapes sequences, viz., \n for newline character, \t for tab etc.

3 ACCC summary results are returned in the Python list smry. The various values available in this list are accessed by their names as an attribute of the returned list. The API documentation provide the list of values available in the returned list.

The following Python commands print some of these values.

- Value ierr gives the error condition.

```
print(smry.ierr)
```

- Value acccsize.nmline gives the number of monitored elements (branches + interfaces).

```
print(smry.acccsize.nmline)
```

- Value file.acc gives the .acc input filename.

```
print(smry.file.acc)
```

- Value smry.melement gives the monitored element labels.

```
print(smry.melement)
```

- Value smry.mvbuslabel gives the monitored bus labels.

```
print(smry.mvbuslabel)
```

18.2.2. Automation File

The examples above can be combined into one Python file, as shown below, and run as a PSS®E automation file to report on the basic data contained within an ACCC output file. Any Python code can be added to this code to further process the returned values.

```
import pssarrays
```


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```
smry = pssarrays.accc_summary(r"savnw.acc")
print(smry.ierr)
print(smry.acccsize.nmline)
print(smry.file.acc)
print(smry.melement)
print(smry.mvbuslabel)
```

18.3. pssarrays.accc_solution

The following Python code example shows how to get ACCC analysis results and write them to a report.

```
import pssarrays
reptfile = 'accc_output.txt'
reptnam = open(reptfile,'w')


smry = pssarrays.accc_summary(accfile=r'savnw.acc')
rate = smry.rating.a
stype = 'con'


for lbl in smry.colabel:
    soln  = pssarrays.accc_solution(accfile=r'savnw.acc',
            colabel=lbl,stype=stype,busmsm=0.5,sysmsm=5.0)
    if  soln == POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal:
        continue # contingency solution not found,
                                # move to next
    if soln.ierr !=0: continue # return any non-zero ierr
    reptnam.write("CONTINGENCY EVENTS\n")


    # (1) contingency events summary
    cnvflag  =  soln.cnvflag
    cnvcond  =  soln.cnvcond
    island   = ("%d" %  soln.island).center(7)
    mvaworst =  soln.mvaworst
    mvatotal =  soln.mvatotal
    for jj in range(len( soln.codesc)):
        desc =  soln.codesc[jj]
        if jj == 0:
            reptnam.write("%(lbl)s %(desc)s %(cnvflag)s \
    %(cnvcond)s %(island)s %(mvaworst)11.4f\
    %(mvatotal)11.4f\n" % vars())
        else:
            tmp = 12*' '
            reptnam.write("%(tmp)s %(desc)s\n" % vars())


    if not cnvflag:
        reptnam.write("\n================================\n")
        continue    # consider solution for converged cases only
```


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```
    # (2) flows
    txtstr = ''
    for jj in range(smry.acccsize.nmline+smry.acccsize.ninter):
        mvaflow = "%9.2f" %  soln.mvaflow[jj]
        if jj < smry.acccsize.nmline:
            ampflow = "%9.2f" %  soln.ampflow[jj]
            pctflow = abs( soln.ampflow[jj])
        else:
            ampflow = 9*' '
            pctflow = abs( soln.mvaflow[jj])
        if rate[jj]:
            pctflow = "%6.2f" % (pctflow*100.0/rate[jj])
        else:
            pctflow = 6*' '


    txtstr += " %(mvaflow)s %(ampflow)s %(pctflow)s " %vars()
    if txtstr:
        reptnam.write("\nMONITORED ELEMENT FLOWS\n")
        reptnam.write("%(lbl)-12s%(txtstr)s\n" % vars())


    # (3) voltages
    if smry.acccsize.nmvbus:
        reptnam.write("\nMONITORED BUS VOLTAGES\n")
        txtstr = ''
        for eachv in  soln.volts:
            txtstr += " %8.5f" % eachv
        if txtstr:
            reptnam.write("%(lbl)-12s%(txtstr)s\n" % vars())


    # (4) loads shed
    if len( soln.lshedbus):
        reptnam.write("\nLOADS SHED\n")
        for jj in range(len(soln.lshedbus)):
            if stype in ['caction','corrective action','cor']:
                reptnam.write("%25s %10.2f %10.2f %12s\n" % \
                    (soln.lshedbus[jj], soln.loadshed[0][jj],
                     soln.loadshed[1][jj], lbl))
            else:
                reptnam.write("%25s %10.2f %12s\n" % \
                    (soln.lshedbus[jj], soln.loadshed[jj], lbl))


    # (5) generator dispatch
    if stype in ['caction','corrective action','cor'] and \
       len( soln.gdispbus):
        reptnam.write("\nGENERATOR DISPATCH\n")
        for jj in range(len( soln.gdispbus)):
            reptnam.write("%25s %11.2f %11.2f %12s\n" % \
                (soln.gdispbus[jj],soln.gendisp[0][jj],soln.gendisp[1][jj],lbl))
```


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```
    # (6) phase shifter
    if stype in ['caction','corrective action','cor'] and \
       len( soln.phsftr):
        reptnam.write("\nPHASE SHIFTER ANGLE\n")
        for jj in range(len( soln.phsftr)):
            reptnam.write("%54s %12.2f %11.2f %12s\n" % \
                    (soln.phsftr[jj], soln.phsftrang[0][jj],
                     soln.phsftrang[1][jj],lbl))


    reptnam.write("\n================================\n")


reptnam.close()
print("ACCC Solution saved in file %s." % reptfile)
```

18.4. pssarrays.accc_violations_report

This function creates the following reports:

- monitored element flow violations (percent flow greater than flow limit). The flow limit is specified as an argument of this function.
- monitored bus voltage violations (bus voltage <minimum voltage bound or >maximum voltage bound). The voltage bound values are taken from monitored element data file (.mon).
- loads shed
- generator dispatch
- phase shifter angle adjustments

The following command, executed from either line mode or a Python automation file, produces ACCC violations reports for corrective action solution and reports on monitored branches where loading is above 95%.

```
ierr = pssarrays.accc_violations_report(accfile=r"savnw.acc",
          stype="cor", busmsm=0.5,sysmsm=5,rating='a',
          flowlimit=95,rptfile=r"savnw_violations.txt")
```

18.5. pssexcel.accc

Using the pssarrays module, ACCC analysis results are retrieved from an *.acc input file and then, using the excelpy module, these results are written to Excel Spreadsheets.

The Python commands below create a 'savnw_acc.xls' file and the following worksheets populated with ACCC Corrective Action solution data from the "savnw.acc" file for contingencies 'base case' and 'trip1nuclear'. If a `colabel` value is not provided, all contingencies are considered. Also the extended bus names of monitored branches and buses are split into three Excel spread columns (`namesplit=True`). The argument `show=True` displays the workbook when it is being populated.

- test Summary

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- test Contingency Events
- test Branch Flow
- test Interface Flow
- test Bus Voltage
- Test Load Shed
- test Generator Dispatch
- test Phase Shifter Angle

```
import pssexcel
pssexcel.accc(accfile   = r"savnw.acc",
              string    = ['b','e','g','i','l','p','s','v'],
              colabel   = ['base case','trip1nuclear'],
              stype     = 'con',
              busmsm    = 0.5,
              sysmsm    = 5.0,
              rating    = 'a',
              namesplit = True,
              xlsfile   = r'savnw_acc.xls',
              sheet     = 'test',
              overwritesheet = True,
              show      = True,
              ratecon   = 'b',
              baseflowvio = True,
              basevoltvio = True,
              flowlimit   = 100.0,
              flowchange  = 0.0,
              voltchange  = 0.0,
              swdrating    = 'a',
              swdratecon   = 'b',
              baseswdflowvio = False,
              baseswdvoltvio = False,
              overloadreport = False
              )
```

In the command above, `accfile` and `string` are the only mandatory arguments. The worksheets to create are defined by the `string` argument.

18.6. pssexcel.pv

Using the pssarrays module, PV solution results are retrieved from a *.pv input file and then, using the excelpy module, these results are written to Excel Spreadsheets.

The Python commands below create a 'savnw_pv.xls' file and the following worksheets populated with PV solution results from the "savnw.pv" file for contingencies 'base case' and 'trip1nuclear'. If a `colabel` value is not provided, all contingencies are considered. Also the extended bus names of monitored branches and

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buses are split into three Excel spread columns (`namesplit=True`). The argument `show=True` displays the workbook when it is being populated.

- testpv Summary
- testpv Bus Voltage
- testpv Mismatch
- testpv Generator Dispatch
- testpv Bus Load
- testpv Branch Flow
- testpv Interface Flow

```
import pssexcel
pssexcel.pv(pvfile    = r"savnw.pv",
            string    = ['s','m','v','g','l','b','i'],
            colabel   = ['base case', 'trip1nuclear'],
            namesplit = True,
            xlsfile   = r'savnw_pv.xls',
            sheet     = 'testpv',
            overwritesheet = True,
            show      = True
            )
```

In the command above, `pvfile` and `string` are the only mandatory arguments. The worksheets to create are defined by the `string` argument.

18.7. excelpy Examples

18.7.1. Export QV Solution to Excel Spreadsheet

The following Python code writes QV solution summary and bus voltage results from "savnw.qv" file to the Excel spreadsheet 'testqv.xls'.

- Using the pssexcel.qv function

```
import pssexcel
pssexcel.qv(qvfile    = r"savnw.qv",
            string    = ['s','v'],
            colabel   = ['base case', 'trip1nuclear'],
            namesplit = False,
            xlsfile   = r'testqv.xls',
            sheet     = '',
            overwritesheet = True,
            show      = True
            )
```

- Using the pssarrays and excelpy modules

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The following example illustrates how the `pssarrays` and `excelpy` modules can be used to export PSS®E activity data and results to Excel spreadsheets.

```
import excelpy
import pssarrays


qvfile  = r"savnw.qv"
colabel = ['base case', 'trip1nuclear']
xlsfile = 'testqv.xls'


# create Excel workbook, with sheet 'qvsummary'
qvxls = excelpy.workbook(xlsfile=xlsfile,sheet='qvsummary',
                         overwritesheet=True)
qvxls.show()        # show workbook
qvxls.font_sheet()  # set default font for'qvsummary' worksheet


# add other sheets as required
qvxls.worksheet_add_end(sheet='qvvolts',overwritesheet=True)
qvxls.font_sheet()  # set default font for ''qvvolts' worksheet


# Get Summary and validate contingency labels provided
smry = pssarrays.qv_summary(qvfile)


# populate qvsummary worksheet
qvxls.set_active_sheet('qvsummary')
row, col = 1, 1
qvxls.set_cell((row,col),"QV SOLUTION RESULTS",fontStyle="bold",
               fontSize=14, fontColor="blue")


tmplst=[
    smry.casetitle.line1,
    smry.casetitle.line2,
    'QV output file  = %s' % smry.file.qv,
    'Saved Case file = %s' % smry.file.sav,
    'DFAX file       = %s' % smry.file.dfx,
    'Subsystem file  = %s' % smry.file.sub,
    'Monitored Element file  = %s' % smry.file.mon,
    'Contingency Description file = %s' % smry.file.con,
    ' ',                                              # blank row
    'Number of Contingencies+Base Case = %d' % smry.qvsize.ncase,
    'Number of Monitored Generators(Plants)= %d' % smry.qvsize.nmgnbus,
    'Number of Voltage Monitored Buses = %d' % smry.qvsize.nmvbus,
    'Number of Voltage Monitored Records = %d' % smry.qvsize.nmvrec,
    'Number of max.volt. setpoint changes = %d' % smry.qvsize.nmxvstp,
    ]


if smry.file.thr:
```


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```
    tmplst.insert(4,'Load throwover file = %s' % smry.file.thr)


row += 2
bottomRow,rightCol = qvxls.set_range(row,col,tmplst,transpose=True)
qvxls.font_color((row,col,row+1,col),'brown')


row = bottomRow+2
qvxls.set_cell((row,col),"QV Contingencies",fontStyle="Bold",
               fontSize=12, fontColor="red")


conlst = [['CON#', 'LABEL', 'Min Vstp', 'Max Vstp', 'Min MVAR',
           'Max MVAR', 'Max Mismatch', 'DESCRIPTION']]
for i in range(smry.qvsize.ncase):
    if i==0:
        srnum = ' '
    else:
        srnum = str(i)
    nam  = smry.colabel[i]
    minvstp = smry.minvstp[i]
    maxvstp = smry.maxvstp[i]
    minmvar = smry.minmvar[i]
    maxmvar = smry.maxmvar[i]
    maxmsm  = smry.maxmsm[i]
    for j in range(len(smry.codesc[i])):
        dsc = smry.codesc[i][j]
        if j==0:
            conlst.append([srnum,nam,minvstp,maxvstp,minmvar,
                           maxmvar,maxmsm,dsc])
        else:
            conlst.append(['' ,'' ,'' ,'' ,'' ,'' ,'' ,dsc])


row += 1
bottomRow,rightCol = qvxls.set_range(row,col,conlst)
qvxls.font_color((row,col,row,rightCol),"dgreen")
qvxls.font((row,col+2,bottomRow,col+3),numberFormat='0.00')
qvxls.font((row,col+4,bottomRow,col+6),numberFormat='0.000')
qvxls.align((row,col),'right')
qvxls.font((row,col,row,rightCol),fontStyle=('Bold',))
qvxls.autofit_columns((row,col+1,row,rightCol)) # done qvsummary worksheet


# populate qvvolts worksheet
mvbuslabel = list(smry.mvbuslabel)
mvbuslabel.insert(0,'VOLTAGE SETPOINT->')


qvxls.set_active_sheet('qvvolts')
row, col = 1, 1
```


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```
for lbl in colabel:
    soln =  pssarrays.qv_solution(qvfile,lbl)
    if soln==POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal: continue
    #     contingency solution not found
    if soln.ierr !=0: continue # solution with error


    contitle = 'CONTINGENCY: ' + lbl.strip()


    # assemble data in columns: 1st=MW Transfer, rest=bus voltages
    tmplst    = [ mvbuslabel ]
    for i in range(len(soln.vsetpoint)):
        t = list(soln.volts[i])
        t.insert(0,soln.vsetpoint[i])
        tmplst.append(t)


    col = 1
    qvxls.set_cell((row,col+1),contitle,fontStyle='bold',
                   fontSize=12,fontColor="dgreen")
    row += 1
    bottomRow,rightCol = qvxls.set_range(row,col,tmplst,transpose=True)
    qvxls.font((row,col,row,rightCol),fontColor="red",fontStyle='bold')
    qvxls.font((row,col+1,bottomRow,rightCol),numberFormat="0.000")
    qvxls.align((row,col,row,rightCol),'h_center')
    qvxls.align((row,col),'right')
    qvxls.font((row+1,col,bottomRow,col),fontColor="blue",
               fontStyle='bold')
    row = bottomRow + 2 # one blank row


# format qvvolts worksheet qvxls.autofit_columns((1,1))


# Save the workbook and close the Excel
application
xlsfile = qvxls.save(xlsfile)
print("\n Excel workbook saved to file %(xlsfile)s.\n" % vars())
```

18.7.2. Write Data to Excel Spreadsheet

Following example code shows how to use excelpy functions and methods to create, populate and format Excel workbook from Python.

```
import excelpy


testxls = excelpy.workbook(r"test.xls","TestExcelPy")
testxls.show()
testxls.font_sheet()


testdata=[
```


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```
    ['CON#', 'LABEL',      'Max MW', 'DESCRIPTION'],
    [' ',    'BASE CASE',    756.25, 'BASE CASE    '],
    ['1',    'TRIP1NUCLEAR', 1000.0, 'REMOVE UNIT 1'],
    ['2',    'TRIP2NUCLEAR',    0.0, 'REMOVE UNIT 1'],
    ['',                 '',     '', 'REMOVE UNIT 2'],
    ['3',    'ADDLARGELOAD',    0.0, 'INCREASE LOAD'],
    ['4',    'LOSEWESTGEN',   625.0, 'REMOVE UNIT 3'],
    ['5',    'LOSEWESTBIGT',  312.5, 'TRIP LINE 1  '],
    ]


row=1
col=1
testxls.set_cell(value="How to use EXCELPY module?",
     address=(row,col),fontName="Arial",fontSize=12,
     fontColor='red',fontStyle=('Bold','italic'))


row = row+2 #add one blank row
bottomRow, rightCol = testxls.set_range(row,col,testdata,transpose=False)
testxls.font((row,1,row,rightCol),fontStyle='bold',fontColor='blue')
testxls.align((row,1,row,rightCol),alignv='h_center')
testxls.align((row,1,bottomRow,2),alignv='h_center')
testxls.autofit_columns((row,2,row,rightCol))
testxls.font((row,3,bottomRow,3),numberFormat='0.000')
xlsfile = testxls.save(xlsfile)
print("Excel workbook saved to '%s' file.\n"% xlsfile)
```


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Chapter 19
Implementing the Powerflow Customization Interface (PCI)

The PCI is implemented by creating a Python module. This module must have the name pssuserpf. See Section 16.4.5, Finding user-created modules about where this file may be located and found by PSS®E.

19.1. Contents of pssuserpf

Normally you will create the pssuserpf module by creating a file named pssuserpf.py. The following general comments apply:

- The function names shown in the sections below are specific; only these exact names will be called via the PCI.
- It is not necessary to define all of these functions.
- Those that you do define must specify exactly the number of arguments shown above. The names of the arguments are not important, but those names will be used in other parts of the documentation.
- Functions described below that return a value must return a value of the type shown. Values should not be returned by functions described below that do not return values.
- You may define other functions than these, if you wish, along with other constants and variables.

In the following, the pssuserpf functions are shown in their simplest form, i.e. taking no action, other than returning the default value if needed.

19.1.1. on_case

```
def on_case(casename):
    pass
```

It is called for CASE, just before the saved case file is closed.

The argument contains the filename of the saved case file, in the form that it was given, e.g. if, in python, you entered psspy.case('myfile.sav') then casename would contain 'myfile.sav'; if you entered psspy.case(r'd:\stuff\afile') then casename will contain r'd:\stuff\afile'. When running interactively through the PSSE GUI, full file paths are supplied.

It does not return a value.

19.1.2. pre_accc

```
def pre_accc(opts,sub,mon,con):
    pass
```


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It is called for ACCC, >ACCC_WITH_DSP, and MACCC after edit checking of arguments and before normal processing starts. Before this routine is called, the Distribution Factor File is opened to retrieve the input filenames.

The arguments are a list containing the first 7 options (OPTACC), the Subsystem Description filename, the Monitored Element Description filename, and the Contingency Description filename. Note that the filenames are in the form that they were entered, e.g. if, in python, you entered 'myfile.con' for the Contingency Description filename, then that will be the value of the argument 'con'. When running interactively through the PSSE GUI, full file paths are supplied.

It does not return a value.

19.1.3. post_accc

```
def post_accc():
    pass
```

It is called immediately before the termination of the activity.

There are no arguments.

It does not return a value.

19.1.4. pre_solution

```
def pre_solution():
    pass
```

It is called for FDNS, FNSL, RUNLASTSOLUTION (if the last solution was fdns or fnsl), and GIC (if the option to Solve Power Flow is set to 'fdns' or fnsl') after edit checking of arguments and before normal processing starts. If the Connectivity Check option is on, the connectivity check is performed before this call.

There are no arguments.

It does not return a value.

19.1.5. post_solution

```
def post_solution(errcode,solved):
    pass
```

It is called for FDNS, FNSL, RUNLASTSOLUTION (if the last solution was fdns or fnsl), and GIC (if the option to Solve Power Flow is set to 'fdns' or fnsl') after the solution is completed (but before interrupt options are cleared).

The arguments contain the values of the return code FDNS or FNSL and the value of the convergence flag at that time.

It does not return a value.

19.1.6. pre_ca_solution

```
def pre_ca_solution(conlabel):
```


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```
    pass
```

It is called for ACCC, ACCC_WITH_DSP, and MACCC after the contingency has been applied but before any processing of that contingency has begun.

The argument contains the label of the current contingency.

It does not return a value.

19.1.7. post_ca_soln_eval

```
def post_ca_soln_eval(conlabel,solved):
    return True
```

It is called for ACCC, ACCC_WITH_DSP, and MACCC after each solution for the contingency has been completed but before the restoration of the base case, provided that the limit for the number of contingency iterations has not been reached. There is a maximum number of 20 contingency iterations allowed per contingency.

The arguments contain the label of the current contingency and the value of the convergence flag at that time.

It returns a logical value. Return False to request that post_ca_soln_adj be called to make changes to the contingency case, and then rerun the contingency with those changes. Return True to continue with normal termination of the current contingency.

For MACCC implementation is not complete in this release. The call to post_ca_soln_eval will take place, but the contingency will not be repeated.

19.1.8. post_ca_soln_adj

```
def post_ca_soln_adj(conlabel,solved):
    pass
```

It is called for ACCC, ACCC_WITH_DSP, and MACCC if post_ca_soln_eval returns False.

The arguments contain the label of the current contingency and the value of the convergence flag at that time.

It does not return a value.

19.1.9. pre_iteration

```
def pre_iteration():
    pass
```

It is called during initialization of a solution iteration for FDNS, FNSL, and any of the other functions that utilize those solutions (e.g. ACCC,MACCC, GIC, etc.).

There are no arguments.

It does not return a value.

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19.1.10. post_iteration

```
def post_iteration():
    return True
```

It is called during evaluation of the solution results for FDNS, FNSL, and any of the other functions that utilize those solutions (e.g. ACCC, MACCC, GIC, etc.).

There are no arguments.

It returns a logical value. Return False to request another iteration. Return True to allow the solution to determine whether another iteration is needed.

19.2. Importing pssuserpf and Progress information

The pssuserpf module is not loaded until the first time an attempt is made to call one of the functions that it can define.

The pssuserpf module is not required, therefore there is no message if it fails to load. A message is written to the Progress output device if the pssuserpf module is loaded successfully.

The first time each of the functions that can be defined in pssuserpf for the PCI is called, a message is written to the Progress output device.

You can import pssuserpf at the command line in PSS®E. If you suspect an import failure - say, a syntax error in your module - this will cause a traceback to printed to the Progress output device. Importing the module in this way makes it available to the command input environment; it does not make the module available to the PCI. However, even if the module is successfully imported into both the PCI and the command input environment, only one copy of the module will be in memory.

Only one attempt will be made to load the pssuserpf module through the PCI.

Once the pssuserpf module has been successfully loaded, you can use the Python reload command from the PSS®E command line to replace it, subject to the usual restrictions on this ability. See Python's documentation of the reload command.

If you are not sure that the pssuserpf module (or any other python module) that is loaded is the correct one (say you have multiple projects that you are working on and different modules defined for each) you can use the showLoadedModules function in psspy to display all the loaded Python modules and the file that they were loaded from. Enter:

```
psspy.showLoadeModules()
```

from the PSSE command line.

19.3. Use of the psspy module in pssuserpf

You can import psspy and use the functions defined there, with some limitations and considerations.

Certain of the pssuserpf functions are considered "outer" functions, others are "inner" functions. The "outer" functions are: on_case, pre_accc, post_accc, pre_solution, post_solution, . The others are "inner" functions.

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"Outer" functions can use any of the functions defined in the module psspy, except for those that conflict with the current API function in use (e.g. on_case cannot call CASE or READ).

"Inner" functions can only use those functions defined in the module psspy that do not change the state of the program, i.e. those that strictly retrieve data.

19.4. Using other PSSE modules from pssuserpf

19.4.1. pfcontrol

It is critical to the intended purpose of the PCI that certain values are modifiable from the "inner" functions, but the PSS®E API functions in the psspy module to do so are not available to be called.

You can import pfcontrol and use the functions defined there to modify the case during solutions or contingencies. There is an important difference in the impact of changes that are made in the different "inner" functions:

pre_iteration, post_iteration - changes made here are permanent. This is consistent with the expectation that the network case at the end of the solution represents a solved case.

pre_ca_solution, post_ca_soln_adj - changes made here affect the contingency case only. This is consistent with the expectation that the network case at the end of the contingency analysis is unchanged. These changes are reflected in the incremental saved case saved for that contingency.

post_ca_soln_eval - changes cannot be made from this function

Not all the functions in pfcontrol can be used in all of the "inner" functions. The pfcontrol functions cannot be used in the "outer" functions.

19.4.2. ppsppe

The read function in the module ppsppe was designed for reading parameters into a pssuserpf module.

19.4.3. others

In general other modules that are available through the interface to PSS®E that you are using can also be used in pssuserpf. pssarrays will not be able to access the acc file being written in the contingency analysis functions.

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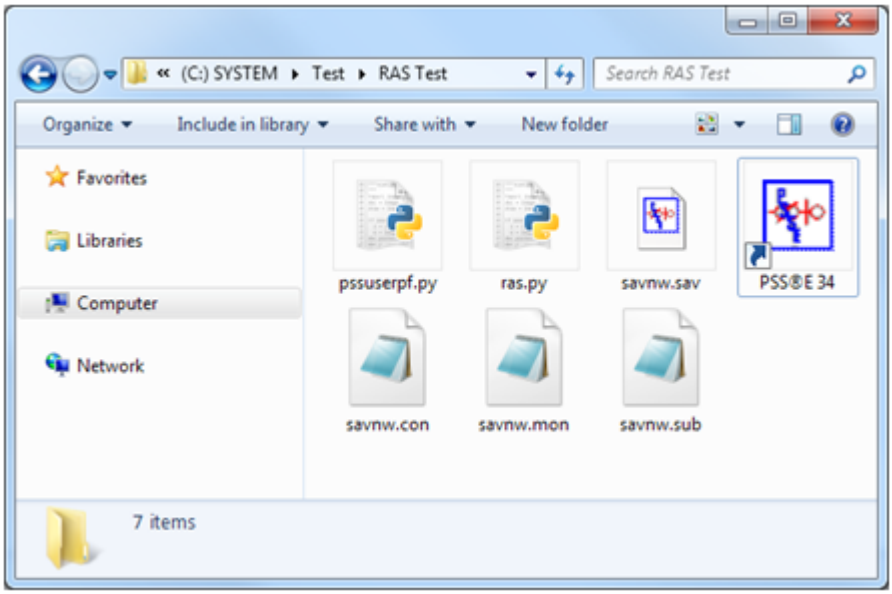
Chapter 20
Using the Remedial Action Scheme API

The Python-based RAS API allows the user to define arbitrarily complex remedial action schemes (also known as system adjustments, operating procedures or mitigation procedures) that can be simulated within the PSS®E ACCC activity. The API is based on the PCI (Powerflow Control Interface) described in the previous section. In order to use the RAS API, a custom RAS definition module and pssuserpf module have to be included in the working directory (or Python Path) as shown below.



The contents of these files are described in the following sections.

20.1. Minimal Contents of the User-defined pssuserpf.py Module for RAS Simulation

The following code shows the minimum content required in the pssuserpf.py file in order to simulate RAS behavior using the PCI-based system. Note that additional code may be added to this file based on other uses of the pssuserpf.py interface (such as custom control models or reports). Note also, that the "import ras" line of code in the below file contents must in general import the actual RAS definition file. For example, if the RAS definitions were included in a module "myras.py", then the import statement should read "import myras".

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pssuserpf.py - C:\Program Files (x86)\PTI\PSSE33\EXAMPLE\RAS\pssuserpf.py (2.7.9)

File Edit Format Run Options Windows Help

```
import pssras
import psspy

# ============================ EDIT HERE ====================================

# import your RAS definitions. The module does not have to be named ras:
import ras

#===========================================================================


def pre_ca_solution(conlabel):
    """This function is called before a contingency is applied in ACCC"""
    pssras.pre_ca_solution(conlabel)


def post_ca_soln_eval(conlabel, solved):
    """This function is called during the contingency solution after a....
    conlabel: The contingency label of the contingency that has been applied
    """

    # call the pssras function that checks all of the user defined conditions:
    solved = pssras.post_ca_soln_eval(conlabel, solved)

    # return the flag from the condition check. False means we need to
    # re-solve:
    return solved


def post_ca_soln_adj(conlabel, solved):
    """This function is called after conditions have been checked in the
    post_ca_soln_eval function. If the return from that function call was
    False, then system adjustments may be required and the system ....
    conlabel: The contingency label of the contingency that has been applied
    """

    # call the ras perform_actions function to implement the triggered actions:
    pssras.post_ca_soln_adj(conlabel, solved)


# you may want to write a message to PSSE in order to announce that this file
# was loaded and therefore RAS definitions are enabled:

psspy.report("RAS functionality enabled.\n\n")
psspy.report(pssras.ras_summaries())
```

Ln: 36 Col: 4

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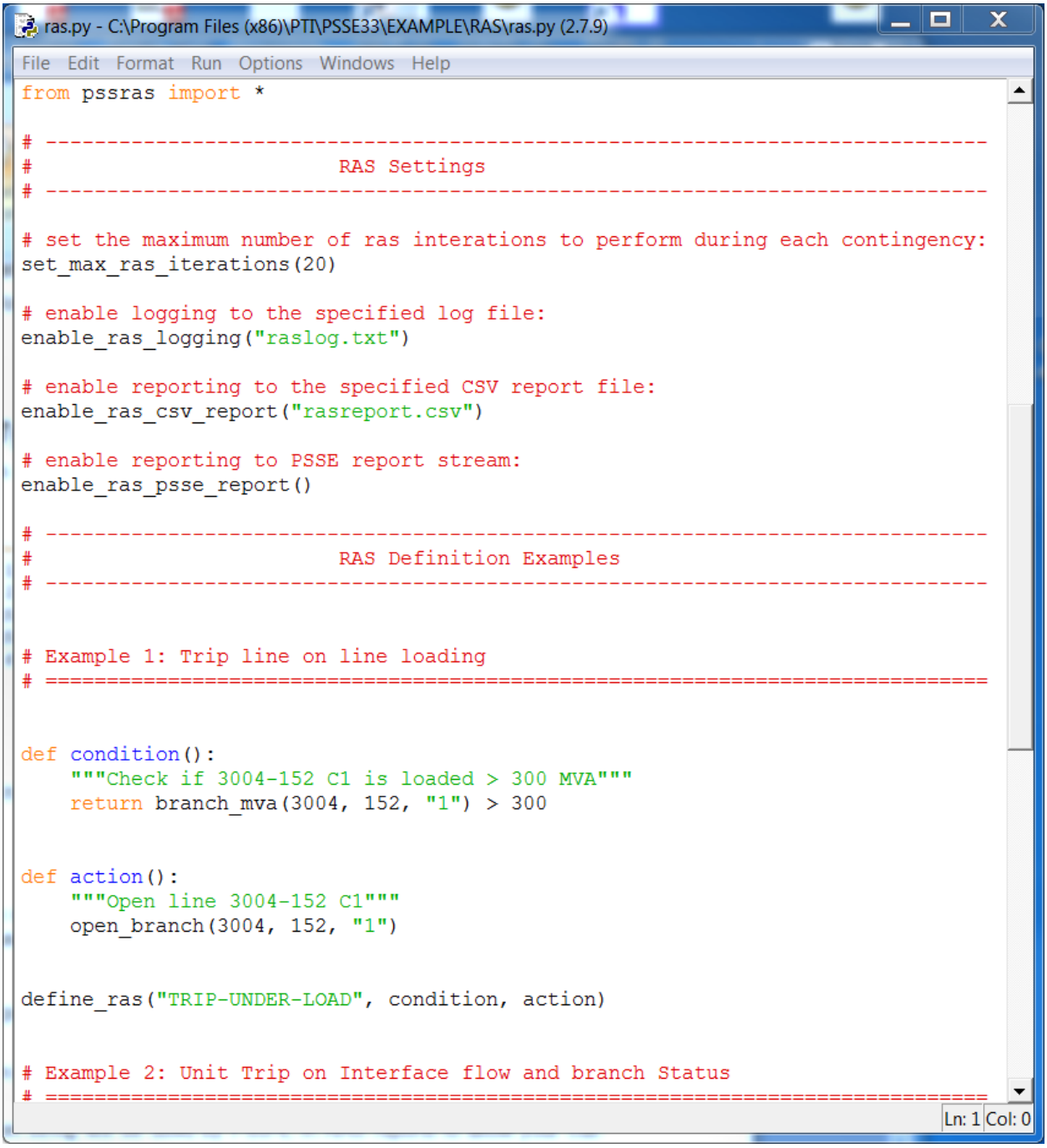
20.2. Contents of the User-defined ras.py File

Below are the contents of an example RAS definition file.



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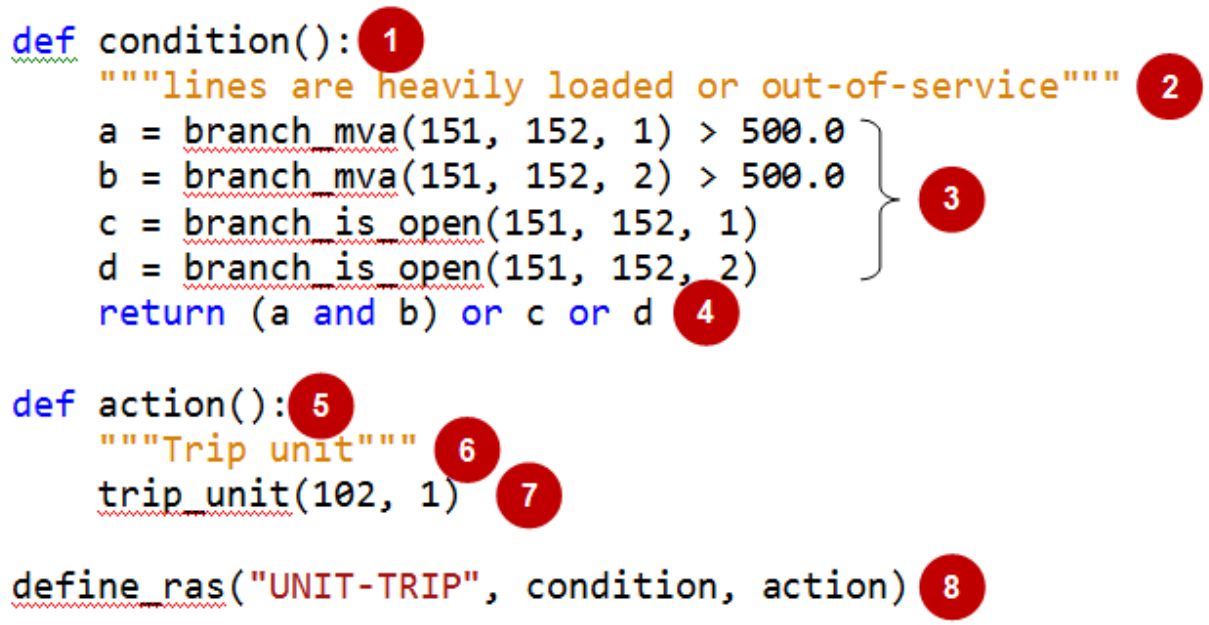
20.3. Explanation of the Typical RAS Definition

The construction of an example RAS is described below, and consists of three major parts: the condition definition, the action definition, and the RAS definition.



1. Define the Condition

   a. The condition function does not have to be called "condition", but whatever name you choose, you need to use the same name in the RAS definition function (see item 8 below).

   b. You cannot include arguments in the condition function definition (the parentheses must be empty).

2. Define the Condition Description (optional)

   a. This optional description string will be used by PSS®E in RAS reports to allow your custom description to be including in the log and report messages.

   b. You can use single quotes, but you must use triple quotes if the description spans multiple lines (based on Python multi-line string rules)

3. Define True/False Conditional Statement(s)

   a. This is where you put statements that result in True or False

   b. You can make logical combinations of True/False expressions by assigning each Boolean expression to a variable (as shown in this example).

4. Return the Result of the Condition Function

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a. You must include a return statement as the last line of the condition function that returns True (RAS triggered) or False (RAS not triggered). For compound expressions, you can return a Boolean expression using a combinations of Boolean variables using "or", "and" and parentheses to define the order of nested operations.

5. Define the Action

   a. The action function does not have to be called "action", but whatever name you choose, you need to use the same name in the RAS definition function (see item 8 below).

   b. You cannot have arguments in the action function definition (the parentheses must be empty).

6. Define the Action Description (optional)

   a. This optional description string will be used by PSS ®E in RAS reports

   b. You can use single quotes, but you must use triple quotes if the description spans multiple lines (based on Python multi-line string rules)

7. Define Action Statements

   a. This is where you put statements that perform adjustments to the system as desired.

   b. You can use functions from PTI-provided pssras, psspy, or pfcontrol modules. Note that you can only make adjustments with the functions in pssras or pfcontrol modules, as data-changing functions in psspy will have unexpected results.

8. Define the RAS

   a. Finally, the RAS definition is created by calling the define_ras function, passing in the desired name of the RAS, along with the names of the condition and action functions defined in the previous steps.

20.4. Python Operators

For your reference, the following table contains a subset of the math, assignment, comparison and logical operators in the Python language. These will be useful is defining conditional expressions based on systems conditions determined in the Condition Function.

Math	+ - * / % **
Assignment	= += -= *= /= %= **=
Comparison	== != >= <= > <
Logical	and or not

20.5. Available RAS Getter Functions

The below functions are useful in assessing system conditions that trigger RAS actions. Note that functions that return float values must be used in combination with comparison operators in order to provide boolean (True/False) results needed to trigger a RAS action.

Function	Argument(s)	Return Value(s)
bus_get_typecode(ibus)	ibus: Bus Number (int)	Bus Typecode (int)

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Function	Argument(s)	Return Value(s)
bus_is_disconnected(ibus)	ibus: Bus Number (int)	True/False
branch_status(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)	True/False
branch_is_open(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)	True/False
branch_is_closed(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)	True/False
breaker_status(subnum, inode, jnode, ckt)	subnum: Sub Number (int) inode: From Node Num (int) jnode: To Node Num (int) ckt: SWD ID (int or string)	True/False
fixed_shunt_status(ibus, shntid)	ibus: Bus Number (int) shntid: Shunt ID (int or string)	True/False
fixed_shunt_mvar(ibus, shntid)	ibus: Bus Number (int) shntid: Shunt ID (int or string)	MVAR (float)
sw_shunt_status(ibus)	ibus: Bus Number (int)	True/False
sw_shunt_qinit(ibus)	ibus: Bus Number (int)	MVAR (float)
sw_shunt_vmax(ibus)	ibus: Bus Number (int)	Vpu (float)
sw_shunt_vmin(ibus)	ibus: Bus Number (int)	Vpu (float)
load_status(ibus, loadid)	ibus: Bus Number (int) loadid: Load ID (int or string)	True/False
branch_mw(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)	MW (float)
branch_mvar(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)	MVAR (float)
branch_amps(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int)	Amps (pu) (float)

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Function	Argument(s)	Return Value(s)
	ckt: Circuit ID (int or string)	
branch_mva(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)	MVA (float)
branch_loading(ibus, jbus, ckt, rating=1, amps=False)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string) rating: Rating to use (1-12) amps: Return equiv. amp loading	Loading (pu) (float)
bus_voltage(ibus)	ibus: Bus Number (int)	Vpu (float)
bus_kv(ibus)	ibus: Bus Number (int)	kV (actual) (float)
bus_kvbase(ibus)	ibus: Bus Number (int)	kV (base) (float)
bus_angle(ibus, degrees=False)	ibus: Bus Number (int) degrees: True/False	angle (float)
bus_vrange_normal(ibus)	ibus: Bus Number (int)	(v1, v2) (float, float)
bus_vrange_emergency(ibus)	ibus: Bus Number (int)	(v1, v2) (float, float)
area_load_mw(area, bus_assignment=True)	area: Area Number (int) bus_assignment: Use Bus Assig-ment (True/False)	MW (float)
area_load_mvar(area, bus_assignment=True)	area: Area Number (int) bus_assignment: Use Bus Assig-ment (True/False)	MVAR (float)
area_gen_mw(area)	area: Area Number (int)	Total MW (float)
area_gen_mvar(area)	area: Area Number (int)	Total MVAR (float)
area_loss_mw(area)	area: Area Number (int)	Total MW (float)
area_loss_mvar(area)	area: Area Number (int)	Total MVAR (float)
area_interchange_mw(area)	area: Area Number (int)	Net MW (float)
area_interchange_mvar(area)	area: Area Number (int)	Net MVAR (float)
dc_line_current(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	Amps (pu) (float)
dc_line_setpoint(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	Setpoint (float)
dc_converter_mw(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	MW (float)
dc_converter_mvar(name, ibus)	name: DC Line Name (string)	MVAR (float)

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Function	Argument(s)	Return Value(s)
	ibus: Bus Number (int)	
dc_converter_mva(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	MVA (float)
dc_converter_amps(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	Amps (pu) (float)
dc_converter_kv(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	kV (float)
dc_converter_angled(name, ibus)	name: DC Line Name (string) ibus: Bus Number (int)	Angle (degree) (float)
dc_line_control_mode(name)	name: DC Line Name (string)	Control Mode (int)
dc_nterm_firing_angle_status(name)	name: DC Line Name (string)	True/False
dc_nterm_tap_status(name)	name: DC Line Name (string)	True/False
plant_mw(ibus)	ibus: Bus Number (int)	Total MW (float)
plant_mvar(ibus)	ibus: Bus Number (int)	Total MVAR (float)
plant_mva(ibus)	ibus: Bus Number (int)	Total MVA (float)
machine_pgen(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MW (float)
machine_pmax(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MW (float)
machine_pmin(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MW (float)
machine_qgen(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MVAR (float)
machine_qmax(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MVAR (float)
machine_qmin(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MVAR (float)
machine_mva(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MVA (float)
machine_mbase(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	MW (base) (float)
machine_status(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)	True/False

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Function	Argument(s)	Return Value(s)
load_mw(ibus, loadid, actual=False)	ibus: Bus Number (int) loadid: Load ID (int or string)	MW (float)
load_mvar(ibus, loadid, actual=False)	ibus: Bus Number (int) loadid: Load ID (int or string)	MVAR (float)
load_mva(ibus, loadid, actual=False)	ibus: Bus Number (int) loadid: Load ID (int or string)	MVA (float)

20.6. Available Setter Functions

These functions can be used to define system adjustments required in a RAS action definition.

Action Function	Argument(s)
trip_unit(ibus, machid)	ibus: Bus Number (int) machid: Mach ID (int or string)
open_branch(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)
close_branch(ibus, jbus, ckt)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string)
disconnect_bus(ibus)	ibus: Bus Number (int)
set_branch_status(ibus, jbus, ckt, status)	ibus: Bus Number (int) jbus: To Bus Number (int) ckt: Circuit ID (int or string) status: True/False
set_breaker_status(subnum, inode, jnode, ckt, status)	subnum: Sub Number (int) inode: From Node Num (int) jnode: To Node Num (int) ckt: SWD ID (int or string) status: True/False
set_fixed_shunt_status(ibus, shntid, status)	ibus: Bus Number (int) shntid: Shunt ID (int or string) status: True/False

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Action Function	Argument(s)
set_fixed_shunt_mvar(ibus, shntid, mvar)	ibus: Bus Number (int) shntid: Shunt ID (int or string) mvar: Nominal MVAR
set_sw_shunt_status(ibus, status)	ibus: Bus Number (int) status: True/False
set_sw_shunt_qinit(ibus, qinit)	ibus: Bus Number (int) qinit: Nominal Q Init MVAR
set_sw_shunt_vmax(ibus, vmax)	ibus: Bus Number (int) vmax: Vpu Max
set_sw_shunt_vmin(ibus, vmin)	ibus: Bus Number (int) vmin: Vpu Min
set_load_status(ibus, loadid, status)	ibus: Bus Number (int) loadid: Load ID (int or string) status: True/False
set_machine_status(ibus, machid, status)	ibus: Bus Number (int) machid: Mach ID (int or string) status: True/False
set_machine_mw(ibus, machid, mw)	ibus: Bus Number (int) machid: Mach ID (int or string) mw: Machine MW (pgen)
set_machine_qmin(ibus, machid, mvar)	ibus: Bus Number (int) machid: Mach ID (int or string) mvar: Machine MVAR Min (Qmin)
set_machine_qmax(ibus, machid, mvar)	ibus: Bus Number (int) machid: Mach ID (int or string) mvar: Machine MVAR Max (Qmax)
set_scheduled_voltage(ibus, machid, vpu)	ibus: Bus Number (int) machid: Mach ID (int or string) vpu: Vpu Setpoint
set_vcsdc_mw(name, cnvbus, mw)	name: DC Line Name (string) cnvbus: Converter Bus (int)

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Action Function	Argument(s)
	mw: MW to set
set_vcsdc_qmin(name, cnvbus, mvar)	name: DC Line Name (string) cnvbus: Converter Bus (int) mvar: MVAR Min to Set
set_vcsdc_qmax(name, cnvbus, mvar)	name: DC Line Name (string) cnvbus: Converter Bus (int) mvar: MVAR Max to Set

20.7. Settings and other Utility Functions

These functions provide access to settings and other utility operations.

Function	Description
set_max_ras_iterations(n)	Set the max number of RAS iterations (system adjustment iterations)
enable_ras_psse_report(enable=True, to_progress=False)	Enable verbose RAS reporting. enabled: If True, verbose reporting will be sent to the PSS®E report stream during the set up and operation of the RAS system in contingency analysis. If False, no reporting will be provided. to_progress: If True, reporting will be sent to the defined progress output device. If False, reporting will be sent to the defined report output device.
enable_ras_csv_report(filepath=POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal, enable=True)	Enable a csv output report for programmatic post-processing. filepath: csv filepath to which the report messages will be sent. enabled: If True, verbose reporting will be sent to the PSS®E report stream during the set up and operation of the RAS system in contingency analysis. If False, no reporting will be provided.
enable_ras_logging(filepath="raslog.txt", enable=True, overwrite=True)	Enable verbose RAS reporting. filepath: Log filepath to which the log messages will be sent. enabled: If True, verbose reporting will be sent to the PSS®E report stream during the set up and operation of the RAS system in contingency analysis. If False, no reporting will be provided.

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Function	Description
	overwrite: If True, and existing file will be overwritten with each run of the contingency engine.
report(text, newline=True, timestamp=False)	Send a message to the PSS®E report stream. Has no effect if REPORT flag is False. To enable reporting, call the function enable_ras_reporting() in your RAS definition script. text: The text to send to the report. newline: (optional, default=True) add a newline at the end of the message. timestamp: (optional, default=False) add a timestamp at the beginning of the message.
log(text, newline=True, timestamp=True)	Send a message to the log file. Has no effect if LOG flag is False. To enable logging and to define the log file path, call the function enable_ras_logging(path) in your RAS definition script. text: The text to send to the log. newline: (optional, default=True) add a newline at the end of the message. timestamp: (optional, default=False) add a timestamp at the beginning of the message.
csv_report(contingency, iteration, ras_name, notes="")	Write a row to the CSVreport contingency: Label of current contingency iteration: System adjustment iteration limit ras_name: Name of triggered RAS scheme
report_and_log(text, newline=True, report_timestamp=False, log_timestamp=True)	Automatically sends a message to report and log streams (see report and log functions for argument detail)
clear_log()	Clears the log file (makes the file empty).
ras_summaries(*ras_names)	Provides a string containing summaries of RAS definitions. ras_names: (optional) arg list of RAS definitions to include in the summary. If empty, all define schemes will be included.
enable_ras(name)	Enable (arm) a RAS. name: RAS name (case-sensitive)
disable_ras(name)	Disable (disarm) a RAS. name: RAS name (case-sensitive)

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Function	Description
set_ras_enabled(name, enabled)	Enable (arm) or disable (disarm) a RAS. name: RAS name (case-sensitive) enabled: True (arm) or False (disarm)
clear_all_ras()	Un-define all defined RAS schemes.
define_ras(name, condition, action, enabled=True, report=POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal, description=POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal)	Defines a RAS. name: The user-defined name of the scheme. condition: The name of a defined function that will be called each ras iteration to determine if this ras is triggered. This function should return True for triggered or False otherwise. action: The name of the function that will be called if this ras is triggered. enabled: (optional) True for enabled (armed) or False for disabled (disarmed). report: (optional) a function that returns a string for custom reporting. description: (optional) detailed description of the RAS. returns: The Scheme object created.

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Chapter 21
Using the Vendor Neutral Remedial Action Scheme Format

21.1. Overview

The Vendor Neutral RAS module is created to support the new and evolving Contingency and Remedial Action Scheme standard used in the Western United States. This standard defines a way to identify contingencies, Remedial Action Schemes (RASs) and their triggering conditions in a vendor neutral manner.

This standard is currently being adopted by the Western Electricity Coordinating council, WECC, in the United States.

The RAS specification is based on the Vendor Neutral RAS format developed and supported by PowerWorld. The RAS module currently supports Version 19 of the "PowerWorld Simulator, Auxiliary File Format Description for Specifying Contingency Definitions and Remedial Actions". The complete document describing the specification can be found here: https://www.powerworld.com/files/PowerWorld_RASFileFormat.pdf

We're currently in the process of upgrading our implementation to Version 20 of the specification. It is our current policy to continue to support new versions of this standard as it evolves.

21.2. PSSE Specific Additions to the Vendor Neutral RAS file

For the `CTG_Options_Value` object, the following values:

- `PSSESOLUTIONMETHOD` – can be set to `'FNSL', 'FDNS', or 'OPTIMIZEDFDNS'`
- `PSSEDCTAPADJUST` – can be set to `'ENABLE' or 'DISABLE'`
- `PSSENONDIVERGENT` – can be set to `'ENABLE' or 'DISABLE'`
- `PSSEINDUCTIONMACHINELOWVOLT` – can be set to `'STALL' or 'TRIP'`
- `PSSEINDUCTIONMACHINEFAILURE` – can be set to `'NONCONVERGED' or 'CONVERGED'`
- `PSSEDISPATCHMODE` – can be set to `'DISABLE', 'RESERVE', 'PMAX', 'INERTIA', or 'DROOP'`

21.3. How is the Vendor Neutral Contingency and RAS supported in PSSE?

Currently, PSSE uses the SUB/MON/CON files to define contingencies. There are additional files that can be used, such as TRP files (corrective actions) or TRW files (load throwover). Remedial actions are controlled through a Python API (pssuserpf.py, and the PSSRAS module)

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Using the new specification, contingencies, remedial action schemes and their triggering conditions are defined in one or more auxiliary files, commonly called AUX files. These files usually have AUX as their suffix, although WCRF is commonly used as well. AUX is the default suffix recognized by PSSE but WCRF can also be supported and recognized by adding an additional suffix under

```
File > Change File Type Properties....
```

AUX files can be quite complicated and a useful approach to understanding them is to start with a simple AUX file or study some of the AUX files already being used in the WECC. As mentioned before the complete specification can be found here: https://www.powerworld.com/files/PowerWorld_RASFileFormat.pdf

Currently, PSSE supports this standard as a text file, meaning that the AUX files are developed and maintained as text files in a text editor. PSSE 34.6 and 35.0 supports an embedded editor that is used to create, view and modify AUX files. This embedded editor serves as a "window" into the RAS engine, allow the user to view and modify the RAS data currently in memory.

21.4. The Vendor Neutral RAS Model in Memory

As AUX files are read into PSSE, an RAS object model is constructed in the engine. This memory resident RAS model is what is used to drive the actual contingency and RAS analysis. This is similar to "compiling" the SUB, MON and CON files into the DFAX file which is then used to drive the existing ACCC API routines.Processing of the memory resident RAS model is more efficient and optimized for processing the 1000s of contingencies, RASs and triggering conditions commonly found in the AUX files.

The RAS model in memory operates in the following manner:

21.4.1. A new AUX file is created in the Editor

When this file is saved, the contents are used to populate the memory resident RAS model

When edits are made to this file in the Editor, the contents are used to update the memory resident RAS model

When the RAS analysis is run, the memory resident RAS model is used to identify the contingencies, RASs and triggering conditions

21.4.2. An existing AUX file is opened in the editor

When the file is opened, any existing memory resident RAS model is re-initialized, the AUX file is read and the memory resident RAS model is populated

When edits are made to this file in the Editor, the contents are used to update the memory resident RAS model

When the RAS analysis is run, the memory resident RAS model is used to identify the contingencies, RASs and triggering conditions

21.4.3. Appending to an existing memory resident RAS model

Once you have an existing memory resident RAS model, you can add to it by "appending" other AUX files to it. Rather than re-initializing the memory resident RAS model, appending to it will effectively "add" to the model. If existing items in the memory resident RAS also appear in the AUX files used to append to the

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memory resident RAS model, the existing items will be "replaced" in whole or in part by the contents of the appending AUX files.

21.4.4. Notes of Interest

Once an AUX file has been opened in the Editor, appending to memory resident RAS model will NOT update the view of the RAS model in the Editor. The Editor window will still display the contents of the original AUX file opened.

To see the current contents of the memory resident RAS model, the original AUX file plus any edits and any appended AUX file(s), write out the current memory RAS model into an AUX file and then open this AUX file in another Editor. At this point the program operates as described in "An existing AUX file is opened in the editor".

21.5. API routines to Support the Vendor Neutral RAS process

The PSSE API has been enhanced by the development of five new routines to support the RAS functionality described above. The API routines are summarized in the following. For a more detailed description of these routines, please see the API manual in the chapter titled "Using the Vendor Neutral Contingency and RAS Auxiliary File". Note that filenames used in the PSSE API can be full pathnames or relative pathnames (including filenames in the current working directory).

21.5.1. READ_RAS

This API routine is used to populate the memory resident RAS model with a new model from an existing AUX file. Reading in the file initializes the memory resident RAS model, effectively deleting any RAS model currently defined. If the current memory resident RAS model is to be preserved, it must be written out using the WRITE_RAS API routine.

Input to this routine is the filename identifying the AUX file to use to create the memory resident RAS model.

Reading in an AUX file works similar to a compiler, any errors in the formatting of the AUX file are fully identified and called out while reading the AUX file. If working interactively, the AUX file can be edited directly in the Editor, saved and the file is re-read, confirming the "fixing" of the errors or identifying further issues to be corrected.

If errors are encountered, the memory resident RAS model is not created. Error messages are more fully described in a following section.

21.5.2. APPEND_RAS

This API routine is used to add/append to an existing memory resident RAS model. This routine can only be used if there is an existing memory resident RAS model.

Input to this routine is the filename identifying the AUX file to use to add to the current memory resident RAS model.

Appending an AUX file works similar to a compiler, any errors in the formatting of the AUX file are fully identified and called out while reading the AUX file. If working interactively, the appended AUX file is NOT

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opened in an editor and cannot be edited directly in PSSE. If errors are encountered, the AUX file will need to be edited outside of PSSE, saved and then re-appended to the existing memory resident RAS model.

If errors are encountered, the memory resident RAS model is not updated. Error messages are more fully described in a following section.

21.5.3. WRITE_RAS

This API routine is used to write the existing memory resident RAS model to an AUX file. This file contains the entire memory resident RAS model. That is to say, the total RAS model created by READ_RAS and any number of APPEND_RAS operations.

Input to this routine is the filename identifying the AUX file to use to save current memory resident RAS model as well as several output options controlling what to do with bad data.

If errors are encountered, the memory resident RAS model is not output. Error messages are more fully described in a following section.

21.5.4. VALIDATE_RAS

This API routine is used to validate the existing memory resident RAS model against the existing network model. There are many elements of the RAS model that refer directly to elements in the network model. This routine verifies that all network elements referenced in the memory resident RAS model actually exist in the network model. If elements in the memory resident RAS model are NOT found in the current network model, then the ACCC_RAS API routine will operate without those using those elements of the RAS model.

There are no inputs to this routine but is does require the existence of a memory resident RAS model.

If errors are encountered, the ability to execute the ACCC_RAS is disabled. Error messages are more fully described in a following section.

21.5.5. ACCC_RAS

This API routine is used to run the contingency and RAS analysis on the current network model. It is similar to the existing ACCC API routines in PSSE in execution, but differs in that it requires no inputs other than the output ACC file to run. All other inputs concerning solutions settings and configuration files are defined in the memory resident RAS model.

A memory resident RAS model as well as a network model must be present in order for this routine to execute.

The memory resident RAS model must have been validated against the network model, using VALIDATE_RAS, in order for this routine to execute.

Input to this routine is the filename identifying the ACC file to use to save the output from the contingency and RAS analysis.

21.5.6. Error Reporting from the RAS APIs

During input (READ_RAS), the file which generated errors is identified as follows:

```
Messages for file: D:\RAS\Dec-5-17 RAS and associated PSSE files\SCE_Whirlwind_RAS.aux
```


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If the INCLUDE facility is used, each individual file is still identified. An example of an error is the following:

```
Field name "PostCTGAuxFile" undefined, ignored; Line # 115, buffer:
 > Contingency (Name,Category,Skip,Memo,PostCTGAuxFile,SolutionOptionsUse,MonitorException
```

Note that the error identifies the specific error, the line number of the file, and includes the entire input line in which the error was found.

During validation (`VALIDATE_RAS`), the RAS model in memory is validated against the current network case. This flexibility allows the network to be changed, and the current RAS module in memory can be revalidated without needing to read in the RAS file again. An example of an error is the following:

```
Error 283: ModelCondition ["PGAE_012_GATES 500/230KV PRIMARY_OUTAGE"]
 File "D:\RAS\Dec-5-17 RAS and associated PSSE files\PGAE_012_Gates 500-230kV RAS.aux" lin
 ->  Referenced object "Branch '30055' '34607' '11'" in field "Object" is not found.
```

The first line shows the running total of validation errors so far, with the RAS object being validated, and its key fields.

The second line shows the file from which the definition of the object was read, and the line number.

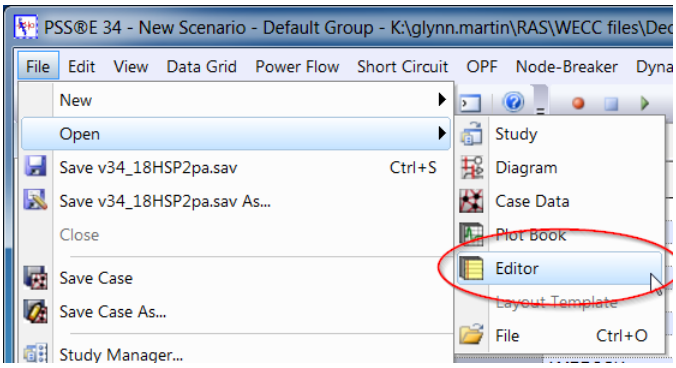
The third line shows the specifics of the error.

21.6. Interactive use of the Vendor Neutral RAS module

Though it is possible to fully automate the Vendor Neutral RAS module using the APIs described above, the entire process can be run interactively as well. The following sections outline the interactive use of the RAS module.

21.6.1. How to read an AUX file

An AUX file is read into PSSE using `File > Open > Editor` as shown in the following image:



An AUX file is then selected as shown in the following image. Note support for WCRF and AUX suffixes.

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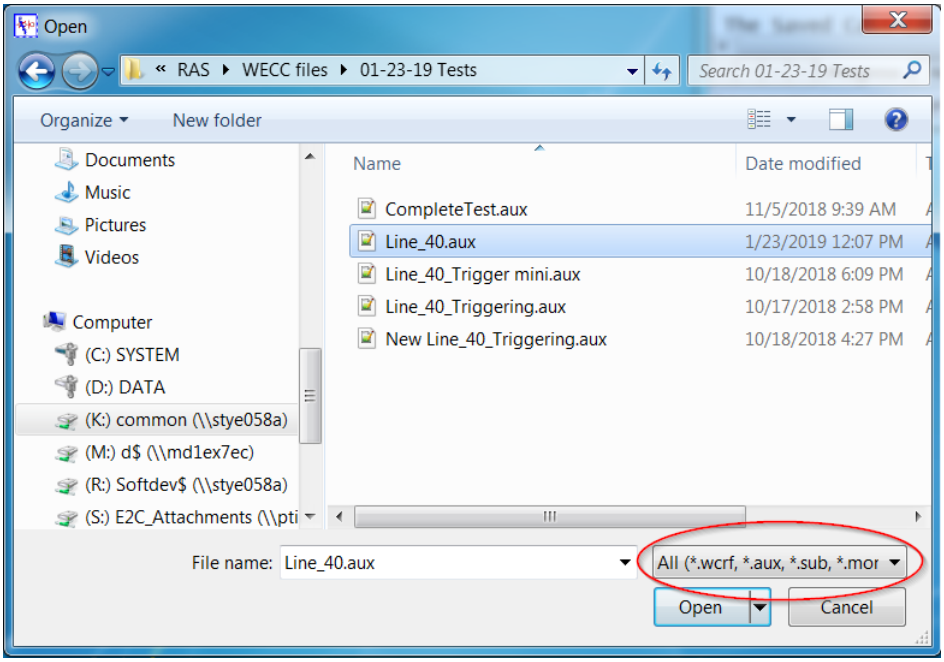
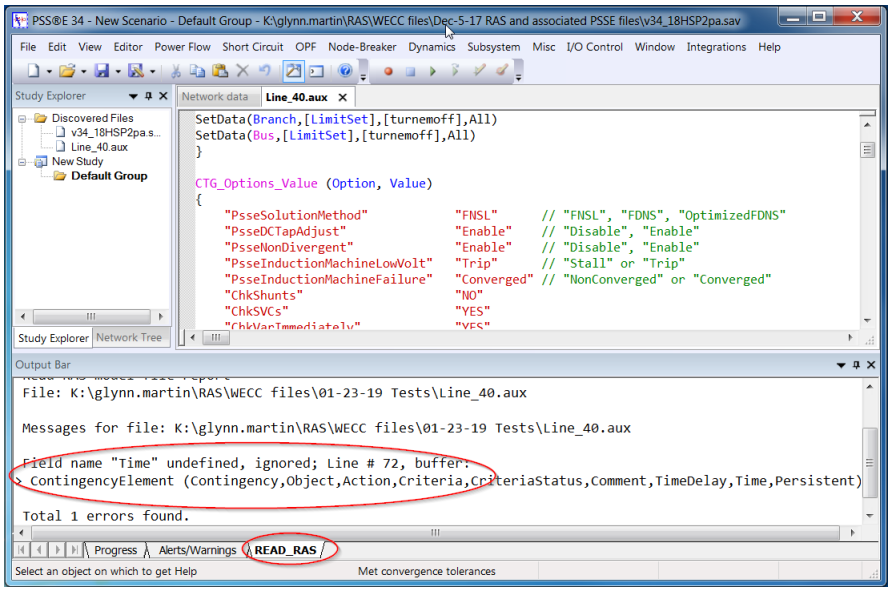


Once the AUX file selected, it is opened in an Editor window in PSSE as shown in the following image. Note the error messages on the READ_RAS tab in the output window.



21.6.2. Chaining/including of files

AUX files can include references to other aux files through the use of INCLUDE statements in the AUX file. In the AUX file shown in the following image, it includes the basic contingencies as well as many AUX files containing RASs. The use of the INCLUDE statement in an AUX file basically creates a giant AUX file that contains the contents of all the INCLUDED AUX files and creates a single memory resident RAS model.

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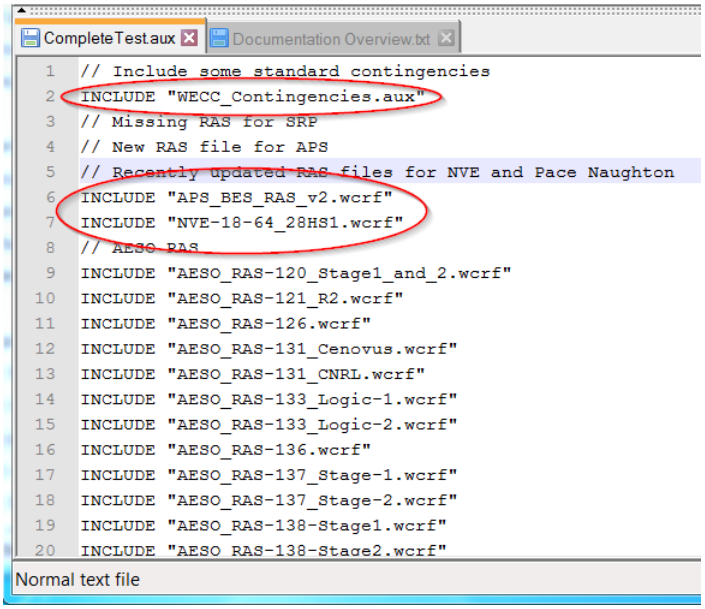

All files in a directory can be specified by an INCLUDE file by using an asterisk for the filename. For example:

```
Include "aux files\test1\*.aux"
```

21.6.3. What happens if I read another AUX file?

Reading in another AUX file effectively re-initializes the memory resident RAS model with the contents of the new AUX file opened as shown in the following image:

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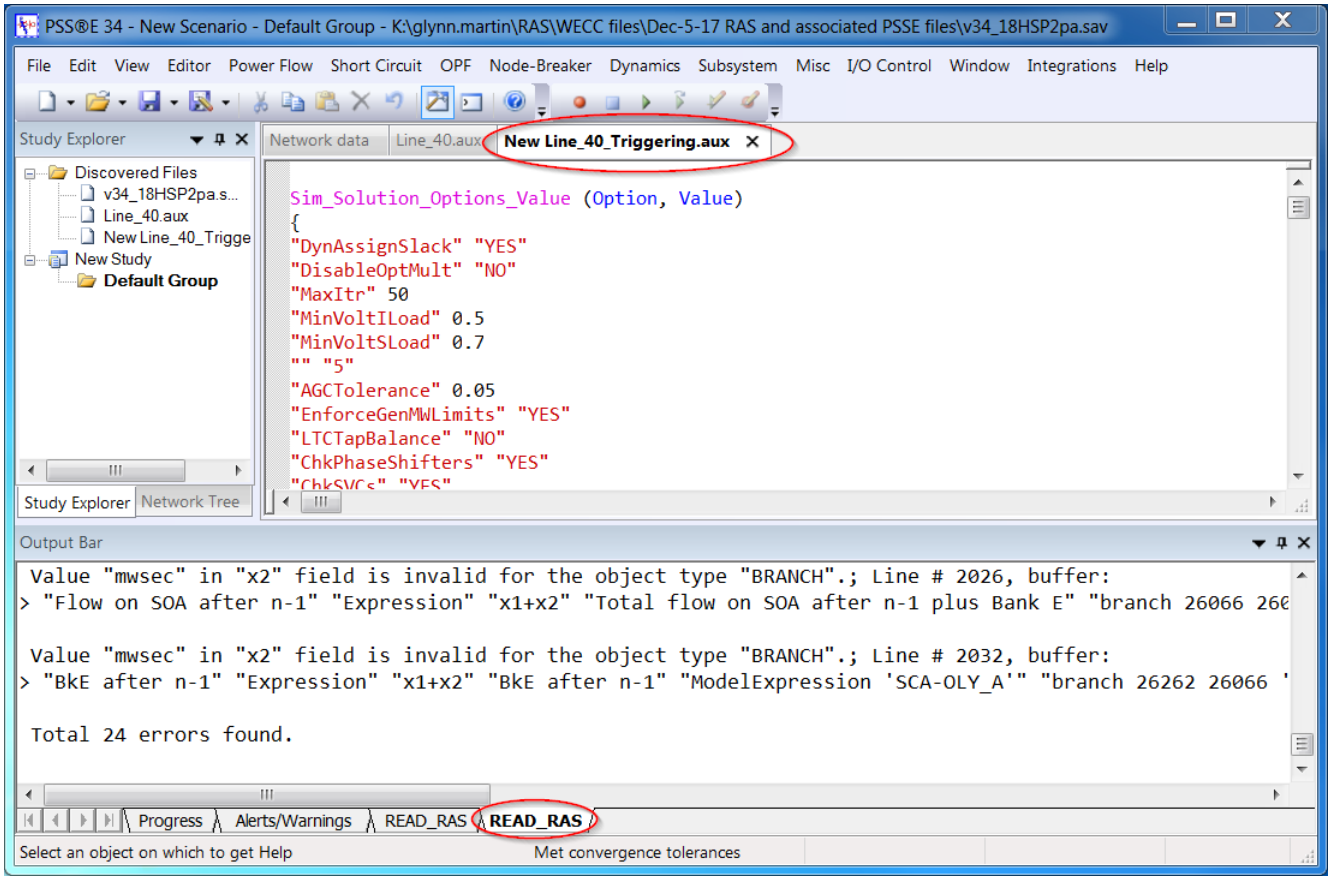
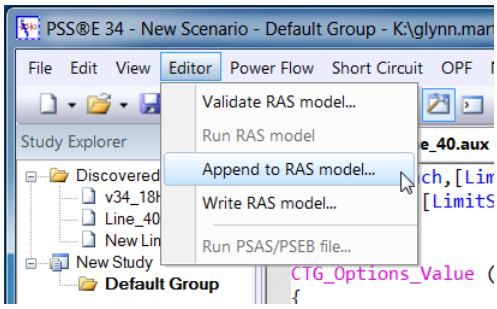


The presence of two or more Editor windows in PSSE can cause confusion. In the above image, selecting the "Line_40.aux" Editor, making a change and saving it will cause the AUX file to be re-read, effectively replacing the memory resident RAS model created by reading in "New Line_40_Tirggering.aux" with the RAS model contained in the updated "Line_40.aux". To avoid confusion, close any Editor windows not currently in use.

21.6.4. How to append an AUX file to the current RAS model

As described above, it is possible to add to an existing memory resident RAS model through the use of the append function. From `Editor > Append to RAS model...` select the AUX file to be added to the memory resident RAS model.



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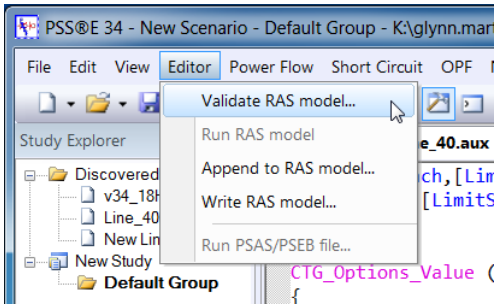
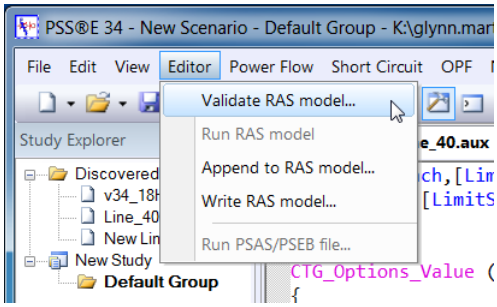
The AUX file will be read and the contents added to current memory resident RAS model. Any warnings / errors encountered when reading the AUX file to append will be output to the Output bar in a tab titled ""APPEND_RAS".



Any warnings / errors encountered when validating the memory resident RAS model will be output to the Output bar in a tab titled "VALIDATE_RAS".

21.6.5. How to validate the current RAS model

As described above, it is possible to validate the existing memory resident RAS model against the existing network model to determine if there are any missing, or incorrectly identified, network elements in the AUX file(s) that would affect the contingency or remedial actions. The network validation can be run by selecting `Editor > Validate RAS model...`.



Any warnings / errors encountered when validating the memory resident RAS model will be output to the Output bar in a tab titled "VALIDATE_RAS".

21.6.6. How to run the RAS analysis against the memory resident RAS model

Once the memory resident RAS model has been validated against the network model, the contingency and RAS analysis can be run by selecting `Editor > ACCC RAS model...`

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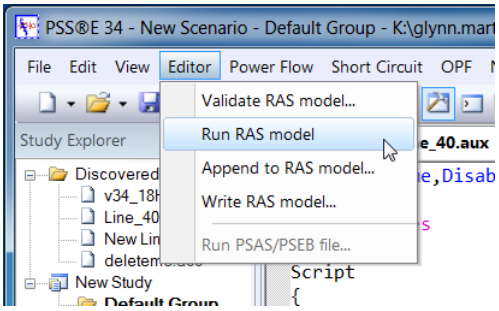
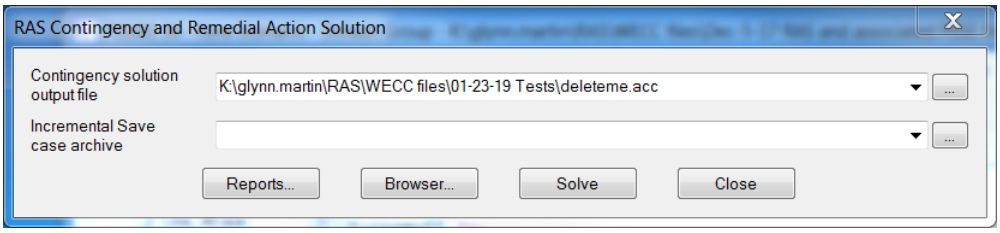


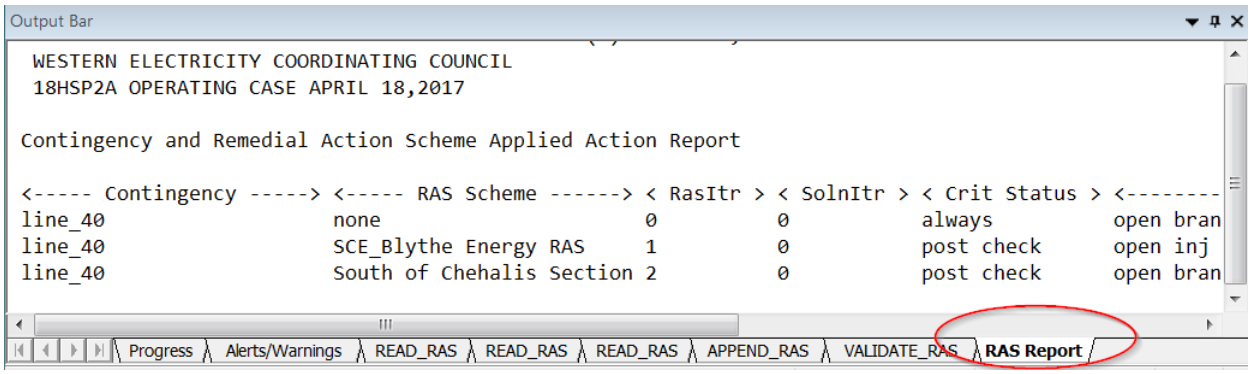
When selected, the following dialog appears. This operates much like the standard ACCC activities in PSSE but requires little input as the complete contingency and RAS setup is contained in the memory resident RAS model. The only input required is the ACC file to capture the output of the analysis.



21.6.7. How to find out what happened when running the RAS analysis

Similar to the existing PSSE APIs, the ACCC_RAS API will output messages to the progress window describing the processing of the memory resident RAS model. In addition, a RAS Report is output to summarize the contingencies and RASs triggered during the analysis.

An example of a RAS Report can be found in the following image:



21.7. Known Issues

All known deviations of the PSSE implementation from the Vendor Neutral RAS specification are listed below. These items could not be implemented in the PSSE-34 engine as they require new data structures and data. These basic changes to the PSSE engine can only be made at the next major release. Currently, the majority of the PSSE user community in WECC is operating on the PSSE-33.X series. MMWG has committed to moving to the PSSE-34.X series in April 2019 and this will be the first opportunity for them to use the RAS engine as currently implemented. PSSE-35.0 is currently planned for Spring 2019.

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21.7.1. Items that Will be Implemented in the PSSE-35 Engine

It was known at the beginning that some items would require commitments to extend the PSSE engine in a future release.

1. Identification of items by Label Identifiers (Sec 2.3.6)
2. Unlimited String and line lengths

These items will appear in PSSE-35.1 currently planned for Fall 2019

21.7.2. Items that Would Require Extensions to the PSSE Engine

It was known at the beginning that supporting items that required extending the PSSE engine would not part of the project.

The following are items requiring extensions to the PSSE engine and are therefore not currently supported in the Vendor Neutral RAS implementation:

1. Contingencies:
   - All multi-terminal DC line contingencies
   - All Line Shunt contingencies
   - Interface contingencies that require merit ordering
   - Injection Group contingencies that require merit ordering
   - Branch contingencies involving BYPASS and NOTBYPASS
   - Shunt VPU CHANGEBY contingencies (requires a new shunt power flow model)
2. Limit monitoring for BRANCHOPEN, BRANCHCLOSE, GENOPEN, LOADOPEN
3. Support for coordinated voltage control at the VoltageControlGroup level
4. Not all solution and contingency settings are meaningful to PSSE (such as DynAssignSlack or ShuntInner, etc.) because of the way the solution is designed. Are we missing some WECC solution options that need additional function to map to RAS solution space.

These items will implemented as resource and project schedules allow. They will require new data structures and data, and so can only be included in PSSE-35 and above.

21.7.3. Items that Cannot be Supported due to Limitations of the Specification

Some items could not be implemented because the specification does not contain the necessary data to adequately identify and/or model them. These are as follows:

1. It is not possible to distinguish between parallel 2-term dc lines in PSSE given the identification information provided in the specification. Work needs to be done in the spec to define these elements in a better way. PowerWorld is aware of these issues concerning DC elements and has thoughts on how to address.

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Chapter 22
Miscellaneous Activity Descriptions

This chapter contains operational descriptions of those PSS®E activities related to file and output device management and other miscellaneous functions used in performing engineering studies in PSS®E. Each section of this chapter documents a single PSS®E activity. Interface references are provided, with links to the *PSS®E GUI Users Guide* and the *PSS®E Command Line Interface (CLI) Users Guide*, where appropriate.

22.1. Terminating PSS®E

STOP

Run Activity STOP - GUI
Run Line Mode Activity STOP - CLI
ACTIVITY? >>**STOP**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The PSS®E termination activity STOP is the normal exit from PSS®E. When selected, activity STOP closes any Response Files which are active (see Sections Section 2.3.4, "Channel Output Files" and Section 17.12, "Running a Response File") and any output files that are open, such as redirected virtual output devices (see Section 4.4, "Virtual Output Devices"). It then returns the user to operating system level.

Activity STOP does not affect the status of any command files or session logging files which might have been opened outside of PSS®E.

If the working case represents a system condition that may be needed in the future, the user should ensure that it is preserved in a Saved Case File prior to selecting activity STOP. A similar argument applies to preserving dynamics working memory in a Snapshot File.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Terminating PSS®E

22.2. Applying a User-Written Activity

USER

Run Line Mode Activity USER - CLI
ACTIVITY? >>**USER**

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Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The PSS®E interface provides for the incorporation of a user-written activity into PSS®E.

Activity USER is not sensitive to any interrupt control code options.

Additional Information
Additional Resources for PSS®E,Section 1.3.1, Dynamics User Model DLLs *PSS®E Command Line Interface (CLI) Users Guide*, Running a User-Written Activity

22.3. Displaying Help Information

HELP

Run Activity HELP - GUI
Run Line Mode Activity HELP - CLI
ACTIVITY? >>**HELP**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The line mode activity HELP prints brief summary documentation at the Progress device.

For Future PSS®E Release: The GUI Help function is entirely different. It provides information on using the GUI dialogs.

22.4. Selecting Prompt Output Destination

ODEV

Direct Prompt Output - GUI
I/O Control>Direct Prompt output... *[Prompt Output Destination Selector]*
Run Line Mode Activity ODEV - CLI
ACTIVITY? >>**ODEV** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Selecting the prompt output destination controls where output identified in PSS®E as prompt, or dialog, information is sent. The PSS®E GUI default behavior (prompt window) is to display accumulated prompts in

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the general input window that is generated for terminal input requests. Prompts are messages that request input from the PSS®E user. The GUI itself does not generate any such output, but they can be created in automation files.

Prompts generated by using the command line in the GUI are an exception to the above default behavior. Those prompts are forced to the Progress device and appear immediately. When using a console application, such as PSSECMD, redirecting the prompts will affect all the conversational dialog. Care should be taken in this case as the user may not be able the see the questions being asked.

The line mode activity ODEV instructs the user to select the device to be used for these tabulations and presents a menu of choices. Specifying either zero or one to the device selection request assigns report output to the user's terminal, or the prompt window in the PSS®E GUI. ODEV will redirect alert output as well as prompt output. Using ODEV is effectively the same as redirecting both prompt and alert output.

The destination specified for prompts may be the same file or printer that is used for any or all the other virtual output devices. Output may be directed to the Progress or Report device, in which case prompt output will be handled as part of that output.

When a printer is specified for the prompt device, the output is normally not printed until either prompt output is reassigned elsewhere or PSS®E is terminated. The PSS®E GUI and the prompt_output API routine provide a print immediate option that will send the output to the printer at the end of every activity.

Additional Information
PSS®E GUI Users Guide, Alerts and Prompts Destination *PSS®E Command Line Interface (CLI) Users Guide*, Specifying Alert and Prompt Destination See also: Section 4.4, "Virtual Output Devices"

22.5. Selecting Alert Output Destination

ODEV

Direct Alert Output - GUI
I/O Control>Direct Alert output... *[Alert Output Destination Selector]*
Run Line Mode Activity ODEV - CLI
ACTIVITY? >>**ODEV** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Selecting the alert output destination controls where output identified in PSS®E as alert information is sent. The PSS®E GUI default behavior (alert box) is to display alert information in a MessageBox, and a copy of that message in both the Progress window and the Alerts/Warnings window. Alerts are messages that inform the PSS®E user of an error in program execution. This should be understood to mean situations where the program was unable to perform the assigned task, and specifically not situations where results might or might not meet some standard, such as a solution failing to converge.

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Alerts generated by using the command line in the GUI and during Automation File execution are exceptions to the above default behavior. Those alerts are forced to the Progress device and appear immediately.

The line mode activity ODEV will redirect alert output as well as prompt output. Using ODEV is effectively the same as redirecting both prompt and alert output.

The destination specified for alerts may be the same file or printer that is used for any or all the other virtual output devices. Output may be directed to the Progress or Report device, in which case alert output will be handled as part of that output.

When a printer is specified for the alert device, the output is normally not printed until either alert output is reassigned elsewhere or PSS®E is terminated. The PSS®E GUI and the alert_output API routine provide a print immediate option that will send the output to the printer at the end of every activity.

Additional Information
PSS®E GUI Users Guide,Alerts and Prompts Destination *PSS®E Command Line Interface (CLI) Users Guide*, Specifying Alert and Prompt Destination See also: Section 4.4, "Virtual Output Devices"

22.6. Selecting Progress Output Destination

PDEV

Direct Progress Output - GUI
I/O Control>Direct Progress output (PDEV)... *[Progress Output Destination Selector]*
Run Line Mode Activity PDEV - CLI
ACTIVITY? >>**PDEV** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Selecting the progress output destination controls where output identified in PSS®E as progress information is sent. The PSS®E GUI default behavior (Progress window) is to display progress information in the Progress tab of the output bar. Progress output are messages that inform the PSS®E user of intermediate results, confirmation of instructions, supplemental information, and usage history. These include such tabulations as the network solution convergence monitors, the machine initial conditions summary of activities STRT, ESTR, and GSTR, and the output channel tabulations of activities RUN, ERUN, and GRUN.

The line mode activity PDEV instructs the user to select the device to be used for these tabulations and presents a menu of choices. Specifying either zero or one to the device selection request assigns progress output to the user's terminal, or the Progress window in the PSS®E GUI.

The destination specified for progress may be the same file or printer that is used for any or all the other virtual output devices. Output may be directed to the Report device, in which case progress output will be handled as part of that output.

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When a printer is specified for the progress device, the output is normally not printed until either progress output is reassigned elsewhere or PSS®E is terminated. The PSS®E GUI and the progress_output API routine provide a print immediate option that will send the output to the printer at the end of every activity.

Additional Information
PSS®E GUI Users Guide, Progress Destination *PSS®E Command Line Interface (CLI) Users Guide*, Specifying Progress Report Destination See also: Section 4.4, "Virtual Output Devices"

22.7. Selecting Report Output Destination

OPEN

Direct Report Output - GUI
I/O Control>Direct Report output (OPEN)... *[Report Output Destination Selector]*
Run Line Mode Activity OPEN - CLI
ACTIVITY? >>**OPEN** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Selecting the report output destination controls where output identified in PSS®E as report information is sent. The PSS®E GUI default behavior (Report window) is to display report information in the Report tab (or tabs) of the output bar. Report output are the results of calculations or analysis that are specifically requested by the PSS®E user.

The line mode activity OPEN instructs the user to select the device to be used for these tabulations and presents a menu of choices. Specifying either zero or one to the device selection request assigns report output to the user's terminal, or the Report window in the PSS®E GUI.

The destination specified for reports may be the same file or printer that is used for any or all the other virtual output devices. Output may be directed to the progress device, in which case report output will be handled as part of that output.

When a printer is specified for the report device, the output is normally not printed until either report output is reassigned elsewhere or PSS®E is terminated. The PSS®E GUI and the report_output API routine provide a print immediate option that will send the output to the printer at the end of every activity.

Selecting report output destination has a secondary effect in the line mode. Until the first time that report output is explicitly directed by the user, line mode activities that produce report output produce a menu requesting where the report output for that one activity should be directed. Once report output has been explicitly designated by the user (via the GUI, the line mode OPEN activity, or the API routine report_output) this mode of operation is terminated. It can be restored by using the line mode activity CLOS (see Section 22.8, "Closing Report Destination").

Additional Information
PSS®E GUI Users Guide, Global Report Destination

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Specifying Global Report Destination See also: Section 4.4, "Virtual Output Devices"

22.8. Closing Report Destination

CLOS

Run Line Mode Activity CLOS - CLI
ACTIVITY? >>**CLOS**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The report output device closing activity CLOS terminates output to the previously OPENed selection (see Section 22.7, "Selecting Report Output Destination"), and returns to the operating mode in which each reporting activity initiated using the line mode (see Section 4.2.2, "Command Line Interface") requests the user to select the destination for its report. As a result, the report output destination is set back to the default, or standard, behavior.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Closing Report Destination

22.9. Setting the Path for Use with & Filenames

PATH

Run Activity PATH - GUI
I/O Control>Set path for use with "&" filenames (PATH)... *[Define PATH by selecting any file in the desired directory]*
Run Line Mode Activity PATH - CLI
ACTIVITY? >>**PATH** ENTER PATHNAME PREFIX FOR USE WITH '&' FILE NAMES >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The directory path specification activity PATH allows the user to specify a directory name. Then, whenever a simple filename preceded by an ampersand (e.g., *&myfile*) is specified to a PSS®E file accessing activity, the file is opened in the directory specified to activity PATH rather than in the user's current directory. Refer to Section 2.4.2, "Specifying Filenames".

Additional Information
PSS®E GUI Users Guide, Setting the Path for Use with & Filenames

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Setting the Path for Use with & Filenames

22.10. Changing PSS®E Program Settings

OPTN

Run Activity OPTN - GUI
Misc>Change program settings (OPTN) ... *[Program Settings]*
Run Line Mode Activity OPTN - CLI
ACTIVITY? >>**OPTN** PRESENT OPTION SETTINGS ARE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The PSS®E option setting activity OPTN enables the user to override most of the default option settings as established when PSS®E is installed on the system (refer to Section 3.3.3, "Program Run-Time Option Settings"). It also provides for saving the current option settings in a PSS®E Options File for access on subsequent entries into PSS®E.

When saved, activity OPTN preserves the current settings of most PSS®E options in a PSS®E Options File (see Section 22.10.1, "Application Notes", Section 2.3.7, "IPLAN Program Files", and Section 3.3.3, "Program Run-Time Option Settings").

When invoked in line mode, activity OPTN displays the current settings of all options which may be modified. The user may re-specify any option. Following such a change, the option menu is again displayed with any new setting shown.

PSS®E options are listed in Table 3.2, "Saved Case Specific PSS®E Program Option Settings ", Table 3.3, "User Specific PSS®E Program Option Settings", and Table 3.4, "Installation Dependent Option Settings".

22.10.1. Application Notes

Whenever PSS®E is started up, the default option settings as established when PSS®E is installed on the system are in effect. These are overridden by the settings contained in a PSS®E Options File if one is found in the user's current or home directory or in the PSS®E master directory PSSLIB; PSS®E follows the directory search sequence described in Section 2.1, "Directories and Files Overview" when looking for the PSS®E Options File.

Most PSS®E option settings are stored with the working case when it is saved with activity SAVE. When a power flow case is retrieved with activity CASE, the option settings contained in it override those then set in working memory. Any time activity CASE changes an option setting, an appropriate message is printed.

Number of contingency processor defines the number of processors in using for AC contingency analysis solutions and N-1-1 contingency solutions. The option is computer system dependent, and its maximum is the number of processors of a computer up to 24 (1 by default).

Additional Information
PSS®E GUI Users Guide Introduction,, Changing Program Settings

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Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Changing Program Settings

22.11. Returning Activity Selection to Dynamic Simulation

RTRN

Run Line Mode Activity RTRN - CLI
ACTIVITY? >>**RTRN**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

Note: Starting with PSS®E-31, mode switching is not apparent to the user and need not be done explicitly. It is retained only for backwards compatibility.

The return to dynamics activity RTRN allows the user to execute any dynamics activity after having been operating in power flow mode.

Activity RTRN coordinates dynamics model data and network data by performing the following steps:

1. It coordinates the power flow machine electrical power arrays and the dynamic simulation machine electrical power arrays.

2. It sets up a translation array (NUMTRM) indexed by internal machine sequence number and pointing to the associated network bus. Out-of-service machines have their corresponding NUMTRM entries set to a negative number.

3. It sets up a translation array (NUMLOD) indexed by internal load sequence number and pointing to the associated network bus. Out-of-service loads have their corresponding NUMLOD entries set to a negative number.

4. It sets up a translation array (RINCNX) indexed by internal double entry branch index and pointing to the associated branch relay model connection table entry.

5. It updates internal pointer arrays used in associating dynamic simulation models with the machine and load indexing contained in the power flow working case.

6. It updates output channel addresses to reflect any changes in the internal sequence numbering of buses, machines, and loads in the working case.

7. It builds translation arrays DC2CNX, DCMCNX, VSCCNX, and FCTCNX for two-terminal dc lines, multi-terminal dc lines, VSC dc lines, and FACTS devices, respectively. These arrays are indexed by the internal device index in the power flow working case and point to the dynamic model associated with the corresponding two-terminal dc line, multi-terminal dc line, VSC dc line, or FACTS device, as appropriate.

8. It sets up a translation array (AUXLOC) indexed by internal auxiliary signal model number and pointing to the internal device index of the two-terminal dc line, multi-terminal dc line, VSC dc line, or FACTS device associated with the auxiliary signal model.

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9. It sets up a translation array (SWSCNX) indexed by internal switched shunt sequence number and pointing to the dynamic model associated with the switched shunt.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Returning Analytical Output to Dynamic Simulation

22.12. Editing the Case Heading or Long Title

EDTR

Run Line Mode Activity EDTR - CLI
ACTIVITY? >>**EDTR** EDITOR OPTIONS ARE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The title editing activity EDTR allows the user to perform editing operations on the two line case heading and on the sixteen line long title. It is accessible only as a command line activity. In the GUI the Case Titles dialog (activity CHTI) provides convenient editing ability. See also activity CHNG.

When either the two line case heading or the sixteen line long title is selected, the user specifies a valid line number and appropriate editing commands.

The general form of each editing command is XnA where:

X	Is a single letter editing command as described below.
n	Is a number in the range of 1 through 32000 (default is 1) which specifies the number of times that the command is to be executed. Alternatively, n may be specified as an asterisk (*), indicating that the given command is to be executed as many times (perhaps zero times) as possible for the displayed line.
A	Are the argument(s), if any, required for a given command.

As an example, consider the command C2/WX/YZ/, where C specifies the change command, 2 specifies that the change is to be performed twice, and /WX/YZ/ is the argument to the change command. This command searches the displayed title line for the first two occurrences of WX and replaces them with YZ.

The / characters used in this example are delimiters and are used to separate the two strings required by the change command. Valid delimiters are: . / $ % ! : ' " and &. Any one of these delimiters may be used as long as it does not occur in either of the specified strings. (The starting, middle, and ending delimiter characters must, of course, all be the same.)

Several edit commands can be strung together in a single editing command input line. Prior to executing each editing command input line, an internal edit pointer is initialized to point to column one of the line

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being edited. As each edit command of the command input line is executed, this pointer moves toward the right end of line being edited. In other words, the editing proceeds from left to right, with each new editing command in the command input line starting where the previous one ended. For example, the command C/AB/CD/C/WX/TY/ would cause the first occurrence of AB to be replaced with CD and the first occurrence of WX *after* the original AB to be replaced with TY.

Whenever an error condition is encountered in performing the operations specified in an editing command input line, an appropriate error message is printed and the line is unchanged.

The editing commands are:

C or S	The change (or substitute) command. This command searches for one or more occurrences of a given string and substitutes a second string for the original string. The format of this command is one of the following:
	
	`Cn/string1/string2/`
	`Sn/string1/string2/`
	
	where n and the / delimiters have been described above. Here, string1 is the original string and string2 is the string to be substituted. Note that string2 can be the null string (//), so that the change command can be used to delete one or more occurrences of a given string. Normally, n is not specified and only the first occurrence of string1 is changed.

F	The find command. This command moves the edit pointer to the position in the line being edited where a specific string is located. The format of this command is: `Fn/string/` where n and the / delimiters have been described above. This command moves the edit pointer to the first character of the nth occurrence (or last occurrence if an asterisk was specified) of string. Normally, n is not specified, so the find command moves the edit pointer to the first character of the first occurrence of string.
M	The move command. This command moves the edit pointer a specified number of character positions to the right. The format of this command is: `Mn` here n indicates the number of columns to move the edit pointer (relative to its current location). An asterisk is not allowed for n in this command.

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D	The delete command. Starting at the current position, this command deletes a specified number of characters. The format of this command is: `Dn` where n indicates the number of characters to be deleted.
I	The insert command. This command inserts a given string of characters into the title line, starting just before the current position. The format of this command is: `I/string/` where the / delimiters have been described above and string is the string of characters to be inserted. The edit pointer is positioned to the first location following the inserted string. Characters that are pushed past the end of the title line by the insert command are lost, and no error message is generated.
O or R	The overlay command. Starting at the current location, this command overlays (replaces) characters in the title line. The format of this command is one of the following: `O/string/ or R/string/` where the / delimiters have been described above and string is the string of characters that are to overlay (replace) characters in the displayed title line. The edit pointer is positioned to the first location following string.
A	The append command. This command appends a specified string to the title line, starting immediately after the last non-blank character in the line. The format of this command is: `A/string/` where the / delimiters have been described above and string is the string of characters that is to be appended. The edit pointer is positioned to the first location following the appended string.

Following are several examples of using the editing commands.

Example 1: Simple C(hange) commands.

Assume that a title line originally contains ABCADEF. The edit command C/B/XY/ changes it to AXYCADEF, while the command C2/A/MN/ results in a line of MNBCMNDEF. Note that the second change command could

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have been written as C*:A:MN:, where an asterisk is used instead of an explicit count and a different delimiter is being used.

Two (or more) change commands can be specified as one EDTR command. For example, assume that the title line originally contains ABCABCABC and that the edit command C/B/X/C/A/Q/ is given. The resulting line is AXCQBCABC. Note that the first A *was not* changed, because the second C(hange) command started where the first C(hange) command left off (and *not* at the beginning of the line).

Example 2: The F(ind), I(nsert), and D(elete) commands.

Assume that a title line originally contains ABCDEF. The edit command F/C/I/NEW/ results in the title line containing ABNEWCDEF. Note that the string NEW is inserted directly *before* the character that was located with the F(ind) command. Starting with the same original line, the edit command F/C/D2I/NEW/ results in the title line containing ABNEWEF, where the D(elete) command has removed the original CD characters.

Example 3: The O(verlay), A(ppend), and M(ove) commands.

Assume once more that the line originally contains ABCDEF. The edit command F/C/O/xxx/A/stuff/ results in a line containing ABxxxFstuff. Starting with the original line, the edit command M3O/bigstuff/ results in the title line ABCbigstuff. Note that this command would have failed if there were not enough room in the title line to hold the additional characters.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Editing the Two-Line Case Title or the Long Title

22.13. Listing Directory Contents

CATA

Run Line Mode Activity CATA - CLI
ACTIVITY? >>**CATA** ENTER OUTPUT DEVICE CODE:
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The directory listing activity CATA tabulates an alphabetical listing of the names of files contained in the current directory.

The names of all files in the current directory which appropriately contain a specified character string as a part of their name are listed, including imbedded asterisks (*) which are treated as wild characters matching zero, one or more characters.

Additional Information
PSS®E Command Line Interface (CLI) Users Guide, Directory Contents

22.14. Listing Saved Case or Snapshot Files

SHOW

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Run Activity SHOW - GUI
File>File information (SIZE/SHOW/BUSN)... *[File Information]* OPTION: List Save case and Snapshot files
Run Line Mode Activity SHOW - CLI
ACTIVITY? >>**SHOW** ENTER INPUT FILE NAME (0 TO EXIT, 1 FOR TERMINAL): >>
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The Saved Case and Snapshot File summary activity SHOW lists the case headings of Saved Case and Snapshot Files contained in the current directory. This list is printed to the *Progress* device (see Section 4.4, "Virtual Output Devices"). Separate lists of Saved Case or Snapshot Files may be specified.

Each file listed is followed by the report of activity SIZE if the file is a Saved Case File, or a tabulation of the number of dynamics data array elements stored in the file if it is a Snapshot File. In addition, on some host computers activity SHOW allows the specification of pathnames as well as simple filenames. The actual implementation of activity SHOW differs among the various host computers of PSS®E.

Additional Information
PSS®E GUI Users Guide, Listing Saved Case FilenamesListing Snapshot Filenames *PSS®E Command Line Interface (CLI) Users Guide*, Saved Case or Snapshot Filenames

22.15. Displaying Timing Statistics

TIME

Run Activity TIME - GUI
Misc>Reset timing statistics to zero (TIME,INIT) *Misc>Display timing statistics (TIME)*
Run Line Mode Activity TIME - CLI
ACTIVITY? >>**TIME, INIT** DAY, MMM DD YYYY H:M - TIMER INITIALIZED
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The timing statistics activity TIME allows the user to obtain execution time statistics during a PSS®E work session.

On subsequent executions of activity TIME it prints a summary of elapsed time, in seconds, since the previous execution of activity TIME, and cumulative times from the point at which the timers were last initialized. It

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also prints two additional system dependent timing statistics, at least one of which indicates CPU utilization. This tabulation, which is printed to the *Progress* device, is of the form:

```
FRI, MAY 23 2003  15:21   ELAPSE   heading2  heading3
  SINCE LAST "TIME"       XX.XXX    XX.XXX    XX.XXX
  CUMULATIVE              XX.XXX    XX.XXX    XX.XXX
```

Additional Information
PSS®E GUI Users Guide, Overview, Displaying and Resetting Timing Statistics *PSS®E Command Line Interface (CLI) Users Guide*, Timing Statistics

22.16. Inserting Comments into the Progress Output

TEXT

Run Activity TEXT - GUI
Misc>Insert text into the Progress stream (TEXT) *[Insert Text into the Progress Stream]*
Run Line Mode Activity TEXT - CLI
ACTIVITY? >>**TEXT,message>**
Interrupt Control Codes
POM-Data-ReadingPowerFlowDataAdditionsFromTheTerminal

The null activity TEXT doesn't manipulate power flow data. However, it does provide a mechanism by which the user may insert descriptive comments in a Response File at any point at which the next activity to be executed is specified. In addition, it enables the user to insert comments in the *Progress* output of the PSS®E interface and into a recording file, if one is being used.

Additional Information
PSS®E GUI Users Guide, Inserting Comments into the Progress Output *PSS®E Command Line Interface (CLI) Users Guide*, Inserting Comments into the Progress Output

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Chapter 23
Dynamic Simulation Setup and Procedures

The user of PSS®E is able to run dynamic simulations when, after opening PSS®E, the user selects the dynamics tab. This will give user access to various dynamic simulation activities. The individual dynamic simulation activities are described in detail in Chapter 15 of this manual. The purpose of this section is to describe the sequencing of these activities in constructing a valid system model, and the use of this model in simulation calculations.

23.1. Simulation Overview

The objective in performing a dynamic simulation of a physical system is to determine the response of the system to prescribed stimuli. The process of dynamic simulation is quite straightforward conceptually.

At an instant in time T, it is known "where you are". From "where you are" and the differential equations describing the behavior of the system, it can be determined "where you're going". Then "go there", advance time to T+t (where t is the time step or integration step), and run it again.

Stated in slightly more formal terms, the behavior of a system is described by a set of differential equations. At every time step of the simulation, the time derivative of each state variable in the system is calculated, with the constant and variable parameters, which describe the condition of the system at that time instant in detail, as initial conditions. From the present value of each state variable and its rate of change (i.e., its time derivative), the state variable values at the next time step are determined. Simulation time is advanced and the process is repeated.

23.1.1. Program Structure

The user of the dynamic simulation section of PSS®E requires a working knowledge of the dynamics modeling structure. The principal steps of dynamic simulation as applied to electric power systems are shown in Figure 23.1, "Dynamic Simulation Basic Logic Flow". The dynamic simulation procedure outlined above is complicated by the presence of the electrical network which is described by a large set of simultaneous algebraic equations.

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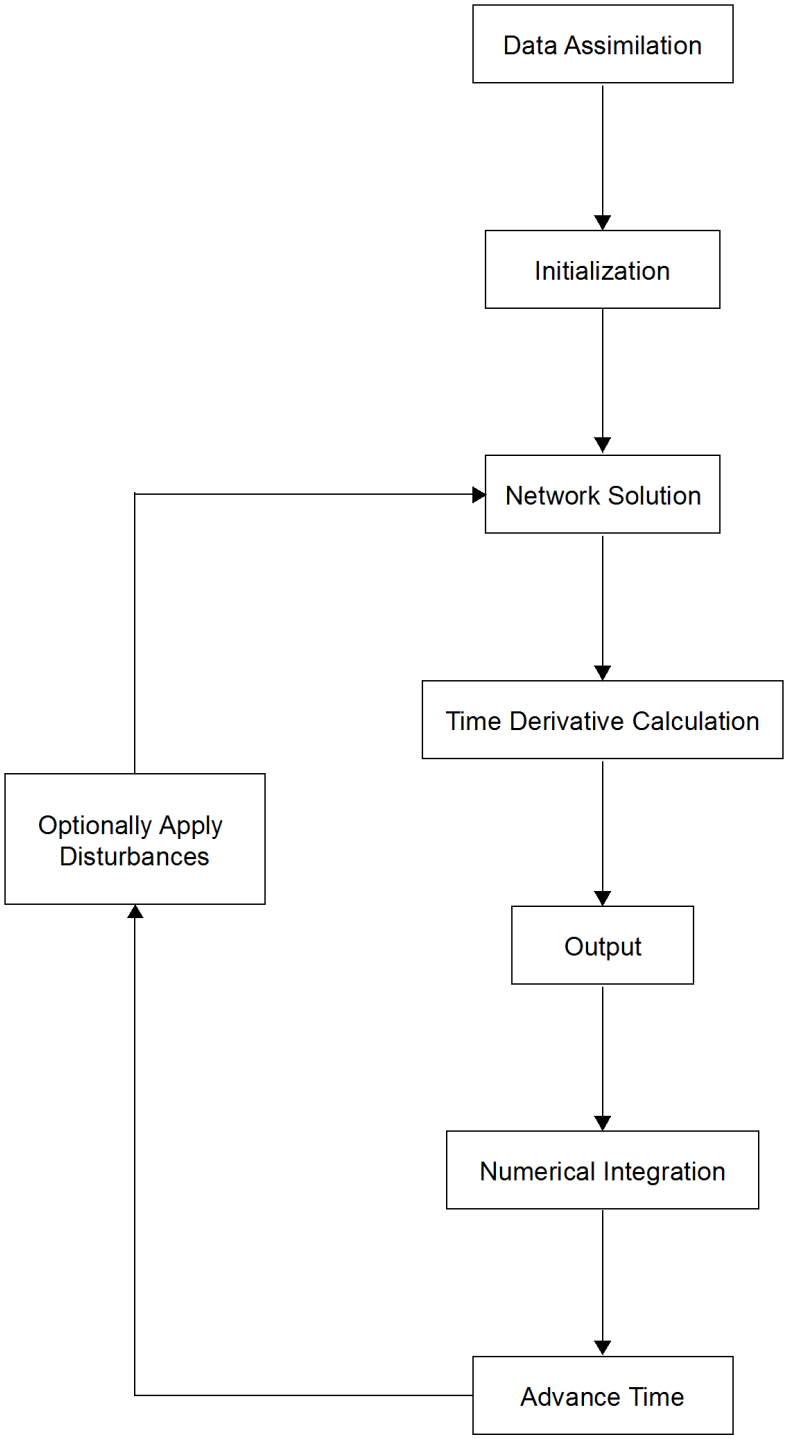

Figure 23.1. Dynamic Simulation Basic Logic Flow

The activities which initialize a dynamic simulation (activity STRT) and calculate the instantaneous state of the system at each time step (activity RUN) contain the basic elements of a general purpose dynamic simu-

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lation calculation (e.g., numerical integration, time stepping, output). The following calculation phases are required to extend this dynamic simulation control structure to that required for the simulation of electric power systems:

1. The solution of the electric network given the machine internal flux linkages and the load boundary conditions.

2. The calculation of the time derivative of each state variable used in modeling equipment, given the present values of all state variables and of all generator stator currents. This phase includes calculating the values of all algebraic variables needed in the course of obtaining numerical values of the state variable time derivatives.

3. The modeling of equipment in which there is an algebraic relationship between the voltage at a bus and the current drawn by, or key variable n by, the device. These include such devices as induction motors, relays, meters and certain loads.

The requirement given in (1) above is handled by the subroutine SITER which is supplied as a part of PSS®E. The equipment modeling requirements of (2) and (3) above are satisfied through the use of specialized equipment model subroutines, each of which contains the coded equations for modeling one specific type of equipment. The PSS®E Model Library contains a large number of such models. In addition, PSS®E is structured such that users may develop and use their own model subroutines whenever there is no standard library model which accurately models a given item of equipment (see Chapter 25, *Model Writing*).

After equipment models for the system under study are selected, the user must supply the intelligence connecting these models to the machines, buses, dc lines, and other equipment in the electric network. This information is imbedded within ten subroutines:

TBLCNC	Is supplied as a part of PSS®E and is responsible for the requirements given in (2) above for machines, their control systems, loads, auxiliary signal models, FACTS devices, and dc lines. This subroutine utilizes a set of tables which associate models and their data with machines, loads, FACTS device, dc lines, and switched shunt in the working case.
CONEC	Is responsible for the state variable requirements given in (2) above for all equipment in the network that are not in the table-driven form.
TBLCNT	Is supplied as part of PSS®E and is responsible for the requirements given in (3) above for machines, loads, load relays, line relays, FACTS devices, dc lines, and switched shunt. This subroutine utilizes a set of tables that associate their data with machines, loads, line relays, FACTS devices, dc lines, and switched shunt in the working case.
CONET	Is responsible for the modeling of the network-related equipment described in (3) above for all equipment in the network that are not in the table-driven form.
USRXXX	Is responsible for calling user-written plant and wind related models. Calls to USRXXX are initiated from TBLCNC and TBLCNT. Staring PSS®E version 31, the presence of this subroutines is not mandatory. If the USRXXX subroutine is not present, PSS®E has the pro-

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	vision to call user written plant and wind related models directly from TBLCNC and TBLCNT.
USRLOD	Is responsible for calling user-written load-related models. Calls to USRLOD are initiated from TBLCNC and TBLCNT. Staring PSS®E version 31, the presence of this subroutines is not mandatory. If the USRLOD subroutine is not present, PSS®E has the provision to call user written load-related models directly from TBLCNC and TBLCNT.
USRREL	Is responsible for calling user-written line relay models. Calls to USRREL are initiated from TBLCNT. Staring PSS®E version 31, the presence of this subroutines is not mandatory. If the USRREL subroutine is not present, PSS®E has the provision to call user written line relay models directly from TBLCNT.
USRAUX	Is responsible for calling user-written auxiliary signal models that could be associated with dc line and FACTS device models. Calls to USRAUX are initiated from TBLCNC. Staring PSS®E version 31, the presence of this subroutines is not mandatory. If the USRAUX subroutine is not present, PSS®E has the provision to call user written auxiliary signal models directly from TBLCNC.
USRFCT	Is responsible for calling user-written FACTS device models. Calls to USRFCT are initiated from TBLCNC and TBLCNT. Staring PSS®E version 31, the presence of this subroutines is not mandatory. If the USRFCT subroutine is not present, PSS®E has the provision to call user written FACTS device models directly from TBLCNC and TBLCNT.
USRDCL	Is responsible for calling user-written dc line models. Calls to USRDCL are initiated from TBLCNC and TBLCNT. Staring PSS®E version 31, the presence of this subroutines is not mandatory. If the USRDCL subroutine is not present, PSS®E has the provision to call user written dc line models directly from TBLCNC and TBLCNT.

It is the user's responsibility to supply the subroutines CONEC and CONET if there are user model calls in these two subroutines. While subroutines CONEC and CONET bear modeling responsibility, often they contain no user-written code at all, but are simply a sequence of FORTRAN CALL statements connecting models from the PSS®E Model Library to network components. Presence of USRXXX, USRLOD, USRREL, USRAUX, USRFCT, and USRDCL subroutines as explained above is not mandatory.

23.1.2. Data Structure

The types of data involved in dynamic simulation calculations may be separated into four categories:

- *Constants* are parameters that do not vary during the simulation.
- *State variables* are quantities for which the instantaneous values are determined by differential equations.

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- *Algebraic variables* are quantities for which the values at any instant can be determined if all state variables and constants are known.
- *Input variables* are quantities for which the values at any instant are specified by logic outside of the dynamic simulation.

The PSS®E dynamic simulation data structure contains a group of named arrays, related to buses, machines and loads, that comprise quantities that are frequently needed as input and output in setting up and displaying the results of dynamic simulations. These include arrays such as VOLT, BSFREQ, MBASE, PELEC, EFD, TPLOAD etc. PSS®E supplied dynamic models as well as user-written models gain access to these quantities through the bus, machine, load, and branch numbering sequences.

In addition, there are four large general purpose storage arrays:

CON	Contains constants.
STATE	Contains state variables.
VAR	Contains algebraic variables.
ICON	Contains integer quantities which may be either constants or algebraic variables. Corresponding to each ICON is an entry in the CHRICN array which may contain character quantities.

These four general purpose arrays are treated as large storage bins. Each reference to a model from the PSS®E Model Library and to a user-written model may be assigned a small contiguous block of locations in one or more of these arrays. The block of consecutive storage locations in these arrays is allocated for each model type (i.e., plant-related model, load related model etc.) on a first come, first served basis.

The filling of the model connection tables used by PSS ®E supplied as well as user defined models, then, requires enumeration of the CONs, STATEs, VARs and/or ICONs used for each model reference. The number of locations that would be allocated in each of these arrays for all PSS®E supplied models is shown in data sheets; refer to PSS®E Model Library.

Although the user must understand this process, the bookkeeping chore of assigning blocks of storage locations for each model reference is handled by the PSS®E activities DYRE, CHAN, and CHSB, and by the *Add/Edit constants* Add/Replace functions. These activities also construct the model connection tables and the connection subroutines (CONEC and CONET), and transfer the constant parameters associated with each model call into the CON and ICON blocks assigned to the model.

Table 23.1, "Dynamic Simulation Arrays" tabulates the principal arrays used in the dynamic simulation section of PSS ®E.

Table 23.1. Dynamic Simulation Arrays

Array	Contents	Indexed By
Constants:		
CON	General constants (real)	CON number
ICON	General constants (integer)	ICON number
CHRICN	General constants (character) (maximum 12 characters)	ICON number
MBASE	Machine base MVA	Machine index
ZSORCE	Machine impedance (complex)	Machine index

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Array	Contents	Indexed By
Constants:		
XTRAN	Step-up transformer impedance (complex)	Machine index
GENTAP	Step-up transformer tap ratio	Machine index
State Variables:		
STATE	General state variable array (state-space)	STATE number
Algebraic Variables:		
VAR	General algebraic variable array	VAR number
VOLT	Bus pu voltages (complex)	Bus sequence number
BSFREQ	Bus pu frequency deviations	Bus sequence number
ANGLE	Machine relative rotor angle (degrees)	Machine index
PELEC	Machine electrical power (pu on SBASE)	Machine index
QELEC	Machine reactive power (pu on SBASE)	Machine index
ETERM	Machine terminal voltage (pu)	Machine index
EFD	Generator main field voltage (pu)	Machine index
PMECH	Turbine mechanical power (pu on MBASE)	Machine index
SPEED	Machine speed deviation from nominal (pu)	Machine index
XADIFD	Machine field current (pu)	Machine index
ECOMP	Voltage regulator compensated voltage (pu)	Machine index
VOTHSG	Stabilizer output signal (pu)	Machine index
VUEL	Minimum excitation limiter output signal (pu)	Machine index
VOEL	Maximum excitation limiter output signal (pu)	Machine index
TPLOAD	Effective MW load (pu on SBASE)	Load index
TQLOAD	Effective Mvar load (pu on SBASE)	Load index
DC2SIG	Auxiliary-signal input for 2-terminal dc line models	auxiliary signal index and 2-terminal dc line index
DCMSIG	Auxiliary-signal input for multi-terminal dc line models	auxiliary signal index and multi-terminal dc line index
VSCSIG	Auxiliary-signal input for VSC dc line models	Auxiliary signal index and VSC index
FCTSIG	Auxiliary-signal input for FACTS device models	Auxiliary signal index and FACTS device index
VRFDBK	AVR rate feedback output	Machine Index
VFEOUT	AVR VFE signal	Machine index

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Algebraic Variables:		
WVLCTY [a]	Wind velocity, m/s	Wind machine index
WTRBSP *	Wind turbine rotor speed deviation (pu)	Wind machine index
WPITCH *	Pitch angle (degree)	Wind machine index
WAEROT *	aerodynamic torque (pu on MBASE)	Wind machine index
WROTRV *	Rotor voltage (pu on MBASE)	Wind machine index
WROTRI *	Rotor current (pu on MBASE)	Wind machine index
WPCMND *	Active power command from wind control (pu on MBASE)	Wind machine index
WQCMND *	Reactive power command from wind control (pu on MBASE)	Wind machine index
WAUXSG *	Output of wind auxiliary control (pu on MBASE)	Wind machine index
WROTVD *	Rotor voltage d-component (pu on MBASE)	Wind machine index
WROTVQ *	Rotor voltage q-component (pu on MBASE)	Wind machine index
WROTID *	Rotor current d-component (pu on MBASE)	Wind machine index
WROTIQ *	Rotor current q-component (pu on MBASE)	Wind machine index
WIPCMD *	Active current command from wind control (pu on MBASE)	Wind machine index
WIQCMD *	Reactive current command from wind control (pu on MBASE)	Wind machine index
WEQCMD *	q-axis voltage command from wind control (pu on MBASE)	Wind machine index
Input Variables:		
VREF	Voltage regulator voltage setpoint (pu)	Machine index
GREF	Turbine governor reference	Machine index
LCREF	Turbine Load Control reference	Machine index
SWSREF	Switched shunt reference	Switched shunt index
Internal Arrays:		
DSTATE	General state variable time derivatives (state-space)	STATE number
STORE	General state variable integrator memory	STATE number
STORMT	General memory (extended term)	2 * STATE number - 1 and 2 * STATE number
CURNT	Network solution current injection (complex)	Bus sequence number

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Internal Arrays:		
BSFMEM	Memory for frequency calculation	Bus sequence number
STRTIN	Starting array indices for plant-related models	Array allocation table index
NUMTRM	Pointer to bus sequence number	Machine index
NUMBUS	External bus number	Bus sequence number
MACHID	Machine identifier	Machine index
WNDNUM	0 if the machine is not a wind machine, else contains the wind machine index	Machine index
WSTRTIN	Starting array indices for wind-related models	Wind array allocation table index
LDSTRT	Starting shared data array indices for load-related models	Load array allocation table index 1
LDSTR2	Starting private data array indices for load-related models	Load array allocation table index 2
NUMLOD	Pointer to bus sequence number	Load index
LOADID	Load identifier	Load index
RLSTRT	Starting array indices for line relay models	Line relay array allocation table index
RINCNX	Branch relay model connection table index	Double entry branch index
INTICN	Integer memory array	ICON number
STRTAU	Starting array indices for auxiliary-signal models	Auxiliary-signal array allocation table index
STRTFC	Starting array indices for FACTS device models	FACTS device connection index
STRT2D	Starting array indices for two-terminal dc line models	two-terminal dc line connection index
STRTMD	Starting array indices for multi-terminal dc line models	multi-terminal dc line connection index
STRTVS	Starting array indices for VSC dc line models	VSC dc line connection index
FCTCNX	FACTS device connection table index	FACTS device index
DC2CNX	Two-terminal dc line connection table index	Two-terminal dc line index
DCMCNX	Multi-terminal dc line connection table index	Multi-terminal dc line index
VSCCNX	VSC dc line connection table index	VSC dc line index
STRTSS	Starting array indices for switched shunt models	Switched shunt connection index
STRTCCT	Starting array indices for CCT type models (the family of models that are called CCT type models are as	CCT array allocation index

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Internal Arrays:		
	follows: branch device and other models, 2-winding transformer device and other models, 3-winding transformer device, protection and other models, bus protection and other models, machine protection and other models, 2-terminal dc line pro-taction and other models, n-terminal dc line protec-tion and other models, VSC dc line protection and other models, FACTS device protection and other models, switched shunt protection and other mod-els, induction machine protection and other models, and miscellaneous other models)	
NUMSWS	Bus Sequence Number	Switched Shunt Index

[a]Can be used only for wind machine models (i.e., models for machines that are designated as wind machines in power flow).

23.2. Linkage to Power Flow

PSS®E allows the user to execute power flow activities (i.e., those activities described in Chapter 6, *Power Flow Solution Activity Descriptions*) at any point during a dynamic simulation run. This is accomplished by selecting the network tab on the PSS®E GUI.

The interface elements for the Dynamics study elements of PSS®E are contained directly within the application. Switching between the two modes of analysis, steady-state and stability, is done automatically, there is no need to worry about the mode of operation.

Moving from the dynamics to the power flow window has the effect of transferring the generator power outputs, PELEC and QELEC of Table 23.1, "Dynamic Simulation Arrays", into the plant and machine output arrays in the working case, and making the power flow related menu items accessible.

The return back to the dynamic simulation window is accomplished by selecting the dynamics tab on the PSS®E GUI. Moving from the power flow to dynamics window has the effect of setting up the translation arrays NUMTRM, NUMLOD, RINCNX, FCTCNX, DC2CNX, DCMCNX, VSCCNX etc. (see Table 23.1, "Dynamic Simulation Arrays"), updating the plant-related, load-related, line relay model, FACTS device, VSC (Voltage Source Converter) dc line connection tables etc., (see Section 23.1.1, "Program Structure" and Section 23.4, "The Dynamic Model"), updating the output channel addresses (see Section 23.6, "Simulation Outputs"), and transferring control back to dynamic simulation activity selection.

The principal applications of the power flow linkage are to:

1. Set up the power flow working case and to establish the admittance matrix temporary file (if the factored matrix is not constructed prior to running the dynamic simulation, PSS®E will run the factorization).

2. Allow examination of the network conditions existing at any instant of a simulation.

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The required sequencing of PSS®E activities in these two applications are given in Section 23.3, "The Network Model" and Section 23.7.2, "Applying Disturbances", respectively.

Care must be taken while the power flow linkage is in effect not to modify the power flow working case unless a system disturbance or reinitialization of the dynamic simulation is intended. The power flow output reporting activities and activity SAVE may be used freely without risk of disturbing the simulation. Activities such as CASE, CHNG, DSCN, and SCAL should be used with care in the midst of a simulation run. Additional comments are given in Section 23.7.2, "Applying Disturbances".

23.3. The Network Model

Every dynamic simulation is based upon a power flow Saved Case that provides it with the required transmission network data, load data, machine positive sequence characteristic data (i.e., MBASE, ZSORCE, XTRAN and GENTAP), and the initial network condition from which the dynamic models are initialized. Thus, the dynamics data structure with its corresponding connection subroutines described in Section 23.1.2, "Data Structure" is augmented by the power flow working case to form a system model of the form required for dynamic simulations.

The power flow Saved Case must be present in the working case and be converted from its standard power flow form to the form required for dynamic simulations. In power flow, most generators are represented by specifying their desired bus voltage magnitudes and active power outputs; their voltage phase angles and reactive power outputs, within specified limits, are free variables which are determined by the power flow solution activities. In dynamics, generators are represented by their Norton equivalents as current sources. This conversion of machines from their power flow model to their Norton equivalent is handled by activity CONG. The determination of the initial source current (ISORCE) by activity CONG is based upon the machine power outputs and bus voltages, along with the impedance data in the working case characterizing each machine. Activity CONG converts all power flow Type 3 (swing) buses to Type 2.

The constant MVA load representation used in conventional power flow solutions is usually not an acceptable load characteristic for dynamic simulations. The basic conversion of loads from their constant MVA representation to an arbitrary combination of constant MVA, current and admittance is handled by activity CONL. The modeling of loads beyond this level is handled by applying models from the library of load characteristic models.

It is then recommended that activities FACT and TYSL be executed to fine tune the solution, and that the converted case be preserved in a Saved Case File. The file specified to activity SAVE *should not* be the same one from which the original (unconverted) case was picked up because the use of activity CONG renders the working case and its Saved Case image invalid for the standard power flow solution activities, and activity CONG is not reversible.

To summarize, then, the following sequence should be followed in setting up the converted network model for use in dynamic simulations:

1. Pick up a standard power flow case with activity CASE.

2. Using the power flow solution activities, solve the working case to an acceptable mismatch level. The bus boundary conditions should represent the desired initial network condition from which the dynamic simulation is to be initialized.

3. Ensure that the machine characteristic data MBASE, ZSORCE, XTRAN and GENTAP are properly entered for all machines in the working case (see Generator Data and activities MCRE and CONL).

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4. Preserve the unconverted initial network condition in a power flow Saved Case File with activity SAVE.

5. Execute activities CONG and CONL.

6. Optionally, execute activities FACT and TYSL. Although not essential, this step is recommended.

7. Save the converted case in a Saved Case File other than the one specified in (4).

After the converted Saved Case is created, it can then be accessed each time it is needed in constructing the dynamics model as described in Section 23.4, "The Dynamic Model" and in executing dynamic simulation calculations as described in Section 23.7, "Running a Simulation".

23.4. The Dynamic Model

From the preceding paragraphs of this section, then, the system model for dynamic simulation calculations in PSS®E is comprised of:

1. The power flow working case, giving a positive sequence representation of the electric network. A standard power flow case must be augmented by the machine impedance characteristic data MBASE, ZSORCE, XTRAN and GENTAP, and the load and generator representations must be converted from their power flow modeling to that required by the dynamic simulation activities. Refer to Section 23.3, "The Network Model" and Section 6.3.13, "Load", Generator Data, and activities MCRE, CONG, and CONL.

2. Dynamics data working memory which includes: (1) the constants associated with equipment models from the PSS®E Model Library and user-written models which are to be used in modeling the system; and (2) the model connection table arrays defining the model connection and data array entries associated with plant-related, load-related, line relay, auxiliary signal, FACTS devices, dc line, and switched shunt models. Refer to Section 23.1.1, "Program Structure" and Section 23.1.2, "Data Structure".

3. A set of connection subroutines, CONEC and CONET, which link together equipment models and their data with network elements.

The remainder of this section describes the development of the second and third items listed above, and the incorporation of the connection subroutines into the PSS®E program structure.

23.4.1. Subroutines TBLCNC and CONEC

As described in Section 23.1.1, "Program Structure", subroutines TBLCNC and CONEC have the principal responsibility of processing those equipment models involving state variables and differential equations. The majority of such models are those used in the modeling of generating plant equipment (generators, power system stabilizers, compensators, excitation limiters, excitation systems, and turbine governors), some types of load characteristic models, auxiliary signal models, FACTS device models, dc line models, and switched shunt models, and are called through subroutine TBLCNC.

The form of each model reference in these subroutines is that of a FORTRAN CALL statement referencing the model name and an argument list. The arguments generally specify the location of the model in the network and the starting locations in the general purpose storage arrays CON, STATE, VAR and/or ICON of the block of storage locations reserved for that model reference.

Although implemented differently, plant-related models, load characteristic models, auxiliary signal models, FACTS device models, dc line models, and switched shunt models called through subroutine TBLCNC receive

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the same information via their argument lists whenever they are called; see Section 25.5, "Writing the Basic Model" for details.

The construction of the model connection tables for subroutine TBLCNC and of subroutine CONEC, then, involves the following steps:

1. A model must be selected from the *PSS®E Model Library* for each piece of equipment to be represented in the simulation, and the appropriate constant parameters required must be recorded.
2. Storage locations in the general purpose storage arrays must be assigned and numeric values must be determined for all subroutine arguments for each model selected in (1).
3. A model connection table entry must be made for each TBLCNC model reference.
4. The subroutine CONEC model calls must be collected into the form of FORTRAN subroutines.
5. Each model's constant parameters must be placed into its CON and/or ICON locations assigned in (2).

Although steps 2, 4 and 5 above could be done manually by the user, this process would be both time consuming and error prone. Rather, the PSS®E activity DYRE is normally used to accomplish the bookkeeping and data handling functions in 2 through 5. The use of activity DYRE is further discussed in Section 23.4.3, "Data Assimilation ".

23.4.2. Subroutines TBLCNT and CONET

Subroutines TBLCNT and CONET contain two general types of models:

1. Those for which there is an algebraic relationship between the voltage at a bus and the current drawn by the device being modeled, and other models which inject a current at each iteration of the network solution.
2. Those for which input or output is a quantity determined solely by network quantities, such as relays and monitoring models.

The construction of these subroutines involves the same type of approach as in the construction of subroutines TBLCNC and CONEC described in the previous section. The majority of CONET models are processed by activity DYRE as described in the following section.

There is a group of models (e.g., static VAR system models, and certain load characteristic models) which involve both state variables and network current injections. Such models are handled by a coordinated model pair in which there is a call in subroutine TBLCNC or CONEC to process the differential equation part of the modeling, and a call in subroutine TBLCNT or CONET to handle the network current injection. For example, when the series reactor model CRANI is called from subroutine CONEC, its CONET counterpart, TRANI, must be called from subroutine CONET with the same set of arguments.

23.4.3. Data Assimilation

Activity DYRE reads data records from a Dynamics Data Input File, and in its processing performs the following functions:

1. It assigns storage locations in the general purpose arrays for each model reference.
2. It places the constant parameters associated with each model reference into the assigned locations in the CON and/or ICON arrays.

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3. It builds the model connection table arrays for plant-related, load-related, line relay, FACTS device, dc line, and switched shunt models.

4. It builds the CONEC and CONET subroutines containing the model calls with their appropriate argument lists for each remaining model.

The user should first create a Dynamics Data Input File of the form described in Section 16.1.1, "Dynamics Model Raw Data File Contents". One record should be entered for each PSS®E Model Library reference and each user-written model reference used in modeling the system.

The following points should be noted in constructing this file:

1. Each record must have the model name enclosed in single quotes and *must* be terminated with a slash.

2. A record may occupy more than one physical line in the file as long as only the final line of the record is ended with the slash.

3. Data records may appear in any order.

4. Any equipment which is modeled by a coordinated call model pair (e.g., CDC4T / TDC4T and CSVGN1/ TSVGN1) should have only one record entered for it with the TBLCNC or CONEC model name specified.

After a Dynamics Data Input File has been created, the following procedure is used:

1. Initiate PSS®E.

2. Retrieve the converted power flow Saved Case representing the network to be modeled.

3. Read dynamics data file, and preserve the dynamics working memory in a snapshot file. Preserve dynamics working memory in a Snapshot File and terminate PSS®E.

4. Incorporate the connection subroutines into the PSS®E program structure as described in Section 23.4.4, "Program Set-Up".

Error messages tabulated during the execution of activity DYRE are displayed at the *Progress* tab. Any such messages should be resolved prior to proceeding with further. This may involve an editing operation on the data input file to correct typographical errors or to add data records to the file, and a subsequent re-execution of activity DYRE.

The progress report output of activity DYRE and the consistency check may be preserved in hard copy form and retained with the user's study notes. This may be done either via activity PDEV, the computer system's session logging facility, or a cut and paste operation from the *Progress* tab of the PSS®E interface.

Those models which are not automatically processed by activity DYRE (see Section Section 16.1.5, "Application Notes") are either set up by activity CHAN (see Section Section 16.2.2, "Application Notes" and Section 23.6, "Simulation Outputs"), they may be treated as User-Written Models by activity DYRE, or their calls may be manually entered into the connection subroutines by the user (see Section 23.9, "Advanced Uses of CONEC and CONET ").

23.4.4. Program Set-Up

Prior to compiling the connection subroutines, the user may add any desired control logic to them; this is discussed in more detail in Section 23.9, "Advanced Uses of CONEC and CONET ". There is however a

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situation in which the connection subroutines as generated by activity DYRE *require* statement insertion by the user. It occurs when dc lines are being modeled (using one of the following models that are still not in the table-driven form: CDCRL, CASEA1, CDCVUP, CEELRI, CMDWS2, CMFORD, CHIGAT, CMDWAS) and one or more of the dc line supplementary signal models are being used. The insertion involves placing the auxiliary signal into the *Other signals VAR*, VAR(L), of the dc line model. The procedure is illustrated by example.

Suppose dc line 1 is sensitive to a supplementary signal which is formed by summing the outputs of a pair of PAUX1T models. Suppose the supplementary signal model PAUX1T model stores the model output in VAR(9) and VAR(20). Note that because PAUX1T is a table-driven model, activity DYRE does not generate PAUX1T model calls in CONEC. Suppose the dc line is modeled using the dc line model CMFORDT. Because CMFORD model is still not in the table-driven form, activity DYRE will generate CMFORD model call in subroutine CONEC. A portion of subroutine CONEC as constructed by activity DYRE appears as follows:

```
      CALL CMFORD  (    1,  206,   73,   31,
 3)
C
      RETURN
```

Referring to the CMFORD data sheets (see *PSS®E Model Library*, Generic Renewable Plant Control Models), the fifth argument in the CMFORD calling sequence is the index of the first VAR used by it, and this VAR contains the supplementary signal used as input to the model.

Thus, immediately preceding the call of the CMFORD model, the following statement must be inserted:

```
VAR(3) = VAR(9) + VAR(20)
```

If extended term simulations are to be performed (see Section 23.10, "Extended Term Simulations"), the supplementary signal model output and the setting of the dc line model's *Other signals VAR* must be brought over into subroutine CONET as well. Using the above example, the following code fragment results when the above code is inserted before the TMFORD model call in subroutine CONET corresponding to the CMFORD call in subroutine CONEC:

```
   IF (MIDTRM)
   .  VAR(3) = VAR(9) + VAR(20)
   ...FIN
C
   CALL TMFORD  (    1,  206,   73,   31,    3)
```

With valid connection subroutines prepared, the user must compile them and link the resulting object code with the rest of PSS®E. This procedure is described in Section 4.10, "Loading the Connection Subroutines".

23.5. Model Verification

Having gone through the process of collecting data and feeding it through activity DYRE into PSS®E dynamics working memory, prior to executing simulation runs the user must ensure himself that the system model thus constructed faithfully models the physical power system. A model setup with unchecked and unreliable data usually produces inaccurate results in system simulations.

There are thousands of data parameters from a wide variety of sources used in a large scale system simulation study. The checking of each of these for accuracy is no easy task. PSS®E provides for three levels of model verification:

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1. Data parameters are checked for reasonableness. This detects many simple errors such as transcription errors.

2. The performance of generating units are tested in hypothetical isolated operating conditions where the effects of excitation and governor action are not masked by the electrical synchronization effect, but are clearly and unambiguously visible.

3. The initialization of the simulation at the predisturbance network operating condition is checked for model variables initialized outside of prescribed limits and for steady-state operation.

The following paragraphs discuss these levels of data verification in greater detail.

23.5.1. Parameter Range Checking

Activity DOCU provides documentation of the dynamic simulation setup. For each model, it tabulates the equivalent of a filled in model data sheet. This consists of a listing of the constant data read from the Dynamics Data Input File along with a statement of the locations assigned in the general purpose storage arrays by activity DYRE.

However, critical examination of this report for a large system is difficult because of its volume. It is unlikely that an engineer studying this report would find all errors and it is therefore not a practical way of locating errors.

Through its interactive dialog, activity DOCU may be operated in a mode which provides an initial editing of the report and restricts it to those models for which the values of one or more constant parameters are suspect. Parameters are checked against typical ranges of values. In addition, certain relational checks between parameters are made (e.g., machine leakage reactance must be less than sub-transient reactance). The suspect parameters are tabulated followed by the standard model output block.

It must be noted that being flagged by activity DOCU in its data checking mode does not indicate that parameter values are necessarily wrong; it simply indicates that they deserve to be checked. Conversely, the absence of warning messages from activity DOCU in this mode *does not* indicate that all data is valid. Most power system parameters can fall within reasonable typical value ranges and still be totally invalid.

Thus, the data checking mode of activity DOCU is intended to detect gross errors such as misplaced decimal points and as such provides only a rough initial screening for typing and stenographic errors.

As a final point, the preprogrammed typical parameter ranges are set on the assumption that plant equipment data is specified on actual machine base MVA. The practice of specifying all machine data on a common base (say, 100 MVA) is discouraged because this often results in the spurious flagging of generator and governor models in the data checking mode of activity DOCU, thus partially defeating its report editing intent. It is strongly recommended, then, that the machine quantity MBASE in the power flow, on which machine model data is specified, be set to the actual machine base MVA for each machine, and that model parameters be specified accordingly.

23.5.2. Performance Verification

The proper criterion for judging the correctness of generator, excitation system, and turbine governor data is not that all parameter values lie within typical ranges, but rather that the parameter values modeling each item of equipment correspond to correct performance of the unit under test conditions. Therefore, after data has been checked for gross errors using the reasonable data check option of activity DOCU, the user should verify that the sets of parameter values in the system model lead to:

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1. Correct steady-state values of all quantities that are normally measured or documented in normal operation of a generating unit.

2. Correct reproduction of dynamic response tests that may be performed on generating units and their control systems in isolation.

The following paragraphs describe the tools available to the PSS®E user for assessing the correctness of a system model.

Generator Reactances and Saturation Data

The PSS®E machine models require the specification of unsaturated synchronous machine reactances together with two points on the machine's open circuit saturation curve.

Generator saturation data is often among the least checked of the constant parameters describing a power system. The correct specification of the saturation data is necessary in order for the generator field voltage, EFD, to take on its correct value at all loadings. The correct initial value of EFD is of major importance in any simulation run in which the excitation system ceiling is a significant parameter (e.g., in virtually all transient stability runs). Any error in the initial value of EFD in such simulations produces a corresponding error in the maneuvering range available to raise the field flux linkages while the exciter is at ceiling during and following a fault. Because this field forcing effort is known to be a key factor in determining transient stability limits, a correct initial value of EFD is just as important as correct specification of excitation ceiling.

The best form of verification of generator reactances and saturation data is the machine V-curves which plot generator terminal current versus excitation voltage, EFD, over its whole operating range. These curves are commonly provided by generator manufacturers along with open circuit saturation curves as standard design documentation.

The PSS®E auxiliary program VCV produces such a plot for a specified set of machine reactances and saturation data. It also tabulates the steady-state excitation voltage corresponding to a given loading. The use of program VCV is described in *Additional Resources for PSS® E*, VCV.

Exciter Response Ratio Test

The data describing the rotating exciter may be checked by simulating an exciter response ratio test. In this test, the exciter is set up to run at the output voltage and current corresponding to rated main generator operating point. The voltage regulator setpoint is then raised suddenly by a large amount to drive the exciter to its ceiling as rapidly as possible.

The procedure for carrying out this test is as follows:

1. Initiate PSS®E.

2. Restore the Snapshot containing the dynamics data of the system being modeled (activity RSTR).

3. Retrieve the corresponding converted power flow Saved Case. Note that the generators must be converted. However, there is no need to factorize the matrices (i.e., activity FACT is not required) because the transmission network is ignored in this test, with all machines operating in isolation.

4. From the menu, select the option to perform exciter response ratio simulation. After initializing, run the response ratio simulation to zero seconds (specifying TPAUSE equal to 0). Then continue the simulation to TPAUSE set at 2 seconds.

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The response ratio test is executed simultaneously on either all units in the system or all machines at a designated bus. Any user-specified output channels (see Section 23.6, "Simulation Outputs") are overridden and the exciter outputs, EFD, of all online machines being tested are placed in output channels.

Activity ESTR overrides the initial generator loadings as contained in the working case and initializes each unit to rated MVA at a user-specified power factor. Following completion of activity ESTR, it is advisable to check the value of EFD for each machine to ensure that it is as determined by program VCV or from generator test data for rated load. An unacceptable value of full load excitation usually indicates an erroneous specification of the *generator* saturation data; all generator data, in particular SE(1.2), should be checked before proceeding.

Activity ERUN reveals both the ceiling value of EFD and the excitation system response ratio for each unit. The user should examine these values and adjust the excitation loop data if they do not correspond to the known ceiling and response ratio.

Because the machine initialization to rated MVA is based on the value specified as MBASE for each machine, the response ratio test is valid only for those machines where data is specified on their actual MVA base. For operational and additional application details, refer to activities ESTR and ERUN.

Excitation System Open Circuit Step Response Test

The response ratio test described above provides information on the modeling of exciters but gives no indication of the correctness of the voltage regulator gains and time constants. These are checked by ensuring that the excitation system gives stable and effective control of generator terminal voltage when the machine is operating at rated speed on open circuit (e.g., just before synchronizing).

This aspect of excitation system performance is readily tested by applying a simple step change of about five percent to the voltage regulator reference and observing the resulting responses of field voltage (EFD) and generator terminal voltage (ETERM). The values of excitation system parameters should be such that a well damped but not overly sluggish response is obtained.

The procedure for carrying out this test is as follows:

1. Initiate PSS®E.

2. Restore the Snapshot containing the dynamics data of the system being modeled (activity RSTR).

3. Retrieve the corresponding converted power flow Saved Case. Note that the generators must be converted. However, there is no need to factorize the matrices (i.e., activity FACT is not required) because the transmission network is ignored in this test, with all machines operating in isolation.

4. From the menu, select the option to perform exciter simulation test. After initializing, run the open circuit simulation to zero seconds (specifying TPAUSE equal to 0). Then continue the simulation to TPAUSE set at 2 seconds.

The open circuit response test is executed simultaneously on either all units in the system or all machines at a designated bus. Any user-specified output channels (see Section 23.6, "Simulation Outputs") are overridden and the exciter outputs, EFD, and machine terminal voltages, ETERM, of all online machines being tested are placed in output channels.

Activity ESTR overrides the initial generator loadings as contained in the working case and initializes each unit to the rated terminal voltage on open circuit. Following completion of activity ESTR, the initial value of EFD for each unit should be slightly greater than the air gap line value of unity due to saturation in the main

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generator. An initial open circuit value of EFD exceeding about 1.2 pu is a fair indication that the *generator* saturation data is erroneous.

The response to a setpoint step of about 0.05 pu should normally show a slight overshoot but be well damped with no persistent oscillations. For operational and additional application details, refer to activities ESTR and ERUN.

Governor Response Test

The principal purpose of the governor response test is to ensure that the governor gain and time constant parameters correspond to a correctly tuned well damped response. In this test, each machine is initialized to a given load and the response of the governors to a step change in the load is measured. The load electrical power is held constant after the step so that the response indicates the damping due to the turbine and governor loop only.

The procedure for carrying out this test is as follows:

1. Initiate PSS®E.
2. Restore the Snapshot containing the dynamics data of the system being modeled (activity RSTR).
3. Retrieve the corresponding converted power flow Saved Case. Note that the generators must be converted. However, there is no need to factorize the matrices (i.e., activity FACT is not required) because the transmission network is ignored in this test, with all machines operating in isolation.
4. From the menu, select the option to perform governor response simulation. After initializing, run the simulation to zero seconds (specifying TPAUSE equal to 0). Then continue the simulation to TPAUSE set at 2 seconds.

The governor test is executed simultaneously on either all units in the system or all machines at a designated bus. Any user-specified output channels (see Section 23.6, "Simulation Outputs") are overridden and the speed deviations, SPEED, and turbine mechanical powers, PMECH, of all online machines being tested are placed in output channels.

It is recommended that the governors be initialized to about 0.8 pu load and that the load step be approximately 0.1 pu. Note that the damping of hydro governing loops is usually decreased with increasing load and hence, unless the objective is to compare the calculated response with a test result taken at a low load level, governor response tests should normally be made near full load.

Activity GSTR overrides the initial generator loadings as contained in the working case and initializes each unit to a specified loading at unity power factor. This test reveals the transient variation of turbine power. All units should have a well damped response. Hydro governors will generally show a somewhat greater overshoot than steam turbine governors but should still be well damped, showing no persistent oscillations. Note that a small negative change in hydro turbine power before it follows a positive change of load power is normal and does not indicate incorrect governor tuning.

Because the machine initialization is based on the value specified as MBASE for each machine, the governor response test is valid only for those machines where data is specified on their actual MVA base. For operational and additional application details, refer to activities GSTR and GRUN.

Application Notes

The response tests described in Exciter Response Ratio Test, Excitation System Open Circuit Step Response Test, and Governor Response Test are special PSS®E simulations of standard test conditions. Steady-state and

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transient results are observed and compared with either actual test results or with typical unit performance. Because the form of the test result needed for satisfactory performance of the unit is well understood in practice, these simulations assist in detecting those situations where parameter values appear to be reasonable but are actually incorrect.

These response tests are able to calculate the response of all generating units in the system simultaneously. Because machines are operating in isolation, no network solution is required in the simulation. Furthermore, only the generator and their excitation system or governor models perform any calculations in these simulations; all other models are inactive. Thus, these simulations require substantially less computer time than complete system simulations.

In the second and subsequent passes through one of the above response tests, even further execution time advantages can be realized. Suppose, for example, that on previous response test simulations all units except the single machine at bus 201 have shown satisfactory performance. One additional test is to be run with modified data for this machine and only this machine is of interest, and hence the excitation open circuit test need be performed for this machine only.

The response test subsequently performed simulates only the response of the machines at bus 201, and the channel output file written during the simulation contains only the variables of interest.

Note again that activities ESTR and GSTR override any output channel selections previously made by the user (see Section 23.6, "Simulation Outputs"). Thus, following execution of either of these activities, Snapshot Files containing these channel specifications should not be overwritten with activity SNAP if the channel selections are to be preserved.

23.5.3. Simulation Initialization

The verification procedures heretofore described are used to assess the correctness of the equipment model parameters comprising the system model. However, a dynamic simulation model must be correct with respect to both its equipment parameters and its initial condition operating point.

The checking of the initial condition operating point should begin before setting up the converted network for use in the dynamic simulation activities (see Section 23.3, "The Network Model"). Most importantly, machine terminal conditions should be examined to check for acceptable voltage, loading and power factor, and to ensure that machine loadings are within turbine governor limits. Activity GEOL is useful for this check. Other operating point checks that can be made include: a cross check of the operating point of dc transmission lines with the dynamics data parameters specified for the dc line model, particularly the voltage dependent current limit; acceptably small mismatches from the power flow solution; and the selection of load characteristics in establishing the converted case.

These checks can only verify operating point variables outward from the generator terminals. The checking of the initial condition with respect to variables within the equipment dynamic models is assisted by the dynamic simulation initialization activity STRT.

Beginning with the generator terminal conditions as contained in the working case, activity STRT works backward through all dynamic simulation models to determine the initial condition values of all state, algebraic, and input variables along with those of several of PSS®E's internal arrays (refer to Table 23.1, "Dynamic Simulation Arrays"). These quantities represent the conditions inside the generators, excitation systems and other equipment modeled by the PSS®E Model Library subroutines that comprise the system model.

Activity STRT also performs a number of data consistency and operating limit checks. This checking is handled in conjunction with this backward initialization process. Each equipment model checks all variables to which

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limits are applied to ensure that the calculated initial value of the variable falls between the limits. Limit violations generate an alarm and the calculated values are overridden by the appropriate limits.

All limit violations should be resolved before proceeding with dynamic simulation runs. This may involve either simply correcting an erroneously specified limit, redoing of power flow work to modify the initial condition operating point, or tracking down some other parameter in error in the model.

It is possible for the initial condition operating point to be such that a variable is initialized exactly at one of its limits. Such a situation may occur, for example, if a machine is fully loaded and its turbine mechanical power initializes precisely at its ceiling. This condition, although it has generated an alarm, is acceptable.

A network solution is used during the initialization process and the number of iterations it requires is tabulated by activity STRT. In the vast majority of cases, particularly if the network solution in the converted power flow case was refined with activity TYSL as suggested in Section 23.3, "The Network Model", only one iteration should be required. When more than one iteration is needed, this usually indicates a serious modeling error (e.g., omitting the dc line model for a dc line represented in the working case) which must be resolved.

A useful aid in tracking down the cause of this type of error is to execute activity STRT with the network solution convergence monitor turned on (i.e., select activity STRT,CM; activity STRT). The bus experiencing the largest voltage change on the first iteration is a good starting point for detecting the cause of the problem.

Any limit violation errors and the causes of excessive iteration during the initial condition network solution should be reconciled first. These errors usually result in a number of steady-state violation errors (i.e., nonzero time derivatives) described next which are eliminated upon correction of these first two categories of errors.

Upon completing the calculation of the system initial condition, activity STRT performs a test calculation of the time derivatives of all state variables using the same calculation sequence that is used during dynamic simulation runs. In the absence of limit violations, the network solution should converge in one iteration and all time derivatives should be close to zero. (Some will not be exactly zero due to the precision and rounding errors inherent in the computer's finite word length). Any state variable for which the time derivative differs significantly from zero in relation to the value of the state variable itself generates an alarm. The STATE number and the values of the state variable and its derivative are printed. For any plant-related, load-related, dc line, FACTS, and switched shunt models with a state variable included in the tabulation, the name and location of the model are also listed. For other models, individual STATEs may be correlated with specific model calls by examining the output of activity DOCU,CN and DOCU,CT.

These steady-state violation errors should be checked out before the system model is accepted as a valid starting point for dynamic simulations. Perseverance in tracking down the causes of these errors is essential if good quality simulation results are to be obtained; they are often symptoms of errors or incompatibilities in the power flow and dynamic simulation data on which the system model is based.

It is possible for a state variable to generate an alarm because of the precision limitations of the computer referred to above rather than due to a modeling error. Such STATEs are usually characterized by a high gain and a short time constant. The voltage regulator output, STATE(K+1), of the IEEET1 model, for example, often possesses these characteristics. Usually, these errors may be accepted if the time derivative, DSTATE, is less than about five or six percent of the value of the STATE.

In these cases, and, in fact, for all simulation setups, a simulation run with no disturbance applied is often a useful mechanism for revealing previously undetected data errors and/or numerical integration instability (i.e., a time step too large in relation to the shortest time constant in the process). Such a simulation should remain completely steady with no deviation from the initial condition regardless of the number of integration steps taken.

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Another useful test run involves initiating some movement in the system with a minor perturbation. This could be the tripping of a small machine, the shedding of a small amount of load, or the tripping of a non-critical transmission line. Such a simulation should require only two or three network solution iterations per time step following the disturbance and should approach a new steady state fairly quickly. There should be no undamped or persistent oscillations.

In both the steady-state and minor disturbance simulations, observing the network solution convergence monitor as described above often gives a good indication of any problems. For operational and additional application details, refer to activity STRT.

23.6. Simulation Outputs

PSS®E contains no standard set of simulation variables that are recorded to document the results of a simulation run. Rather, any power system variable that can be represented by an element of the following dynamics data arrays may be selected as an output of a dynamic simulation run (refer to Table 23.1, "Dynamic Simulation Arrays"):

ANGLE	PELEC	QELEC	ETERM	EFD	PMECH
SPEED	XADIFD	ECOMP	VOTHSG	VREF	VUEL
VOEL	VAR	STATE	BSFREQ	TPLOAD	TQLOAD
GREF	LCREF	WVLCTY	WTRBSP	WPITCH	WAEROT
WROTRV	WROTRI	WPCMND	WQCMND	WAUXSG	

This is accomplished by assigning the desired value to an *output channel*. PSS®E places the set of output channel values into a Channel Output File at specified regular intervals during the simulation run. This file, then, records a comprehensive history of the event being simulated.

23.6.1. Output Channel Selection

The assignment of simulation variables to output channels is handled by the PSS®E activities CHAN and CHSB. As each output channel variable is assigned, an entry is made in the following output channel control arrays:

IPRINT	Contains the addresses of the output variables in the dynamic simulation array space of PSS®E.
IDENT	Contains the thirty-two character alphanumeric channel identifiers assigned to each output channel.
CHNTYP	Contains coded descriptions of the type of quantity assigned to each output channel.
CHNBUS	Contains descriptions of the physical location of each output channel quantity.

These arrays are preserved with the dynamics data Snapshot as it is saved and retrieved with activities SNAP and RSTR, respectively.

The arrays IPRINT and IDENT are used during activity STRT and RUN in the process of filling the Channel Output File. Each variable address is formed by summing the appropriate *array base address* and an *offset*. The array base addresses are installation dependent; they are a function of the capacity limits for which PSS®E is dimensioned. (They are displayed in activity ALTR when output channel data changes are selected.) The

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offset for VARs and STATEs is simply the VAR or STATE number; for the remaining quantities, the offset is determined by the internal numbering scheme of machines, buses and loads in the power flow working case.

When activity CHAN is used to assign a simulation variable to an output channel, the channel identifier may be specified by the user. If no identifier is entered, activity CHAN supplies a default identifier (see Table 15-2 Activity CHAN Summary). Activity CHSB always supplies channel identifiers.

The arrays CHNTYP and CHNBUS are used in describing the quantity assigned to an output channel in activities DLST and ALTR. They are also used by activities RSTR, SRRS, and RTRN for recomputing the channel address, IPRINTs, for the situation where the internal numbering of buses, machines and loads in the working case differs from that contained in the working case at the time activities CHAN and /or CHSB were executed.

Several of the network quantities available in the menu of activities CHAN and CHSB are not directly available from any of the arrays listed at the beginning of Section 23.6, "Simulation Outputs". To calculate and place such variables into output channels, activities CHAN and CHSB generate appropriate references to PSS®E monitoring models which place their results into VARs during the simulation. These model call references are located in the model connection table arrays, and the corresponding models are called automatically at each time step of the dynamic simulation. Refer to Section 16.2.2, "Application Notes" and Section 16.25.2, "Application Notes".

Activities CHAN and CHSB are used to establish the set of output channels for which values are to be recorded during the simulation. They may also be used to append additional output channel selections to those previously specified, and to modify the simulation variable assignment of individual output channels.

Although activity ALTR may be used to add or change output channel assignments, this is not recommended because this process involves the determination of the channel addresses in dynamics memory by the user. This, in turn, requires the knowledge of the internal numbering of buses, machines and loads in the working case to determine the offset to be used in forming the channel address. Because activities CHAN and CHSB automatically handle the look-up of these numbers, their use, rather than through the line model activity ALTR, reduces the risk of error in specifying output channels.

Two separate index pointers used in the processing of output channels are maintained. The variable NXTCHN points to the next available channel number and is used in setting up the default channel number responses in subsequent executions of activities SNAP, CHAN and CHSB. The variable NCHAN, which is normally equal to NXTCHN-1, is sensed by activities STRT and RUN and defines the number of output channels to be monitored during dynamic simulation runs. NCHAN may be set to any nonnegative number less than NXTCHN by the user. Both of these pointer variables may be changed either using the GUI or by running the line model activity ALTR. For operational and additional application details, refer to activities CHAN and CHSB, respectively.

23.6.2. Channel Output File

Other than as described in Section 23.7, "Running a Simulation" , the PSS®E simulation activities do not plot the set of output channels during the execution of the simulation run. Rather, the values of the system model variables assigned to output channels are deposited into a Channel Output File at regular intervals during the simulation.

The Channel Output File is specified in response to an instruction during activity STRT. Activity STRT creates a file with the specified name (if it does not already exist) and initializes it by writing the case heading and the channel identifiers at the beginning of the file. Values of simulation TIME and the variables designated as output channels are written into successive records of the file by activity RUN. This occurs every NPLT time steps, where NPLT is specified by the user following the selection of activity RUN.

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The user may specify (or change) the Channel Output File to be used during a simulation run. This ability is useful in the following situations:

1. The user inadvertently bypasses the specification of a Channel Output File during activity STRT. A file may be specified prior to entering activity RUN.

2. Following activity STRT and before commencing execution of the simulation with activity RUN, additional simulation variables are to be added to the set of output channels previously assigned. These are added with activity CHAN or CHSB, a Channel Output File is specified, and activity RUN is then entered.

3. The Snapshot and power flow Saved Case used as the initial condition operating point for the simulation were preserved immediately following a previous execution of activity STRT, and activity STRT is not to be used again to initialize this simulation. Working memory is set up in the standard manner (activities RSTR,LOFL, CASE, FACT, and RTRN), a Channel Output File is specified either using the GUI or with activity ALTR, and activity RUN is entered.

Unless some specific purpose is to be served, the user should not normally change the Channel Output File used in a simulation in the middle of a run.

The Channel Output File is a binary file of variable length records. The results of a simulation run, as recorded in the Channel Output File, may be plotted and/or tabulated upon completion of the simulation calculation. The processing of this file is handled by PSS®E plot package as well as the by the auxiliary program PSSPLT.

23.6.3. CRT Plot Channels

PSS®E allows a subset of up to six of the main simulation output channels to be plotted on the user's graphical CRT workstation as the simulation is taking place. To enable this feature:

1. The selected output channels must be assigned to CRT Plot Channels (see below).

2. The control variable CRTPLT, which is specified at the beginning of activity RUN, must be specified with a value greater than zero.

3. A Channel Output File must be active (see Section 23.6.2, "Channel Output File").

4. The graphics output device option setting (see Section 3.3.3, "Program Run-Time Option Settings" and OPTN) must specify a graphics CRT.

The specification of CRT plot channels, which are numbered one through six, is made either using the GUI or with activity ALTR after the main simulation output channels have been designated. The user has to specify the number of the main output channel to be assigned to the selected CRT channel, along with the scales to be used on the rough plot. Specifying a main channel number of zero for any of the six CRT plot channels deactivates that CRT channel.

The user is free to modify the CRT channel assignments and/or scales at any time during a simulation run.

23.7. Running a Simulation

A dynamic simulation study involves two general phases: the setup and check out of the system model, and the execution of simulation runs using the system model. The setup phase, as described in the preceding

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sections, is the more critical and involves the greater amount of work. However, after a valid system model is constructed, an extensive series of simulation runs can be made with relatively little incremental effort.

23.7.1. Initialization

As described previously, the process of setting up working memory in preparation for a dynamic simulation calculation requires the following steps:

1. Ensure that the CONEC, and CONET subroutines have been compiled and linked into the PSS®E program structure as described in Section 4.10, "Loading the Connection Subroutines".

2. Start up PSS®E.

3. Open the power flow case and the dynamics snapshot files.

The above sequence of activities assumes that the Saved Case and the Snapshot files are present, and that the Snapshot File contains the desired output channel assignments for the simulation run.

Whenever network data is present in the working case and dynamics data is present in PSS®E working memory, when moving between the power flow and dynamics data spreadsheets, PSS®E takes care to update the dynamics model connection table array pointers and output channel addresses to reflect changes in the internal bus, machine, load, and branch numbering of the working case. These steps correctly set up these arrays in preparation for the dynamic simulation initialization activity, STRT which initializes all simulation models comprising the system model.

The user may preserve the initial conditions of the dynamic models by taking a Snapshot. The attainment of a clean STRT is a key point in a simulation study and such a Snapshot often provides a useful reference in analyzing simulation results. Although such a Snapshot may be used as the initial condition for future simulations with the same initial condition without repeating activity STRT, the inclusion of activity STRT in each simulation run is strongly recommended because it provides for:

1. A cross check for compatibility of the Snapshot and power flow Saved Case; and

2. The specification of a new Channel Output File; one common mistake in using such a Snapshot and omitting activity STRT in subsequent simulations occurs when neglecting to change the Channel Output File, thus unintentionally overwriting the results of a previous simulation.

The execution of activity STRT contributes only a small fraction of the total job time in a system simulation; including activity STRT in each simulation to provide the checks described above is well worth the investment. For operational and additional application details, refer to activity STRT and Section 23.5.3, "Simulation Initialization".

23.7.2. Applying Disturbances

The process of making dynamic simulation runs involves executing activity RUN. Activity RUN starts from the present value of simulation TIME, and calculates the state of the system, advancing TIME until it reaches the value specified as TPAUSE. TPAUSE is designated upon entering activity RUN.

Activity STRT initializes the value of simulation TIME to minus two time steps. This provides for a period of steady-state simulation prior to initiating any disturbance. This serves as a further check on the formulation of the system model and the initial condition operating point, and produces a corresponding flat segment at

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the beginning of plots generated by the Output Channel File processing program, PSSPLT, executed following the completion of the simulation.

Thus, following an error-free execution of activity STRT, activity RUN should be selected with its first TPAUSE set for TIME equals zero seconds. Some PSS®E users have adopted the practice of routinely including a longer period of steady-state running in each simulation, typically to TIME equal to 0.05 or 0.1 seconds.

System disturbances and switchings may be specified via activity ALTR, via the power flow data editing spreadsheet, or via the functions accessible from the *Disturbance* menu entry in the Dynamics spreadsheet view.

If activity ALTR is used, ALTR first allows the user to modify dynamics data (such as CON and ICON values) and then gives the option of applying network changes. Network switching changes such as fault application, fault clearance, and branch tripping may be applied either to the current working case or to a new set of network data that is retrieved from a converted power flow Saved Case. A reply of 1 (yes) to the question PICK UP NEW SAVE CASE? enables the user to specify the name of a Saved Case File which is then brought into the working case. In this situation, the bus voltages contained in working memory are preserved rather than being replaced by the voltages from the Saved Case File.

The new Saved Case feature is useful in undoing the application of a fault or other disturbance by returning directly to the predisturbance network data. It may also be used to establish an often used complicated switching operation by imposing the switchings on the original converted case beforehand and saving the resulting network in a Saved Case File. However, this application should be used with great care in any simulations in which relays are active and hence machines, branches, or loads may have been automatically tripped.

Furthermore, several PSS®E load characteristic models modify the load data arrays in the working case during the execution of activity STRT. This, in turn, requires that any such Saved Case be established following the execution of activity STRT when any of these types of simulation models are being used. Therefore, the new Saved Case feature of activity ALTR should be used with caution during a simulation run.

Activity CHNG is automatically entered and network switching operations may be applied in the standard manner. Prior to running simulation, PSS®E will automatically run activity FACT (to set up the admittance matrix temporary file) if needed. If FACT detects that a new ordering calculation is required, activity ORDR is also automatically executed.

Three phase faults are applied in PSS®E by placing a large shunt susceptance at the faulted bus. The typical value recommended is B = -2.E9 Mvar. This is usually large enough to drive the voltage at the faulted bus to zero yet avoid floating point arithmetic underflow. Unbalanced bus faults are simulated by adding the positive sequence equivalent fault admittance, as tabulated by activity SCMU, to the bus shunt (see Section 11.6.8, “Unbalance Condition Summary Report”). Transmission line unbalances are simulated by replacing the branch impedance, charging and line shunts of the branch experiencing the unbalance with those tabulated by activity SPCB (see Section 11.13.5, “Output of Activity SPCB”).

The network data changes required to impose disturbances or switchings may also be implemented by going to the network spreadsheet, and either making the appropriate changes on the spreadsheet itself, or selecting the appropriate power flow activities. In either of these cases, PSS®E will automatically run activity FACT if needed.

While the power flow linkage is in effect, power flow reporting activities such as POUT and GENS may be executed to examine network conditions at that instant of the simulation. Note that, while the power flow activities do recognize the basic load characteristics as implemented by activity CONL, they do not have a knowledge of other boundary condition modeling as implemented by load characteristic models. Thus, net-

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work modeling in a simulation such as loads modeled beyond the level provided by the basic load characteristics or frequency sensitivity of loads and network parameters shows as mismatches in power flow output reports. Any dc lines are correctly reported.

During simulations, particularly while a fault is applied or the system voltages are generally depressed, the network solution may fail to converge within the prescribed number of iterations. It is also possible for convergence failure to result from the system reaching an unrealistic state of stress which relay action in the physical system, which is unmodeled in the simulation, would not allow.

Any convergence failure is reported with an appropriate message, including the final convergence monitor line in the same form as is used in activities SOLV, MSLV, and TYSL. The simulation then continues as if convergence had been achieved.

Convergence failure can often be corrected by decreasing the acceleration factor used and increasing the number of iterations permitted during the network solution with activity ALTR. It is advisable to observe the convergence monitor when the iteration is failing to converge and following a change in acceleration.

23.7.3. Snapshots

Dynamic simulation working memory may be preserved in the form of a *Snapshot* which is placed into a Snapshot File by activity SNAP. Among other data items, a Snapshot contains both the constant parameters associated with the dynamic equipment models and those arrays which specify the instantaneous condition of the equipment models. It does *not*, however, contain any information pertaining to the electric network. As in power flow and fault analysis applications, network data (i.e., the working case) is preserved in a Saved Case File.

Thus, a system dynamic model is completely specified by the combination of a Snapshot and its corresponding power flow Saved Case. It is strongly recommended, particularly for the novice user of PSS®E, that in the dynamic simulation portions of PSS®E Snapshots and Saved Cases be saved and retrieved in coordinated pairs.

Note that although the Saved Case does not contain the elements of the factorized admittance matrix, PSS®E will automatically execute activity FACT and generate the factored admittance matrix, if needed, before a dynamic simulation is performed.

23.7.4. Application Notes

It is permissible in PSS®E to stop a simulation run and examine in detail the results to date, and then restart the run from that point just as if it had not been interrupted. This is accomplished by first recording the instantaneous state of the system in Snapshot and Saved Case form. PSS®E may then be terminated and the Channel Output File of the simulation examined with the plot package provided with PSS®E. The user should however be careful not to overwrite the original Snapshot and Saved Case Files on which future simulations may be based.

To continue such an interrupted simulation, PSS®E is started up again, the Snapshot and Saved Case files (which were preserved when the simulation was halted) are read, and the simulation is then continued.

This preserving of mid-run conditions in a Snapshot and Saved Case is also useful for the case in which alternative switching options at, say, three seconds into a simulation are to be compared. The identical first part of the run need not be repeated.

In these applications, the user does need to exercise care with regard to the run's Channel Output File. The name of this file and its position pointer are preserved with the Snapshot. In either of the above applications, care must be taken not to delete this file or otherwise modify its contents.

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In the case of the comparison simulations described above, the user often wishes to retain the Channel Output File for both of the simulations for future plotting. Upon completion of the first simulation run, the Channel Output File containing its results should be copied into another file for retention. The second of the comparison runs may then be commenced with its results placed into the original Channel Output File.

Care must be taken with regards to power flow Saved Cases for any simulation setups using load characteristic models. During the execution of activity STRT, these models modify the load arrays contained in working memory. Therefore, the pre-STRT converted Saved Case *should not* be overwritten by activity SAVE following execution of activity STRT. With such a system model, successive STRTs are permitted because the pre-STRT load arrays are returned to their original working case values prior to the simulation initialization calculations.

23.8. Procedural Summary

This section describes those procedures which are used frequently in the dynamic simulation portion of PSS®E. Many of the operations are described in detail elsewhere in this manual and are reviewed here for reference.

23.8.1. The Executable Run File

If the dynamic case has user models, the user dll (usually the dsusr.dll) has to be created. The user dll will contain references to the user models and to any connection routines that may have been generated as a result of running activity DYRE.

To create the user dll, the create usrdll program can be used. The create usrdll is GUI driven method that allows users to compile the model source code and create the dll. An alternative way to create the user dll is to use the CLOAD4 command which is run via the DOS prompt.

Any time a system model change is implemented by a change to either of the connection subroutine files, the new versions must be compiled and the user dll file must be rebuilt by again running the create usrdll tool or by running the CLOAD4 command. See *Additional Resources for PSS®E*, CreateUsrDll.

23.8.2. The Converted Saved Case

The dynamic simulation activities require that the network in the working case be converted from its power flow representation to the form required for dynamic simulations. The converted case may initially be set up by:

1. Picking up the appropriate power flow case which must be solved to an acceptable mismatch level. The solved unconverted case should be preserved in a Saved Case File.

2. Ensuring that the machine characteristic data MBASE, ZSORCE, XTRAN and GENTAP are correctly specified for all in-service machines. These machine attributes are a part of the power flow data that is kept with the Saved Case.

3. Converting the generator representation by executing activity CONG.

4. Converting the load representation by executing activity CONL. Selective load conversion is accommodated by specifying the appropriate bus subsystem, area, zone, owner, or basekV when invoking activity CONL.

5. Calculating the triangular factors of the system admittance matrix and storing them in the admittance matrix temporary file by executing activity FACT.

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6. Refining the voltage solution by executing activity TYSL.

7. Preserving the converted power flow case in a Saved Case File by executing activity SAVE. This *should not* be the same Saved Case File in which the unconverted (i.e., power flow style) Saved Case is located.

This procedure need be done only once for each initial condition operating point.

Note that the results of the conversion process of steps 3 and 4, and the refined network solution of step 6 are reflected in the Saved Case of step 7. The triangular factors determined in step 5, however, are located in the admittance matrix temporary file and *are not* in the Saved Case File.

23.8.3. Initial Model Setup

In constructing a new system model from source data, the following steps are used:

1. Set up the converted Saved Case as described in Section 23.8.2, "The Converted Saved Case".

2. Using the model data sheets for reference, prepare a Dynamics Data Input File in the form given in Section 16.1.1, "Dynamics Model Raw Data File Contents".

3. Initiate PSS®E

4. Retrieve the converted Saved Case with activity CASE.

5. Execute activity DYRE as described in Section 16.1.2, "Operation of Activity DYRE". Specify the filenames into which the CONEC and CONET subroutines are to be placed, making sure not to overwrite existing files which the user wishes to retain. Specify starting addresses of one for each of the general purpose storage arrays. Specify a filename to be used as a compiling file.

6. Use the consistency check for plant models via GUI (available from *Dynamics > Model maintenance...> Consistency Check*) to check for modeling inconsistencies as described in Section 15.7.6 Consistency Check. Resolve any error messages printed by activity DYRE and the consistency check. If necessary, edit the Dynamics Data Input File as required and repeat steps 3 through 6.

7. Preserve dynamics working memory in a Snapshot File by executing activity SNAP.

8. Exit PSS®E with activity STOP.

9. If any supplementary signal models are being used with one of the dc line models that is not in the table-driven form, modify subroutines CONEC and CONET as detailed in Section 23.4.4, "Program Set-Up". Then compile the connection subroutines, and create the user dll.

10. Verify any unchecked data using the procedures given in Section 23.5.2, "Performance Verification ". Correct erroneous data, and update the Snapshot created in (7) with activity SNAP.

23.8.4. Retrieving System Model Data

The converted network may be accessed, along with the Snapshot containing the system model dynamics data, during subsequent dynamic simulation runs with the following steps:

1. Start up PSS®E.

2. Open the converted network.

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3. Open the Snapshot file.

4. Run the simulation. It may be noted that if setting up the factored matrix (i.e., running activity FACT) is not done by user, PSS®E will take care of doing this before running activities like ASTR, STRT, RUN,MSTR, and MRUN. Setting up of the factored matrix is not required if only activities such as DYRE, CHAN, DOCU, and MLST are to be executed.

23.8.5. Selecting Output Channels

To add the user's selection of those system model variables to be monitored during simulation runs to the dynamics data contained in the Snapshot, the following steps may be used:

1. Retrieve the system model data into working memory by the procedure given in Section 23.8.4, "Retrieving System Model Data".

2. Select activity CHAN and/or CHSB to designate those variables which are to be placed into output channels. Specify the default starting addresses in the VAR, ICON, and output channel arrays. See also Section 23.6, "Simulation Outputs".

3. Preserve dynamics working memory, which now contains the output channel specification arrays, by executing activity SNAP. The Snapshot File specified would normally be the same one referenced in Step 7 of Section 23.8.3, "Initial Model Setup" and Step 3 of Section 23.8.4, "Retrieving System Model Data".

23.8.6. Running a Simulation

With the system model constructed, the procedure used in executing dynamic simulation runs is as follows:

1. Retrieve the system model data into working memory by the procedure given in Section 23.8.4, "Retrieving System Model Data".

2. Initialize the dynamic models to correspond to the initial condition operating point with activity STRT. Specify the name of the Channel Output File in response to the prompt from activity STRT. Any error conditions should be reconciled before continuing with the simulation (see also Section 23.5.3, "Simulation Initialization").

3. Simulate a period of steady-state operation with activity RUN, specifying a TPAUSE of zero (or 0.05 or 0.1) seconds.

4. Apply a disturbance or switching.

5. Execute activity RUN, specifying as TPAUSE the next switching time.

6. Repeat steps 4 and 5 for each switching operation.

7. Optionally, preserve the end of run system conditions by taking a Snapshot and Saved Case.

23.8.7. Modifying the Network Model

A change in the initial condition operating point for a simulation run is handled by executing power flow activities on the base case network condition from which the original converted Saved Case of Section 23.8.2, "The Converted Saved Case" was constructed. Because this involves power flow solutions, the unconverted case must be used. One procedure is as follows:

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1. Initiate PSS®E, and retrieve the appropriate unconverted power flow Saved Case.

2. Set up and solve the power flow to represent the modified system condition, saving the resulting network condition in a new Saved Case File with activity SAVE.

3. Set up and preserve the converted Saved Case as in Section 23.8.2, "The Converted Saved Case".

4. Retrieve the dynamic snapshot file.

5. If desired, add output channels. Then execute dynamic simulations as described in Section 23.8.6, "Running a Simulation ".

This approach assumes that there are no dynamic modeling changes required to model the new initial condition operating point. Changes to the dynamic model are described in the following section.

23.8.8. Modifying the Dynamic Modeling

The majority of dynamic modeling changes fall into one of the following two categories:

1. A change to the constant parameters of a model for which the reference already exists.

2. The addition of a new model reference to an existing dynamics setup.

The first type of modeling change is straightforward and requires no modification of the model connection table arrays or the connection subroutines. Working memory is setup as described in Section 23.8.4, "Retrieving System Model Data", and the appropriate data changes are entered on the Dynamics spreadsheet, or with activities CCON, ALTR, or PSAS. Dynamics working memory, with the modified data items, is then preserved with activity SNAP, either overwriting the original Snapshot or in a new Snapshot File as appropriate.

The second type of modeling change requires the modification of the intelligence connecting specific models to network elements: the model connection tables in the case of plant-related, load-related, line relay models, auxiliary-signal models, FACTS device models, dc line models, switched shunt models or the connection subroutines for models related to other network elements (see Section 23.1.1, "Program Structure"). Furthermore, it may involve either adding a model for an equipment item which was previously not modeled (e.g., adding a governor to a machine) or replacing one model with another (e.g., changing an IEEET1 model to an SCRX).

The majority of plant-related, load-related, line relay, auxiliary-signal, FACTS, dc line, and switched shunt models may be introduced into the dynamic simulation setup using the appropriate spreadsheet in the Dynamics or Models tab.

All other models must be introduced via activity DYRE (plant-related, load-related, line relay, auxiliary-signal, FACTS, and dc line models may also be introduced in this manner). One way to do this is to edit the original Dynamics Data Input File (or its current version as output by activity DYDA), and reconstruct the entire dynamics setup. Alternatively, activity DYRE may be used in an *add models* mode (see Section 16.1.3, "Adding Models"); the following approach may be used:

1. Prepare a Dynamics Data Input File in the form shown in Section 16.1.1, "Dynamics Model Raw Data File Contents". Only data records for the new models should be included.

2. Initiate PSS®E, and retrieve the data for the current system model into working memory as described in Section 23.8.4, "Retrieving System Model Data" (step 4 may be omitted).

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3. Run activity DYRE,ADD, specifying the file created in (1), and files into which the connection subroutine calls are to be placed. They *should not* be the same files containing the CONEC and CONET subroutines for the current version of the system model. Specify the default next available locations in the dynamics data general purpose storage arrays.

4. Preserve dynamics working memory as updated in (3) by executing activity SNAP.

If the changes in modeling included new model calls in subroutines CONEC, or CONET, the following additional steps are required:

5. Terminate PSS®E.

6. Merge the CONEC and CONET model calls as generated in the above execution of activity DYRE with the versions of these subroutines containing the rest of the system model calls. If models are being replaced, the user *must* either delete or comment out the model calls which were replaced so that the resulting connection subroutine contains the new call and not the old reference.

7. Compile and link to create the user dll.

23.8.9. A Comment

The experienced user of the dynamic simulation activities of PSS®E will find that many of the procedures described above, several of which may be required in a given application, may be combined into a single operating procedure. This is perfectly valid and encouraged, but requires that the user gain an understanding of, and be comfortable with:

1. The prerequisites of the various PSS®E activities involved.

2. The proper sequencing of these activities required for a given application.

3. The contents of working memory in the context of the recent sequencing of PSS®E activities.

4. The relationship among the various files involved in the data handling and simulation phases referenced above.

23.9. Advanced Uses of CONEC and CONET

As described in Section 23.4, "The Dynamic Model", the connection subroutines CONEC and CONET are a vital component of the system model. They are, however, simply FORTRAN subroutines and the experienced PSS®E user may add any desired code to them to handle a wide variety of simulation control and modeling requirements.

This section describes some typical examples of user-written code in these subroutines. It is not intended to be an all-encompassing treatment on the subject, but rather to suggest approaches that might be used to meet specific requirements.

23.9.1. Manual Model Addition

The process of manually adding models to the connection subroutines involves filling out a model sheet for each model to be added, incorporating the model CALL statement into the connection subroutine(s), and entering the model's constant data parameters into the PSS®E dynamics data general purpose storage arrays. The procedure is illustrated by example.

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Suppose that in a simulation setup, the user wishes to place the branch flow exclusive of line connected shunt elements for circuit 1 from bus 100 to bus 110 into output channels. The model FLOW2 may be used to calculate these quantities and place them into VARs. The VARs may then be assigned to output channels. The following procedure may be used:

1. Enter the constant parameters for the FLOW2 model on its data sheet in the *Value* column of the ICON block (see Figure 23.2, "Branch Flow Model").

2. Start up PSS®E, restore the Snapshot containing the original simulation data and, using the dialog for solution parameter changes (available from the GUI menu *Dynamics>Simulation>Solution Parameters...*), note the next available locations in the ICON and VAR arrays (FLOW2 does not use any CONs or STATEs). Assume that these are listed as 101 and 601 respectively.

3. Assign storage locations in the ICON and VAR arrays for the FLOW2 model. For the ICON array, we assign locations starting at 101 (the next available location), and ending at 103 (i.e., I+2 from the data sheet). Similarly, in the VAR array, we assign locations 601 and 602 which will hold the MW and Mvar flow; the MVA flow is not needed so L will be set to zero. The model call can now be written from the completely filled in model data sheet as:

   ```
   CALL FLOW2 ( 101, 601, 602 0 )
   ```

   At this point, the next available ICON and VAR indices are 104 and 603, respectively.

4. Modify the CONET subroutine by adding the FLOW2 model call listed in (3). Compile the revised CONET subroutine and link it to create the user dll.

5. Initiate PSS®E and retrieve the original system model data as detailed in Section 23.8.4, "Retrieving System Model Data".

6. From the GUI menu *Dynamics>Simulation>Solution Parameters...*, or using the solution parameter changes of activity ALTR, update the next available location pointers for the ICON and VAR arrays to correspond to the values determined at the end of step 3.

7. Referring to Figure 23.2, "Branch Flow Model", set the constant parameters for the new model by setting the ICON values in the dynamics data spreadsheet for ICON, or with activity ALTR, or by executing an appropriate Response File generated by the PSS®E simulation run assembler activity, PSAS.

8. Using activity CHAN, assign VARs 601 and 602 to the next available output channels.

9. Preserve the modified dynamics data in Snapshot form with activity SNAP.

Models called from subroutines CONEC and CONET may be added either via activity DYRE,ADD or using the approach described above. Refer to Section 23.1.1, "Program Structure", Section 23.4.1, "Subroutines TBLCNC and CONEC ", and Section 16.1.3, "Adding Models". See also User-Written Models.

The first use of any user-written plant-related, wind related, load-related, line relay, auxiliary signal, FACTS device, dc line, and switched shunt models in a simulation setup must be added to it by modifying the model connection table arrays through activity DYRE,ADD. Instances of plant-related, wind related, load-related, line relay, auxiliary signal, FACTS device, dc line, and switched shunt models contained in the PSS®E Model Library, or of user-written plant-related, wind related, load related, line relay, auxiliary signal, FACTS device, dc line, and switched shunt models already in use in the simulation setup, may be introduced either via activity DYRE,ADD or by using the appropriate Dynamics spreadsheets.

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FLOW2

Branch Flow Model

CALL FLOW2 (I,N,M,L) from CONET		
This model ICONs starting with	#_____	I.
It stores the real component of flow in VAR	#_____	N,
the imaginary component in VAR	#_____	M,
and the VA flow in VAR	#_____	L
Flow is calculated out of the bus number contained in ICON(I).		
N, M, and/or L may be zero to bypass storing of the respective quantity		

VARs	#	Description
N	601	MW
M	602	Mvar
L	0	MVA

ICONs	#	Value	Description
I	101	100	from bus number
I+1	102	110	to bus number
I+2	103	1	Circuit identifier [a]

[a]Enter circuit identifier of -1 to sum flows of all parallel circuits between the two buses.

Note: Flows do not include the line shunt components.

Model call must be manually inserted into subroutine CONET after the IFLAG test and data must be entered.

Figure 23.2. Branch Flow Model

23.9.2. Accessing Machine and Load Data

There are a number of arrays in the PSS®E data structure containing machine variables. These are identified in Table 21-1 Dynamic Simulation Arrays as being indexed by machine index. The machine index for a given machine is an internal sequence number assigned by PSS®E to the machine as it is introduced into the power flow case with activities READ, TREA, RDCH, or MCRE (see Plant and Machine Sequence Numbers) or the machine data spreadsheet.

PSS®E includes the subroutine GENCHK which may be called by the user in subroutine CONEC to determine the machine index of a given machine. Its calling sequence is:

```
CALL GENCHK (IBUS, 'I', K, 'string')
```

where:

IBUS	Is the number of the bus.
I	Is the machine identifier of the machine for which the index is to be found.

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K	Is returned as the machine index, or zero if the machine is not found.
string	Is text which is included in any error message printed whenever GENCHK cannot find the specified machine.

Suppose, for example, the user wished to calculate the accelerating power (ignoring stator losses) in MW of machine one at bus 100 and place it in a VAR for monitoring during a series of simulation runs. This could be accomplished as follows:

1. Start up PSS®E, restore the Snapshot containing the simulation data and, using the GUI menu *Dynamics>Simulation>Solution parameters...*, note the next available location in the VAR array. In this example, assume it is 714. The coding to be added to subroutine CONEC will place the calculated quantity in this VAR. Change the next available VAR pointer to 715, and then, with activity CHAN, assign VAR 714 to an output channel. Preserve these changes by taking a Snapshot and terminate PSS®E.

2. Edit the source code of subroutine CONEC to include the following FORTRAN statements just before the RETURN statement:

```
IF (MODE.NE.2) RETURN
CALL GENCHK ( 100, '1', II, 'IN CONEC' )
IF (II.EQ.0) RETURN
VAR(714) = PMECH(II)*MBASE(II) - PELEC(II)*SBASE
```

   These calculations need be done only in MODE two, hence the logical IF statement. Note that the PMECH array elements are in pu on machine base, and that the PELEC array elements are in pu on system base. Thus, the calculated quantity is expressed in MW.

3. Compile the modified CONEC subroutine and create the user dll.

Table 23.1, "Dynamic Simulation Arrays" also identifies several arrays as being indexed by load index. The load index for a given load is an internal sequence number assigned by PSS®E to the load.

PSS®E includes the subroutine LODCHK, which may be called by the user in subroutine CONET to determine the load index of a given load. Its calling sequence is:

```
CALL LODCHK (IBUS, 'I', K)
```

where:

IBUS	Is the number of the bus.
I	Is the load identifier of the load for which the index is to be found.
K	Is the returned load index, or zero if the load is not found.

23.9.3. Getting Bus Sequence Numbers

PSS®E includes the subroutine BSSEQN which may be called by the user in subroutine CONEC or CONET to determine the bus sequence number of a given bus. Its calling sequence is:

```
CALL BSSEQN (IBUS, I, *alt )
```


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where:

IBUS	Is the number of the bus for which the bus sequence number is to be found.
I	Is returned as the bus sequence number; it is unchanged if the bus is not found.
alt	Is the FORTRAN statement label to which BSSEQN returns if the bus is not found.

Suppose, for example, the user wished to convert the pu bus frequency deviation at a given bus to bus frequency in Hertz. This involves a determination of the bus sequence number (see Bus Sequence Numbers) corresponding to a given bus. The FLECS code fragment shown in Figure 23.3, "Bus Frequency Example", inserted before the RETURN statement in subroutine CONEC, determines the bus sequence number of the bus for which the external number is contained in ICON 101, and stores the corresponding bus frequency in VAR 401.

```
      IF (MODE.GT.2) RETURN
C
      WHEN (MODE.EQ.1 .OR. KPAUSE.EQ.2)
      .  IBUS=ICON(101)
      .  IS=0
      .  CALL BSSEQN( IBUS, IS, *10 )
   10 .  INTICN(101)=IS
      ...FIN
      ELSE IS=INTICN(101)
C
      WHEN (IS.EQ.0) VAR(401)=0.
      ELSE
VAR(401)=60.*(1.+BSFREQ(IS))
```

Figure 23.3. Bus Frequency Example

This fragment of code takes advantage of the following points concerning the PSS®E data structure and the logic sequence of activity RUN to execute efficiently:

1. It only needs to be executed when the value of the control variable MODE is one (initialization) or two (time derivative calculation following the network solution).

2. The array INTICN is indexed in the same manner as the array ICON and is available to the PSS®E model writer (or, in this case, the CONEC twiddler) as a memory location. Its content is preserved from one entry of subroutine CONEC to the next and in Snapshots.

3. The bus sequence number remains unchanged for a given execution of activity RUN. The range of the WHEN block of the WHEN/ELSE construction looks up the bus sequence number, IS, only during activity STRT (when MODE is one) and on the first time step derivative calculation following a pause (when KPAUSE is two). It stores it in INTICN 101 for use during subsequent time steps.

This code provides for the case in which ICON 101 contains an invalid bus number or specifies a bus not contained in the power flow working case; VAR 401 is set to zero in these cases. The user may augment

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this logic to print an appropriate error message in this situation, writing to the logical unit specified by the variable LPDEV.

An obvious extension of the coding in Figure 23.3, "Bus Frequency Example" is to repeat the calculation for a group of buses. Figure 23.4, "Multiple Bus Frequency Example" shows this logic expanded to calculate the bus frequencies of the buses designated in ICONs 101 through 105 and store them in VARs 401 through 405.

```
      IF (MODE.GT.2) RETURN
C
      IF (MODE.EQ.1 .OR. KPAUSE.EQ.2)
C     .
      .  DO (INDXIC=101,105)
      .  .  IBUS=ICON(INDXIC)
      .  .  IS=0
      .  .  CALL BSSEQN( IBUS, IS, *10 )
   10 .  .  INTICN(INDXIC)=IS
      .  ...FIN
C     .
      ...FIN
C
      INDXVR=401
C
      DO (INDXIC=101,105)
      .  IS=INTICN(INDXIC)
C     .
      .  WHEN (IS.EQ.0) VAR(INDXVR)=0.
      .  ELSE           VAR(INDXVR)=60.*(1.+BSFREQ(IS))
C     .
      .  INDXVR=INDXVR+1
      ...FIN
```

Figure 23.4. Multiple Bus Frequency Example

23.9.4. Run Termination

Often in a simulation, it is desired to terminate the run based upon some system condition such as a convergence failure of the network solution or the machine angle spread exceeding some threshold. This is particularly useful for simulations executed in batch mode.

This example shows an approach for stopping a simulation upon convergence failure, with the conditions at the point of termination preserved in Snapshot and Saved Case Files.

First, with the text editor create a Response File named stop.idv containing the following PSS®E activity commands and responses:

```
snap,temp.snp

save,temp.sav
```


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```
stop
```

Note: The blank line following the selection of activity SNAP activates the default response for the *number of elements to save* field in activity SNAP. This sequence of commands assumes that, if either of the files temp.snp or temp.sav already exists, PSS®E is being operated in its file overwrite mode (see Section 3.3.3, "Program Run-Time Option Settings").

Then, in subroutine CONET, insert the following statement after any relay and monitoring model calls and immediately before statement label 9000:

```
IF (ITER.GT.ITMXDS) CALL UPAUSE('stop.idv')
```

The effect of this statement is to force a pause upon detection of convergence failure. Subsequent dialog input is then taken from the Response File stop.idv, which causes a Snapshot and power flow Saved Case to be written to the files temp.snp and temp.sav, respectively, and PSS®E to be terminated.

The UPAUSE run termination routine (supplied with PSS®E) may be called from CONET as in the above example; it may also be called from CONEC when the value of the variable MODE is three (see Section 25.3, "Model Requirements").

23.10. Extended Term Simulations

A class of dynamic simulation calculations termed extended term simulations is characterized by:

1. Simulation times ranging from many seconds to several minutes.
2. The need to model additional effects not normally considered in stability runs of several seconds.
3. Significant periods of simulated time during which the higher frequency effects associated with system disturbances have subsided.

While such simulations could be performed with the state-space simulation activities STRT and RUN, this is usually not done because:

1. The user would need to develop code to model phenomena such as boiler effects, tap changer and phase shifter action, and maximum excitation limiters.
2. Computer time requirements would be excessive.
3. The accumulation of round off errors after many thousands of time steps using explicit integration techniques may cause the simulation to drift.

The extended term simulation option of PSS®E provides a simulation calculation more suitable for simulations of up to several minutes. It includes:

1. The augmentation of most equipment models containing state variables to include an implicit integration algorithm for use in performing extended term simulations in addition to the explicit integration technique used in state-space simulations.
2. The ability to change the simulation time step during the simulation.
3. Several additions to the PSS®E model library.

Synchronous machine models operate in one of three modes during extended term simulations:

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1. A *small time step* mode, in which all state variables are handled in the same manner.

2. An *intermediate time step* or *midterm dynamics* mode in which the higher frequency modes of the synchronous machine models GENROU, GENROE,GENSAL, GENSAE and GENCLS have some extra damping.

3. A *large time step* or *uniform island frequency* mode in which, for each electrical island in the system, an average frequency is determined and all machines in the island take on the island frequency as their SPEEDs.

Two solution parameters are used to determine which of these three modes is being used: an *intermediate mode time step threshold* (DLTBKW) and an *island frequency time step threshold* (DLTEXT). The simulation time step (DELT) is compared to these threshold values and the simulation mode is set to:

- *Small time step* mode if DELT < DLTBKW.
- *Intermediate time step* mode if DLTBKW ≤ DELT ≤ DLTEXT.
- *Large time step* mode if DELT ≥ DLTEXT.

Default values of the time step thresholds are established when PSS®E is started up; they are set as follows:

- DLTBKW = 3 cycles.
- DLTEXT = 7 cycles.

In the *island frequency* mode, a separate acceleration factor and convergence tolerance is applied to the average island frequency calculation during the network solution. When PSS®E is started up, these parameters are set to 1.0 and 0.0005 respectively. In this mode, the network solution iteration continues until both the largest voltage change is below its convergence tolerance *and* the largest machine angle change is below the *island frequency* mode convergence tolerance.

The time step thresholds, acceleration factor and convergence tolerance may be changed using *Dynamics>Simulation>Solution Parameters...* in the GUI, or with activity ALTR and are preserved with the Snapshot when it is saved with activity SNAP. Care must be taken in changing the time step thresholds during an extended term simulation (see Section 16.23.1, "Application Notes").

The state-space simulation activities utilize the following arrays in their second order explicit integration calculations:

STATE	Contains the values of the state variables.
DSTATE	Contains their time derivatives.
STORE	Contains the integrator memory.

The extended term simulation activities use the Z-form expression of the trapezoidal integration algorithm, an implicit integration algorithm (see the *PSS®E Program Application Guide*). The same STORE array described above is used in extended term simulations; the STATE and DSTATE arrays are not used in extended term simulations. In their place, the array STORMT, dimensioned twice as large as the STATE array, is used.

Referring to the model data sheets, for each STATE allocated to a model reference, the following entries in the above arrays are reserved:

State-Space	Extended Term
STATE(K)	STORE(K)

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State-Space	Extended Term
STORE(K)	STORMT(2*K-1)
DSTATE(K)	STORMT(2*K)

where K is the STATE number.

In PSS®E's address space, the STORMT array occupies the same locations as the STATE and DSTATE arrays. In state-space simulations, models do not reference their STORMT array elements, and in extended term simulations, they do not reference their STATE and DSTATE array elements.

Clearly, in extended term simulations, STATE values placed in output channels or tabulated with activity DLST are not useful.

23.11. Creating and Viewing the Results of Dynamic Simulations

The channels captured during the Dynamic Simulation can be viewed and plotted directly from within PSS®E. Multiple channel output files can be open simultaneously and the channels from these multiple channel files combined on a single plot. Currently, the plotting functionality is only a subset of the plotting capabilities found in PSS™PLT. In future releases, the full functionality of PSS®PLT will be incorporated into PSS®E. Therefore, PSS®PLT is still installed and can be used in the same manner as in previous releases of PSS®E.

23.11.1. General Workflow to View Dynamic Results

The results of Dynamic Simulations are the plot channels of various PSS®E quantities that are selected prior to performing the Dynamic Simulation. The plot channels captured during any Dynamic Simulation are contained in a plot channel output file which is in a binary form. The plotting functions contained within PSS®E can be used to read these binary channel files and display the plots.

In order to view the plots, a Plot Book has to be created. The Plot Book is configured like an Excel Worksheet with multiple Plot Pages each capable of displaying several Plots. A Plot in turn can contain several channels/curves/traces. A Plot Book by default will contain on Page with one Plot on that Page.

After a Plot Book is created, a channel output file has to be opened. The channel output file will appear in the Tree View, which, when expanded, will display all the channel outputs that were selected prior to the Dynamic Simulation run.

Any channel output quantity can be viewed by dragging and dropping into the Plot Page.

23.11.2. Recording

Many of the interactions involved in creating and customizing the Plot Book are recordable and can be played back through the use of Automation files. In future releases, most of the Plot Book interactions will be recordable and available for playback.

Additional Information
PSS®E GUI Users Guide, Reproducing User Actions

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Chapter 24
Example Data Files

24.1. SAVNW Case Data Input Files

SAVNW case comprises a simple power system network. It shows how to build a case in PSS®E and the kind of data input required to study power flow, fault analysis and dynamic behavior of power system network. A PSS®E File Planning Sheet is shown on next page.

These example files are available in the Example folder in the PSS®E product installation folder:

```
\\Program Files\PTI\PSSE33\EXAMPLE
```

24.2. Sample Case Data Input Files

Sample Case shows the capabilities of PSS®E in modelling of various power system components. It depicts all the power system components that can be modelled in PSS®E, however it does not represent a realistic power system configuration and therefore may not provide the most accurate results as other analysis features available in PSS®E.

PSS®E File Planning Sheet	
Name of Study:_SAVNW Case Directory Name: \\Program Files\PTI\PSSE33\EXAMPLE	
Filename	Description
Power Flow Raw Data Files (input data, source)	
a. __ savnw.raw_____ _ PSS®E savnw case_____	
b. _____ _____	
Diagram Data Files (input data, source and/or binary)	
a. __savnw.sld_____ _Oneline diagram._____	
b. _____ _____	
Dynamics Data Files (input data, source)	
a. __savnw.dyr_____ _ GENROU, GENSAL, TGOV1, HYGOV, IEEET1,__	
b. _____ _SCRX, SEXS_____	
Power Flow Saved Case Files (saved case, binary)	
a. __savnw.sav_____ _____	
b. __savcnv.sav_____ _ All loads converted to 100% Constant I and B._____	
c. _____ _All generators converted using ZSOURCE._____	
d. _____ _____	
Simulation Snapshot Files (snapshot, binary)	

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Filename	Description
a. __ savnw.snp_____ _____	
b. _____ _____	
Sequence Data Files (input data, source)	
a. __savnw.seq_____ _____	
b. _____ _____	
Optimal Power Flow Data Files (input data, source)	
a. _____ _____	
b. _____ _____	
Other Auxiliary Files (input data, source)	
a. __ savnw.bkd_____ _Generated using RWDY._____	
b. _____ _____	
c. _____ _____	
d. _____ _____	
Linear Network Analysis Data Files (input data, source)	
a. __savnw.sub_____ _ACCC files_____	
b. __savnw.mon_____ _____	
c. __savnw.con_____ _____	
d. _____ _____	
Channel Output Files (output, binary)	
a. __savnw.out_____ _____	
b. _____ _____	
CONEC and CONET Subroutine Files (FORTRAN, source)	
a. __ conec.flx_____ _____	
b. __conet.flx_____ _____	

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Chapter 25
Model Writing

This chapter gives an introduction to the fundamentals of writing equipment models for use in the dynamic simulation of PSS®E.

25.1. Background

The objective of dynamic simulations is to accurately simulate the response of a physical system to some event. This, in turn, requires that each component affecting the response be faithfully modeled over the time frame of interest. In PSS®E terms, this means that, for each equipment being represented, a model which implements the differential equations describing the dynamic behavior of the equipment must be supplied.

The PSS®E Model Library contains a wide variety of equipment models which satisfy this requirement for the vast majority of generating plant equipment. However, situations may arise in which there is no library model which corresponds to the differential equations needed to model a given piece of equipment. To handle this situation, engineers historically have tried to modify the data characterizing the equipment to fit the block diagram of an existing model. Rather than resorting to this approach, the PSS®E user is encouraged to write a model that accurately models the equipment.

The purpose of this section is to describe the basic approach to writing PSS®E models at the level needed to get something working. It is not the all-encompassing treatment on the subject needed to make the user a PSS®E model writing expert; that would require a book by itself. However, it should allow the user to develop models for excitation systems, stabilizers, excitation limiters, turbine governors and dc line supplementary signals which are not contained in the Model Library. Models of other system components generally have a direct impact on the network balance portion of the dynamic simulation calculation. This, in turn, requires a much more intimate familiarity with the internal design of PSS®E than is necessary for these basic modeling functions. Section 25.9, "Writing a Basic Load Model" outlines the basic requirements of load-related models.

While the discussion here is limited to the categories of models listed above, the principles given in this section apply to all types of PSS®E simulation models.

25.2. Prerequisites

Section 23.1, "Simulation Overview" gives an overview of the dynamic simulation process and the dynamic simulation data structure as implemented in PSS®E. It is recommended that all of Chapter 23, *Dynamic Simulation Setup and Procedures*, particularly Section 21.1, be reviewed before continuing. Chapter GIC of the *PSS®E Program Application Guide* also contains useful background material.

The remainder of this section assumes that the user:

1. Knows the physics of the equipment to be modeled or has been given the block diagram and/or the differential and algebraic equations describing the equipment.
2. Has sufficient calculus and block diagram algebra background to transform the equations to the form required of a PSS®E model.
3. Is familiar with the PSS®E data arrays given in Table 23.1, "Dynamic Simulation Arrays".

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4. Knows the FORTRAN programming language.

Before attempting to write any model, the user should first ensure that the modeling requirements cannot be met by a standard library model; perhaps setting a gain to one or a time constant to zero in an existing model reduces it to the required form.

25.3. Model Requirements

Having concluded that a new model is required, the user must gain an understanding of the requirements of PSS®E models. Each model must make different types of computations at different stages in the dynamic simulation process.

A set of scalar variables which are present in PSS®E COMMON memory are used to communicate between several of the PSS ®E dynamic simulation activities and equipment models. The PSS®E activities which invoke equipment models set seven flags before calling these modules. These flags are the variables MODE, KPAUSE, MSTATE, MIDTRM, ITER, IFLAG and IBDOCU. Activities STRT, RUN, MSTR and MRUN sense the variable KTRIP which must be set by any equipment models that implement network data changes which affect the network admittance matrix.

At the most basic model writing level, the MODE flag is the most critical of these variables. In state-space simulations the significance of MODE is as follows:

MODE=1	The model must initialize all of its state variables and algebraic variables. Prior to the first network solution iteration (i.e., when the variable ITER is zero), load characteristic models which are not coordinated call models (see Section 23.4.1, "Subroutines TBLCNC and CONEC ") must set the initial values of any variables needed in the current injection calculations of subsequent model calls issued during network solution iterations.
MODE=2	The model must make all computations needed to place time derivatives into the DSTATE array; each stabilizer model must compute the present value of its output signal and place it in the appropriate entry in the VOTHSG array; each minimum excitation limiter model must compute the present value of its output signal and place it in the appropriate entry in the VUEL array; each maximum excitation limiter model must compute the present value of its output signal and place it in the appropriate entry in the VOEL array.
MODE=3	Each stabilizer model must compute the present value of its output signal and place it in the appropriate entry in the VOTHSG array; each minimum excitation limiter model must compute the present value of its output signal and place it in the appropriate entry in the VUEL array; each maximum excitation limiter model must compute the present value of its output signal and place it in the appropriate entry in the VOEL array; an excitation system model must compute the present value of excitation system out-

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	put voltage and place it in the EFD array; a turbine governor model must compute the present value of turbine mechanical power and place it in the PMECH array; other models written by users normally have no requirements in MODE three.
MODE=4	The model must update the PSS®E variable NINTEG indicating the highest numbered STATE being used. Load characteristic models which are coordinated call models (see Section 23.4.1, "Subroutines TBLCNC and CONEC ") must set the initial values of any variables needed in the current injection calculations of subsequent model calls issued during network solution iterations. Other user-written models normally have no other duties in MODE four.
MODE=5	The model is being called by activity DOCU in its reporting mode and must write out the model data report.
MODE=6	The model is being called by activity DYDA and must write out the appropriate data record.
MODE=7	The model is being called by activity DOCU in its data checking mode and must perform appropriate model constant data checks; if any data exceptions are detected they must be reported followed by the normal DOCU report.
MODE=8	The model is being called to return descriptions of each CON and ICON data input value used by the model.

The variable KPAUSE is meaningful in MODEs one through four and indicates the type of time step calculation being performed by activities RUN, MRUN, ERUN, or GRUN:

KPAUSE=0	Models are being called to make their normal time step calculations.
KPAUSE=1	Models are being called for the value of simulation TIME equal to TPAUSE −, just prior to a pause. Any models which perform local integrations or accumulations, or which update local transport delay tables, should bypass these calculations.
KPAUSE=2	Models are being called at the first time step following a pause (i.e., for the same value of simulation TIME as on the last time step calculation).

Most user-written models need not be sensitive to the variable KPAUSE.

The variable MSTATE is meaningful in MODEs one through four and indicates the type of simulation being performed by activities STRT/RUN, MSTR/MRUN, ESTR/ERUN, GSTR/GRUN or ASTR:

MSTATE=0	Standard state-space dynamic simulation via activities STRT and RUN, or activity ASTR.

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MSTATE=1	Excitation system response ratio test via activities ESTR and ERUN. Models other than generator and excitation system models are not called.
MSTATE=2	Excitation system open circuit step response test via activities ESTR and ERUN. Models other than generator and excitation system models are not called.
MSTATE=3	Governor response test via activities GSTR and GRUN. Models other than generator and turbine governor models are not called.
MSTATE=4	Extended term dynamic simulation via activities MSTR and MRUN.
MSTATE=5	Dynamics data is present but no initialization activity has been successfully executed.

The LOGICAL variable MIDTRM is meaningful in MODEs one through four and indicates whether state-space or extended term simulations are being performed:

MIDTRM=.FALSE.	State-space simulation.
MIDTRM=.TRUE.	Extended term simulation.

The variable ITER is meaningful only in MODEs one and three during the network solution calculation, and it indicates the number of iterations completed during the network solution for the present value of simulation time. It is usually sensed in MODE one by load characteristic models which are not implemented as coordinated call models (see Section 23.4.1, "Subroutines TBLCNC and CONEC ") and which have initialization requirements which must be performed prior to any network solution iterations (i.e., when ITER is zero).

The variable IFLAG is meaningful only in MODEs one and three during the network solution calculation and is usually sensed by models called from subroutines TBLCNT and CONET. These subroutines are called once each iteration, and once more following convergence of the network solution. IFLAG indicates the status of the network solution calculation:

IFLAG=.FALSE.	The network solution has not achieved convergence.
IFLAG=.TRUE.	The network solution has converged or reached its maximum number of iterations.

The variable IBDOCU is meaningful only in MODEs five through seven and indicates the mode of operation of activities DOCU or DYDA:

IBDOCU=0	Process all models called.
IBDOCU>0	External bus number; only process models called at bus IBDOCU. Only models called from subroutines CONEC or CONET need to sense IBDOCU.

The variable KTRIP, as well as the variable KPAUSE (see above), must be set by any equipment model, or by user logic in CONEC or CONET, which imposes network switchings or other data changes that require a dual time step calculation (i.e., one time step calculation before the switching at TIME = t-, and another after the switching at TIME = t+). This may only be done when KPAUSE is not 2. The model library subroutines which trip buses (i.e., BSDSCN), machines (i.e., GENTRP), loads (e.g., LDSHxx), and branches (e.g., LINTRP) set KPAUSE and KTRIP as required, so that user code needs to be concerned with setting these variables only when it changes PSS®E network data array values directly. To force a dual time step calculation, KPAUSE must be set to 1 and KTRIP to the appropriate non-zero value:

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KTRIP=0	No dual time step calculation required.
KTRIP=1	In activities RUN and MRUN one or more data changes requiring a dual time step calculation were implemented, but none of these affect the network admittance matrix.
KTRIP=-1	In activities STRT and MSTR, one or more load models changed the value of the constant admittance component of a load. In activities RUN and MRUN, one or more data changes affecting the admittance matrix have been implemented; no zero impedance line switchings were implemented.
KTRIP=-3	In activities RUN and MRUN, one or more data changes affecting the admittance matrix have been implemented; at least one zero impedance line switching was implemented.

When KTRIP of 1 (or -1) is required in RUN and MRUN, care must be taken to ensure that some other model has not already set KTRIP to -1 (or -3). Therefore, the statement for setting KTRIP is one of the following:

```
IF (KTRIP.EQ.0) KTRIP=1
IF (KTRIP.GE.0) KTRIP=-1
KTRIP=-3
```

25.4. Model Calling Sequence Rules

When activity DYRE is used to introduce user-written models into a simulation setup, the corresponding records in the Dynamics Data Input File are required to be in the format of the special USRMDL, USRLOD, USRREL, USRAUX, USRFCT, and USRDCL records described in User-Written Models. In addition, these models must be written so as to be compatible with the calling sequences assumed when these models are called from subroutines TBLCNC, TBLCNT, CONEC or CONET (see Section 23.1.1, “Program Structure”).

The SUBROUTINE statements for plant-related models must be of the form:

```
SUBROUTINE modelname (MC, ISLOT)
```

where:

MC	Is the internal PSS®E machine array index for the machine at which the model is being called.
ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each plant-related model reference are set as follows:

STRTIN(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTIN(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTIN(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.

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STRTIN(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The SUBROUTINE statements for load-related models must be of the form:

```
SUBROUTINE modelname (LD, ISLOT, ISLOT2)
```

where:

LD	Is the internal PSS®E load array index for the load at which the model is being called.
ISLOT	Is the internal PSS®E array allocation table index for the shared data for this model call.
ISLOT2	Is the internal PSS®E array allocation table index for the private data for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each load-related model reference are set as follows:

LDSTRT(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
LDSTRT(2,ISLOT)	Contains the index of the first of NI shared ICONs used by the model, or zero if NI is zero.
LDSTR2(1,ISLOT2)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
LDSTR2(2,ISLOT2)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
LDSTR2(3,ISLOT2)	Contains the index of the first of NRI reserved (i.e., private) ICONs used by the model, or zero if NRI is zero.

The SUBROUTINE statements for line relay models must be of the form:

```
SUBROUTINE modelname (KM, RS, ISLOT)
```

where:

KM	Is the internal PSS®E double entry line array index for the line at which the model is being called.
RS	Is the relay slot (1 or 2) as specified on the USRREL data record.
ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each line relay model reference are set as follows:

RLSTRT(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
RLSTRT(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.

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RLSTRT(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
RLSTRT(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The SUBROUTINE statements for auxiliary-signal models must be of the form:

```
SUBROUTINE  modelname (IDVX, IDVT, ISGX, ISLOT)
```

where:

IDVX	Is the device index (the device number in case of two-terminal dc line, multi-terminal dc line and FACTS device, and the internal PSS®E array index in the case of VSC dc line).
IDVT	Is the device type with which the auxiliary-signal is associated: 1: for two-terminal dc line 2: for multi-terminal dc line 3: for VSC dc line 4: for FACTS device
ISGX	Is the signal injection point number (or the auxiliary-signal index). This is any number 1 through maximum number of auxiliary-signals that can be associated with the device type under consideration (see Table 3.1, "Standard Maximum PSS®E Program Capacities").
ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each auxiliary-signal model reference are set as follows:

STRTAU(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTAU(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTAU(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero
STRTAU(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The auxiliary-signal model output has to be calculated and stored in the first VAR location VAR(L) where L is the starting VAR index for that auxiliary-signal model (i.e., L=STRTAU(3,ISLOT)). Subsequently, depending on the type of device with which the auxiliary-signal model is associated with, the model output contained in VAR(L) is then transferred by PSS®E into one of the PSS®E standard arrays, DC2SIG or DCMSIG or VSCSIG or FCTSIG. The array locations in DC2SIG, DCMSIG, VSCSIG, and FCTSIG into which VAR(L) is transferred,

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corresponds to the device number (internal VSC device index for VSC dc lines) and the signal index (i.e., the signal injection point number) for which the auxiliary-signal model is called.

The SUBROUTINE statements for FACTS device models must be of the form:

```
SUBROUTINE  modelname (IDVX, ISLOT)
```

where:

IDVX	Is the device index.
ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each FACTS device model reference are set as follows:

STRTFC(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTFC(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTFC(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRTFC(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The SUBROUTINE statements for dc line models (for all those dc line models that are called from internal PSS®E tables) must be of the form:

```
SUBROUTINE  modelname (IDVX, ISLOT)
```

where:

IDVX	Is the device index.
ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each two-terminal dc line, multi-terminal dc line, and VSC dc line model reference are set as follows.

Two-terminal dc line allocation tables:

STRT2D(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRT2D(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRT2D(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRT2D(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

Multi-terminal dc line allocation tables:

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STRTMD(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTMD(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTMD(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRTMD(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

VSC dc line allocation tables:

STRTVS(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTVS(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTVS(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRTVS(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The SUBROUTINE statements for switched shunt models must be of the form:

```
SUBROUTINE  modelname (ISW, ISLOT)
```

where:

ISW	Is the internal switched shunt index.
ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each switched shunted device model reference are set as follows:

STRTSS(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTSS(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTSS(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRTSS(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The SUBROUTINE statements for induction machine models must be of the form:

```
SUBROUTINE  modelname (IMC, ISLOT)
```

where:

IMC	Is the internal PSS®E induction machine array index for the machine at which the model is being called.

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ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for each induction machine model reference are set as follows:

STRTIM(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTIM(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTIM(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRTIM(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The SUBROUTINE statements for all the following models (collectively called the CCT models) - the branch device and other models, two-winding transformer device and other models, three-winding transformer device, protection and other models, bus protection and other models, machine protection and other models, dc line (2-terminal, n-terminal and VSC dc) protection and other models, FACTS device protection and other models, switched shunt protection and other models, induction machine protection and other models, and the miscellaneous other models must be of the form:

```
SUBROUTINE  modelname (KM, ISLOT)
```

where:

KM	For all CCT type models except the miscellaneous other model, KM is the in-ternal PSS®E device of that device to which the CCT model is attached.

In the case of miscellaneous other models, KM is the model instance of the miscellaneous other model

ISLOT	Is the internal PSS®E array allocation table index for this model call.

Using the notation of User-Written Models, at the completion of activity DYRE, the array allocation table entries for CCT models are set as follows.

STRTCCT(1,ISLOT)	Contains the index of the first of NC CONs used by the model, or zero if NC is zero.
STRTCCT(2,ISLOT)	Contains the index of the first of NS STATEs used by the model, or zero if NS is zero.
STRTCCT(3,ISLOT)	Contains the index of the first of NV VARs used by the model, or zero if NV is zero.
STRTCCT(4,ISLOT)	Contains the index of the first of NI ICONs used by the model, or zero if NI is zero.

The CALL statements for the two-terminal dc line models that are not yet in a table-driven form (i.e., not called from internal PSS®E tables - CDCRL, CASEA1, CDCVUP, CEELRIT, CMDWS2T, CMFORDT, CHIGATT, and CMDWAST), are generated as:

```
CALL modelname (IDC, I, J, K, L)
```


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where IDC is the dc line number, and, again using the notation of User-Written Models:

I	Is the index of the first of NI ICONs used by the model, or zero if NI is zero.
J	Is the index of the first of NC CONs used by the model, or zero if NC is zero.
K	Is the index of the first of NS STATEs used by the model, or zero if NS is zero.
L	Is the index of the first of NV VARs used by the model, or zero if NV is zero.

For all other model categories (i.e., those that are not in any of the following categories: plant related, load related, line-relay models, auxiliary-signal models, dc line models, and FACTS device models), the CALL statements are generated as:

```
CALL modelname (I, J, K, L)
```

where I, J, K and L are as described above.

25.5. Writing the Basic Model

The steps in writing a PSS®E model are:

1. Determine the block diagram and/or the differential and algebraic equations of the equipment to be modeled.
2. Identify the state variables associated with the model and determine a procedure for computing their time derivatives.
3. Identify those quantities needed as inputs to the model.
4. Allocate locations in the CON, STATE, VAR and/or ICON arrays as required.
5. Write the model subroutine in FORTRAN or FLECS.

This procedure is best illustrated by an example. Consider the simple excitation system shown in Figure 25.1, "Simple Excitation System". (Note that this example is for illustrative purposes only. In fact, this model is not likely to be suitable for modeling any excitation system.)

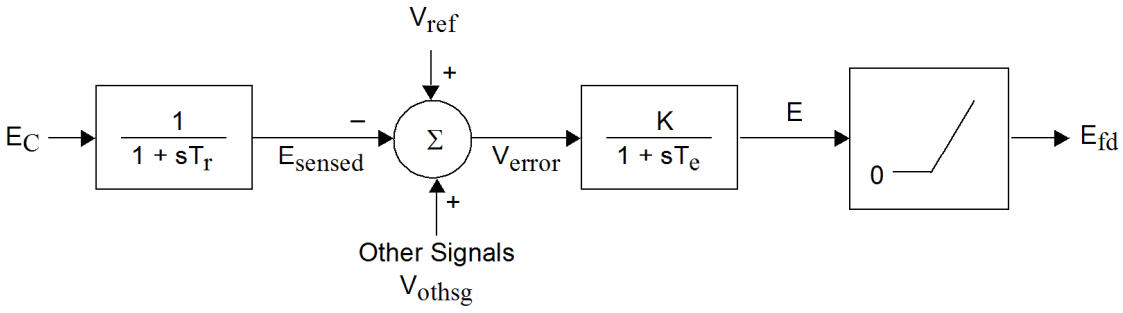

Figure 25.1. Simple Excitation System

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The first transfer function block of this model involves one state variable which is shown as E_{sensed}, the voltage transducer output. This transfer function gives the equation:

$$E_{sensed} = \frac{1}{1 + sT_r} \times E_C$$

Cross multiplying and rearranging gives the required expression for the time derivative as:

$$sE_{sensed} = \frac{dE_{sensed}}{dt} = \frac{E_C - E_{sensed}}{T_r}$$

Similarly, for the second transfer function, the exciter:

$$E = \frac{K}{1 + sT_e} \times V_{error}$$

resulting in:

$$sE = \frac{dE}{dt} = \frac{K \times V_{error} - E}{T_e}$$

The limit block is a simple clip (windup limit) prohibiting a negative exciter output voltage. Note that, because this is an excitation system model, the model output, E_{fd}, must be placed in the EFD array for use by the generator model.

We may now allocate memory locations in the various dynamics data arrays (refer to Table 23.1, "Dynamic Simulation Arrays").

Model inputs:

ECOMP(I)	E_c, the compensated voltage input
VOTHSG(I)	V_{othsg}, a possible supplementary input signal from a stabilizer model

Model outputs:

EFD(I)	E_{fd}, the excitation system output voltage

Other model variables:

CON(J)	= T_r VREF(I) = V_{ref}
CON(J+1)	= K STATE(K) = E_{sensed}
CON(J+2)	= T_e STATE(K+1) = E

V_{error} may be treated as a local variable in our model subroutine because its value need not be preserved from one time step to the next. The model writer could elect to assign such a quantity to a VAR if it is desired to place it in an output channel.

Thus, our model requires three CONs (J through J+2), two STATEs (K and K+1), and no VARs or ICONs. The index I shown above is the machine array index assigned to the machine at which the model is to be called.

Following the model calling conventions of PSS®E plant-related models (see Section 25.4, "Model Calling Sequence Rules"), the CALL statement for this model will be:

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```
      CALL DEMOEX ( I, ISLOT )
```

The model subroutine may now be written. In MODEs one through four, the bus sequence number must be determined. If the machine is off-line, or the generator model at the machine is either a static VAR system, an induction machine, a DSMES, CSMES battery energy storage CBES, or a static condenser CSTATT, (none of which allow excitation systems), the model does no further calculations. The starting array indices in the CON and STATE arrays for use in this model call are determined. The coding is:

```
C
C  BUS SEQUENCE NUMBER NEGATIVE IF MACHINE
C    IS OFF-LINE, SVS OR INDUCTION MACHINE
C
      IB=NUMTRM(I)
      IF (IB.LE.0) RETURN
C
C  GET STARTING 'CON' AND 'STATE' INDICES
C
      J=STRTIN(1,ISLOT)
      K=STRTIN(2,ISLOT)
```

In the above code fragment, I is the machine array index and IB is the bus sequence number at which machine I is connected (see Bus Sequence Numbers and Plant and Machine Sequence Numbers).

In MODE one, the model must initialize its STATEs, along with its entry in the VREF array. (The values of EFD and ECOMP are initialized by the generator model prior to entering an excitation system model, and VOTHSG is initialized by activity STRT before any stabilizer models are called.) This initialization is done on the basis of known model input and output values, and the assumption that time derivatives of all STATE variables are zero:

```
      STATE(K)=ECOMP(I)
      STATE(K+1)=EFD(I)
      VREF(I)=ECOMP(I) + EFD(I)/CON(J+1)
```

It is advisable to alarm if the STATE variable E is initialized outside of its permissible range. The quantities written out are the external identifiers (bus number and machine identifier) which are indexed by variables which were described above:

```
      IF (EFD(I).LT.0.) WRITE(LPDEV,307) NUMBUS(IB),MACHID(I)
 307  FORMAT(' DEMOEX AT BUS',I6,' MACHINE ',A,
     *       ' INITIALIZED OUT OF LIMITS')
```

In MODE two, the model must calculate the values of the time derivatives as derived above:

```
      DSTATE(K)=(ECOMP(I)-STATE(K))/CON(J)
      VERROR=VREF(I)+VOTHSG(I)-STATE(K)
      DSTATE(K+1)=(CON(J+1)*VERROR-STATE(K+1))/CON(J+2)
```

In MODE three, the model must transfer the STATE variable E into the EFD array with the appropriate clipping:

```
      EFD(I)=MAX(STATE(K+1),0.)
```

In MODE four, the model must update NINTEG if the highest numbered STATE which it uses is greater than the present value of NINTEG:

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```
NINTEG=MAX(NINTEG,K+1)
```

MODE's five through six handle the model's responsibilities during activities DOCU and DYDA. In activities DOCU and DYDA, all model output should be written to the FORTRAN unit number contained in the variable IPRT. In MODE eight, definitions of CON and ICON input values are placed in the arrays CON_DSCRPT and ICON_DSCRPT, respectively.

The complete basic model subroutine may now be assembled from the above code fragments as shown in Figure 25.3, "Basic DEMOEX Model Routine (Part 2)". This code references the FLECS procedure shown in Figure 25.5, "FLECS Procedure", which has to be inserted just before the END statement of Figure 25.3, "Basic DEMOEX Model Routine (Part 2)".

The following points on this model coding should be noted.

1. The file specified in the INCLUDE statement provides access to the main PSS®E data arrays and COMMON variables. This syntax can be used for model source codes written in FLECS as well as pure FORTRAN.

   The 'I' of the INCLUDE is in Column 7 for both FLECS and FORTRAN models.

2. The code in Figure 25.3, "Basic DEMOEX Model Routine (Part 2)" contains declaration of FORTRAN intrinsic functions, EXTERNAL subprogram declarations, and type declaration for local variables.

3. This model has not as yet been coded for use in extended term simulations (see Section 25.9, "Writing a Basic Load Model"). When MIDTRM is.TRUE., activity MSTR is being executed; the model calls BADMID which prints an appropriate message and sets a flag which does not allow activity MRUN to be executed. The model then RETURNs.

   The three arguments of subroutine BADMID are the machine array index, the bus sequence number, and the model name enclosed in single quotes. It is used in plant-related models which have not been written for use in extended term simulations.

   A similar subroutine, BADMDL, is used in load-related models which have not been expanded for use in extended term simulations. Its three arguments are the load array index, the bus sequence number, and the model name enclosed in single quotes.

   Finally, a subroutine, BADMD2, is used in all other models which have not been expanded for use in extended term simulations. It has a single argument: the model name enclosed in single quotes.

4. Any error or progress report messages written in MODEs one through four should be written to the FORTRAN unit number specified by the variable LPDEV.

5. When this model is used at a machine, the output signal of any excitation limiter model called at the machine is ignored. Some excitation system models use the sum:

6. VOTHSG(I) + VUEL(I) + VOEL(I)

7. as their single supplementary signal input.

8. ISLOT is undefined in MODE eight. Hence the code for MODE eight appears before the use of ISLOT.

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```
      SUBROUTINE DEMOEX(I,ISLOT)
      C
          INCLUDE 'COMON4.INS'

      C
          INTEGER  I,ISLOT
      C
      C    I     = MACHINE ARRAY INDEX
      C    ISLOT = ARRAY ALLOCATION TABLE INDEX
      C          J = STRTIN(1,ISLOT) [ USES CON(J) THROUGH CON(J+2) ]
      C          K = STRTIN(2,ISLOT) [ USES STATE(K) AND STATE(K+1) ]
      C
          INTRINSIC MAX, ABS
          EXTERNAL  BADMID, DOCUHD
      C
          INTEGER   IB,   J,  K, IBUS, JJ
          REAL      VERROR
          LOGICAL   NEW
          CHARACTER IM*2
      C
          IF (MODE.EQ.8)
          .  CON_DSCRPT(1)='Tr'
          .  CON_DSCRPT(2)='K'
          .  CON_DSCRPT(3)='Te'
          .  RETURN
          ...FIN
      C
      C  GET STARTING 'CON' AND 'STATE' INDICES
      C
          J=STRTIN(1,ISLOT)
          K=STRTIN(2,ISLOT)
      C
          IF (MODE .GT. 4) GO TO 1000
      C
      C  BUS SEQUENCE NUMBER NEGATIVE IF MACHINE
      C    IS OFF-LINE, SVS OR INDUCTION MACHINE
      C
          IB=NUMTRM(I)
          IF (IB.LE.0) RETURN
      C
          IF (MIDTRM) GO TO 900

          GO TO (100,200,300,400), MODE
      C
      C  MODE = 1 - INITIALIZE
      C
```

Figure 25.2. Basic DEMOEX Model Routine (Part 1)

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```
100   STATE(K)=ECOMP(I)
       STATE(K+1)=EFD(I)
       VREF(I)=ECOMP(I) + EFD(I)/CON(J+1)
       IF (EFD(I).LT.0.) WRITE(LPDEV,307) NUMBUS(IB),MACHID(I)
       RETURN
C
C  MODE = 2 - CALCULATE DERIVATIVES
C
 200   DSTATE(K)=(ECOMP(I)-STATE(K))/CON(J)
       VERROR=VREF(I)+VOTHSG(I)-STATE(K)
       DSTATE(K+1)=(CON(J+1)*VERROR-STATE(K+1))/CON(J+2)
       RETURN
C
C  MODE = 3 - SET EFD
C
 300   EFD(I)=MAX(STATE(K+1),0.)
       RETURN
C
C  MODE = 4 - SET NINTEG
C
 400   NINTEG=MAX(NINTEG,K+1)
       RETURN
C
C  MODE > 4
C
 1000 IM=MACHID(I)
       IB=ABS(NUMTRM(I))
       IBUS=NUMBUS(IB)
C
       IF (MODE.EQ.6) GO TO 2000
C
C MODE = 5 OR 7--ACTIVITY DOCU
C
IF (MODE.EQ.5)
. CALL DOCUHD(*1900)
. GO TO 1100
...FIN
C
C DATA CHECKING CODE
C
NEW=.FALSE.
C
UNLESS (CON(J).GT.2.*DELT .AND. CON(J).LT.0.2)
. PRINT-HEADING
. WRITE(IPRT,107) CON(J)
...FIN
C
UNLESS (NEW) RETURN
C
C DATA TABULATION CODE
```

Figure 25.3. Basic DEMOEX Model Routine (Part 2)

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```
C
1100 JJ=J+2
WRITE(IPRT,17) IBUS,IM,J,JJ,K,K+1
WRITE(IPRT,27) (CON(K),K=J,JJ)
1900 RETURN
C
C MODE = 6--ACTIVITY DYDA
C
2000 WRITE(IPRT,507) IBUS,IM,(CON(K),K=J,J+2)
RETURN
C
900 CALL BADMID(I, IB, 'DEMOEX')
C
17 FORMAT(//6X,'** DEMOEX ** BUS MACH C O N S S T A T E S'/
* I23,3X,A2,3X,2(I7,'-',I6))
27 FORMAT(/21X,'TR K TE'/F24.3,F8.1,F8.3)
37 FORMAT(' DEMOEX AT BUS' I7,' MACHINE ',A,
* ' INITIALIZED OUT OF LIMITS')
97 FORMAT(//' BUS',I7,' MACHINE ',A,':')
107 FORMAT(' TR=',F10.4)
507 FORMAT(I7,' ''DEMOEX''',2X,A,1X,3G13.5,'/')
END
```

Figure 25.4. Basic DEMOEX Model Routine (Part 3)

```
      TO PRINT-HEADING
C     .
      .  UNLESS (NEW)
      .  .  NEW=.TRUE.
      .  .  CALL DOCUHD(*1900)
      .  .  WRITE(IPRT,97) IBUS,IM
      .  ...FIN
C     .
      ...FIN
```

Figure 25.5. FLECS Procedure

25.6. Model Writing Notes

To incorporate a model into a dynamics setup after it has been written, compile the model along with the CONEC and CONET files (if any) and create a dll (typically the dsusr.dll).

Activity DYRE recognizes user-written models and processes them correctly only if:

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1. They are specified on the special USRMDL, USRLOD, USRREL, USRAUX, USRFCT, and USRDCL data records in the Dynamics Data Input File (see User-Written Models). and

2. They are written assuming the subroutine calling sequences given in Section 25.4, "Model Calling Sequence Rules".

The model developed in Section 25.5, "Writing the Basic Model" was written with the calling sequence required for plant-related models. The corresponding DYRE data record might be:

```
101  'USRMDL' 1 'DEMOEX'  4 0  0 3 2 0  0.05 100.
0.4 /
```

User-written models for generating plant equipment, load characteristics and relays, line relays, auxiliary-signal, FACTS devices, and dc lines must be written so as to be compatible with the USRMDL, USRLOD, USRREL, USRAUX, USRFCT, and USRDCL data records, respectively, given in User-Written Models. It is strongly recommended that other equipment models also be compatible with the USRMDL data record requirements; however, user-written models for nonplant equipment which do not conform to the above constraints may be introduced into a simulation using the approach described in Section 23.9.1, "Manual Model Addition ".

While the excitation system model developed in Section 25.5, "Writing the Basic Model"was required to initialize an entry in the VREF array, turbine governor, stabilizer and excitation limiter models have no initialization duties other than initializing their own STATEs and VARs.

Plant-related models use the approach shown in the previous example to obtain machine and bus array indices.

For user-written cross-compound governor models (i.e., IC is eleven on the USRMDL data record), the machine index of the low pressure unit is obtained as in the following example:

```
I2=IBITS(MACNUM(ISLOT),16,16)
```

where IBITS is the FORTRAN intrinsic function for extracting a sequence of bits, and ISLOT is the second argument of the model's SUBROUTINE statement as in the DEMOEX model.

The DEMOEX model is written under the assumption that all of its constant parameters have nonzero values. If a constant is allowed to be zero, care must be taken in the model to avoid division by that constant. A common case is the transfer function of a transducer such as the first STATE in the DEMOEX model.

If T_r is allowed to be zero, statement 200 and the statement following would be replaced by the following code fragment:

```
200  IF (CON(J).GT.2.*DELT) GO TO 210
     STATE(K)=ECOMP(I)
     STORE(K)=ECOMP(I)
     GO TO 220
210  DSTATE(K)=(ECOMP(I)-STATE(K))/CON(J)
220  VERROR=VREF(I)+VOTHSG(I)-STATE(K)
```

This is somewhat more general in that the IF statement checks for a small time constant and makes the transfer function algebraic if the time constant is less than two time steps. This approach reduces the risk of numerical integration instability. This coding technique is recommended whenever transfer functions are likely to be characterized by small time constants.

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The proper handling of limits is critical to the accurate modeling of equipment. Generally, there are two types of limits:

1. A clip (or a windup limit). The STATE variable may be driven beyond its limits, but the value used downstream is the limited value.

2. A hard limit on the transfer function (or a limit without windup). The STATE variable itself is limited and its time derivative is not allowed to drive it beyond its limit.

Both types of limits occur in practice and the user must determine which is appropriate in a given situation.

The limit in the DEMOEX model is a clip. To model it as a limit without windup, the MODE three calculation at statement 300 should be changed to:

```
300   IF (STATE(K+1).LT.0.) STATE(K+1)=0.
      EFD(I)=STATE(K+1)
```

In addition, after calculating DSTATE(K+1) and before the RETURN statement in MODE two, the following statements should be added:

```
      IF (STATE(K+1).GT.0.) RETURN
      STORE(K+1)=0.
      IF (DSTATE(K+1).LT.0.) DSTATE(K+1)=0.
```

Note that the output of a transfer function block must be limited *before* it is used in any calculations. This is the case in our example because STATE(K+1) is limited in MODE three; recall that at each simulation time step, MODE three calculations are done first, followed by the network solution and then the MODE two calls of subroutines CONEC and TBLCNC.

The data checking code shown in Figure 23-3 requires that CON(J) (i.e., T_r) be greater than two simulation time steps and less than 0.2 seconds. Otherwise, T_r is tabulated along with the standard activity DOCU report of the model. The remaining model constants would be checked using similar UNLESS blocks.

Coding such as that shown in this example is suitable for all plant-related models. Models called from subroutines CONEC and CONET are handled in essentially the same manner but have two additional duties in activities DOCU and DYDA:

1. The model must determine if selective mode is active, and if so, if it should do anything or simply RETURN. The following code fragment, assuming a bus type model, with the bus number contained in ICON(I), handles this function:

   ```
   IF (IBDOCU.GT.0)
   .  IF (ICON(I).NE.IBDOCU) RETURN
   ...FIN
   ```

2. For line type models, it becomes a little more complicated. In activity DOCU, both ends of the branch need to be looked at and, if either is equal to IBDOCU, the model should act. In activity DYDA, however, only one end (e.g., the from bus of an ac branch) should be checked so that the record appears only once in the output.

3. In activity DYDA, the model must check for the AB interrupt code before writing its output record. Code such as in the following code fragment must be inserted before the first executable statement:

   ```
   CHARACTER IVEC(1)*2
   ```


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```
SAVE      IVEC
DATA      IVEC/'AB'/
```

Then the following code is needed immediately preceding the activity DYDA WRITE statement:

```
CALL CONINT(IVEC,1,JJ)
IF (JJ.NE.0) RETURN
```

25.7. Extended Term Simulation

The differential equation treatment shown in the above example illustrates that required for the state-space explicit integration method used in the STRT/RUN, ESTR/ERUN and GSTR/GRUN activity sequences. Additional model coding for the Z-form trapezoidal integration is required before using the extended term simulation MSTR/MRUN activity sequence; details are given in the *PSS®E Program Application Guide*.

It is strongly recommended that models first be developed, tested and used with activities STRT and RUN using the modeling techniques described in this section. For users with the Extended Term Simulation Section of PSS®E, the model should be augmented to be valid with activities MSTR and MRUN only after accurate modeling has been achieved with activities STRT and RUN.

25.8. Coordinated Call Models

Many equipment models are called during the network solution to calculate current injections which are dependent on the voltage at the bus to which they are connected. Models that contain both differential equation responsibilities and current injection responsibilities meet their modeling requirements in PSS®E by being implemented as coordinated call models. Such a model is called from subroutine TBLCNC or CONEC (see Section 23.4.1, "Subroutines TBLCNC and CONEC ") for calculations related to its differential equations, and from subroutine TBLCNT or CONET (see Section 23.4.2, "Subroutines TBLCNT and CONET") for current injection calculations. Models for equipment such as static Var systems, dc lines, FACTS devices, and loads are usually implemented as coordinated call models (see activity DYRE).

In a coordinated call model, the model is called at its primary entry point (i.e., the name specified on its Dynamics Data Input File data record and on the model's SUBROUTINE statement) for the standard MODE one through four state variable calculations. The model is also called at a supplementary entry point at each iteration of the network solution to inject the model's contribution into the CURNT array, and once more following network solution convergence for any post-solution housekeeping required (e.g., updating one or more of the model's VARs). The supplementary ENTRY point name of a coordinated call model is formed by replacing the first character of the SUBROUTINE name with a T. Clearly, coordinated call models, both those from the PSS®E Model Library as well as user-written models, must have their primary entry point name beginning with a character other than T.

25.9. Writing a Basic Load Model

A user-written load model must be written as a coordinated call model (see Section 23.8 Coordinated Call Models). The fundamental steps in writing such a model are identical to those outlined in Section 25.5, "Writing the Basic Model". The primary entry point is written using an approach similar to that used for DEMOEX earlier in this chapter.

A simple example is used to illustrate the procedure. Consider a load model which is to apply the same frequency sensitivity factor to the three standard load characteristics (see Section 6.3.13 Load). If PQ is the

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total complex load (i.e., the sum of the three characteristics) at the present value of voltage, the frequency sensitive load is calculated as: PQ * (1. + K * pu frequency deviation).

We may now allocate memory locations in the various dynamics data arrays (refer to Table 23.1, "Dynamic Simulation Arrays").

Model inputs:

VOLT(IB)	the complex voltage at bus IB
BSFREQ(IB)	pu frequency deviation at bus IB

Model outputs:

CURNT(IB)	complex current injection array used during the network solution
TPLOAD(I)	effective MW load
TQLOAD(I)	effective Mvar load

Other model variables:

CON(J)	K, the frequency sensitivity factor
CLODFR(1,I)	complex nominal constant MVA load at load I
CLODFR(2,I)	complex nominal constant current load at load I
CLODFR(3,I)	complex nominal constant admittance load at load I

Thus, our model requires one CON (J), and no STATEs, VARs or ICONs. The index I shown above is the index array index assigned to the load at which the model is to be called, and the index IB is the bus sequence number of the bus to which the load is connected.

Following the model naming and calling conventions of PSS®E load-related models (see Load-Related Models and Section 25.4, "Model Calling Sequence Rules"), the CALL statement for this model will be:

```
      CALL DEMOBL ( I, ISLOT, ISLOT2 )
```

The model subroutine may now be written. Because no state variables are involved and no special initialization calculations are required (the initial condition bus frequency deviation is zero), no calculations are required in MODEs one through four of the model's primary entry point and no special coding is required for handling extended term simulations (see Section 25.7, "Extended Term Simulation").

In the model's supplementary entry point, the bus sequence number must be determined. If the load is out-of-service, the model does no further calculations. The starting array index in the CON array for use in this model call is determined. The coding is:

```
C
C  BUS SEQUENCE NUMBER NEGATIVE IF LOAD IS OUT-OF-SERVICE
C
      IB=NUMLOD(I)
      IF (IB.LE.0) RETURN
```


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```
C
C  GET STARTING 'CON' INDEX
C
      J=LDSTRT(1,ISLOT)
```

In the above code fragment, I is the load array index and IB is the bus sequence number at which load I is connected (see Bus Sequence Numbers).

In load characteristic models, it is important to recognize that the network solution subroutine SITER has already made an injection into the CURNT array corresponding to the values present in the standard load characteristic array CLODFR. It is the model's responsibility, therefore, to inject the difference between the total load as calculated by the model and the standard load characteristic already injected. This may be done either by injecting the incremental amount, or by first making an injection to cancel out the standard injection from CURNT and then injecting the new total load model injection.

The following code fragment calculates the total load injected by SITER expressed in MVA (PQOLD), and the total effective load as corrected for frequency sensitivity (PQNEW). It places the active and reactive components of the effective load into the TPLOAD and TQLOAD arrays, respectively. Finally, it injects the load increment into the CURNT array. The coding is:

```
C
C  CALCULATE TOTAL LOAD FOR STANDARD CHARACTERISTICS
C
      VM=ABS(VOLT(IB))
C
      PQOLD=CLODFR(1,I)
      IF (VM.LT.PQBRAK) PQOLD=PQOLD*CNSTPQ(VM)
C
      S=VM*CLODFR(2,I)
      IF (VM.LT.0.5) S=S*CNSTCR(VM)
      PQOLD=PQOLD+S
C
      PQOLD=PQOLD + VM*VM*CONJG(CLODFR(3,I))
C
C  CALCULATED DESIRED TOTAL LOAD INJECTION
C
      PQNEW=PQOLD*(1.+CON(J)*BSFREQ(IB))
C
      TPLOAD(I)=REAL (PQNEW)
      TQLOAD(I)=AIMAG(PQNEW)
C
C  INJECT LOAD INCREMENT (MODEL INJECTION - STANDARD INJECTION)
C
      CURNT(IB)=CURNT(IB)-CONJG((PQNEW-PQOLD)/VOLT(IB))
```

The complete basic model subroutine may now be assembled from the above code fragments as shown in Figure 25.6, "Basic DEMOBL Model Routine".

Augmenting the model to handle the requirements of MODEs five through eight follows the approach as outlined for plant-related models (see Figure 25.3, "Basic DEMOEX Model Routine (Part 2)"), except that the LOADID and NUMLOD arrays are used in place of the MACHID and NUMTRM arrays.

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```
      SUBROUTINE DEMOBL(I,ISLOT,ISLOT2)
C
      INCLUDE 'COMONFOR.INS'
C
      INTEGER          I,ISLOT,ISLOT2
C
C I      = LOAD ARRAY INDEX
C ISLOT  = SHARED  DATA ARRAY ALLOCATION TABLE INDEX
C ISLOT2 = PRIVATE DATA ARRAY ALLOCATION TABLE INDEX
C
C DO NOTHING IN 'DOCU', 'DYDA', 'ADD/EDIT CONSTANTS' FUNCTIONS FOR NOW
C
      REAL     CNSTCR, CNSTPQ
      EXTERNAL  CNSTCR, CNSTPQ
C
      INTRINSIC ABS,    AIMAG,  CONJG,  REAL
C
      INTEGER   IB,     J
      REAL      VM
      COMPLEX   PQNEW,  PQOLD,  S
C
      IF (MODE.GE.4) RETURN
C
C NO STATE VARIABLE CODE NEEDED IN MODES 1 THROUGH 4
C
      RETURN
C
C SUPPLEMENTARY ENTRY POINT
C
      ENTRY TEMOBL(I,ISLOT,ISLOT2)
C
C BUS SEQUENCE NUMBER NEGATIVE IF LOAD IS OUT-OF-SERVICE
C
      IB=NUMLOD(I)
      IF (IB.LE.0) RETURN
C
C GET STARTING 'CON' INDEX|
C
      J=LDSTRT(1,ISLOT)
C
C CALCULATE TOTAL LOAD FOR STANDARD CHARACTERISTICS
C
      VM=ABS(VOLT(IB))
C
```

Figure 25.6. Basic DEMOBL Model Routine

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Chapter 26
Event Studies

Run Activity Event Studies - GUI
Dynamics Tree View>Dynamics>Event Studies>Add Event Study [Open] Event Study file (*.evs)

26.1. Overview

Event Studies are a user-defined series of events that can be replayed over and over with data values and settings changed interactively. Event Studies consist of one or more Event Items. Event Items currently consist of network equipment connection and disconnection Events and Fault Events.

When the Event Study is run as a Dynamics Event Study, Event Items are implemented as Disturbance APIs when a match is available, and as general Network APIs when no defined disturbance API exists e.g. Load connect/disconnect events. Event Items in a Dynamics Event Study have a time element associated with them and are performed in ascending time sequence.

When the Event Study is run as a Power Flow Study, Fault Event Items are mapped to SCMU APIs and network Event Items are mapped to Network APIs. Currently, simultaneous Fault Events are not supported and so any simultaneous Events in the Event Study are de-activated and only the first encountered Fault Event is performed.

Dynamics and Power Flow event studies can be accessed in several different ways. Event studies can be added and viewed by accessing the Dynamics tab in the Tree View. Individual event items can be added to the active study from the tree view, from the diagram, and from the event spreadsheet. The event spreadsheet enables the user to edit the events in the active study.

Event Studies are preserved in an external file with an *.evs suffix. Through the use of the Event Study file, defined Event Studies maybe used with more than one set of files.

26.1.1. Event Item Type

The following table describes the supported Event Types and their implementation when run in Dynamics and Power Flow Study modes.

Table 26.1. Supported Event Types

Event Type	Dynamics Implementation	Power Flow Implementation
Connect Bus	Re-connect Bus - RECN	Re-connect Bus - RECN
Disconnect Bus	Disconnect Bus - DSCN	Disconnect Bus -DSCN
Connect Machine	Machine Status switch	Machine Status switch
Disconnect Machine	Machine Status switch	Machine Status switch
Connect Load	Load Status switch	Load Status switch

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Event Type	Dynamics Implementation	Power Flow Implementation
Disconnect Load	Load Status switch	Load Status switch
Connect/Close Line	Branch/Transformer Close Disturbance	Branch/Transformer Status switch
Disconnect/Trip Line	Branch/Transformer Trip Disturbance	Branch/Transformer Status switch
Bus Fault	Bus Fault Disturbance	Multiple unbalanced network solution - SCMU
Line Fault	Branch/Transformer Line Fault Disturbance	Multiple unbalanced network solution - SCMU
Unbalanced Bus Fault	Calculate and Apply Unbalanced Bus Fault	Multiple unbalanced network solution - SCMU
Unbalanced Line Fault	Calculate and Apply Branch Fault	Multiple unbalanced network solution - SCMU
Vref Change	Vref Change Disturbance	(not applicable)

26.1.2. Event Study Properties

The user specifies the general parameters which will be used when the event study is run. All fields are initially filled with default values, however the user may edit them all.

The following fields are used only if the Event Study is run as a Dynamics Event Study. The values in these fields have no significance if the Event Study is run as a Power Flow Event Study.

Table 26.2. Dynamics Event Study Properties

Property	Description
Study length	Total duration of the event study in seconds.
Integration step	Time interval, in seconds, between steps.
Tolerance	PSS®E dynamic simulation convergence tolerance.
Iteration limit	PSS®E dynamic simulation iteration limit.
Print interval	Number of steps between text outputs.
Graph interval	Number of steps between channel file outputs.
Plot interval	Number of steps between plot results output.

The *active study* is the only study to which event items can be added from the diagram and from the network tree. There are two ways in which an event study becomes the *active study*.

- The event study currently being added automatically becomes the *active study*.
- Selecting the event study pop up menu displays an option to set a particular event study to be the *active study*.

The *active study* is the one which will be executed when the option to *Run Dynamic Study* or *Run Power Flow Study* is selected. When you exit the program, the studies are saved to the named file. A flag indicating which of the studies is the *active study* is also saved, so that when the file is read back in, the same event study will still be the *active study*. If the *active study* is deleted from the tree view, no study is active until another is added or a remaining one is made active by selecting the *Make Active Study* choice on the popup menu.

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If there is an event which should not be included when an event study is executed, it can easily be deactivated in the event spreadsheet. This eliminates having to delete and reconfigure events which may be needed at another time.

26.1.3. Adding an Event Item

The user specifies the Event Items which will be part of the selected study.

Table 26.3. Event Item Properties

Property	Description
Name	User-specified; can be edited to suit the user.
Event Type	User-specified event type: • Unbalanced Bus fault • Unbalanced Line fault • Line fault • Bus fault VREF change • Disconnect Load • Connect Load • Disconnect Bus • Connect Bus • Disconnect Machine • Connect Machine • Line trip • Close Line
Event initiation time (sec)	For Dynamics Event Study only. The time at which this event item occurs; this value must be less than the total time duration of the entire event study.
Admittance/Fault impedance	R (ohms); default = 0
	X (ohms); default = 0
Network Items	User-specified network elements: • Bus • Load • Machine • Branch

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Property	Description
	• 3-Winding Transformer

26.1.4. Running an Event Study

There are two types of event studies, a dynamic study and a power flow study. In order to run a Dynamic Study, dynamics data must be present. If an event study file has been read into the program, event items in the active study will also be indicated on the one-line diagram. More that one Event Item can be associated with any Network Item and only one Event symbol will be displayed on the Network Item in the diagram.

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Chapter 27
Scenarios

Run Activity Scenarios - GUI
File>Scenarios>New Scenarios...
File>Scenarios>Open Scenarios...
File>Scenarios>Save Scenarios...
File>Scenarios>Close Scenarios...
File>Scenarios>Edit Scenarios...
Scenario Files (*.pssxml)

27.1. Overview

The Scenario Manager is designed to eliminate the file management complexities that arise when using PSS®E by keeping track of and organizing the files that are used with PSS®E. Scenarios are groups of files that are used to perform a study. They can consist of any files on either local or remote machines, and can be of any type, including network files, both Power Flow and Dynamics, in both the raw and binary forms, Slider Diagrams, Bus Location and Subsystem configuration files, Contingency analysis files (subsystem Description files, Monitored Element file, Contingency file, Distribution Factors file) and Sequence and OPF data files. All the files used in the study are kept track of in a Scenario file, and, through the use of zipping functionality, all files can be stored in a single zip file. The Scenario file format is based on XML, so it is easily readable by all text editors (see Figure section / Sample Scenario File). When a Scenario file is opened in PSS®E two things happen, all the files designated to load in the Scenario file are opened and the file selectors on all dialogs will be pre-populated with all files in the Scenario that are of the type required by the selector.

Figure 27.1. Sample Scenario File

27.2. Scenario Creation

A Scenario can be created through the GUI or through the API. See GUI, API. When done through the GUI a default group named "Group" will be added to the new Scenario automatically and will be set as the active group and the startup group in order to easily facilitate the population and usage of the Scenario. When creating a New Scenario through the API there will be no automatic group creation and configuration, additional API calls will be needed to create a new group.

No file name is required when a new Scenario is created, however a file name can be supplied through the API call. If a file name is supplied, the new Scenario will be saved to that location after it is initialized.

The default root path for a new Scenario will be the current working directory, unless a valid file name was supplied through the API, in which case the root path will be the directory location of the supplied file name.

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A newly created Scenario will also mark its creation date, the username of the creator, the current version of PSS®E, and the current version of Scenario Manager.

27.3. Scenario Population and Configuration

Once a Scenario is loaded into PSS®E, either through opening a Scenario file or creating a new one, there are several methods that can be used to add files to the Scenario. Files may be added one at a time through the API (See API) or through the GUI by means of selector dialog, drag and drop, or file tracking (See GUI). When a file is added to the Scenario that file item will have several values associated with it. These values are the path, type, and load. The path tells the Scenario where the file is located. The type attribute defines what the file type is. Since PSS®E does not enforce specific file extensions the type attribute is used to associate the files with defined PSS®E file types defined here. The load attribute is used to determine if and when a file will be loaded when the Scenario is opened.

Many file adding methods in the GUI will allow the user to add multiple files of different types at once. However, since there are no standard extensions for PSS®E, an attempt will be made to associate default extensions to their related PSS®E types. If a file is added with a non-default extension its type attribute will be left blank for the user to define later. If a file has an extension that is the default extension of a different PSS®E type the attribute can be modified later, as can all attributes of the files, through the API or GUI.

To allow for better flexibility Groups have been introduced. Groups are nothing more than a collection of files, but they can allow a user to organize a Scenario in whatever manner they see fit. A file can be placed in multiple Groups, so while some Groups can be completely different sets of files, others could be slight variations of each other, possibly having a slightly different Saved Case but the same automation files. New Groups can be created through the API or GUI. Several methods are available for copying and moving files from Group to Group as well as making duplicate Groups or combining Groups in the Scenario through the API and GUI.

27.4. Using a Scenario in PSSE

Using Scenarios in PSS®E presents several alternative methods for using the PSS®E GUI, however, some functions will require an active group to work. One of the main features that will become active with or without an active group is the population of the file selector fields in the GUI. Each file selector will be populated with a list of files of the type the field is looking for. If the file selector field is for an input file and there is a defined active group, and an order has been designated for the desired file type, then the file at the top of the order will automatically be selected. However, no automatic selection will be made if the file selector field item will be used for output, or if there is no active group.

Another difference is the open file and save file dialogs. If there is an active group then an alternative open/save file dialog will be shown which lists the files contained in the Scenario in order to facilitate the quick opening of files already defined in the Scenario. If the user would like to open or save a file not in the Scenario there is a button to bring up the standard open/save dialog.

Lastly, as files are accessed in PSS®E any files not already present in the active group will be kept track of and the user will be prompted to add the new files if desired. However, if there is no active group then the user will not be prompted until and active group is selected.

27.5. Saving and Packing a Scenario

A Scenario can be saved in two different forms. The first form is the Scenario XML File, which contains the Scenario definitions and paths to all files in the Scenario, but not the files themselves. The second form is a

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Zip file, which is an archive of all of the files in the Scenario as well as the Scenario XML File. The Zip file is in essence a single file solution to containing all data used in PSS®E allowing a user to easily hand an entire study or other piece of work to another user. Either file type can be read into PSS®E, however when a zip file is read in, PSS®E will use the data files from within the zip file and not the original file. This will be reflected in the path of each file in the scenario.

27.6. Opening and Unpacking a Scenario

As stated in the previous section, there are two forms a saved Scenario can take. If a Scenario XML File is opened, then the file is read and Scenario makes note of the locations of all of the files. If a Zip file is opened, then the Scenario Manager will look for a Scenario XML File with the same name as the Zip file. If no file inside the Zip file matches that criteria the Scenario Manager will then look for any XML File. If this fails it will then attempt to open each file as a Scenario XML File. If no Scenario XML file is found the Zip file will be ignored. While Scenario Manager can now read and write directly to the zip file there may come a time when the files need to be unzipped. When Scenario Manager is told to unzip the archive it will attempt to place files in locations based on several unzipping options and the file paths stored in the Scenario XML File.

When unzipping, each file in the archive is categorized as either a Root File a Local File or a Remote File. Each category of file has an option for resolving file conflicts which can be set to "skip", "ask", or "overwrite". If the conflict resolution option for a particular file category was set to "skip" then when an attempt is made to extract a file of that category and another file is already in its destination, the file will not be extracted and no note of the conflict will be made. If the option is set to "ask", then when a conflict occurs the file will not be extracted, but a notification of the conflict will be made so the user can make a decision on what action should be done to resolve the conflict. If the option is set to "overwrite" then the file is extracted and the file at the destination will be replaced with the file from the archive, if a file was previously there.

In addition to these three options there are options for the placement of Local Files and Remote Files. The Scenario Manager will always attempt to extract Root Files to the same location relative to the Root Path. However, there are several options for the placement of the other two categories which are "do not unzip", "place in root directory", and "place by absolute path", and Local Files have the additional option for "place by relative path". The "do not unzip" option means any file in that category will not be extracted. The "place in root directory" option means an attempt will be made to extract the file to the root path defined in the Scenario. Selecting the "place by absolute path" means an attempt will be made to extract the files to the same path the files were in before being placed in the archive. Selecting "place files by relative path" means the extraction process will attempt to place the files in the same location they were in before, relative to the root path. For example if the root path at the time the archive was created was "C:\Users\Joe\Documents\Psse\Study5" and the file was located at "C:\Users\ Joe \Documents\Psse\Models\joemodels.dll" and the archive was extracted at "C:\Users\Jack\Documents\Psse\Joe5" joemodels.dll would be extracted to "C:\Users\Jack\Documents\Psse \Models\joemodels.dll", or in relative form "..\Models\mymodels.dll".

The GUI will prompt the user if any conflicts are found and allow the user to choose a course of action for each file. The user can choose to "ignore", "retry", or "overwrite" a conflict. Ignoring will result in the file not being extracted, whereas overwrite will result in the file being extracted and the file at the destination being deleted. Retry can be used to try the extraction again using a different file name, or can be used to try again with the same name after the user may have moved or renamed the file at the destination. If a conflict occurs when a retry is attempted the user will be prompted again.

There are several API calls that can be used through python to check and resolve differences. There is also a batch API call to resolve differences, but since there is no method for returning data though batch commands there is no way to know what conflicts occurred.

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27.7. Scenario Terms

File - The file is the simplest item of a Scenario. These items can be considered data sources, though they do not contain the actual data. Instead these items contain the address of a data source, such as a file on the local system, a file on a remote computer, or even a database. If the data source is moved or deleted the file item will become invalid as it will no longer point to a valid data source.

A PSS®E file type is assigned to each file to associate the file with the relevant PSS®E routines. For example a .cnv file can be associated with the Saved Case file type so it will show up in a selector for Saved Cases. Each file type association will be given by a unique 3 character file type designation. The file types and their corresponding unique identifiers can be found on this table (here).

Each file will also have a load order attribute which will determine if and when the file is loaded when the group the file is part of is loaded. Each file can also have an argument string which will be used to pass additional arguments to the opening process when the file is being automatically loaded. Some files have different argument strings and the GUI provides a simple way to define the appropriate string for each file.

Group - A group is a collection of files and can be thought of as the organizational level of the Scenario. Groups can contain files that are in other groups of the same Scenario, but cannot contain the same file twice. Each group has its own load order which defines what files should be opened when the group is opened, what order to open the files in, and what order the files should be listed in when a selection box is populated. One group in a Scenario can be designated the startup group.

Scenario - The scenario is the highest level of organization for the Scenario Manager tool. It can be seen as a kind of suitcase, allowing the user to place many items inside of it and keep track of them all from a single item.

The scenario can contain any number of groups and one of the groups can be designated the startup group. The scenario will also keep track of its creation and modification dates and the users associated with those dates.

Another important piece of information for the scenario is the root path which plays an important role when using the zip functions of the Scenario Manager.

Load Order - This is the order in which files will be opened and/or arranged when making a selection.

Only files that are "memory resident" will be loaded. Some examples of "memory resident" files are Saved Case, Snapshot, and Slider files. These files contain data about the system that will be stored in PSS®E while the program is running. Many of these file types will only be allowed a single file of that type to be selected for loading since opening multiple files of the same time would result in the overwriting of data that was loaded from the previous file, e.g. Saved Case files. Some files, such as Saved Case and Power Flow Raw Data files, also contain similar information. Since loading both would once again overwrite previously loaded data, only a single file from that category is allowed to be selected for automatic loading. In this case a single Saved Case or a single Power Flow Raw Data file can be selected for automatic loading, but not both. Several other file types have similar groupings, such as Snapshot and Dynamics Data files. However, the number of files that can be automatically loaded through Power Flow Raw Change Data or Dynamics Add Data files is not limited since these files augment the data in PSS®E.

Examples of files that are not "memory resident" would be Subsystem Description Data, Monitored Element Data, Contingency Description Data, Distribution Factor Data, AC Contingency Solution Output, PV Solution Output, and QV Solution Output files. These files are the inputs and/or outputs of certain PSS®E activities, but the information in them does not stay in PSS®E memory once the activity is finished. These files may

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be assigned an order, with each non "memory resident" PSS®E file type having its own order. This order specification will force a specific arrangement of the files when populating a selection box with that file type.

Root Path - This defines the top level of the tree where root files will be located. This path is also where the zip file will be created when executing the scenario zip function with the default arguments. When a zipped scenario is moved and then unzipped, the tree that was below the root path when the scenario zip was created will be recreated at the current location of the zip file. This new location of the zip file will then define the new root path.

Startup Group - The group whose load order will be enacted upon opening of the scenario file.

Active Group - The group whose load order was last run. The startup group (if defined) will always be the first active group under the default scenario opening procedure. All file tracking operations will interact with this group. If a Scenario does not have a startup group, or was opened in a way that skipped the running of the startup group, then no group will be the active group when the Scenario is initially opened.

Root File - A file that is located at or below the root path. These files could be on a local machine or a remote machine depending on what the root path is.

Local File - A file that is located on a local machine, i.e. access to that file is available from the current machine even if no network connection is available. This does not include files at or below the Root Path.

Remote File - A file that is located on a machine that is not the local machine, i.e. a network connection would be needed to access the file. This does not include files at or below the Root Path.

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Appendix A
Appendix A - Activity Summary

Each PSS®E activity performs a specific single function. Each activity requires that certain data be present in working memory and/or in one or more of the PSS®E temporary files, and may influence the contents of working memory and/or the temporary files. The following table summarizes many of PSS®E activities, together with their prerequisites and brief descriptions of their functions.

Table A.1. PSS® E Activities

Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	File>New... *[New]*		Create a new case, case with diagram, diagram, or plot book.	- - -	- - -
	File>Open... *[Open]*	CASE	Open power flow Saved Case file and establish it as the working case.	Previous contents of the working case overwritten.	Saved Case file (*.sav)
	File>Open... *[Open]*	DYRE	Read a Dynamics Data File	Validly specified power flow case, solved to an acceptable mismatch level.	Generators represented as current sources (activity CONG must have been executed). Dynamics Model Raw Data File (*.dyr)
	File>Open... *[Open]*	MCRE	Add machines to an existing plant, enter machine impedance data, and split the plant loading among individual machines.	Validly specified power flow case.	Machine Impedance Data file (*.rwm)
	File>Open... *[Open]*	RDCH	Read power flow source data records from a file	Validly specified power flow case.	Power Flow Raw Data file (*.raw)

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	File>Open... *[Open]*	READ	Read power flow source data records from a file.	If reading change data, the working case must contain a validly specified power flow case.	- - -
	File>Open... *[Open]*	REMM	Read transaction raw data from a Transactions Raw Data File, replacing all existing transaction data.	Validly specified power flow case.	Transactions Raw Data File (*.mwm)
	File>Open... *[Open]*	RESQ	Read sequence data for fault analysis	Positive sequence network where sequence data is to be read must be present in the working case.	Sequence Data file (*.seq)
	File>Open... *[Open]*	ROPF	Read optimal power flow data.	Validly specified power flow case.	Optimal Power Flow Data File (*.rop)
	File>Open... *[Open]*	RSTR	Restore dynamics working memory from a binary Snapshot File	- - -	Dynamics Snapshot Data File (*.snp)
	File>Open... *[Open]*	SRRS	Read a Source Form Snapshot File into dynamics working memory. The previous contents of dynamics working memory are destroyed.	Validly specified power flow case, solved to an acceptable mismatch level.	Generators represented as current sources (activity CONG must have been executed). Dynamics Snapshot Raw Data file (*.srs).
	File>Save... *[Save Network Data]* Case Data	SAVE	Create Saved Case file (*.sav)	Validly specified power flow case.	- - -

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	File>Save... *[Save / Show Dynamics Data]* Dynamics Model Data	DYDA	Write simulation model data in the form of a Dynamics Data File.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	File>Save... *[Save Network Data]* Power Flow Raw Data	RAWD	Create Power Flow Raw Data file.	Validly specified power flow case.	- - -
	File>Save... *[Save Network Data]* IEEE Format Power Flow Data	RWCM	Write working case as IEEE Common Format for the Exchange of Solved Load Flow Cases. The 132-character record tape format is used.	Validly specified power flow case.	- - -
	File>Save... *[Save / Show Dynamics Data]* Machine Parametric Source Data	RWDY	Write dynamic model data of plant-related models: • Inertia and Governor Response Data File (for activity INLF) • Breaker Duty Data File (for activity BKDY) • branch impedance data for branches with relay models (for	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	- - -

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			the PSSPLT activity RELY)		
	File>Save... *[Save Network Data]* Machine Impedance Data	RWMA	Create Machine Impedance Data file.	Validly specified power flow case. If generator reactive powers are to be used in calculations, the case must be solved to an acceptable mismatch level.	Machine impedance data must be correctly specified for those machines to be processed.
	File>Save... *[Save Network Data]* Transaction Data	RWMM	Write transaction data in format suitable for activity REMM.	Validly specified power flow case.	- - -
	File>Save... *[Save Network Data]* Optimal Power Flow Data	RWOP	Create Optimal Power Flow Raw Data File	Validly specified power flow case with optimal power flow data appended to it.	- - -
	File>Save... *[Save Network Data]* Sequence Data	RWSQ	Create Sequence Data File	Validly specified power flow case with sequence data appended to it.	- - -
	File>Save... *[Save / Show Dynamics Data]* Snapshot Data	SNAP	Save a Snapshot of dynamics working memory in a binary file.	Dynamics working memory must contain the appropriate dynamics data.	- - -
	File>Save... *[Save / Show Dynamics Data]* Dump Output Channels	DMPC	Write channel selections from an existing simulation setup to a new Snapshot in Response File format.	Dynamics data must be present in dynamics working memory.	- - -
	File>Scenarios>New Scenario...		Create a new PSS®E study.		

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Save As]				
	File>Scenarios>Open Scenario... *[Open]*		Open an existing PSS®E study.		
	File>Scenarios>Save Scenario...		Save an existing PSS®E study.		
	File>Scenarios>Close Scenario		Close an existing PSS®E study without saving changes.		
	File>Scenarios>Edit Scenario... *[PSS® E Scenario Editor]*		Edit an existing PSS®E study.		
- - -	*File>Compare...* Case Totals	CMPR	Compare selected totals of power flow data and solution results contained in the working case and a designated Saved Case.	Validly specified power flow case. Both cases solved to an acceptable mismatch level if comparing losses, mismatches, or interchange.	Saved Case file (*.sav)
- - -	*File>Compare...* Powerflow Cases	DIFF	Compare selected power flow data, fault analysis data, and solution results contained in the working case and a designated Saved Case.	Validly specified power flow case. Both cases solved to an acceptable mismatch level if comparing bus voltages, line flows, or line losses.	Saved Case file (*.sav)
- - -	*File>Compare...* *[Compare]* Tie Lines	DFTI	Compare selected power flow data, fault analysis data, and solution results of subsystem tie lines contained in the working case with a Saved Case.	Validly specified power flow case. Both cases solved to an acceptable mismatch level if comparing line flows or line losses.	Saved Case file (*.sav)

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
- - -	*File>File information (SIZE/SHOW/BUSN)...* *[File Information]* List unused bus numbers in a range	BUSN	List unused bus numbers in a specified bus number range	Validly specified power flow case.	
- - -	*File>File information (SIZE/SHOW/BUSN)...* *[File Information]* List Save case and Snapshot files	SHOW	List Saved Case and Snapshot files.	None	
- - -	*File>File information (SIZE/SHOW/BUSN)...* *[File Information]* List the number of system components	SIZE	List the number of system components.	Non-null power flow case.	
- - -	*File>Case titles, short & long ...* *[Case Titles]*	CHTI	Re-enter individual lines of the 16 line long title.	Non-null power flow case.	
- - -	*File>Import>Long Title...* *[Select file containing long case title]*	RETI	Accept source data records from a file or the and place them into the 16 line long title.	Non-null power flow case.	
- - -	*File>Import>ECDI File* *[Economic dispatch data file]*				
- - -	*File>Import>DRAW file...*	DRAW	Import a one-line diagram of	If bus voltages and/or line flows	Diagram View

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Open]		a portion of the working case from prior version of PSS®E.	are to be printed, validly specified power flow case solved to an acceptable mismatch tolerance.	Draw Data file (*.drw)
- - -	*File>Import>Image file...* *[Open]*				Diagram View Bitmap file (*.bmp)
- - -	*File>Renumber buses in auxiliary files (RNFI)...* *[Renumber Buses in Auxiliary Files]* or *File>Renumber buses in Diagram (RNFI)...* *[Bus number translation file]*	RNFI	List changes in bus numbers created by activity BSNM for the auxiliary data input files used by activities RESQ, GDIF, DYRE, MCRE, ECDI, INLF, BKDY, ASCC, GCAP, ACCC, and ANSI.	Validly specified power flow case.	Bus Number Translation file (*.trn) containing records of the form: old bus # new bus #
- - -	*File>Exit*	STOP	Close all PSS®E files and exit	None	
- - -	*Edit>Preferences...* *[Program Preferences]*		Specify GUI performance and file management.	None	
	Diagram>Annotation... *[Powerflow Data Annotation]* *[Diagram Range Checking]*		Annotate and color-code one-line diagram.	Validly specified power flow case solved to an acceptable mismatch level.	Diagram View
	Diagram>Generate graphical report (GRPG)... *[Graphical Report Data file]*	GRPG	Send a specified drawing to a specified graphics output device.	Validly specified power flow case solved to an acceptable mismatch level if network data is	Graphical Report Data file (*.grp)

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Graphical Report Data file (*.grp) *[Terminal Read]*			to be included on the drawing.	
	Diagram>Generate graphical power flow bus display (GOUT/GEXM)... *[Select Bus]*	GOUT/GEXM	Display power flow solution results in graphical form for one bus at a time.	Validly specified power flow case solved to an acceptable mismatch level.	
	Locate bus (icon only) *[Select Bus]*		Locate a specified bus on one-line diagram.	Validly specified power flow case solved to an acceptable mismatch level.	Diagram View
	Diagram>Results>Powerflow results		Display power flow solution results on one-line diagram.	Validly specified power flow case solved to an acceptable mismatch level.	Diagram View
	Diagram>Results>Impedance data		Display impedance data from power flow solution results on one-line diagram	Validly specified power flow case solved to an acceptable mismatch level.	Diagram View
	Diagram>Results>Graphical difference data... *[Compare Cases on a Diagram]*	GDIF	Compare differences in power flow solution results and power flow boundary condition data with a Saved Case.	Validly specified power flow case. Both cases solved to an acceptable mismatch level.	Diagram View
	Diagram>Results>ASCC Fault analysis results... *[Display Fault Analysis Results]*		Display three phase and single line to ground faults on one-line diagram.		Diagram View
	Diagram>Results>IEC Fault Analysis results		Display IEC fault analysis results on one-line diagram.		Diagram View
	Diagram>Results>Reliability Analysis results		Display probabilistic reliability		Diagram View

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			results on one-line diagram.		
	Diagram>Results>Dynamics Analysis results		Display dynamic analysis results on one-line diagram.		Diagram View
	Power Flow>Solution> Parameters... *[Solution Parameters]*				
	Power Flow>Solution>Solve (NSOL/FNSL/FDNS/SOLV/ MSLV) ... *[Loadflow solutions]* Newton	NSOL	Apply a decoupled Newton-Raphson power flow iteration to the network modeled in the working case.	Validly specified power flow case with voltages that are a reasonable estimate of a solution.	
	Power Flow>Solution>Solve (NSOL/FNSL/FDNS/SOLV/ MSLV) ... *[Loadflow solutions]* Newton	FNSL	Apply the Newton-Raphson power flow iteration to the network modeled in the working case.	Validly specified power flow case with voltages that are a reasonable estimate of a solution.	
	Power Flow>Solution>Solve (NSOL/FNSL/FDNS/SOLV/ MSLV) ... *[Loadflow solutions]* Newton	FDNS	Apply a fixed slope decoupled Newton-Raphson power flow iteration to the network.	Validly specified power flow case with voltages that are a reasonable estimate of a solution.	
	Power Flow>Solution>Solve (NSOL/FNSL/FDNS/SOLV/ MSLV) ...	SOLV	Apply Gauss-Seidel power flow iteration to the network.	Validly specified power flow case with voltages that are a reasonable esti-	No series capacitors.

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Loadflow solutions] Gauss			mate of a solution.	
	Power Flow>Solution>Solve (NSOL/FNSL/FDNS/SOLV/ MSLV) ... *[Loadflow solutions]* Gauss	MSLV	Apply modified Gauss-Seidel power flow iteration to the network modeled in the working case.	Validly specified power flow case with voltages that are a reasonable estimate of a solution.	The working case may contain series capacitors only between Type 1 buses.
	Power Flow>Solution>AC contingency solution (ACCC)... *[AC Contingency Solution]*	ACCC	Calculate and report contingency case power flow solutions.	Validly specified power flow case solved to an acceptable mismatch tolerance.	Distribution Factor Data File corresponding to the network condition.
	Power Flow>Solution>Multi-Level AC contingency solution... *[Multi-Level AC Contingency Solution]*				
	Power Flow>Solution>Implement Generation dispatch... *[Implement Generation Dispatch]*				
	Power Flow>Solution>AC corrective actions... *[AC Corrective Actions]*				
	Power Flow>Solution>NR solution with	INLF	Solve the network contained	Validly specified power flow case, solved in	Data changes specified.

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	inertial governor dispatch (INLF)... *[N-R Solution with Inertial / Governor Redispatch]*		in the working case. Activity INLF may be used either as an inertial power flow or as a governor response power flow.	the pre-event condition.	No prior power flow solutions in working case. Inertia and Governor Response Data File (*.inl) containing machine data for all in-service machines.
	Power Flow>Solution>Order network for matrix operations (ORDR)... *[Order Network]*	ORDR	Perform the optimal ordering function, setting up an internal ordering of system buses to optimize the sparsity of the Jacobian and triangularized admittance matrices used in the network solution activities.	Validly specified power flow case.	
	Power Flow>Solution>Factorize admittance matrix (FACT)	FACT	Decompose the admittance matrix of the network contained in the working case into its triangular factors for subsequent use in activity TYSL and in the network solution used in dynamic simulations.	Validly specified power flow case.	Activities CONG and CONL (normally) executed.
	Power Flow>Solution>Solution for switching studies (TYSL)... *[Solution for Switching Studies]*	TYSL	Solution intended for the solution of network conditions at the instant following the execution of a switching opera-	Validly specified power flow case with voltages corresponding to the preswitching network condition, and with	Activities CONG, CONL (normally), and ORDR executed. Switching applied and activi-

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			tion or fault application.	impedance data specified for all machines.	ty FACT executed.
	Power Flow>Solution>PV analysis... *[PV Analysis]*				
	Power Flow>Solution>PV analysis using previous results... *[PV Parameters]*				
	Power Flow>Solution>Implement PV transfer... *[Implement PV Transfer]*				
	Power Flow>Solution>QV analysis... *[QV Analysis]*				
	Power Flow>Solution>QV analysis using previous results... *[QV Parameters]*				
	Power Flow>Changing>Reconnect bus (DSCN/RECN)... *[Disconnect / Reconnect Bus]*	DSCN *Disconnect/*	Perform data changes necessary to electrically disconnect selected buses in the working case. For each bus specified, the bus type code is set to four and all branches connected to the	Validly specified specified power flow case.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			bus are placed out-of-service.		
	Power Flow>Changing>Reconnect bus (DSCN/RECN)... *[Disconnect / Reconnect Bus]*	RECN *Disconnect/*	Perform the data changes necessary to electrically reconnect selected buses in the working case.	Validly specified power flow case.	
	Power Flow>Changing>buses (JOIN)... *[Join Buses]*	JOIN *Join*	Joins pairs of buses, retaining the identity of one of the two buses and removing the other completely from the working case.	Validly specified power flow case.	
	Power Flow>Changing>buses (SPLT)... *[Split Buses]*	SPLT *Split*	Split buses, introducing an additional bus for each split bus.	Validly specified power flow case.	
	Power Flow>Changing>line (LTAP)... *[Tap Line]*	LTAP *Tap*	Tap any non-transformer ac branch at a designated location along the line, introducing a new bus.	Validly specified power flow case.	
	Power Flow>Changing>network elements (MOVE)... *[Move Network Elements]*	MOVE *Move*	Disconnect the to bus ends of selected branches and reconnect them to different buses. Move selected loads, switched shunts, machines and plants.	Validly specified power flow case.	
	Power Flow>Changing>network elements (PURG/EXTR)...	PURG *Delete*	Remove outaged equipment from the working case, packing all tables to	Validly specified power flow case.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Delete Buses and Outaged Network Elements]		remove gaps in numeric sequence.		
	Power Flow>Changing> network elements (PURG/ EXTR)... *[Delete Buses and Outaged Network Elements]* Remove Buses from Case	EXTR *Delete*	Remove buses and their connected equipment from the working case, packing all tables to remove gaps in numeric sequence.	Validly specified power flow case.	
	Power Flow>Changing> generation, load, shunt (SCAL)... *[Scale Powerflow Data]*	SCAL *Scale*	Modify total load, generation (positive generator powers), motor load (negative generator powers), bus connected capacitors, and/ or bus connected reactors in a specified subsystem of the working case.	Validly specified power flow case. If generation is to be scaled, and a Type 3 bus is in the subsystem being scaled, the working case must be solved to an acceptable mismatch level.	
	Power Flow>Changing> adjustment flags (TFLG) *[Transformer Adjustment Flags]*	TFLG *Transformer*	Set the tap adjustment enable flags of all adjustable transformers connecting specified buses to either zero (disable) or one (enable).	Validly specified power flow case.	
	Power Flow>Changing> Dispatch (ECDI) *[Economic Dispatch]*	ECDI *Economic*	Perform a unit commitment and economic dispatch calculation for machines residing	Validly specified power flow case.	Economic Dispatch Data File (*.ecd)

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			in the working case.		
	Power Flow>Changing>	ImOutage	Allows the user to change the status of induction machines that are in-service, but that have been set to the "stalled" (for motors) or "tripped" (for generators) state by the power flow solution, to out-of-service.	Validly specified power flow case.	Economic Dispatch Data File (*.ecd)
	Outage "stalled"/"tripped" induction machines				
	Power Flow>Reports>Area/zone based reports... *[Area / Zone Based Reports]* Inter area/zone flow>Output for Areas	INTA	Summarize area interchange between an area and every other area. Interchange is evaluated on the basis of the metering points designated for each area tie branch.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Area/zone based reports... *[Area / Zone Based Reports]* Inter area/zone flow>Output for Zones	INTZ	Summarize zone interchange between a zone and every other zone. Interchange is evaluated on the basis of the metering points designated for each zone tie branch.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Area/zone based reports... *[Area / Zone Based Reports]*	TIES	List a summary of the loadings on all ties from each interchange area	Validly specified power flow case solved to an acceptable mismatch tolerance.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Tie line>Output for Areas				
	Power Flow>Reports> Area/zone based reports... *[Area / Zone Based Reports]* Tie line>Output for Zones	TIEZ	List a summary of the loadings on all ties from each zone	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports> Area / owner / zone totals... *[Area / Owner / Zone Totals]* Report on Areas	AREA	Report generation, load, bus shunt, line shunt, line charging, losses, net interchange, desired net interchanges for areas.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports> Area / owner / zone totals... *[Area / Owner / Zone Totals]* Report on Owners	OWNR	Report summary of owner totals of generation, load, bus shunt, line shunt, line charging, and losses in the working case.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports> Area / owner / zone totals... *[Area / Owner / Zone Totals]* Report on Zones	ZONE	Report zone totals	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Bus based reports... *[Bus Based Reports]*	LAMP	Wide format power flow solution output activity, including branch current loadings in amps	Validly specified power flow case solved to an acceptable mismatch tolerance.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Powerflow output>Wide format output>Output with amps		Activity LAMP lists the mismatch at a bus only when it exceeds 0.5 MVA.		
	Power Flow>Reports>Bus based reports... *[Bus Based Reports]* Powerflow output>Wide format output	LOUT	Wide format power flow solution output activity. Activity LOUT lists the mismatch at a bus only when it exceeds 0.5 MVA.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Bus based reports... *[Bus Based Reports]* Powerflow output	POUT	Narrow format power flow solution output activity. List bus mismatch only when it exceeds 0.5 MVA.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Bus based reports... *[Bus Based Reports]* Subsystem summary / totals	SUBS	Summarize conditions of power flow working case.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Reactive capability	GCAP	Read machine capability curve data and check var loadings and limits against Machine Capability Curve Data File. Machine reactive power limits may be updated.	Validly specified power flow case.	Machine Capability Data File (*.gcp)
	Power Flow>Reports>Limit	GENS	List generating plant loading data includ-	Validly specified power flow case. Plant	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	checking reports... *[Limit Checking Reports]* Generator bus		ing power output, reactive power limits, and scheduled and actual voltage. The report may be limited to plants at a reactive power limit, in-service plants, or all plants.	outputs and voltages are valid only if the working case is solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Machine terminal	GEOL	Print a summary of in-service machines including power output, terminal voltage, and current. The report may be restricted to overloaded machines.	Validly specified power flow case. Machine outputs and voltages are valid only if the working case is solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Branches	OLTL	List transmission lines where current loading is greater than a specified percentage of the selected rating.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Branches	OLTR	List transformers where MVA loading is greater than a specified percentage of the selected rating.	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports...	RAT3	Check current flow in each ac series branch for which at least one nonzero rating is specified	Validly specified power flow case solved to an acceptable mismatch tolerance.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Limit Checking Reports] Branches				
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Branches	RATE	Check current flow in each ac series branch for which a nonzero rating is specified	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Regulated buses	REGB	List voltage controlled buses and their controlling equipment: local and remote generators, switched shunts, transformers, and FACTS devices.	Validly specified power flow case. Bus voltages are valid only if the working case is solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Controlling transformer	TLST	List controlling transformer summary where ratio or phase shift angle may be adjusted by the power flow solution activities	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>Limit checking reports... *[Limit Checking Reports]* Out-of-limit bus voltage	VCHK	List buses where per unit voltage magnitude is outside a specified range	Validly specified power flow case solved to an acceptable mismatch tolerance.	
	Power Flow>Reports>AC Contingency reports...				

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[AC Contingency Reports]				
	Power Flow>Reports>Append to AC Contingency solution output file... *[AC Contingency Reports]*				
	Power Flow>Reports>Multiple AC Contingency run report... *[Multiple AC Contingency Run Report]*				
	Power Flow>Reports>Multi-terminal DC line solution output (MTDC)	section / MTDC	Report network conditions within the dc network formed by each unblocked multiterminal dc line. Results listed include the dc voltage at each dc bus, flows between the dc network and the ac converter station buses, and flows on each dc link in the network.	Validly specified power flow case solved to an acceptable mismatch tolerance. Unblocked multiterminal dc lines must be present in the working case.	
	Power Flow>Reports>Export ACCC, PV/QV results to Excel... *[Export PSS® E Data/Result to Excel]*				
	Power Flow>Convert Loads and Generators...	CONG	Convert all online machines from their power flow repre-	Validly specified power flow case solved to an ac-	Machine impedance data must be correctly specified for

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Convert / Reconstruct Loads and Generators] Convert Generators>Use ZSOURCE		sentation to the Norton current source model used in dynamic simulations and switching studies.	ceptable mismatch level.	all online machines.
	Power Flow>Convert Loads and Generators... *[Convert / Reconstruct Loads and Generators]* Convert/Reconstruct Loads>Convert constant MVA loads	CONL	Convert constant MVA loads to a specified mixture of constant MVA, constant current, and constant admittance loads. A different split among the three characteristics may be specified for the real and reactive components of load.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Convert Loads and Generators... *[Convert / Reconstruct Loads and Generators]* Convert/Reconstruct Loads	RCNL	Reconstruct power, current, and impedance loads at selected buses and then convert to a new specified mix of constant power, current, and impedance	Validly specified power flow case. If the existing voltage vector is used in the load reconstruction, the working case must be solved to an acceptable mismatch.	
	Power Flow>Equivalence Networks...>Net boundary bus mismatches (BGEN) *[Equivalence Networks]* Net boundary bus mismatches (BGEN)	BGEN	Model the net inflow at boundary buses as equivalent generation and/or load. A bus is assumed to be a boundary bus if its mismatch exceeds 0.5 MVA.	Validly specified subsystem of a solved power flow case.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Power Flow>Equivalence Networks...>Build electrical equivalent (EEQV) *[Equivalence Networks]* Build electrical equivalent (EEQV)	EEQV	Build an electrical equivalent of Type 1 buses in the specified subsystem of the working case.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Equivalence Networks...>Equivalence radial / 2-point buses (EQRD/RDEQ) *[Equivalence Networks]* Equivalence radial / 2-point buses (EQRD/RDEQ)	EQRD	Build an electrical equivalent of all radial and, optionally, two-point Type 1 buses, except those connected to three-winding transformers.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Equivalence Networks...>Equivalence radial / 2-point buses (EQRD/RDEQ) *[Equivalence Networks]* Equivalence radial / 2-point buses (EQRD/RDEQ)	RDEQ	Build an electrical equivalent of radial and, optionally, two-point Type 1 buses, except those connected to three-winding transformers. When sequence data is contained in the working case, the zero sequence network is also calculated as an electrical equivalent.	Validly specified power flow case solved to an acceptable mismatch level.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Power Flow>Equivalence Networks...>Net generation with load (NETG/ GNET/IMNET) *[Equivalence Networks]* Net generation with load (NETG/ GNET)	NETG	Net generation with the load at all but specified generator buses. The type code of any bus where generation is netted is changed to one.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Equivalence Networks...>Net generation with load (NETG/ GNET/ImNet) *[Equivalence Networks]* Net generation with load (NETG/ GNET)	GNET	Net the generation with the load at specified generator buses. The type code of any bus where generation is netted is changed to one.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Equivalence Networks...>Build three sequence equivalent (SCEQ) *[Equivalence Networks]*	SCEQ	Build a three sequence equivalent of a subsystem of the working case under classical fault analysis assumptions.	Validly specified power flow case reflecting classical fault analysis assumptions.	Sequence data included in the case (activity RESQ).
	Power Flow>Equivalence Networks...>Net Generation With load (NETG/GNET/ImNet) *[Equivalence Networks]*	ImNet	Net induction motors with the load at specified generator buses. The type code of any bus where generation is netted is changed to one.	Validly specified power flow case solved to an acceptable mismatch level.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Power Flow>Equivalence Networks...>Net Generation With load (NETG/GNET/ImNet) *[Equivalence Networks]*	NetIm	Net induction motors with the load at all but specified generator buses. The type code of any bus where generation is netted is changed to one.	Validly specified power flow case solved to an acceptable mismatch level.	
	Power Flow>Linear Network>DC network solution and report (DCLF)... *[DC Network Solution and Report]*	DCLF	Apply the dc analogy network solution algorithm to the network in the working case and, optionally, to that network with the status of a specified branch changed.	Validly specified power flow case, normally solved to an acceptable mismatch tolerance.	
	Power Flow>Linear Network>Single contingency ranking (RANK) ... *[Single Contingency Ranking]*	RANK	Estimate the severity of single branch outage contingencies on the basis of a linearized network model.	Validly specified power flow case solved to an acceptable mismatch tolerance.	When ranking contingencies with respect to line overloading, Subsystem Description Data File and Monitored Element Data File
	Power Flow>Linear Network>Create/modify SUB, MON and CON configuration files... *[Configuration File Builder]*				
	Power Flow>Linear Network>Build distribution fac-	DFAX	Build a Distribution Factor Data File in the form required by the linear network	Validly specified power flow case solved to an acceptable	For subsystem definition and monitored element specification: Subsys-

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	tor data file (DFAX)... *[Build Distribution Factor Data File]*		analysis activities OTDF, DCCC, TLTG, SPIL, POLY, IMPC, and LLRF, and by the AC Contingency Solution.	mismatch tolerance.	tem Description Data File, Monitored Element Data File, Contingency Description Data File or manual entry of data.
	Power Flow>Linear Network>DC contingency checking (DCCC)... *[DC Contingency Checking]*	DCCC	Estimate contingency case line flows for single or multiple event contingency cases on the basis of a linearized network model.	Validly specified power flow case solved to an acceptable mismatch tolerance.	Distribution Factor Data File corresponding to the network condition in the working case.
	Power Flow>Linear Network>DC corrective actions... *[DC Corrective Actions]*				
	Power Flow>Linear Network>Calculate and print distribution factors (OTDF)... *[Select Distribution Factor data file to calculate factors for]*	OTDF	Report distribution factors corresponding to single event contingency cases for a set of monitored lines and interfaces.	Validly specified power flow case solved to an acceptable mismatch tolerance.	Distribution Factor Data File (*.dfx) corresponding to the network condition.
	Power Flow>Linear Network>Transmission interchange limits calculation (TLTG)... *[Transmission Interchange Limits Calculation]*	TLTG	Transmission interchange limits calculation	Validly specified power flow case solved to an acceptable mismatch tolerance.	Distribution Factor Data File corresponding to the network condition in the working case

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Power Flow>Linear Network>Sequential participation interchange limit (SPIL)... *[Sequential Participation Interchange Limit]*	SPIL	Estimate the transmission interchange limits of a specified subsystem on the basis of a linearized network model. Interchange limits are calculated for the base case and for single or multiple event contingency cases.	Validly specified power flow case solved to an acceptable mismatch tolerance.	Distribution Factor Data File corresponding to the network condition in the working case. Subsystem Participation Data File corresponding to the subsystem definitions in the Distribution Factor Data File.
	Power Flow>Linear Network>Interchange limits with two opposing systems (POLY)... *[Interchange Limits with Two Opposing Systems]*	POLY	Estimate the transmission interchange limits of a specified subsystem with two opposing systems on the basis of a linearized network model.	Validly specified power flow case solved to an acceptable mismatch tolerance.	Distribution Factor Data File corresponding to the network condition.
	Power Flow>Linear Network>Interchange limits calculations (POLY) - previous results... *[Interchange limits with two opposing systems (POLY) - parameters]*				
	Power Flow>Linear Network>Midwest MW-mile calculation (MWMI)...	MWMI	Supports Midwest ISO planning modules.	Contact Midwest ISO (http://www.midwestiso.org/home).	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Midwest MW-miles calculation]				
	Power Flow>Reliability>Reliability assessment ... *[Probabilistic Reliability Assessment]*				
	Power Flow>Reliability>Substation reliability assessment ... *[Substation Reliability Assessment]*				
	Power Flow>List Data... *[List Data]* Bus names	ALPH	Report buses in the working case in alphabetical order.	Validly specified power flow case.	
	Power Flow>List data... *[List Data]* Examine Powerflow / sequence data	EXAM	List power flow data: all components connected to the bus including all transmission lines, transformers, dc lines, and FACTS devices originating or terminating on that bus.	Non-null power flow case.	
	Power Flow>List Data... *[List Data]* Powerflow>All data	LIST	Report power flow data contained in the working case.	Non-null power flow case.	
	Power Flow>List Data... *[List Data]*	LSTO	Report optimal power flow data contained in the working case.	Validly specified power flow case with optimal power flow	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	OPF			data appended to it.	
	Power Flow>List Data... *[List Data]* Outaged equipment	OUTS	List outaged equipment.	Non-null power flow case.	
	Power Flow>List Data... *[List Data]* Powerflow>Fixed bus shunts Powerflow>Switched bus shunts or Powerflow>Fixed bus shunts	SHNT	List bus shunts.	Validly specified power flow case.	
	Power Flow>List Data... *[List Data]* Examine Powerflow / sequence data>Sequence data	SQEX	List all sequence data, including transmission lines, transformers, dc lines, and FACTS devices originating or terminating on that bus	Non-null power flow case.	Sequence data must be contained in the working case (i.e., activity RESQ must have been executed).
	Power Flow>List Data... *[List Data]* Sequence data	SQLI	List specified category of sequence data	Validly specified power flow case with sequence data appended to it.	
	Power Flow>Check Data>Branch parameters (BRCH)... *[Check Branch Parameters]*	BRCH	Perform several checks on the parameters associated with ac branches in the working case.	Non-null power flow case.	
	Power Flow>Check Data>Check /	CNTB	Display local and remote generators, switched	Validly specified power flow case. Bus volt-	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Change controled bus scheduled voltages (CNTB)... *[Check / Change Controlled Bus Scheduled Voltages]*		shunts, transformers, and FACTS devices that control voltage at buses in a designated subsystem of the working case during power flow solutions.	ages listed are valid only if the case is solved to an acceptable mismatch level.	
	Power Flow>Check Data>Check / Change transformer adjustment data (TPCH)... *[Check / Change Transformer Adjustment Data]*	TPCH	Change transformer adjustment data associated with voltage and flow controlling transformers.	Validly specified power flow case.	
	Power Flow>Check Data>Buses not in swing bus tree (TREE)	TREE	List buses not in the tree of a swing bus by island, along with island load, shunt and generation totals.	Validly specified power flow case.	
	Power Flow>Renumbering Areas / Owners / Zones... *[Renumbering Areas / Owners / Zones]* Area assignments>Reassign Buses	ARNM	Re-assign buses to a specified area	Validly specified specified power flow case.	
	Power Flow>Renumbering Areas / Owners / Zones...	LDAR	Change the area numbers of selected loads in the working case.	Validly specified power flow case.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Renumbering Areas / Owners / Zones] Area assignments>Reassign Buses/Loads				
	Power Flow>Renumbering Areas / Owners / Zones... *[Renumbering Areas / Owners / Zones]* Reassign ...	OWNM	Change the ownership assignments of selected buses, loads, machines, branches, FACTS devices, and/or VSC dc lines in the working case.	Validly specified power flow case.	
	Power Flow>Renumbering Areas / Owners / Zones... *[Renumbering Areas / Owners / Zones]* Zone assignments>Reassign Buses/Loads	LDZO	Change the zone numbers of selected loads in the working case.	Validly specified power flow case.	
	Power Flow>Renumbering Areas / Owners / Zones... *[Renumbering Areas / Owners / Zones]* Zone assignments>Reassign Buses	ZONM	Change zone numbers of buses	Validly specified power flow case.	
	Power Flow>Renumber buses>By Bus Number...	BSNM	Change the numbers of selected buses in the working case. A file recording the	Validly specified power flow case.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Renumber Buses by Bus Number] *Power Flow>Renumber buses>By Bus Name...* *[Renumber Buses by Bus Name]* *Power Flow>Renumber buses>By Packing...* *[Renumber Buses by Bus Packing]* *Power Flow>Renumber buses>By Subsystem...* *[Renumber Buses by Subsystem]*		bus number changes may be generated.		
	Fault>Setup for special fault calculations (FLAT)... *[Setup for Special Fault Calculations]*	FLAT	Set the voltages of all buses in the working case to one per unit at zero phase angle.	Validly specified power flow case.	
	Fault>Setup network for unbalanced solution (SEQD)... *[Setup Network for Unbalanced Solution]*	SEQD	Build and factorize the three sequence admittance matrices in preparation for the unbalanced network solution activity SCMU and for the branch un-	Validly specified power flow case, with sequence data appended, solved to an acceptable mismatch level with bus voltages corresponding to pre-	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			balance activity SPCB. Setup network for unbalanced solution may also be performed from the SCMU and SPCB dialogs.	fault network condition.	
	Fault>ANSI fault calculation (ANSI)... *[ANSI Fault Current Calculation]*	ANSI	Calculate fault currents according to the ANSI standard C37.5-1979.	Validly specified specified power flow case with sequence data appended to it.	
	Fault>IEC 60909 fault calculation (IECS)... *[IEC 60909 Fault Calculation]*	IECS	IEC 60909 fault calculation		IEC Fault Calculation Data file (*.iec) Fault Control Data file (*.fcd)
	Fault>Automatic sequence fault calculation (ASCC)... *[Automatic Sequence Fault Calculation]*	ASCC	Calculate and report three phase and single line to ground faults at all buses in a specified subsystem.	Validly specified power flow case with sequence data appended to it.	
	Fault>Circuit breaker interrupting duty (BKDY)... *[Circuit Breaker Interrupting Duty]*	BKDY	Calculate and report circuit breaker interrupting duty for three phase faults at all buses in a specified subsystem of the working case.	Validly specified power flow case must reflect the pre-fault network condition.	Generators represented as current sources (activity CONG must have been executed). Normally, the load characteristics must be modified with CONL prior to execution. Breaker Duty Data File.

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Fault>Separate pole circuit breaker (SPCB)... *[Separate Pole Circuit Breaker]*	SPCB	Calculate the positive sequence pi-equivalent of a single branch unbalance.	Validly specified power flow case with sequence data appended to it. It must be solved to an acceptable mismatch level with bus voltages corresponding to the preunbalance condition.	The branch to have the unbalance must be removed from service.
	Fault>Solve and report network with unbalances (SCMU/SCOP)... *[Multiple Simultaneous Unbalances]*	SCMU	Solve the network in the presence of unbalanced conditions and print a summary report.	Validly specified power flow case with sequence data appended to it.	Activity SEQD must be successfully executed.
	Fault>Unbalanced network tabular output (SCOP)... *[Multiple Simultaneous Unbalances]*	SCOP	Solve the network in the presence of unbalanced conditions and print a summary report.	Validly specified power flow case.	Activity SCMU must have successfully solved the network in the presence of unbalanced conditions.
	OPF>Solve... *[OPF Solution]*	NOPF	Apply interior point solution to the nonlinear power flow network.	Validly specified power flow case with optimal power flow data appended to it.	
	OPF>Data...				
	OPF>Data tables... *[OPF Data Tables]*				
	OPF>Parameters... *[OPF - Change Parameters]*	NOPF	Modify any of the optimal power flow solution option settings currently in effect. Settings can be	Validly specified power flow case with optimal power flow data appended to it.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			saved in an Optimal Power Flow Options File (*pssopf.opt*).		
	Trans Access>Data...				
	Trans Access>Allocations> Megawatt shift factors... *[Megawatt Shift Factors]* *Trans Access>Allocations> Vector Absolute MW-mile* *Trans Access>Allocations> Vector Absolute MW-ohm* *Trans Access>Allocations> Vector Sum MW-ohm*	ALOC	Generate transmission pricing reports: Vector Absolute MW-mile allocation matrix, Vector Absolute MW-mile generation shift factors, Vector Absolute MW-ohm allocation matrix, Vector Absolute MW-ohm generation shift factors, Vector Sum MW-ohm allocation matrix, branch mileage summary.	Validly specified power flow case.	Transaction events must have been previously introduced into working memory via activity REMM or the transaction editor.
	Trans Access>Calculators>Impact on monitored elements... *[Impact on Monitored Elements]*	IMPC	Report incremental MW flow impact on monitored elements due to a transaction event.	Validly specified power flow case.	Transaction events in working memory via activity REMM or the transaction editor. Distribution Factor Data File (*.dfx) corresponding to the network condition.
	Trans Access>Calculators>Line loading relief by transaction adjustment...	LLRF	Report line loading relief, including: incremental changes in transaction event curtail-	Validly specified power flow case.	Transaction events previously introduced into working memory via activity REMM or

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Line Loading Relief by Transaction Adjustment]		ments required to achieve a target monitored element MW flow adjustment, distribution factors of monitored element flow to transaction event magnitude, and modification of the distribution factor tolerance.		the transaction editor. Distribution Factor Data File corresponding to the network condition.
	Trans Access>Summaries>Summary of branch mileage by owner				
	Dynamics>Simulation>Solution parameters... *[Dynamic Solution Parameters]*				
	Dynamics>Simulation>Simulation options... *[Dynamic Simulation Options]*				
	Dynamics>Simulation>Perform simulation (STRT/RUN)...	STRT	Initialize models for state-space simulation and test initial conditions	Validly specified power flow case, solved to an acceptable mismatch level. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>Simulation>Perform simulation (STRT/RUN)...	RUN	Advance dynamic simulation through time, solving the network, calculating time deriv-	Validly specified power flow case. Dynamics data must be present	Generators represented as current sources (activity CONG must have been executed).

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			atives, and performing numerical integration at each time step.	in dynamics working memory.	Activity STRT successfully executed.
	Dynamics>Simulation>Perform exciter simulation (ESTR/ERUN)... *[Perform Exciter Simulation Test]*	ESTR	Initialize the dynamics models for testing excitation system response with individual units operating in isolation. Activity ESTR overrides manual selections of output channel assignments.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>Simulation>Perform exciter response ratio simulation (ESTR/ERUN)... *[Perform Exciter Response Ratio Simulation Test]*	ERUN	Execute simulation of excitation system response with individual units operating in isolation.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed). ERUN successfully executed.
	Dynamics>Simulation>Perform extended term simulation (MSTR/MRUN)...	MSTR	Initialize all dynamics models from the network conditions contained in the working case in preparation for extended term simulation calculations (i.e., for activity MRUN).	Validly specified power flow case, solved to an acceptable mismatch level. Dynamics data must be present in dynamics working memory. All equipment models used in the system dynamic model must have been upgraded for use in the extended	Generators represented as current sources (activity CONG must have been executed).

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
				term simulation mode.	
	Dynamics>Simulation>Perform extended term simulation (MSTR/MRUN)... *[Perform Extended Term Dynamic Simulation]*	MRUN	Advance the extended term dynamic simulation through time, solving the network, calculating time derivatives, and performing numerical integration at each time step.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed). Activity MSTR successfully executed.
	Dynamics>Simulation>Perform governor response simulation (GSTR/GRUN)... *[Perform Governor Response Simulation]*	GSTR	Initialize the dynamics models for testing the governing response of individual units in isolation. Output channel assignments are overridden, and the speed deviation and mechanical power of each machine with a connected governor model are placed into output channels.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>Simulation>Perform governor response simulation (GSTR/GRUN)... *[Perform Governor Response Simulation]*	GRUN	Execute a simulation of the governing response of individual units operating in isolation.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed). Activity GSTR successfully executed.
	Dynamics>Channel Setup Wizard				

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Channel Setup Wizard]				
	Dynamics>Define simulation output (CHAN)>Bus quantity *[Assign Channels for Bus Quantities]* *Dynamics>Define simulation output (CHAN)>Line quantity* *[Assign Channels for Line Quantities]* *Dynamics>Define simulation output (CHAN)>Load quantity* *[Assign Channels for Load Quantities]* *Dynamics>Define simulation output (CHAN)>Machine quantity* *[Assign Channels for Machine Quantities]* *Dynamics>Define simulation output (CHAN)>Misc quantity*	CHAN	Designate quantities which are to be monitored during dynamic simulation calculations	Validly specified power flow case, solved to an acceptable mismatch level. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	[Assign Channels for Misc. Quantities]				
	Dynamics>Define simulation output by subsystem (CHSB) *[Select Channels by Subsystem]*	CHSB	Designate quantities which are to be monitored during dynamic simulation calculations.	Validly specified power flow case, solved to an acceptable mismatch level. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>Model maintenance... *[Model Maintenance]*	DYCH	Perform table maintenance functions for plant-related dynamics models	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>List>Models and data (DOCU)... *[List Dynamics Model Data]*	DOCU	Report models referenced in dynamics setup. The report may be restricted to those models for which constant data fails a reasonableness check.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>List>Dynamics data (DLST)... *[List Dynamics Data Common]*	DLST	Report contents of any or all of the following: • CON array • VAR array • STATE array • ICON array • Main output channel selections	Dynamics data must be present in dynamics working memory.	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			• CRT plot channel selections		
	Dynamics>List>Model storage locations... *[List Model Storage Locations]*	MLST	List plant-related models referenced in dynamics setup.	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Dynamics>Launch NEVA Eigenvalue analysis				
	Force Generator Conversion (icon only)				
	Update Plot Book (icon only)				
	Update Dynamic Diagram (icon only)				
	Dynamics>Build matrices for LSYSAN program (ASTR)... *[Build Matrices for LSYSAN]*	ASTR	Initialize all dynamics models from the network conditions contained in the working case and construct the state variable system matrices required by the Linear Dynamic Analysis program, LSYSAN.	Validly specified power flow case, solved to an acceptable mismatch level. Dynamics data is located in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	I/O Control>Direct Report output (OPEN)... *[Report Output Destination Selector]*	OPEN	Preselect report destination: terminal, file, or high-speed printing device.	None	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	I/O Control>Direct Progress output (PDEV)... *[Progress Output Destination Selector]*	PDEV	Assign destination progress report output to the terminal, a file, or a hard copy printing device.	None	
	I/O Control>Direct Prompt output... *[Prompt Output Destination Selector]*	ODEV	Assign destination of the dialog output of a Response File or an IPLAN program to the terminal, a file, or a hard copy printing device.		None, although in practice, activity ODEV would be executed only from a Response File or an IPLAN program.
	I/O Control>Direct Alert output... *[Alert Output Destination Selector]*	ODEV	Assign destination of the dialog output of a Response File or an IPLAN program to the terminal, a file, or a hard copy printing device.		None, although in practice, activity ODEV would be executed only from a Response File or an IPLAN program.
	I/O Control>Set path for use with "&" filenames (PATH)... *[Define PATH by selecting any file in the desired directory]*	PATH	Designate a pathname to be used as a prefix for all filenames to be entered with an ampersand (&) preceding them.	None	
 	I/O Control>Start recording... *[Select Program Automation File to Record]* *I/O Control>Stop recording*		Record user action for part or all of work session	None	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	I/O Control>Run program Automation file... *[Select Program Automation File to Run]* IPLAN File (*.irf)	EXEC	Execute IPLAN Executable Program File		IPLAN Executable Program File generated by IPLAN language compiler program.
	I/O Control>Run program Automation file... *[Select Program Automation File to Run]* Response file (*.idv)	IDEV	Run Response File		Response File, if a filename is specified
	I/O Control>Run program Automation file... *[Select Program Automation File to Run]* PSAS Command file (*.psa)	PSAS	Specify a dynamic simulation run in a high-level language. Output is a Response File that is automatically executed.	None	PSAS Command file (*.psa)
	I/O Control>Run program Automation file... *[Select Program Automation File to Run]* PSEB Command file (*.pse)	PSEB	Specify a power flow run in a high-level language. Output is a Response File that is automatically executed.	None	PSEB Command File (*.pse)
	I/O Control>Run program Automation file... *[Select Program Automation File to Run]*		Run Python (*.py) file		Python file (*.py)

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Python file (*.py)				
	Misc>Change program settings (OPTN) ... *[Program Settings]*	OPTN	Modify PSS®E program settings. New program settings remain in effect unless reset with activity OPTN or overridden with the settings from a Saved Case file.	None	
	Misc>Display timing statistics (TIME)	TIME	Display timing statistics	None	
	Misc>Reset timing statistics to zero (TIME)	TIME, INIT	Reset timing statistics	None	
	Misc>Insert text into the Progress stream (TEXT) *[Insert Text into the Progress Stream]*	TEXT	Insert text into the Progress stream of an active terminal session logging file.	None	
	Misc>Select extended bus name input format				
	Dynamics spreadsheet	ALTR	Modify data contained in dynamics working memory, including model constants and variables, output channel selections, solution parameters used during the simulation, and the two line case title.	Dynamics data must be present in dynamics working memory.	
	Line mode only	CATA	Report files contained in the	none	

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			current directory in alphabetical order		
	Line mode only	CCON	Modify the constant data values associated with plant-related dynamic simulation model references	Validly specified power flow case. Dynamics data must be present in dynamics working memory.	Generators represented as current sources (activity CONG must have been executed).
	Network spreadsheet	CHNG	Power flow data modification.		
	Command Line Interface only	CLOS	Terminate output to the previous action initiated by activity OPEN.	None	
	Command Line Interface only	ECHO	Copies all terminal input to a file.		
	Line mode only	EDTR	Modify the 2-line case heading and the 16-line long title	Validly specified power flow case.	
	Command Line Interface only	FIND	Find bus number from bus name or partial bus name.		
	Command Line Interface only	LOFL	Link to power flow activity selection, enabling execution of any Command Line input power flow activity.	PSS®E must have been initiated at its dynamic simulation entry point, PSSDS4.	
	Command Line Interface only	MBID	Change equipment identifier (machines, loads, fixed bus shunts, ac branches, multi-section line groupings, dc lines, FACTS de-		

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
			vices, and inter-area transfers).		
	Command Line Interface only	MODR	Modify line resistances of nontransformer branches. All in-service branches that have one or both of their ends in the specified subsystem are processed.	Validly specified power flow case solved to an acceptable mismatch level.	
	Line mode only	PRTI	Print the 16 line long title	Non-null power flow case.	
	Command Line Interface only	RTRN	Return to dynamic simulation activity selection from power flow activity selection. Establish connectionsbetween models and network elements, and update output channel addresses.		
	Network spreadsheet	SQCH	Sequence data modification activity.		
	Line mode only	TREA	Power flow data input from the terminal.		
	Line mode only	TRSQ	Sequence data input from terminal.		
	Command Line Interface only	USER	Power flow user-written activity.		
	Line Mode only	Version	Set/display current Line Mode language version being interpreted.		

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Icon	Navigation, [GUI], Tab Selection	Activity ID	Function	Working Case Requirement(s)	Activity Prerequisite(s)
	Line Mode only	HELP	Prints brief summary documentation of line mode activities at progress device		

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Appendix B
Appendix B - PSS®E Documentation

PSS®E and its auxiliary programs are documented in a set of manuals. Your installation may not include all of the documents listed below, depending upon which of the optional program sections are included in your lease of PSS®E.

PSS®E Program Operation Manual

This manual is a comprehensive working guide to PSS®E, independent of interface differences, documenting operational procedures and providing characteristic results.

PSS®E Program Application Guide

discusses the engineering aspects of formulating problems for PSS®E and interpreting its results. The two part guide can be considered an engineering textbook specifically related to system modeling and techniques within PSS®E for engineers responsible for planning and operations studies.

The *Program Operation Manual* and the *Program Application Guide* discuss the use of PSS®E from different viewpoints and hence complement each other. The user is encouraged to become familiar with both of these manuals.

PSS®E Installation Guide

contains installation instructions.

PSS®E GUI Users Guide

describes the use of the power flow functions, Optimal Power Flow program, and dynamic simulation functions of PSS®E using the Graphical User Interface (GUI) introduced at PSS®E-30.

PSS®E Command Line Interface (CLI) Users Guide

describes the structure and general operation of PSS®E for those customers who prefer operation using command line input.

PSS®E Model Library

provides data sheets and model descriptions for all dynamic models supported by PSS®E dynamic simulation activities.

PSS®E Application Program Interface (API)

describes the interfaces to the various engineering functions of PSS®E. Included are the syntax details for accessing APIs in the batch command facility introduced at PSS®E-25 and in the Python programming language introduced at PSS®E-30.

PSS®E Compatibility Reference

complete documentation of compatibility of the current release with prior releases of PSS®E and prior Optimal Power Flow (OPF) releases as a separate application. This document also contains version histories.

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PSAS User's Ready Reference and PSEB User's Ready Reference

describe the Command File structures that allow the PSS®E user to specify many routine power flow and dynamic simulation runs, activity PSAS and activity PSEB respectively, in English sentence form. These command files are created by the user with a text editor before starting up PSS®E. The PSAS and PSEB commands are translated into a PSS®E Response File for immediate execution or for subsequent processing as a standard PSS®E Response File.

Graphic Report Generator GRPG User's Ready Reference

document summarizing the commands recognized by activity GRPG.

PSSPLT Program Manual

describes the use of the simulation channel output file processing program used for plotting dynamics study results.

IPLAN Program Manual

describes the Siemens PTI IPLAN programming language and the use of the IPLAN compiler. Compiled IPLAN programs may be executed by PSS®E activity EXEC.

PSS®E–WECC Data Conversion Manual

describes the auxiliary programs WECCLF, WECCDS, and RAWECC. These auxiliary programs convert data between the Western Electricity Coordinating Council (WECC) power flow and stability program data formats and PSS®E input data file formats.

UCTE Data Exchange Format Application Guide and Manual

document summarizing conversion from UCTE data exchange format 2.0 (UCTEDEF) for power flow and three-phase short circuit studies into PSS®E-31 raw file format, and viceversa.

Transmission Line Characteristics (TMLC) Program Manual and *PSS®E* Line Properties Calculator Manual

document a pair of auxiliary programs available to calculate line impedances from a transmission corridor definition.

Saved Case Data Extraction Subroutines Programming Manual

describes a programming interface whereby users may develop FORTRAN programs that access power flow data directly from PSS®E Saved Case Files.

Additional Resources for PSS®E

a collection of additional resource materials, including auxiliary and utility applications, printing assistance, the FLECS Fortran-to-Fortran Translator, the latest compiler information, etc.

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Appendix C
Appendix C - Legacy Activities

C.1. DRED

The coordinate data editing activity DRED is no longer supplied with PSS®E. It has been replaced by functions in the Diagram View of the Graphical User Interface (GUI) which provide for the modification of the one-line diagram. The *PSS®E GUI Users Guide*, Chapter 3 describes the Diagram View.

C.2. GRED

The graphical report data editing activity GRED is no longer supplied with PSS®E.

C.3. SCGR

The graphical fault analysis activity SCGR, which applied a sequence of single faults at buses in the working case, is no longer supported. The Line Mode dialog of activity SCGR is implemented in the Line Mode interpreter and the results have been mapped to the active Slider Diagram.

Additional Information
PSS®E GUI Users Guide, Displaying ASCC Fault Analysis Results

C.4. DRAW

Implement Activity DRAW - CLI

```
ACTIVITY?
>>DRAW
ENTER COORDINATE FILE NAME, BINARY OPTION
(-1 TO EXIT):
>>
```

The one-line diagram drawing activity DRAW has been replaced by the Diagram View of the Graphical User Interface (GUI). The *PSS®E GUI Users Guide*, Chapter 3 describes the Diagram View.

The old Line Mode dialog of activity DRAW is implemented in the Line Mode interpreter and results in the execution of the Diagram View's one-line diagram function.

Additional Information
PSS®E GUI Users Guide, Importing a DRAW File *PSS®E Command Line Interface (CLI) Users Guide*, Drawing a One-Line Diagram from a Drawing Coordinate File

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C.4.1. Drawing Coordinate Data File Contents

The selection and position of network elements on each one-line diagram is specified via a user-supplied Drawing Coordinate Data File. The records in the Drawing Coordinate Data File define the elements to be drawn, their location on the diagram, and annotation options. A Drawing Coordinate Data File may be constructed and/or modified manually via the host computer's text editor.

The Drawing Coordinate Data File consists of records of the following general form:

```
record type , option , data values
```

The record type is a two-character code describing the type and significance of the data contained on the remainder of the record. The option field (which may be a blank or B) specifies a one- or two-character code, which designates options to be used in drawing the equipment or graphical element specified by the record. The occurrence of the single character B in the option field is treated as a blank option field. The option field should always be preceded and followed by commas. The remainder of each record specifies positional data and other drawing information.

There are two categories of records: control records and graphical element records. The control records (types HD, EN, VO, RA, VL, AN, CO, CL, and GR) are used to control the reading of the file and the specification of diagram options such as character size. The graphical element records describe the network elements (i.e., the buses, branches, and other equipment), as well as elements such as text strings, symbols and straight lines, which are to be drawn on the diagram.

Records in the file may appear in any order with the exception of the header record (type HD), which must be the first line, and the terminator record (type EN), which must be the last line. In addition, only the GN, BL, and LF summation participation record may be specified between a SU record and the corresponding EN record of a summation block structure (refer to Summation Block Structure). In addition, only LI branch records may be specified between TT record and the corresponding EN record of a three-winding transformer group. (refer to Two-Terminal dc Transmission Line Record - DC and Multi-Terminal dc Transmission Line Record - MD).

The x- and y-axis coordinate values *must not* be specified as negative numbers on any data record.

On any data record on which buses are specified, the bus identifier is taken to be the bus number when the *numbers* input option is in effect, and the extended bus name, enclosed in single quotes, using the *names* input option (refer to Section 3.3.3 Program Run-Time Option Settings and activity OPTN).

Header Record - HD

The header record, which must be the first record in the Drawing Coordinate Data File, is of the form:

```
HD, O, DRAWX, DRAWY, CHRX, XSIZE, 'f', TITLEY
```

where:

O	Is a valid option (blank, S, N, A, or L); below.
DRAWX	Is the length of the horizontal (x-axis) side of the drawing in inches. If DRAWX is less than 7.5 activity DRAW automatically changes it to 7.5 in.
DRAWY	Is the length of the vertical (y-axis) side of the drawing in inches exclusive of the banner block at the bottom of the drawing.

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CHRX	Is the character height in inches; default is 0.05 in.
XSIZE	Is a scaling factor specified in inches; default is zero. If XSIZE is greater than zero, activity DRAW does not ask the user if the drawing size is to be changed immediately preceding the drawing of the diagram. Rather, the drawing size is automatically scaled by the ratio XSIZE/DRAWX.
'f'	Is a single-character orientation flag enclosed in single quotes. It controls the orientation of text written on lines perpendicular to the title block. If this flag is specified as 'R', such text is positioned to be read when viewing the drawing from the right side (relative to the title block being at the bottom). An 'L' positions such text to be read from the left side. The flag default is 'R'.
TITLEY	Is the length of the vertical (y-axis) side of the banner block at the bottom of the drawing in inches; default is 0.5 in.; maximum is 2.0 in.

DRAWX and DRAWY define the size of the box that is drawn around the one-line diagram. If these quantities are not entered, activity DRAW calculates the dimensions of the box.

The valid entries in the option field are as follows:

blank	Draw a one-line diagram of the portion of the working case as specified in the Drawing Coordinate Data File.
N	No printing on the one-line diagram. This option suppresses all text on the drawing. This option is useful in the initial debugging of a new Drawing Coordinate Data File.
A	Accept the network specified in the Drawing Coordinate Data File. This option is called the *accept* mode and is used to draw a diagram of a network not necessarily part of the working case. If bus numbers are selected in the dialog portion of activity DRAW, they are printed; if the name option is selected, a dummy name is printed for each bus. Bus voltages and tap ratios are printed as one per unit. If branch flows are selected, they are printed as zero; if the impedance option is selected, impedances are printed as zero.
L	Same as the A option, but with annotation suppressed.

Terminator Record - EN

A terminator record, EN, is required both as the last record of each summation block structure (refer to Summation Block Structure), as the last record of each three-winding transformer group, and as the last record of the Drawing Coordinate Data File. If the Drawing Coordinate File is not concluded with this record, activity DRAW proceeds as if the EN record was encountered.

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Voltage Level Record - VO

The voltage level specification record is used to specify three voltage level breakpoints. If voltage levels are selected during the range checking selection portion of the dialog, the voltage levels specified on the VO data record are taken as the default base voltage breakpoints.

The voltage level specification record is of the form:

```
VO, O, V1, V2, V3
```

The option field O is either a blank or F. The F option forces the voltage level annotation option on the one-line diagram even if this range checking option is not specifically selected in the interactive dialog of activity DRAW. V1, V2, and V3 are specified in kV in ascending order.

By default, the line used in drawing the bus and its connected branches is generated as follows:

	On Monochromatic Devices	On Pen Plotters	On Color CRTs
Base kV ≤ V1	One dot wide	Pen # 1	White
V1 < base kV ≤ V2	Two dots wide	Pen # 2	Red
V2 < base kV ≤ V3	Three dots wide	Pen # 3	Green
V3 < base kV	Four dots wide	Pen # 4	Blue

On monochromatic devices, when only two voltage levels are specified, it is suggested that they be drawn using one and three dots. When any of the three voltage breakpoints is specified as zero, or is less than or equal to the preceding voltage breakpoint, the corresponding line width or color is not used to distinguish among buses at different voltage levels. Therefore, if the user wishes to distinguish between buses up to 138 kV and those greater than 138 kV, the VO data record could be entered as VO, ,138,138 or VO, ,138,0.

The default line widths, pen numbers, or colors for designating voltage levels may be overridden by specifying a CO record (refer to Color Specification Record - CO).

Bus Records - BU and BN

Buses to be included on the one-line diagram are specified in bus data records in the Drawing Coordinate Data File. Bus data records are of the form:

```
BU, O, IB, X1, Y1, X2, Y2, I1, I2
```

or:

```
BN, O, IB, X1, Y1, X2, Y2, I1, I2
```

where:

O	Is a valid option (blank, R, N, V, or S); below.
IB	Is the bus identifier (number or name).
X1,Y1	Is the (x,y) coordinate of one end of the bus, specified in inches.

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X2,Y2	Is the (x,y) coordinate of the other end of the bus, specified in inches.
I1	Is an integer variable used to control the position of the bus annotation (bus name and/or number or voltage and/or angle) that is printed at the left end of a horizontal bus or the top of a vertical bus. I1 is the number of character widths to move the annotation to the right. A negative value of I1 moves the annotation to the left. I1 is limited to the integers between -10 and 10 inclusive, and has a default value of zero.
I2	Is an integer variable used to control the position of the bus annotation that is printed at the right end of a horizontal bus or the bottom of a vertical bus. I2 is the number of character widths to move the annotation to the right. A negative value of I2 moves the annotation to the left. I2 is limited to the integers between -10 and 10 inclusive, and has a default value of zero.

Bus data record types BU and BN behave identically except in the component checking option of activity DRAW.

Buses *must* be either horizontal (Y1 = Y2) or vertical (X1 = X2).

For vertical buses, I1 and I2 values of zero place the respective annotation adjacent to the busbar on the right side. A value of -1 places the annotation adjacent to the busbar on the left side; values of -2 through -10 move it additional character widths to the left.

The valid entries in the option field are as follows:

blank	Print bus voltage and/or angle at the right (if a horizontal bus) or bottom (if a vertical bus), and the bus identifier (number and/or name) at the opposite end.
R	Reverse identifier and voltage fields (identifier at right or bottom, as appropriate).
N	Print bus name/number but not voltage or angle.
V	Print bus voltage and/or angle but not the bus identifier.
S	Suppress all annotation for the bus.

The R option may be used in combination with the N or V option. For example, specifying RN in the option field causes the voltage annotation to be suppressed and the bus identifier to be written at the right end of a horizontal bus or the bottom of a vertical bus.

By default, Type 4 buses and their generation, loads, and shunts are drawn with dotted lines and annotation values other than the bus identifier(s) are either zeros or not displayed. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

The minimum information necessary in a bus data record is:

```
BU, ,IB, X1, Y1, X2, Y2
```


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Branch Record - LI

The ac transmission lines and transformers to be included on the one-line diagram are specified in branch data records in the Drawing Coordinate Data File. Branch data records are of the form:

```
LI, O, IFROM, ITO, ICKT, XF, YF, XT, YT, SDF, SF,
SDT, ST
```

where:

O	Is a valid option (blank, F, T, SP, N, A, FA, TA, BX, FX, TX, SX, BN, FN, TN, S, SN, C, FC, TC, SC, R, RN, X, XA, E, EN, EX, RC, or EC); below.
IFROM	Is the from bus identifier.
ITO	Is the to bus identifier.
ICKT	Is the circuit or multi-section line grouping identifier; default value is '1'.
XF,YF	Is the (x,y) coordinate of the line at the from bus end, specified in inches.
XT,YT	Is the (x,y) coordinate of the line at the to bus end, specified in inches.
SDF	Is the single character L, R, U, or D (for left, right, up, and down, respectively) indicating the direction of the stub drawn at the from bus.
SF	Is the length of the line stub drawn at the from bus. It is specified in inches and may be entered as zero. The default value is 0.125 in.
SDT	Is the single character L, R, U, or D (for left, right, up, and down, respectively) indicating the direction of the stub drawn at the to bus.
ST	Is the length of the line stub drawn at the to bus. It is specified in inches and may be entered as zero. The default value is 0.125 in.

Each branch data record causes a line to be drawn in three line segments: a line from each of the from and to buses where direction and length are characterized by the parameters SDF and SF at the from bus and SDT and ST at the to bus, and a line connecting the end points of these two stub segments. It is recommended that the coordinates be specified such that the length of the line is at least 0.5 in. to accommodate the line annotation.

A value of zero for SF or ST suppresses the drawing of the stub segment at the corresponding bus' (x,y) coordinates as specified on the data record; in this case, the corresponding stub direction code, SDF or SDT, is ignored. This is useful for those situations where it is desirable to have a branch drawn with more than three line segments. To do this, two or more branch data records are entered in the Drawing Coordinate Data File for the branch, with the appropriate coordination of the (x,y) coordinates and use of option field specifications. The following records illustrate an application of this approach:

```
BU,  ,10  1.0  0.5  1.0  1.5
BU,  ,20  4.5  5.5  5.5  5.5
LI,F ,10 20 1  1.  1.  3.  1.   R  .125  L  0.
```


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```
LI,SN,10 20 1  3.  1.  3.  6.   U   0.   D  0.
LI,T ,10 20 1  3.  6.  5.  5.5  R   0.   U  .5
```

ST should also be specified as zero when the R, RN, X, XA, E, EN, EX, RC, or EC option is specified in the option field (and therefore the to bus is not to be drawn). SF and ST *should not* be zero if the corresponding bus is included in the Drawing Coordinate Data File and the corresponding (x,y) coordinates coincide with coordinates on the busbar. Otherwise, the branch annotation will not be printed in the proper position.

Branch coordinates should not be specified such that the two stub endpoints coincide or overlap (i.e., the line connecting the two stubs has zero length or retraces the stub paths).

A pair of tables summarizes the valid entries in the option fields of branch coordinate data records. Table C-1 applies to branches for which bus coordinate data records for both of its endpoint buses are included in the Drawing Coordinate Data File. Its columns correspond to the various levels of line annotation available in activity DRAW:

(1) Print the branch flow at the from bus end (bus IFROM). (2) Print the branch flow at the to bus end (bus ITO). (3) Draw a transformer symbol. (4) Print the tap ratio and, if it is non-zero, the phase shift angle. (5) Print the phase shift angle but not the tap ratio. (6) Draw a series capacitor symbol.

When one of the series capacitor options C, FC, TC, or SC is specified on a branch coordinate data record and the branch is either a transformer, a positive reactance branch or being treated as a zero impedance line, the branch is annotated as if the option blank, F, T, or SN, respectively, had been specified. For the remaining options listed in the table that include transformer annotation, if the branch is not a transformer, the transformer annotation columns are ignored and the from and to buses are annotated as indicated in Table C-1.

Table C.1. Options For Branch Coordinate Data Records

Option	(1)	(2)	(3)	(4)	(5)	(6)
	From Bus Flow	To Bus Flow	Transformer Symbol	Tap Ratio and Phase Shift Angle	Phase Shift Angle	Series Capacitor Symbol
blank	√	√	√	√		
F	√		√	√		
T		√	√	√		
SP			√	√		
N	[a]	*				
A	√	√	√		√	
FA	√		√		√	
TA		√	√		√	
BX	√	√	√			
FX	√		√			
TX		√	√			
SX			√			
BN	√	√				
FN	√					
TN		√				

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Option	(1)	(2)	(3)	(4)	(5)	(6)
	From Bus Flow	To Bus Flow	Transformer Symbol	Tap Ratio and Phase Shift Angle	Phase Shift Angle	Series Capacitor Symbol
S or SN						
C	√	√				√
FC	√					√
TC		√				√
SC						√

[a]If the branch is a transformer, flow is printed on the tapped side; the transformer symbol, tap ratio, and phase shift angle are not printed. If the branch is not a transformer, flow is printed at both ends.

Table C-2 lists those option field values that must be used on branch coordinate data records for open ended radial lines, where the from bus (bus IFROM) has a bus record included in the Drawing Coordinate Data File and the to bus (bus ITO) does not. Its columns correspond to the various levels of line annotation available for such branches:

(1) Print the branch flow at the from bus end (bus IFROM). (2) Draw a transformer symbol. (3) Print the tap ratio and, if it is non-zero, the phase shift angle. (4) Print the phase shift angle but not the tap ratio. (5) Draw a series capacitor symbol.

The to bus number and/or name is printed at the to bus end of the line in accordance with the bus annotation option as selected during the dialog of activity DRAW, as specified on the bus annotation specification record, or as set to default values (refer to Bus Annotation Record).

When one of the series capacitor options RC or EC is specified on a branch coordinate data record and the branch is either a transformer, a positive reactance branch or being treated as a zero impedance line, the branch is annotated as if the option R or EX, respectively, had been specified. For the remaining options listed in the table that include transformer annotation, if the branch is not a transformer, the transformer annotation columns are ignored and the from bus is annotated as indicated in Table C-2.

By default, the North American transformer symbol is used where transformer symbols are drawn. A European transformer symbol may be selected by specifying a transformer annotation specification record (refer to Transformer Annotation Record).

By default, out-of-service branches are drawn with dotted lines and no flows are printed. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

In an impedance diagram, the specified option is ignored. The branch impedance is printed at one end of the line; total line charging and, optionally, a selected line rating are printed at the other end. For branches with one of the option field values listed in Table C-2, the impedance is printed at the from bus end and the charging and rating annotation is suppressed.

Table C.2. Special Options for Open Ended Radial Lines

Option	(1)	(2)	(3)	(4)	(5)
	From Bus Flow	Transformer Symbol	Tap Ratio and Phase Shift Angle	Phase Shift Angle	Series Capacitor Symbol
R or RN	√				

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Option	(1)	(2)	(3)	(4)	(5)
	From Bus Flow	Transformer Symbol	Tap Ratio and Phase Shift Angle	Phase Shift Angle	Series Capac-itor Symbol
X	√	√	√		
XA	√	√		√	
E or EN					
EX		√			
RC	√				√
EC					√

The minimum information necessary in a branch data record is:

```
LI, , IFROM, ITO,, XF, YF, XT, YT, SDF,, SDT
```

Transformer Record - TR

A special two-winding transformer record type is provided to specify transformers when the A or L option is entered on the header record or the suffix AC is specified in invoking activity DRAW. Because these options indicate that the network to be drawn is not necessarily in the working case, the TR record provides the user with the ability to designate that a network branch is a transformer. When a transformer record is entered, activity DRAW assumes that the tapped side of the transformer is the from bus (bus IFROM) and the untapped or impedance side is the to bus (bus ITO). (When executed in *non-accept* mode (refer to Header Record - HD), DRAW accesses the working case to determine whether or not a branch is a transformer, as well as determining each transformer's tapped impedance side relationship.)

The format and options of the transformer data record are exactly the same as those of the branch data record (type LI) described in Branch Record - LI. The TR record behaves as does the LI record when the diagram is coordinated with the working case.

Three-Winding Transformer Record - TT

Three-winding transformers to be included on the one-line diagram are specified in three-winding transformer groups in the Coordinate Data File. Each of three-winding transformer groups consists of:

- One three-winding transformer record TT (see below).
- Followed by branch records LI (see below).
- Followed by one terminator record EN (refer to Terminator Record - EN).

Three-winding transformer records are of the form:

```
TT, O, IFROM, ITO, ILAST, ICKT, X, Y
```

where:

O	Is a valid option (blank, or S); below.
IFROM	Is the Winding 1 bus identifier.
ITO	Is the Winding 2 bus identifier.

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ILAST	Is the Winding 3 bus identifier.
ICKT	Is the three-winding transformer circuit identifier; default value is '1'.
X,Y	Is the (X,Y) coordinate of the reference point of the star point bus, specified in inches.

The valid entries in the option field are as follows:

blank	Print the tap ratio for all windings of the three-winding transformer.
S	Suppress all annotation for the three-winding transformer.

The three-winding transformer record in the three-winding transformer group are followed by branch records LI. Refer to Branch Record - LI for the correct form for branch records. There are two limitations for these records:

- IFROM should be 0 to indicated that the branch is a part of three-winding transformer.
- The valid options are TN and SN only.

The following records illustrate an example of three-winding transformer group:

```
TT,'   ', 160   152     153 1    5.875   4.000
LI,'TN',    0   153 1    5.875   3.875   5.875   3.250 D  0.125
U  0.125
LI,'TN',    0   152 1    6.000   4.125   6.000   4.750 U  0.125
D  0.125
LI,'TN',    0   160 1    5.750   4.125   4.250   4.250 L  1.500
D  0.125
EN
```

Two-Terminal dc Transmission Line Record - DC

Two-terminal dc transmission lines to be included on the one-line diagram are specified in dc transmission line data records in the Drawing Coordinate Data File. The dc line data records are of the form:

```
DC, O, NDC, X1, Y1, X2, Y2, SDF, SF, SDT, ST
```

where:

O	Is a valid option (blank, N, F, FN, T, TN, or S); below.
NDC	Is the two-terminal dc line number.
X1,Y1	Is the (x,y) coordinate of the dc line at the rectifier converter station bus, specified in inches.
X2,Y2	Is the (x,y) coordinate of the dc line at the inverter converter station bus, specified in inches.
SDF	Is the single character L, R, U, or D (for left, right, up, and down, respectively) indicating the direction of the stub drawn at the rectifier bus.

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SF	Is the length of the line stub drawn at the rectifier bus. It is specified in inches and may be entered as zero. The default value is 0.125 in.
SDT	Is the single character L, R, U, or D (for left, right, up, and down, respectively) indicating the direction of the stub drawn at the inverter bus.
ST	Is the length of the line stub drawn at the inverter bus. It is specified in inches and may be entered as zero. The default value is 0.125 in.

Each two-terminal dc line data record causes a line to be drawn in three line segments in a similar manner in which ac branches are drawn. Refer to Branch Record - LI for details on the use of the data parameters SDF, SF, SDT, and ST.

As with ac branches, it is not permissible to specify a dc line's coordinates and stub lengths such that the two stub endpoints coincide or overlap.

The valid entries in the option field are as follows:

blank	Print flow at each end. The converter tap settings and dc symbols are drawn.
N	No printing of dc symbols or converter tap settings. Flows are printed at both ends.
F	Print flow at rectifier bus end only. The converter tap setting and dc symbol are drawn.
FN	No printing of dc symbol or converter tap setting. Print flow at rectifier bus end only.
T	Print flow at inverter bus end only. The converter tap setting and dc symbol are drawn.
TN	No printing of dc symbol or converter tap setting. Print flow at inverter bus end only.
S	Suppress all printing on the dc line.

The power flowing into a dc line is the apparent ac system complex load as n at the converter bus (i.e., PACR and QACR at the rectifier end, and PACI and QACI at the inverter end).

By default, blocked two-terminal dc lines are drawn with dotted lines and no flows are printed; the dc symbols are drawn. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

For an impedance diagram, the specified option is ignored and the commutating reactances, in ohms, are printed at each end of the line. The dc line resistance, in ohms, is printed above the commutating reactance at the rectifier end of the line, and the minimum alpha and gamma angles are printed at the corresponding dc symbols.

If the A or L option is specified on the header record, or AC is specified in invoking activity DRAW using line mode, the working case is ignored and the dc line coordinates are not checked against the coordinates of the appropriate buses because the buses involved are normally determined from the working case and not from the dc data record.

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The minimum information necessary in a dc line data record is:

```
DC, , NDC, X1, Y1, X2, Y2, SDF,, SDT
```

Multi-Terminal dc Transmission Line Record - MD

Multi-terminal dc transmission line converters to be included on the one-line diagram are specified in multi-terminal dc line records in the Drawing Coordinate Data File. Multi-terminal dc data records are of the form:

```
MD, NDC, IB, X1, Y1, X2, Y2
```

where:

NDC	Is the multi-terminal dc line number.
IB	Is the bus identifier. Bus IB must be a converter station bus of multi-terminal dc line NDC.
X1,Y1	Is the (x,y) coordinate of the line on the busbar, specified in inches.
X2,Y2	Is the (x,y) coordinate of the end point of the straight line segment, specified in inches.

The length of the line specified by the coordinates (X1,Y1) and (X2,Y2) should be at least 1 in. The line specified by the coordinates *must* be either horizontal (Y1 = Y2) or vertical (X1 = X2).

At the end of the line connected to the bus, the power flowing into the dc line (i.e., the apparent ac system complex load as n at the converter bus) is printed. The dc symbol is drawn at the midpoint of the line, and the converter alpha or gamma angle, as appropriate, and tap position are printed at the end of the line segment.

By default, at each converter of a blocked multi-terminal dc line for which an MD record is specified, the multi-terminal dc line is drawn with a dotted line and the flow, angle and tap are shown as zero; the dc symbol is drawn. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

Multi-terminal dc line converters are annotated identically in impedance and flow diagrams.

While the dc link connections within multi-terminal dc lines *are not* handled automatically by activity DRAW, the ends of these line segments may be appropriately connected by using straight line records (refer to Straight Line Record - SL).

Load Records - LO, LP, LC, and LY

Bus loads to be included on the one-line diagram are specified in load data records in the Drawing Coordinate Data File. Load data records are of the form:

```
La, O, IB, X1, Y1, X2, Y2
```

where:

La	Is either LO, LP, LC, or LY and defines the load characteristic; below.

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O	Is blank or the load identifier; below.
IB	Is the bus identifier.
X1,Y1	Is the (x,y) coordinate of the line on the busbar, specified in inches.
X2,Y2	Is the (x,y) coordinate of the end point of the straight line segment, specified in inches. This is the end at which the arrow head is drawn.

The length of the line specified by the coordinates (X1,Y1) and (X2,Y2) should be at least 0.25 in.; a longer length may be needed depending on the magnitude of the load and the character size specified on the HD record. The line specified by the coordinates *must* be either horizontal (Y1 = Y2) or vertical (X1 = X2).

The LP, LC, and LY data records are used to display the constant power, constant current, and constant admittance load characteristics, respectively, of the load at bus IB for which the identifier is contained in the option field of the data record. If the option field contains a blank, the sum of the appropriate load characteristic for all in-service loads at bus IB is printed.

The LO data record is used to display the sum of the three load characteristics for the load at bus IB for which the identifier is contained in the option field. If the option field contains a blank, the total in-service load at bus IB is printed. The specified load identifier is printed inside of the load symbol.

By default, out-of-service loads are drawn with dotted lines and annotation values are shown as zero. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

In an impedance diagram, nominal loads are printed (i.e., at one pu voltage).

Shunt Record - SH

Bus shunts to be included on the one-line diagram are specified in shunt data records in the Drawing Coordinate Data File. Shunt data records are of the form:

```
SH, O, IB, X1, Y1, X2, Y2
```

where:

O	Is a valid option (blank, L, S or R); below.
IB	Is the bus identifier.
X1,Y1	Is the (x,y) coordinate of the line on the busbar, specified in inches.
X2,Y2	Is the (x,y) coordinate of the end point of the straight line segment, specified in inches. This is the end at which the capacitor or reactor symbol is drawn.

The length of the line specified by the coordinates (X1,Y1) and (X2,Y2) should be at least 0.25 in.; a longer length may be needed depending on the magnitude of the shunt and the character size specified on the HD record. The line specified by the coordinates *must* be either horizontal (Y1 = Y2) or vertical (X1 = X2).

The valid entries in the option field are as follows:

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blank	The shunt printed is the sum of the fixed shunt and switched shunt devices.
L	The shunt printed is the fixed shunt (refer to Bus Data).
S	The shunt printed is the switched shunt (refer to Switched Shunt Data).
R	When the reactive component of the shunt is zero or when in *accept* mode, a reactor symbol is drawn; under these conditions when the R option is not specified, a capacitor symbol is drawn.

The R option may be used in combination with the L or S options (i.e., RL or RS).

Constant admittance load *is not* included as bus shunt; these are drawn as loads (refer to Load Records - LO, LP, LC, and LY).

By default, shunts at Type 4 buses are drawn with dotted lines and annotation values are shown as zero. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

On a flow diagram, shunt MW and Mvar are printed; on an impedance diagram, nominal shunt values are printed. The sign of the reactive component of shunt admittance determines whether a capacitor or reactor symbol is drawn. When this quantity is zero, a reactor symbol is drawn if the R option is specified; otherwise, a capacitor symbol is drawn.

Generator Record - GE

Generators to be included on the one-line diagram are specified in generator data records in the Drawing Coordinate Data File. Generator data records are of the form:

```
GE, O, IB, X1, Y1, X2, Y2
```

where:

O	Is blank or the machine identifier; below.
IB	Is the bus identifier.
X1,Y1	Is the (x,y) coordinate of the line on the busbar, specified in inches.
X2,Y2	Is the (x,y) coordinate of the end point of the straight line segment, specified in inches. This is the end at which the circle is drawn.

The length of the line specified by the coordinates (X1,Y1) and (X2,Y2) should be at least 0.25 in.; a longer length may be needed depending on the magnitude of the generation and the character size specified on the HD record. The line specified by the coordinates must be either horizontal (Y1 = Y2) or vertical (X1 = X2).

By default, out-of-service plants and machines are drawn with dotted lines and annotation values are shown as zero. This default manner of drawing out-of-service equipment may be overridden by specifying a CO record (refer to Color Specification Record - CO).

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On a flow diagram, generator MW and Mvar are printed; on an impedance diagram, the generator reactive power limits replace the Mvar output on the drawing.

When the option field is left blank, plant totals are printed. Otherwise, the option field specifies the identifier of a machine in a multimachine plant for which the data is to be printed. The machine identifier is printed inside the generator circular symbol. Thus, if several machines at a bus are to be drawn, one generator data record in the Drawing Coordinate Data File is required for each one.

To specify a machine with a machine identifier of B, the option field must be specified as BB (refer to comments in Drawing Coordinate Data File Contents above).

Facts Device Record - FD

FACTS devices to be included on the one-line diagram are specified in FACTS device data records in the Drawing Coordinate Data File. FACTS device data records are of the form:

```
FD, O, FACTID, X1, Y1, X2, Y2, STBDIR, STBLTH,
DEVTYP
```

where:

O	Is a valid option (blank, B, N, S, T); below.
FACTID	Is the facts device identifier.
X1,Y1	Is the (x,y) coordinate of the line connection at the sending bus, specified in inches. For UPFC and SSSC devices, this is the coordinate point associated with the series element line connection that connects the send and terminal buses. For a STATCON device, this is the coordinate point associated with the shunt device connection line.
X2,Y2	Is the (x,y) coordinate endpoint of the line connection, specified in inches. For UPFC and SSSC devices, this coordinate is located on the terminal bus and terminates the line connection between the send and terminal buses. For STATCON devices, this is the coordinate point associated with the end of the shunt device stub line. The line specified by the coordinates must be either horizontal (Y1 = Y2) or vertical (X1 = X2).
STBDIR	Is the single-character L, R, U, or D (for left, right, up, and down, respectively) indicating the direction of the shunt stub drawing relative to the send bus (X1,Y1) coordinate. For SSSC and STATCON FACTS devices, this field may be blank and is ignored.
	For horizontal send bus bars, only L,R values are allowed with L assigned as the default. For vertical send bus bars, only U,D values are allowed with D assigned as the default. STBDIR is ignored for all SSSC and STATCON devices.

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STBLTH	Is the length of the shunt stub offset relative to the (X1,Y1) coordinate of the send bus. It is specified in inches with 0.125 in. as default. For SSSC and STATCON FACTS devices, this field may be blank and is ignored. For UPFC FACTS devices, the shunt stub line is drawn starting at the point defined by the shunt stub drawing direction and the shunt stub offset length relative to (X1,Y1). The shunt stub line is drawn with a length of 0.2 in. The shunt FACTS device rectangle and ground symbols are drawn at the end of the shunt stub line.
DEVTYP	Is the FACTS device type (1-UPFC device, 2-SSSC device, 3-STATCON). The default type is 1. This field may be blank and *is ignored* when DRAW is executed in *non-accept* mode.

The valid entries in the option field are as follows:

blank, B	Display flows on series, bridge, and shunt elements.
N	Do not display any flows.
S	Display flows only at send bus and shunt element end.
T	Display flow only at terminal bus.

For UPFC and SSSC FACTS devices, a circle with an arrow depicting power flow direction is drawn midway on the line connecting the sending and terminal buses. For UPFC FACTS devices, two lines are drawn connecting the shunt rectangle and the circle. By default, out-of-service FACTS devices are drawn with dotted lines, and annotation values are blanked.

Straight Line Record - SL

Arbitrarily located straight lines to be included on the one-line diagram are specified in straight line data records in the Drawing Coordinate Data File. Straight line records are of the form:

```
SL, O, X1, Y1, X2, Y2
```

where:

O	Is a valid option (blank, D, H, n, Dn, or Hn); below.
X1,Y1	Is the (x,y) coordinate of one end of the line, specified in inches.
X2,Y2	Is the (x,y) coordinate of the other end of the line segment, specified in inches.

Straight lines may be used, for example, to designate area boundaries, or to show the dc link connections within multi-terminal dc lines.

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The valid entries in the option field are as follows:

blank	Draw solid straight line segment.
D	Draw dotted straight line segment.
H	Draw dashed straight line segment.

By combining any of these options with the numbers one through eight (e.g., 2, D5), a line width, pen number or color is designated for the drawing of the straight line. It is the user's responsibility to ensure that the designated number is appropriate for the graphics output device being used (e.g., 6 would be invalid for a pen plotter with only four pens). If the numeric option is invalid, a device dependent default is used.

Text Specification Record - TX

The text specification record allows strings of characters to be written anywhere on the one-line diagram. Text specification records are of the form:

```
TX, O, X, Y, TSIZ, TANG, '...text...'
```

where:

O	Is a valid option (blank, or 1 through 8; blank and 1 are equivalent); below.
X,Y	Is the (x,y) coordinate of the first character of the text string, specified in inches.
TSIZ	Is the character height specified in inches. TSIZ must be between 0.05 and 1.5 in.
TANG	Is the angle of rotation of the character string, specified in degrees between 0 and 360; 0 degrees is the angle at which the title lines are written on the diagram.
text	Is the character string. It must be enclosed in single quotes and is limited to 40 characters.

The option field is used to designate a line width, pen number or color for the drawing of the text. It is the user's responsibility to ensure that the designated number is appropriate for the graphics output device being used (e.g., 6 would be invalid for a pen plotter with only four pens). If the numeric option is invalid, a device dependent default is used.

Clip Specification Record - CL

The clip specification record defines the size of the clipping window where the user may zoom in or out on a selected portion of the diagram. The clip specification record is of the form:

```
CL, , X, Y
```

where:

X,Y	Define the size of the window along the x- and y-axes, respectively, and are specified in inches.

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The clipping window defines the portion of the diagram, which is drawn in pages when zooming is enabled.

There are no options recognized on the clip specification data record.

Grid Record - GR

The grid record enables the overlaying of a coordinate grid upon the one-line diagram. The grid record is of the form:

```
GR, , GRDRES
```

where GRDRES is the grid resolution specified in inches. Horizontal and vertical lines composing the grid are spaced GRDRES inches apart. GRDRES is restricted to the range 0.1 through 1.0 in. and has a default value of 0.125 in. If it is less than 0.1 in., lines are drawn 1/10 of an inch apart; if it is greater than 1.0, lines are drawn 1 in. apart.

There are no options recognized on the grid data record.

Color Specification Record - CO

The color specification record allows the user to override the default line width, pen number, or color conventions for denoting voltage levels, heavily loaded and out-of-service branches, and buses where voltages are out-of-limits. The color specification record is of the form:

```
CO, , EQOUT, LNOVR, BSUND, BSOVR, V1, V2, V3, V4
```

where:

EQOUT	Defines the line width, pen number, or color to be used in drawing out-of-service equipment. EQOUT must be between zero and eight. If the default of zero is specified for EQOUT, out-of-service equipment is drawn with dotted lines.
LNOVR	Defines the line width, pen number, or color to be used in drawing heavily loaded lines. LNOVR must be between zero and eight. If the default of zero is specified for LNOVR, overloaded lines are drawn with dashed lines.
BSUND	Defines the line width, pen number, or color to be used in drawing buses where voltage magnitudes are below a specified threshold. BSUND must be between zero and eight. If the default of zero is specified for BSUND, undervoltage buses are drawn with a dashed line on the bottom portion of the busbar.
BSOVR	Defines the line width, pen number, or color to be used in drawing buses where voltage magnitudes are above a specified threshold. BSOVR must be between zero and eight. If the default of zero is specified for BSOVR, overvoltage buses are drawn with a dotted line on the top portion of the busbar.

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V1,V2,V3,V4	Define the line width, pen number, or color to be used in denoting voltage levels (refer to Voltage Level Record - VO). The values specified must be between zero and eight. If not entered, V1 through V4 default to one through four, respectively.

On color CRTs, the use of color to designate the conditions specified by EQOUT, LNOVR, BSUND, and/or BSOVR takes precedence over the voltage level color if that option is also enabled.

It is the user's responsibility to ensure that the numbers designated on the CO record are appropriate for the graphics output device being used (e.g., 6 would be invalid for a pen plotter with only four pens). If any value is invalid, a device dependent default is used.

Symbol Record - SY

The symbol record allows symbols from the graphics character set to be drawn anywhere on the one-line diagram. The symbol record is of the form:

```
SY, O, ISYM, X, Y, SIZE, ANGLE
```

where:

O	Is a valid option (blank, or 1 through 8; blank and 1 are equivalent); below.
ISYM	Is the symbol number assigned to the symbol to be drawn. Figure C-1 shows the symbols contained in the graphics character set. ISYM must be between 1 and 155.
X,Y	Is the (x,y) coordinate of the reference point of the symbol, specified in inches. The intersections of the dotted lines in Figure C-1 show the reference points for the symbols contained in the graphics character set.
SIZE	Is the symbol height specified in inches. SIZE must be between 0.05 and 1.5 in.
ANGLE	Is the angle of rotation of the symbol, specified in degrees between 0 and 360; 0 degrees is the angle at which the title lines are written on the diagram. The symbols in Figure C-1 are drawn at zero degrees.

The option field is used to designate a line width, pen number, or color for the drawing of the symbol. It is the user's responsibility to ensure that the designated number is appropriate for the graphics output device being used (e.g., 6 would be invalid for a pen plotter with only four pens). If the numeric option is invalid, a device dependent default is used.

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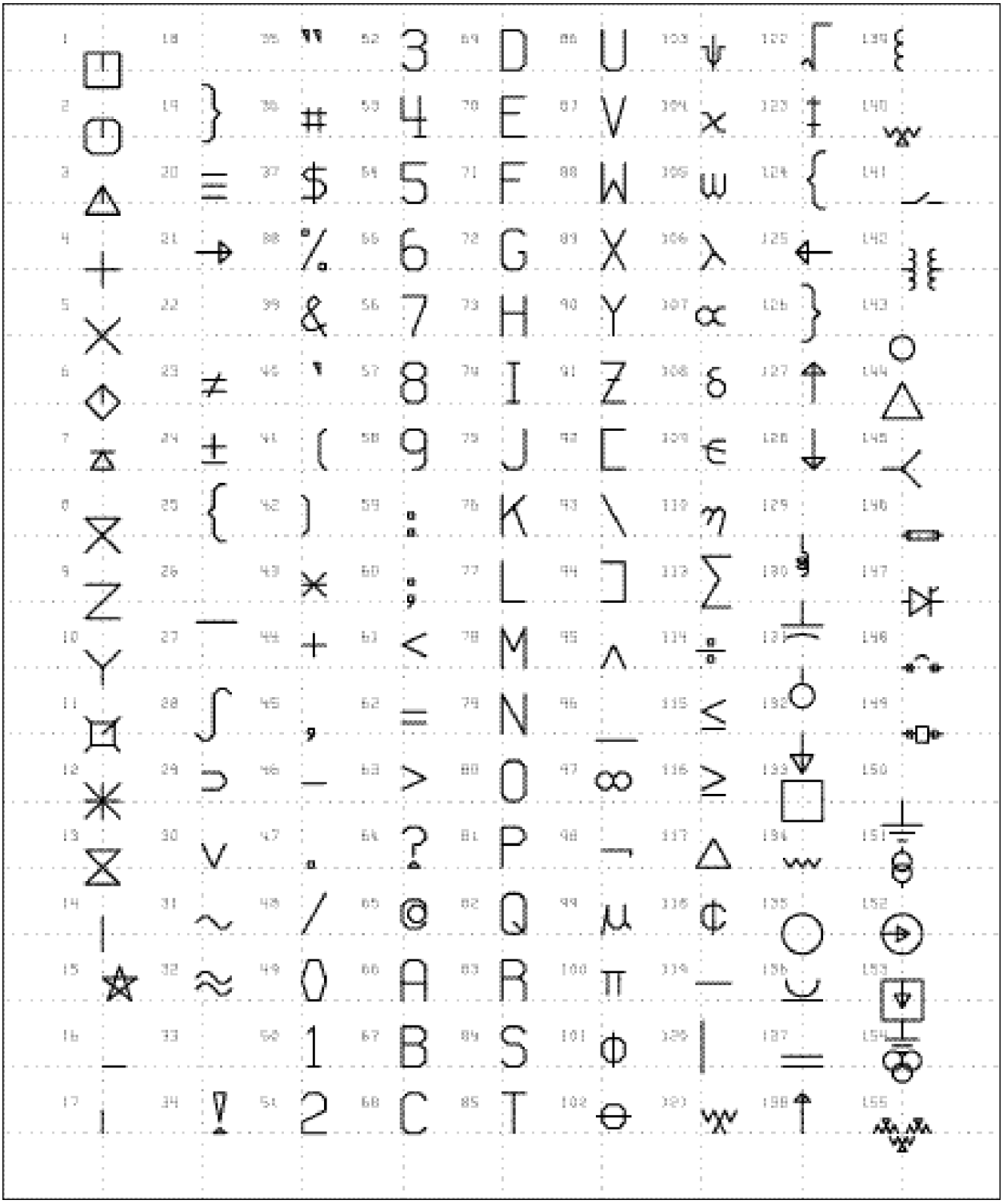

Figure C.1. Graphics Character Set

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Rating Specification Record - RA

The branch rating specification record specifies the rating set and percentage loading threshold to be used in identifying heavily loaded branches on the one-line diagram. Using this option, all branches to be drawn that have a non-zero value for the selected rating and where current loading at either end exceeds the specified percentage of the selected rating are flagged by being drawn with dashed lines. This default manner of drawing heavily loaded lines may be overridden by specifying a CO record (refer to Color Specification Record - CO).

If either the line rating *range checking* option or one of the *% RATING* line flow annotation options is selected, the rating set and percentage loading threshold specified on the RA record are taken as the default values.

The rating specification record is of the form:

```
RA, O, RAT, PCTRAT
```

where:

O	Is a valid option (blank or F; below).
RAT	Is one of the strings RATE, RATEA, RATEB, or RATEC. If RATE is specified, the rating set established as the default rating set program option is selected (refer to Section 3.3.3 Program Run-Time Option Settings and activity OPTN). Otherwise, the designated rating set is selected.
PCTRAT	Is the percentage loading threshold. The default value is 100%.

The valid entries in the option field are as follows:

blank	The line rating *range checking* option is initially disabled and may be activated in the interactive dialog of activity DRAW.
F	Force the line rating *range checking* option on the one-line diagram even if it is not specifically selected in the interactive dialog.

Voltage Limits Specification Record - VL

The bus voltage limits specification record specifies the voltage range to be used in identifying buses with high or low voltages. Using this option, buses with a voltage below the specified minimum have the lower portion of their bus box drawn with a dotted line. Buses with a voltage above the specified maximum have the upper portion of their bus box drawn with a dashed line. This default manner of drawing buses outside of voltage limits may be overridden by specifying a CO record (refer to Color Specification Record - CO).

If the voltage limits *range checking* option is selected during the interactive dialog of activity DRAW, the voltage limits specified on the VL record are taken as the default limits.

The voltage limits specification record is of the form:

```
VL, O, VMIN, VMAX
```


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where:

O	Is a valid option (blank or F; below).
VMIN	Is the low-voltage threshold limit specified in per unit.
VMAX	Is the high-voltage threshold limit specified in per unit.

The valid entries in the option field are as follows:

blank	The voltage limits *range checking* option is initially disabled and may be activated in the interactive dialog of activity DRAW.
F	Force the voltage limits *range checking* option on the one-line diagram even if it is not specifically selected in the interactive dialog.

Diagram Annotation Record - AN

The one-line diagram annotation specification records specify branch, bus, and equipment annotation options. Selections specified on AN records are taken as the default selections during the interactive dialog of activity DRAW.

As shown in the following paragraphs, the option field on the AN record defines the diagram annotation option being specified.

Line Annotation Record

The line annotation specification record is of the form:

```
AN, LI, ANNOPT, SGNOPT
```

The valid entries in the ANNOPT field are as follows:

MW-MVAR	Display MW and Mvar line flows; this is the default line annotation option.
MVA	Display MVA line flows.
MVA-PCRATE	Display MVA line flows and current loading as a percentage of rating.
MW-PCRATE	Display MW line flows and current loading as a percentage of rating.
AMP-PCRATE	Display current loadings in amps and as a percentage of rating.
AMP-PUCUR	Display current loadings in amps and per unit.
Z	Display line impedances and charging.
Z-RATE	Display line impedances, charging, and the selected rating.
NONE	Suppress branch annotation; transformer and dc symbols that would be drawn using the other branch annotation options are still drawn.

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The valid entries in the SGNOPT field are as follows:

SIGN	Flow direction is indicated using the standard PSS®E sign convention: positive for power leaving the bus, and negative for power arriving at the bus.
ARROW	Flows are printed unsigned and arrows in front of the flow values indicate flow direction.

In calculating percentage loading, current expressed as MVA (refer to Non-Transformer Branch Data and activity RATE) is calculated at the appropriate end(s) of the branch and is used in calculating the loading percentage. The rating set used is as defined on any RA record (refer to Rating Specification Record - RA), as defined by the rating set program option setting (refer to activity OPTN), or as overridden in the program dialog.

When current in amps is selected, the current in per unit is printed for any branch for which the required bus base voltage has not been entered (i.e., if it is zero).

On an impedance diagram, unless NONE is specified for ANNOPT, the ANNOPT entry is ignored and branches are annotated as described in Branch Record - LI and Two-Terminal dc Transmission Line Record - DC.

Bus Annotation Record

The bus annotation specification record is of the form:

```
AN, BU, NAMOPT, VLTOPT
```

The valid entries in the NAMOPT field are as follows:

NAME	Display the twelve-character bus name. If a bus has a blank name, no identifier is printed for the bus. This is the default bus identification option when the *names* output option is in effect.
NUMBER	Display the bus number. This is the default bus identification option when the *numbers* output option is in effect.
BOTH	Display the bus name above the bus number.
NONE	Suppress bus identification annotation.

The selection of bus names, which are up to twelve characters, may require adjustment of coordinates in the Drawing Coordinate Data File if the diagram was set up using the *numbers* output option (bus numbers are at most six digits).

The valid entries in the VLTOPT field are as follows:

VPU	Display bus voltage magnitude in per unit. This is the default bus voltage annotation option.
VPU-ANGLE	Display bus voltage magnitude in per unit and phase angle in degrees.
VPU-KV	Display bus voltage magnitude in per unit and kV.
KV	Display bus voltage magnitude in kV.

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KV-ANGLE	Display bus voltage magnitude in kV and phase angle in degrees.
ANGLE	Display bus voltage phase angle in degrees.
NONE	Suppress bus voltage annotation.

On a flow diagram, when voltage in kV is selected, the voltage in per unit is printed for any bus where base voltage has not been entered (i.e., if it is zero).

On a flow diagram, voltage and/or angle are printed as zero for any bus where type code is four (i.e., for disconnected buses).

On an impedance diagram, for any VLTOPT specification except NONE, bus base voltages in kV are printed.

Equipment Annotation Record

The equipment annotation specification record specifies the annotation option for load, generator, shunt, and multi-terminal dc line elements and is of the form:

```
AN, EQ, ANNOPT
```

The valid entries in the ANNOPT field are as follows:

MW-MVAR	Display MW and Mvar values. This is the default equipment annotation option.
MVA	Display MVA values for load, generator and shunt elements. Multi-terminal dc lines have MW and Mvar displayed.
NONE	Suppress equipment annotation.

On an impedance diagram, unless NONE is specified for ANNOPT, the ANNOPT entry is ignored and equipment is annotated as described in Multi-Terminal dc Transmission Line Record - MD, Load Records - LO, LP, LC, and LY, Shunt Record - SH, and Generator Record - GE.

Activity Annotation Record

Activity annotation specification records define the branch annotation conventions to be used when the Drawing Coordinate Data File is specified to activity GDIF. These records are of the following form:

```
AN, AC, ACTNAM, ACTOPT
```

The valid entries in the ACTNAM field are GDIF and SCGR. An activity annotation record is ignored unless its ACTNAM field matches the name of the activity being executed.

The valid entries in the ACTOPT field are as follows:

IGNORE	Ignore the option field on each branch record and display data at both ends of the branch. Branches where LI or TR coordinate data records contain one of the *radial line* option field entries (see Table C-2) are annotated at the from bus end only. The use of multiple branch records to define the branch path is detected using this option and such branches are annotated on

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	their first and last line segments. This is the default convention used by activity SCGR.
HONOR	Annotate each branch according to the option field on its branch record. This is the default convention used by activity GDIF.

Transformer Annotation Record

The transformer annotation specification record is of the form:

```
AN, TR, ANNOPT
```

The valid entries in the ANNOPT field are as follows:

NA-SYM	Use the North American transformer symbol (symbol number 140 of Figure C-1 for two-winding transformer and symbol number 155 of Figure C-1 for three-winding transformer) and display only primary winding for two-winding transformer; this is the default transformer annotation option.
EU-SYM	Use the European transformer symbol (symbol number 151 of Figure C-1 for two-winding transformer and symbol number 154 of Figure C-1 for three-winding transformer) and display only primary winding for two-winding transformer.
NA-WN2	Use the North American transformer symbol (symbol number 140 of Figure C-1 for two-winding transformer and symbol number 155 of Figure C-1 for three-winding transformer) and display both windings of two-winding transformer.
EU-WN2	Use the European transformer symbol (symbol number 151 of Figure C-1 for two-winding transformer and symbol number 154 of Figure C-1 for three-winding transformer) and display both windings of two-winding transformer.

Summation Block Structure

The summation block structure provides for:

- Initializing a summing variable; usually, the initial value of the summing variable is zero, but a non-zero value may be specified.
- Specifying individual ac branches, generator buses, and/or load buses where active power components of line flow, generation, and load, respectively, are added to or subtracted from the summing variable.
- Specifying a character string to be printed in front of the calculated value of the summing variable anywhere on the one-line diagram.

Each summation block structure consists of:

- One summation record, SUM or SU (see below).

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- Optionally followed by one or more summation participation records GN, BL, and LF (see below).
- Followed by one terminator record, EN (refer to Terminator Record - EN).

Summation records are of the form:

```
SU, O, '...text...', X, Y, TSIZ, TANG, CNST
```

where:

O	Is a valid option (blank, or 1 through 8; blank and 1 are equivalent); below.
text	Is the character string to be displayed in front of the summing variable's value. It must be enclosed in single quotes and is limited to 40 characters.
X,Y	Is the (x,y) coordinate of the first character of the text string, specified in inches.
TSIZ	Is the character height specified in inches. TSIZ must be between 0.05 and 1.5 in.
TANG	Is the angle of rotation of the character string, specified in degrees between 0 and 360; 0 degrees is the angle at which the title lines are written on the diagram.
CNST	Is the initial value to be assigned to the summing variable. The default value is zero.

The option field is used to designate a line width, pen number, or color for the drawing of the text. It is the user's responsibility to ensure that the designated number is appropriate for the graphics output device being used (e.g., 6 would be invalid for a pen plotter with only four pens). If the numeric option is invalid, a device dependent default is used.

The summation record in the summation block structure may be followed by summation participation records. These define the generator buses, load buses, and ac branches where active power components are to be included in the summation. The summation participation records may be entered in any order and are of the following forms:

```
GN, O, IB                  (for bus generation)
BL, O, IB                  (for bus load)
LF, O, IFR, ITO, ID, DIR   (for ac branch flow)
```

where:

O	Is a valid option (blank, + or -). Blank or + indicates that the active power of the corresponding network element is to be added to the summing variable; - indicates that it is to be subtracted from the summing variable.
IB	Is a bus identifier.
IFR	Is the from bus identifier.
ITO	Is the to bus identifier.

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ICKT	Is the circuit or multi-section line grouping identifier; default value is '1'.
DIR	Is either F or T and indicates flow direction. Line flow is always calculated at the bus IFR end of the line; if DIR is F, flow is in the bus IFR to bus ITO direction, and if it is T, flow is in the bus ITO to bus IFR direction. DIR is F by default.

If an element specified on a summation participation record is out-of-service, it does not contribute to the summation.

The following records illustrate an example of the summation block structure:

```
SUM, ,'AREA 1 = ', 4.8  2.4  0.12  45.  0.
 GN,+,101
 GN,+,102
 LF,-,151 201 1 F
 LF,-,152 202 1 F
 BL,-,151
 EN
```

C.5. BMAT

Activity BMAT and the PTI network loss coefficient program, BMATRIX are no longer supplied with PSS®E.

C.6. MENU

Activity MENU is no longer supplied with PSS®E.

C.7. WORK

At PSS®E-28, the use of a power flow working file by PSS®E was eliminated. Therefore, the working case retrieval activity WORK no longer exists.

C.8. XLIS

The windows-based power flow data modification activity XLIS is no longer supplied with PSS®E. It has been replaced by similar functions in the Spreadsheet View of the Graphical User Interface (GUI). Chapter 2 of the *PSS®E GUI Users Guide* describes the Spreadsheet View.

C.8.1. Equipment Status Changes and the Spreadsheet View

In addition to modifying parametric data, the service status of equipment in the working case may be modified from the Spreadsheet View.

Associated with each ac series branch is a status flag which is initially set to one (for in-service) or zero (for out-of-service) as specified on the branch data record or in the transformer record block when a new branch is introduced into the working case with activities READ, TREA, or RDCH. The status of a branch may be modified by changing this flag. Out-of-service branches and their data remain with the case as it is saved

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and retrieved with activities SAVE and CASE, and are simply ignored during the execution of other PSS®E activities such as SOLV or POUT.

If branch status changes isolate a bus, the user *must* change the bus type code to four. Activities TREE and OUTS and the power flow solution network connectivity check option (see Section 3.3.3 Program Run-Time Option Settings and activity SOLV) aid the user in identifying isolated buses and islands.

When the specified branch is a member of a multi-section line grouping (see Multi-Section Line Grouping Data), the same status change is automatically implemented in the working case for all branches in the multi-section line. In addition, the bus type codes of the dummy buses are changed to:

4	When the branch is being removed from service.
1	If the branch is being placed in-service and the bus has no generator data associated with it.
2	If the branch is being placed in-service and the bus has generator data associated with it. In this case, the individual machine status flags, which are not changed in this status processing of multi-section lines, determine which machines are returned to service.

The bus type codes of the endpoint buses of the multi-section line grouping are not changed in the status processing of a multi-section line. As with single section lines, it is the user's responsibility to ensure that branch statuses and bus type codes are coordinated.

When changing the status of a three-winding transformer, status values of 0 through 4 are permitted (see Transformer Data); the status flags of the three two-winding transformers which comprise the three-winding transformer are set to the appropriate values. As with multi-section line groupings and two terminal branches, the bus type codes of the endpoint buses of the three-winding transformer are not changed in the branch status processing.

FACTS devices may be blocked in a similar manner. A value of zero for MODE, the control mode flag, indicates a blocked FACTS device (see FACTS Device Data).

The status of individual machines at a generator bus may be modified by changing the machine status; plant total power output and power limits are automatically updated following a change of machine power or status. A plant may be taken out-of-service *either* by changing the status of all units at the bus to zero *or* by changing the bus type code to one or four (this is the recommended approach because power flow solution activities will run slightly faster). Conversely, for a machine to be treated as in-service, *both* its status must have the value one *and* the bus type code must be either two or three.

Similarly, the status of an individual load at a bus may be modified by changing its load status, and the status of an individual fixed bus shunt may be modified by changing its shunt status.

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Appendix D

Appendix D - Command Line Options

Exe-cutable	Command Line Options								
PSSE		-inifile	-embed	-rspfile / -pyfile	-buses -pyver	-argstr	{startup} -resetwrk		
AcccBr-wsGrid		-inifile							
Conver-tRaw		-inifile		-inpfile / {infile} {inver} {outfile} {outver}	-nolog				
CreateR-aw		-inifile		-inpfile / {infile} {outfile} {outver}	-nolog				
LineProp									
Scenari-oMgrCn-vt									
WECCLF		-inifile							
PSSPLT		-inifile	-gnikool	-inpdev					
IMD		-inifile	-gnikool						
ASPNLF		-inifile	-nogui	-inpdev					
CMDYRE		-inifile	-nogui	-inpdev					
CNVxx		-inifile	-nogui	-inpdev					
CNVDRW		-inifile	-nogui	-inpdev					
CNVRSQ		-inifile	-nogui	-inpdev					
COMDAT		-inifile	-nogui	-inpdev					
COMFOR		-inifile	-nogui	-inpdev					
DBUILD		-inifile	-nogui	-inpdev					
RAWxx		-inifile	-nogui	-inpdev					
LSYSAN		-inifile		-inpdev					
PLINC		-inifile		-inpdev					
PSAP4		-inifile		-inpdev					
PsseCmd		-inifile		-inpdev	-buses -pyver				
TMLC		-inifile							

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Exe-cutable	Command Line Options								
VCV		-inifile							
IPLAN	<file-name>	-inifile	-loop	-comp / -run / -crun	-[no]inc	-[no]arg	-[no]list	-[no]stat	-search
WECCDS		-inifile	(-input -dyre -rwm -rdch)	-msg	-motor	-gnet	-gwind	-bfreq	
psseinit		-inifile			-buses				

Meaning of symbols in table	
-name	a command line option specified as "name", followed by a value
-name	named option with no value
/	set of options, only one allowed
[]	option modifier
<>	positional option (no option name)
{}	unnamed option (order matters, but not position)
()	all required if any are used

Meaning of command line options		
options that take arguments (e.g. -inpdev filename):		
	-inifile	specify an ini file (i.e. override default)
	-inpdev	specify a response file and arguments
	-rspfile	specify a response file
	-pyfile	specify a python file
	-argstr	specify arguments for -rspfile or -pyfile
	-inpfile	specify batch file; a series of sets of command line options
	-gnikool	set GUI mode (on or off)
	-buses	specify number of buses (i.e. override default)
Named options that do not take arguments (e.g. -embed)		
	-nogui	suppress GUI; run from console
	-embed	suppress GUI and run in background while batch commands are supplied
	-nolog	suppress log file (log file generated by default)
	-resetwrk	reset the program workspace to system defaults

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Meaning of command line options		
Unnamed arguments (i.e. specify values only)		
	{startup}	one of the PSS®E startup files, see list below
	{infile}	input filename
	{outfile}	output file name
	{inver}	input file version
	{outver}	output file version
Notes:	see IPLAN manual for those options that are unique to IPLAN	
	the options that are unique to WECCDS correspond to the interactive dialog	
	-pyver	specify the release of the python interpreter to be loaded

PSS®E startup files	
These files either define a case or can be read in without a case. They are recognized by file extension:	
*.pssxml, *.zip	Scenario file
*.sav	Saved case file
*.raw	Power Flow raw data file
*.snp, *.srs	Snapshot file
*.py, *.idv, *.irf, *.pse, *.psa, *.pyc, *.pyw	Automation file
*.sld, *.sldxml, *.slt	Slider file
*.out	Channel output file
*.evs	Event study file
*.dll	Dynamic link library
*.cfg	Comtrade file
*.drw	Coordinate data file
*.xml	PSSE XML configuration file

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